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5	Brakes
6	Clutch
7	Cooling
8A	Audio/Video
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8Q	Vehicle Theft Security
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INTRODUCTION

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VEHICLE IDENTIFICATION NUMBER

DESCRIPTION

The Vehicle Identification Number (VIN) plate is located on the lower windshield fence near the left A-pillar. The VIN contains 17 characters that provide data concerning the vehicle. Refer to the VIN decoding chart to determine the identification of a vehicle.

The Vehicle Identification Number is also imprinted on the:

- Vehicle Safety Certification Label.
- Frame rail.

To protect the consumer from theft and possible fraud the manufacturer is required to include a Check Digit at the ninth position of the Vehicle Identification Number. The check digit is used by the manufacturer and government agencies to verify the authenticity of the vehicle and official documentation. The formula to use the check digit is not released to the general public.

VEHICLE IDENTIFICATION NUMBER DECODING CHART

POSITION	INTERPRETATION	CODE = DESCRIPTION
1	Country of Origin	1 = Manufactured By DaimlerChrysler Corporation
2	Make	J = Jeep
3	Vehicle Type	4 = MPV
4	Gross Vehicle Weight Rating	E = 3001-4000 lbs. F = 4001-5000 lbs.
5	Vehicle Line	A = Wrangler 4X4 (LHD) 4 = Wrangler 4X4 (RHD)
6	Series	3 = X 4 = Sport/Unlimited 5 = Sahara 6 = Rubicon
7	Body Style	4 = Extended Open Body 9 = Open Body
8	Engine	1 = 2.4L 4 cyl DOHC Gasoline S = 4.0L 6 cyl Gasoline

VEHICLE IDENTIFICATION NUMBER (Continued)

POSITION	INTERPRETATION	CODE = DESCRIPTION
9	Check Digit	0 through 9 or X
10	Model Year	5=2005
11	Assembly Plant	P = Toledo #2
12 thru 17	Vehicle Build Sequence	

VEHICLE EMISSION CONTROL INFORMATION (VECI) LABEL

DESCRIPTION

All models have a Vehicle Emission Control Information (VECI) Label. DaimlerChrysler permanently attaches the label in the engine compartment (Fig. 1). It cannot be removed without defacing information and destroying the label.

The label contains the vehicle's emission specifications and vacuum hose routings. All hoses must be connected and routed according to the label.

The VECI label contains the following:

- Engine family and displacement
- Evaporative family
- Emission control system schematic
- Certification application
- Engine timing specifications (if adjustable)
- Idle speeds (if adjustable)
- Spark plug and gap

The label also contains an engine vacuum schematic. These labels are permanently attached and cannot be removed without defacing information and destroying label.

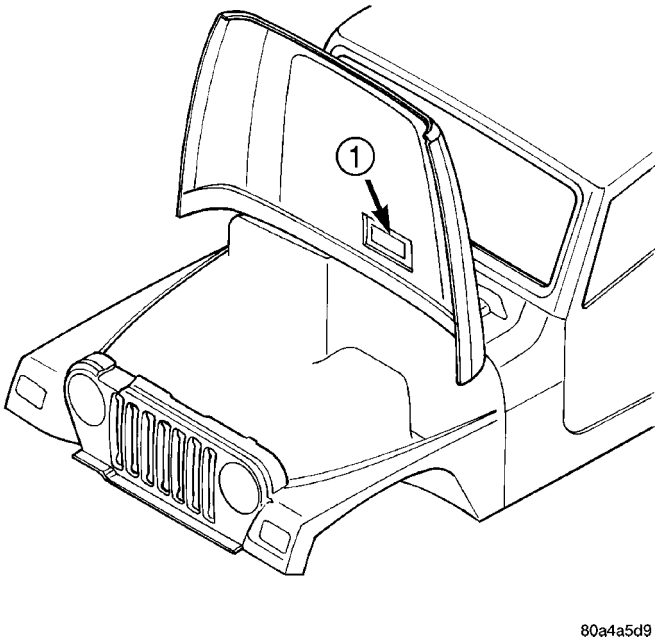

Fig. 1 VECI Label Location

1 - VECI LABEL

The label is located on the driver-side door shutface.

VEHICLE CERTIFICATION LABEL

DESCRIPTION

A vehicle certification label (Fig. 2) is attached to every DaimlerChrysler Corporation vehicle. The label certifies that the vehicle conforms to all applicable Federal Motor Vehicle Standards. The label also lists:

- Month and year of vehicle manufacture.
- Gross Vehicle Weight Rating (GVWR). The gross front and rear axle weight ratings (GAWR's) are based on a minimum rim size and maximum cold tire inflation pressure.
- Vehicle Identification Number (VIN).
- Type of vehicle.
- Type of rear wheels.
- Bar code.
- Month, Day and Hour (MDH) of final assembly.
- Paint and Trim codes.
- Country of origin.

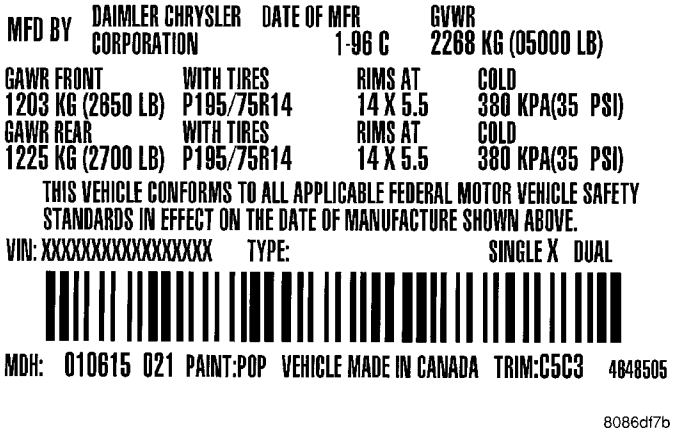

Fig. 2 VEHICLE CERTIFICATION LABEL - TYPICAL

BODY CODE PLATE

DESCRIPTION

BODY CODE PLATE

A metal body code plate is attached to the floor pan under the drivers seat (Fig. 3). Disengage the snaps attaching the carpet to the floor pan to read the information. There are seven lines of information on the body code plate. Lines 4, 5, 6, and 7 are not used to define service information. Information reads from left to right, starting with line 3 in the center of the plate to line 1 at the bottom of the plate (Fig. 4).

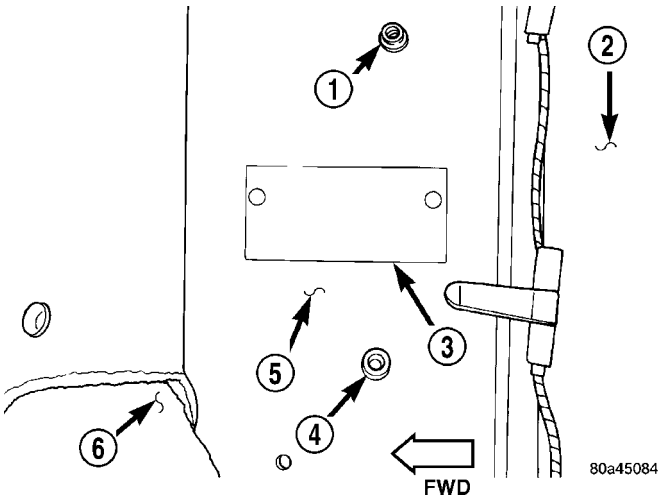

Fig. 3 Body Code Plate Location

1 - SNAP
2 - REAR CARPET
3 - BODY CODE PLATE
4 - SNAP
5 - FLOOR PAN
6 - FRONT CARPET

The last code imprinted on a vehicle code plate will be followed by the imprinted word END. When two vehicle code plates are required, the last available spaces on the first plate will be imprinted with the letters CTD (for continued).

When a second vehicle code plate is necessary, the first four spaces on each row will not be used because of the plate overlap.

BODY CODE PLATE—LINE 3

DIGITS 1 THROUGH 12

Vehicle Order Number

DIGITS 13, 14, AND 15

Roof

- VJB = Soft Top Medium Gray
- VJC = Soft Top Medium Khaki
- VJG = Soft Top Dark Green
- VJK = Soft Top Dark Khaki

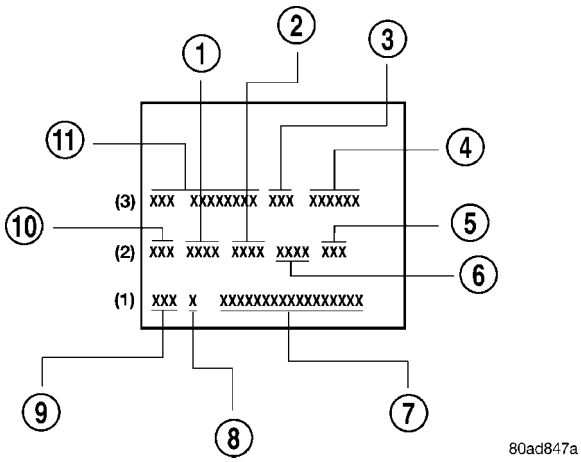

Fig. 4 Body Code Plate Decoding

1 - PRIMARY PAINT
2 - SECONDARY PAINT
3 - ROOF
4 - CAR LINE SHELL
5 - ENGINE
6 - TRIM
7 - VIN
8 - MARKET
9 - TRANSMISSION
10 - PAINT PROCEDURE
11 - VEHICLE ORDER NUMBER

- VKB = Hard Top Medium Gray
- VKC = Hard Top Medium Khaki
- VKK = Hard Top Dark Khaki

DIGITS 16, 17, AND 18

Car Line Shell

- TJJ = Wrangler (LHD)
- TJU = Wrangler (RHD)

DIGIT 19

Price Class

- L = Wrangler (All)

DIGITS 20 AND 21

Body Type

- 77 = Wheel Base (93.4 in.)
- 78 = Wheel Base (103.4 in.)

BODY CODE PLATE—LINE 2

DIGITS 1,2, AND 3

Paint Procedure

DIGIT 4

Open Space

DIGITS 5 THROUGH 8

Primary Paint

(Refer to 23 - BODY/PAINT - SPECIFICATIONS) for color codes.

DIGIT 9

Open Space

BODY CODE PLATE (Continued)

DIGITS 10 THROUGH 13

Secondary Paint

DIGIT 14

Open Space

DIGITS 15 THROUGH 18

Interior Trim Code

DIGIT 19

Open Space

DIGITS 20, 21, AND 22

Engine Code

- ED1 = 2.4L 4 cyl. MPI Gasoline
- ERH = 4.0L 6 cyl. MPI Gasoline

BODY CODE PLATE—LINE 1

DIGITS 1, 2, AND 3

Transmission Codes

- DEH = NSG370 6 - speed Manual
- DG6 = 42RLE 4 - speed Automatic

DIGIT 4

Open Space

DIGIT 5

Market Code

- B = International

DIGIT 6

Open Space

DIGITS 7 THROUGH 23

Vehicle Identification Number (VIN)

(Refer to VEHICLE DATA/VEHICLE INFORMATION/VEHICLE IDENTIFICATION NUMBER - DESCRIPTION) for breakdown of VIN code.

INTERNATIONAL SYMBOLS

DESCRIPTION

The graphic symbols illustrated in the following International Control and Display Symbols Chart (Fig. 5) are used to identify various instrument controls. The symbols correspond to the controls and displays that are located on the instrument panel.

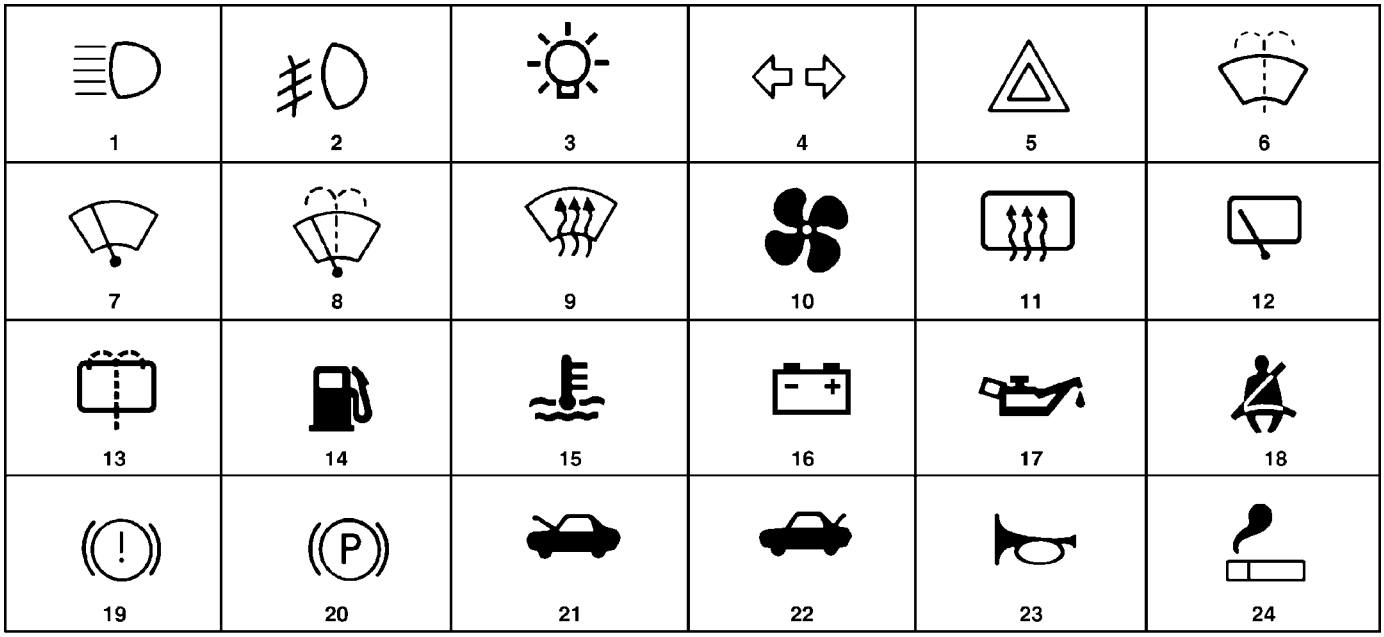

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Fig. 5 INTERNATIONAL CONTROL AND DISPLAY SYMBOLS

1	High Beam	13	Rear Window Washer
2	Fog Lamps	14	Fuel
3	Headlamp, Parking Lamps, Panel Lamps	15	Engine Coolant Temperature
4	Turn Warning	16	Battery Charging Condition
5	Hazard Warning	17	Engine Oil
6	Windshield Washer	18	Seat Belt
7	Windshield Wiper	19	Brake Failure
8	Windshield Wiper and Washer	20	Parking Brake
9	Windscreen Demisting and Defrosting	21	Front Hood
10	Ventilating Fan	22	Rear hood (Decklid)
11	Rear Window Defogger	23	Horn
12	Rear Window Wiper	24	Lighter

FASTENER IDENTIFICATION

DESCRIPTION

The SAE bolt strength grades range from grade 2 to grade 8. The higher the grade number, the greater the bolt strength. Identification is determined by the line marks on the top of each bolt head. The actual bolt strength grade corresponds to the number of line marks plus 2. The most commonly used metric bolt strength classes are 9.8 and 10.9. The metric strength class identification number is imprinted on the head of the bolt. The higher the class number, the greater the bolt strength. Some metric nuts are imprinted with a single-digit strength class on the nut face. Refer to the Fastener Identification and Fastener Strength Charts (Fig. 6).

BOLT MARKINGS AND TORQUES - METRIC

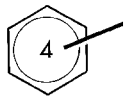
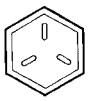
Bolt Markings	8.8/8.9		10.9		12.9	
Bolt Dia.	N·m	Ft. Lbs.	N·m	Ft. Lbs.	N·m	Ft. Lbs.
6	12	105*	14	120*	16	12
8	25	250*	32	23	38	28
10	54	40	60	45	74	55
12	95	70	108	80	135	100
14	155	115	175	130	216	160
16	243	180	324	210	324	240
* Inch Lbs.						

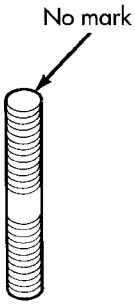
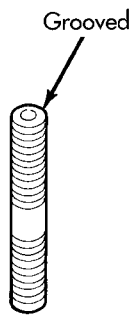
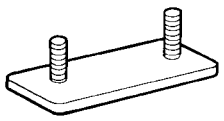
BOLT MARKINGS AND TORQUES - U. S. CUSTOMARY

Bolt Markings	Grade 5		Grade 8	
Bolt Dia.	N·m	Ft. Lbs	N·m	Ft. Lbs
1/4 - 20	10	95*	14	125*
1/4 - 28	10	95*	17	150*
5/16 - 18	22	200*	30	270*
5/16 - 24	26	240*	33	300*
3/8 - 16	40	30	55	40
3/8 - 24	47	35	60	45
7/16 - 14	68	50	88	65
7/16 - 20	74	55	95	70
1/2 - 13	101	75	135	100
1/2 -20	115	85	150	110
9/16 - 12	135	105	182	135
9/16 - 18	155	115	202	150
5/8 - 11	202	150	263	195
5/8 - 18	215	160	284	210
3/4 - 10	230	170	297	220
3/4 - 16	236	175	304	225
7/8 - 14	405	300	540	400
* Inch Lbs.				

FASTENER IDENTIFICATION (Continued)

HOW TO DETERMINE BOLT STRENGTH

	Mark	Class
Hexagon head bolt	 Bolt head No. 4 — 4T; 5 — 5T; 6 — 6T; 7 — 7T; 8 — 8T; 9 — 9T; 10 — 10T; 11 — 11T	
	No mark	4T
Hexagon flange bolt w/washer hexagon bolt	No mark	4T
Hexagon head bolt	Two protruding lines	5T
Hexagon flange bolt w/washer hexagon bolt	Two protruding lines	6T
Hexagon head bolt	 Three protruding lines	7T
Hexagon head bolt	Four protruding lines	8T

	Mark	Class
Stud bolt		4T
		6T
Welded bolt		4T

95IN-4

Fig. 6 FASTENER STRENGTH

FASTENER USAGE

DESCRIPTION

DESCRIPTION - FASTENER USAGE

WARNING: USE OF AN INCORRECT FASTENER MAY RESULT IN COMPONENT DAMAGE OR PERSONAL INJURY.

Fasteners and torque specifications references in this Service Manual are identified in metric and SAE format.

During any maintenance or repair procedures, it is important to salvage all fasteners (nuts, bolts, etc.) for reassembly. If the fastener is not salvageable, a fastener of equivalent specification must be used.

DESCRIPTION - THREADED HOLE REPAIR

Most stripped threaded holes can be repaired using a Helicoil®. Follow the vehicle or Helicoil® recommendations for application and repair procedures.

METRIC SYSTEM

DESCRIPTION

The metric system is based on quantities of one, ten, one hundred, one thousand and one million.

The following chart will assist in converting metric units to equivalent English and SAE units, or vise versa.

CONVERSION FORMULAS AND EQUIVALENT VALUES

MULTIPLY	BY	TO GET	MULTIPLY	BY	TO GET
in-lbs	x 0.11298	= Newton Meters (N·m)	N·m	x 8.851	= in-lbs
ft-lbs	x 1.3558	= Newton Meters (N·m)	N·m	x 0.7376	= ft-lbs
Inches Hg (60° F)	x 3.377	= Kilopascals (kPa)	kPa	x 0.2961	= Inches Hg
psi	x 6.895	= Kilopascals (kPa)	kPa	x 0.145	= psi
Inches	x 25.4	= Millimeters (mm)	mm	x 0.03937	= Inches
Feet	x 0.3048	= Meters (M)	M	x 3.281	= Feet
Yards	x 0.9144	= Meters	M	x 1.0936	= Yards
mph	x 1.6093	= Kilometers/Hr. (Km/h)	Km/h	x 0.6214	= mph
Feet/Sec	x 0.3048	= Meters/Sec (M/S)	M/S	x 3.281	= Feet/Sec
mph	x 0.4470	= Meters/Sec (M/S)	M/S	x 2.237	= mph
Kilometers/Hr. (Km/h)	x 0.27778	= Meters/Sec (M/S)	M/S	x 3.600	Kilometers/Hr. (Km/h)

COMMON METRIC EQUIVALENTS

1 inch = 25 Millimeters	1 Cubic Inch = 16 Cubic Centimeters
1 Foot = 0.3 Meter	1 Cubic Foot = 0.03 Cubic Meter
1 Yard = 0.9 Meter	1 Cubic Yard = 0.8 Cubic Meter
1 Mile = 1.6 Kilometers	

Refer to the Metric Conversion Chart to convert torque values listed in metric Newton- meters (N·m). Also, use the chart to convert between millimeters (mm) and inches (in.) (Fig. 7).

METRIC SYSTEM (Continued)

in-lbs to N•m

in- lb	N•m	in-lb	N•m	in-lb	N•m	in-lb	N•m	in-lb	N•m
2	.2260	42	4.7453	82	9.2646	122	13.7839	162	18.3032
4	.4519	44	4.9713	84	9.4906	124	14.0099	164	18.5292
6	.6779	46	5.1972	86	9.7165	126	14.2359	166	18.7552
8	.9039	48	5.4232	88	9.9425	128	14.4618	168	18.9811
10	1.1298	50	5.6492	90	10.1685	130	14.6878	170	19.2071
12	1.3558	52	5.8751	92	10.3944	132	14.9138	172	19.4331
14	1.5818	54	6.1011	94	10.6204	134	15.1397	174	19.6590
16	1.8077	56	6.3270	96	10.8464	136	15.3657	176	19.8850
18	2.0337	58	6.5530	98	11.0723	138	15.5917	178	20.1110
20	2.2597	60	6.7790	100	11.2983	140	15.8176	180	20.3369
22	2.4856	62	7.0049	102	11.5243	142	16.0436	182	20.5629
24	2.7116	64	7.2309	104	11.7502	144	16.2696	184	20.7889
26	2.9376	66	7.4569	106	11.9762	146	16.4955	186	21.0148
28	3.1635	68	7.6828	108	12.2022	148	16.7215	188	21.2408
30	3.3895	70	7.9088	110	12.4281	150	16.9475	190	21.4668
32	3.6155	72	8.1348	112	12.6541	152	17.1734	192	21.6927
34	3.8414	74	8.3607	114	12.8801	154	17.3994	194	21.9187
36	4.0674	76	8.5867	116	13.1060	156	17.6253	196	22.1447
38	4.2934	78	8.8127	118	13.3320	158	17.8513	198	22.3706
40	4.5193	80	9.0386	120	13.5580	160	18.0773	200	22.5966

N•m to in-lbs

N•m	in-lb	N•m	in-lb	N•m	in-lb	N•m	in-lb	N•m	in-lb
.2	1.7702	4.2	37.1747	8.2	72.5792	12.2	107.9837	16.2	143.3882
.4	3.5404	4.4	38.9449	8.4	74.3494	12.4	109.7539	16.4	145.1584
.6	5.3107	4.6	40.7152	8.6	76.1197	12.6	111.5242	16.6	146.9287
.8	7.0809	4.8	42.4854	8.8	77.8899	12.8	113.2944	16.8	148.6989
1	8.8511	5	44.2556	9	79.6601	13	115.0646	17	150.4691
1.2	10.6213	5.2	46.0258	9.2	81.4303	13.2	116.8348	17.2	152.2393
1.4	12.3916	5.4	47.7961	9.4	83.2006	13.4	118.6051	17.4	154.0096
1.6	14.1618	5.6	49.5663	9.6	84.9708	13.6	120.3753	17.6	155.7798
1.8	15.9320	5.8	51.3365	9.8	86.7410	13.8	122.1455	17.8	157.5500
2	17.7022	6	53.1067	10	88.5112	14	123.9157	18	159.3202
2.2	19.4725	6.2	54.8770	10.2	90.2815	14.2	125.6860	18.5	163.7458
2.4	21.2427	6.4	56.6472	10.4	92.0517	14.4	127.4562	19	168.1714
2.6	23.0129	6.6	58.4174	10.6	93.8219	14.6	129.2264	19.5	172.5970
2.8	24.7831	6.8	60.1876	10.8	95.5921	14.8	130.9966	20	177.0225
3	26.5534	7	61.9579	11	97.3624	15	132.7669	20.5	181.4480
3.2	28.3236	7.2	63.7281	11.2	99.1326	15.2	134.5371	21	185.8736
3.4	30.0938	7.4	65.4983	11.4	100.9028	15.4	136.3073	22	194.7247
3.6	31.8640	7.6	67.2685	11.6	102.6730	15.6	138.0775	23	203.5759
3.8	33.6342	7.8	69.0388	11.8	104.4433	15.8	139.8478	24	212.4270
4	35.4045	8	70.8090	12	106.2135	16	141.6180	25	221.2781

ft-lbs to N•m

ft-lb	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	N•m
1	1.3558	21	28.4722	41	55.5885	61	82.7049	81	109.8212
2	2.7116	22	29.8280	42	56.9444	62	84.0607	82	111.1770
3	4.0675	23	31.1838	43	58.3002	63	85.4165	83	112.5328
4	5.4233	24	32.5396	44	59.6560	64	86.7723	84	113.8888
5	6.7791	25	33.8954	45	61.0118	65	88.1281	85	115.2446
6	8.1349	26	35.2513	46	62.3676	66	89.4840	86	116.6004
7	9.4907	27	36.6071	47	63.7234	67	90.8398	87	117.9562
8	10.8465	28	37.9629	48	65.0793	68	92.1956	88	119.3120
9	12.2024	29	39.3187	49	66.4351	69	93.5514	89	120.6678
10	13.5582	30	40.6745	50	67.7909	70	94.9073	90	122.0236
11	14.9140	31	42.0304	51	69.1467	71	96.2631	91	123.3794
12	16.2698	32	43.3862	52	70.5025	72	97.6189	92	124.7352
13	17.6256	33	44.7420	53	71.8583	73	98.9747	93	126.0910
14	18.9815	34	46.0978	54	73.2142	74	100.3316	94	127.4468
15	20.3373	35	47.4536	55	74.5700	75	101.6862	95	128.8026
16	21.6931	36	48.8094	56	75.9258	76	103.0422	96	130.1586
17	23.0489	37	50.1653	57	77.2816	77	104.3980	97	131.5144
18	24.4047	38	51.5211	58	78.6374	78	105.7538	98	132.8702
19	25.7605	39	52.8769	59	79.9933	79	107.1196	99	134.2260
20	27.1164	40	54.2327	60	81.3491	80	108.4654	100	135.5820

N•m to ft-lbs

N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb
1	.7376	21	15.9888	41	30.2400	61	44.9913	81	59.7425
2	1.4751	22	16.2264	42	30.9776	62	45.7289	82	60.4801
3	2.2127	23	16.9639	43	31.7152	63	46.4664	83	61.2177
4	2.9502	24	17.7015	44	32.4527	64	47.2040	84	61.9552
5	3.6878	25	18.4391	45	33.1903	65	47.9415	85	62.6928
6	4.4254	26	19.1766	46	33.9279	66	48.6791	86	63.4303
7	5.1629	27	19.9142	47	34.6654	67	49.4167	87	64.1679
8	5.9005	28	20.6517	48	35.4030	68	50.1542	88	64.9545
9	6.6381	29	21.3893	49	36.1405	69	50.8918	89	65.6430
10	7.3756	30	22.1269	50	36.8781	70	51.6293	90	66.3806
11	8.1132	31	22.8644	51	37.6157	71	52.3669	91	67.1181
12	8.8507	32	23.6020	52	38.3532	72	53.1045	92	67.8557
13	9.5883	33	24.3395	53	39.0908	73	53.8420	93	68.5933
14	10.3259	34	25.0771	54	39.8284	74	54.5720	94	69.3308
15	11.0634	35	25.8147	55	40.5659	75	55.3172	95	70.0684
16	11.8010	36	26.5522	56	41.3035	76	56.0547	96	70.8060
17	12.5386	37	27.2898	57	42.0410	77	56.7923	97	71.5435
18	13.2761	38	28.0274	58	42.7786	78	57.5298	98	72.2811
19	14.0137	39	28.7649	59	43.5162	79	58.2674	99	73.0187
20	14.7512	40	29.5025	60	44.2537	80	59.0050	100	73.7562

in. to mm

in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
.01	.254	.21	5.334	.41	10.414	.61	15.494	.81	20.574
.02	.508	.22	5.588	.42	10.668	.62	15.748	.82	20.828
.03	.762	.23	5.842	.43	10.922	.63	16.002	.83	21.082
.04	1.016	.24	6.096	.44	11.176	.64	16.256	.84	21.336
.05	1.270	.25	6.350	.45	11.430	.65	16.510	.85	21.590
.06	1.524	.26	6.604	.46	11.684	.66	16.764	.86	21.844
.07	1.778	.27	6.858	.47	11.938	.67	17.018	.87	22.098
.08	2.032	.28	7.112	.48	12.192	.68	17.272	.88	22.352
.09	2.286	.29	7.366	.49	12.446	.69	17.526	.89	22.606
.10	2.540	.30	7.620	.50	12.700	.70	17.780	.90	22.860
.11	2.794	.31	7.874	.51	12.954	.71	18.034	.91	23.114
.12	3.048	.32	8.128	.52	13.208	.72	18.288	.92	23.368
.13	3.302	.33	8.382	.53	13.462	.73	18.542	.93	23.622
.14	3.556	.34	8.636	.54	13.716	.74	18.796	.94	23.876
.15	3.810	.35	8.890	.55	13.970	.75	19.050	.95	24.130
.16	4.064	.36	9.144	.56	14.224	.76	19.304	.96	24.384
.17	3.318	.37	9.398	.57	14.478	.77	19.558	.97	24.638
.18	4.572	.38	9.652	.58	14.732	.78	19.812	.98	24.892
.19	4.826	.39	9.906	.59	14.986	.79	20.066	.99	25.146
.20	5.080	.40	10.160	.60	15.240	.80	20.320	1.00	25.400

mm to in.

mm	in.	mm	in.	mm	in.	mm	in.	mm	in.
.01	.00039	.21	.00827	.41	.01614	.61	.02402	.81	.03189
.02	.00079	.22	.00866	.42	.01654	.62	.02441	.82	.03228
.03	.00118	.23	.00906	.43	.01693	.63	.02480	.83	.03268
.04	.00157	.24	.00945	.44	.01732	.64	.02520	.84	.03307
.05	.00197	.25	.00984	.45	.01772	.65	.02559	.85	.03346
.06	.00236	.26	.01024	.46	.01811	.66	.02598	.86	.03386
.07	.00276	.27	.01063	.47	.01850	.67	.02638	.87	.03425
.08	.00315	.28	.01102	.48	.01890	.68	.02677	.88	.03465
.09	.00354	.29	.01142	.49	.01929	.69	.02717	.89	.03504
.10	.00394	.30	.01181	.50	.01969	.70	.02756	.90	.03543
.11	.00433	.31	.01220	.51	.02008	.71	.02795	.91	.03583
.12	.00472	.32	.01260	.52	.02047	.72	.02835	.92	.03622
.13	.00512	.33	.01299	.53	.02087	.73	.02874	.93	.03661
.14	.00551	.34	.01339	.54	.02126	.74	.02913	.94	.03701
.15	.00591	.35	.01378	.55	.02165	.75	.02953	.95	.03740
.16	.00630	.36	.01417	.56	.02205	.76	.02992	.96	.03780
.17	.00669	.37	.01457	.57	.02244	.77	.03032	.97	.03819
.18	.00709	.38	.01496	.58	.02283	.78	.03071	.98	.03858
.19	.00748	.39	.01535	.59	.02323	.79	.03110	.99	.03898
.20	.00787	.40	.01575	.60	.02362	.80	.03150	1.00	.03937

J901N-10

Fig. 7 METRIC CONVERSION CHART

TORQUE REFERENCES

DESCRIPTION

Individual Torque Charts appear within many or the Groups. Refer to the Standard Torque Specifications Chart for torque references not listed in the individual torque charts (Fig. 8).

SPECIFIED TORQUE FOR STANDARD BOLTS

Class	Diameter mm	Pitch mm	Specified torque					
			Hexagon head bolt			Hexagon flange bolt		
			N•m	kgf-cm	ft-lbf	N•m	kgf-cm	ft-lbf
4T	6	1	5	55	48 in.-lbf	6	60	52 in.-lbf
	8	1.25	12.5	130	9	14	145	10
	10	1.25	26	260	19	29	290	21
	12	1.25	47	480	35	53	540	39
	14	1.5	74	760	55	84	850	61
	16	1.5	115	1,150	83	—	—	—
5T	6	1	6.5	65	56 in.-lbf	7.5	75	65 in.-lbf
	8	1.25	15.5	160	12	17.5	175	13
	10	1.25	32	330	24	36	360	26
	12	1.25	59	600	43	65	670	48
	14	1.5	91	930	67	100	1,050	76
	16	1.5	140	1,400	101	—	—	—
6T	6	1	8	80	69 in.-lbf	9	90	78 in.-lbf
	8	1.25	19	195	14	21	210	15
	10	1.25	39	400	29	44	440	32
	12	1.25	71	730	53	80	810	59
	14	1.5	110	1,100	80	125	1,250	90
	16	1.5	170	1,750	127	—	—	—
7T	6	1	10.5	110	8	12	120	9
	8	1.25	25	260	19	28	290	21
	10	1.25	52	530	38	58	590	43
	12	1.25	95	970	70	105	1,050	76
	14	1.5	145	1,500	108	165	1,700	123
	16	1.5	230	2,300	166	—	—	—
8T	8	1.25	29	300	22	33	330	24
	10	1.25	61	620	45	68	690	50
	12	1.25	110	1,100	80	120	1,250	90
9T	8	1.25	34	340	25	37	380	27
	10	1.25	70	710	51	78	790	57
	12	1.25	125	1,300	94	140	1,450	105
10T	8	1.25	38	390	28	42	430	31
	10	1.25	78	800	58	88	890	64
	12	1.25	140	1,450	105	155	1,600	116
11T	8	1.25	42	430	31	47	480	35
	10	1.25	87	890	64	97	990	72
	12	1.25	155	1,600	116	175	1,800	130

Fig. 8 TORQUE SPECIFICATIONS

LUBRICATION & MAINTENANCE

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INTERNATIONAL SYMBOLS

DESCRIPTION

DaimlerChrysler Corporation uses international symbols to identify engine compartment lubricant and fluid inspection and fill locations (Fig. 1).

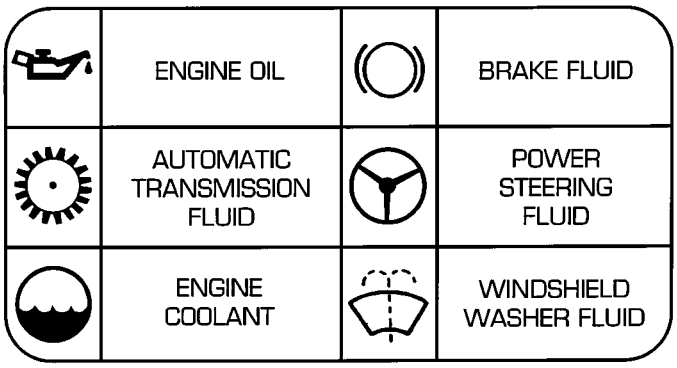

Fig. 1 INTERNATIONAL SYMBOLS

PARTS & LUBRICANT RECOMMENDATION

DESCRIPTION

LUBRICANT RECOMMENDATIONS

Chassis

Component	Fluid, Lubricant, or Genuine Part
Steering Gear & Linkage, Ball Joints, Prop Shafts & Yokes, Wheel Bearings	Mopar® Multi-Purpose Grease NLGI Grade 2 EP, GC-LB

PARTS & LUBRICANT RECOMMENDATION (Continued)

Body

Component	Fluid, Lubricant, or Genuine Part
Hinges:	
Door And Hood	Mopar® Engine Oil
Liftgate	Mopar® Multi-Purpose Grease NLGI Grade 2 EP, GC-LB
Latches:	
Door, Hood/Safety Catch, Liftgate	Mopar® Multi-Purpose Grease NLGI Grade 2 EP, GC-LB
Seat Regulator & Track	Mopar® Multi-Purpose Grease NLGI Grade 2 EP, GC-LB
Window System Components	Mopar® Spray White Lubricant
Lock Cylinders	Mopar® Lock Cylinder Lubricant
Parking Brake Mechanism	Mopar® Wheel Bearing Grease NLGI Grade 1, GC-LBB
Soft Top	Mopar® Soft Top Zipper Cleaner & Lubricant

FLUID TYPES

When service is required, DaimlerChrysler Corporation recommends that only Mopar® brand parts, lubricants and chemicals be used. Mopar® provides the best engineered products for servicing DaimlerChrysler Corporation vehicles.

Only lubricants bearing designations defined by the following organization should be used to service a Chrysler Corporation vehicle.

- Society of Automotive Engineers (SAE)
- American Petroleum Institute (API) (Fig. 2)
- National Lubricating Grease Institute (NLGI)

API QUALITY CLASSIFICATION

This symbol on the front of an oil container means that the oil has been certified by the American Petroleum Institute (API) to meet all the lubrication requirements specified by DaimlerChrysler Corporation.

9400-9

Fig. 2 API Symbol

GEAR LUBRICANTS

SAE ratings also apply to multigrade gear lubricants. In addition, API classification defines the lubricants usage. Such as API GL-5 and SAE 75W-90.

FLUID TYPES

DESCRIPTION

ENGINE OIL

WARNING: NEW OR USED ENGINE OIL CAN BE IRRITATING TO THE SKIN. AVOID PROLONGED OR REPEATED SKIN CONTACT WITH ENGINE OIL. CONTAMINANTS IN USED ENGINE OIL, CAUSED BY INTERNAL COMBUSTION, CAN BE HAZARDOUS TO YOUR HEALTH. THOROUGHLY WASH EXPOSED SKIN WITH SOAP AND WATER. DO NOT WASH SKIN WITH GASOLINE, DIESEL FUEL, THINNER, OR SOLVENTS, HEALTH PROBLEMS CAN RESULT. DO NOT POLLUTE, DISPOSE OF USED ENGINE OIL PROPERLY. CONTACT YOUR DEALER OR GOVERNMENT AGENCY FOR LOCATION OF COLLECTION CENTER IN YOUR AREA.

Only lubricants bearing designations defined by the following organization should be used.

- Society of Automotive Engineers (SAE)
- American Petroleum Institute (API)
- National Lubricating Grease Institute (NLGI)
- Association des Constructeurs Européens d' Automobiles (European Automobile Manufacturers Association) (ACEA)

API SERVICE GRADE CERTIFIED

Use an engine oil that is API Certified. MOPAR® provides engine oils, that meet or exceed this requirement.

FLUID TYPES (Continued)

SAE VISCOSITY

An SAE viscosity grade is used to specify the viscosity of engine oil. Use only engine oils with multiple viscosities such as 5W-30 or 10W-30. These are specified with a dual SAE viscosity grade which indicates the cold-to-hot temperature viscosity range. Select an engine oil that is best suited to your particular temperature range and variation (Fig. 3).

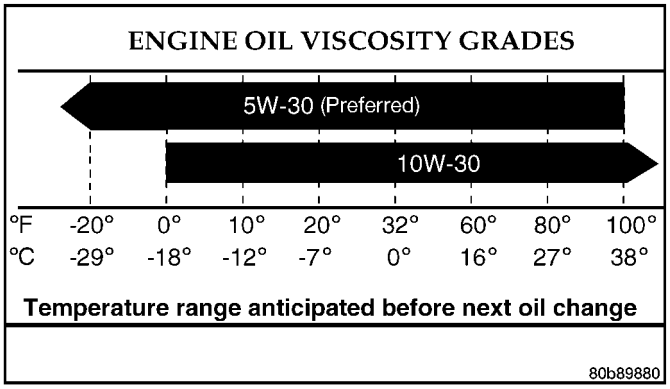

Fig. 3 Temperature/Engine Oil Viscosity

ACEA Categories

For countries that use the ACEA European Oil Categories for Service Fill Oils, use engine oils that meet the requirements of ACEA A1/B1, A2/B2, or A3/B3.

ENERGY CONSERVING OIL

An Energy Conserving type oil is recommended for gasoline engines. The designation of ENERGY CONSERVING is located on the label of an engine oil container.

CONTAINER IDENTIFICATION

Standard engine oil identification notations have been adopted to aid in the proper selection of engine oil. The identifying notations are located on the front label of engine oil plastic bottles and the top of engine oil cans (Fig. 4).

This symbol means that the oil has been certified by the American Petroleum Institute (API). DaimlerChrysler only recommend API Certified engine oils. Use Mopar® engine oil or equivalent.

9400-9

Fig. 4 API Certification Mark

FUEL REQUIREMENTS

Your engine is designed to meet all emissions regulations and provide excellent fuel economy and performance when using high quality unleaded “regular” gasoline having an octane rating of 87. The routine use of premium gasoline is not recommended. Under normal conditions the use of premium fuel will not provide a benefit over high quality regular gasolines and in some circumstances may result in poorer performance.

Light spark knock at low engine speeds is not harmful to your engine. However, continued heavy spark knock at high speeds can cause damage and immediate service is required. Engine damage resulting from operation with a heavy spark knock may not be covered by the new vehicle warranty.

Poor quality gasoline can cause problems such as hard starting, stalling and hesitations. If you experience these symptoms, try another brand of gasoline before considering service for the vehicle.

Over 40 auto manufacturers world-wide have issued and endorsed consistent gasoline specifications (the Worldwide Fuel Charter, WWFC) to define fuel properties necessary to deliver enhanced emissions, performance and durability for your vehicle. We recommend the use of gasolines that meet the WWFC specifications if they are available.

REFORMULATED GASOLINE

Many areas of the country require the use of cleaner burning gasoline referred to as “reformulated” gasoline. Reformulated gasoline contain oxygenates, and are specifically blended to reduce vehicle emissions and improve air quality.

We strongly support the use of reformulated gasoline. Properly blended reformulated gasoline will provide excellent performance and durability for the engine and fuel system components.

GASOLINE/OXYGENATE BLENDS

Some fuel suppliers blend unleaded gasoline with oxygenates such as 10% ethanol, MTBE, and ETBE. Oxygenates are required in some areas of the country during the winter months to reduce carbon monoxide emissions. Fuels blended with these oxygenates may be used in your vehicle.

CAUTION: DO NOT use gasoline containing METHANOL. Gasoline containing methanol may damage critical fuel system components.

MMT IN GASOLINE

MMT is a manganese-containing metallic additive that is blended into some gasoline to increase octane. Gasoline blended with MMT provide no performance

FLUID TYPES (Continued)

advantage beyond gasoline of the same octane number without MMT. Gasoline blended with MMT reduce spark plug life and reduce emission system performance in some vehicles. We recommend that gasolines free of MMT be used in your vehicle. The MMT content of gasoline may not be indicated on the gasoline pump; therefore, you should ask your gasoline retailer whether or not his/her gasoline contains MMT.

It is even more important to look for gasoline without MMT in Canada because MMT can be used at levels higher than allowed in the United States. MMT is prohibited in Federal and California reformulated gasoline.

SULFUR IN GASOLINE

If you live in the northeast United States, your vehicle may have been designed to meet California low emission standards with Cleaner-Burning California reformulated gasoline with low sulfur. If such fuels are not available in states adopting California emission standards, your vehicles will operate satisfactorily on fuels meeting federal specifications, but emission control system performance may be adversely affected. Gasoline sold outside of California is permitted to have higher sulfur levels which may affect the performance of the vehicle's catalytic converter. This may cause the Malfunction Indicator Lamp (MIL), Check Engine or Service Engine Soon light to illuminate. We recommend that you try a different brand of unleaded gasoline having lower sulfur to determine if the problem is fuel related prior to returning your vehicle to an authorized dealer for service.

CAUTION: If the Malfunction Indicator Lamp (MIL), Check Engine or Service Engine Soon light is flashing, immediate service is required; see on-board diagnostics system section.

MATERIALS ADDED TO FUEL

All gasoline sold in the United States and Canada are required to contain effective detergent additives. Use of additional detergents or other additives is not needed under normal conditions.

FUEL SYSTEM CAUTIONS

CAUTION: Follow these guidelines to maintain your vehicle's performance:

- The use of leaded gas is prohibited by Federal law. Using leaded gasoline can impair engine performance, damage the emission control system, and could result in loss of warranty coverage.
- An out-of-tune engine, or certain fuel or ignition malfunctions, can cause the catalytic converter to overheat. If you notice a pungent burning odor or some light smoke, your engine may be out of tune or malfunctioning and may require immediate service. Contact your dealer for service assistance.
- When pulling a heavy load or driving a fully loaded vehicle when the humidity is low and the temperature is high, use a premium unleaded fuel to help prevent spark knock. If spark knock persists, lighten the load, or engine piston damage may result.
- The use of fuel additives which are now being sold as octane enhancers is not recommended. Most of these products contain high concentrations of methanol. Fuel system damage or vehicle performance problems resulting from the use of such fuels or additives is not the responsibility of DaimlerChrysler Corporation and may not be covered under the new vehicle warranty.

NOTE: Intentional tampering with emissions control systems can result in civil penalties being assessed against you.

HOAT COOLANT

WARNING: ANTIFREEZE IS AN ETHYLENE-GLYCOL BASE COOLANT AND IS HARMFUL IF SWALLOWED OR INHALED. IF SWALLOWED, DRINK TWO GLASSES OF WATER AND INDUCE VOMITING. IF INHALED, MOVE TO FRESH AIR AREA. SEEK MEDICAL ATTENTION IMMEDIATELY. DO NOT STORE IN OPEN OR UNMARKED CONTAINERS. WASH SKIN AND CLOTHING THOROUGHLY AFTER COMING IN CONTACT WITH ETHYLENE-GLYCOL. KEEP OUT OF REACH OF CHILDREN. DISPOSE OF GLYCOL BASE COOLANT PROPERLY, CONTACT YOUR DEALER OR GOVERNMENT AGENCY FOR LOCATION OF COLLECTION CENTER IN YOUR AREA. DO NOT OPEN A COOLING SYSTEM WHEN THE ENGINE IS AT OPERATING TEMPERATURE OR HOT UNDER PRESSURE, PERSONAL INJURY CAN RESULT. AVOID RADIATOR COOLING FAN WHEN ENGINE COMPARTMENT RELATED SERVICE IS PERFORMED, PERSONAL INJURY CAN RESULT.

CAUTION: Use of Propylene-Glycol based coolants is not recommended, as they provide less freeze protection and less corrosion protection.

The cooling system is designed around the coolant. The coolant must accept heat from engine metal, in the cylinder head area near the exhaust valves and engine block. Then coolant carries the heat to the

FLUID TYPES (Continued)

radiator where the tube/fin radiator can transfer the heat to the air.

The use of aluminum cylinder blocks, cylinder heads, and water pumps requires special corrosion protection. Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (MS-9769), or the equivalent ethylene-glycol base coolant with organic corrosion inhibitors (called HOAT, for Hybrid Organic Additive Technology) is recommended. This coolant offers the best engine cooling without corrosion when mixed with 50% ethylene-glycol and 50% distilled water to obtain a freeze point of -37°C (-35°F). If it loses color or becomes contaminated, drain, flush, and replace with fresh properly mixed coolant solution.

CAUTION: Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (MS-9769) may not be mixed with any other type of antifreeze. Mixing of coolants other than specified (non-HOAT or other HOAT), may result in engine damage that may not be covered under the new vehicle warranty, and decreased corrosion protection.

COOLANT PERFORMANCE

The required ethylene-glycol (antifreeze) and water mixture depends upon climate and vehicle operating conditions. The coolant performance of various mixtures follows:

Pure Water- Water can absorb more heat than a mixture of water and ethylene-glycol. This is for purpose of heat transfer only. Water also freezes at a higher temperature and allows corrosion.

100 percent Ethylene-Glycol - The corrosion inhibiting additives in ethylene-glycol need the presence of water to dissolve. Without water, additives form deposits in system. These act as insulation causing temperature to rise to as high as 149°C (300°F). This temperature is hot enough to melt plastic and soften solder. The increased temperature can result in engine detonation. In addition, 100 percent ethylene-glycol freezes at -22°C (-8°F).

50/50 Ethylene-Glycol and Water - Is the recommended mixture, it provides protection against freezing to -37°C (-34°F). The antifreeze concentration **must always** be a minimum of 44 percent, year-round in all climates. If percentage is lower, engine parts may be eroded by cavitation. Maximum protection against freezing is provided with a 68 percent antifreeze concentration, which prevents freezing down to -67.7°C (-90°F). A higher percentage will freeze at a warmer temperature. Also, a higher percentage of antifreeze can cause the engine to overheat because specific heat of antifreeze is lower than that of water.

CAUTION: Richer antifreeze mixtures cannot be measured with normal field equipment and can cause problems associated with 100 percent ethylene-glycol.

COOLANT SELECTION AND ADDITIVES

The use of aluminum cylinder blocks, cylinder heads and water pumps requires special corrosion protection. Only Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (glycol base coolant with corrosion inhibitors called HOAT, for Hybrid Organic Additive Technology) is recommended. This coolant offers the best engine cooling without corrosion when mixed with 50% distilled water to obtain to obtain a freeze point of -37°C (-35°F). If it loses color or becomes contaminated, drain, flush, and replace with fresh properly mixed coolant solution.

CAUTION: Do not use coolant additives that are claimed to improve engine cooling.

TRANSFER CASE - NV231

Recommended lubricant for the NV231 transfer case is Mopar® ATF +4, Automatic Transmission Fluid.

TRANSFER CASE - NV241

Recommended lubricant for the NV241 transfer case is Mopar® ATF +4, Automatic Transmission Fluid.

AXLE LUBRICATION

NOTE: DiamlerChrysler recommends using Mopar® lubricants or lubricants of equal quality.

FRONT AXLE

- 181 FBI (Model 30) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)
- RUBICON 216 FBI (Model 44) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)

REAR AXLE

- 194 RBI (Model 35) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)
- 194 RBI (Model 35) 4.56 Ratio - 2.4 L Enigne and 42 RLE Automatic Transmission - Mopar® Synthetic Gear Lubricant 75W-140
- 226 RBI (Model 44) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)

FLUID TYPES (Continued)

- RUBICON 226 RBI (Model 44) - Mopar® Synthetic Gear Lubricant 75W-140

NOTE: Trac-lok® equipped axles require 118 ml (4 ounces) of Limited Slip Additive in the lubricant.

MANUAL TRANSMISSION

NOTE: DaimlerChrysler recommends using Mopar® lubricants or lubricants of equal quality.

- NSG370 - Mopar® Manual Transmission Lubricant MS-9224

AUTOMATIC TRANSMISSION FLUID

NOTE: Refer to Service Procedures in this group for fluid level checking procedures.

Mopar® ATF +4, Automatic Transmission Fluid is the recommended fluid for DaimlerChrysler automatic transmissions.

Dexron II fluid IS NOT recommended. Clutch chatter can result from the use of improper fluid.

Mopar® ATF +4, Automatic Transmission Fluid when new is red in color. The ATF is dyed red so it can be identified from other fluids used in the vehicle such as engine oil or antifreeze. The red color is not permanent and is not an indicator of fluid condition. As the vehicle is driven, the ATF will begin to look darker in color and may eventually become brown. **This is normal.** ATF+4 also has a unique odor that may change with age. Consequently, odor and color cannot be used to indicate the fluid condition or the need for a fluid change.

FLUID ADDITIVES

DaimlerChrysler strongly recommends against the addition of any fluids to the transmission, other than those automatic transmission fluids listed above. Exceptions to this policy are the use of special dyes to aid in detecting fluid leaks.

Various “special” additives and supplements exist that claim to improve shift feel and/or quality. These additives and others also claim to improve converter clutch operation and inhibit overheating, oxidation, varnish, and sludge. These claims have not been supported to the satisfaction of DaimlerChrysler and these additives **must not be used.** The use of transmission “sealers” should also be avoided, since they may adversely affect the integrity of transmission seals.

POWER STEERING FLUID

The recommended fluid for the power steering system is Mopar® ATF +4.

Mopar® ATF+4, when new is red in color. The ATF+4 is dyed red so it can be identified from other fluids used in the vehicle such as engine oil or antifreeze. The red color is not permanent and is not an indicator of fluid condition, As the vehicle is driven, the ATF+4 will begin to look darker in color and may eventually become brown. **THIS IS NORMAL.** ATF+4 also has a unique odor that may change with age. Consequently, odor and color cannot be used to indicate the fluid condition or the need for a fluid change.

OPERATION - AUTOMATIC TRANSMISSION FLUID

The automatic transmission fluid is selected based upon several qualities. The fluid must provide a high level of protection for the internal components by providing a lubricating film between adjacent metal components. The fluid must also be thermally stable so that it can maintain a consistent viscosity through a large temperature range. If the viscosity stays constant through the temperature range of operation, transmission operation and shift feel will remain consistent. Transmission fluid must also be a good conductor of heat. The fluid must absorb heat from the internal transmission components and transfer that heat to the transmission case.

FLUID CAPACITIES

SPECIFICATIONS - FLUID CAPACITIES

DESCRIPTION	SPECIFICATION
FUEL TANK	19 U.S. Gallons (71.9 Liters)****
ENGINE OIL	
Engine Oil - with Filter - 2.4L	3.8 L (4.0 qts.)
Engine Oil - with Filter - 4.0L	5.7 L (6.0 qts.)
ENGINE COOLANT	
Cooling System - 2.4 L	8.5 L (9.0 qts.)
Cooling System - 4.0 L	9.9 L (10.5 qts.)
POWER STEERING SYSTEM	
Power steering fluid capacities are dependent on engine/chassis options as well as steering gear/cooler options. Depending on type and size of internal cooler, length and inside diameter of cooler lines, or use of an auxiliary cooler, these capacities may vary. Refer to 19, Steering for proper fill and bleed procedures.	
AUTOMATIC TRANSMISSION	
Service Fill - 42RLE	3.8 L (8.0 pts)
O-haul Fill - 42RLE	8.3 L (17.6 pts.)
Dry fill capacity Depending on type and size of internal cooler, length and inside diameter of cooler lines, or use of an auxiliary cooler, these figures may vary. (Refer to 21 - TRANSMISSION/AUTOMATIC - 42RLE/FLUID - STANDARD PROCEDURE)	
TRANSFER CASE	
NV231	1.0 L (2.2 pts.)
NV241	2.0 L (4.2 pts.)
MANUAL TRANSMISSION	
NSG 370 Approximate dry fill or fill to bottom edge of the fill plug hole.	1.5 L (3.17 pts.)
FRONT AXLE ± .03 L (1 oz.)	
181 FBI (Model 30)	1.2 L (2.5 pts.)
216 RBI (Model 44)	1.89 L (4.0 pts.)
REAR AXLE ± .03 L (1 oz.)	
194 RBI (Model 35)	1.66 L (3.5 pts.)*
216 RBI (Model 44)	1.89 L (4.0 pts.)*
* With Trac-lok add 118 ml (4.0 oz.) of Limited Slip Additive.	
****Nominal refill capacities are shown. A variation may be observed from vehicle to vehicle due to manufacturing tolerance and refill procedure.	

MAINTENANCE SCHEDULES

DESCRIPTION

MAINTENANCE SCHEDULES

There are two maintenance schedules that show the **required** service for your vehicle.

First is Schedule **"B"**. It is for vehicles that are operated under the conditions that are listed below and at the beginning of the schedule.

- Day or night temperatures are below 32°F (0°C)
- Stop and go driving
- Excessive engine idling
- Driving in dusty conditions
- Short trips of less than 10 miles (16.2 km)
- More than 50% of your driving is at sustained high speeds during hot weather, above 90°F (32°C)
- Trailer towing
- Taxi, police, or delivery service (commercial service)
- Off-road or desert driving
- **If equipped for and operated with E-85 (ethanol) fuel.**

NOTE: If ANY of these apply to you then change your engine oil every 3,000 miles (5 000 km) or 3 months, whichever comes first and follow "Schedule B" of the "Maintenance Schedules" section of this manual.

NOTE: If ANY of these apply to you then flush and replace your engine coolant/anti-freeze every 102,000 miles (163 000 km) or 60 months, whichever comes first, and follow "Schedule B" of the "Maintenance Schedules" section of this manual.

NOTE: Most vehicles are operated under the conditions listed for Schedule "B."

Second is Schedule **"A"**. It is for vehicles that are not operated under any of the conditions listed under Schedule "B."

Use the schedule that best describes your driving conditions. Where time and mileage are listed, follow the interval that occurs first.

CAUTION: Failure to perform the required maintenance items may result in damage to the vehicle.

MAINTENANCE SCHEDULES (Continued)

At Each Stop for Fuel

- Check the engine oil level about 5 minutes after a fully warmed engine is shut off. Checking the oil level while the vehicle is on level ground will improve the accuracy of the oil level reading. Add oil only when the level is at or below the ADD or MIN mark.
- Check the windshield washer solvent, add as required.

Once a Month

- Check the tire pressure and look for unusual wear or damage.
- Inspect the battery and clean and tighten the terminals as required.
- Check the fluid levels of the coolant reservoir, brake master cylinder, power steering, and transmission, and add as needed.
- Check all lights and all other electrical items for correct operation.

At Each Oil Change

- Change the engine oil filter.
- Inspect the exhaust system.
- Inspect brake hoses.
- Check the coolant level, hoses, and clamps.
- Rotate the tires.
- Inspect manual transmission fluid level - if equipped.
- After completion of off-road operation, the underside of the vehicle should be thoroughly inspected. Examine threaded fasteners for looseness.

Schedule "B"

Follow this schedule if you usually operate your vehicle under one or more of the following conditions.

- Day or night temperatures are below 0°C (32°F)
- Stop and go driving
- Excessive engine idling
- Driving in dusty conditions
- Short trips of less than 16.2 km (10 miles)
- More than 50% of your driving is at sustained high speeds during hot weather, above 32°C (90°F)
- Trailer towing
- Taxi, police, or delivery service (commercial service)
- Off-road or desert driving
- **If equipped for and operated with E-85 (ethanol) fuel.**

NOTE: If ANY of these apply to you then change your engine oil every 3,000 miles (5 000 km) or 3 months, whichever comes first and follow "Schedule B" of the "Maintenance Schedules" section of this manual.

NOTE: If ANY of these apply to you then flush and replace your engine coolant/anti-freeze every 102,000 miles (163 000 km) or 60 months, whichever comes first, and follow "Schedule B" of the "Maintenance Schedules" section of this manual.

Miles (Kilometers)	**3,000 (5 000)**	**6,000 (10 000)**	**9,000 (14 000)**	**12,000 (19 000)**	**15,000 (24 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.				X	
Drain and refill the front and rear axle fluid‡				X	

MAINTENANCE SCHEDULES (Continued)

Miles	**18,000**	**21,000**	**24,000**	**27,000**	**30,000**
(Kilometers)	**(29 000)**	**(34 000)**	**(38 000)**	**(43 000)**	**(48 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.			X		
Drain and refill the front and rear axle fluid‡			X		
Inspect the transfer case fluid, add if necessary.					X

Miles	**33,000**	**36,000**	**39,000**	**42,000**	**45,000**
(Kilometers)	**(53 000)**	**(58 000)**	**(62 000)**	**(67 000)**	**(72 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.		X			
Drain and refill the front and rear axle fluid‡		X			
Inspect the drive belt and replace as needed.					X

MAINTENANCE SCHEDULES (Continued)

Miles	**48,000**	**51,000**	**54,000**	**57,000**	**60,000**
(Kilometers)	**(77 000)**	**(82 000)**	**(86 000)**	**(91 000)**	**(96 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Inspect the ignition cables, and replace if necessary (2.4L Only).					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.	X				X
Drain and refill the front and rear axle fluid‡	X				X
Drain and refill the automatic transmission fluid, and change filter.					X
Inspect the drive belt and replace as needed. Not required if belt was previously.					X
Drain and refill the transfer case fluid.					X
Flush and replace the engine coolant/anti-freeze at 60 months, if not done at 102,000 miles (163 000 km).					X

Miles	**63,000**	**66,000**	**69,000**	**72,000**	**75,000**
(Kilometers)	**(101 000)**	**(106 000)**	**(110 000)**	**(115 000)**	**(120 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.				X	
Drain and refill the front and rear axle fluid‡				X	
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X

MAINTENANCE SCHEDULES (Continued)

Miles **(Kilometers)**	**78,000 (125 000)**	**81,000 (130 000)**	**84,000 (134 000)**	**87,000 (139 000)**	**90,000 (144 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Replace the timing belt (2.4L Only) ◇.					X
Inspect the brake linings.			X		
Drain and refill the front and rear axle fluid‡			X		
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Inspect the transfer case fluid, add if necessary.					X

Miles **(Kilometers)**	**93,000 (149 000)**	**96,000 (154 000)**	**99,000 (158 000)**	**102,000 (163 000)**	**105,000 (168 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.		X			
Drain and refill the front and rear axle fluid‡		X			
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Flush and replace the engine coolant/anti-freeze, if not done at 60 months.				X	

MAINTENANCE SCHEDULES (Continued)

Miles (Kilometers)	**108,000 (173 000)**	**111,000 (178 000)**	**114,000 (182 000)**	**117,000 (187 000)**	**120,000 (192 000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Inspect the ignition cables, and replace if necessary (2.4L Only).					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.	X				X
Drain and refill the front and rear axle fluid‡	X				X
Drain and refill the automatic transmission fluid, and replace main sump filter.					X
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Drain and refill the transfer case fluid.					X
Flush and replace the engine coolant/anti-freeze at 120 months, if not replaced at 102,000 miles (163 000 km).					X

Inspection and service should also be performed anytime a malfunction is observed or suspected. Retain all receipts.

◇ This maintenance is recommended by the manufacturer to the owner, but is not required to maintain emissions warranty.

‡Off-highway operation, trailer towing, taxi, limousine, bus, snow plowing, or other types of commercial service or prolonged operation with heavy loading, especially in hot weather, require front and rear axle service indicated with a ‡ in Schedule “B”. Perform these services if the vehicle is usually operated under these conditions.

MAINTENANCE SCHEDULES (Continued)

Schedule "A"

Miles	**6,000**	**12,000**	**18,000**	**24,000**	**30,000**
(Kilometers)	**(10 000)**	**(19 000)**	**(29 000)**	**(38 000)**	**(48 000)**
[Months]	**[6]**	**[12]**	**[18]**	**[24]**	**[30]**
Change the engine oil and engine oil filter.	X	X	X	X	X
Rotate the tires.	X	X	X	X	X
Inspect the engine air filter element, and replace if necessary.					X
Replace the spark plugs.					X
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.			X		
Inspect the transfer case fluid.					X

Miles	**36,000**	**42,000**	**48,000**	**54, 000**
(Kilometers)	**(58 000)**	**(67 000)**	**(77 000)**	**(86 000)**
[Months]	**[36]**	**[42]**	**[48]**	**[54]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Inspect the brake linings.	X			X

Miles	**60,000**	**66,000**	**72,000**	**78,000**
(Kilometers)	**(96 000)**	**(106 000)**	**(115 000)**	**(125 000)**
[Months]	**[60]**	**[66]**	**[72]**	**[78]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Inspect the engine air filter element, and replace if necessary.	X			
Inspect the PCV Valve, and replace if necessary. ◇	X			
Replace the ignition cables (2.4L Only).	X			
Replace the spark plugs.	X			
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Inspect the brake linings.			X	
Inspect the drive belt, and replace as needed.	X			
Inspect the drive belt, and replace as needed. Not required if belt was previously replaced.			X	
Flush and replace the engine coolant/anti-freeze at 60 months, if not done at 102,000 miles (163 000 km).	X			
Inspect transfer case fluid.	X			

MAINTENANCE SCHEDULES (Continued)

Miles **(Kilometers)** **[Months]**	**84,000** **(134 000)** **[84]**	**90,000** **(144 000)** **[90]**	**96,000** **(154 000)** **[96]**	**102, 000** **(163 000)** **[102]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Inspect the engine air filter element, and replace if necessary.		X		
Inspect the PCV Valve, and replace if necessary. ◊		X		
Replace the spark plugs.		X		
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Inspect the brake linings.		X		
Inspect the drive belt, and replace as needed. Not required if previously replaced.		X		X
Flush and replace the engine coolant/anti-freeze if not done at 60 months.				X
Inspect the transfer case fluid.		X		

Miles **(Kilometers)** **[Months]**	**108,000** **(173 000)** **[108]**	**114,000** **(182 000)** **[114]**	**120,000** **(192 000)** **[120]**
Change the engine oil and engine oil filter.	X	X	X
Rotate the tires.	X	X	X
Inspect the engine air filter element, and replace if necessary.			X
Replace the ignition cables (2.4L Only).			X
Inspect the PCV Valve, and replace if necessary. ◊			X
Replace the spark plugs.			X
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X
Lubricate the steering and suspension ball joints.	X		X
Inspect the brake linings.	X		
Replace the timing belt (2.4L Only). ◊			X
Inspect the drive belt, and replace as needed. Not required if previously replaced.			X
Drain and refill the transfer case fluid.			X
Flush and replace the engine coolant/anti-freeze, if not done at 102,000 miles (163 000 km).			X

Inspection and service should also be performed anytime a malfunction is observed or suspected. Retain all receipts.

◊ This maintenance is recommended by the manufacturer to the owner, but is not required to maintain emissions warranty.

MAINTENANCE SCHEDULES (Continued)

MAINTENANCE SCHEDULES – EXPORT

There are two maintenance schedules that show the **required** service for your vehicle.

First is Schedule **"A"**. It is for vehicles that are not operated under any of the conditions listed under Schedule "B."

Second is Schedule **"B"**. It is for vehicles that are operated under the conditions that are listed below and at the beginning of the schedule.

- Day or night temperatures are below 0°C (32°F)
- Stop and go driving
- Excessive engine idling
- Driving in dusty conditions
- Short trips of less than 16.2 km (10 miles)
- More than 50% of your driving is at sustained high speeds during hot weather, above 32°C (90°F)
- Trailer towing
- Taxi, police, or delivery service (commercial service)
- Off-road or desert driving
- **If equipped for and operated with E-85 (ethanol) fuel.**

NOTE: Most vehicles are operated under the conditions listed for Schedule "B."

Use the schedule that best describes your driving conditions. Where time and mileage are listed, follow the interval that occurs first.

CAUTION: Failure to perform the required maintenance items may result in damage to the vehicle.

At Each Stop for Fuel

- Check the engine oil level about 5 minutes after a fully warmed engine is shut off. Checking the oil level while the vehicle is on level ground will improve the accuracy of the oil level reading. Add oil only when the level is at or below the ADD or MIN mark.
- Check the windshield washer solvent, add as required.

Once a Month

- Check the tire pressure and look for unusual wear or damage.
- Inspect the battery and clean and tighten the terminals as required.
- Check the fluid levels of the coolant reservoir, brake master cylinder, power steering, and transmission, and add as needed.
- Check all lights and all other electrical items for correct operation.

At Each Oil Change

- Change the engine oil filter.
- Inspect the exhaust system.
- Inspect brake hoses.
- Check the coolant level, hoses, and clamps.
- Rotate the tires.
- Inspect manual transmission fluid level - if equipped.
- After completion of off-road operation, the underside of the vehicle should be thoroughly inspected. Examine threaded fasteners for looseness.

Schedule "A"

Kilometers **(Miles)** **[Months]**	**10 000** **(6,000)** **[6]**	**19 000** **(12,000)** **[12]**	**29 000** **(18,000)** **[18]**	**38 000** **(24,000)** **[24]**	**48 000** **(30,000)** **[30]**
Change the engine oil and engine oil filter.	X	X	X	X	X
Rotate the tires.	X	X	X	X	X
Inspect the engine air filter element, and replace if necessary.					X
Replace the spark plugs.					X
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.			X		
Inspect the transfer case fluid.					X

MAINTENANCE SCHEDULES (Continued)

Kilometers **(Miles)** **[Months]**	**58 000** **(36,000)** **[36]**	**67 000** **(42,000)** **[42]**	**77 000** **(48,000)** **[48]**	**86 000** **(54, 000)** **[54]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Inspect the brake linings.	X			X

Kilometers **(Miles)** **[Months]**	**96 000** **(60,000)** **[60]**	**106 000** **(66,000)** **[66]**	**115 000** **(72,000)** **[72]**	**125 000** **(78,000)** **[78]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Inspect the engine air filter element, and replace if necessary.	X			
Inspect the PCV Valve, and replace if necessary. ◇	X			
Replace the ignition cables (2.4L Only).	X			
Replace the spark plugs.	X			
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Inspect the brake linings.			X	
Inspect the drive belt, and replace as needed.	X			
Inspect the drive belt, and replace as needed. Not required if belt was previously replaced.			X	
Flush and replace the engine coolant/anti-freeze at 60 months, if not done at 163 000 km (102,000 miles).	X			
Inspect transfer case fluid.	X			

MAINTENANCE SCHEDULES (Continued)

Kilometers **(Miles)** **[Months]**	**134 000** **(84,000)** **[84]**	**144 000** **(90,000)** **[90]**	**154 000** **(96,000)** **[96]**	**163 000** **(102, 000)** **[102]**
Change the engine oil and engine oil filter.	X	X	X	X
Rotate the tires.	X	X	X	X
Inspect the engine air filter element, and replace if necessary.		X		
Inspect the PCV Valve, and replace if necessary. ◊		X		
Replace the spark plugs.		X		
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X	
Inspect the brake linings.		X		
Inspect the drive belt, and replace as needed. Not required if previously replaced.		X		X
Flush and replace the engine coolant/anti-freeze if not done at 60 months.				X
Inspect the transfer case fluid.		X		

Kilometers **(Miles)** **[Months]**	**173 000** **(108,000)** **[108]**	**182 000** **(114,000)** **[114]**	**192 000** **(120,000)** **[120]**
Change the engine oil and engine oil filter.	X	X	X
Rotate the tires.	X	X	X
Inspect the engine air filter element, and replace if necessary.			X
Replace the ignition cables (2.4L Only).			X
Inspect the PCV Valve, and replace if necessary. ◊			X
Replace the spark plugs.			X
Lubricate the steering linkage joints and outer tie rod ends.	X	X	X
Lubricate the steering and suspension ball joints.	X		X
Inspect the brake linings.	X		
Replace the timing belt (2.4L Only). ◊			X
Inspect the drive belt, and replace as needed. Not required if previously replaced.			X
Drain and refill the transfer case fluid.			X
Flush and replace the engine coolant/anti-freeze, if not done at 163 000 km (102,000 miles).			X

Inspection and service should also be performed anytime a malfunction is observed or suspected. Retain all receipts.

◊ This maintenance is recommended by the manufacturer to the owner, but is not required to maintain emissions warranty.

MAINTENANCE SCHEDULES (Continued)

Schedule "B"

Follow schedule "B" if you usually operate your vehicle under one or more of the following conditions.

- Day or night temperatures are below 0°C (32°F).
- Stop and go driving.
- Extensive engine idling.
- Driving in dusty conditions.
- Short trips of less than 16.2 km (10 miles).
- More than 50% of your driving is at sustained high speeds during hot weather, above 32°C (90°F).
- Trailer towing.
- Taxi, police, or delivery service (commercial service).
- Off-road or desert driving.
- **If equipped for and operated with E-85 (ethanol) fuel.**

Kilometers	**5 000**	**10 000**	**14 000**	**19 000**	**24 000**
(Miles)	**(3,000)**	**(6,000)**	**(9,000)**	**(12,000)**	**(15,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.				X	
Drain and refill the front and rear axle fluid‡				X	

Kilometers	**29 000**	**34 000**	**38 000**	**43 000**	**48 000**
(Miles)	**(18,000)**	**(21,000)**	**(24,000)**	**(27,000)**	**(30,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.			X		
Drain and refill the front and rear axle fluid‡			X		
Inspect the transfer case fluid, add if necessary.					X

MAINTENANCE SCHEDULES (Continued)

Kilometers	**53 000**	**58 000**	**62 000**	**67 000**	**72 000**
(Miles)	**(33,000)**	**(36,000)**	**(39,000)**	**(42,000)**	**(45,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.		X			
Drain and refill the front and rear axle fluid‡		X			
Inspect the drive belt and replace as needed.					X

Kilometers	**77 000**	**82 000**	**86 000**	**91 000**	**96 000**
(Miles)	**(48,000)**	**(51,000)**	**(54,000)**	**(57,000)**	**(60,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Inspect the ignition cables, and replace if necessary (2.4L Only).					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.	X				X
Drain and refill the front and rear axle fluid‡	X				X
Drain and refill the automatic transmission fluid, and change filter.					X
Inspect the drive belt and replace as needed. Not required if belt was previously.					X
Drain and refill the transfer case fluid.					X
Flush and replace the engine coolant/anti-freeze at 60 months, if not done at 163 000 km (102,000 miles).					X

MAINTENANCE SCHEDULES (Continued)

Kilometers	**101 000**	**106 000**	**110 000**	**115 000**	**120 000**
(Miles)	**(63,000)**	**(66,000)**	**(69,000)**	**(72,000)**	**(75,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.				X	
Drain and refill the front and rear axle fluid‡				X	
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X

Kilometers	**125 000**	**130 000**	**134 000**	**139 000**	**144 000**
(Miles)	**(78,000)**	**(81,000)**	**(84,000)**	**(87,000)**	**(90,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Replace the timing belt (2.4L Only) ◇.					X
Inspect the brake linings.			X		
Drain and refill the front and rear axle fluid‡			X		
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Inspect the transfer case fluid, add if necessary.					X

MAINTENANCE SCHEDULES (Continued)

Kilometers	**149 000**	**154 000**	**158 000**	**163 000**	**168 000**
(Miles)	**(93,000)**	**(96,000)**	**(99,000)**	**(102,000)**	**(105,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.		X		X	
Inspect the engine air filter element, replace if necessary.					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.		X		X	
Inspect the brake linings.		X			
Drain and refill the front and rear axle fluid‡		X			
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Flush and replace the engine coolant/anti-freeze, if not done at 60 months.				X	

Kilometers	**173 000**	**178 000**	**182 000**	**187 000**	**192 000**
(Miles)	**(108,000)**	**(111,000)**	**(114,000)**	**(117,000)**	**(120,000)**
Change the engine oil and engine oil filter, if not replaced at 3 months.	X	X	X	X	X
Rotate the tires.	X		X		X
Inspect the engine air filter element, replace if necessary.					X
Inspect the PCV Valve, and replace if necessary. ◇					X
Replace the spark plugs.					X
Inspect the ignition cables, and replace if necessary (2.4L Only).					X
Lubricate the steering linkage tie rod ends.	X	X	X	X	X
Lubricate the steering and suspension ball joints.	X		X		X
Inspect the brake linings.	X				X
Drain and refill the front and rear axle fluid‡	X				X
Drain and refill the automatic transmission fluid, and replace main sump filter.					X
Inspect the drive belt and replace as needed. Not required if belt was previously replaced.					X
Drain and refill the transfer case fluid.					X
Flush and replace the engine coolant/anti-freeze at 120 months, if not replaced at 163 000 km (102,000 miles).					X

Inspection and service should also be performed anytime a malfunction is observed or suspected. Retain all receipts.

◇ This maintenance is recommended by the manufacturer to the owner, but is not required to maintain emissions warranty.

‡Off-highway operation, trailer towing, taxi, limousine, bus, snow plowing, or other types of commercial service or prolonged operation with heavy loading, especially in hot weather, require front and rear axle service indicated with a ‡ in Schedule “B”. Perform these services if the vehicle is usually operated under these conditions.

HOISTING

STANDARD PROCEDURE - HOISTING RECOMMENDATIONS

Refer to the Owner's Manual for emergency vehicle lifting procedures.

When properly positioned, a floor jack can be used to lift a Jeep vehicle (Fig. 5). Support the vehicle in the raised position with jack stands at the front and rear ends of the frame rails.

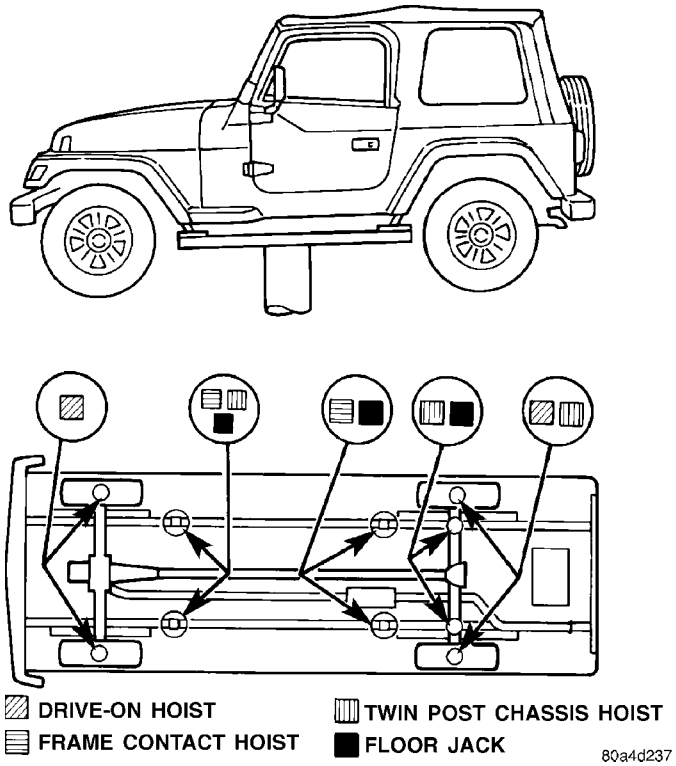

Fig. 5 Vehicle Lifting Locations

CAUTION: Do not attempt to lift a Jeep vehicle with a floor jack positioned under:

- An axle tube.
- A body side sill.
- A steering linkage component.
- A drive shaft.
- The engine or transmission oil pan.
- The fuel tank.
- A front suspension arm.
- Transfer case.

NOTE: Use the correct sub-frame rail or frame rail lifting locations only.

HOIST

Refer to the Owner's Manual for emergency vehicle lifting procedures.

A vehicle can be lifted with:

- A single-post, frame-contact hoist.
- A twin-post, chassis hoist.
- A ramp-type, drive-on hoist.

NOTE: When a frame-contact type hoist is used, verify that the lifting pads are positioned properly.

WARNING: THE HOISTING AND JACK LIFTING POINTS PROVIDED ARE FOR A COMPLETE VEHICLE. WHEN A CHASSIS OR DRIVETRAIN COMPONENT IS REMOVED FROM A VEHICLE, THE CENTER OF GRAVITY IS ALTERED MAKING SOME HOISTING CONDITIONS UNSTABLE. PROPERLY SUPPORT OR SECURE VEHICLE TO HOISTING DEVICE WHEN THESE CONDITIONS EXIST.

TOWING

STANDARD PROCEDURE - TOWING RECOMMENDATIONS

DaimlerChrysler Corporation recommends that a 4WD vehicle be transported on a flat-bed device. A Wheel-lift or front end attached Sling-type device can be used provided all the wheels are lifted off the ground using tow dollies (Fig. 6) and (Fig. 7).

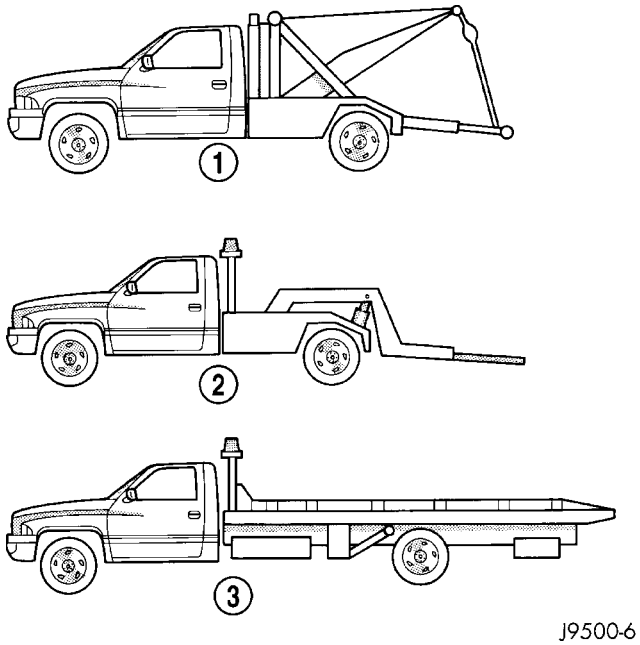

Fig. 6 Tow Vehicles With Approved Equipment

1 - SLING TYPE
2 - WHEEL LIFT
3 - FLAT BED

SAFETY PRECAUTIONS

- Secure loose and protruding parts.

TOWING (Continued)

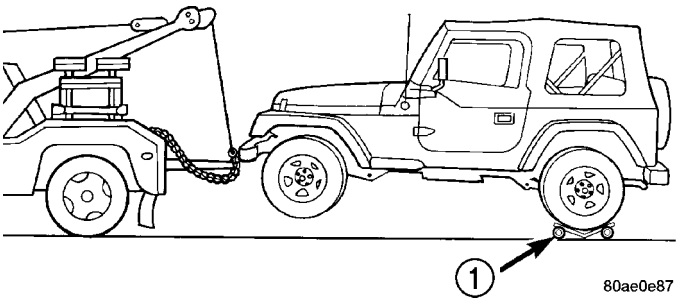

Fig. 7 Towing With Tow Dollies

1 - TOW DOLLY

- Always use a safety chain system that is independent of the lifting and towing equipment.
- Do not allow towing equipment to contact the disabled vehicle's fuel tank.
- Do not allow anyone under the disabled vehicle while it is lifted by the towing device.
- Do not allow passengers to ride in a vehicle being towed.
- Always observe state and local laws regarding towing regulations.
- Do not tow a vehicle in a manner that could jeopardize the safety of the operator, pedestrians or other motorists.
- Do not attach tow chains, T-hooks, J-hooks, or a tow sling to a bumper, steering linkage, drive shafts or a non-reinforced frame hole.

GROUND CLEARANCE

CAUTION: If vehicle is towed with wheels removed, install lug nuts to retain brake drums.

A towed vehicle should be raised until lifted wheels are a minimum 100 mm (4 in) from the ground. Be sure there is adequate ground clearance at the opposite end of the vehicle, especially when towing over rough terrain or steep rises in the road. If necessary, remove the wheels from the lifted end of the vehicle and lower the vehicle closer to the ground, to increase the ground clearance at the opposite end of the vehicle. Install lug nuts on wheel attaching studs to retain brake drums.

FLAT-BED TOWING RAMP ANGLE

If a vehicle with flat-bed towing equipment is used, the approach ramp angle should not exceed 15 degrees.

VEHICLE TOWING

WARNING: WHEN TOWING A DISABLED VEHICLE AND THE DRIVE WHEELS ARE SECURED IN A WHEEL LIFT OR TOW DOLLIES, ENSURE THE TRANSMISSION IS IN THE PARK POSITION (AUTOMATIC TRANSMISSION) OR A FORWARD DRIVE GEAR (MANUAL TRANSMISSION).DO NOT ATTACH SLING-TYPE TOWING EQUIPMENT TO THE REAR OF A TJ.

TOWING-FRONT END LIFTED (WHEEL LIFT)

(1) Raise the rear of the vehicle off the ground and install tow dollies under rear wheels.

(2) Attach the wheel lift to the front wheels.

TOWING-REAR END LIFTED (WHEEL LIFT ONLY)

(1) Raise the front of the vehicle off the ground and install tow dollies under front wheels.

(2) Attach the wheel lift to the rear wheels.

TOWING-FRONT END LIFTED (SLING-TYPE)

(1) Raise the rear of the vehicle off the ground and install tow dollies under rear wheels.

(2) Attach T-hooks to the access holes on the outboard side of the frame rails (Fig. 8).

(3) Before tightening the chain, position a protective pad between the chain and the bumper.

(4) Attach the safety chains to the vehicle (Fig. 9).

(5) Turn the ignition switch to the OFF position to unlock the steering wheel.

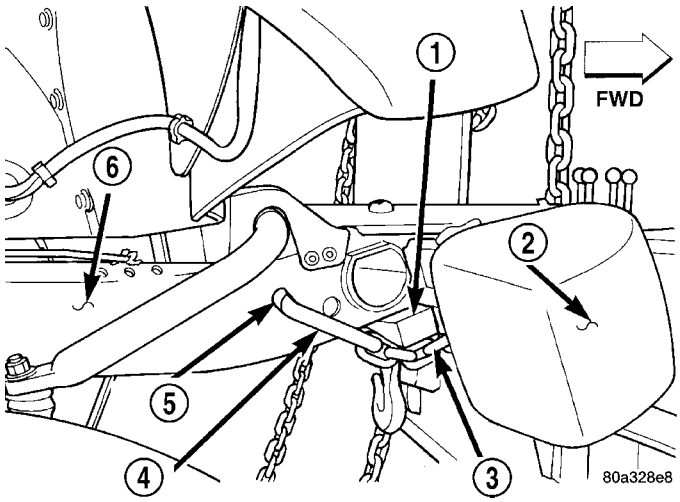

Fig. 8 T-Hook Attachment

1 - PROTECTIVE PAD
2 - BUMPER
3 - CHAIN
4 - T-HOOK
5 - ACCESS HOLE
6 - FRAME

TOWING (Continued)

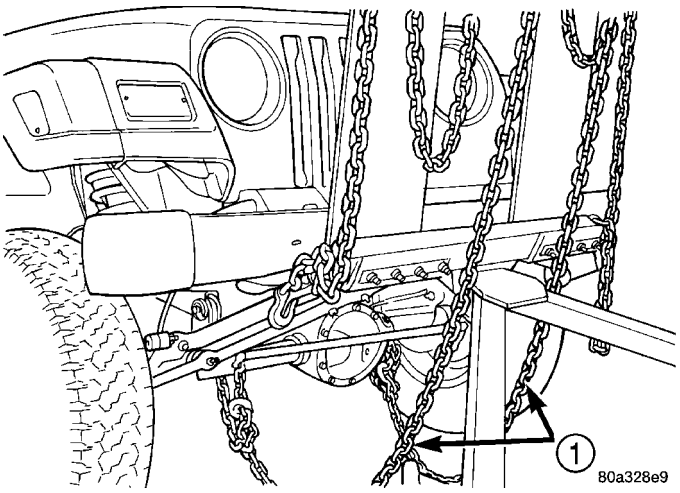

Fig. 9 Safety Chain Attachment

1 - SAFETY CHAIN

RECREATIONAL TOWING

Refer to the Owners Manual for towing procedures.

EMERGENCY TOW HOOKS

WARNING: REMAIN AT A SAFE DISTANCE FROM A VEHICLE THAT IS BEING TOWED VIA ITS TOW HOOKS. THE TOW STRAPS/CHAINS COULD BREAK AND CAUSE SERIOUS INJURY.

Some Jeep vehicles are equipped with front emergency tow hooks. The tow hooks should be used for **EMERGENCY** purposes only.

CAUTION: DO NOT use emergency tow hooks for tow truck hook-up or highway towing.

JUMP STARTING

STANDARD PROCEDURE - JUMP STARTING PROCEDURE

WARNING: REVIEW ALL SAFETY PRECAUTIONS AND WARNINGS IN THE BATTERY SYSTEM SECTION OF THE SERVICE MANUAL. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - STANDARD PROCEDURE)

- **DO NOT JUMP START A FROZEN BATTERY, PERSONAL INJURY CAN RESULT.**
- **IF EQUIPPED, DO NOT JUMP START WHEN MAINTENANCE FREE BATTERY INDICATOR DOT IS YELLOW OR BRIGHT COLOR.**
- **DO NOT JUMP START A VEHICLE WHEN THE BATTERY FLUID IS BELOW THE TOP OF LEAD PLATES.**
- **DO NOT ALLOW JUMPER CABLE CLAMPS TO TOUCH EACH OTHER WHEN CONNECTED TO A BOOSTER SOURCE.**
- **DO NOT USE OPEN FLAME NEAR BATTERY.**
- **REMOVE METALLIC JEWELRY WORN ON HANDS OR WRISTS TO AVOID INJURY BY ACCIDENTAL ARCING OF BATTERY CURRENT.**
- **WHEN USING A HIGH OUTPUT BOOSTING DEVICE, DO NOT ALLOW BATTERY VOLTAGE TO EXCEED 16 VOLTS. REFER TO INSTRUCTIONS PROVIDED WITH DEVICE BEING USED.**

FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN PERSONAL INJURY.

CAUTION: When using another vehicle as a booster, do not allow vehicles to touch. Electrical systems can be damaged on either vehicle.

TO JUMP START A DISABLED VEHICLE:

(1) Raise hood on disabled vehicle and visually inspect engine compartment for:

- Generator drive belt condition and tension.
- Fuel fumes or leakage, correct if necessary.
- Frozen battery.
- Yellow or bright color test indicator, if equipped.
- Low battery fluid level.

CAUTION: If the cause of starting problem on disabled vehicle is severe, damage to booster vehicle charging system can result.

(2) When using another vehicle as a booster source, turn off all accessories, place gear selector in park or neutral, set park brake or equivalent and operate engine at 1200 rpm.

(3) On disabled vehicle, place gear selector in park or neutral and set park brake or equivalent. Turn OFF all accessories.

(4) Connect jumper cables to booster battery. RED clamp to positive terminal (+). BLACK clamp to negative terminal (-). DO NOT allow clamps at opposite end of cables to touch, electrical arc will result (Fig. 10). Review all warnings in this procedure.

(5) On disabled vehicle, connect RED jumper cable clamp to battery positive (+) terminal. Connect BLACK jumper cable clamp to the engine as close to the ground cable connection as possible (Fig. 10).

CAUTION: Do not crank starter motor on disabled vehicle for more than 15 seconds, starter will overheat and could fail.

JUMP STARTING (Continued)

(6) Allow battery in disabled vehicle to charge to at least 12.4 volts (75% charge) before attempting to start engine. If engine does not start within 15 seconds, stop cranking engine and allow starter to cool (15 min.), before cranking again.

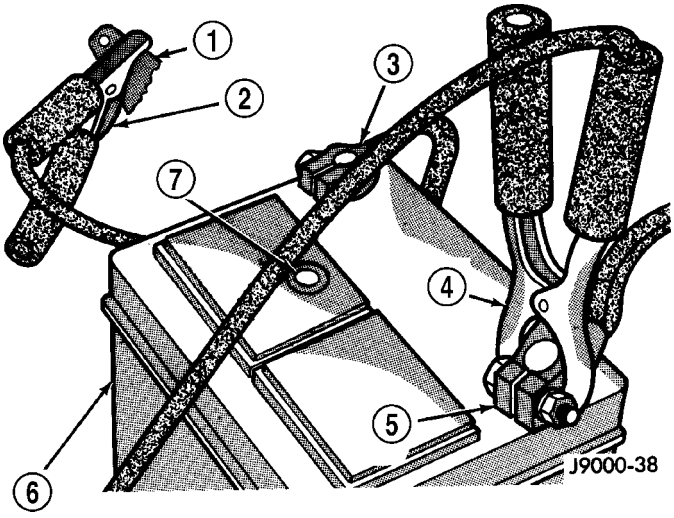

Fig. 10 Jumper Cable Clamp Connections

1 - ENGINE GROUND
2 - NEGATIVE JUMPER CABLE
3 - BATTERY NEGATIVE CABLE
4 - POSITIVE JUMPER CABLE
5 - BATTERY POSITIVE CABLE
6 - BATTERY
7 - TEST INDICATOR (IF EQUIPPED)

DISCONNECT CABLE CLAMPS AS FOLLOWS:

- Disconnect BLACK cable clamp from engine ground on disabled vehicle.
- When using a Booster vehicle, disconnect BLACK cable clamp from battery negative terminal. Disconnect RED cable clamp from battery positive terminal.
- Disconnect RED cable clamp from battery positive terminal on disabled vehicle.

EMERGENCY TOW HOOKS

DESCRIPTION — EMERGENCY TOW HOOKS

WARNING: REMAIN AT A SAFE DISTANCE FROM A VEHICLE THAT IS BEING TOWED VIA ITS TOW HOOKS. THE TOW STRAPS/CHAINS COULD BREAK AND CAUSE SERIOUS INJURY.

Some Jeep vehicles are equipped with front emergency tow hooks. The tow hooks should be used for **EMERGENCY** purposes only.

CAUTION: DO NOT use emergency tow hooks for tow truck hook-up or highway towing.

SUSPENSION

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WHEEL ALIGNMENT

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WHEEL ALIGNMENT

DESCRIPTION

Wheel alignment involves the correct positioning of the wheels in relation to the vehicle. The positioning is accomplished through suspension and steering linkage adjustments. An alignment is considered essential for efficient steering, good directional stability and to minimize tire wear. The most important measurements of an alignment are caster, camber and toe position (Fig. 1).

CAUTION: Never attempt to modify suspension or steering components by heating or bending.

CAUTION: Components attached with a nut and cotter pin must be torqued to specification. Then if the slot in the nut does not line up with the cotter pin hole, tighten nut until it is aligned. Never loosen the nut to align the cotter pin hole.

NOTE: Periodic lubrication of the front suspension/ steering system components may be required. Rubber bushings must never be lubricated. Refer to Lubrication And Maintenance for the recommended maintenance schedule.

OPERATION

- **CASTER** is the forward or rearward tilt of the steering knuckle from vertical. Tilting the top of the knuckle rearward provides positive caster. Tilting the top of the knuckle forward provides negative caster. Caster is a directional stability angle. This angle enables the front wheels to return to a straight ahead position after turns (Fig. 1)
- **CAMBER** is the inward or outward tilt of the wheel relative to the center of the vehicle. Tilting the top of the wheel inward provides negative camber. Tilting the top of the wheel outward provides positive camber. Incorrect camber will cause wear on the inside or outside edge of the tire. The angle is not adjustable, damaged component(s) must be replaced to correct the camber angle (Fig. 1)
- **WHEEL TOE POSITION** is the difference between the leading inside edges and trailing inside edges of the front tires. Incorrect wheel toe position is the most common cause of unstable steering and uneven tire wear. The wheel toe position is the **final** front wheel alignment adjustment (Fig. 1)
- **STEERING AXIS INCLINATION ANGLE** is measured in degrees and is the angle that the steering knuckles are tilted. The inclination angle has a fixed relationship with the camber angle. It will not change except when a spindle or ball stud is damaged or bent. The angle is not adjustable, damaged

WHEEL ALIGNMENT (Continued)

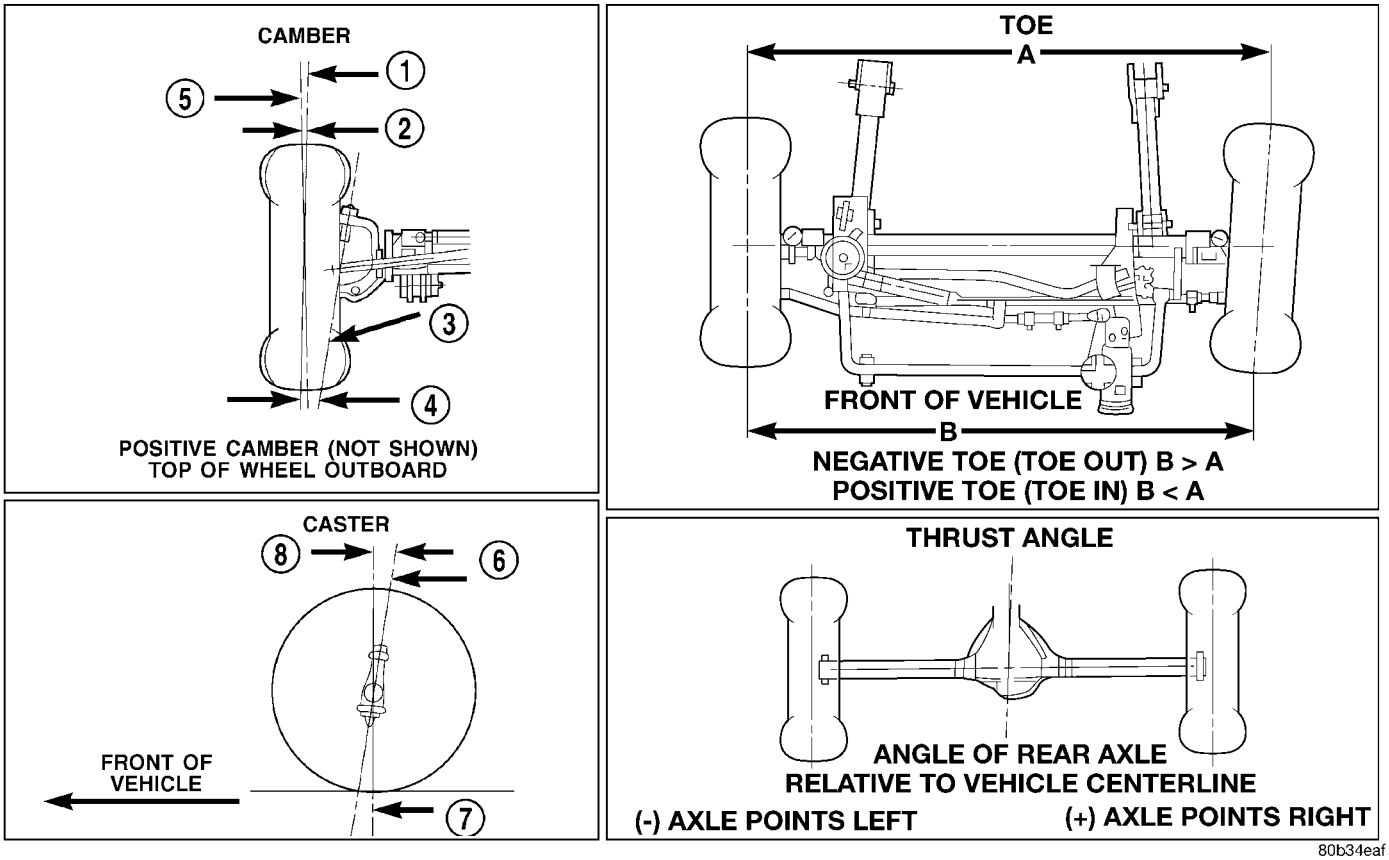

Fig. 1 Wheel Alignment Measurements

1 - WHEEL CENTERLINE
2 - NEGATIVE CAMBER ANGLE
3 - PIVOT CENTERLINE
4 - SCRUB RADIUS
5 - TRUE VERTICAL
6 - KING PIN
7 - VERTICAL
8 - POSITIVE CASTER

component(s) must be replaced to correct the steering axis inclination angle (Fig. 1)

- **THRUST ANGLE** is the angle of the rear axle relative to the centerline of the vehicle. Incorrect thrust angle can cause off-center steering and excessive tire wear. This angle is not adjustable, damaged component(s) must be replaced to correct the thrust angle (Fig. 1)

DIAGNOSIS AND TESTING - SUSPENSION AND STEERING SYSTEM

CONDITION	POSSIBLE CAUSES	CORRECTION
FRONT END NOISE	1. Loose or worn wheel bearings.	1. Adjust or replace wheel bearings.
	2. Loose or worn steering or suspension components.	2. Tighten or replace components as necessary.
EXCESSIVE PLAY IN STEERING	1. Loose or worn wheel bearings.	1. Adjust or replace wheel bearings.
	2. Loose or worn steering or suspension components.	2. Tighten or replace components as necessary.
	3. Loose or worn steering gear.	3. Adjust or replace steering gear.

WHEEL ALIGNMENT (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
FRONT WHEELS SHIMMY	1. Loose or worn wheel bearings.	1. Adjust or replace wheel bearings.
	2. Loose or worn steering or suspension components.	2. Tighten or replace components as necessary.
	3. Tires worn or out of balance.	3. Replace or balance tires.
	4. Alignment.	4. Align vehicle to specifications.
	5. Leaking steering dampener.	5. Replace steering dampener.
VEHICLE INSTABILITY	1. Loose or worn wheel bearings.	1. Adjust or replace wheel bearings.
	2. Loose or worn steering or suspension components.	2. Tighten or replace components as necessary.
	3. Tire pressure.	3. Adjust tire pressure.
	4. Alignment.	4. Align vehicle to specifications.
EXCESSIVE STEERING EFFORT	1. Loose or worn steering gear.	1. Adjust or replace steering gear.
	2. Power steering fluid low.	2. Add fluid and repair leak.
	3. Column coupler binding.	3. Replace coupler.
	4. Tire pressure.	4. Adjust tire pressure.
	5. Alignment.	5. Align vehicle to specifications.
VEHICLE PULLS TO ONE SIDE DURING BRAKING	1. Uneven tire pressure.	1. Adjust tire pressure.
	2. Worn brake components.	2. Repair brakes as necessary.
	3. Air in brake line.	3. Repair as necessary.
VEHICLE LEADS OR DRIFTS FROM STRAIGHT AHEAD DIRECTION ON UNCROWNED ROAD	1. Radial tire lead.	1. Cross front tires.
	2. Brakes dragging.	2. Repair brake as necessary.
	3. Weak or broken spring.	3. Replace spring.
	4. Uneven tire pressure.	4. Adjust tire pressure.
	5. Wheel Alignment.	5. Align vehicle.
	6. Loose or worn steering or suspension components.	6. Repair as necessary.
	7. Cross caster out of spec.	7. Align vehicle.
KNOCKING, RATTLING OR SQUEAKING	1. Worn shock bushings.	1. Replace shock.
	2. Loose, worn or bent steering/ suspension components.	2. Inspect, tighten or replace components as necessary.
	3. Shock valve.	3. Replace shock.
IMPROPER TRACKING	1. Loose, worn or bent track bar.	1. Inspect, tighten or replace component as necessary.
	2. Loose, worn or bent steering/ suspension components.	2. Inspect, tighten or replace components as necessary.

WHEEL ALIGNMENT (Continued)

STANDARD PROCEDURE

STANDARD PROCEDURE - CAMBER

Before each alignment reading the vehicle should be jounced (rear first, then front). Grasp each bumper at the center and jounce the vehicle up and down three times. Always release the bumper in the down position.

The wheel camber angle is preset. This angle is not adjustable and cannot be altered.

STANDARD PROCEDURE - CASTER

Before each alignment reading the vehicle should be jounced (rear first, then front). Grasp each bumper at the center and jounce the vehicle up and down three times. Always release the bumper in the down position.

Check the caster of the front axle for correct angle. Be sure the axle is not bent or twisted. Road test the vehicle and observe the steering wheel return-to-center position. Low caster will cause poor steering wheel returnability.

During the road test, turn the vehicle to both the left and right. If the steering wheel returns to the center position unassisted, the caster angle is correct. However, if steering wheel does not return toward the center position unassisted, a low caster angle is probable.

Caster can be adjusted by installing cam bolts and rotating the cams on the lower suspension arm (Fig. 2).

NOTE: Changing caster angle will also change the front propeller shaft angle. The propeller shaft angle has priority over caster. Refer to Group 3, Differential and Driveline for additional information.

STANDARD PROCEDURE - TOE POSITION

Before each alignment reading the vehicle should be jounced (rear first, then front). Grasp each bumper at the center and jounce the vehicle up and down three times. Always release the bumper in the down position.

NOTE: The wheel toe position adjustment is the final adjustment. This adjustment must be performed with the engine running, if the vehicle is equipped with power steering.

(1) Start the engine and turn wheels both ways before straightening the steering wheel. Center and secure the steering wheel.

(2) Loosen the adjustment sleeve clamp bolts (Fig. 3).

(3) Adjust the right wheel toe position with the drag link (Fig. 4). Turn the sleeve until the right wheel is at the correct positive TOE-IN position. Position the clamp bolts as shown (Fig. 3) and tighten to 49 N·m (36 ft. lbs.). **Make sure the toe setting does not change during clamp tightening.**

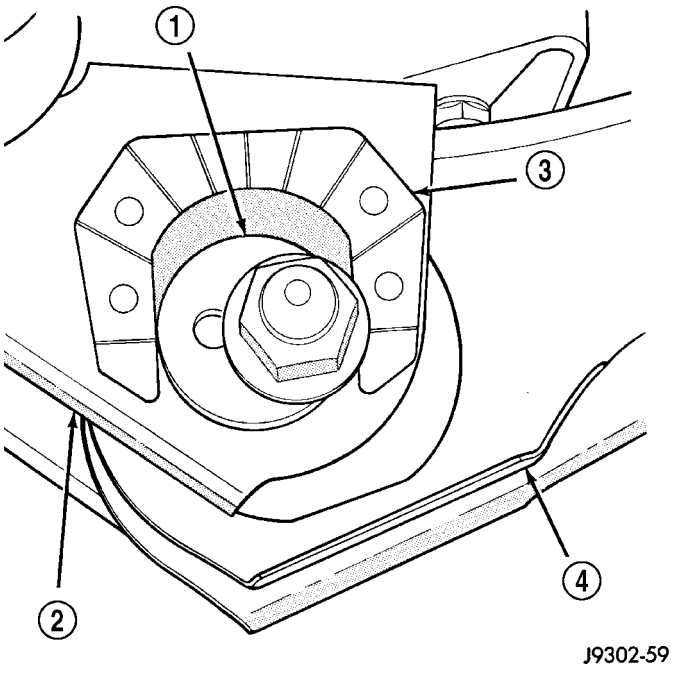

Fig. 2 Cam Adjuster

1 - ADJUSTMENT CAM
2 - AXLE BRACKET
3 - BRACKET REINFORCEMENT
4 - LOWER SUSPENSION ARM

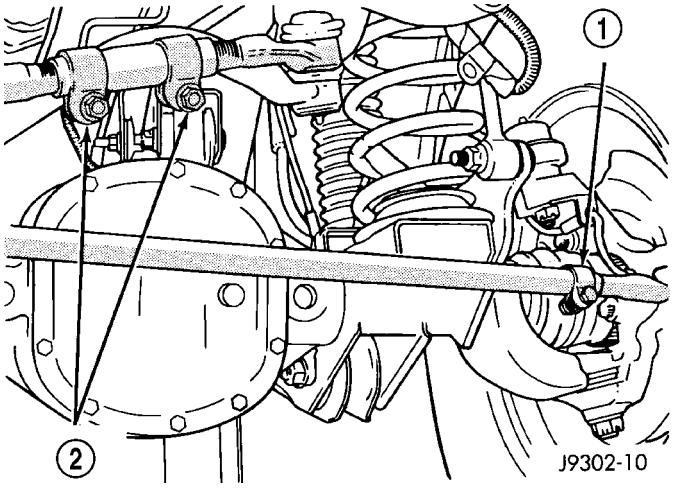

Fig. 3 Drag Link and Tie Rod Clamp

1 - TIE ROD CLAMP
2 - DRAG LINK CLAMPS

(4) Adjust the left wheel toe position with the tie rod. Turn the sleeve until the left wheel is at the same TOE-IN position as the right wheel. Position the clamp bolts as shown (Fig. 3) and tighten to 27

WHEEL ALIGNMENT (Continued)

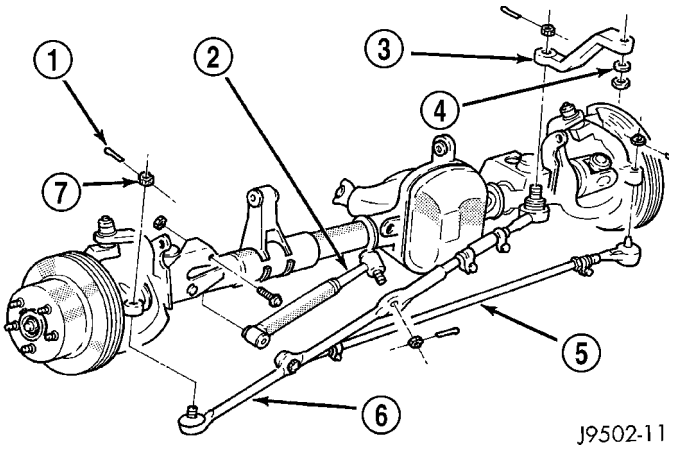

Fig. 4 Steering Linkage

1 - COTTER PIN
2 - DAMPENER
3 - PITMAN ARM
4 - WASHER
5 - TIE ROD
6 - DRAG LINK
7 - NUT

N·m (20 ft. lbs.). **Make sure the toe setting does not change during clamp tightening.**

(5) Verify the right toe specifications and turn off the engine.

SPECIFICATIONS

ALIGNMENT SPECIFICATIONS

NOTE: Alignment specifications are in degrees.

SPECIFICATIONS

DESCRIPTION	SPECIFICATION		
PREFERRED	CASTER + 7.0°	CAMBER (fixed angle) − 0.25°	TOTAL TOE-IN + 0.15° (each front wheel)
RANGE	±1.0°	± 0.63°	±0.06°
MAX RT/LT DIFFERENCE	0.65°	±1.0°	.06°
REAR SPECIFICATION			
PREFERRED	N/A	REAR CAMBER –0.25°	TOTAL TOE-IN +0.25°
RANGE	N/A	0° to –.50°	0° to .5°
THRUST ANGLE 0° ± 0.25°			

FRONT

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FRONT

DESCRIPTION

FRONT SUSPENSION

The front suspension is a link/coil design comprised of:

- Shock absorbers
- Jounce Bumper
- Coil springs
- Upper and lower suspension arms
- Stabilizer bar
- Track bar

CAUTION: Components attached with a nut and cotter pin must be torqued to specification. Then if the slot in the nut does not line up with the cotter pin hole, tighten nut until it is aligned. Never loosen the nut to align the cotter pin hole.

CAUTION: Suspension components with rubber/urethane bushings (except stabilizer bar) should be tightened with the vehicle at normal ride height. It is important to have the springs supporting the weight of the vehicle when the fasteners are torqued. If springs are not at their normal ride position, vehicle ride comfort could be affected and premature bushing wear may occur.

STANDARD PROCEDURE - LUBRICATION

Periodic lubrication of the suspension system is required. Refer to Lubrication And Maintenance for the recommended maintenance schedule.

The following component must be lubricated:

- Track bar

FRONT (Continued)

SPECIFICATIONS

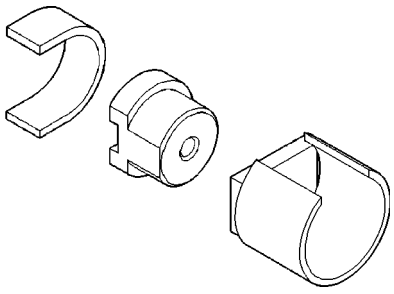
TORQUE CHART

TORQUE SPECIFICATIONS

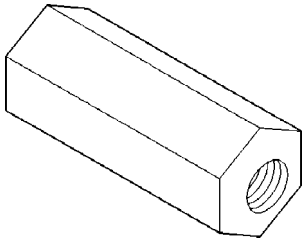
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Shock Absorber Upper Nut	23	17	—
Shock Absorber Lower Nut	28	—	250
Suspension Arm Lower Axle Bracket Nut	176	130	—
Suspension Arm Lower Frame Bracket Nut	176	130	—
Suspension Arm Upper Axle Bracket Nut	81	60	—
Suspension Arm Upper Frame Bracket Bolt	81	60	—
Stabilizer Bar Retainer Bolts	61	45	—
Stabilizer Bar Link Upper Nut	61	45	—
Stabilizer Bar Link Lower Bolt	102	75	—
Track Bar Ball Stud Nut	81	60	—
Track Bar Axle Bracket Bolt	47	40	—
Hub/Bearing Bolts	102	75	—
Hub/Bearing Axle Nut	237	175	—

SPECIAL TOOLS

FRONT SUSPENSION

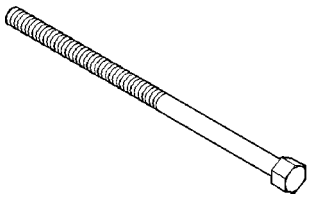


Remover/Installer Suspension Bushing 7932

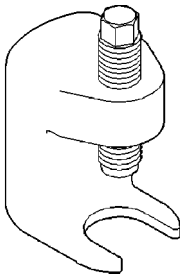


Nut, Long 7603

FRONT (Continued)



Bolt, Special 7604



Remover C-4150A

BUSHINGS

REMOVAL

(1) Remove the upper suspension arm from axle.

(2) Position Spacer 7932-3 over the axle bushing on a 4x2 vehicle and right side on a 4x4 vehicle.

(3) Place Receiver 7932-1 over flanged end of the bushing. (Fig. 1).

(4) Place small end of Remover/Install 7932-2 against other side of the bushing.

(5) Install bolt 7604 through remover, bushing and receiver.

(6) Install Long Nut 7603 and tighten nut too pull bushing out of the axle bracket.

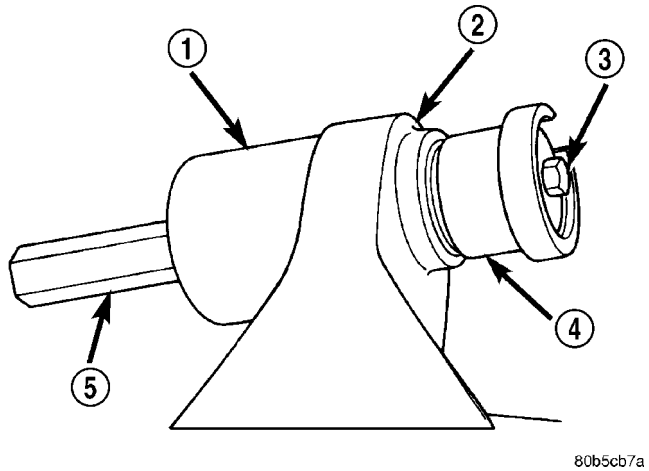


Fig. 1 Bushing Removal

1 - RECEIVER
2 - AXLE BRACKET
3 - BOLT
4 - REMOVER/INSTALLER
5 - LONG NUT

(7) Remove nut, bolt, receiver, remover and bushing.

NOTE: On 4x2 vehicle and right side of 4x4 vehicle, leave Spacer 7932-3 in position for bushing installation.

INSTALLATION

(1) Place Receiver 7932-1 on the other side of the axle bracket.

(2) Position new bushing up to the axle bracket., and large end of Remover/Install 7932-2 against the bushing (Fig. 2).

(3) Install bolt 7604 through receiver, bushing and installer.

(4) Install Long Nut 7603 and tighten nut to draw the bushing into the axle bracket.

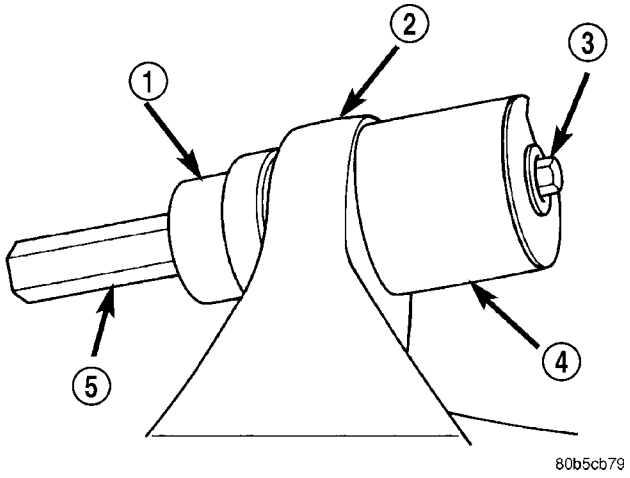


Fig. 2 Bushing Installation

1 - REMOVER/INSTALLER
2 - AXLE BRACKET
3 - BOLT
4 - RECEIVER
5 - LONG NUT

(5) Remove tools and install the upper suspension arm.

HUB / BEARING

DESCRIPTION

The bearing used on the front hub of this vehicle is the combined hub and bearing unit type assembly. This unit assembly combines the front wheel mounting hub (flange) and the front wheel bearing into a one piece unit. The wheel mounting studs are the only replaceable component of the hub/bearing assembly.

HUB / BEARING (Continued)

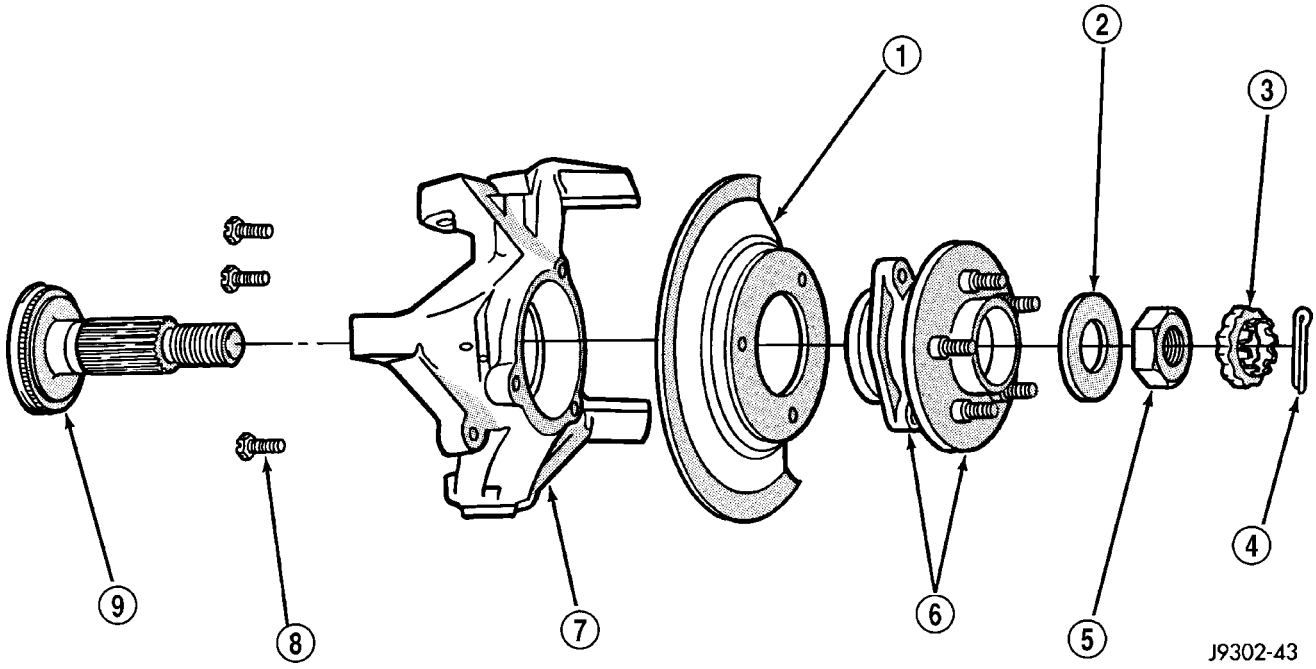


Fig. 3 Hub Bearing & Knuckle

1 - BRAKE SHIELD
2 - WASHER
3 - RETAINER
4 - COTTER PIN
5 - NUT
6 - HUB AND BEARING ASSEMBLY
7 - STEERING KNUCKLE
8 - BOLT
9 - TONE WHEEL (ABS)

OPERATION

The hub/bearing assembly is mounted to the steering knuckle and is retained by three mounting bolts accessible from the back of the steering knuckle. The hub/bearing unit is not serviceable and must be replaced as an assembly if the bearing or the hub is determined to be defective.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the wheel and tire assembly.

(3) Remove the brake caliper, rotor and ABS wheel speed sensor, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - REMOVAL).

(4) Remove the cotter pin, nut retainer and axle hub nut (Fig. 3).

(5) Remove the hub bearing mounting bolts from the back of the steering knuckle. Remove hub bearing from the steering knuckle and off the axle shaft.

INSTALLATION

(1) Install the hub bearing and brake dust shield to the knuckle.

(2) Install the hub bearing to knuckle bolts and tighten to 102 N·m (75 ft. lbs.).

(3) Install the hub washer and nut. Tighten the hub nut to 237 N·m (175 ft. lbs.). Install the nut retainer and a new cotter pin.

(4) Install the brake rotor, caliper and ABS wheel speed sensor, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - INSTALLATION).

(5) Install the wheel and tire assembly. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(6) Remove support and lower the vehicle.

JOUNCE BUMPER

DESCRIPTION

The jounce bumpers are mounted under the frame rails inside of the coil springs.

KNUCKLE

REMOVAL

Ball stud service procedures below require removal of the hub bearing and axle shaft. Removal and installation of upper and lower ball studs require the use of Tool Kit 6289.

(1) Remove hub bearing and axle shaft. (Refer to 2 - SUSPENSION/FRONT/HUB / BEARING - REMOVAL) (Refer to 3 - DIFFERENTIAL & DRIVELINE/FRONT AXLE - 181FBI/AXLE SHAFTS - REMOVAL).

(2) Disconnect the tie-rod or drag link from the steering knuckle arm,(Refer to 19 - STEERING/LINKAGE/TIE ROD END - REMOVAL) OR (Refer to 19 - STEERING/LINKAGE/DRAG LINK - REMOVAL).

(3) Remove the cotter pins from the upper and lower ball studs.

(4) Remove the upper and lower ball stud nuts.

(5) Using special tool C-4150A seperate the ball joints from the steering knuckle. Remove knuckle from ball studs (Fig. 4).

INSTALLATION

Ball stud service procedures below require removal of the hub bearing and axle shaft. Removal and installation of upper and lower ball studs require the use of Tool Kit 6289.

(1) Position the steering knuckle on the ball studs.

(2) Install and tighten the bottom retaining nut to 109 N·m (80 ft. lbs.) torque. Install new cotter pin.

(3) Install and tighten the top retaining nut to 101 N·m (75 ft. lbs.) torque. Install new cotter pin.

(4) Install the hub bearing and axle shaft. (Refer to 2 - SUSPENSION/FRONT/HUB / BEARING - INSTALLATION) (Refer to 3 - DIFFERENTIAL & DRIVELINE/FRONT AXLE - 181FBI/AXLE SHAFTS - INSTALLATION).

(5) Connect the tie-rod or drag link end to the steering knuckle arm,(Refer to 19 - STEERING/LINKAGE/TIE ROD END - INSTALLATION) OR (Refer to 19 - STEERING/LINKAGE/DRAG LINK - INSTALLATION).

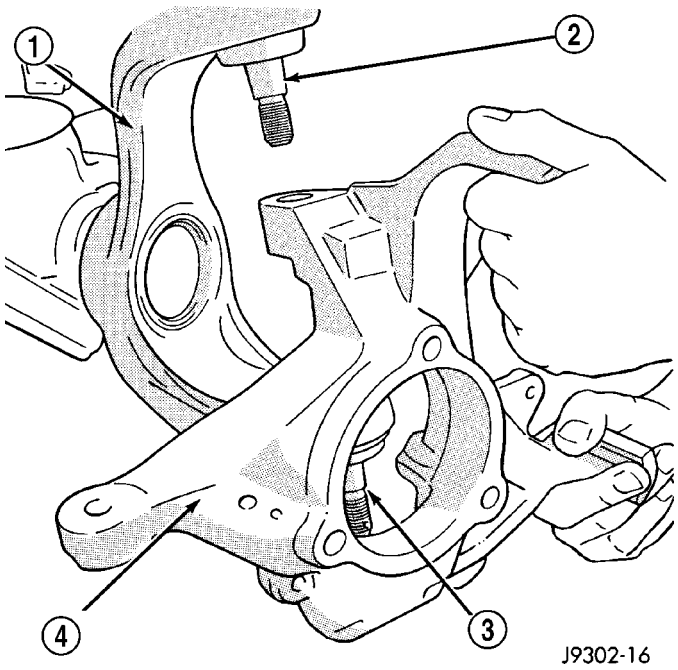


Fig. 4 Steering Knuckle Removal/Installation

1 - AXLE YOKE
2 - UPPER BALL STUD
3 - LOWER BALL STUD
4 - STEERING KNUCKLE

LOWER BALL JOINT

REMOVAL

Ball stud service procedures below require removal of the hub bearing and axle shaft. (Refer to 2 - SUSPENSION/FRONT/HUB / BEARING - REMOVAL) (Refer to 3 - DIFFERENTIAL & DRIVELINE/FRONT AXLE - 181FBI/AXLE SHAFTS - REMOVAL). Removal and installation of upper and lower ball studs require the use of Tool Kit 6289.

(1) Position tools as shown to remove and install ball stud (Fig. 5).

LOWER BALL JOINT (Continued)

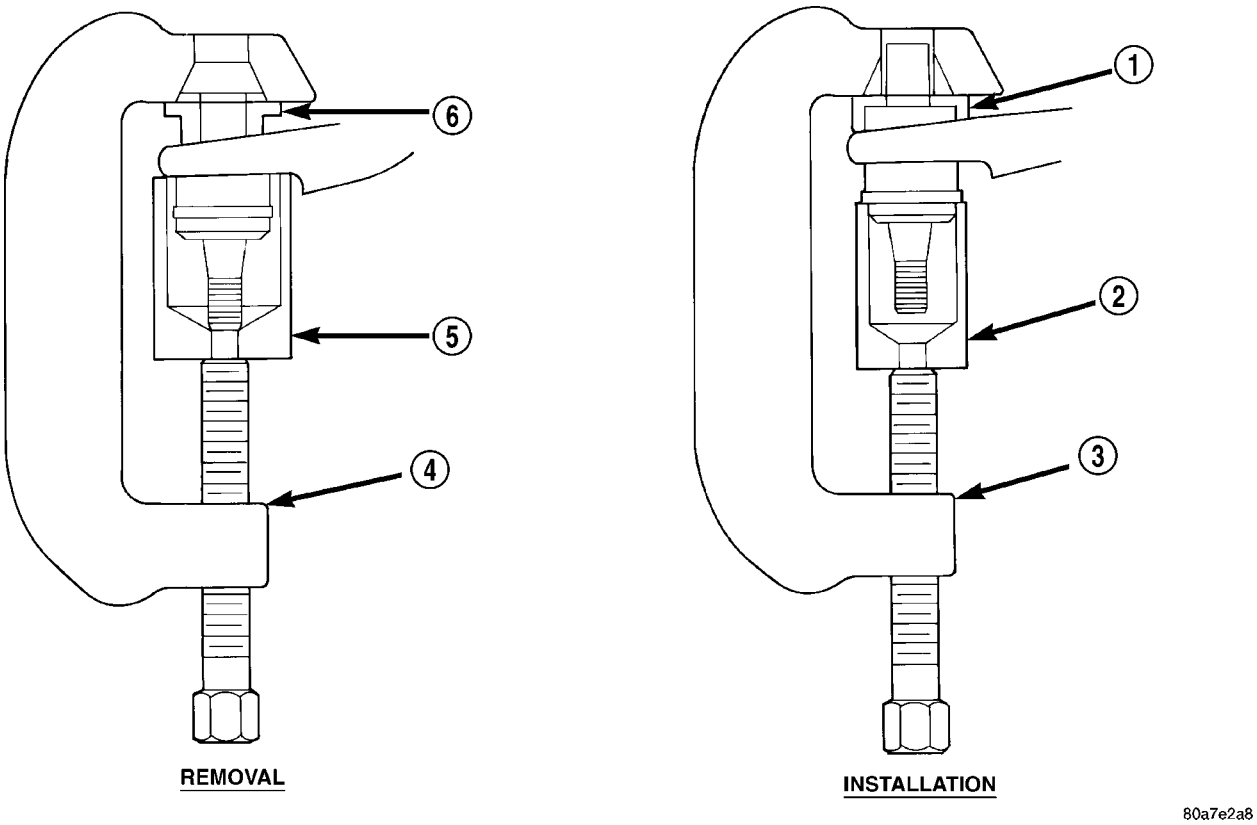


Fig. 5 Lower Ball Stud Remove/Install

1 - SPECIAL TOOL 6289–12
2 - SPECIAL TOOL 6289–4
3 - SPECIAL TOOL 4212F
4 - SPECIAL TOOL 4212F
5 - SPECIAL TOOL 6289–1
6 - SPECIAL TOOL 6289–3

LOWER CONTROL ARM

DESCRIPTION

The lower suspension arms are steel and use bushings at both ends of the arm. The arms mount to the frame rail bracket and the axle brackets.

OPERATION

The lower suspension arm bushings provide isolation from the axle. The arm and bushings provide location and react to loads from the axle. The lower suspension arms can be used to adjust caster and pinion angle by installing a cam bolt service package.

REMOVAL

(1) Raise and support the vehicle.

(2) If equipped with ABS brakes remove sensor wire from the inboard side of the arm.

(3) If the vehicle is equipped with a cam bolt service package paint or scribe alignment marks on the cam adjusters and suspension arm for installation reference (Fig. 6).

(4) Remove the lower suspension arm nut and bolt from the axle (Fig. 7).

(5) Remove the nut and bolt/cam bolt from the frame rail bracket and remove the lower suspension arm (Fig. 7).

INSTALLATION

(1) Position the lower suspension arm in the axle bracket and frame rail bracket.

NOTE: Small holes in the side of the arm face inboard.

(2) Install the rear bolt and nut finger tighten.

(3) Install bolt/cam bolt and new nut finger tighten in the axle and align the reference marks.

(4) If equipped with ABS brakes install sensor wire to the inboard side of the arm with new clips.

(5) Lower the vehicle.

(6) Tighten axle bracket nut to 176 N·m (130 ft. lbs.).

(7) Tighten frame bracket nut to 176 N·m (130 ft. lbs.).

LOWER CONTROL ARM (Continued)

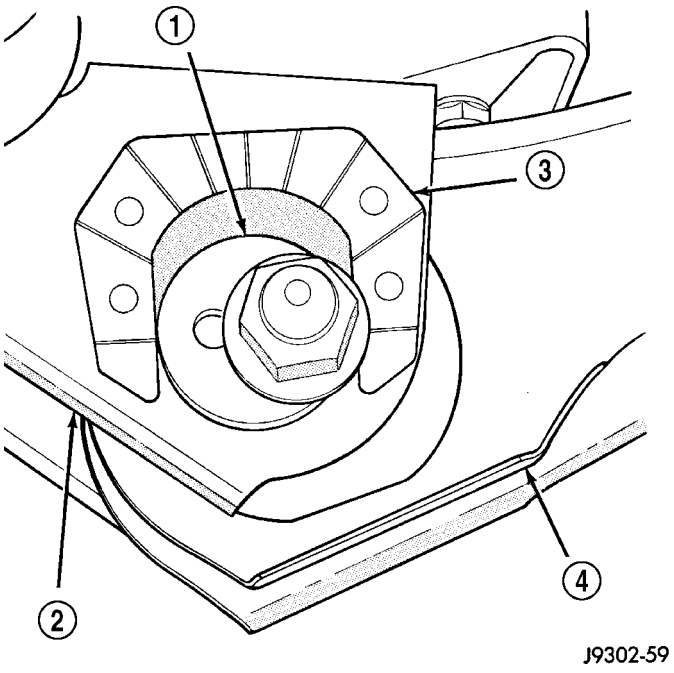


Fig. 6 Cam Bolt Service Package

1 - ADJUSTMENT CAM
2 - AXLE BRACKET
3 - BRACKET REINFORCEMENT
4 - LOWER SUSPENSION ARM

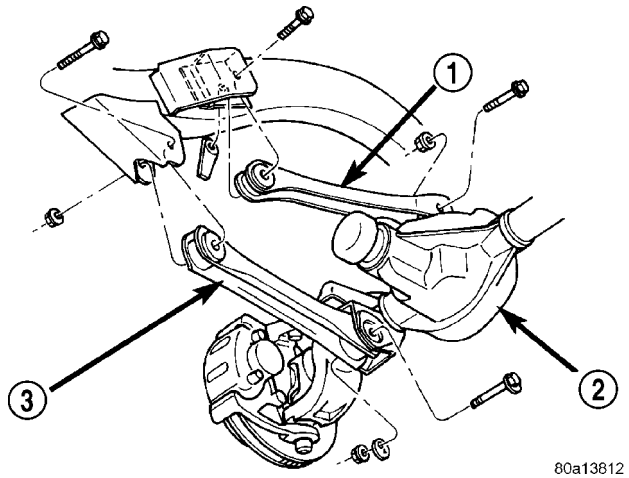


Fig. 7 Upper & Lower Suspension Arms

1 - UPPER SUSPENSION ARM
2 - FRONT AXLE
3 - LOWER SUSPENSION ARM

(8) Align vehicle to specifications. (Refer to 2 - SUSPENSION/WHEEL ALIGNMENT - STANDARD PROCEDURE).

SHOCK

DESCRIPTION

The top of the shock absorbers are bolted to a frame bracket. The bottom of the shocks are bolted to the axle brackets.

OPERATION

The shock absorbers dampen jounce and rebound motion of the vehicle over various road conditions and limit suspension rebound travel.

DIAGNOSIS AND TESTING - SHOCK ABSORBER

A knocking or rattling noise from a shock absorber may be caused by movement between mounting bushings and metal brackets or attaching components. These noises can usually be stopped by tightening the attaching nuts. If the noise persists, inspect for damaged and worn bushings, and attaching components. Repair as necessary if any of these conditions exist.

A squeaking noise from the shock absorber may be caused by the hydraulic valving and may be intermittent. This condition is not repairable and the shock absorber must be replaced.

The shock absorbers are not refillable or adjustable. If a malfunction occurs, the shock absorber must be replaced. To test a shock absorber, hold it in an upright position and force the piston in and out of the cylinder four or five times. The action throughout each stroke should be smooth and even.

The shock absorber bushings do not require any type of lubrication. Do not attempt to stop bushing noise by lubricating them. Grease and mineral oil-base lubricants will deteriorate the bushing.

REMOVAL

(1) Remove the nut, retainer and grommet from the upper stud through engine compartment access hole (Fig. 8).

(2) Remove the lower nuts and bolts from the axle bracket and remove the shock absorber.

INSTALLATION

(1) Position the lower retainer and grommet on the upper stud. Insert the shock absorber through the shock bracket hole.

(2) Install the lower bolts and nuts. Tighten nuts to 28 N·m (250 in. lbs.).

SHOCK (Continued)

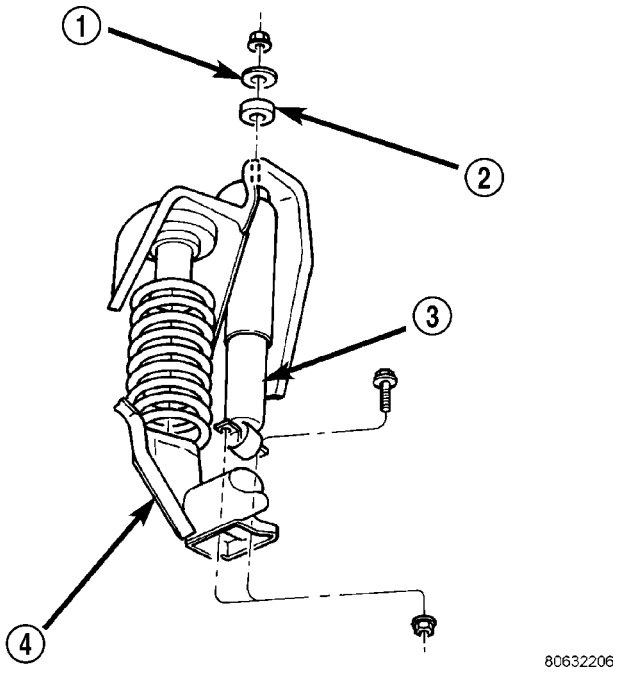


Fig. 8 Coil Spring & Shock Absorber

1 - RETAINER
2 - GROMMET
3 - SHOCK
4 - FRONT AXLE

(3) Install the upper grommet and retainer on the stud and install the nut and tighten to 23 N·m (17 ft. lbs.).

SPRING

DESCRIPTION

The coil springs mount up in the wheelhouse which is part of the unitized body bracket. A rubber doughnut isolator is located between the top of the spring and the bracket. The bottom of the spring seats on a axle pad.

OPERATION

The coil springs control ride quality and maintain proper ride height. The isolators provide road noise isolation.

REMOVAL

(1) Raise and support the vehicle.
(2) Remove the wheel and tire assemblies.
(3) Position a hydraulic jack under the axle to support it.
(4) Remove the front shocks at the lower mountings, (Refer to 2 - SUSPENSION/FRONT/SHOCK - REMOVAL).
(5) Remove the ABS wire mounting brackets at the axle. (if equipped)
(6) Remove lower suspension arms mounting nuts and bolts from the frame, (Refer to 2 - SUSPENSION/FRONT/LOWER CONTROL ARM - REMOVAL).
(7) Remove the track bar from the axle bracket, (Refer to 2 - SUSPENSION/FRONT/TRACK BAR - REMOVAL).
(8) Remove the right side of the drag link from the right side knuckle, (Refer to 19 - STEERING/LINKAGE/DRAG LINK - REMOVAL).
(9) Lower the axle until the spring is free from the upper mount.

NOTE: Rotation of the spring and prying down slightly on the axle will aid in removal.

(10) Remove the coil spring retainer clip and remove the spring.
(11) Remove the upper spring isolator. (if needed)
(12) Pull jounce bumper out of mount. (if needed)

INSTALLATION

(1) Install jounce bumper into mount.
(2) Install the spring isolator.

NOTE: Rotation of the spring and prying down slightly on the axle will aid in installation.

(3) Position the coil spring on the axle pad. It may be necessary to rotate the spring while installing.
(4) Install the spring retainer clip and bolt. Tighten bolt to 21 N·m (16 ft. lbs.).
(5) Raise the axle into position until the spring seats in the upper mount.
(6) Install the shock at the axle, (Refer to 2 - SUSPENSION/FRONT/SHOCK - INSTALLATION).
(7) Install the ABS wire mounting brackets at the axle (if equipped).
(8) Install the track bar to the axle bracket, (Refer to 2 - SUSPENSION/FRONT/TRACK BAR - INSTALLATION).
(9) Install the lower suspension arms to the frame. Install mounting bolts and nuts finger tight, (Refer to 2 - SUSPENSION/FRONT/LOWER CONTROL ARM - INSTALLATION).
(10) Install the drag link to the right side knuckle, (Refer to 19 - STEERING/LINKAGE/DRAG LINK - INSTALLATION).
(11) Remove the hydraulic jack from under the axle.
(12) Install the wheel and tire assemblies, (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(13) Remove the supports and lower the vehicle.
(14) Tighten the lower suspension arms nuts to 115 N·m (85 ft. lbs.) at normal ride height with the vehicle weight.

STABILIZER BAR

DESCRIPTION

The spring steel bar extends across the top of the chassis frame rails. Links are connected from the bar to the axle brackets. The stabilizer bar and links are isolated by rubber bushings.

OPERATION

The stabilizer bar is used to control vehicle body roll during turns. The bar helps to control the vehicle body in relationship to the suspension.

REMOVAL

(1) Remove upper link nuts (Fig. 9) and separate the links from the stabilizer bar with Remover MB-991113.

(2) Remove front bumper valence.

(3) Remove stabilizer retainer bolts (Fig. 9)and remove retainers.

(4) Remove stabilizer bar.

(5) Remove lower link nuts and bolts and remove links (Fig. 9).

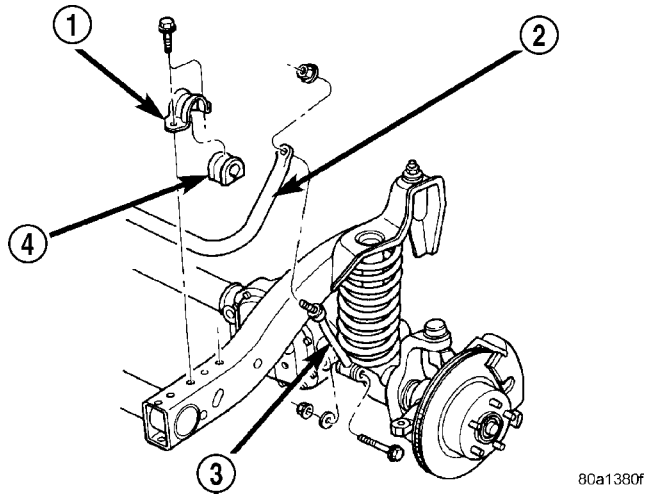


Fig. 9 Stabilizer Bar

1 - RETAINER
2 - STABILIZER BAR
3 - LINK
4 - BUSHING

INSTALLATION

(1) Center stabilizer bar on top of the frame rails and install retainers and bolts. Tighten bolts to 61 N·m (45 ft. lbs.).

(2) Position links on axle brackets and into the stabilizer bar. Install lower link bolts and nuts and tighten to 102 N·m (75 ft. lbs.).

(3) Install upper link nuts and tighten to 61 N·m (45 ft. lbs.).

(4) Install bumper valence.

TRACK BAR

DESCRIPTION

The bar is attached to a frame rail bracket with a ball stud and an axle bracket with a bushing. The bar is forged and has non replaceable isolator bushing and ball stud.

OPERATION

The track bar is used to control front axle lateral movement and provides cross car location of the axle assembly.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the cotter pin and nut from the ball stud end at the frame rail bracket (Fig. 10).

(3) Use a universal puller tool to separate the track bar ball stud from the frame rail bracket.

(4) Remove the bolt and flag nut from the axle bracket (Fig. 10). Remove the track bar.

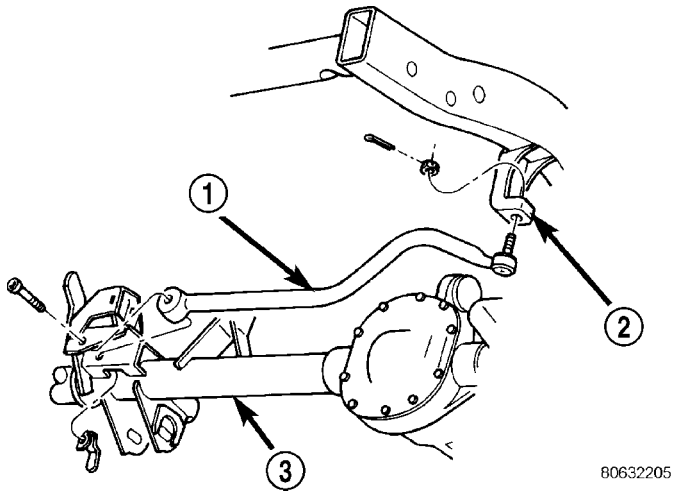


Fig. 10 Track Bar

1 - TRACK BAR
2 - FRAME BRACKET
3 - FRONT AXLE

INSTALLATION

(1) Install the track bar at axle tube bracket. Loosely install the retaining bolt and flag nut.

(2) It may be necessary to pry the axle assembly over to install the track bar at the frame rail. Install track bar at the frame rail bracket. Install the retaining nut on the stud.

(3) Tighten the ball stud nut to 81N·m (60 ft. lbs.) and install a new cotter pin.

(4) Remove the supports and lower the vehicle.

(5) Tighten the bolt at the axle bracket to 47 N·m (40 ft. lbs.).

TRACK BAR (Continued)

(6) Check alignment if a new track bar was installed. (Refer to 2 - SUSPENSION/WHEEL ALIGNMENT - STANDARD PROCEDURE).

UPPER BALL JOINT

REMOVAL

Ball stud service procedures below require removal of the hub bearing and axle shaft. (Refer to 2 - SUSPENSION/FRONT/HUB / BEARING - REMOVAL) (Refer to 3 - DIFFERENTIAL & DRIVELINE/ FRONT AXLE - 181FBI/AXLE SHAFTS - REMOVAL). Removal and installation of upper and lower ball studs require the use of Tool Kit 6289.

(1) Position tools as shown to remove and install ball stud (Fig. 11).

UPPER CONTROL ARM

DESCRIPTION

The upper suspension arms are steel and use rubber bushings at each end of the arm. The arms mount to the frame rail bracket and the axle brackets.

OPERATION

The arm and bushings provide location and react to loads from the axle. The bushings provide isolation from the axle.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the upper suspension arm nut and bolt at the axle bracket (Fig. 7).

(3) Remove the nut and bolt at the frame rail and remove the upper suspension arm.

INSTALLATION

(1) Position the upper suspension arm at the axle and frame rail.

(2) Install the bolts and finger tighten the nuts.

(3) Remove the supports and lower the vehicle.

(4) Tighten the nut at the axle and frame brackets to 81 N·m (60 ft. lbs.).

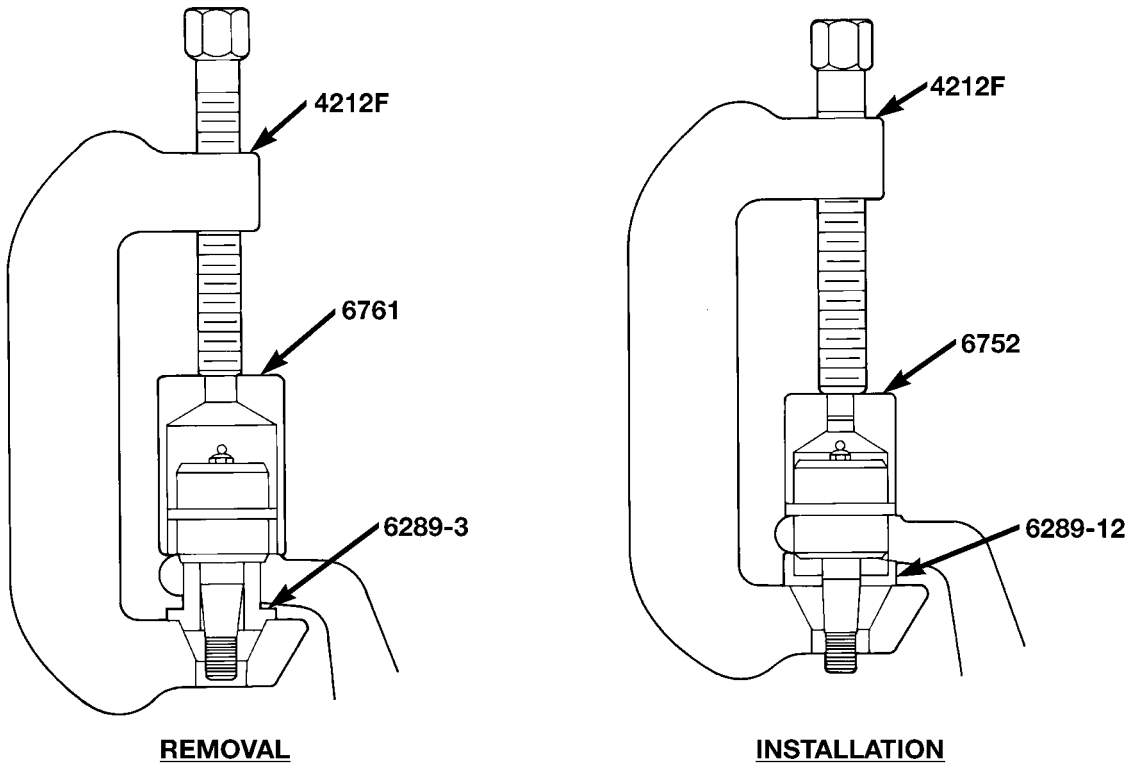


Fig. 11 Upper

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REAR

DESCRIPTION

REAR SUSPENSION

The rear suspension is link/coil design comprised of:

- Shock absorbers
- Coil springs
- Upper and lower suspension arms
- Stabilizer bar
- Track bar

CAUTION: Suspension components with rubber/urethane bushings (except stabilizer bar) should be tightened with the vehicle at normal ride height. It is important to have the springs supporting the weight of the vehicle when the fasteners are torqued. This will maintain vehicle ride comfort and prevent premature bushing wear.

SPECIFICATIONS

TORQUE CHART

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Shock Absorber Upper Bolts	35	26	—
Shock Absorber Lower Nut	100	74	—
Suspension Arm Lower Axle Bracket Nut	203	150	—

REAR (Continued)

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Suspension Arm Lower Frame Bracket Nut	203	150	—
Suspension Arm Upper Axle Bracket Nut	75	55	—
Suspension Arm Upper Frame Bracket Bolt	75	55	—
Stabilizer Bar Retainer Bolts	54	40	—
Stabilizer Bar Link Nut/Bolt	54	40	—
Track Bar Frame Bracket Nut	100	74	—
Track Bar Axle Bracket Bolt	100	74	—

JOUNCE BUMPER

DESCRIPTION

The jounce bumpers are mounted inside the coil spring to the frame rail.

OPERATION

The jounce bumpers are used to limit suspension travel in compression.

LOWER CONTROL ARM

DESCRIPTION

The lower suspension arms are steel and use bushings at each end of the arm. The arms are mounted from the frame to the axle brackets.

OPERATION

The bushings isolation axle and road noise. The arm and bushings provide location and react to loads from the axle.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the lower suspension arm nut and bolt at the axle bracket (Fig. 1).

(3) Remove the nut and bolt at the frame rail mount (Fig. 2) and remove the lower suspension arm.

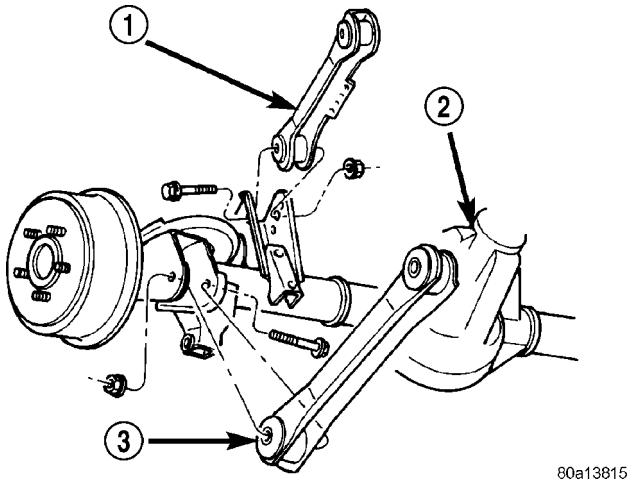

Fig. 1 Upper & Lower Suspension Arms

1 - UPPER SUSPENSION ARM
2 - REAR AXLE
3 - LOWER SUSPENSION ARM

INSTALLATION

(1) Position the lower suspension arm in the axle bracket and frame rail mount.

(2) Install the mounting bolts and finger tighten the nuts.

(3) Remove the supports and lower the vehicle.

(4) Tighten the lower suspension arm nuts to 203 N·m (150 ft. lbs.).

LOWER CONTROL ARM (Continued)

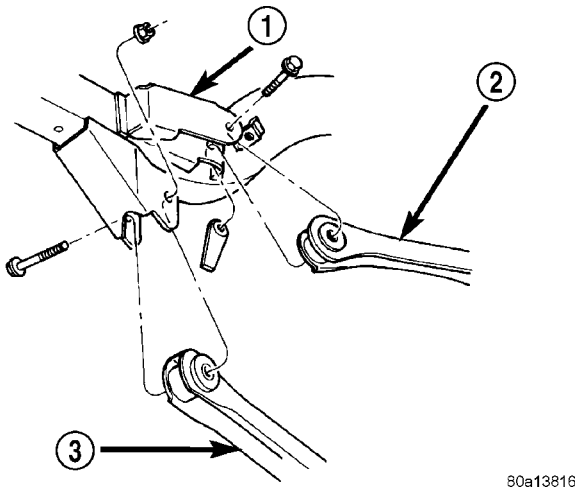

Fig. 2 Upper & Lower Suspension Arms

1 - FRAME MOUNT
2 - UPPER SUSPENSION ARM
3 - LOWER SUSPENSION ARM

SHOCK

DESCRIPTION

The top of the shock absorbers are bolted to the frame. The bottom of the shocks are bolted to the axle brackets.

OPERATION

The shock absorbers dampen jounce and rebound motion of the vehicle over various road conditions and limit suspension rebound travel.

DIAGNOSIS AND TESTING - SHOCK ABSORBER

A knocking or rattling noise from a shock absorber may be caused by movement between mounting bushings and metal brackets or attaching components. These noises can usually be stopped by tightening the attaching nuts. If the noise persists, inspect for damaged and worn bushings, and attaching components. Repair as necessary if any of these conditions exist.

A squeaking noise from the shock absorber may be caused by the hydraulic valving and may be intermittent. This condition is not repairable and the shock absorber must be replaced.

The shock absorbers are not refillable or adjustable. If a malfunction occurs, the shock absorber must be replaced. To test a shock absorber, hold it in an upright position and force the piston in and out of the cylinder four or five times. The action throughout each stroke should be smooth and even.

The shock absorber bushings do not require any type of lubrication. Do not attempt to stop bushing noise by lubricating them. Grease and mineral oil-base lubricants will deteriorate the bushing.

REMOVAL

(1) Raise and support the vehicle and the axle.
(2) Remove the upper mounting bolts (Fig. 3).
(3) Remove the lower nut and bolt from the axle bracket. Remove the shock absorber.

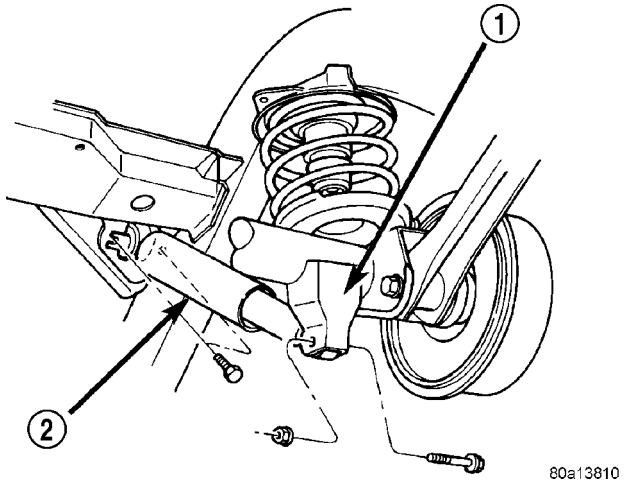

Fig. 3 Shock Absorber

1 - AXLE BRACKET
2 - SHOCK

INSTALLATION

(1) Install the shock absorber on the upper frame rail and install mounting bolts.
(2) Tighten the upper bolts to 31 N·m (23 ft. lbs.).
(3) Install lower bolt and nut finger tight.
(4) Remove the supports and lower the vehicle.
(5) Tighten the lower nut to 100 N·m (74 ft. lbs.).

SPRING

DESCRIPTION

The coil springs mount between the bottom of the frame rail and the top of the axle. A rubber doughnut isolator is located between the top of the spring and the frame rail. A plastic isolator is located between the bottom of the spring and the axle.

OPERATION

The coil springs control ride quality and maintain proper ride height. The isolators are used to isolate road noise.

REMOVAL

(1) Raise and support the vehicle. Position a hydraulic jack under the axle to support it.
(2) Disconnect the stabilizer bar links and shock absorbers from the axle brackets. (Refer to 2 - SUS-

SPRING (Continued)

PENSION/REAR/STABILIZER BAR - REMOVAL) (Refer to 2 - SUSPENSION/REAR/SHOCK - REMOVAL).

(3) Disconnect the track bar from the frame rail bracket. (Refer to 2 - SUSPENSION/REAR/TRACK BAR - REMOVAL).

(4) Lower the axle until the spring is free from the upper mount seat and remove the spring.

INSTALLATION

NOTE: Springs can be install with either end up.

(1) Position the coil spring on the axle pad isolator.

(2) Raise the axle into position until the spring seats on the upper isolator.

(3) Connect the stabilizer bar links and shock absorbers to the axle bracket. Connect the track bar to the frame rail bracket.

(4) Remove the supports and lower the vehicle.

(5) Tighten the stabilizer bar links, shock absorbers and track bar to specified torque.

STABILIZER BAR

DESCRIPTION

The spring steel bar extends across the axle and mounts to bracket on the axle. Links are connected from the bar to the side of the frame rail. The stabilizer bar and links are isolated by rubber bushings.

OPERATION

The stabilizer bar is used to control vehicle body roll during turns. The bar helps to control the vehicle body in relationship to the suspension.

REMOVAL

(1) Raise and support the vehicle.

(2) Remove the stabilizer bar link bolts from the frame mounts (Fig. 4).

(3) Remove the link bolts from the stabilizer bar.

(4) Remove the stabilizer bar retainer bolts and retainers from the axle mounts (Fig. 5) and remove the bar.

INSTALLATION

(1) Install the stabilizer bar on the axle mounts and install the retainers and bolts.

NOTE: Ensure the bar is centered with equal spacing on both sides and is positioned above the differential housing (Fig. 5).

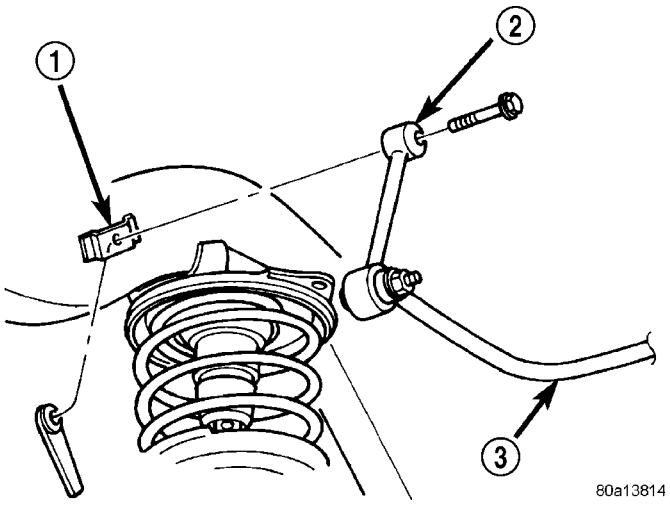

Fig. 4 Stabilizer Bar Link

1 - FRAME MOUNT
2 - LINK
3 - STABILIZER BAR

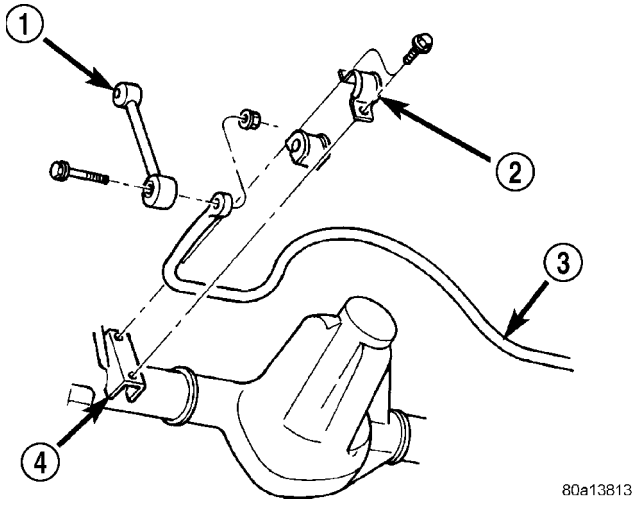

Fig. 5 Stabilizer Bar

1 - LINK
2 - RETAINER
3 - STABILIZER BAR
4 - AXLE MOUNT

(2) Tighten the retainer bolts to 54 N·m (40 ft. lbs.).

(3) Install the links onto the stabilizer bar and frame mounts. Install the bolts and nuts finger tight.

(4) Remove support and lower vehicle.

(5) Tighten the link nuts/bolts to 54 N·m (40 ft. lbs.).

TRACK BAR

DESCRIPTION

The bar is attached to a frame rail bracket and axle bracket. The bar has bushings at both ends.

OPERATION

The track bar is used to control rear axle lateral movement.

REMOVAL

(1) Raise and support the vehicle.
(2) Remove the bolt and nut from the frame rail bracket (Fig. 6).
(3) Remove the bolt from the axle bracket (Fig. 6) and remove the track bar.

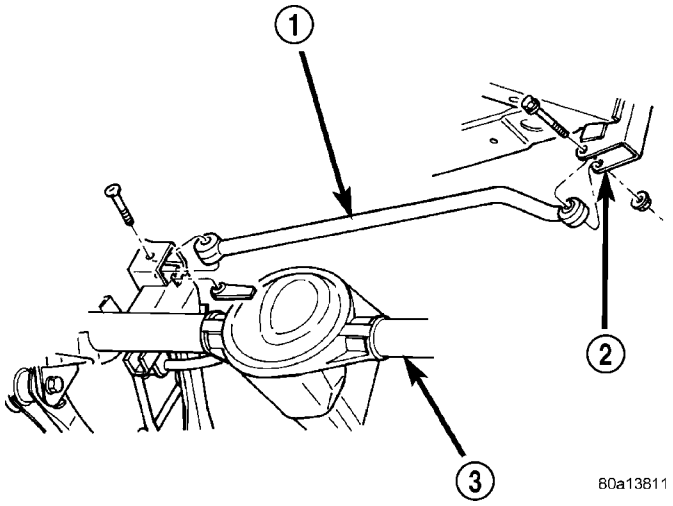

Fig. 6 Rear Track Bar

1 - TRACK BAR
2 - FRAME BRACKET
3 - REAR AXLE

INSTALLATION

(1) Install the track bar in the axle bracket and install the bolt loosely.
(2) Install the track bar in the frame rail bracket and loosely install the bolt and nut.

NOTE: It may be necessary to pry the axle assembly over to install the track bar.

(3) Remove supports and lower the vehicle.
(4) Tighten the track bar nut/bolt at both ends to 100 N·m (74 ft. lbs.).

UPPER CONTROL ARM

REMOVAL

(1) Raise and support the vehicle.
(2) Remove the parking brake cable/bracket and ABS wiring bracket from the arm if equipped (Fig. 7).

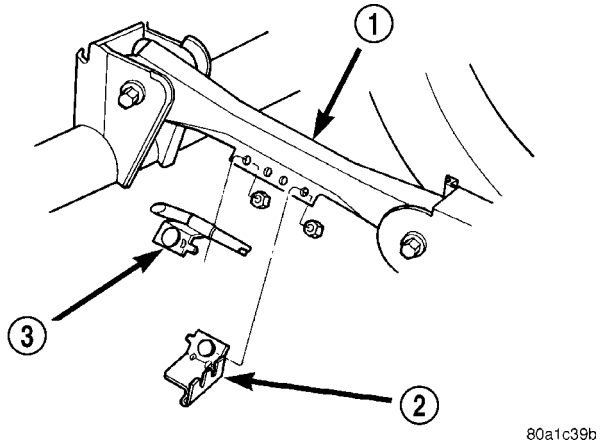

Fig. 7 Parking Brake Cable/Bracket And Wiring Bracket

1 - UPPER SUSPENSION ARM
2 - WIRING BRACKET
3 - PARKING BRAKE CABLE BRACKET

(3) Remove the upper suspension arm nut and bolt from the axle bracket (Fig. 1).
(4) Remove the nut and bolt from the frame rail bracket (Fig. 2)and remove the upper suspension arm.

INSTALLATION

(1) Position the upper suspension arm in the axle bracket and frame rail bracket.
(2) Install the bolts and finger tighten the nuts.
(3) Install the parking brake cable/bracket and ABS wiring bracket on the arm if equipped.
(4) Remove the supports and lower the vehicle.
(5) Tighten the upper suspension arm frame rail bracket bolt to 75 N·m (55 ft. lbs.).
(6) Tighten the upper suspension arm axle bracket nut to 75 N·m (55 ft. lbs.).

DIFFERENTIAL & DRIVELINE

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PROPELLER SHAFT

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PROPELLER SHAFT

DIAGNOSIS AND TESTING

VIBRATION

Tires that are out-of-round, or wheels that are unbalanced, will cause a low frequency vibration. Refer to Group 22, Tires and Wheels, for additional information.

Brake drums that are unbalanced will cause a harsh, low frequency vibration. Refer to Group 5, Brakes, for additional information.

Driveline vibration can also result from loose or damaged engine mounts. Refer to Group 9 Engines for additional information.

Propeller shaft vibration increases as the vehicle speed is increased. A vibration that occurs within a specific speed range is not usually caused by a propeller shaft being unbalanced. Defective universal joints, or an incorrect propeller shaft angle, are usually the cause of such a vibration.

PROPELLER SHAFT (Continued)

DRIVELINE VIBRATION

Drive Condition	Possible Cause	Correction
Propeller Shaft Noise	1) Undercoating or other foreign material on shaft.	1) Clean exterior of shaft and wash with solvent.
	2) Loose U-joint clamp screws.	2) Install new clamps and screws and tighten to proper torque.
	3) Loose or bent U-joint yoke or excessive runout.	3) Install new yoke.
	4) Incorrect driveline angularity.	4) Measure and correct driveline angles.
	5) Rear spring center bolt not in seat.	5) Loosen spring u-bolts and seat center bolt.
	6) Worn U-joint bearings.	6) Install new U-joint.
	7) Propeller shaft damaged or out of balance.	7) Installl new propeller shaft.
	8) Broken rear spring.	8) Install new rear spring.
	9) Excessive runout or unbalanced condition.	9) Re-index propeller shaft, test, and evaluate.
	10) Excessive drive pinion gear shaft runout.	10) Re-index propeller shaft and evaluate.
	11) Excessive axle yoke deflection.	11) Inspect and replace yoke if necessary.
	12) Excessive transfer case runout.	12) Inspect and repair as necessary.
Universal Joint Noise	1) Loose U-joint clamp screws.	1) Install new clamps and screws and tighten to proper torque.
	2) Lack of lubrication.	2) Replace U-joints as necessary.

PROPELLER SHAFT (Continued)

BALANCE

NOTE: Removing and re-indexing the propeller shaft 180° relative to the yoke may eliminate some vibrations.

If propeller shaft is suspected of being unbalanced, it can be verified with the following procedure:

(1) Raise the vehicle.

(2) Clean all the foreign material from the propeller shaft and the universal joints.

(3) Inspect the propeller shaft for missing balance weights, broken welds, and bent areas. **If the propeller shaft is bent, it must be replaced.**

(4) Inspect the universal joints to ensure that they are not worn, are properly installed, and are correctly aligned with the shaft.

(5) Check the universal joint clamp screws torque.

(6) Remove the wheels and tires. Install the wheel lug nuts to retain the brake drums or rotors.

(7) Mark and number the shaft six inches from the yoke end at four positions 90° apart.

(8) Run and accelerate the vehicle until vibration occurs. Note the intensity and speed the vibration occurred. Stop the engine.

(9) Install a screw clamp at position 1 (Fig. 1).

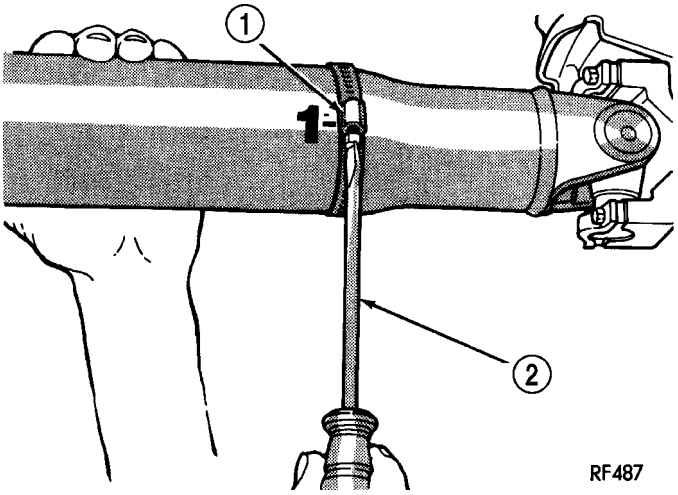


Fig. 1 CLAMP SCREW AT POSITION 1

1 - CLAMP
2 - SCREWDRIVER

(10) Start the engine and re-check for vibration. If there is little or no change in vibration, move the clamp to one of the other three positions. Repeat the vibration test.

(11) If there is no difference in vibration at the other positions, the source of the vibration may not be propeller shaft.

(12) If the vibration decreased, install a second clamp (Fig. 2)and repeat the test.

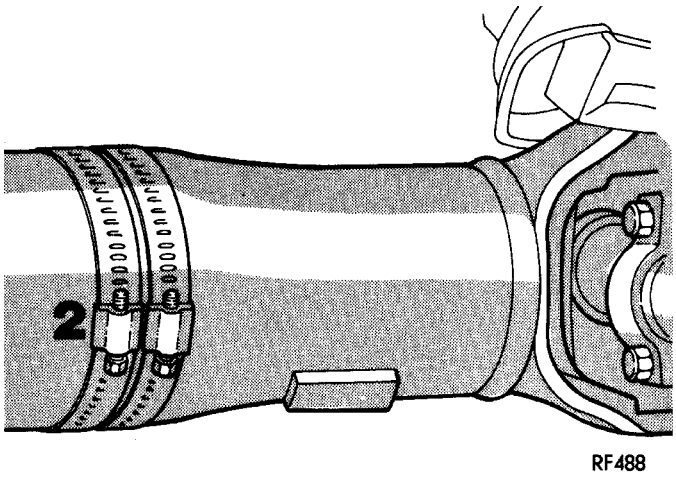


Fig. 2 TWO CLAMP SCREWS

(13) If the additional clamp causes an additional vibration, separate the clamps (1/4 inch above and below the mark). Repeat the vibration test (Fig. 3).

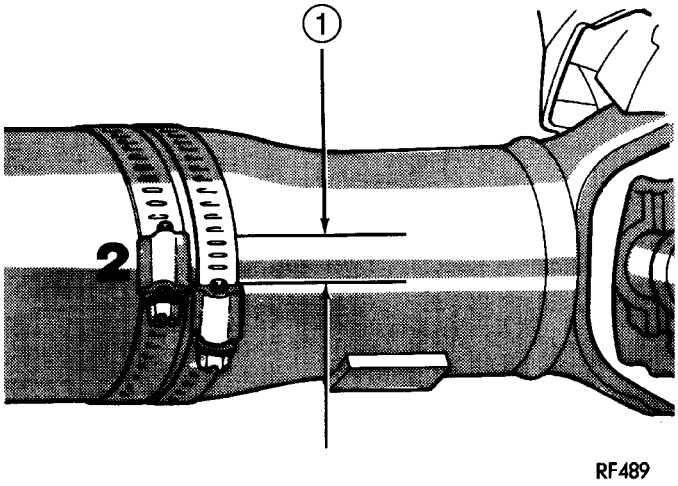


Fig. 3 CLAMP SCREWS SEPARATED

1 - 1/2 INCH

(14) Increase distance between the clamp screws and repeat the test until the amount of vibration is at the lowest level. Bend the slack end of the clamps so the screws will not loosen.

(15) If the vibration remains unacceptable, apply the same steps to the front end of the propeller shaft.

(16) Install the wheel and tires. Lower the vehicle.

RUNOUT

(1) Remove dirt, rust, paint, and undercoating from the propeller shaft surface where the dial indicator will contact the shaft.

(2) The dial indicator must be installed perpendicular to the shaft surface.

(3) Measure runout at the center and ends of the shaft sufficiently far away from weld areas to ensure

PROPELLER SHAFT (Continued)

that the effects of the weld process will not enter into the measurements.

(4) Refer to Runout Specifications chart.

(5) If the propeller shaft runout is out of specification, remove the propeller shaft, index the shaft 180°, and re-install the propeller shaft. Measure shaft runout again.

(6) If the propeller shaft runout is now within specifications, mark the shaft and yokes for proper orientation.

(7) If the propeller shaft runout is not within specifications, verify that the runout of the transmission/transfer case and axle are within specifications. Correct as necessary and re-measure propeller shaft runout.

(8) Replace the propeller shaft if the runout still exceeds the limits.

RUNOUT SPECIFICATIONS

Front of Shaft	0.020 in. (0.50 mm)
Center of Shaft	0.025 in. (0.63 mm)
Rear of Shaft	0.020 in. (0.50 mm)
note: Measure front/rear runout approximately 3 inches (76 mm) from the weld seam at each end of the shaft tube for tube lengths over 30 inches. For tube lengths under 30 inches, the maximum allowed runout is 0.020 in. (0.50 mm) for the full length of the tube.	

STANDARD PROCEDURE

PROPELLER SHAFT ANGLE

(1) Place transmission in netural.

(2) Raise and support the vehicle at the axles as level as possible.

(3) Remove any external bearing snap rings from universal joint so protractor base sits flat.

(4) Rotate the shaft until transmission/transfer case output yoke bearing is facing downward.

NOTE: Always make measurements from front to rear and from the same side of the vehicle.

(5) Place Inclinometer 7663 (J-23498A) on yoke bearing (A) parallel to the shaft (Fig. 4). Center bubble in sight glass and record measurement.

NOTE: This measurement will give you the Output Yoke Angle (A).

(6) Rotate propeller shaft 90 degrees and place Inclinometer on yoke bearing parallel to the shaft (Fig. 5). Center bubble in sight glass and record measurement. This measurement can also be taken at the rear end of the shaft.

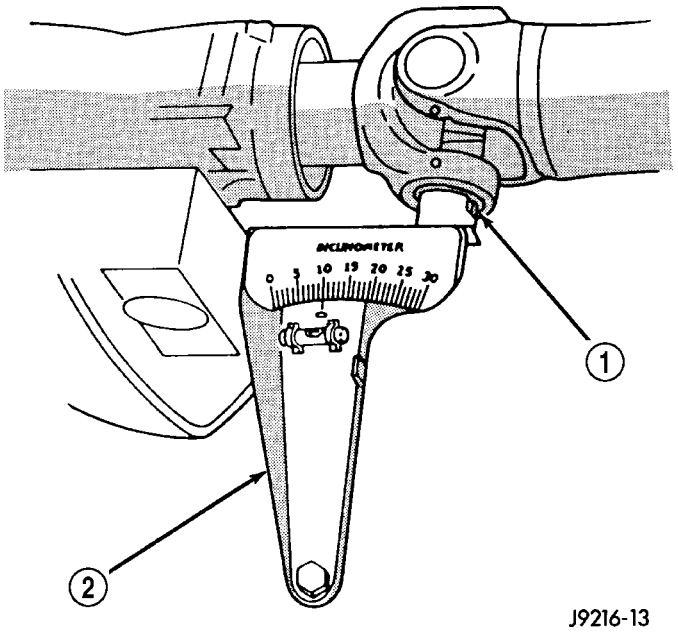


Fig. 4 OUTPUT YOKE ANGLE (A)

1 - SLIP YOKE BEARING CAP
2 - INCLINOMETER

NOTE: This measurement will give you the Propeller Shaft Angle (C).

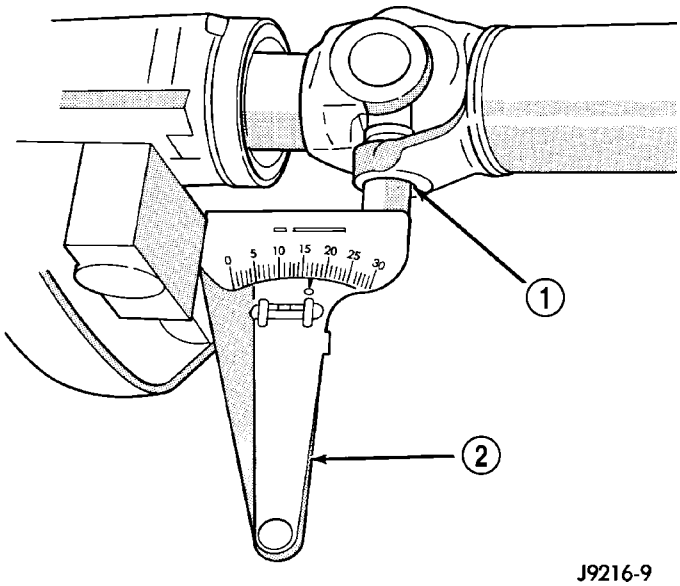


Fig. 5 PROPELLER SHAFT ANGLE (C)

1 - SHAFT YOKE BEARING CAP
2 - INCLINOMETER

(7) Subtract smaller figure from larger (C minus A) to obtain Transmission Output Operating Angle.

(8) Rotate propeller shaft 90 degrees and place Inclinometer on pinion yoke bearing parallel to the shaft (Fig. 6). Center bubble in sight glass and record measurement.

PROPELLER SHAFT (Continued)

NOTE: This measurement will give you the pinion shaft or Input Yoke Angle (B).

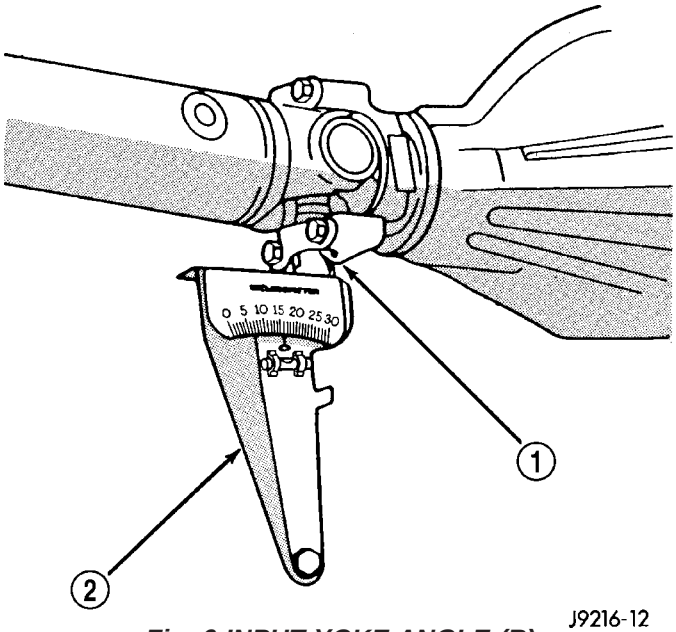


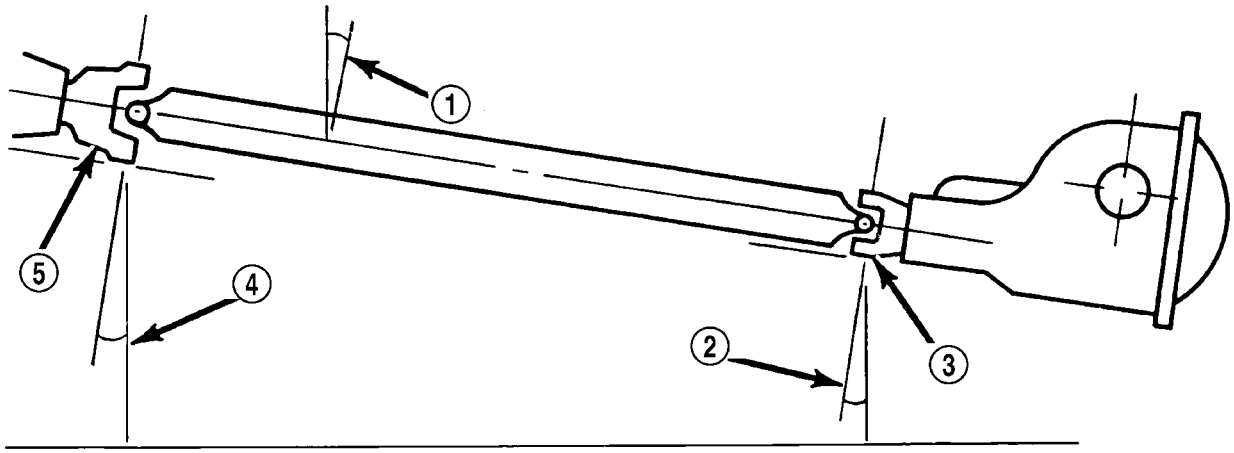
Fig. 6 INPUT YOKE ANGLE (B)

1 - PINION YOKE BEARING CAP
2 - INCLINOMETER

(9) Subtract smaller figure from larger (C minus B) to obtain axle Input Operating Angle.

Refer to rules given below and the example in (Fig. 7) for additional information.

- Good cancellation of U-joint operating angles (within 1°).
- Operating angles less than 3°.
- At least 1/2 of one degree continuous operating (propeller shaft) angle.



(A) Output Yoke = 3.0°	or 4.9°	(B) Axle Input Yoke = 3.2°	or 4.9°
(C) Prop. Shaft = 4.9°	-3.0°	(C) Prop. Shaft = 4.9°	-3.2°
Transmission Output Operating Angle	1.9°	Axle Input Operating Angle	1.7°

Trans. Output Operating Angle	1.9°
Axle Input Operating Angle	-1.7°
Amount of U-Joint Cancellation	0.2°

J9316-3

Fig. 7 UNIVERSAL JOINT ANGLE EXAMPLE

1 - 4.9° Angle (C)
2 - 3.2° Angle (B)
3 - Input Yoke
4 - 3.0° Angle (A)
5 - Output Yoke

PROPELLER SHAFT (Continued)

ADJUSTMENTS

The pinion angle of the front axle can be adjusted by the use of adjustment cams in the lower suspension arms (Fig. 8). The primary function of the cams is to adjust caster angle for alignment of the front suspension. When using the cams to adjust pinion angle, make sure that both cams are moved equally. After pinion angle is adjusted, the front suspension alignment should be checked to ensure that side-to-side caster is with in acceptable range. Having the correct pinion angle does have priority over having the preferred caster angle.

A cam kit is available for the rear axle upper suspension arms in order to provide adjustablity of the pinion angle. Follow the procedures supplied with the kit in order to ensure a safe installation.

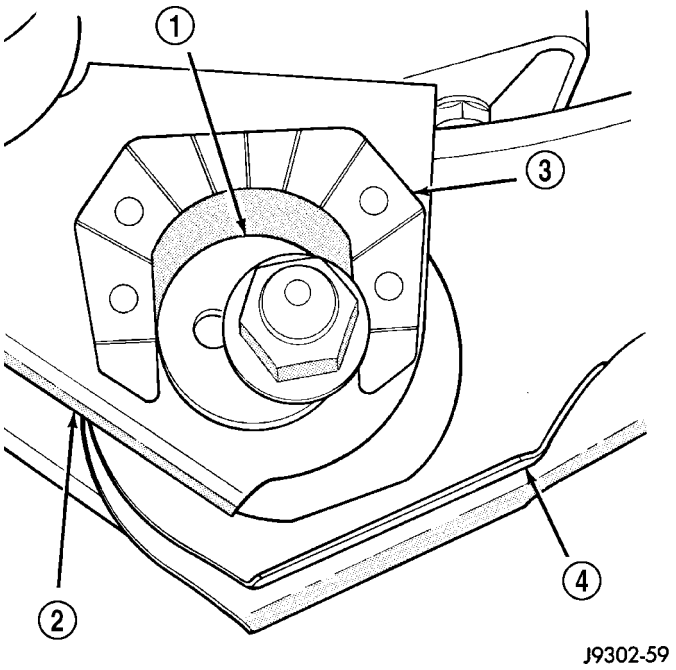


Fig. 8 ADJUSTMENT CAM

1 - ADJUSTMENT CAM
2 - AXLE BRACKET
3 - BRACKET REINFORCEMENT
4 - LOWER SUSPENSION ARM

SPECIFICATIONS

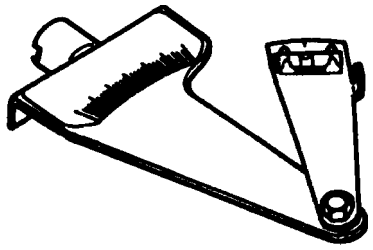
PROPELLER SHAFT

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Front Shaft Front Yoke Bolts	19	14	-
Front Shaft Rear Flange Bolts	27	20	-
Rear Saft Rear Yoke Bolts	19	14	-
Rear Shaft Transfer Case Flange Bolts	115	85	-

PROPELLER SHAFT (Continued)

SPECIAL TOOLS



INCLINOMETER 7663

PROPELLER SHAFT - FRONT

REMOVAL

(1) With vehicle in neutral positon vehicle on hoist.

(2) Remove skid plate if equipped to gain access to the propeller shaft.

(3) Mark a line across the transfer case yoke, link yoke and propeller shaft yoke for installation reference (Fig. 9).

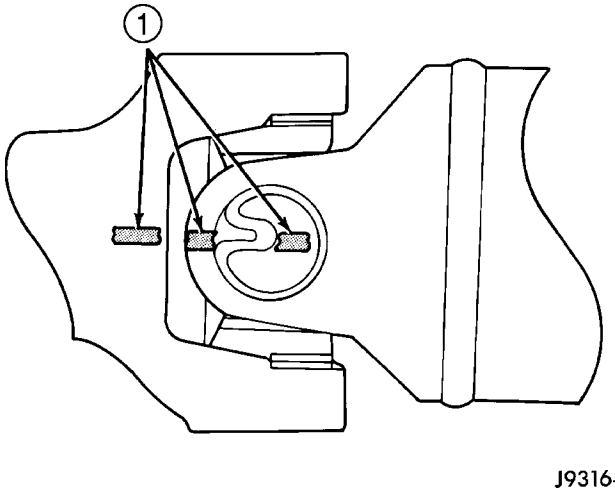


Fig. 9 YOKE REFERENCE MARKS

1 - REFERENCE MARKS

(4) Remove U-joint strap bolts at the pinion shaft yoke (Fig. 10).

(5) Remove transfer case yoke bolts.

(6) Push rear of propeller shaft forward to clear transfer case yoke.

(7) Remove propeller shaft from vehicle.

INSTALLATION

(1) Install propeller shaft with reference marks aligned.

(2) Loosely install bolts to hold universal joint to transfer case yoke.

(3) Install U-joint strap bolts at the axle yoke and tighten to 19 N·m (14 ft. lbs.).

(4) Install universal joint to transfer case bolts and tighten to 27 N·m (20 ft. lbs.).

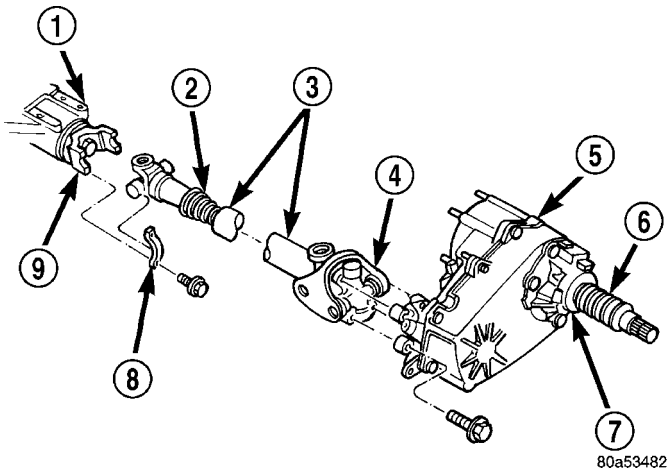


Fig. 10 FRONT PROPELLER SHAFT

1 - FRONT AXLE
2 - BOOT
3 - PROPELLER SHAFT
4 - CV-JOINT
5 - TRANSFER CASE
6 - BOOT
7 - SLINGER
8 - CLAMP
9 - YOKE

PROPELLER SHAFT - REAR

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Mark a line across the shaft and at each end of the propeller shaft for installation reference.

(3) Remove U-joint strap bolts, at pinion shaft yoke.

(4) Pry open clamp, holding dust boot to propeller shaft yoke (Fig. 11).

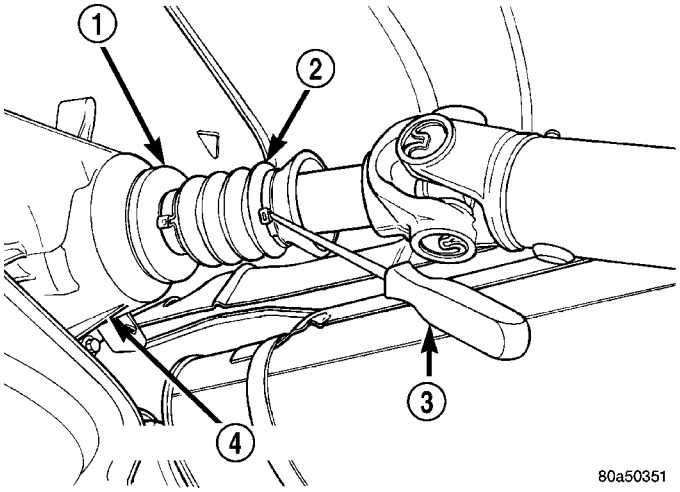


Fig. 11 DUST BOOT CLAMP

1 - SLINGER
2 - BOOT
3 - AWL
4 - TRANSFER CASE

PROPELLER SHAFT - REAR (Continued)

(5) Slide slip yoke off of transmission/transfer case output shaft and remove propeller shaft (Fig. 12).

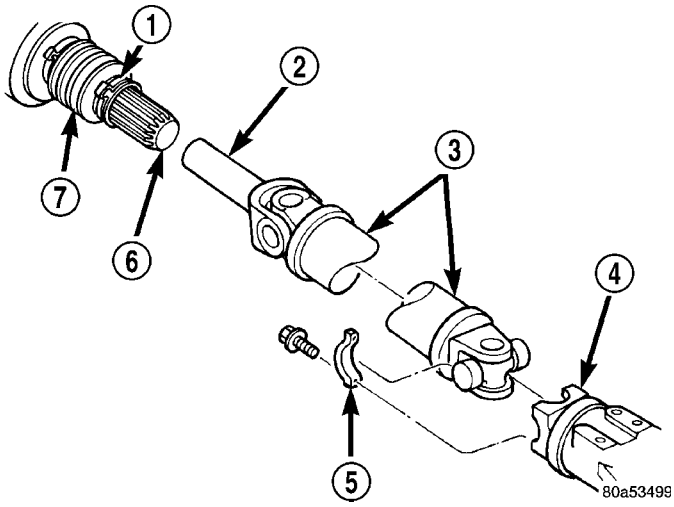


Fig. 12 REAR PROPELLER SHAFT

1 - CLAMP
2 - YOKE
3 - PROPELLER SHAFT
4 - AXLE YOKE
5 - CLAMP
6 - OUTPUT SHAFT
7 - BOOT

INSTALLATION

(1) Slide slip yoke on transmission/transfer case output shaft. Align installation reference marks at axle yoke and install propeller shaft.

(2) Install U-joint strap bolts at axle yoke and tighten to 19 N·m (14 ft. lbs.).

(3) Tighten clamp with Clamp Tool C-4975A to hold dust boot to propeller shaft yoke (Fig. 13).

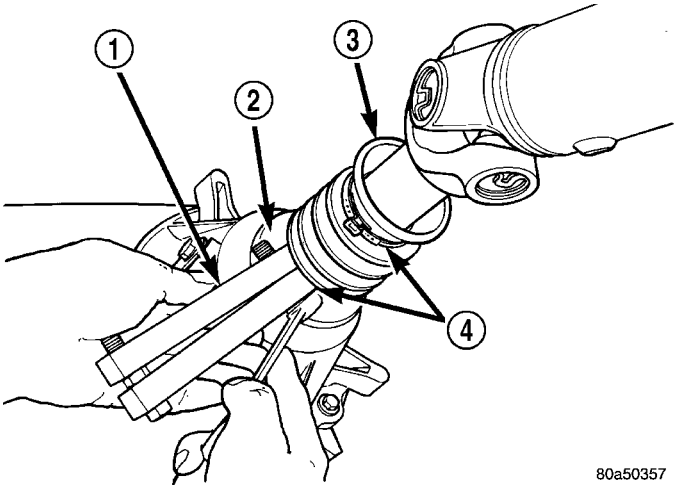


Fig. 13 CRIMPING DUST BOOT CLAMP

1 - CLAMP TOOL
2 - SLINGER
3 - BOOT
4 - CLAMP

PROPELLER SHAFT - REAR RUBICON

REMOVAL

(1) With vihicle in neutral, postion vehicle on hoist.

(2) Mark reference lines across the pinion yoke, transfer case flange and propeller shaft for installation reference.

(3) Remove transfer case flange bolts (Fig. 14) and U-joint strap bolts at the pinion shaft yoke.

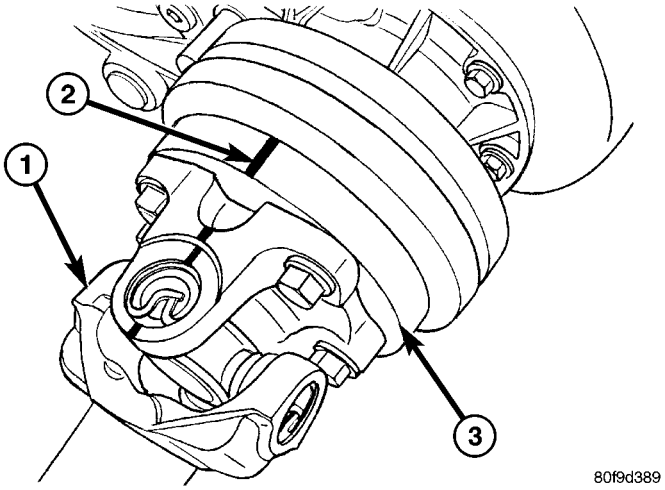


Fig. 14 TRANSFER CASE FLANGE

1 - REAR PROPELLER SHAFT
2 - REFERENCE MARK
3 - TRANSFER CASE FLANGE

(4) Remove propeller shaft from vehicle.

PROPELLER SHAFT - REAR RUBICON (Continued)

INSTALLATION

(1) Align installation reference marks at the pinion yoke, transfer case flange and propeller shaft.

(2) Install U-joint strap and tighten bolts to 19 N·m (14 ft. lbs.).

(3) Install transfer case flange bolts (Fig. 15) and tighten to 115 N·m (85 ft. lbs.).

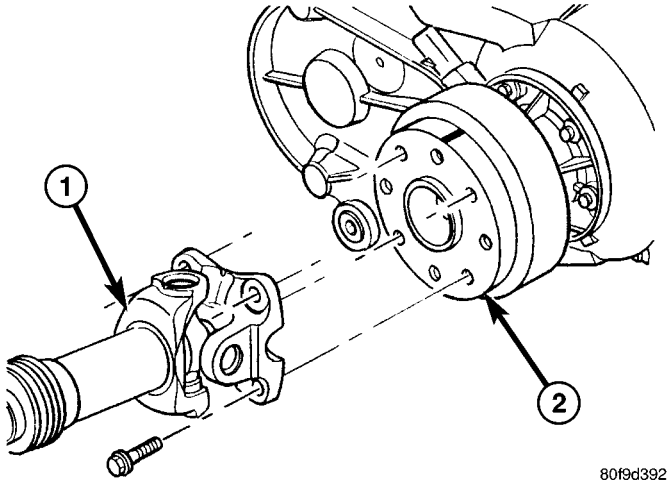


Fig. 15 TRANSFER CASE FLANGE BOLTS

1 - REAR PROPELLER SHAFT
2 - TRANSFER CASE FLANGE

SINGLE CARDAN UNIVERSAL JOINTS

DISASSEMBLY

NOTE: Individual components of cardan universal joints are not serviceable, they must be replaced as an assembly.

(1) Tap the outside of the bearing cap assembly with a drift to loosen the snap rings.

(2) Remove snap rings from both sides of yoke (Fig. 16).

(3) Position a socket with a inside diameter large enough to receive the bearing cap beneath the yoke on the press.

(4) Position yoke with the grease fitting if equipped, pointing up.

(5) Place another socket with an outside diameter smaller than bearing cap on the upper bearing cap and press the lower cap through the yoke (Fig. 17).

(6) Pull bearing cap of the yoke.

NOTE: If bearing cap will not come out, tap the yoke ear near the bearing cap to dislodge the cap.

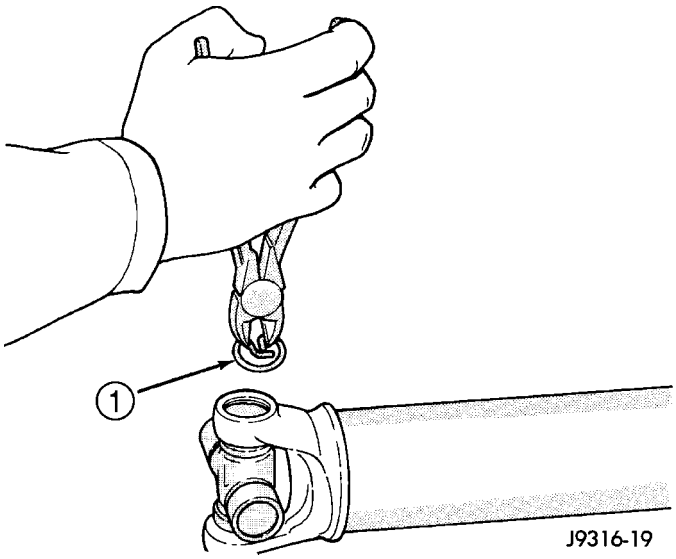


Fig. 16 YOKE SNAP RING

1 - SNAP RING

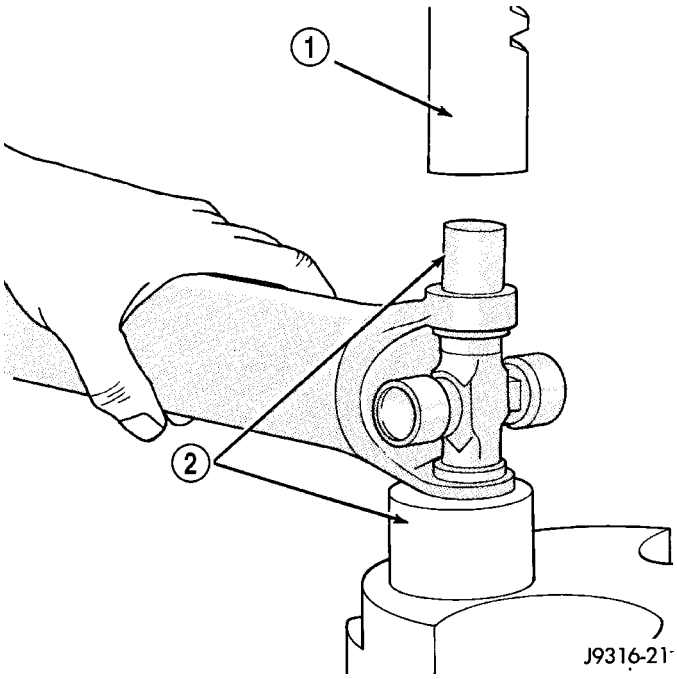


Fig. 17 PRESS OUT BEARING CAP

1 - PRESS
2 - SOCKET

SINGLE CARDAN UNIVERSAL JOINTS (Continued)

(7) Turn yoke over and straighten the cross in the open hole. Then carefully press the end of the cross until the other bearing cap can be removed (Fig. 18).

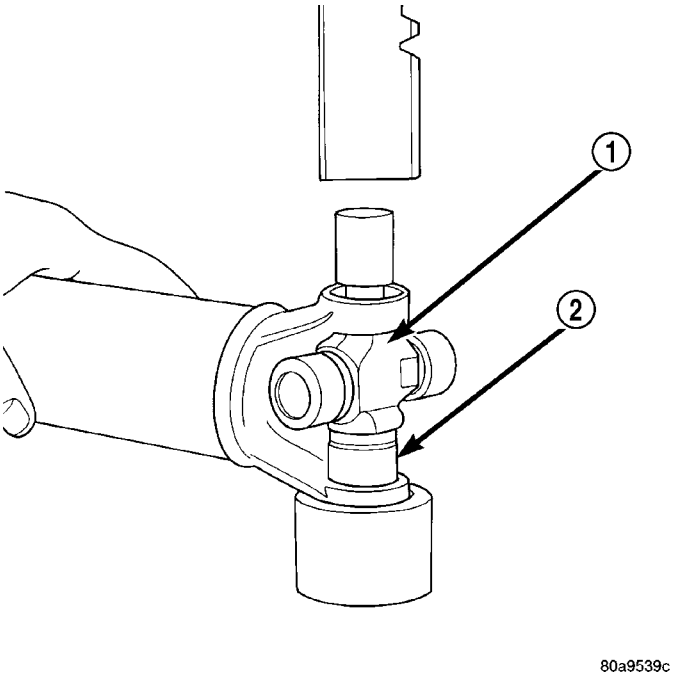


Fig. 18 PRESS OUT BEARING CAP

1 - CROSS
2 - BEARING CAP

ASSEMBLY

CAUTION: Keep cross and bearing cap straight during installation. Failure to heed caution may result in damage.

(1) Apply extreme pressure (EP) N.L.G.I. Grade 1 or 2 grease to inside of yoke bores.
(2) Position cross in the yoke with lube fitting if equipped, pointing up (Fig. 19).
(3) Place a bearing cap over the trunnion and align cap with yoke bore (Fig. 20).
(4) Press bearing cap into the yoke bore enough to clear snap ring groove.
(5) Install a snap ring.
(6) Repeat Step 3 and Step 4 to install the other bearing cap.

NOTE: If joint is stiff or binding, strike the yoke with a soft hammer to seat the needle bearings.

(7) Add grease to lube fitting, if equipped.

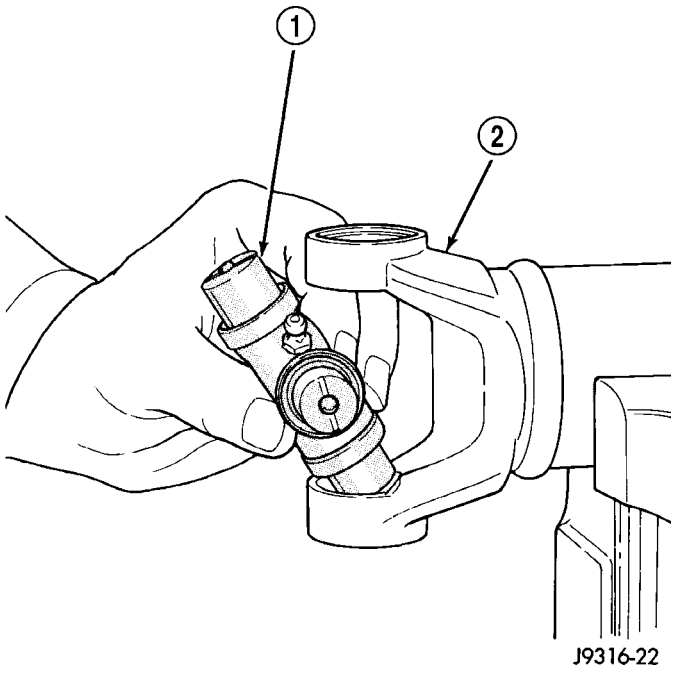


Fig. 19 U-JOINT CROSS

1 - CROSS
2 - YOKE

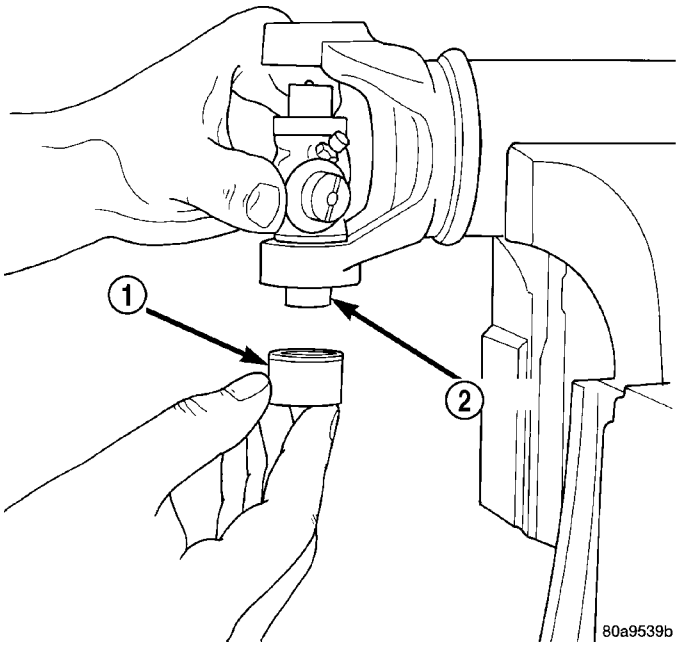


Fig. 20 BEARING AND CROSS

1 - BEARING CAP
2 - TRUNNION

DOUBLE CARDAN UNIVERSAL JOINTS

DISASSEMBLY

NOTE: Individual components of cardan universal joints are not serviceable they must be replaced as an assembly.

(1) Mark propeller shaft yoke and link yoke for assembly reference.
(2) Tap the outside of the bearing cap assembly with drift to loosen snap rings.
(3) Remove all bearing cap snap rings (Fig. 21).

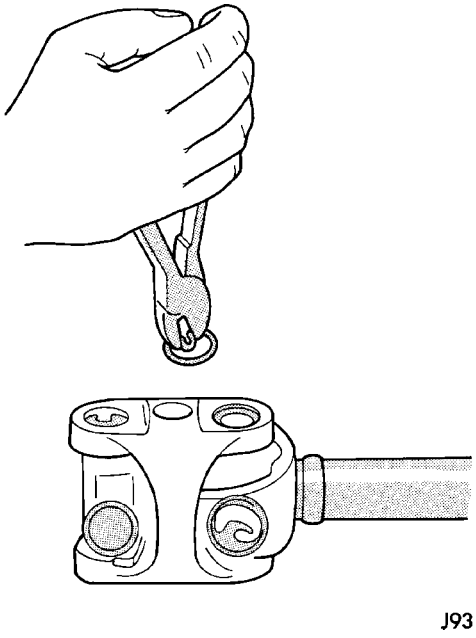


Fig. 21 SNAP RINGS

(4) Remove any grease fittings if equipped.
(5) Position a socket on the press with an inside diameter large enough to receive the bearing cap under the link yoke.
(6) Place another socket with an outside diameter smaller than the bearing cap on the upper bearing cap.
(7) Press one bearing cap from the outboard side of the link yoke enough to grasp the cap with vise jaws (Fig. 22).
(8) Grasp protruding bearing cap with vise jaws and tap link yoke with a mallet and drift to remove bearing cap (Fig. 23).

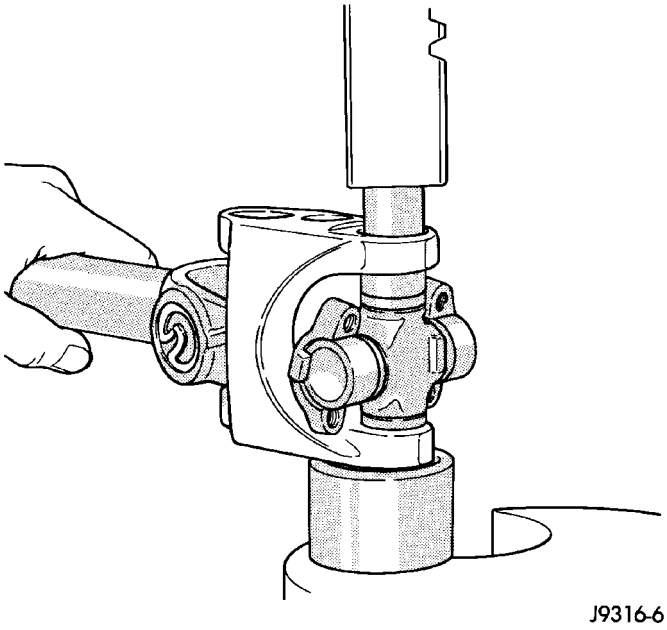


Fig. 22 PRESS OUT BEARING

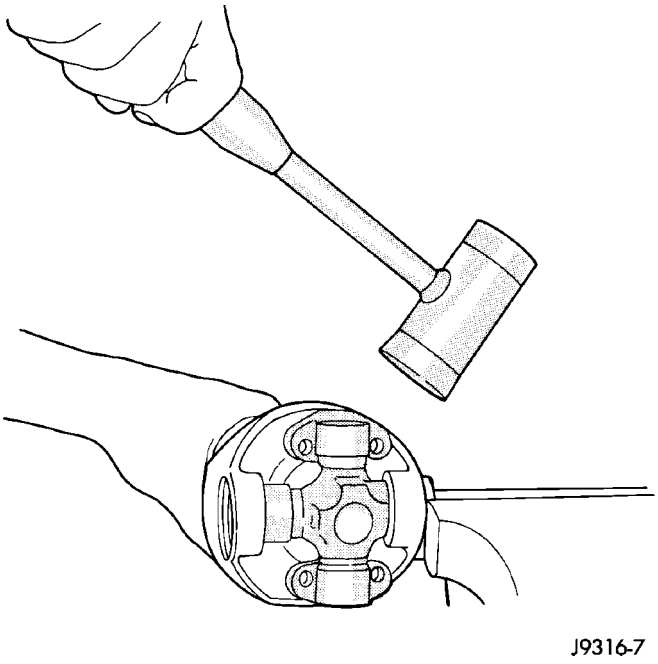


Fig. 23 REMOVE BEARING FROM YOKE

DOUBLE CARDAN UNIVERSAL JOINTS (Continued)

(9) Flip assembly and repeat Step 5, Step 6, Step 7 and Step 8 to remove the opposite bearing cap.

(10) Remove cross centering kit assembly and spring (Fig. 24).

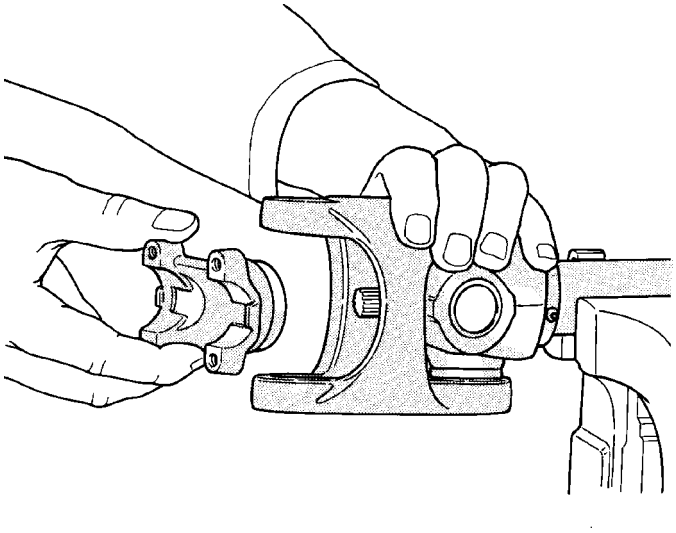


Fig. 24 CENTERING KIT

(11) Press remaining bearing caps out the other end of the link yoke, as described above to complete the disassembly.

ASSEMBLY

CAUTION: Alignment marks on link yoke and propeller shaft yoke must be aligned during assembled. Keep needle bearings upright in the bearing cap. Failure to heed caution may result in damage.

(1) Apply extreme pressure (EP) N.L.G.I. Grade 1 or 2 grease to inside of yoke bores.

(2) Fit a cross into propeller shaft yoke (Fig. 25).

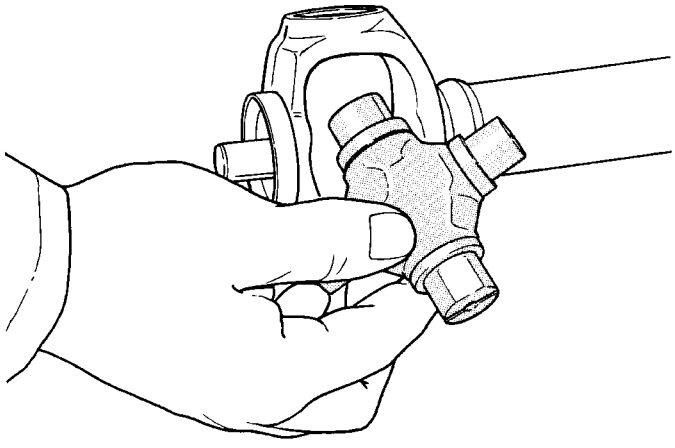


Fig. 25 INSTALL CROSS IN YOKE

(3) Place a bearing cap over the trunnion and align cap with the yoke bore (Fig. 26).

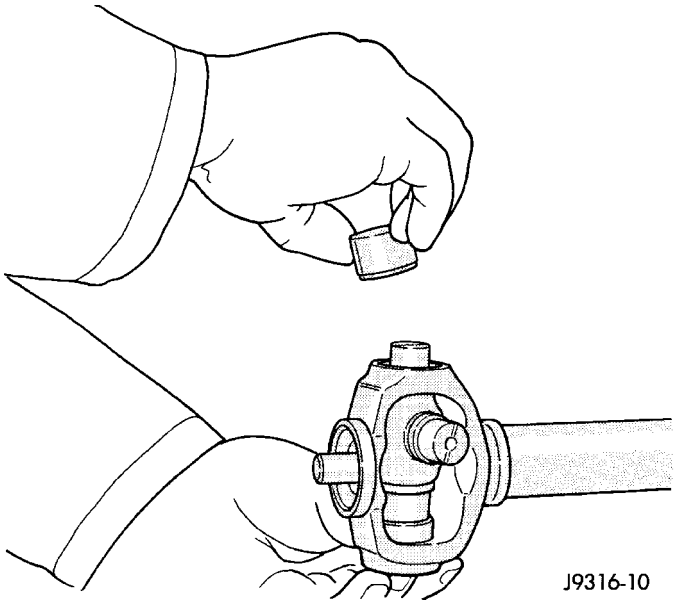


Fig. 26 INSTALL BEARING CAP

(4) Press bearing cap into the yoke bore enough to clear snap ring groove (Fig. 27).

(5) Install a snap ring.

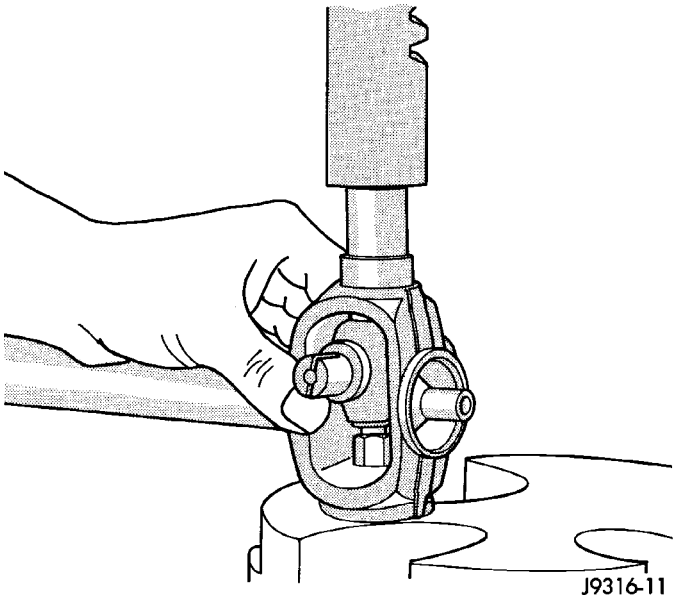


Fig. 27 PRESS BEARING CAP

DOUBLE CARDAN UNIVERSAL JOINTS (Continued)

(6) Flip propeller shaft yoke and install other bearing cap onto the opposite trunnion and install a snap ring (Fig. 28).

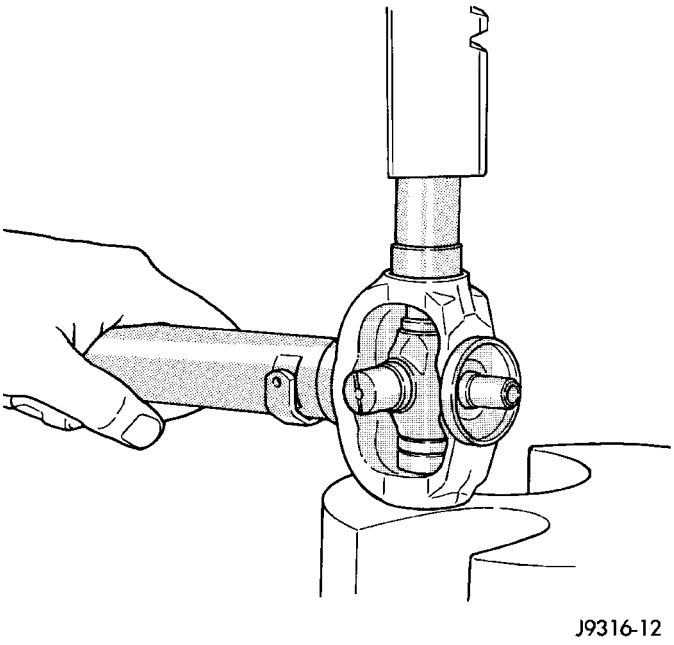


Fig. 28 PRESS BEARING CAP

(7) Fit link yoke onto the remaining trunnions and press both bearing caps into place and install snap rings (Fig. 29).

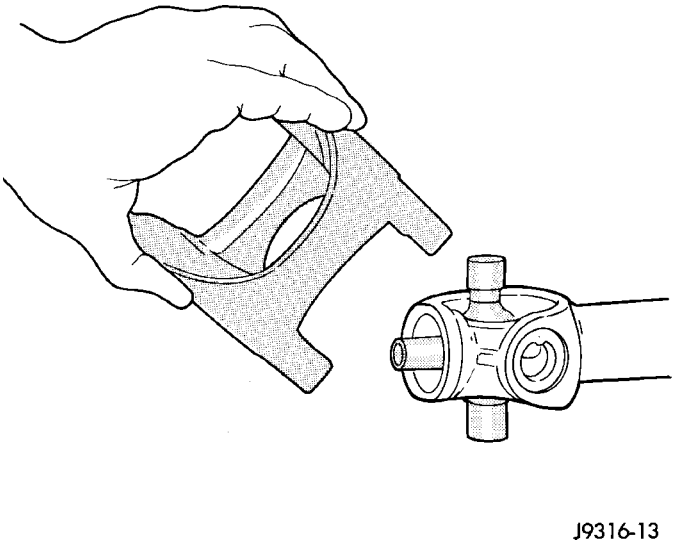


Fig. 29 INSTALL LINK YOKE

(8) Install centering kit assembly inside the link yoke (Fig. 30).

NOTE: Making sure the spring is properly positioned.

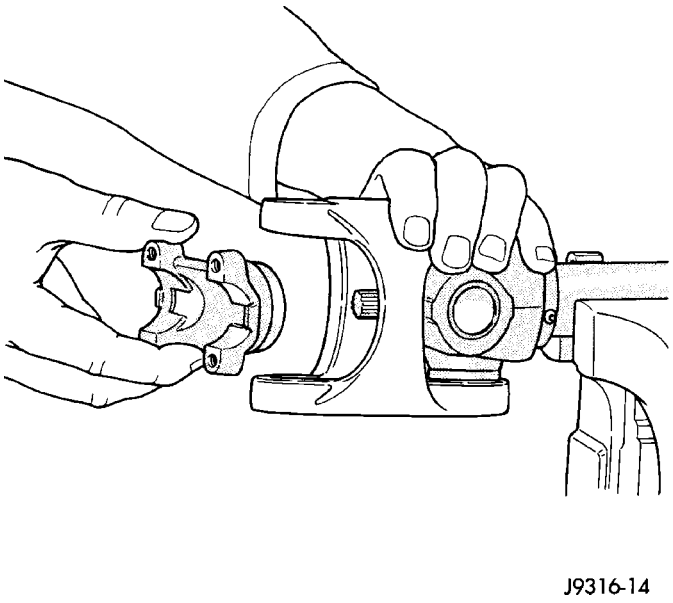


Fig. 30 CENTERING KIT

(9) Place two bearing caps on opposite trunnions of the remaining cross. Fit open trunnions into the link yoke bores and the bearing caps into the centering kit (Fig. 31).

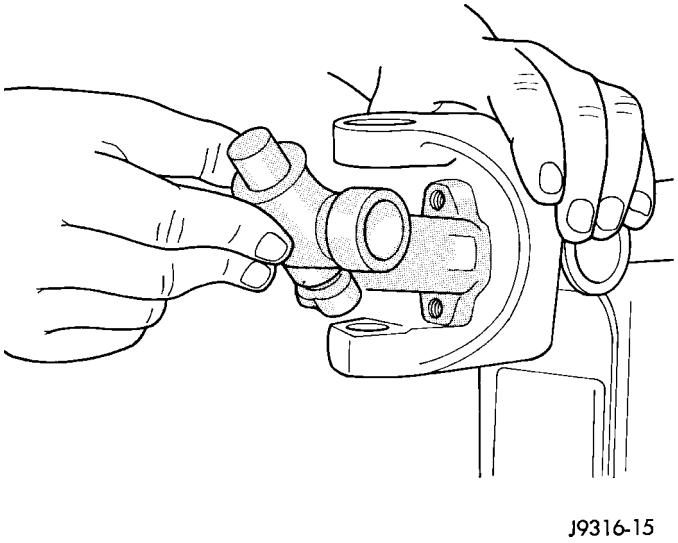


Fig. 31 REMAINING CROSS

DOUBLE CARDAN UNIVERSAL JOINTS (Continued)

(10) Press remaining two bearing caps into place and install snap rings (Fig. 32).

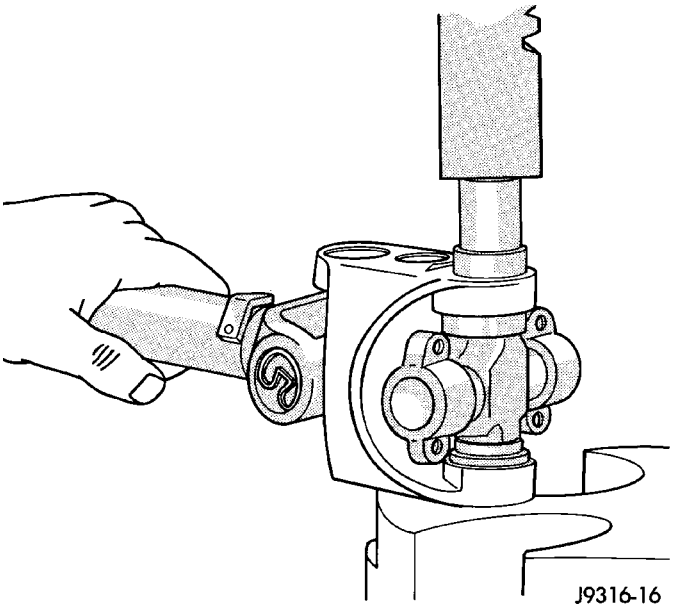


Fig. 32 PRESS BEARING CAP

(11) Tap snap rings to seat them into the grooves (Fig. 33).

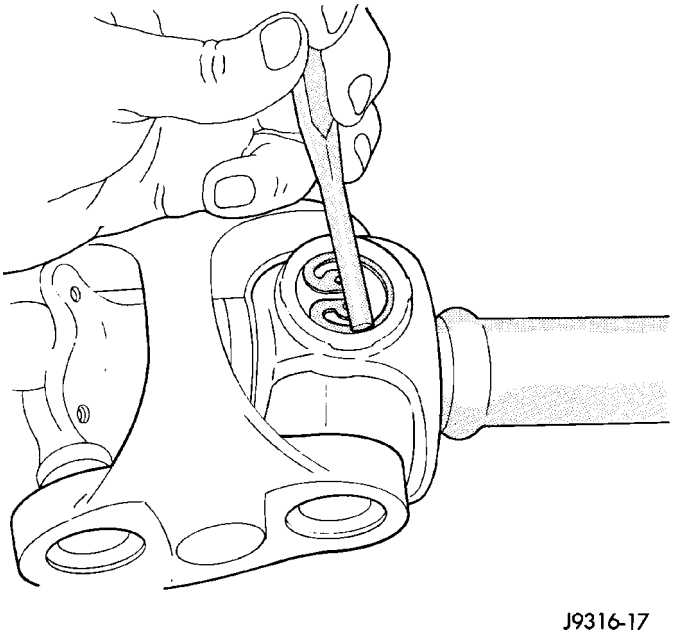


Fig. 33 SEAT SNAP RINGS

(12) Flexing joint beyond center, the joint should snap over-center in both directions if assembled correctly (Fig. 34).

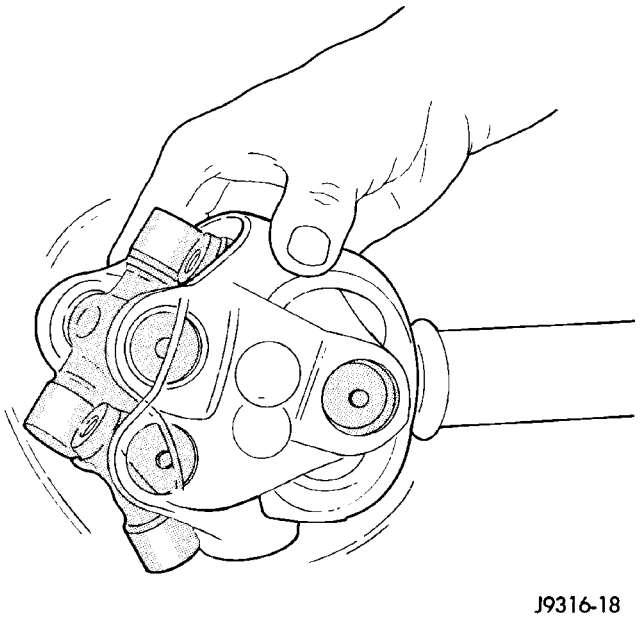


Fig. 34 VERIFY ASSEMBLY

FRONT AXLE - 181FBI

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FRONT AXLE - 181FBI

DIAGNOSIS AND TESTING

GEAR NOISE

Axle gear noise can be caused by insufficient lubricant, incorrect backlash, tooth contact, worn/damaged gears or the carrier housing not having the proper offset and squareness.

Gear noise usually happens at a specific speed range. The noise can also occur during a specific type of driving condition. These conditions are acceleration, deceleration, coast, or constant load.

When road testing, first warm-up the axle fluid by driving the vehicle at least 5 miles and then accelerate the vehicle to the speed range where the noise is the greatest. Shift out-of-gear and coast through the peak-noise range. If the noise stops or changes greatly:

- Check for insufficient lubricant.
- Incorrect ring gear backlash.
- Gear damage.

Differential side gears and pinions can be checked by turning the vehicle. They usually do not cause noise during straight-ahead driving when the gears are unloaded. The side gears are loaded during vehicle turns. A worn pinion mate shaft can also cause a snapping or a knocking noise.

BEARING NOISE

The axle shaft, differential and pinion bearings can all produce noise when worn or damaged. Bearing noise can be either a whining, or a growling sound.

Pinion bearings have a constant-pitch noise. This noise changes only with vehicle speed. Pinion bearing noise will be higher pitched because it rotates at a faster rate. Drive the vehicle and load the differential. If bearing noise occurs, the rear pinion bearing is the source of the noise. If the bearing noise is heard during a coast, the front pinion bearing is the source.

Worn or damaged differential bearings usually produce a low pitch noise. Differential bearing noise is similar to pinion bearing noise. The pitch of differential bearing noise is also constant and varies only with vehicle speed.

Axle shaft bearings produce noise and vibration when worn or damaged. The noise generally changes when the bearings are loaded. Road test the vehicle. Turn the vehicle sharply to the left and to the right. This will load the bearings and change the noise level. Where axle bearing damage is slight, the noise is usually not noticeable at speeds above 30 mph.

FRONT AXLE - 181FBI (Continued)

LOW SPEED KNOCK

Low speed knock is generally caused by a worn U-joint or by worn side-gear thrust washers. A worn pinion shaft bore will also cause low speed knock.

VIBRATION

Vibration at the rear of the vehicle is usually caused by:

- Damaged drive shaft.
- Missing drive shaft balance weight(s).
- Worn or out of balance wheels.
- Loose wheel lug nuts.
- Worn U-joint(s).
- Loose/broken springs.
- Damaged axle shaft bearing(s).
- Loose pinion gear nut.
- Excessive pinion yoke run out.
- Bent axle shaft(s).

Check for loose or damaged front end components or engine/transmission mounts. These components can contribute to what appears to be a rear end vibration. Do not overlook engine accessories, brackets and drive belts.

All driveline components should be examined before starting any repair.

DRIVELINE SNAP

A snap or clunk noise when the vehicle is shifted into gear (or the clutch engaged) can be caused by:

- High engine idle speed.
- Transmission shift operation.
- Loose engine/transmission/transfer case mounts.
- Worn U-joints.
- Loose spring mounts.
- Loose pinion gear nut and yoke.
- Excessive ring gear backlash.
- Excessive side gear to case clearance.

The source of a snap or a clunk noise can be determined with the assistance of a helper. Raise the vehicle on a hoist with the wheels free to rotate. Instruct the helper to shift the transmission into gear. Listen for the noise, a mechanics stethoscope is helpful in isolating the source of a noise.

DIAGNOSTIC CHART

Condition	Possible Causes	Correction
Wheel Noise	1. Wheel loose.	1. Tighten loose nuts.
	2. Faulty, brinelled wheel bearing.	2. Replace bearing.
Axle Shaft Noise	1. Misaligned axle tube.	1. Inspect axle tube alignment. Correct as necessary.
	2. Bent or sprung axle shaft.	2. Inspect and correct as necessary.
	3. End-play in pinion bearings.	3. Refer to pinion pre-load information and correct as necessary.
	4. Excessive gear backlash between the ring gear and pinion.	4. Check adjustment of the ring gear and pinion backlash. Correct as necessary.
	5. Improper adjustment of pinion gear bearings.	5. Adjust the pinion bearings pre-load.
	6. Loose pinion yoke nut.	6. Tighten the pinion yoke nut.
	7. Scuffed gear tooth contact surfaces.	7. Inspect and replace as necessary.

FRONT AXLE - 181FBI (Continued)

Condition	Possible Causes	Correction
Axle Shaft Broke	1. Misaligned axle tube.	1. Replace the broken shaft after correcting tube mis-alignment.
	2 Vehicle overloaded.	2. Replace broken shaft and avoid excessive weight on vehicle.
	3. Erratic clutch operation.	3. Replace broken shaft and avoid or correct erratic clutch operation.
	4. Grabbing clutch.	4. Replace broken shaft and inspect and repair clutch as necessary.
Differential Cracked	1. Improper adjustment of the differential bearings.	1. Replace case and inspect gears and bearings for further damage. Set differential bearing pre-load properly.
	2. Excessive ring gear backlash.	2. Replace case and inspect gears and bearings for further damage. Set ring gear backlash properly.
	3. Vehicle overloaded.	3. Replace case and inspect gears and bearings for further damage. Avoid excessive vehicle weight.
	4. Erratic clutch operation.	4. Replace case and inspect gears and bearings for further damage. Avoid erratic use of clutch.
Differential Gears Scored	1. Insufficient lubrication.	1. Replace scored gears. Fill differential with the correct fluid type and quantity.
	2. Improper grade of lubricant.	2. Replace scored gears. Fill differential with the correct fluid type and quantity.
	3. Excessive spinning of one wheel/tire.	3. Replace scored gears. Inspect all gears, pinion bores, and shaft for damage. Service as necessary.
Loss Of Lubricant	1. Lubricant level too high.	1. Drain lubricant to the correct level.
	2. Worn axle shaft seals.	2. Replace seals.
	3. Cracked differential housing.	3. Repair as necessary.
	4. Worn pinion seal.	4. Replace seal.
	5. Worn/scored yoke.	5. Replace yoke and seal.
	6. Axle cover not properly sealed.	6. Remove, clean, and re-seal cover.
Axle Overheating	1. Lubricant level low.	1. Fill differential to correct level.
	2. Improper grade of lubricant.	2. Fill differential with the correct fluid type and quantity.
	3. Bearing pre-loads too high.	3. Re-adjust bearing pre-loads.
	4. Insufficient ring gear backlash.	4. Re-adjust ring gear backlash.

FRONT AXLE - 181FBI (Continued)

Condition	Possible Causes	Correction
Gear Teeth Broke	1. Overloading.	1. Replace gears. Examine other gears and bearings for possible damage.
	2. Erratic clutch operation.	2. Replace gears and examine the remaining parts for damage. Avoid erratic clutch operation.
	3. Ice-spotted pavement.	3. Replace gears and examine remaining parts for damage.
	4. Improper adjustments.	4. Replace gears and examine remaining parts for damage. Ensure ring gear backlash is correct.
Axle Noise	1. Insufficient lubricant.	1. Fill differential with the correct fluid type and quantity.
	2. Improper ring gear and pinion adjustment.	2. Check ring gear and pinion contact pattern.
	3. Unmatched ring gear and pinion.	3. Replace gears with a matched ring gear and pinion.
	4. Worn teeth on ring gear and/or pinion.	4. Replace ring gear and pinion.
	5. Loose pinion bearings.	5. Adjust pinion bearing pre-load.
	6. Loose differential bearings.	6. Adjust differential bearing pre-load.
	7. Mis-aligned or sprung ring gear.	7. Measure ring gear run-out. Replace components as necessary.
	8. Loose differential bearing cap bolts.	8. Inspect differential components and replace as necessary. Ensure that the bearing caps are torqued tot he proper specification.
	9. Housing not machined properly.	9. Replace housing.

FRONT AXLE - 181FBI (Continued)

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Position a lift under the axle and secure to the axle.
(3) Remove brake components.
(4) Remove vent hose from axle shaft tube.
(5) Remove propeller shaft.
(6) Remove stabilizer bar links at the axle (Fig. 1).

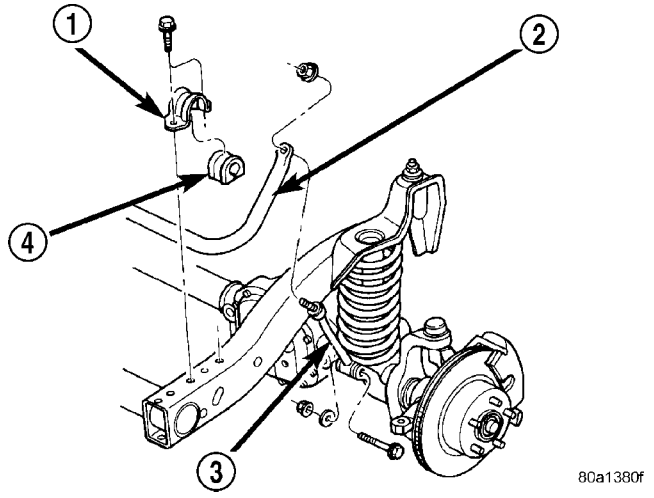

Fig. 1 STABILIZER BAR

1 - RETAINER
2 - STABILIZER BAR
3 - LINK
4 - BUSHING

(7) Remove shock absorbers from axle brackets (Fig. 2).
(8) Remove track bar from axle bracket (Fig. 3).
(9) Remove tie rod and drag link from steering knuckle.
(10) Remove steering damper from axle bracket.
(11) Remove upper and lower suspension arms from axle brackets.
(12) Lower lift enough to remove the axle. The coil springs will drop with the axle.
(13) Remove coil springs from the axle.

INSTALLATION

CAUTION: The weight of the vehicle must be supported by the springs before suspension arms and track bar fasteners can be tightened. Failure to heed caution may result in damage.

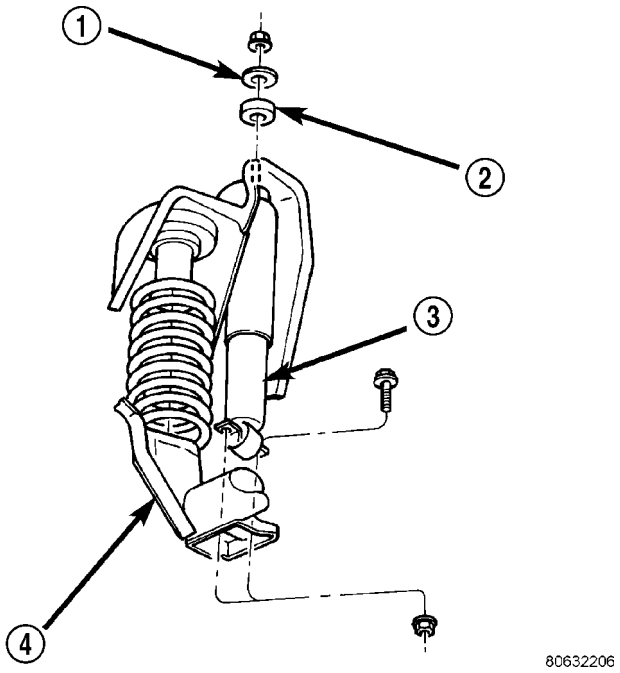

Fig. 2 COIL SPRING & SHOCK ABSORBER

1 - RETAINER
2 - GROMMET
3 - SHOCK
4 - FRONT AXLE

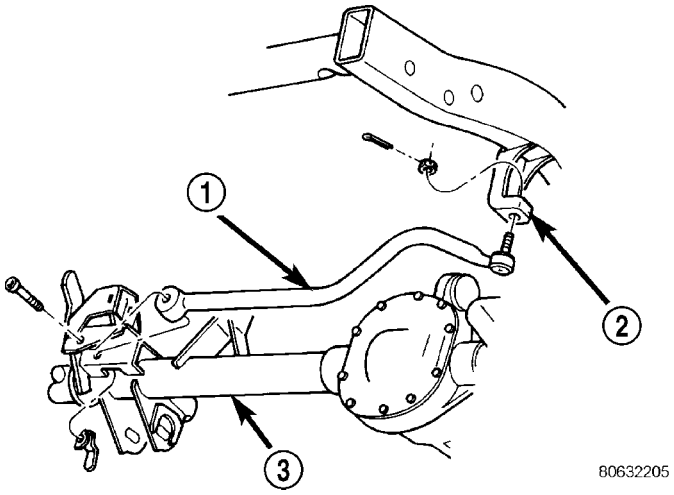

Fig. 3 TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - FRONT AXLE

(1) Install springs and retainers and tighten retainer bolts to 21 N·m (16 ft. lbs.).
(2) Position axle under vehicle and align it with the spring pads.

FRONT AXLE - 181FBI (Continued)

(3) Install upper and lower suspension arms in the axle brackets and loosely install bolts and nuts (Fig. 4).

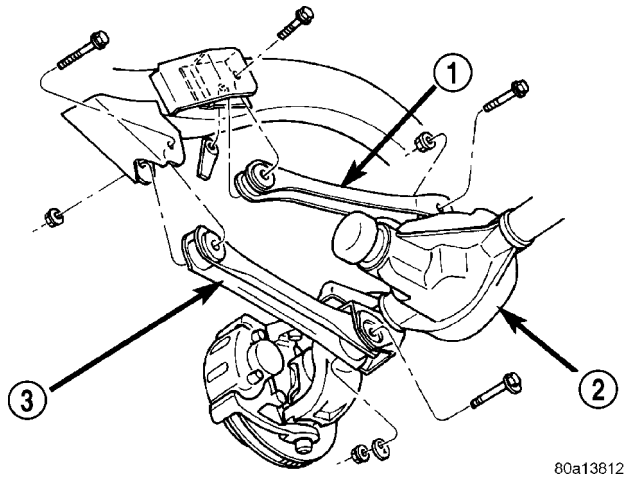

Fig. 4 UPPER & LOWER SUSPENSION ARMS

1 - UPPER SUSPENSION ARM
2 - FRONT AXLE
3 - LOWER SUSPENSION ARM

(4) Connect the vent hose to the axle shaft tube.

(5) Install track bar to the axle bracket and loosely install bolt (Fig. 5).

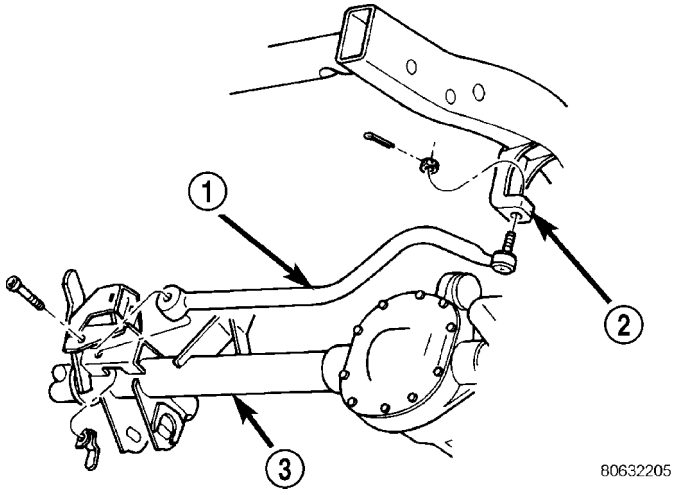

Fig. 5 TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - FRONT AXLE

(6) Install shock absorbers and tighten the bolts (Fig. 6) to torque specification.

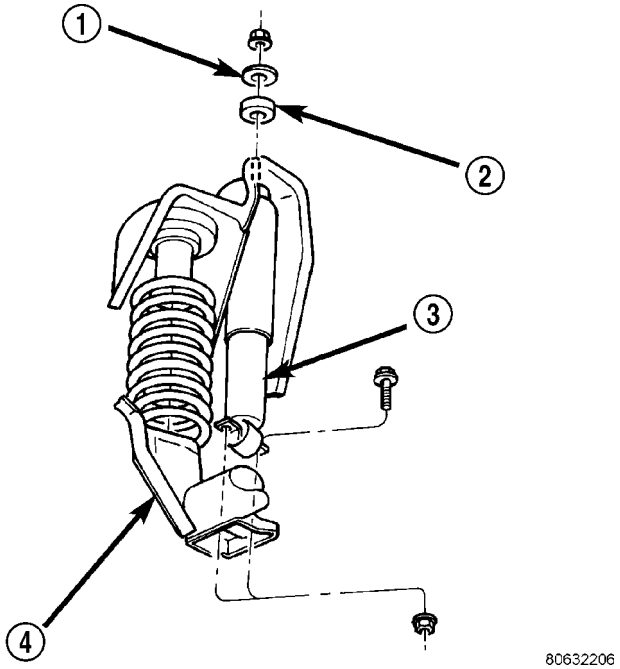

Fig. 6 COIL SPRING & SHOCK ABSORBER

1 - RETAINER
2 - GROMMET
3 - SHOCK
4 - FRONT AXLE

(7) Install stabilizer bar links to the axle brackets and tighten nuts (Fig. 7) to torque specification.

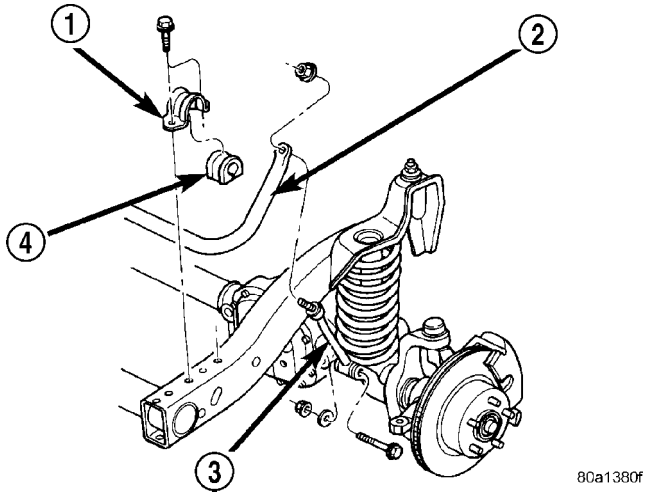

Fig. 7 STABILIZER BAR

1 - RETAINER
2 - STABILIZER BAR
3 - LINK
4 - BUSHING

(8) Install drag link and tie rod and tighten nut to torque specification

(9) Install steering damper to the axle bracket and tighten nut to torque specification.

(10) Install brake components.

(11) Install propeller shaft.

(12) Remove lift from the axle and lower vehicle.

(13) Tighten upper and lower control arm nuts to torque specification.

(14) Tighten track bar bolt at the axle bracket to torque specification.

(15) Check the front wheel alignment.

ADJUSTMENTS

Ring and pinion gears are supplied as matched sets only. The identifying numbers for the ring and pinion gear are etched/marked onto each gear (Fig. 8). A plus (+) number, minus (–) number or zero (0) is etched/marked on the pinion gear. This number is the amount (in thousandths of an inch) the depth varies from the standard depth setting of a pinion etched/marked with a (0). The standard setting from the center line of the ring gear to the back face of the pinion is 92.08 mm (3.625 in.). The standard depth provides the best gear tooth contact pattern.

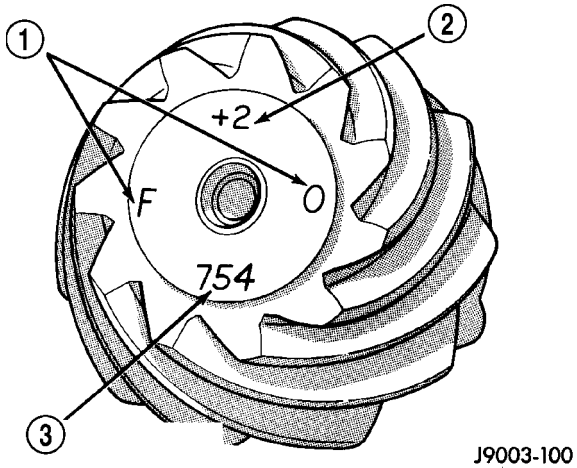

Fig. 8 PINION GEAR ID NUMBERS

1 - PRODUCTION NUMBERS
2 - PINION GEAR DEPTH VARIANCE
3 - GEAR MATCHING NUMBER

Compensation for pinion depth variance is achieved with a select shim/oil slinger. The shims are placed between the rear pinion bearing and the pinion gear head (Fig. 9).

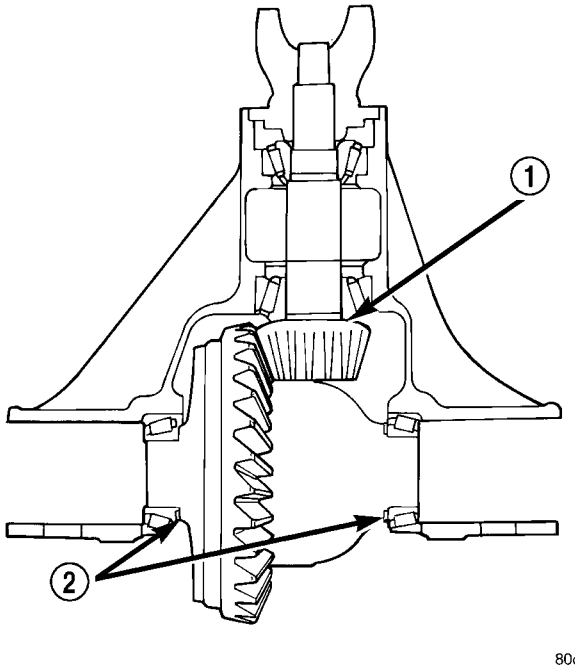

Fig. 9 SHIM LOCATIONS

1 - PINION GEAR DEPTH SHIM/OIL SLINGER
2 - DIFFERENTIAL BEARING SHIM

If a new gear set is being installed, note the depth variance etched into both the original and replacement pinion. Add or subtract this number from the thickness of the original depth shim/oil slinger to compensate for the difference in the depth variances. Refer to the Pinion Gear Depth Variance chart.

Note where Old and New Pinion Marking columns intersect. Intersecting figure represents plus or minus the amount needed.

Note the etched number on the face of the pinion gear head (–1, –2, 0, +1, +2, etc.). The numbers represent thousands of an inch deviation from the standard. If the number is negative, add that value to the required thickness of the depth shims. If the number is positive, subtract that value from the thickness of the depth shim. If the number is 0 no change is necessary.

FRONT AXLE - 181FBI (Continued)

PINION GEAR DEPTH VARIANCE

Original Pinion Gear Depth Variance	Replacement Pinion Gear Depth Variance								
	−4	−3	−2	−1	0	+1	+2	+3	+4
+4	+0.008	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0
+3	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001
+2	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002
+1	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003
0	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004
−1	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005
−2	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006
−3	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007
−4	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007	−0.008

PINION DEPTH MEASUREMENT AND ADJUSTMENT

Measurements are taken with pinion bearing cups and pinion bearings installed in the housing. Take measurements with Pinion Gauge Set and Dial Indicator C-3339 (Fig. 10).

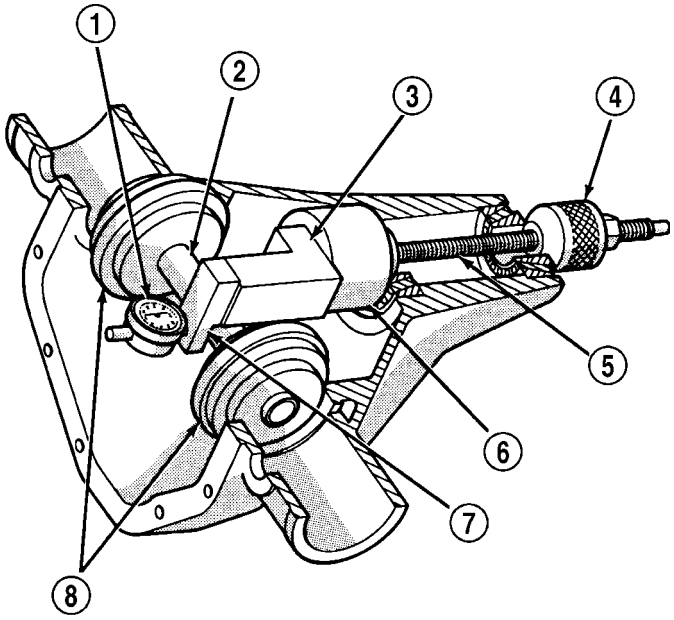

Fig. 10 PINION GEAR DEPTH GAUGE TOOLS

1 - DIAL INDICATOR
2 - ARBOR
3 - PINION HEIGHT BLOCK
4 - CONE
5 - SCREW
6 - PINION BLOCK
7 - SCOOTER BLOCK
8 - ARBOR DISC

(1) Assemble Pinion Height Block 6739, Pinion Block 6733 and rear pinion bearing onto Screw 6741 (Fig. 10).

(2) Insert assembled height gauge components, rear bearing and screw into the housing through pinion bearing cups (Fig. 11).

(3) Install front pinion bearing and Cone-nut 6740 hand tight.

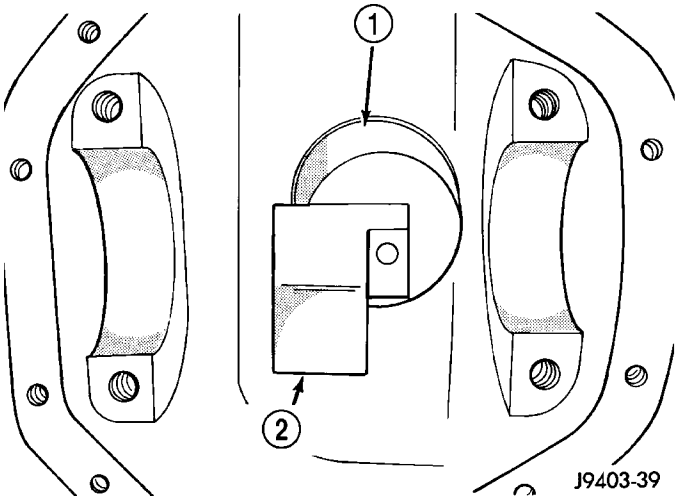

Fig. 11 PINION HEIGHT BLOCK

1 - PINION BLOCK
2 - PINION HEIGHT BLOCK

FRONT AXLE - 181FBI (Continued)

(4) Place Arbor Disc 6732 on Arbor D-115-3 in position in axle housing side bearing cradles (Fig. 12). Install differential bearing caps on arbor discs and tighten cap bolts to 41 N·m (30 ft. lbs.).

NOTE: Arbor Discs 6732 has different step diameters to fit other axles. Choose proper step for axle being serviced.

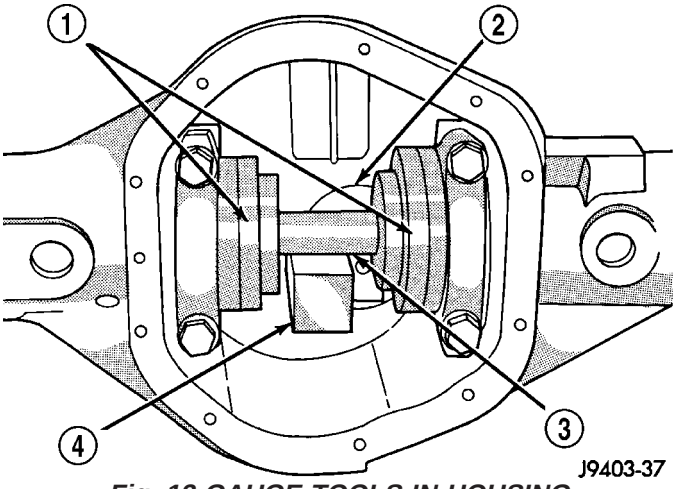

Fig. 12 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(5) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(6) Position Scooter Block/Dial Indicator flush on the pinion height block. Hold scooter block and zero the dial indicator.

(7) Slowly slide the scooter block across the pinion height block over to the arbor (Fig. 13). Move the scooter block till the dial indicator probe crests the arbor and record the highest reading.

(8) Select a shim/oil baffle equal to the dial indicator reading plus the pinion depth variance number etched in the face of the pinion (Fig. 8). For example, if the depth variance is –2, add +0.002 in. to the dial indicator reading.

DIFFERENTIAL SIDE BEARING PRELOAD AND GEAR BACKLASH

Differential side bearing preload and gear backlash is achieved by selective shims positioned behind the differential side bearing cones. The proper shim thickness can be determined using slip-fit Dummy Bearings D-348 in place of the differential side bearings and a Dial Indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 14). Differential shim measurements are performed with spreader W-129-B removed.

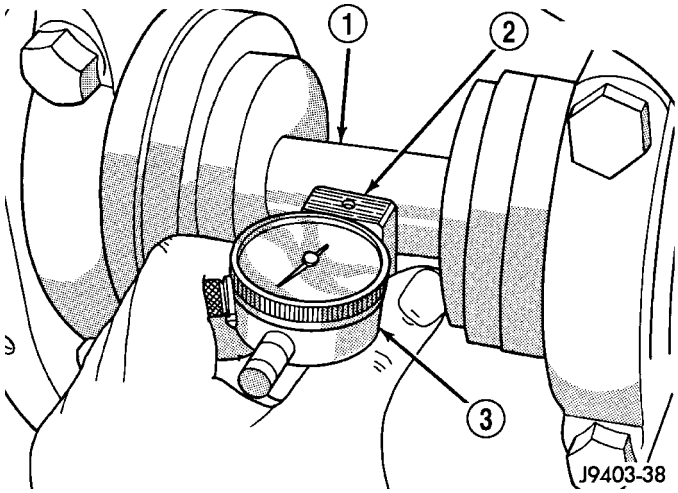

Fig. 13 PINION DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

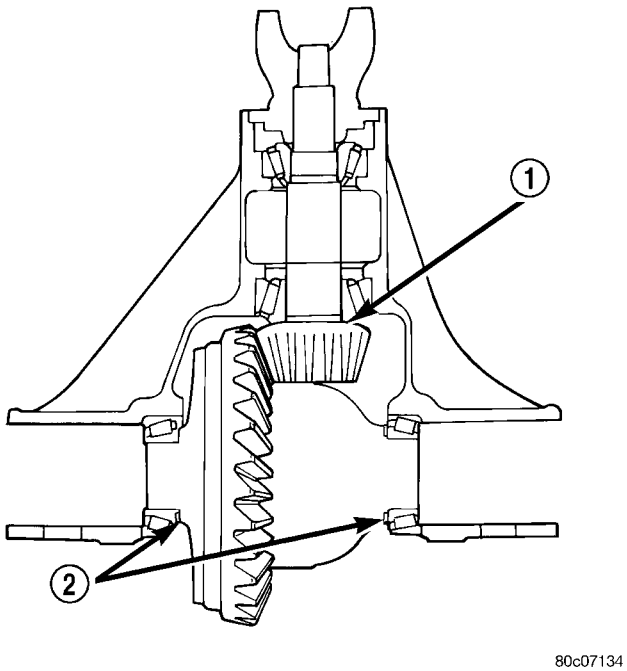

Fig. 14 SHIM LOCATIONS

1 - PINION GEAR DEPTH SHIM/OIL SLINGER
2 - DIFFERENTIAL BEARING SHIM

FRONT AXLE - 181FBI (Continued)

PRELOAD SHIM SELECTION

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove differential side bearings from differential case.
(2) Remove factory installed shims from differential case.
(3) Install ring gear on differential case and tighten bolts to specification.
(4) Install dummy side bearings D-348 on differential case.
(5) Install differential case in the housing.
(6) Install marked bearing caps in their correct positions. Install and snug the bolts (Fig. 15).

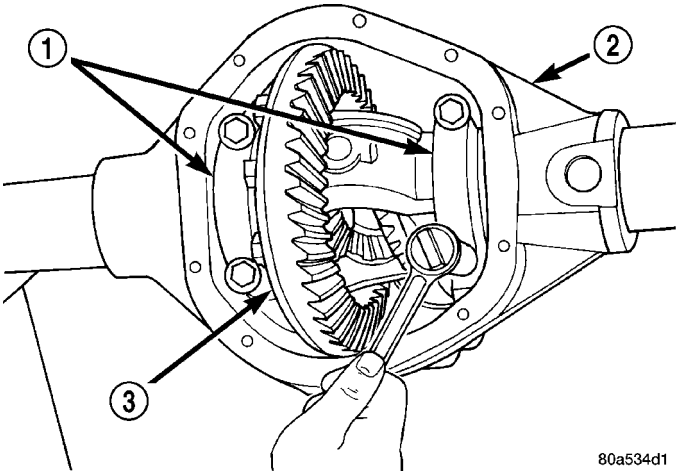

Fig. 15 BEARING CAP BOLTS

1 - BEARING CAP
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

(7) With a dead-blow hammer, seat the differential dummy bearings to each side of the housing (Fig. 16) and (Fig. 17).
(8) Thread Pilot Stud C-3288-B into rear cover bolt hole below ring gear (Fig. 18).
(9) Attach the Dial Indicator C-3339 to pilot stud. Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 18).
(10) Push and hold differential case to pinion gear side of the housing and zero dial indicator (Fig. 19).
(11) Push and hold differential case to ring gear side of the housing and record the dial indicator reading (Fig. 20).
(12) Add 0.008 in. (0.2 mm) to the zero end play total. This new total represents the thickness of shims to compress or preload the new bearings when the differential is installed.
(13) Rotate dial indicator out of the way on the pilot stud.

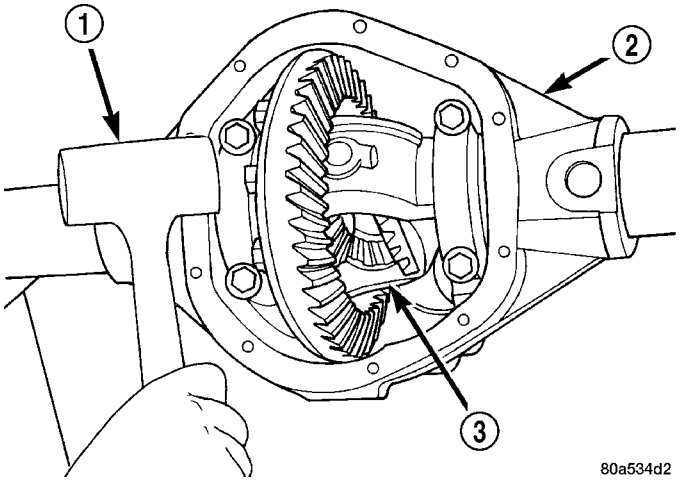

Fig. 16 SEAT DUMMY BEARINGS PINION SIDE

1 - MALLET
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

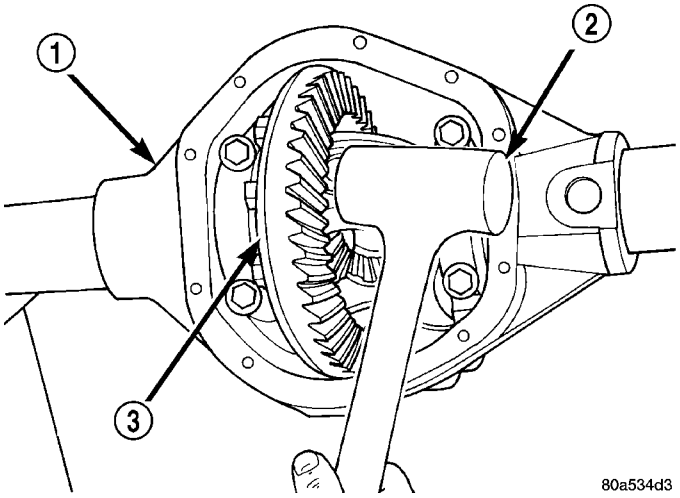

Fig. 17 SEAT DUMMY BEARING RING GEAR SIDE

1 - DIFFERENTIAL HOUSING
2 - MALLET
3 - DIFFERENTIAL CASE

(14) Remove differential case and dummy bearings from the housing.
(15) Install the pinion gear in the housing. Install the pinion yoke and establish the correct pinion rotating torque.
(16) Install differential case and dummy bearings D-348 in the housing (without shims), install bearing caps and tighten bolts snug.
(17) Seat ring gear side dummy bearing (Fig. 17).

FRONT AXLE - 181FBI (Continued)

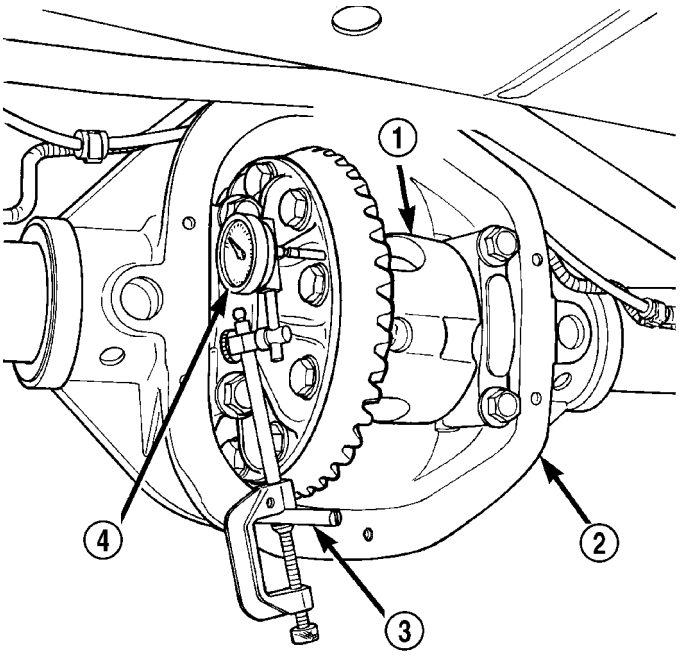

Fig. 18 DIFFERENTIAL SIDE PLAY MEASUREMENT

1 - DIFFERENTIAL CASE
2 - DIFFERENTIAL HOUSING
3 - PILOT STUD
4 - DIAL INDICATOR

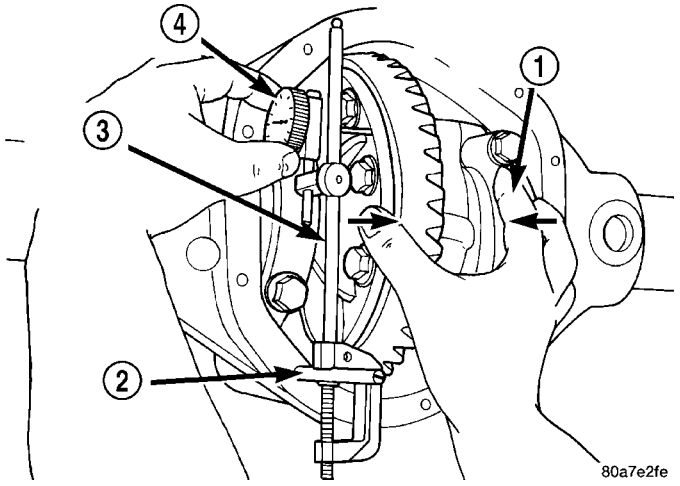

Fig. 19 DIAL INDICATOR LOCATION

1 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
2 - PILOT STUD
3 - DIAL INDICATOR
4 - ZERO DIAL INDICATOR FACE

(18) Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 18).

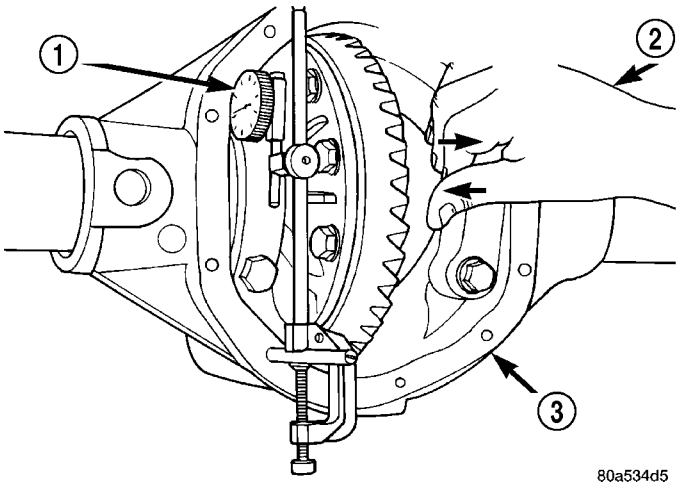

Fig. 20 READ DIAL INDICATOR

1 - READ DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(19) Push and hold differential case toward pinion gear and zero the dial indicator (Fig. 21).

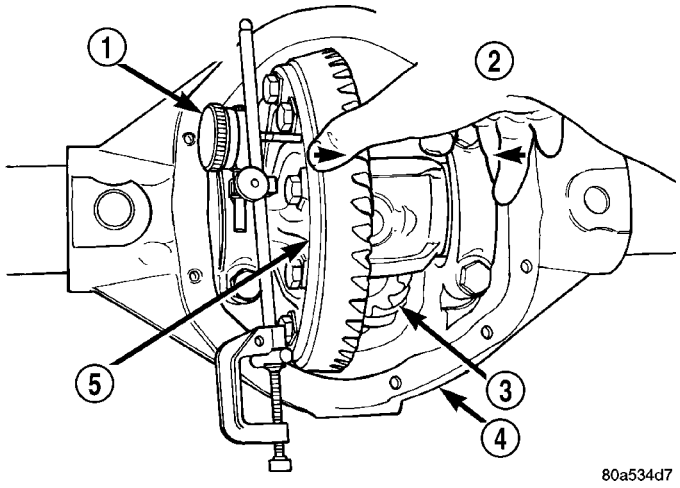

Fig. 21 ZERO DIAL INDICATOR

1 - ZERO DIAL INDICATOR FACE
2 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

FRONT AXLE - 181FBI (Continued)

(20) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 22).

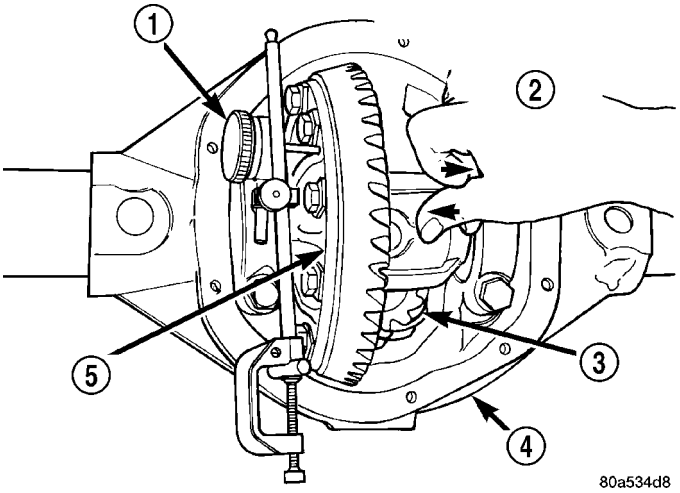

Fig. 22 DIFFERENTIAL CASE RING GEAR SIDE

1 - READ DIAL INDICATOR
2 - FORCE DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(21) Subtract 0.05 mm (0.002 in.) from the dial indicator reading to compensate for backlash between ring and pinion gears. This total is the thickness shim required to achieve proper backlash.

(22) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

(23) Rotate dial indicator out of the way on pilot stud.

(24) Remove differential case and dummy bearings from the housing.

(25) Install the selected shims onto the differential case hubs.

(26) Install side bearings and cups on differential case.

(27) Install spreader W-129-B and adaters from Adapter Set 6987, onto housing. Spread the housing to receive differential case.

(28) Install differential case into the housing.

(29) Remove spreader from the housing.

(30) Install bearing caps and tighten bolts to 61 N·m (45 ft. lbs.).

(31) Rotate the differential case several times to seat the side bearings.

(32) Position the indicator plunger against a ring gear tooth (Fig. 23).

(33) Push and hold ring gear upward while not allowing the pinion gear to rotate.

(34) Zero dial indicator face to pointer.

(35) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 24).

(36) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform Gear Contact Pattern procedure.

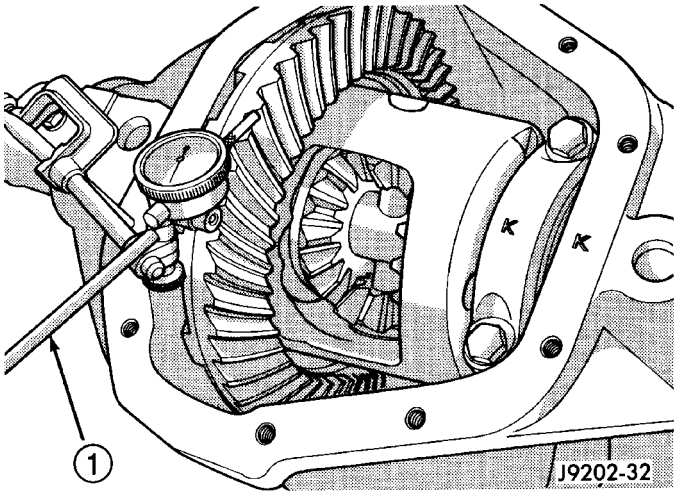

Fig. 23 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

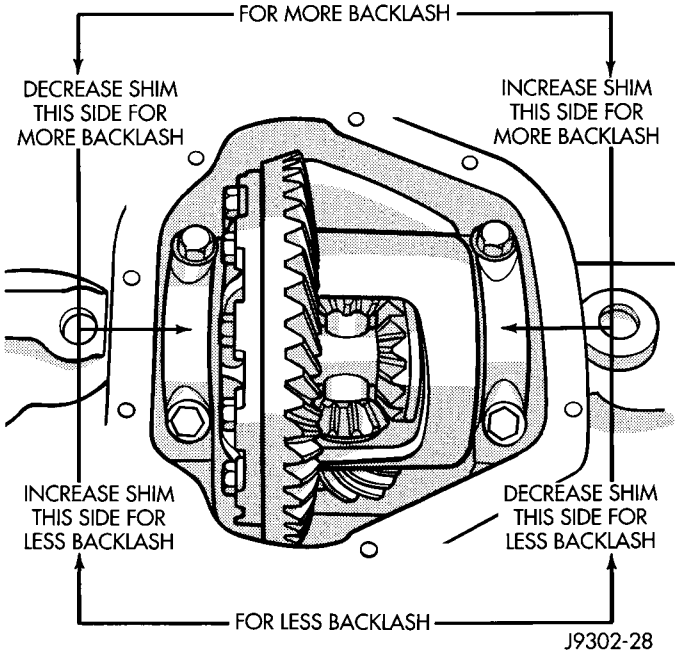

Fig. 24 BACKLASH SHIM

FRONT AXLE - 181FBI (Continued)

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth at the outer-end. The ROOT of the gear tooth is the lowest portion of the tooth (Fig. 25).

NOTE: If the PROFILE across the tooth is the same it is a 3 Axis cut gear. If the PROFILE across the tooth is tapered it is a 2 Axis cut gear.

(1) Apply a thin coat of hydrated ferric oxide or equivalent to the drive and coast side of the ring gear teeth.

(2) Wrap, twist and hold a shop towel around the pinion yoke to increase the turning resistance of the pinion. This will provide a more distinct contact pattern.

(3) With a boxed end wrench on the ring gear bolt, rotate the differential case one complete revolution in both directions while a load is being applied from shop towel.

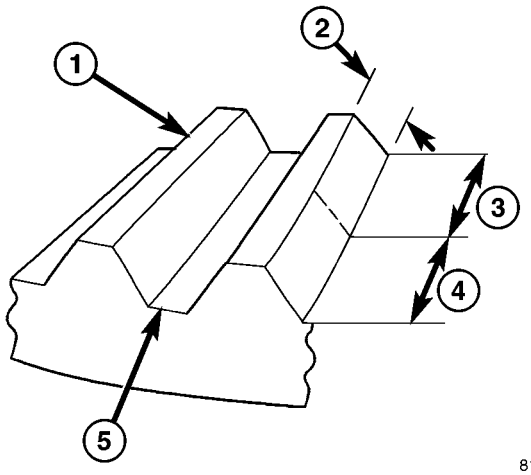

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Fig. 25 GEAR DESCRIPTION

1 - TOP LAND
2 - PROFILE
3 - TOE
4 - HEEL
5 - ROOT

The areas on the ring gear teeth with the greatest degree of contact against the pinion teeth will squeeze the compound to the areas with the least amount of contact. Note and compare patterns on the ring gear teeth to Gear Tooth Contact Patterns chart (Fig. 26) and (Fig. 27) and adjust pinion depth and gear backlash as necessary.

DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top heel contact		**Top toe contact**	Backlash correct. Thicker pinion position shim required.
	Root toe contact		**Root heel contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top heel contact**	Pinion position shim correct. Decrease backlash.
	Root toe contact		**Root toe contact**	Pinion position shim correct. Increase backlash.

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Fig. 26 PATTERN INTERPRETATION (GEAR CUT 2 AXIS)

FRONT AXLE - 181FBI (Continued)

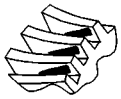
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top toe contact		**Top heel contact**	Backlash correct. Thicker pinion position shim required.
	Root heel contact		**Root toe contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top toe contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root heel contact**	Pinion position shim correct. Increase backlash.

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Fig. 27 PATTERN INTERPRETATION (GEAR CUT 3 AXIS)

DIFFERENTIAL BEARING PRELOAD CHECK

The final check on the differential assembly before installing the axles is torque to rotate pinion and differential combined. This will verify the correct differential bearing preload.

Torque to rotate the differential and pinion should be the torque to rotate the pinion plus 0.79-1.24 N·m (7-11 in. lbs.).

FRONT AXLE - 181FBI (Continued)

SPECIFICATIONS

FRONT AXLE

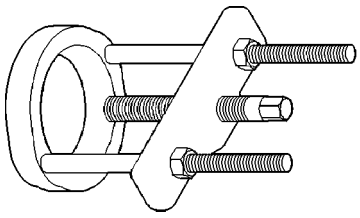
AXLE SPECIFICATIONS

DESCRIPTION	SPECIFICATION
Axle Ratio	3.07, 3.73, 4.10, 4.56
Ring Gear Backlash	0.12-0.20 mm (0.005-0.008 in.)
Pinion Bearing Preload - Original Bearings	1.13-2.26 N·m (10-20 in. lbs.)
Pinion Bearing Preload - New Bearings	2-3.4 N·m (15-30 in. lbs.)

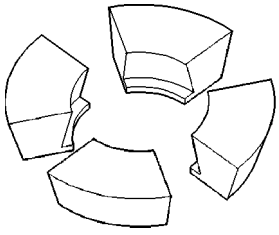
TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Fill Hole Plug	34	25	-
Differential Cover Bolts	41	30	-
Bearing Cap Bolts	61	45	-
Ring Gear Bolts	108	80	-
Pinion Bearing Nut Min - Max	217 - 678	160 - 500	-
Axle Nut	237	175	-
Hub Bearing Bolts	102	75	-

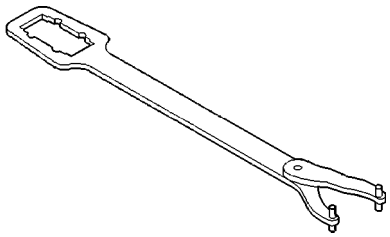
SPECIAL TOOLS



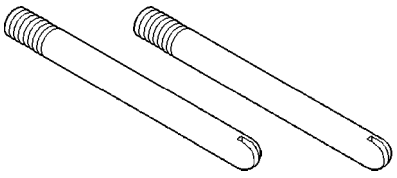
PULLER C-293-PA



ADAPTER C-293-39

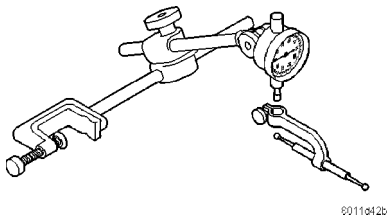


WRENCH C-3281

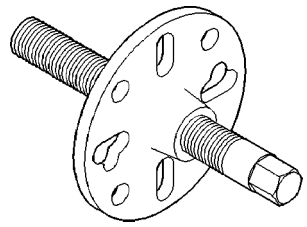


PILOT STUD C-3288-B

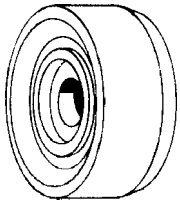
FRONT AXLE - 181FBI (Continued)



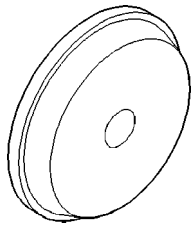
DIAL INDICATOR C-3339



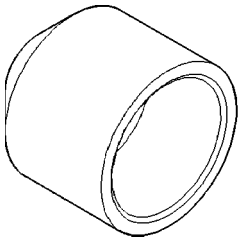
PULLER C-452



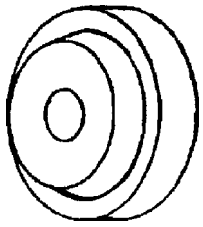
INSTALLER C-3716-A



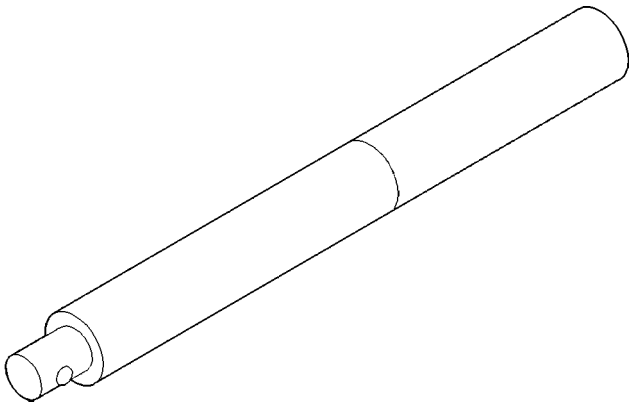
INSTALLER D-130



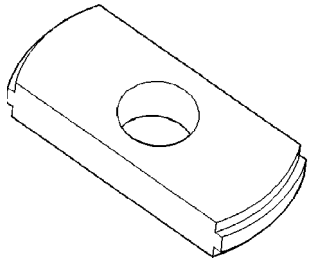
INSTALLER C-3972-A



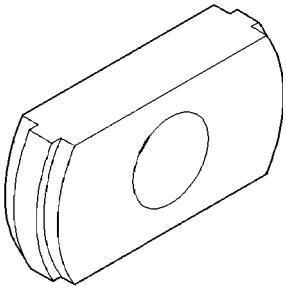
INSTALLER D-146



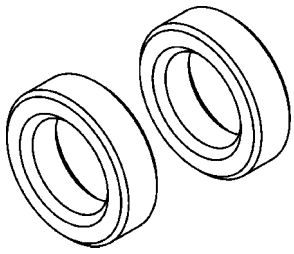
HANDLE C-4171



REMOVER D-149

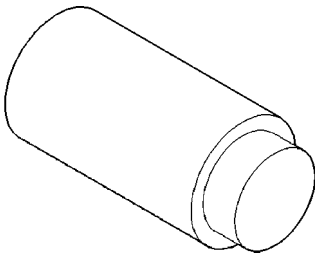


REMOVER C-4345

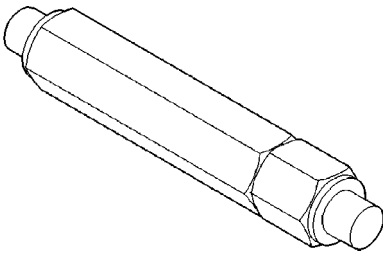


DUMMY BEARINGS D-348

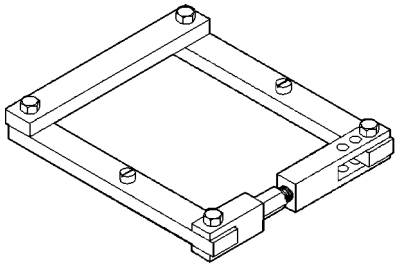
FRONT AXLE - 181FBI (Continued)



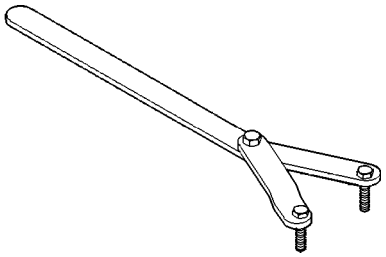
PLUG SP-3289



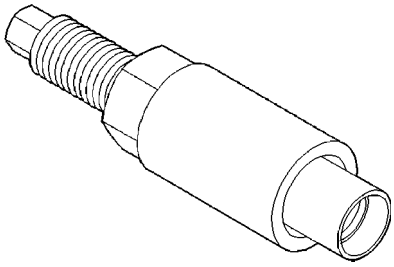
TURNBUCKLE 6797



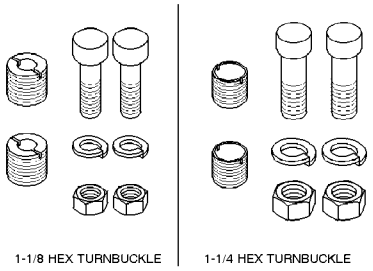
SPREADER W-129-B



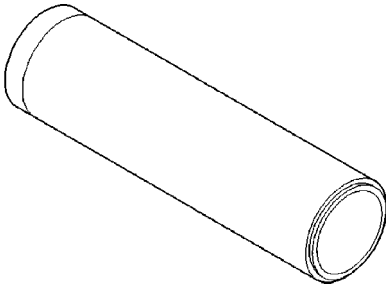
SPANNER WRENCH 6958



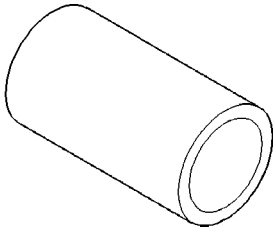
INSTALLER W-162-D



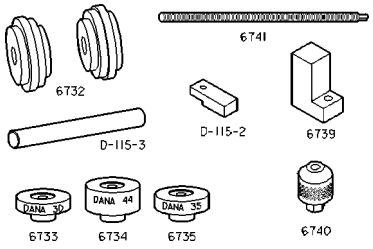
ADAPTER KIT 6987B



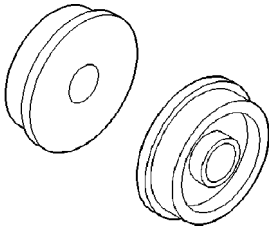
INSTALLER W-262



CUP 8109



PINION DEPTH SET 6774



INSTALLER DISCS 8110

AXLE SHAFTS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake components.
(3) Remove cotter pin, nut retainer and axle hub nut.
(4) Remove hub bearing bolts (Fig. 28).

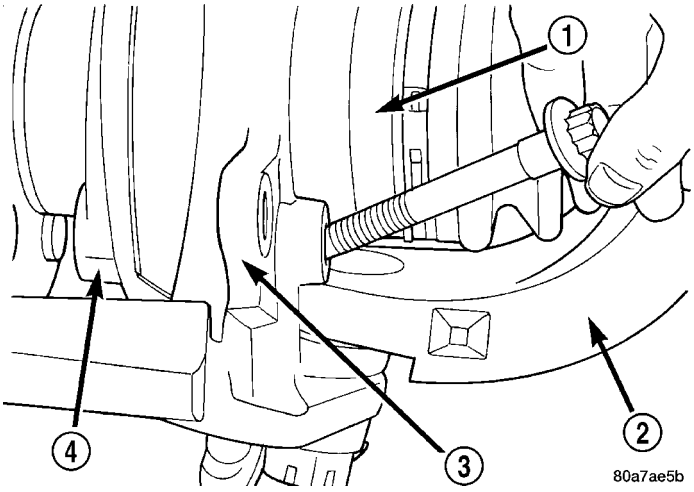


Fig. 28 HUB BEARING BOLTS

1 - AXLE SHAFT
2 - AXLE
3 - KNUCKLE
4 - HUB BEARING

(5) Remove hub bearing from steering knuckle.
(6) Remove axle shaft assembly (Fig. 29) from axle

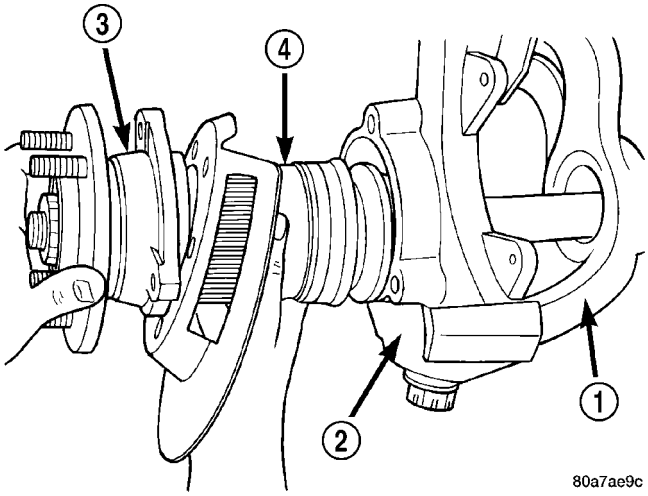


Fig. 29 HUB BEARING AND AXLE ASSEMBLY

1 - AXLE
2 - KNUCKLE
3 - HUB BEARING
4 - AXLE SHAFT

(7) Remove brake rotor shield from the hub bearing or knuckle.

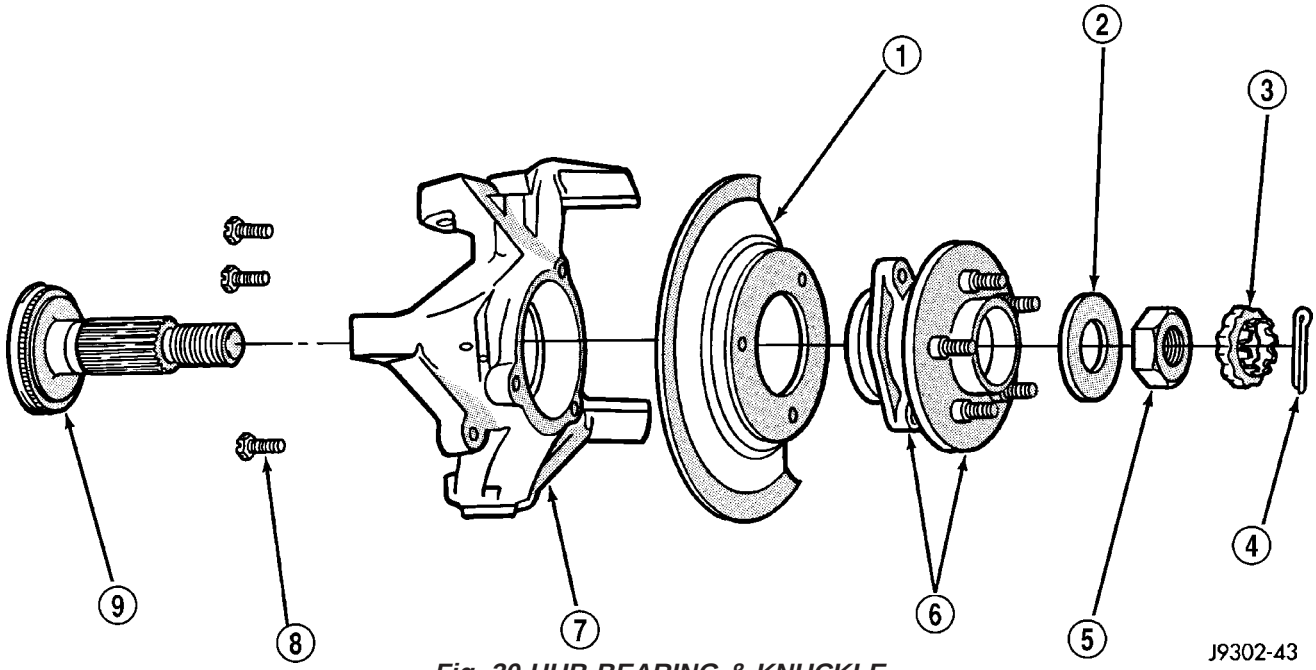


Fig. 30 HUB BEARING & KNUCKLE

1 - BRAKE SHIELD
2 - WASHER
3 - RETAINER
4 - COTTER PIN
5 - NUT
6 - HUB AND BEARING ASSEMBLY
7 - STEERING KNUCKLE
8 - BOLT
9 - TONE WHEEL (ABS)

AXLE SHAFTS (Continued)

INSTALLATION

(1) Clean axle shaft and apply a thin film of Mopar Wheel Bearing Grease or equivalent to the shaft splines, seal contact surface and hub bore.
(2) Install brake rotor shield on knuckle.
(3) Install axle shaft into differential side gears.
(4) Install hub bearing and tighten bolts to 102 N·m (75 ft. lbs.).
(5) Install axle washer and nut. Tighten nut to 237 N·m (175 ft. lbs.) and install nut retainer and cotter pin (Fig. 30).
(6) Install brake components.

AXLE SHAFT SEALS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove axle shafts (Fig. 31).

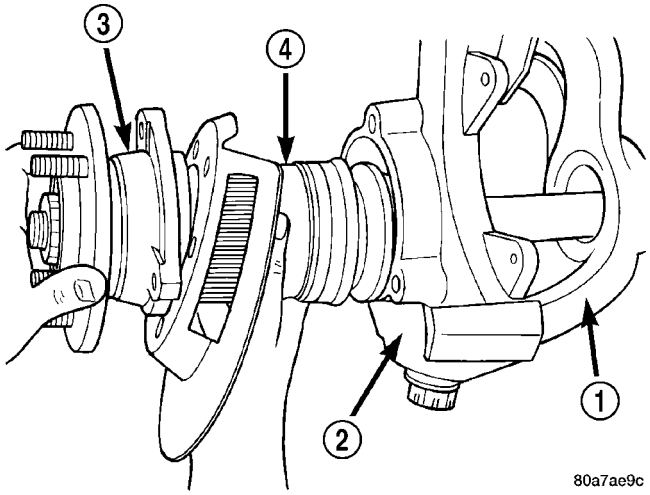


Fig. 31 HUB BEARING AND AXLE ASSEMBLY

1 - AXLE
2 - KNUCKLE
3 - HUB BEARING
4 - AXLE SHAFT

(3) Remove differential assembly (Fig. 32).
(4) Remove inner axle shaft seals with a pry bar.

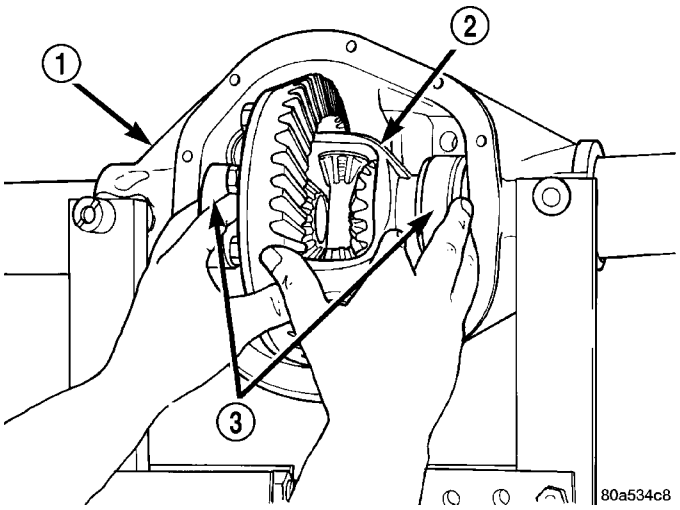


Fig. 32 DIFFERENTIAL CASE

1 - AXLE HOUSING
2 - DIFFERENTIAL CASE
3 - BEARING CUPS

INSTALLATION

(1) Remove any sealer remaining from original seals.
(2) Install oil seals with Discs 8110 and Turnbuckle 6797 (Fig. 33). Tighten tool until disc bottoms in housing.

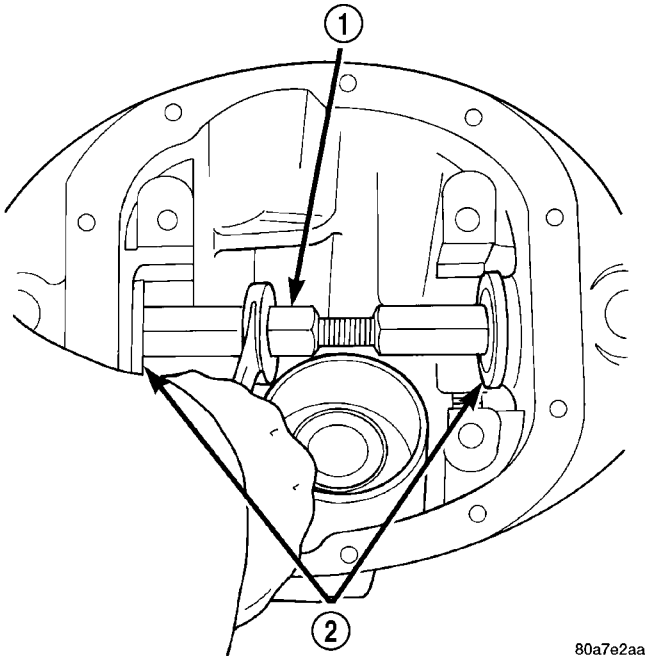


Fig. 33 AXLE SEAL INSTALLATION

1 - TURNBUCKLE
2 - DISCS

(3) Install differential and axle shafts.

SINGLE CARDAN UNIVERSAL JOINT

REMOVAL

CAUTION: Clamp only the narrow forged portion of the yoke in the vise. Failure to heed caution may result in damage.

(1) Remove axle shaft.
(2) Remove bearing cap retaining snap rings (Fig. 34).

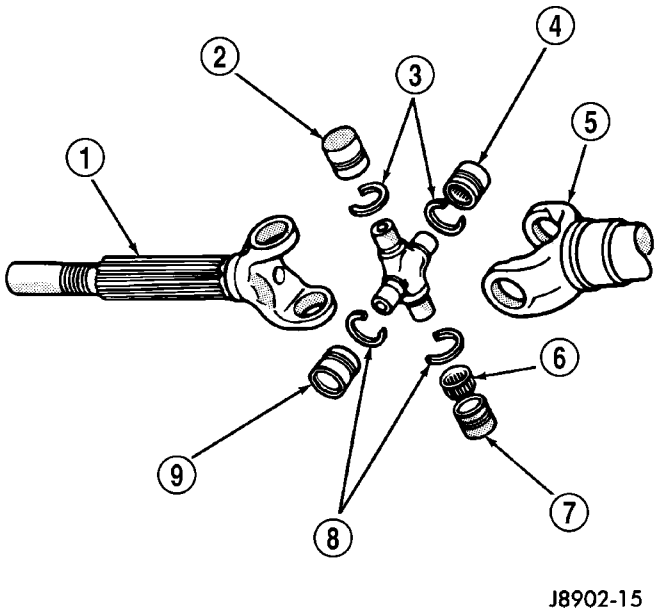


Fig. 34 AXLE SHAFT OUTER U-JOINT

1 - SHAFT YOKE
2 - BEARING CAP
3 - SNAP RINGS
4 - BEARING CAP
5 - SPINDLE YOKE
6 - BEARING
7 - BEARING CAP
8 - SNAP RINGS
9 - BEARING CAP

NOTE: Saturate the bearing caps with penetrating oil prior to removal.

(3) Place a socket (receiver) with an inside diameter larger than the bearing cap against the yoke and the perimeter of the bearing cap to be removed.
(4) Place a socket (driver) with an outside diameter smaller than the bearing cap against the opposite bearing cap.
(5) Position yoke with the sockets in a vise (Fig. 35).

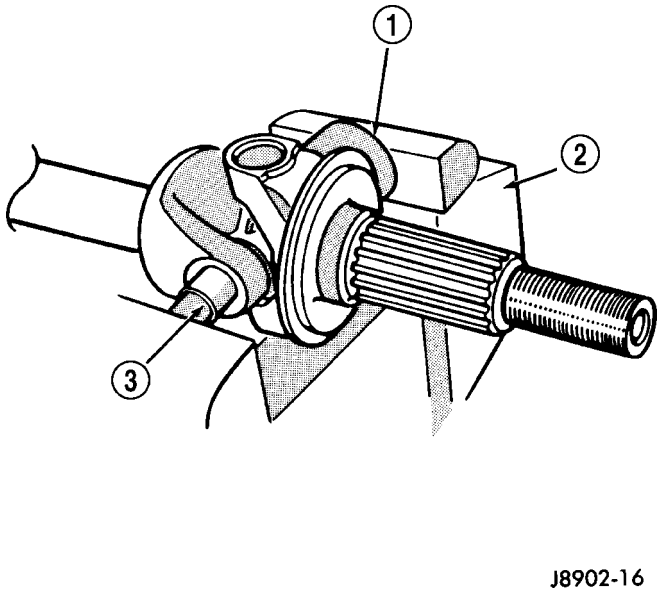


Fig. 35 YOKE BEARING CAP REMOVAL

1 - LARGE-DIAMETER SOCKET
2 - VISE
3 - SMALL-DIAMETER SOCKET

(6) Tighten vise jaws, to force bearing cap into the socket (receiver).
(7) Release vise jaws. Remove sockets and bearing cap forced out of the yoke.
(8) Repeat above procedure for the remaining bearing cap and remove cross from propeller shaft yoke.

SINGLE CARDAN UNIVERSAL JOINT (Continued)

INSTALLATION

CAUTION: Keep cross and bearing cap straight during installation. Failure to heed caution may result in damage.

(1) Pack bearing caps 1/3 full of wheel bearing lubricant (Fig. 36). Apply extreme pressure (EP), lithium-base lubricant to aid in installation.

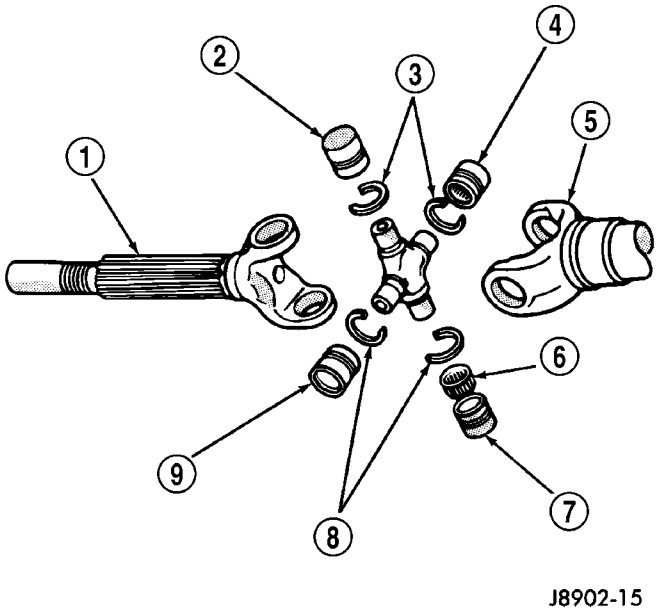


Fig. 36 AXLE SHAFT OUTER U-JOINT

1 - SHAFT YOKE
2 - BEARING CAP
3 - SNAP RINGS
4 - BEARING CAP
5 - SPINDLE YOKE
6 - BEARING
7 - BEARING CAP
8 - SNAP RINGS
9 - BEARING CAP

(2) Position cross in the yoke. Insert seals and bearings, then tap bearing caps into the yoke bores far enough to hold spider in position.

(3) Place socket (driver) against one bearing cap. Position yoke with the socket in a vise.

(4) Tighten vise to force bearing caps into the yoke. Force the caps enough to install retaining clips.

(5) Install bearing cap retaining clips.

(6) Install axle shaft.

PINION SEAL

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Remove brake rotors and calipers.

(3) Remove propeller shaft from the yoke.

(4) Rotate pinion gear three or four times.

(5) Record rotating torque of the pinion gear with an inch pound torque wrench, for installation reference.

(6) Hold pinion yoke with Wrench 6958 and remove pinion nut and washer.

(7) Remove pinion yoke with Remover C-452 and Wrench C-3281 (Fig. 37).

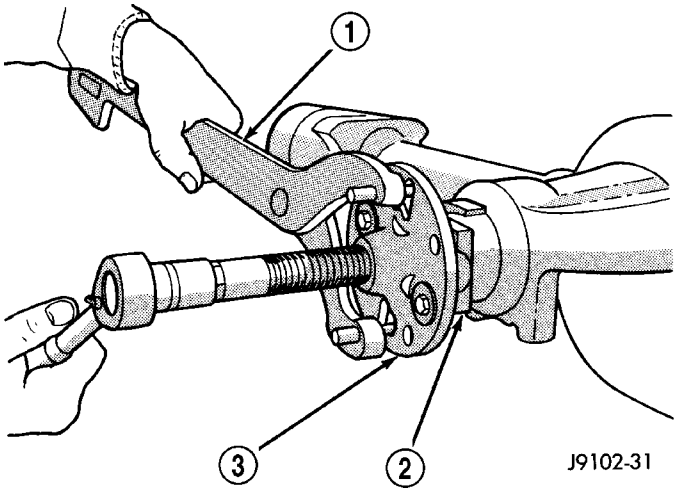


Fig. 37 PINION YOKE REMOVER

1 - WRENCH
2 - PINION YOKE
3 - REMOVER

(8) Remove seal with a pry tool or a slide hammer mounted screw.

INSTALLATION

(1) Apply a light coating of gear lubricant on the lip of pinion seal. Install seal with an appropriate installer (Fig. 38).

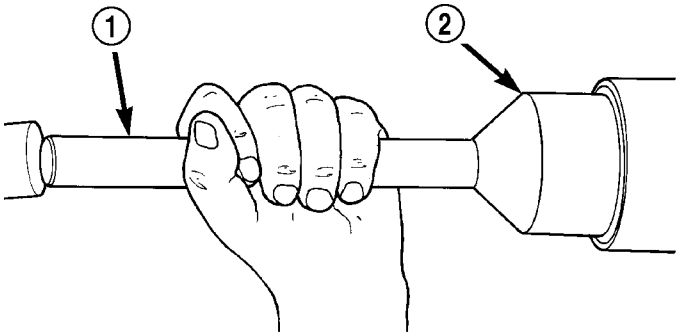


Fig. 38 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

PINION SEAL (Continued)

(2) Install yoke on pinion gear with Installer W-162-D, Cup 8109 and Holder 6958 (Fig. 39).

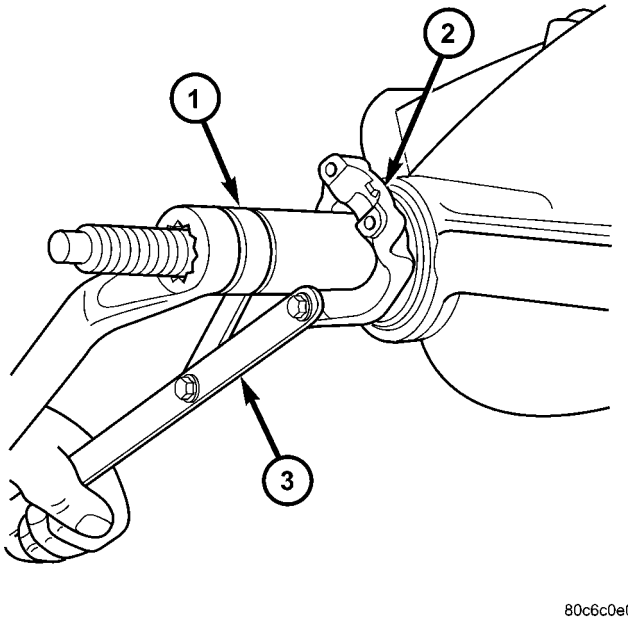


Fig. 39 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

CAUTION: Do not exceed the minimum tightening torque when installing the pinion yoke retaining nut at this point. Failure to heed caution may result in damage

(3) Install pinion washer and a **new** nut on the pinion gear shaft.

NOTE: Tighten nut only enough to remove the shaft end play.

(4) Tighten pinion nut to 217 N·m (160 ft. lbs.).

(5) Rotate pinion shaft using an inch pound torque wrench. Rotating torque should be equal to the reading recorded during removal, plus an additional 0.56 N·m (5 in. lbs.) (Fig. 40).

(6) If rotating torque is low, use Spanner 6958 (Fig. 41) to hold the pinion yoke and tighten pinion shaft nut in 6.8 N·m (5 ft. lbs.) increments until proper rotating torque is achieved.

(7) Install propeller shaft with reference marks aligned.

(8) Install brake rotors and calipers.

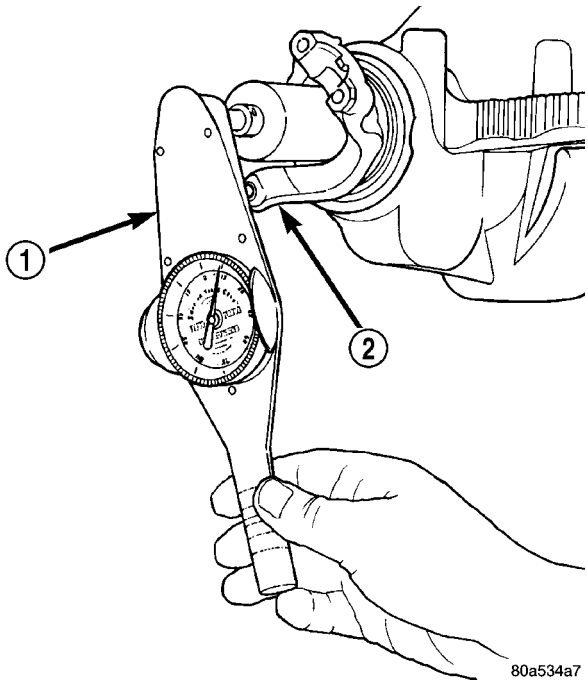


Fig. 40 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

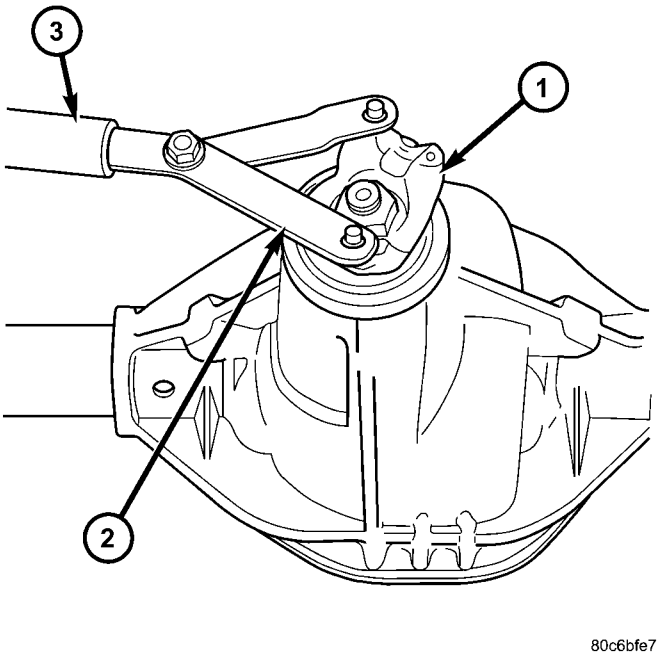


Fig. 41 YOKE HOLDER

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

COLLAPSIBLE SPACER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake rotors and calipers.
(3) Remove propeller shaft from axle yoke.
(4) Rotate pinion gear three or four times.
(5) Record rotating torque of the pinion gear with an inch pound torque wrench.
(6) Hold pinion yoke with Spanner Wrench 6958 and remove pinion nut and washer.
(7) Remove pinion yoke with Remover C-452 and Flange Wrench C-3281 (Fig. 42).

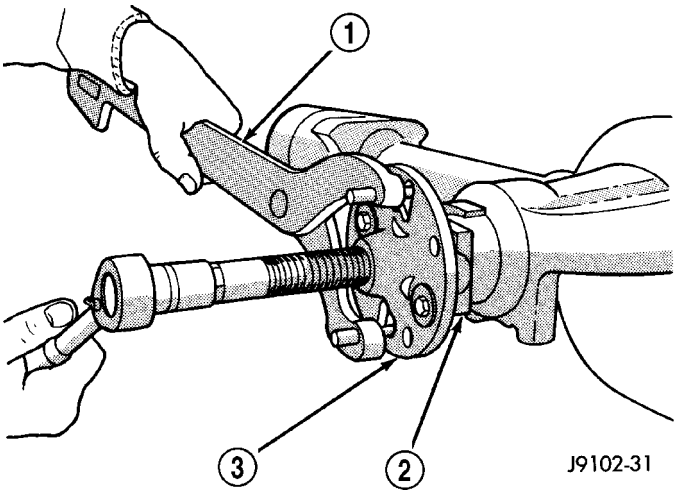


Fig. 42 PINION YOKE REMOVER

1 - WRENCH
2 - PINION YOKE
3 - REMOVER

(8) Remove pinion seal with a suitable pry tool or a slide hammer mounted screw.
(9) Remove front pinion bearing off pinion gear shaft, with a pair of pick tools. If bearing becomes bound on the shaft, lightly tap the end of the pinion gear with a rubber hammer.
(10) Remove collapsible spacer.

INSTALLATION

(1) Install **new** collapsible spacer on pinion shaft.
(2) Install front pinion bearing.
(3) Apply a light coating of gear lubricant on the lip of pinion seal and install seal with an appropriate installer (Fig. 43).
(4) Install yoke with Installer W-162-D, Cup 8109 and Spanner Wrench 6958 (Fig. 44).
(5) Install pinion washer and **new** nut and tighten nut to 217 N·m (160 ft. lbs.).

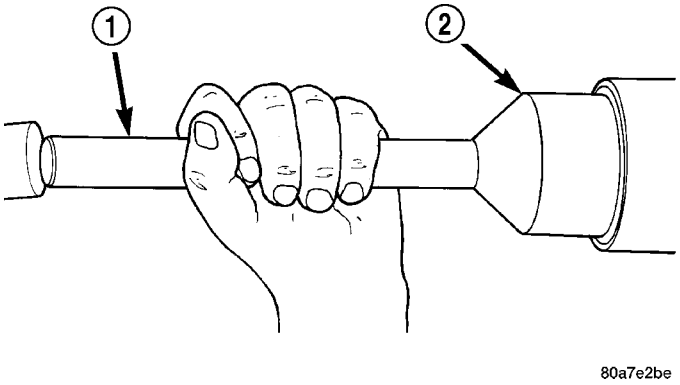


Fig. 43 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

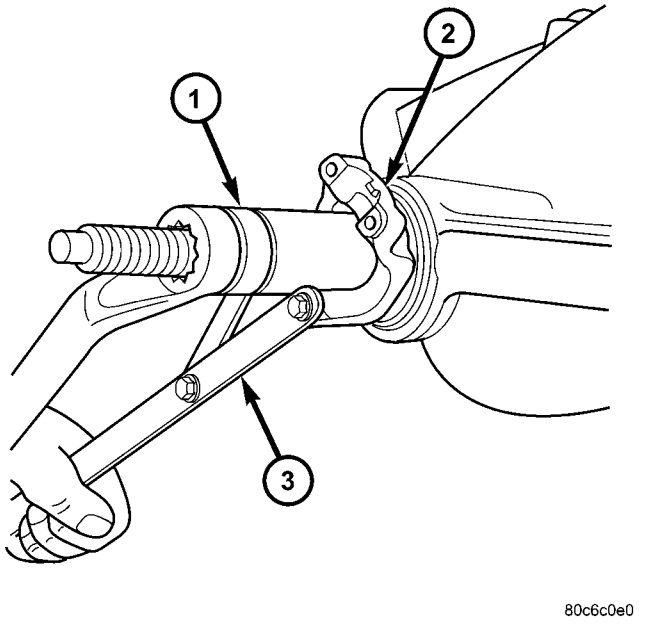


Fig. 44 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

CAUTION: Never loosen pinion gear nut to decrease pinion gear bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

COLLAPSIBLE SPACER (Continued)

(6) With yoke Spanner Wrench 6958 and a torque wrench set at 678 N·m (500 ft. lbs.), slowly tighten nut (Fig. 45) in 6.8 N·m (5 ft. lbs.) increments until rotating torque is achieved. Measure rotating torque with inch pound torque wrench frequently to avoid over crushing the collapsible spacer (Fig. 46).

CAUTION: If more than 687 N·m (500 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

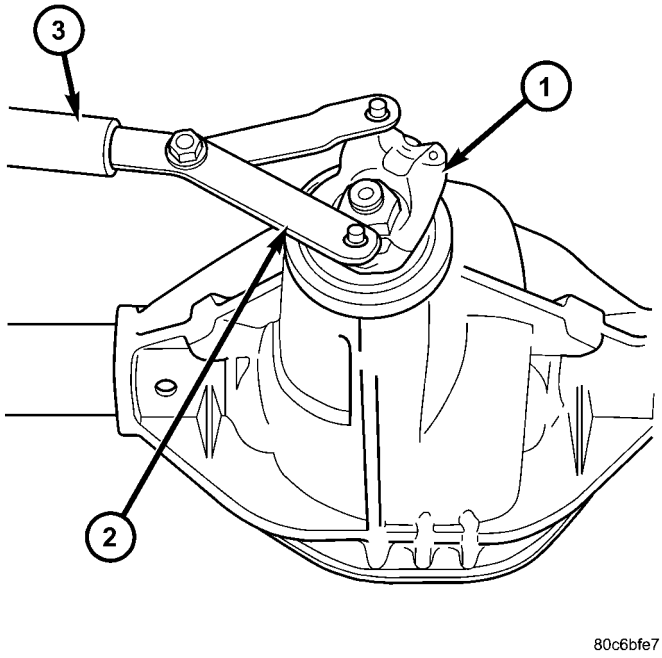


Fig. 45 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(7) Rotating torque should be the recorded reading during removal, plus an additional 0.56 N·m (5 in. lbs.).

(8) Install propeller shaft.

(9) Install brake rotors and calipers.

DIFFERENTIAL COVER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Remove drain plug.

(3) Remove cover bolts.

(4) Remove cover and drain lubricant.

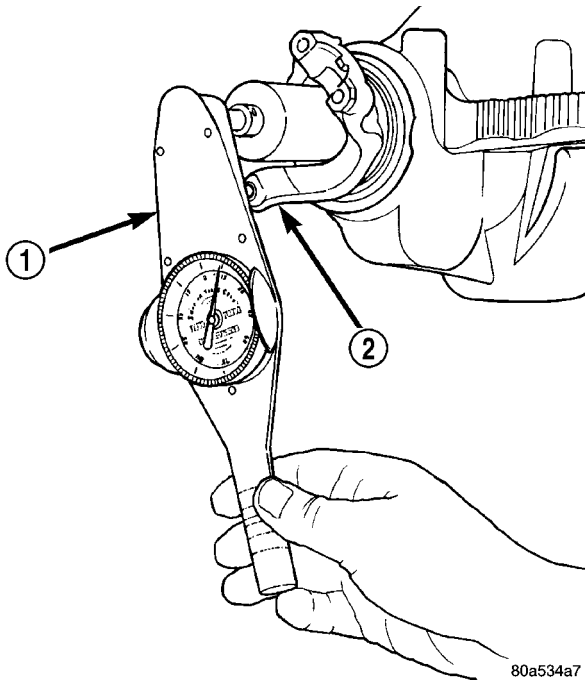


Fig. 46 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

INSTALLATION

(1) Apply a 6.35mm (1/4 in.) bead of Mopar Silicone Rubber Sealant or equivalent to the housing cover (Fig. 47).

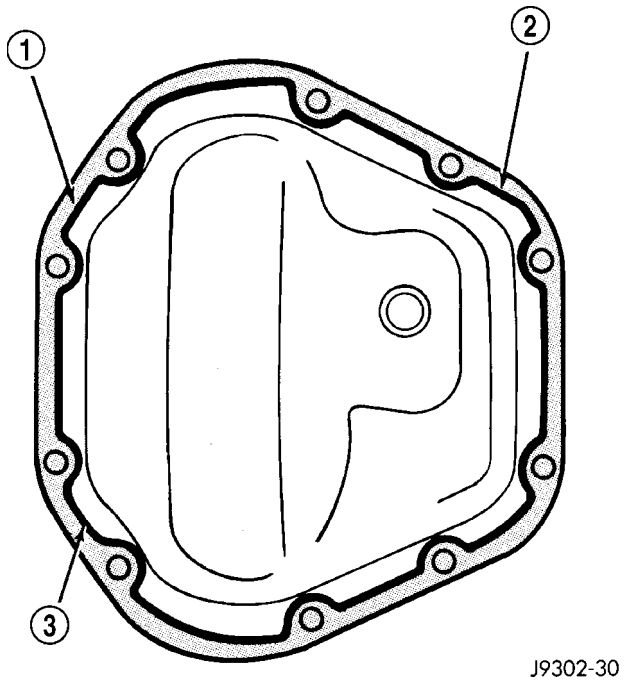


Fig. 47 HOUSING COVER - TYPICAL

1 - SEALANT SURFACE
2 - SEALANT
3 - SEALANT THICKESS

DIFFERENTIAL COVER (Continued)

CAUTION: If housing cover is not installed within 3 to 5 minutes, the cover must be cleaned and new RTV applied. Failure to heed caution may result in damage.

(2) Install cover and identification tag. Tighten cover bolts in a criss-cross pattern to 41 N·m (30 ft. lbs.).
(3) Fill differential to specifications.
(4) Install fill plug and tighten to 34 N·m (25 ft. lbs.).

DIFFERENTIAL

REMOVAL

(1) Remove differential cover and fluid to drain.
(2) Remove hub bearings and axle shafts.
(3) Note the installation reference letters stamped on the bearing caps and housing machined sealing surface (Fig. 48).

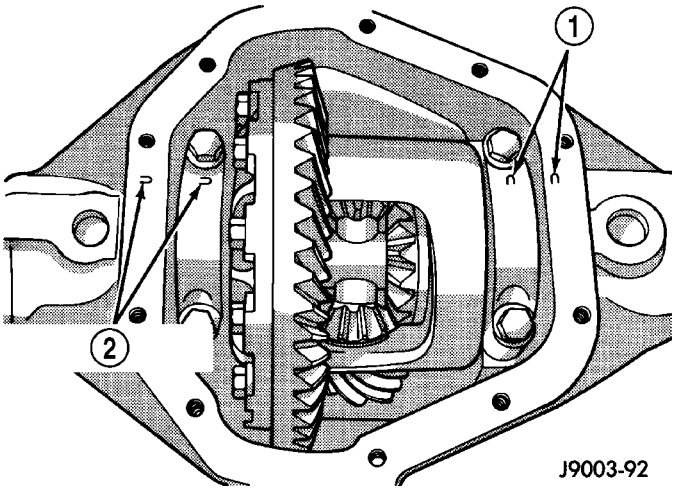


Fig. 48 BEARING CAP IDENTIFICATION

1 - INSTALLATION REFERENCE LETTERS
2 - INSTALLATION REFERENCE LETTERS

(4) Loosen the differential bearing cap bolts.
(5) Position Spreader W-129-B and Adapter from Kit 6987, with the tool dowel pins seated in the locating holes (Fig. 49). Install holddown clamps and tighten the tool turnbuckle finger-tight.
(6) Install Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to guide pin. Load indicator plunger against the opposite side of the housing (Fig. 50) and zero indicator.

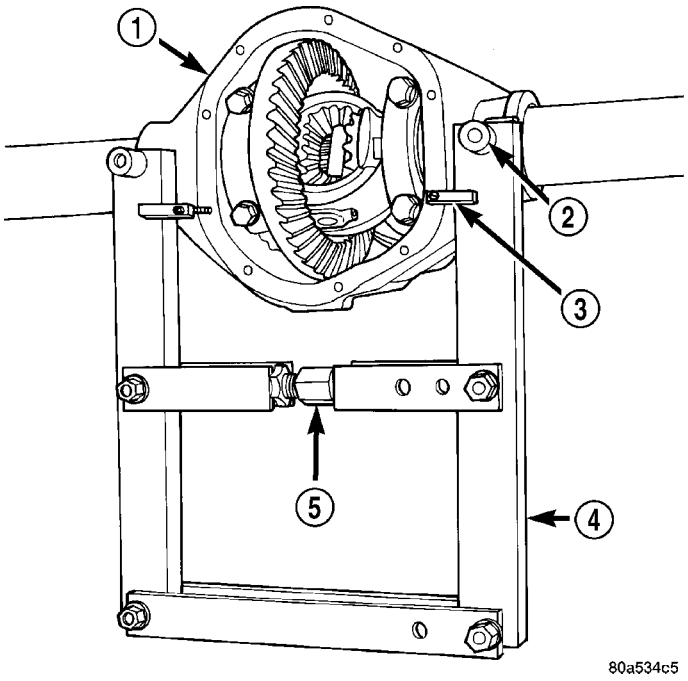


Fig. 49 SPREADER LOCATION

1 - AXLE HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

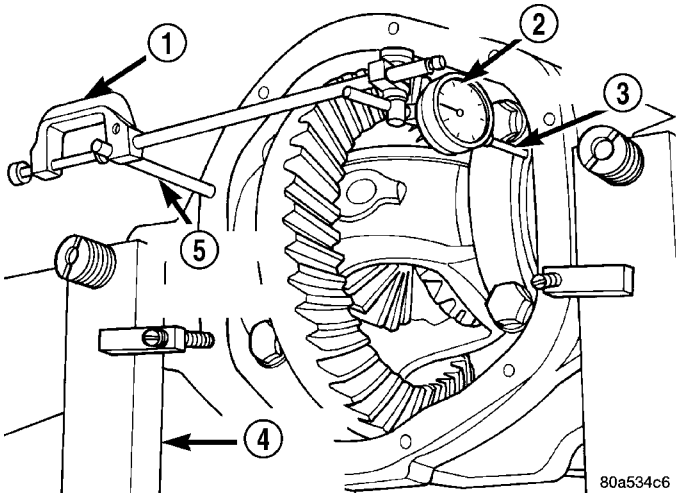


Fig. 50 DIAL INDICATOR LOCATION

1 - INDICATOR CLAMP
2 - DIAL INDICATOR
3 - LEVER ADAPTER
4 - SPREADER
5 - PILOT STUD

DIFFERENTIAL (Continued)

(7) Spread housing while measuring the distance with the dial indicator (Fig. 51).

CAUTION: Do not spread over 0.50 mm (0.020 in). Failure to heed caution may result in damage.

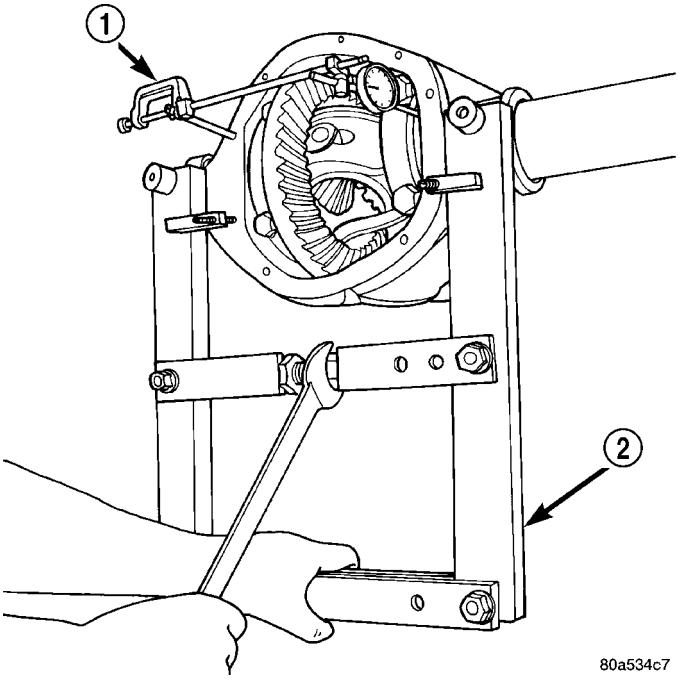


Fig. 51 SPREAD DIFFERENTIAL HOUSING

1 - INDICATOR
2 - SPREADER

(8) Remove dial indicator.

(9) Hold differential case in position while removing differential bearing caps.

(10) Remove differential from the housing (Fig. 52).

(11) Remove spreader from housing.

(12) Clean housing cavity with a flushing oil, light engine oil or lint free cloth.

NOTE: Do not use water, steam, kerosene or gasoline for cleaning.

DISASSEMBLY

(1) Remove ring gear.

(2) Drive out roll pin holding pinion gear mate shaft with a punch (Fig. 53).

(3) Remove pinion gear mate shaft from the differential case and pinion mate gears.

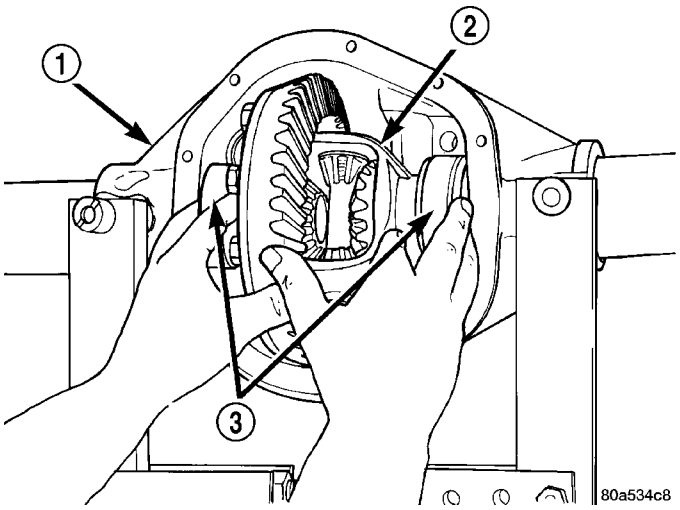


Fig. 52 DIFFERENTIAL CASE

1 - AXLE HOUSING
2 - DIFFERENTIAL CASE
3 - BEARING CUPS

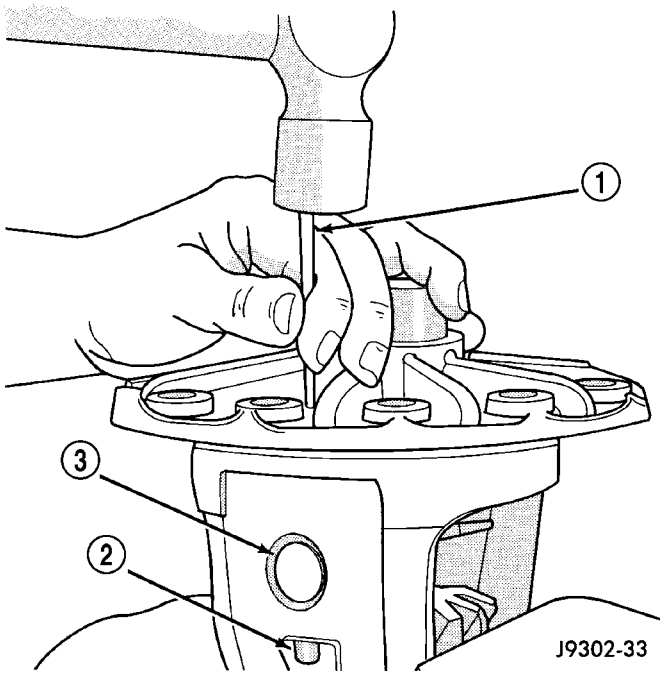


Fig. 53 MATE SHAFT ROLL PIN

1 - DRIFT
2 - LOCKPIN
3 - MATE SHAFT

DIFFERENTIAL (Continued)

(4) Rotate differential side gears and remove pinion mate gears and thrust washers (Fig. 54).

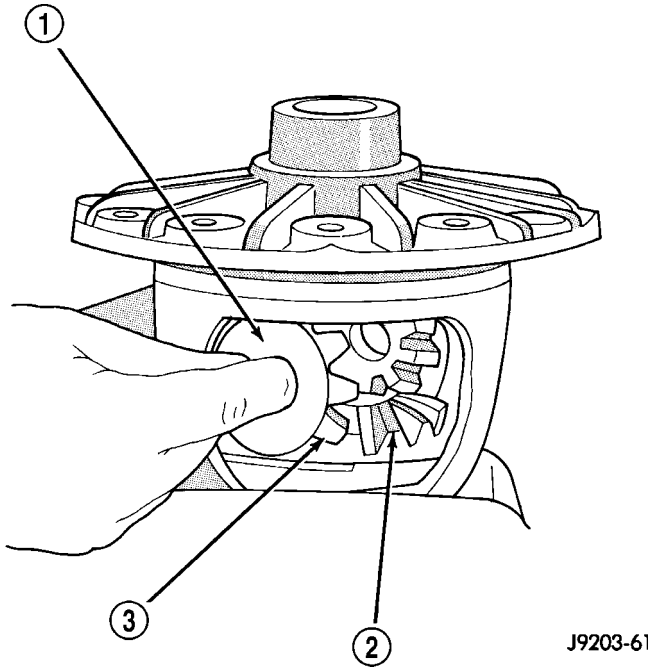


Fig. 54 PINION MATE GEAR

1 - THRUST WASHER
2 - SIDE GEAR
3 - PINION MATE GEAR

(5) Remove differential side gears and thrust washers.

ASSEMBLY

(1) Install differential side gears and thrust washers.

(2) Install pinion mate gears and thrust washers.

(3) Install pinion gear mate shaft. Align roll pin holes in shaft and the differential case.

(4) Install pinion mate shaft roll pin in the differential case (Fig. 55).

(5) Install the ring gear.

INSTALLATION

NOTE: If replacement differential bearings or differential case are being installed, differential side bearing shim requirements may change. Refer to Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Position Spreader W-129-B and Adapter Kit 6987, with the tool dowel pins seated in the locating holes (Fig. 56). Install holddown clamps and tighten turnbuckle finger-tight.

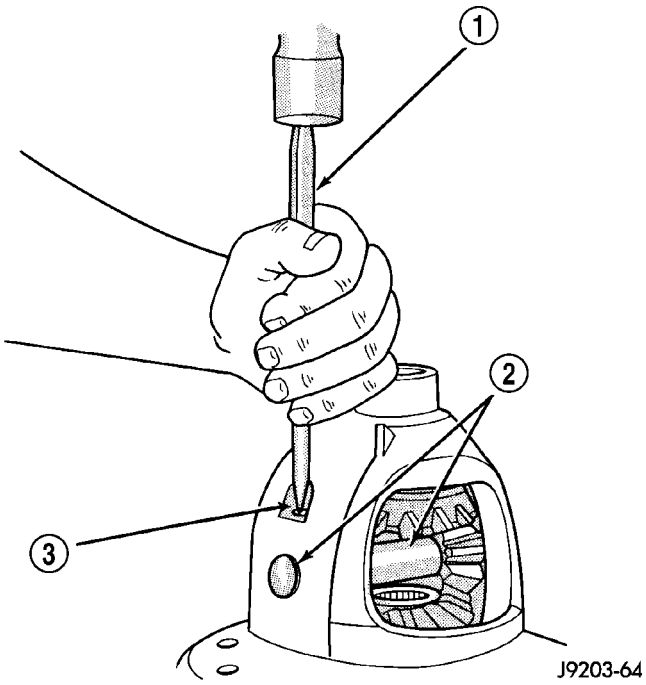


Fig. 55 MATE SHAFT ROLL PIN

1 - PUNCH
2 - PINION MATE SHAFT
3 - MATE SHAFT LOCKPIN

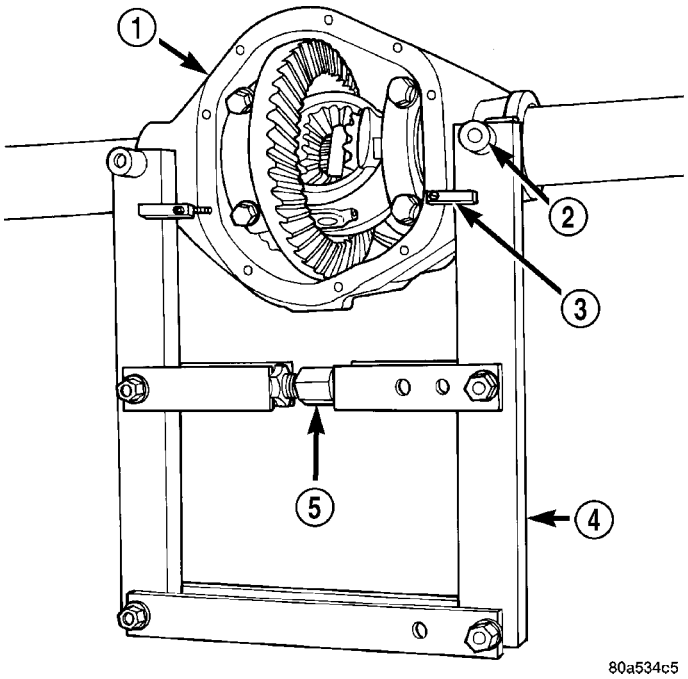


Fig. 56 SPREADER LOCATION

1 - AXLE HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

DIFFERENTIAL (Continued)

(2) Install a Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load the lever adapter against the opposite side of the housing (Fig. 57) and zero indicator.

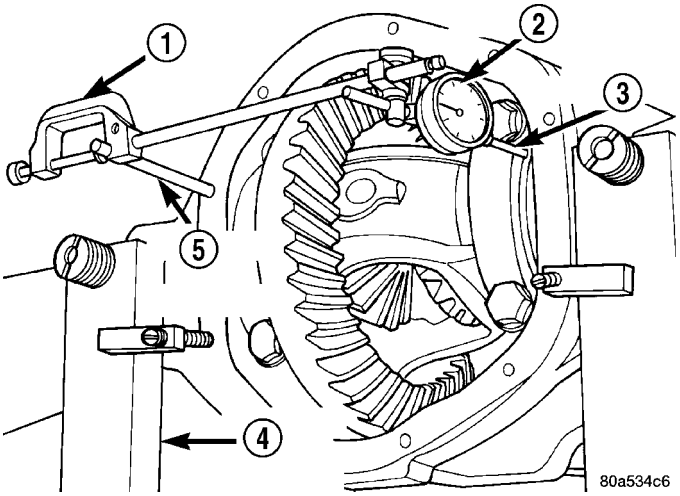


Fig. 57 DIAL INDICATOR LOCATION

1 - CLAMP
2 - DIAL INDICATOR
3 - LEVER ADAPTER
4 - SPREADER
5 - PILOT STUD

(3) Spread housing while measuring the distance with the dial indicator (Fig. 58).

CAUTION: Do not spread over 0.50 mm (0.020 in). Failure to heed caution may result in damage.

(4) Remove dial indicator.
(5) Install differential case in the housing. Tap differential case to seat bearings cups in the housing.
(6) Install bearing caps at their original locations (Fig. 59).
(7) Loosely install differential bearing cap bolts.
(8) Remove axle housing spreader.
(9) Tighten bearing cap bolts to 61 N·m (45 ft. lbs.).
(10) Install hub bearings and axle shafts.
(11) Install differential cover.

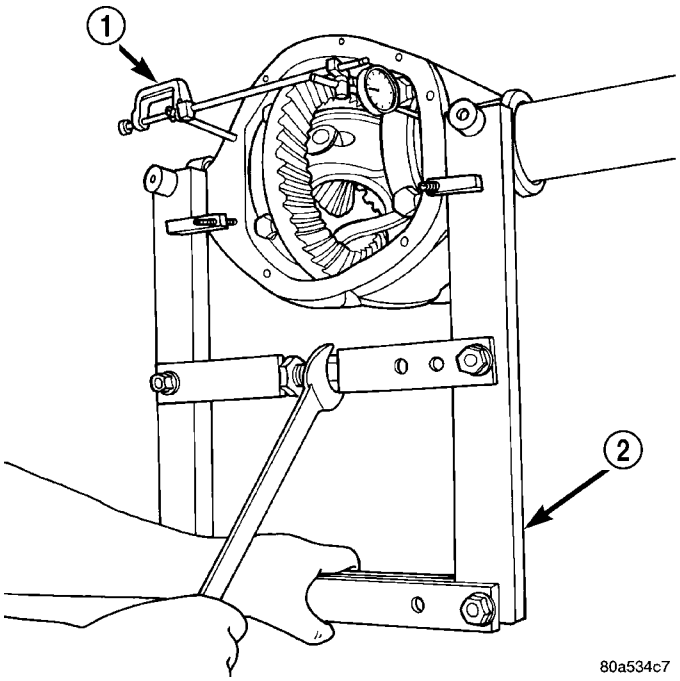


Fig. 58 SPREAD DIFFERENTIAL HOUSING

1 - INDICATOR
2 - SPREADER

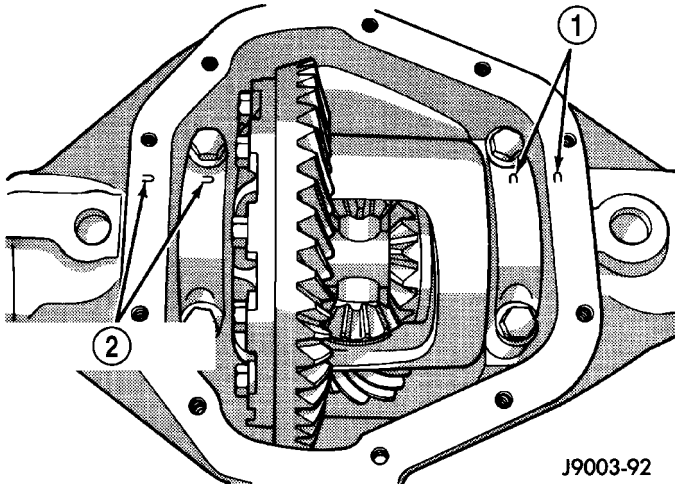


Fig. 59 BEARING CAP IDENTIFICATION

1 - INSTALLATION REFERENCE LETTERS
2 - INSTALLATION REFERENCE LETTERS

DIFFERENTIAL CASE BEARINGS

REMOVAL

(1) Remove bearings from the differential case with Puller/Press C-293-PA, C-293-39 Adapter and Plug SP-3289 (Fig. 60).

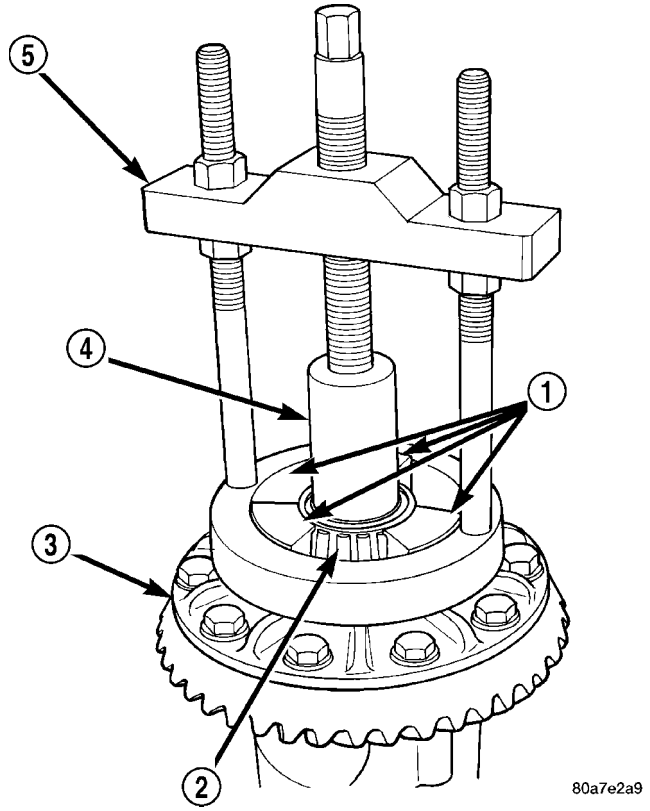


Fig. 60 DIFFERENTIAL BEARING

1 - ADAPTERS
2 - BEARING
3 - DIFFERENTIAL
4 - PLUG
5 - PULLER

INSTALLATION

NOTE: If replacement differential side bearings or differential case are being installed, differential side bearing shim requirements may change. Refer to Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Install differential side bearing shims onto differential case hubs.

(2) Install differential side bearings with Installer C-3716-A and Handle C-4171 (Fig. 61).

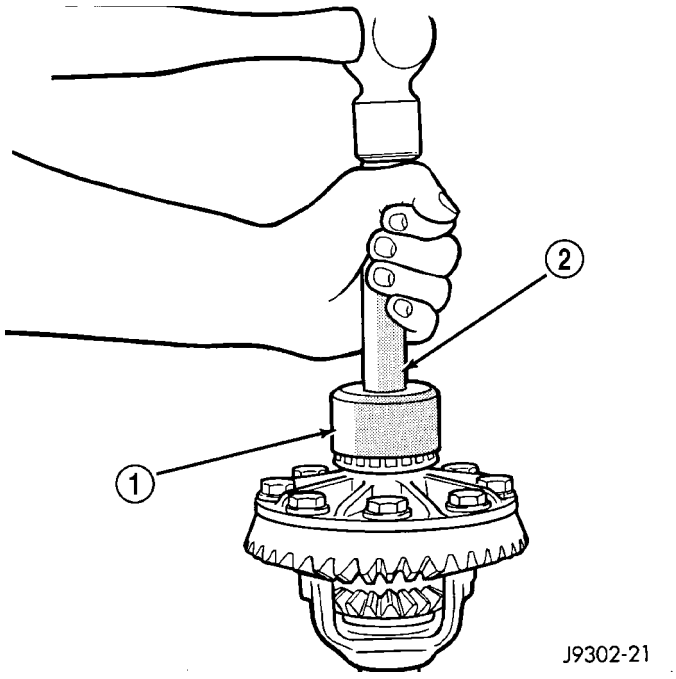


Fig. 61 DIFFERENTIAL CASE BEARING

1 - INSTALLER
2 - HANDLE

PINION GEAR/RING GEAR

REMOVAL

NOTE: The ring and pinion gears are serviced as a matched set. Never replace one gear without replacing the other matched gear.

(1) Remove differential from axle housing.
(2) Secure differential case in a vise with soft metal jaw (Fig. 62).
(3) Remove ring gear bolts from the differential case.
(4) Drive ring gear off the differential case with a dead-blow hammer (Fig. 62).

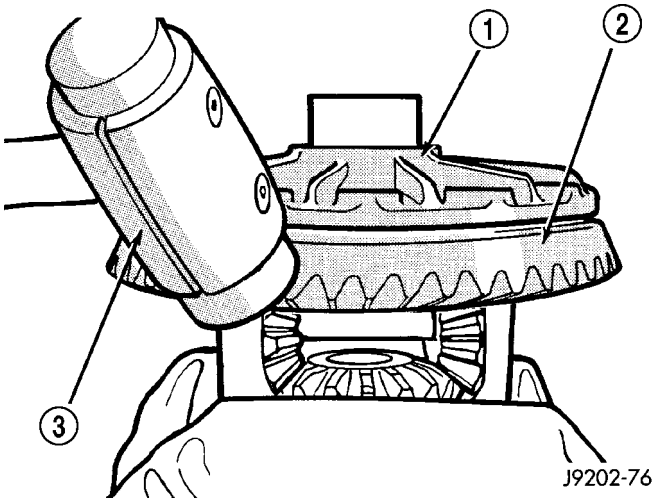


Fig. 62 RING GEAR

1 - DIFFERENTAIL CASE
2 - RING GEAR
3 - HAMMER

(5) Hold yoke with Spanner Wrench 6958 and remove the pinion nut and washer (Fig. 63).
(6) Remove pinion yoke with Remover C-452 and Flange Wrench C-3281 (Fig. 64).

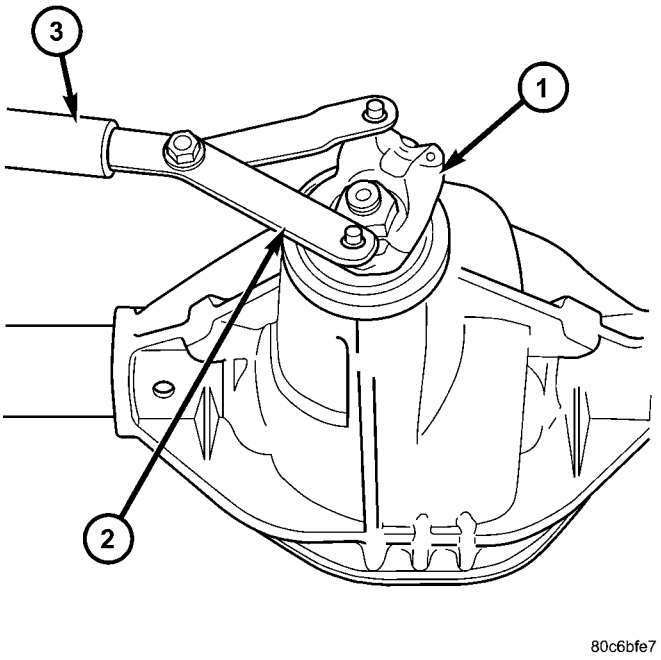


Fig. 63 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

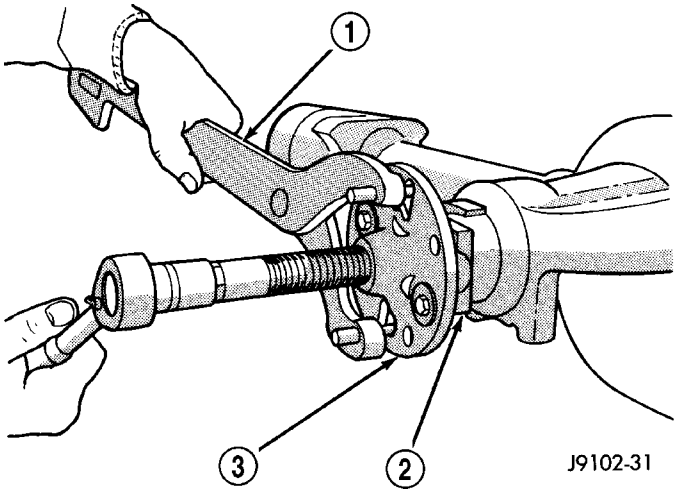


Fig. 64 PINION YOKE

1 - FLANGE WRENCH
2 - PINION YOKE
3 - REMOVER

PINION GEAR/RING GEAR (Continued)

(7) Remove pinion and collapsible spacer from the housing (Fig. 65).

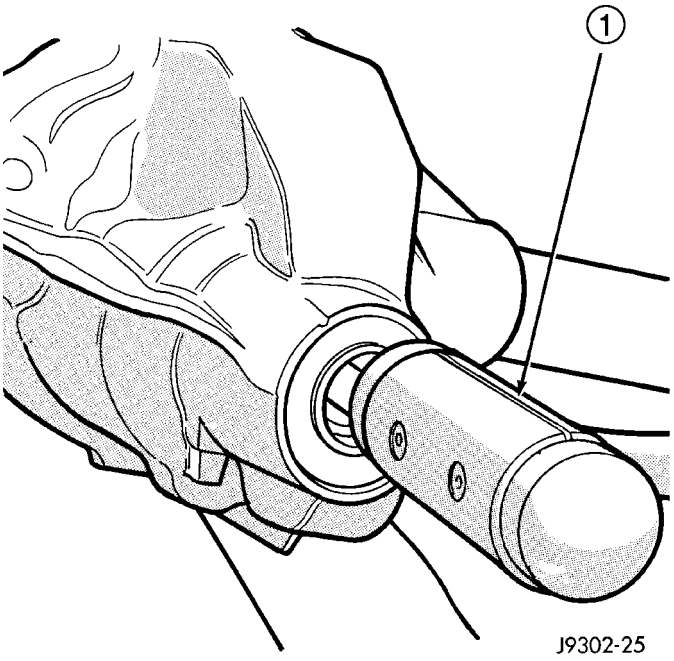


Fig. 65 PINION GEAR

1 - DEAD-BLOW HAMMER

(8) Remove front pinion bearing cup, bearing, oil slinger and pinion seal with Remover C-4345 and Handle C-4171 (Fig. 66).

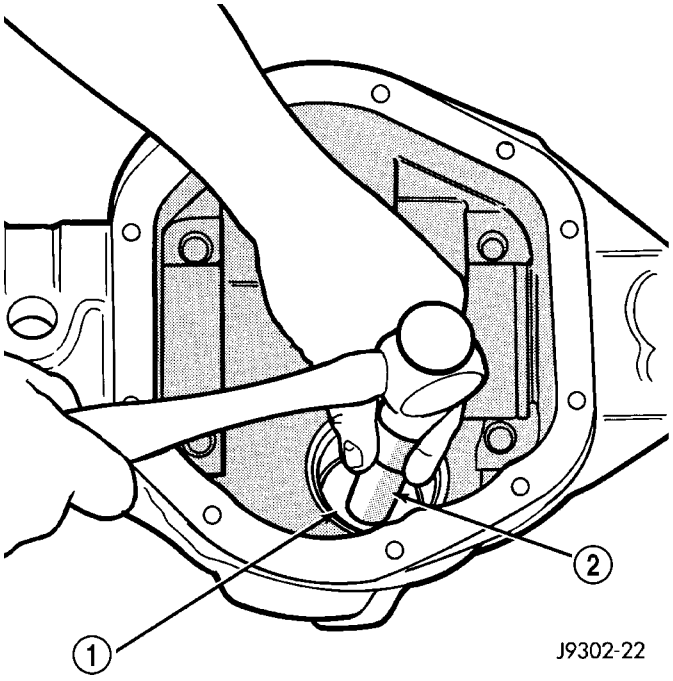


Fig. 66 FRONT BEARING CUP

1 - REMOVER
2 - HANDLE

(9) Remove rear pinion bearing cup from the housing (Fig. 67) with Remover D-149 and Handle C-4171.

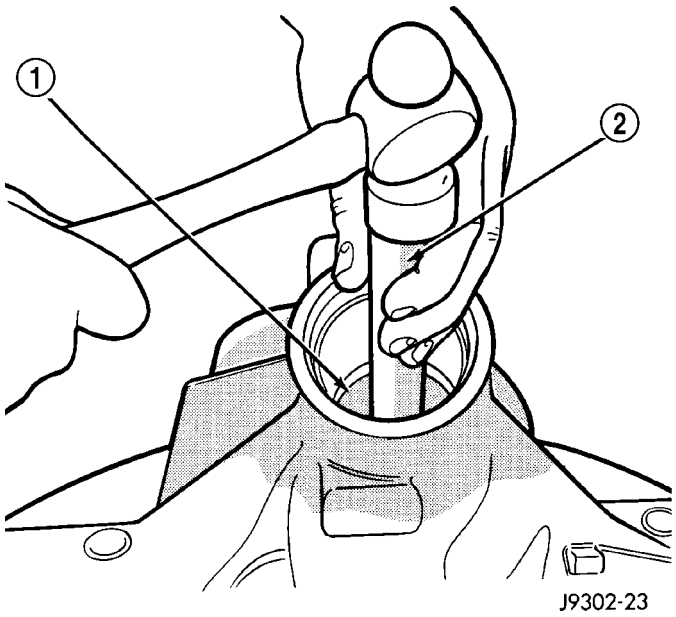


Fig. 67 REAR BEARING CUP

1 - REMOVER
2 - HANDLE

(10) Remove collapsible spacer from pinion shaft (Fig. 68).

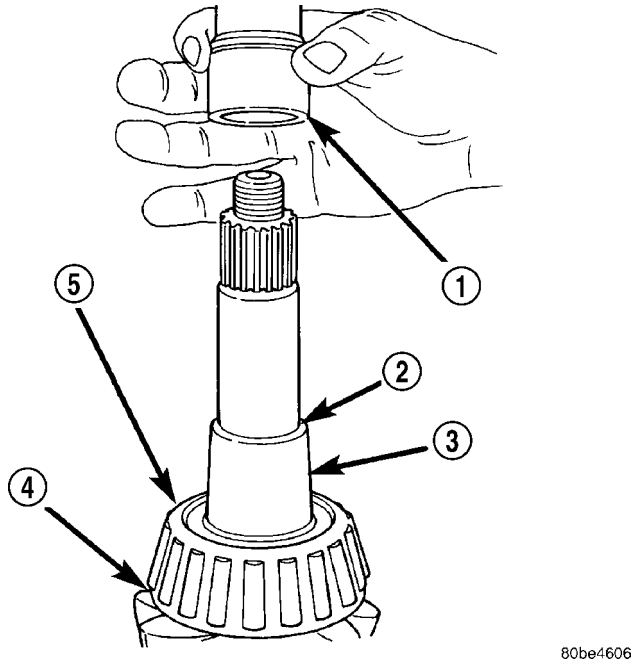


Fig. 68 COLLAPSIBLE SPACER

1 - COLLAPSIBLE SPACER
2 - SHOULDER
3 - PINION
4 - PINION DEPTH SHIM
5 - REAR BEARING

PINION GEAR/RING GEAR (Continued)

(11) Remove rear pinion bearing with Puller/Press C-293-PA and Adapters C-293-39 (Fig. 69).

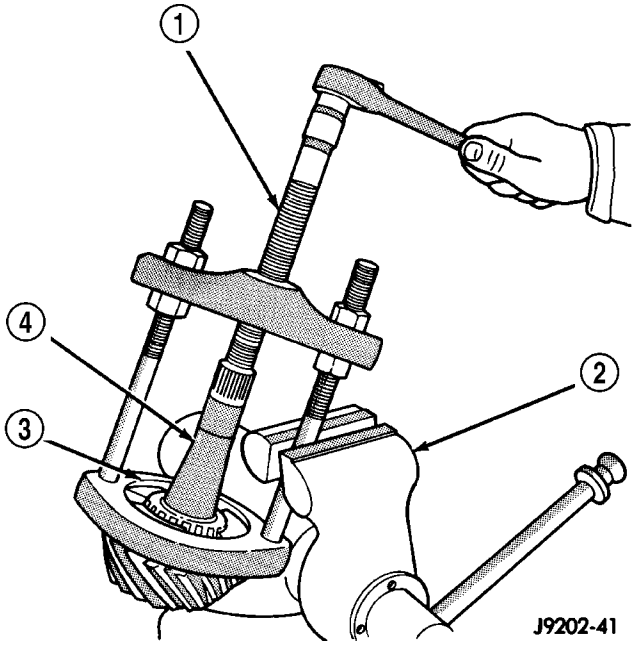


Fig. 69 REAR PINION BEARING

1 - PULLER
2 - VISE
3 - ADAPTERS
4 - PINION GEAR SHAFT

(12) Remove pinion depth shim/oil slinger from the pinion shaft and record thickness.

INSTALLATION

NOTE: A pinion depth shim/oil slinger is placed between the rear pinion bearing cone and the pinion head to achieve proper ring gear and pinion mesh. If ring gear and pinion are reused, the pinion depth shim/oil slinger should not require replacement. Refer to Adjustment (Pinion Gear Depth) to select the proper thickness shim/oil slinger if ring and pinion gears are replaced.

(1) Apply Mopar® Door Ease or equivalent lubricant to outside surface of the pinion bearing cups. Install rear bearing cup with Installer D-146 and Driver Handle C-4171 (Fig. 70) and verify cup is seated.

(2) Install front bearing cup with Installer D-130 and Handle C-4171 (Fig. 71) and verify cup is seated.

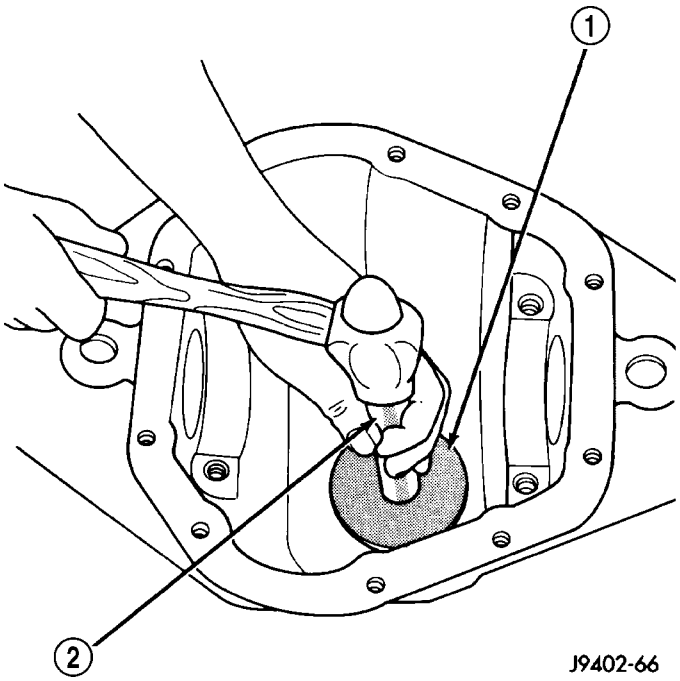


Fig. 70 REAR PINION BEARING CUP

1 - INSTALLER
2 - HANDLE

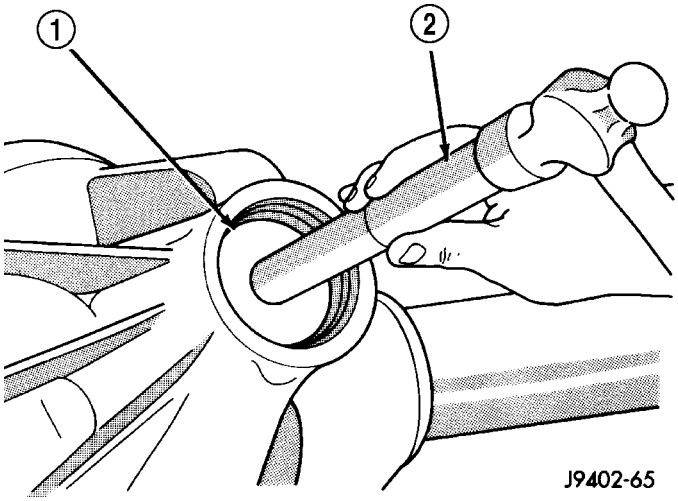


Fig. 71 FRONT PINION BEARING CUP

1 - INSTALLER
2 - HANDLE

PINION GEAR/RING GEAR (Continued)

(3) Install front pinion bearing, and oil slinger.

(4) Apply a light coating of gear lubricant on the lip of pinion seal and install seal with an appropriate installer (Fig. 72).

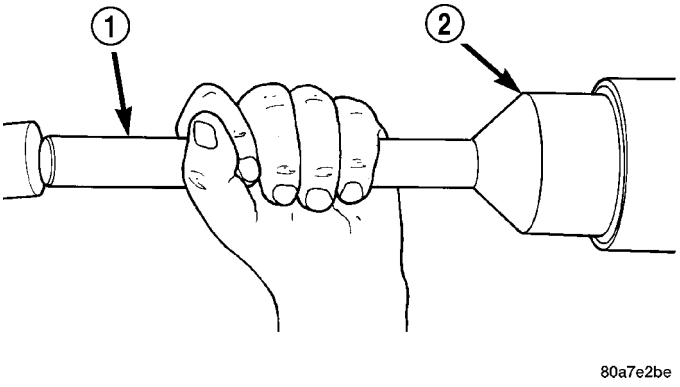


Fig. 72 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(5) Install rear pinion depth shim/oil slinger and bearing on the pinion shaft with Installer W-262 and a press (Fig. 73).

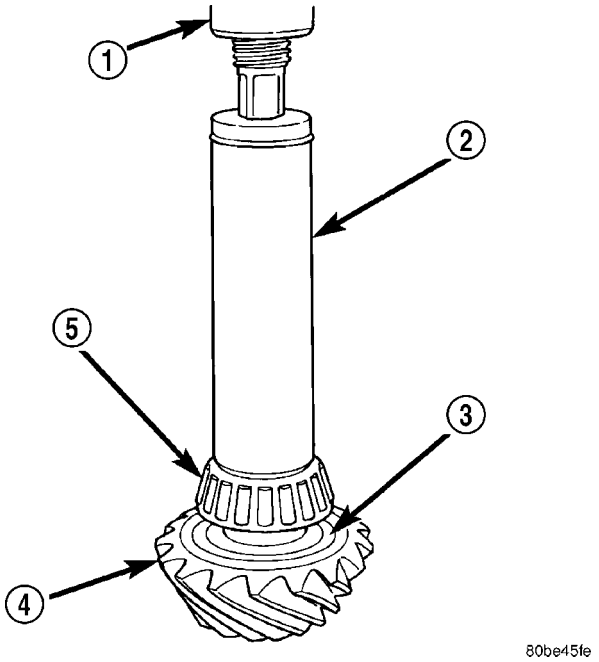


Fig. 73 REAR PINION BEARING

1 - PRESS
2 - INSTALLATER
3 - PINION DEPTH SHIM/OIL SLINGER
4 - DRIVE PINION
5 - REAR PINION BEARING

(6) Install **new** collapsible spacer on pinion shaft and install pinion into the housing (Fig. 74).

(7) Install yoke with Installer W-162-B, Cup 8109 and Spanner Wrench 6958 (Fig. 75).

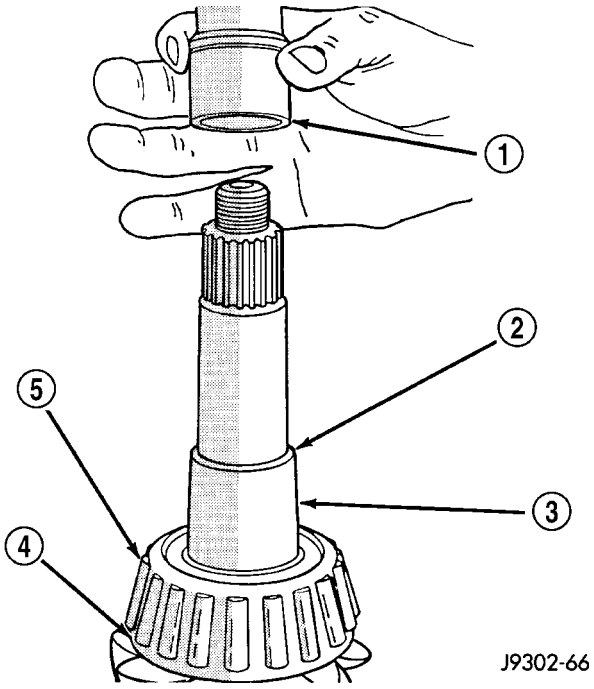


Fig. 74 COLLAPSIBLE PRELOAD SPACER

1 - COLLAPSIBLE SPACER
2 - SHOULDER
3 - PINION GEAR
4 - OIL SLINGER
5 - REAR BEARING

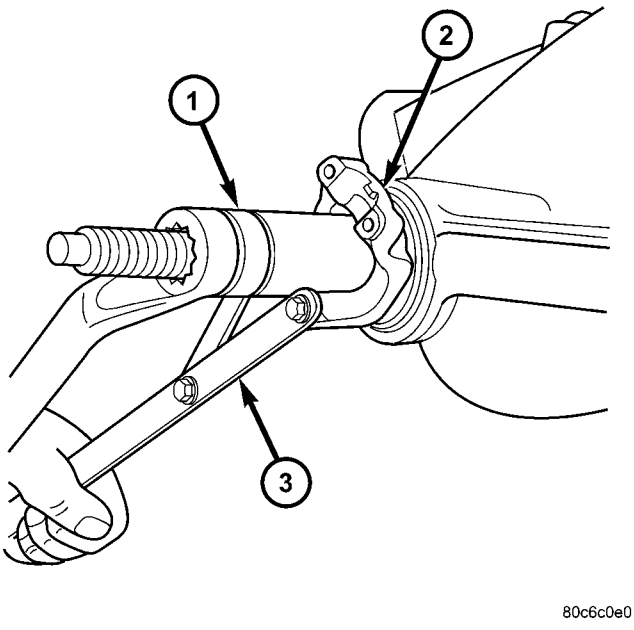


Fig. 75 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

(8) Install pinion washer and a **new** nut onto the pinion. Tighten the nut to 216 N·m (160 ft. lbs.).

PINION GEAR/RING GEAR (Continued)

CAUTION: Never loosen the pinion nut to decrease pinion bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(9) Using Spanner Wrench 6958 and torque wrench set at 678 N·m (500 ft. lbs.) (Fig. 76). Slowly tighten the nut in 6.8 N·m (5 ft. lb.) increments until the rotating torque is achieved. Measure rotating torque frequently to avoid over crushing the collapsible spacer (Fig. 77).

CAUTION: If more than 678 N·m (500 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

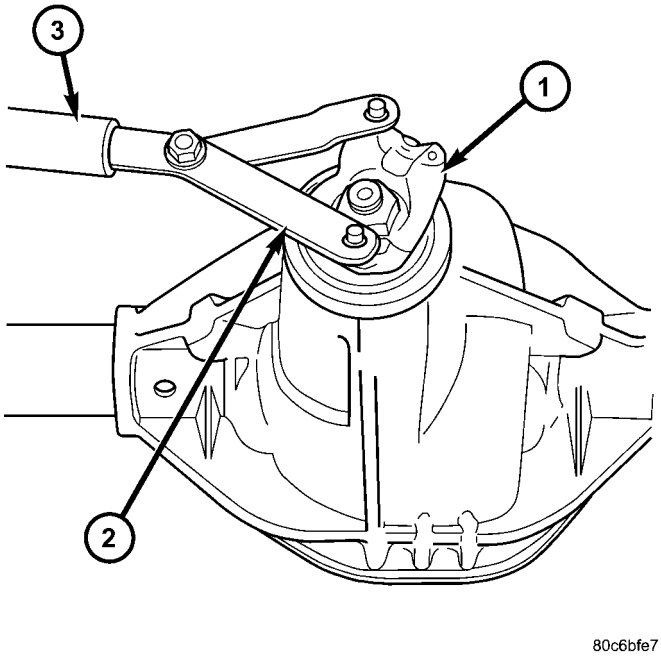


Fig. 76 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(10) Check bearing rotating torque with an inch pound torque wrench (Fig. 77). The torque necessary to rotate the pinion should be:

- Original Bearings: 1.13 to 2.26 N·m (10 to 20 in. lbs.).
- New Bearings: 1.7 to 3.4 N·m (15 to 30 in. lbs.).

(11) Invert the differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

(12) Invert the differential case in the vise and install **new** ring gear bolts and alternately tighten to 108 N·m (80 ft. lbs.) (Fig. 78).

CAUTION: Never reuse ring gear bolts. Failure to heed caution may result in damage.

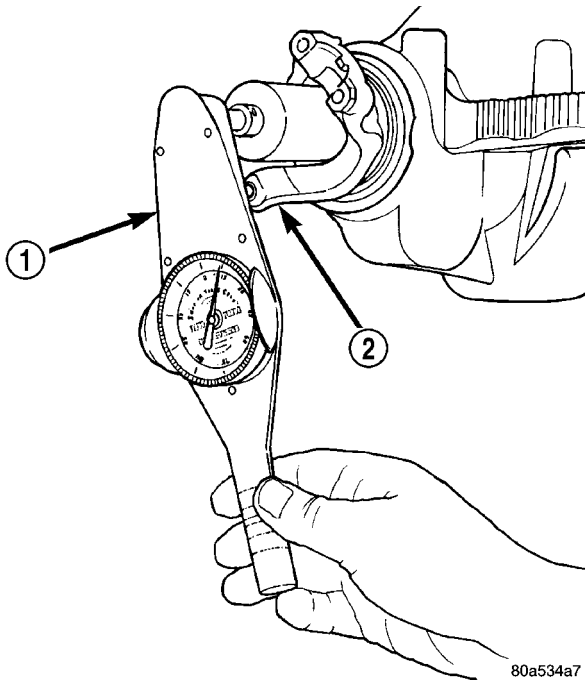


Fig. 77 PINION ROTATION TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

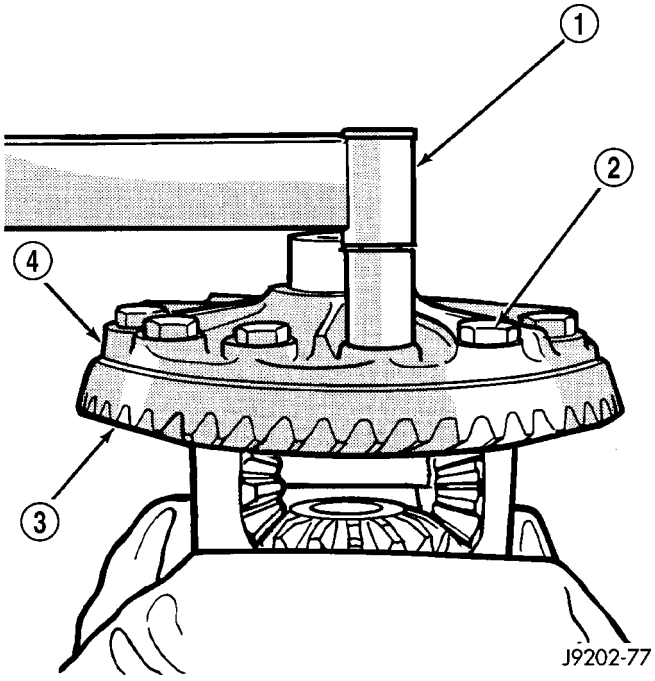


Fig. 78 RING GEAR BOLT

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(13) Install differential and verify differential bearing preload, gear mesh and contact pattern.

FRONT AXLE - 216FBI

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FRONT AXLE - 216FBI

DIAGNOSIS AND TESTING

GEAR NOISE

Axle gear noise can be caused by insufficient lubricant, incorrect backlash, tooth contact, worn/damaged gears or the carrier housing not having the proper offset and squareness.

Gear noise usually happens at a specific speed range. The noise can also occur during a specific type of driving condition. These conditions are acceleration, deceleration, coast, or constant load.

When road testing, first warm-up the axle fluid by driving the vehicle at least 5 miles and then accelerate the vehicle to the speed range where the noise is the greatest. Shift out-of-gear and coast through the peak-noise range. If the noise stops or changes greatly:

- Check for insufficient lubricant.
- Incorrect ring gear backlash.
- Gear damage.

Differential side gears and pinions can be checked by turning the vehicle. They usually do not cause noise during straight-ahead driving when the gears are unloaded. The side gears are loaded during vehicle turns. A worn pinion mate shaft can also cause a snapping or a knocking noise.

BEARING NOISE

The axle shaft, differential and pinion bearings can all produce noise when worn or damaged. Bearing noise can be either a whining, or a growling sound.

Pinion bearings have a constant-pitch noise. This noise changes only with vehicle speed. Pinion bearing noise will be higher pitched because it rotates at a faster rate. Drive the vehicle and load the differential. If bearing noise occurs, the rear pinion bearing is the source of the noise. If the bearing noise is heard during a coast, the front pinion bearing is the source.

Worn or damaged differential bearings usually produce a low pitch noise. Differential bearing noise is similar to pinion bearing noise. The pitch of differential bearing noise is also constant and varies only with vehicle speed.

Axle shaft bearings produce noise and vibration when worn or damaged. The noise generally changes when the bearings are loaded. Road test the vehicle. Turn the vehicle sharply to the left and to the right. This will load the bearings and change the noise level. Where axle bearing damage is slight, the noise is usually not noticeable at speeds above 30 mph.

FRONT AXLE - 216FBI (Continued)

LOW SPEED KNOCK

Low speed knock is generally caused by a worn U-joint or by worn side-gear thrust washers. A worn pinion shaft bore will also cause low speed knock.

VIBRATION

Vibration at the rear of the vehicle is usually caused by:

- Damaged drive shaft.
- Missing drive shaft balance weight(s).
- Worn or out of balance wheels.
- Loose wheel lug nuts.
- Worn U-joint(s).
- Loose/broken springs.
- Damaged axle shaft bearing(s).
- Loose pinion gear nut.
- Excessive pinion yoke run out.
- Bent axle shaft(s).

Check for loose or damaged front end components or engine/transmission mounts. These components can contribute to what appears to be a rear end vibration. Do not overlook engine accessories, brackets and drive belts.

All driveline components should be examined before starting any repair.

DRIVELINE SNAP

A snap or clunk noise when the vehicle is shifted into gear (or the clutch engaged) can be caused by:

- High engine idle speed.
- Transmission shift operation.
- Loose engine/transmission/transfer case mounts.
- Worn U-joints.
- Loose spring mounts.
- Loose pinion gear nut and yoke.
- Excessive ring gear backlash.
- Excessive side gear to case clearance.

The source of a snap or a clunk noise can be determined with the assistance of a helper. Raise the vehicle on a hoist with the wheels free to rotate. Instruct the helper to shift the transmission into gear. Listen for the noise, a mechanics stethoscope is helpful in isolating the source of a noise.

DIAGNOSTIC CHART

Condition	Possible Causes	Correction
Wheel Noise	1. Wheel loose.	1. Tighten loose nuts.
	2. Faulty, brinelled wheel bearing.	2. Replace bearing.
Axle Shaft Noise	1. Misaligned axle tube.	1. Inspect axle tube alignment. Correct as necessary.
	2. Bent or sprung axle shaft.	2. Inspect and correct as necessary.
	3. End-play in pinion bearings.	3. Refer to pinion pre-load information and correct as necessary.
	4. Excessive gear backlash between the ring gear and pinion.	4. Check adjustment of the ring gear and pinion backlash. Correct as necessary.
	5. Improper adjustment of pinion gear bearings.	5. Adjust the pinion bearings pre-load.
	6. Loose pinion yoke nut.	6. Tighten the pinion yoke nut.
	7. Scuffed gear tooth contact surfaces.	7. Inspect and replace as necessary.

FRONT AXLE - 216FBI (Continued)

Condition	Possible Causes	Correction
Axle Shaft Broke	1. Misaligned axle tube.	1. Replace the broken shaft after correcting tube mis-alignment.
	2 Vehicle overloaded.	2. Replace broken shaft and avoid excessive weight on vehicle.
	3. Erratic clutch operation.	3. Replace broken shaft and avoid or correct erratic clutch operation.
	4. Grabbing clutch.	4. Replace broken shaft and inspect and repair clutch as necessary.
Differential Cracked	1. Improper adjustment of the differential bearings.	1. Replace case and inspect gears and bearings for further damage. Set differential bearing pre-load properly.
	2. Excessive ring gear backlash.	2. Replace case and inspect gears and bearings for further damage. Set ring gear backlash properly.
	3. Vehicle overloaded.	3. Replace case and inspect gears and bearings for further damage. Avoid excessive vehicle weight.
	4. Erratic clutch operation.	4. Replace case and inspect gears and bearings for further damage. Avoid erratic use of clutch.
Differential Gears Scored	1. Insufficient lubrication.	1. Replace scored gears. Fill differential with the correct fluid type and quantity.
	2. Improper grade of lubricant.	2. Replace scored gears. Fill differential with the correct fluid type and quantity.
	3. Excessive spinning of one wheel/tire.	3. Replace scored gears. Inspect all gears, pinion bores, and shaft for damage. Service as necessary.
Loss Of Lubricant	1. Lubricant level too high.	1. Drain lubricant to the correct level.
	2. Worn axle shaft seals.	2. Replace seals.
	3. Cracked differential housing.	3. Repair as necessary.
	4. Worn pinion seal.	4. Replace seal.
	5. Worn/scored yoke.	5. Replace yoke and seal.
	6. Axle cover not properly sealed.	6. Remove, clean, and re-seal cover.
Axle Overheating	1. Lubricant level low.	1. Fill differential to correct level.
	2. Improper grade of lubricant.	2. Fill differential with the correct fluid type and quantity.
	3. Bearing pre-loads too high.	3. Re-adjust bearing pre-loads.
	4. Insufficient ring gear backlash.	4. Re-adjust ring gear backlash.

FRONT AXLE - 216FBI (Continued)

Condition	Possible Causes	Correction
Gear Teeth Broke	1. Overloading.	1. Replace gears. Examine other gears and bearings for possible damage.
	2. Erratic clutch operation.	2. Replace gears and examine the remaining parts for damage. Avoid erratic clutch operation.
	3. Ice-spotted pavement.	3. Replace gears and examine remaining parts for damage.
	4. Improper adjustments.	4. Replace gears and examine remaining parts for damage. Ensure ring gear backlash is correct.
Axle Noise	1. Insufficient lubricant.	1. Fill differential with the correct fluid type and quantity.
	2. Improper ring gear and pinion adjustment.	2. Check ring gear and pinion contact pattern.
	3. Unmatched ring gear and pinion.	3. Replace gears with a matched ring gear and pinion.
	4. Worn teeth on ring gear and/or pinion.	4. Replace ring gear and pinion.
	5. Loose pinion bearings.	5. Adjust pinion bearing pre-load.
	6. Loose differential bearings.	6. Adjust differential bearing pre-load.
	7. Mis-aligned or sprung ring gear.	7. Measure ring gear run-out. Replace components as necessary.
	8. Loose differential bearing cap bolts.	8. Inspect differential components and replace as necessary. Ensure that the bearing caps are torqued tot he proper specification.
	9. Housing not machined properly.	9. Replace housing.

FRONT AXLE - 216FBI (Continued)

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Position a lift under the axle and secure to the axle.
(3) Remove brake components.
(4) Remove vent hose from axle shaft tube.
(5) Remove propeller shaft.
(6) Remove stabilizer bar links at the axle (Fig. 1).

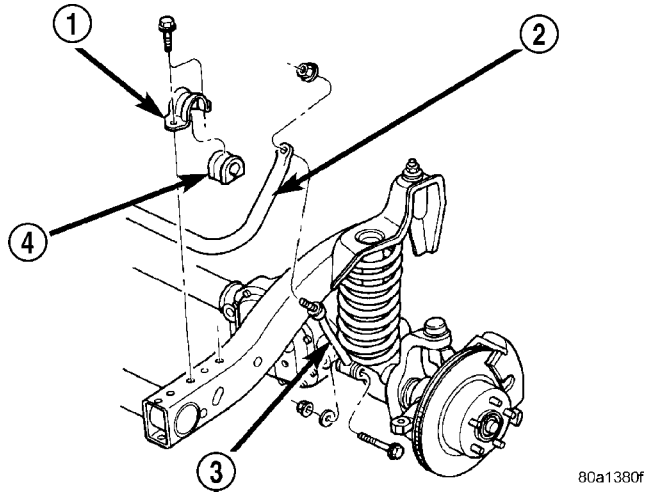


Fig. 1 STABILIZER BAR

1 - RETAINER
2 - STABILIZER BAR
3 - LINK
4 - BUSHING

(7) Remove shock absorbers from axle brackets (Fig. 2).
(8) Remove track bar from axle bracket (Fig. 3).
(9) Remove tie rod and drag link from steering knuckle.
(10) Remove steering damper from axle bracket.
(11) Remove upper and lower suspension arms from axle brackets.
(12) Lower lift enough to remove the axle. The coil springs will drop with the axle.
(13) Remove coil springs from the axle.

INSTALLATION

CAUTION: The weight of the vehicle must be supported by the springs before suspension arms and track bar fasteners can be tightened. Failure to heed caution may result in damage.

(1) Install springs and retainers and tighten retainer bolts to 21 N·m (16 ft. lbs.).

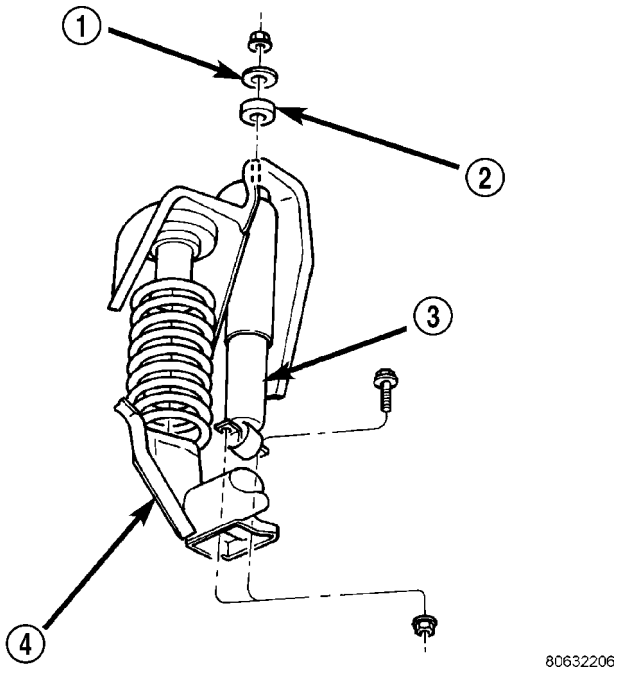


Fig. 2 COIL SPRING AND SHOCK ABSORBER

1 - RETAINER
2 - GROMMET
3 - SHOCK
4 - FRONT AXLE

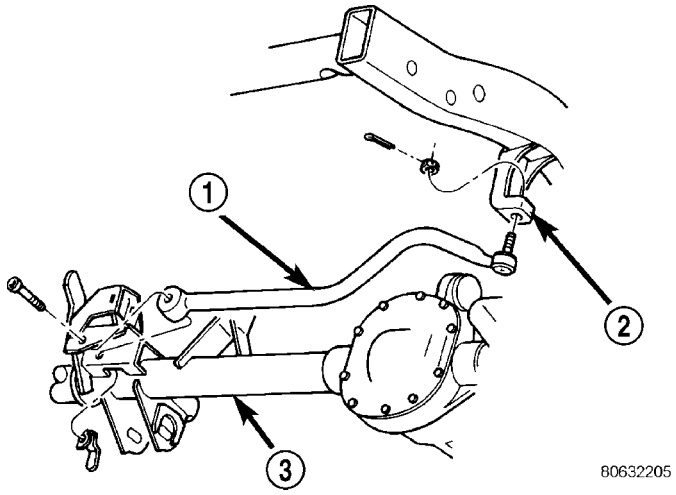


Fig. 3 TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - FRONT AXLE

(2) Position axle under vehicle and align it with the spring pads.

FRONT AXLE - 216FBI (Continued)

(3) Install upper and lower suspension arms in the axle brackets and loosely install bolts and nuts (Fig. 4).

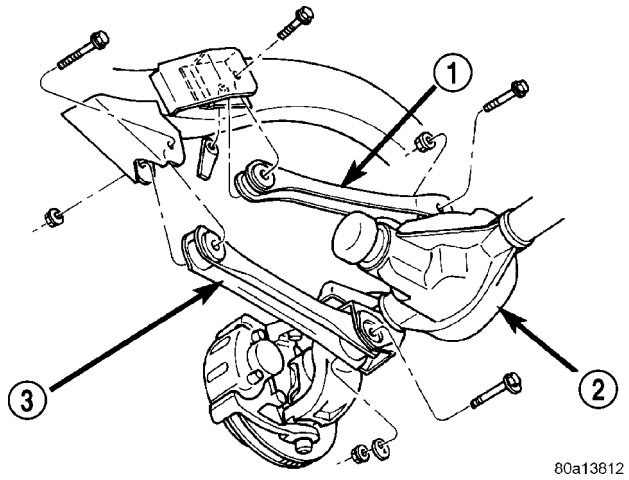


Fig. 4 UPPER AND LOWER SUSPENSION ARMS

1 - UPPER SUSPENSION ARM
2 - FRONT AXLE
3 - LOWER SUSPENSION ARM

(4) Connect the vent hose to the axle shaft tube.

(5) Install track bar to the axle bracket and loosely install bolt (Fig. 5).

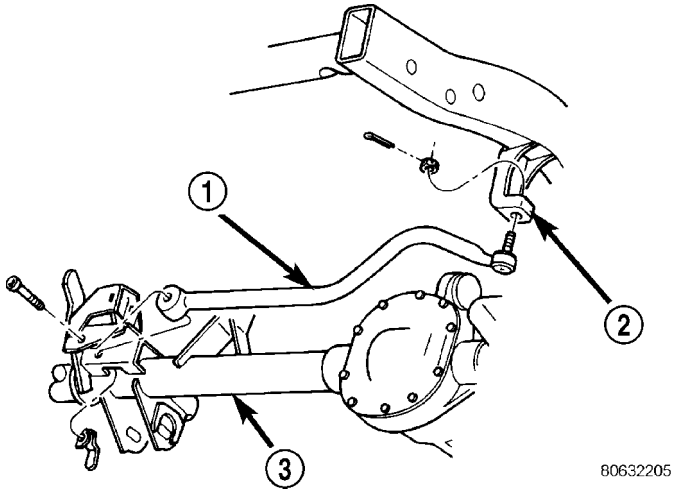


Fig. 5 TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - FRONT AXLE

(6) Install shock absorbers and tighten the bolts (Fig. 6) to torque specification.

(7) Install stabilizer bar links to the axle brackets and tighten nuts (Fig. 7) to torque specification.

(8) Install drag link and tie rod and tighten nut to torque specification

(9) Install steering damper to the axle bracket and tighten nut to torque specification.

(10) Install brake components.

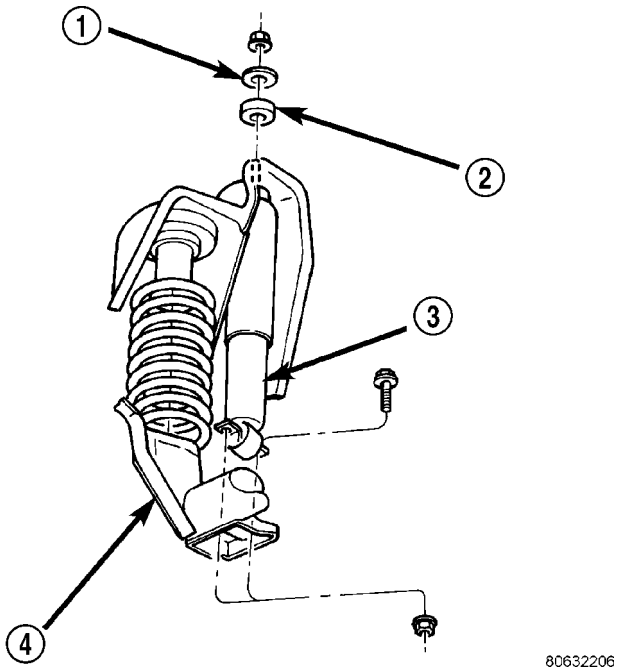


Fig. 6 COIL SPRING AND SHOCK ABSORBER

1 - RETAINER
2 - GROMMET
3 - SHOCK
4 - FRONT AXLE

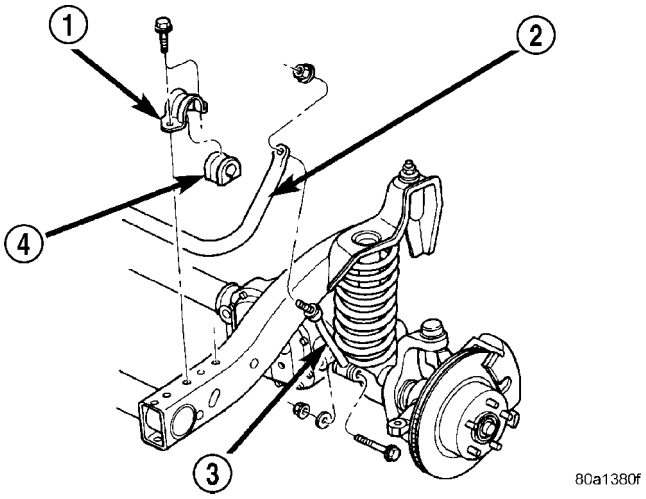


Fig. 7 STABILIZER BAR

1 - RETAINER
2 - STABILIZER BAR
3 - LINK
4 - BUSHING

(11) Install propeller shaft.

(12) Remove lift from the axle and lower vehicle.

(13) Tighten upper and lower control arm nuts to torque specification.

(14) Tighten track bar bolt at the axle bracket to torque specification.

(15) Check the front wheel alignment.

FRONT AXLE - 216FBI (Continued)

ADJUSTMENTS

Ring and pinion gears are supplied as matched sets only. The identifying numbers for the ring and pinion gear are etched into the face of each gear (Fig. 8). A plus (+) number, minus (–) number or zero (0) is etched into the face of the pinion gear. This number is the amount (in thousandths of an inch) the depth varies from the standard depth setting of a pinion etched with a (0). The standard depth provides the best gear tooth contact pattern. Refer to Backlash and Contact Pattern in this section for additional information.

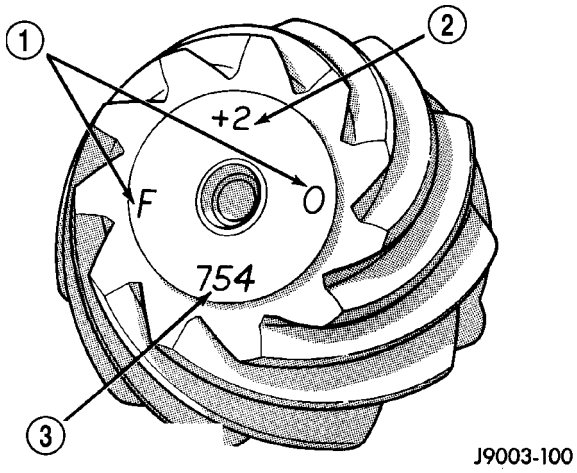


Fig. 8 PINION GEAR ID NUMBERS

1 - PRODUCTION NUMBERS
2 - PINION GEAR DEPTH VARIANCE
3 - GEAR MATCHING NUMBER

Compensation for pinion depth variance is achieved with a select shim/oil slinger. The shims are placed between the rear pinion bearing and the pinion gear head (Fig. 9).

If a new gear set is being installed, note the depth variance etched into both the original and replacement pinion. Add or subtract this number from the thickness of the original depth shim/oil slinger to compensate for the difference in the depth variances. Refer to the Depth Variance chart.

Note where Old and New Pinion Marking columns intersect. Intersecting figure represents plus or minus the amount needed.

Note the etched number on the face of the pinion gear head (–1, –2, 0, +1, +2, etc.). The numbers represent thousands of an inch deviation from the standard. If the number is negative, add that value to the required thickness of the depth shims. If the number is positive, subtract that value from the thickness of the depth shim. If the number is 0 no change is necessary.

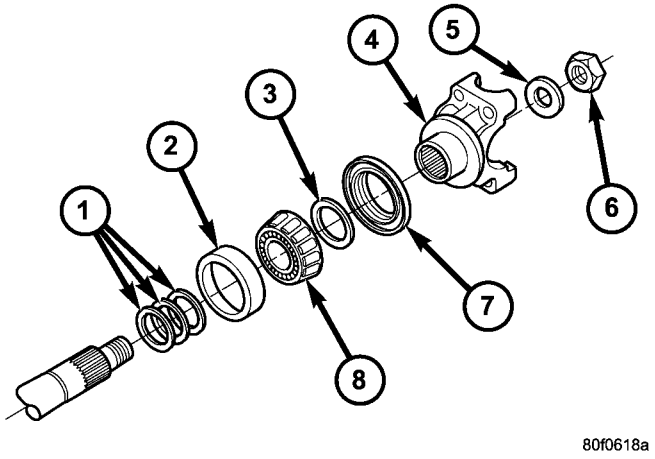


Fig. 9 PINION PRELOAD SHIMS

1 - PRELOAD SHIMS
2 - FRONT BEARING CUP
3 - SLINGER
4 - PINION YOKE
5 - WASHER
6 - PINION NUT
7 - PINION OIL SEAL
8 - FRONT PINION BEARING

FRONT AXLE - 216FBI (Continued)

PINION GEAR DEPTH VARIANCE

Original Pinion Gear Depth Variance	Replacement Pinion Gear Depth Variance								
	−4	−3	−2	−1	0	+1	+2	+3	+4
+4	+0.008	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0
+3	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001
+2	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002
+1	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003
0	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004
−1	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005
−2	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006
−3	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007
−4	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007	−0.008

PINION DEPTH MEASUREMENT AND ADJUSTMENT

Measurements are taken with pinion bearing cups and pinion bearings installed in the housing. Take measurements with Pinion Gauge Set and Dial Indicator C-3339 (Fig. 10).

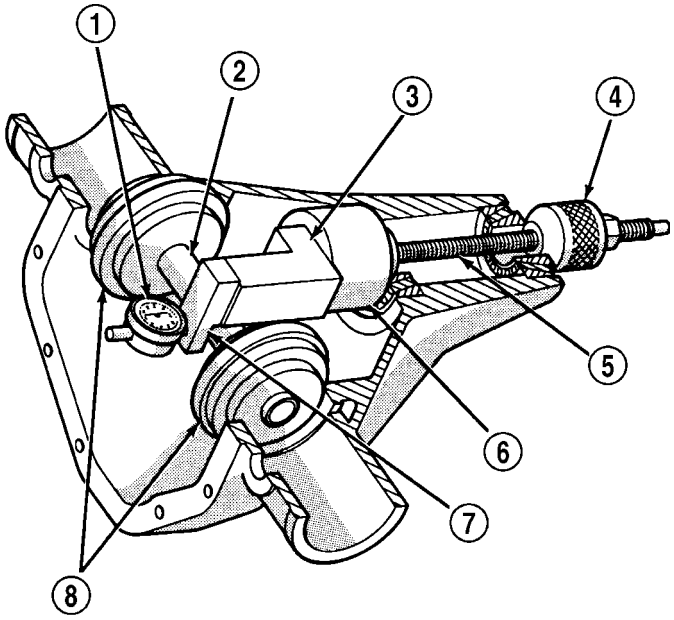


Fig. 10 PINION GEAR DEPTH GAUGE TOOLS

1 - DIAL INDICATOR
2 - ARBOR
3 - PINION HEIGHT BLOCK
4 - CONE
5 - SCREW
6 - PINION BLOCK
7 - SCOOTER BLOCK
8 - ARBOR DISC

(1) Assemble Pinion Height Block 6739, Pinion Block 6734 and rear pinion bearing onto Screw 6741 (Fig. 10).

(2) Insert assembled height gauge components, rear bearing and screw into the housing through pinion bearing cups (Fig. 11).

(3) Install front pinion bearing and Cone-nut 6740 hand tight.

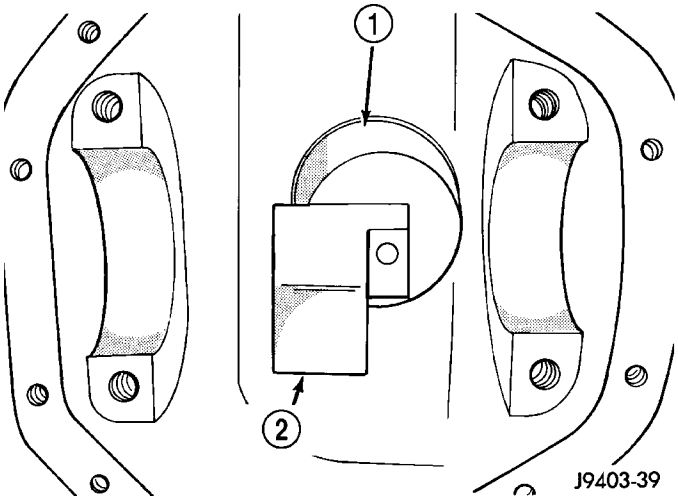


Fig. 11 PINION HEIGHT BLOCK

1 - PINION BLOCK
2 - PINION HEIGHT BLOCK

FRONT AXLE - 216FBI (Continued)

(4) Place Arbor Disc 6732 on Arbor D-115-3 in position in the housing side bearing cradles (Fig. 12).

NOTE: Arbor Discs 6732 has different step diameters to fit other axles. Choose proper step for axle being serviced.

(5) Install differential bearing caps on arbor discs and install bearing cap bolts. Tighten bearing cap bolts to 108 N·m (80 ft. lbs.).

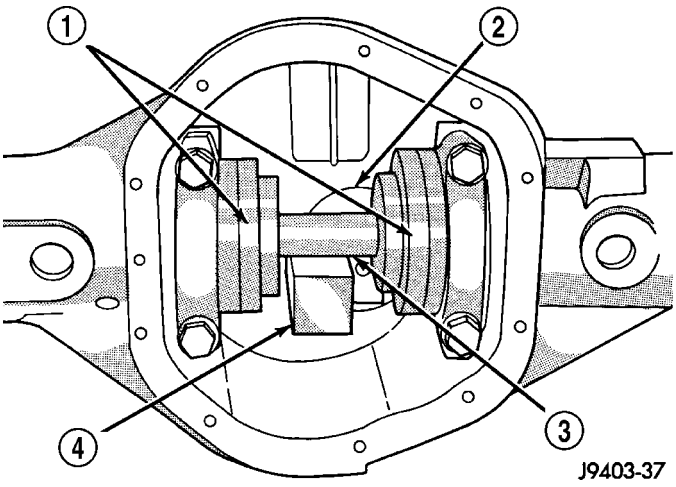


Fig. 12 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(6) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(7) Position Scooter Block/Dial Indicator flush on the pinion height block. Hold scooter block and zero the dial indicator.

(8) Slowly slide the scooter block across the pinion height block over to the arbor (Fig. 13). Move the scooter block till dial indicator crests the arbor, then record the highest reading.

(9) Select a shim/oil slinger equal to the dial indicator reading plus the pinion depth variance number etched in the face of the pinion (Fig. 8). For example, if the depth variance is –2, add +0.002 in. to the dial indicator reading.

PRELOAD SHIM SELECTION

Differential side bearing preload and gear backlash is achieved by selective shims positioned between the differential side bearing cups and the housing. The proper shim thickness is determined using slip-fit Dummy Bearings D-345 in place of the differential side bearings and a Dial Indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion for installation. Establishing proper pinion gear

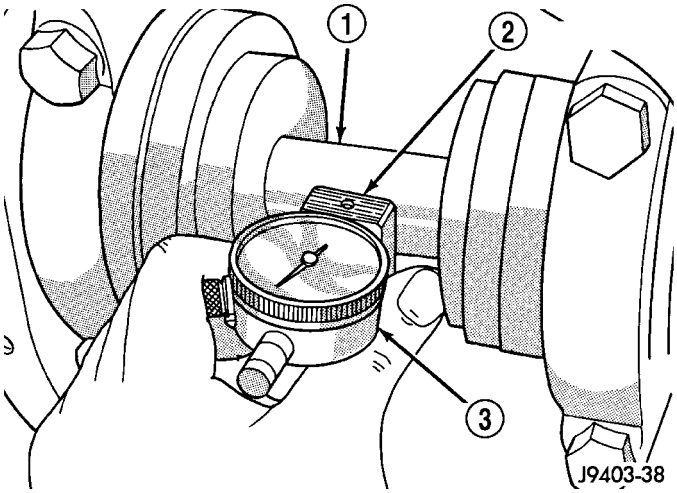


Fig. 13 PINION DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 14). Differential shim measurements are performed with spreader W-129-B removed.

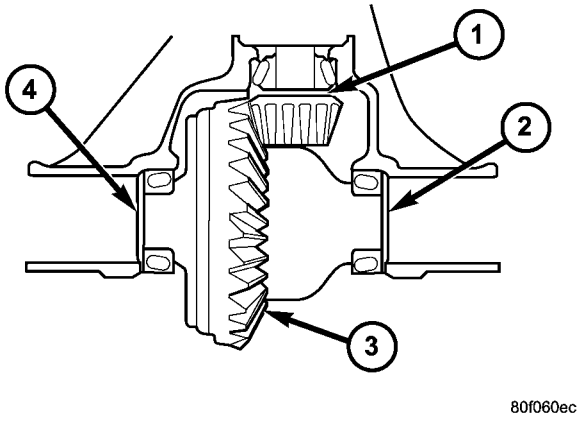


Fig. 14 SHIM LOCATION

1. PINION DEPTH SHIM
2. DIFFERENTIAL SHIM PINION GEAR SIDE
3. RING GEAR
4. DIFFERENTIAL SHIM RING GEAR SIDE

FRONT AXLE - 216FBI (Continued)

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove differential side bearings from differential case.
(2) Remove factory installed shims from differential case.
(3) Install ring gear on differential case and tighten bolts to specification.
(4) Install dummy side bearings D-345 on differential case.
(5) Install differential case in the housing.
(6) Install the marked bearing caps in their correct positions and snug the bolts (Fig. 15).

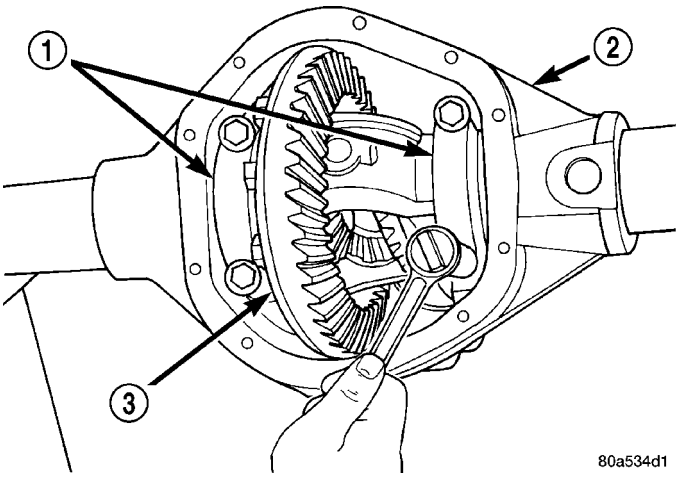


Fig. 15 BEARING CAP BOLTS

1 - BEARING CAP
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

(7) With a dead-blow hammer, seat the differential dummy bearings to each side of the housing (Fig. 16) and (Fig. 17).
(8) Thread Pilot Stud C-3288-B into rear cover bolt hole below ring gear (Fig. 18).

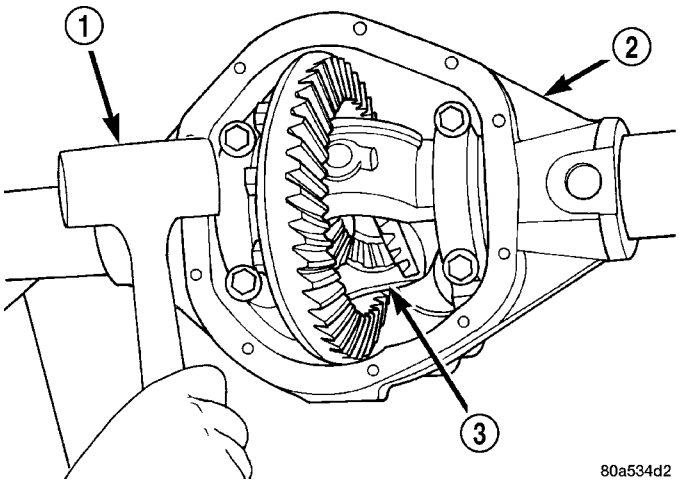


Fig. 16 SEAT DUMMY BEARING PINION SIDE

1 - DEAD-BLOW HAMMER
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

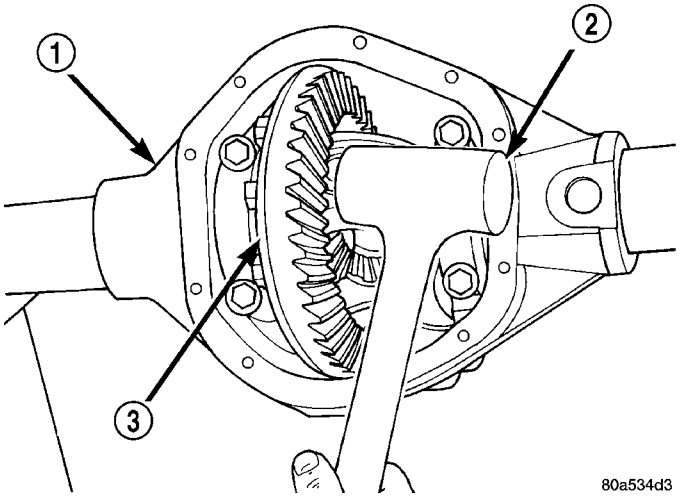


Fig. 17 SEAT DUMMY BEARING RING GEAR SIDE

1 - DIFFERENTIAL HOUSING
2 - DEAD-BLOW HAMMER
3 - DIFFERENTIAL CASE

FRONT AXLE - 216FBI (Continued)

(9) Attach the Dial Indicator C-3339 to pilot stud. Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 18).

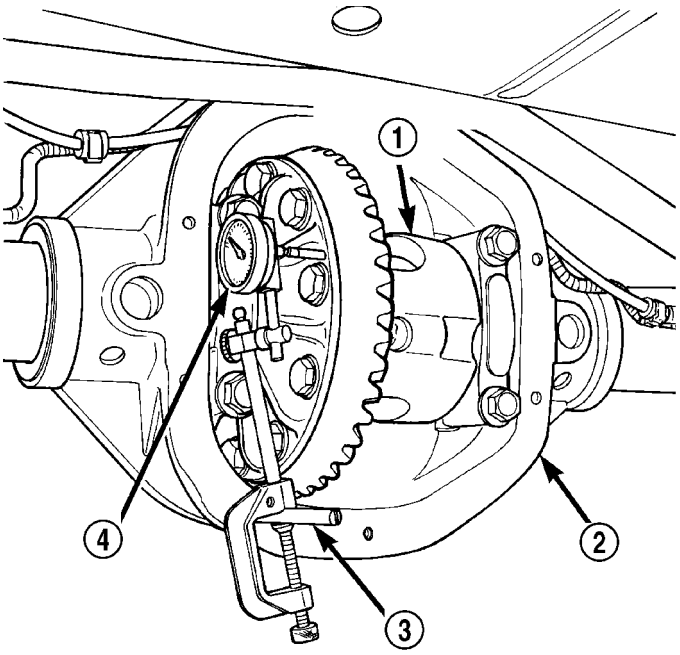


Fig. 18 DIFFERENTIAL SIDE PLAY MEASUREMENT

1 - DIFFERENTIAL CASE
2 - DIFFERENTIAL HOUSING
3 - PILOT STUD
4 - DIAL INDICATOR

(10) Push and hold differential case to pinion gear side of the housing and zero dial indicator (Fig. 19).

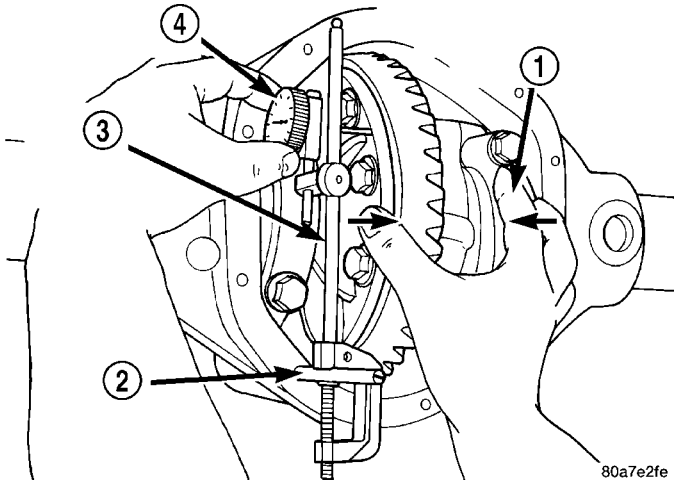


Fig. 19 DIAL INDICATOR LOCATION

1 - DIFFERENTIAL CASE TO PINION GEAR SIDE
2 - PILOT STUD
3 - DIAL INDICATOR ARM
4 - DIAL INDICATOR FACE

(11) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 20).

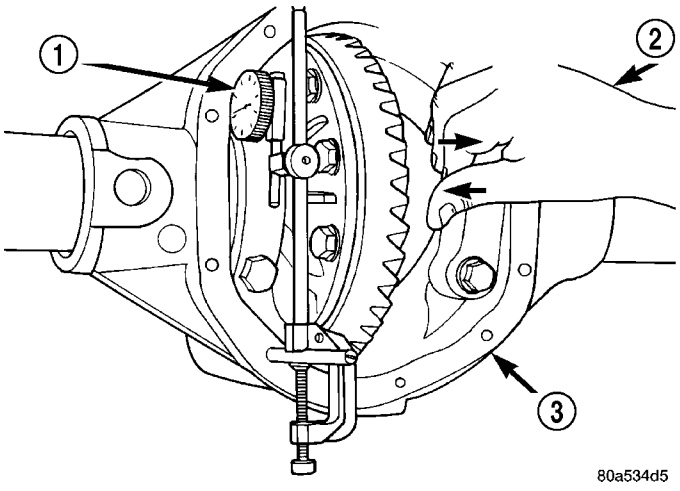


Fig. 20 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(12) Add 0.25 mm (0.010 in.) to the zero end play total. This total represents the thickness of shims needed to preload the new bearings when the differential is installed.

(13) Rotate dial indicator out of the way on the pilot stud.

(14) Remove differential case and dummy bearings from the housing.

(15) Install the pinion gear in the housing. Install the pinion yoke and establish the correct pinion rotating torque.

(16) Install differential case and dummy bearings D-345 in the housing (without shims), install bearing caps and tighten bolts snug.

(17) Seat ring gear side dummy bearing (Fig. 17).

(18) Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 18).

(19) Push and hold differential case toward pinion gear and zero dial indicator (Fig. 21).

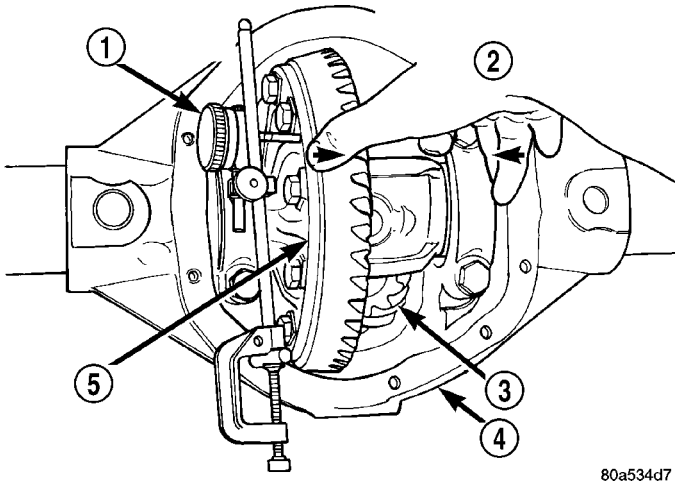


Fig. 21 ZERO DIAL INDICATOR

1 - DIAL INDICATOR FACE
2 - DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(20) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 22). Subtract 0.05 mm (0.002 in.) from this reading. This is the shim thickness for the ring gear side.

NOTE: This is the shim needed on the ring gear side for proper backlash.

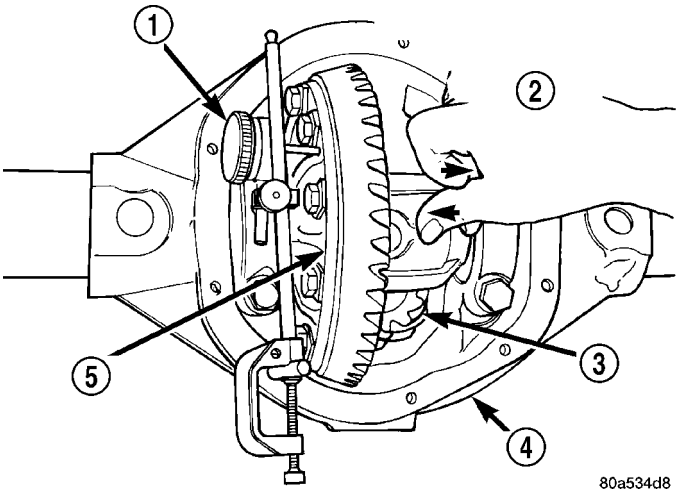


Fig. 22 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(21) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

(22) Rotate dial indicator out of the way on pilot stud.

(23) Remove differential case and dummy bearings from the housing.

(24) Install the selected shims onto the differential case hubs.

(25) Install side bearings on differential case hubs with Install C-3716-A and Handle C-4171.

(26) Install bearing cups on differential.

(27) Install Spreader W-129-B and some items from Adapter Set 6987 on the housing and spread open enough to receive differential case.

CAUTION: Do not spread housing over 0.38 mm (0.015 in.). Failure to heed caution may result in damage.

(28) Install differential case into the housing.

(29) Remove spreader from the housing.

(30) Rotate the differential case several times to seat the side bearings.

(31) Position the indicator plunger against a ring gear tooth (Fig. 23).

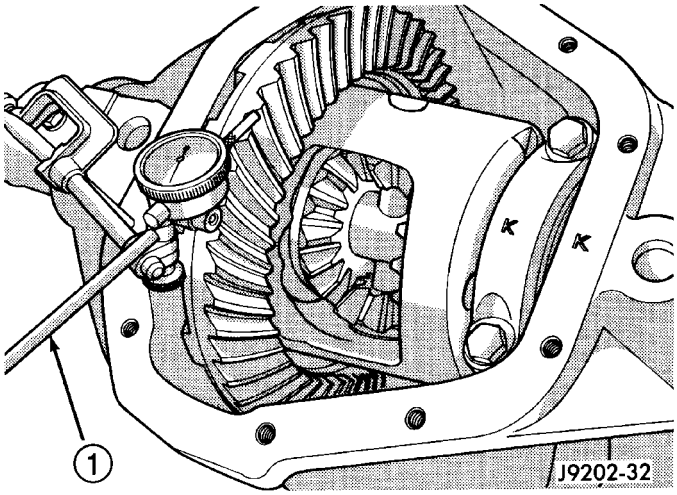


Fig. 23 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

(32) Push and hold ring gear upward while not allowing the pinion gear to rotate.

(33) Zero dial indicator face to pointer.

(34) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 24).

FRONT AXLE - 216FBI (Continued)

(35) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform Gear Contact Pattern Analysis procedure.

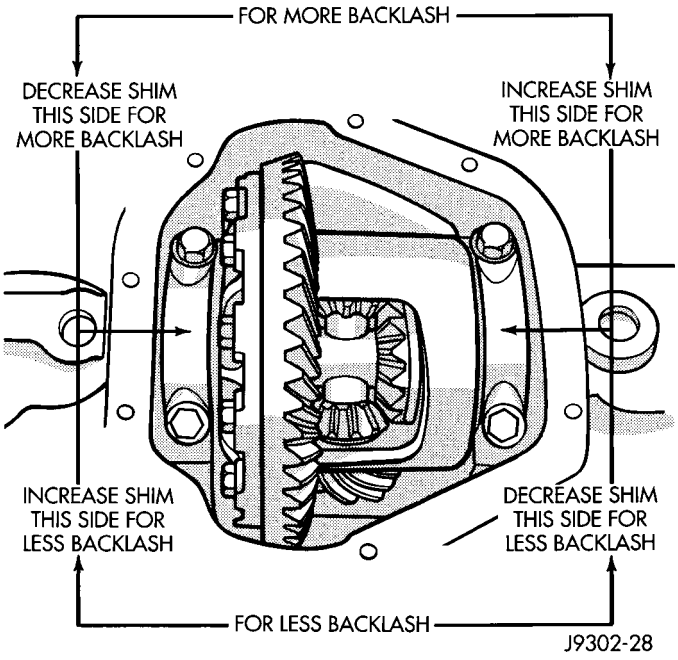


Fig. 24 BACKLASH SHIM

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth at the outer-end. The ROOT of the gear tooth is the lowest portion of the tooth (Fig. 25).

NOTE: If the PROFILE across the tooth is the same it is a 3 Axis cut gear. If the PROFILE across the tooth is tapered it is a 2 Axis cut gear.

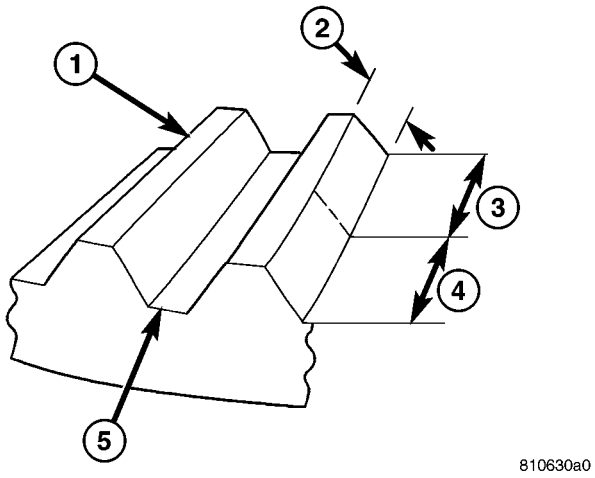


Fig. 25 GEAR DESCRIPTION

1 - TOP LAND
2 - PROFILE
3 - TOE
4 - HEEL
5 - ROOT

(1) Apply a thin coat of hydrated ferric oxide or equivalent to the drive and coast side of the ring gear teeth.

(2) Wrap, twist and hold a shop towel around the pinion yoke to increase the turning resistance of the pinion. This will provide a more distinct contact pattern.

(3) With a boxed end wrench on the ring gear bolt, rotate the differential case one complete revolution in both directions while a load is being applied from shop towel.

The areas on the ring gear teeth with the greatest degree of contact against the pinion teeth will squeegee the compound to the areas with the least amount of contact. Note and compare patterns on the ring gear teeth to Gear Tooth Contact Patterns chart (Fig. 26) and (Fig. 27) and adjust pinion depth and gear backlash as necessary.

DIFFERENTIAL BEARING PRELOAD CHECK

The final check on the differential assembly before installing the axles is torque to rotate pinion and differential combined. This will verify the correct differential bearing preload.

Torque to rotate the differential and pinion should be the torque to rotate the pinion plus 0.79-1.24 N·m (7-11 in. lbs.).

FRONT AXLE - 216FBI (Continued)

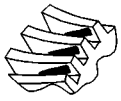
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top heel contact		**Top toe contact**	Backlash correct. Thicker pinion position shim required.
	Root toe contact		**Root heel contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top heel contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root toe contact**	Pinion position shim correct. Increase backlash.

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Fig. 26 PATTERN INTERPRETATION (GEAR CUT 2 AXIS)

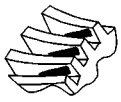
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top toe contact		**Top heel contact**	Backlash correct. Thicker pinion position shim required.
	Root heel contact		**Root toe contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top toe contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root heel contact**	Pinion position shim correct. Increase backlash.

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Fig. 27 PATTERN INTERPRETATION (GEAR CUT 3 AXIS)

FRONT AXLE - 216FBI (Continued)

SPECIFICATIONS

FRONT AXLE

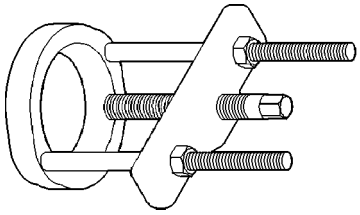
AXLE SPECIFICATIONS

DESCRIPTION	SPECIFICATION
Axle Ratio	4.10
Ring Gear Backlash	0.12-0.20 mm (0.005-0.008 in.)
Pinion Bearing Preload - Original Bearing	1-2 N·m (10-20 in. lbs.)
Pinion Bearing Preload - New Bearing	2.3-4.5 N·m (20-40 in. lbs.)

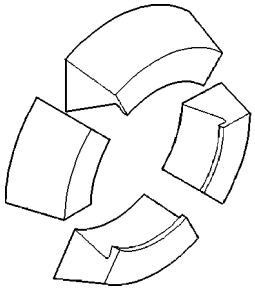
TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Fill Plug	34	25	-
Drain Plug	34	25	-
Differential Cover Bolts	41	30	-
Bearing Cap Bolts	108	80	-
Pinion Nut	217-271	160-200	-
Ring Gear Bolts	136	100	-
Axle Nut	237	175	-

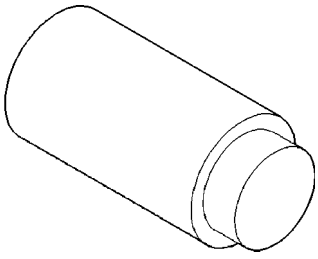
SPECIAL TOOLS



PULLER- C-293-PA

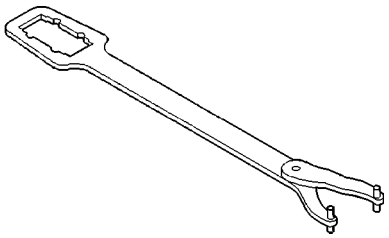


ADAPTERS C-293-39

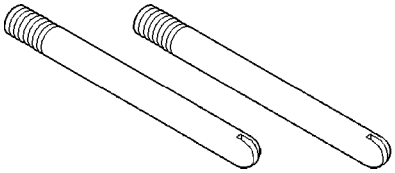


PLUG C-293-3

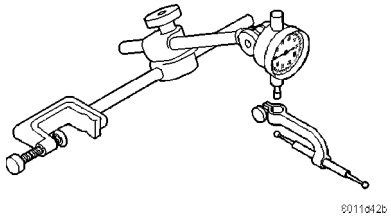
FRONT AXLE - 216FBI (Continued)



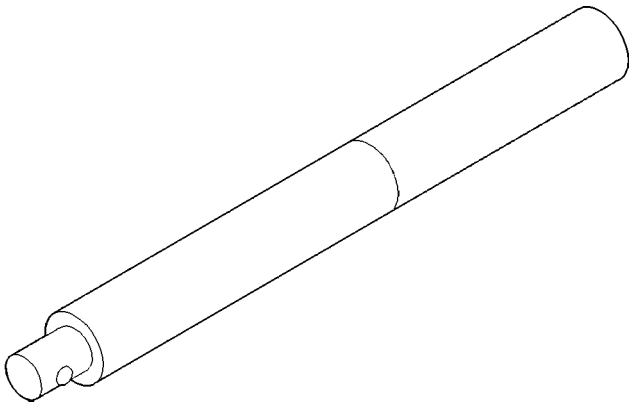
FLANGE WRENCH C-3281



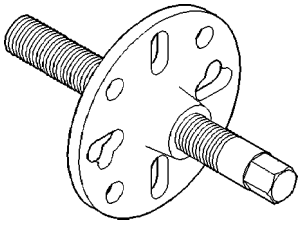
PILOT STUDS C-3288-B



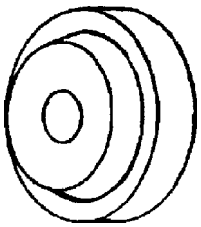
DIAL INDICATOR C-3339



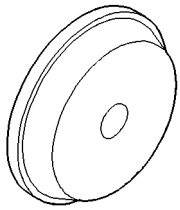
HANDLE C-4171



REMOVER C-452



INSTALLER D-144



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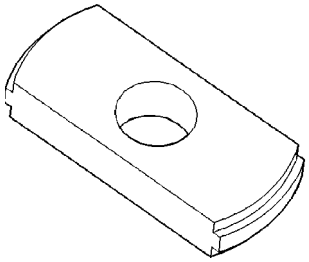

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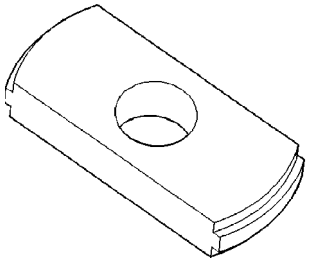

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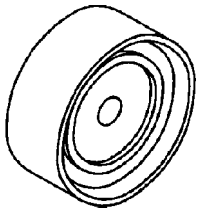
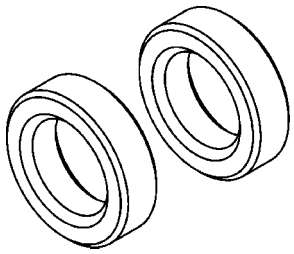
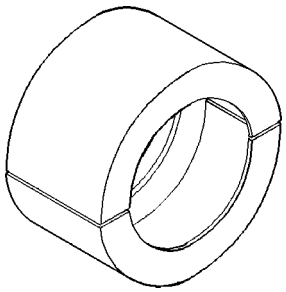


INSTALLER D-156

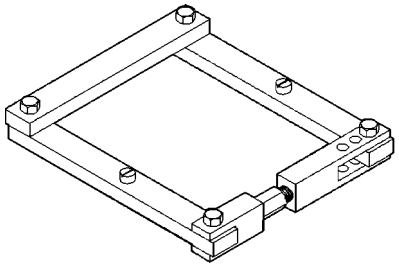
FRONT AXLE - 216FBI (Continued)



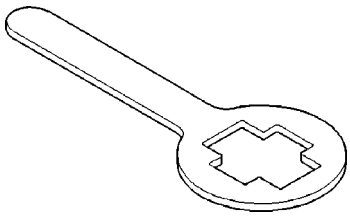
DUMMY BEARINGS D-345



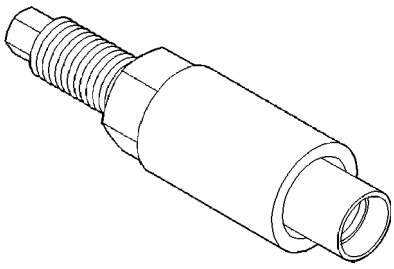
JAWS 6447



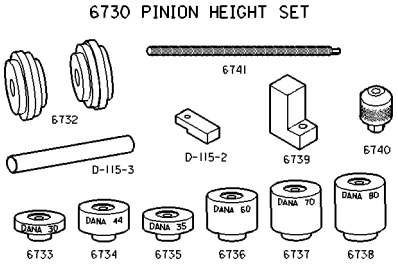
SPREADER W-129-B



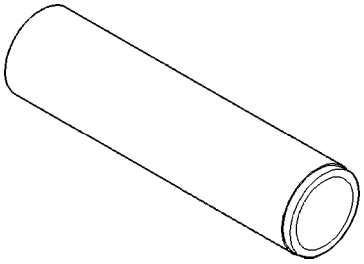
HOLDER YOKE 6719A



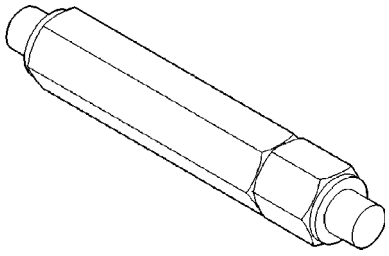
INSTALLER W-162-D



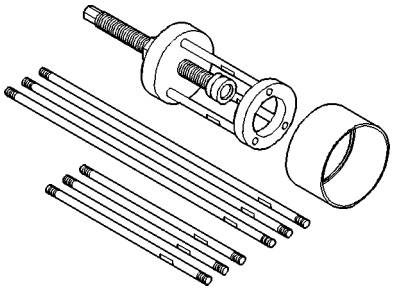
PINION DEPTH SET 6730



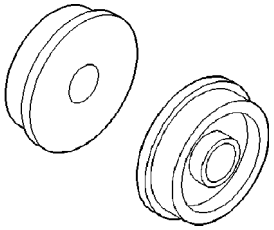
INSTALLER W-262



TURNBUCKLE 6797



PULLER 6444



INSTALLER DISCS 8110

AXLE SHAFTS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake components.
(3) Remove cotter pin, nut retainer and axle hub nut.
(4) Remove hub bearing bolts (Fig. 28).

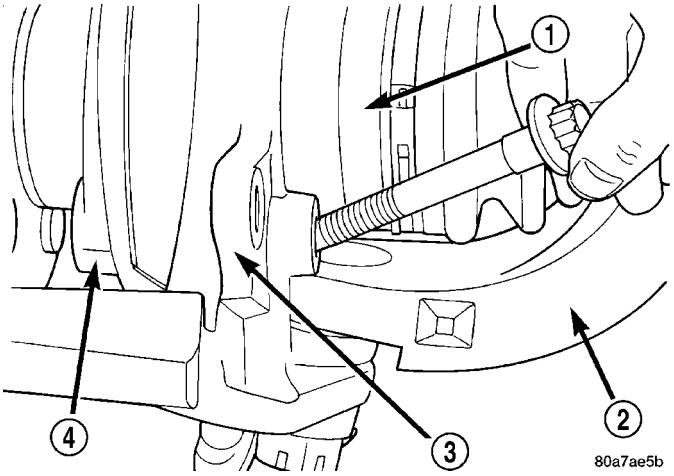


Fig. 28 HUB BEARING BOLTS

1 - AXLE SHAFT
2 - AXLE
3 - KNUCKLE
4 - HUB BEARING

(5) Remove hub bearing from steering knuckle.

(6) Remove axle shaft assembly (Fig. 29) from axle

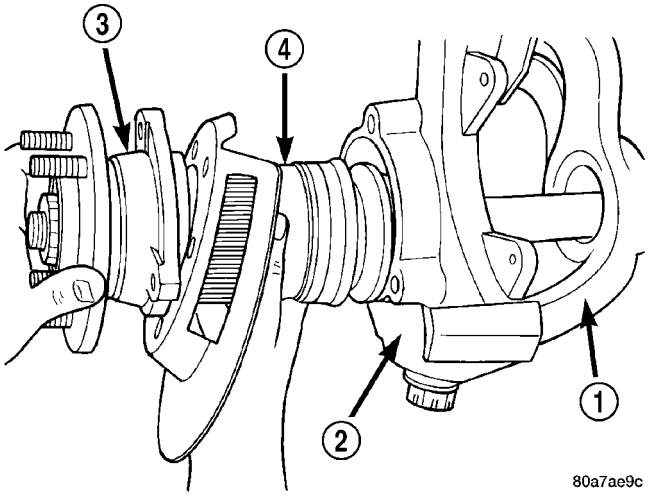


Fig. 29 HUB BEARING AND AXLE

1 - AXLE
2 - KNUCKLE
3 - HUB BEARING
4 - AXLE SHAFT

(7) Remove brake rotor shield from the hub bearing or knuckle.

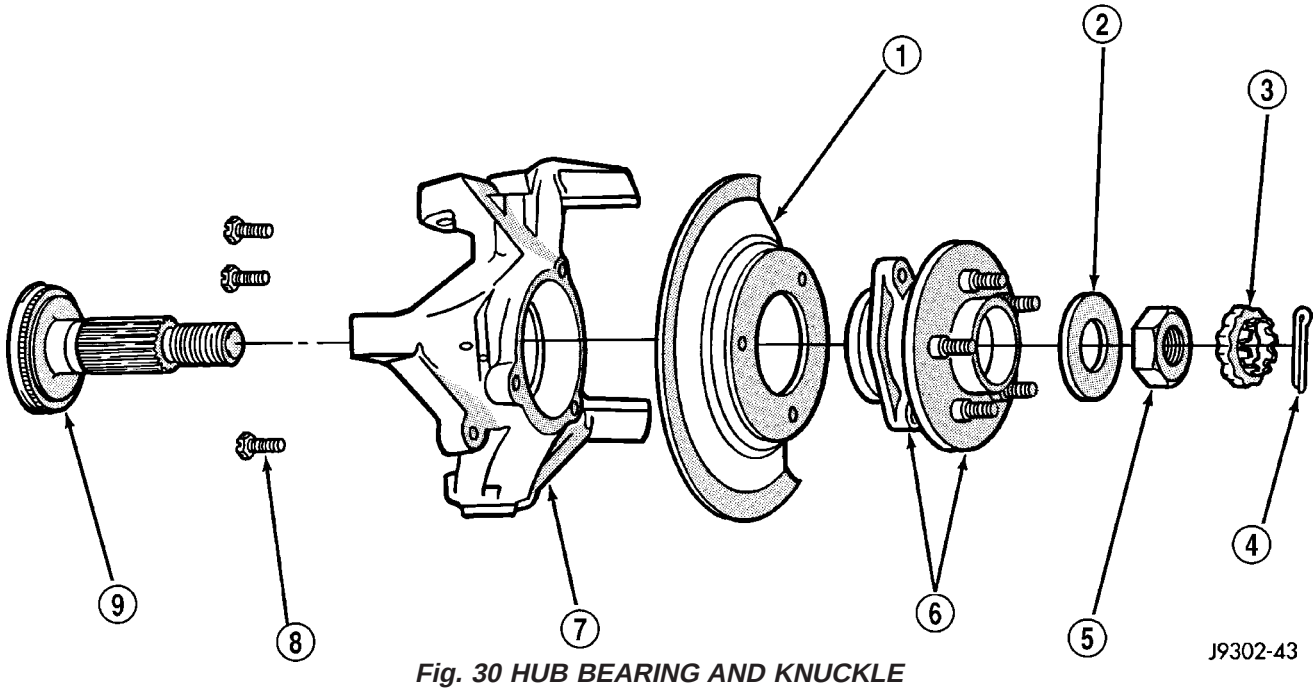


Fig. 30 HUB BEARING AND KNUCKLE

1 - BRAKE SHIELD
2 - WASHER
3 - RETAINER
4 - COTTER PIN
5 - NUT
6 - HUB AND BEARING ASSEMBLY
7 - STEERING KNUCKLE
8 - BOLT
9 - TONE WHEEL (ABS)

AXLE SHAFTS (Continued)

INSTALLATION

(1) Clean axle shaft and apply a thin film of Mopar Wheel Bearing Grease or equivalent to the shaft splines, seal contact surface and hub bore.
(2) Install brake rotor shield on knuckle.
(3) Install axle shaft into differential side gears.
(4) Install hub bearing and tighten bolts to 102 N·m (75 ft. lbs.).
(5) Install axle washer and nut. Tighten nut to 237 N·m (175 ft. lbs.) and install nut retainer and cotter pin (Fig. 30).
(6) Install brake components.

AXLE SHAFT SEALS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove axle shafts (Fig. 31).

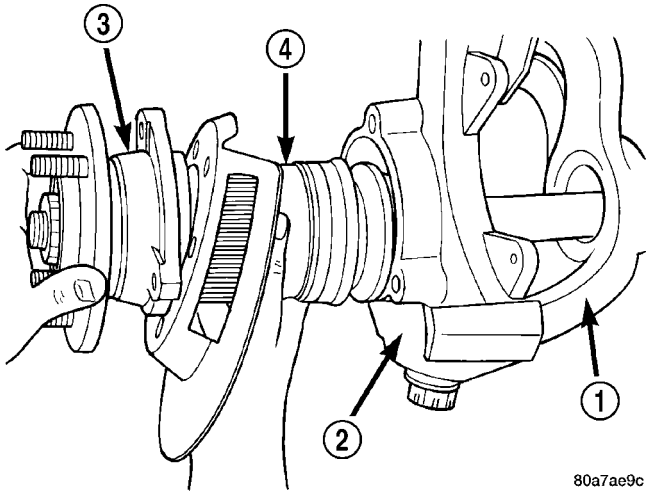


Fig. 31 HUB BEARING AND AXLE ASSEMBLY

1 - AXLE
2 - KNUCKLE
3 - HUB BEARING
4 - AXLE SHAFT

(3) Remove differential assembly (Fig. 32).
(4) Remove inner axle shaft seals with a pry bar.

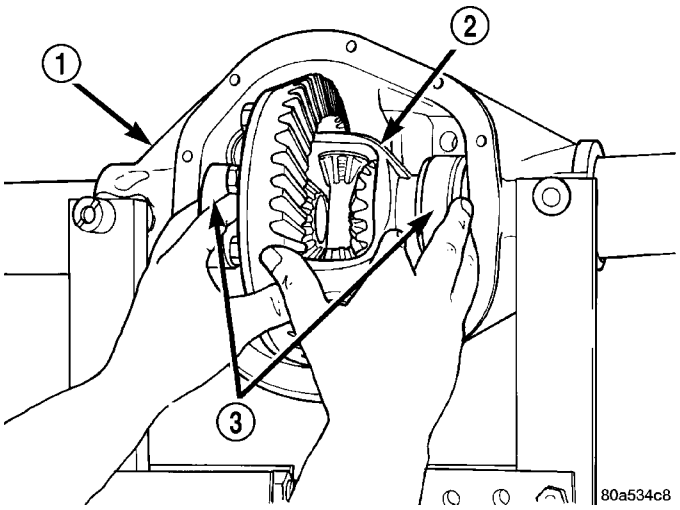


Fig. 32 DIFFERENTIAL CASE

1 - AXLE HOUSING
2 - DIFFERENTIAL CASE
3 - BEARING CUPS

INSTALLATION

(1) Remove any sealer remaining from original seals.
(2) Install oil seals with Discs 8110 and Turnbuckle 6797 (Fig. 33). Tighten tool until disc bottoms in housing.

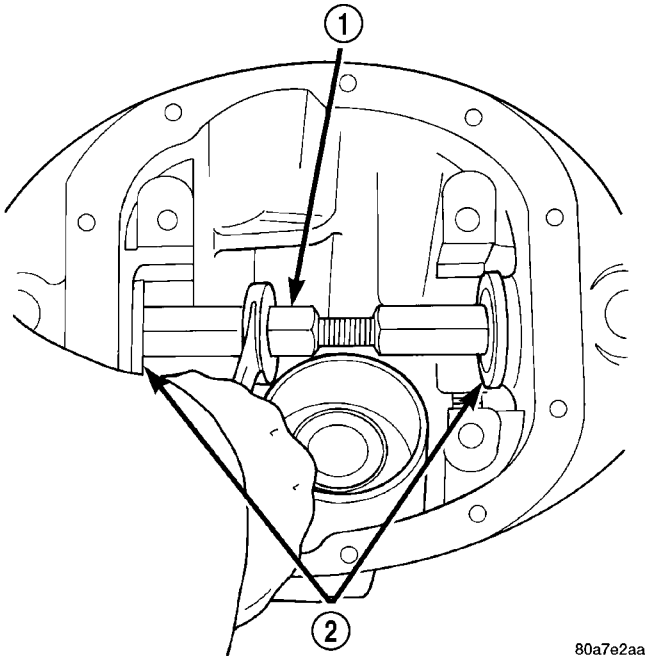


Fig. 33 AXLE SEAL INSTALLER

1 - TURNBUCKLE
2 - DISCS

(3) Install differential and axle shafts.

SINGLE CARDAN UNIVERSAL JOINT

REMOVAL

CAUTION: Clamp only the narrow forged portion of the yoke in the vise. Failure to heed caution may result in damage.

(1) Remove axle shaft.
(2) Remove bearing cap retaining snap rings (Fig. 34).

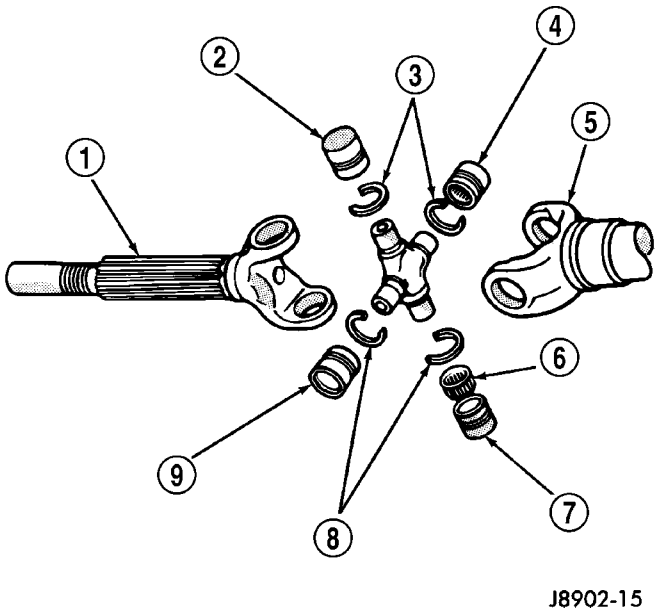


Fig. 34 AXLE SHAFT OUTER U-JOINT

1 - SHAFT YOKE
2 - BEARING CAP
3 - SNAP RINGS
4 - BEARING CAP
5 - SPINDLE YOKE
6 - BEARING
7 - BEARING CAP
8 - SNAP RINGS
9 - BEARING CAP

NOTE: Saturate the bearing caps with penetrating oil prior to removal.

(3) Place a socket (receiver) with an inside diameter larger than the bearing cap against the yoke and the perimeter of the bearing cap to be removed.
(4) Place a socket (driver) with an outside diameter smaller than the bearing cap against the opposite bearing cap.
(5) Position yoke with the sockets in a vise (Fig. 35).

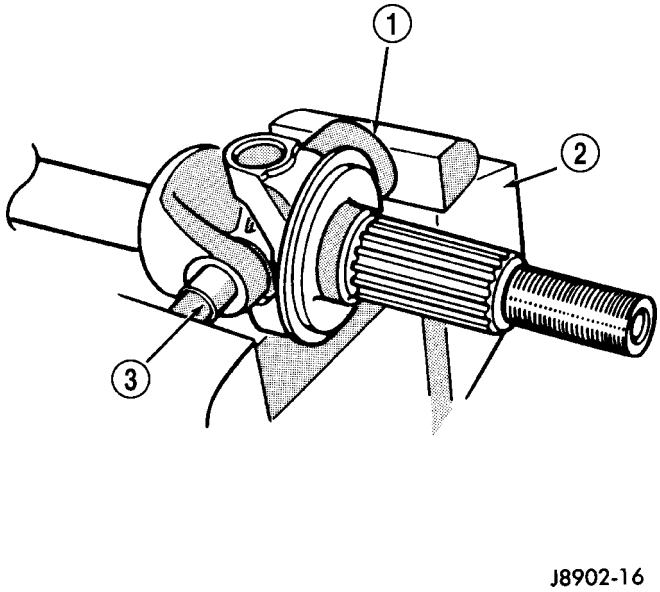


Fig. 35 YOKE BEARING CAP

1 - LARGE-DIAMETER SOCKET
2 - VISE
3 - SMALL-DIAMETER SOCKET

(6) Tighten vise jaws, to force bearing cap into the socket (receiver).
(7) Release vise jaws. Remove sockets and bearing cap forced out of the yoke.
(8) Repeat above procedure for the remaining bearing cap and remove cross from propeller shaft yoke.

INSTALLATION

CAUTION: Keep cross and bearing cap straight during installation. Failure to heed caution may result in damage.

(1) Pack bearing caps 1/3 full of wheel bearing lubricant (Fig. 36). Apply extreme pressure (EP), lithium-base lubricant to aid in installation.
(2) Position cross in the yoke. Insert seals and bearings, then tap bearing caps into the yoke bores far enough to hold spider in position.
(3) Place socket (driver) against one bearing cap. Position yoke with the socket in a vise.
(4) Tighten vise to force bearing caps into the yoke. Force the caps enough to install retaining clips.
(5) Install bearing cap retaining clips.
(6) Install axle shaft.

SINGLE CARDAN UNIVERSAL JOINT (Continued)

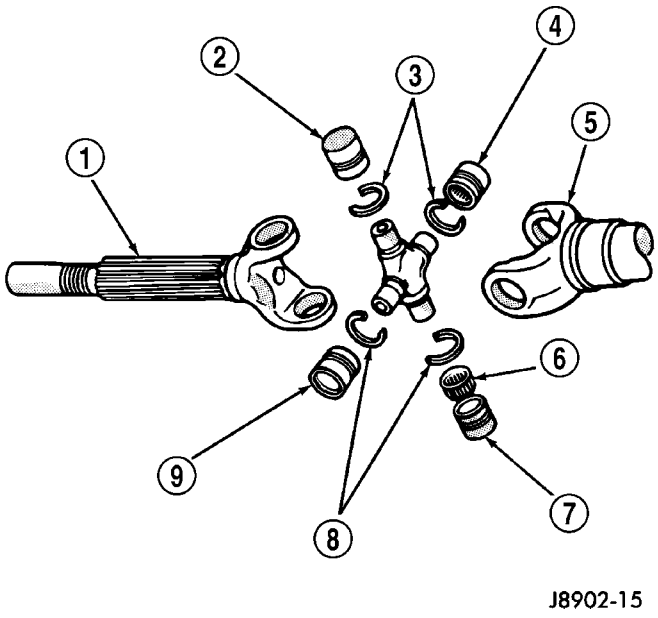


Fig. 36 AXLE SHAFT OUTER U-JOINT

1 - SHAFT YOKE
2 - BEARING CAP
3 - SNAP RINGS
4 - BEARING CAP
5 - SPINDLE YOKE
6 - BEARING
7 - BEARING CAP
8 - SNAP RINGS
9 - BEARING CAP

PINION SEAL

REMOVAL

(1) Mark the propeller shaft and pinion yoke for installation reference.

(2) Remove propeller shaft from the yoke.

(3) Rotate pinion gear three or four times and verify it rotates smoothly.

(4) Remove pinion yoke nut and washer with Remover C-452 and Flange Wrench C-3281 (Fig. 37).

(5) Remove pinion shaft seal with a pry tool or slide hammer mounted screw.

INSTALLATION

(1) Apply a light coating of gear lubricant on the lip of pinion seal. Install seal with an appropriate installer (Fig. 38).

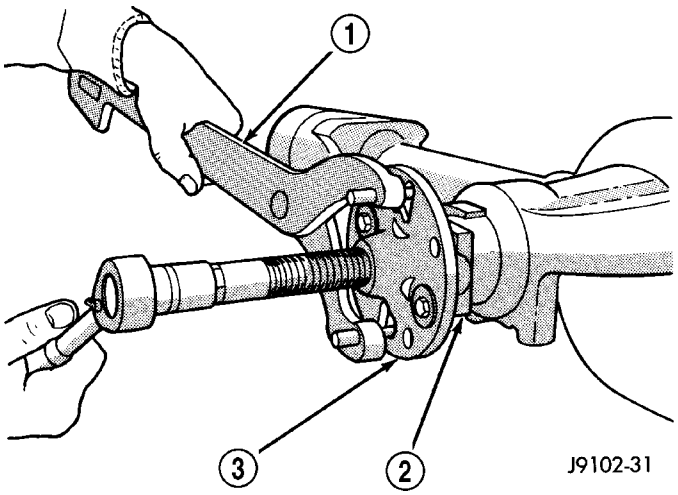


Fig. 37 PINION YOKE REMOVER

1 - WRENCH
2 - YOKE
3 - PULLER

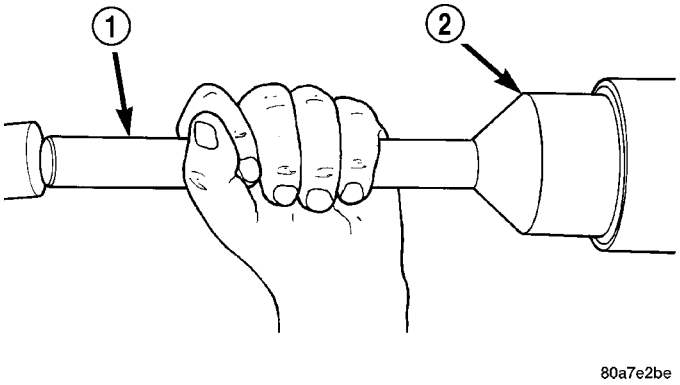


Fig. 38 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(2) Install yoke on the pinion gear with Installer W-162-D and Spanner Wrench 6958 (Fig. 39).

(3) Install a **new** nut on the pinion gear and tighten the nut to 217-271 N·m (160-200 in. lbs.) (Fig. 40).

(4) Installation propeller shaft with reference marks aligned.

(5) Check and add gear lubricant to axle if necessary.

PINION SEAL (Continued)

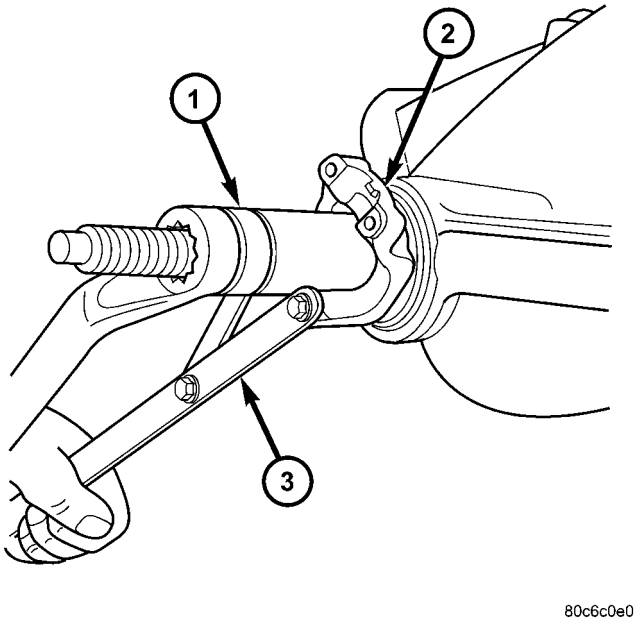


Fig. 39 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

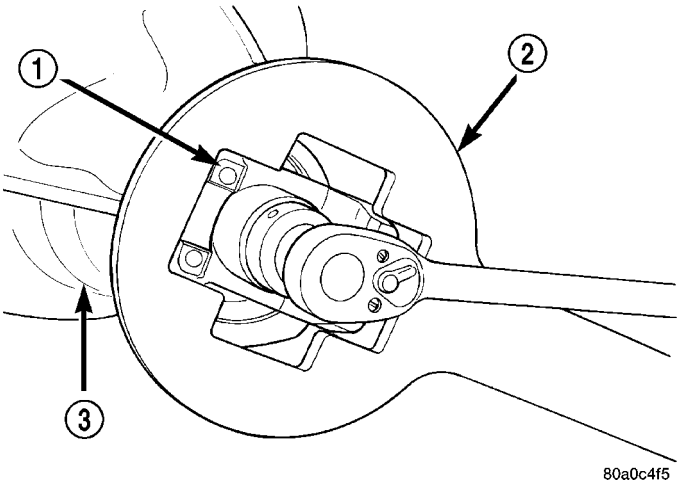


Fig. 40 PINION SHAFT NUT

1 - PINION FLANGE
2 - HOLDING TOOL
3 - DIFFERENTIAL HOUSING

DIFFERENTIAL COVER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Remove drain plug.

(3) Disconnect indicator switch harness and remove connector from differential cover (Fig. 41).

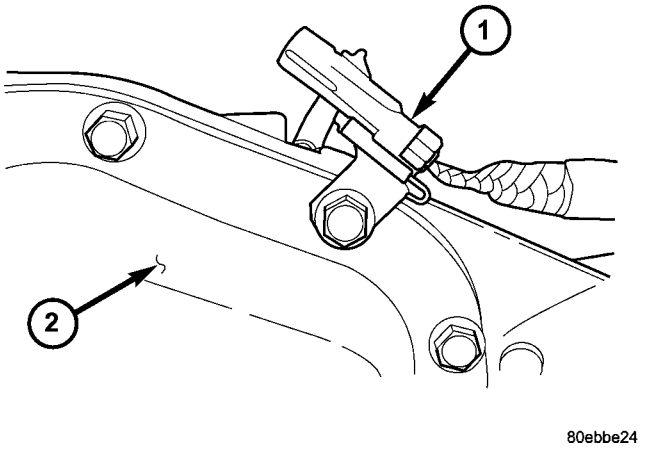


Fig. 41 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(4) Remove cover bolts.

(5) Remove cover and drain lubricant.

INSTALLATION

(1) Apply a 6.35mm (1/4 in.) bead of Mopar Silicone Rubber Sealant or equivalent to the housing cover (Fig. 42).

CAUTION: If housing cover is not installed within 3 to 5 minutes, the cover must be cleaned and new RTV applied. Failure to heed caution may result in damage.

(2) Install cover, identification tag and indicator switch connector (Fig. 43). Tighten cover bolts in a criss-cross pattern to 41 N·m (30 ft. lbs.).

(3) Connect indicator switch harness.

(4) Fill differential to specifications.

(5) Install fill plug and tighten to 34 N·m (25 ft. lbs.).

DIFFERENTIAL COVER (Continued)

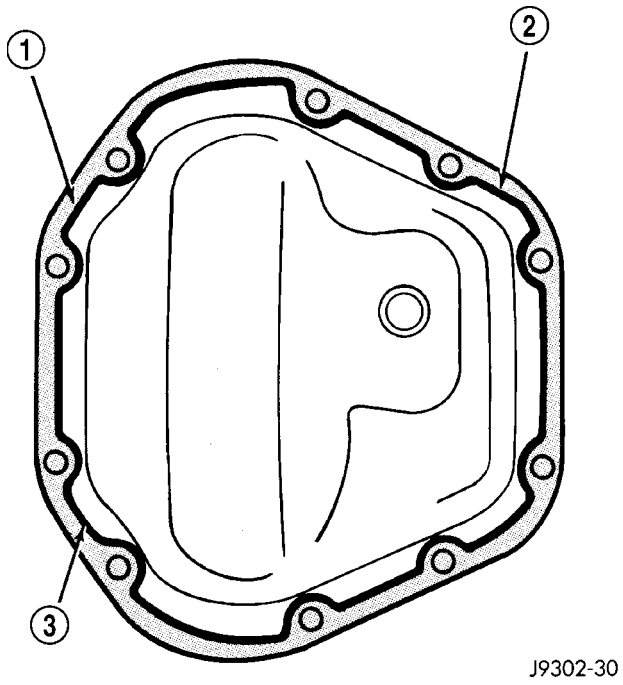


Fig. 42 HOUSING COVER - TYPICAL

1 - SEALANT SURFACE
2 - SEALANT
3 - SEALANT THICKESS

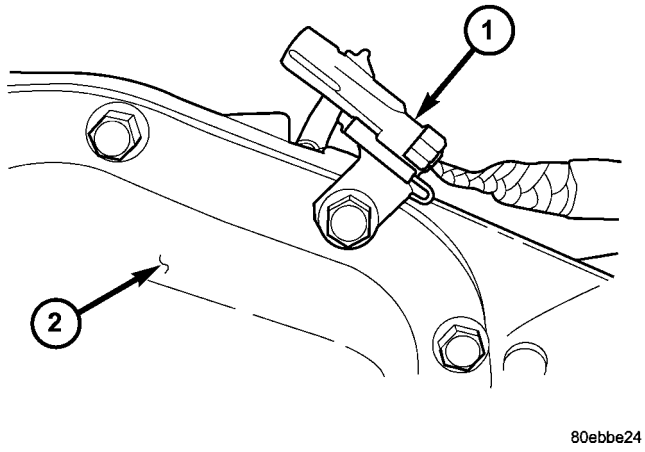


Fig. 43 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

DIFFERENTIAL - TRU-LOK

DESCRIPTION

The differential is a locking differential, that provides a positive mechanical connection between the right and left axle when engauged. The differential uses a dog clutch to connection the right and left axle.

OPERATION

The Tru-lok differential is activated by the axle lock switch located on the dash panel. When the switch is activated, an air pump with a built-in pressure regulator sends 5 PSI of air pressure to a accuator diaphragm in the differential housing. The diaphragm then engages a dog clutch and a position switch. The dog clutch has one gear attached to the differential case and another gear attached to a differential side gear. When the dog clutch is engaged the right and left wheels turn at the same speed. The position switch lights a lamp on the dash to indicate the system has been engaged. The differential works as standard differential when not engaged.

NOTE: The differential is serviced as an assembly, the diaphragm and indicator switch are serviced separately. The differential case must be removed to service the diaphragm actuator and indicator switch.

DIAGNOSIS AND TESTING

UNLOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the OFF position.

(5) Rotate one tire by hand. The other tire should spin in the opposite direction.

NOTE: If wheel cannot be rotated the differential must be repaired/replaced.

DIFFERENTIAL - TRU-LOK (Continued)

LOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the ON position.

(5) Try to rotate one tire by hand. You should not be able to rotate the tire.

NOTE: If wheel does rotate verify locker pump operation. If the pump is operating properly the differential must be repaired/replaced.

TRU-LOK INDICATOR SWITCH

(1) Turn ignition switch off.

(2) Disconnect indicator switch harness (Fig. 44) from the differential housing.

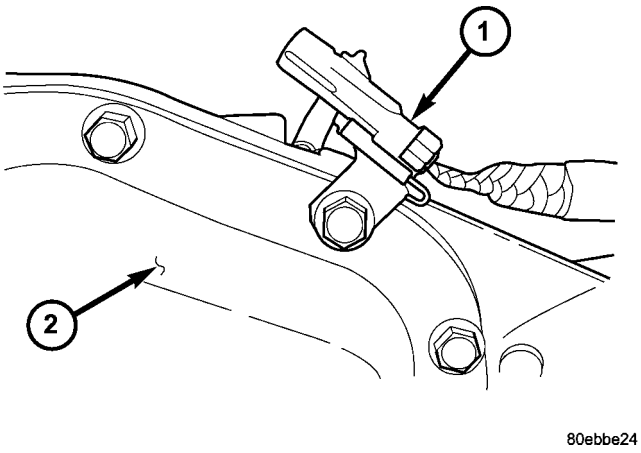


Fig. 44 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(3) Measure electrical continuity across the switch terminals. Circuit should be closed (zero resistance).

NOTE: If circuit is not closed remove differential and replace locker indicator switch.

TRU-LOK PUMPS

(1) Connect a pressure gauge to the front pump.

(2) Place transfer case in 4WD Low and automatic transmission in Park (1st gear if manual transmission).

(3) Turn ignition to the ON position and push the dash switch twice to activate the front pump.

NOTE: If pump is not running, verify pump has (Fig. 45) 12 volts and a ground.

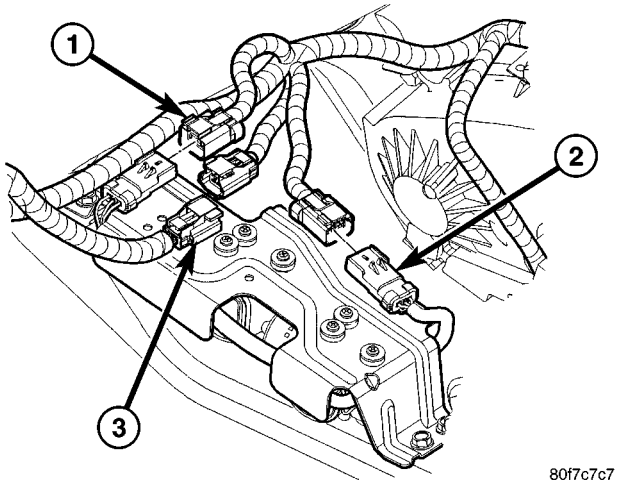


Fig. 45 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

(4) With the pump running the pressure gauge should show 5 psi..

NOTE: If pump does not produce 5 psi. replace the pump.

REMOVAL

(1) Remove drain plug from the differential housing.

(2) Disconnect indicator switch harness and remove connector from differential cover (Fig. 46).

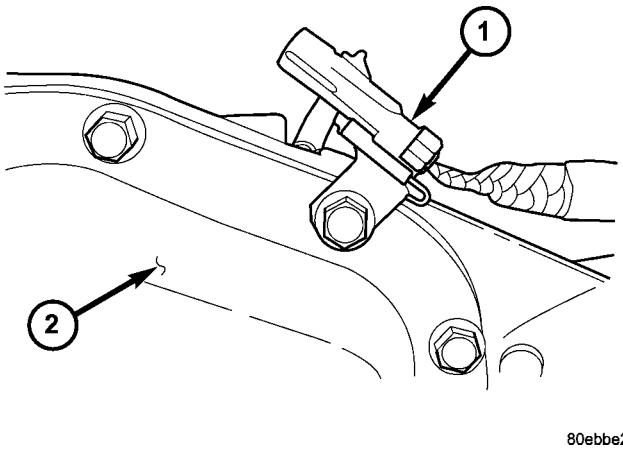


Fig. 46 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(3) Remove differential housing cover.

DIFFERENTIAL - TRU-LOK (Continued)

(4) Remove axle shafts.

(5) Remove pressure hose from actuator assembly (Fig. 47).

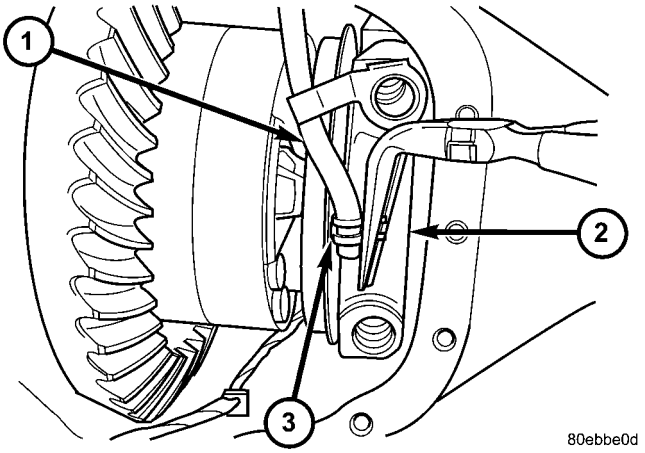


Fig. 47 PRESSURE HOSE

1 - PRESSURE HOSE
2 - BEARING CAP
3 - PRESSURE HOSE CLAMP

(6) Note the reference letters stamped on the bearing caps and housing machined sealing surface (Fig. 48).

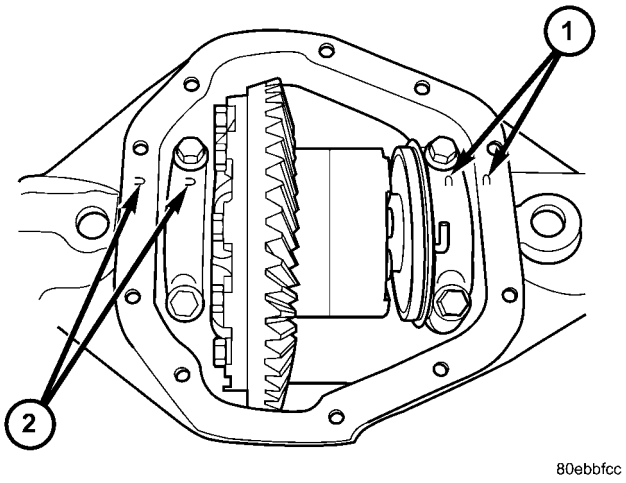


Fig. 48 BEARING CAP REFERENCE MARKS

1 - REFERENCE MARKS
2 - REFERENCE MARKS

(7) Loosen differential bearing cap bolts.

(8) Position Spreader W-129-B with Adapter Kit 6987B on differential locating holes (Fig. 49). Install hold-down clamps and tighten the turnbuckle finger-tight.

(9) Install a Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud (Fig. 50). Load indicator plunger against the opposite side of the housing and zero the indicator.

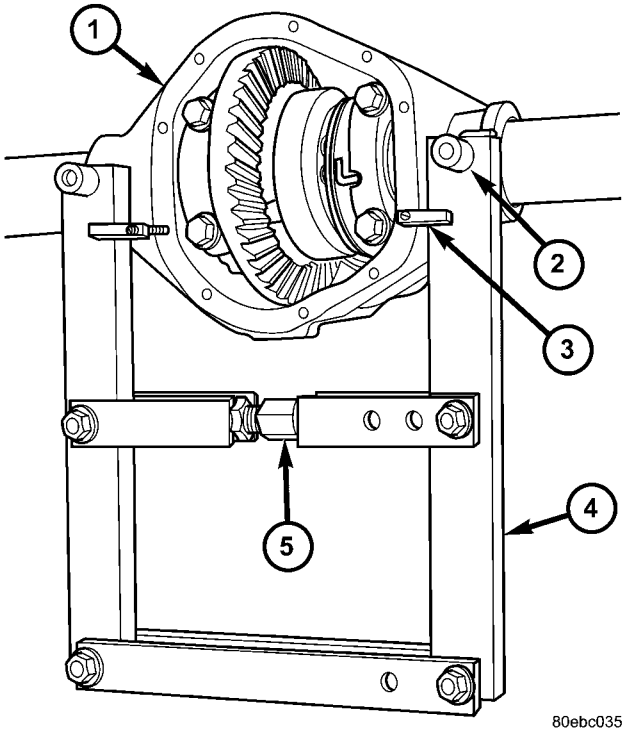


Fig. 49 SPREADER LOCATION

1 - DIFFERENTIAL CASE
2 - ADAPTER
3 - HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

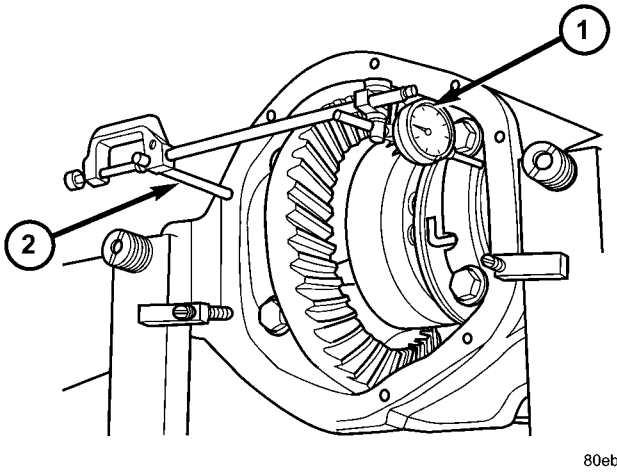


Fig. 50 DIAL INDICATOR LOCATION

1 - DIAL INDICATOR
2 - PILOT STUD

CAUTION: Never spread the housing over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

DIFFERENTIAL - TRU-LOK (Continued)

(10) Spread housing while measuring the distance with the dial indicator (Fig. 51).

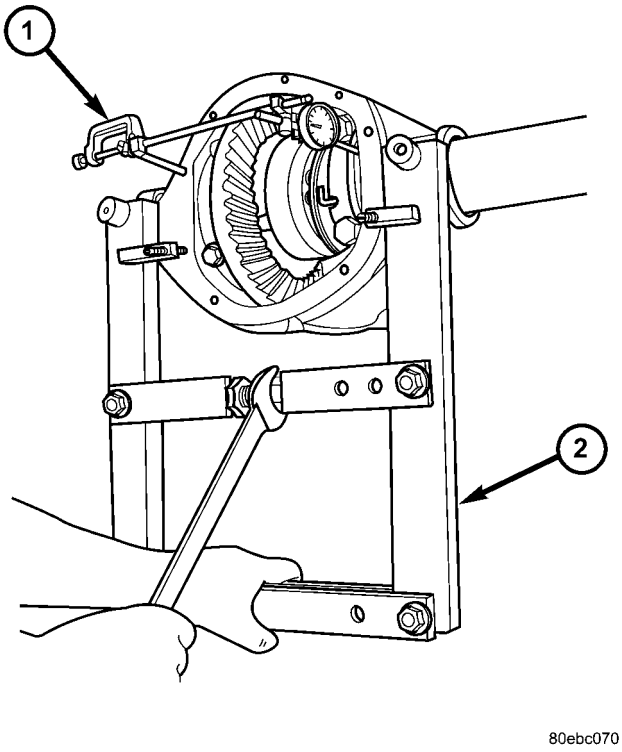


Fig. 51 SPREAD DIFFERENTIAL CASE

1 - DIAL INDICATOR
2 - SPREADER

(11) Remove dial indicator.

(12) Remove differential bearing cap bolts and ring gear side bearing cap.

(13) Remove differential from housing with pinion gear side bearing cap (Fig. 52) and tag differential bearing cups and preload shims to indicate location.

CAUTION: Do not bend actuator mounting tabs, during differential removal. Failure to heed caution may result in damage.

(14) Remove spreader from housing.

(15) Remove indicator switch (Fig. 53) from differential housing.

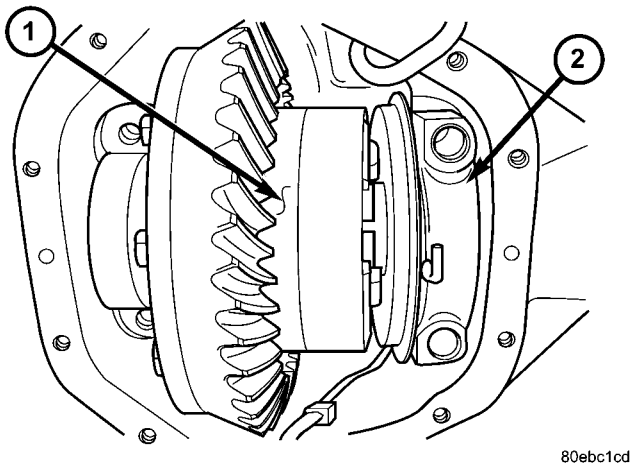


Fig. 52 DIFFERENTIAL

1 - DIFFERENTIAL
2 - BEARING CAP

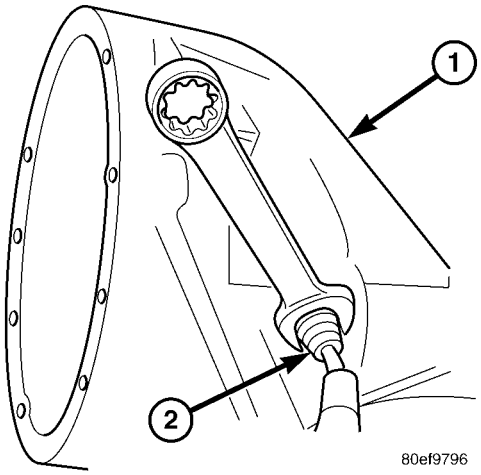


Fig. 53 LOCKER INDICATOR SWITCH

1 - DIFFERENTIAL CASE
2 - SWITCH

(16) Clean the housing cavity with flushing oil, light engine oil or lint free cloth.

NOTE: Do not use water, steam, kerosene or gasoline for cleaning.

DIFFERENTIAL - TRU-LOK (Continued)

DISASSEMBLY

(1) Install Plug C-293-3 into the differential axle shaft hole.

(2) Remove differential case bearings with Puller 6444, Puller Rods 6444-3 and Puller Flange 6444-1. Position puller (Fig. 54) on the differential.

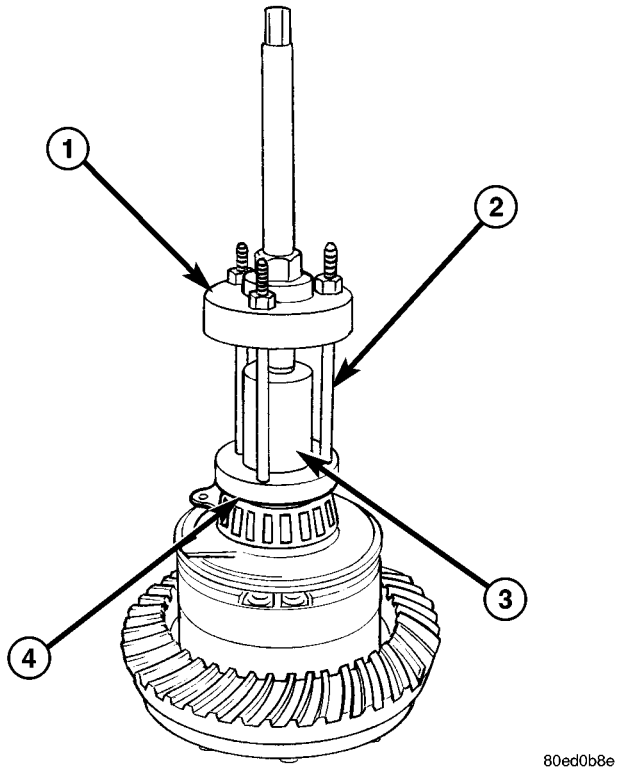


Fig. 54 PULLER AND FLANGE

1 - PULLER
2 - ROD
3 - PLUG
4 - FLANGE

(3) Position Puller Jaws 6444-7 (Fig. 55) around the case bearing and puller flange.

(4) Position Puller Collar 6444-8 (Fig. 56) around the puller jaws.

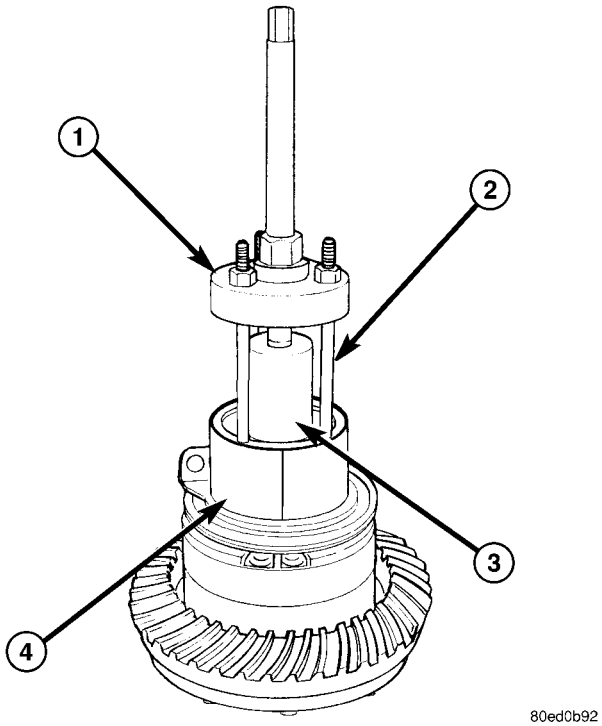


Fig. 55 PULLER AND JAWS

1 - PULLER
2 - ROD
3 - PLUG
4 - JAW

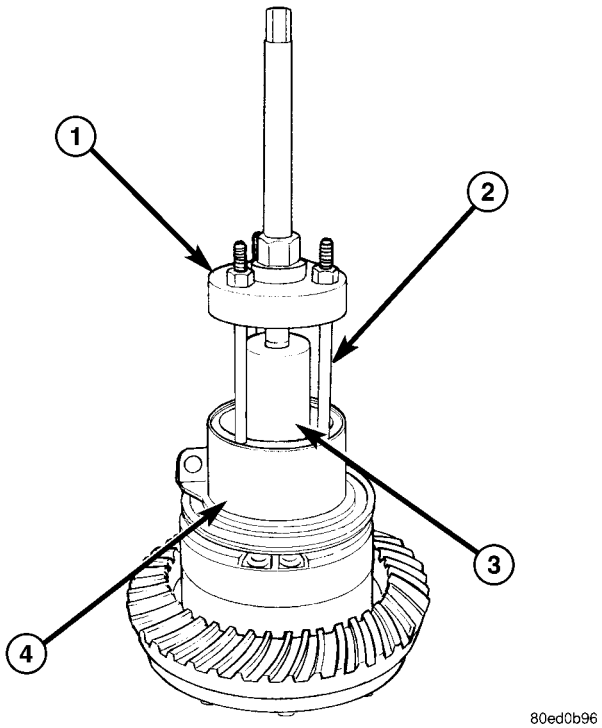


Fig. 56 PULLER AND COLLAR

1 - PULLER
2 - ROD
3 - PLUG
4 - COLLAR

DIFFERENTIAL - TRU-LOK (Continued)

(5) Tighten the puller nut and remove differential case bearing.

(6) Remove actuator (Fig. 57).

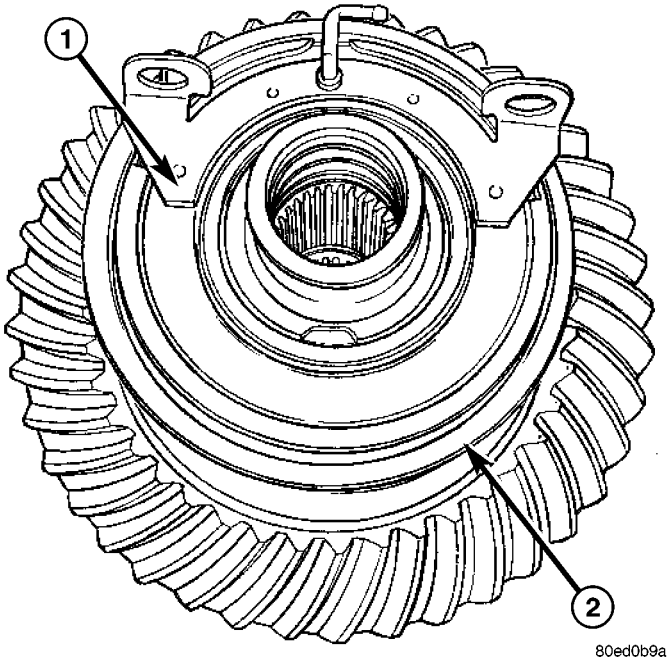


Fig. 57 LOCKING ACTUATOR

1 - ACTUATOR
2 - PRESSURE PLATE

ASSEMBLY

(1) Verify pressure plate tabs (Fig. 58) are seated on the dog clutch slots.

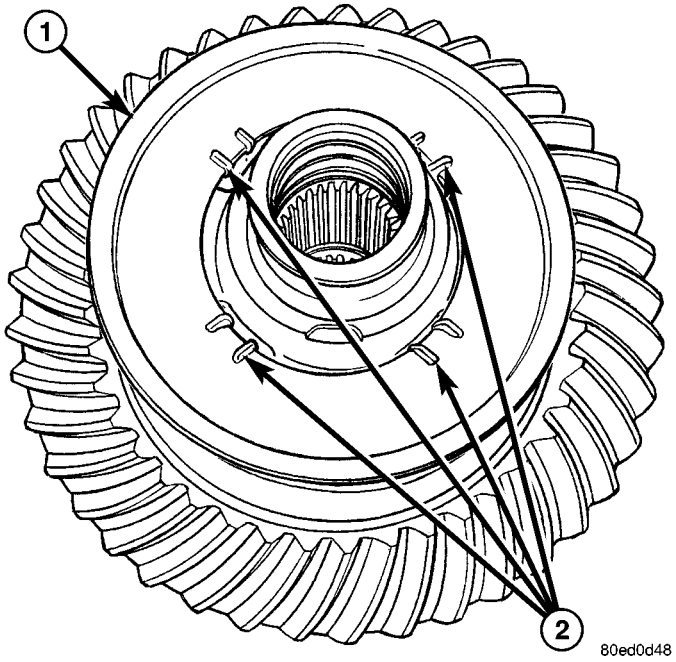


Fig. 58 PRESSURE PLATE

1 - PLATE
2 - PLATE TABS

(2) Place actuator on the pressure plate and case bearing on the case (Fig. 59).

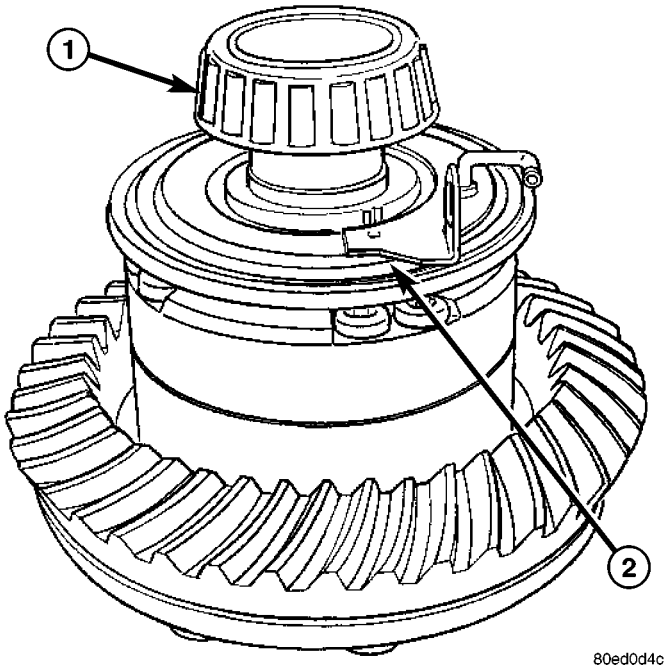


Fig. 59 ACTUATOR AND CASE BEARING

1 - CASE BEARING
2 - ACTUATOR

(3) Install differential case bearings with Installer D-156 and Handle C-4171.

INSTALLATION

NOTE: If replacement differential bearings or differential case are being installed, differential side bearing shim requirements may change. Refer Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Install indicator switch into the housing.

(2) Position Spreader W-129-B and adapters from Adapter set 6987, with the tool dowel pins seated in the locating holes (Fig. 60). Install hold-down clamps and tighten the tool turnbuckle finger-tight.

(3) Install Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load indicator plunger against the opposite side of the housing and zero indicator.

(4) Spread housing while measuring the distance with the dial indicator.

CAUTION: Never spread over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

(5) Remove dial indicator.

(6) Install differential case with pinion gear side bearing cap and differential preload shims in the

DIFFERENTIAL - TRU-LOK (Continued)

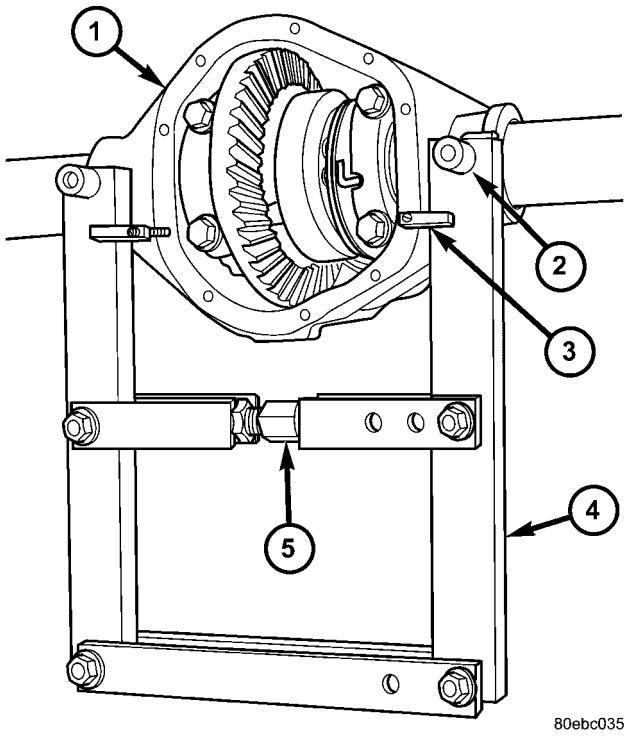


Fig. 60 SPREADER LOCATION

1 - DIFFERENTIAL CASE
2 - ADAPTER
3 - HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

housing. Tap differential case to seat bearings cups in the housing.

CAUTION: Ensure indicator switch plunger head is positioned under the actuator. Failure to heed caution may result in damage.

(7) Install ring gear side bearing cap with reference marks aligned (Fig. 61).

(8) Loosely install differential bearing cap bolts.

(9) Remove axle housing spreader.

(10) Tighten bearing cap bolts to 108 N·m (80 ft. lbs.).

(11) With a 1/4 inch drill bit check the clearance between the actuator and actuator pressure plate at the top and the bottom.

CAUTION: If clearance is not correct, indicator switch plunger may be on top of the actuator or actuator mounting tabs may be bent. Failure to heed caution may result in damage.

(12) Install pressure hose on the actuator assembly.

(13) Install axle shafts.

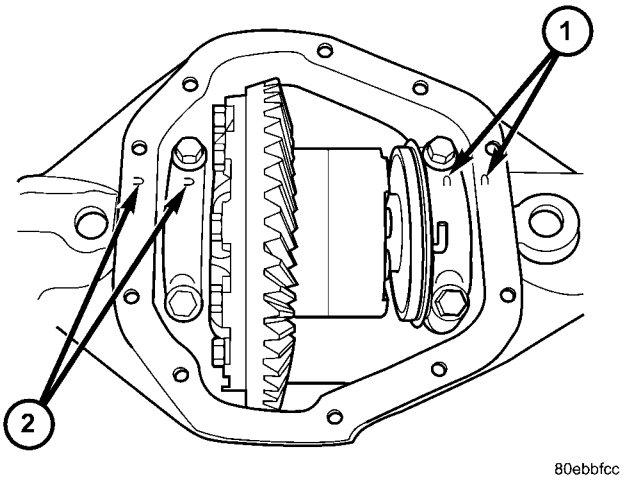


Fig. 61 BEARING CAP REFERENCE MARKS

1 - REFERENCE MARKS
2 - REFERENCE MARKS

(14) Install differential cover, identification tag and indicator switch connector (Fig. 62).

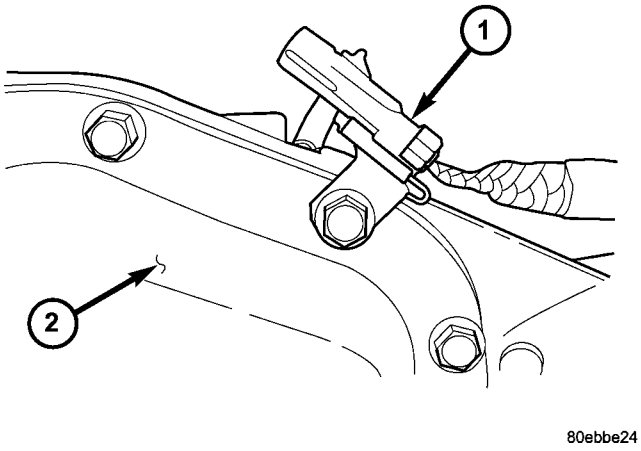


Fig. 62 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

DIFFERENTIAL - TRU-LOK PUMP

REMOVAL

(1) Disconnect pumps and axle harness connectors (Fig. 63).

(2) Remove pressure hoses from the front and rear pumps.

(3) Remove pump mounting bracket bolts (Fig. 64).

(4) Remove pumps and bracket assembly from the vehicle.

(5) Remove pump connectors from the mounting bracket.

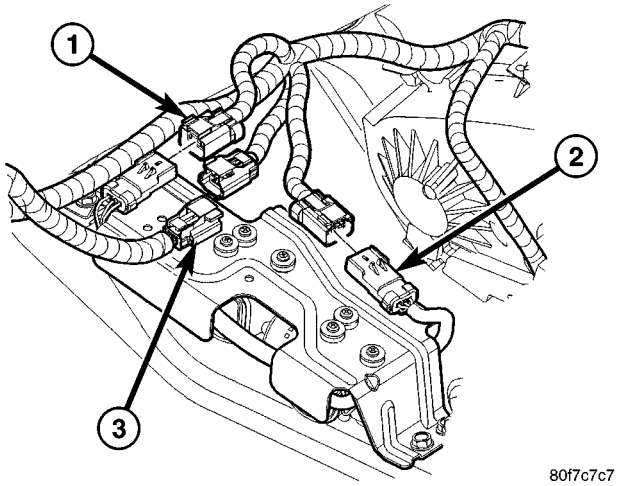


Fig. 63 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

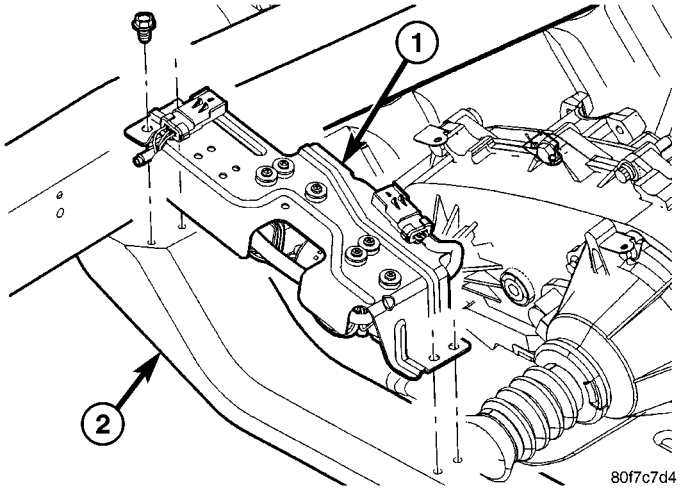


Fig. 64 PUMP BRACKET

1 - PUMP BRACKET
2 - CROSSMEMBER/SKID PLATE

(6) Remove pump mounting screws (Fig. 65) from the bracket and remove pump/pumps.

INSTALLATION

(1) Align pump with screw holes in pump bracket.
(2) Install and tighten pump mounting screws (Fig. 66).
(3) Install pump bracket assembly on the crossmember/skid plate and install pump bracket mounting bolts (Fig. 67).
(4) Tighten pump bracket bolts to 14 N·m (125 in. lbs.).

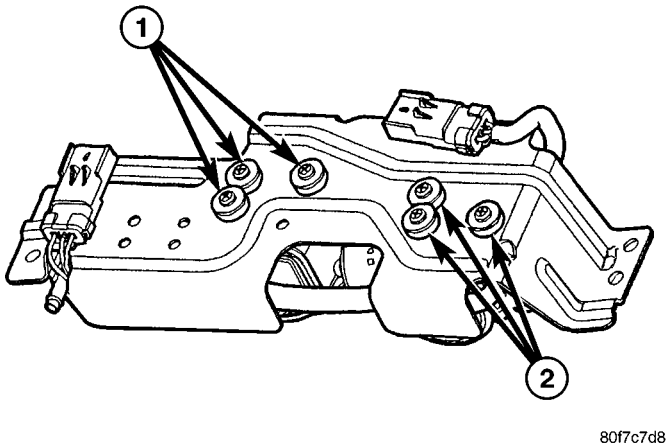


Fig. 65 PUMP SCREWS

1 - REAR PUMP SCREWS
2 - FRONT PUMP SCREWS

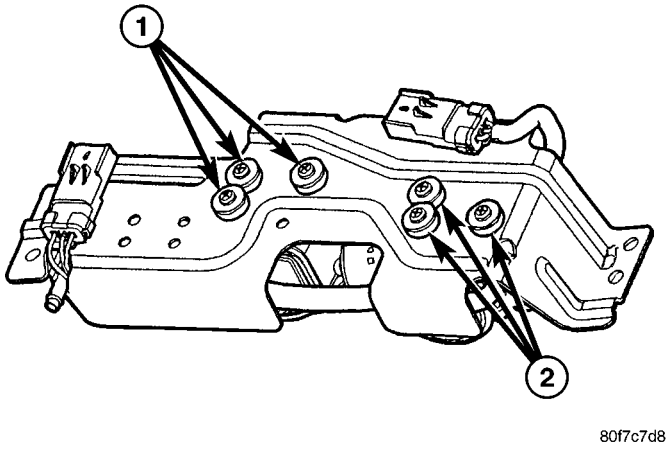


Fig. 66 PUMP SCREWS

1 - REAR PUMP SCREWS
2 - FRONT PUMP SCREWS

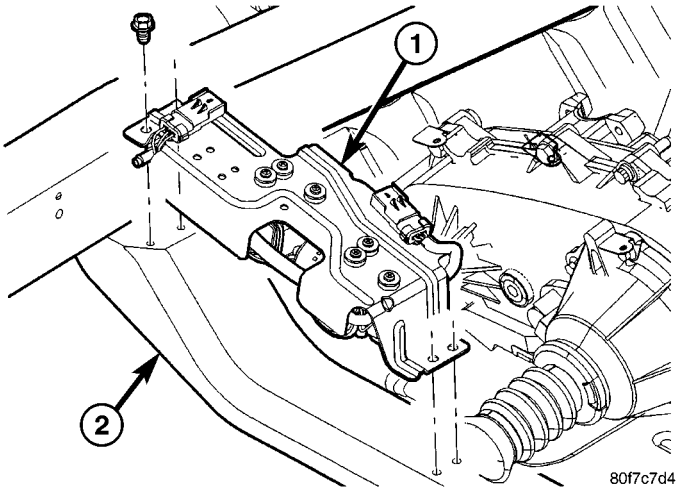


Fig. 67 PUMP BRACKET

1 - PUMP BRACKET
2 - CROSSMEMBER/SKID PLATE

DIFFERENTIAL - TRU-LOK PUMP (Continued)

(5) Connector pump and axle harness connectors (Fig. 68).

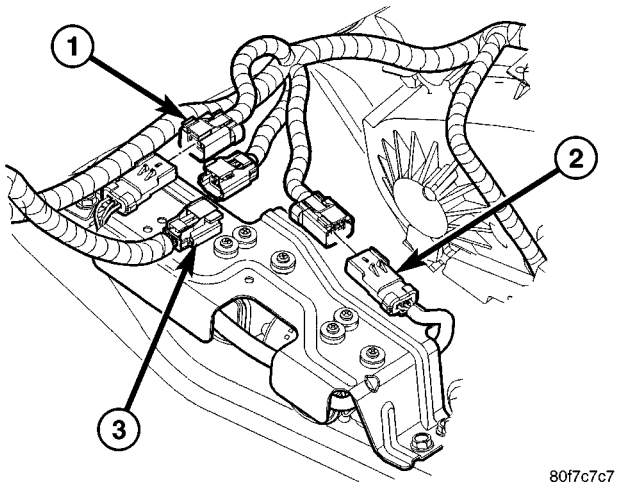


Fig. 68 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

(6) Install pressure hoses on the pumps.
(7) Verify pump operation.

PINION GEAR/RING GEAR

REMOVAL

NOTE: The ring and pinion gears are serviced as a matched set. Never replace one gear without replacing the other gear.

(1) Remove differential from housing.
(2) Secure differential case in a vise with soft metal jaw.
(3) Remove ring gear bolts from the differential case.
(4) Drive ring gear off the differential case with a dead-blow hammer (Fig. 69).
(5) Hold yoke with Holder 6719A and remove pinion nut and washer.
(6) Remove pinion yoke from the pinion shaft with Puller C-452 and Flange Wrench C-3281 (Fig. 70).

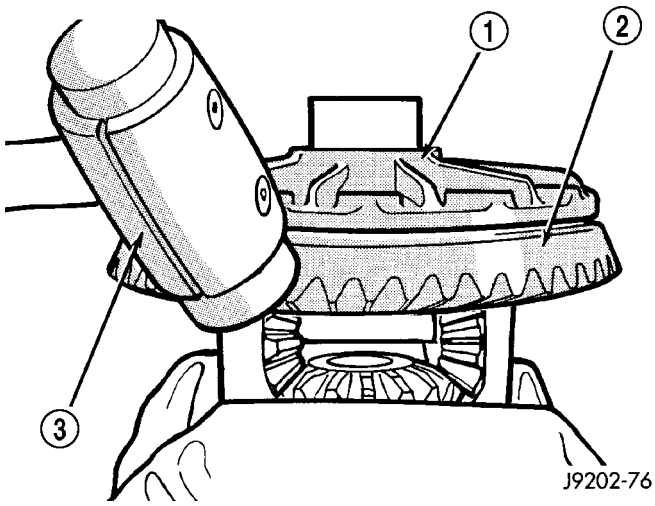


Fig. 69 RING GEAR REMOVAL

1 - CASE
2 - RING GEAR
3- HAMMER

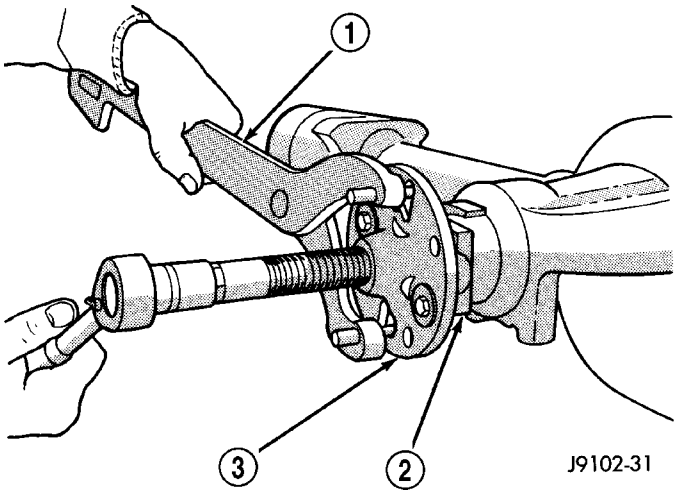


Fig. 70 PINION YOKE REMOVAL

1 - FLANGER WRENCH
2 - YOKE
3 - PULLER

PINION GEAR/RING GEAR (Continued)

(7) Remove pinion gear and preload shims from housing (Fig. 71).

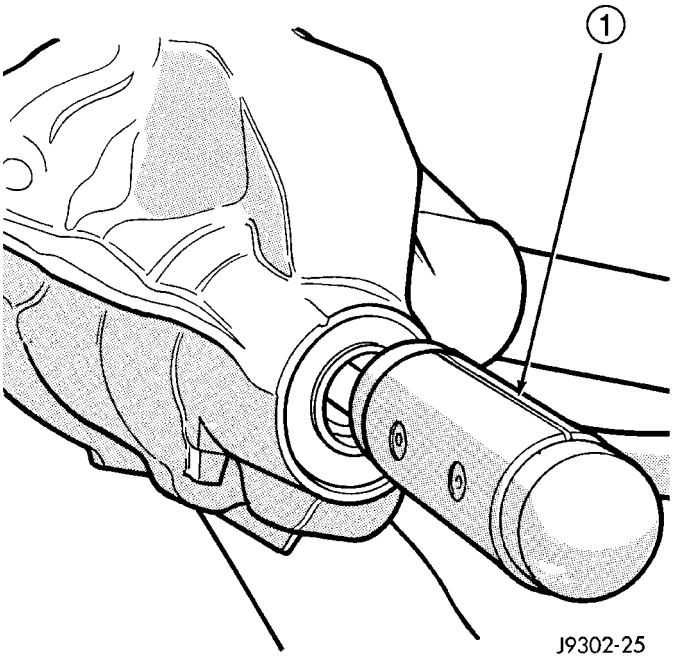


Fig. 71 REMOVE PINION GEAR

1 - DEAD-BLOW HAMMER

(8) Remove pinion seal with a pry bar or slide hammer mounted screw.

(9) Remove front pinion bearing cup with Remover D-147 and Handle C-4171 (Fig. 72).

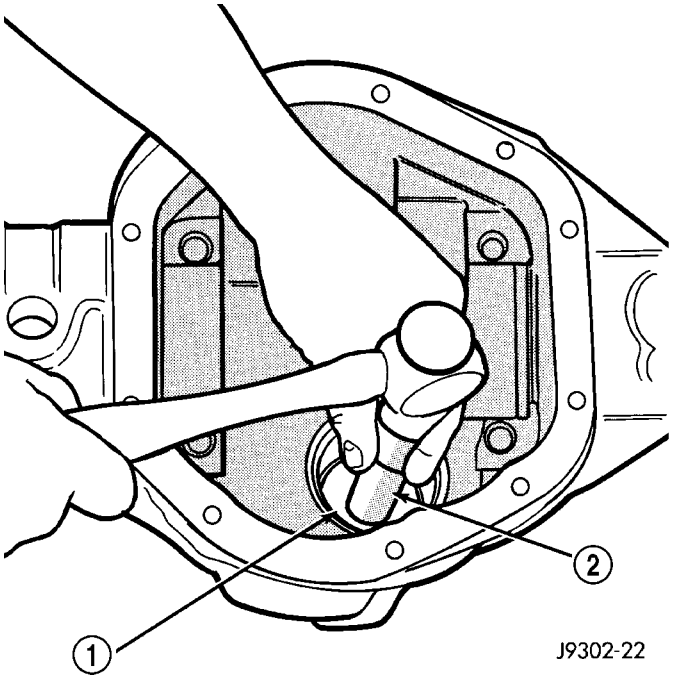


Fig. 72 FRONT BEARING CUP REMOVAL

1 - REMOVER
2 - HANDLE

(10) Remove rear bearing cup (Fig. 73) with Remover D-148 and Handle C-4171.

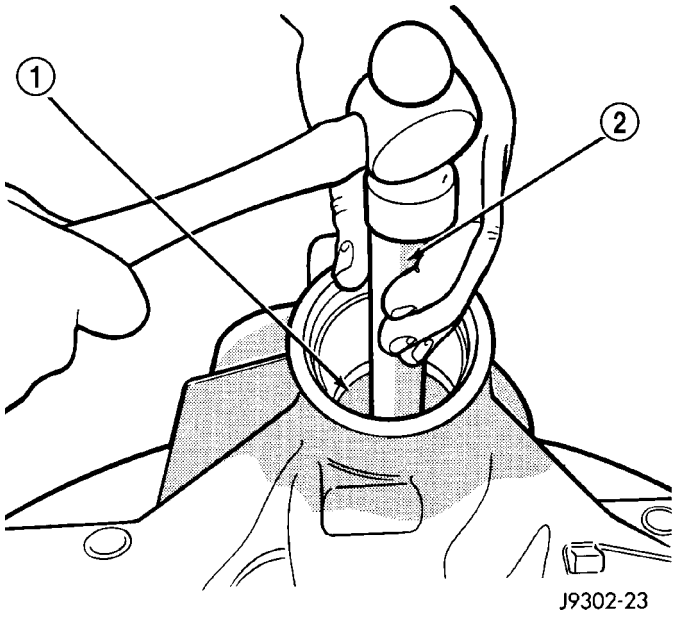


Fig. 73 REAR BEARING CUP REMOVAL

1 - DRIVER
2 - HANDLE

(11) Remove rear pinion bearing with Puller C-293-PA and Adapters C-293-39 (Fig. 74).

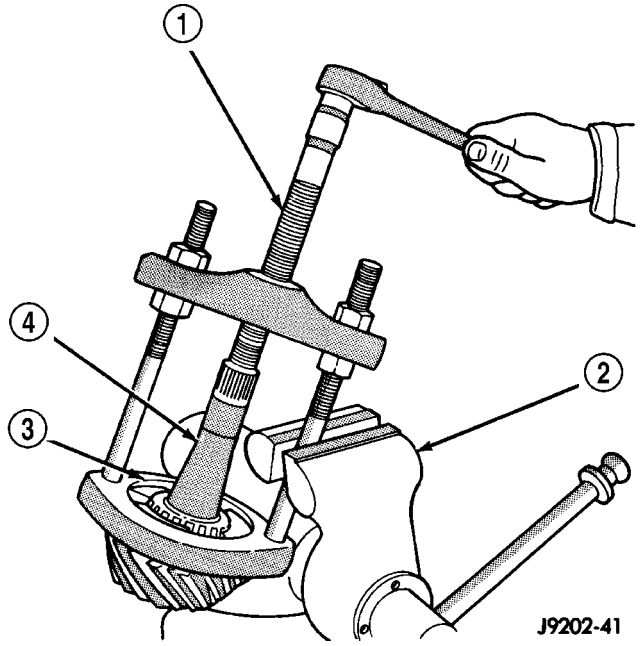


Fig. 74 REAR BEARING REMOVAL

1 - PULLER
2 - VISE
3 - ADAPTERS
4 - PINION SHAFT

PINION GEAR/RING GEAR (Continued)

(12) Remove pinion depth shim/oil slinger from pinion shaft and record thickness.

INSTALLATION

NOTE: Pinion depth shims are placed between the rear pinion bearing cone and pinion gear to achieve proper ring and pinion gear mesh. If ring and pinion gears are reused, the original pinion depth shim can be used. Refer to Adjustments (Pinion Gear Depth) to select the proper shim thickness if ring and pinion gears are replaced.

(1) Apply Mopar Door Ease stick or equivalent lubricant to outside surface of bearing cups.

(2) Install rear pinion bearing cup with Installer D-146 and Handle C-4171 (Fig. 75).

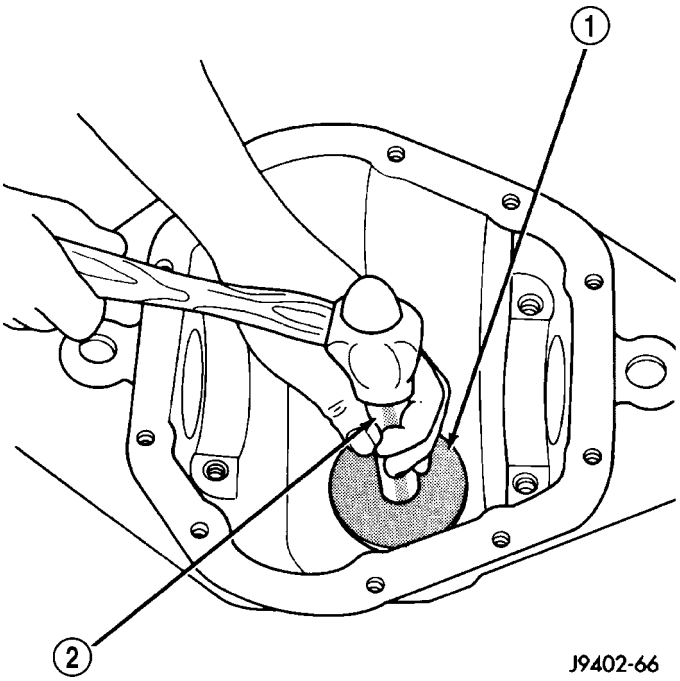


Fig. 75 REAR PINION BEARING CUP

1 - INSTALLER
2 - HANDLE

(3) Install front pinion bearing cup with Installer D-144 and Handle C-4171 (Fig. 76).

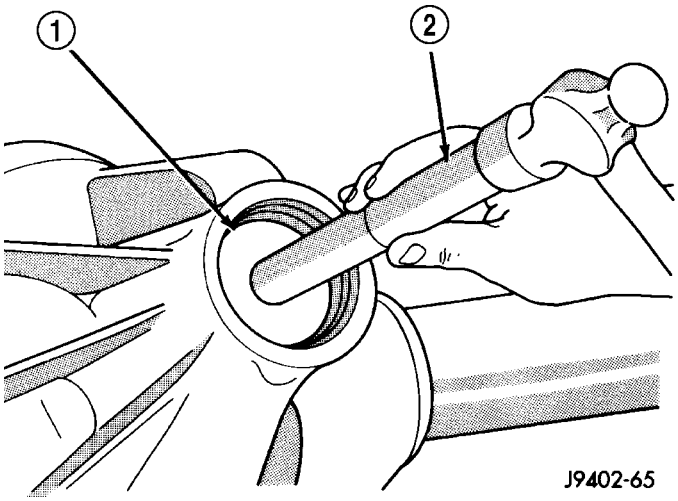


Fig. 76 FRONT PINION BEARING CUP INSTALLER

1 - INSTALLER
2 - HANDLE

(4) Install pinion front bearing, oil slinger. Apply a light coating of gear lubricant on the lip of pinion seal.

(5) Install pinion seal with an appropriate installer (Fig. 77).

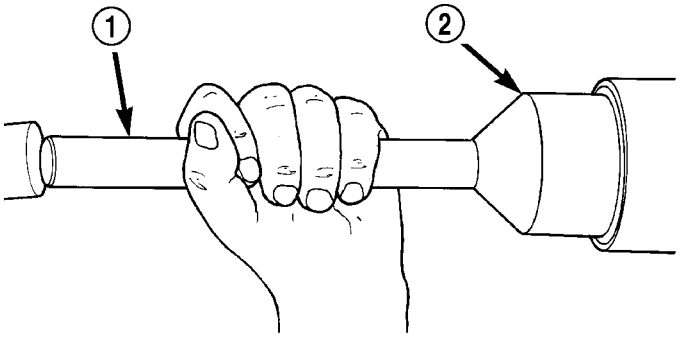


Fig. 77 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(6) Install proper thickness depth shim on the pinion gear.

PINION GEAR/RING GEAR (Continued)

(7) Install rear bearing and oil slinger, if equipped on pinion gear with Installer W-262 (Fig. 78).

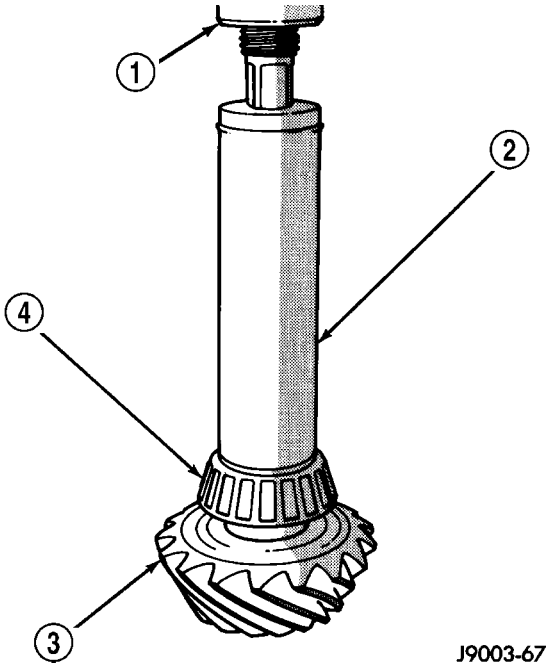


Fig. 78 REAR PINION BEARING

1 - PRESS
2 - INSTALLER
3 - PINION GEAR
4 - PINION BEARING

(8) Install pinion gear in housing.
(9) Install pinion preload shims (Fig. 79)

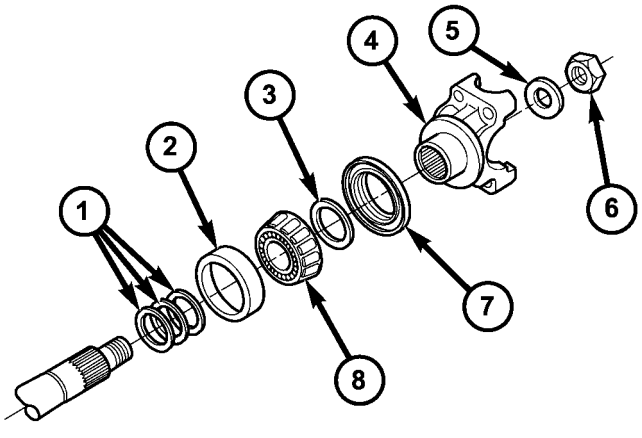


Fig. 79 PINION PRELOAD SHIMS

1 - PRELOAD SHIMS
2 - FRONT BEARING CUP
3 - SLINGER
4 - PINION YOKE
5 - WASHER
6 - PINION NUT
7 - PINION OIL SEAL
8 - FRONT PINION BEARING

(10) Install yoke with Installer W-162-D and Yoke Holder 6719A (Fig. 80).

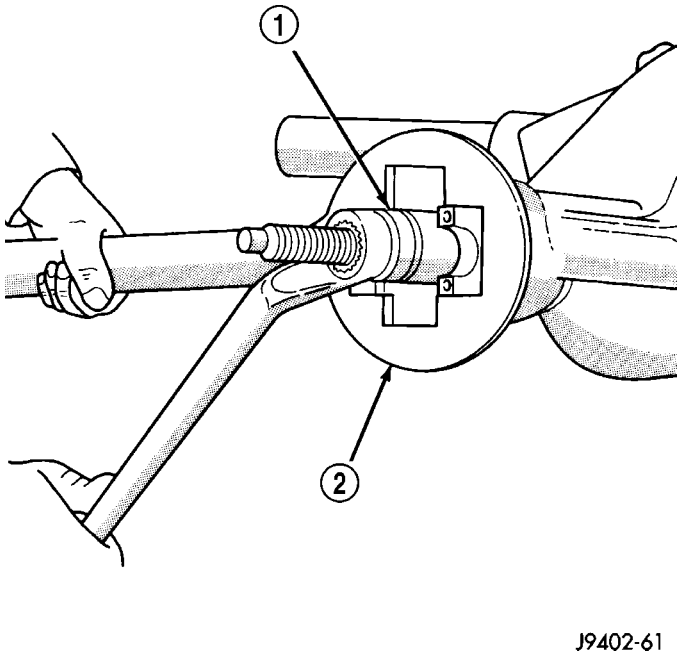


Fig. 80 PINION YOKE INSTALLER

1 - INSTALLER
2 - YOKE HOLDER

(11) Install yoke washer and a **new** nut on the pinion gear. Tighten the nut to specification (Fig. 81).

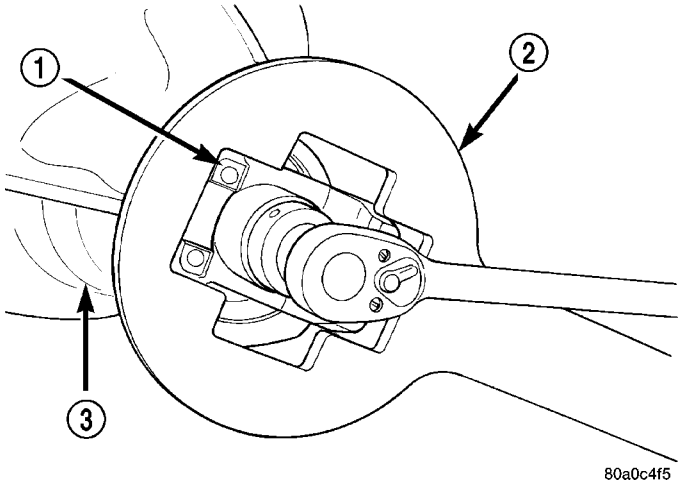


Fig. 81 TIGHTENING PINION NUT

1 - PINION YOKE
2 - YOKE HOLDING
3 - DIFFERENTIAL HOUSING

PINION GEAR/RING GEAR (Continued)

(12) Check bearing preload torque with an inch pound torque wrench (Fig. 82). The torque to rotate the pinion gear should be:

- Original Bearings: 1 to 2 N·m (10 to 20 in. lbs.).
- New Bearings: 2.3 to 4.5 N·m (20 to 40 in. lbs.).

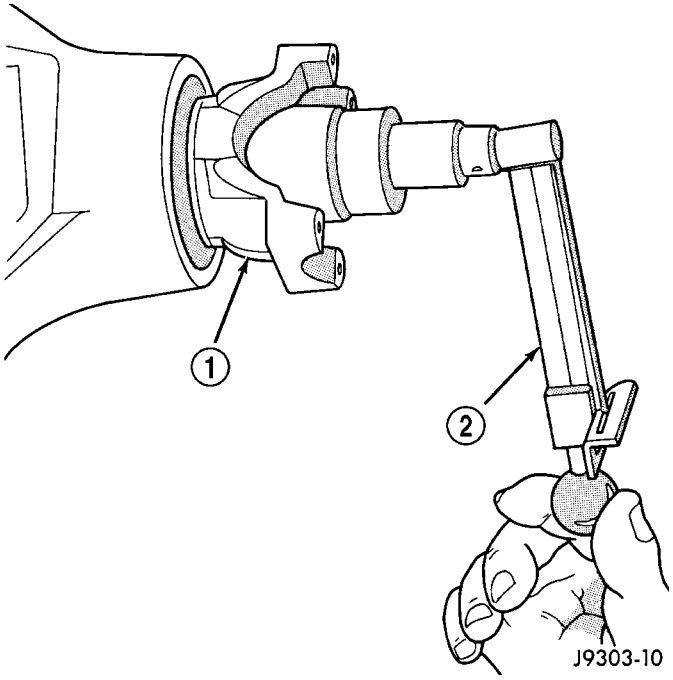


Fig. 82 PINION ROTATION TORQUE

1 - PINION YOKE
2 - INCH POUND TORQUE WRENCH

(13) Invert differential case in a vise and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

(14) Install **new** ring gear bolts and alternately tighten to 136 N·m (100 ft. lbs.). (Fig. 83).

CAUTION: Never reuse the ring gear bolts. Failure to heed caution may result in damage.

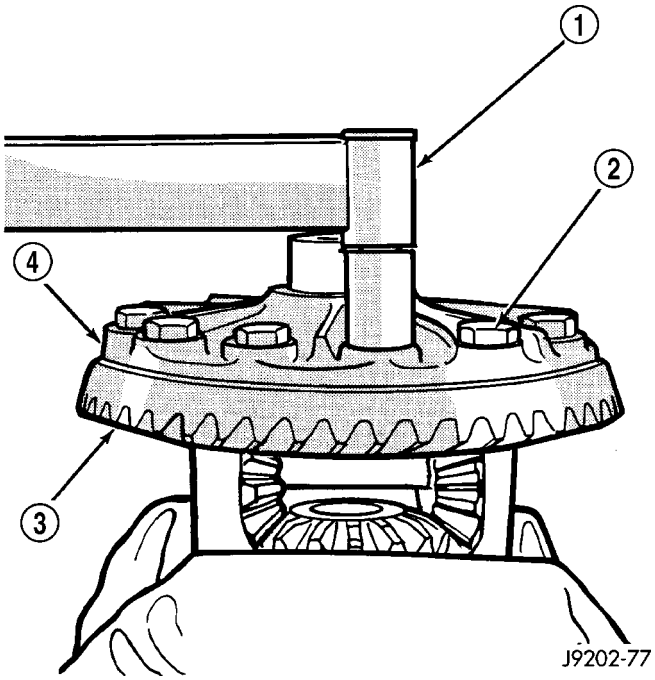


Fig. 83 RING GEAR INSTALLATION

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(15) Install differential in axle housing and verify gear mesh and contact pattern. Refer to Adjustments (Gear Contact Pattern).

REAR AXLE - 194RBI

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REAR AXLE - 194RBI

DIAGNOSIS AND TESTING

GEAR NOISE

Axle gear noise can be caused by insufficient lubricant, incorrect backlash, incorrect pinion depth, tooth contact, worn/damaged gears, or the carrier housing not having the proper offset and squareness.

Gear noise usually happens at a specific speed range. The noise can also occur during a specific type of driving condition. These conditions are acceleration, deceleration, coast, or constant load.

When road testing, first warm-up the axle fluid by driving the vehicle at least 5 miles and then accelerate the vehicle to the speed range where the noise is the greatest. Shift out-of-gear and coast through the peak-noise range. If the noise stops or changes greatly:

- Check for insufficient lubricant.
- Incorrect ring gear backlash.
- Gear damage.

Differential side gears and pinions can be checked by turning the vehicle. They usually do not cause noise during straight-ahead driving when the gears are unloaded. The side gears are loaded during vehicle turns. A worn pinion shaft can also cause a snapping or a knocking noise.

BEARING NOISE

The axle shaft, differential and pinion bearings can all produce noise when worn or damaged. Bearing noise can be either a whining, or a growling sound.

Pinion bearings have a constant-pitch noise. This noise changes only with vehicle speed. Pinion bearing noise will be higher pitched because it rotates at a faster rate. Drive the vehicle and load the differential. If bearing noise occurs, the rear pinion bearing is the source of the noise. If the bearing noise is heard during a coast, the front pinion bearing is the source.

Worn or damaged differential bearings usually produce a low pitch noise. Differential bearing noise is similar to pinion bearing noise. The pitch of differential bearing noise is also constant and varies only with vehicle speed.

Axle shaft bearings produce noise and vibration when worn or damaged. The noise generally changes when the bearings are loaded. Road test the vehicle. Turn the vehicle sharply to the left and to the right. This will load the bearings and change the noise

level. Where axle bearing damage is slight, the noise is usually not noticeable at speeds above 30 mph.

LOW SPEED KNOCK

Low speed knock is generally caused by a worn U-joint or by worn side–gear thrust washers. A worn pinion shaft bore will also cause low speed knock.

VIBRATION

Vibration at the rear of the vehicle is usually caused by a:

- Damaged drive shaft.
- Missing drive shaft balance weight(s).
- Worn or out-of-balance wheels.
- Loose wheel lug nuts.
- Worn U-joint(s).
- Loose/broken springs.
- Damaged axle shaft bearing(s).
- Loose pinion gear nut.
- Excessive pinion yoke run out.
- Bent axle shaft(s).

Check for loose or damaged front-end components or engine/transmission mounts. These components can contribute to what appears to be a rearend vibration. Do not overlook engine accessories, brackets and drive belts.

NOTE: All driveline components should be examined before starting any repair.

DRIVELINE SNAP

A snap or clunk noise when the vehicle is shifted into gear (or the clutch engaged), can be caused by:

- High engine idle speed.
- Transmission shift operation.
- Loose engine/transmission/transfer case mounts.
- Worn U-joints.
- Loose spring mounts.
- Loose pinion gear nut and yoke.
- Excessive ring gear backlash.
- Excessive side gear to case clearance.

The source of a snap or a clunk noise can be determined with the assistance of a helper. Raise the vehicle on a hoist with the wheels free to rotate. Instruct the helper to shift the transmission into gear. Listen for the noise, a mechanics stethoscope is helpful in isolating the source of a noise.

DIAGNOSTIC CHART

Condition	Possible Causes	Correction
Wheel Noise	1. Wheel loose.	1. Tighten loose nuts.
	2. Faulty, brinelled wheel bearing.	2. Replace bearing.
Axle Shaft Noise	1. Misaligned axle tube.	1. Inspect axle tube alignment. Correct as necessary.
	2. Bent or sprung axle shaft.	2. Inspect and correct as necessary.
Axle Shaft Broke	1. Misaligned axle tube.	1. Replace the broken shaft after correcting tube mis-alignment.
	2 Vehicle overloaded.	2. Replace broken shaft and avoid excessive weight on vehicle.
	3. Erratic clutch operation.	3. Replace broken shaft and avoid or correct erratic clutch operation.
	4. Grabbing clutch.	4. Replace broken shaft and inspect and repair clutch as necessary.

REAR AXLE - 194RBI (Continued)

Condition	Possible Causes	Correction
Differential Cracked	1. Improper adjustment of the differential bearings.	1. Replace case and inspect gears and bearings for further damage. Set differential bearing pre-load properly.
	2. Excessive ring gear backlash.	2. Replace case and inspect gears and bearings for further damage. Set ring gear backlash properly.
	3. Vehicle overloaded.	3. Replace case and inspect gears and bearings for further damage. Avoid excessive vehicle weight.
	4. Erratic clutch operation.	4. Replace case and inspect gears and bearings for further damage. Avoid erratic use of clutch.
Differential Gears Scored	1. Insufficient lubrication.	1. Replace scored gears. Fill differential with the correct fluid type and quantity.
	2. Improper grade of lubricant.	2. Replace scored gears. Fill differential with the correct fluid type and quantity.
	3. Excessive spinning of one wheel/tire.	3. Replace scored gears. Inspect all gears, pinion bores, and shaft for damage. Service as necessary.
Loss Of Lubricant	1. Lubricant level too high.	1. Drain lubricant to the correct level.
	2. Worn axle shaft seals.	2. Replace seals.
	3. Cracked differential housing.	3. Repair as necessary.
	4. Worn pinion seal.	4. Replace seal.
	5. Worn/scored yoke.	5. Replace yoke and seal.
	6. Axle cover not properly sealed.	6. Remove, clean, and re-seal cover.
Axle Overheating	1. Lubricant level low.	1. Fill differential to correct level.
	2. Improper grade of lubricant.	2. Fill differential with the correct fluid type and quantity.
	3. Bearing pre-loads too high.	3. Re-adjust bearing pre-loads.
	4. Insufficient ring gear backlash.	4. Re-adjust ring gear backlash.

REAR AXLE - 194RBI (Continued)

Condition	Possible Causes	Correction
Gear Teeth Broke	1. Overloading.	1. Replace gears. Examine other gears and bearings for possible damage.
	2. Erratic clutch operation.	2. Replace gears and examine the remaining parts for damage. Avoid erratic clutch operation.
	3. Ice-spotted pavement.	3. Replace gears and examine remaining parts for damage.
	4. Improper adjustments.	4. Replace gears and examine remaining parts for damage. Ensure ring gear backlash is correct.
Axle Noise	1. Insufficient lubricant.	1. Fill differential with the correct fluid type and quantity.
	2. Improper ring gear and pinion adjustment.	2. Check ring gear and pinion contact pattern. Adjust backlash or pinion depth.
	3. Unmatched ring gear and pinion.	3. Replace gears with a matched ring gear and pinion.
	4. Worn teeth on ring gear and/or pinion.	4. Replace ring gear and pinion.
	5. Loose pinion bearings.	5. Adjust pinion bearing pre-load.
	6. Loose differential bearings.	6. Adjust differential bearing pre-load.
	7. Mis-aligned or sprung ring gear.	7. Measure ring gear run-out. Replace components as necessary.
	8. Loose differential bearing cap bolts.	8. Inspect differential components and replace as necessary. Ensure that the bearing caps are torqued tot he proper specification.
	9. Housing not machined properly.	9. Replace housing.

REAR AXLE - 194RBI (Continued)

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Position a lift device under the axle and secure axle to device.

(3) Remove brake components.

(4) Disconnect the vent hose from the axle shaft tube.

(5) Remove propeller shaft.

(6) Disconnect stabilizer bar links (Fig. 1).

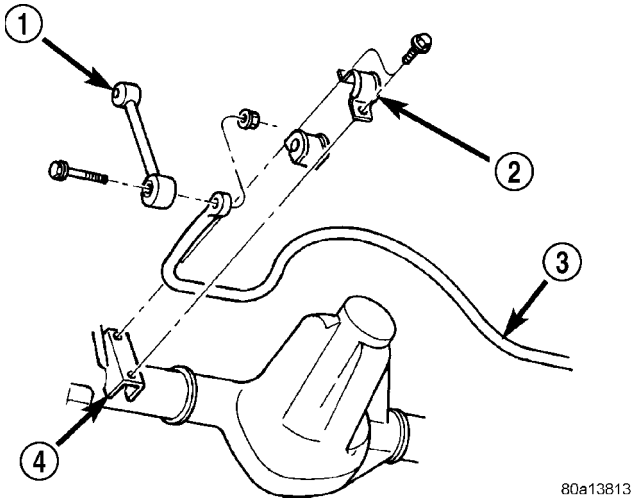


Fig. 1 STABILIZER BAR

1 - LINK
2 - RETAINER
3 - STABILIZER BAR
4 - AXLE MOUNT

(7) Remove shock absorbers from axle bracket (Fig. 2).

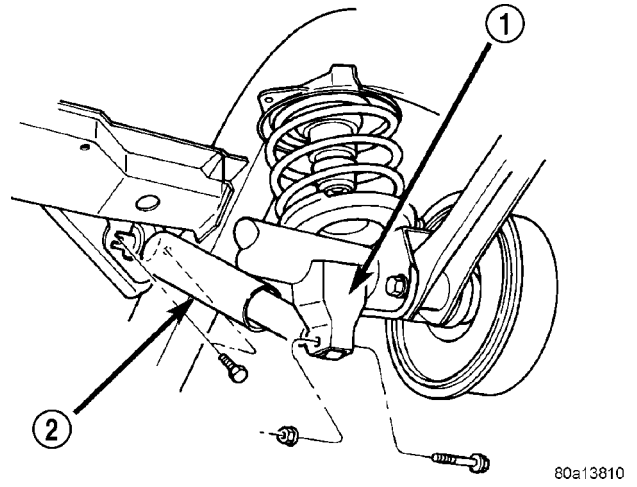


Fig. 2 SHOCK ABSORBER

1 - AXLE BRACKET
2 - SHOCK

(8) Remove track bar from axle bracket (Fig. 3).

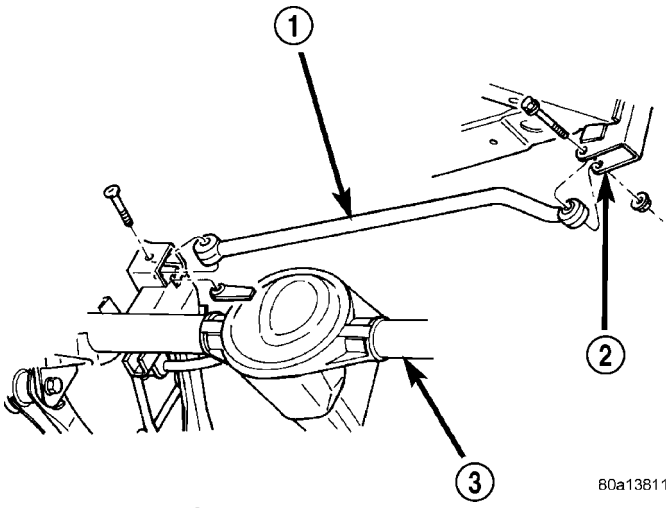


Fig. 3 REAR TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - REAR AXLE

(9) Remove upper and lower control arms from axle brackets.

(10) Lower axle and remove springs.

INSTALLATION

CAUTION: The weight of the vehicle must be supported by the springs before suspension arms and track bar fasteners can be tightened. Failure to heed caution may result in damage.

(1) Raise axle and align the spring centering bolts with mating holes in the axle spring perch.

(2) Install upper and lower control arms in the axle brackets (Fig. 4) and loosely install the mounting bolts.

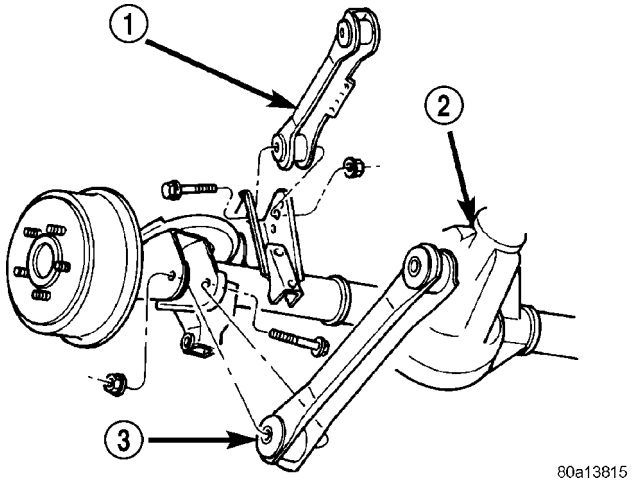


Fig. 4 UPPER AND LOWER SUSPENSION ARMS

1 - UPPER SUSPENSION ARM
2 - REAR AXLE
3 - LOWER SUSPENSION ARM

(3) Install shock absorbers and tighten nuts to torque specification.

(4) Install stabilizer bar links and tighten nuts to torque specification.

(5) Install brake components.

(6) Install axle vent hose.

(7) Install propeller shaft (Fig. 5).

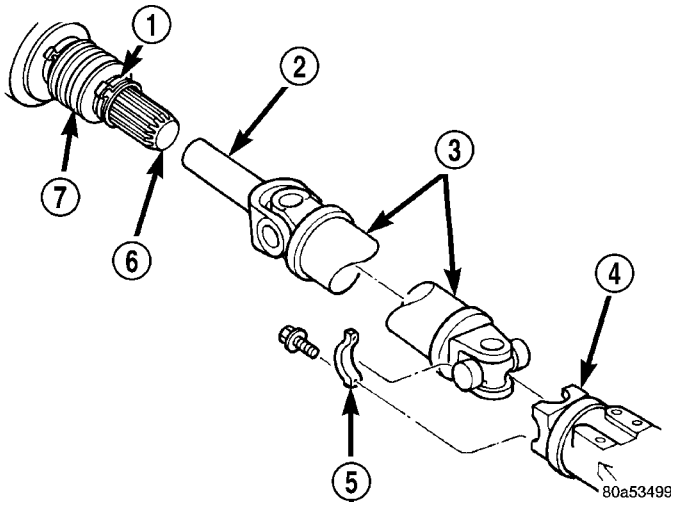


Fig. 5 REAR PROPELLER SHAFT

1 - CLAMP
2 - YOKE
3 - PROPELLER SHAFT
4 - AXLE YOKE
5 - CLAMP
6 - OUTPUT SHAFT
7 - BOOT

(8) Tighten upper and lower control arms nuts to torque specification.

ADJUSTMENTS

RING AND PINION GEAR

Ring and pinion gears are supplied as matched sets only. The identifying numbers for the ring and pinion gear are etched into the face of each gear (Fig. 6). A plus (+) number, minus (–) number or zero (0) is etched into the face of the pinion gear. This number is the amount (in thousandths of an inch) the depth varies from the standard depth setting of a pinion etched with a (0). The standard depth provides the best teeth contact pattern. Refer to Backlash and Contact Pattern Analysis Paragraph in this section for additional information.

Compensation for pinion depth variance is achieved with select shims. The shims are placed under the inner pinion bearing cone (Fig. 7).

If a new gear set is being installed, note the depth variance etched into both the original and replacement pinion gear. Add or subtract the thickness of the original depth shims to compensate for the difference in the depth variances. Refer to the Depth Variance charts.

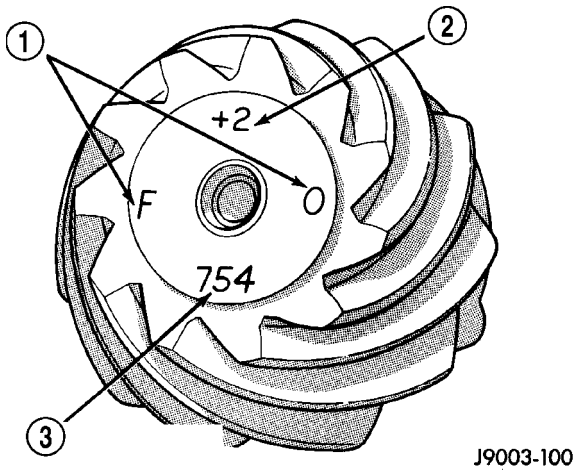


Fig. 6 PINION GEAR ID NUMBERS

1 - PRODUCTION NUMBERS
2 - DRIVE PINION GEAR DEPTH VARIANCE
3 - GEAR MATCHING NUMBER

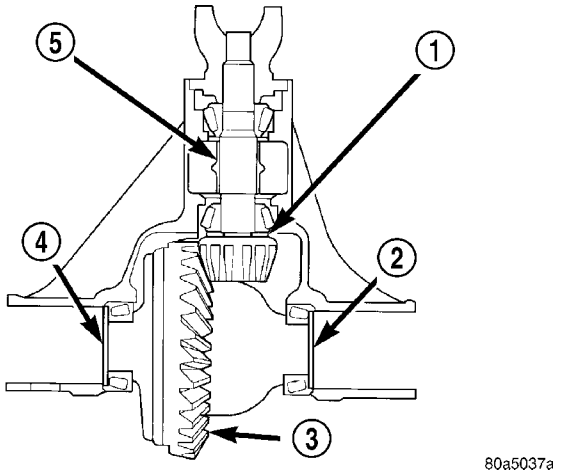


Fig. 7 ADJUSTMENT SHIM LOCATIONS

1 - PINION GEAR DEPTH SHIM
2 - DIFFERENTIAL SHIM-PINION GEAR SIDE
3 - RING GEAR
4 - DIFFERENTIAL SHIM-RING GEAR SIDE
5 - COLLAPSIBLE SPACER

Note where Old and New Pinion Marking columns intersect. Intersecting figure represents plus or minus amount needed.

Note the etched number on the face of the drive pinion gear (–1, –2, 0, +1, +2, etc.). The numbers represent thousands of an inch deviation from the standard. If the number is negative, add that value to the required thickness of the depth shim(s). If the number is positive, subtract that value from the thickness of the depth shim(s). If the number is 0 no change is necessary. Refer to the Pinion Gear Depth Variance Chart.

REAR AXLE - 194RBI (Continued)

PINION GEAR DEPTH VARIANCE

Original Pinion Gear Depth Variance	Replacement Pinion Gear Depth Variance								
	−4	−3	−2	−1	0	+1	+2	+3	+4
+4	+0.008	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0
+3	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001
+2	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002
+1	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003
0	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004
−1	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005
−2	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006
−3	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007
−4	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007	−0.008

PINION DEPTH MEASUREMENT

Measurements are taken with pinion cups and pinion bearings installed in the housing. Take measurements with a Pinion Gauge Set, Pinion Block 6735, Arbor Discs 6732 and Dial Indicator C-3339 (Fig. 8).

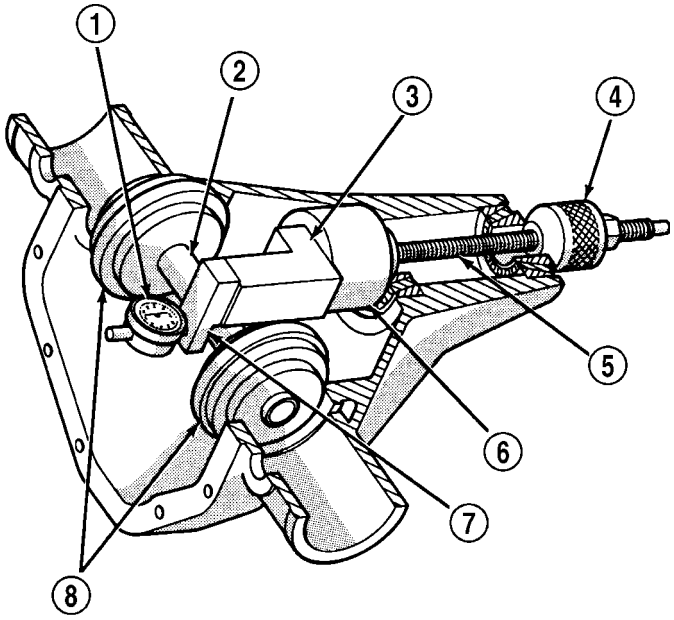


Fig. 8 PINION GEAR DEPTH TOOLS

1 - DIAL INDICATOR
2 - ARBOR
3 - PINION HEIGHT BLOCK
4 - CONE
5 - SCREW
6 - PINION BLOCK
7 - SCOOTER BLOCK
8 - ARBOR DISC

(1) Assemble Pinion Height Block 6739, Pinion Block 6735 and rear pinion bearing onto Screw 6741 (Fig. 8).

(2) Insert assembled height gauge components, rear bearing and screw into the housing through pinion bearing cups (Fig. 9).

(3) Install front pinion bearing and Cone 6740 hand tight.

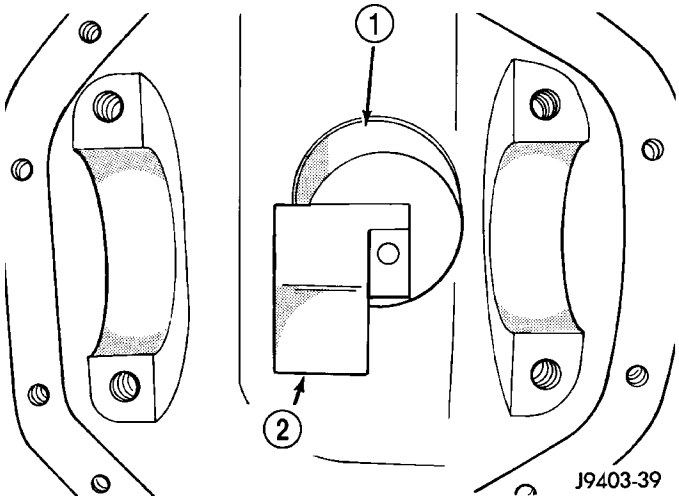


Fig. 9 PINION HEIGHT BLOCK

1 - PINION BLOCK
2 - PINION HEIGHT BLOCK

REAR AXLE - 194RBI (Continued)

(4) Place Arbor Disc 6732 on Arbor D-115-3 in position in the housing side bearing cradles (Fig. 10). Install differential bearing caps on Arbor Discs and tighten cap bolts. Refer to the Torque Specifications in this section.

NOTE: Arbor Discs 6732 have different step diameters to fit other axle sizes. Pick correct size step for axle being serviced.

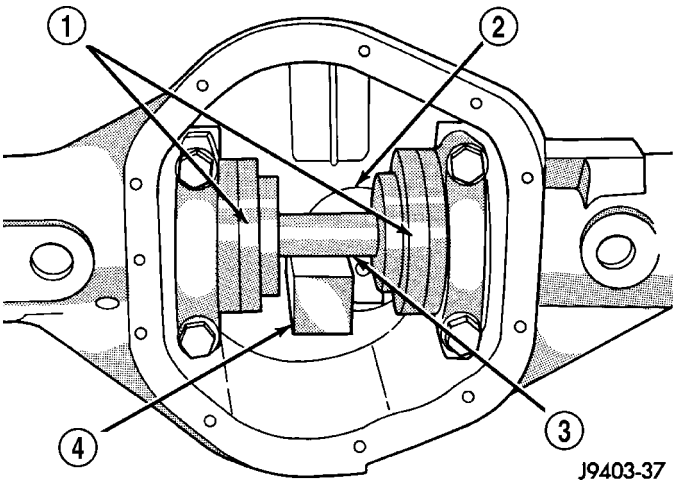


Fig. 10 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(5) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(6) Position Scooter Block/Dial Indicator flush on the pinion height block. Hold scooter block and zero the dial indicator.

(7) Slowly slide the scooter block across the pinion height block over to the arbor (Fig. 11). Move the scooter block till the dial indicator probe crests the arbor, then record the highest reading.

(8) Select a shim equal to the dial indicator reading plus the drive pinion gear depth variance number etched in the face of the pinion gear (Fig. 6) using the opposite sign on the variance number. For example, if the depth variance is –2, add +0.002 in. to the dial indicator reading.

(9) Remove the pinion depth gauge components from the housing

DIFFERENTIAL BEARING PRELOAD

Differential side bearing preload and gear backlash is achieved by selective shims inserted between the bearing cup and the housing. The proper shim thickness can be determined using slip-fit Dummy Bearings D-348 in place of the differential side bearings and a Dial Indicator C-3339. Before proceeding with the differential bearing preload and gear backlash

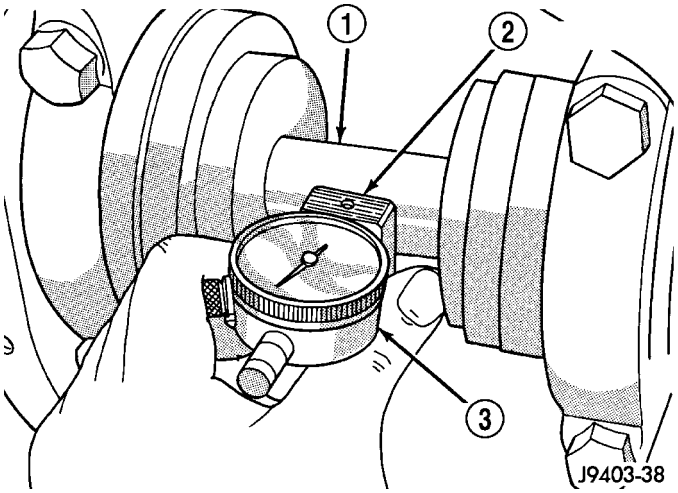


Fig. 11 PINION GEAR DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

measurements, measure the pinion gear depth and prepare the pinion gear for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion gear is installed and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading, starting point shim thickness and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 12).

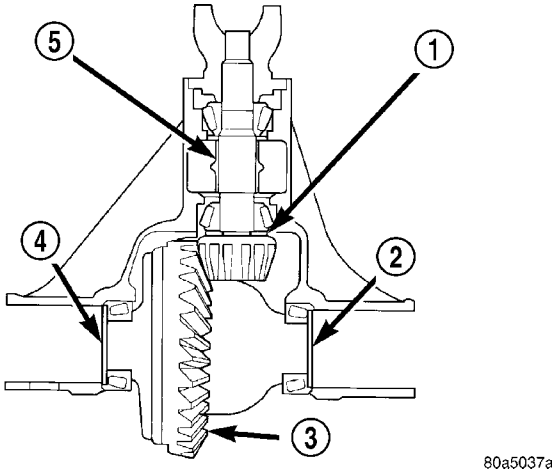


Fig. 12 SHIM LOCATIONS

1 - PINION GEAR DEPTH SHIM
2 - DIFFERENTIAL SHIM-PINION GEAR SIDE
3 - RING GEAR
4 - DIFFERENTIAL SHIM-RING GEAR SIDE
5 - COLLAPSIBLE SPACER

REAR AXLE - 194RBI (Continued)

PRELOAD SHIM SELECTION

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove side bearings from differential case.
(2) Install ring gear, if necessary, on differential case and tighten bolts to specification.
(3) Install Dummy Bearings D-348 on differential case.
(4) Install differential case in the housing.
(5) Insert Dummy Shims 8107 (3.0 mm / 0.118 in.) starting point shims between the dummy bearing and the housing (Fig. 13).

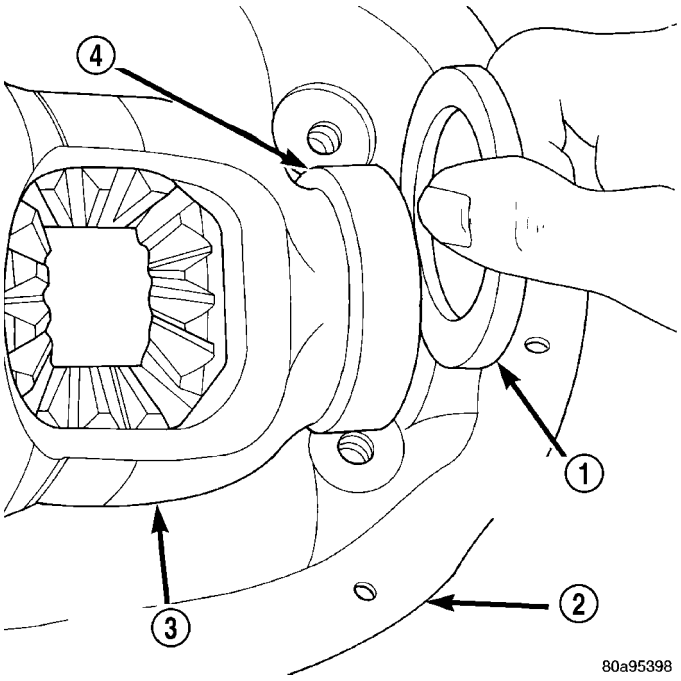


Fig. 13 DUMMY SHIM LOCATION

1 - DUMMY SHIM
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE
4 - DUMMY BEARING

(6) Install bearing caps in their correct positions and snug the bolts.
(7) Using a dead-blow hammer to seat the differential dummy bearings to each side of the housing (Fig. 14) and (Fig. 15).

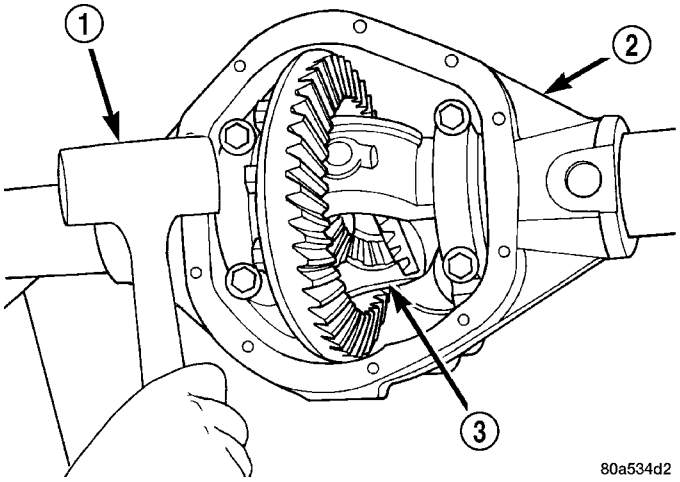


Fig. 14 SEAT DUMMY BEARING PINION GEAR SIDE

1 - MALLET
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

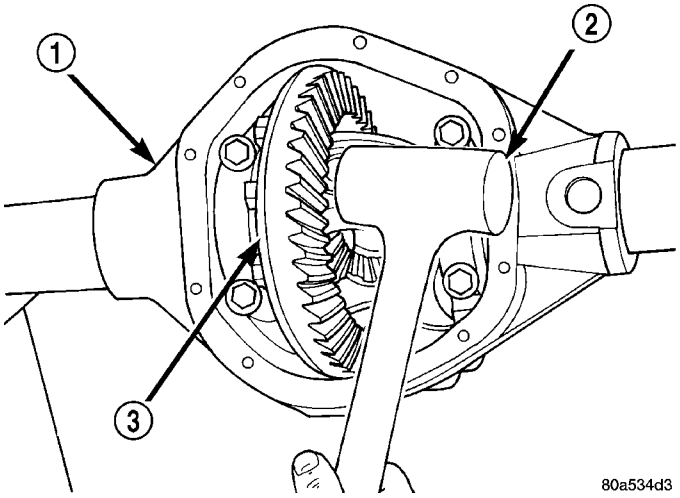


Fig. 15 SEAT DUMMY BEARING RING GEAR SIDE

1 - DIFFERENTIAL HOUSING
2 - MALLET
3 - DIFFERENTIAL CASE

REAR AXLE - 194RBI (Continued)

(8) Thread Pilot Stud C-3288-B into rear cover bolt hole below ring gear (Fig. 16).

(9) Attach Dial Indicator C-3339 to the pilot stud and position indicator plunger on a flat surface of the ring gear bolt head (Fig. 16).

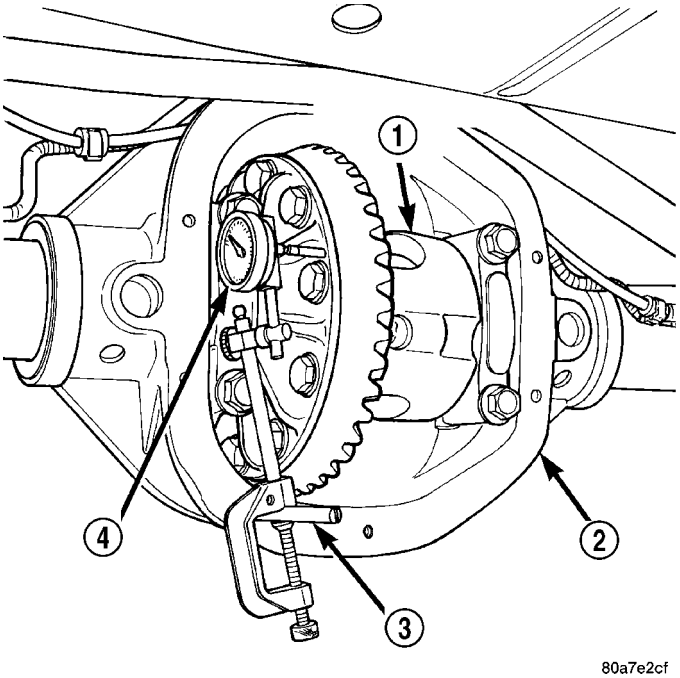


Fig. 16 DIFFERENTIAL SIDE PLAY MEASUREMENT

1 - DIFFERENTIAL CASE
2 - DIFFERENTIAL HOUSING
3 - PILOT STUD
4 - DIAL INDICATOR

(10) Push differential case to the pinion gear side of the housing (Fig. 17) and zero dial indicator.

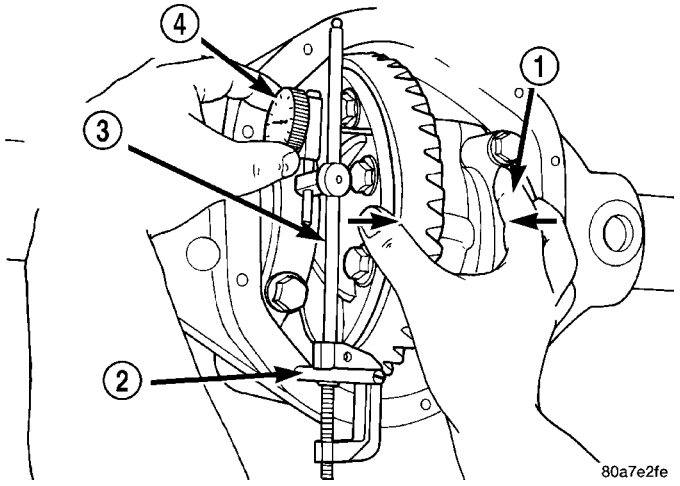


Fig. 17 ZERO DIAL INDICATOR

1 - DIFFERENTIAL CASE TO PINION GEAR SIDE
2 - PILOT STUD
3 - DIAL INDICATOR EXTENSION
4 - ZERO DIAL INDICATOR FACE

(11) Push differential case to the ring gear side and record dial indicator reading (Fig. 18).

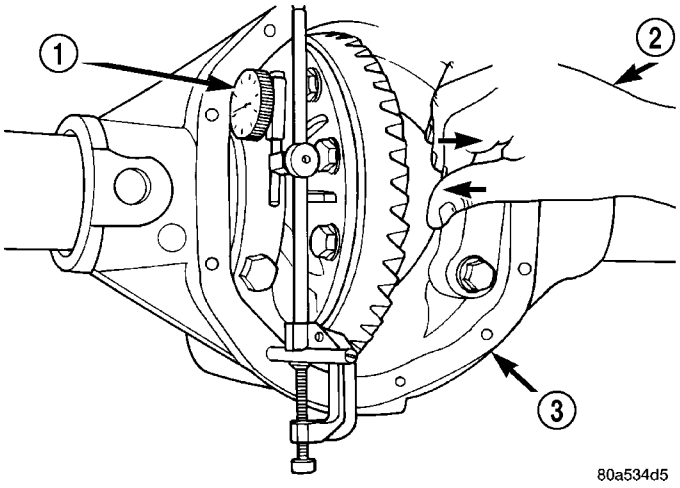


Fig. 18 RECORD DIAL INDICATOR READING

1 - READ DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(12) Add the dial indicator reading to the starting point shim thickness to determine total shim thickness to achieve zero differential end play.

(13) Add 0.008 in. (0.2 mm) to the zero end play total. This new total represents the thickness of shims to compress or preload the new bearings when the differential is installed.

(14) Rotate dial indicator out of the way.

(15) Remove differential case, dummy bearings and starting point shims from the housing.

(16) Install pinion gear in the housing. Install the yoke and establish the correct pinion rotating torque.

(17) Install differential case and dummy bearings in the housing (without shims) and tighten retaining cap bolts.

(18) Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 16).

(19) Push and hold differential case toward pinion gear.

(20) Zero dial indicator face to pointer.

(21) Push and hold differential case to ring gear side of the housing and record dial indicator reading.

(22) Subtract 0.002 in. (0.05 mm) from the dial indicator reading to compensate for backlash between ring and pinion gears. This total is the thickness of shim required to achieve proper backlash.

(23) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

(24) Rotate dial indicator out of the way on pilot stud.

(25) Remove differential case and dummy bearings from the housing.

REAR AXLE - 194RBI (Continued)

(26) Install new side bearing cones and cups on differential case.

(27) Install spreader W-129-B and some components of Adapter Set 6987 on differential housing and spread axle opening enough to receive differential case.

(28) Place side bearing shims into the housing against the differential bearing bore.

(29) Install differential case in the housing.

(30) Rotate the differential case several times to seat the side bearings.

(31) Position the indicator plunger against a ring gear tooth (Fig. 19).

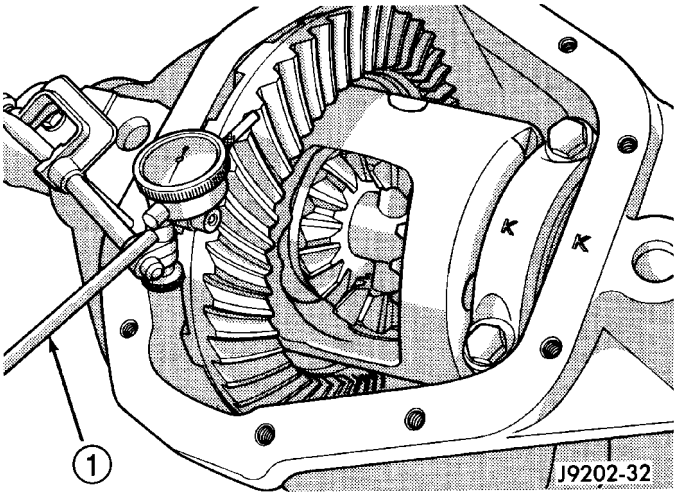


Fig. 19 RING GEAR BACKLASH

1 - DIAL INDICATOR

(32) Push and hold ring gear upward while not allowing the pinion gear to rotate.

(33) Zero dial indicator face to pointer.

(34) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the differential housing to the other (Fig. 20).

(35) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform the Gear Contact Pattern Analysis procedure.

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted

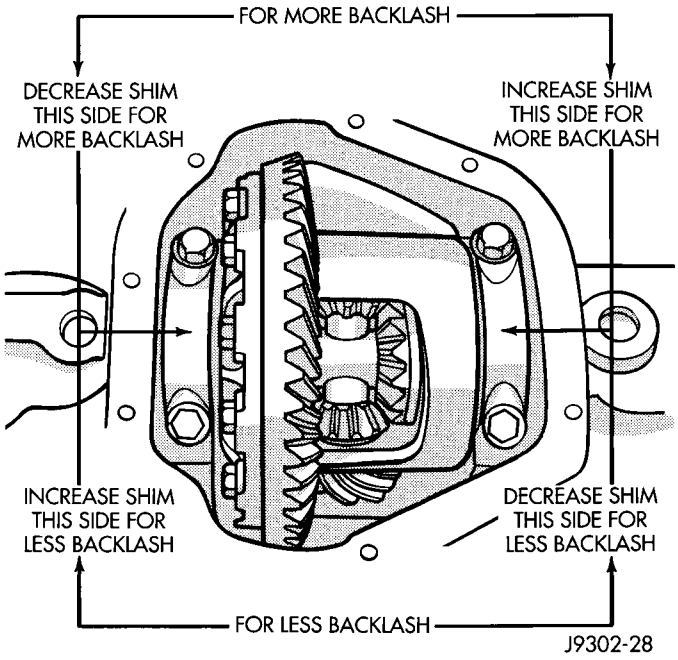


Fig. 20 BACKLASH SHIM ADJUSTMENT

within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth at the outer-end. The ROOT of the gear tooth is the lowest portion of the tooth (Fig. 21).

NOTE: If the PROFILE across the tooth is the same it is a 3 Axis cut gear. If the PROFILE across the tooth is tapered it is a 2 Axis cut gear.

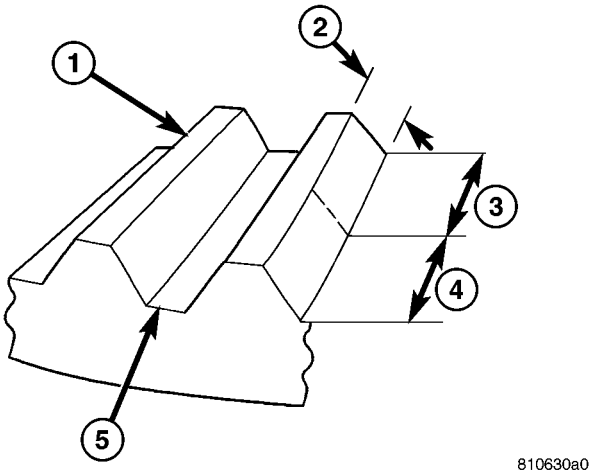


Fig. 21 GEAR DESCRIPTION

1 - TOP LAND
2 - PROFILE
3 - TOE
4 - HEEL
5 - ROOT

REAR AXLE - 194RBI (Continued)

(1) Apply a thin coat of hydrated ferric oxide or equivalent to the drive and coast side of the ring gear teeth.

(2) Wrap, twist, and hold a shop towel around the pinion yoke to increase the turning resistance of the pinion. This will provide a more distinct contact pattern.

(3) With a boxed end wrench on a ring gear bolt, rotate the differential case one complete revolution in both directions while a load is being applied from shop towel.

The areas on the ring gear teeth with the greatest degree of contact against the pinion teeth will squeeze the compound to the areas with the least amount of contact. Note and compare patterns on the ring gear teeth to Gear Tooth Contact Patterns charts (Fig. 22) and (Fig. 23) and adjust pinion depth and gear backlash as necessary.

DIFFERENTIAL BEARING PRELOAD CHECK

The final check on the differential assembly before installing the axles is torque to rotate pinion and differential combined. This will verify the correct differential bearing preload.

Torque to rotate the differential and pinion should be the torque to rotate the pinion plus 0.79-1.24 N·m (7-11 in. lbs.).

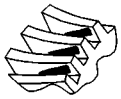
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top heel contact		**Top toe contact**	Backlash correct. Thicker pinion position shim required.
	Root toe contact		**Root heel contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top heel contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root toe contact**	Pinion position shim correct. Increase backlash.

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Fig. 22 PATTERN INTERPRETATION (GEAR CUT 2 AXIS)

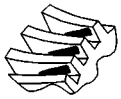
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top toe contact		**Top heel contact**	Backlash correct. Thicker pinion position shim required.
	Root heel contact		**Root toe contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top toe contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root heel contact**	Pinion position shim correct. Increase backlash.

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Fig. 23 PATTERN INTERPRETATION (GEAR CUT 3 AXIS)

SPECIFICATIONS

REAR AXLE

AXLE SPECIFICATIONS

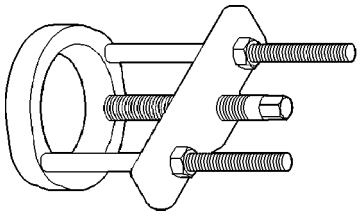
DESCRIPTION	SPECIFICATION
Axle Ratio	3.07, 3.73, 4.10, 4.56
Ring Gear Backlash	0.12-0.20 mm (0.005-0.008 in.)
Pinion Bearing Preload - Original Bearings	1-2 N·m (10-20 in. lbs.)
Pinion Bearing Preload - New Bearings	2-4 N·m (20-35 in. lbs.)

TORQUE SPECIFICATIONS

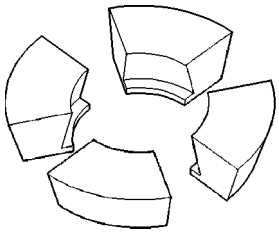
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Differential Cover Bolts	41	30	-
Bearing Cap Bolts	77	57	-
Ring Gear Bolts	136	100	-
Pinion Nut Min / Max	271-475	200-350	-
Pinion Mate Shaft Screw	16.25	12	-

REAR AXLE - 194RBI (Continued)

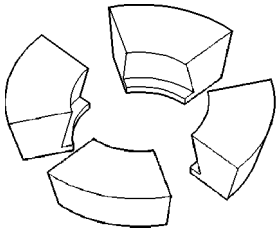
SPECIAL TOOLS



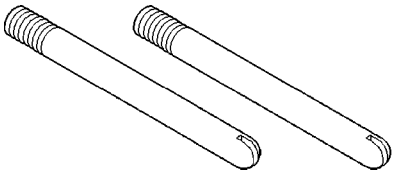
PULLER C-293-PA



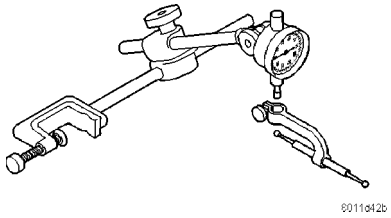
ADAPTER C-293-39



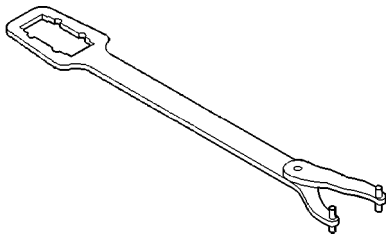
ADAPTER C-293-40



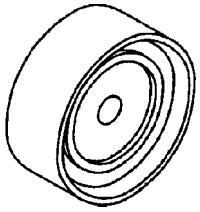
PILOT STUDS C-3288-B



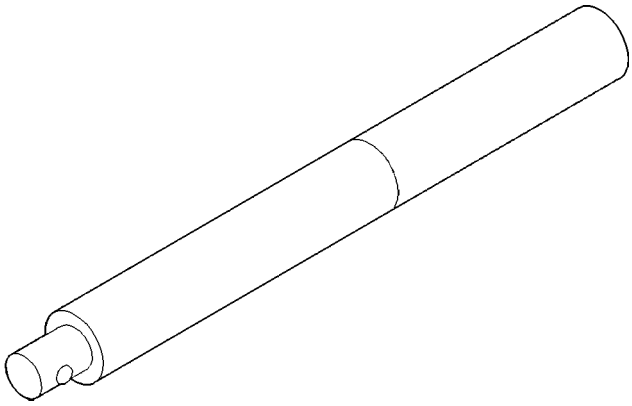
DIAL INDICATOR C-3339



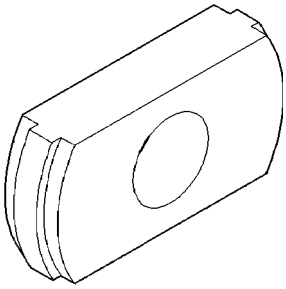
WRENCH C-3281



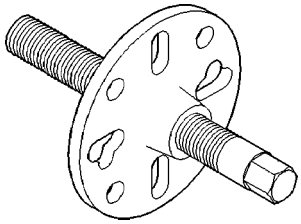
INSTALLER C-3716-A



HANDLE C-4171

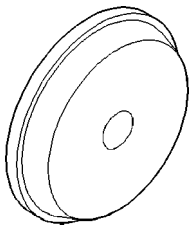


REMOVER C-4345

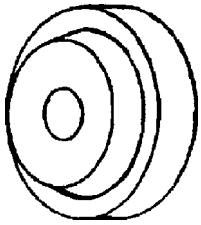


PULLER C-452

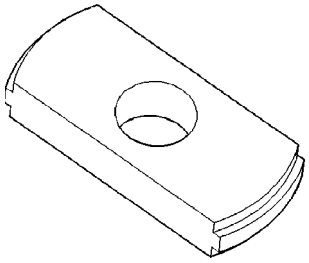
REAR AXLE - 194RBI (Continued)



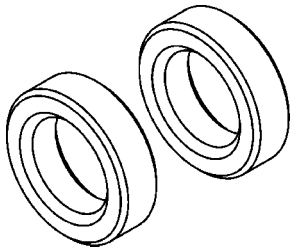
INSTALLER D-130



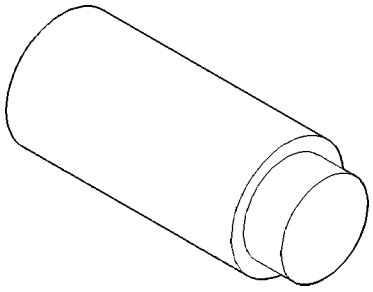
INSTALLER D-146



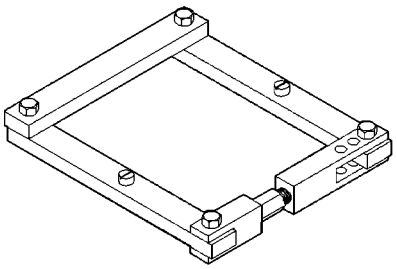
REMOVER D-149



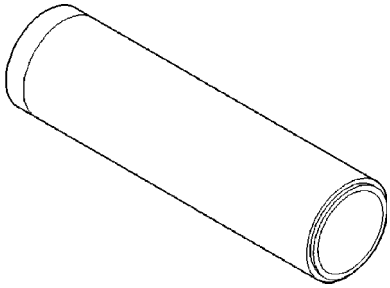
DUMMY BEARINGS D-348



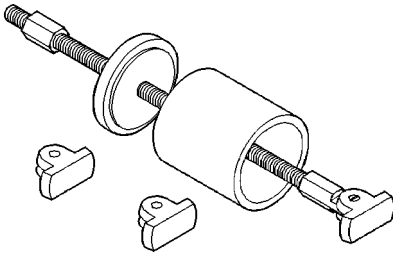
ADAPTER PLUG SP-3289



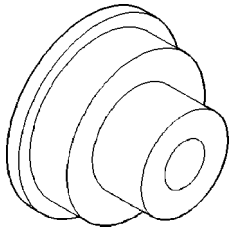
SPREADER W-129-B



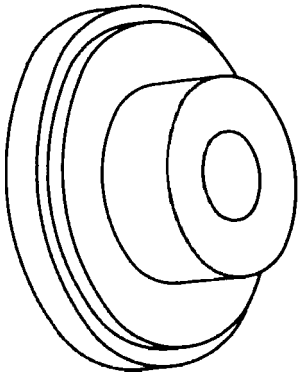
INSTALLER W-262



BEARING REMOVER 6310

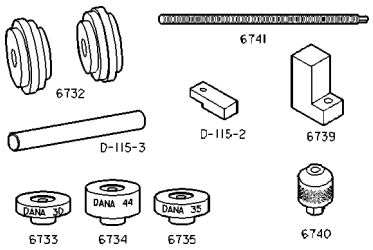


INSTALLER 6436

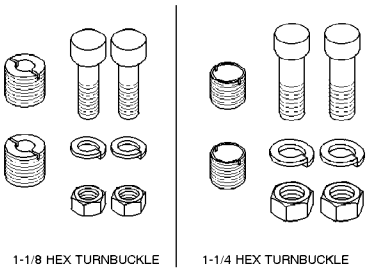


INSTALLER 6437

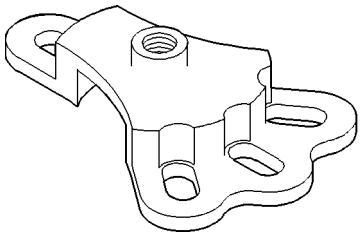
REAR AXLE - 194RBI (Continued)



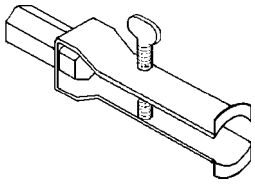
PINION DEPTH 6774



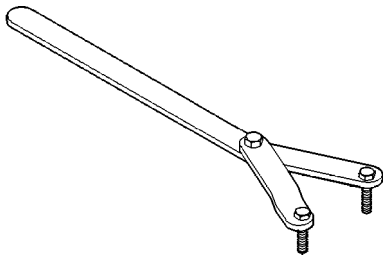
ADAPTER KIT 6987



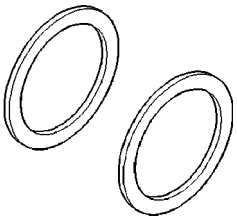
HUB PULLER 6790



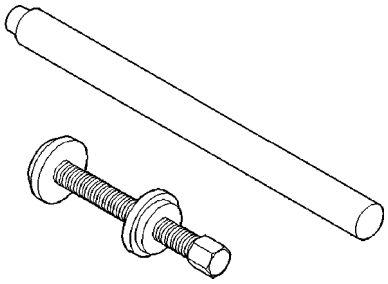
PULLER 7794-A



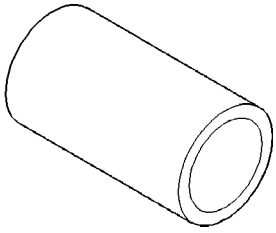
WRENCH SPANNER 6958



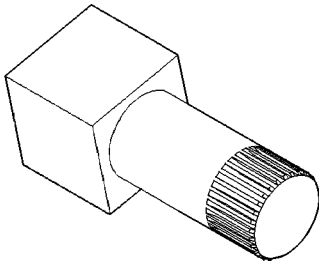
SHIM DUMMY 8107



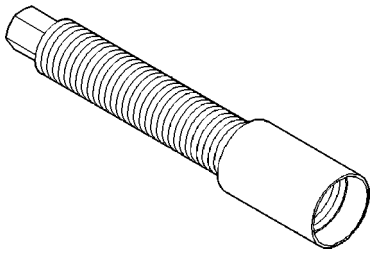
TRAC-LOK TOOLS 6960



CUP 8109



HOLDER FIXTURE 6965



INSTALLER SCREW 8112

AXLE SHAFTS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake drum.
(3) Remove differential cover and drain fluid.
(4) Rotate differential case to access pinion mate gear shaft lock screw. Remove lock screw and shaft from differential case (Fig. 24).

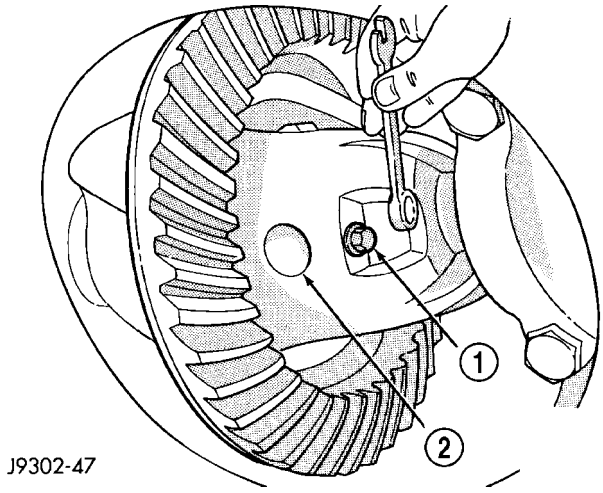


Fig. 24 MATE SHAFT LOCK SCREW

1 - LOCK SCREW
2 - PINION GEAR MATE SHAFT

(5) Push axle shaft inward and remove axle C-clip lock (Fig. 25).

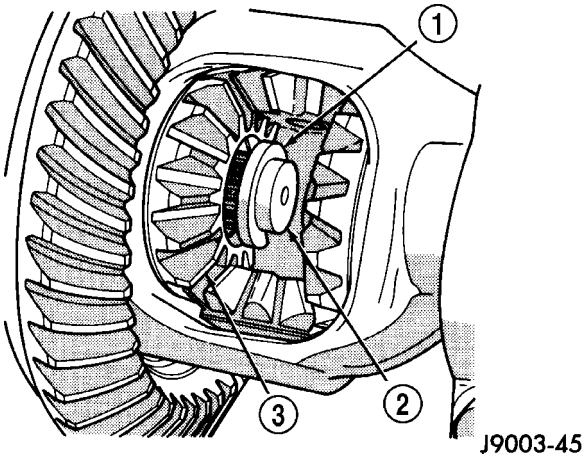


Fig. 25 AXLE SHAFT C-CLIP

1 - C-CLIP LOCK
2 - AXLE SHAFT
3 - SIDE GEAR

(6) Remove axle shaft.

INSTALLATION

(1) Lubricate bearing bore and seal lip with gear lubricant. Insert axle shaft through seal, bearing, and engage it into side gear splines.
(2) Install C-clip lock on the axle shaft, then push axle outward to seat C-clip lock in side gear.
(3) Insert pinion mate shaft into differential case and through thrust washers and pinion gears.
(4) Align hole in shaft with hole in the differential case and install lock screw with Loctite® on the threads. Tighten lock screw to 19 N·m (14 ft. lbs.).
(5) Install differential cover.

AXLE SHAFT SEALS

REMOVAL

(1) Remove the axle shaft.
(2) Remove axle shaft seal from the end of the axle shaft tube with a small pry bar.
(3) Inspect the axle shaft tube bore for roughness and burrs and remove as necessary.

INSTALLATION

(1) Wipe the axle shaft tube bore clean.
(2) Install **new** axle shaft seal with Installer 6437 and Handle C-4171 (Fig. 26).

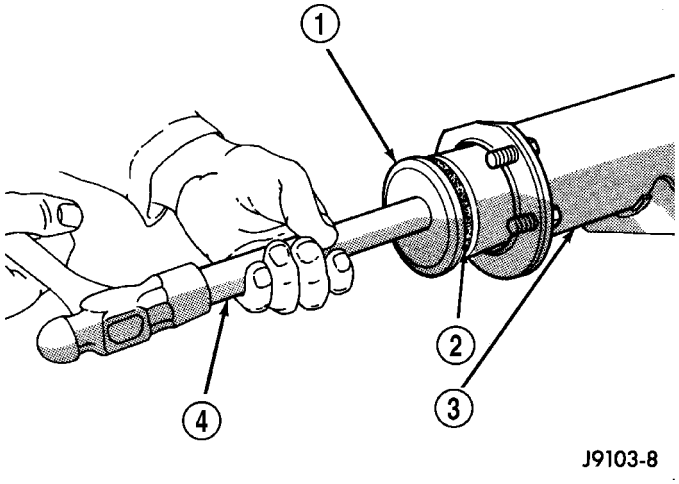


Fig. 26 AXLE SHAFT SEAL

1 - INSTALLER
2 - SEAL
3 - AXLE SHAFT TUBE
4 - HANDLE

(3) Install the axle shaft.

AXLE BEARINGS

REMOVAL

(1) Remove axle shaft.
(2) Remove axle shaft seal from axle shaft tube with a small pry bar.
(3) Remove axle shaft bearing from the axle tube with Bearing Removal Tool Set 6310 and Adapter Foot 6310-5 (Fig. 27).

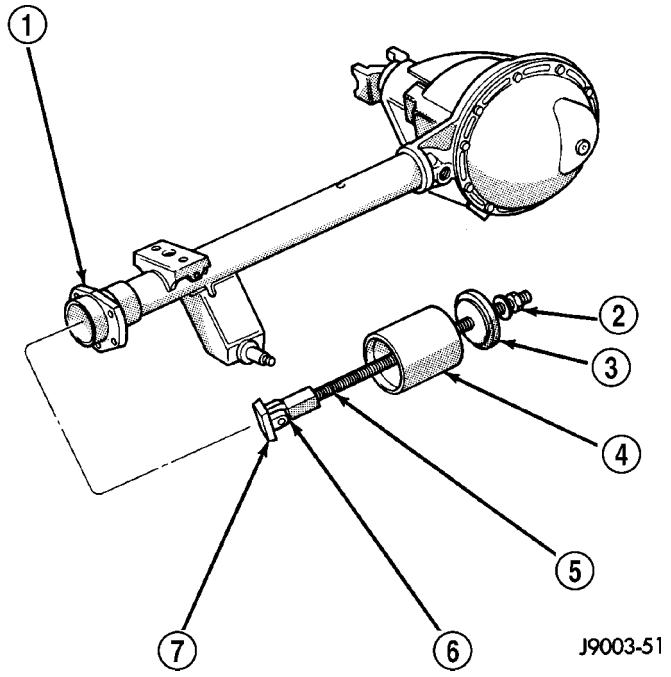


Fig. 27 AXLE SHAFT BEARING TOOLS

1 - AXLE SHAFT TUBE
2 - NUT
3 - GUIDE PLATE
4 - GUIDE
5 - THREADED ROD
6 - ADAPTER
7 - FOOT

INSTALLATION

(1) Wipe axle shaft tube bore clean.
(2) Install axle shaft bearing with Installer 6436 and Handle C-4171.

NOTE: Part number on the bearing must be against the installer.

(3) Install **new** axle shaft seal with Installer 6437 and Handle C-4171 (Fig. 28).
(4) Install axle shaft.

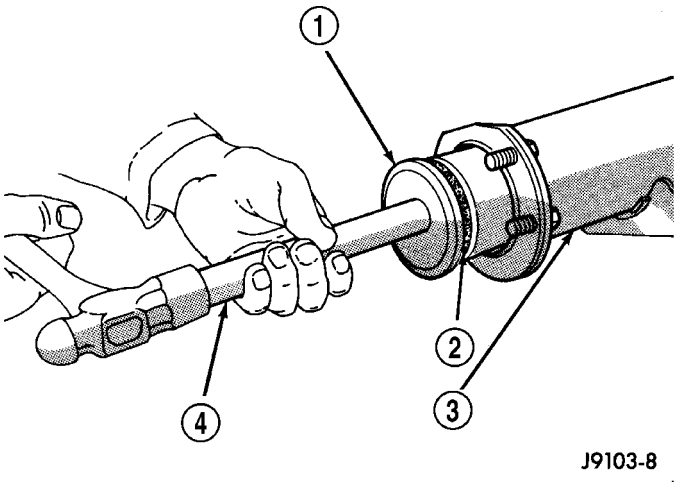


Fig. 28 AXLE SHAFT SEAL INSTALLER

1 - INSTALLER
2 - SEAL
3 - AXLE SHAFT TUBE
4 - HANDLE

PINION SEAL

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake drums.
(3) Remove propeller shaft from the yoke.
(4) Rotate pinion gear three or four times.
(5) Record torque necessary to rotate the pinion gear with an inch pound dial-type torque wrench.
(6) Hold the yoke with Wrench 6958 and remove pinion nut and washer.
(7) Remove pinion yoke with Remover C-452 and Wrench C-3281 (Fig. 29).

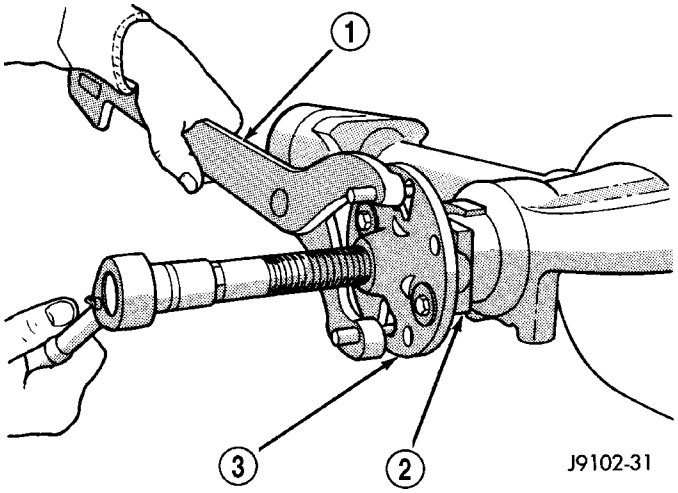


Fig. 29 PINION YOKE REMOVER

1 - WRENCH
2 - YOKE
3 - REMOVER

(8) Remove pinion seal with a pry tool or slide hammer mounted screw.

PINION SEAL (Continued)

INSTALLATION

(1) Apply a light coating of gear lubricant on the lip of pinion seal and install seal with an appropriate installer (Fig. 30).

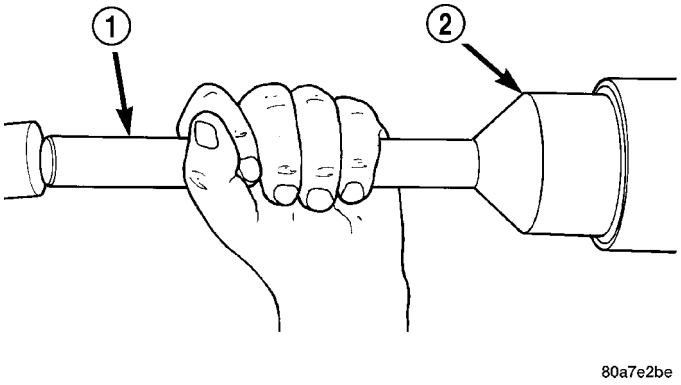


Fig. 30 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(2) Install pinion yoke Screw 8112, Cup 8109 and Spanner Wrench 6958 (Fig. 31).

CAUTION: Do not exceed the minimum tightening torque 271 N·m (200 ft. lbs.) when installing the pinion yoke at this point. Failure to heed caution may result in damage.

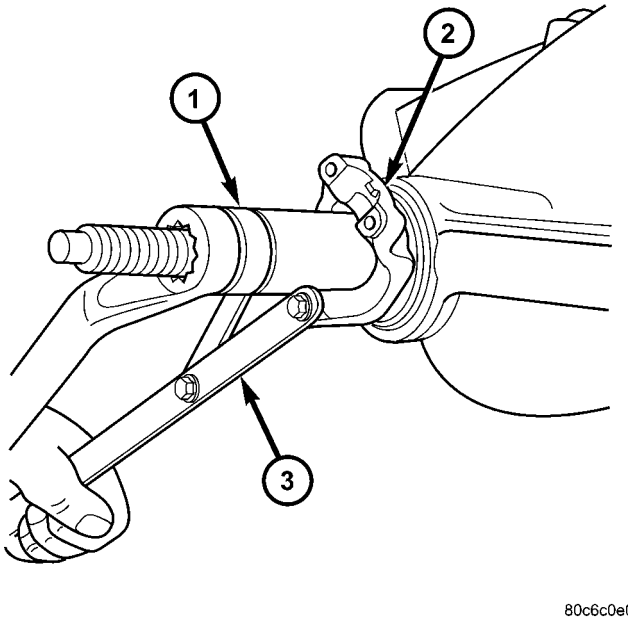


Fig. 31 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

(3) Install yoke washer and **new** nut on pinion gear and tighten nut until there is zero bearing endplay.

(4) Tighten pinion nut to 271 N·m (200 ft. lbs.).

CAUTION: Never loosen pinion gear nut to decrease pinion gear bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(5) Rotate pinion shaft using an inch pound torque wrench. Rotating torque should be equal to the reading recorded during removal plus an additional 0.56 N·m (5 in. lbs.) (Fig. 32).

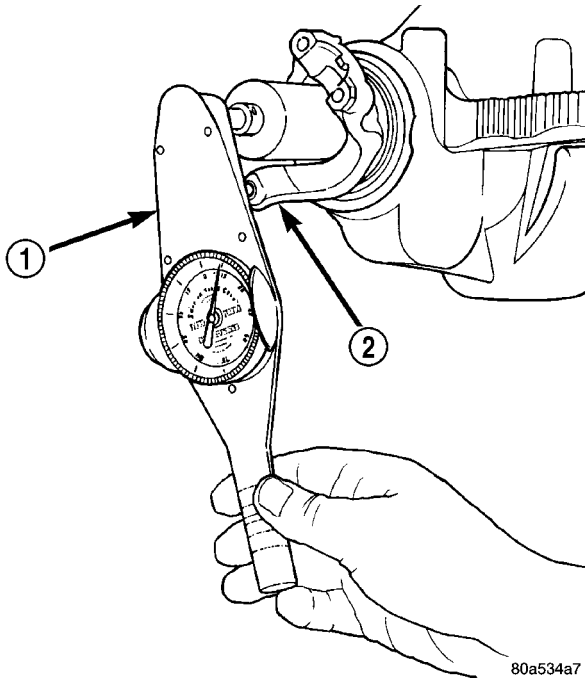


Fig. 32 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

PINION SEAL (Continued)

(6) If rotating torque is low, use Wrench 6958 to hold pinion yoke (Fig. 33), and tighten pinion nut in 6.8 N·m (5 ft. lbs.) increments until proper rotating torque is achieved.

CAUTION: If maximum tightening torque 475 N·m (350 ft. lbs.) is reached prior to reaching the required rotating torque, the collapsible spacer may have been damaged. Failure to heed caution may result in damage.

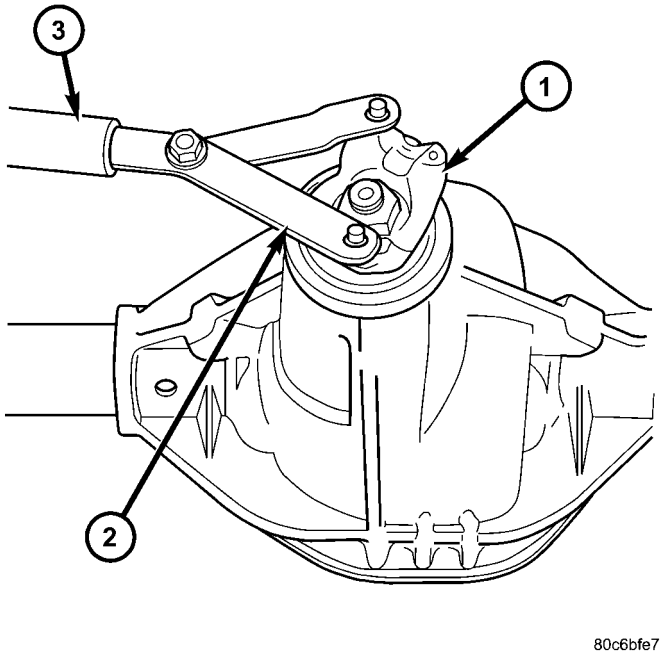


Fig. 33 YOKE HOLDER

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(7) Install propeller shaft.
(8) Install the brake drums.

COLLAPSIBLE SPACER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove rear brake drums.
(3) Remove propeller shaft.
(4) Rotate pinion gear three or four times.
(5) Record torque to rotate the pinion gear with an inch pound dial-type torque wrench.
(6) Hold pinion yoke with Spanner Wrench 6958 and remove pinion nut and washer.
(7) Remove pinion yoke with Remover C-452 and Flange Wrench C-3281 (Fig. 34).
(8) Remove pinion shaft seal with a pry tool or a slide hammer mounted screw.

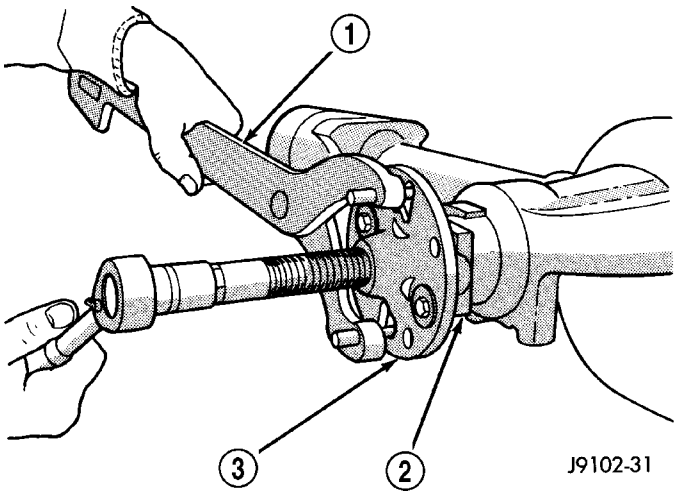


Fig. 34 PINION YOKE REMOVER

1 - WRENCH
2 - YOKE
3 - REMOVER

(9) Remove front pinion bearing using a pair of pick tools to pull the bearing straight off the pinion gear shaft.

NOTE: If bearing becomes bound on pinion shaft, lightly tap the pinion shaft with a rawhide/rubber hammer.

(10) Remove collapsible spacer.

INSTALLATION

(1) Install a **new** collapsible preload spacer on pinion shaft.
(2) Install pinion front bearing.
(3) Apply a light coating of gear lubricant on the lip of pinion seal and install seal with an appropriate installer (Fig. 35).

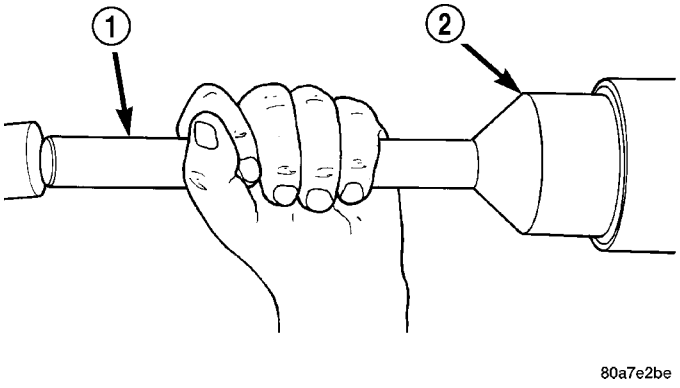


Fig. 35 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

COLLAPSIBLE SPACER (Continued)

(4) Install yoke with Screw 8112, Cup 8109 and Spanner Wrench 6958 (Fig. 36).

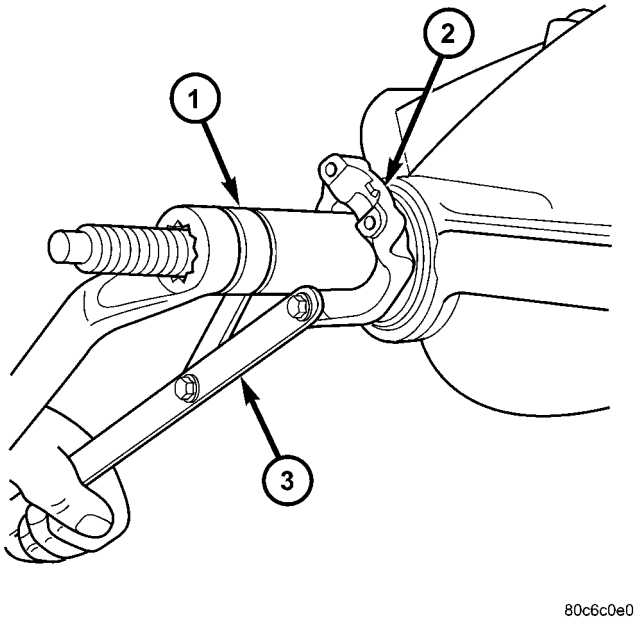


Fig. 36 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

(5) Install yoke washer and **new** nut on the pinion gear. Tighten the nut to 271 N·m (200 ft. lbs.).

CAUTION: Never loosen pinion gear nut to decrease pinion gear bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(6) Using yoke with Spanner Wrench 6958 and a torque wrench set at 475 N·m (350 ft. lbs.), (Fig. 37) slowly tighten the nut in 6.8 N·m (5 ft. lbs.) increments until the rotating torque is achieved. Measure rotating torque frequently to avoid over crushing the collapsible spacer (Fig. 38).

CAUTION: If more than 475 N·m (350 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

(7) Check rotating torque with an inch pound torque wrench (Fig. 38). The rotating torque of the pinion should be, the reading recorded during removal plus an additional 0.56 N·m (5 in. lbs.).

(8) Install propeller shaft.

(9) Install rear brake drums.

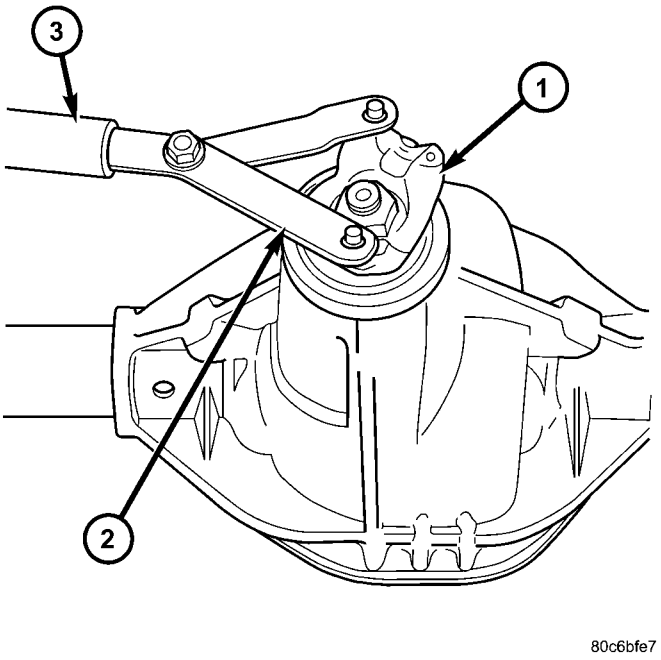


Fig. 37 YOKE HOLDER

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

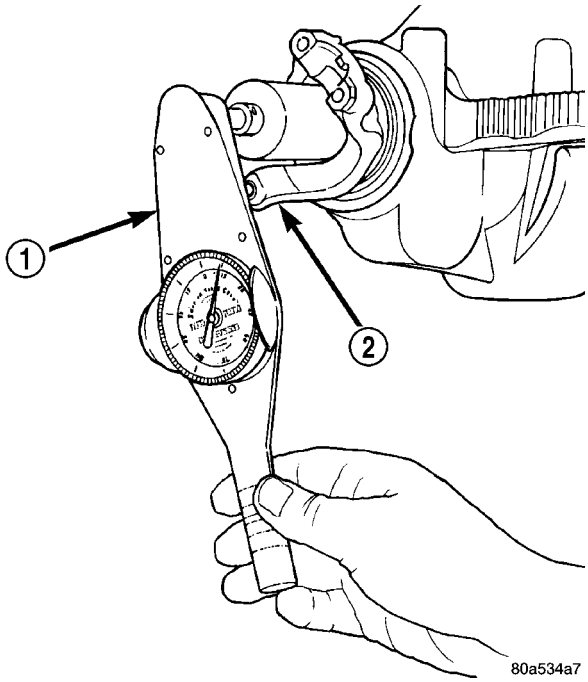


Fig. 38 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

DIFFERENTIAL COVER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove drain plug.
(3) Remove cover bolts.
(4) Remove cover and drain lubricant.

INSTALLATION

(1) Apply a 6.35mm (1/4 in.) bead of Mopar Silicone Rubber Sealant or equivalent to the housing cover (Fig. 39).

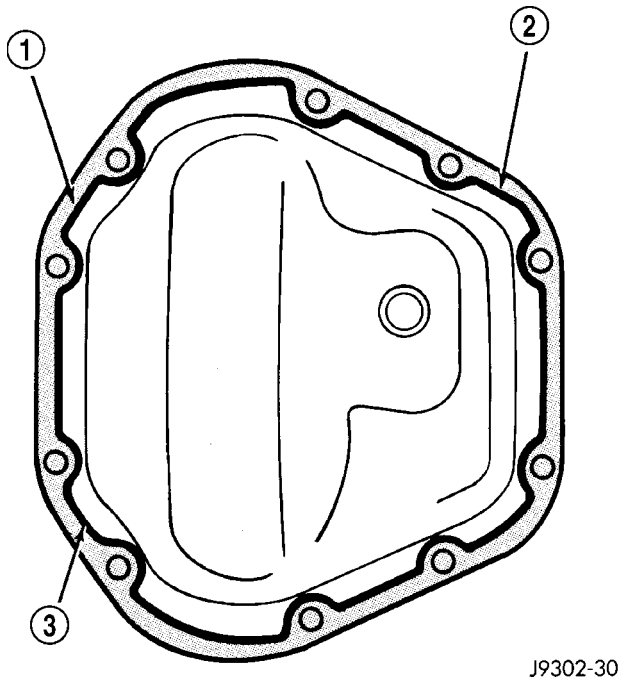


Fig. 39 HOUSING COVER - TYPICAL

1 - SEALANT SURFACE
2 - SEALANT
3 - SEALANT THICKESS

CAUTION: If housing cover is not installed within 3 to 5 minutes, the cover must be cleaned and new RTV applied. Failure to heed caution may result in damage.

(2) Install cover and identification tag. Tighten cover bolts in a criss-cross pattern to 41 N·m (30 ft. lbs.).
(3) Fill differential to specifications.
(4) Install fill plug and tighten to 34 N·m (25 ft. lbs.).

DIFFERENTIAL

REMOVAL

(1) Remove differential fill plug.
(2) Remove differential cover and drain fluid.
(3) Remove axle shafts.
(4) Note reference letters stamped on the bearing caps and housing machined sealing surface (Fig. 40).

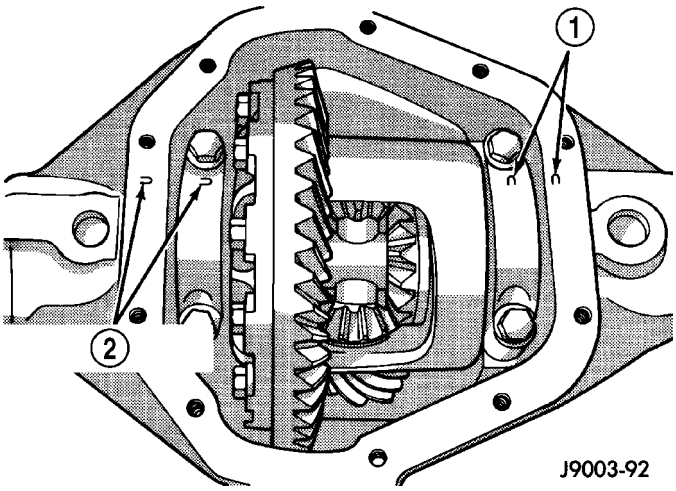


Fig. 40 BEARING CAP REFERENCE

1 - REFERENCE LETTERS
2 - REFERENCE LETTERS

(5) Loosen differential bearing cap bolts.
(6) Position Spreader W-129-B with Adapter Kit 6987B on differential locating holes (Fig. 41). Install hold-down clamps and tighten the turnbuckle finger-tight.

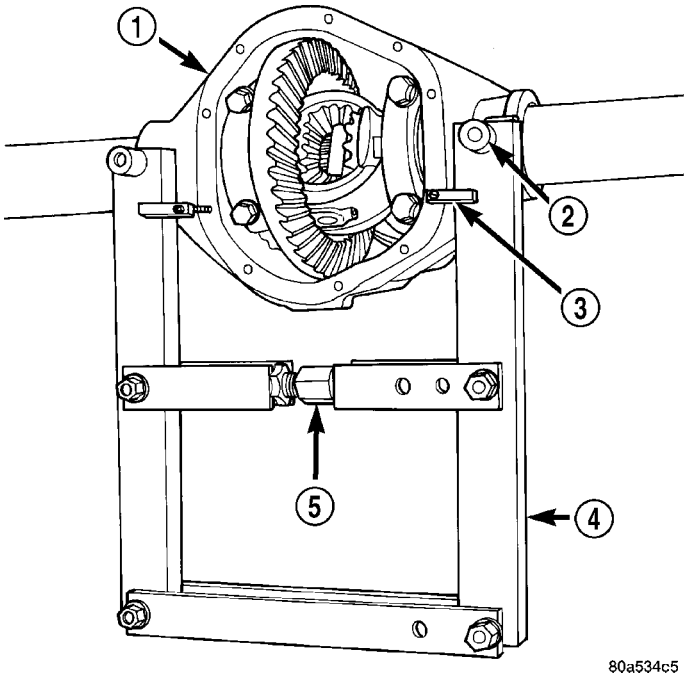


Fig. 41 SPREADER LOCATION

1 - DIFFERENTIAL HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

DIFFERENTIAL (Continued)

(7) Install a Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load indicator plunger against the opposite side of the housing (Fig. 42) and zero the indicator.

CAUTION: Never spread the housing over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

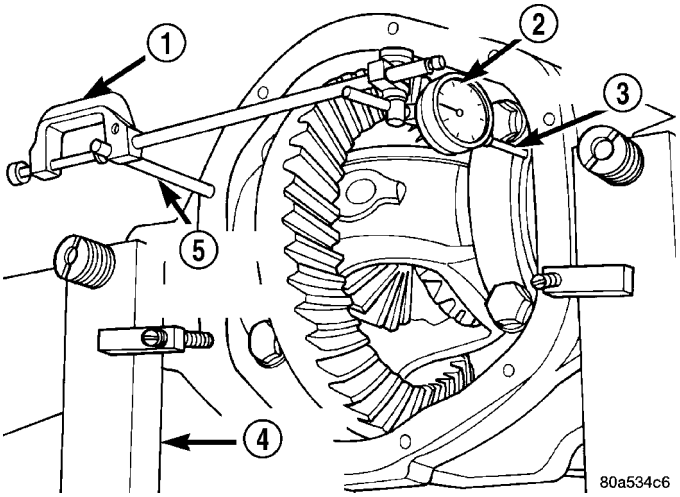


Fig. 42 DIAL INDICATOR LOCATION

1 - CLAMP
2 - DIAL INDICATOR
3 - LEVER ADAPTER
4 - SPREADER
5 - PILOT STUD

(8) Spread housing while measuring the distance with the dial indicator (Fig. 43).

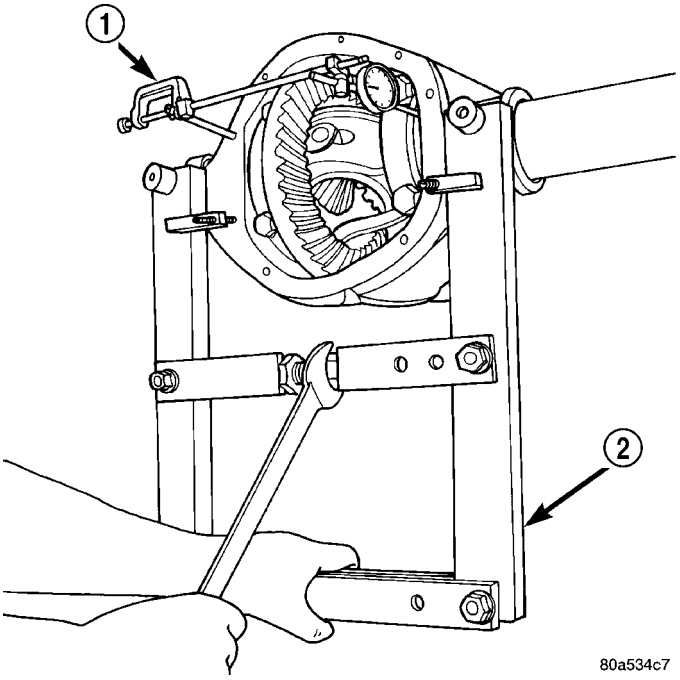


Fig. 43 SPREAD DIFFERENTIAL HOUSING

1 - DIAL INDICATOR
2 - SPREADER

(9) Remove dial indicator.

(10) While holding differential case in position, remove differential bearing caps.

(11) Remove differential from housing and tag differential bearing cups to indicate location (Fig. 44).

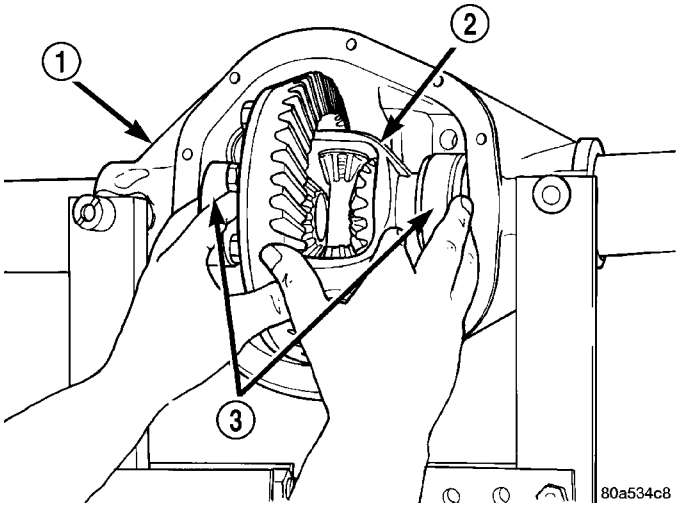


Fig. 44 DIFFERENTIAL CASE REMOVAL

1 - DIFFERENTIAL HOUSING
2 - DIFFERENTIAL CASE
3 - BEARING CUPS

(12) Remove spreader from housing.

(13) Clean the housing cavity with flushing oil, light engine oil or lint free cloth.

NOTE: Do not use water, steam, kerosene or gasoline for cleaning.

DISASSEMBLY

(1) Remove pinion shaft.

(2) Rotate differential side gears and remove differential pinions and thrust washers (Fig. 45).

(3) Remove differential side gears and thrust washers.

ASSEMBLY

(1) Lubricate all differential components with hypoid gear lubricant.

(2) Install differential side gears and thrust washers.

(3) Install differential pinion gears and thrust washers.

(4) Install the pinion mate shaft.

(5) Align hole in the pinion mate shaft with the hole in the differential case and install pinion mate shaft lock screw finger tight.

DIFFERENTIAL (Continued)

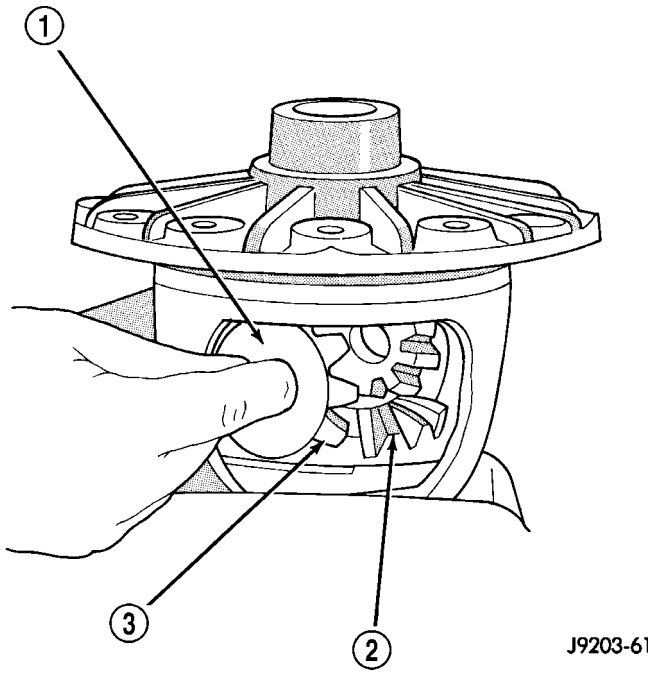


Fig. 45 DIFFERENTIAL GEARS

1 - THRUST WASHER
2 - SIDE GEAR
3 - DIFFERENTIAL PINION

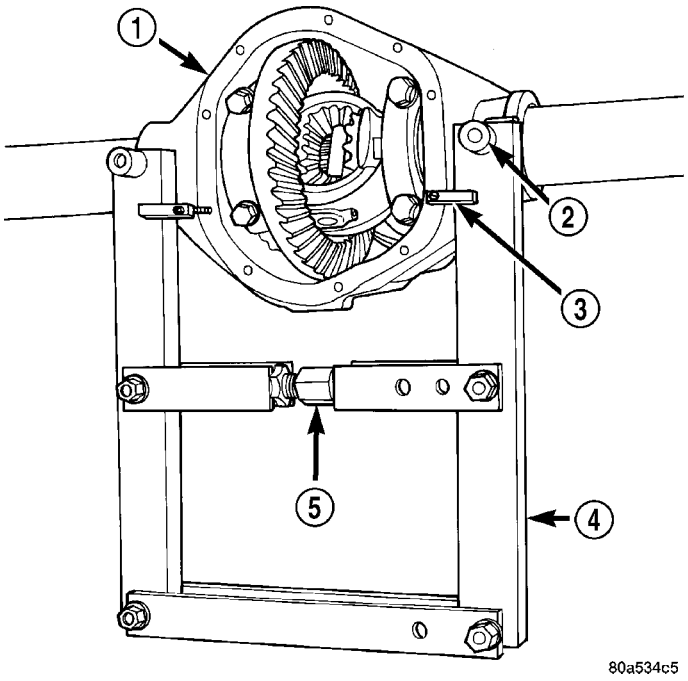


Fig. 46 SPREADER LOCATION

1 - DIFFERENTIAL HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

INSTALLATION

NOTE: If replacement differential bearings or differential case are being installed, differential side bearing shim requirements may change. Refer Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Position Spreader W-129-B and adapters from Adapter set 6987, with the tool dowel pins seated in the locating holes (Fig. 46). Install hold-down clamps and tighten turnbuckle finger-tight.

(2) Install Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load the indicator plunger against the opposite side of the housing and zero the indicator.

CAUTION: Never spread over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

(3) Spread the housing enough to install the case in the housing. Measure the distance with the dial indicator.

(4) Remove the dial indicator.

(5) Install differential case in the housing. Ensure that the differential bearing cups remain in position on the differential bearings. Tap the differential case to ensure the bearings cups are seated in the housing.

(6) Install bearing caps in their original locations (Fig. 47).

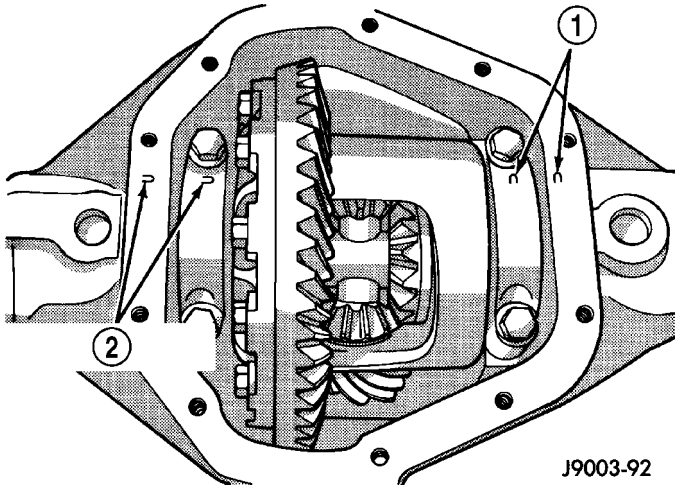


Fig. 47 BEARING CAP REFERENCE

1 - REFERENCE LETTERS
2 - REFERENCE LETTERS

(7) Loosely install differential bearing cap bolts.

(8) Remove axle housing spreader.

(9) Tighten bearing cap bolts to 77 N·m (57 ft. lbs.).

(10) Install axle shafts.

(11) Install cover and identification tag.

DIFFERENTIAL - TRAC-LOK

DESCRIPTION

The Trac-Lok® differential has a one-piece differential case, and similar internal components as a standard differential, plus two clutch disc packs. Differential bearing preload and ring gear backlash are adjusted with shims located between the differential case bearing cups and housing. Pinion bearing preload is set and maintained by the use of a collapsible spacer.

OPERATION

This differentials clutches are engaged by two concurrent forces. The first being the preload force exerted through Belleville spring washers within the clutch packs. The second is the separating forces generated by the side gears as torque is applied through the ring gear (Fig. 48).

This design provides the differential action needed for turning corners and for driving straight ahead during periods of unequal traction. When one wheel looses traction, the clutch packs transfer additional torque to the wheel having the most traction. This differential resist wheel spin on bumpy roads and provide more pulling power when one wheel looses traction. Pulling power is provided continuously until both wheels loose traction. If both wheels slip due to unequal traction, Trac-lok® operation is normal. In extreme cases of differences of traction, the wheel with the least traction may spin.

DIAGNOSIS AND TESTING

The most common problem is a chatter noise when turning corners. Before removing the unit for repair, drain, flush and refill the axle with the specified lubricant. A container of Mopar Trac-lok® Lubricant (friction modifier) should be added after repair service or during a lubricant change.

After changing the lubricant, drive the vehicle and make 10 to 12 slow, figure-eight turns. This maneuver will pump lubricant through the clutches. This will correct the condition in most instances. If chatter persists, clutch damage could have occurred.

DIFFERENTIAL TEST

The differential can be tested without removing the differential case by measuring rotating torque. Make sure brakes are not dragging during this measurement.

(1) Place blocks in front and rear of both front wheels.

(2) Raise one rear wheel until it is completely off the ground.

(3) Engine off, transmission in neutral, and parking brake off.

(4) Remove wheel and bolt Special Tool 6790 or equivalent tool to studs.

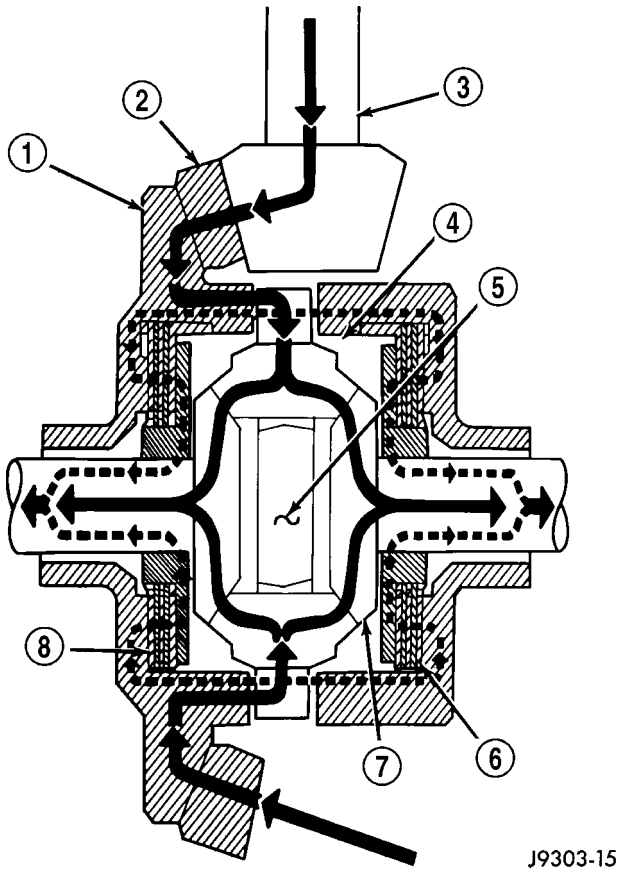


Fig. 48 TRAC-LOK LIMITED SLIP DIFFERENTIAL

1 - CASE
2 - RING GEAR
3 - DRIVE PINION
4 - PINION GEAR
5 - MATE SHAFT
6 - CLUTCH PACK
7 - SIDE GEAR
8 - CLUTCH PACK

DIFFERENTIAL - TRAC-LOK (Continued)

(5) Use torque wrench on special tool to rotate wheel and read rotating torque (Fig. 49).

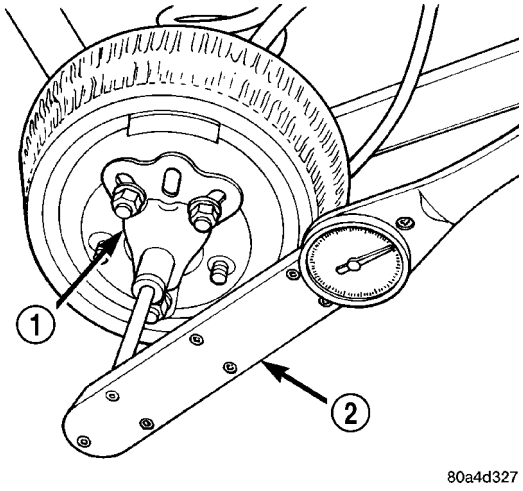


Fig. 49 ROTATING TORQUE TEST

1 - SPECIAL TOOL WITH BOLT IN CENTER HOLE
2 - TORQUE WRENCH

(6) If rotating torque is less than 41 N·m (30 ft. lbs.) or more than 271 N·m (200 ft. lbs.) on either wheel the unit should be serviced.

DISASSEMBLY

(1) Clamp Holding Fixture 6965 in vise and set differential case on fixture (Fig. 50).

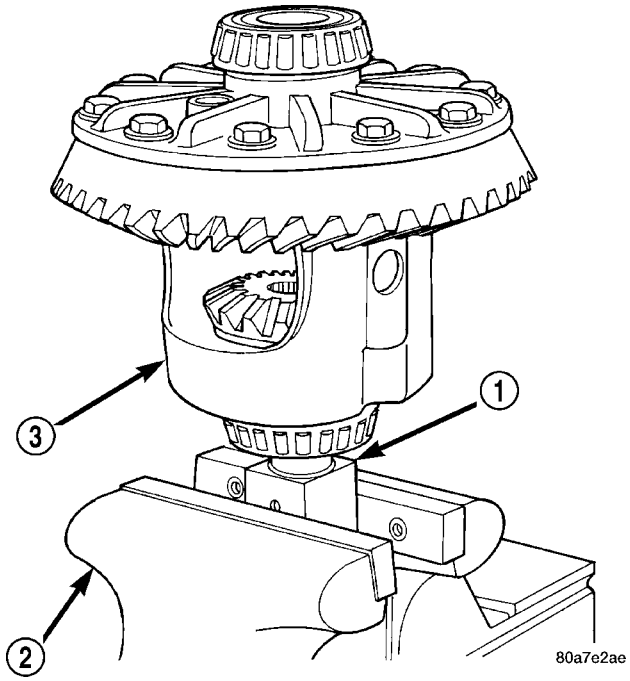


Fig. 50 DIFFERENTIAL CASE FIXTURE

1 - FIXTURE
2 - VISE
3 - DIFFERENTIAL

(2) Remove pinion gear mate shaft lock screw.
(3) Remove pinion gear mate shaft.
(4) Lubricate and install disc without threaded hole from Trac-Lok® Tool Kit 6960 into lower side gear (Fig. 51).

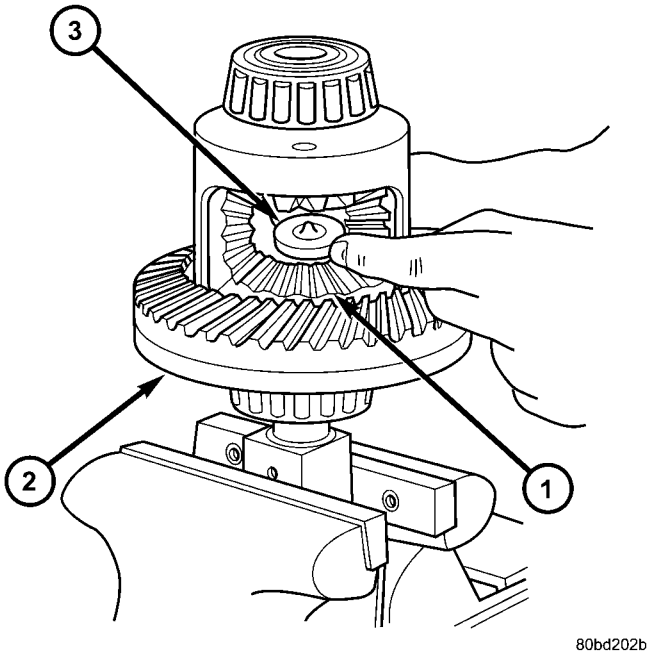


Fig. 51 DISC PLATE

1 - LOWER SIDE GEAR
2 - DIFFERENTIAL CASE
3 - DISC

(5) Install disc with threaded hole in the upper side gear. Thread forcing screw from Trac-Lok® Tool Kit 6960 through the upper disc until it comes in contact with lower disc.

DIFFERENTIAL - TRAC-LOK (Continued)

(6) Insert a screw driver in slot of upper disc (Fig. 52) to prevent disc from turning.

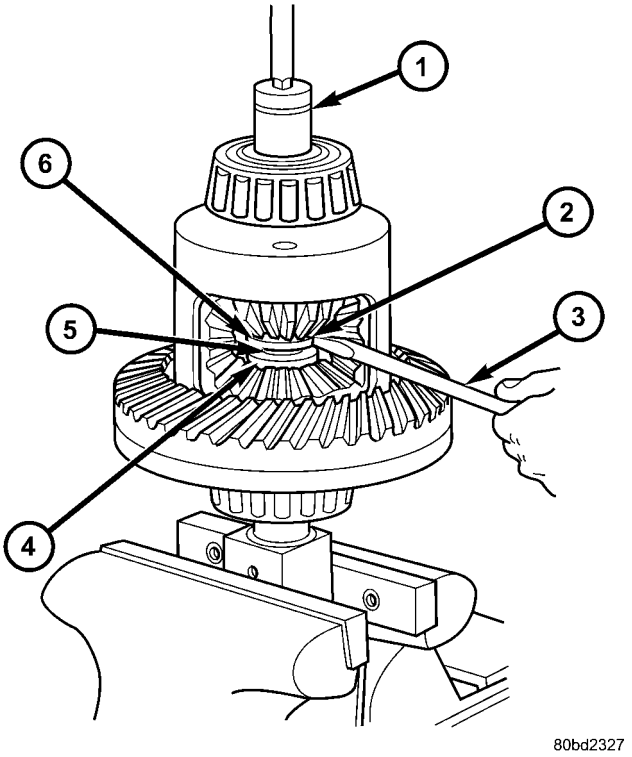


Fig. 52 THREAD ADAPTER DISC

1 - SOCKET
2 - SLOT IN DISC
3 - SCREWDRIVER
4 - LOWER DISC
5 - FORCING SCREW
6 - UPPER DISC

(7) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress Belleville springs in clutch packs (Fig. 53).

(8) With a feeler gauge remove thrust washers from behind the pinion gears (Fig. 54).

(9) Insert turning bar from tool kit into the pinion mate shaft hole in the case (Fig. 55).

(10) Loosen forcing screw in small increments until clutch pack tension is relieved and the differential case can be rotated with turning bar.

(11) Rotate differential case until pinion gears can be removed.

(12) Remove pinion gears from differential case.

(13) Remove forcing screw and discs.

(14) Remove top side gear, clutch pack retainers and clutch pack (Fig. 56).

NOTE: Keep plates in order during removal.

(15) Remove differential case from the fixture.

(16) Remove side gear, clutch pack retainer and clutch pack.

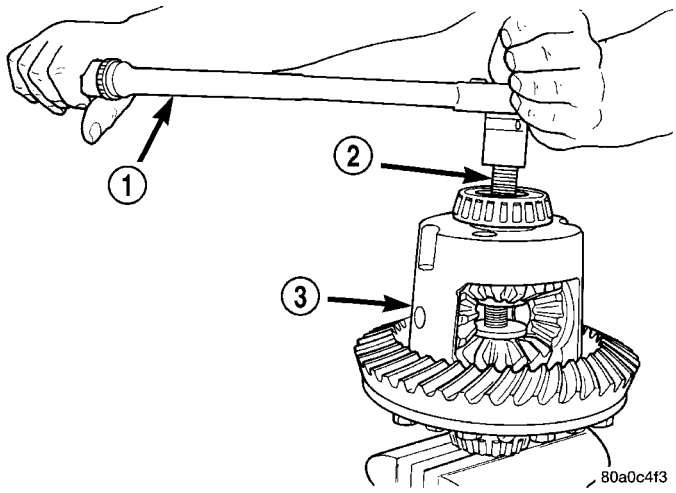


Fig. 53 TIGHTEN FORCING SCREW

1 - TORQUE WRENCH
2 - FORCING SCREW
3 - DIFFERENTIAL CASE

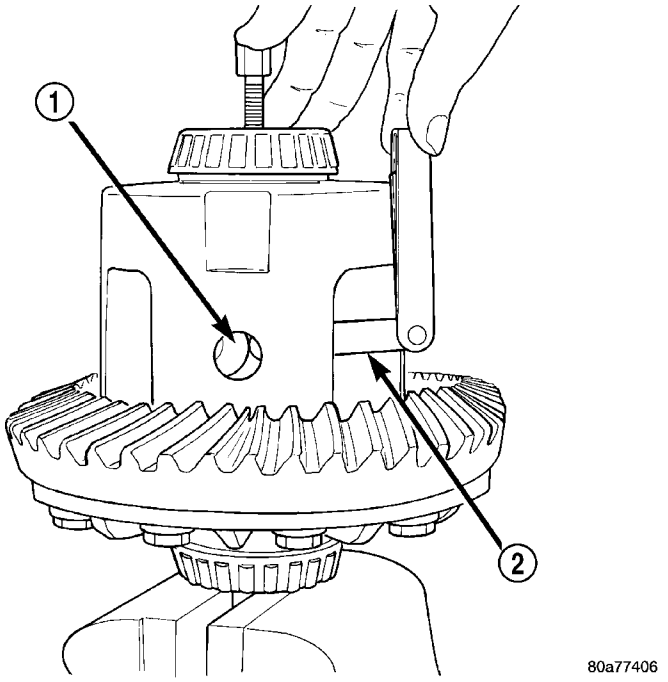


Fig. 54 PINION GEAR THRUST WASHER

1 - THRUST WASHER
2 - FEELER GAUGE

NOTE: Keep plates in order during removal.

ASSEMBLY

NOTE: New Plates and discs with fiber coating (no grooves or lines) must be presoaked in Friction Modifier before assembly. Soak plates and discs for a minimum of 20 minutes.

DIFFERENTIAL - TRAC-LOK (Continued)

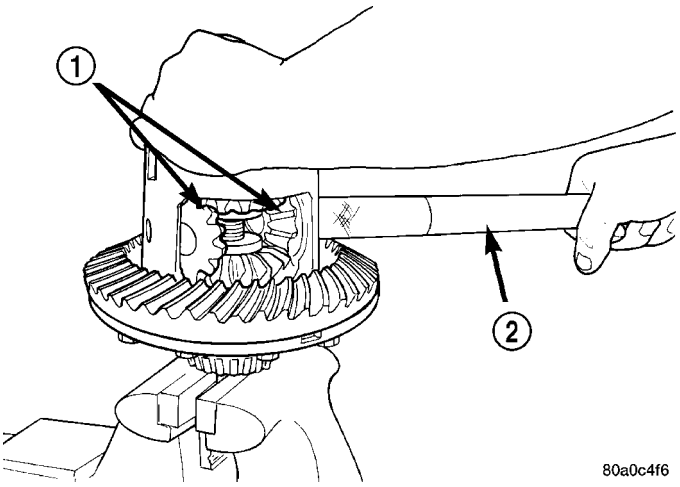


Fig. 55 TURNING BAR

1 - PINION GEARS
2 - TURNING BAR

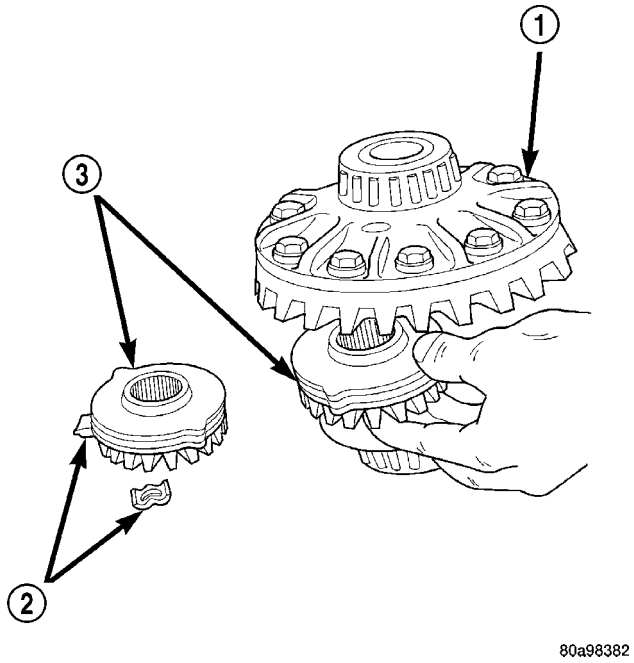


Fig. 56 SIDE GEAR & CLUTCH DISC

1 - DIFFERENTIAL CASE
2 - RETAINER
3 - SIDE GEAR AND CLUTCH DISC PACK

(1) Assemble clutch discs into packs and secure disc packs with retaining clips (Fig. 57).

(2) Install assembled clutch disc packs on side gear hubs.

(3) Install clutch pack and lower side gear in differential case (Fig. 58).

NOTE: Verify clutch pack retaining clips are in position and seated in case pockets.

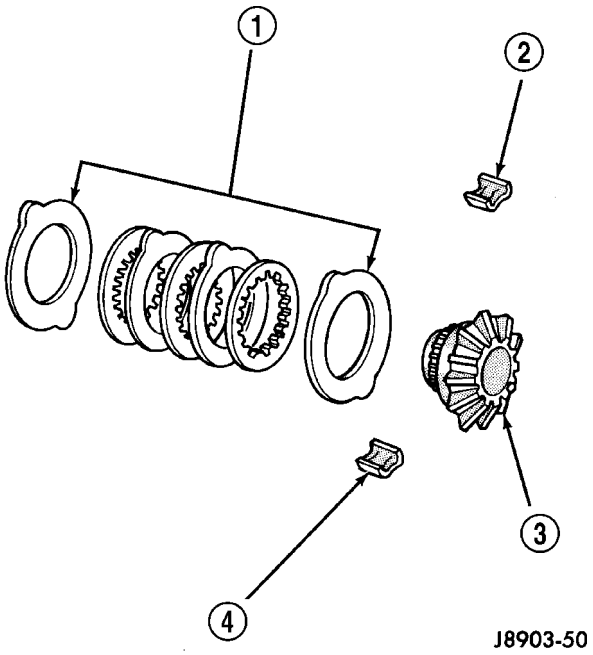


Fig. 57 CLUTCH DISC PACK

1 - CLUTCH PACK
2 - RETAINER
3 - SIDE GEAR
4 - RETAINER

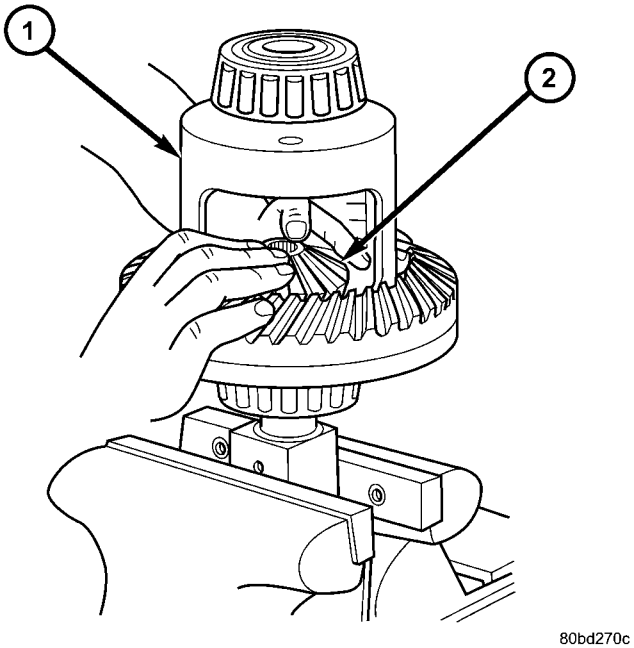


Fig. 58 CLUTCH PACK AND LOWER SIDE GEAR

1 - DIFFERENTIAL CASE
2 - LOWER SIDE GEAR AND CLUTCH PACK

DIFFERENTIAL - TRAC-LOK (Continued)

(4) Lubricated and install disc without the hole from Trac-Lok® Tool Kit 6960 into lower side gear.

(5) Install upper side gear and clutch disc pack (Fig. 59).

NOTE: Verify clutch pack retaining clips are in position and seated in case pockets.

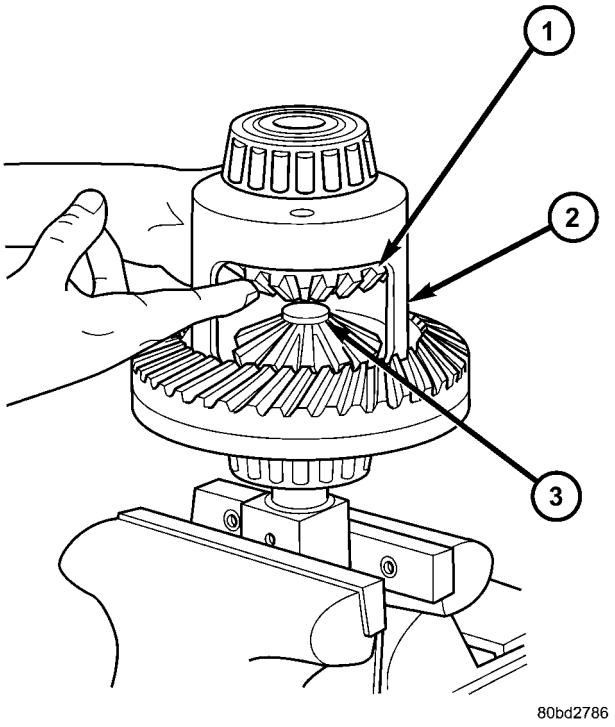


Fig. 59 CLUTCH PACK AND UPPER SIDE GEAR

1 - SIDE GEAR AND CLUTCH PACK
2 - DIFFERENTIAL CASE
3 - LOWER DISC

(6) Lubricate and install disc with threaded hole into top side gear.

(7) Thread forcing screw from tool kit through the top disc and until screw slightly compress clutch disc.

(8) Place pinion gears in position in side gears and verify pinion mate shaft hole is aligned.

(9) Rotate case with turning bar until pinion mate shaft holes in pinion gears align with holes in case.

NOTE: It may be necessary to slightly tighten the forcing screw in order to install the pinion gears.

(10) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress the Belleville springs.

(11) Lubricate and install thrust washers behind pinion gears and align washers with a small screw driver. Insert mate shaft into each pinion gear to verify alignment.

(12) Remove forcing screw and discs.

(13) Install pinion gear mate shaft and align holes in shaft and case.

(14) Install pinion mate shaft lock screw finger tight to hold shaft during differential installation.

(15) Lubricate all differential components with hypoid gear lubricant.

DIFFERENTIAL CASE BEARINGS

REMOVAL

(1) Remove differential from housing.

(2) Remove bearings from differential case with Puller/Press C-293-PA , Adapters C-293-39 and Plug SP-3289 (Fig. 60).

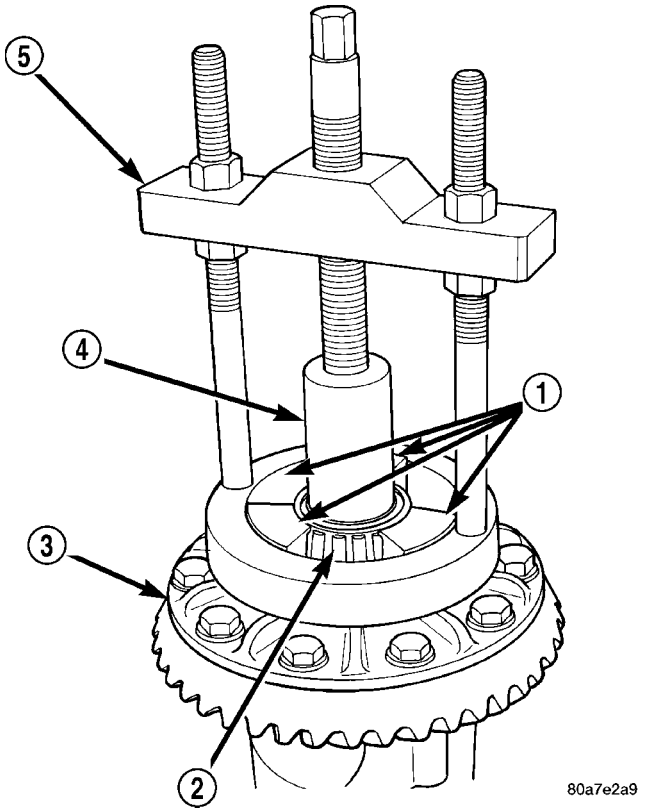


Fig. 60 BEARING REMOVAL

1 - ADAPTERS
2 - BEARING
3 - DIFFERENTIAL
4 - PLUG
5 - PULLER

DIFFERENTIAL CASE BEARINGS (Continued)

INSTALLATION

(1) Install differential side bearings with Installer C-3716-A with Handle C-4171 (Fig. 61).

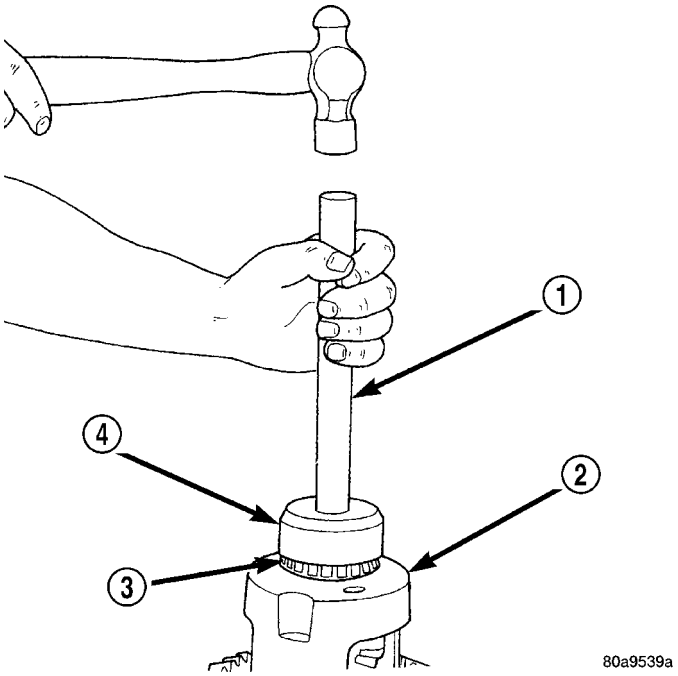


Fig. 61 DIFFERENTIAL CASE BEARINGS

1 - HANDLE
2 - DIFFERENTIAL
3 - BEARING
4 - INSTALLER

(2) Install differential in the housing.

PINION GEAR/RING GEAR

REMOVAL

NOTE: The ring and pinion gears are serviced in a matched set. Never replace one gear without replacing the other gear.

(1) Remove differential from housing.
(2) Place differential case in a vise with soft metal jaw.
(3) Remove ring gear bolts from the differential case.
(4) Drive ring gear off the differential case with a dead-blow hammer (Fig. 62).

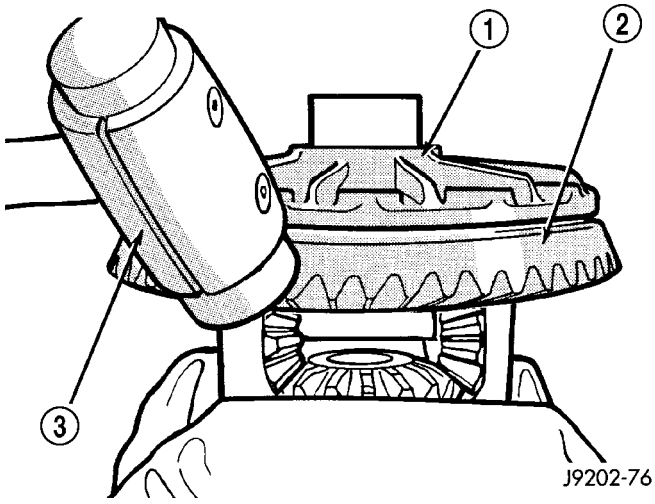


Fig. 62 RING GEAR

1 - CASE
2 - RING GEAR
3 - DEAD-BLOW HAMMER

(5) Hold pinion yoke with Wrench 6958 and remove pinion yoke nut and washer.
(6) Remove pinion yoke from pinion shaft with Remover C-452 and Wrench C-3281 (Fig. 63).

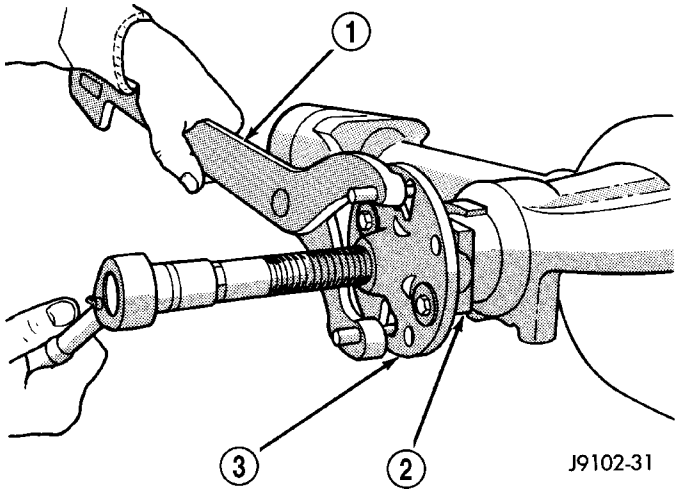


Fig. 63 PINION YOKE REMOVER

1 - WRENCH
2 - YOKE
3 - PULLER

PINION GEAR/RING GEAR (Continued)

(7) Remove pinion gear from housing with a dead-blow hammer (Fig. 64).

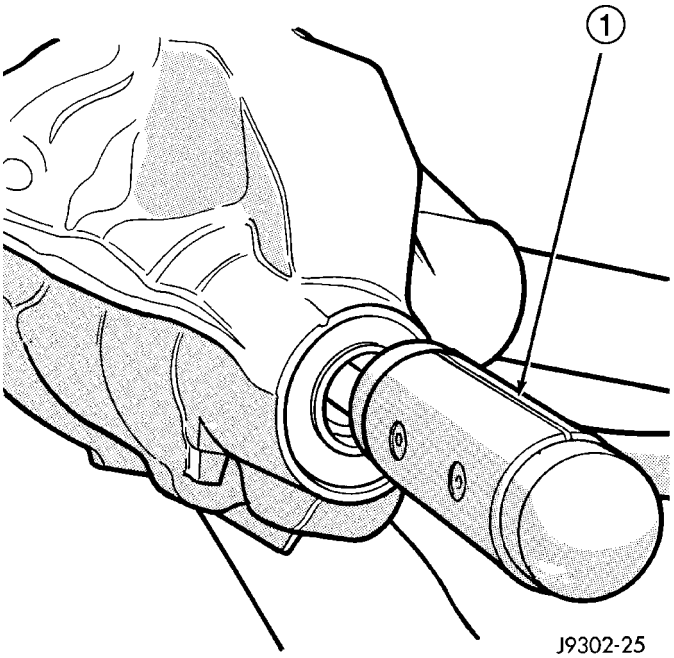


Fig. 64 REMOVE PINION GEAR

1 - DEAD-BLOW HAMMER

(8) Remove pinion shaft seal with a pry tool or a slide hammer mounted screw.

(9) Remove oil slinger, if equipped, and front pinion bearing.

(10) Remove front pinion bearing cup with Remover C-4345 and Handle C-4171 (Fig. 65).

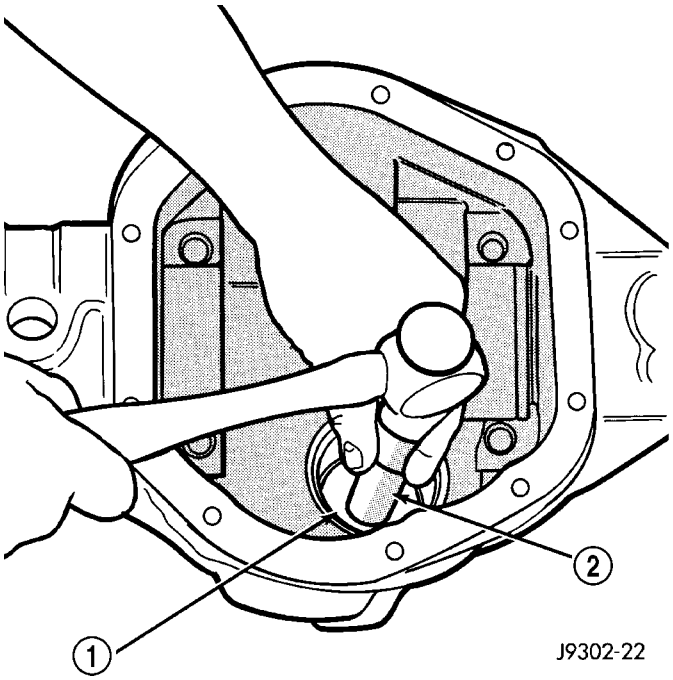


Fig. 65 FRONT PINION BEARING CUP

1 - REMOVER
2 - HANDLE

(11) Remove rear bearing cup from housing (Fig. 66) with Remover D-149 and Handle C-4171.

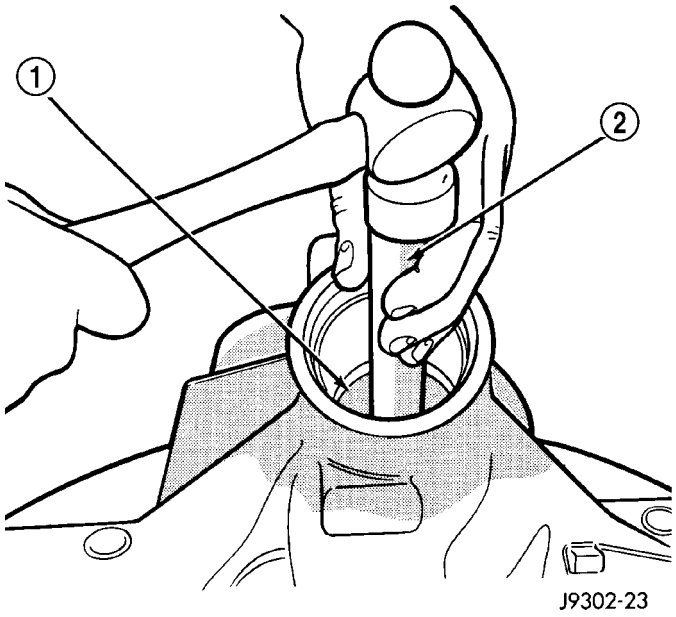


Fig. 66 REAR PINION BEARING CUP

1 - DRIVER
2 - HANDLE

(12) Remove collapsible preload spacer (Fig. 67).

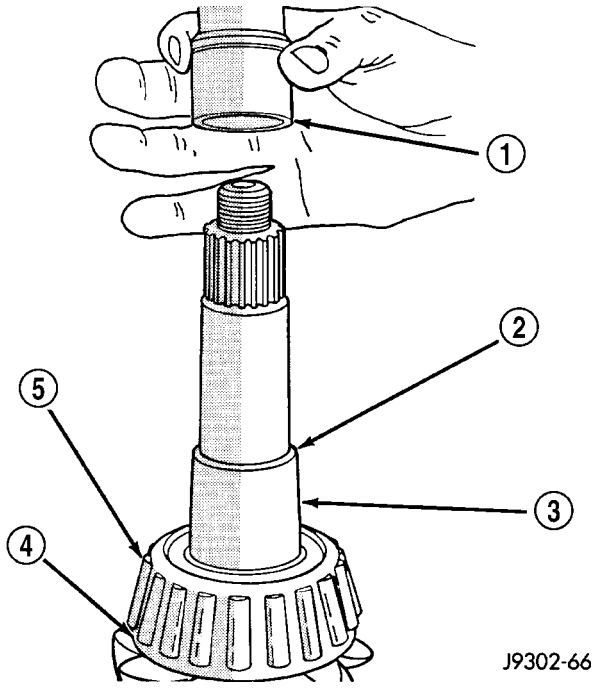


Fig. 67 COLLAPSIBLE SPACER

1 - COLLAPSIBLE SPACER
2 - SHOULDER
3 - PINION GEAR
4 - SHIM
5 - REAR BEARING

PINION GEAR/RING GEAR (Continued)

(13) Remove rear bearing from the pinion shaft with Puller/Press C-293-PA and Adapters C-293-40 (Fig. 68).

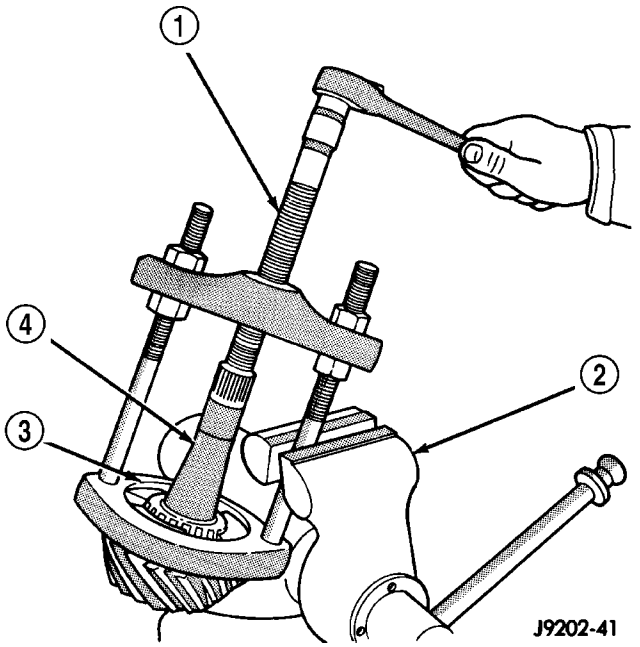


Fig. 68 REAR PINION BEARING

1 - PULLER
2 - VISE
3 - ADAPTERS
4 - DRIVE PINION GEAR SHAFT

(14) Remove depth shims from the pinion shaft and record the shims thickness.

INSTALLATION

NOTE: A pinion depth shim/oil slinger is placed between the rear pinion bearing cone and the pinion head to achieve proper ring gear and pinion mesh. If ring gear and pinion are reused, the pinion depth shim/oil slinger should not require replacement. Refer to Adjustment (Pinion Gear Depth) to select the proper thickness shim/oil slinger if ring and pinion gears are replaced.

(1) Apply Mopar Door Ease or equivalent lubricant to outside surface of bearing cups.

(2) Install pinion rear bearing cup with Installer D-146 and Driver Handle C-4171 (Fig. 69) and verify cup is seated.

(3) Install pinion front bearing cup with Installer D-130 and Handle C-4171 (Fig. 70) and verify cup is seated.

(4) Install pinion front bearing and shim.

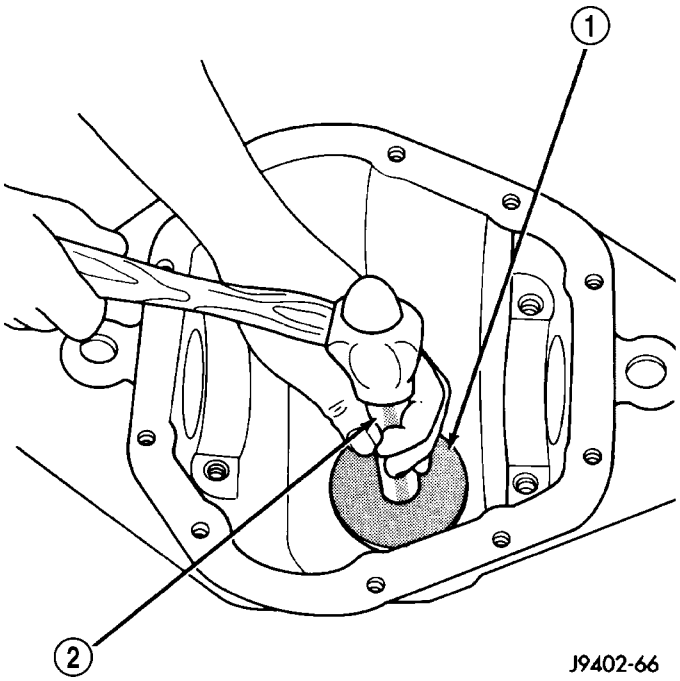


Fig. 69 REAR PINION BEARING CUP

1 - INSTALLER
2 - HANDLE

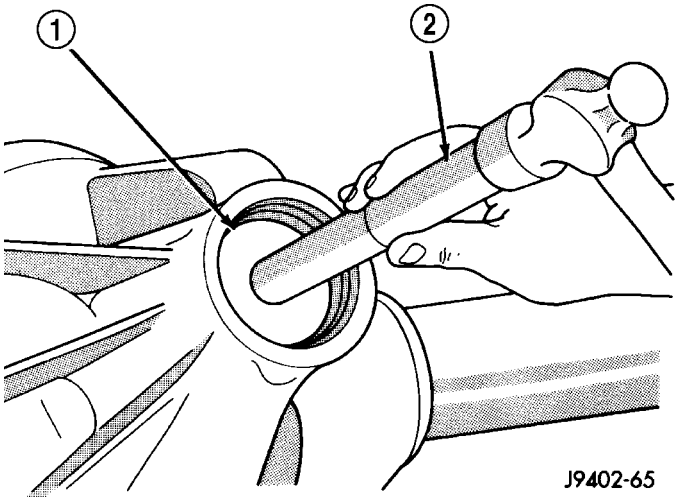


Fig. 70 FRONT PINION BEARING CUP

1 - INSTALLER
2 - HANDLE

PINION GEAR/RING GEAR (Continued)

(5) Apply a light coating of gear lubricant on the lip of pinion seal and install seal with an appropriate installer (Fig. 71).

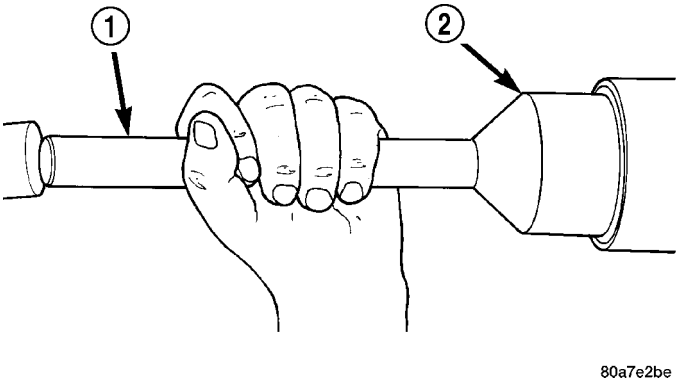


Fig. 71 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(6) Install depth shim on the pinion gear.

(7) Install rear bearing and shim on the pinion gear with Installer W-262 and a press (Fig. 72).

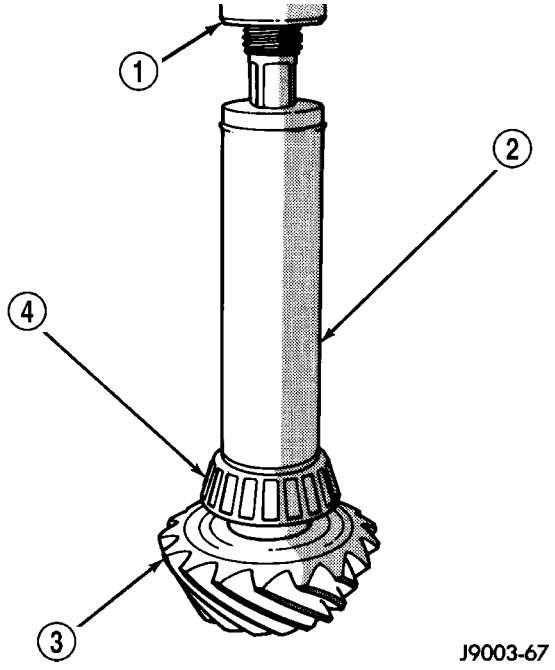


Fig. 72 REAR PINION BEARING

1 - PRESS
2 - INSTALLER
3 - DRIVE PINION GEAR
4 - REAR PINION BEARING

(8) Install a **new** collapsible preload spacer on pinion shaft and install pinion gear in the housing (Fig. 73).

(9) Install yoke with Installer Screw 8112, Cup 8109 and Wrench 6958 (Fig. 74).

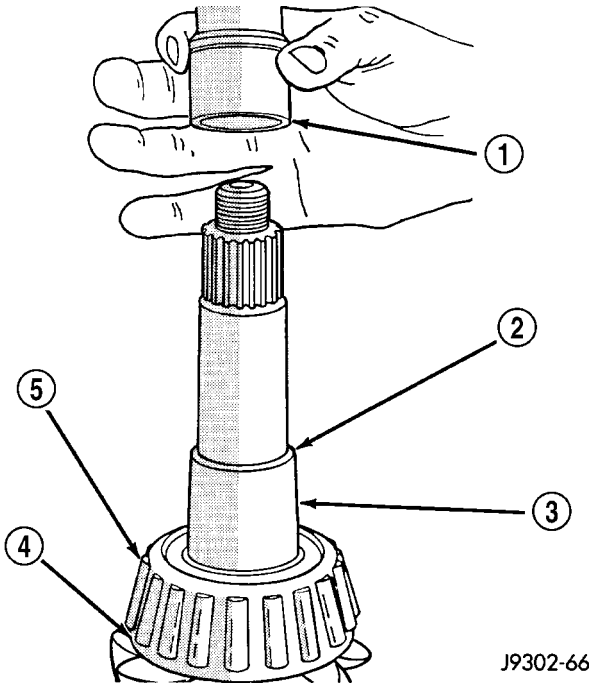


Fig. 73 COLLAPSIBLE SPACER

1 - COLLAPSIBLE SPACER
2 - SHOULDER
3 - PINION GEAR
4 - SHIM
5 - REAR BEARING

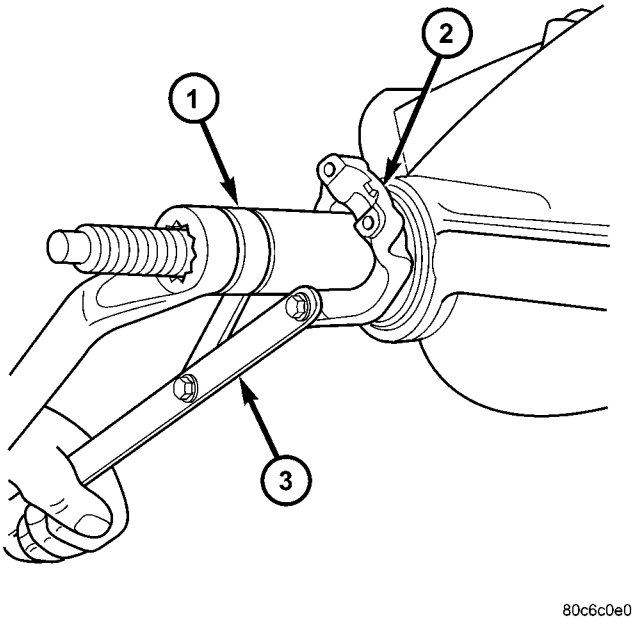


Fig. 74 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

PINION GEAR/RING GEAR (Continued)

(10) Install yoke washer and a **new** nut on the pinion gear and tighten the pinion nut until there is zero bearing end-play.

(11) Tighten the nut to 271 N·m (200 ft. lbs.).

CAUTION: Never loosen pinion gear nut to decrease pinion gear bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(12) Using Spanner Wrench 6958 and a torque wrench set at 475 N·m (350 ft. lbs.), (Fig. 75) slowly tighten the nut in 6.8 N·m (5 ft. lbs.) increments until the rotating torque is achieved. Measure the rotating torque frequently to avoid over crushing the collapsible spacer (Fig. 76).

CAUTION: If more than 475 N·m (350 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

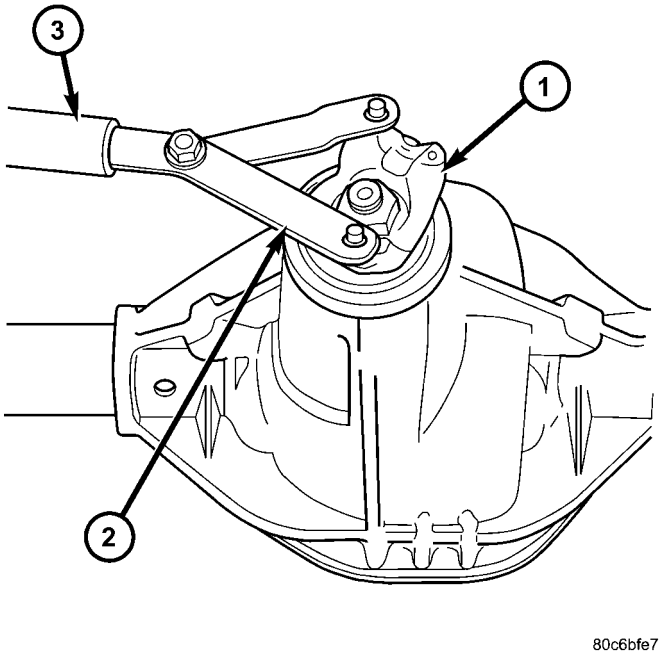


Fig. 75 PINION YOKE WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(13) Check bearing rotating torque with a inch pound torque wrench (Fig. 76). The pinion gear rotating torque should be:

- Original Bearings: 1 to 2 N·m (10 to 20 in. lbs.).
- New Bearings: 2 to 4 N·m (20 to 35 in. lbs.).

(14) Invert the differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

(15) Invert the differential case in the vise.

(16) Install new ring gear bolts and alternately tighten to 136 N·m (100 ft. lbs.) (Fig. 77).

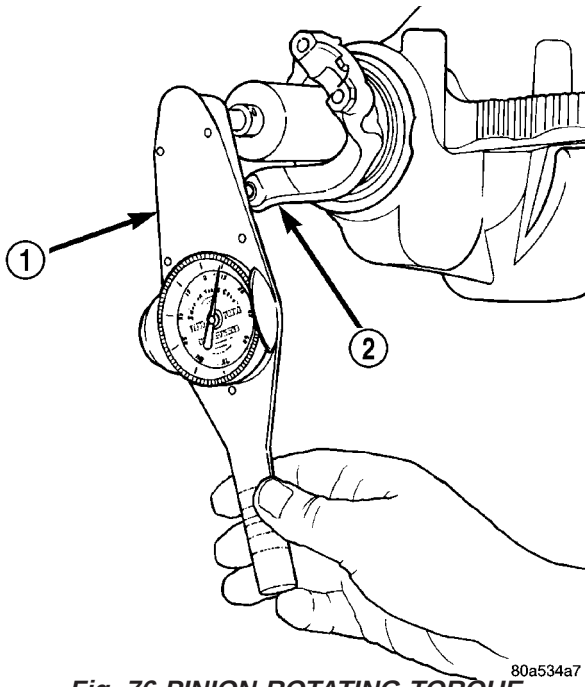


Fig. 76 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

CAUTION: Do not reuse ring gear bolts, the bolts can fracture causing extensive damage. Failure to heed caution may result in damage.

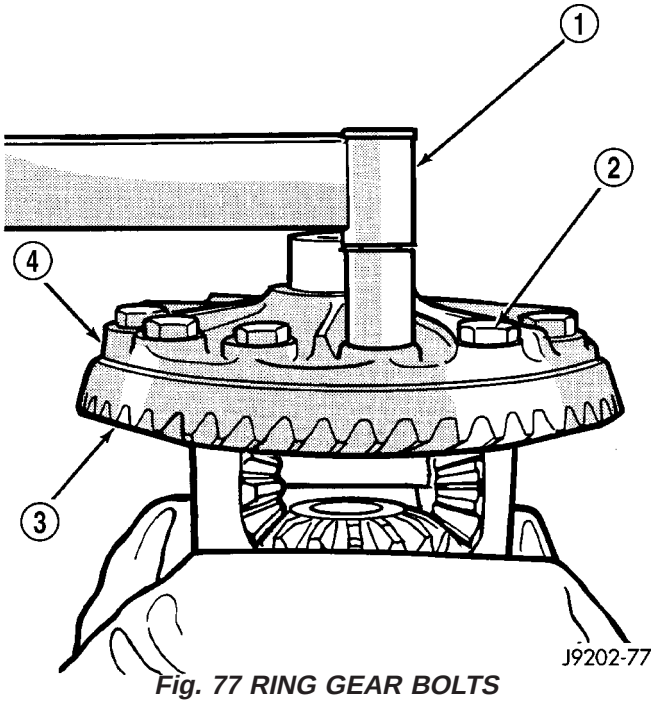


Fig. 77 RING GEAR BOLTS

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(17) Install differential in axle housing and verify gear mesh and contact pattern. Refer to Adjustments (Gear Contact Pattern) for procedure.

REAR AXLE - 216RBI

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REAR AXLE - 216RBI

DIAGNOSIS AND TESTING

GEAR NOISE

Axle gear noise can be caused by insufficient lubricant, incorrect backlash, incorrect pinion depth, tooth contact, worn/damaged gears, or the carrier housing not having the proper offset and squareness.

Gear noise usually happens at a specific speed range. The noise can also occur during a specific type of driving condition. These conditions are acceleration, deceleration, coast, or constant load.

When road testing, first warm-up the axle fluid by driving the vehicle at least 5 miles and then accelerate the vehicle to the speed range where the noise is the greatest. Shift out-of-gear and coast through the peak-noise range. If the noise stops or changes greatly:

- Check for insufficient lubricant.
- Incorrect ring gear backlash.
- Gear damage.

Differential side gears and pinions can be checked by turning the vehicle. They usually do not cause noise during straight-ahead driving when the gears are unloaded. The side gears are loaded during vehicle turns. A worn pinion shaft can also cause a snapping or a knocking noise.

BEARING NOISE

The axle shaft, differential and pinion bearings can all produce noise when worn or damaged. Bearing noise can be either a whining, or a growling sound.

Pinion bearings have a constant-pitch noise. This noise changes only with vehicle speed. Pinion bearing noise will be higher pitched because it rotates at a faster rate. Drive the vehicle and load the differential. If bearing noise occurs, the rear pinion bearing is the source of the noise. If the bearing noise is heard during a coast, the front pinion bearing is the source.

Worn or damaged differential bearings usually produce a low pitch noise. Differential bearing noise is similar to pinion bearing noise. The pitch of differential bearing noise is also constant and varies only with vehicle speed.

REAR AXLE - 216RBI (Continued)

Axle shaft bearings produce noise and vibration when worn or damaged. The noise generally changes when the bearings are loaded. Road test the vehicle. Turn the vehicle sharply to the left and to the right. This will load the bearings and change the noise level. Where axle bearing damage is slight, the noise is usually not noticeable at speeds above 30 mph.

LOW SPEED KNOCK

Low speed knock is generally caused by a worn U-joint or by worn side-gear thrust washers. A worn pinion shaft bore will also cause low speed knock.

VIBRATION

Vibration at the rear of the vehicle is usually caused by a:

- Damaged drive shaft.
- Missing drive shaft balance weight(s).
- Worn or out-of-balance wheels.
- Loose wheel lug nuts.
- Worn U-joint(s).
- Loose/broken springs.
- Damaged axle shaft bearing(s).
- Loose pinion gear nut.
- Excessive pinion yoke run out.
- Bent axle shaft(s).

Check for loose or damaged front-end components or engine/transmission mounts. These components can contribute to what appears to be a rearend vibration. Do not overlook engine accessories, brackets and drive belts.

NOTE: All driveline components should be examined before starting any repair.

DRIVELINE SNAP

A snap or clunk noise when the vehicle is shifted into gear (or the clutch engaged), can be caused by:

- High engine idle speed.
- Transmission shift operation.
- Loose engine/transmission/transfer case mounts.
- Worn U-joints.
- Loose spring mounts.
- Loose pinion gear nut and yoke.
- Excessive ring gear backlash.
- Excessive side gear to case clearance.

The source of a snap or a clunk noise can be determined with the assistance of a helper. Raise the vehicle on a hoist with the wheels free to rotate. Instruct the helper to shift the transmission into gear. Listen for the noise, a mechanics stethoscope is helpful in isolating the source of a noise.

DIAGNOSTIC CHART

Condition	Possible Causes	Correction
Wheel Noise	1. Wheel loose.	1. Tighten loose nuts.
	2. Faulty, brinelled wheel bearing.	2. Replace bearing.
Axle Shaft Noise	1. Misaligned axle tube.	1. Inspect axle tube alignment. Correct as necessary.
	2. Bent or sprung axle shaft.	2. Inspect and correct as necessary.
Axle Shaft Broke	1. Misaligned axle tube.	1. Replace the broken shaft after correcting tube mis-alignment.
	2 Vehicle overloaded.	2. Replace broken shaft and avoid excessive weight on vehicle.
	3. Erratic clutch operation.	3. Replace broken shaft and avoid or correct erratic clutch operation.
	4. Grabbing clutch.	4. Replace broken shaft and inspect and repair clutch as necessary.

REAR AXLE - 216RBI (Continued)

Condition	Possible Causes	Correction
Differential Cracked	1. Improper adjustment of the differential bearings.	1. Replace case and inspect gears and bearings for further damage. Set differential bearing pre-load properly.
	2. Excessive ring gear backlash.	2. Replace case and inspect gears and bearings for further damage. Set ring gear backlash properly.
	3. Vehicle overloaded.	3. Replace case and inspect gears and bearings for further damage. Avoid excessive vehicle weight.
	4. Erratic clutch operation.	4. Replace case and inspect gears and bearings for further damage. Avoid erratic use of clutch.
Differential Gears Scored	1. Insufficient lubrication.	1. Replace scored gears. Fill differential with the correct fluid type and quantity.
	2. Improper grade of lubricant.	2. Replace scored gears. Fill differential with the correct fluid type and quantity.
	3. Excessive spinning of one wheel/tire.	3. Replace scored gears. Inspect all gears, pinion bores, and shaft for damage. Service as necessary.
Loss Of Lubricant	1. Lubricant level too high.	1. Drain lubricant to the correct level.
	2. Worn axle shaft seals.	2. Replace seals.
	3. Cracked differential housing.	3. Repair as necessary.
	4. Worn pinion seal.	4. Replace seal.
	5. Worn/scored yoke.	5. Replace yoke and seal.
	6. Axle cover not properly sealed.	6. Remove, clean, and re-seal cover.
Axle Overheating	1. Lubricant level low.	1. Fill differential to correct level.
	2. Improper grade of lubricant.	2. Fill differential with the correct fluid type and quantity.
	3. Bearing pre-loads too high.	3. Re-adjust bearing pre-loads.
	4. Insufficient ring gear backlash.	4. Re-adjust ring gear backlash.

REAR AXLE - 216RBI (Continued)

Condition	Possible Causes	Correction
Gear Teeth Broke	1. Overloading.	1. Replace gears. Examine other gears and bearings for possible damage.
	2. Erratic clutch operation.	2. Replace gears and examine the remaining parts for damage. Avoid erratic clutch operation.
	3. Ice-spotted pavement.	3. Replace gears and examine remaining parts for damage.
	4. Improper adjustments.	4. Replace gears and examine remaining parts for damage. Ensure ring gear backlash is correct.
Axle Noise	1. Insufficient lubricant.	1. Fill differential with the correct fluid type and quantity.
	2. Improper ring gear and pinion adjustment.	2. Check ring gear and pinion contact pattern. Adjust backlash or pinion depth.
	3. Unmatched ring gear and pinion.	3. Replace gears with a matched ring gear and pinion.
	4. Worn teeth on ring gear and/or pinion.	4. Replace ring gear and pinion.
	5. Loose pinion bearings.	5. Adjust pinion bearing pre-load.
	6. Loose differential bearings.	6. Adjust differential bearing pre-load.
	7. Mis-aligned or sprung ring gear.	7. Measure ring gear run-out. Replace components as necessary.
	8. Loose differential bearing cap bolts.	8. Inspect differential components and replace as necessary. Ensure that the bearing caps are torqued tot he proper specification.
	9. Housing not machined properly.	9. Replace housing.

REAR AXLE - 216RBI (Continued)

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Position a suitable lifting device under the axle.
(3) Secure axle to device.
(4) Remove brake components.
(5) Remove brake hose at the axle junction block.
(6) Remove vent hose from axle tube.
(7) If equipped with Locker differential disconnect pressure hose from differential housing and locker indicator connector (Fig. 1).

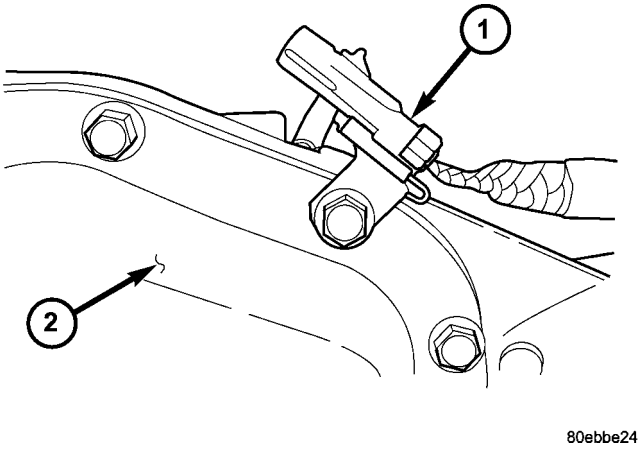


Fig. 1 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(8) Remove propeller shaft.
(9) Remove stabilizer bar links from axle.
(10) Remove shock absorbers from axle (Fig. 2).

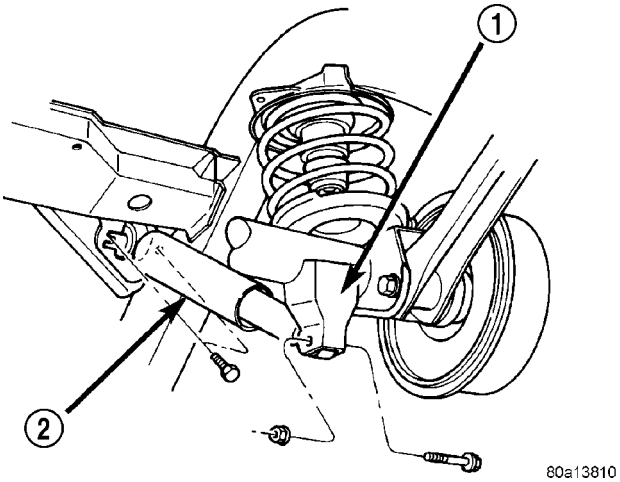


Fig. 2 SHOCK ABSORBER

1 - AXLE BRACKET
2 - SHOCK

(11) Remove track bar from axle (Fig. 3).

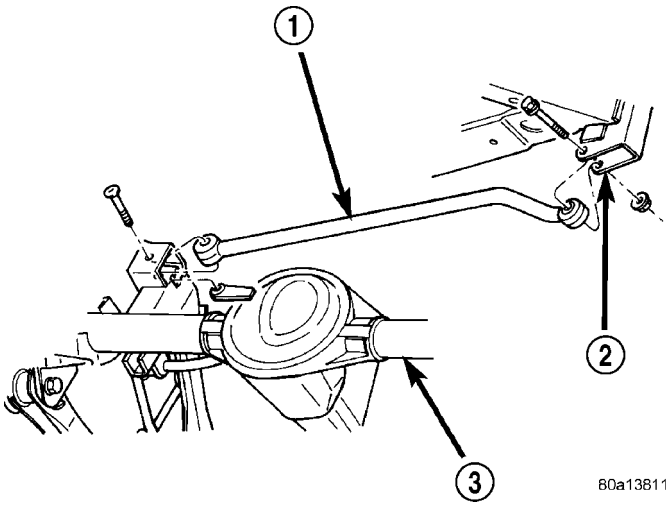


Fig. 3 REAR TRACK BAR

1 - TRACK BAR
2 - FRAME BRACKET
3 - REAR AXLE

(12) Remove upper and lower suspension arms from the axle brackets.
(13) Separate the axle from the vehicle.

INSTALLATION

CAUTION: The weight of the vehicle must be supported by the springs before suspension arms and track bar fasteners can be tightened. Failure to heed caution may result in damage.

(1) Raise axle with lifting device and align coil springs.
(2) Position upper and lower suspension arms on the axle brackets. Install nuts and bolts, do not tighten bolts at this time.
(3) Install track bar and attachment bolts, do not tighten bolts at this time.
(4) Install shock absorbers and tighten nuts to torque specification.
(5) Install stabilizer bar links and tighten nuts to torque specification.
(6) Install the wheel speed sensors, if necessary.
(7) Install brake components.
(8) Install brake hose to the axle junction block and bleed brakes.
(9) Install axle vent hose.
(10) If equipped with Locker differential connect pressure hose and Locker indicator swith.
(11) Install propeller shaft.
(12) Remove lifting device from axle and lower the vehicle.
(13) Tighten upper and lower control arms nuts to torque specification (Fig. 4).
(14) Tighten track bar bolts to torque specification.

REAR AXLE - 216RBI (Continued)

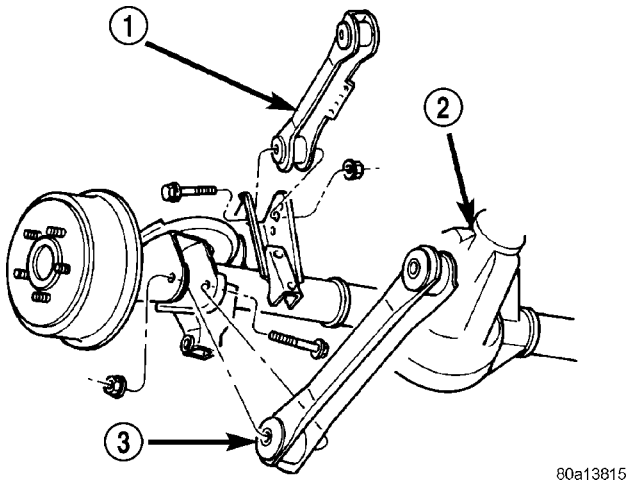


Fig. 4 UPPER AND LOWER SUSPENSION ARMS

1 - UPPER SUSPENSION ARM
2 - REAR AXLE
3 - LOWER SUSPENSION ARM

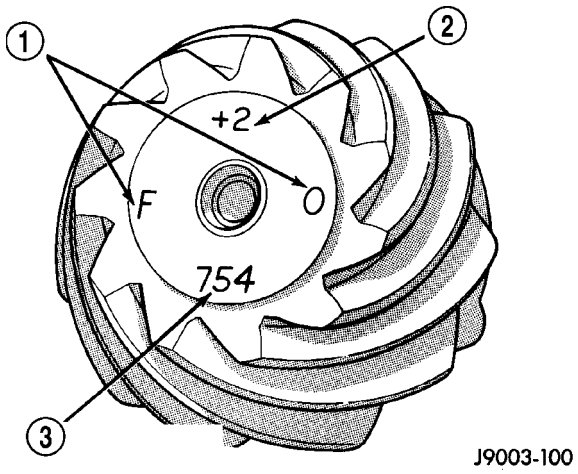


Fig. 5 PINION GEAR ID NUMBERS

1 - PRODUCTION NUMBERS
2 - DRIVE PINION GEAR DEPTH VARIANCE
3 - GEARS MATCHING NUMBER

ADJUSTMENTS

RING AND PINION GEAR

Ring and pinion gears are supplied as matched sets only. The identifying numbers for the ring and pinion gear are etched into the face of each gear (Fig. 5). A plus (+) number, minus (–) number or zero (0) is etched into the face of the pinion gear. This number is the amount (in thousandths of an inch) the depth varies from the standard depth setting of a pinion etched with a (0). The standard depth provides the best teeth contact pattern. Refer to Backlash and Contact Pattern Analysis Paragraph in this section for additional information.

Compensation for pinion depth variance is achieved with select shims. The shims are placed under the inner pinion bearing on the pinion shaft.

If a new gear set is being installed, note the depth variance etched into both the original and replacement pinion gear. Add or subtract the thickness of the original depth shims to compensate for the difference in the depth variances. Refer to the Depth Variance charts.

Note where Old and New Pinion Marking columns intersect. The intersecting figure represents plus or minus the amount needed.

Note the etched number on the face of the drive pinion gear (–1, –2, 0, +1, +2, etc.). The numbers represent thousands of an inch deviation from the standard. If the number is negative, add that value to the required thickness of the depth shim(s). If the number is positive, subtract that value from the thickness of the depth shim(s). If the number is 0 no change is necessary. Refer to the Pinion Gear Depth Variance Chart.

REAR AXLE - 216RBI (Continued)

PINION GEAR DEPTH VARIANCE

Original Pinion Gear Depth Variance	Replacement Pinion Gear Depth Variance								
	−4	−3	−2	−1	0	+1	+2	+3	+4
+4	+0.008	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0
+3	+0.007	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001
+2	+0.006	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002
+1	+0.005	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003
0	+0.004	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004
−1	+0.003	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005
−2	+0.002	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006
−3	+0.001	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007
−4	0	−0.001	−0.002	−0.003	−0.004	−0.005	−0.006	−0.007	−0.008

PINION DEPTH MEASUREMENT AND ADJUSTMENT

Measurements are taken with pinion cups and pinion bearings installed in housing. Take measurements with Pinion Gauge Set 6730 and Dial Indicator C-3339 (Fig. 6).

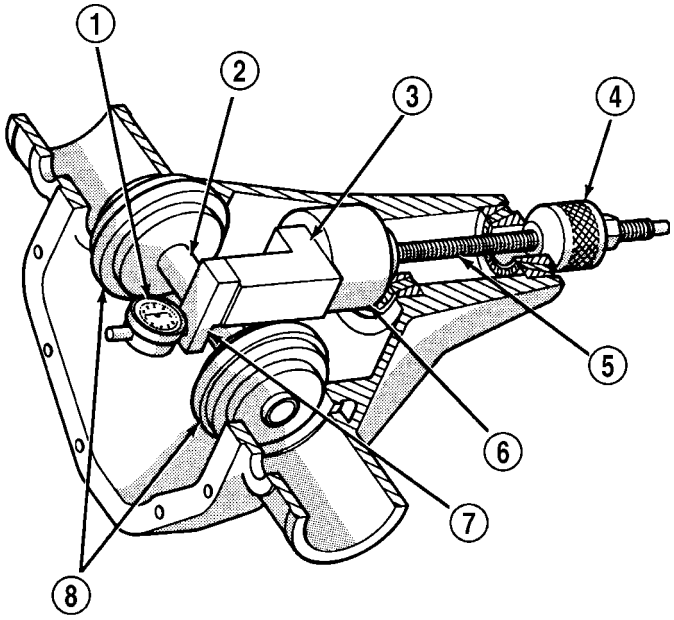


Fig. 6 PINION DEPTH GAUGE TOOLS

1 - DIAL INDICATOR
2 - ARBOR
3 - PINION HEIGHT BLOCK
4 - CONE
5 - SCREW
6 - PINION BLOCK
7 - SCOOTER BLOCK
8 - ARBOR DISC

(1) Assemble Pinion Height Block 6739, Pinion Block 6734 and rear pinion bearing onto Screw 6741 (Fig. 6).

(2) Insert assembled height gauge components, rear bearing and screw into axle housing through pinion bearing cups (Fig. 7).

(3) Install front pinion bearing and Cone 6740 hand tight.

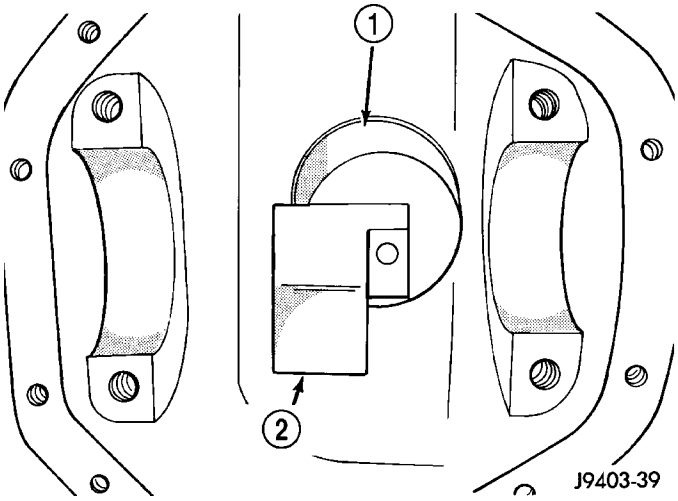


Fig. 7 PINION HEIGHT BLOCK

1 - PINION BLOCK
2 - PINION HEIGHT BLOCK

REAR AXLE - 216RBI (Continued)

(4) Place Arbor Disc 6732 on Arbor D-115-3 in position in the housing side bearing cradles (Fig. 8). Install differential bearing caps on Arbor Discs and tighten cap bolts to 41 N·m (30 ft. lbs.).

NOTE: Arbor Discs 6732 have different step diameters to fit other axle sizes. Pick correct size step for axle being serviced.

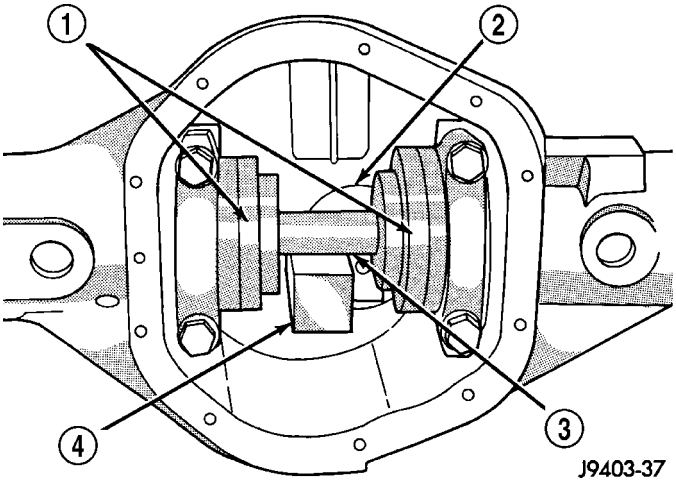


Fig. 8 GAUGE TOOLS IN HOUSING

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(5) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(6) Place Scooter Block/Dial Indicator in position in axle housing so dial probe and scooter block are flush against the rearward surface of the pinion height block. Hold scooter block in place and zero the dial indicator face to the pointer. Tighten dial indicator face lock screw.

(7) Slowly slide the Scooter Block/Dial Indicator across the pinion height block over to the arbor bar (Fig. 9). Move the Scooter Block/Dial Indicator to the crest of the arbor bar and record the highest reading.

(8) Select a shim equal to the thickest dial indicator reading plus or minus the drive pinion gear depth variance number etched in the face of the pinion gear (Fig. 5).

(9) Remove the pinion depth gauge components from the axle housing.

PRELOAD SHIM SELECTION

Differential side bearing preload and gear backlash is achieved by selective shims inserted between the bearing cups and the housing. The proper shim thickness can be determined using slip-fit dummy bearings D-345 in place of the differential side bearings and a dial indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion gear for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion gear is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 10).

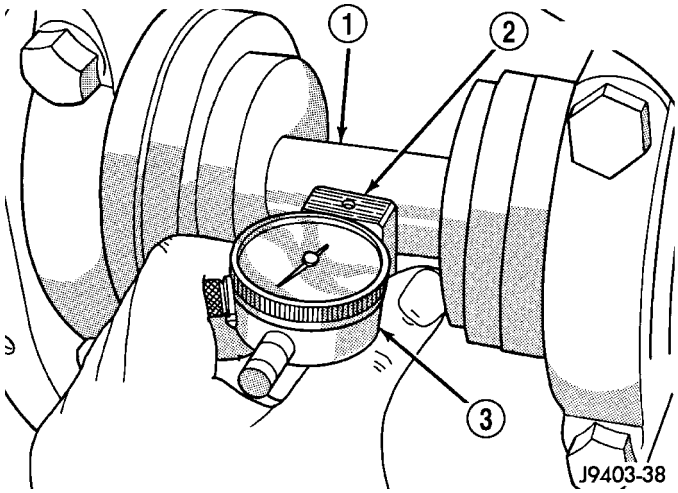


Fig. 9 PINION DEPTH MEASUREMENT

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

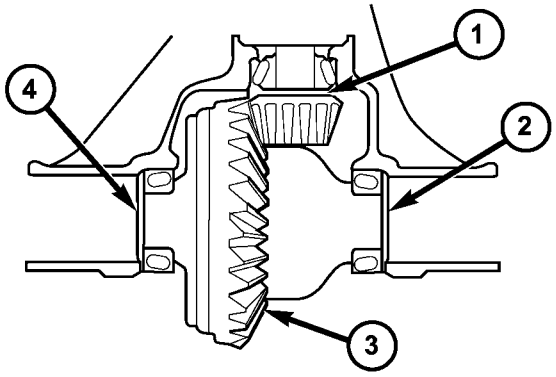


Fig. 10 ADJUSTMENT SHIM LOCATIONS

1 - PINION DEPTH SHIM
2 - DIFFERENTIAL BEARING PRELOAD SHIM
3 - RING GEAR
4 - DIFFERENTIAL BEARING PRELOAD SHIM

REAR AXLE - 216RBI (Continued)

NOTE: It is difficult to salvage the differential side bearings during the removal procedure. Install replacement bearings if necessary.

(1) Remove differential side bearings from differential case.
(2) Remove factory installed shims from differential case.
(3) Install ring gear on differential case and tighten bolts to specification.
(4) Install dummy side bearings D-345 on differential case.
(5) Install differential case in axle housing (Fig. 11).
(6) Install the marked bearing caps in their correct positions. Install and snug the bolts (Fig. 12).

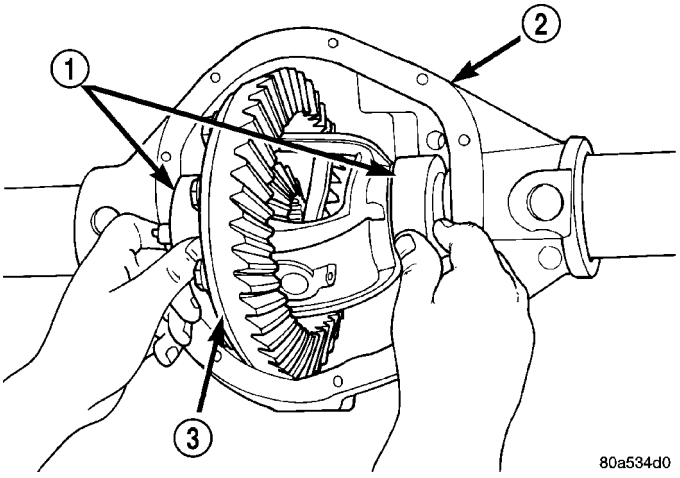


Fig. 11 DIFFERENTIAL DUMMY BEARINGS

1 - DUMMY BEARINGS
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

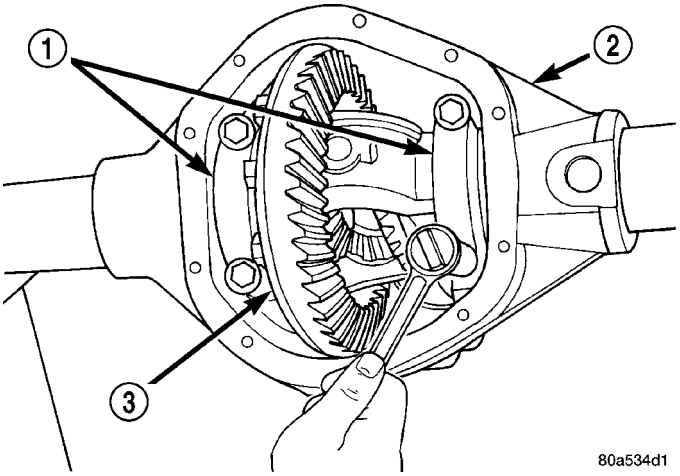


Fig. 12 BEARING CAP BOLTS

1 - BEARING CAP
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

(7) Using a dead-blow hammer seat the differential dummy bearings to each side of the housing (Fig. 13) and (Fig. 14).

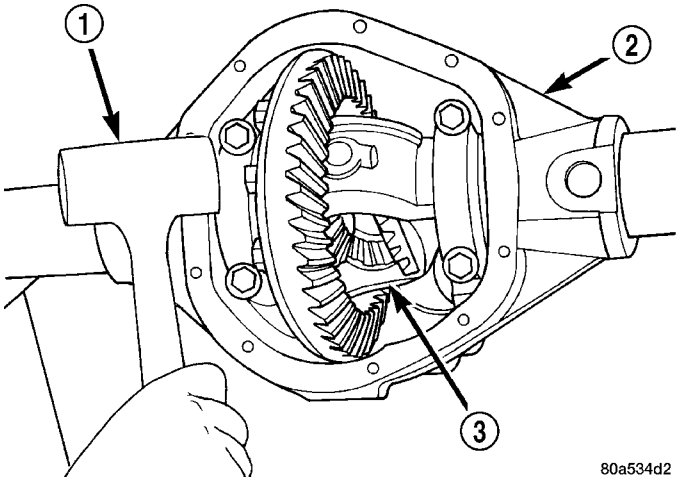


Fig. 13 SEAT DUMMY BEARING PINION SIDE

1 - MALLET
2 - DIFFERENTIAL HOUSING
3 - DIFFERENTIAL CASE

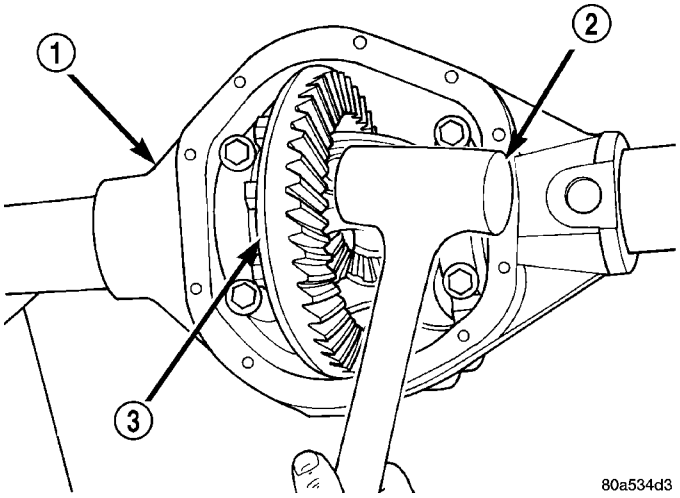


Fig. 14 SEAT DUMMY BEARING RING GEAR SIDE

1 - DIFFERENTIAL HOUSING
2 - MALLET
3 - DIFFERENTIAL CASE

REAR AXLE - 216RBI (Continued)

(8) Thread Pilot Stud C-3288-B into rear cover bolt hole below ring gear (Fig. 15).

(9) Attach a dial indicator C-3339 to Pilot Stud C-3288-B. Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 15).

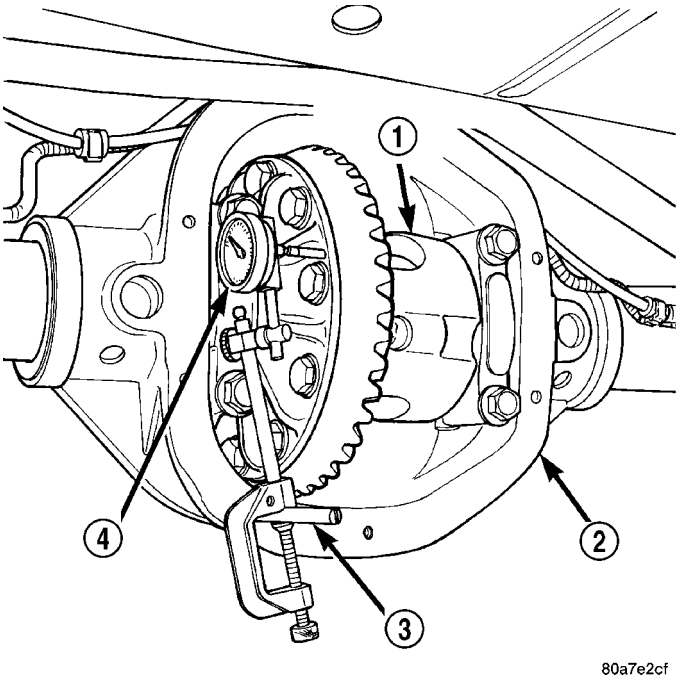


Fig. 15 DIFFERENTIAL SIDE PLAY MEASUREMENT

1 - DIFFERENTIAL CASE
2 - DIFFERENTIAL HOUSING
3 - PILOT STUD
4 - DIAL INDICATOR

(10) Push and hold differential case to pinion gear side of the housing and zero dial indicator (Fig. 16).

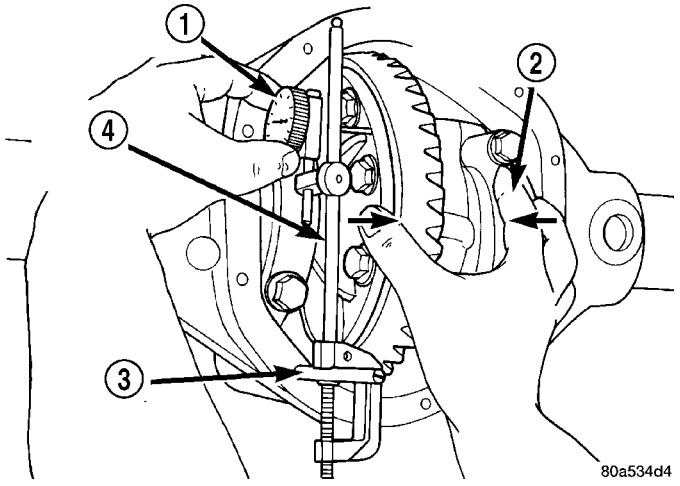


Fig. 16 ZERO DIAL INDICATOR

1 - DIAL INDICATOR
2 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PILOT STUD
4 - EXTENSION

(11) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 17).

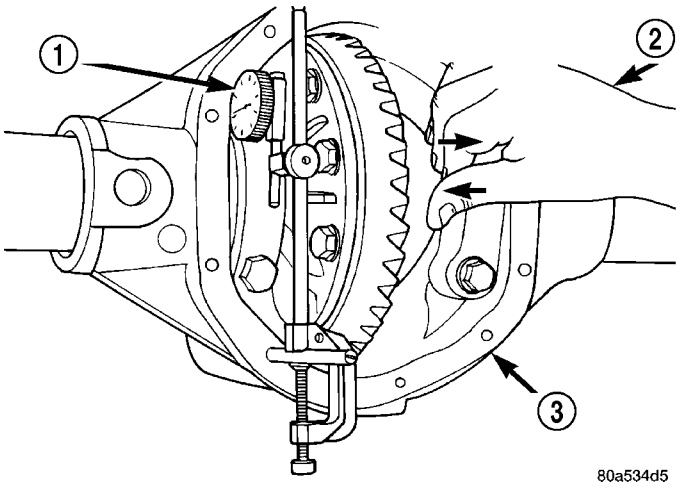


Fig. 17 RECORD DIAL INDICATOR READING

1 - READ DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(12) Add dial indicator reading to the starting point shim thickness. This is the total shim thickness to achieve zero differential end play.

(13) Add 0.254 mm (0.010 in.) to the zero end play total. This new total represents the thickness of shims to compress, or preload the new bearings when the differential is installed.

(14) Rotate dial indicator out of the way on pilot stud.

(15) Remove differential case and dummy bearings from the housing.

(16) Install pinion gear in the housing. Install the pinion yoke and establish the correct pinion rotating torque.

(17) Install differential case and dummy bearings D-345 in the housing (without shims), install bearing caps and tighten bolts snug.

(18) Seat ring gear side dummy bearing (Fig. 14).

(19) Position the dial indicator plunger on a flat surface between the ring gear bolt heads. (Fig. 15).

(20) Push and hold differential case toward the pinion gear and zero dial indicator (Fig. 18).

(21) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 19).

(22) Subtract 0.05 mm (0.002 in.) from the dial indicator reading to compensate for backlash between ring and pinion gears. This total is the thickness shim required to achieve proper backlash.

(23) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

REAR AXLE - 216RBI (Continued)

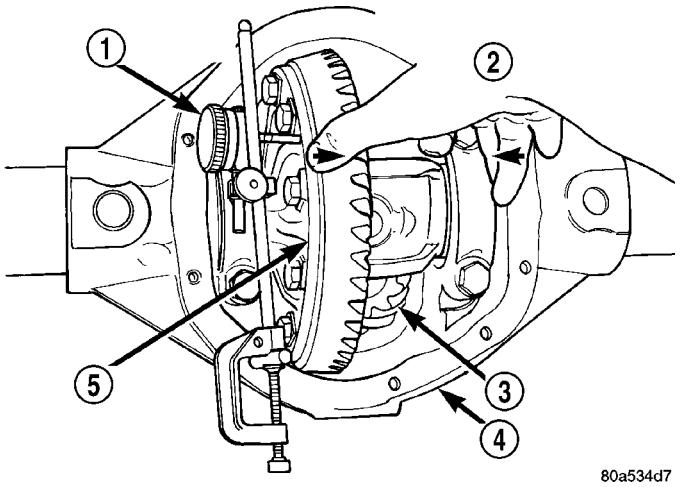


Fig. 18 ZERO DIAL INDICATOR

1 - DIAL INDICATOR FACE
2 - DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

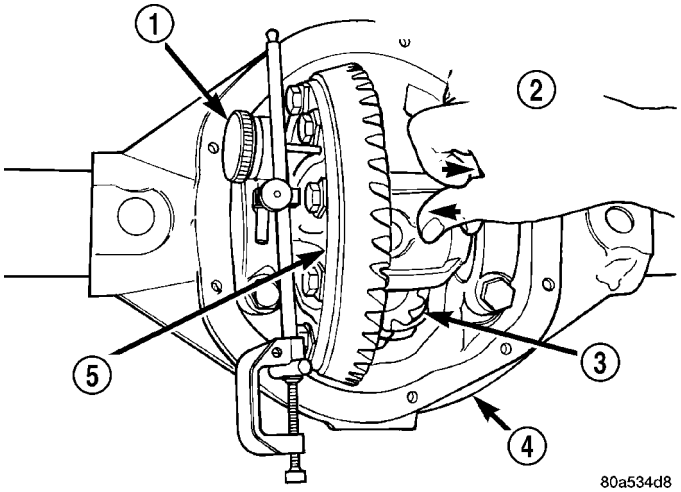


Fig. 19 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(24) Rotate dial indicator out of the way on pilot stud.

(25) Remove differential case and dummy bearings from the housing.

(26) Install side bearing shims on differential case hubs.

(27) Install new side bearing cones and cups on differential case.

(28) Install spreader W-129-B on the housing and spread axle opening enough to receive differential case.

(29) Install differential case in the housing.

(30) Install differential bearing caps loosely.

(31) Remove spreader from the housing.

(32) Tighten bearing caps bolts to 108 N·m (80 ft. lbs.).

(33) Rotate the differential case several times to seat the side bearings.

(34) Position the indicator plunger against a ring gear tooth (Fig. 20).

(35) Push and hold ring gear upward while not allowing the pinion gear to rotate and zero dial indicator.

(36) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 21).

(37) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform Gear Contact Pattern Analysis procedure.

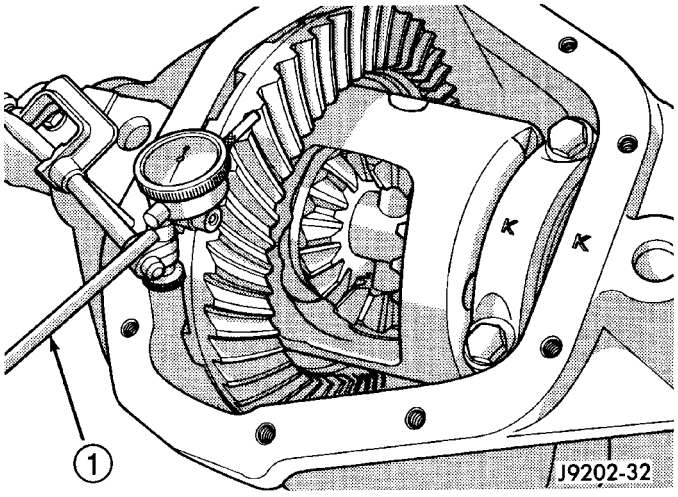


Fig. 20 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth

REAR AXLE - 216RBI (Continued)

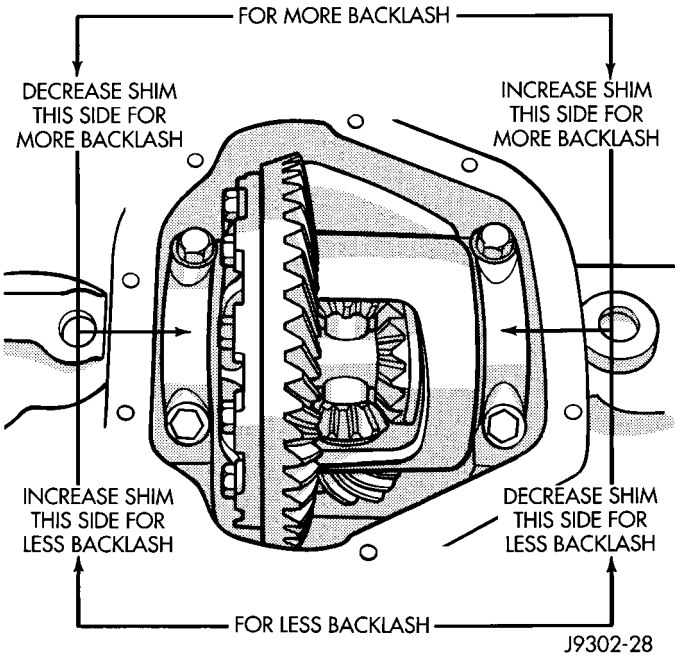


Fig. 21 BACKLASH SHIM ADJUSTMENT

at the outer-end. The ROOT of the gear tooth is the lowest portion of the tooth (Fig. 22).

NOTE: If the PROFILE across the tooth is the same it is a 3 Axis cut gear. If the PROFILE across the tooth is tapered it is a 2 Axis cut gear.

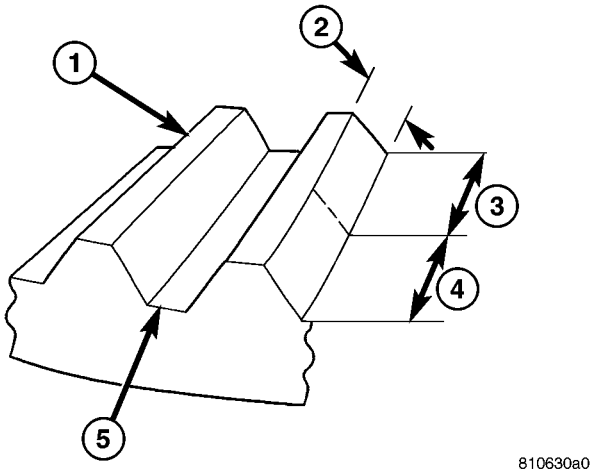


Fig. 22 GEAR DESCRIPTION

1 - TOP LAND
2 - PROFILE
3 - TOE
4 - HEEL
5 - ROOT

(1) Apply a thin coat of hydrated ferric oxide or equivalent to the drive and coast side of the ring gear teeth.

(2) Wrap, twist, and hold a shop towel around the pinion yoke to increase the turning resistance of the pinion. This will provide a more distinct contact pattern.

(3) Using a boxed end wrench on a ring gear bolt, Rotate the differential case one complete revolution in both directions while a load is being applied from shop towel.

The areas on the ring gear teeth with the greatest degree of contact against the pinion teeth will squeegee the compound to the areas with the least amount of contact. Note and compare patterns on the ring gear teeth to Gear Tooth Contact Patterns chart (Fig. 23) and (Fig. 24) and adjust pinion depth and gear backlash as necessary.

DIFFERENTIAL BEARING PRELOAD CHECK

The final check on the differential assembly before installing the axles is torque to rotate pinion and differential combined. This will verify the correct differential bearing preload.

Torque to rotate the differential and pinion should be the torque to rotate the pinion plus 0.79-1.24 N·m (7-11 in. lbs.).

REAR AXLE - 216RBI (Continued)

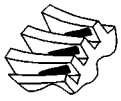
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	Desirable pattern. The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top heel contact		**Top toe contact**	Backlash correct. Thicker pinion position shim required.
	Root toe contact		**Root heel contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top heel contact**	Pinion position shim correct. Decrease backlash.
	**Root toe contact**		**Root toe contact**	Pinion position shim correct. Increase backlash.

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Fig. 23 PATTERN INTERPRETATION (GEAR CUT 2 AXIS)

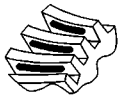
DRIVE SIDE HEEL TOE	CONDITION	COAST SIDE TOE HEEL	CONDITION	ACTION REQUIRED
	**Desirable pattern.** The drive pattern should be centered on the tooth. There should be some clearance between the pattern and the top of the tooth.		**Desirable pattern.** The coast pattern should be centered on the tooth, but may be slightly toward the toe. There should be some clearance between the pattern and the top of the tooth.	None
	Top toe contact		**Top heel contact**	Backlash correct. Thicker pinion position shim required.
	Root heel contact		**Root toe contact**	Backlash correct. Thinner pinion position shim required.
	Top heel contact		**Top toe contact**	Pinion position shim correct. Decrease backlash.
	Root toe contact		**Root heel contact**	Pinion position shim correct. Increase backlash.

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Fig. 24 PATTERN INTERPRETATION (GEAR CUT 3 AXIS)

REAR AXLE - 216RBI (Continued)

SPECIFICATIONS

REAR AXLE

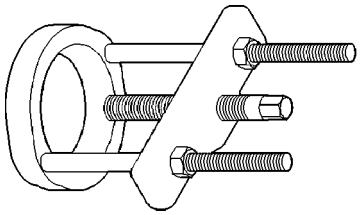
AXLE SPECIFICATIONS

DESCRIPTION	SPECIFICATION
Axle Ratio	3.07, 3.73, 4.10
Ring Gear Backlash	0.12-0.20 mm (0.005-0.008 in.)
Pinion Bearing Preload - Original Bearings	1-2 N·m (10-20 in. lbs.)
Pinion Bearing Preload - New Bearings	2-4.5 N·m (20-40 in. lbs.)

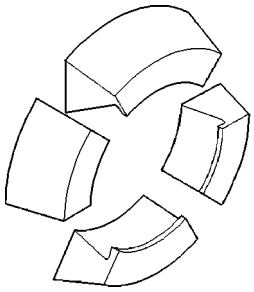
TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Fill Hole Plug	34	25	-
Differential Cover Bolts	41	30	-
Bearing Cap Bolts	108	80	-
Ring Gear Bolts	136	100	-
Pinion Nut	217-271	160-200	-

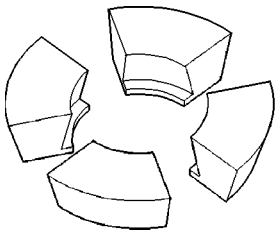
SPECIAL TOOLS



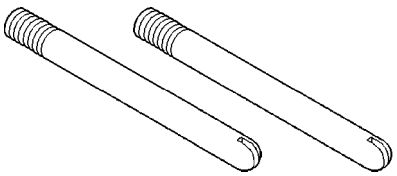
PULLER C-293-PA



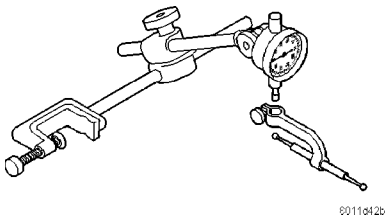
ADAPTERS C-293-18



ADAPTERS C-293-39

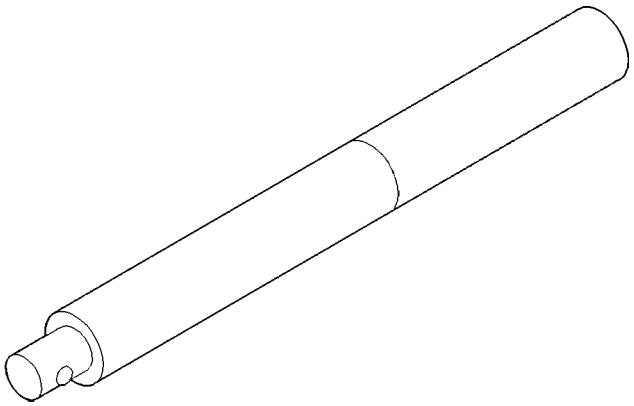


PILOT STUDS C-3288-B

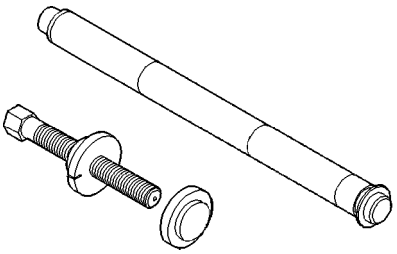


DIAL INDICATOR C-3339

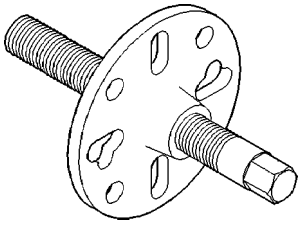
REAR AXLE - 216RBI (Continued)



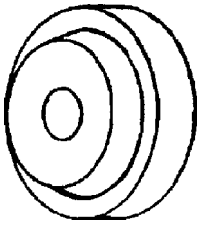
HANDLE C-4171



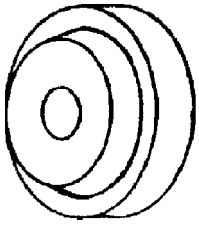
TRAC-LOK TOOLS C-4487



PULLER C-452



INSTALLER D-144



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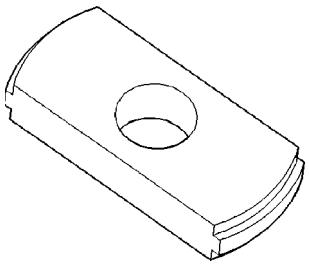
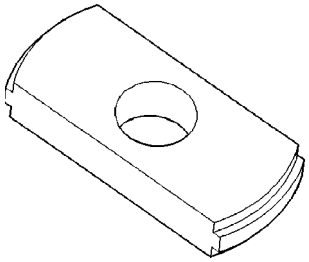
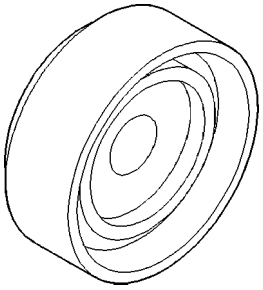


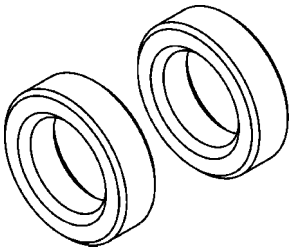
REMOVER D-147



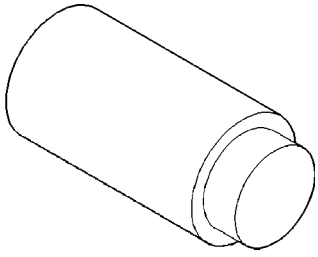
REMOVER D-148



INSTALLER D-156

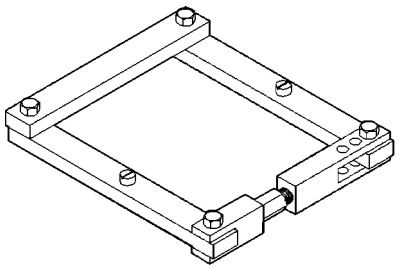


DUMMY BEARINGS D-345

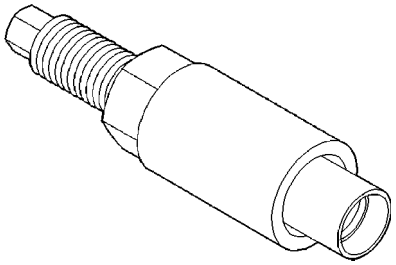


PLUG SP-3289

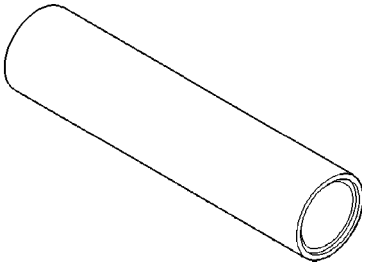
REAR AXLE - 216RBI (Continued)



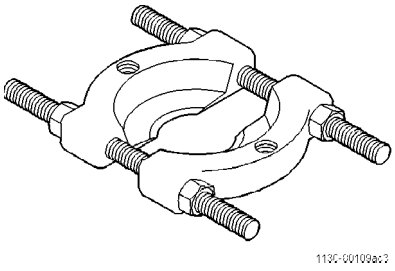
SPREADER W-129-B



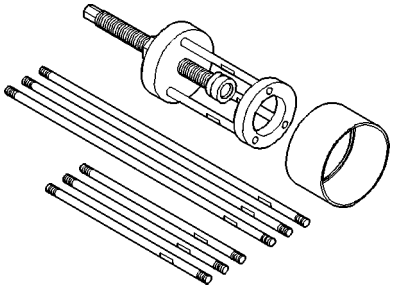
INSTALLER W-162-D



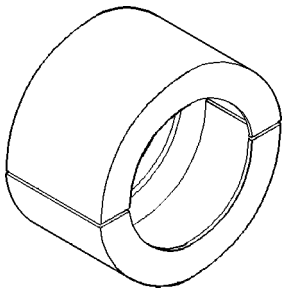
INSTALLER W-262



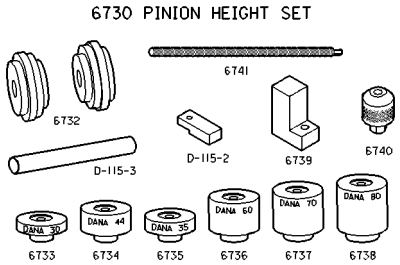
SPLITTER 1130



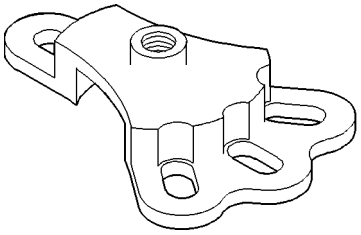
PULLER 6444



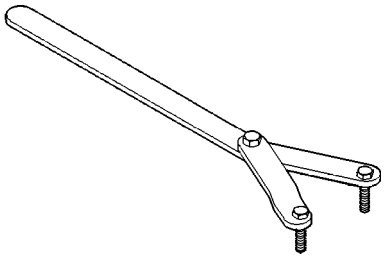
JAWS 6447



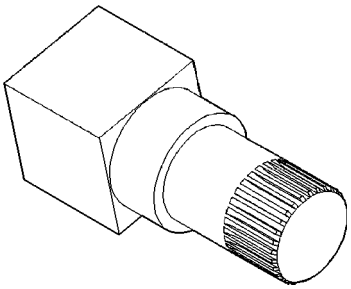
PINION DEPTH SET 6730



PULLER 6790

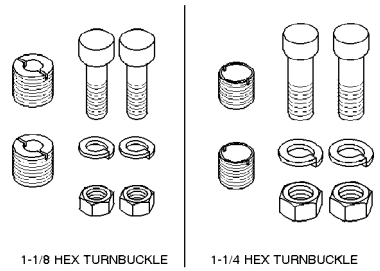


SPANNER WRENCH 6958

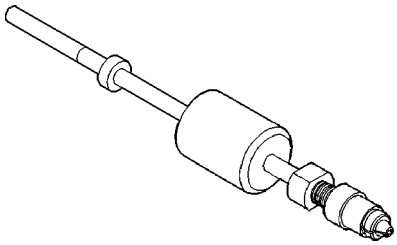


FIXTURE 6963

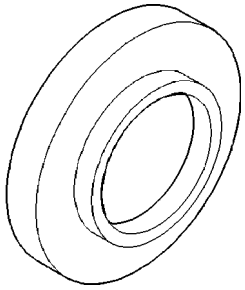
REAR AXLE - 216RBI (Continued)



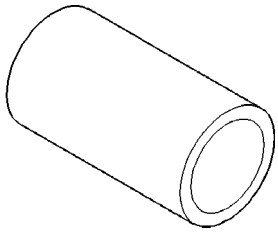
ADAPTER KIT 6987B



SLIDE HAMMER 7420



INSTALLER 7913-A



CUP 8109

AXLE SHAFTS

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Remove brake caliper and rotor.

(3) Remove axle retainer plate nuts through access hole in axle flange (Fig. 25).

(4) Pull axle shaft from the axle. If axle will not come out use Slide Hammer 7420 and Adapter 6790.

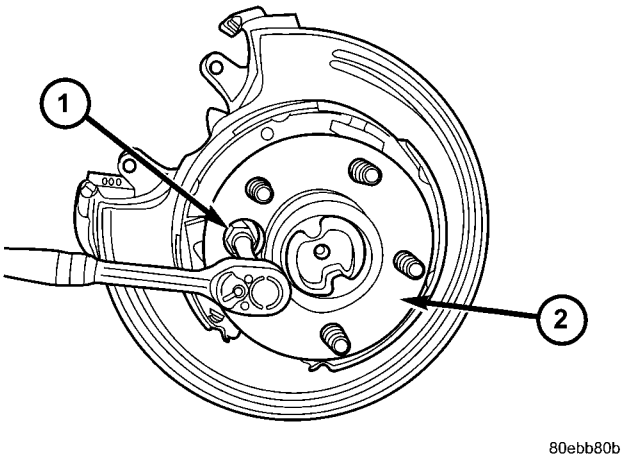


Fig. 25 AXLE SHAFT BOLTS

1 - ACCESS HOLE
2 - AXLE FLANGE

INSTALLATION

WARNING: NEVER REUSE AXLE RETAINING BOLTS AND NUTS. USED TORQUE NUT CAN LOOSEN. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN PERSONAL INJURY.

(1) Install axle into axle tube with the flat area of the retainer plate upward (Fig. 26).

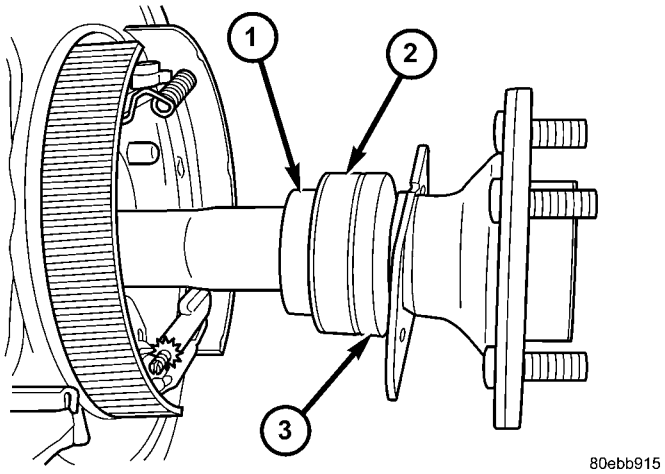


Fig. 26 AXLE SHAFT

1 - RETAINER RING
2 - AXLE BEARING
3 - AXLE SEAL

(2) Install **new** axle retaining bolts.

(3) Install **new** retaining nuts and tighten to 61 N·m (45 ft. lbs.).

(4) Install brake rotor and caliper.

AXLE BEARINGS

REMOVAL

(1) Remove axle shaft from vehicle.

NOTE: It is normal that the axle bearing race is loose in the axle tube.

(2) Drill a shallow hole into soft steel axle bearing retaining ring with a 3/8 in. drill bit (Fig. 27). Use a drill depth stop to avoid marking the axle.

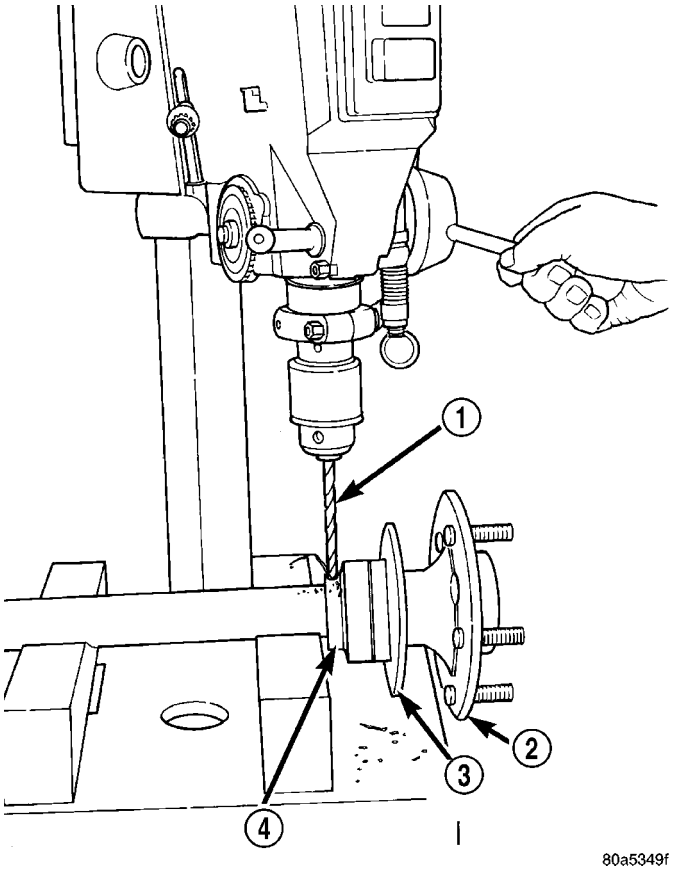


Fig. 27 DRILL RETAINING RING

1 - 3/8 in. DIA. DRILL BIT
2 - AXLE
3 - RETAINING PLATE
4 - SOFT METAL RETAINING RING

(3) With a cold chisel cut the retaining ring across drilled hole. (Fig. 28)

(4) Slide retaining ring from axle shaft.

(5) Remove axle bearing from the shaft with, a press and Splitter 1130 placed between the seal and bearing (Fig. 29).

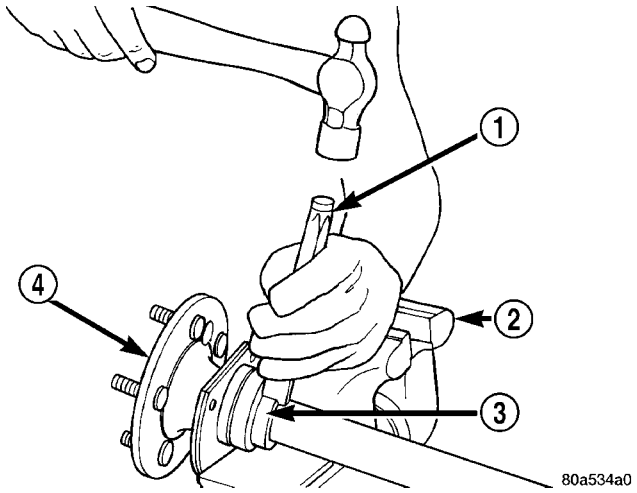


Fig. 28 CUT RETAINING RING

1 - COLD CHISEL
2 - VISE
3 - RETAINING RING
4 - AXLE

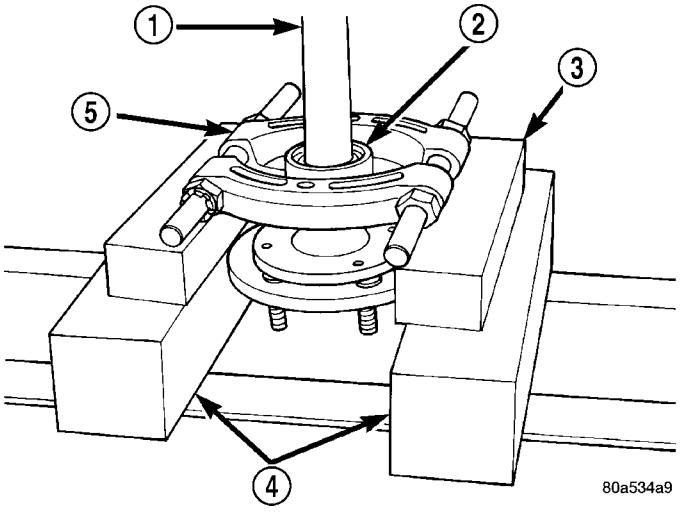


Fig. 29 AXLE BEARING AND SEAL

1 - AXLE
2 - UNIT BEARING
3 - PRESS PLATES
4 - BLOCKS
5 - SPLITTER

(6) Remove seal from axle.

(7) Remove retaining plate from axle shaft.

AXLE BEARINGS (Continued)

INSTALLATION

(1) Verify axle shaft retaining plate is flat with a straight edge.

NOTE: Replace the retaining plate if warped.

(2) Install retaining plate on axle (Fig. 30).
(3) Apply a coat of multi-purpose grease on sealing surface of axle seal.
(4) Install seal on axle with cavity away from retaining plate (Fig. 30).

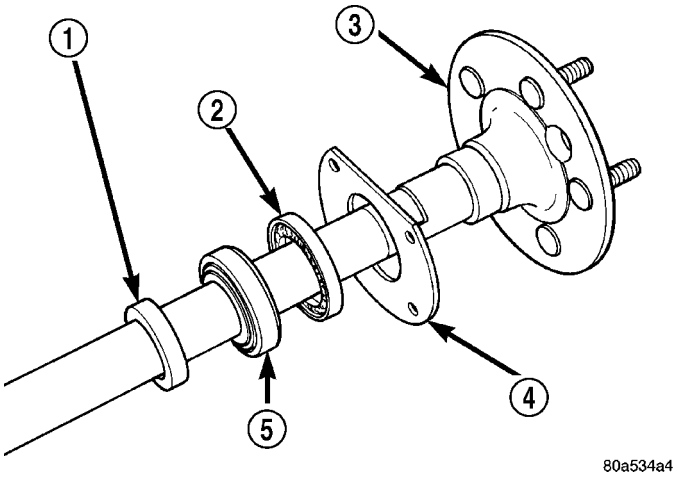


Fig. 30 AXLE BEARING AND SEAL COMPONENTS

1 - RETAINING RING
2 - SEAL
3 - AXLE
4 - RETAINING PLATE
5 - BEARING

(5) Lubricate bearing with Mopar Wheel Bearing Grease, or equivalent. Wipe excess grease from outside of bearing.
(6) Install bearing on the axle shaft with Installer 7913 and a press (Fig. 31).

NOTE: Install bearing with groove on the outer surface toward the seal.

(7) Press soft metal retaining ring onto axle shaft with Installer 7913 and a press (Fig. 32).
(8) Install axle in vehicle.

PINION SEAL

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove brake drums/calipers.
(3) Remove propeller shaft.
(4) Rotate pinion gear three or four times and verify that pinion rotates smoothly.

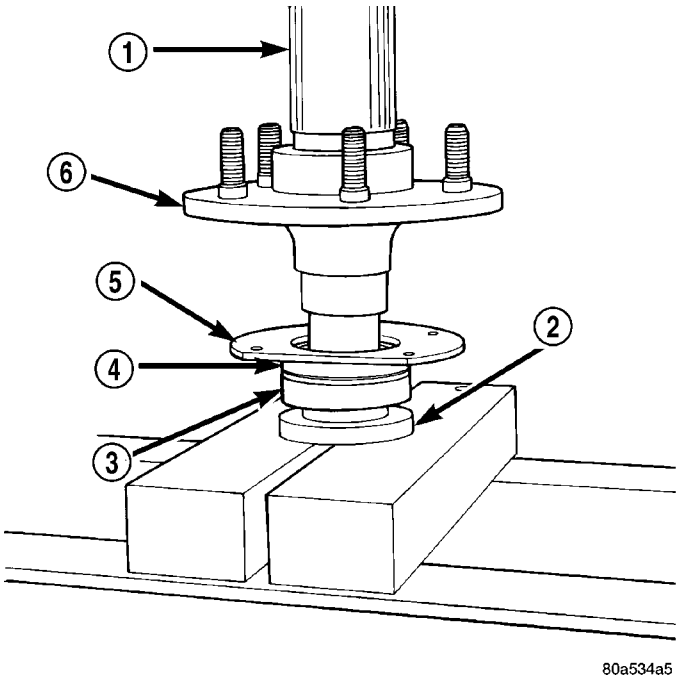


Fig. 31 PRESS BEARING ON AXLE

1 - PRESS RAM
2 - INSTALLER
3 - UNIT BEARING
4 - SEAL
5 - RETAINING PLATE
6 - AXLE

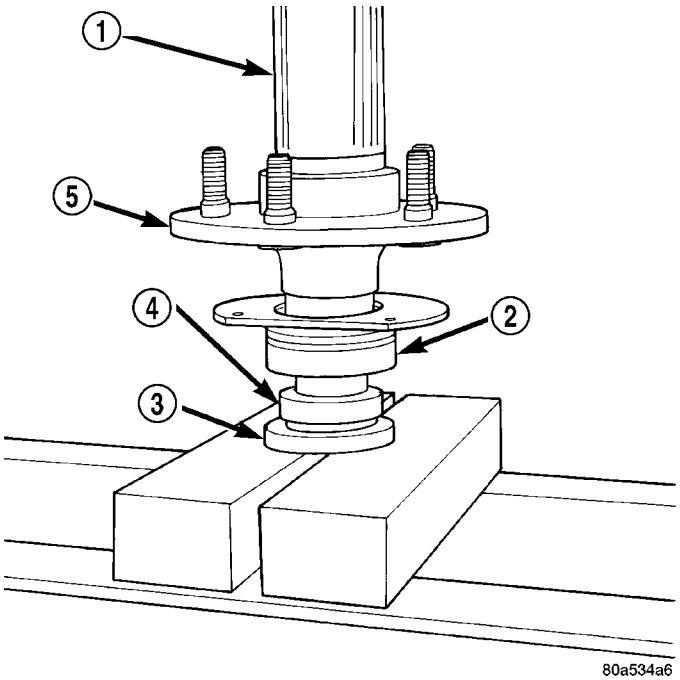


Fig. 32 PRESS RETAINING RING ON AXLE

1 - PRESS RAM
2 - UNIT BEARING
3 - INSTALLER
4 - METAL RETAINING RING
5 - AXLE

PINION SEAL (Continued)

(5) Using Spanner Wrench 6958 to hold the pinion yoke, remove the pinion nut and washer.

(6) Remove pinion yoke with Remover C-452 and Flange Wrench C-3281 (Fig. 33).

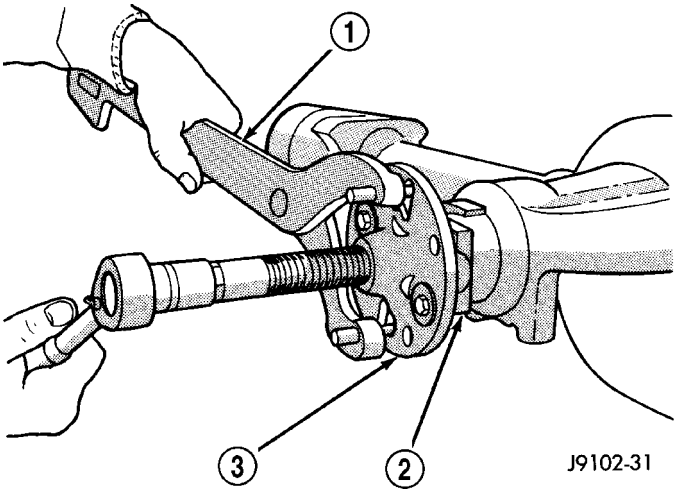


Fig. 33 PINION YOKE REMOVER

1 - FLANGE WRENCH
2 - PINION YOKE
3 - REMOVER

(7) Remove pinion shaft seal with a pry tool or slide hammer mounted screw.

INSTALLATION

(1) Apply a light coating of gear lubricant on the lip of pinion seal. Install seal with Installer 8681 (Fig. 34).

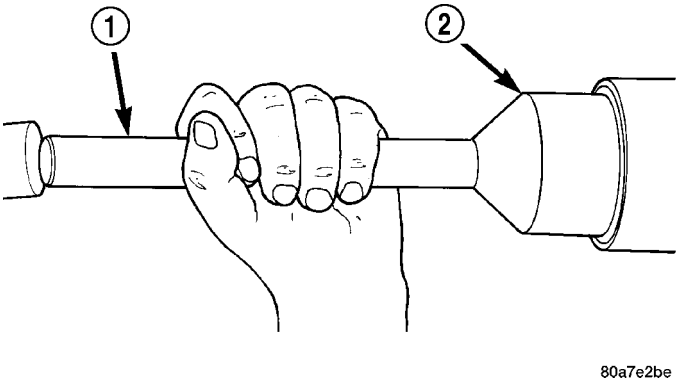


Fig. 34 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

(2) Install yoke on pinion gear with Installer W-162–D, Cup 8109 and Spanner Wrench 6958 (Fig. 35).

(3) Install pinion washer and a **new** nut on the pinion gear and tighten the nut until there is zero bearing end-play.

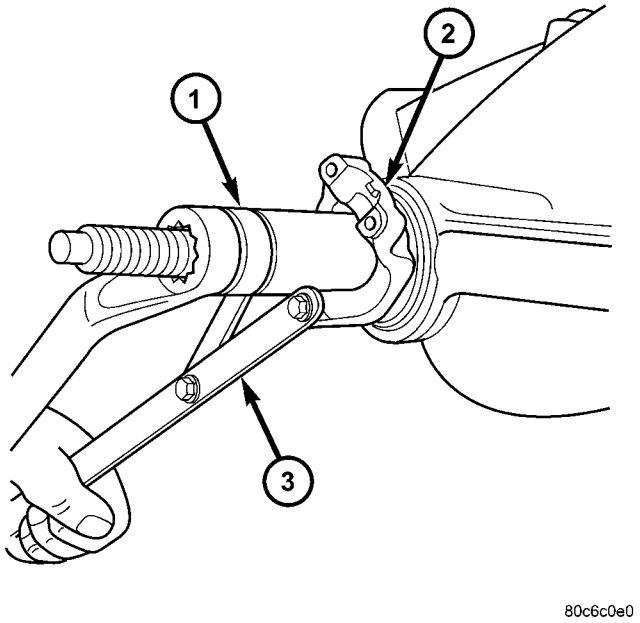


Fig. 35 YOKE SPANNER WRENCH

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

(4) Hold pinion yoke with Spanner Wrench 6958 (Fig. 36) and tighten pinion nut to 217 to 271N·m (160 to 200 ft. lbs.).

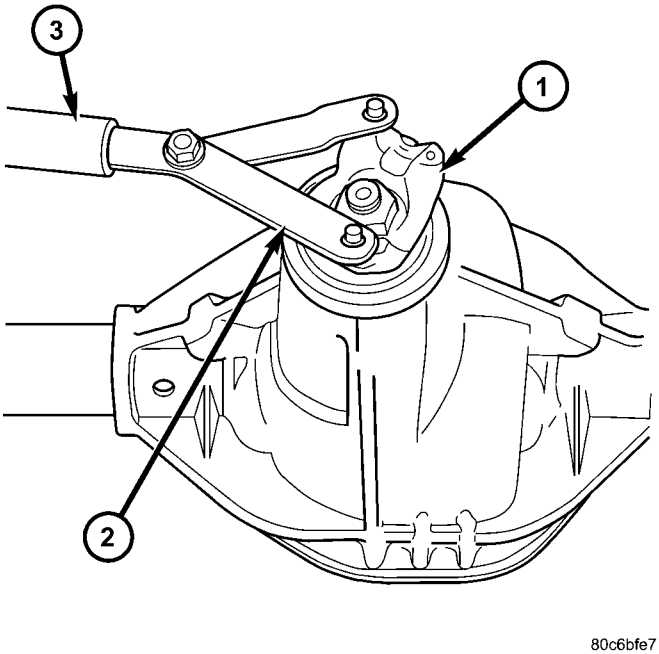


Fig. 36 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(5) Install propeller shaft.

(6) Install brake drums/calipers.

DIFFERENTIAL COVER

REMOVAL

(1) With vehicle in neutral, position vehicle on hoist.

(2) Remove drain plug.

(3) If equipped with Tru-Lok, disconnect indicator switch harness and remove connector from differential cover (Fig. 37).

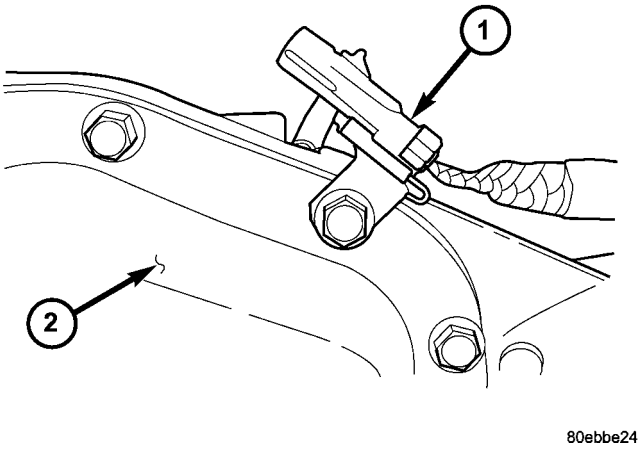


Fig. 37 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(4) Remove cover bolts.

(5) Remove cover and drain lubricant.

INSTALLATION

(1) Apply a 6.35mm (1/4 in.) bead of Mopar Silicone Rubber Sealant or equivalent to the housing cover (Fig. 38).

CAUTION: If housing cover is not installed within 3 to 5 minutes, the cover must be cleaned and new RTV applied. Failure to heed caution may result in damage.

(2) Install cover, identification tag and indicator switch connector if equipped with Tru-Lok (Fig. 39). Tighten cover bolts in a criss-cross pattern to 41 N·m (30 ft. lbs.).

(3) Connect indicator switch harness if equipped with Tru-Lok.

(4) Fill differential to specifications.

(5) Install fill plug and tighten to 34 N·m (25 ft. lbs.).

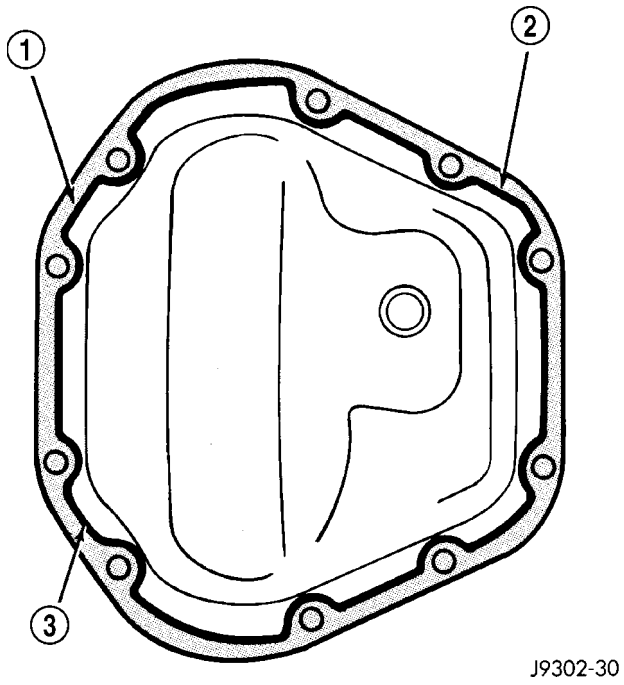


Fig. 38 HOUSING COVER - TYPICAL

1 - SEALANT SURFACE
2 - SEALANT
3 - SEALANT THICKESS

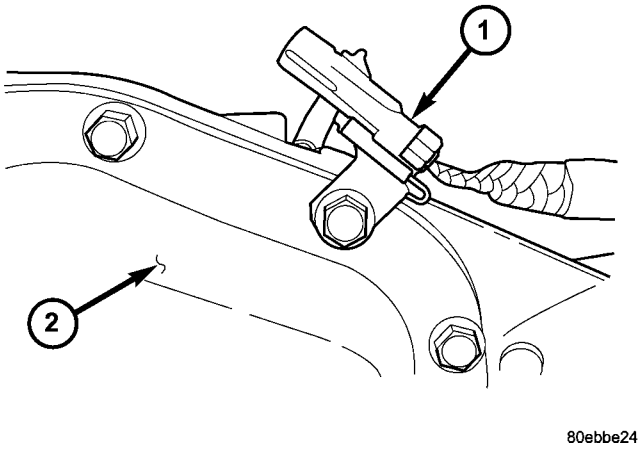


Fig. 39 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

DIFFERENTIAL

REMOVAL

(1) Remove differential housing cover and drain fluid.

(2) Remove axle shafts.

(3) Note the reference letters stamped on the bearing caps and housing machined sealing surface (Fig. 40).

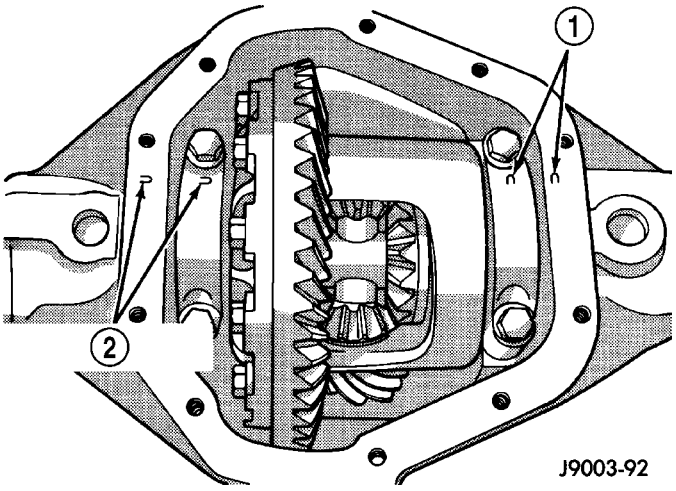


Fig. 40 BEARING CAP REFERENCE

1 - REFERENCE LETTERS
2 - REFERENCE LETTERS

(4) Loosen the differential bearing cap bolts.

(5) Position Spreader W-129-B with Adapter Kit 6987B on differential locating holes (Fig. 41). Install hold-down clamps and tighten the turnbuckle finger-tight.

(6) Install a Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load indicator plunger against the opposite side of the housing (Fig. 42) and zero the indicator.

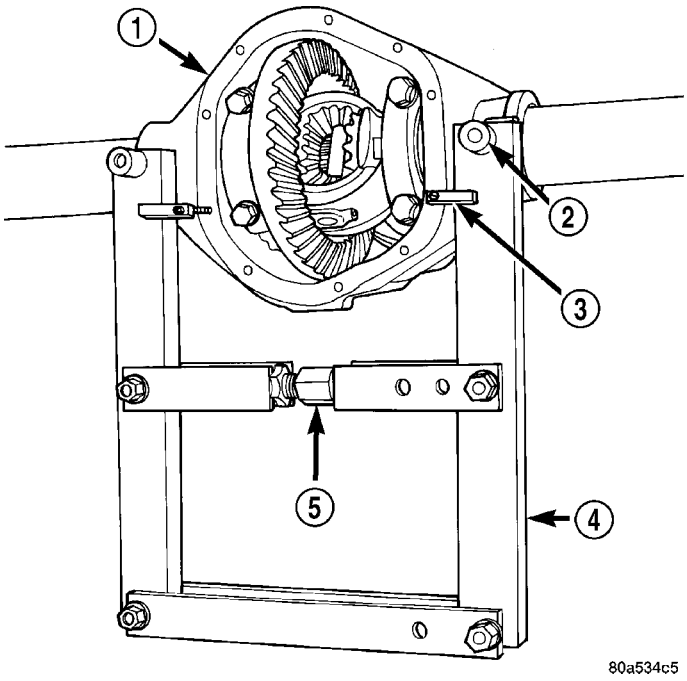


Fig. 41 SPREADER LOCATION

1 - DIFFERENTIAL HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

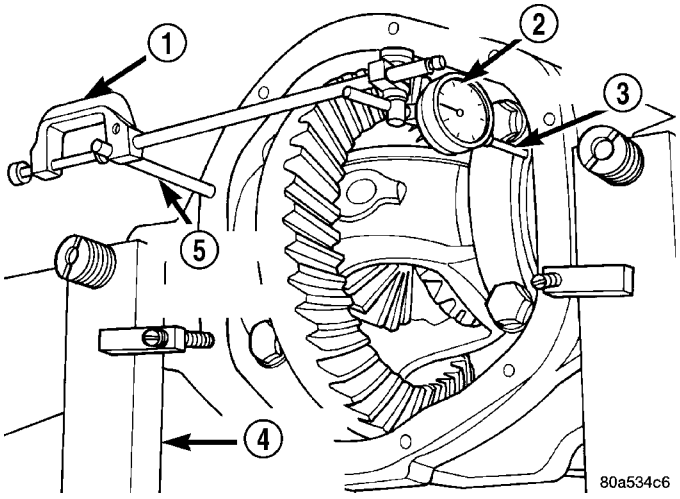


Fig. 42 DIAL INDICATOR LOCATION

1 - CLAMP
2 - DIAL INDICATOR
3 - LEVER ADAPTER
4 - SPREADER
5 - PILOT STUD

DIFFERENTIAL (Continued)

(7) Spread housing while measuring the distance with the dial indicator (Fig. 43).

CAUTION: Never spread the housing over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

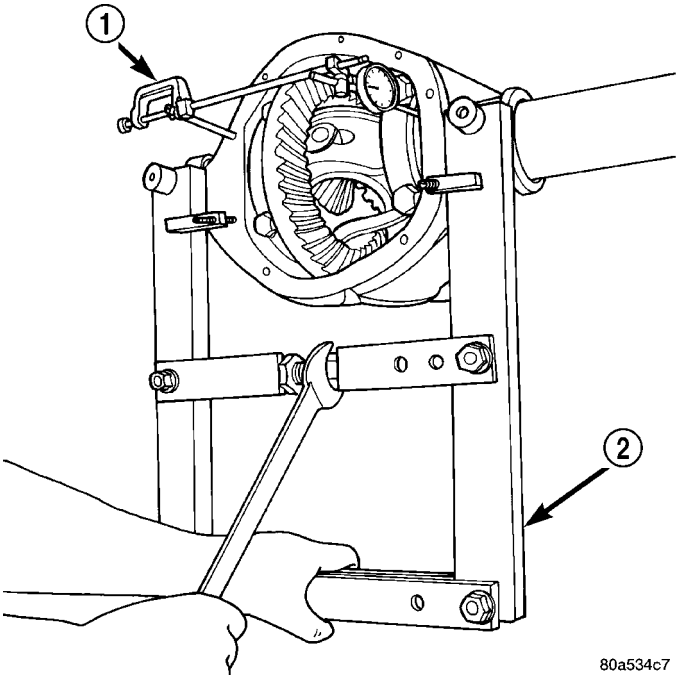


Fig. 43 SPREAD DIFFERENTIAL HOUSING

1 - DIAL INDICATOR
2 - SPREADER

(8) Remove dial indicator.

(9) Hold differential case in position and remove differential bearing caps.

(10) Remove differential from housing and tag differential bearing cups to indicate location (Fig. 44).

(11) Remove spreader from housing.

(12) Clean housing cavity with flushing oil, light engine oil or lint free cloth.

NOTE: Do not use water, steam, kerosene or gasoline for cleaning.

DISASSEMBLY

(1) Remove pinion shaft lock roll pin with a hammer and punch.

(2) Remove pinion shaft.

(3) Rotate differential side gears and remove differential pinions and thrust washers (Fig. 45).

(4) Remove differential side gears and thrust washers.

ASSEMBLY

(1) Install differential side gears and thrust washers.

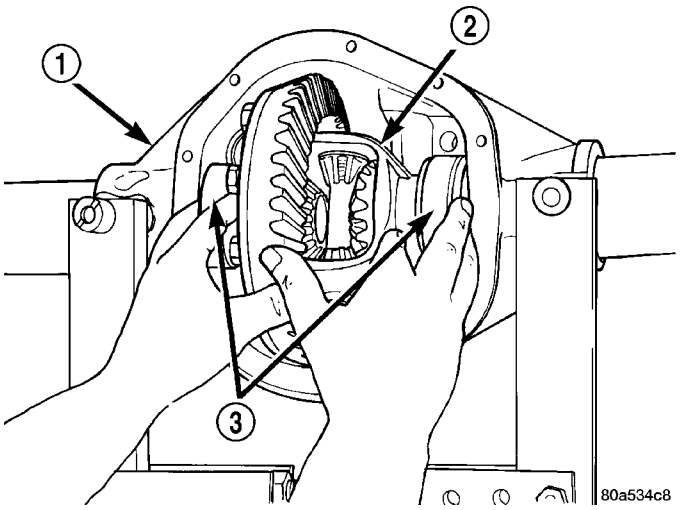


Fig. 44 DIFFERENTIAL CASE REMOVAL

1 - DIFFERENTIAL HOUSING
2 - DIFFERENTIAL CASE
3 - BEARING CUPS

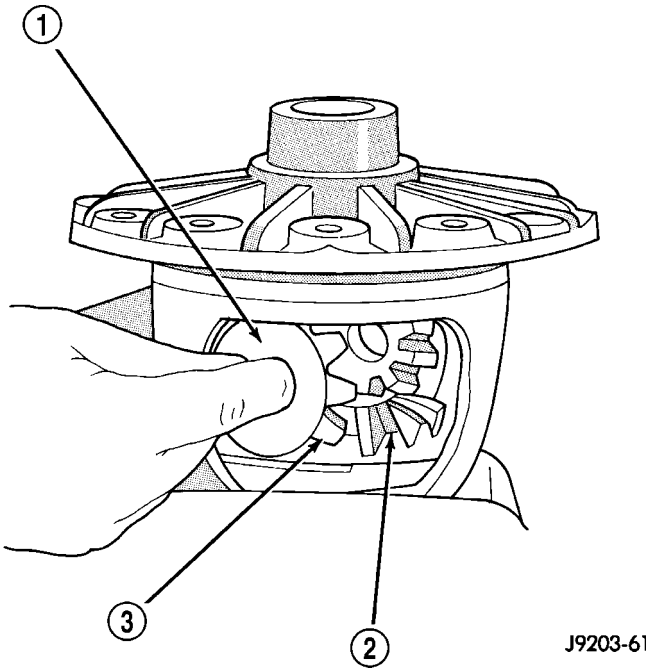


Fig. 45 DIFFERENTIAL GEARS

1 - THRUST WASHER
2 - SIDE GEAR
3 - DIFFERENTIAL PINION

(2) Install differential pinion gears and thrust washers.

(3) Install pinion mate shaft.

(4) Align hole in the pinion mate shaft with the hole in the differential case and install pinion mate shaft roll pin. Stake (peen) metal of case over pin in two places 180 degrees apart.

(5) Lubricate all differential components with hypoid gear lubricant.

DIFFERENTIAL (Continued)

INSTALLATION

NOTE: If replacement differential bearings or differential case are being installed, differential side bearing shim requirements may change. Refer Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Position Spreader W-129-B and adapters from Adapter set 6987, with the tool dowel pins seated in the locating holes (Fig. 46). Install hold-down clamps and tighten turnbuckle finger-tight.

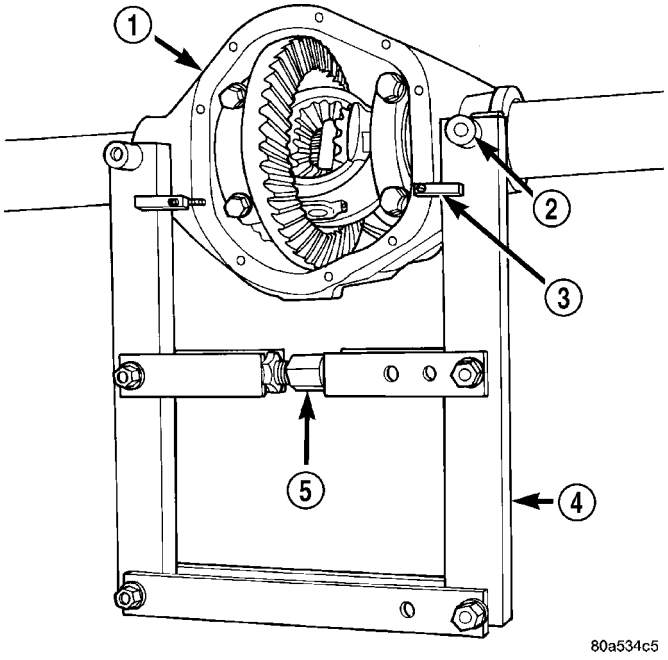


Fig. 46 SPREADER LOCATION

1 - DIFFERENTIAL HOUSING
2 - DOWEL
3 - SAFETY HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

(2) Install Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load indicator plunger against the opposite side of the housing and zero the indicator.

(3) Spread housing while measuring the distance with the dial indicator.

CAUTION: Never spread over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

(4) Remove dial indicator.

(5) Install differential case in the housing. Tap differential case to seat bearings cups in the housing.

(6) Install bearing caps in their original locations (Fig. 47).

(7) Loosely install differential bearing cap bolts.

(8) Remove axle housing spreader.

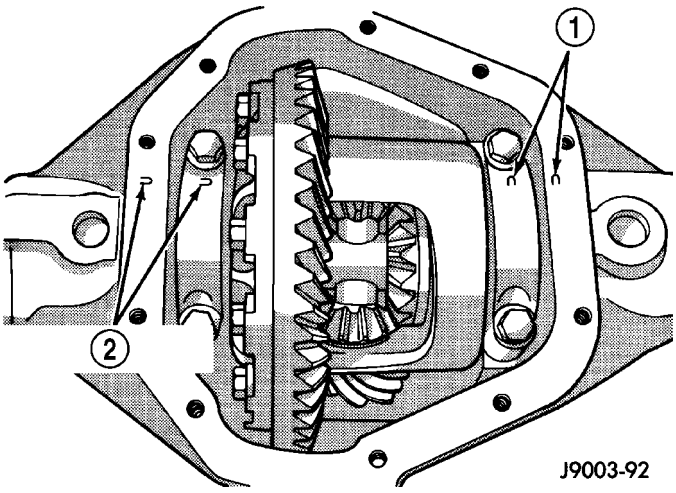


Fig. 47 BEARING CAP REFERENCE

1 - REFERENCE LETTERS
2 - REFERENCE LETTERS

(9) Tighten bearing cap bolts to 108 N·m (80 ft. lbs.).

(10) Install axle shafts.

(11) Install cover and identification tag.

DIFFERENTIAL -TRAC-LOK

DESCRIPTION

The Trac-Lok® differential has a one-piece differential case, and similar internal components as a standard differential, plus two clutch disc packs. Differential bearing preload and ring gear backlash are adjusted with shims located between the differential case bearing cups and housing. Pinion bearing preload is set and maintained by the use of a collapsible spacer.

OPERATION

This differentials clutches are engaged by two concurrent forces. The first being the preload force exerted through Belleville spring washers within the clutch packs. The second is the separating forces generated by the side gears as torque is applied through the ring gear (Fig. 48).

This design provides the differential action needed for turning corners and for driving straight ahead during periods of unequal traction. When one wheel looses traction, the clutch packs transfer additional torque to the wheel having the most traction. This differential resist wheel spin on bumpy roads and provide more pulling power when one wheel looses traction. Pulling power is provided continuously until both wheels loose traction. If both wheels slip due to unequal traction, Trac-lok® operation is normal. In extreme cases of differences of traction, the wheel with the least traction may spin.

DIFFERENTIAL -TRAC-LOK (Continued)

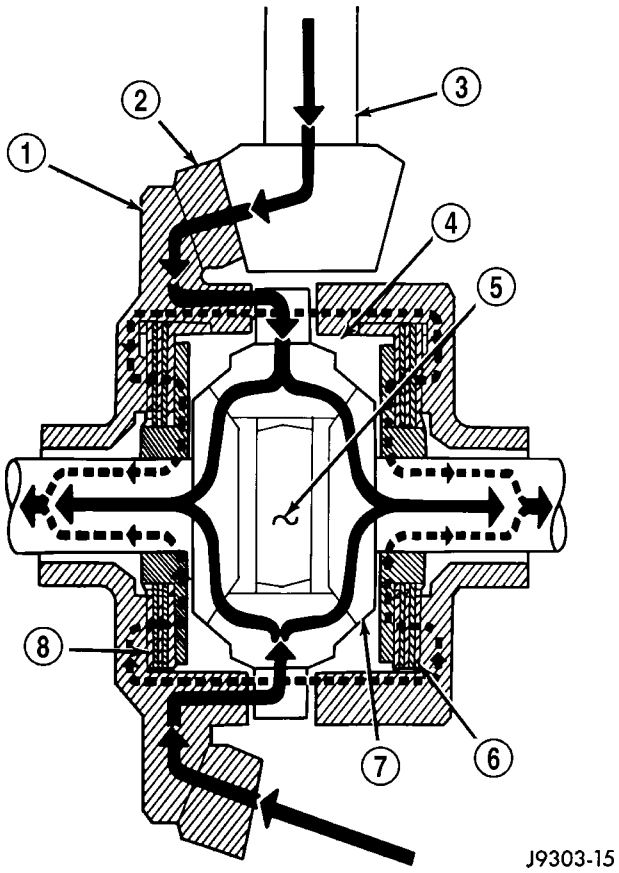


Fig. 48 TRAC-LOK LIMITED SLIP DIFFERENTIAL

1 - CASE
2 - RING GEAR
3 - DRIVE PINION
4 - PINION GEAR
5 - MATE SHAFT
6 - CLUTCH PACK
7 - SIDE GEAR
8 - CLUTCH PACK

DIAGNOSIS AND TESTING

The most common problem is a chatter noise when turning corners. Before removing the unit for repair, drain, flush and refill the axle with the specified lubricant. A container of Mopar Trac-lok® Lubricant (friction modifier) should be added after repair service or during a lubricant change.

After changing the lubricant, drive the vehicle and make 10 to 12 slow, figure-eight turns. This maneuver will pump lubricant through the clutches. This will correct the condition in most instances. If the chatter persists, clutch damage could have occurred.

DIFFERENTIAL TEST

The differential can be tested without removing the differential case by measuring rotating torque. Make sure brakes are not dragging during this measurement.

(1) Place blocks in front and rear of both front wheels.

(2) Raise one rear wheel until it is completely off the ground.

(3) Engine off, transmission in neutral, and parking brake off.

(4) Remove wheel and bolt Special Tool 6790 or equivalent tool to studs.

(5) Use torque wrench on special tool to rotate wheel and read rotating torque (Fig. 49).

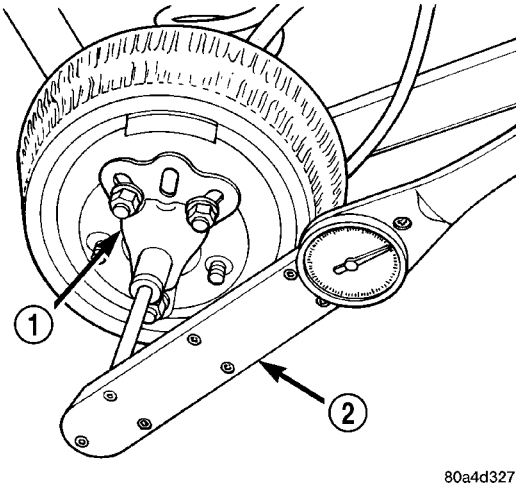


Fig. 49 ROTATING TORQUE TEST

1 - SPECIAL TOOL WITH BOLT IN CENTER HOLE
2 - TORQUE WRENCH

(6) If rotating torque is less than 41 N·m (56 ft. lbs.) or more than 271 N·m (200 ft. lbs.) on either wheel the unit should be serviced.

DISASSEMBLY

(1) Clamp side gear Fixture 6965 in a vise and set differential case on the fixture (Fig. 50).

NOTE: The Trac-lok® differential can be serviced with the ring gear installed unless the gear hangs over the pinion shaft, then it must be removed.

(2) Remove pinion gear mate shaft roll pin with a hammer and punch.

(3) Remove pinion gear mate shaft with a drift and hammer.

(4) Lubricate and install disc without threaded hole from Trac-lok® Tool C-4487 into lower side gear (Fig. 51).

(5) Lubricate and install disc with threaded hole from Trac-lok® Tool C-4487 into upper side gear. Thread forcing screw from tool into threaded disc until it becomes centered in lower disc.

(6) Position a small screw driver in slot of the threaded disc (Fig. 52) to prevent adapter from turning.

(7) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress Belleville springs in clutch packs (Fig. 53).

DIFFERENTIAL -TRAC-LOK (Continued)

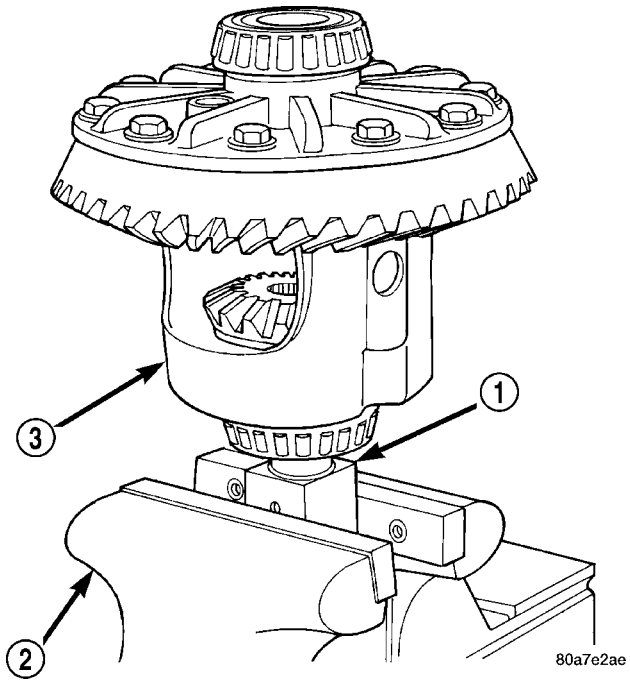


Fig. 50 DIFFERENTIAL CASE FIXTURE

1 - HOLDING FIXTURE
2 - VISE
3 - DIFFERENTIAL

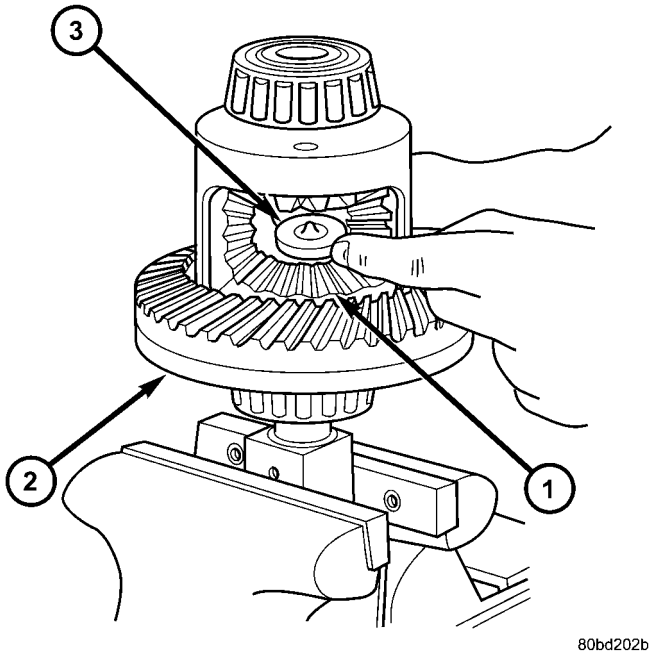


Fig. 51 TRAC-LOK DISC

1 - LOWER SIDE GEAR
2 - DIFFERENTIAL CASE
3 - DISC

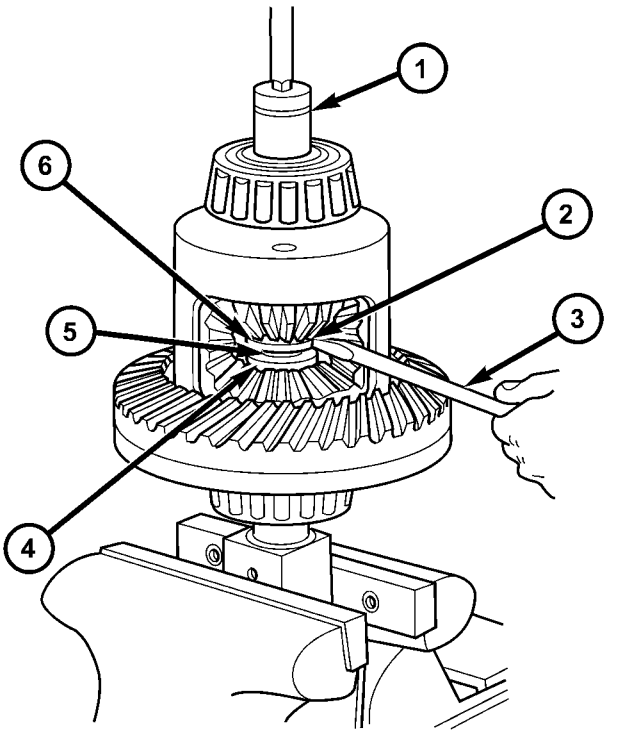


Fig. 52 TRAC-LOK TOOLS ASSEMBLY

1 - SOCKET
2 - SLOT IN DISC
3 - SCREWDRIVER
4 - STEP PLATE
5 - THREADED ROD
6 - DISC

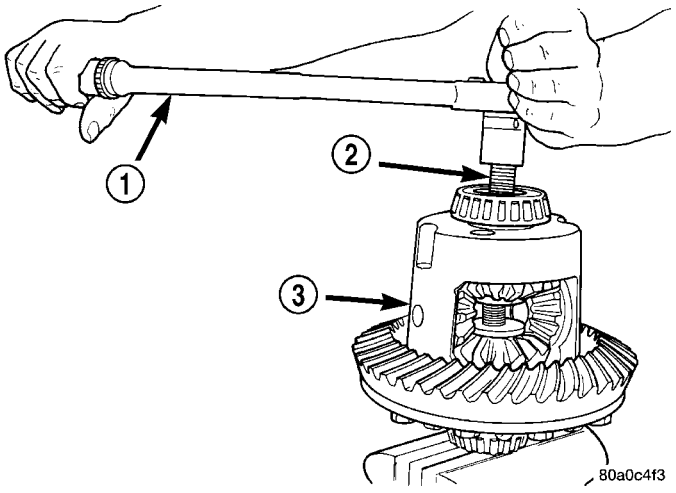


Fig. 53 COMPRESS BELLEVILLE SPRING

1 - TORQUE WRENCH
2 - FORCING SCREW
3 - DIFFERENTIAL CASE

DIFFERENTIAL -TRAC-LOK (Continued)

(8) With a feeler gauge remove thrust washers from behind the pinion gears (Fig. 54).

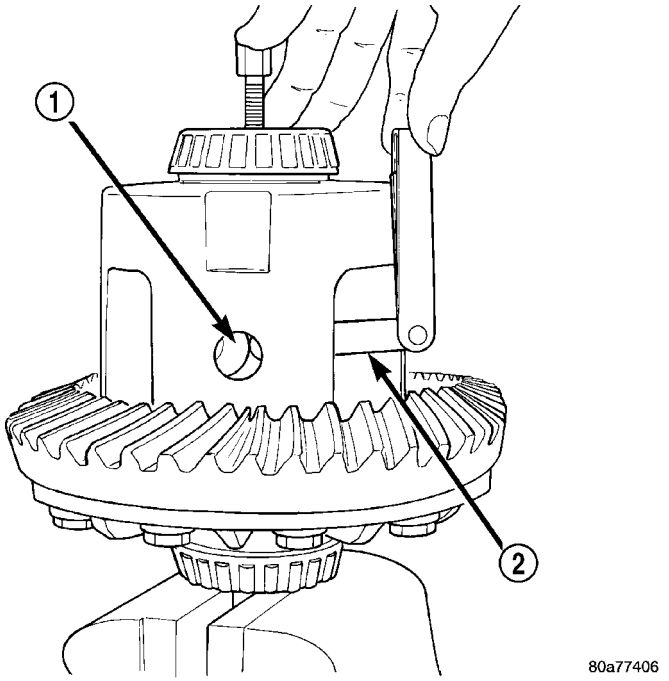


Fig. 54 PINION GEAR THRUST WASHER

1 - THRUST WASHER
2 - FEELER GAUGE

(9) Insert turning bar from tool into the pinion mate shaft hole in the case (Fig. 55).

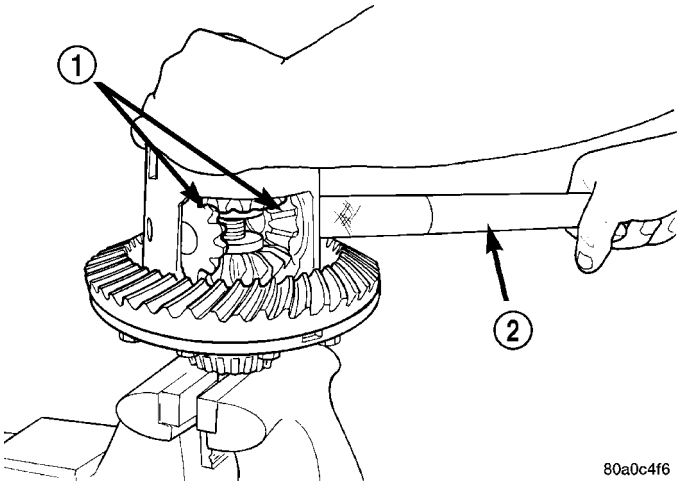


Fig. 55 PINION GEARS

1 - PINION GEARS
2 - TURNING BAR

(10) Loosen forcing screw in small increments until clutch pack tension is relieved and the differential case can be turned using turning bar.

(11) Rotate differential case until the pinion gears can be removed.

(12) Remove pinion gears from differential case.

(13) Remove Forcing Screw, Step Plate and Threaded Adapter.

(14) Remove top side gear, clutch pack retainer and clutch pack. Keep plates in order during removal (Fig. 56).

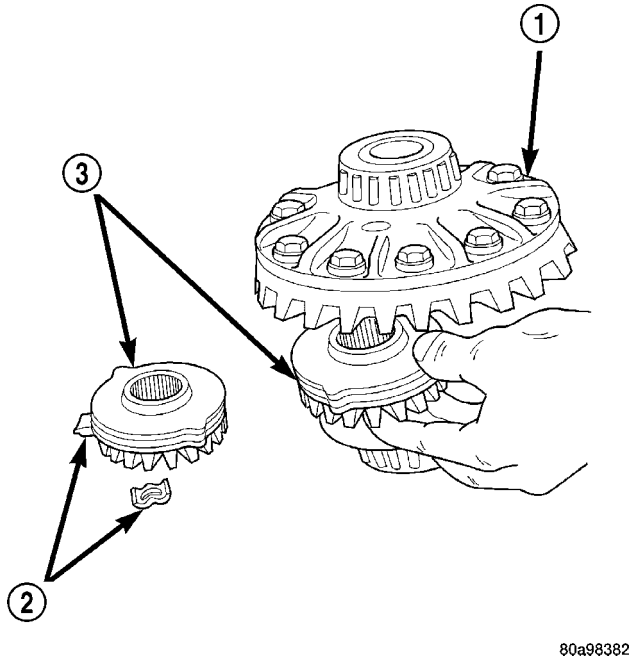


Fig. 56 SIDE GEARS AND CLUTCH DISCS

1 - DIFFERENTIAL CASE
2 - RETAINER
3 - SIDE GEAR AND CLUTCH DISC PACK

(15) Remove differential case from the Holding Fixture. Remove side gear, clutch pack retainer and clutch pack. Keep plates in order during removal.

ASSEMBLY

NOTE: New Plates and discs with fiber coating (no grooves or lines) must be presoaked in Friction Modifier before assembly. Soak plates and discs for a minimum of 20 minutes.

(1) Lubricate components with gear lubricant.

(2) Assemble clutch discs into packs and secure disc packs with retaining clips (Fig. 57).

NOTE: Dished plate is position with the convex side against the side gear.

(3) Position assembled clutch disc packs on the side gear hubs.

(4) Install clutch pack and side gear in the ring gear side of the differential case, with retaining clips seated in the case pockets (Fig. 58).

(5) Position the differential case on the Holding Fixture 6965.

DIFFERENTIAL -TRAC-LOK (Continued)

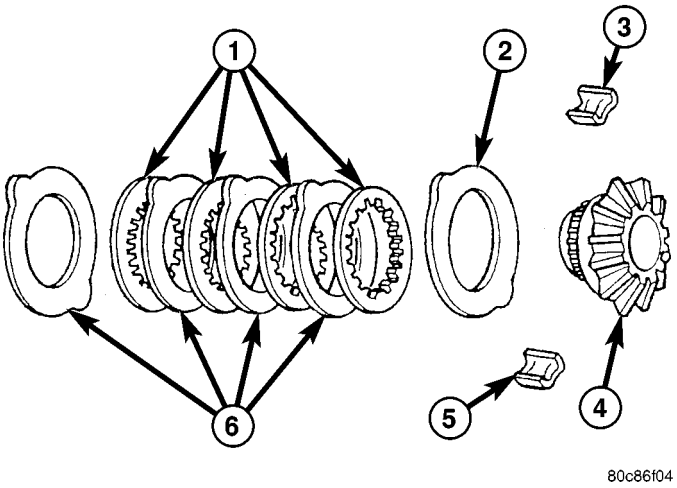


Fig. 57 CLUTCH PACK

1 - DISCS
2 - DISHED PLATE
3 - RETAINER
4 - SIDE GEAR
5 - RETAINER
6 - PLATES

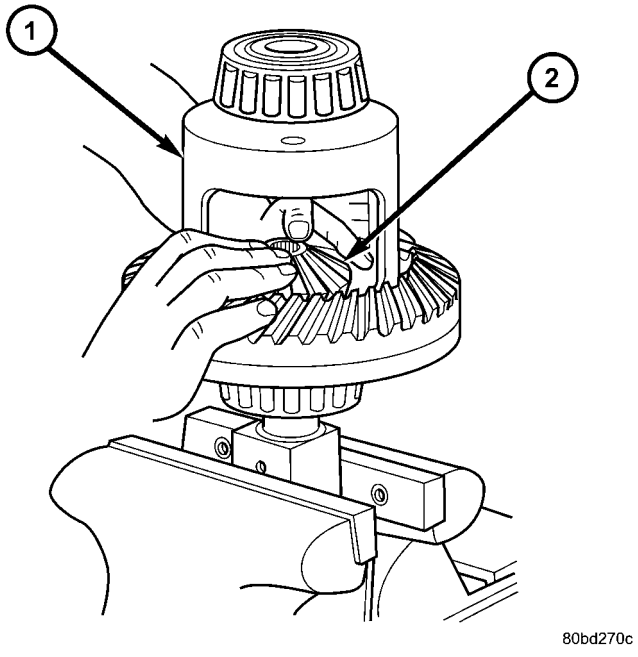


Fig. 58 CLUTCH PACK AND LOWER SIDE GEAR

1 - DIFFERENTIAL CASE
2 - SIDE GEAR AND CLUTCH PACK

(6) Lubricated and install disc without threaded hole from Trac-Lok® Tool C-4487 into lower side gear.

(7) Install upper side gear and clutch disc pack with retaining clips seated in the case pockets (Fig. 59).

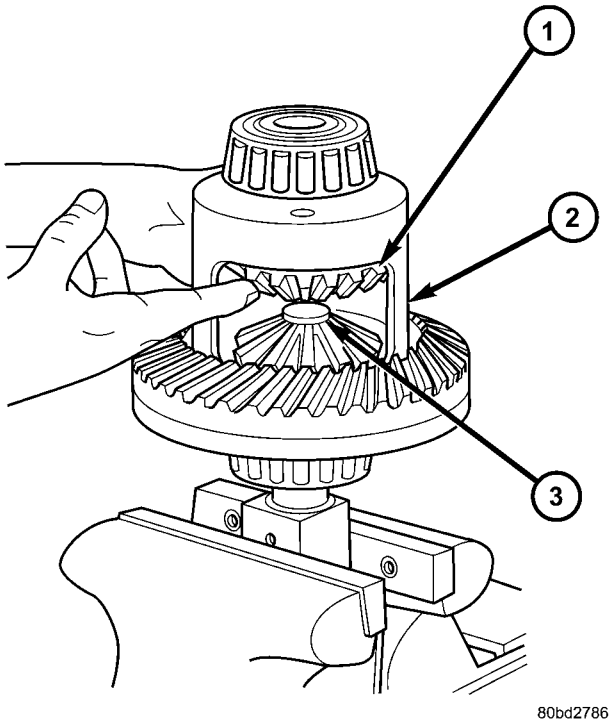


Fig. 59 CLUTCH PACK AND UPPER SIDE GEAR

1 - SIDE GEAR AND CLUTCH PACK
2 - DIFFERENTIAL CASE
3 - DISC

(8) Hold assembly in position and install threaded disc from Trac-Lok® Tool C-4487 into top side gear.

(9) Install forcing screw and tighten screw to slightly compress clutch disc.

(10) Place pinion gears in position in side gears and verify that the pinion mate shaft hole is aligned.

(11) Rotate case with turning bar C-4487-2 until pinion mate shaft holes in pinion gears align with holes in case. It may be necessary to slightly tighten the forcing screw in order to install the pinion gears.

(12) Tighten forcing screw to 122 N·m (90 ft. lbs.) maximum to compress the Belleville springs.

(13) Lubricate and install thrust washers behind pinion gears and align washers with a small screw driver. Insert mate shaft into each pinion gear to verify alignment.

(14) Remove Forcing Screw, Step Plate and Threaded Adapter.

(15) Install pinion gear mate shaft and align holes in shaft and case.

(16) Install pinion mate shaft roll pin. Stake (peen) case over pin in two places 180 degrees apart.

(17) Lubricate all differential components with hypoid gear lubricant.

DIFFERENTIAL - TRU-LOK

DESCRIPTION

The differential is a locking differential, that provides a positive mechanical connection between the right and left axle when engauged. The differential uses a dog clutch to connection the right and left axle.

OPERATION

The Tru-lok differential is activated by the axle lock switch located on the dash panel. When the switch is activated, an air pump with a built-in pressure regulator sends 5 PSI of air pressure to a accuator diaphragm in the differential housing. The diaphragm then engages a dog clutch and a position switch. The dog clutch has one gear attached to the differential case and another gear attached to a differential side gear. When the dog clutch is engaged the right and left wheels turn at the same speed. The position switch lights a lamp on the dash to indicate the system has been engaged. The differential works as standard differential when not engaged.

NOTE: The differential is serviced as an assembly, the diaphragm and indicator switch are serviced separately. The differential case must be removed to service the diaphragm actuator and indicator switch.

DIAGNOSIS AND TESTING

UNLOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the OFF position.

(5) Rotate one tire by hand. The other tire should spin in the opposite direction.

NOTE: If wheel cannot be rotated the differential must be repaired/replaced.

LOCKED

(1) Block tires opposite the axle to be tested to prevent the vehicle from rolling.

(2) Place transfer case in 4WD Low and automantic transmission in Park (1st gear if manual transmission).

(3) Raise both wheels of the axle to be tested off the ground.

(4) Turn ignition to the ON position and dash switch to the ON position.

(5) Try to rotate one tire by hand. You should not be able to rotate the tire.

NOTE: If wheel does rotated verify locker pump operation. If the pump is fuctional the differential must be repaired/replaced.

TRU-LOK INDICATOR SWITCH

(1) Turn ignition switch off.

(2) Disconnect indicator switch harness (Fig. 60) from the differential housing.

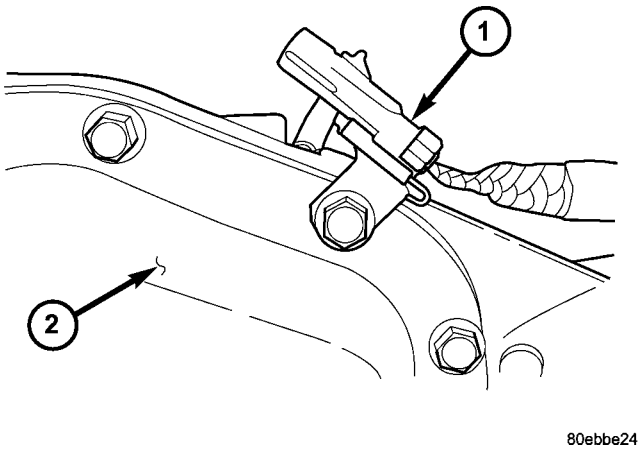


Fig. 60 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

(3) Measure electrical continuity across the switch terminals. Circuit should be closed (zero resistance).

NOTE: If circuit is not closed replace indicator switch.

DIFFERENTIAL - TRU-LOK (Continued)

TRU-LOK PUMPS

(1) Connect a pressure gauge to the rear pump.
(2) Place transfer case in 4WD Low and automatic transmission in Park (1st gear if manual transmission).
(3) Turn ignition to the ON position and push the dash switch to activate the rear pump.

NOTE: If pump is not running, verify pump has (Fig. 61) 12 volts and ground.

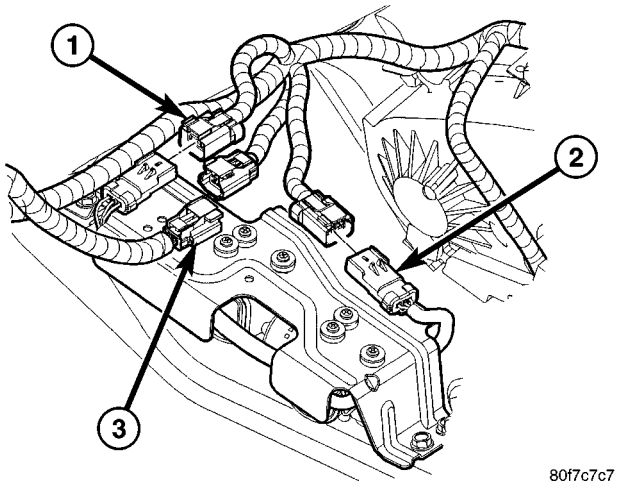


Fig. 61 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

(4) With the pump running the pressure gauge should show 5 psi..

NOTE: If pump does not produce 5 psi. replace the pump.

REMOVAL

(1) Remove drain plug from the differential housing.
(2) Disconnect indicator switch harness and remove connector from differential cover (Fig. 62).
(3) Remove differential housing cover.
(4) Remove axle shafts.
(5) Remove pressure hose from actuator assembly (Fig. 63).

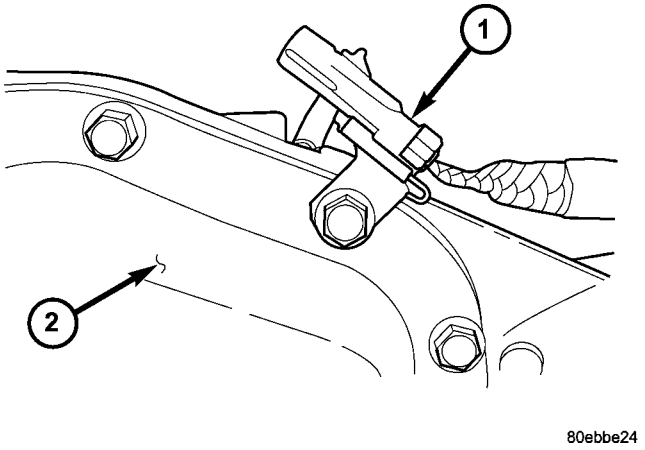


Fig. 62 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

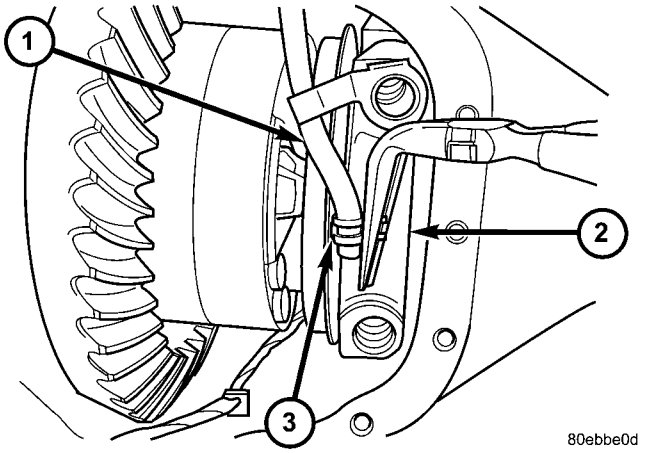


Fig. 63 PRESSURE HOSE

1 - PRESSURE HOSE
2 - BEARING CAP
3 - PRESSURE HOSE CLAMP

DIFFERENTIAL - TRU-LOK (Continued)

(6) Note the reference letters stamped on the bearing caps and housing machined sealing surface (Fig. 64).

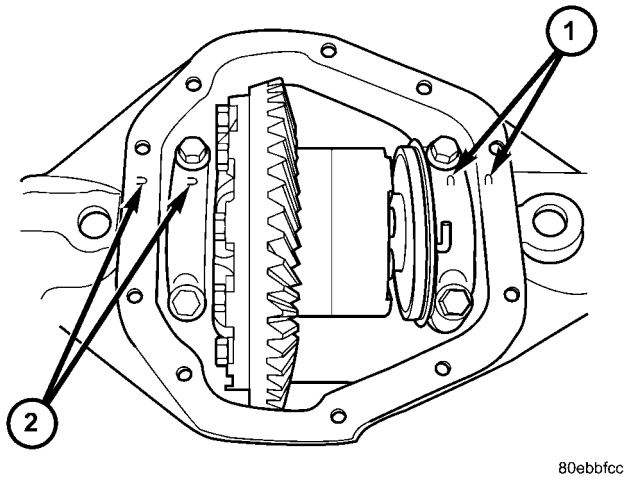


Fig. 64 LOCATION MARKS

1 - REFERENCE MARKS
2 - REFERENCE MARKS

(7) Loosen differential bearing cap bolts.

(8) Position Spreader W-129-B with Adapter Kit 6987B on differential locating holes (Fig. 65). Install hold-down clamps and tighten the turnbuckle finger-tight.

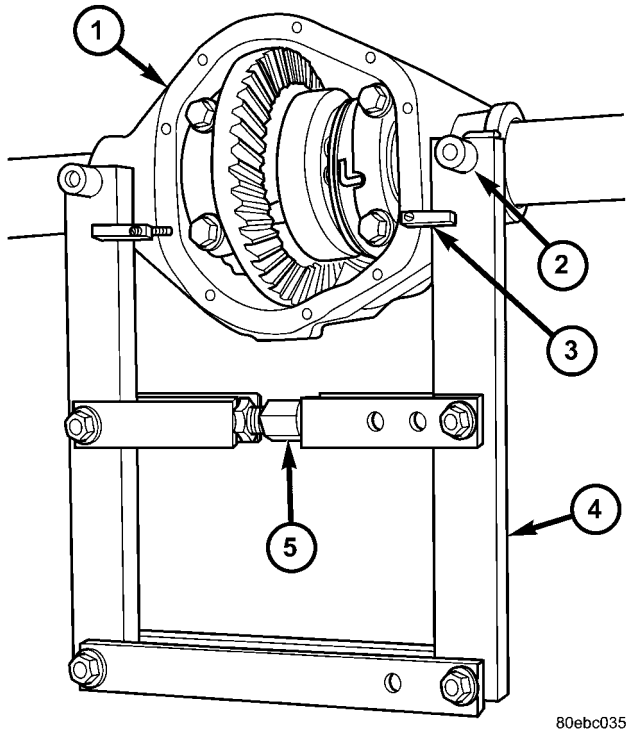


Fig. 65 SPREADER LOCATION

1 - DIFFERENTIAL CASE
2 - ADAPTER
3 - HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

(9) Install a Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud (Fig. 66). Load indicator plunger against the opposite side of the housing and zero the indicator.

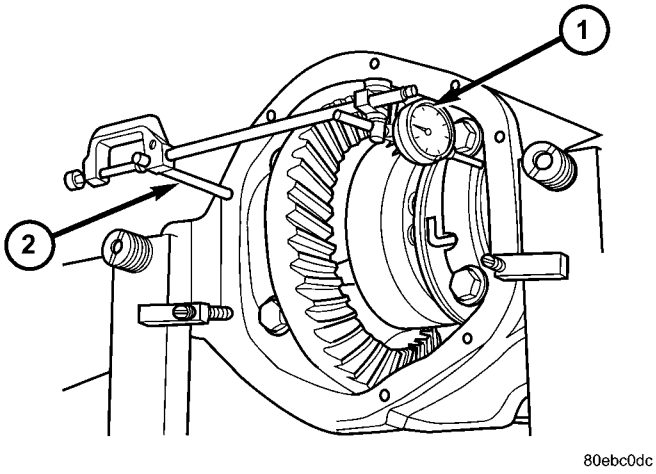


Fig. 66 DIAL INDICATOR LOCATION

1 - DIAL INDICATOR
2 - PILOT STUD

CAUTION: Never spread the housing over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

(10) Spread housing while measuring the distance with the dial indicator (Fig. 67).

(11) Remove dial indicator.

(12) Remove differential bearing cap bolts and ring gear side bearing cap.

(13) Remove differential from housing with pinion gear side bearing cap (Fig. 68) and tag differential bearing cups and preload shims to indicate location.

CAUTION: Do not bend actuator mounting tabs, during differential removal. Failure to heed caution may result in damage.

(14) Remove spreader from housing.

(15) Remove indicator switch (Fig. 69) from differential housing.

(16) Clean the housing cavity with flushing oil, light engine oil or lint free cloth.

NOTE: Do not use water, steam, kerosene or gasoline for cleaning.

DISASSEMBLY

(1) Install Plug C-293-3 into the differential axle shaft hole.

DIFFERENTIAL - TRU-LOK (Continued)

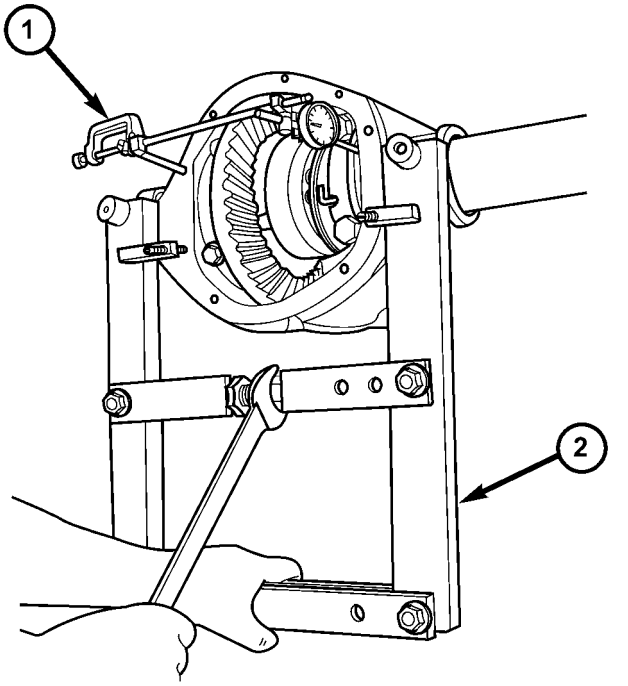


Fig. 67 SPREAD DIFFERENTIAL CASE

1 - DIAL INDICATOR
2 - SPREADER

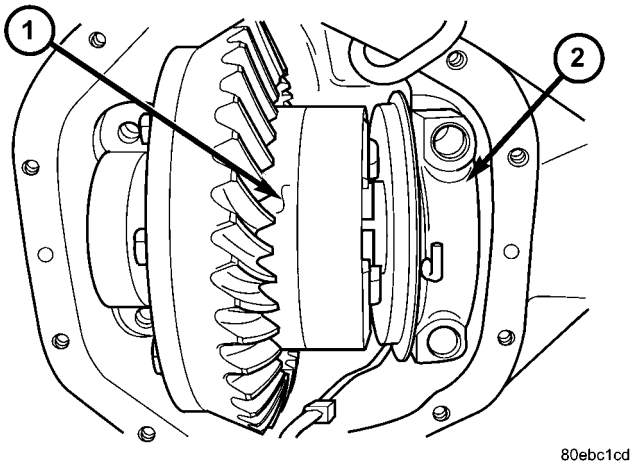


Fig. 68 DIFFERENTIAL

1 - DIFFERENTIAL
2 - BEARING CAP

(2) Remove differential case bearings with Puller 6444, Puller Rods 6444-3 and Puller Flange 6444-1. Position puller (Fig. 70) on the differential.

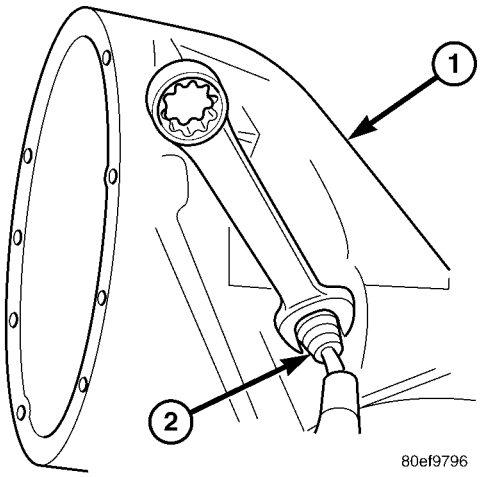


Fig. 69 INDICATOR SWITCH

1 - DIFFERENTIAL CASE
2 - SWITCH

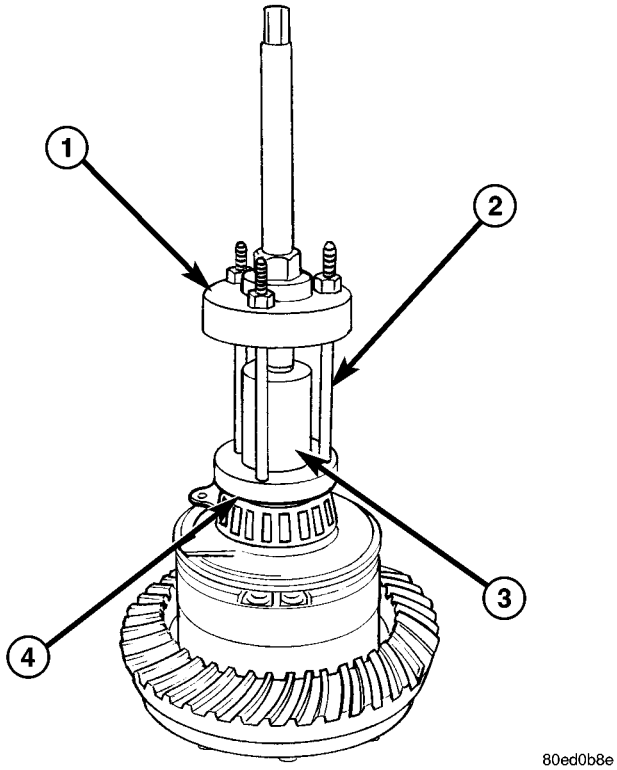


Fig. 70 PULLER AND FLANGE

1 - PULLER
2 - ROD
3 - PLUG
4 - FLANGE

DIFFERENTIAL - TRU-LOK (Continued)

(3) Position Puller Jaws 6444-7 (Fig. 71) around the case bearing and puller flange.

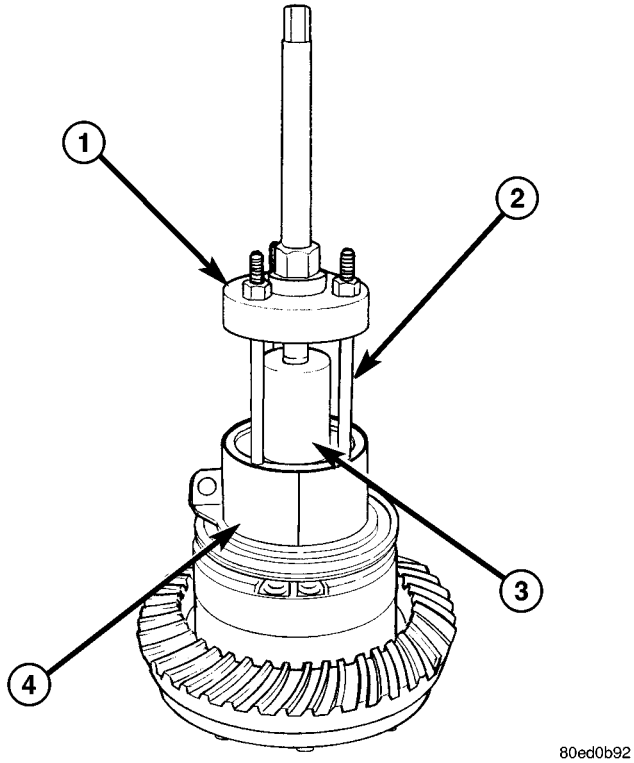


Fig. 71 PULLER AND JAWS

1 - PULLER
2 - ROD
3 - PLUG
4 - JAW

(4) Position Puller Collar 6444-8 (Fig. 72) around the puller jaws.

(5) Tighten the puller nut and remove differential case bearing.

(6) Remove locking actuator (Fig. 73).

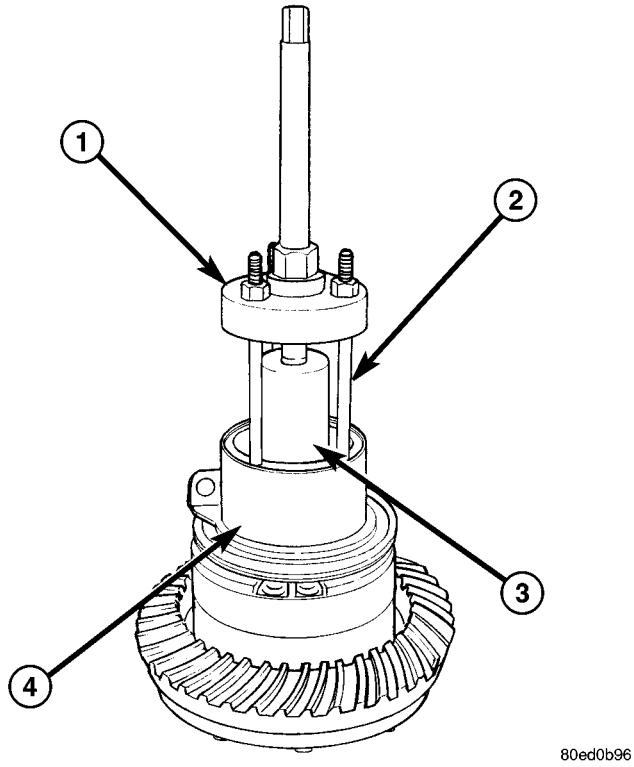


Fig. 72 PULLER AND COLLAR

1 - PULLER
2 - ROD
3 - PLUG
4 - COLLAR

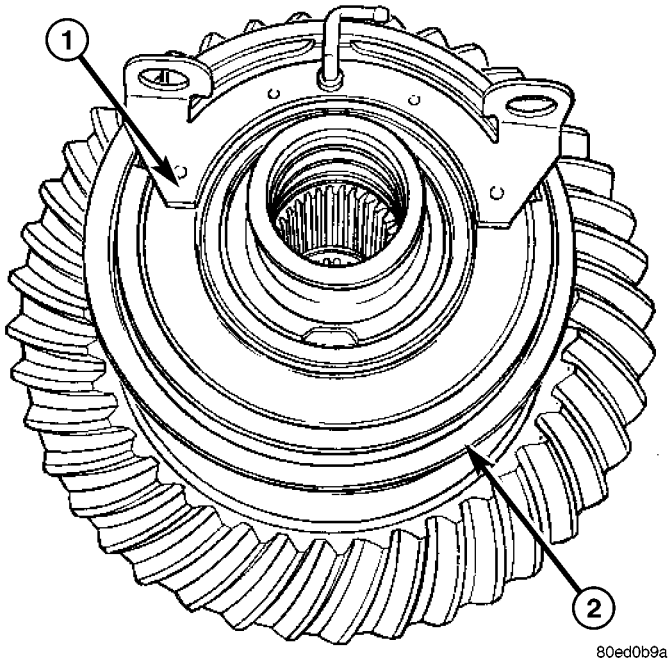


Fig. 73 ACTUATOR

1 - ACTUATOR
2 - PRESSURE PLATE

DIFFERENTIAL - TRU-LOK (Continued)

ASSEMBLY

(1) Verify pressure plate tabs are seated (Fig. 74) in the dog clutch slots.

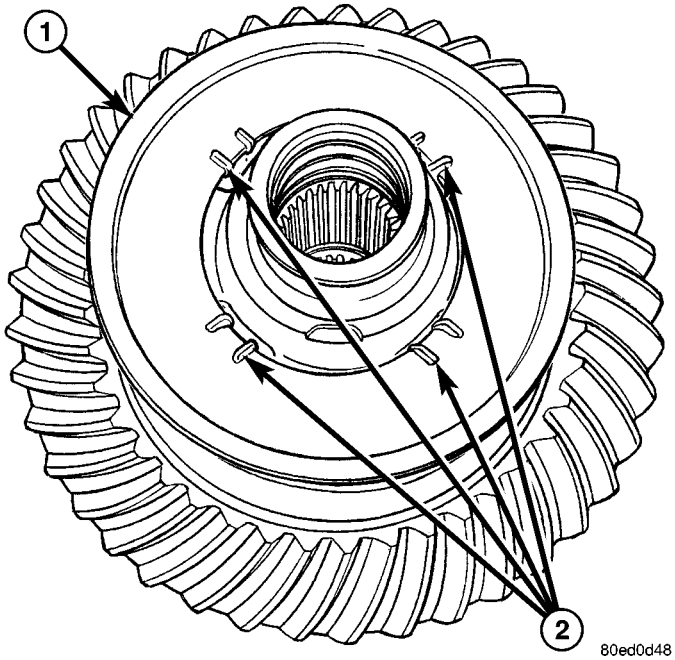


Fig. 74 PRESSURE PLATE

1 - PLATE
2 - PLATE TABS

(2) Place actuator on the pressure plate and case bearing on the case (Fig. 75).

(3) Install differential case bearings with Installer D-156 and Handle C-4171.

INSTALLATION

NOTE: If replacement differential bearings or differential case are being installed, differential side bearing shim requirements may change. Refer Adjustments (Differential Bearing Preload and Gear Backlash) to determine the proper shim selection.

(1) Install indicator switch into the housing.

(2) Position Spreader W-129-B and adapters from Adapter set 6987, with the tool dowel pins seated in the locating holes (Fig. 76). Install hold-down clamps and tighten the tool turnbuckle finger-tight.

(3) Install Pilot Stud C-3288-B at the left side of the differential housing. Attach Dial Indicator C-3339 to pilot stud. Load indicator plunger against the opposite side of the housing and zero indicator.

(4) Spread housing while measuring the distance with the dial indicator.

CAUTION: Never spread over 0.38 mm (0.015 in). Failure to heed caution may result in damage.

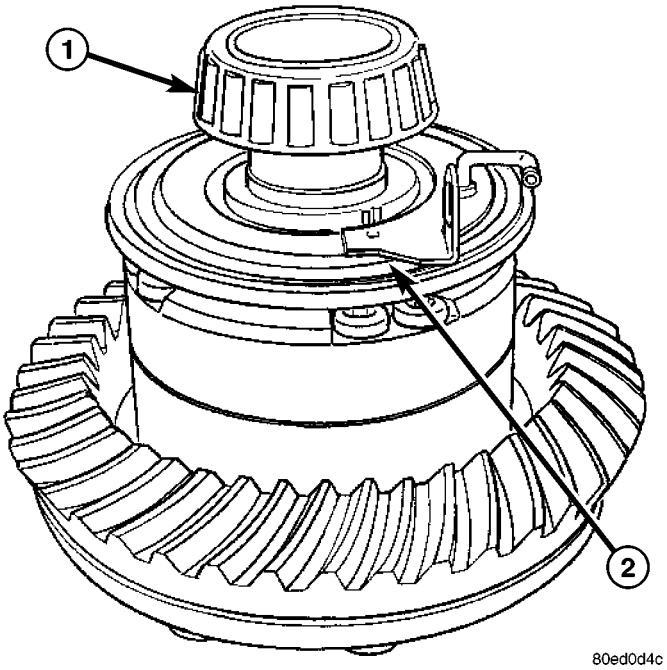


Fig. 75 ACTUATOR AND CASE BEARING

1 - CASE BEARING
2 - ACTUATOR

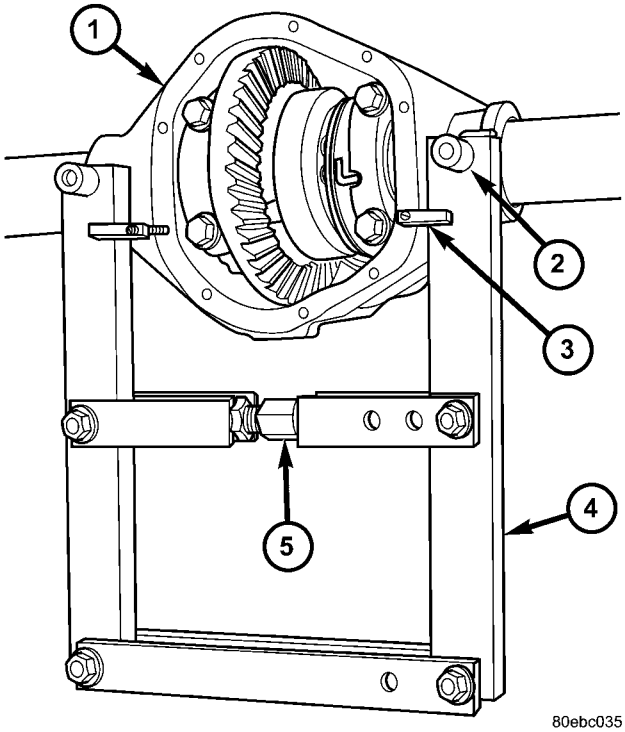


Fig. 76 SPREADER LOCATION

1 - DIFFERENTIAL CASE
2 - ADAPTER
3 - HOLD DOWN
4 - SPREADER
5 - TURNBUCKLE

DIFFERENTIAL - TRU-LOK (Continued)

(5) Remove dial indicator.

(6) Install differential case with pinion gear side bearing cap and differential preload shims in the housing. Tap differential case to seat bearings cups in the housing.

CAUTION: Ensure indicator switch plunger head is positioned under the actuator. Failure to heed caution may result in damage.

(7) Install ring gear side bearing cap with reference marks aligned (Fig. 77).

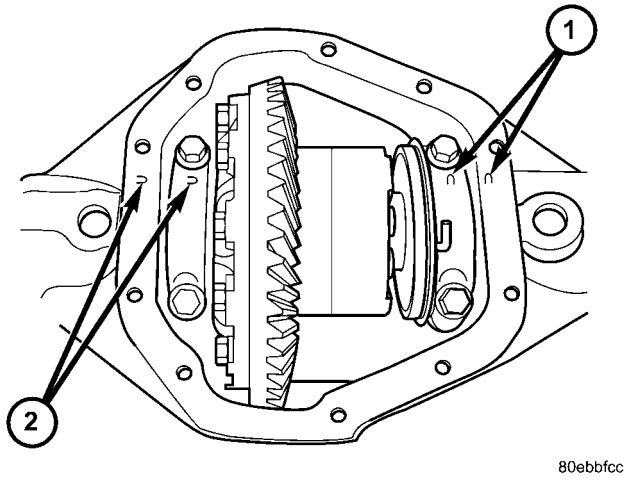


Fig. 77 BEARING CAP REFERENCE MARKS

1 - REFERENCE MARKS
2 - REFERENCE MARKS

(8) Loosely install differential bearing cap bolts.

(9) Remove axle housing spreader.

(10) Tighten bearing cap bolts to 108 N·m (80 ft. lbs.).

(11) With a 1/4 inch drill bit check the clearance between the actuator and actuator pressure plate at the top and the bottom.

CAUTION: If clearance is not correct, indicator switch plunger may be on top of the actuator or actuator mounting tabs may be bent. Failure to heed caution may result in damage.

(12) Install pressure hose on the actuator assembly.

(13) Install axle shafts.

(14) Install differential cover, identification tag and indicator switch connector (Fig. 78).

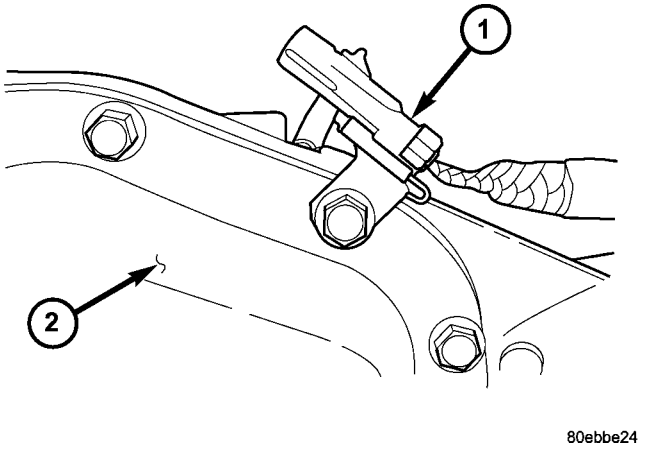


Fig. 78 INDICATOR SWITCH CONNECTOR

1 - CONNECTOR
2 - DIFFERENTIAL COVER

DIFFERENTIAL - TRU-LOK PUMP

REMOVAL

(1) Disconnect pumps and axle harness connectors (Fig. 79).

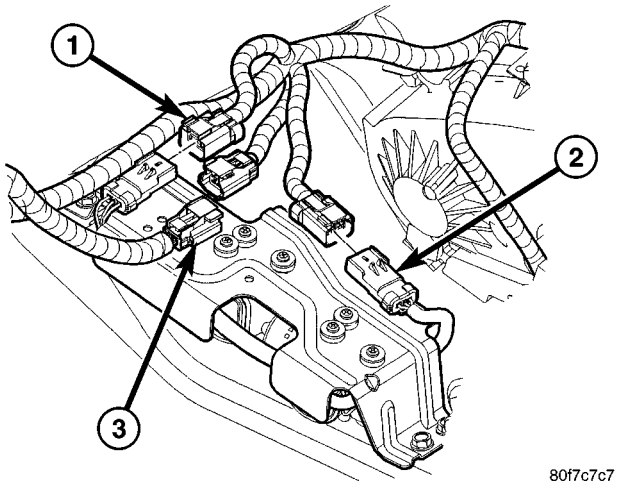


Fig. 79 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

DIFFERENTIAL - TRU-LOK PUMP (Continued)

(2) Remove pressure hoses from the front and rear pumps.

(3) Remove pump mounting bracket bolts (Fig. 80).

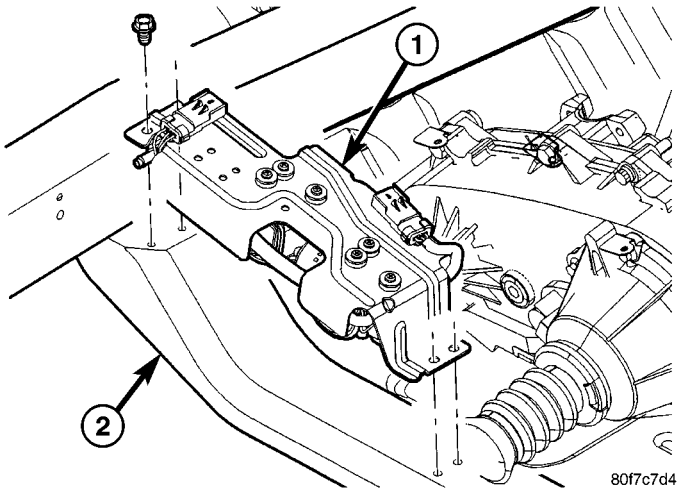


Fig. 80 PUMP BRACKET

1 - PUMP BRACKET
2 - CROSSMEMBER/SKID PLATE

(4) Remove pumps and bracket assembly from the vehicle.

(5) Remove pump connectors from the mounting bracket.

(6) Remove pump mounting screws (Fig. 81) from the bracket and remove pump/pumps.

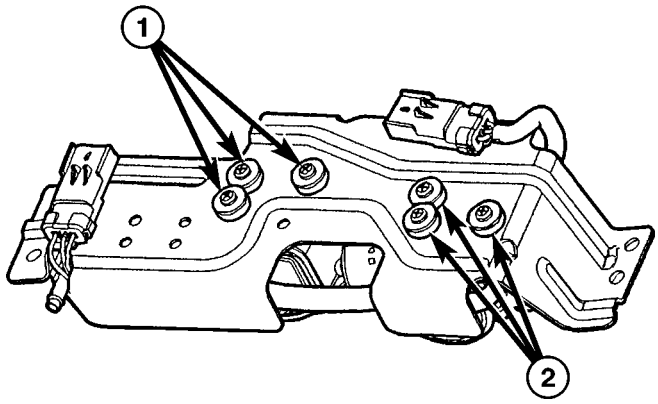


Fig. 81 PUMP SCREWS

1 - REAR PUMP SCREWS
2 - FRONT PUMP SCREWS

INSTALLATION

(1) Align pump with screw holes in pump bracket.

(2) Install and tighten pump mounting screws (Fig. 82).

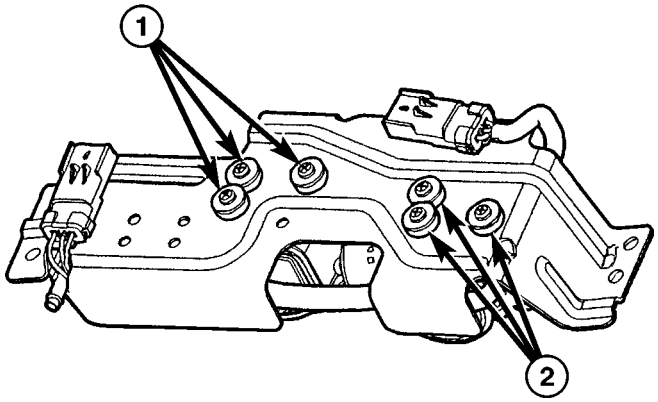


Fig. 82 PUMP SCREWS

1 - REAR PUMP SCREWS
2 - FRONT PUMP SCREWS

(3) Install pump bracket assembly on the crossmember/skid plate and install pump bracket mounting bolts (Fig. 83).

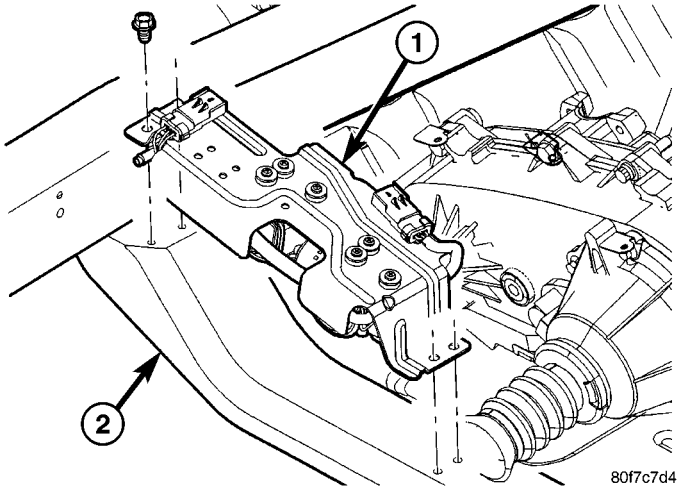


Fig. 83 PUMP BRACKET

1 - PUMP BRACKET
2 - CROSSMEMBER/SKID PLATE

DIFFERENTIAL - TRU-LOK PUMP (Continued)

(4) Tighten pump bracket bolts to 14 N·m (125 in. lbs.).

(5) Connector pump and axle harness connectors (Fig. 84).

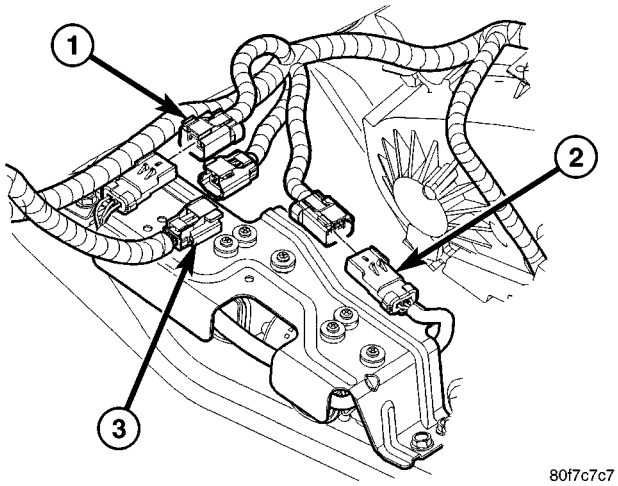


Fig. 84 PUMP CONNECTORS

1 - REAR PUMP CONNECTOR
2 - FRONT PUMP CONNECTOR
3 - AXLE HARNESS

(6) Install pressure hoses on the pumps.

(7) Verify pump operation.

DIFFERENTIAL CASE BEARINGS

REMOVAL

(1) Remove differential case from axle housing.

(2) Remove bearings from the differential case with Puller/Press C-293-PA, Adapters C-293-18 and Plug SP-3289 (Fig. 85).

INSTALLATION

NOTE: If replacement differential side bearings or differential case are being installed, differential side bearing shim requirements may change. Refer to Adjustments for (Differential Bearing Preload and Gear Backlash) procedures.

(1) Install differential side bearings with Installer D-156 and Handle C-4171 (Fig. 86).

(2) Install differential into the housing.

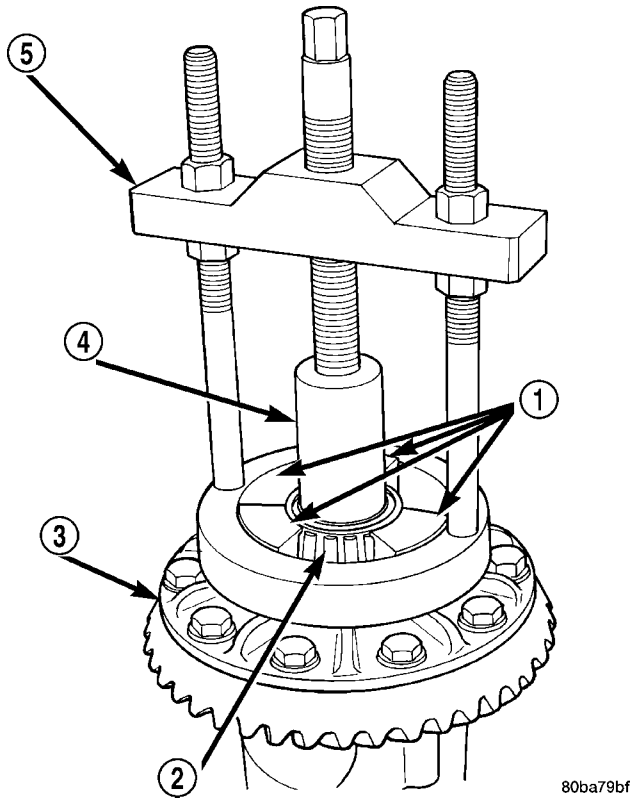


Fig. 85 DIFFERENTIAL CASE BEARING

1 - ADAPTERS
2 - BEARING
3 - DIFFERENTIAL
4 - PLUG
5 - PULLER

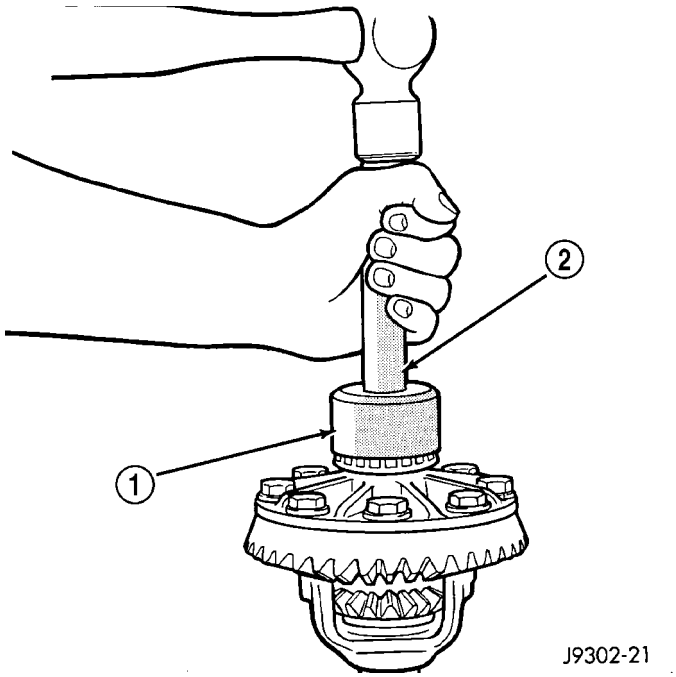


Fig. 86 DIFFERENTIAL SIDE BEARING

1 - INSTALLER
2 - HANDLE

PINION GEAR/RING GEAR/ TONE RING

REMOVAL

NOTE: The ring gear and pinion are serviced as a matched set. Never replace one gear without replacing the other matched gear.

(1) With vehicle in neutral, position vehicle on hoist.
(2) Remove differential assembly from axle housing.
(3) Place differential case in a vise with soft metal jaw.
(4) Remove bolts holding ring gear to differential case.
(5) Driver ring gear off the differential case with a dead-blow hammer (Fig. 87).

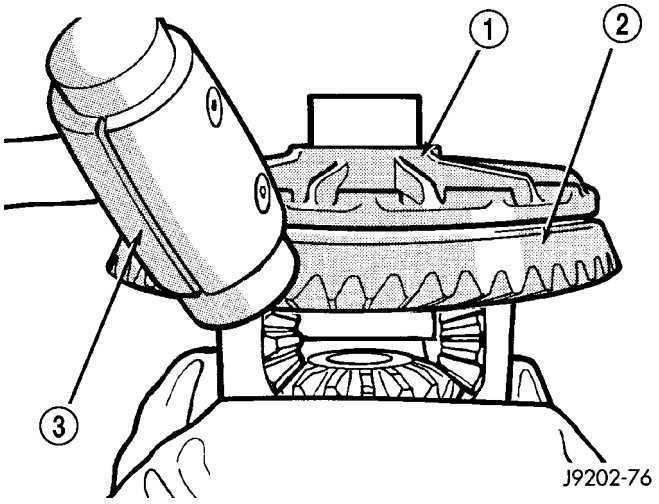


Fig. 87 RING GEAR

1 - CASE
2 - RING GEAR
3 - DEAD-BLOW HAMMER

(6) Using Spanner Wrench 6958 to hold yoke, remove the pinion nut and washer (Fig. 88).
(7) Remove pinion yoke from pinion shaft with Remover C-452 and Flange Wrench C-3281 (Fig. 89).

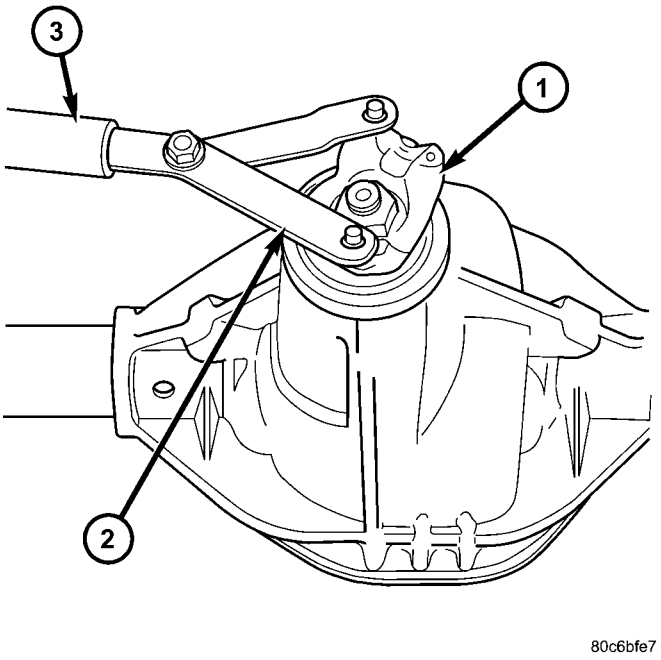


Fig. 88 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

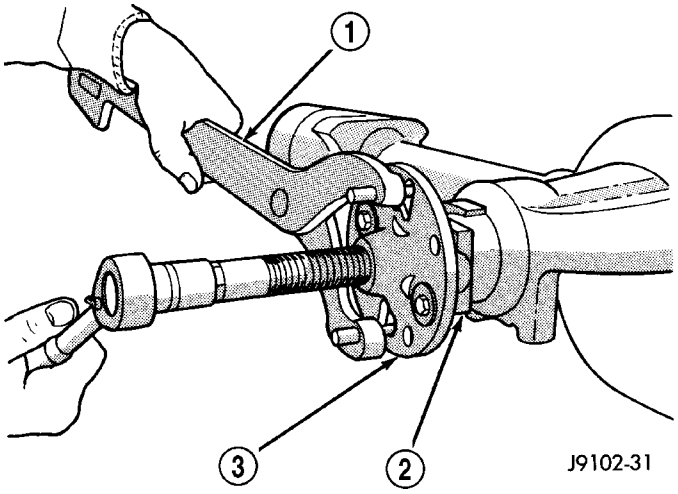


Fig. 89 PINION YOKE REMOVER

1 - REMOVER
2 - PINION YOKE
3 - FLANGE WRENCH

PINION GEAR/RING GEAR/TONE RING (Continued)

(8) Remove pinion gear from the housing (Fig. 90).

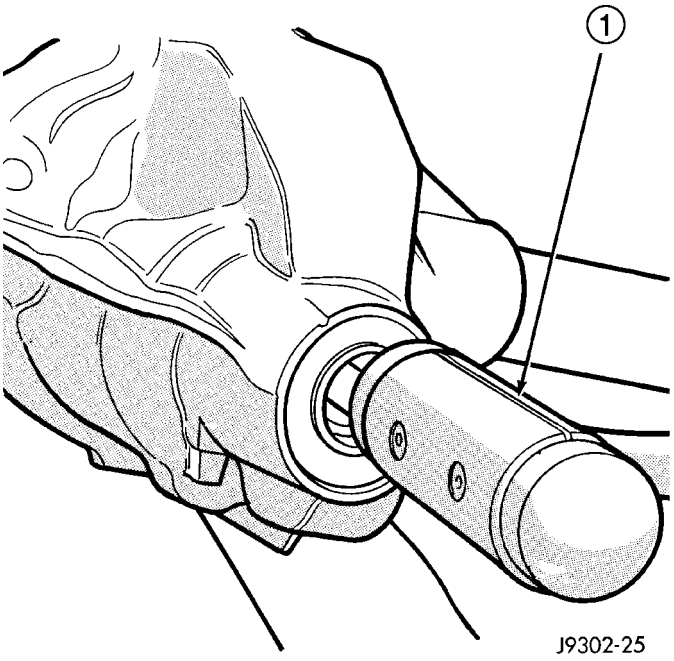


Fig. 90 PINION GEAR

1 - DEAD-BLOW HAMMER

(9) Remove front pinion bearing cup, bearing and pinion seal with Remover D-147 and Handle C-4171 (Fig. 91).

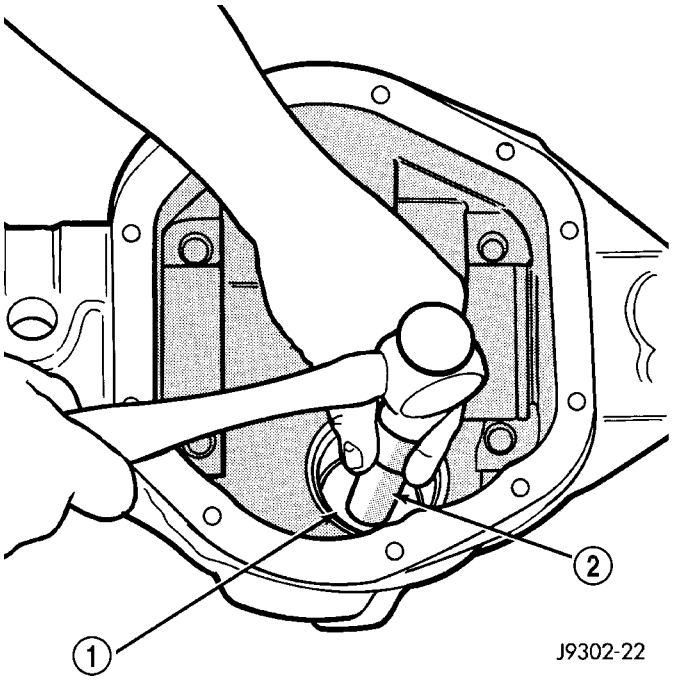


Fig. 91 FRONT BEARING CUP

1 - REMOVER
2 - HANDLE

(10) Remove rear pinion bearing cup from axle housing with remover D-148 and Handle C-4171 (Fig. 92).

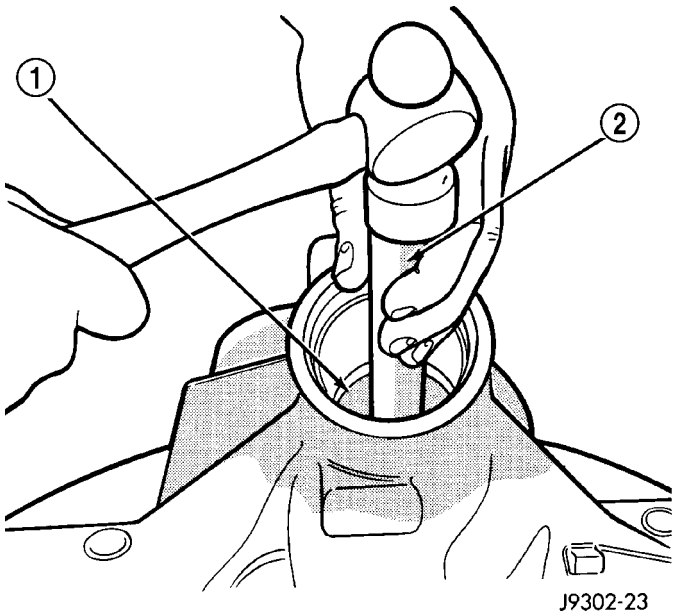


Fig. 92 REAR BEARING CUP

1 - DRIVER
2 - HANDLE

(11) Remove rear pinion bearing from the pinion with Puller/Press C-293-PA and Adapters C-293-39 (Fig. 93).

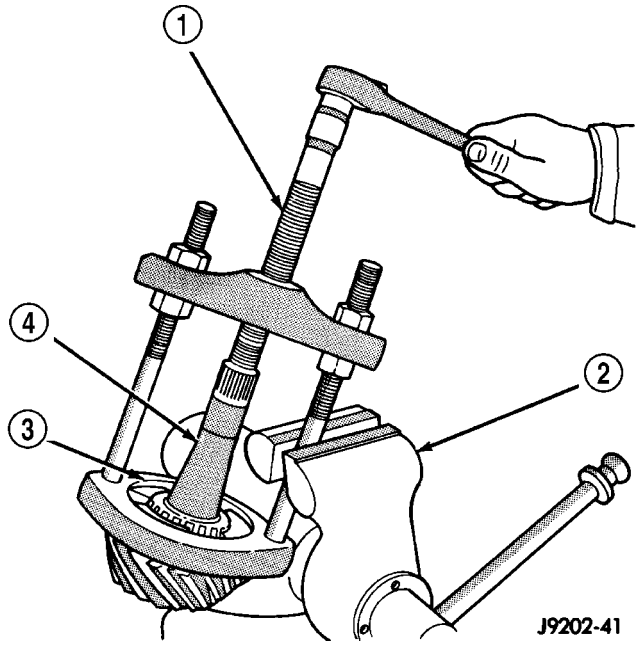


Fig. 93 REAR PINION BEARING

1 - PULLER
2 - VISE
3 - ADAPTERS
4 - PINION GEAR SHAFT

PINION GEAR/RING GEAR/TONE RING (Continued)

(12) Remove and record pinion depth shims from the pinion gear shaft.

INSTALLATION

NOTE: Pinion depth shims are placed between the rear pinion bearing and pinion gear head to achieve proper ring and pinion gear mesh. If the ring and pinion gears are reused, the original pinion depth shim can be used. Refer to Adjustments (Pinion Gear Depth) to select the proper shim thickness.

(1) Install pinion depth shim in rear pinion bearing cup bore.

(2) Apply Mopar Door Ease or equivalent lubricant to the outside surface of pinion bearing cups. Install rear bearing cup with Installer D-145 and Handle C-4171 (Fig. 94) and verify cup is seated.

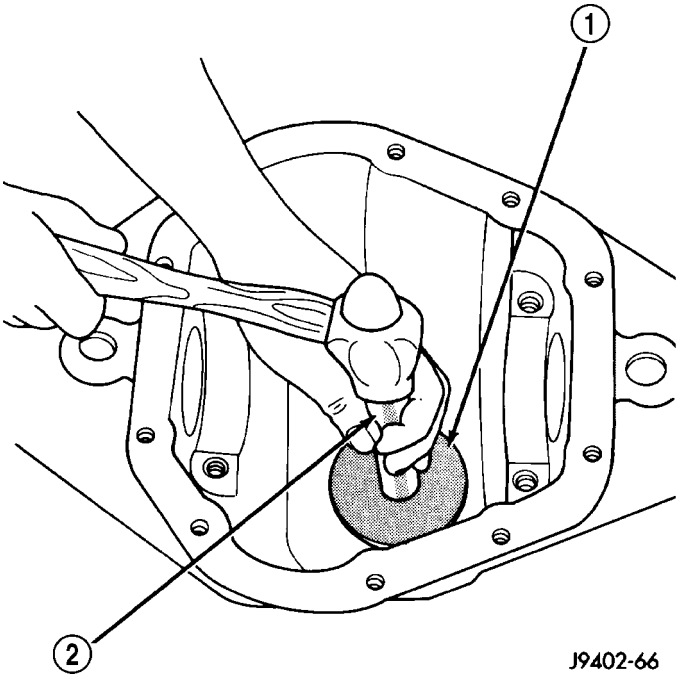


Fig. 94 REAR BEARING CUP

1 - INSTALLER
2 - HANDLE

(3) Install front bearing cup with Installer D-144 and Handle C-4171 (Fig. 95) and verify cup is seated.

(4) Install front pinion bearing and oil slinger, if equipped.

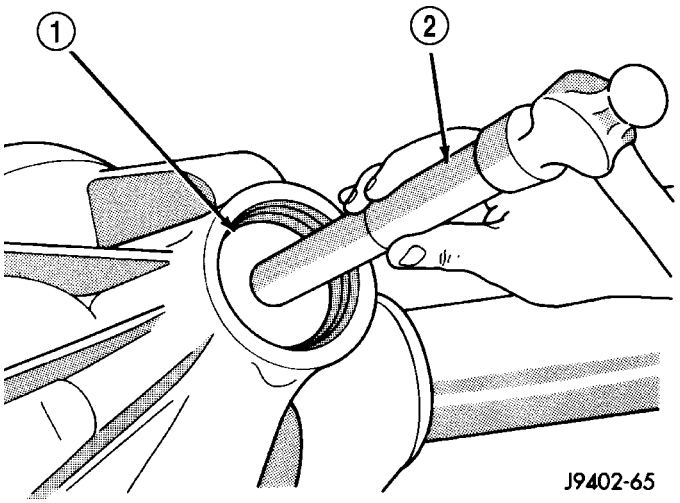


Fig. 95 FRONT BEARING CUP

1 - INSTALLER
2 - HANDLE

(5) Apply a light coating of gear lubricant on the lip of pinion seal. Install seal with Installer 8681 (Fig. 96).

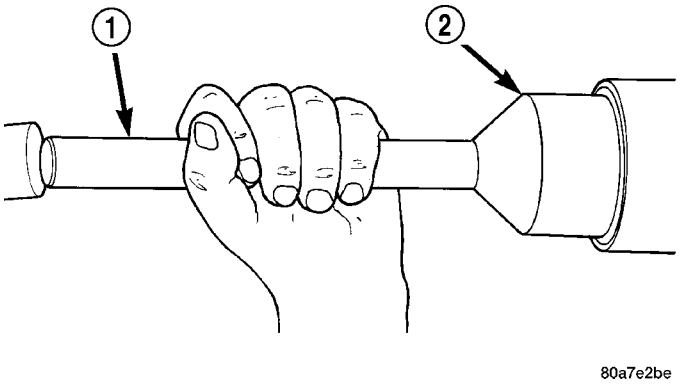


Fig. 96 PINION SEAL INSTALLER

1 - HANDLE
2 - INSTALLER

PINION GEAR/RING GEAR/TONE RING (Continued)

(6) Install pinion depth shim on the pinion shaft. Install rear pinion bearing on the pinion gear with Installer W-262 and a press (Fig. 97).

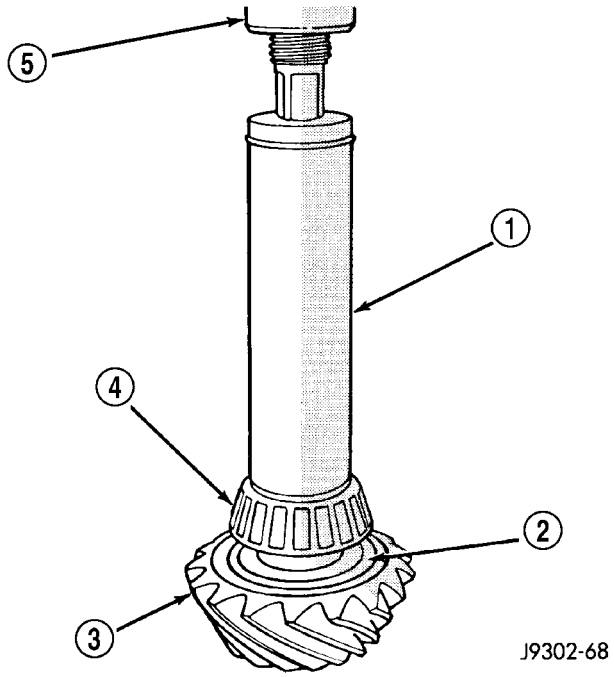


Fig. 97 REAR PINION BEARING

1 - INSTALLER
2 - SHIM
3 - DRIVE PINION GEAR
4 - REAR PINION BEARING
5 - PRESS

(7) Install pinion gear into the housing.

(8) Install yoke with Installer W-162-B, Cup 8109 and Spanner Wrench 6958 (Fig. 98).

(9) Install pinion washer and a **new** nut hold pinion yoke Spanner Wrench 6958 (Fig. 99) and tighten nut to 217-271 N·m (160-200 ft. lbs.).

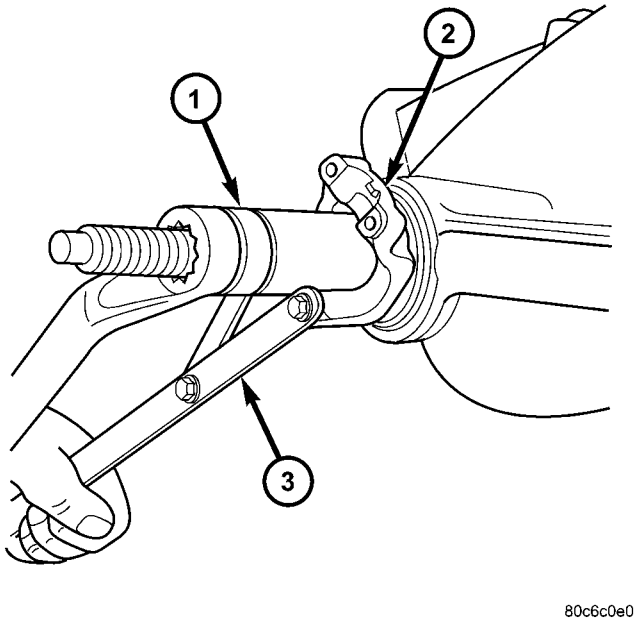


Fig. 98 PINION YOKE INSTALLER

1 - INSTALLER
2 - PINION YOKE
3 - SPANNER WRENCH

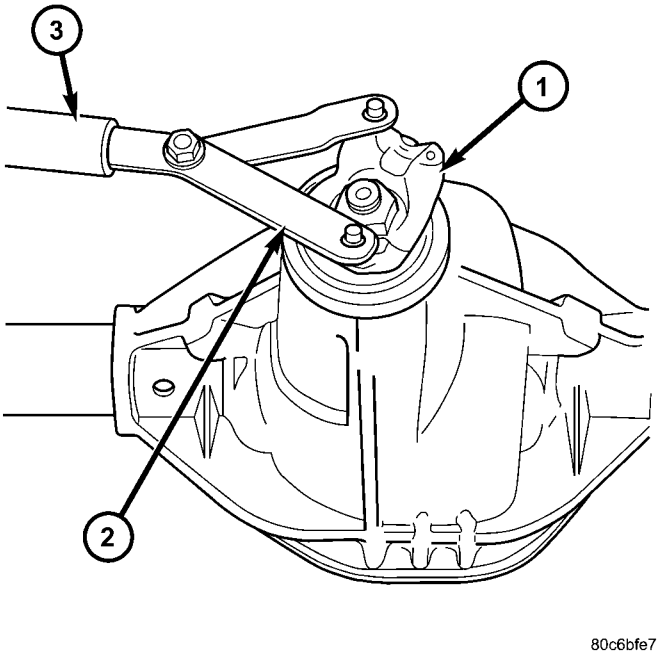


Fig. 99 YOKE SPANNER WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

PINION GEAR/RING GEAR/TONE RING (Continued)

(10) Check bearing preload torque with an inch pound torque wrench (Fig. 100). The torque necessary to rotate the pinion gear should be:

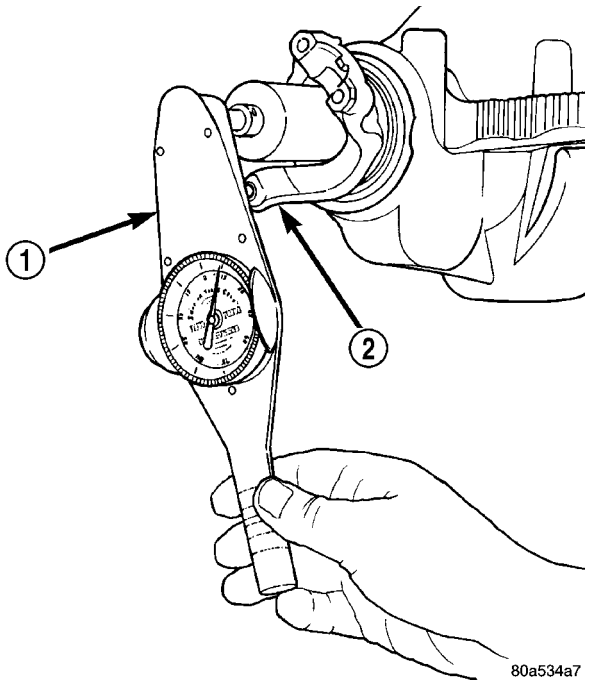


Fig. 100 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

- Original Bearings: 1 to 2.26 N·m (10 to 20 in. lbs.).
- New Bearings: 2.26 to 4.52 N·m (20 to 40 in. lbs.).

(11) If rotating torque is above the desired amount, remove the pinion yoke and increase the preload shim pack thickness. Increasing the shim pack thickness 0.025 mm (0.001 in.) will decrease the rotating torque approximately 0.9 N·m (8 in. lbs.).

(12) If the maximum tightening torque is reached prior to achieving the desired rotating torque, remove the pinion yoke and decrease the thickness of the preload shim pack. Decreasing the shim pack thickness 0.025 mm (0.001 in.) will increase the rotating torque approximately 0.9 N·m (8 in. lbs.).

(13) Invert differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

CAUTION: Never reuse ring gear bolts, the bolts can fracture causing extensive damage. Failure to heed caution may result in damage.

(14) Invert differential case in the vise.

(15) Install **new** ring gear bolts and alternately tighten to 108 N·m (100 ft. lbs.) (Fig. 101).

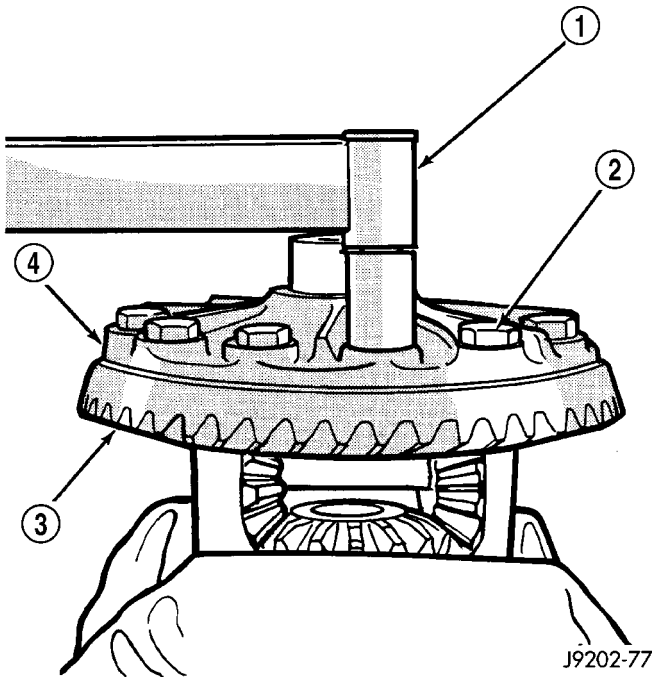


Fig. 101 RING GEAR

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(16) Install differential in axle housing and verify gear mesh and contact pattern. Refer to Adjustment (Gear Contact Pattern).

BRAKES

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BRAKES - BASE

DESCRIPTION

DESCRIPTION - BRAKE SYSTEM

Power assist front disc and rear drum brakes are standard equipment. Disc brake components consist of single piston calipers and ventilated rotors. Rear drum brakes are dual shoe units with cast brake drums.

The parking brake mechanism is lever and cable operated. The cables are attached to levers on the rear drum brake secondary shoes. The parking brakes are operated by a hand lever.

A dual diaphragm vacuum power brake booster is used for all applications. All models have an aluminum master cylinder with plastic reservoir.

All models are equipped with a combination valve. The valve contains a pressure differential valve and switch and a fixed rate rear proportioning valve.

Factory brake lining on all models consists of an organic base material combined with metallic particles. The original equipment linings do not contain asbestos.

DESCRIPTION - SERVICE WARNINGS & CAUTIONS

WARNING: DUST AND DIRT ACCUMULATING ON BRAKE PARTS DURING NORMAL USE MAY CONTAIN ASBESTOS FIBERS FROM PRODUCTION OR AFTERMARKET LININGS. BREATHING EXCESSIVE CONCENTRATIONS OF ASBESTOS FIBERS CAN CAUSE SERIOUS BODILY HARM. EXERCISE CARE WHEN SERVICING BRAKE PARTS. DO NOT CLEAN BRAKE PARTS WITH COMPRESSED AIR OR BY DRY BRUSHING. USE A VACUUM CLEANER SPECIFICALLY DESIGNED FOR THE REMOVAL OF ASBESTOS FIBERS FROM BRAKE COMPONENTS. IF A SUITABLE VACUUM CLEANER IS NOT AVAILABLE, CLEANING SHOULD BE DONE WITH A WATER DAMPENED CLOTH. DO NOT SAND, OR GRIND BRAKE LINING UNLESS EQUIPMENT USED IS DESIGNED TO CONTAIN THE DUST RESIDUE. DISPOSE OF ALL RESIDUE CONTAINING ASBESTOS FIBERS IN SEALED BAGS OR CONTAINERS TO MINIMIZE EXPOSURE TO YOURSELF AND OTHERS. FOLLOW PRACTICES PRESCRIBED BY THE OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION AND THE ENVIRONMENTAL PROTECTION AGENCY FOR THE HANDLING, PROCESSING, AND DISPOSITION OF DUST OR DEBRIS THAT MAY CONTAIN ASBESTOS FIBERS.

CAUTION: Never use gasoline, kerosene, alcohol, motor oil, transmission fluid, or any fluid containing mineral oil to clean the system components. These fluids damage rubber cups and seals. Use only fresh brake fluid or Mopar brake cleaner to clean or flush brake system components. These are the only cleaning materials recommended. If system contamination is suspected, check the fluid for dirt, discoloration, or separation into distinct layers. Also check the reservoir cap seal for distortion. Drain and flush the system with new brake fluid if contamination is suspected.

CAUTION: Use Mopar brake fluid, or an equivalent quality fluid meeting SAE/DOT standards J1703 and DOT 3. Brake fluid must be clean and free of contaminants. Use fresh fluid from sealed containers only to ensure proper antilock component operation.

CAUTION: Use Mopar multi-mileage or high temperature grease to lubricate caliper slide surfaces, drum brake pivot pins, and shoe contact points on the backing plates. Use multi-mileage grease or GE 661 or Dow 111 silicone grease on caliper slide pins to ensure proper operation.

DIAGNOSIS AND TESTING - BASE BRAKE SYSTEM

Base brake components consist of the brake shoes, calipers, wheel cylinders, brake drums, rotors, brake lines, master cylinder, booster, and parking brake components.

Brake diagnosis involves determining if the problem is related to a mechanical, hydraulic, or vacuum operated component.

The first diagnosis step is the preliminary check.

PRELIMINARY BRAKE CHECK

(1) Check condition of tires and wheels. Damaged wheels and worn, damaged, or underinflated tires can cause pull, shudder, vibration, and a condition similar to grab.

(2) If complaint was based on noise when braking, check suspension components. Jounce front and rear of vehicle and listen for noise that might be caused by loose, worn or damaged suspension or steering components.

(3) Inspect brake fluid level and condition. Note that the front disc brake reservoir fluid level will decrease in proportion to normal lining wear. **Also note that brake fluid tends to darken over time. This is normal and should not be mistaken for contamination.**

(a) If fluid level is abnormally low, look for evidence of leaks at calipers, wheel cylinders, brake lines, and master cylinder.

(b) If fluid appears contaminated, drain out a sample. System will have to be flushed if fluid is separated into layers, or contains a substance other than brake fluid. The system seals and cups will also have to be replaced after flushing. Use clean brake fluid to flush the system.

(4) Check parking brake operation. Verify free movement and full release of cables and pedal. Also note if vehicle was being operated with parking brake partially applied.

(5) Check brake pedal operation. Verify that pedal does not bind and has adequate free play. If pedal lacks free play, check pedal and power booster for being loose or for bind condition. Do not road test until condition is corrected.

(6) If components checked appear OK, road test the vehicle.

ROAD TESTING

(1) If complaint involved low brake pedal, pump pedal and note if it comes back up to normal height.

BRAKES - BASE (Continued)

(2) Check brake pedal response with transmission in Neutral and engine running. Pedal should remain firm under constant foot pressure.

(3) During road test, make normal and firm brake stops in 25-40 mph range. Note faulty brake operation such as low pedal, hard pedal, fade, pedal pulsation, pull, grab, drag, noise, etc.

PEDAL FALLS AWAY

A brake pedal that falls away under steady foot pressure is generally the result of a system leak. The leak point could be at a brake line, fitting, hose, or caliper/wheel cylinder. Internal leakage in the master cylinder caused by worn or damaged piston cups, may also be the problem cause.

If leakage is severe, fluid will be evident at or around the leaking component. However, internal leakage in the master cylinder may not be physically evident.

LOW PEDAL

If a low pedal is experienced, pump the pedal several times. If the pedal comes back up, worn lining, rotors, or drums are the most likely causes.

SPONGY PEDAL

A spongy pedal is most often caused by air in the system. However, thin brake drums or substandard brake lines and hoses can also cause a spongy pedal. The proper course of action is to bleed the system, or replace thin drums and suspect quality brake lines and hoses.

HARD PEDAL OR HIGH PEDAL EFFORT

A hard pedal or high pedal effort may be due to lining that is water soaked, contaminated, glazed, or badly worn. The power booster or check valve could also be faulty.

PEDAL PULSATION

Pedal pulsation is caused by components that are loose, or beyond tolerance limits.

The primary cause of pulsation are disc brake rotors with excessive lateral runout or thickness variation, or out of round brake drums. Other causes are loose wheel bearings or calipers and worn, damaged tires.

NOTE: Some pedal pulsation may be felt during ABS activation.

BRAKE DRAG

Brake drag occurs when the lining is in constant contact with the rotor or drum. Drag can occur at one wheel, all wheels, fronts only, or rears only.

Drag is a product of incomplete brake shoe release. Drag can be minor or severe enough to overheat the linings, rotors and drums.

Minor drag will usually cause slight surface charring of the lining. It can also generate hard spots in rotors and drums from the overheat-cool down process. In most cases, the rotors, drums, wheels and tires are quite warm to the touch after the vehicle is stopped.

Severe drag can char the brake lining all the way through. It can also distort and score rotors and drums to the point of replacement. The wheels, tires and brake components will be extremely hot. In severe cases, the lining may generate smoke as it chars from overheating.

Possible causes for brake drag condition are:

- Seized or improperly adjusted parking brake cables.
- Loose/worn wheel bearing.
- Seized caliper or wheel cylinder piston.
- Caliper binding on corroded bushings or rusted slide surfaces.
- Loose caliper mounting bracket.
- Drum brake shoes binding on worn/damaged support plates.
- Mis-assembled components.

If brake drag occurs at all wheels, the problem may be related to a blocked master cylinder return port, or faulty power booster (binds-does not release).

BRAKE FADE

Brake fade is usually a product of overheating caused by brake drag. However, brake overheating and resulting fade can also be caused by riding the brake pedal, making repeated high deceleration stops in a short time span, or constant braking on steep mountain roads. Refer to the Brake Drag information in this section for causes.

BRAKE PULL

Possible causes for front brake pull condition are:

- Contaminated lining in one caliper.
- Seized caliper piston.
- Binding caliper.
- Loose caliper.
- Rusty adapter/caliper slide surfaces.
- Improper brake shoes.
- Damaged rotor.

A worn, damaged wheel bearing or suspension component are further causes of pull. A damaged front tire (bruised, ply separation) can also cause pull.

A common and frequently misdiagnosed pull condition is where direction of pull changes after a few stops. The cause is a combination of brake drag followed by fade at one of the brake units.

As the dragging brake overheats, efficiency is so reduced that fade occurs. Since the opposite brake unit is still functioning normally, its braking effect is magnified. This causes pull to switch direction in favor of the normally functioning brake unit.

An additional point when diagnosing a change in pull condition concerns brake cool down. Remember that pull will return to the original direction, if the dragging brake unit is allowed to cool down (and is not seriously damaged).

REAR BRAKE GRAB OR PULL

Rear grab or pull is usually caused by improperly adjusted or seized parking brake cables, contaminated lining, bent or binding shoes and support plates, or improperly assembled components. This is particularly true when only one rear wheel is involved. However, when both rear wheels are affected, the master cylinder or proportioning valve could be at fault.

BRAKES DO NOT HOLD AFTER DRIVING THROUGH DEEP WATER PUDDLES

This condition is generally caused by water soaked lining. If the lining is only wet, it can be dried by driving with the brakes very lightly applied for a mile or two. However, if the lining is both soaked and dirt contaminated, cleaning and/or replacement will be necessary.

BRAKE SQUEAK/SQUEAL

Brake squeak or squeal may be due to linings that are wet or contaminated with brake fluid, grease, or oil. Glazed linings and rotors with hard spots can also contribute to squeak. Dirt and foreign material embedded in the brake lining will also cause squeak/squeal.

A very loud squeak or squeal is frequently a sign of severely worn brake lining. If the lining has worn through to the brake shoes in spots, metal-to-metal contact occurs. If the condition is allowed to continue, rotors and drums can become so scored that replacement is necessary.

BRAKE CHATTER

Brake chatter is usually caused by loose or worn components, or glazed/burnt lining. Rotors with hard spots can also contribute to chatter. Additional causes of chatter are out-of-tolerance rotors, brake lining not securely attached to the shoes, loose wheel bearings and contaminated brake lining.

THUMP/CLUNK NOISE

Thumping or clunk noises during braking are frequently **not** caused by brake components. In many cases, such noises are caused by loose or damaged steering, suspension, or engine components. However, calipers that bind on the slide surfaces can generate a thump or clunk noise. In addition, worn out, improperly adjusted, or improperly assembled rear brake shoes can also produce a thump noise.

BRAKE LINING CONTAMINATION

Brake lining contamination is mostly a product of leaking calipers or wheel cylinders, worn seals, driving through deep water puddles, or lining that has become covered with grease and grit during repair. Contaminated lining should be replaced to avoid further brake problems.

WHEEL AND TIRE PROBLEMS

Some conditions attributed to brake components may actually be caused by a wheel or tire problem.

A damaged wheel can cause shudder, vibration and pull. A worn or damaged tire can also cause pull.

Severely worn tires with very little tread left can produce a grab-like condition as the tire loses and recovers traction. Flat-spotted tires can cause vibration and generate shudder during brake operation. A tire with internal damage such as a severe bruise, cut, or ply separation can cause pull and vibration.

STANDARD PROCEDURE

STANDARD PROCEDURE - MANUAL BLEEDING

Use Mopar brake fluid, or an equivalent quality fluid meeting SAE J1703-F and DOT 3 standards only. Use fresh, clean fluid from a sealed container at all times.

(1) Remove reservoir filler caps and fill reservoir.

(2) If calipers, or wheel cylinders were overhauled, open all caliper and wheel cylinder bleed screws. Then close each bleed screw as fluid starts to drip from it. Top off master cylinder reservoir once more before proceeding.

(3) Attach one end of bleed hose to bleed screw and insert opposite end in glass container partially filled with brake fluid (Fig. 1). Be sure end of bleed hose is immersed in fluid.

(4) Open up bleeder, then have a helper press down the brake pedal. Once the pedal is down close the bleeder. Repeat bleeding until fluid stream is clear and free of bubbles. Then move to the next wheel.

STANDARD PROCEDURE - PRESSURE BLEEDING

Use Mopar brake fluid, or an equivalent quality fluid meeting SAE J1703-F and DOT 3 standards only. Use fresh, clean fluid from a sealed container at all times.

BRAKES - BASE (Continued)

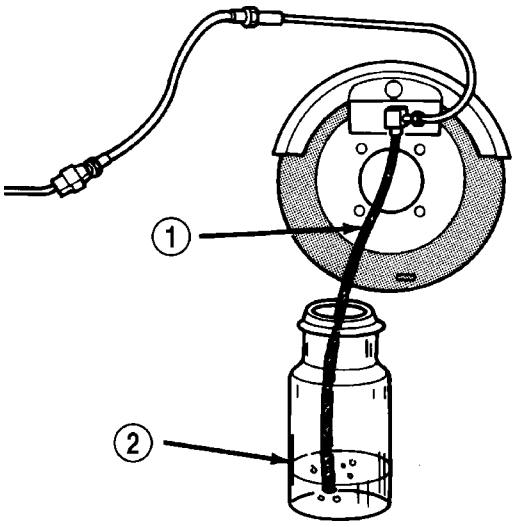


Fig. 1 Bleed Hose Setup

1 - BLEED HOSE
2 - FLUID CONTAINER PARTIALLY FILLED WITH FLUID

Follow the manufacturers instructions carefully when using pressure equipment. Do not exceed the tank manufacturers pressure recommendations. Generally, a tank pressure of 15-20 psi is sufficient for bleeding.

Fill the bleeder tank with recommended fluid and purge air from the tank lines before bleeding.

Do not pressure bleed without a proper master cylinder adapter. The wrong adapter can lead to leakage, or drawing air back into the system. Use adapter provided with the equipment or Adapter 6921.

SPECIFICATIONS

BRAKE COMPONENTS

SPECIFICATIONS

DESCRIPTION	SPECIFICATION
Disc Brake Caliper Type	Sliding
Disc Brake Rotor Type Front	Ventilated
Disc Brake Rotor Type Rear	Solid
Disc Brake Rotor Diameter Front	279.4 x 23.876 mm (11 x 0.94 in.)
Disc Brake Rotor Diameter Rear	285 x 12 mm (11 x 0.472 in.)
Disc Brake Rotor Ventilated Front	Max. Runout 0.12 mm (0.005 in.)
Disc Brake Rotor Solid Rear	Max. Runout 0.102 mm (0.004 in)
Disc Brake Rotor Ventilated Front	Max. Thickness Variation 0.013 mm (0.0005 in.)
Disc Brake Rotor Solid Rear	Max Thickness Variation 0.018 mm (0.0007 in)
Disc Brake Rotor Ventilated Front	Min. Thickness 22.7 mm (0.8937 in.)
Disc Brake Rotor Solid Rear	Min. Thickness 11.00 mm (0.433 in.)
Brake Drum Diameter	228.6 x 63.5 mm (9 x 2.5 in.)
Brake Booster Type	Tandem Diaphragm

BRAKES - BASE (Continued)

TORQUE CHART

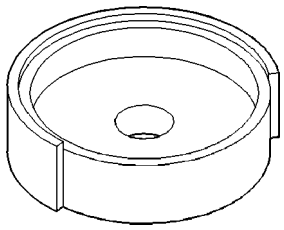
TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Brake Pedal Support Bolt	28	21	—
Brake Booster Mounting Nuts	39	29	—
Master Cylinder Mounting Nuts	17	13	—
Master Cylinder Brake Lines	19	14	—
Combination Valve Mounting Nuts	20	15	—
Combination Valve Brake Lines	19	14	—
Caliper Mounting Bolts Front	15	11	—
Caliper Mounting Bolts Rear	25	—	220
Caliper Brake Hose Banjo Bolt Front	31	23	—
Caliper Brake Hose Banjo Bolt Rear	31	23	—
Wheel Cylinder Mounting Bolts	10	7	—
Wheel Cylinder Brake Line	16	12	—
Parking Brake Lever Bolts	12	9	—
Parking Brake Lever Bracket Bolts	12	9	—
Parking Brake Cable Retainer Nut	1.5	—	14

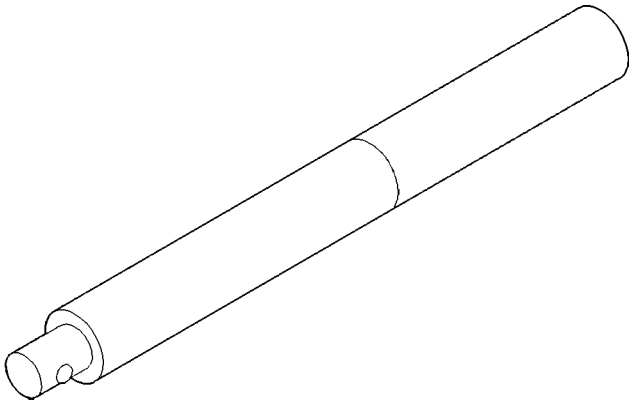
BRAKES - BASE (Continued)

SPECIAL TOOLS

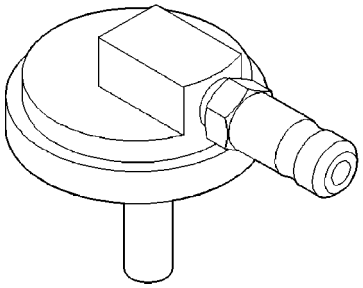
BASE BRAKES



Installer Caliper Dust Boot C-4842



Handle C-4171



Adaptor Cap Pressure Bleeder 6921

BRAKE LINES

DESCRIPTION

Flexible rubber hose is used at both front brakes and at the rear axle junction block. Double walled steel tubing is used to connect the master cylinder to the major hydraulic braking components and then to the flexible rubber hoses. Double inverted style and ISO style flares are used on the brake lines.

OPERATION

The hoses and lines transmit the brake fluid hydraulic pressure to the calipers and or wheel cylinders.

DIAGNOSIS AND TESTING - BRAKE LINE AND HOSES

Flexible rubber hose is used at both front brakes and at the rear axle junction block. Inspect the hoses whenever the brake system is serviced, at every engine oil change, or whenever the vehicle is in for service.

Inspect the hoses for surface cracking, scuffing, or worn spots. Replace any brake hose immediately if the fabric casing of the hose is exposed due to cracks or abrasions.

Also check brake hose installation. Faulty installation can result in kinked, twisted hoses, or contact with the wheels and tires or other chassis components. All of these conditions can lead to scuffing, cracking and eventual failure.

The steel brake lines should be inspected periodically for evidence of corrosion, twists, kinks, leaks, or other damage. Heavily corroded lines will eventually rust through causing leaks. In any case, corroded or damaged brake lines should be replaced.

Factory replacement brake lines and hoses are recommended to ensure quality, correct length and superior fatigue life. Care should be taken to make sure that brake line and hose mating surfaces are clean and free from nicks and burrs. Also remember that right and left brake hoses are not interchangeable.

Use new copper seal washers at all caliper connections. Be sure brake line connections are properly made (not cross threaded) and tightened to recommended torque.

STANDARD PROCEDURE

STANDARD PROCEDURE - BRAKE TUBE FLARING

A preformed metal brake tube is recommended and preferred for all repairs. However, double-wall steel tube can be used for emergency repair when factory replacement parts are not readily available.

Special bending tools are needed to avoid kinking or twisting of metal brake tubes. Special flaring tools are needed to make a double inverted flare or ISO flare (Fig. 2).

STANDARD PROCEDURE - DOUBLE INVERTED FLARING

A preformed metal brake tube is recommended and preferred for all repairs. However, double-wall steel tube can be used for emergency repair when factory replacement parts are not readily available.

(1) Cut off damaged tube with Tubing Cutter.

(2) Ream cut edges of tubing to ensure proper flare.

BRAKE LINES (Continued)

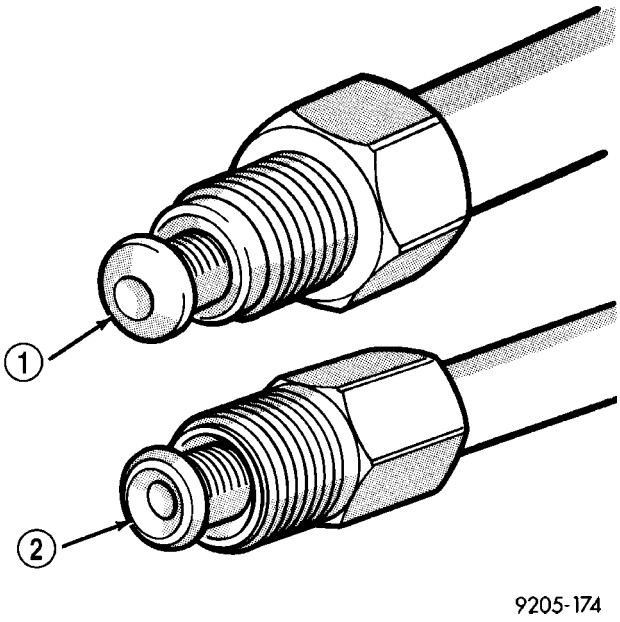


Fig. 2 Inverted Flare And ISO Flare

1 - ISO-STYLE FLARE
2 - DOUBLE INVERTED-STYLE FLARE

(3) Install replacement tube nut on the tube.
(4) Insert tube in flaring tool.
(5) Place gauge form over the end of the tube.
(6) Push tubing through flaring tool jaws until tube contacts recessed notch in gauge that matches tube diameter.
(7) Tighten the tool bar on the tube
(8) Insert plug on gauge in the tube. Then swing compression disc over gauge and center tapered flaring screw in recess of compression disc (Fig. 3).
(9) Tighten tool handle until plug gauge is squarely seated on jaws of flaring tool. This will start the inverted flare.
(10) Remove the plug gauge and complete the inverted flare.

STANDARD PROCEDURE - ISO FLARING

A preformed metal brake tube is recommended and preferred for all repairs. However, double-wall steel tube can be used for emergency repair when factory replacement parts are not readily available.

To make a ISO flare use a Flaring Tool kit.
(1) Cut off damaged tube with Tubing Cutter.
(2) Remove any burrs from the inside of the tube.
(3) Install tube nut on the tube.
(4) Position the tube in the flaring tool flush with the top of the tool bar (Fig. 4). Then tighten the tool bar on the tube.
(5) Install the correct size adaptor on the flaring tool yoke screw.
(6) Lubricate the adaptor.
(7) Align the adaptor and yoke screw over the tube (Fig. 4).

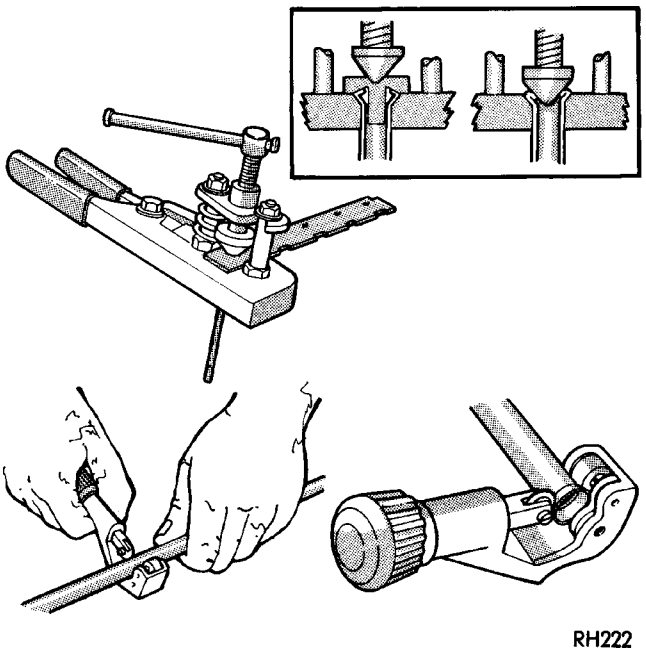


Fig. 3 Inverted Flare Tools

(8) Turn the yoke screw in until the adaptor is squarely seated on the tool bar.

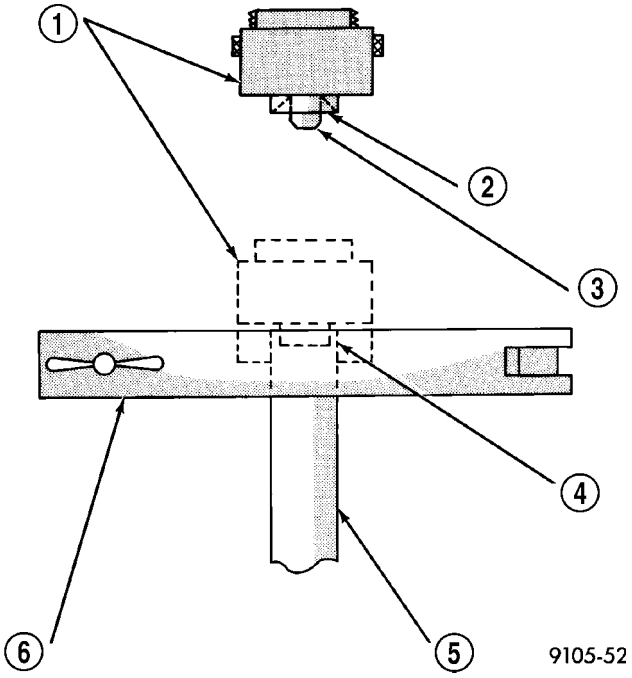


Fig. 4 ISO Flaring

1 - ADAPTER
2 - LUBRICATE HERE
3 - PILOT
4 - FLUSH WITH BAR
5 - TUBING
6 - BAR ASSEMBLY

DISC BRAKE CALIPERS

DESCRIPTION

The calipers are a single piston type. The calipers are free to slide laterally, this allows continuous compensation for lining wear.

OPERATION

When the brakes are applied fluid pressure is exerted against the caliper piston. The fluid pressure is exerted equally and in all directions. This means pressure exerted against the caliper piston and within the caliper bore will be equal (Fig. 5).

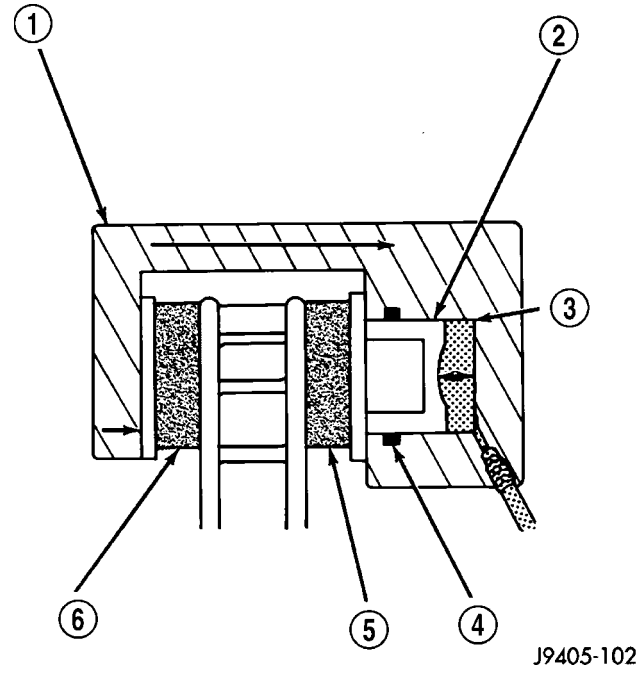


Fig. 5 Brake Caliper Operation

1 - CALIPER
2 - PISTON
3 - PISTON BORE
4 - SEAL
5 - INBOARD SHOE
6 - OUTBOARD SHOE

Fluid pressure applied to the piston is transmitted directly to the inboard brake shoe. This forces the shoe lining against the inner surface of the disc brake rotor. At the same time, fluid pressure within the piston bore forces the caliper to slide inward on the mounting bolts. This action brings the outboard brake shoe lining into contact with the outer surface of the disc brake rotor.

In summary, fluid pressure acting simultaneously on both piston and caliper, produces a strong clamping action. When sufficient force is applied, friction will attempt to stop the rotors from turning and bring the vehicle to a stop.

Application and release of the brake pedal generates only a very slight movement of the caliper and piston. Upon release of the pedal, the caliper and piston return to a rest position. The brake shoes do not retract an appreciable distance from the rotor. In fact, clearance is usually at, or close to zero. The reasons for this are to keep road debris from getting between the rotor and lining and in wiping the rotor surface clear each revolution.

The caliper piston seal controls the amount of piston extension needed to compensate for normal lining wear.

During brake application, the seal is deflected outward by fluid pressure and piston movement (Fig. 6). When the brakes (and fluid pressure) are released, the seal relaxes and retracts the piston.

The amount of piston retraction is determined by the amount of seal deflection. Generally the amount is just enough to maintain contact between the piston and inboard brake shoe.

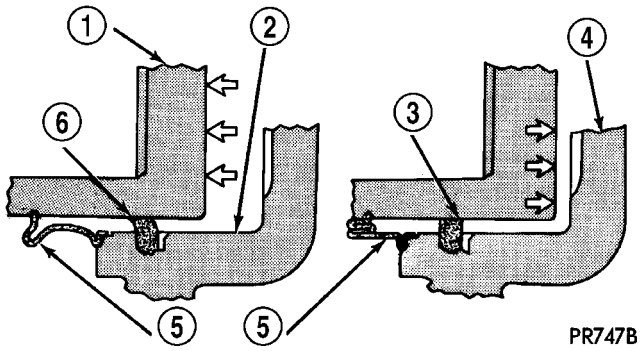


Fig. 6 Lining Wear Compensation By Piston Seal

1 - PISTON
2 - CYLINDER BORE
3 - PISTON SEAL BRAKE PRESSURE OFF
4 - CALIPER HOUSING
5 - DUST BOOT
6 - PISTON SEAL BRAKE PRESSURE ON

REMOVAL

REMOVAL - FRONT

(1) Raise and support vehicle.
(2) Remove front wheel and tire assembly.
(3) Drain small amount of fluid from master cylinder brake reservoir with suction gun.
(4) Bottom caliper piston in bore with C-clamp. Position clamp screw on outboard brake shoe and clamp frame on rear of caliper (Fig. 7). **Do not allow clamp screw to bear directly on outboard shoe retainer spring. Use wood or metal spacer between shoe and clamp screw.**

DISC BRAKE CALIPERS (Continued)

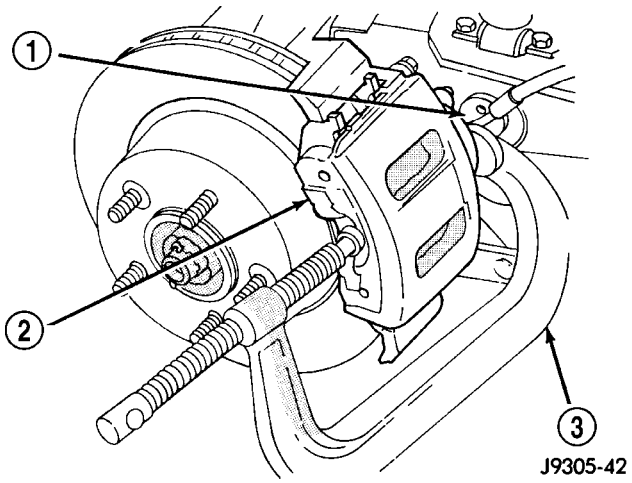


Fig. 7 Bottoming Caliper Piston With C-Clamp

1 - CALIPER BOSS
2 - OUTBOARD BRAKESHOE
3 - C-CLAMP

(5) Remove brake hose mounting bolt and discard washers (Fig. 8).

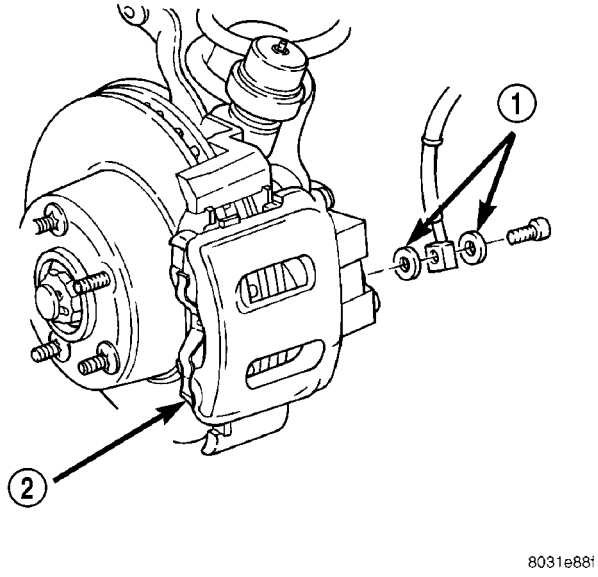


Fig. 8 Brake Hose And Bolt

1 - FITTING WASHERS
2 - CALIPERS

(6) Remove caliper mounting bolts (Fig. 9).

(7) Tilt top of caliper outward with pry tool if necessary (Fig. 10) and remove caliper.

(8) Remove caliper from vehicle.

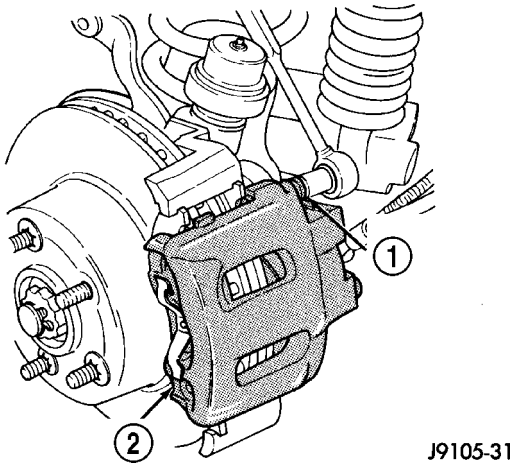


Fig. 9 Caliper Mounting Bolts

1 - CALIPER MOUNTING BOLT (2)
2 - CALIPER

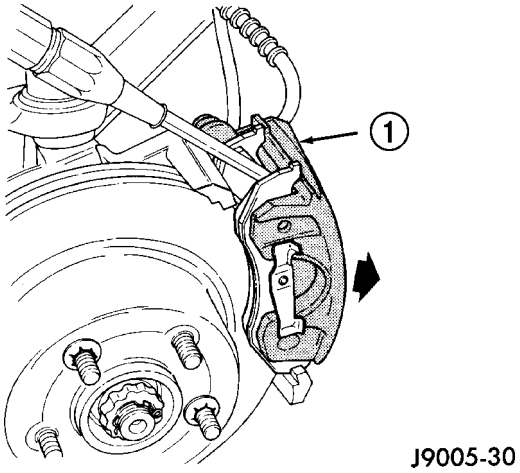


Fig. 10 Caliper Removal

1 - TILT CALIPER OUTBOARD TO REMOVE

REMOVAL - REAR

(1) Install prop rod on the brake pedal to keep pressure on the brake system.

(2) Raise and support vehicle.

(3) Remove the wheel and tire assembly.

(4) Remove the brake hose banjo bolt if replacing caliper.

(5) Remove the caliper mounting slide pin bolts (Fig. 11).

(6) Remove the caliper from vehicle.

DISC BRAKE CALIPERS (Continued)

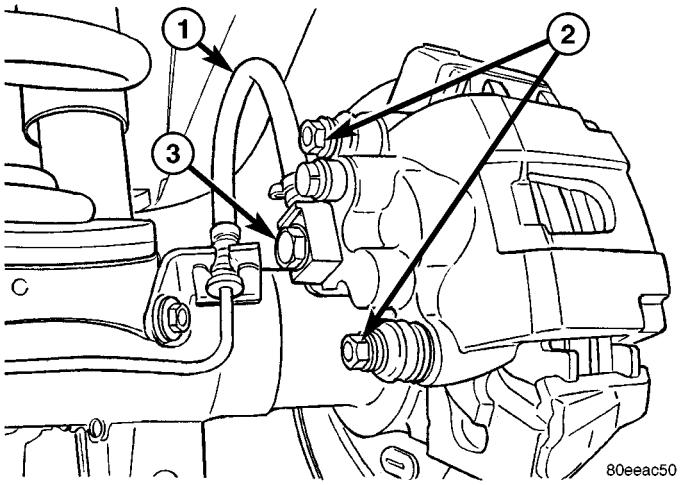


Fig. 11 CALIPER MOUNTING

1 - BRAKE HOSE
2 - CALIPER MOUNTING BOLTS
3 - BANJO BOLT

DISASSEMBLY

(1) Remove brake shoes from caliper.
(2) Drain brake fluid out of caliper.
(3) Take a piece of wood and pad it with one-inch thickness of shop towels. Place this piece in the outboard shoe side of the caliper in front of the piston. This will cushion and protect caliper piston during removal (Fig. 12).

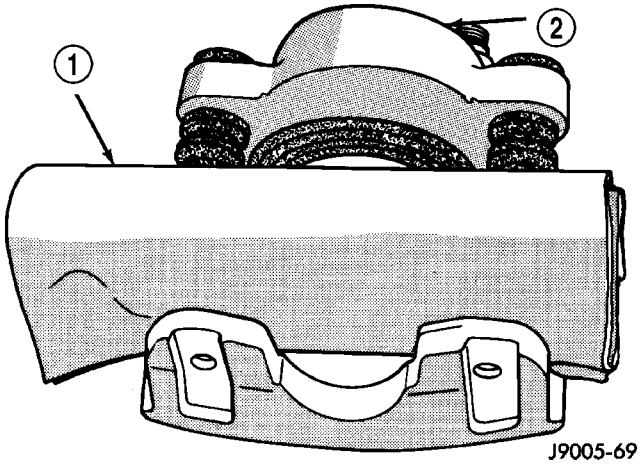


Fig. 12 Padding Caliper Interior

1 - SHOP TOWELS OR CLOTHS
2 - CALIPER

(4) Remove caliper piston with **short bursts** of low pressure compressed air. Direct air through fluid inlet port and ease piston out of bore (Fig. 13).

CAUTION: Do not blow the piston out of the bore with sustained air pressure. This could result in a cracked piston. Use only enough air pressure to ease the piston out.

WARNING: NEVER ATTEMPT TO CATCH THE PISTON AS IT LEAVES THE BORE. THIS MAY RESULT IN PERSONAL INJURY.

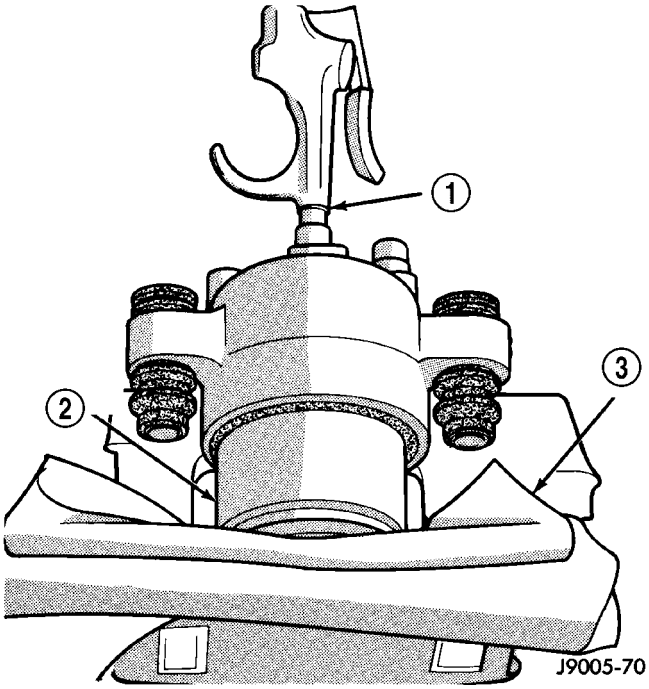


Fig. 13 Caliper Piston Removal

1 - AIR GUN
2 - CALIPER PISTON
3 - PADDING MATERIAL

(5) Remove caliper piston dust boot with suitable pry tool (Fig. 14).

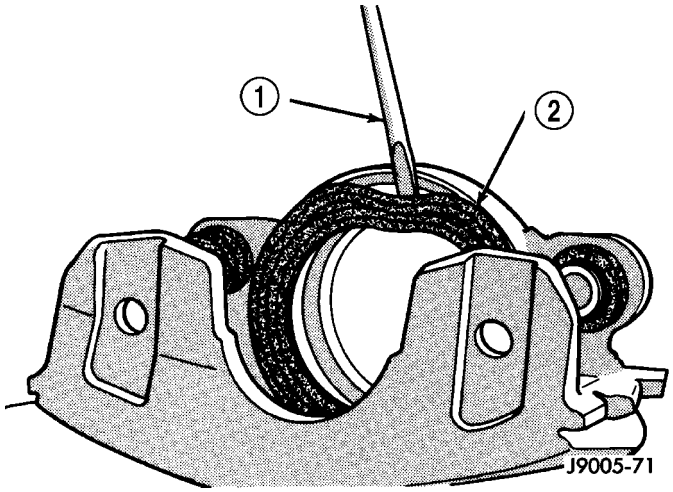


Fig. 14 Caliper

1 - COLLAPSE BOOT WITH PUNCH OR SCREWDRIVER
2 - PISTON DUST BOOT

(6) Remove caliper piston seal with wood or plastic tool (Fig. 15). Do not use metal tools as they will scratch piston bore.

DISC BRAKE CALIPERS (Continued)

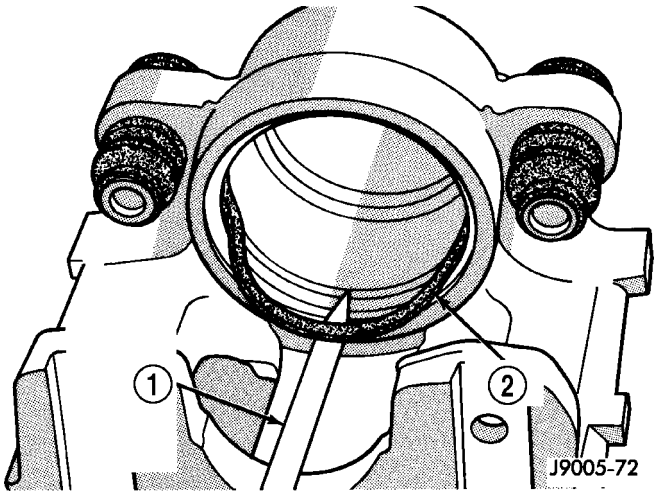


Fig. 15 Piston Seal Removal

1 - REMOVE SEAL WITH WOOD PENCIL OR SIMILAR TOOL
2 - PISTON SEAL

(7) Remove caliper mounting bolt bushings and boots (Fig. 16).

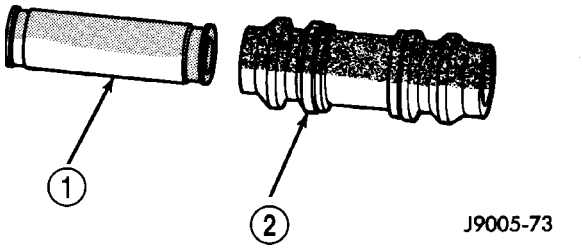


Fig. 16 Mounting Bolt Bushing And Boot

1 - CALIPER SLIDE BUSHING
2 - BOOT

CLEANING

Clean the caliper components with clean brake fluid or brake clean only. Wipe the caliper and piston dry with lint free towels or use low pressure compressed air.

CAUTION: Do not use gasoline, kerosene, paint thinner, or similar solvents. These products may leave a residue that could damage the piston and seal.

INSPECTION

The piston is made from a phenolic resin (plastic material) and should be smooth and clean.

The piston must be replaced if cracked or scored. Do not attempt to restore a scored piston surface by sanding or polishing.

CAUTION: If the caliper piston is replaced, install the same type of piston in the caliper. Never interchange phenolic resin and steel caliper pistons. The pistons, seals, seal grooves, caliper bore and piston tolerances are different.

The bore can be **lightly** polished with a brake hone to remove very minor surface imperfections (Fig. 17). The caliper should be replaced if the bore is severely corroded, rusted, scored, or if polishing would increase bore diameter more than 0.025 mm (0.001 inch).

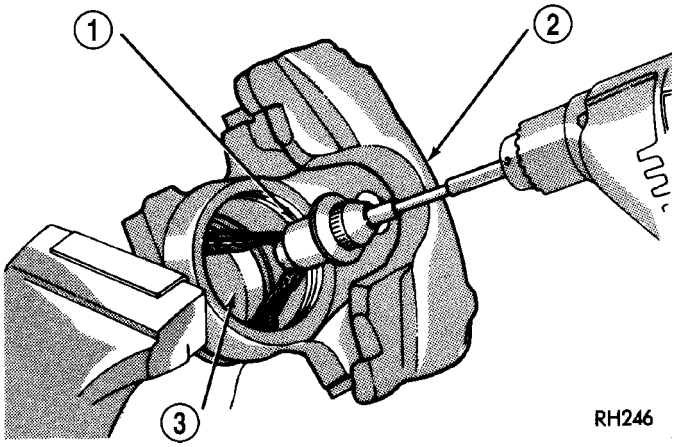


Fig. 17 Polishing Piston Bore

1 - SPECIAL HONE
2 - CALIPER
3 - PISTON BORE

ASSEMBLY

CAUTION: Dirt, oil, and solvents can damage caliper seals. Insure assembly area is clean and dry.

(1) Lubricate caliper piston bore, new piston seal and piston with clean brake fluid.

(2) Lubricate caliper bushings and interior of bushing boots with silicone grease.

DISC BRAKE CALIPERS (Continued)

(3) Install bushing boots in caliper, then insert bushing into boot and push bushing into place (Fig. 18).

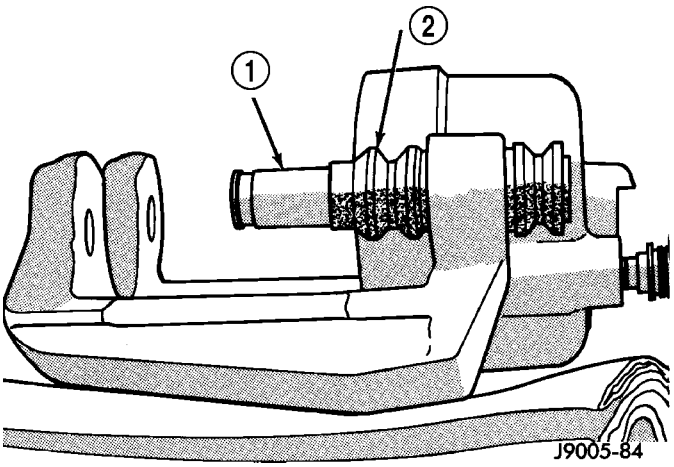


Fig. 18 Bushings And Boots Installation

1 - BUSHING
2 - BOOT

(4) Install new piston seal into seal groove with finger (Fig. 19).

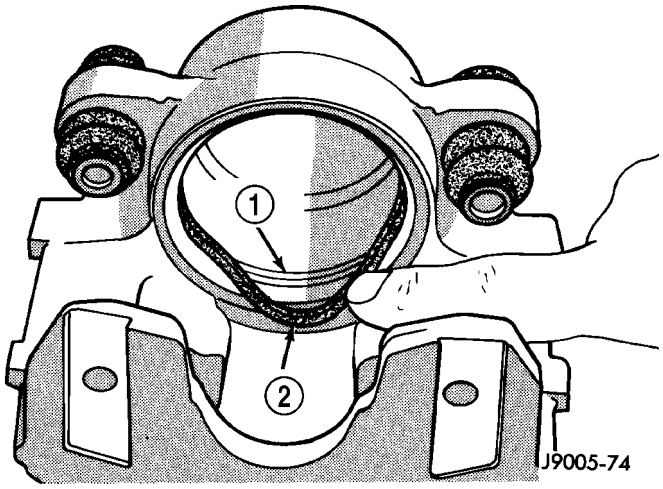


Fig. 19 Piston Seal Installation

1 - SEAL GROOVE
2 - PISTON SEAL

(5) Install new dust boot on caliper piston and seat boot in piston groove (Fig. 20).

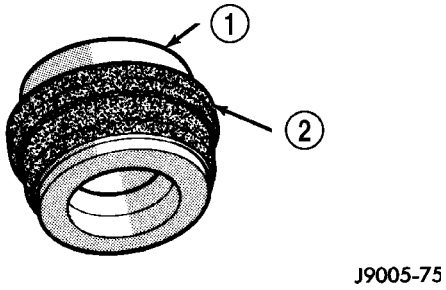


Fig. 20 Dust Boot On Piston

1 - PISTON
2 - DUST BOOT

(6) Press piston into caliper bore by hand, use a turn and push motion to work piston into seal (Fig. 21).

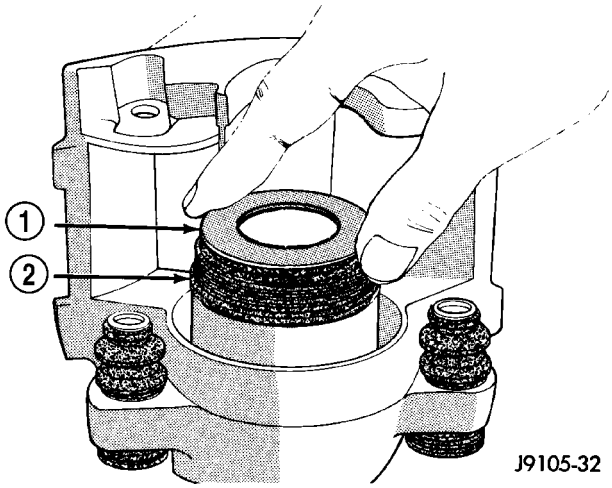


Fig. 21 Caliper Piston Installation

1 - PISTON
2 - BOOT

(7) Press caliper piston to bottom of bore.

(8) Seat dust boot in caliper with Installer Tool C-4842 and Tool Handle C-4171 (Fig. 22).

(9) Replace caliper bleed screw if removed.

DISC BRAKE CALIPERS (Continued)

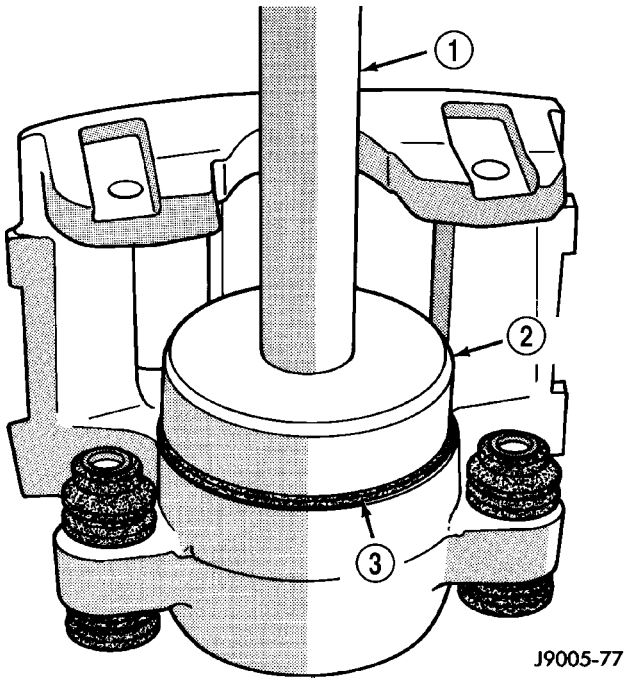


Fig. 22 Piston Dust Boot Installation

1 - HANDLE C-4171
2 - INSTALLER C-4842
3 - DUST BOOT

INSTALLATION

INSTALLATION - FRONT

(1) Clean brake shoe mounting ledges with wire brush and apply light coat of Mopar multi-mileage grease to surfaces (Fig. 23).

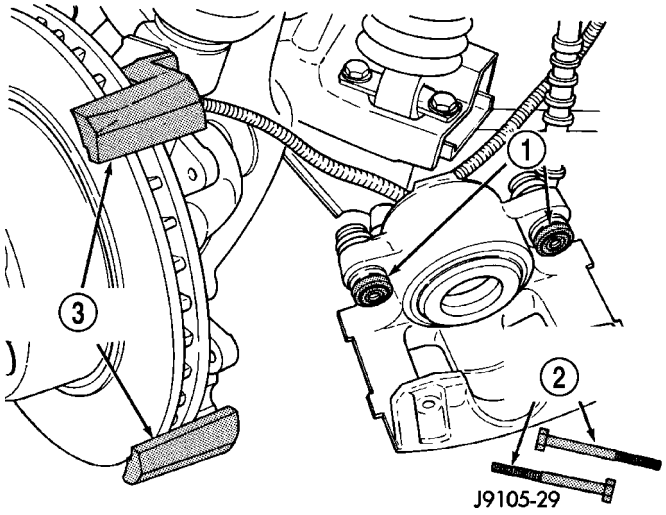


Fig. 23 Caliper Lubrication Points

1 - BUSHINGS
2 - CALIPER MOUNTING BOLTS
3 - MOUNTING LEDGES

(2) Install caliper by position notches at lower end of brake shoes on bottom mounting ledge. Then rotate caliper over rotor and seat notches at upper end of shoes on top mounting ledge (Fig. 24).

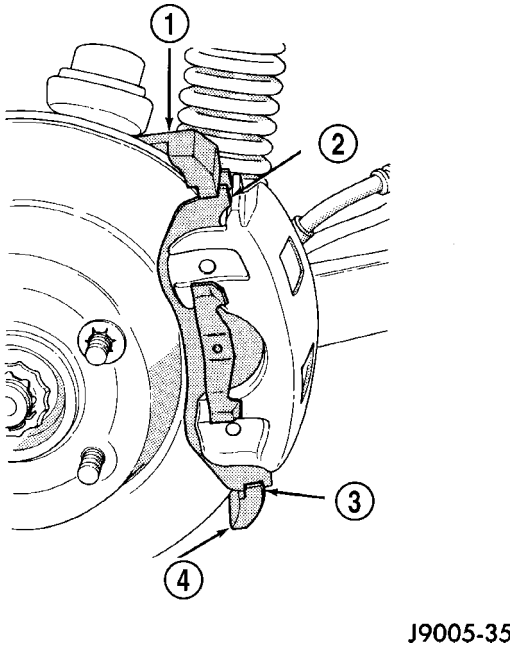


Fig. 24 Caliper Installation

1 - TOP LEDGE
2 - BRAKESHOE TAB ON LEDGE OUTER SURFACE
3 - LEDGE SEATED IN BRAKESHOE NOTCH
4 - BOTTOM LEDGE

(3) Coat caliper mounting bolts with silicone grease. Then install and tighten bolts to 15 N·m (11 ft. lbs.).

CAUTION: If new caliper bolts are being installed, or if the original reason for repair was a drag/pull condition, check caliper bolt length before proceeding. Bolts must not have a shank length greater than 67.6 mm (2.66 in.) (Fig. 25).

CORRECT SHANK LENGTH:

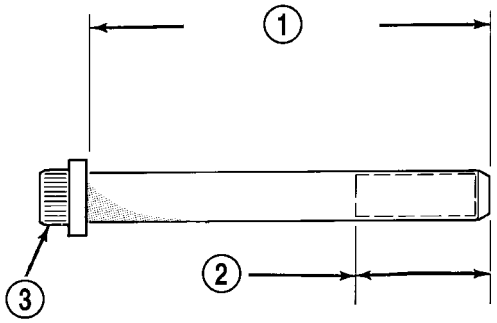


Fig. 25 Mounting Bolt Dimensions

1 - 67 mm (± 0.6 mm) 2.637 in. (± 0.0236 in.)
2 - 22 mm (0.866 in.) THREAD LENGTH
3 - CALIPER BOLT

DISC BRAKE CALIPERS (Continued)

(4) Install brake hose to caliper with **new seal washers** and tighten fitting bolt to 31 N·m (23 ft. lbs.).

CAUTION: Verify brake hose is not twisted or kinked before tightening fitting bolt.

(5) Bleed base brake system. (Refer to 5 - BRAKES - STANDARD PROCEDURE).
(6) Install wheel and tire assemblies. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(7) Remove supports and lower vehicle.
(8) Verify firm pedal before moving vehicle.

INSTALLATION - REAR

(1) Install the brake pads if removed.
(2) Lubricate ant-rattle clips for the disc brake pads (Fig. 26).

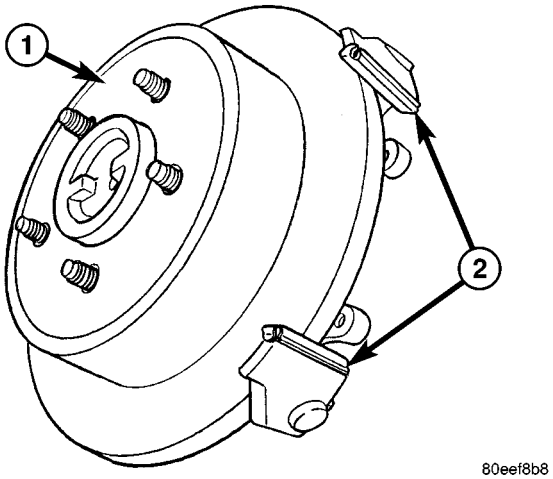


Fig. 26 ANTI-RATTLE CLIPS

1 - ROTOR
2 - ANTI-RATTLE CLIPS

(3) Install caliper to the caliper adapter.
(4) Coat the caliper mounting slide pin bolts with silicone grease. Then install and tighten the bolts to 15 N·m (11 ft. lbs.).
(5) Install the brake hose banjo bolt if removed (Fig. 27).
(6) Install the brake hose to the caliper with **new seal washers** and tighten fitting bolt to 31 N·m (23 ft. lbs.).

CAUTION: Verify brake hose is not twisted or kinked before tightening fitting bolt.

(7) Remove the prop rod from the vehicle.
(8) Bleed the base brake system, (Refer to 5 - BRAKES - STANDARD PROCEDURE) OR (Refer to 5 - BRAKES - STANDARD PROCEDURE).

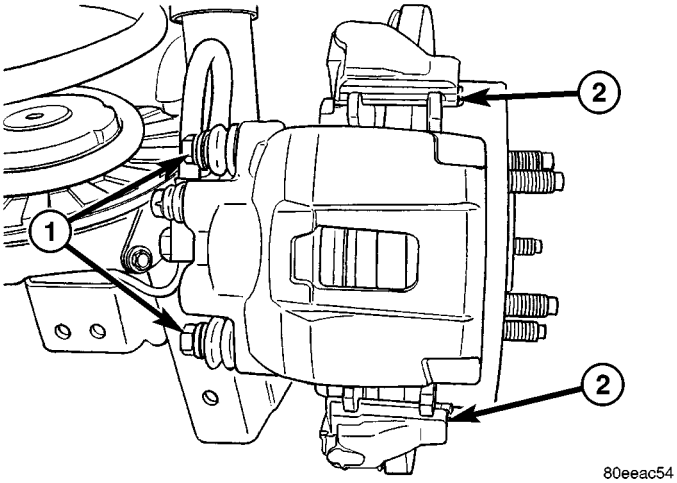


Fig. 27 CALIPER INSTALLED

1 - CALIPER MOUNTING BOLTS
2 - CALIPER SLIDES

(9) Install the wheel and tire assemblies (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(10) Remove the supports and lower the vehicle.
(11) Verify a firm pedal before moving the vehicle.

BRAKE PADS/SHOES

REMOVAL

REMOVAL - FRONT PADS

(1) Raise and support vehicle.
(2) Remove wheel and tire assembly.
(3) Remove caliper. (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL).
(4) Pressing one end of outboard shoe inward to disengage shoe lug. Then rotate shoe upward until retainer spring clears caliper. Press opposite end of shoe inward to disengage shoe lug and rotate shoe up and out of caliper (Fig. 28).
(5) Grasp ends of inboard shoe and tilt shoe outward to release springs from caliper piston (Fig. 29) and remove shoe from caliper.

NOTE: If original brake shoes will be used, keep them in sets left and right. They are not interchangeable.

(6) Secure caliper to nearby suspension part with wire. **Do not allow brake hose to support caliper weight.**
(7) Wipe caliper off with shop rags or towels.

BRAKE PADS/SHOES (Continued)

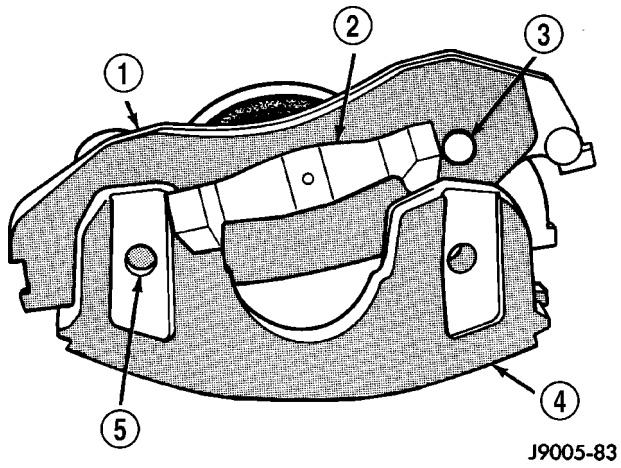


Fig. 28 Outboard Brake Shoe Removal

1 - OUTBOARD BRAKESHOE
2 - SHOE SPRING
3 - LOCATING LUG
4 - CALIPER
5 - LOCATING LUG

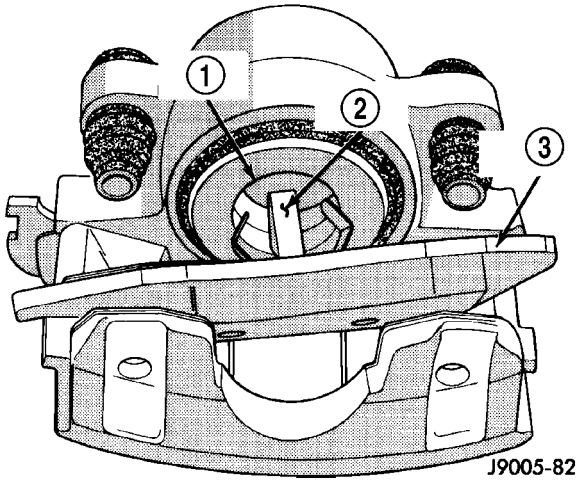


Fig. 29 Inboard Brake

1 - CALIPER PISTON
2 - SHOE SPRINGS
3 - INBOARD BRAKESHOE

CAUTION: Do not use compressed air, this can unseat dust boot and force dirt into piston bore.

REMOVAL - DRUM BRAKE SHOES

(1) Raise vehicle and remove rear wheels.
(2) Remove and discard spring nuts securing drums to wheel studs.
(3) Remove brake drums.

NOTE: If drums are difficult to remove, back off adjuster through support plate access hole with brake tool and screwdriver.

(4) Remove U-clip and washer securing adjuster cable to parking brake lever (Fig. 30).
(5) Remove primary and secondary return springs from anchor pin with brake spring pliers.
(6) Remove hold-down springs, retainers and pins with standard retaining spring tool.
(7) Install spring clamps on wheel cylinders to hold pistons in place.
(8) Remove adjuster lever, adjuster screw and spring.
(9) Remove adjuster cable and cable guide.
(10) Remove brake shoes and parking brake strut.
(11) Disconnect cable from parking brake lever and remove lever.

REMOVAL - REAR DISC BRAKE PADS

(1) Raise and support vehicle.
(2) Remove the wheel and tire assemblies.
(3) Compress the caliper.
(4) Remove the caliper, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL).
(5) Remove the caliper by tilting the top up and off the caliper adapter.

NOTE: Do not allow brake hose to support caliper assembly.

(6) Support and hang the caliper.
(7) Remove the inboard brake pad from the caliper adapter.
(8) Remove the outboard brake pad from the caliper adapter.

INSTALLATION

INSTALLATION - FRONT PADS

(1) Install inboard shoe in caliper and verify shoe retaining is fully seated into the piston.
(2) Starting one end of outboard shoe in caliper and rotating shoe downward into place. Verify shoe locating lugs and shoe spring are seated.
(3) Install caliper. (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).
(4) Install wheel and tire assembly. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(5) Remove support and lower vehicle.
(6) Pump brake pedal until caliper pistons and brake shoes are seated.
(7) Top off brake fluid level if necessary.

INSTALLATION - DRUM BRAKE SHOES

(1) Clean support plate with brake cleaner.

BRAKE PADS/SHOES (Continued)

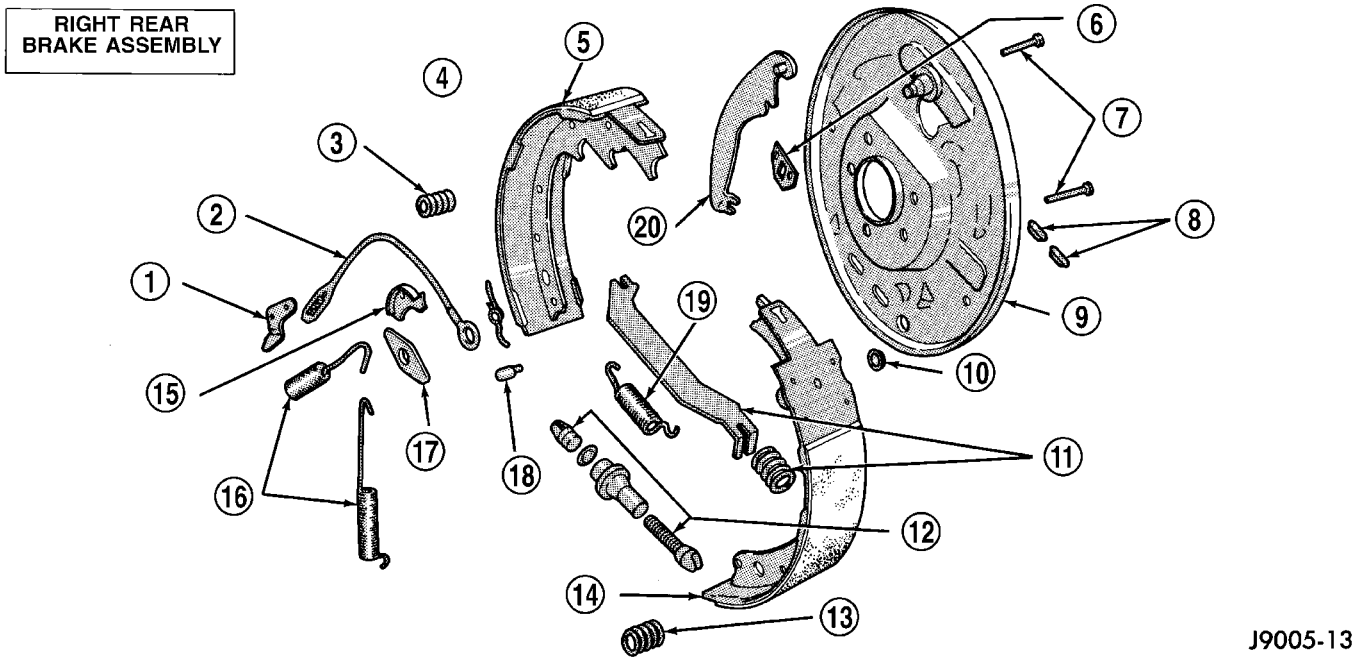


Fig. 30 Drum Brake Components—Typical

1 - ADJUSTER LEVER
2 - ADJUSTER CABLE
3 - HOLDDOWN SPRING AND RETAINERS
4 - ADJUSTER LEVER SPRING
5 - TRAILING SHOE
6 - CYLINDER-TO-SUPPORT SEAL
7 - HOLDDOWN PINS
8 - ACCESS PLUGS
9 - SUPPORT PLATE
10 - CABLE HOLE PLUG
11 - PARK BRAKE STRUT AND SPRING
12 - ADJUSTER SCREW ASSEMBLY
13 - HOLDDOWN SPRING AND RETAINERS
14 - LEADING SHOE
15 - CABLE GUIDE
16 - SHOE RETURN SPRINGS
17 - SHOE GUIDE PLATE
18 - PIN
19 - SHOE SPRING
20 - PARK BRAKE LEVER

(2) If new drums are being installed, remove protective coating with carburetor cleaner or brake cleaner.

(3) Apply multi-purpose grease to brake shoe contact surfaces of support plate (Fig. 31).

(4) Lubricate adjuster screw threads and pivot with spray lube.

(5) Attach parking brake lever to secondary brake shoe. Use new washer and U-clip to secure lever.

(6) Remove wheel cylinder clamps.

(7) Attach parking brake cable to lever.

(8) Install brake shoes on support plate. Secure shoes with new hold-down springs, pins and retainers.

(9) Install parking brake strut and spring.

(10) Install guide plate and adjuster cable on anchor pin.

(11) Install primary and secondary return springs.

(12) Install adjuster cable guide on secondary shoe.

(13) Lubricate and assemble adjuster screw.

(14) Install adjuster screw, spring and lever and connect to adjuster cable.

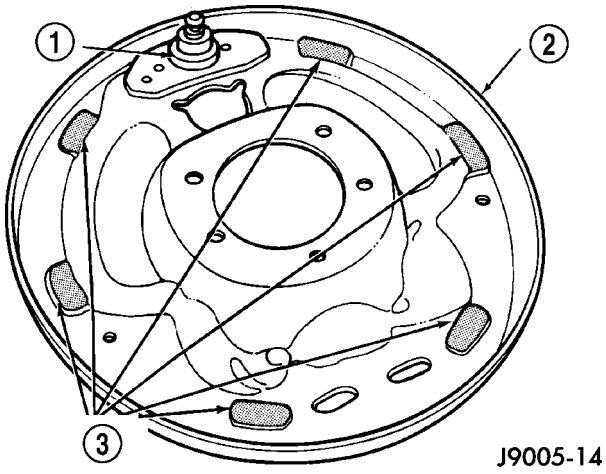


Fig. 31 Shoe Contact Surfaces

1 - ANCHOR PIN
2 - SUPPORT PLATE
3 - SHOE CONTACT SURFACES

(15) Adjust shoes to drum. (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DRUM - ADJUSTMENTS).

BRAKE PADS/SHOES (Continued)

(16) Install wheel/tire assemblies and lower vehicle. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(17) Verify firm brake pedal before moving vehicle.

INSTALLATION - REAR DISC BRAKE PADS

(1) Bottom pistons in caliper bore with C-clamp. Place an old brake shoe between a C-clamp and caliper piston.

(2) Clean caliper mounting adapter and anti-rattle springs.

(3) Lubricate anti-rattle springs with Mopar brake grease.

(4) Install anti-rattle springs.

NOTE: Anti-rattle springs are not interchangeable.

(5) Install inboard brake pad in adapter.

(6) Install outboard brake pad in adapter.

(7) Tilt the top of the caliper over rotor and under adapter. Then push the bottom of the caliper down onto the adapter.

(8) Install caliper, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).

(9) Install wheel and tire assemblies and lower vehicle, (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(10) Apply brakes several times to seat caliper pistons and brake shoes and obtain firm pedal.

(11) Top off master cylinder fluid level.

DRUM

DESCRIPTION

The brake systems use a leading shoe (primary) and trailing shoe (secondary). The mounting hardware is similar but not interchangeable (Fig. 30).

OPERATION

When the brake pedal is depressed hydraulic pressure pushes the rear brake wheel cylinder pistons outward. The wheel cylinder push rods then push the brake shoes outward against the brake drum. When the brake pedal is released return springs attached to the brake shoes pull the shoes back to there original position. (Fig. 30)

DIAGNOSIS AND TESTING - BRAKE DRUM RUNOUT

The maximum allowable diameter of the drum braking surface is indicated on the drum outer edge. Generally, a drum can be machined to a maximum of 1.52 mm (0.060 in.) oversize. Always replace the drum if machining would cause drum diameter to exceed the size limit indicated on the drum.

BRAKE DRUM RUNOUT

Measure drum diameter and runout with an accurate gauge. The most accurate method of measurement involves mounting the drum in a brake lathe and checking variation and runout with a dial indicator.

Variations in drum diameter should not exceed 0.069 mm (0.0028 in.). Drum runout should not exceed 0.18 mm (0.007 in.) out of round. Machine the drum if runout or variation exceed these values. Replace the drum if machining causes the drum to exceed the maximum allowable diameter.

STANDARD PROCEDURE - BRAKE DRUM MACHINING

The brake drums can be machined on a drum lathe when necessary. Initial machining cuts should be limited to 0.12 - 0.20 mm (0.005 - 0.008 in.) at a time as heavier feed rates can produce taper and surface variation. Final finish cuts of 0.025 to 0.038 mm (0.001 to 0.0015 in.) are recommended and will generally provide the best surface finish.

Be sure the drum is securely mounted in the lathe before machining operations. A damper strap should always be used around the drum to reduce vibration and avoid chatter marks.

The maximum allowable diameter of the drum braking surface is stamped or cast into the drum outer edge.

CAUTION: Replace the drum if machining will cause the drum to exceed the maximum allowable diameter.

CLEANING

Clean the individual brake components, including the support plate and wheel cylinder exterior, with a water dampened cloth or with brake cleaner. Do not use any other cleaning agents. Remove light rust and scale from the brake shoe contact pads on the support plate with fine sandpaper.

INSPECTION

As a general rule, riveted brake shoes should be replaced when worn to within 0.78 mm (1/32 in.) of the rivet heads. Bonded lining should be replaced when worn to a thickness of 1.6 mm (1/16 in.).

Examine the lining contact pattern to determine if the shoes are bent or the drum is tapered. The lining should exhibit contact across its entire width. Shoes exhibiting contact only on one side should be replaced and the drum checked for runout or taper.

DRUM (Continued)

Inspect the adjuster screw assembly. Replace the assembly if the star wheel or threads are damaged, or the components are severely rusted or corroded.

Discard the brake springs and retainer components if worn, distorted or collapsed. Also replace the springs if a brake drag condition had occurred. Overheating will distort and weaken the springs.

Inspect the brake shoe contact pads on the support plate, replace the support plate if any of the pads are worn or rusted through. Also replace the plate if it is bent or distorted (Fig. 32).

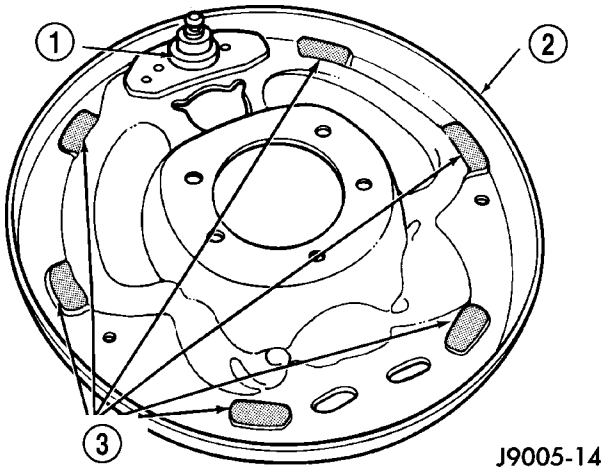


Fig. 32 Shoe Contact Surfaces

1 - ANCHOR PIN
2 - SUPPORT PLATE
3 - SHOE CONTACT SURFACES

ADJUSTMENTS - REAR DRUM BRAKE

The rear drum brakes are equipped with a self-adjusting mechanism. Under normal circumstances, the only time adjustment is required is when the shoes are replaced, removed for access to other parts, or when one or both drums are replaced.

Adjustment can be made with a standard brake gauge or with adjusting tool. Adjustment is performed with the complete brake assembly installed on the backing plate.

ADJUSTMENT WITH BRAKE GAUGE

(1) Be sure parking brakes are fully released.

(2) Raise rear of vehicle and remove wheels and brake drums.

(3) Verify that left and right automatic adjuster levers and cables are properly connected.

(4) Insert brake gauge in drum. Expand gauge until gauge inner legs contact drum braking surface. Then lock gauge in position (Fig. 33).

(5) Reverse gauge and install it on brake shoes. Position gauge legs at shoe centers as shown (Fig. 34). If gauge does not fit (too loose/too tight), adjust shoes.

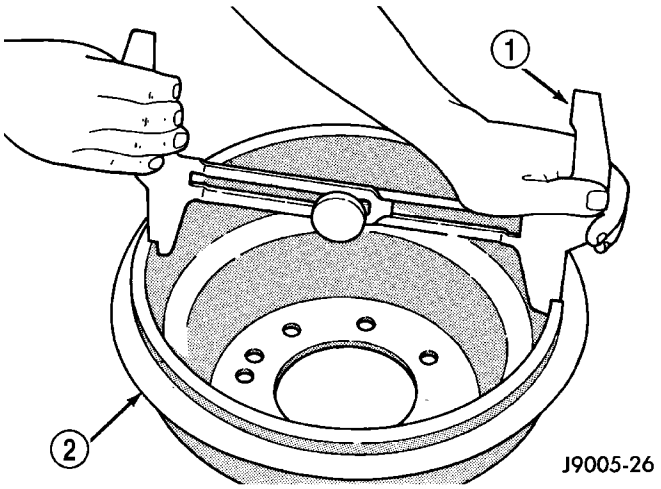


Fig. 33 Adjusting Gauge On Drum

1 - BRAKE GAUGE
2 - BRAKE DRUM

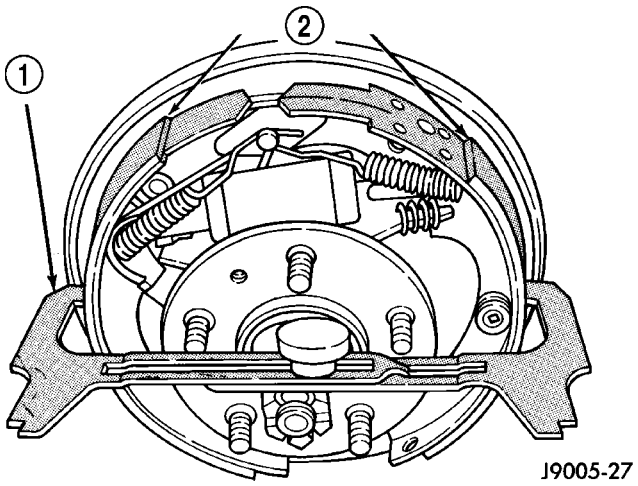


Fig. 34 Adjusting Gauge On Brake Shoes

1 - BRAKE GAUGE
2 - BRAKE SHOES

(6) Pull shoe adjuster lever away from adjuster screw star wheel.

(7) Turn adjuster screw star wheel (by hand) to expand or retract brake shoes. Continue adjustment until gauge outside legs are light drag-fit on shoes.

(8) Install brake drums and wheels and lower vehicle.

(9) Drive vehicle and make one forward stop followed by one reverse stop. Repeat procedure 8-10 times to operate automatic adjusters and equalize adjustment.

NOTE: Bring vehicle to complete standstill at each stop. Incomplete, rolling stops will not activate automatic adjusters.

DRUM (Continued)

ADJUSTMENT WITH ADJUSTING TOOL

(1) Be sure parking brake lever is fully released.

(2) Raise vehicle so rear wheels can be rotated freely.

(3) Remove plug from each access hole in brake support plates.

(4) Loosen parking brake cable adjustment nut until there is slack in front cable.

(5) Insert adjusting tool through support plate access hole and engage tool in teeth of adjusting screw star wheel (Fig. 35).

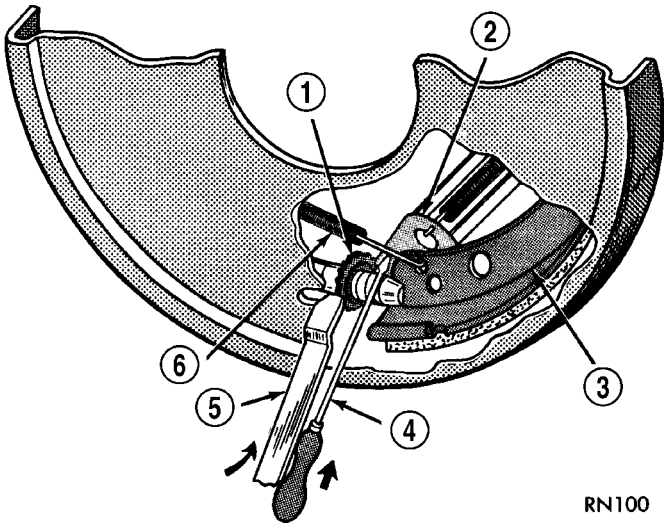


Fig. 35 Brake Adjustment

1 - STAR WHEEL
2 - LEVER
3 - BRAKE SHOE WEB
4 - SCREWDRIVER
5 - ADJUSTING TOOL
6 - ADJUSTER SPRING

(6) Rotate adjuster screw star wheel (move tool handle upward) until slight drag can be felt when wheel is rotated.

(7) Push and hold adjuster lever away from star wheel with thin screwdriver.

(8) Back off adjuster screw star wheel until brake drag is eliminated.

(9) Repeat adjustment at opposite wheel. Be sure adjustment is equal at both wheels.

(10) Install support plate access hole plugs.

(11) Adjust parking brake cable and lower vehicle.

(12) Drive vehicle and make one forward stop followed by one reverse stop. Repeat procedure 8-10 times to operate automatic adjusters and equalize adjustment.

NOTE: Bring vehicle to complete standstill at each stop. Incomplete, rolling stops will not activate automatic adjusters.

FLUID

DIAGNOSIS AND TESTING - BRAKE FLUID CONTAMINATION

Indications of fluid contamination are swollen or deteriorated rubber parts.

Swollen rubber parts indicate the presence of petroleum in the brake fluid.

To test for contamination, put a small amount of drained brake fluid in clear glass jar. If fluid separates into layers, there is mineral oil or other fluid contamination of the brake fluid.

If brake fluid is contaminated, drain and thoroughly flush system. Replace master cylinder, proportioning valve, caliper seals, wheel cylinder seals, Antilock Brakes hydraulic unit and all hydraulic fluid hoses.

STANDARD PROCEDURE - BRAKE FLUID LEVEL

Always clean the master cylinder reservoir and caps before checking fluid level. If not cleaned, dirt could enter the fluid.

The fluid fill level is indicated on the side of the master cylinder reservoir (Fig. 36).

The correct fluid level is to the FULL indicator on the side of the reservoir. If necessary, add fluid to the proper level.

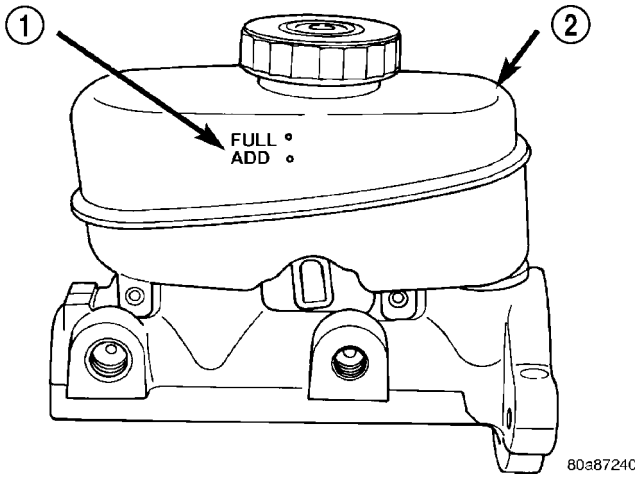


Fig. 36 Master Cylinder Fluid

1 - INDICATOR
2 - RESERVOIR

SPECIFICATIONS

BRAKE FLUID

The brake fluid used in this vehicle must conform to DOT 3 specifications and SAE J1703 standards. No other type of brake fluid is recommended or approved for usage in the vehicle brake system. Use

FLUID (Continued)

only Mopar brake fluid or an equivalent from a tightly sealed container.

CAUTION: Never use reclaimed brake fluid or fluid from an container which has been left open. An open container of brake fluid will absorb moisture from the air and contaminate the fluid.

CAUTION: Never use any type of a petroleum-based fluid in the brake hydraulic system. Use of such type fluids will result in seal damage of the vehicle brake hydraulic system causing a failure of the vehicle brake system. Petroleum based fluids would be items such as engine oil, transmission fluid, power steering fluid, etc.

FLUID RESERVOIR

REMOVAL

(1) Remove reservoir cap and empty fluid into drain container.

(2) Remove pins that retain reservoir to master cylinder. Use hammer and pin punch to remove pins (Fig. 37).

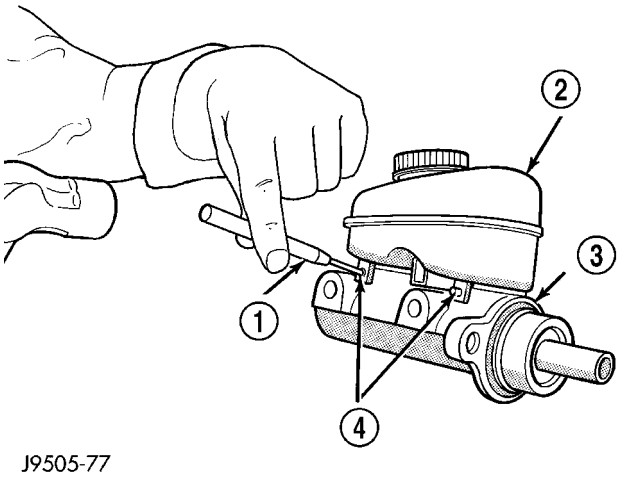


Fig. 37 Reservoir Retaining Pins

1 - PIN PUNCH
2 - RESERVOIR
3 - BODY
4 - ROLL PINS

(3) Clamp cylinder body in vise with brass protective jaws.

(4) Loosen reservoir from grommets with pry tool (Fig. 38).

(5) Remove reservoir by rocking it to one side and pulling free of grommets (Fig. 39).

(6) Remove old grommets from cylinder body (Fig. 40).

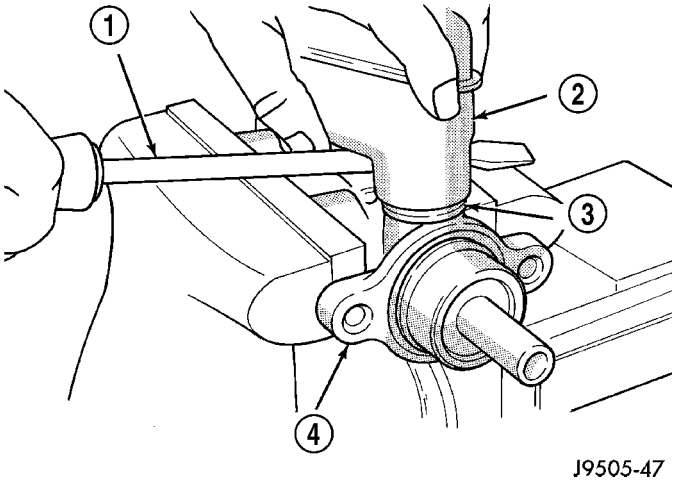


Fig. 38 Loosening Reservoir

1 - PRY TOOL
2 - RESERVOIR
3 - GROMMET
4 - MASTER CYLINDER BODY

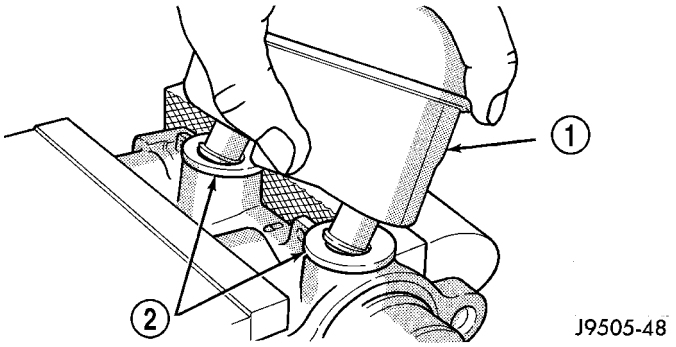


Fig. 39 Reservoir Removal

1 - RESERVOIR
2 - GROMMETS

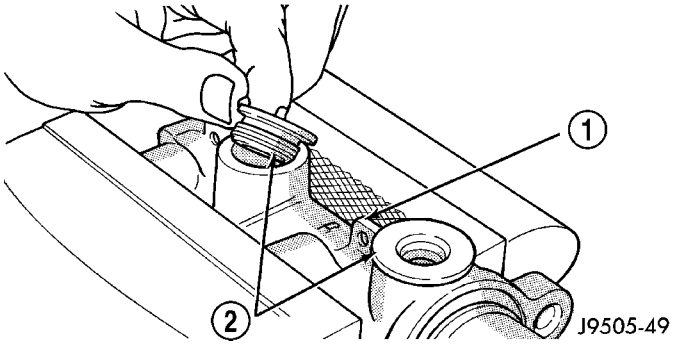


Fig. 40 Grommet Removal

1 - MASTER CYLINDER BODY
2 - GROMMETS

FLUID RESERVOIR (Continued)

INSTALLATION

CAUTION: Do not use any type of tool to install the grommets. Tools may cut, or tear the grommets creating a leak problem after installation. Install the grommets using finger pressure only.

(1) Lubricate new grommets with clean brake fluid and Install new grommets in cylinder body (Fig. 41). Use finger pressure to install and seat grommets.

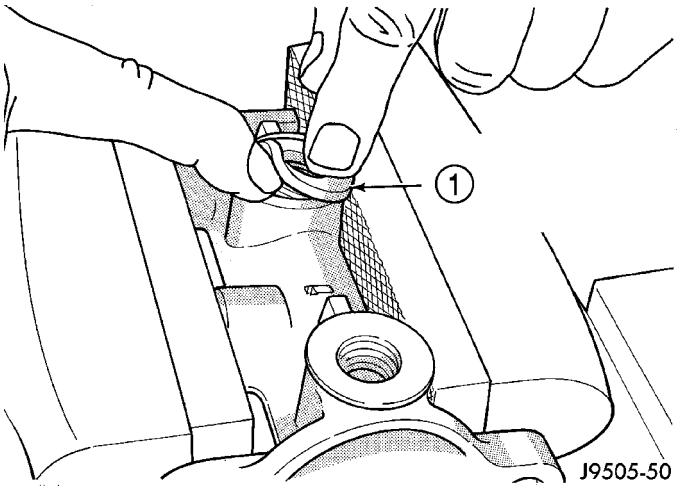


Fig. 41 Grommet Installation

1 - WORK NEW GROMMETS INTO PLACE USING FINGER PRESSURE ONLY

(2) Start reservoir in grommets. Then rock reservoir back and forth while pressing downward to seat it in grommets.

(3) Install pins that retain reservoir to cylinder body.

(4) Fill and bleed master cylinder on bench before installation in vehicle.

MASTER CYLINDER

DESCRIPTION

The master cylinder has a removable nylon reservoir. The cylinder body is made of aluminum and contains a primary and secondary piston assembly. The cylinder body including the piston assemblies are not serviceable. If diagnosis indicates an internal problem with the cylinder body, it must be replaced as an assembly. The reservoir and grommets are the only replaceable parts on the master cylinder.

OPERATION

The master cylinder bore contains a primary and secondary piston. The primary piston supplies hydraulic pressure to the front brakes. The secondary piston supplies hydraulic pressure to the rear brakes. The master cylinder reservoir stores reserve brake fluid for the hydraulic brake circuits.

DIAGNOSIS AND TESTING - MASTER CYLINDER/POWER BOOSTER

(1) Start engine and check booster vacuum hose connections. A hissing noise indicates vacuum leak. Correct any vacuum leak before proceeding.

(2) Stop engine and shift transmission into Neutral.

(3) Pump brake pedal until all vacuum reserve in booster is depleted.

(4) Press and hold brake pedal under light foot pressure. The pedal should hold firm, if the pedal falls away master cylinder is faulty (internal leakage).

(5) Start engine and note pedal action. It should fall away slightly under light foot pressure then hold firm. If no pedal action is discernible, power booster, vacuum supply, or vacuum check valve is faulty. Proceed to the POWER BOOSTER VACUUM TEST.

(6) If the POWER BOOSTER VACUUM TEST passes, rebuild booster vacuum reserve as follows: Release brake pedal. Increase engine speed to 1500 rpm, close the throttle and immediately turn off ignition to stop engine.

(7) Wait a minimum of 90 seconds and try brake action again. Booster should provide two or more vacuum assisted pedal applications. If vacuum assist is not provided, booster is faulty.

POWER BOOSTER VACUUM TEST

(1) Connect vacuum gauge to booster check valve with short length of hose and T-fitting (Fig. 42).

(2) Start and run engine at curb idle speed for one minute.

(3) Observe the vacuum supply. If vacuum supply is not adequate, repair vacuum supply.

(4) Clamp hose shut between vacuum source and check valve.

(5) Stop engine and observe vacuum gauge.

(6) If vacuum drops more than one inch HG (33 millibars) within 15 seconds, booster diaphragm or check valve is faulty.

POWER BOOSTER CHECK VALVE TEST

(1) Disconnect vacuum hose from check valve.

(2) Remove check valve and valve seal from booster.

(3) Use a hand operated vacuum pump for test.

(4) Apply 15-20 inches vacuum at large end of check valve (Fig. 43).

(5) Vacuum should hold steady. If gauge on pump indicates vacuum loss, check valve is faulty and should be replaced.

MASTER CYLINDER (Continued)

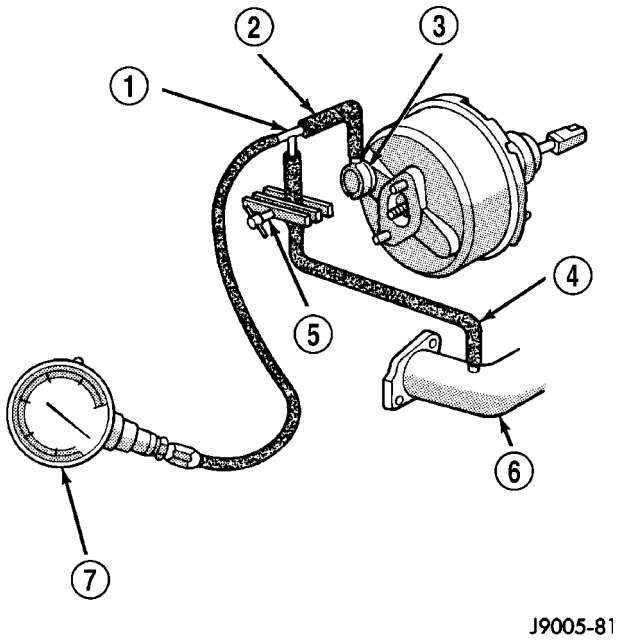


Fig. 42 Typical Booster Vacuum Test Connections

1 - TEE FITTING
2 - SHORT CONNECTING HOSE
3 - CHECK VALVE
4 - CHECK VALVE HOSE
5 - CLAMP TOOL
6 - INTAKE MANIFOLD
7 - VACUUM GAUGE

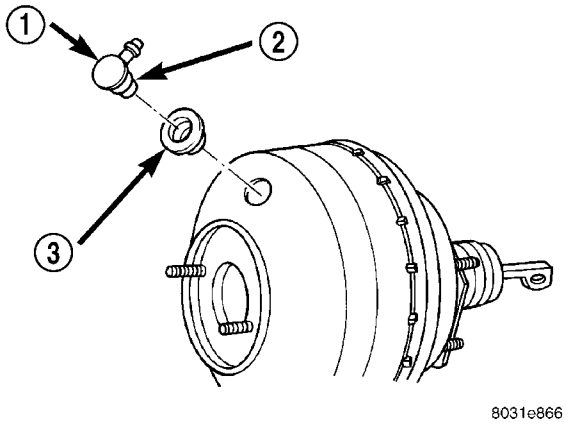


Fig. 43 Vacuum Check Valve And Seal

1 - BOOSTER CHECK VALVE
2 - APPLY TEST VACUUM HERE
3 - VALVE SEAL

STANDARD PROCEDURE - MASTER CYLINDER BLEEDING

A new master cylinder should be bled before installation on the vehicle. Required bleeding tools include bleed tubes and a wood dowel to stroke the pistons. Bleed tubes can be fabricated from brake line.

(1) Mount master cylinder in vise.

(2) Attach bleed tubes to cylinder outlet ports. Then position each tube end into the reservoir (Fig. 44).

(3) Fill reservoir with fresh brake fluid.

(4) Press cylinder pistons inward with wood dowel. Then release pistons and allow them to return under spring pressure. Continue bleeding operations until air bubbles are no longer visible in fluid.

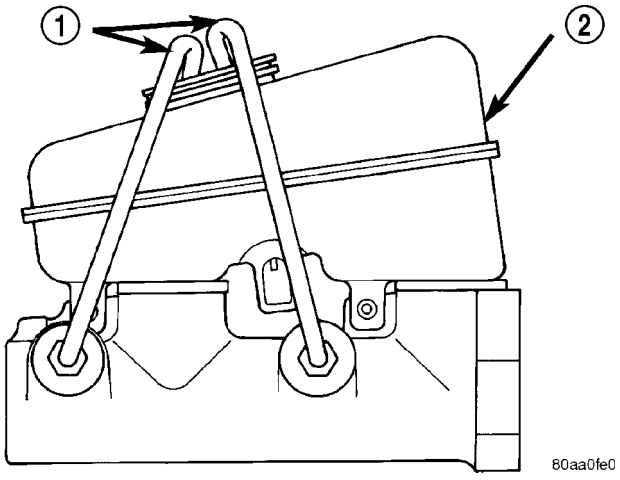


Fig. 44 Master Cylinder Bleeding

1 - BLEEDING TUBES
2 - RESERVOIR

REMOVAL

(1) Remove evaporative canister.

(2) Disconnect brake lines to master cylinder and combination valve (Fig. 45).

(3) Remove combination valve bracket mounting nuts and remove valve.

(4) Remove master cylinder mounting nuts and remove master cylinder.

(5) Remove cylinder cover and drain fluid.

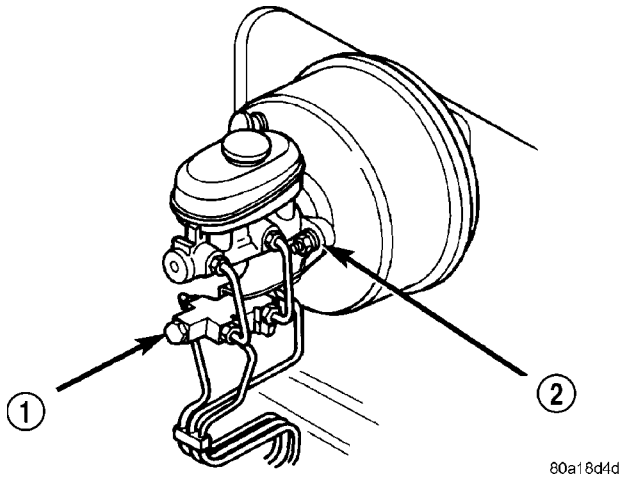


Fig. 45 Master

1 - COMBINATION VALVE
2 - MASTER CYLINDER

MASTER CYLINDER (Continued)

INSTALLATION

NOTE: If master cylinder is replaced, bleed cylinder before installation.

(1) Remove protective sleeve from primary piston shank on new master cylinder.
(2) Check condition of seal at rear of cylinder body. Reposition seal if dislodged. Replace seal if cut, or torn.
(3) Install master cylinder onto brake booster studs and tighten mounting nuts to 17 N·m (13 ft. lbs.).

NOTE: Use only original or factory replacement nuts.

(4) Install combination valve onto brake booster studs and tighten mounting nuts to 20 N·m (15 ft. lbs.).
(5) Install brake lines to master cylinder and combination valve by hand to avoid cross threading.
(6) Tighten master cylinder brake lines to 19 N·m (14 ft. lbs.).
(7) Tighten combination valve brake lines to 19 N·m (14 ft. lbs.).
(8) Install evaporative canister.
(9) Bleed base brake system. (Refer to 5 - BRAKES - STANDARD PROCEDURE).

PEDAL

DESCRIPTION

A suspended-type brake pedal is used, the pedal pivots on a shaft mounted in the pedal support bracket. The bracket is attached to the dash panel.

The brake pedal assembly and pedal pad are the only serviceable component.

OPERATION

The brake pedal is attached to the booster push rod. When the pedal is depressed, the primary booster push rod is depressed which moves the booster secondary rod. The booster secondary rod depresses the master cylinder piston.

REMOVAL

(1) Remove negative battery cable.
(2) Remove brake lamp switch.
(3) Remove ABS controller if equipped.
(4) Remove retainer clip securing booster push rod to pedal (Fig. 46) and clutch rod retainer clip if equipped.
(5) Remove bolts from brake pedal support and booster mounting nuts. Remove mounting stud plate nuts or clutch cylinder mounting nuts if equipped.
(6) Slid brake booster/master cylinder assembly forward.
(7) Remove mounting stud plate or slid clutch cylinder forward if equipped.
(8) Tilt the pedal support down to gain shaft clearance.
(9) Remove pedal shaft C–clip from passenger side of the shaft.
(10) Slide the pedal shaft toward the drivers side and remove the remaining C-clip.
(11) Slid the shaft out of the pedal bracket and remove the pedal.
(12) Remove pedal bushings if they are to be replaced.

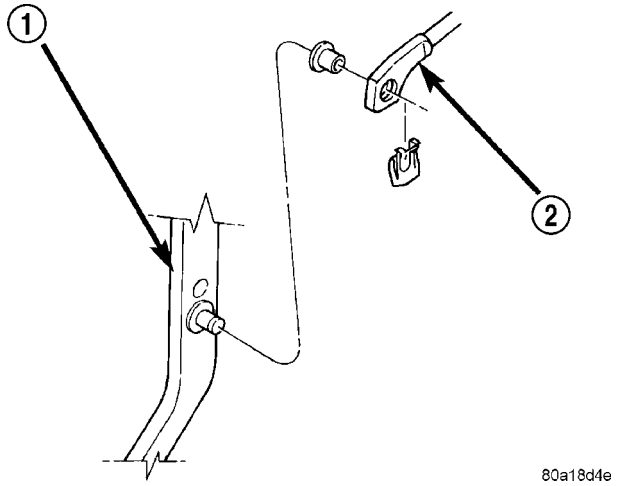


Fig. 46 Push Rod Attachment

1 - BRAKE PEDAL
2 - BOOSTER ROD

INSTALLATION

(1) Install new bushings in pedal. Lubricate bushings and shaft with multi-purpose grease.
(2) Position pedal in bracket and install shaft.
(3) Install new pivot pin C-clip.
(4) Position pedal support and install support bolts and tighten to 28 N·m (21 ft. lbs.).
(5) Slid the booster/master cylinder assembly into place, install mounting nuts and tighten to 39 N·m (29 ft. lbs.).
(6) Install stud plate or clutch cylinder if equipped and tighten mounting nut to 28 N·m (21 ft. lbs.).

Install retainer clip securing booster push rod to pedal (Fig. 46) and clutch rod retainer clip if equipped.

(7) Install ABS controller if equipped.
(8) Install and connect brake lamp switch.
(9) Install negative battery cable.

POWER BRAKE BOOSTER

DESCRIPTION

The booster assembly consists of a housing divided into separate chambers by two internal diaphragms. The outer edge of each diaphragm is attached to the booster housing.

Two push rods are used in the booster. The primary push rod connects the booster to the brake pedal. The secondary push rod connects the booster to the master cylinder to stroke the cylinder pistons.

OPERATION

The atmospheric inlet valve is opened and closed by the primary push rod. Booster vacuum supply is through a hose attached to an intake manifold fitting at one end and to the booster check valve at the other. The vacuum check valve in the booster housing is a one-way device that prevents vacuum leak back.

Power assist is generated by utilizing the pressure differential between normal atmospheric pressure and a vacuum. The vacuum needed for booster operation is taken directly from the engine intake manifold. The entry point for atmospheric pressure is through a filter and inlet valve at the rear of the housing (Fig. 47).

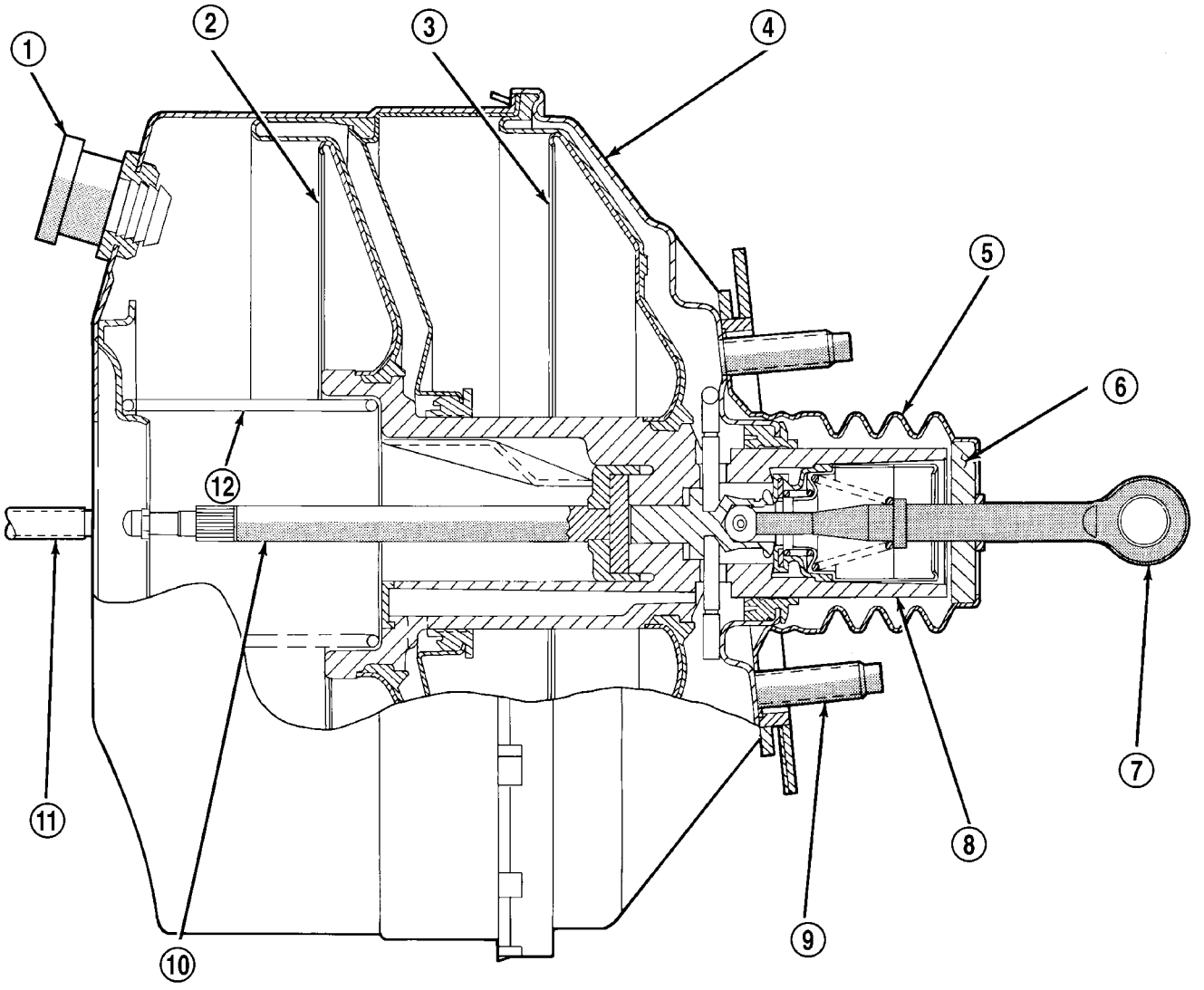


Fig. 47 Power Brake Booster–Typical

1 - VACUUM CHECK VALVE
2 - FRONT DIAPHRAGM
3 - REAR DIAPHRAGM
4 - HOUSING
5 - SEAL
6 - AIR FILTER
7 - PRIMARY PUSH ROD (TO BRAKE PEDAL)
8 - ATMOSPHERIC INLET VALVE ASSEMBLY
9 - BOOSTER MOUNTING STUDS (4)
10 - SECONDARY PUSH ROD (TO MASTER CYLINDER)
11 - MASTER CYLINDER MOUNTING STUD (2)
12 - SPRING

POWER BRAKE BOOSTER (Continued)

The chamber areas forward of the booster diaphragms are exposed to vacuum from the intake manifold. The chamber areas to the rear of the diaphragms, are exposed to normal atmospheric pressure of 101.3 kilopascals (14.7 pounds/square in.).

Brake pedal application causes the primary push rod to open the atmospheric inlet valve. This exposes the area behind the diaphragms to atmospheric pressure. The resulting pressure differential provides the extra apply force for power assist.

The booster check valve, check valve grommet and booster seals are serviceable.

REMOVAL

(1) Remove combination valve and master cylinder.

(2) Disconnect vacuum hose from booster check valve.

(3) Remove retaining clip that secures booster push rod to brake pedal (Fig. 48) and slide the rod off the pin.

(4) Remove four nuts attaching booster to front cowl panel (Fig. 49).

(5) In engine compartment, slide booster studs out of cowl panel, and remove the booster from engine compartment.

(6) Remove dash seal from booster.

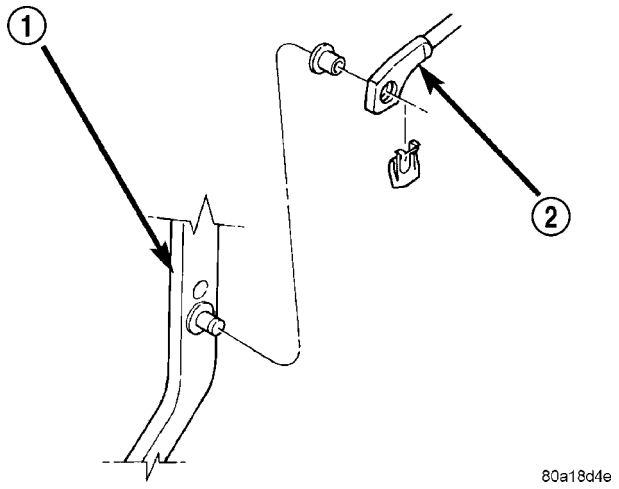


Fig. 48 Push Rod & Clip

1 - BRAKE PEDAL
2 - BOOSTER ROD

INSTALLATION

(1) Clean the booster mounting surface.

(2) Install dash seal on booster.

(3) Align and position booster on the front cowl panel.

(4) In passenger compartment, install nuts that attach booster to dash panel. Tighten nuts just enough to hold booster in place.

(5) Lubricate the pedal pin and bushing with Mopar multi-mileage grease. Then slid the booster push rod onto brake pedal pin and secure with retaining clip.

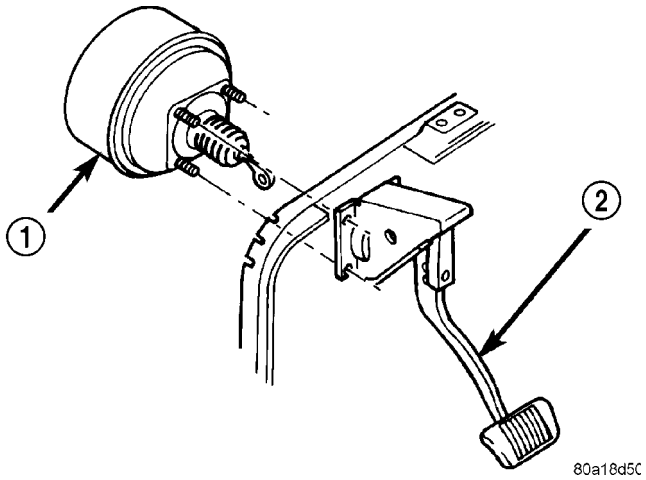


Fig. 49 Booster Mounting Nuts

1 - BOOSTER
2 - BRAKE PEDAL

(6) Tighten booster mounting nuts to 39 N·m (29 ft. lbs.).

(7) Connect vacuum hose to booster check valve.

(8) Install master cylinder and combination valve.

(9) Top off master cylinder fluid level and bleed base brakes.

COMBINATION VALVE

DESCRIPTION

The combination valve contains a pressure differential valve and switch and a rear brake proportioning valve. The valve is not repairable and must be replaced as an assembly if diagnosis indicates this is necessary.

OPERATION

PRESSURE DIFFERENTIAL VALVE

The pressure differential switch is connected to the brake warning light. The switch is actuated by movement of the switch valve. The switch monitors fluid pressure in the separate front/rear brake hydraulic circuits.

A decrease or loss of fluid pressure in either hydraulic circuit will cause the switch valve to shuttle to the low pressure side. Movement of the valve pushes the switch plunger upward. This action closes the switch internal contacts completing the electrical circuit to the red warning light. The switch valve will remain in an actuated position until repairs to the brake system are made.

COMBINATION VALVE (Continued)

PROPORTIONING VALVE

The proportioning valve is used to balance front-rear brake action at high decelerations. The valve allows normal fluid flow during moderate braking. The valve only controls fluid flow during high decelerations brake stops.

DIAGNOSIS AND TESTING - COMBINATION VALVE

Pressure Differential Switch

(1) Have helper sit in drivers seat to apply brake pedal and observe red brake warning light.

(2) Raise vehicle on hoist.

(3) Connect bleed hose to a rear wheel cylinder and immerse hose end in container partially filled with brake fluid.

(4) Have helper press and hold brake pedal to floor and observe warning light.

(a) If warning light illuminates, switch is operating correctly.

(b) If light fails to illuminate, check circuit fuse, bulb, and wiring. The parking brake switch can be used to aid in identifying whether or not the brake light bulb and fuse is functional. Repair or replace parts as necessary and test differential pressure switch operation again.

(5) If warning light still does not illuminate, switch is faulty. Replace combination valve assembly, bleed brake system and verify proper switch and valve operation.

REMOVAL

(1) Remove brake lines that connect master cylinder to combination valve (Fig. 50).

(2) Disconnect brake lines that connect combination valve to front and rear brakes.

(3) Disconnect wire from combination valve switch terminal. Be careful when separating wire connector as lock tabs are easily damaged if not fully disengaged.

(4) Remove nuts attaching combination valve bracket to booster studs and remove valve bracket off booster studs (Fig. 51).

INSTALLATION

(1) Position valve bracket on booster studs and tighten bracket attaching nuts to 20 N·m (15 ft. lbs.).

(2) Align and start brake line fittings in combination valve and master cylinder by hand to avoid cross threading.

(3) Tighten brake line fittings at combination valve to 19 N·m (14 ft. lbs.).

(4) Tighten brake line fittings at master cylinder to 19 N·m (14 ft. lbs.).

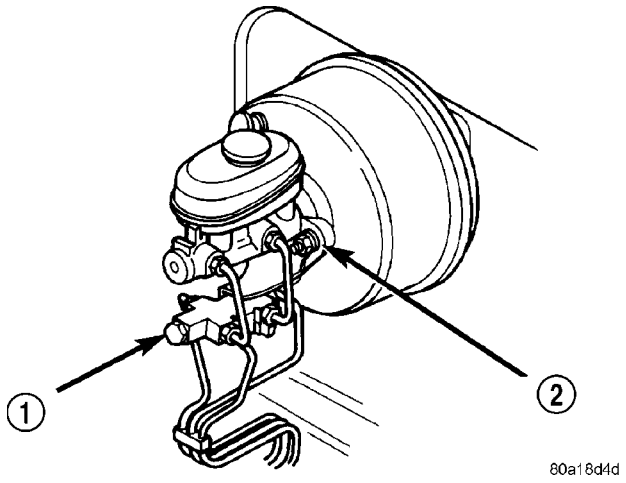


Fig. 50 Combination Valve/Master Cylinder

1 - COMBINATION VALVE
2 - MASTER CYLINDER

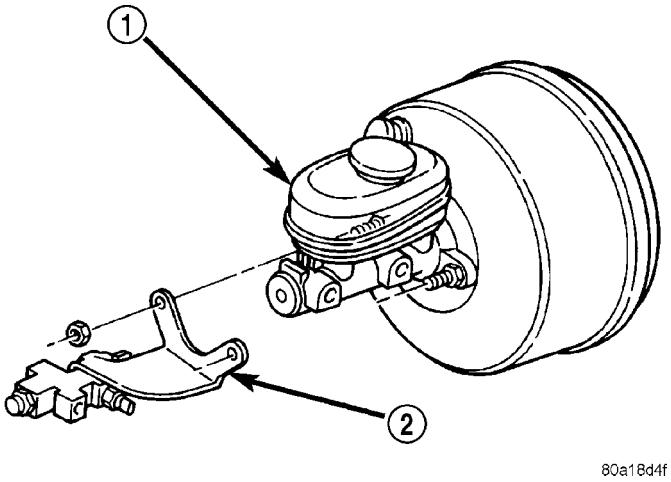


Fig. 51 Combination Valve Bracket

1 - MASTER CYLINDER
2 - COMBINATION VALVE BRACKET

(5) Connect wire to differential pressure switch in combination valve.

(6) Bleed base brake system. (Refer to 5 - BRAKES - STANDARD PROCEDURE).

ROTORS

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - DISC BRAKE ROTOR

The rotor braking surfaces should not be refinished unless necessary.

Light surface rust and scale can be removed with a lathe equipped with dual sanding discs. The rotor

ROTORS (Continued)

surfaces can be restored by machining in a disc brake lathe if surface scoring and wear are light.

Replace the rotor under the following conditions:

- Severely Scored
- Tapered
- Hard Spots
- Cracked
- Below Minimum Thickness

ROTOR MINIMUM THICKNESS

Measure rotor thickness at the center of the brake shoe contact surface. Replace the rotor if worn below minimum thickness, or if machining would reduce thickness below the allowable minimum.

Rotor minimum thickness is usually specified on the rotor hub. The specification is either stamped or cast into the hub surface.

ROTOR RUNOUT

Check rotor lateral runout with dial indicator C-3339 (Fig. 52). Excessive lateral runout will cause brake pedal pulsation and rapid, uneven wear of the brake shoes. Position the dial indicator plunger approximately 25.4 mm (1 in.) inward from the rotor edge.

NOTE: Be sure wheel bearing has zero end play before checking rotor runout.

Maximum allowable rotor runout is 0.102 mm (0.004 in.).

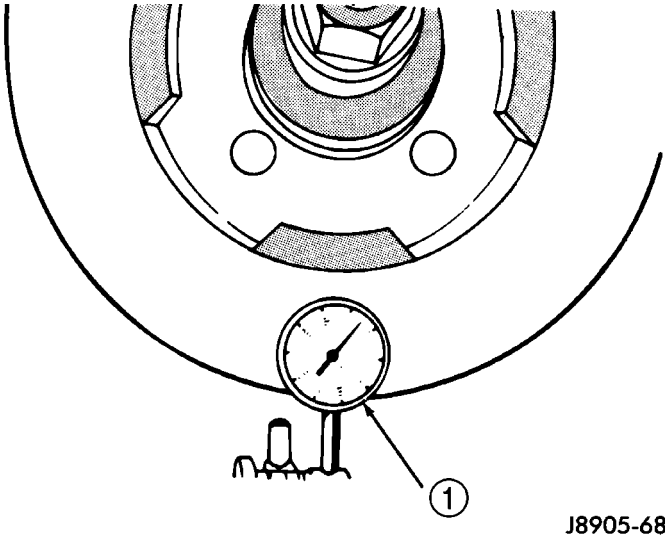


Fig. 52 Checking Rotor Runout And Thickness Variation

1 - DIAL INDICATOR

ROTOR THICKNESS VARIATION

Variations in rotor thickness will cause pedal pulsation, noise and shudder.

Measure rotor thickness at 6-to-12 points around the rotor face (Fig. 53).

Position the micrometer approximately 25.4 mm (1 in.) from the rotor outer circumference for each measurement.

Thickness should not **vary** by more than 0.013 mm (0.0005 in.) from point-to-point on the rotor. Machine or replace the rotor if necessary.

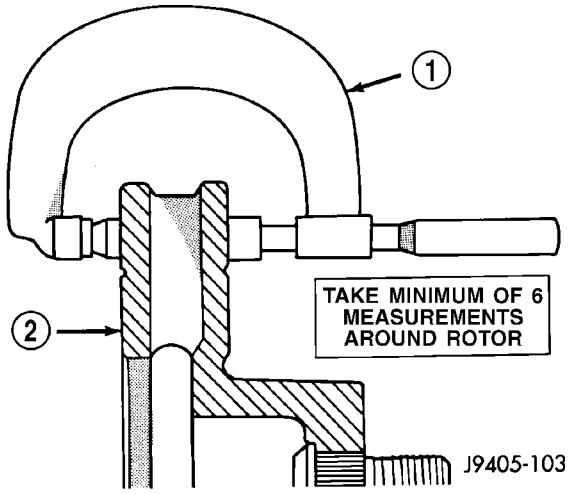


Fig. 53 Measuring Rotor Thickness

1 - MICROMETER
2 - ROTOR

DIAGNOSIS AND TESTING - BRAKE DRUM IN HAT ROTOR

The maximum allowable diameter of the drum braking surface is indicated on the drum outer edge. Always replace the drum if machining would cause drum diameter to exceed the size limit indicated on the drum in hat.

BRAKE DRUM RUNOUT

Measure drum diameter and runout with an accurate gauge. The most accurate method of measurement involves mounting the drum in a brake lathe and checking variation and runout with a dial indicator.

Machine the drum if runout or variation exceed values. Replace the drum in hat rotor if machining causes the drum in hat rotor to exceed the maximum allowable diameter.

STANDARD PROCEDURE

STANDARD PROCEDURE - DISC ROTOR MACHINING

The disc brake rotor can be machined if scored or worn. The lathe must machine both sides of the rotor

ROTORS (Continued)

simultaneously with dual cutter heads. The rotor mounting surface must be clean before placing on the lathe. Equipment capable of machining only one side at a time may produce a tapered rotor.

NOTE: A hub mounted on-vehicle lathe is recommended. This type of lathe trues the rotor to the vehicles hub/bearing.

CAUTION: Brake rotors that do not meet minimum thickness specifications before or after machining must be replaced.

STANDARD PROCEDURE - BRAKE DRUM IN HAT ROTOR MACHINING

The brake drum in hat rotor can be machined on a drum lathe when necessary. Initial machining cuts should be limited to 0.12 - 0.20 mm (0.005 - 0.008 in.) at a time as heavier feed rates can produce taper and surface variation. Final finish cuts of 0.025 to 0.038 mm (0.001 to 0.0015 in.) are recommended and will generally provide the best surface finish.

Be sure the drum in hat rotor is securely mounted in the lathe before machining operations. A damper strap should always be used around the drum to reduce vibration and avoid chatter marks.

The maximum allowable diameter of the drum braking surface is stamped or cast into the drum in hat rotor.

CAUTION: Replace the drum in hat rotor if machining will cause the drum to exceed the maximum allowable diameter.

REMOVAL

REMOVAL - FRONT

(1) Remove wheel and tire assemble.

(2) Remove caliper. (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL).

(3) Remove retainers securing rotor to hub studs (Fig. 54).

(4) Remove rotor from hub.

(5) If rotor shield requires service, remove front hub and bearing assembly.

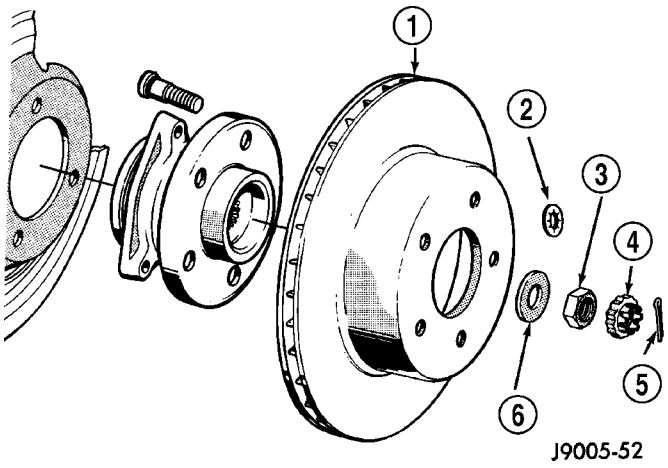


Fig. 54 Rotor & Hub

1 - ROTOR
2 - RETAINER
3 - BEARING NUT
4 - NUT LOCK
5 - COTTER PIN
6 - WASHER

REMOVAL - REAR

(1) Raise and support the vehicle

(2) Remove the tire and wheel assembly (Fig. 55).

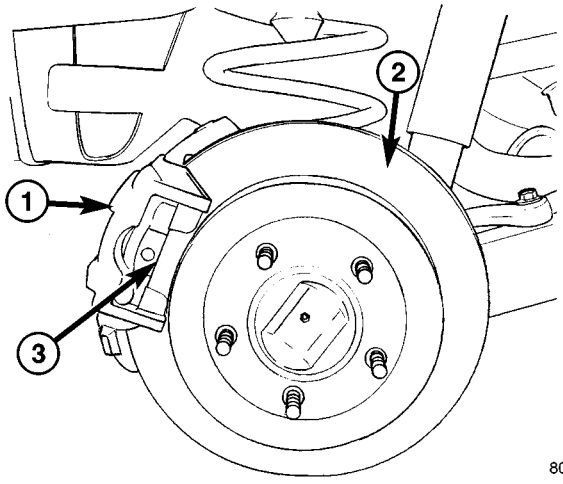


Fig. 55 ROTOR/CALIPER

1 - CALIPER
2 - ROTOR
3 - OUTBOARD DISC BRAKE PAD

(3) Remove the disc brake caliper and pads, (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL) (Fig. 56).

(4) Remove the retianing clips and rotor assembly.

ROTORS (Continued)

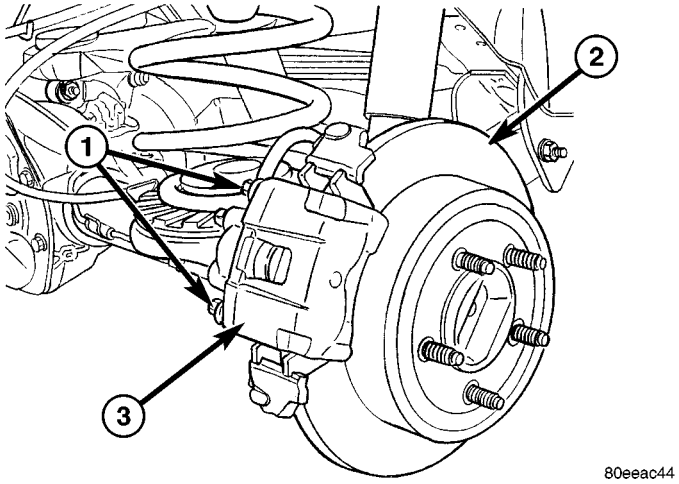


Fig. 56 ROTOR

1 - CALIPER MOUNTING BOLTS
2 - ROTOR
3 - CALIPER

INSTALLATION

INSTALLATION - FRONT

(1) If new rotor is being installed, remove protective coating from rotor surfaces with carburetor cleaner.

(2) Install rotor on hub.

(3) Install caliper. (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).

(4) Install wheel and tire assembly. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

INSTALLATION - REAR

(1) Install the rotor to the axleshaft.

(2) Install the disc brake caliper and pads (Fig. 56), (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).

(3) Install the tire and wheel assembly (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(4) Lower the vehicle.

WHEEL CYLINDERS

REMOVAL

(1) Remove wheel and tire assembly.

(2) Remove brake drum.

(3) Remove wheel cylinder brake line.

(4) Remove brake shoe return springs and move shoes out of engagement with cylinder push rods.

(5) Remove cylinder attaching bolts and remove cylinder from support plate.

DISASSEMBLY

(1) Remove push rods and boots (Fig. 57).

(2) Press pistons, cups and spring and expander out of cylinder bore.

(3) Remove bleed screw.

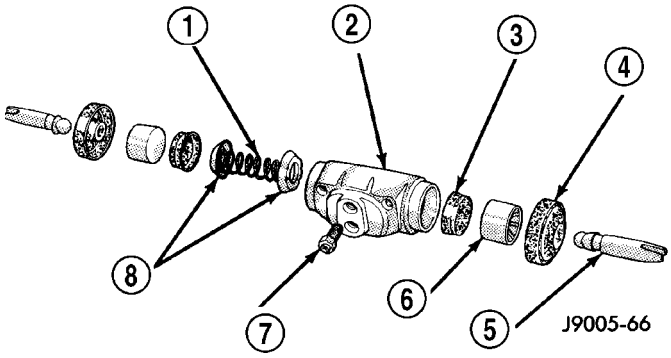


Fig. 57 Wheel Cylinder Components–Typical

1 - SPRING
2 - CYLINDER
3 - PISTON CLIP
4 - BOOT
5 - PUSH ROD
6 - PISTON
7 - BLEED SCREW
8 - CUP EXPANDERS

CLEANING

Clean the cylinder and pistons with clean brake fluid or brake cleaner only. Do not use any other cleaning agents.

Dry the cylinder and pistons with compressed air. Do not use rags or shop towels to dry the cylinder components. Lint from cloth material will adhere to the cylinder bores and pistons.

INSPECTION

Inspect the cylinder bore. Light discoloration and dark stains in the bore are normal and will not impair cylinder operation.

The cylinder bore can be lightly polished but only with crocus cloth. Replace the cylinder if the bore is scored, pitted or heavily corroded. Honing the bore to restore the surface is not recommended.

Inspect the cylinder pistons. The piston surfaces should be smooth and free of scratches, scoring and corrosion. Replace the pistons if worn, scored, or corroded. Do attempt to restore the surface by sanding or polishing.

Discard the old piston cups and the spring and expander. These parts are not reusable. The original dust boots may be reused but only if they are in good condition.

ASSEMBLY

(1) Lubricate wheel cylinder bore, pistons, piston cups and spring and expander with clean brake fluid.

WHEEL CYLINDERS (Continued)

(2) Install first piston in cylinder bore. Then install first cup in bore and against piston. **Be sure lip of piston cup is facing inward (toward spring and expander) and flat side is against piston.**

(3) Install spring and expander followed by remaining piston cup and piston.

(4) Install boots on each end of cylinder and insert push rods in boots.

(5) Install cylinder bleed screw.

INSTALLATION

(1) Apply bead of silicone sealer around cylinder mounting surface of support plate.

(2) Install cylinder mounting bolts and tighten to 10 N·m (7 ft. lbs.).

(3) Install brake line to cylinder and tighten to 16 N·m (12 ft. lbs.).

(4) Install brake shoe return spring.

(5) Install brake drum.

(6) Install wheel and tire assembly. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(7) Bleed base brake system. (Refer to 5 - BRAKES - STANDARD PROCEDURE).

SUPPORT PLATE

REMOVAL

(1) Remove wheel and tire assembly.

(2) Remove the disc brake caliper (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL).

(3) Remove the rotor (Fig. 58), (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - REMOVAL).

(4) Remove the axle shaft (Fig. 59), (Refer to 3 - DIFFERENTIAL & DRIVELINE/REAR AXLE - AXLE SHAFTS - REMOVAL).

(5) Remove the park brake shoes (Fig. 59), (Refer to 5 - BRAKES/PARKING BRAKE/SHOES - REMOVAL).

(6) Remove the parking brake cable from the brake lever.

(7) Remove the bolts attaching the support plate to the axle and remove the support plate.

INSTALLATION

(1) Install support plate on axle flange. Tighten attaching bolts to 115 N·m (85 ft. lbs.).

(2) Install the park brake shoes (Fig. 59), (Refer to 5 - BRAKES/PARKING BRAKE/SHOES - INSTALLATION).

(3) Install parking brake cable in the brake lever.

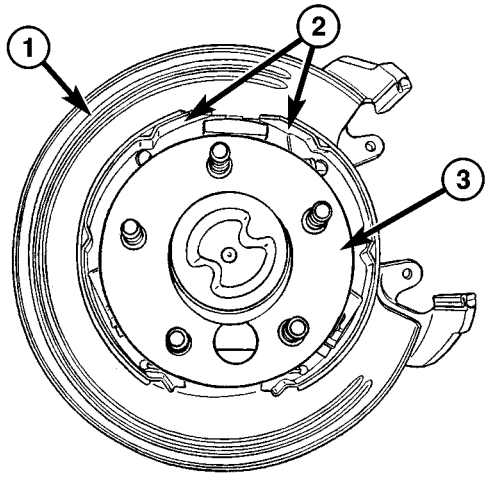


Fig. 58 PARK BRAKE SHOES INSTALLED

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - AXLE

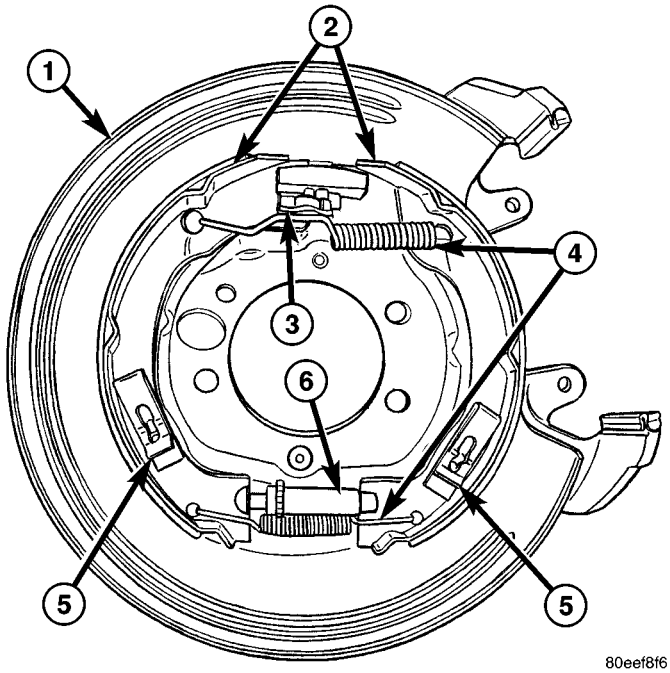


Fig. 59 BRAKE SHOES

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - EQUALIZER
4 - SPRINGS
5 - HOLD DOWN CLIPS
6 - ADJUSTER

(4) Install axle shaft, (Fig. 58), (Refer to 3 - DIFFERENTIAL & DRIVELINE/REAR AXLE - AXLE SHAFTS - INSTALLATION).

(5) Adjust brake shoes to drum with brake gauge (Refer to 5 - BRAKES/PARKING BRAKE/SHOES - ADJUSTMENTS).

SUPPORT PLATE (Continued)

(6) Install the rotor (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - INSTALLATION).

(7) Install the caliper (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).

(8) Install the wheel and tire assembly (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

PARKING BRAKE

DESCRIPTION

The parking bake is a hand lever and cable operated system used to apply the rear brakes.

OPERATION

A hand operated lever in the passenger compartment is the main application device. The front cable is connected between the hand lever and the tensioner. The tensioner rod is attached to the equalizer which is the connecting point for the rear cables (Fig. 60).

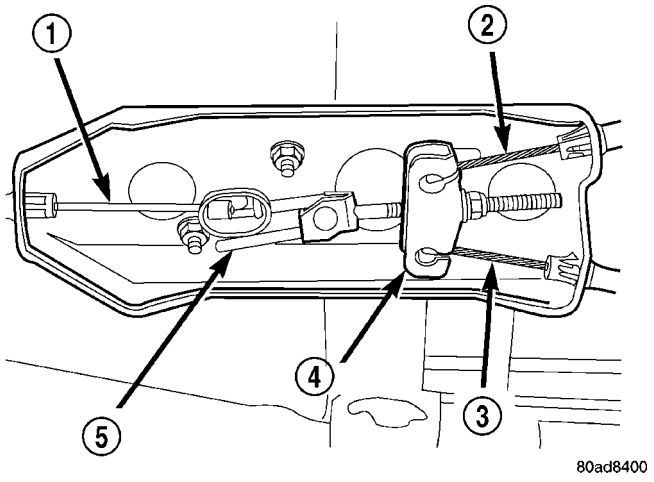


Fig. 60 Parking Brake Components

1 - FRONT CABLE
2 - L.R. CABLE
3 - R.R. CABLE
4 - EQUALIZER
5 - TENSIONER ROD

The rear cables are connected to the actuating lever on each secondary brake shoe. The levers are attached to the brake shoes by a pin either pressed into, or welded to the lever. A clip is used to secure the pin in the brake shoe. The pin allows each lever to pivot independently of the brake shoe.

To apply the parking brakes, the hand lever is pulled upward. This pulls the rear brake shoe actuating levers forward, by means tensioner and cables. As the actuating lever is pulled forward, the parking brake strut (which is connected to both shoes), exerts a linear force against the primary brake shoe. This action presses the primary shoe into contact with the drum. Once the primary shoe contacts the drum, force is exerted through the strut. This force is transferred through the strut to the secondary brake shoe causing it to pivot into the drum as well.

A gear type ratcheting mechanism is used to hold the lever in an applied position. Parking brake release is accomplished by the hand lever release button.

A parking brake switch is mounted on the parking brake lever and is actuated by movement of the lever. The switch, which is in circuit with the red warning light in the dash, will illuminate the warning light whenever the parking brakes are applied.

Parking brake adjustment is controlled by a cable tensioner mechanism. The cable tensioner, once adjusted at the factory, should not need further adjustment under normal circumstances. Adjustment may be required if a new tensioner, or cables are installed, or disconnected.

DIAGNOSIS AND TESTING - PARKING BRAKE

NOTE: Parking brake adjustment is controlled by a cable tensioner. Once the tensioner is adjusted at the factory, it should not require further attention. However, there are two instances when adjustment will be required. The first is when a new tensioner, or cables have been installed. And the second, is when the tensioner and cables are disconnected for access to other brake components.

The parking brake switch is in circuit with the red warning lamp in the dash. The switch will cause the lamp to illuminate only when the parking brakes are applied. If the lamp remains on after parking brake release, the switch or wires are faulty, or cable tensioner adjustment is incorrect.

In most cases, the actual cause of an improperly functioning parking brake (too loose/too tight/won't hold), can be traced to a parking brake component.

The leading cause of improper parking brake operation, is excessive clearance between the parking brake shoes and the shoe braking surface. Excessive clearance is a result of lining and/or drum wear, drum surface machined oversize, or inoperative adjuster components.

Excessive parking brake lever travel (sometimes described as a loose lever or too loose condition), is the result of worn brake shoes, improper brake shoe adjustment, or improperly assembled brake parts.

A condition where the parking brakes do not hold, will most probably be due to a wheel brake component.

PARKING BRAKE (Continued)

Items to look for when diagnosing a parking brake problem, are:

- Rear brake shoe wear.
- Drum surface machined oversize.
- Front cable not secured to lever.
- Rear cable not attached to lever.
- Rear cable seized.
- Brake shoes reversed.
- Parking brake strut not seated in shoes.
- Parking brake lever not seated.
- Parking brake lever bind.
- Adjuster screws seized.
- Adjuster screws reversed.

Parking brake adjustment and parts replacement procedures are described in the Parking Brake section.

SHOES

DESCRIPTION

Drum in hat park brakes are dual shoe, internal expanding units with an automatic self adjusting mechanism (Fig. 61).

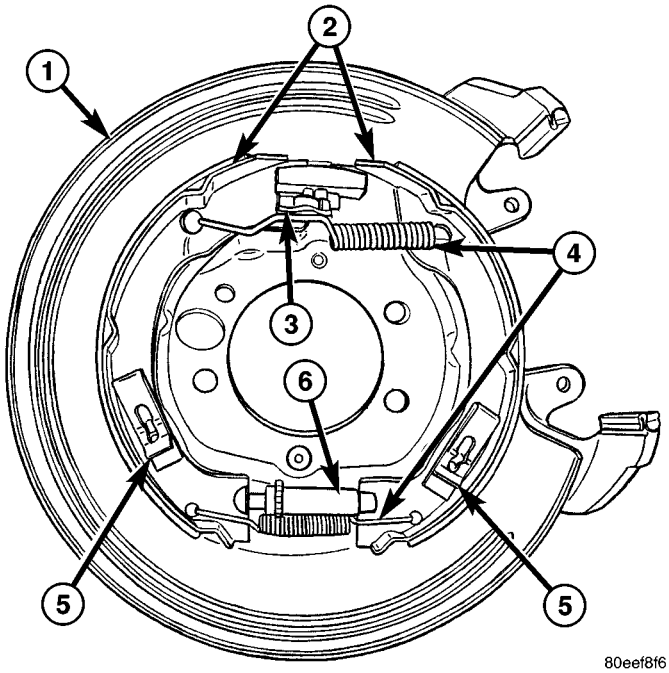


Fig. 61 BRAKE SHOES

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - EQUALIZER
4 - SPRINGS
5 - HOLD DOWN CLIPS
6 - ADJUSTER

OPERATION

When the parking brake pedal is depressed the brake cable pulls the brake shoes outward against the brake drum. When the brake pedal is released the return springs attached to the brake shoes pull the shoes back to there original position.

REMOVAL

(1) Raise and support the vehicle.
(2) Remove the tire and wheel assembly.
(3) Remove the disc brake caliper, (Fig. 62), (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - REMOVAL).
(4) Remove the disc brake rotor, (Fig. 62), (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - REMOVAL).

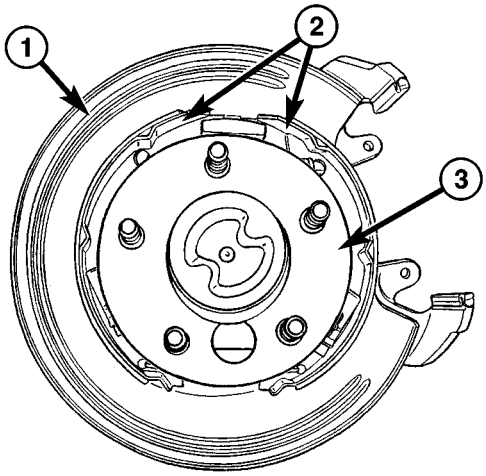


Fig. 62 PARK BRAKE SHOES INSTALLED

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - AXLE

(5) Disassemble the rear park brake shoes (Fig. 63).

CLEANING - REAR DRUM IN HAT BRAKE

Clean the individual brake components, including the support plate exterior, with a water dampened cloth or with brake cleaner. Do not use any other cleaning agents. Remove light rust and scale from the brake shoe contact pads on the support plate with fine sandpaper.

INSPECTION - REAR DRUM IN HAT BRAKE

As a general rule, riveted brake shoes should be replaced when worn to within 0.78 mm (1/32 in.) of the rivet heads. Bonded lining should be replaced when worn to a thickness of 1.6 mm (1/16 in.).

Examine the lining contact pattern to determine if the shoes are bent or the drum is tapered. The lining should exhibit contact across its entire width. Shoes exhibiting contact only on one side should be replaced and the drum checked for runout or taper (Fig. 64).

SHOES (Continued)

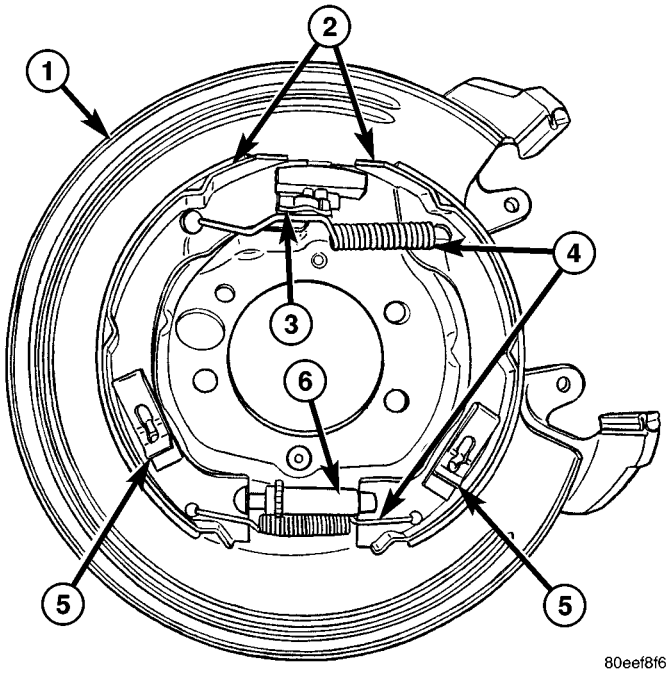


Fig. 63 BRAKE SHOES

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - EQUALIZER
4 - SPRINGS
5 - HOLD DOWN CLIPS
6 - ADJUSTER

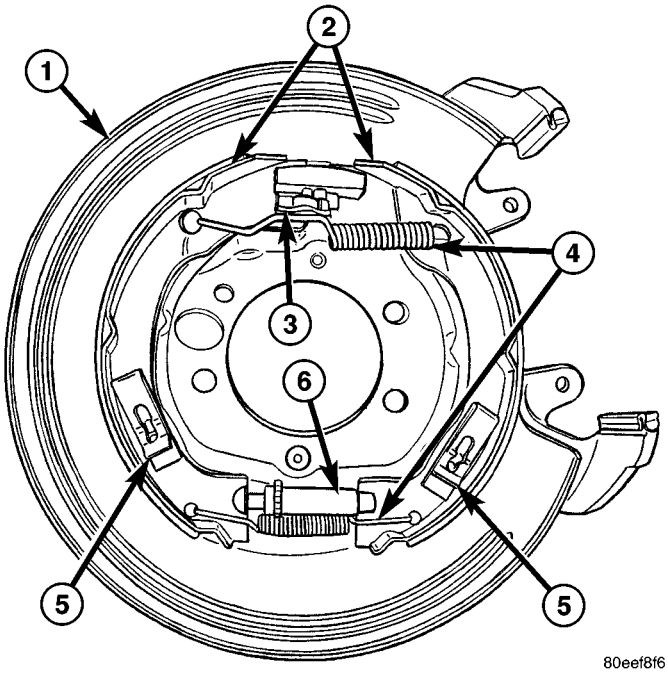


Fig. 64 BRAKE SHOES

1 - SUPPORT PLATE
2 - PARK BRAKE SHOES
3 - EQUALIZER
4 - SPRINGS
5 - HOLD DOWN CLIPS
6 - ADJUSTER

Inspect the adjuster screw assembly. Replace the assembly if the star wheel or threads are damaged, or the components are severely rusted or corroded (Fig. 64).

Discard the brake springs and retainer components if worn, distorted or collapsed. Also replace the springs if a brake drag condition had occurred. Overheating will distort and weaken the springs.

Inspect the brake shoe contact pads on the support plate, replace the support plate if any of the pads are worn or rusted through. Also replace the plate if it is bent or distorted (Fig. 64).

INSTALLATION

NOTE: On a new vehicle or after parking brake lining replacement, it is recommended that the parking brake system be conditioned prior to use. This is done by making one stop from 25 mph on dry pavement or concrete using light to moderate force on the parking brake lever.

(1) Reassemble the rear park brake shoes (Fig. 63).

(2) Adjust the rear brake shoes (Refer to 5 - BRAKES/PARKING BRAKE/SHOES - ADJUSTMENTS).

(3) Install the disc brake rotor (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - INSTALLATION).

(4) Install the disc brake caliper (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/DISC BRAKE CALIPERS - INSTALLATION).

(5) Install the tire and wheel assembly (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).

(6) Lower the vehicle.

ADJUSTMENTS

ADJUSTMENT - REAR DRUM IN HAT PARK BRAKE (ROTOR REMOVED)

Under normal circumstances, the only time adjustment is required is when the shoes are replaced, removed for access to other parts, or when one or both rotors are replaced.

Adjustment can be made with a standard brake gauge or with adjusting tool. Adjustment is performed with the complete brake assembly installed on the backing plate.

CAUTION: Before adjusting the park brake shoes be sure that the park brake pedal is in the fully released position. If park brake pedal is not in the fully released position, the park brake shoes can not be accurately adjusted.

SHOES (Continued)

(1) Raise vehicle.
(2) Remove tire and wheel.
(3) Remove disc brake caliper from caliper adapter (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ DISC BRAKE CALIPERS - REMOVAL).
(4) Remove rotor from the axleshaft (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - REMOVAL).

NOTE: When measuring the brake drum diameter, the diameter should be measured in the center of the area in which the park brake shoes contact the surface of the brake drum.

(5) Using Brake Shoe Gauge, Special Tool C-3919, or equivalent, **accurately** measure the inside diameter of the park brake drum portion of the rotor (Fig. 65).

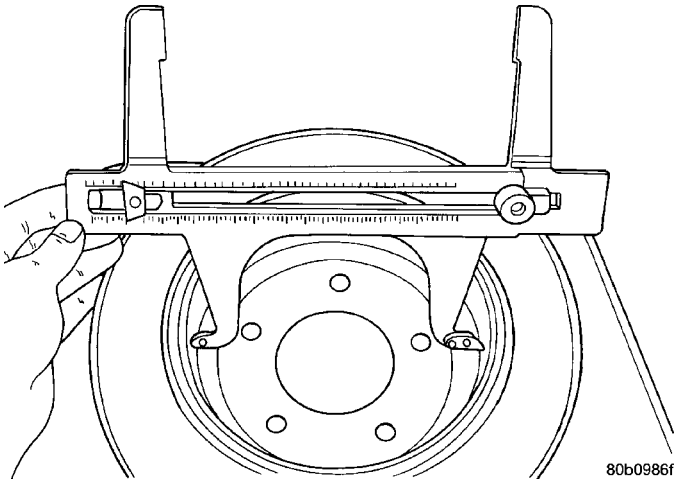


Fig. 65 MEASURING PARK BRAKE DRUM DIAMETER

(6) Using a ruler that reads in 64th of an inch, accurately read the measurement of the inside diameter of the park brake drum from the special tool (Fig. 66).
(7) Reduce the inside diameter measurement of the brake drum that was taken using Special Tool C-3919 by 1/64 of an inch. Reset Gauge, Brake Shoe, Special Tool C-3919 or the equivalent used, so that the outside measurement jaws are set to the reduced measurement (Fig. 67).
(8) Place Gauge, Brake Shoe, Special Tool C-3919, or equivalent over the park brake shoes. The special tool must be located diagonally across at the top of one shoe and bottom of opposite shoe (widest point) of the park brake shoes.
(9) Using the star wheel adjuster, adjust the park brake shoes until the lining on the park brake shoes just touches the jaws on the special tool.
(10) Repeat step 8 above and measure shoes in both directions.

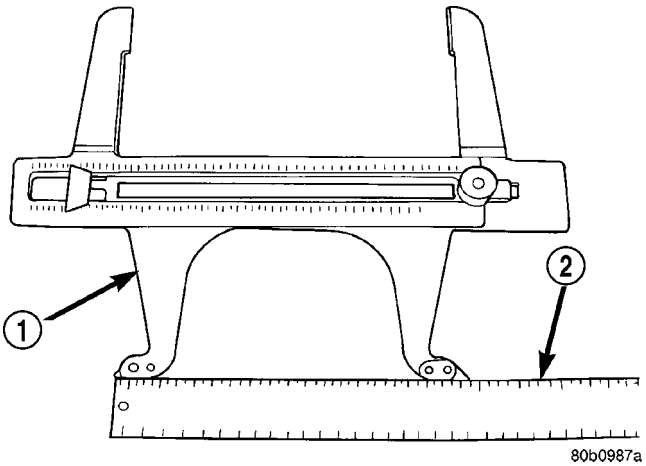


Fig. 66 READING PARK BRAKE DRUM DIAMETER

1 - SPECIAL TOOL C-3919
2 - RULER

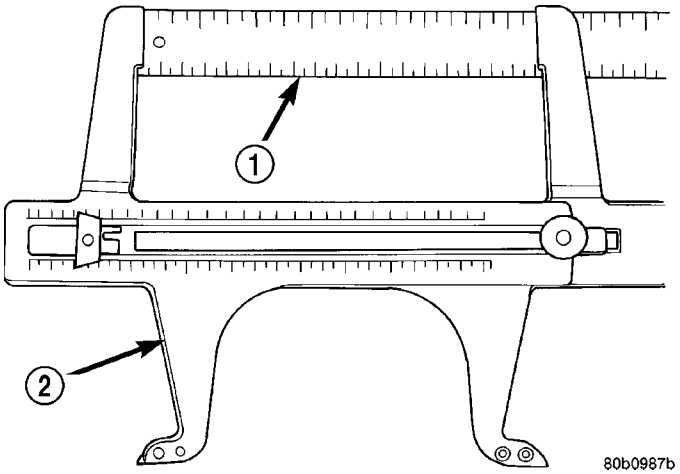


Fig. 67 SETTING GAUGE TO PARK BRAKE SHOE MEASUREMENT

1 - RULER
2 - SPECIAL TOOL C-3919

(11) Install brake rotor on the axleshaft (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ROTORS - INSTALLATION).
(12) Rotate rotor to verify that the park brake shoes are not dragging on the brake drum. If park brake shoes are dragging, remove rotor and back off star wheel adjuster one notch and recheck for brake shoe drag against drum. Continue with the previous step until brake shoes are not dragging on brake drum.
(13) Install disc brake caliper on caliper adapter (Refer to 5 - BRAKES/HYDRAULIC/MECHANICAL/ DISC BRAKE CALIPERS - INSTALLATION).
(14) Install wheel and tire.
(15) Tighten the wheel mounting nuts in the proper sequence until all nuts are torqued to half the specified torque. Then repeat the tightening sequence to the full specified torque of 129 N·m (95 ft. lbs.).
(16) Lower vehicle.

SHOES (Continued)

CAUTION: Before moving vehicle, pump brake pedal several times to ensure the vehicle has a firm enough pedal to stop the vehicle.

NOTE: After parking brake lining replacement, it is recommended that the parking brake system be conditioned prior to use. This is done by making one stop from 25 mph on dry pavement or concrete using light to moderate force on the parking brake hand lever.

(17) Road test the vehicle to ensure proper function of the vehicle's brake system.

ADJUSTMENT - WITH ADJUSTING TOOL

Adjustment can be made with a standard brake gauge or with adjusting tool. Adjustment is performed with the complete brake assembly installed on the backing plate.

(1) Be sure parking brake lever is fully released.

(2) Raise vehicle so rear wheels can be rotated freely.

(3) Remove plug from each access hole in brake support plates.

(4) Loosen parking brake cable adjustment nut until there is slack in front cable.

(5) Insert adjusting tool through support plate access hole and engage tool in teeth of adjusting screw star wheel (Fig. 68).

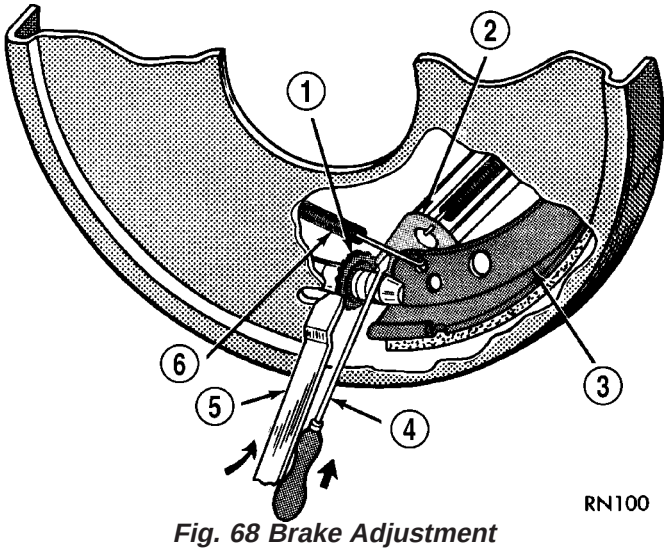


Fig. 68 Brake Adjustment

1 - STAR WHEEL
2 - LEVER
3 - BRAKE SHOE WEB
4 - SCREWDRIVER
5 - ADJUSTING TOOL
6 - ADJUSTER SPRING

(6) Rotate adjuster screw star wheel (move tool handle upward) until slight drag can be felt when wheel is rotated.

(7) Push and hold adjuster lever away from star wheel with thin screwdriver.

(8) Back off adjuster screw star wheel until brake drag is eliminated.

(9) Repeat adjustment at opposite wheel. Be sure adjustment is equal at both wheels.

(10) Install support plate access hole plugs.

(11) Adjust parking brake cable and lower vehicle.

(12) Depress park brake lever and make sure park brakes hold the vehicle staionary.

(13) Release park brake lever.

CABLES

REMOVAL

REMOVAL - DRUM BRAKES

(1) Raise vehicle and loosen equalizer nuts until rear cables are slack.

(2) Disengage cable from equalizer and remove cable.

(3) Remove cable bracket from upper suspension arm (Fig. 69).

(4) Remove rear wheel and brake drum.

(5) Remove secondary brake shoe and disconnect cable from lever on brake shoe.

(6) Compress cable retainer with worm drive hose clamp (Fig. 70) and remove cable from backing plate.

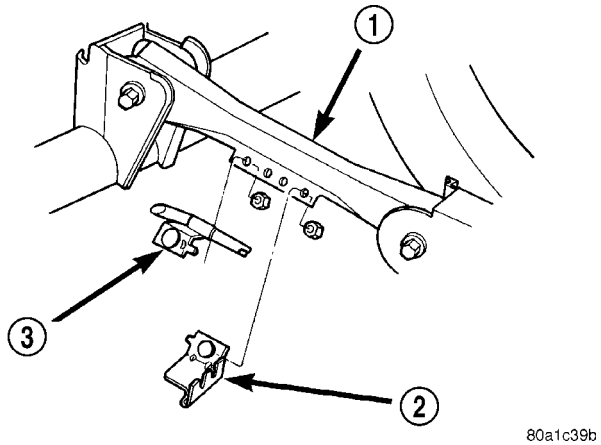


Fig. 69 Parking Brake Cable Bracket

1 - UPPER SUSPENSION ARM
2 - WIRING BRACKET
3 - PARKING BRAKE CABLE BRACKET

CABLES (Continued)

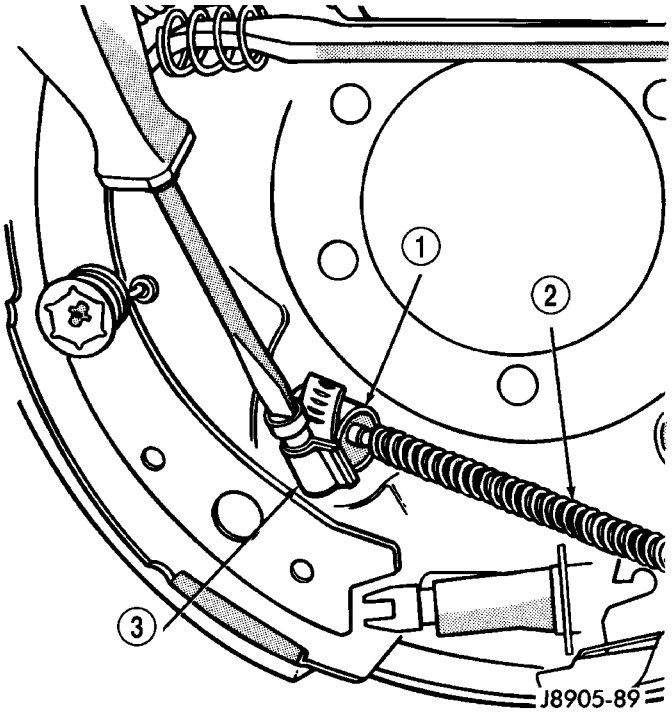


Fig. 70 Cable Retainer

1 - CABLE RETAINER
2 - REAR CABLE
3 - WORM DRIVE HOSE CLAMP

REMOVAL - DISC BRAKES

(1) Raise and support the vehicle.
(2) Lockout the parking brake cable.
(3) Loosen the brake cable at the equalizer and adjuster nut.
(4) Remove the cable from the front cable.
(5) Remove the cable from the equalizer.
(6) Remove the cable from the frame bracket.
(7) Remove the cable from the axle bracket.
(8) Remove the brake cable from the brake lever. (Fig. 71)

INSTALLATION

INSTALLATION - DRUM BRAKES

(1) Install new cable in backing plate. Be sure cable retainer is seated.
(2) Attach cable to lever on brake shoe and install brake shoe on backing plate.
(3) Adjust brake shoes to drum with brake gauge.
(4) Install brake drum and wheel.
(5) Install cable/bracket on upper suspension arm.
(6) Engage cable in equalizer and install equalizer nuts.
(7) Adjust parking brakes.

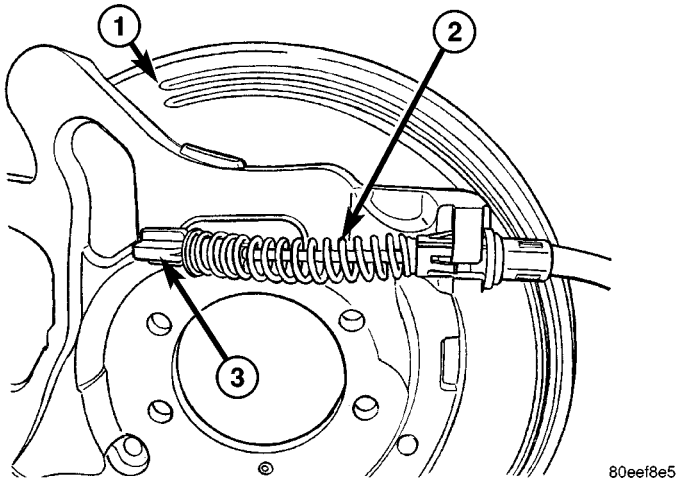


Fig. 71 CABLE REMOVAL/INSTALLATION

1 - SUPPORT PLAT
2 - PARK BRAKE CABLE
3 - EQUALIZER

INSTALLATION - DISC BRAKES

(1) Install the brake cable to the brake lever.
(2) Install the cable to the axle bracket (Fig. 72).
(3) Install the cable to the equalizer (Fig. 72).
(4) Install the cable to the front cable.
(5) Adjust the brake cable at the equalizer and using the adjuster nut.

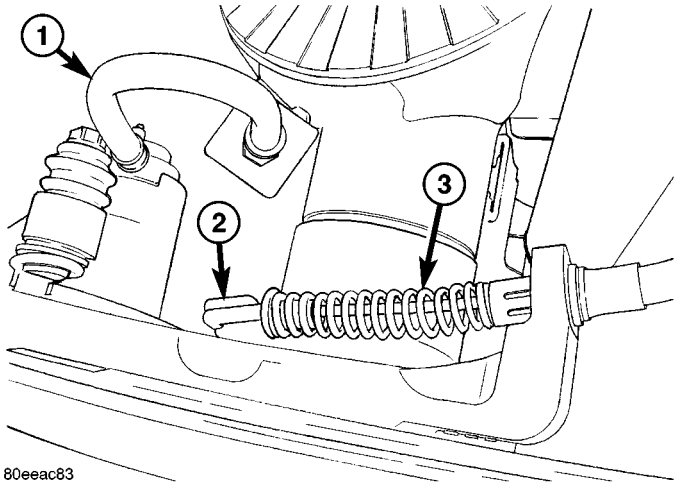


Fig. 72 REAR CABLE CONNECTED

1 - BRAKE HOSE
2 - ACTUATOR LEVER
3 - PARK BRAKE CABLE

BRAKES - ABS

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BRAKES - ABS

DESCRIPTION

The purpose of the antilock system is to prevent wheel lockup during periods of high wheel slip. Preventing lockup helps maintain vehicle braking action and steering control.

The antilock CAB activates the system whenever sensor signals indicate periods of high wheel slip. High wheel slip can be described as the point where wheel rotation begins approaching 20 to 30 percent of actual vehicle speed during braking. Periods of high wheel slip occur when brake stops involve high pedal pressure and rate of vehicle deceleration.

Battery voltage is supplied to the CAB ignition terminal when the ignition switch is turned to Run position. The CAB performs a system initialization procedure at this point. Initialization consists of a static and dynamic self check of system electrical components.

The static check occurs after the ignition switch is turned to Run position. The dynamic check occurs when vehicle road speed reaches approximately 30 kph (18 mph). During the dynamic check, the CAB briefly cycles the pump and solenoids to verify operation.

If an ABS component exhibits a fault during initialization, the CAB illuminates the amber warning light and registers a fault code in the microprocessor memory.

OPERATION

During normal braking, the master cylinder, power booster and wheel brake units all function as they would in a vehicle without ABS. The HCU components are not activated.

During antilock braking fluid pressure is modulated according to wheel speed, degree of slip and rate of deceleration. A sensor at each wheel converts wheel speed into electrical signals. These signals are transmitted to the CAB for processing and determination of wheel slip and deceleration rate.

The ABS system has three fluid pressure control channels. The front brakes are controlled separately and the rear brakes in tandem. A speed sensor input signal indicating a high slip condition activates the CAB antilock program. Two solenoid valves are used in each antilock control channel. The valves are all located within the HCU valve body and work in pairs to either increase, hold, or decrease apply pressure as needed in the individual control channels. The solenoid valves are not static during antilock braking. They are cycled continuously to modulate pressure. Solenoid cycle time in antilock mode can be measured in milliseconds.

DIAGNOSIS AND TESTING - ANTILOCK BRAKES

The ABS brake system performs several self-tests every time the ignition switch is turned on and the vehicle is driven. The CAB monitors the systems input and output circuits to verify the system is oper-

ating correctly. If the on board diagnostic system senses that a circuit is malfunctioning the system will set a trouble code in its memory.

NOTE: An audible noise may be heard during the self-test. This noise should be considered normal.

NOTE: The MDS or DRB III scan tool is used to diagnose the ABS system. For additional information refer to the Antilock Brake section in Group 8W. For test procedures refer to the Chassis Diagnostic Manual.

STANDARD PROCEDURE - BLEEDING ABS BRAKE SYSTEM

ABS system bleeding requires conventional bleeding methods plus use of the DRB scan tool. The procedure involves performing a base brake bleeding, followed by use of the scan tool to cycle and bleed the HCU pump and solenoids. A second base brake bleeding procedure is then required to remove any air remaining in the system.

(1) Perform base brake bleeding. Refer to base brake section for procedure.

(2) Connect scan tool to the Data Link Connector.

(3) Select ANTILOCK BRAKES, followed by MISCELLANEOUS, then ABS BRAKES. Follow the instructions displayed. When scan tool displays TEST COMPLETE, disconnect scan tool and proceed.

(4) Perform base brake bleeding a second time. Refer to base brake section for procedure.

(5) Top off master cylinder fluid level and verify proper brake operation before moving vehicle.

SPECIFICATIONS

T0RQUE CHART

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
G-Sensor Sensor Bolt	4-5	—	35-45
G-Sensor Bracket Bolt	8-13	—	75-115
Hydraulic Control Unit Bracket to HCU Bolts	6.5	—	57
Hydraulic Control Unit Body Bracket Bolts	16-24	—	142-212
Hydraulic Control Unit HCU to Body Bracket Bolts	9-13	—	80-115
Hydraulic Control Unit Brake Lines	15-18	—	130-160
Controller Anitlock Brakes Mounting Bolt	7-9	—	60-80
Wheel Speed Sensors Front Mounting Bolt	4-6	—	34-50
Wheel Speed Sensors Rear Mounting Bolt	12-14	—	106-124

FRONT WHEEL SPEED SENSOR

DESCRIPTION

A speed sensor is used at each wheel. The front sensors are mounted to the steering knuckles. The rear sensors are mounted to the rear brake backing plate.

OPERATION

The sensors convert wheel speed into a small AC electrical signal. This signal is transmitted to the CAB. The CAB convert the AC signal into a digital signal for each wheel. This voltage is generated by magnetic induction when a tone wheel passes by the stationary magnetic of the wheel speed sensor.

A gear type tone ring serves as the trigger mechanism for each sensor. The tone rings are mounted at the outboard ends of the front and rear axle shafts.

Different sensors are used at the front and rear wheels (Fig. 1). The front/rear sensors have the same electrical values but are not interchangeable. The sensors have a resistance between 900 and 1300 ohms.

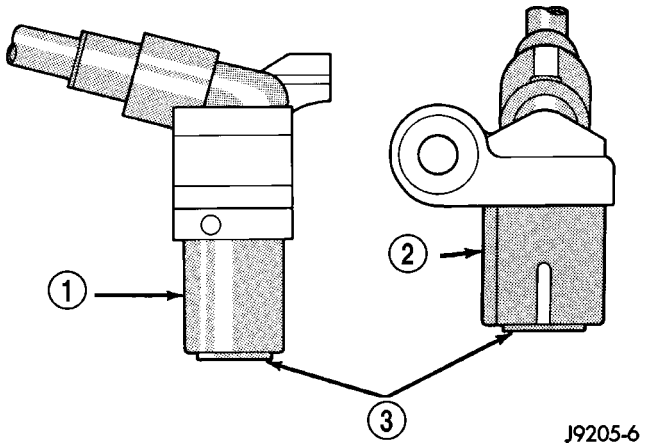

Fig. 1 Typical Wheel Speed Sensors

1 - FRONT SENSOR
2 - REAR SENSOR
3 - PICKUP FACE

FRONT SENSOR AIR GAP

Front sensor air gap is fixed and not adjustable. Only rear sensor air gap is adjustable.

Although front air gap is not adjustable, it can be checked if diagnosis indicates this is necessary. Front air gap should be 0.40 to 1.3 mm (0.0157 to 0.051 in.). If gap is incorrect, the sensor is either loose, or damaged.

REAR SENSOR AIR GAP

A rear sensor air gap adjustment is only needed when reinstalling an original sensor. Replacement sensors have an air gap spacer attached to the sensor pickup face. The spacer establishes correct air gap when pressed against the tone ring during installation. As the tone ring rotates, it peels the spacer off the sensor to create the required air gap. Rear sensor air gap is 0.28-1.5 mm (0.011-0.059 in.).

Sensor air gap measurement, or adjustment procedures are provided in this section. Refer to the front, or rear sensor removal and installation procedures as required.

REMOVAL

(1) Raise vehicle and turn wheel outward to access the sensor.

(2) Disconnect sensor wire connector at harness plug.

(3) Remove sensor wire from mounting retainers.

(4) Clean sensor and surrounding area with shop towel before removal.

(5) Remove bolt attaching sensor to steering knuckle and remove sensor (Fig. 2).

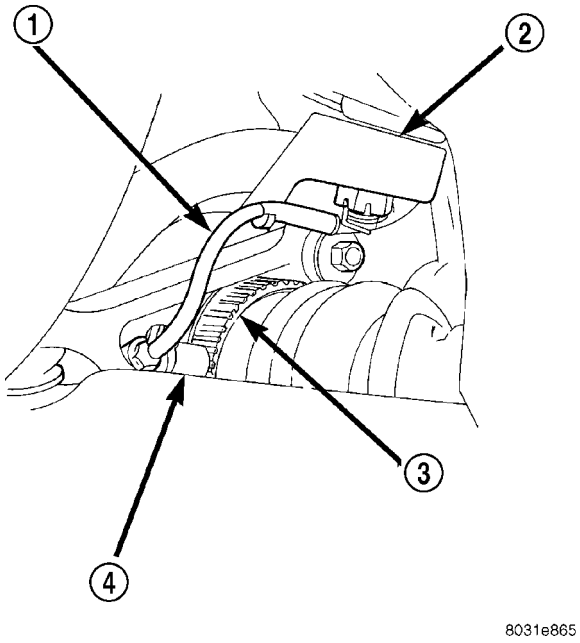

Fig. 2 Front Wheel Speed Sensor

1 - WHEEL SPEED SENSOR PIGTAIL
2 - STEERING KNUCKLE
3 - TONE WHEEL
4 - FRONT WHEEL SPEED SENSOR

INSTALLATION

(1) If **original** sensor will be installed, wipe all traces of old spacer material off sensor pickup face. Use a dry shop towel for this purpose.

(2) Apply Mopar Lock N' Seal or Loctite ® 242 on bolt that secures sensor in steering knuckle. Use new sensor bolt if original bolt is worn or damaged.

FRONT WHEEL SPEED SENSOR (Continued)

(3) Position sensor on steering knuckle. Seat sensor locating tab in hole in knuckle and install sensor attaching bolt finger tight.

(4) Tighten sensor attaching bolt to 4-6 N·m (34-50 in. lbs.).

(5) If original sensor has been installed, check sensor air gap. Air gap should be 0.40 to 1.3 mm (0.0157 to 0.051 in.). If gap is incorrect, sensor is either loose, or damaged.

(6) Route sensor wire and install into mounting retainers.

(7) Connect sensor wire to harness.

G-SWITCH

DESCRIPTION

The G-switch is located in front of the console/shifter mounted to a bracket on the floor pan. The switch has directional arrow and must be mounted with the arrow pointing towards the front of the vehicle.

OPERATION

The switch (Fig. 3) , provides an additional vehicle deceleration reference during 4-wheel drive operation. The switch is monitored by the CAB at all times. The switch reference signal is utilized by the CAB when all wheels are decelerating at the same speed.

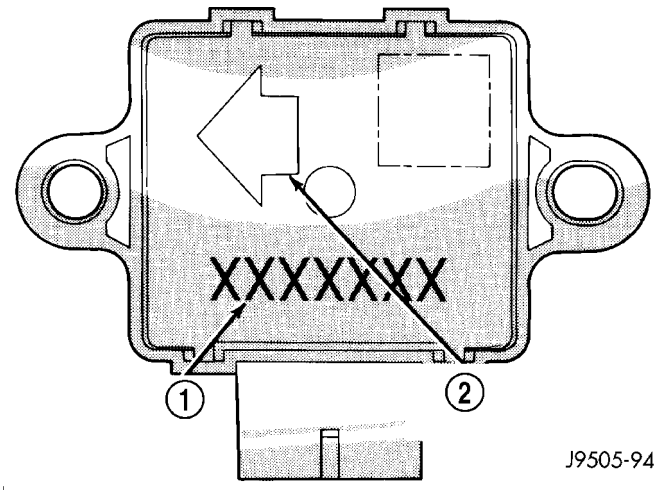

Fig. 3 G-Switch

1 - SWITCH PART NUMBER
2 - ARROW INDICATES FRONT OF SWITCH FOR PROPER MOUNTING

REMOVAL

(1) From the drivers side lift carpet back in front of the console/shifter.

(2) Disconnect harness for switch.

(3) Remove mounting bolts and remove switch (Fig. 4).

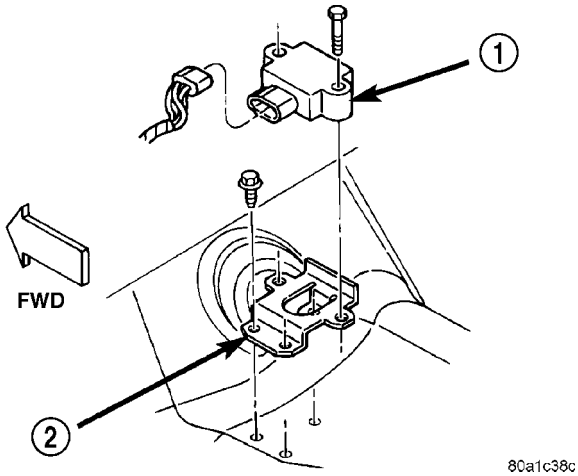

Fig. 4 G-Switch

1 - ACCELERATION SWITCH
2 - MOUNTING BRACKET

INSTALLATION

CAUTION: The mercury switch (inside the G-switch), will not function properly if the switch is installed incorrectly. Verify that the switch locating arrow is pointing to the front of the vehicle (Fig. 3).

(1) Position switch on mounting bracket.

(2) Install mounting bolts and tighten to 4-5 N·m (35-45 in. lbs.)

(3) Connect harness to switch.

(4) Place carpet back into position.

REAR WHEEL SPEED SENSOR

DESCRIPTION

A speed sensor is used at each wheel. The front sensors are mounted to the steering knuckles. The rear sensors are mounted to the rear brake backing plate.

OPERATION

The sensors convert wheel speed into a small AC electrical signal. This signal is transmitted to the CAB. The CAB convert the AC signal into a digital signal for each wheel. This voltage is generated by magnetic induction when a tone wheel passes by the stationary magnetic of the wheel speed sensor.

A gear type tone ring serves as the trigger mechanism for each sensor. The tone rings are mounted at the outboard ends of the front and rear axle shafts.

Different sensors are used at the front and rear wheels (Fig. 1). The front/rear sensors have the same electrical values but are not interchangeable. The

REAR WHEEL SPEED SENSOR (Continued)

sensors have a resistance between 900 and 1300 ohms.

REMOVAL

(1) Disconnect sensors at rear harness connectors.
(2) Remove wheel and tire assembly.
(3) Remove brake drum.
(4) Remove clips securing sensor wires to brake lines, rear axle and, brake hose.
(5) Unseat sensor wire support plate grommet.
(6) Remove bolt attaching sensor to bracket (Fig. 5) and remove sensor.

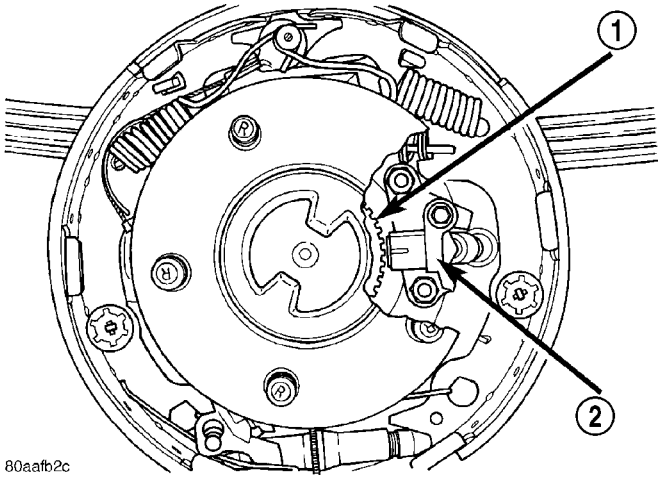

Fig. 5 Wheel Speed Sensor

1 - TONE WHEEL
2 - WHEEL SPEED SENSOR

INSTALLATION

(1) If **original sensor** is being installed, remove any remaining pieces of cardboard spacer from sensor pickup face. Use dry shop towel only to remove old spacer material.
(2) Insert sensor wire through support plate hole. Then seat sensor grommet in support plate.
(3) Apply Mopar Lock N' Seal or Loctite ® 242 to original sensor bolt. Use new bolt if original is worn or damaged.
(4) Install sensor bolt finger tight only at this time.
(5) If **original** rear sensor was installed, adjust sensor air gap to 0.28-1.5 mm (0.011-0.059 in.). Use feeler gauge to measure air gap (Fig. 6). Tighten sensor bolt to 12-14 N·m (106-124 in. lbs.).
(6) If **new** sensor was installed, push cardboard spacer on sensor face against tone ring (Fig. 7). Then tighten sensor bolt to 12-14 N·m (106-124 in. lbs.). Correct air gap will be established as tone ring rotates and peels spacer off sensor face.
(7) Secure the rear sensor wires to the retainer clips. Verify that wire is clear of rotating components.
(8) Connect sensor wire to harness connector.

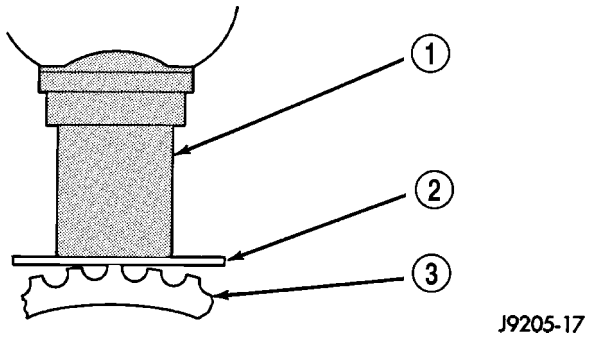

Fig. 6 Setting Air Gap On Original Rear Sensor

1 - WHEEL SPEED SENSOR
2 - BRASS FEELER GAUGE
3 - TONE RING

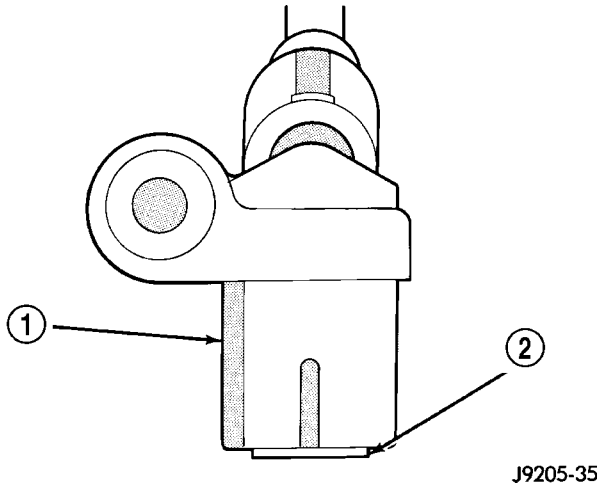

Fig. 7 New Rear Sensor

1 - REAR SENSOR
2 - AIR GAP SPACER ATTACHED TO SENSOR FACE

(9) Install brake drum and wheel and tire assembly. (Refer to 22 - TIRES/WHEELS/WHEELS - STANDARD PROCEDURE).
(10) Lower vehicle.
(11) Connect sensor wire to harness connector.

HCU (HYDRAULIC CONTROL UNIT)

DESCRIPTION

The HCU consists of a valve body, pump motor, and wire harness.

OPERATION

Accumulators in the valve body store extra fluid released to the system for ABS mode operation. The pump provides the fluid volume needed and is operated by a DC type motor. The motor is controlled by the CAB.

HCU (HYDRAULIC CONTROL UNIT) (Continued)

The valves modulate brake pressure during antilock braking and are controlled by the CAB.

The HCU provides three channel pressure control to the front and rear brakes. One channel controls the rear wheel brakes in tandem. The two remaining channels control the front wheel brakes individually.

During antilock braking, the solenoid valves are opened and closed as needed. The valves are not static. They are cycled rapidly and continuously to modulate pressure and control wheel slip and deceleration.

During normal braking, the HCU solenoid valves and pump are not activated. The master cylinder and power booster operate the same as a vehicle without an ABS brake system.

During antilock braking, solenoid valve pressure modulation occurs in three stages, pressure increase, pressure hold, and pressure decrease. The valves are all contained in the valve body portion of the HCU.

PRESSURE DECREASE

The outlet valve is opened and the inlet valve is closed during the pressure decrease cycle.

A pressure decrease cycle is initiated when speed sensor signals indicate high wheel slip at one or more wheels. At this point, the CAB closes the inlet then opens the outlet valve, which also opens the return circuit to the accumulators. Fluid pressure is allowed to bleed off (decrease) as needed to prevent wheel lock.

Once the period of high wheel slip has ended, the CAB closes the outlet valve and begins a pressure increase or hold cycle as needed.

PRESSURE HOLD

Both solenoid valves are closed in the pressure hold cycle. Fluid apply pressure in the control channel is maintained at a constant rate. The CAB maintains the hold cycle until sensor inputs indicate a pressure change is necessary.

PRESSURE INCREASE

The inlet valve is open and the outlet valve is closed during the pressure increase cycle. The pressure increase cycle is used to counteract unequal wheel speeds. This cycle controls re-application of fluid apply pressure due to changing road surfaces or wheel speed.

REMOVAL

(1) Install prop rod on the brake pedal to keep pressure on the brake system.

(2) Remove negative battery cable from the battery.

(3) Pull up on the CAB harness connector release (Fig. 8)and remove connector.

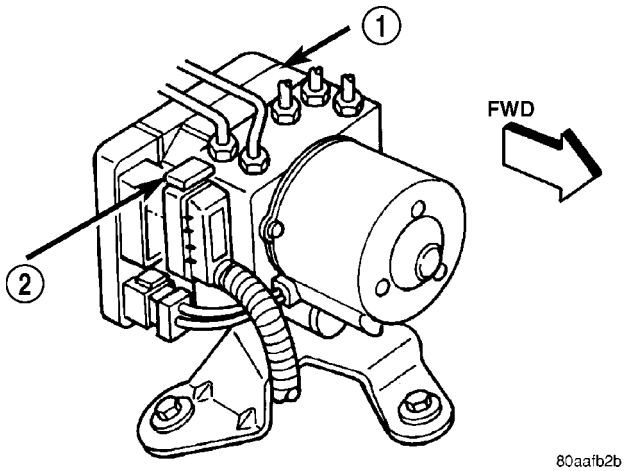

Fig. 8 CAB Harness Connector Release

1 - CAB
2 - CAB HARNESS RELEASE

(4) Remove brake lines from the HCU.

(5) Remove HCU/CAB mounting nuts and bolt (Fig. 9) and remove HCU/CAB.

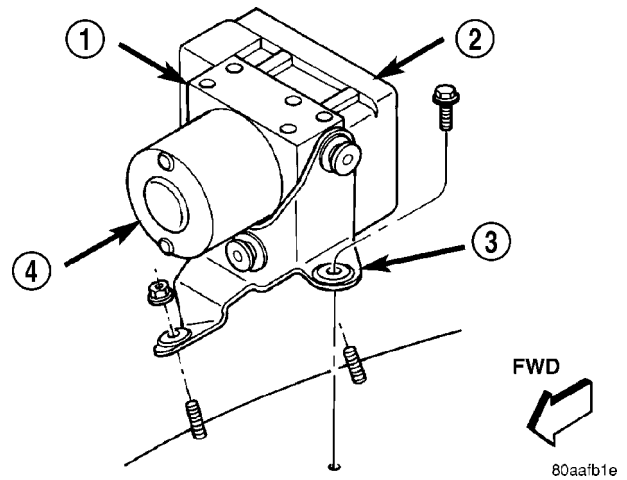

Fig. 9 HCU/CAB Mounting

1 - HCU
2 - CAB
3 - HCU/CAB BRACKET
4 - MOTOR

INSTALLATION

NOTE: If the CAB is being replaced with a new CAB is must be reprogrammed with the use of a DRB III.

(1) Install HCU/CAB on the mounting studs.

(2) Install mounting nuts and bolt. Tighten to 11.5 N·m (102 in. lbs.).

(3) Install brake lines to the HCU and tighten to 19 N·m (170 in. lbs.).

(4) Install wiring harness connector to the CAB and push down on the release to secure the connector.

(5) Install negative battery cable to the battery.

(6) Bleed ABS brake system (Refer to 5 - BRAKES - STANDARD PROCEDURE).

CLUTCH

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CLUTCH

WARNING

WARNING: Exercise care when servicing clutch components. Factory installed clutch discs do not contain asbestos fibers. Dust and dirt on clutch parts may contain asbestos fibers from aftermarket components. Breathing excessive concentrations of these fibers can cause serious bodily harm. Wear a respirator during service and never clean clutch components with compressed air or with a dry brush. Either clean the components with water dampened rags or use a vacuum cleaner specifically designed to remove asbestos fibers and dust. Do not create dust by sanding a clutch discs. Replace the disc if the friction material is damaged. Dispose of all dust and dirt containing asbestos fibers in sealed bags or containers. This will minimize exposure to yourself and to others. Follow all recommended safety practices prescribed by the occupational safety and health administration (OSHA) and the environmental safety agency (EPA), for the handling and disposal of products containing asbestos. Failure to follow these instructions may result in personal injury or death

DIAGNOSIS AND TESTING

Drive the vehicle at normal speeds. Shift the transmission through all gear ranges and observe clutch action. If the clutch chatters, grabs, slips or does not release properly, remove and inspect the clutch components. If the problem is noise or hard shifting, further diagnosis may be needed as the transmission or another driveline component may be at fault.

NOTE: Vehicles equipped with a Dual Mass Flywheel may produce a rattle when the engine is shut off. This noise is considered normal.

CLUTCH CONTAMINATION

Fluid contamination is a frequent cause of clutch malfunctions. Oil, water or clutch fluid on the clutch disc and pressure plate surfaces will cause chatter, slip and grab. Inspect components for oil, hydraulic fluid or water/road splash contamination.

Oil contamination indicates a leak at either the rear main seal or transmission input shaft. Clutch fluid leaks are usually from damaged slave cylinder push rod seals. Heat buildup caused by slippage between the pressure plate, disc and flywheel can bake the oil residue onto the components. The glaze-like residue ranges in color from amber to black.

Road splash contamination is dirt/water entering the clutch housing due to loose bolts, housing cracks. Driving through deep water puddles can force water/road splash into the housing through such openings.

IMPROPER RELEASE OR CLUTCH ENGAGEMENT

Clutch release or engagement problems are caused by wear or damage clutch components. A visual inspection of the release components will usually reveal the problem part.

CLUTCH (Continued)

Release problems can result in hard shifting and noise. Look for leaks at the clutch cylinders and interconnecting line and loose slave cylinder bolts. Also worn/loose release fork, pivot stud, clutch disc, pressure plate or release bearing.

Engagement problems can result in slip, chatter/shudder and noisy operation. The causes may be clutch disc contamination, wear, distortion or flywheel damage. Visually inspect to determine the actual cause of the problem.

CLUTCH MISALIGNMENT

Clutch components must be in proper alignment with the crankshaft and transmission input shaft. Misalignment caused by excessive runout or warpage of any clutch component will cause grab, chatter and improper clutch release.

PRESSURE PLATE AND DISC RUNOUT

Check the clutch disc before installation. Axial (face) runout of a **new** disc should not exceed 0.50 mm (0.020 in.). Measure runout about 6 mm (1/4 in.) from the outer edge of the disc facing. Obtain another disc if runout is excessive.

Check condition of the clutch before installation. A warped cover or diaphragm spring will cause grab and incomplete release or engagement. Be careful when handling the cover and disc. Impact can distort the cover, diaphragm spring, release fingers and the hub of the clutch disc.

Use an alignment tool when positioning the disc on the flywheel. The tool prevents accidental misalignment which could result in cover distortion and disc damage.

A frequent cause of clutch cover distortion (and consequent misalignment) is improper bolt tightening.

FLYWHEEL RUNOUT

Check flywheel runout whenever misalignment is suspected. Flywheel runout should not exceed 0.08 mm (0.003 in.). Measure runout at the outer edge of the flywheel face with a dial indicator. Mount the indicator on a stud installed in place of one of the flywheel bolts.

Common causes of runout are:

- heat warpage
- improper machining
- incorrect bolt tightening
- improper seating on crankshaft flange shoulder
- foreign material on crankshaft flange

Flywheel machining is not recommended. The flywheel clutch surface is machined to a unique contour and machining will negate this feature. Minor flywheel scoring can be cleaned up by hand with 180 grit emery or with surface grinding equipment. Remove only enough material to reduce scoring (approximately 0.001 - 0.003 in.). Heavy stock removal is **not recommended.** Replace the flywheel if scoring is severe and deeper than 0.076 mm (0.003 in.). Excessive stock removal can result in flywheel cracking or warpage after installation; it can also weaken the flywheel and interfere with proper clutch release.

Clean the crankshaft flange before mounting the flywheel. Dirt and grease on the flange surface may cock the flywheel causing excessive runout. Use new bolts when remounting a flywheel and secure the bolts with Mopar Lock And Seal or equivalent. Tighten flywheel bolts to specified torque only. Overtightening can distort the flywheel hub causing runout.

DIAGNOSIS CHART

The diagnosis charts Diagnosis Chart describe common clutch problems, causes and correction. Conditions, causes and corrective action are outlined in the indicated columns.

CLUTCH (Continued)

DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
Disc facing worn out	1. Normal wear.	1. Replace cover and disc.
	2. Driver frequently rides (slips) the clutch. Results in rapid overheating and wear.	2. Replace cover and disc.
	3. Insufficient clutch cover diaphragm spring tension.	3. Replace cover and disc.
Clutch disc facing contaminated with oil, grease, or clutch fluid.	1. Leak at rear main engine seal or transmission input shaft seal.	1. Replace appropriate seal.
	2. Excessive amount of grease applied to the input shaft splines.	2. Remove grease and apply the correct amount of grease.
	3. Road splash, water entering housing.	3. Replace clutch disc. Clean clutch cover and reuse if in good condition.
	4. Slave cylinder leaking.	4. Replace hydraulic clutch linkage.
Clutch is running partially disengaged.	1. Release bearing sticking or binding and does not return to the normal running position.	1. Verify failure. Replace the release bearing and transmission front bearing retainer as necessary.
Flywheel below minimum thickness specification.	1. Improper flywheel machining. Flywheel has excessive taper or excessive material removal.	1. Replace flywheel.
Clutch disc, cover and/or diaphragm spring warped or distorted.	1. Rough handling. Impact bent cover, spring, or disc.	1. Replace disc or cover as necessary.
	2. Improper bolt tightening procedure.	2. Tighten clutch cover using proper procedure.
Facing on flywheel side of disc torn, gouged, or worn.	1. Flywheel surface scored or nicked.	2. Correct surface condition if possible. Replace flywheel and disc as necessary.
	2. Clutch disc sticking or binding on transmission input shaft.	2. Lubricate splines with high temperature graese.
Clutch disc facing burnt. Flywheel and cover pressure plate surfaces heavily glazed.	1. Frequent operation under high loads or hard acceleration conditions.	1. Correct condition of flywheel and pressure plate surface. Replace clutch cover and disc. Alert driver to problem cause.
	2. Driver frequently rides (slips) clutch. Results in rapid wear and overheating of disc and cover.	2. Correct condition of flywheel and pressure plate surface. Replace clutch cover and disc. Alert driver to problem cause.

CLUTCH (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
Clutch disc binds on input shaft splines.	1. Clutch disc hub splines damaged during installation.	1. Clean, smooth, and lubricate hub splines if possible. Replace disc if necessary.
	2. Input shaft splines rough, damaged, or corroded.	2. Clean, smooth, and lubricate shaft splines if possible. Replace input shaft if necessary.
Clutch disc rusted to flywheel and/or pressure plate.	1. Clutch not used for and extended period of time (e.g. long term vehicle storage).	1. Sand rusted surfaces with 180 grit sanding paper. Replace clutch cover and flywheel if necessary.
Pilot bearing seized, loose, or rollers are worn.	1. Bearing cocked during installation.	1. Install and lubricate a new bearing.
	2. Bearing defective.	2. Install and lubricate a new bearing.
	3. Bearing not lubricated.	3. Install and lubricate a new bearing.
	4. Clutch misalignment.	4. Inspect clutch and correct as necessary. Install and lubricate a new bearing.
Clutch will not disengage properly.	1. Low clutch fluid level.	1. Replace hydraulic linkage assembly.
	2. Clutch cover loose.	2. Follow proper bolt tightening procedure.
	3. Clutch disc bent or distorted.	3. Replace clutch disc.
	4. Clutch cover diaphragm spring bent or warped.	4. Replace clutch cover.
	5. Clutch disc installed backwards.	5. Remove and install clutch disc correctly.
	6. Release fork bent or fork pivot loose or damaged.	6. Replace fork or pivot as necessary.
	7. Clutch master or slave cylinder failure.	7. Replace hydraulic linkage assembly.
Clutch pedal squeak.	1. Pivot pin loose.	1. Tighten pivot pin if possible. Replace clutch pedal if necessary.
	2. Master cylinder bushing not lubricated.	2. Lubricate master cylinder bushing.
	3. Pedal bushings worn out or cracked.	3. Replace and lubricate bushings.
Clutch master or slave cylinder plunger dragging andør binding	1. Master or slave cylinder components worn or corroded.	1. Replace clutch hydraulic linkage assembly.
Release bearing is noisy.	1. Release bearing defective or damaged.	1. Replace release bearing.

CLUTCH (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
Contact surface of release bearing damaged.	1. Clutch cover incorrect or release fingers bent or distorted.	1. Replace clutch cover and release bearing.
	2. Release bearing defective or damaged.	2. Replace the release bearing.
	3. Release bearing misaligned.	3. Check and correct runout of clutch components. Check front bearing sleeve for damage/ alignment. Repair as necessary.
Partial engagement of clutch disc. One side of disc is worn and the other side is glazed and lightly worn.	1. Clutch pressure plate position incorrect.	1. Replace clutch disc and cover.
	2. Clutch cover, spring, or release fingers bent or distorted.	2. Replace clutch disc and cover.
	3. Clutch disc damaged or distorted.	2. Replace clutch disc.
	4. Clutch misalignment.	4. Check alignment and runout of flywheel, disc, pressure plate, andør clutch housing. Correct as necessary.

SPECIFICATIONS

CLUTCH

TORQUE SPECIFICATIONS

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Clutch Cover Bolts - 2.4L	31	23	-
Clutch Cover Bolts - 4.0L	50	37	-
Clutch Cylinder Bolts	23	-	200
Clutch Housing to Engine Bolts	75	55	-
Clutch Housing to Trans Bolts	46	34	-
Dust Shield Bolts	50	37	-
Flywheel Bolts - 2.4L	95	70	-
Flywheel Bolts - 4.0L	142	105	-
Crossmember Frame Bolts	41	30	-

CLUTCH DISC

REMOVAL

(1) Remove transmission.
(2) Mark position of pressure plate on flywheel for installation reference, if original pressure plate will be reused.
(3) Remove pressure plate bolts and remove cover and disc (Fig. 1).

CAUTION: If original pressure plate will be reused, loosen cover bolts evenly in rotation to relieve spring tension equally. Failure to heed caution may result in damage.

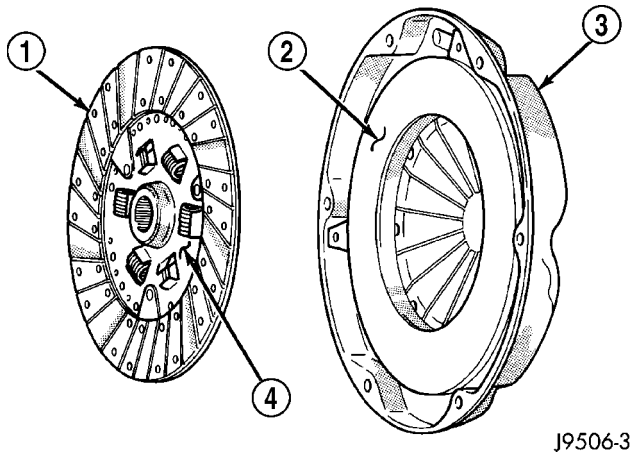

Fig. 1 CLUTCH DISC AND PRESSURE PLATE

1 - DISC
2 - PRESSURE PLATE
3 - PRESSURE PLATE COVER
4 - "FLYWHEEL SIDE" STAMPED ON THIS SURFACE

INSTALLATION

(1) Lightly sand flywheel face with 180 grit emery cloth. Then clean surface with a wax and grease remover.
(2) Lubricate pilot bearing with Mopar high temperature bearing grease.
(3) Position clutch disc on flywheel with side marked flywheel positioned against flywheel.

NOTE: If disc is not marked, flat side of disc hub goes towards the flywheel.

(4) Insert alignment tool through disc and into the pilot bearing on the flywheel (Fig. 2).
(5) Position pressure plate over disc and on flywheel (Fig. 2).

NOTE: Align reference marks if pressure plate is reused.

(6) Install clutch cover bolts finger tight.

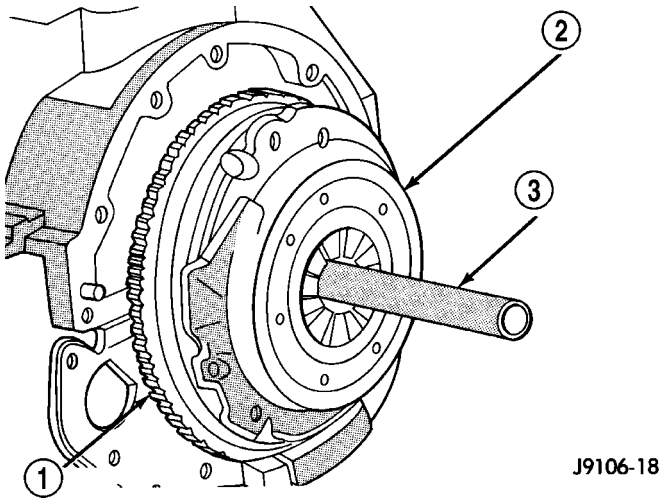

Fig. 2 ALIGNING CLUTCH DISC

1 - FLYWHEEL
2 - PRESSURE PLATE AND DISC
3 - ALIGNMENT TOOL

(7) Tighten cover bolts evenly and in rotation a few threads at a time. Tightening bolts to:
- 2.5L Engine - 31 N·m (23 ft. lbs.)
- 4.0L Engine - 50 N·m (37 ft. lbs.)

CAUTION: Cover bolts must be tightened evenly and to specified torque. Failure to heed caution may result in damage.

(8) Lightly coat clutch disc hub and transmission input shaft splines with Mopar high temperature bearing grease or equivalent.

CAUTION: Over lubricating shaft splines will result in grease contamination of disc. Failure to heed caution may result in damage.

(9) Install transmission.

CLUTCH RELEASE BEARING

REMOVAL

(1) Remove transmission.
(2) Disconnect release bearing from release lever and remove bearing (Fig. 3).
(3) Inspect bearing slide surface of transmission front bearing retainer. Replace retainer if slide surface is scored, worn or cracked.
(4) Inspect release fork and fork pivot. Verify pivot is secure and in good condition, fork is not distorted or worn and retainer spring is not bent or damaged.

INSTALLATION

(1) Lubricate crankshaft pilot bearing, input shaft splines, bearing retainer slide surface, fork pivot and

CLUTCH RELEASE BEARING (Continued)

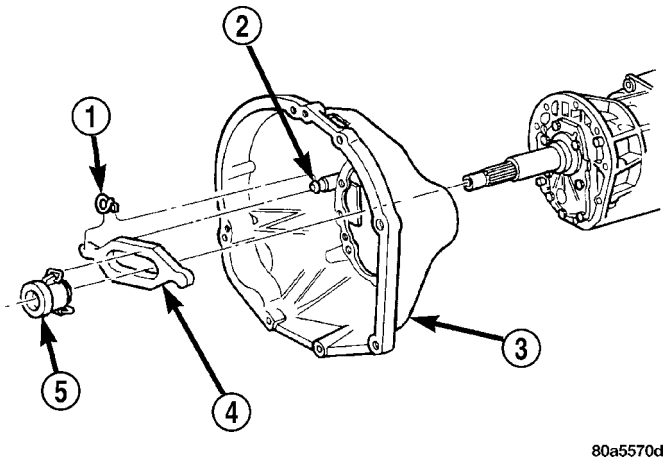

Fig. 3 RELEASE BEARING

1 - RETAINER SPRING
2 - PIVOT BALL STUD
3 - CLUTCH HOUSING
4 - RELEASE FORK
5 - RELEASE BEARING

release fork pivot surface. Lubricate with Mopar high temperature bearing grease or equivalent.

(2) Install new release bearing. Verify bearing is properly secured to release fork.

(3) Install transmission.

FLYWHEEL

DESCRIPTION

STANDARD FLYWHEEL

The standard flywheel is used on the 4.0L engine. The flywheel (Fig. 4) is a heavy plate bolted to the rear of the crankshaft. The flywheel incorporates the ring gear to mesh with the starter. The rear face of the flywheel serves as the driving member to the clutch disc.

DUAL MASS FLYWHEEL

The Dual Mass Flywheel is used on the 2.4L engine (Fig. 5). The flywheel incorporates the ring gear to mesh with the starter. The primary flywheel side is bolted to the crankshaft. The secondary flywheel face serves as the driving member to the clutch disc. Internal springs between the flywheels are use to dampen energy.

OPERATION

The flywheel serves to dampen the engine firing pulses. The heavy weight of the flywheel relative to the rotating mass of the engine components serves to stabilize the flow of power to the remainder of the drivetrain. The crankshaft has the tendency to attempt to speed up and slow down in response to

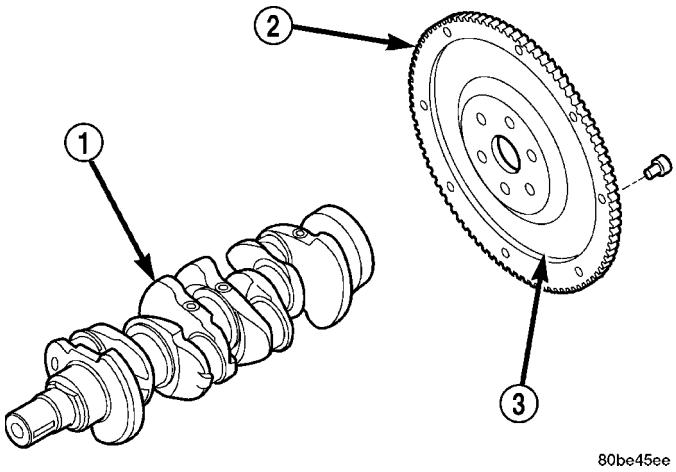

Fig. 4 FLYWHEEL

1 - CRANKSHAFT
2 - RING GEAR
3 - FLYWHEEL

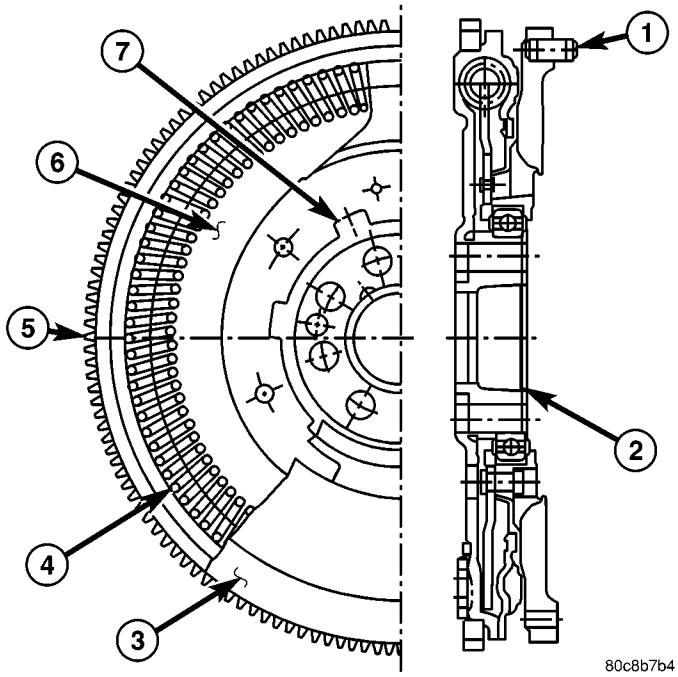

Fig. 5 DUAL MASS FLYWHEEL

1 - LOCATING STUD
2 - BEARING
3 - SECONDARY FLYWHEEL
4 - DAMPER SPRING
5 - RING GEAR
6 - PRIMARY FLYWHEEL
7 - FRICTION DISC

the cylinder firing pulses. The flywheel dampens these impulses by absorbing energy when the crankshaft speeds and releasing the energy back into the system when the crankshaft slows down.

On a Dual Mass Flywheel the additional secondary mass coupled to the transmission lowers the natural frequency of the transmission rotating elements. This

FLYWHEEL (Continued)

decreases the transmission gear rattle. The damper springs between the two flywheel masses replace the clutch disc damper springs and assist in a smooth transfer of torque to the transmission.

CAUTION: The Dual Mass Flywheel is serviced as an assembly only and should never be taken apart. Failure to heed caution may result in damage.

DIAGNOSIS AND TESTING

Check flywheel runout whenever misalignment is suspected. Flywheel runout should not exceed 0.08 mm (0.003 in.). Measure runout at the outer edge of the flywheel face with a dial indicator. Mount the indicator on a stud installed in place of one of the flywheel bolts.

Common causes of runout are:

- heat warpage
- improper machining
- incorrect bolt tightening
- improper seating on crankshaft flange shoulder
- foreign material on crankshaft flange

Flywheel machining is not recommended. The flywheel clutch surface is machined to a unique contour and machining will negate this feature. Minor flywheel scoring can be cleaned up by hand with 180 grit emery or with surface grinding equipment. Remove only enough material to reduce scoring (approximately 0.001 - 0.003 in.). Heavy stock removal is **not recommended.** Replace the flywheel if scoring is severe and deeper than 0.076 mm (0.003 in.). Excessive stock removal can result in flywheel cracking or warpage after installation; it can also weaken the flywheel and interfere with proper clutch release.

Clean the crankshaft flange before mounting the flywheel. Dirt and grease on the flange surface may cock the flywheel causing excessive runout. Use new bolts when remounting a flywheel and secure the bolts with Mopar Lock And Seal or equivalent. Tighten flywheel bolts to specified torque only. Overtightening can distort the flywheel hub causing runout.

PILOT BEARING

REMOVAL

(1) Remove transmission.
(2) Remove pressure plate and clutch disc.
(3) Remove pilot bearing with an internal (blind hole) puller.

INSTALLATION

(1) Lubricate new bearing with Mopar high temperature bearing grease or equivalent.
(2) Start new bearing into crankshaft by hand. Then seat bearing with clutch alignment tool (Fig. 6).

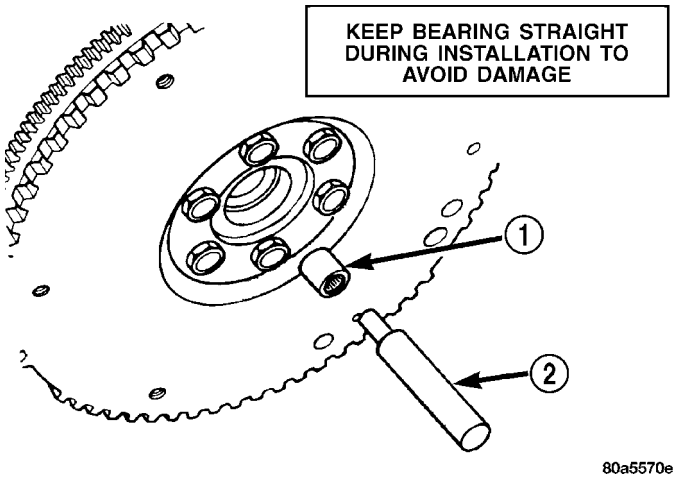

Fig. 6 PILOT BEARING INSTALLER

1 - PILOT BEARING
2 - ALIGNMENT TOOL

(3) Lightly scuff sand flywheel surface with 180 grit emery cloth. Then clean surface with wax and grease remover.
(4) Install clutch disc and pressure plate.
(5) Install transmission.

LINKAGE

REMOVAL

NOTE: Clutch master cylinder, slave cylinder and connecting line are serviced as an assembly only. Components cannot be overhauled or serviced separately. Cylinders and connecting line are sealed units.

Removal/installation procedures for right and left hand drive models are basically the same.
(1) With vehicle in neutral, position vehicle on hoist
(2) Remove fasteners attaching slave cylinder to clutch housing.
(3) Remove slave cylinder from clutch housing (Fig. 7).
(4) Disengage clutch fluid line from body clips.
(5) Lower vehicle.
(6) Verify clutch master cylinder reservoir cap is tight.
(7) Remove clutch master cylinder attaching nuts (Fig. 7) or (Fig. 8).
(8) Disengage captured bushing on clutch master cylinder actuator from pivot pin on pedal arm.
(9) Slide actuator off pivot pin.

LINKAGE (Continued)

(10) Disconnect clutch interlock safety switch wires.

(11) Remove clutch hydraulic linkage through engine compartment.

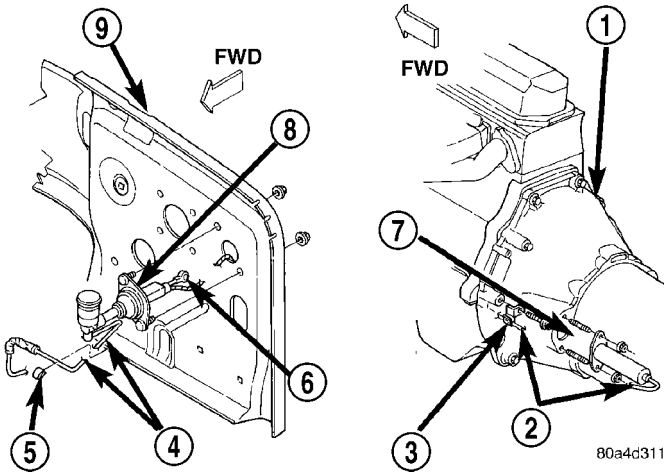

Fig. 7 CLUTCH LINKAGE - LHD

1 - CLUTCH HOUSING
2 - FLUID LINE
3 - BRACKET
4 - FLUID LINE
5 - CLIP
6 - CAPTURED BUSHING
7 - CLUTCH SLAVE CYLINDER
8 - CLUTCH MASTER CYLINDER
9 - DASH PANEL

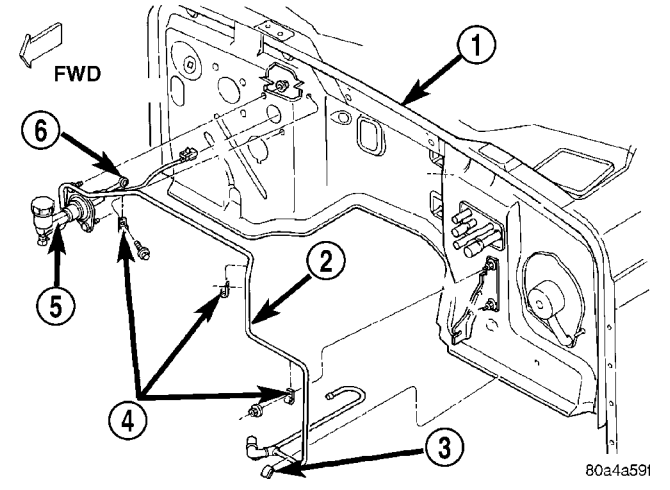

Fig. 8 CLUTCH LINKAGE - RHD

1 - DASH PANEL
2 - FLUID LINE
3 - CLIP
4 - HOLD DOWN STRAP
5 - CLUTCH MASTER CYLINDER
6 - CAPTURED BUSHING

INSTALLATION

NOTE: Clutch master cylinder, slave cylinder and connecting line are serviced as an assembly only. Components cannot be overhauled or serviced separately. Cylinders and connecting line are sealed units.

Removal/installation procedures for right and left hand drive models are basically the same.

(1) Tighten clutch master cylinder cap.

(2) Position connecting line and slave cylinder downward past engine and next to clutch housing.

(3) Position clutch master cylinder on dash panel.

(4) Install clutch master cylinder actuator on clutch pedal pivot pin.

(5) Install clutch master cylinder and tighten nuts to 38 N·m (28 ft. lbs.).

(6) Insert slave cylinder push rod through clutch housing opening and into release lever. Be sure cap on end of rod is securely engaged in lever. Check this before installing cylinder attaching nuts.

(7) Install slave cylinder and tighten nuts to 23 N·m (17 ft. lbs.).

(8) Secure clutch fluid line in body and transmission clips.

(9) Connect clutch interlock safety switch wires.

MASTER CYLINDER

INSPECTION

The clutch fluid reservoir, master cylinder, slave cylinder and fluid lines are pre-filled at the factory. The hydraulic system should not require additional fluid under normal circumstances. The reservoir fluid level increases as clutch wear occurs.

CAUTION: Do not overfill or removing fluid from the reservoir. Failure to heed caution may result in damage.

Wipe reservoir and cover clean. Remove cap and diaphragm, fluid level should not be above indicator ring located on the outside of the reservoir.

CLUTCH PEDAL

REMOVAL

(1) Remove steering column lower cover and knee blocker for access.

(2) Disconnect clutch pedal position switch wires.

(3) Disengage captured bushing lock tabs attaching clutch master cylinder actuator to pedal pivot (Fig. 9) or (Fig. 10).

(4) Remove nuts attaching pedal and bracket to dash panel and upper cowl support (Fig. 9) or (Fig. 10).

(5) Separate pedal assemble from vehicle.

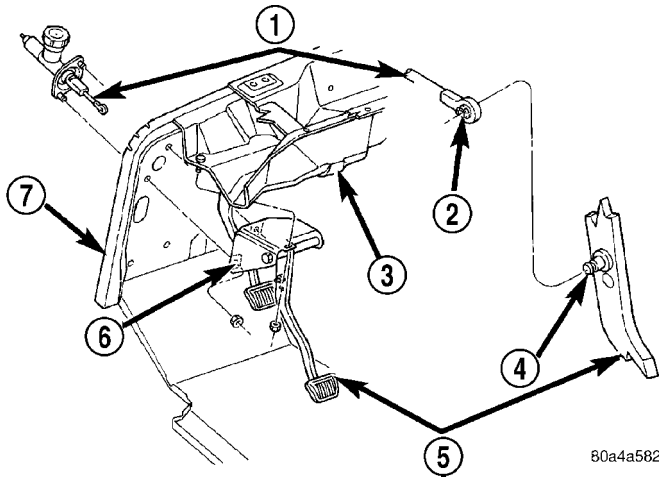

Fig. 9 CLUTCH PEDAL MOUNTING - LHD

1 - CLUTCH CYLINDER ACTUATOR
2 - CAPTURED BUSHING
3 - UPPER COWL SUPPORT
4 - PIVOT
5 - CLUTCH PEDAL
6 - BRACKET
7 - DASH PANEL

INSTALLATION

(1) Place clutch pedal and bracket over studs on dash panel and cowl support. Install nuts and tighten to 39 N·m (29 ft. lbs.).

(2) Engage captured bushing and actuator on brake pedal pivot.

(3) Connect clutch pedal position switch wires.

CLUTCH PEDAL POSITION SWITCH

DESCRIPTION

The Clutch Pedal Position Switch (CPPS) is located under the instrument panel, attached to the clutch master cylinder push rod (Fig. 11). The wiring harness connector is inside of the vehicle under the left side of the instrument panel (Fig. 12).

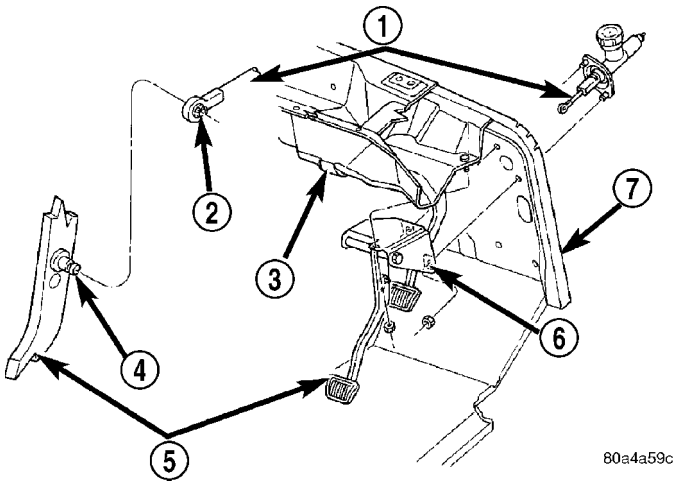

Fig. 10 CLUTCH PEDAL MOUNTING - RHD

1 - CLUTCH CYLINDER ACTUATOR
2 - CAPTURED BUSHING
3 - UPPER COWL SUPPORT
4 - PIVOT
5 - CLUTCH PEDAL
6 - BRACKET
7 - DASH PANEL

NOTE: Switch is serviced with clutch master cylinder only.

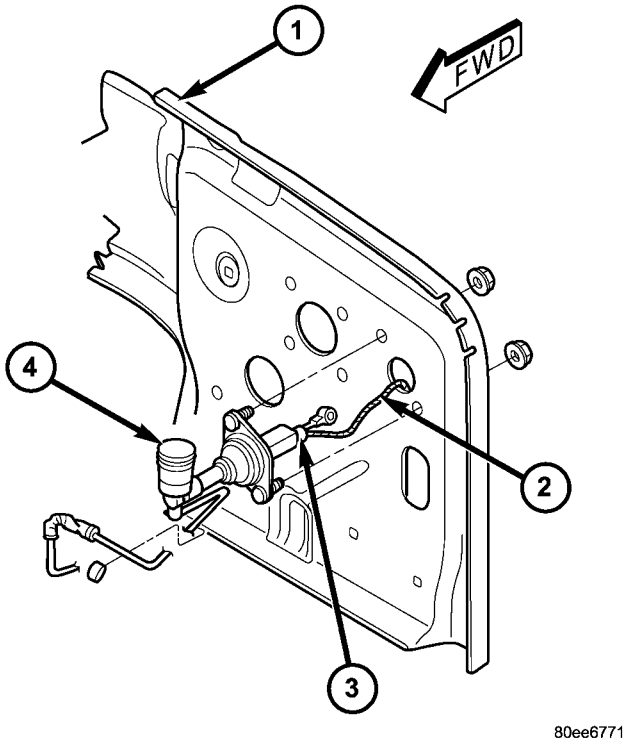

Fig. 11 CLUTCH PEDAL POSITION SWITCH (CPPS)

1 - INSTRUMENT PANEL
2 - SWITCH WIRING HARNESS
3 - CLUTCH PEDAL POSITION SWITCH
4 - CLUTCH MASTER CYLINDER

CLUTCH PEDAL POSITION SWITCH (Continued)

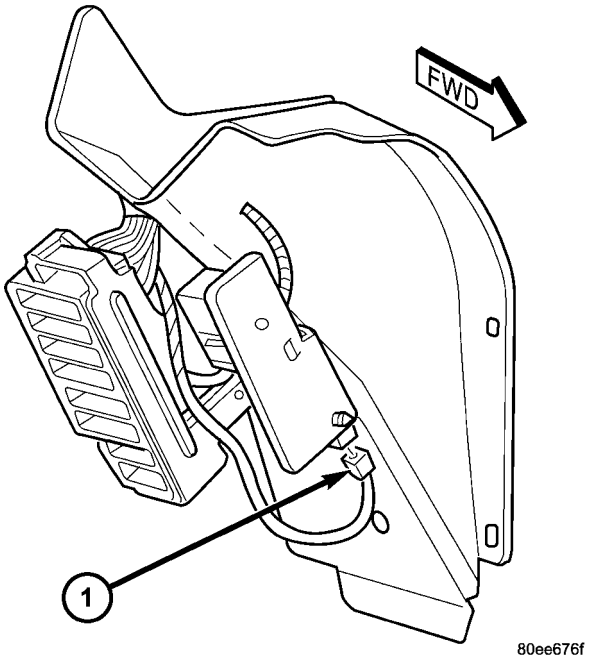

Fig. 12 CPPS ELECTRICAL CONNECTION

1 - CPPS CONNECTION TO MAIN HARNESS

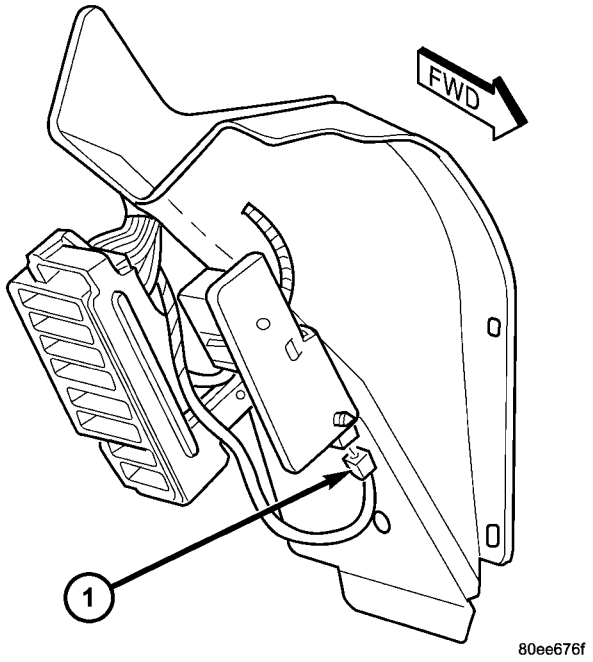

Fig. 13 CPPS ELECTRICAL CONNECTION

1 - CPPS CONNECTION TO MAIN HARNESS

OPERATION

The clutch pedal position switch is used to prevent starter motor engagement unless the clutch pedal is depressed. An input from this switch is also used to either shut down/prevent operation of the speed control system when pedal is depressed.

DIAGNOSIS AND TESTING

(1) Disconnect 2-wire switch connector under instrument panel (Fig. 13).

(2) Check switch continuity with an ohmmeter while operating clutch pedal.

- Pedal Depressed - Continuity
- Pedal Released - No Continuity

(3) If continuity is not present or always present, replace clutch master cylinder. Switch is not serviced separately.

COOLING

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COOLING

DESCRIPTION

DESCRIPTION - COOLING SYSTEM 2.4L ENGINE

The cooling system consists of the following items:

- Electric cooling fan - Standard.
- Radiator
- Hot bottle pressure cap
- Thermostat
- Coolant reserve/overflow system
- Radiator in-tank transmission oil cooler (if equipped with an automatic transmission)
- Coolant
- Water pump
- Hoses and hose clamps

DESCRIPTION - 4.0L ENGINE

The cooling system (Fig. 1) is designed to maintain engine temperature at an efficient level during all engine operating conditions.

The components of the cooling system are:

- A heavy duty radiator
- Cooling fan (mechanical)
- Thermal viscous fan drive
- Fan shroud
- Radiator pressure cap

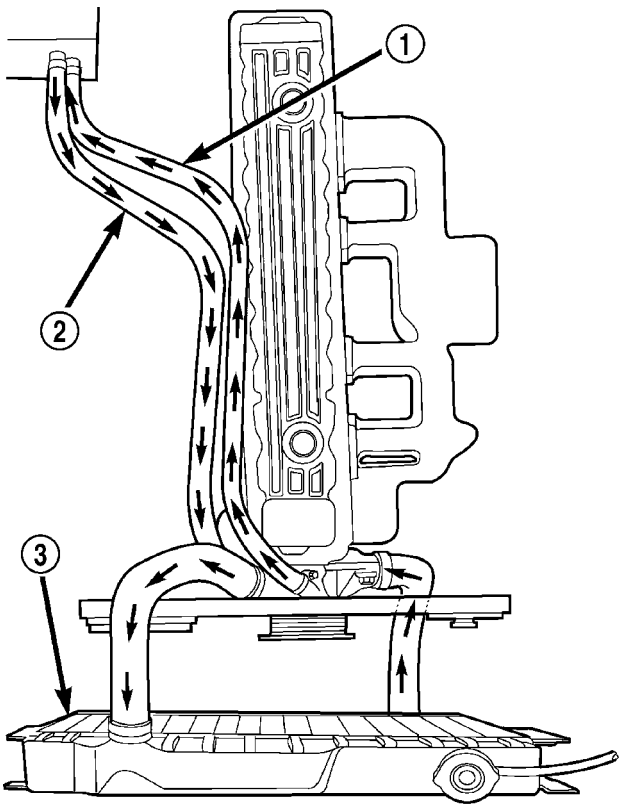


Fig. 1 Coolant Circulation - 4.0L Engine

1 - FROM THERMOSTAT
2 - TO WATER PUMP
3 - RADIATOR

COOLING (Continued)

- Thermostat
- Coolant reserve/overflow system
- Automatic transmission oil cooler (internal to radiator)
- Coolant
- Water pump
- Coolant hoses and clamps

DESCRIPTION - HOSE CLAMPS

The cooling system utilizes both worm drive and spring type hose clamps. If a spring type clamp replacement is necessary, replace with the original Mopar® equipment spring type clamp.

CAUTION: A number or letter is stamped into the tongue of constant tension clamps. If replacement is necessary, use only a original equipment clamp with matching number or letter (Fig. 2).

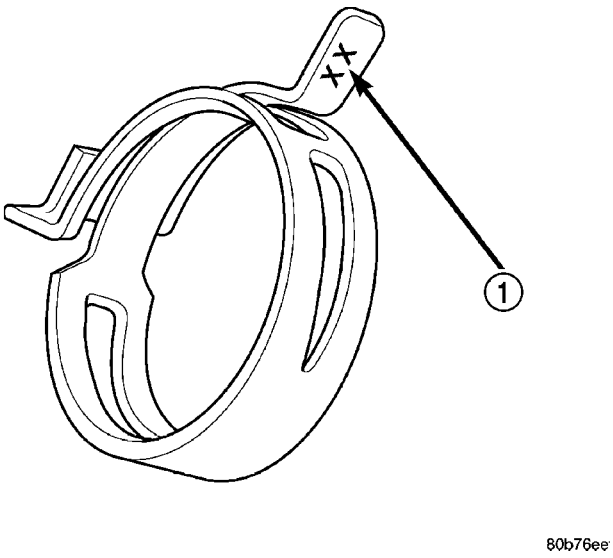


Fig. 2 Spring Clamp Size Location

1 - SPRING CLAMP SIZE LOCATION

OPERATION - HOSE CLAMPS

The worm type hose clamp uses a specified torque value to maintain proper tension on a hose connection.

The spring type hose clamp applies constant tension on a hose connection. To remove a spring type hose clamp, use Special Tool 6094 or equivalent, constant tension clamp pliers (Fig. 3) to compress the hose clamp.

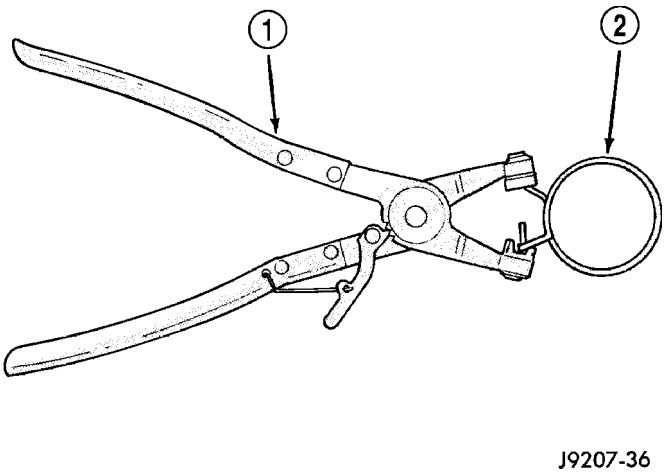


Fig. 3 Hose Clamp Tool

1 - HOSE CLAMP TOOL 6094
2 - HOSE CLAMP

DIAGNOSIS AND TESTING

ON-BOARD DIAGNOSTICS - OBD

COOLING SYSTEM RELATED DIAGNOSTICS

The Powertrain Control Module (PCM) has been programmed to monitor the certain following cooling system components:

- If the engine has remained cool for too long a period, such as with a stuck open thermostat, a Diagnostic Trouble Code (DTC) can be set.
- If an open or shorted condition has developed in the relay circuit controlling the electric radiator fan, a Diagnostic Trouble Code (DTC) can be set.

If the problem is sensed in a monitored circuit often enough to indicate an actual problem, a DTC is stored. The DTC will be stored in the PCM memory for eventual display to the service technician.

ACCESSING DIAGNOSTIC TROUBLE CODES

To read DTC's and to obtain cooling system data, (Refer to 25 - EMISSIONS CONTROL - DESCRIPTION).

DIAGNOSIS AND TESTING - PRELIMINARY CHECKS

ENGINE COOLING SYSTEM OVERHEATING

Establish what driving conditions caused the complaint. Abnormal loads on the cooling system such as the following may be the cause:

- Prolonged idle
- Very high ambient temperature
- Slight tail wind at idle
- Slow traffic
- Traffic jams
- High speed or steep grades

Driving techniques that avoid overheating are:

COOLING (Continued)

- Idle with A/C off when temperature gauge is at end of normal range.
- Increasing engine speed for more air flow is recommended.

TRAILER TOWING:

Consult Trailer Towing section of owners manual. Do not exceed limits.

AIR CONDITIONING; ADD-ON OR AFTER MARKET:

A maximum cooling package should have been ordered with vehicle if add-on or after market A/C is installed. If not, maximum cooling system components should be installed for model involved per manufacturer's specifications.

RECENT SERVICE OR ACCIDENT REPAIR:

Determine if any recent service has been performed on vehicle that may effect cooling system. This may be:

- Engine adjustments (incorrect timing)
- Slipping engine accessory drive belt(s)
- Brakes (possibly dragging)
- Changed parts. Incorrect water pump or pump rotating in wrong direction due to belt not correctly routed
- Reconditioned radiator or cooling system refilling (possibly under filled or air trapped in system).

NOTE: If investigation reveals none of the previous items as a cause for an engine overheating complaint, refer to COOLING SYSTEM DIAGNOSIS CHART BELOW.

These charts are to be used as a quick-reference only. Refer to COOLING SYSTEM DIAGNOSIS CHART

COOLING SYSTEM DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
TEMPERATURE GAUGE READS LOW	1. Has a Diagnostic Trouble Code (DTC) been set indicating a stuck open thermostat?	1. (Refer to 25 - EMISSIONS CONTROL - DESCRIPTION) for On-Board Diagnostics and DTC information. Replace thermostat if necessary.
	2. Is the temperature sending unit connected?	2. Check the temperature sensor connector. (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - SCHEMATIC - ELECTRICAL) Repair connector if necessary.
	3. Is the temperature gauge operating OK?	3. Check gauge operation. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER/ENGINE TEMPERATURE GAUGE - DESCRIPTION). Repair as necessary.
	4. Coolant level low in cold ambient temperatures accompanied with poor heater performance.	4. Check coolant level in the coolant reserve/overflow tank or degas bottle and the radiator. Inspect system for leaks. Repair leaks as necessary. Refer to the Coolant section of the manual text for WARNINGS and CAUTIONS associated with removing the radiator cap.
	5. Improper operation of internal heater doors or heater controls.	5. Inspect heater and repair as necessary. (Refer to 24 - HEATING & AIR CONDITIONING - DIAGNOSIS AND TESTING) for procedures.

COOLING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
TEMPERATURE GAUGE READS HIGH OR THE COOLANT WARNING LAMP ILLUMINATES. COOLANT MAY OR MAY NOT BE LOST OR LEAKING FROM THE COOLING SYSTEM	1. Trailer is being towed, a steep hill is being climbed, vehicle is operated in slow moving traffic, or engine is being idled with very high ambient (outside) temperatures and the air conditioning is on. Higher altitudes could aggravate these conditions.	1. This may be a temporary condition and repair is not necessary. Turn off the air conditioning and attempt to drive the vehicle without any of the previous conditions. Observe the temperature gauge. The gauge should return to the normal range. If the gauge does not return to the normal range, determine the cause for overheating and repair. Refer to Possible Causes (2-18).
	2. Is the temperature gauge reading correctly?	2. Check gauge. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - SCHEMATIC - ELECTRICAL). Repair as necessary.
	3. Is the temperature warning illuminating unnecessarily?	3. (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - SCHEMATIC - ELECTRICAL).
	4. Coolant low in coolant reserve/ overflow tank and radiator?	4. Check for coolant leaks and repair as necessary. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).
	5. Pressure cap not installed tightly. If cap is loose, boiling point of coolant will be lowered. Also refer to the following Step 6.	5. Tighten cap
	6. Poor seals at the radiator cap.	6. (a) Check condition of cap and cap seals. Refer to Radiator Cap. Replace cap if necessary. (b) Check condition of radiator filler neck or degas bottle. If neck is bent or damaged, replace radiator or degas bottle.
	7. Coolant level low in radiator but not in coolant reserve/overflow tank. This means the radiator is not drawing coolant from the coolant reserve/overflow tank as the engine cools	7. (a) Check condition of radiator cap and cap seals. Refer to Radiator Cap in this Group. Replace cap if necessary. (b) Check condition of radiator filler neck. If neck is bent or damaged, replace radiator. (c) Check condition of the hose from the radiator to the coolant tank. It should fit tight at both ends without any kinks or tears. Replace hose if necessary. (d) Check coolant reserve/overflow tank and tanks hoses for blockage. Repair as necessary.
	8. Incorrect coolant concentration	8. Check coolant. (Refer to LUBRICATION & MAINTENANCE/ FLUID TYPES - DESCRIPTION).

COOLING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
	9. Coolant not flowing through system	9. Check for coolant flow at radiator filler neck with some coolant removed, engine warm and thermostat open. Coolant should be observed flowing through radiator. If flow is not observed, determine area of obstruction and repair as necessary.
	10. Radiator or A/C condenser fins are dirty or clogged.	10. Remove insects and debris. (Refer to 7 - COOLING - STANDARD PROCEDURE).
	11. Radiator core is corroded or plugged.	11. Have radiator re-cored or replaced.
	12. Fuel or ignition system problems.	12. Refer to 14 - Fuel System or 8 - Electrical for diagnosis and testing procedures.
	13. Dragging brakes.	13. Check and correct as necessary. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING) for correct procedures.
	14. Bug screen or cardboard is being, reducing airflow.	14. Remove bug screen or cardboard.
	15. Thermostat partially or completely shut.	15. Check thermostat operation and replace as necessary. (Refer to 7 - COOLING/ENGINE/ENGINE COOLANT THERMOSTAT - REMOVAL).
	16. Viscous fan drive not operating properly.	16. Check fan drive operation and replace as necessary. (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL) .
	17. Cylinder head gasket leaking.	17. Check for cylinder head gasket leaks. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).
	18. Heater core leaking.	18. Check heater core for leaks. (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - DIAGNOSIS AND TESTING). Repair as necessary.

COOLING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
TEMPERATURE GAUGE READING IS INCONSISTANT (FLUCTUATES, CYCLES, OR IS ERRATIC	1. During cold weather operation, with the heater in the high position, the gauge reading may drop slightly.	1. A normal condition. No correction is necessary.
	2. Temperature gauge or engine mounted gauge sensor defective or shorted. Also, corroded or loose wiring in this circuit.	2. Check operation of gauge and repair if necessary. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).
	3. Gauge reading rises when vehicle is brought to a stop after heavy use (engine still running)	3. A normal condition. No correction is necessary. Gauge should return to normal range after vehicle is driven.
	4. Gauge reading high after re-starting a warmed up (hot) engine.	4. A normal condition. No correction is necessary. The gauge should return to normal range after a few minutes of engine operation.
	5. Coolant level low in radiator (air will build up in the cooling system causing the thermostat to open late).	5. Check and correct coolant leaks. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).
	6. Cylinder head gasket leaking allowing exhaust gas to enter cooling system causing a thermostat to open late.	6. (a) Check for cylinder head gasket leaks. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING). (b) Check for coolant in the engine oil. Inspect for white steam emitting from the exhaust system. Repair as necessary.
	7. Water pump impeller loose on shaft.	7. Check water pump and replace as necessary. (Refer to 7 - COOLING/ ENGINE/WATER PUMP - REMOVAL).
	8. Loose accessory drive belt. (water pump slipping)	8. (Refer to 7 - COOLING/ ACCESSORY DRIVE/DRIVE BELTS - DIAGNOSIS AND TESTING). Check and correct as necessary.
	9. Air leak on the suction side of the water pump allows air to build up in cooling system causing thermostat to open late.	9. Locate leak and repair as necessary.
PRESSURE CAP IS BLOWING OFF STEAM AND/OR COOLANT TO COOLANT TANK. TEMPERATURE GAUGE READING MAY BE ABOVE NORMAL BUT NOT HIGH. COOLANT LEVEL MAY BE HIGH IN COOLANT RESERVE/OVERFLOW TANK	1. Pressure relief valve in radiator cap is defective.	1. Check condition of radiator cap and cap seals. (Refer to 7 - COOLING/ ENGINE/RADIATOR PRESSURE CAP - DIAGNOSIS AND TESTING). Replace cap as necessary.
COOLANT LOSS TO THE GROUND WITHOUT PRESSURE CAP BLOWOFF. GAUGE READING HIGH OR HOT	1. Coolant leaks in radiator, cooling system hoses, water pump or engine.	1. Pressure test and repair as necessary. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).

COOLING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
DETONATION OR PRE-IGNITION (NOT CAUSED BY IGNITION SYSTEM). GAUGE MAY OR MAY NOT BE READING HIGH	1. Engine overheating.	1. Check reason for overheating and repair as necessary.
	2. Freeze point of coolant not correct. Mixture is too rich or too lean.	2. Check coolant concentration. (Refer to LUBRICATION & MAINTENANCE/FLUID TYPES - DESCRIPTION).
HOSE OR HOSES COLLAPSE WHILE ENGINE IS RUNNING	1. Vacuum created in cooling system on engine cool-down is not being relieved through coolant reserve/ overflow system.	1. (a) Radiator cap relief valve stuck. (Refer to 7 - COOLING/ENGINE/ RADIATOR PRESSURE CAP - DIAGNOSIS AND TESTING). Replace if necessary (b) Hose between coolant reserve/ overflow tank and radiator is kinked. Repair as necessary. (c) Vent at coolant reserve/overflow tank is plugged. Clean vent and repair as necessary. (d) Reserve/overflow tank is internally blocked or plugged. Check for blockage and repair as necessary.
NOISY VISCOUS FAN/DRIVE	1. Fan blades loose.	1. Replace fan blade assembly. (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - REMOVAL)
	2. Fan blades striking a surrounding object.	2. Locate point of fan blade contact and repair as necessary.
	3. Air obstructions at radiator or air conditioning condenser.	3. Remove obstructions and/or clean debris or insects from radiator or A/C condenser.
	4. Thermal viscous fan drive has defective bearing.	4. Replace fan drive. Bearing is not serviceable. (Refer to 7 - COOLING/ ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL).
	5. A certain amount of fan noise may be evident on models equipped with a thermal viscous fan drive. Some of this noise is normal.	5. (Refer to 7 - COOLING/ENGINE/ FAN DRIVE VISCOUS CLUTCH - DESCRIPTION) for an explanation of normal fan noise.

COOLING (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
INADEQUATE HEATER PERFORMANCE. THERMOSTAT FAILED IN OPEN POSITION	1. Has a Diagnostic trouble Code (DTC) been set?	1. (Refer to 25 - EMISSIONS CONTROL - DESCRIPTION) for correct procedures and replace thermostat if necessary
	2. Coolant level low	2. (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).
	3. Obstructions in heater hose/fittings	3. Remove heater hoses at both ends and check for obstructions
	4. Heater hose kinked	4. Locate kinked area and repair as necessary
	5. Water pump is not pumping water to/through the heater core. When the engine is fully warmed up, both heater hoses should be hot to the touch. If only one of the hoses is hot, the water pump may not be operating correctly or the heater core may be plugged. Accessory drive belt may be slipping causing poor water pump operation.	5. (Refer to 7 - COOLING/ENGINE/ WATER PUMP - REMOVAL). If a slipping belt is detected, (Refer to 7 - COOLING/ACCESSORY DRIVE/ DRIVE BELTS - DIAGNOSIS AND TESTING). If heater core obstruction is detected, (Refer to 24 - HEATING & AIR CONDITIONING/ PLUMBING/HEATER CORE - REMOVAL).
STEAM IS COMING FROM THE FRONT OF VEHICLE NEAR THE GRILL AREA WHEN WEATHER IS WET, ENGINE IS WARMED UP AND RUNNING, AND VEHICLE IS STATIONARY. TEMPERATURE GAUGE IS IN NORMAL RANGE	1. During wet weather, moisture (snow, ice or rain condensation) on the radiator will evaporate when the thermostat opens. This opening allows heated water into the radiator. When the moisture contacts the hot radiator, steam may be emitted. This usually occurs in cold weather with no fan or airflow to blow it away.	1. Occasional steam emitting from this area is normal. No repair is necessary.
COOLANT COLOR	1. Coolant color is not necessarily an indication of adequate corrosion or temperature protection. Do not rely on coolant color for determining condition of coolant.	1. (Refer to LUBRICATION & MAINTENANCE/FLUID TYPES - DESCRIPTION). Adjust coolant mixture as necessary.
COOLANT LEVEL CHANGES IN COOLANT RESERVE/OVERFLOW TANK. TEMPERATURE GAUGE IS IN NORMAL RANGE	1. Level changes are to be expected as coolant volume fluctuates with engine temperature. If the level in the tank was between the FULL and ADD marks at normal operating temperature, the level should return to within that range after operation at elevated temperatures.	1. A normal condition. No repair is necessary.

COOLING (Continued)

DIAGNOSIS AND TESTING - COOLING SYSTEM - TESTING FOR LEAKS

ULTRAVIOLET LIGHT METHOD

All Jeep models have a leak detection additive added to the cooling system before they leave the factory. The additive is highly visible under ultraviolet light (black light). If the factory original coolant has been drained, pour one ounce of additive into the cooling system. The additive is available through the parts department. Place the heater control unit in HEAT position. Start and operate the engine until the radiator upper hose is warm to the touch. Aim the commercially available black light tool at the components to be checked. If leaks are present, the black light will cause the additive to glow a bright green color.

The black light can be used along with a pressure tester to determine if any external leaks exist (Fig. 4).

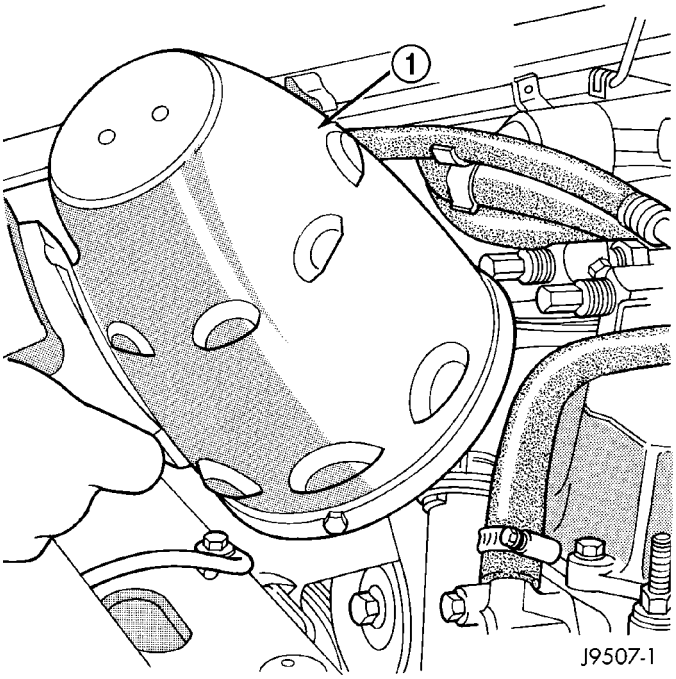


Fig. 4 Leak Detection Using Black Light—Typical

1 - TYPICAL BLACK LIGHT TOOL

PRESSURE TESTER METHOD

The engine should be at the normal operating temperature. Recheck the system cold if the cause of coolant loss is not located during warm engine examination.

WARNING: HOT, PRESSURIZED COOLANT CAN CAUSE INJURY BY SCALDING.

Carefully remove the radiator pressure cap from the filler neck and check the coolant level. Push down on the cap to disengage it from the stop tabs. Wipe the inner part of the filler neck and examine the lower inside sealing seat for nicks, cracks, paint, dirt and solder residue. Inspect the reserve/overflow tank tube for internal obstructions. Insert a wire through the tube to be sure it is not obstructed.

Inspect the cams on the outside part of the filler neck. If the cams are bent, seating of pressure cap valve and tester seal will be affected. Replace cap if cams are bent.

Attach pressure tester 7700 (or an equivalent) to the radiator filler neck (Fig. 5).

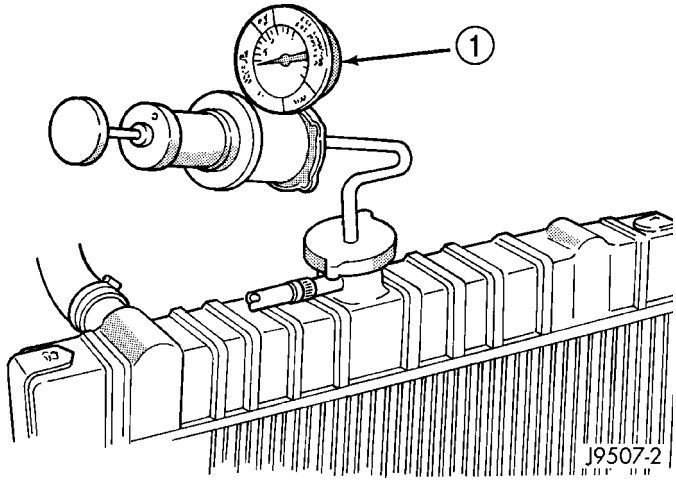


Fig. 5 Pressurizing System - Typical

1 - TYPICAL COOLING SYSTEM PRESSURE TESTER

Operate the tester pump to apply 124 kPa (18 psi) pressure to the system. If the hoses enlarge excessively or bulge while testing, replace as necessary. Observe the gauge pointer and determine the condition of the cooling system according to the following criteria:

- Holds Steady: If the pointer remains steady for two minutes, there are no serious coolant leaks in the system. However, there could be an internal leak that does not appear with normal system test pressure. Inspect for interior leakage or do the Internal Leakage Test. Do this if it is certain that coolant is being lost and no leaks can be detected.
- Drops Slowly: Shows a small leak or seepage is occurring. Examine all connections for seepage or slight leakage with a flashlight. Inspect the radiator, hoses, gasket edges and heater. Seal any small leak holes with a Sealer Lubricant or equivalent. Repair leak holes and reinspect the system with pressure applied.
- Drops Quickly: Shows that a serious leakage is occurring. Examine the system for serious external leakage. If no leaks are visible, inspect for internal

COOLING (Continued)

leakage. Large radiator leak holes should be repaired by a reputable radiator repair shop.

INTERNAL LEAKAGE INSPECTION

Remove the engine oil pan drain plug and drain a small amount of engine oil. Coolant, being heavier than engine oil, will drain first. Another way of testing is to operate the engine and check for water globules on the engine oil dipstick. Also inspect the automatic transmission oil dipstick for water globules. Inspect the automatic transmission fluid cooler for leakage. Operate the engine without the pressure cap on the radiator until thermostat opens.

Attach a pressure tester to the filler neck. If pressure builds up quickly, a leak exists as a result of a faulty cylinder head gasket or crack in the engine. Repair as necessary.

WARNING: DO NOT ALLOW PRESSURE TO EXCEED 124 KPA (18 PSI). TURN THE ENGINE OFF. TO RELEASE THE PRESSURE, ROCK THE TESTER FROM SIDE TO SIDE. WHEN REMOVING THE TESTER, DO NOT TURN THE TESTER MORE THAN 1/2 TURN IF THE SYSTEM IS UNDER PRESSURE.

If there is no immediate pressure increase, pump the pressure tester until the indicated pressure is within the system range. Vibration of the gauge pointer indicates compression or combustion leakage into the cooling system.

WARNING: DO NOT DISCONNECT THE SPARK PLUG WIRES WHILE THE ENGINE IS OPERATING.

CAUTION: Do not operate the engine with a spark plug shorted for more than a minute. The catalytic converter may be damaged.

Isolate the compression leak by shorting each spark plug to the cylinder block. The gauge pointer should stop or decrease vibration when spark plug for leaking cylinder is shorted. This happens because of the absence of combustion pressure.

COMBUSTION LEAKAGE TEST (WITHOUT PRESSURE TESTER)

DO NOT WASTE reusable coolant. If the solution is clean, drain the coolant into a clean container for reuse.

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS OR LOOSEN THE RADIATOR DRAINCOCK WITH THE SYSTEM HOT AND UNDER PRESSURE. SERIOUS BURNS FROM COOLANT CAN OCCUR.

Remove thermostat (Refer to 7 - COOLING/ENGINE/ENGINE COOLANT THERMOSTAT - REMOVAL).

Remove accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

Add coolant to the radiator to bring the level to within 6.3 mm (1/4 in) of the top of the thermostat housing.

CAUTION: Avoid overheating. Do not operate the engine for an excessive period of time. Open the draincock immediately after the test to eliminate boil over of coolant.

Start the engine and accelerate rapidly three times (to approximately 3000 rpm) while observing the coolant. If internal engine combustion gases are leaking into the cooling system, bubbles will appear in the coolant. If bubbles do not appear, there is no internal combustion gas leakage.

DIAGNOSIS AND TESTING - RADIATOR COOLANT FLOW CHECK

The following procedure will determine if coolant is flowing through the cooling system.

If engine is cold, idle engine until normal operating temperature is reached. Then feel the upper radiator hose. If hose is hot, the thermostat is open and water is circulating through cooling system.

STANDARD PROCEDURE

STANDARD PROCEDURE - DRAINING COOLING SYSTEM - 2.4L

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS (Fig. 6) OR LOOSEN THE RADIATOR DRAINCOCK WITH SYSTEM HOT AND UNDER PRESSURE. SERIOUS BURNS FROM COOLANT CAN OCCUR.

(1) DO NOT remove radiator cap first. With engine cold, raise vehicle on a hoist and locate radiator draincock.

NOTE: Radiator draincock is located on the left/lower side of radiator facing to rear of vehicle.

(2) Attach one end of a hose to the draincock. Put the other end into a clean container. Open draincock and drain coolant from radiator. This will empty the coolant reserve/overflow tank. The coolant does not have to be removed from the tank unless the system is being refilled with a fresh mixture. When tank is

empty, remove radiator cap and continue draining cooling system.

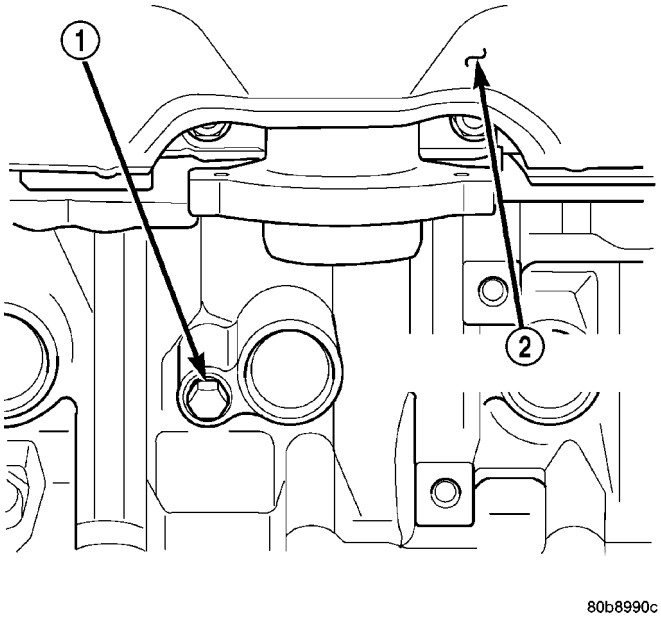


Fig. 6 Drain Plug - 3.7L Engine

1 - CYLINDER BLOCK DRAIN PLUG
2 - EXHAUST MANIFOLD AND HEAT SHIELD

STANDARD PROCEDURE - REFILLING COOLING SYSTEM - 2.4L

(1) Tighten the radiator draincock and the cylinder block drain plug(s) (if removed).

CAUTION: Failure to purge air from the cooling system can result in an overheating condition and severe engine damage.

(2) Fill system using a 50/50 mixture of ethylene-glycol antifreeze and low mineral content water. Fill pressure bottle to service line and install cap.

NOTE: The engine cooling system will push any remaining air into the coolant bottle within about an hour of normal driving. As a result, a drop in coolant level in the pressure bottle may occur. If the engine cooling system overheats and pushes coolant into the overflow side of the coolant bottle, this coolant will be sucked back into the cooling system ONLY IF THE PRESSURE CAP IS LEFT ON THE BOTTLE. Removing the pressure cap breaks the vacuum path between the two bottle sections and the coolant will not return to cooling system.

(3) With heater control unit in the HEAT position, operate engine with pressure bottle cap in place.

(4) Add coolant to pressure bottle as necessary. **Only add coolant to the pressure bottle when the engine is cold. Coolant level in a warm engine will be higher due to thermal expansion.**

NOTE: The coolant bottle has two chambers. Coolant will normally only be in the outboard (larger) of the two. The inboard chamber is only to recover coolant in the event of an overheat or after a recent service fill. The inboard chamber should normally be empty. If there is coolant in the overflow side of the coolant bottle (after several warm/cold cycles of the engine) and coolant level is above cold full when cold, disconnect the end of the overflow hose at the fill neck and lower it into a clean container. Allow coolant to drain into the container until emptied. Reconnect overflow hose to fill neck.

STANDARD PROCEDURE - COOLANT LEVEL CHECK

NOTE: Do not remove radiator cap for routine coolant level inspections. The coolant level can be checked at coolant reserve/overflow tank.

The coolant reserve/overflow system provides a quick visual method for determining coolant level without removing radiator pressure cap. With engine idling and at normal operating temperature, observe coolant level in reserve/overflow tank. The coolant level should be between ADD and FULL marks.

STANDARD PROCEDURE - COOLING SYSTEM - DRAINING

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS OR LOOSEN THE RADIATOR DRAINCOCK WITH SYSTEM HOT AND UNDER PRESSURE. SERIOUS BURNS FROM COOLANT CAN OCCUR.

DO NOT WASTE reusable coolant. If the solution is clean, drain the coolant into a clean container for reuse.

DO NOT remove the radiator cap when draining the coolant from the reserve/overflow tank. Open the radiator draincock and when the tank is empty, remove the radiator cap. The coolant does not have to be removed from the tank unless the system is being refilled with a fresh mixture.

(1) Drain the coolant from the radiator by loosening the draincock.

(2) Drain coolant from engine block by removing drain plug at left rear side of block (Fig. 7).

COOLING (Continued)

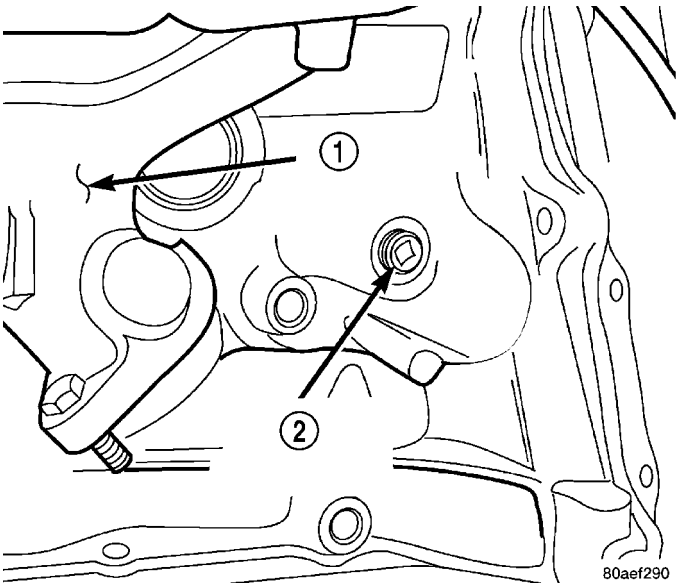


Fig. 7 Draining Coolant From Block - 2.4L/4.0L Engines

1 - EXHAUST MANIFOLD
2 - CYLINDER BLOCK COOLANT DRAIN PLUG

STANDARD PROCEDURE - COOLING SYSTEM - REFILLING

(1) Tighten the radiator draincock and the cylinder block drain plug(s).

(2) Fill system using a 50/50 mixture of water and antifreeze. Fill the radiator to the top and install the radiator cap. Add sufficient coolant to the reserve/overflow tank to raise the level to the FULL mark.

(3) Operate the engine with both the radiator cap and reserve/overflow tank cap in place. After the engine has reached the normal operating temperature, shut the engine off and allow it to cool.

(4) Add coolant to the reserve/overflow tank as necessary. **Only add coolant when the engine is cold. Coolant level in a warm engine will be higher due to thermal expansion.**

STANDARD PROCEDURE - COOLING SYSTEM - REVERSE FLUSHING

CAUTION: The cooling system normally operates at 97 - 110 C.P.A. 14 - 16 -16 psi) pressure. Exceeding this pressure may damage the radiator or hoses.

Reverse flushing of the cooling system is the forcing of water through the cooling system. This is done using air pressure in the opposite direction of normal coolant flow. It is usually only necessary with very dirty systems with evidence of partial plugging.

CHEMICAL CLEANING

If visual inspection indicates the formation of sludge or scaly deposits, use a radiator cleaner (Mopar Radiator Kleen or equivalent) before flushing. This will soften scale and other deposits and aid the flushing operation.

CAUTION: Be sure instructions on the container are followed.

REVERSE FLUSHING RADIATOR

Disconnect the radiator hoses from the radiator fittings. Attach a section of radiator hose to the radiator bottom outlet fitting and insert the flushing gun. Connect a water supply hose and air supply hose to the flushing gun.

CAUTION: The cooling system normally operates at 97 - 110 C.P.A. (14 - 16 psi) pressure. Exceeding this pressure may damage the radiator or hoses.

Allow the radiator to fill with water. When radiator is filled, apply air in short blasts allowing radiator to refill between blasts. Continue this reverse flushing until clean water flows out through rear of radiator cooling tube passages. For more information, refer to operating instructions supplied with flushing equipment. Have radiator cleaned more extensively by a radiator repair shop.

REVERSE FLUSHING ENGINE

Drain the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE). Remove the thermostat housing and thermostat. Install the thermostat housing. Disconnect the radiator upper hose from the radiator and attach the flushing gun to the hose. Disconnect the radiator lower hose from the water pump. Attach a lead away hose to the water pump inlet fitting.

CAUTION: Be sure that the heater control valve is closed (heat off). This is done to prevent coolant flow with scale and other deposits from entering the heater core.

Connect the water supply hose and air supply hose to the flushing gun. Allow the engine to fill with water. When the engine is filled, apply air in short blasts, allowing the system to fill between air blasts. Continue until clean water flows through the lead away hose. For more information, refer to operating instructions supplied with flushing equipment.

Remove the lead away hose, flushing gun, water supply hose and air supply hose. Remove the thermostat housing (Refer to 7 - COOLING/ENGINE/ENGINE COOLANT THERMOSTAT - REMOVAL). Install the thermostat and housing with a replacement gasket (Refer to 7 - COOLING/ENGINE/ENGINE COOLANT THERMOSTAT -

COOLING (Continued)

INSTALLATION). Connect the radiator hoses. Refill the cooling system with the correct antifreeze/water mixture (Refer to 7 - COOLING - STANDARD PROCEDURE).

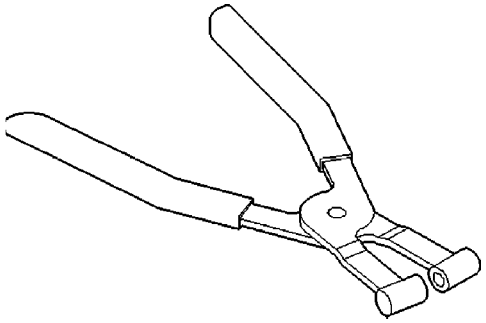
SPECIFICATIONS

TORQUE

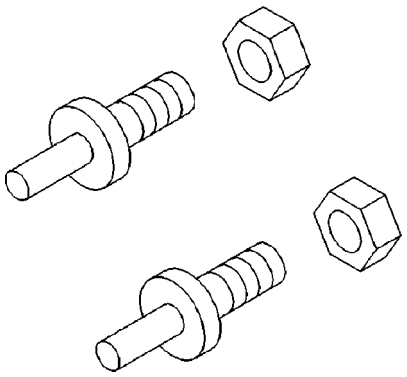
DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Automatic Belt Tensioner to Mounting Bracket - 2.4L	41	30	-
Bolt, Automatic Belt Tensioner Pulley - 2.4L	61	45	-
Bolt, Block Heater - 2.4L	2	-	17
Bolts, Condenser to Radiator			
Bolts, Coolant Overflow Bottle to Plenum - 2.4L only	8.5	-	75
Electric Fan to Fan Shroud bolts	5.5		50
Bolts, 3.7LFan Blade Assy. to Viscous Drive	23	-	210
Bolts, Fan Shroud to Radiator Mounting	8	-	70
Bolts, Radiator Upper Isolator to Crossmember	9.5	-	85
Bolts, Thermostat Housing - 2.4L	28	-	250
Bolts, Water Pump			
2.4L	12	-	105
4.0l	23	-	200

SPECIAL TOOLS

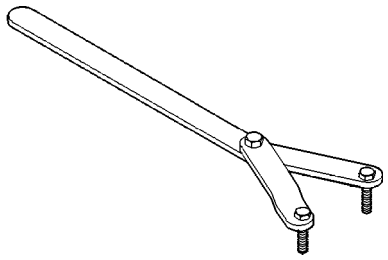
COOLING



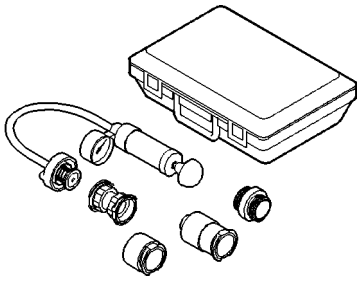
Pliers 6094



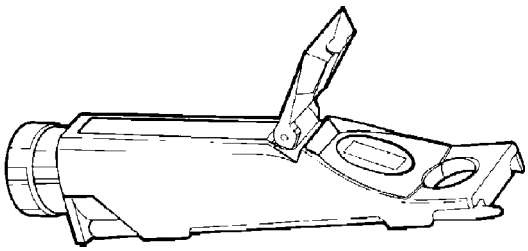
Adapter Pins 8346



Spanner Wrench 6958 with 8346 adapter pins



Pressure Tester 7700–A



Coolant Refractometer 8286

ACCESSORY DRIVE

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ACCESSORY DRIVE

SPECIFICATIONS

BELT TENSION

Belt tension can not be adjusted. Refer to the following Belt Tension chart for specifications.

DESCRIPTION	N·m	Lbs. ft.
New Serpentine Belt*	800-900	180-200
Used Serpentine Belt	623-712	140-160
* Belt is considered new if it has been used 15 minutes or less.		

BELT TENSIONERS

DESCRIPTION

The automatic belt tensioner (Fig. 1) is a spring loaded arm and pulley assembly. The tensioner assembly is designed to apply constant pressure on the accessory drive belt to maintain proper belt tension. There are three marks on the tensioner body, these marks are there to indicate belt wear and belt tension.

NOTE: On 4.0L engines, the tensioner arm has three marks. Upon installation of a new belt, the double line marks close to each other should be very close to the mark on the base. The belt should be replaced if the single line mark lines up with the mark on the base.

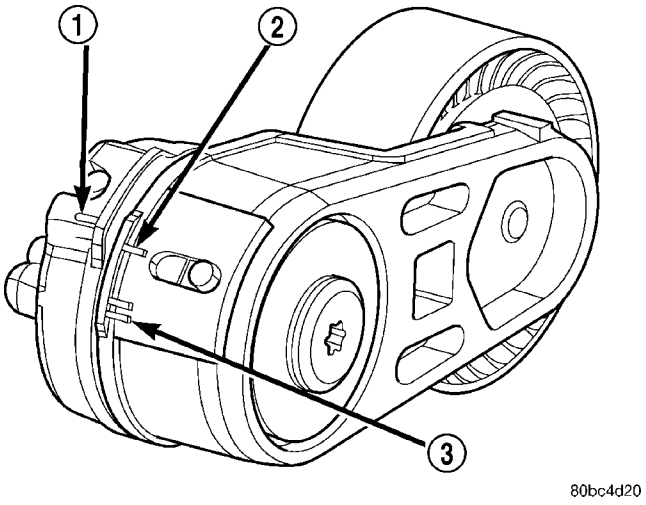

Fig. 1 Accessory Drive Belt Wear Indicator - 4.0L Engine

1 - INDICATOR MARK
2 - MINIMUM TENSION MARK
3 - MAXIMUM TENSION MARK

BELT TENSIONERS (Continued)

REMOVAL

REMOVAL - 2.4L ENGINE

NOTE: The belt routing schematics are published from the latest information available at the time of publication. If anything differs between these schematics and the Belt Routing Label, use the schematics on Belt Routing Label. This label is located in the engine compartment.

CAUTION: DO NOT LET TENSIONER ARM SNAP BACK TO THE FREEARM POSITION, SEVERE DAMAGE MAY OCCUR TO THE TENSIONER.

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Disconnect negative battery cable from battery.

(2) Rotate belt tensioner until it contacts its stop. Remove belt, then slowly rotate the tensioner into the freearm position.

REMOVAL

On 4.0L engines, the tensioner arm has three marks. Upon installation of a new belt, the double line marks close to each other should be very close to the mark on the base. The belt should be replaced if the single line mark lines up with the mark on the base.

If the above specification cannot be met, check for:

- The wrong belt being installed (incorrect length/ width)
- Worn bearings on an engine accessory (A/C compressor, power steering pump, water pump, idler pulley or generator)
- A pulley on an engine accessory being loose
- Misalignment of an engine accessory
- Belt incorrectly routed.

NOTE: A used belt should be replaced if tensioner indexing arrow has moved to the minimum travel indicator. Tensioner travel stops at this point.

(1) Remove accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(2) Remove tensioner assembly from mounting bracket (Fig. 2).

WARNING: BECAUSE OF HIGH SPRING TENSION, DO NOT ATTEMPT TO DISASSEMBLE AUTOMATIC TENSIONER. UNIT IS SERVICED AS AN ASSEMBLY.

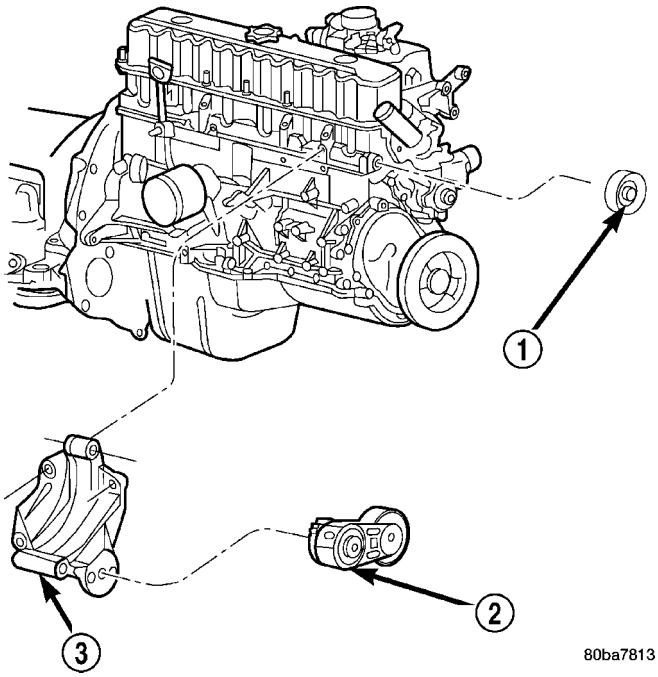

Fig. 2 Automatic Belt Tensioner - 4.0L Engine

1 - IDLER PULLEY TIGHTEN TO 47 N·m (35 FT. LBS.)
2 - AUTOMATIC BELT TENSIONER
3 - GENERATOR MOUNTING BRACKET

INSTALLATION

INSTALLATION - 2.4L ENGINE

NOTE: The belt routing schematics are published from the latest information available at the time of publication. If anything differs between these schematics and the Belt Routing Label, use the schematics on Belt Routing Label. This label is located in the engine compartment.

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Check condition of all pulleys.

CAUTION: When installing the serpentine accessory drive belt, the belt MUST be routed correctly. If not, the engine may overheat due to the water pump rotating in the wrong direction.

(2) Install new belt (Fig. 3) or (Fig. 4). Route the belt around all pulleys except the idler pulley. Rotate the tensioner arm until it contacts its stop position. Route the belt around the idler and slowly let the tensioner rotate into the belt. Make sure the belt is seated onto all pulleys.

(3) With the drive belt installed, inspect the belt wear indicator. On 2.4L Engines the gap between the tang and the housing stop (measurement A) must not exceed 24 mm (.94 inches).

BELT TENSIONERS (Continued)

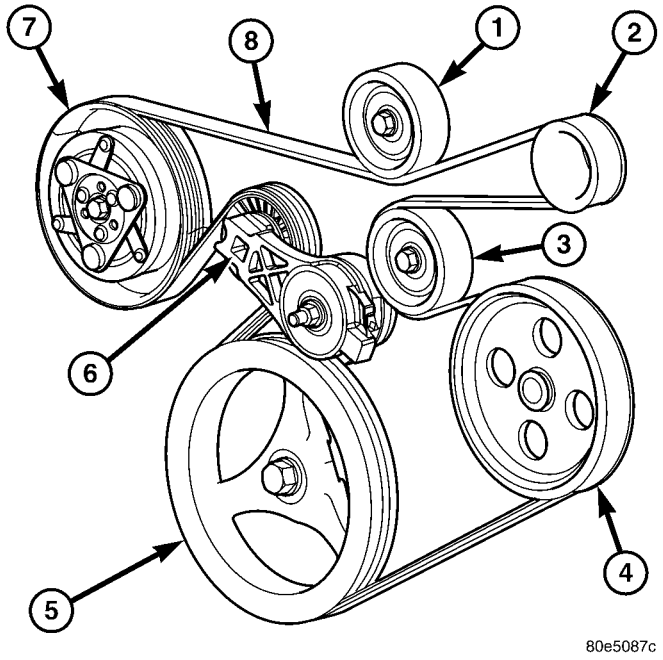

Fig. 3 Belt Routing 2.4L With A/C

1 - IDLER PULLEY
2 - GENERATOR PULLEY
3 - IDLER PULLEY
4 - POWER STEERING PUMP PULLEY
5 -CRANKSHAFT PULLEY
6 - TENSIONER
7 - A/C COMPRESSOR PULLEY
8 - ACCESSORY DRIVE BELT

INSTALLATION

(1) Install tensioner assembly to mounting bracket, align the two dowels on the tensioner with the mounting bracket and hand start the bolt. Tighten bolt to 28 N·m (250 in. lbs.).

(2) Install accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(3) Check belt indexing marks.

DRIVE BELTS - 2.4L

DIAGNOSIS AND TESTING - ACCESSORY DRIVE BELT

VISUAL DIAGNOSIS

When diagnosing serpentine accessory drive belts, small cracks that run across the ribbed surface of the belt from rib to rib (Fig. 5), are considered normal. These are not a reason to replace the belt. However, cracks running along a rib (not across) are **not** normal. Any belt with cracks running along a rib must be replaced (Fig. 5). Also replace the belt if it has excessive wear, frayed cords or severe glazing.

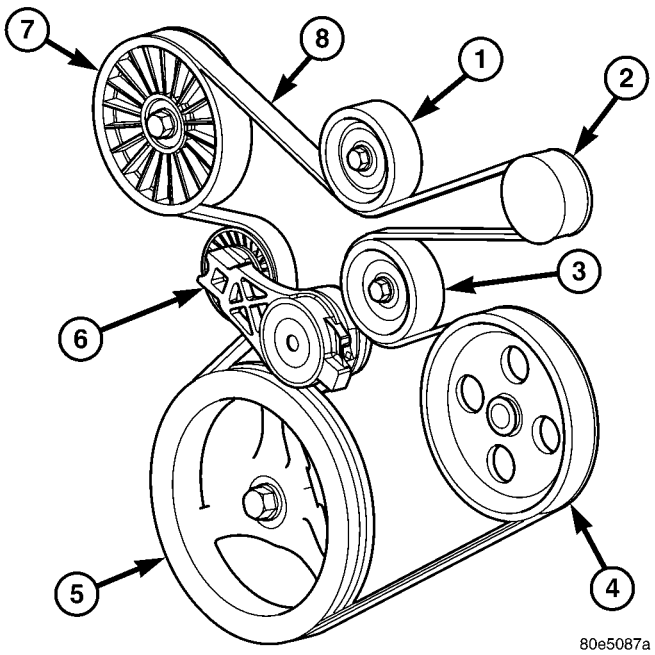

Fig. 4 Belt Routing 2.4L Without A/C

1 - IDLER PULLEY
2 - GENERATOR PULLEY
3 - IDLER PULLEY
4 - POWER STEERING PUMP PULLEY
5 -CRANKSHAFT PULLEY
6 - TENSIONER
7 - NON A/C IDLER PULLEY
8 - ACCESSORY DRIVE BELT

Refer to the ACCESSORY DRIVE BELT DIAGNOSIS CHART for further belt diagnosis.

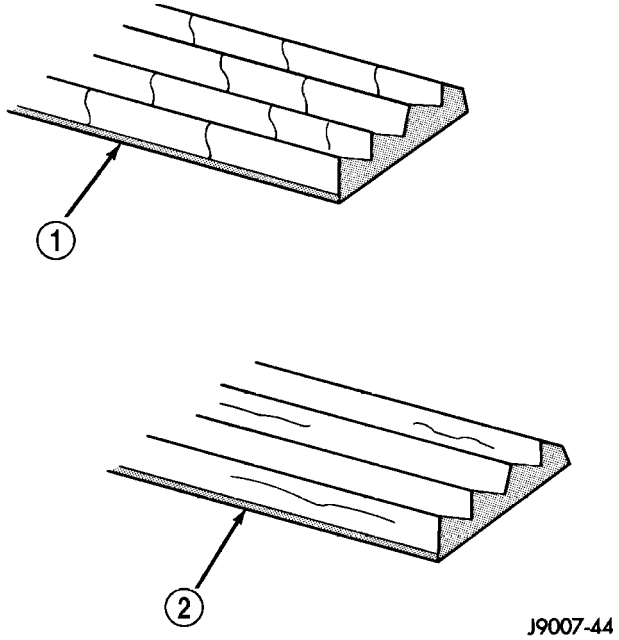

Fig. 5 Belt Wear Patterns

1 - NORMAL CRACKS BELT OK
2 - NOT NORMAL CRACKS REPLACE BELT

DRIVE BELTS - 2.4L (Continued)

NOISE DIAGNOSIS

Noises generated by the accessory drive belt are most noticeable at idle. Before replacing a belt to resolve a noise condition, inspect all of the accessory drive pulleys for alignment, glazing, or excessive end play.

ACCESSORY DRIVE BELT DIAGNOSIS CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
RIB CHUNKING (One or more ribs has separated from belt body)	1. Foreign objects imbedded in pulley grooves.	1. Remove foreign objects from pulley grooves. Replace belt.
	2. Installation damage	2. Replace belt
RIB OR BELT WEAR	1. Pulley misaligned	1. Align pulley(s)
	2. Abrasive environment	2. Clean pulley(s). Replace belt if necessary
	3. Rusted pulley(s)	3. Clean rust from pulley(s)
	4. Sharp or jagged pulley groove tips	4. Replace pulley. Inspect belt.
	5. Belt rubber deteriorated	5. Replace belt
BELT SLIPS	1. Belt slipping because of insufficient tension	1. Adjust tension (2.5L)
	2. Belt or pulley exposed to substance that has reduced friction (belt dressing, oil, ethylene glycol)	2. Replace belt and clean pulleys
	3. Driven component bearing failure (seizure)	3. Replace faulty component or bearing
	4. Belt glazed or hardened from heat and excessive slippage	4. Replace belt.
LONGITUDAL BELT CRACKING	1. Belt has mistracked from pulley groove	1. Replace belt
	2. Pulley groove tip has worn away rubber to tensile member	2. Replace belt
″GROOVE JUMPING″ (Belt does not maintain correct position on pulley)	1. Belt tension either too low or too high	1. Adjust belt tension (2.5L)
	2. Pulley(s) not within design tolerance	2. Replace pulley(s)
	3. Foreign object(s) in grooves	3. Remove foreign objects from grooves
	4. Pulley misalignment	4. Align component
	5. Belt cordline is broken	5. Replace belt

DRIVE BELTS - 2.4L (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
BELT BROKEN (Note: Identify and correct problem before new belt is installed)	1. Excessive tension	1. Replace belt and adjust tension to specification
	2. Tensile member damaged during belt installation	2. Replace belt
	3. Severe misalignment	3. Align pulley(s)
	4. Bracket, pulley, or bearing failure	4. Replace defective component and belt
NOISE (Objectionable squeal, squeak, or rumble is heard or felt while drive belt is in operation)	1. Belt slippage	1. Adjust belt tension (2.5L)
	2. Bearing noise	2. Locate and repair
	3. Belt misalignment	3. Align belt/pulley(s)
	4. Belt to pulley mismatch	4. Install correct belt
	5. Driven component induced vibration	5. Locate defective driven component and repair
	6. System resonant frequency induced vibration	6. Vary belt tension within specifications.
TENSION SHEETING FABRIC FAILURE (Woven fabric on outside, circumference of belt has cracked or separated from body of belt)	1. Tension sheeting contacting stationary object	1. Correct rubbing condition
	2. Excessive heat causing woven fabric to age	2. Replace belt
	3. Tension sheeting splice has fractured	3. Replace belt
CORD EDGE FAILURE (Tensile member exposed at edges of belt or separated from belt body)	1. Excessive tension	1. Adjust belt tension (2.5L)
	2. Belt contacting stationary object	2. Replace belt
	3. Pulley(s) out of tolerance	3. Replace pulley
	4. Insufficient adhesion between tensile member and rubber matrix	4. Replace belt and adjust tension to specifications

DRIVE BELTS - 2.4L (Continued)

REMOVAL

REMOVAL

NOTE: The belt routing schematics are published from the latest information available at the time of publication. If anything differs between these schematics and the Belt Routing Label, use the schematics on Belt Routing Label. This label is located in the engine compartment.

CAUTION: DO NOT LET TENSIONER ARM SNAP BACK TO THE FREEARM POSITION, SEVERE DAMAGE MAY OCCUR TO THE TENSIONER.

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Disconnect negative battery cable from battery.

(2) Rotate belt tensioner until it contacts its stop. Remove belt, then slowly rotate the tensioner into the freearm position.

REMOVAL

NOTE: The belt routing schematics are published from the latest information available at the time of publication. If anything differs between these schematics and the Belt Routing Label, use the schematics on Belt Routing Label. This label is located in the engine compartment.

CAUTION: DO NOT LET TENSIONER ARM SNAP BACK TO THE FREEARM POSITION, SEVERE DAMAGE MAY OCCUR TO THE TENSIONER.

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Disconnect negative battery cable from battery.

(2) Rotate belt tensioner until it contacts its stop. Remove belt, then slowly rotate the tensioner into the freearm position. (Fig. 6) or (Fig. 7).

INSTALLATION

INSTALLATION

NOTE: The belt routing schematics are published from the latest information available at the time of publication. If anything differs between these schematics and the Belt Routing Label, use the schematics on Belt Routing Label. This label is located in the engine compartment.

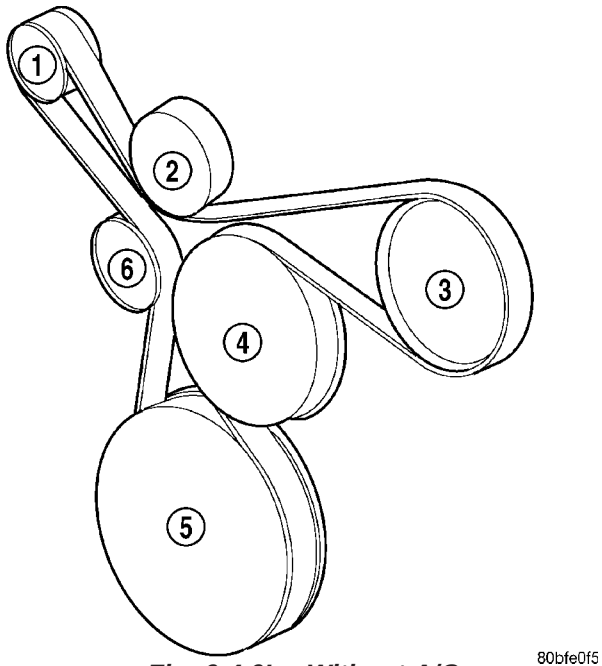

Fig. 6 4.0L - Without A/C

1 - GENERATOR PULLEY
2 - IDLER PULLEY
3 - POWER STEERING PUMP PULLEY
4 - WATER PUMP PULLEY
5 - CRANKSHAFT PULLEY
6 - TENSIONER PULLEY

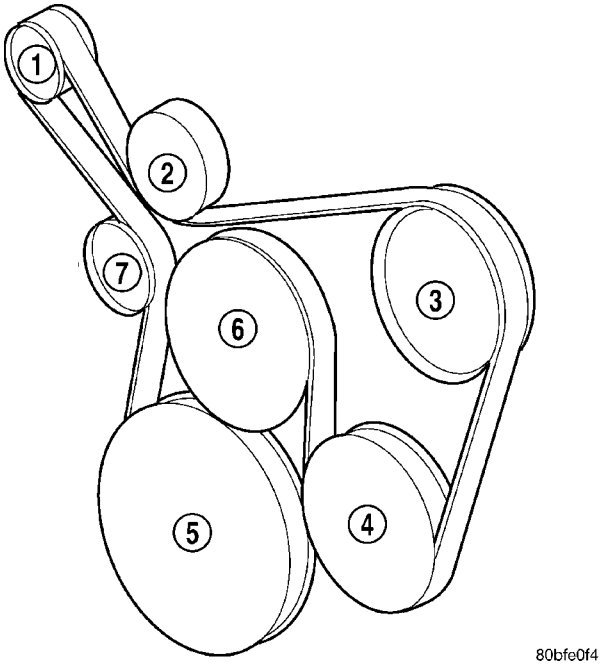

Fig. 7 4.0L Engines - With A/C

1 - GENERATOR PULLEY
2 - IDLER PULLEY
3 - POWER STEERING PUMP PULLEY
4 - AIR CONDITIONING COMPRESSOR PULLEY
5 - CRANKSHAFT PULLEY
6 - WATER PUMP PULLEY
7 - TENSIONER PULLEY

DRIVE BELTS - 2.4L (Continued)

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Check condition of all pulleys.

CAUTION: When installing the serpentine accessory drive belt, the belt MUST be routed correctly. If not, the engine may overheat due to the water pump rotating in the wrong direction.

(2) Install new belt (Fig. 8) or (Fig. 9). Route the belt around all pulleys except the idler pulley. Rotate the tensioner arm until it contacts its stop position. Route the belt around the idler and slowly let the tensioner rotate into the belt. Make sure the belt is seated onto all pulleys.

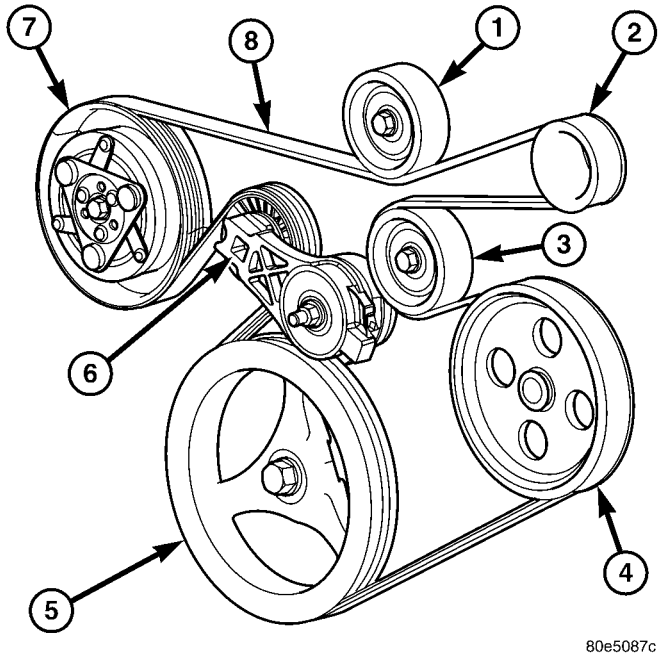

Fig. 8 Belt Routing 2.4L With A/C

1 - IDLER PULLEY
2 - GENERATOR PULLEY
3 - IDLER PULLEY
4 - POWER STEERING PUMP PULLEY
5 -CRANKSHAFT PULLEY
6 - TENSIONER
7 - A/C COMPRESSOR PULLEY
8 - ACCESSORY DRIVE BELT

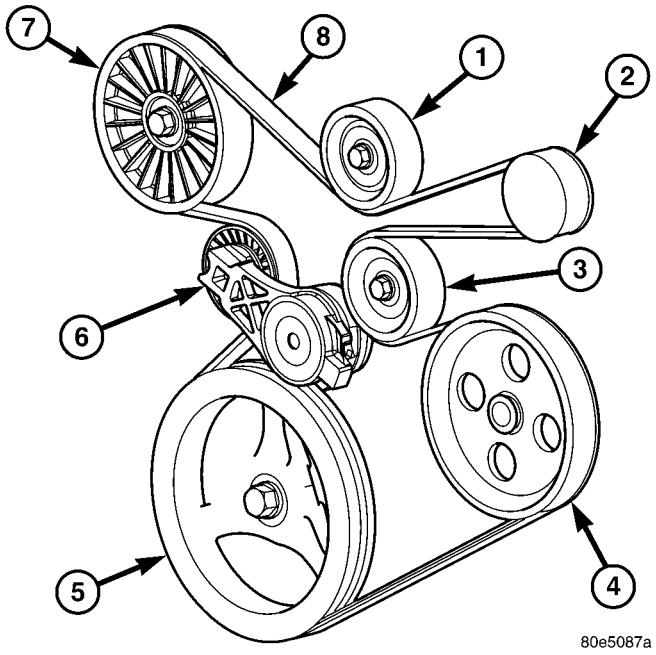

Fig. 9 Belt Routing 2.4L Without A/C

1 - IDLER PULLEY
2 - GENERATOR PULLEY
3 - IDLER PULLEY
4 - POWER STEERING PUMP PULLEY
5 -CRANKSHAFT PULLEY
6 - TENSIONER
7 - NON A/C IDLER PULLEY
8 - ACCESSORY DRIVE BELT

(3) With the drive belt installed, inspect the belt wear indicator. On 2.4L Engines the gap between the tang and the housing stop (measurement A) must not exceed 24 mm (.94 inches).

DRIVE BELTS - 2.4L (Continued)

INSTALLATION

Belt tension is not adjustable. Belt adjustment is maintained by an automatic (spring load) belt tensioner.

(1) Check condition of all pulleys.

CAUTION: When installing the serpentine accessory drive belt, the belt MUST be routed correctly. If not, the engine may overheat due to the water pump rotating in the wrong direction (Fig. 11) (Fig. 12).

(2) Install new belt (Fig. 12) or (Fig. 11). Route the belt around all pulleys except the idler pulley. Rotate the tensioner arm until it contacts its stop position. Route the belt around the idler and slowly let the tensioner rotate into the belt. Make sure the belt is seated onto all pulleys.

(3) With the drive belt installed, inspect the belt wear indicator (Fig. 10). On 4.0L Engines, the indicator mark must be between the minimum and maximum marks. If the measurement exceedes this specification replace the serpentine accessory drive belt.

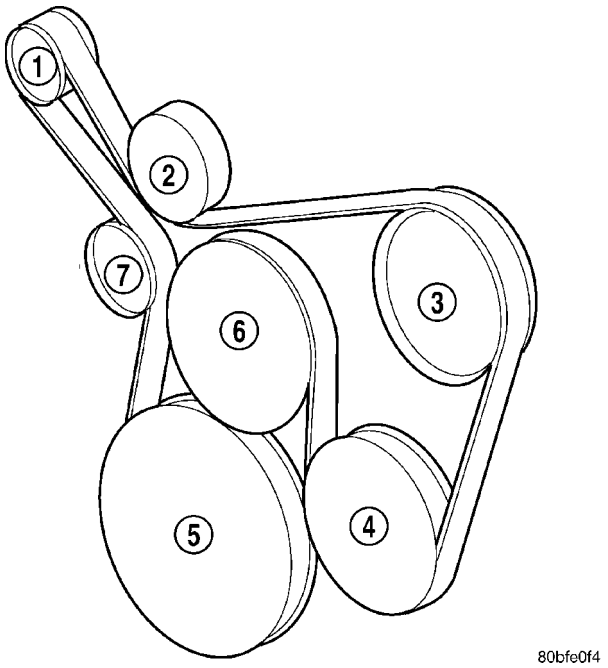

Fig. 10 4.0L Engines - With A/C

1 - GENERATOR PULLEY
2 - IDLER PULLEY
3 - POWER STEERING PUMP PULLEY
4 - AIR CONDITIONING COMPRESSOR PULLEY
5 - CRANKSHAFT PULLEY
6 - WATER PUMP PULLEY
7 - TENSIONER PULLEY

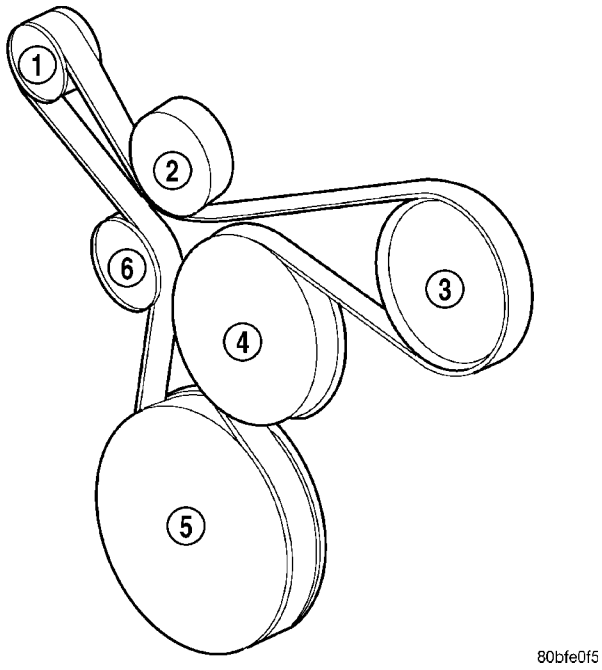

Fig. 11 4.0L - Without/A/C

1 - GENERATOR PULLEY
2 - IDLER PULLEY
3 - POWER STEERING PUMP PULLEY
4 - WATER PUMP PULLEY
5 - CRANKSHAFT PULLEY
6 - TENSIONER PULLEY

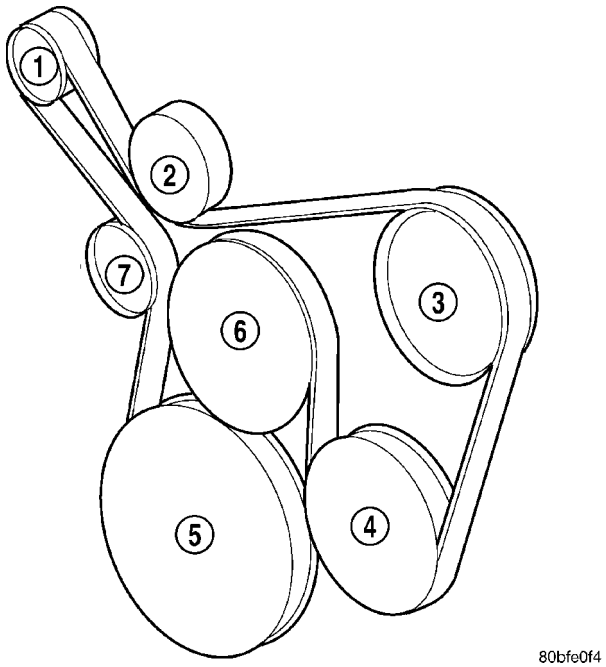

Fig. 12 4.0L Engines - With A/C

1 - GENERATOR PULLEY
2 - IDLER PULLEY
3 - POWER STEERING PUMP PULLEY
4 - AIR CONDITIONING COMPRESSOR PULLEY
5 - CRANKSHAFT PULLEY
6 - WATER PUMP PULLEY
7 - TENSIONER PULLEY

ENGINE

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COOLANT

DESCRIPTION

DESCRIPTION - HOAT COOLANT

WARNING: ANTIFREEZE IS AN ETHYLENE GLYCOL BASE COOLANT AND IS HARMFUL IF SWALLOWED OR INHALED. IF SWALLOWED, DRINK TWO GLASSES OF WATER AND INDUCE VOMITING. IF INHALED, MOVE TO FRESH AIR AREA. SEEK MEDICAL ATTENTION IMMEDIATELY. DO NOT STORE IN OPEN OR UNMARKED CONTAINERS. WASH SKIN AND CLOTHING THOROUGHLY AFTER COMING IN CONTACT WITH ETHYLENE GLYCOL. KEEP OUT OF REACH OF CHILDREN. DISPOSE OF GLYCOL BASE COOLANT PROPERLY, CONTACT YOUR DEALER OR GOVERNMENT AGENCY FOR LOCATION OF COLLECTION CENTER IN YOUR AREA. DO NOT OPEN A COOLING SYSTEM WHEN THE ENGINE IS AT OPERATING TEMPERATURE OR HOT UNDER PRESSURE, PERSONAL INJURY CAN RESULT. AVOID RADIATOR COOLING FAN WHEN ENGINE COMPARTMENT RELATED SERVICE IS PERFORMED, PERSONAL INJURY CAN RESULT.

CAUTION: Use of Propylene Glycol based coolants is not recommended, as they provide less freeze protection and less corrosion protection.

The cooling system is designed around the coolant. The coolant must accept heat from engine metal, in the cylinder head area near the exhaust valves and engine block. Then coolant carries the heat to the radiator where the tube/fin radiator can transfer the heat to the air.

The use of aluminum cylinder blocks, cylinder heads, and water pumps requires special corrosion protection. Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (MS-9769), or the equivalent ethylene glycol base coolant with organic corrosion inhibitors (called HOAT, for Hybrid Organic Additive Technology) is recommended. This coolant offers the best engine cooling without corrosion when mixed with 50% Ethylene Glycol and 50% distilled water to obtain a freeze point of -37°C (-35°F). If it loses color or becomes contaminated, drain, flush, and replace with fresh properly mixed coolant solution.

CAUTION: Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (MS-9769) may not be mixed with any other type of antifreeze. Mixing of coolants other than specified (non-HOAT or other HOAT), may result in engine damage that may not be covered under the new vehicle warranty, and decreased corrosion protection.

COOLANT PERFORMANCE

The required ethylene-glycol (antifreeze) and water mixture depends upon climate and vehicle operating conditions. The coolant performance of various mixtures follows:

Pure Water-Water can absorb more heat than a mixture of water and ethylene-glycol. This is for purpose of heat transfer only. Water also freezes at a higher temperature and allows corrosion.

100 percent Ethylene-Glycol-The corrosion inhibiting additives in ethylene-glycol need the presence of water to dissolve. Without water, additives form deposits in system. These act as insulation causing temperature to rise to as high as 149°C (300°F). This temperature is hot enough to melt plastic and soften solder. The increased temperature can result in engine detonation. In addition, 100 percent ethylene-glycol freezes at -22°C (-8°F).

50/50 Ethylene-Glycol and Water-Is the recommended mixture, it provides protection against freezing to -37°C (-34°F). The antifreeze concentration **must always** be a minimum of 44 percent, year-round in all climates. If percentage is lower, engine parts may be eroded by cavitation. Maximum protection against freezing is provided with a 68 percent antifreeze concentration, which prevents freezing down to -67.7°C (-90°F). A higher percentage will freeze at a warmer temperature. Also, a higher percentage of antifreeze can cause the engine to overheat because specific heat of antifreeze is lower than that of water.

CAUTION: Richer antifreeze mixtures cannot be measured with normal field equipment and can cause problems associated with 100 percent ethylene-glycol.

COOLANT SELECTION AND ADDITIVES

The use of aluminum cylinder blocks, cylinder heads and water pumps requires special corrosion protection. Only Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (glycol base coolant with corrosion inhibitors called HOAT, for Hybrid Organic Additive Technology) is recommended. This coolant offers the best engine cooling without corrosion when mixed with 50% distilled water to obtain to obtain a freeze point of -37°C (-35°F). If it loses color or becomes contaminated, drain, flush, and replace with fresh properly mixed coolant solution.

CAUTION: Do not use coolant additives that are claimed to improve engine cooling.

COOLANT (Continued)

DESCRIPTION - ENGINE COOLANT

ETHYLENE-GLYCOL MIXTURES

CAUTION: Richer antifreeze mixtures cannot be measured with normal field equipment and can cause problems associated with 100 percent ethylene-glycol.

The required ethylene-glycol (antifreeze) and water mixture depends upon the climate and vehicle operating conditions. The recommended mixture of 50/50 ethylene-glycol and water will provide protection against freezing to -37 deg. C (-35 deg. F). The antifreeze concentration **must always** be a minimum of 44 percent, year-round in all climates. **If percentage is lower than 44 percent, engine parts may be eroded by cavitation, and cooling system components may be severely damaged by corrosion.** Maximum protection against freezing is provided with a 68 percent antifreeze concentration, which prevents freezing down to -67.7 deg. C (-90 deg. F). A higher percentage will freeze at a warmer temperature. Also, a higher percentage of antifreeze can cause the engine to overheat because the specific heat of antifreeze is lower than that of water.

Use of 100 percent ethylene-glycol will cause formation of additive deposits in the system, as the corrosion inhibitive additives in ethylene-glycol require the presence of water to dissolve. The deposits act as insulation, causing temperatures to rise to as high as 149 deg. C (300) deg. F). This temperature is hot enough to melt plastic and soften solder. The increased temperature can result in engine detonation. In addition, 100 percent ethylene-glycol freezes at 22 deg. C (-8 deg. F).

PROPYLENE-GLYCOL MIXTURES

It's overall effective temperature range is smaller than that of ethylene-glycol. The freeze point of 50/50 propylene-glycol and water is -32 deg. C (-26 deg. F). 5 deg. C higher than ethylene-glycol's freeze point. The boiling point (protection against summer boil-over) of propylene-glycol is 125 deg. C (257 deg. F) at 96.5 kPa (14 psi), compared to 128 deg. C (263 deg. F) for ethylene-glycol. Use of propylene-glycol can result in boil-over or freeze-up on a cooling system designed for ethylene-glycol. Propylene glycol also has poorer heat transfer characteristics than ethylene glycol. This can increase cylinder head temperatures under certain conditions.

Propylene-glycol/ethylene-glycol Mixtures can cause the destabilization of various corrosion inhibitors, causing damage to the various cooling system components. Also, once ethylene-glycol and propylene-glycol based coolants are mixed in the vehicle, conventional methods of determining freeze point will not be accurate. Both the refractive index and specific gravity differ between ethylene glycol and propylene glycol.

OPERATION

Coolant flows through the engine block absorbing the heat from the engine, then flows to the radiator where the cooling fins in the radiator transfers the heat from the coolant to the atmosphere. During cold weather the ethylene-glycol coolant prevents water present in the cooling system from freezing within temperatures indicated by mixture ratio of coolant to water.

COOLANT RECOVERY CONTAINER

DESCRIPTION

The coolant reserve/overflow system consists of a radiator mounted pressurized cap, a plastic coolant recovery bottle (Fig. 1), a tube (hose) connecting the radiator and recovery bottle, and an overflow tube on the side of the bottle.

The reservoir bottle also has an anti-slosh insert located within the bottle, this insert will aid in reducing coolant loss from splash and spillage.

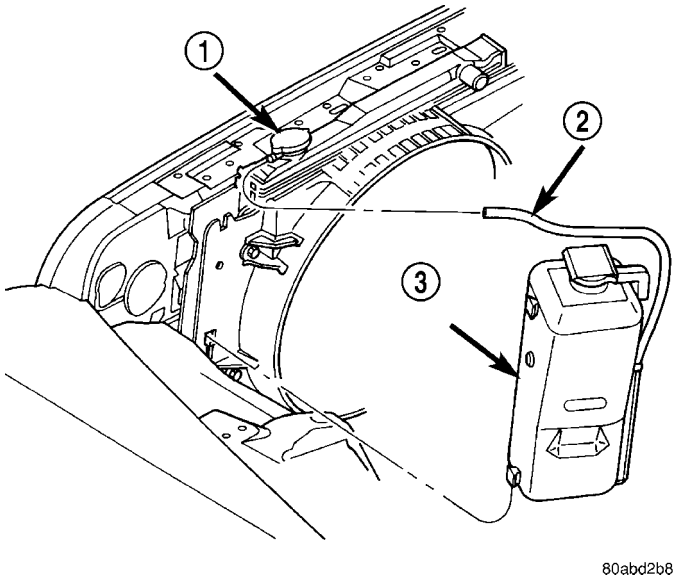

Fig. 1 Coolant Recovery Bottle

1 - RADIATOR PRESSURE CAP
2 - HOSE
3 - COOLANT RECOVERY BOTTLE

COOLANT RECOVERY CONTAINER (Continued)

OPERATION

The system works along with the radiator pressure cap. This is done by using thermal expansion and contraction of the coolant to keep the coolant free of trapped air. It provides:

- A volume for coolant expansion and contraction.
- A convenient and safe method for checking/adjusting coolant level at atmospheric pressure. This is done without removing the radiator pressure cap.
- Some reserve coolant to the radiator to cover minor leaks and evaporation or boiling losses.

As the engine cools, a vacuum is formed in the cooling system of both the radiator and engine. Coolant will then be drawn from the coolant tank and returned to a proper level in the radiator.

REMOVAL

(1) Disconnect the hose from radiator filler neck.
(2) Remove coolant recovery bottle (Fig. 2).

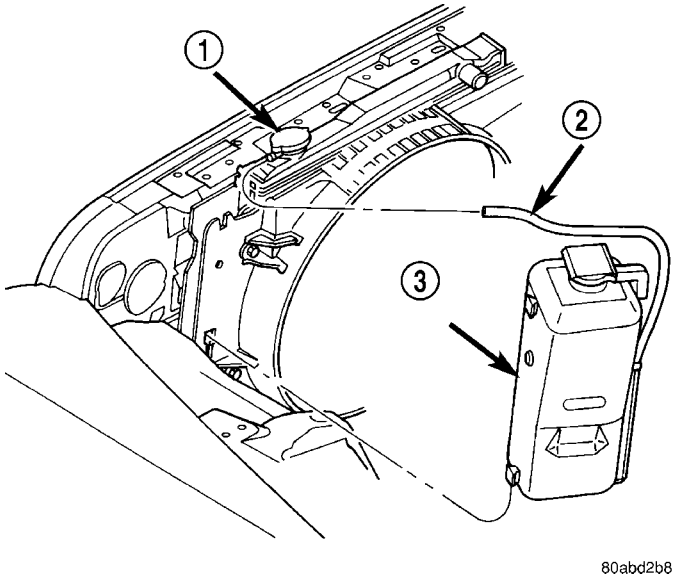

Fig. 2 Coolant Reserve/Overflow Bottle

1 - RADIATOR PRESSURE CAP
2 - HOSE
3 - COOLANT RECOVERY BOTTLE

INSTALLATION

(1) Position the tabs on the overflow bottle into the slots on the fan shroud.
(2) Reconnect the overflow hose onto the radiator filler neck.
(3) Fill resrevoir/overflow bottle.

ENGINE BLOCK HEATER

DESCRIPTION

WARNING: DO NOT OPERATE ENGINE UNLESS BLOCK HEATER CORD HAS BEEN DISCONNECTED FROM POWER SOURCE AND SECURED IN PLACE.

An optional engine block heater (Fig. 3) is available for all models. The heater is equipped with a power cord. The cord is attached to an engine compartment component with tie-straps. The heater warms the engine providing easier engine starting and faster warm-up in low temperatures. The heater is mounted in a core hole of the engine cylinder block (in place of a freeze plug) with the heating element immersed in engine coolant.

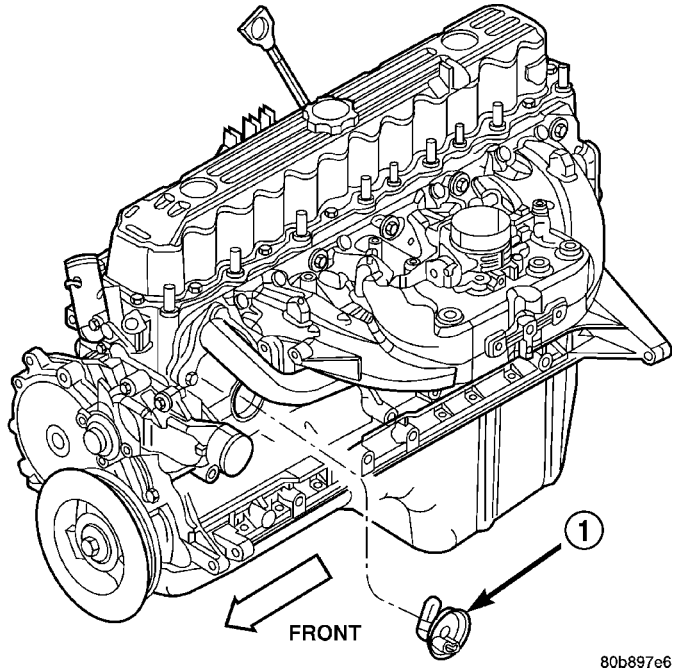

Fig. 3 Block Heater - Typical

1 - ENGINE BLOCK HEATER

OPERATION

Connecting the power cord to a grounded 110-120 volt AC electrical outlet with a grounded, three-wire extension cord, supplies the electricity required to heat the element thus heating the engine coolant.

ENGINE BLOCK HEATER (Continued)

REMOVAL

REMOVAL - 4.0L

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS OR LOOSEN THE RADIATOR DRAINCOCK WITH THE SYSTEM HOT AND PRESSURIZED. SERIOUS BURNS FROM THE COOLANT CAN OCCUR.

DO NOT WASTE reusable coolant. If solution is clean, drain coolant into a clean container for reuse.

(1) Drain coolant (Refer to 7 - COOLING - STANDARD PROCEDURE).

(2) Unplug power cord from block heater.

(3) Loosen screw in center of block heater (Fig. 4).

(4) Remove block heater from cylinder block.

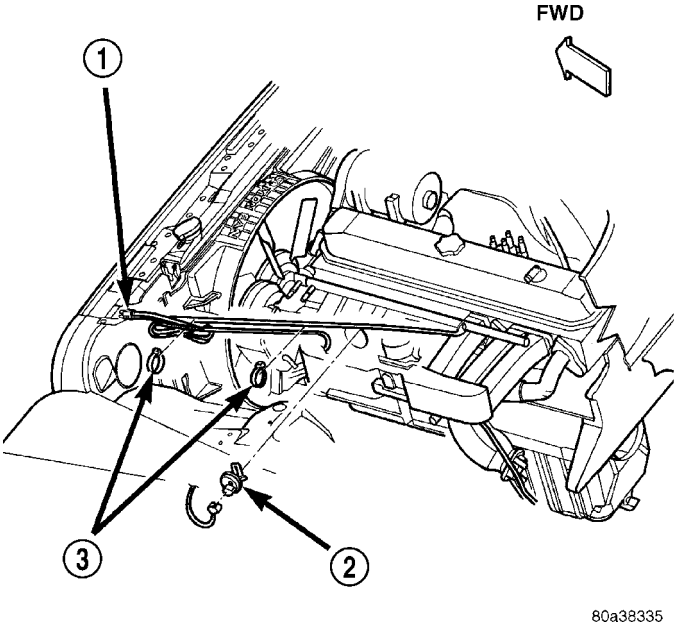

Fig. 4 Block Heater and Cord - 4.0L Engine

1 - POWER CORD
2 - BLOCK HEATER
3 - TIE-STRAPS

REMOVAL - 2.4L

(1) Drain cooling system (Refer to 7 - COOLING/ ENGINE - STANDARD PROCEDURE).

(2) Raise vehicle on hoist.

(3) Detach power cord plug from heater.

(4) Loosen screw in center of heater. Remove heater assembly.

INSTALLATION

INSTALLATION 4.0L

(1) Thoroughly clean the engine core hole and the block heater seat.

(2) Insert block heater assembly into core hole with element loop pointing **Up** (Fig. 4).

(3) Seat block heater flush against block face. Tighten mounting screw to 4 N·m (31 in. lbs.) torque.

(4) Fill cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE), and inspect for leaks (Refer to 7 - COOLING - DIAGNOSIS AND TESTING).

(5) Plug power cord into block heater. Route cord away from moving parts, linkages and exhaust system components. Secure cord in place with tie-straps.

INSTALLATION - 2.4L

(1) Thoroughly clean core hole and heater seat.

(2) Insert heater assembly (Fig. 5) with element loop positioned **upward** .

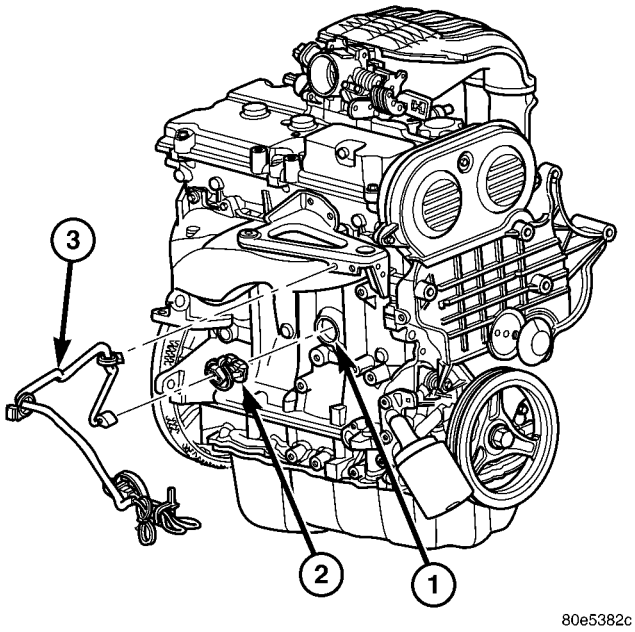

Fig. 5 ENGINE BLOCK HEATER 2.4L

1 - CORE HOLE
2 - BLOCK HEATER
3 - POWER CORD

(3) With heater seated, tighten center screw securely to assure a positive seal.

CAUTION: To prevent damage, the power cord must be secured in it's retaining clips, and not positioned so it could contact linkages or exhaust manifold.

(4) Connect power cord to heater.

(5) Lower vehicle.

(6) Fill cooling system (Refer to 7 - COOLING/ENGINE - STANDARD PROCEDURE).

ENGINE COOLANT TEMPERATURE SENSOR

DESCRIPTION

The Engine Coolant Temperature (ECT) sensor is used to sense engine coolant temperature. The sensor protrudes into an engine water jacket.

The ECT sensor is a two-wire Negative Thermal Coefficient (NTC) sensor. Meaning, as engine coolant temperature increases, resistance (voltage) in the sensor decreases. As temperature decreases, resistance (voltage) in the sensor increases.

OPERATION

At key-on, the Powertrain Control Module (PCM) sends out a regulated 5 volt signal to the ECT sensor. The PCM then monitors the signal as it passes through the ECT sensor to the sensor ground (sensor return).

When the engine is cold, the PCM will operate in Open Loop cycle. It will demand slightly richer air-fuel mixtures and higher idle speeds. This is done until normal operating temperatures are reached.

The PCM uses inputs from the ECT sensor for the following calculations:

- For engine coolant temperature gauge operation through CCD or PCI (J1850) communications
- Injector pulse-width
- Spark-advance curves
- ASD relay shut-down times
- Idle Air Control (IAC) motor key-on steps
- Pulse-width prime-shot during cranking
- O2 sensor closed loop times
- Purge solenoid on/off times
- EGR solenoid on/off times (if equipped)
- Leak Detection Pump operation (if equipped)
- Radiator fan relay on/off times (if equipped)
- Target idle speed

REMOVAL

WARNING: HOT, PRESSURIZED COOLANT CAN CAUSE INJURY BY SCALDING. COOLING SYSTEM MUST BE PARTIALLY DRAINED BEFORE REMOVING ECT (ENGINE COOLANT TEMPERATURE) SENSOR.

(1) Partially drain cooling system until coolant level is below cylinder head. (Refer to 7 - COOLING - STANDARD PROCEDURE).

(2) Disconnect ECT (Engine Coolant Temperature) sensor wire connector (Fig. 6) or (Fig. 7).

(3) Remove sensor from engine.

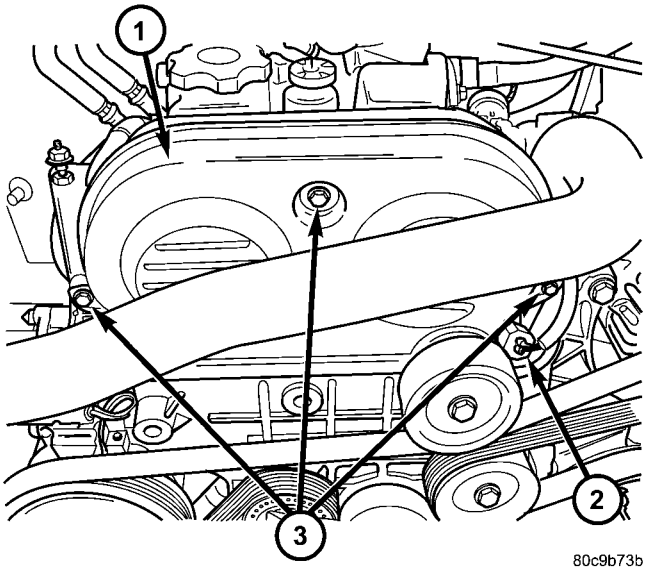

Fig. 6 ECT SENSOR LOCATION- 2.4L

1 - UPPER TIMING BELT COVER
2 - ELECTRICAL CONNECTOR (ECT)
3 - MOUNTING BOLTS (3)

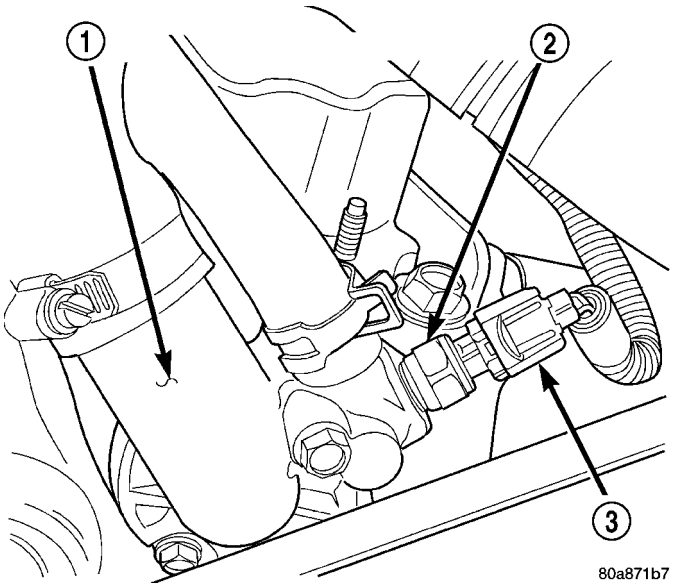

Fig. 7 ECT SENSOR LOCATION - 4.0L

1 - THERMOSTAT HOUSING
2 - ENGINE COOLANT TEMPERATURE SENSOR
3 - ELECTRICAL CONNECTOR

INSTALLATION

(1) Apply sealant to sensor threads (new replacement sensors will have sealant already applied).

(2) Install coolant temperature sensor into thermostat housing. Tighten to 11 N·m (8 ft. lbs.) torque.

(3) Connect wire connector.

(4) Fill cooling system. (Refer to 7 - COOLING - STANDARD PROCEDURE).

ENGINE COOLANT THERMOSTAT

DESCRIPTION

CAUTION: Do not operate an engine without a thermostat, except for servicing or testing.

The thermostat (Fig. 8) on all gas powered engines is located beneath the thermostat housing at the front of the intake manifold.

The thermostat is a wax pellet driven, reverse poppet choke type.

Coolant leakage into the pellet container will cause the thermostat to fail in the open position. Thermostats very rarely stick. Do not attempt to free a thermostat with a prying device.

The same thermostat is used for winter and summer seasons. An engine should not be operated without a thermostat, except for servicing or testing. Operating without a thermostat causes longer engine warmup time, unreliable warmup performance, increased exhaust emissions and crankcase condensation that can result in sludge formation.

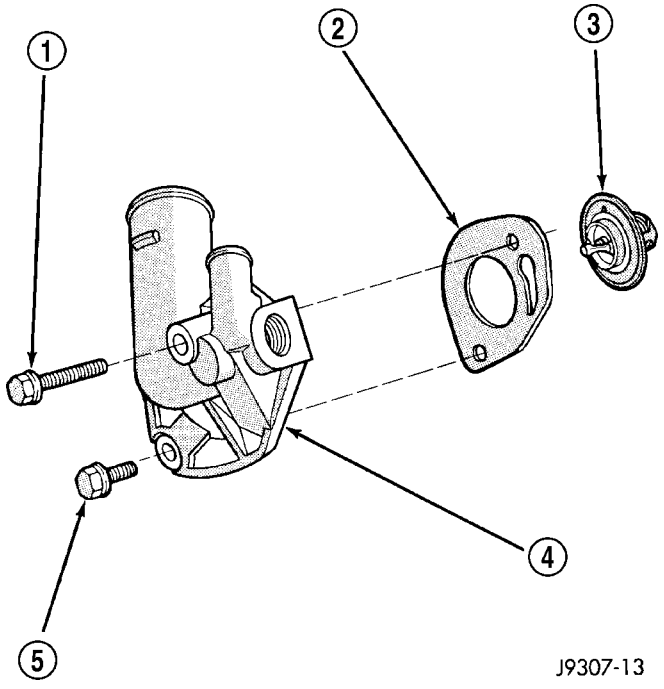

Fig. 8 Thermostat—Typical

1 - LONG BOLT
2 - GASKET
3 - THERMOSTAT
4 - THERMOSTAT HOUSING
5 - SHORT BOLT

OPERATION

The wax pellet is located in a sealed container at the spring end of the thermostat. When heated, the pellet expands, overcoming closing spring tension and water pump pressure to force the valve to open.

REMOVAL

REMOVAL

WARNING: DO NOT LOOSEN THE RADIATOR DRAINCOCK WITH THE SYSTEM HOT AND PRESSURIZED. SERIOUS BURNS FROM THE COOLANT CAN OCCUR.

DO NOT WASTE reusable coolant. If the solution is clean, drain the coolant into a clean container for reuse.

(1) Drain the coolant from the radiator until the level is below the thermostat housing (Refer to 7 - COOLING - STANDARD PROCEDURE).

WARNING: CONSTANT TENSION HOSE CLAMPS ARE USED ON MOST COOLING SYSTEM HOSES. WHEN REMOVING OR INSTALLING, USE ONLY TOOLS DESIGNED FOR SERVICING THIS TYPE OF CLAMP, SUCH AS SPECIAL CLAMP TOOL NUMBER 6094. ALWAYS WEAR SAFETY GLASSES WHEN SERVICING CONSTANT TENSION CLAMPS.

CAUTION: A number or letter is stamped into the tongue of constant tension clamps (Fig. 32). If replacement is necessary, use only an original equipment clamp with matching number or letter.

(2) Remove radiator upper hose and heater hose at thermostat housing.

(3) Disconnect wiring connector at engine coolant temperature sensor.

(4) Remove thermostat housing mounting bolts, thermostat housing, gasket and thermostat (Fig. 9). Discard old gasket.

(5) Clean the gasket mating surfaces.

ENGINE COOLANT THERMOSTAT (Continued)

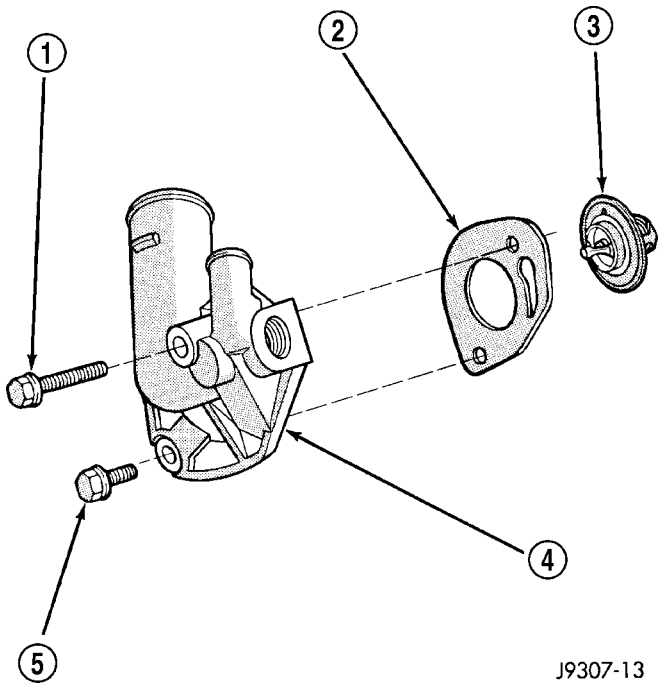

Fig. 9 Thermostat Removal/Installation

1 - LONG BOLT
2 - GASKET
3 - THERMOSTAT
4 - THERMOSTAT HOUSING
5 - SHORT BOLT

REMOVAL

(1) Drain cooling system below thermostat housing level.
(2) Disconnect engine coolant temperature sensor.
(3) Disconnect heater supply hose.
(4) Remove housing attaching bolts (Fig. 10).
(5) Remove housing, gasket and thermostat (Fig. 10).

INSTALLATION

INSTALLATION

(1) Install the replacement thermostat so that the pellet, which is encircled by a coil spring, faces the engine. All thermostats are marked on the outer flange to indicate the proper installed position.
(2) Observe the recess groove in the engine cylinder head (Fig. 11).
(3) Position thermostat into this groove with arrow and air bleed hole on outer flange pointing up.
(4) Install replacement gasket and thermostat housing.

CAUTION: Tightening the thermostat housing unevenly or with the thermostat out of its recess may result in a cracked housing.

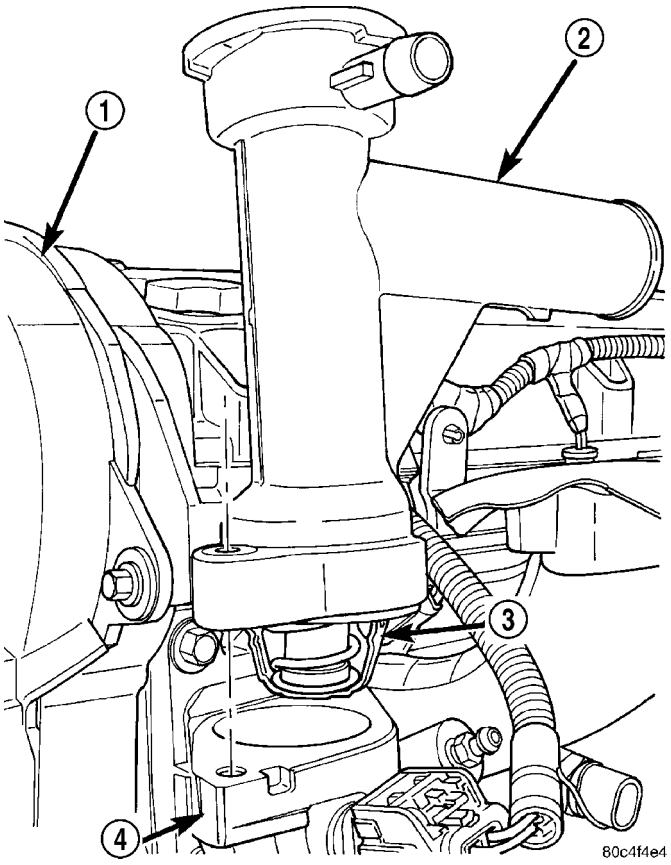

Fig. 10 Thermostat and Coolant Outlet Connector

1 - TIMING BELT COVER
2 - OUTLET CONNECTOR
3 - THERMOSTAT
4 - HOUSING

(5) Tighten the housing bolts to 20 N·m (15 ft. lbs.) torque.
(6) Install hoses to thermostat housing.
(7) Install electrical connector to coolant temperature sensor.
(8) Be sure that the radiator draincock is tightly closed. Fill the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).
(9) Start and warm the engine. Check for leaks.

INSTALLATION

(1) Clean all gasket sealing surfaces.
(2) Place a new gasket (dipped in clean water) on the coolant outlet connector surface. Position thermostat with air bleed at 12 o'clock position in thermostat housing (Fig. 12).
(3) Position the coolant outlet connector and gasket over the thermostat, making sure thermostat is seated in the thermostat housing.
(4) Position outlet connector to thermostat housing and install bolts. Tighten bolts to 28 N·m (20 ft. lbs.).
(5) Install radiator hose to coolant outlet housing.
(6) Connect engine coolant temperature sensor.

ENGINE COOLANT THERMOSTAT (Continued)

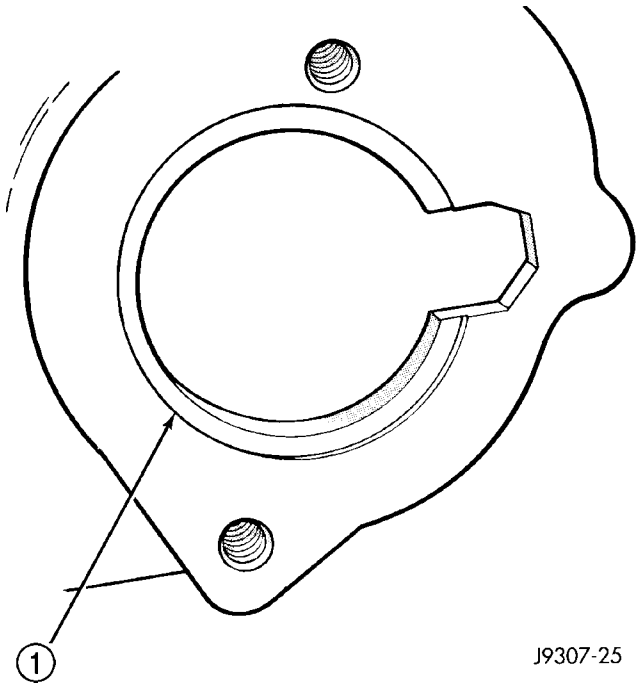

Fig. 11 Thermostat Recess

1 - GROOVE

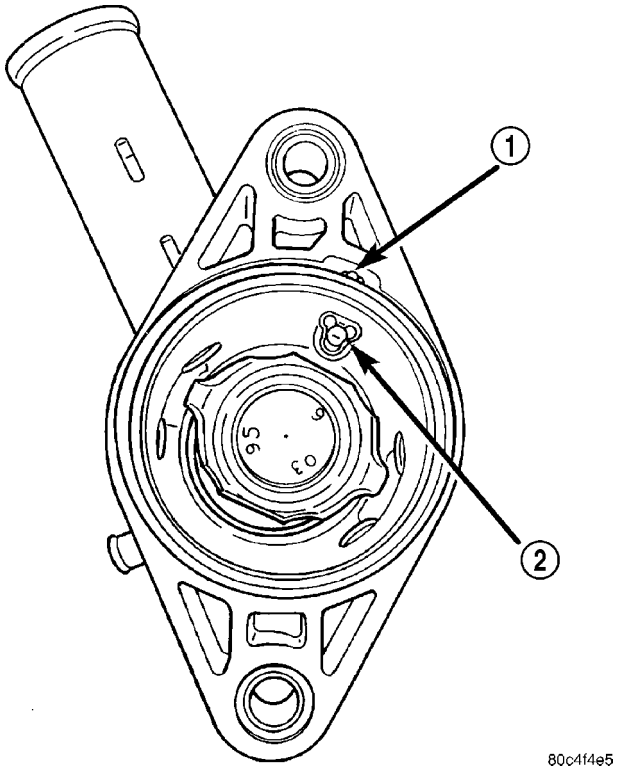

Fig. 12 Thermostat Position

1 - LOCATOR NOTCH
2 - AIR BLEED

(7) Fill cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).

FAN DRIVE VISCOUS CLUTCH

DESCRIPTION

CAUTION: Engines equipped with serpentine drive belts have reverse rotating fans and viscous fan drives. They are marked with the word REVERSE to designate their usage. Installation of the wrong fan or viscous fan drive can result in engine overheating.

The thermal viscous fan drive (Fig. 13) and (Fig. 14) is a silicone-fluid-filled coupling used to connect the fan blades to the water pump shaft. The coupling allows the fan to be driven in a normal manner. This is done at low engine speeds while limiting the top speed of the fan to a predetermined maximum level at higher engine speeds.

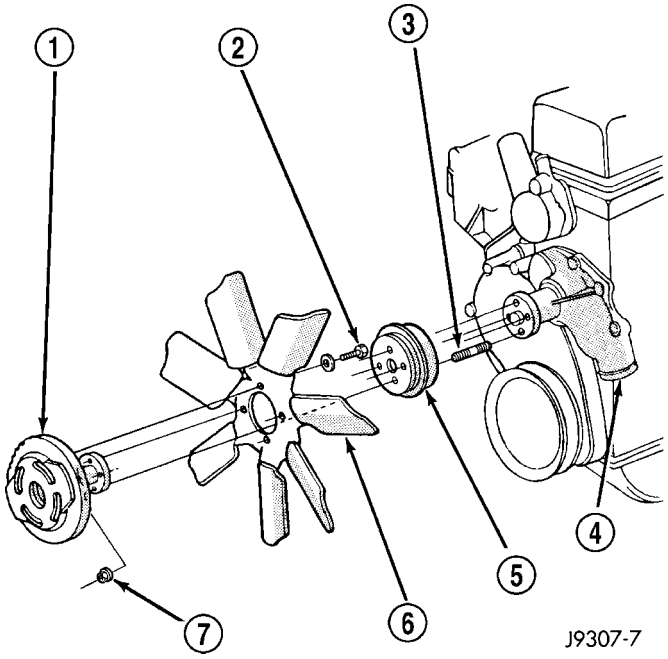

Fig. 13 Water Pump Mounted Fan Drive - 2.5L Engine

1 - THERMAL VISCOUS FAN DRIVE
2 - (4) FAN BLADE-TO-VISCOUS DRIVE BOLTS
3 - (4) FAN HUB-TO-PUMP PULLEY STUDS
4 - WATER PUMP
5 - WATER PUMP PULLEY
6 - FAN BLADE
7 - (4) FAN HUB-TO-PUMP PULLEY NUTS

FAN DRIVE VISCOUS CLUTCH (Continued)

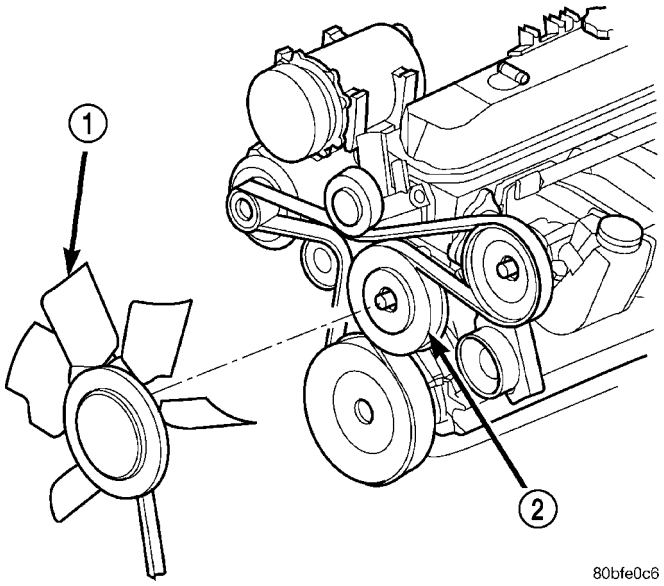

Fig. 14 Water Pump Mounted Fan Drive - 4.0L Engine

1 - FAN AND FAN DRIVE
2 - WATER PUMP PULLEY

OPERATION

A thermostatic bimetallic spring coil is located on the front face of the viscous fan drive unit (a typical viscous unit is shown in (Fig. 15) (Fig. 16). This spring coil reacts to the temperature of the radiator discharge air. It engages the viscous fan drive for higher fan speed if the air temperature from the radiator rises above a certain point. Until additional engine cooling is necessary, the fan will remain at a reduced rpm regardless of engine speed.

Only when sufficient heat is present, will the viscous fan drive engage. This is when the air flowing through the radiator core causes a reaction to the bimetallic coil. It then increases fan speed to provide the necessary additional engine cooling.

Once the engine has cooled, the radiator discharge temperature will drop. The bimetallic coil again reacts and the fan speed is reduced to the previous disengaged speed.

DIAGNOSIS AND TESTING

VISCOUS FAN DRIVE

NOISE

NOTE: It is normal for fan noise to be louder (roaring) when:

- The under hood temperature is above the engagement point for the viscous drive coupling. This may occur when ambient (outside air temperature) is very high.
- Engine loads and temperatures are high such as when towing a trailer.
- Cool silicone fluid within the fan drive unit is being redistributed back to its normal disengaged (warm) position. This can occur during the first 15 seconds to one minute after engine start-up on a cold engine.

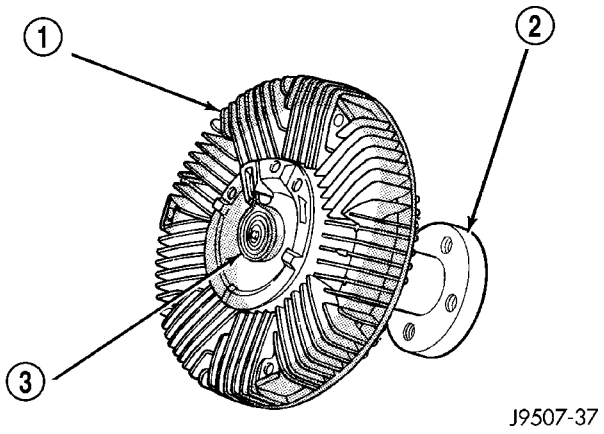

Fig. 15 Viscous Fan Drive - 2.5L Engine

1 - VISCOUS FAN DRIVE
2 - MOUNTING HUB
3 - THERMOSTATIC SPRING

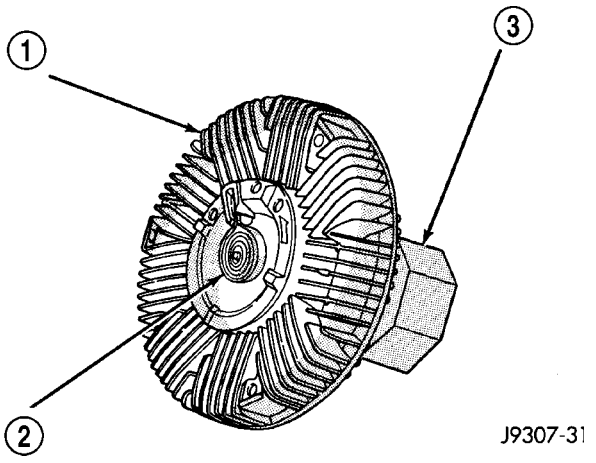

Fig. 16 Viscous Fan Drive - 4.0L Engine

1 - VISCOUS FAN DRIVE
2 - THERMOSTATIC SPRING
3 - MOUNTING NUT TO WATER PUMP HUB

LEAKS

Viscous fan drive operation is not affected by small oil stains near the drive bearing. If leakage appears excessive, replace the fan drive unit.

TESTING

If the fan assembly free-wheels without drag (the fan blades will revolve more than five turns when

FAN DRIVE VISCOUS CLUTCH (Continued)

spun by hand), replace the fan drive. This spin test must be performed when the engine is cool.

For the following test, the cooling system must be in good condition. It also will ensure against excessively high coolant temperature.

WARNING: BE SURE THAT THERE IS ADEQUATE FAN BLADE CLEARANCE BEFORE DRILLING.

(1) Drill a 3.18 mm (1/8 in.) diameter hole in the top center of the fan shroud.

(2) Obtain a dial thermometer with an 8 inch stem (or equivalent). It should have a range of -18° - 105°C (-4°- 220° F). Insert thermometer through the hole in the shroud. Be sure that there is adequate clearance from the fan blades.

(3) Connect a tachometer and an engine ignition timing light (timing light is to be used as a strobe light).

(4) Block the air flow through the radiator. Secure a sheet of plastic in front of the radiator (or air conditioner condenser). Use tape at the top to secure the plastic and be sure that the air flow is blocked.

(5) Be sure that the air conditioner (if equipped) is turned off.

WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING. DO NOT STAND IN A DIRECT LINE WITH THE FAN. DO NOT PUT YOUR HANDS NEAR THE PULLEYS, BELTS OR FAN. DO NOT WEAR LOOSE CLOTHING.

(6) Start the engine and operate at 2400 rpm. Within ten minutes the air temperature (indicated on the dial thermometer) should be up to 88° C (190° F). Fan drive **engagement** should have started to occur at between 74° - 85° C (165° - 185° F). Engagement is distinguishable by a definite **increase** in fan flow noise (roaring). The timing light also will indicate an increase in the speed of the fan.

(7) When the air temperature reaches 88° C (190° F), remove the plastic sheet. Fan drive **disengagement** should have started to occur at between 57° - 82° C (135° - 180° F). A definite **decrease** of fan flow noise (roaring) should be noticed. If not, replace the defective viscous fan drive unit.

CAUTION: Engines equipped with serpentine drive belts have reverse rotating fans and viscous fan drives. They are marked with the word REVERSE to designate their usage. Installation of the wrong fan or viscous fan drive can result in engine overheating.

CAUTION: If the viscous fan drive is replaced because of mechanical damage, the cooling fan blades should also be inspected. Inspect for fatigue cracks, loose blades, or loose rivets that could have resulted from excessive vibration. Replace fan blade assembly if any of these conditions are found. Also inspect water pump bearing and shaft assembly for any related damage due to a viscous fan drive malfunction.

REMOVAL

(1) The thermal viscous fan drive/fan blade assembly is attached (threaded) to water pump hub shaft. Remove fan blade/viscous fan drive assembly from water pump by turning mounting nut counterclockwise as viewed from front. Threads on viscous fan drive are **RIGHT HAND.** Using a suitable fan wrench loosen the fan drive (Fig. 17).

(2) Remove accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(3) Some models with certain engines may require the removal of the fan shroud to remove the viscous fan drive. The fan shroud and fan blade/viscous fan drive should be removed from the vehicle as one assembly.

(4) After removing fan blade/viscous fan drive assembly, **do not** place thermal viscous fan drive in horizontal position. If stored horizontally, silicone fluid in viscous fan drive could drain into its bearing assembly and contaminate lubricant.

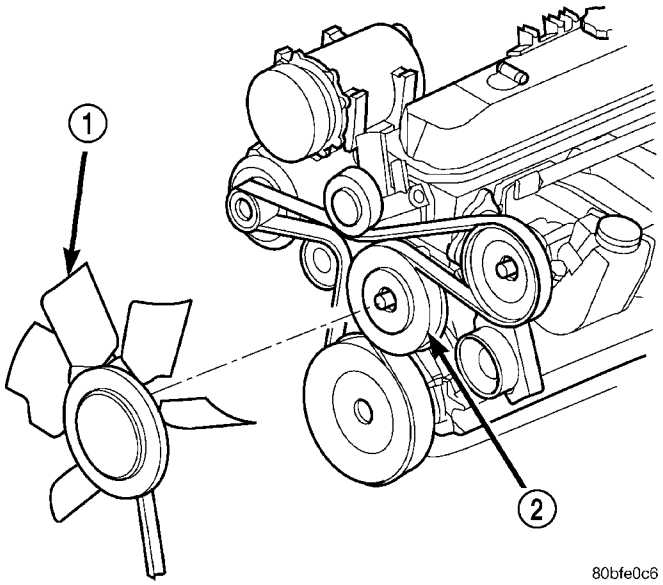

Fig. 17 Fan and Fan Drive - 4.0L Engine

1 - FAN AND FAN DRIVE
2 - WATER PUMP PULLEY

INSTALLATION

(1) Assemble fan blade to viscous fan drive. Tighten mounting bolts to 27 N·m (20 ft. lbs.) torque.

(2) Thread the fan and fan drive onto the water pump pulley.

FAN DRIVE VISCOUS CLUTCH (Continued)

CAUTION: When installing a serpentine accessory drive belt, the belt MUST be routed correctly. If not, the engine may overheat due to the water pump rotating in the wrong direction.

(3) Install accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

RADIATOR

DESCRIPTION

CAUTION: Plastic tanks, while stronger than brass, are subject to damage by impact, such as wrenches, mishandling, etc.

A heavy duty down-flow aluminum/plastic radiator is used (Fig. 18). The radiator consists of an aluminum core and plastic end tanks, which are fastened to the core with clinch tabs and sealed with a high temperature rubber gasket. On automatic transmission equipped vehicles, the lower tank contains a concentric-tube transmission oil cooler.

If the plastic tank has been damaged, individual parts are not available, and the radiator must be replaced.

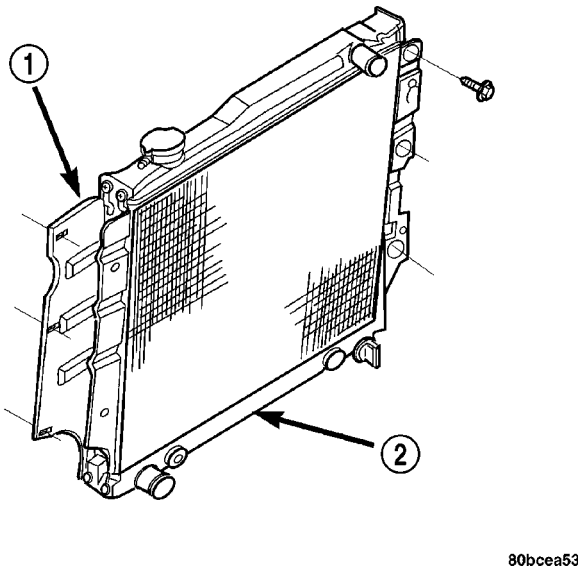

Fig. 18 Downflow Radiator - Typical

1 - DOWNFLOW RADIATOR
2 - INTEGRAL TRANSMISSION OIL COOLER (INTERNAL TO RADIATOR)

OPERATION

As air passes through the radiator core, the heat within the coolant is dissipated into the ambient air.

DIAGNOSIS AND TESTING - RADIATOR COOLANT FLOW

The following procedure will determine if coolant is flowing through the cooling system.

If engine is cold, idle engine until normal operating temperature is reached. Then feel the upper radiator hose. If hose is hot, the thermostat is open and water is circulating through cooling system.

REMOVAL

WARNING: DO NOT REMOVE THE CYLINDER BLOCK DRAIN PLUGS OR LOOSEN THE RADIATOR DRAINCOCK WITH THE SYSTEM HOT AND PRESSURIZED. SERIOUS BURNS FROM THE COOLANT CAN OCCUR.

Do not waste reusable coolant. If the solution is clean, drain the coolant into a clean container for reuse.

WARNING: CONSTANT TENSION HOSE CLAMPS ARE USED ON MOST COOLING SYSTEM HOSES. WHEN REMOVING OR INSTALLING, USE ONLY TOOLS DESIGNED FOR SERVICING THIS TYPE OF CLAMP, SUCH AS SPECIAL CLAMP TOOL NUMBER 6094 (Fig. 19). ALWAYS WEAR SAFETY GLASSES WHEN SERVICING CONSTANT TENSION CLAMPS.

CAUTION: A number or letter is stamped into the tongue of constant tension clamps (Fig. 20). If replacement is necessary, use only an original equipment clamp with matching number or letter.

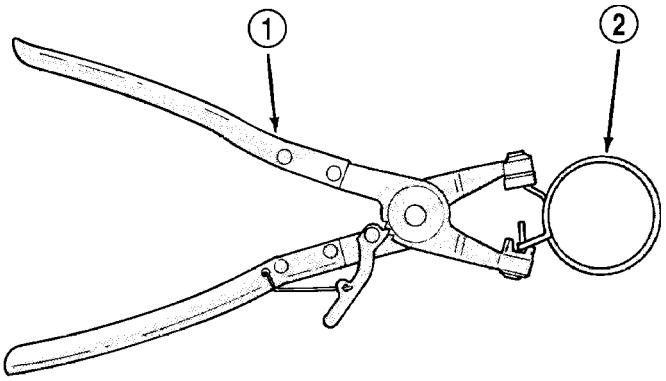

Fig. 19 Hose Clamp Tool - Typical

1 - HOSE CLAMP TOOL 6094
2 - HOSE CLAMP

RADIATOR (Continued)

CAUTION: When removing the radiator or A/C condenser for any reason, note the location of all radiator-to-body and radiator-to-A/C condenser rubber air seals (Fig. 21). These are used at the top, bottom and sides of the radiator and A/C condenser. To prevent overheating, these seals must be installed to their original positions.

(1) Disconnect negative battery cable at battery.

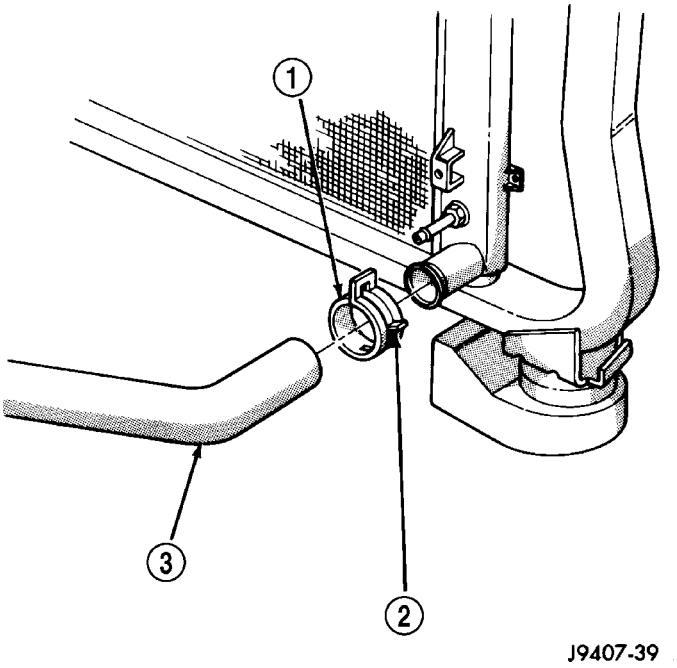

Fig. 20 Clamp Number/Letter Location - Typical

1 - TYPICAL CONSTANT TENSION HOSE CLAMP
2 - CLAMP NUMBER/LETTER LOCATION
3 - TYPICAL HOSE

(2) Observe the previous **WARNING.** Remove the radiator cap.

(3) Remove the condenser lower seal from the lower core support (Fig. 22).

(4) Drain cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE) drain coolant into a clean container for reuse.

(5) Remove radiator upper and lower hose clamps. Remove radiator hoses.

(6) Disconnect coolant reserve/overflow tank hose from radiator.

(7) 2.4L ONLY - Remove the electric cooling fan (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - REMOVAL).

(8) 4.0L ONLY - Remove the reservoir tank and power steering reservoir and position out of the way.

(9) 4.0L ONLY - Remove the four fan shroud mounting bolts (Fig. 23). Position the fan shroud back over the fan blades.

(10) If equipped, disconnect and plug automatic transmission fluid cooler lines.

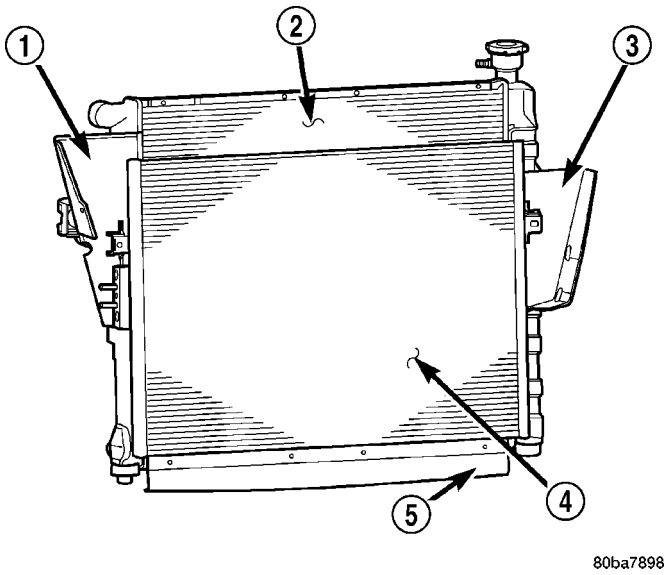

Fig. 21 Air Seals - Typical

1 - AIR DAM
2 - RADIATOR
3 - AIR DAM
4 - A/C CONDENSER
5 - AIR SEAL

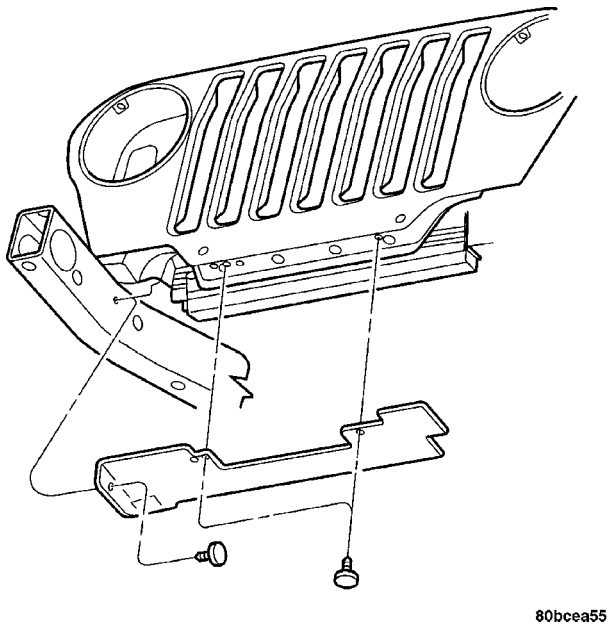

Fig. 22 Condenser Lower Seal

(11) Remove six radiator mounting bolts. Position the front axle vent hose (Fig. 23) to the side.

(12) The lower part of radiator is equipped with two alignment dowel pins (Fig. 24). They are located on the bottom of radiator tank and fit into rubber grommets. These rubber grommets are pressed into the radiator lower crossmember.

(13) Lift radiator straight up and out of vehicle taking care not to damage radiator fins.

RADIATOR (Continued)

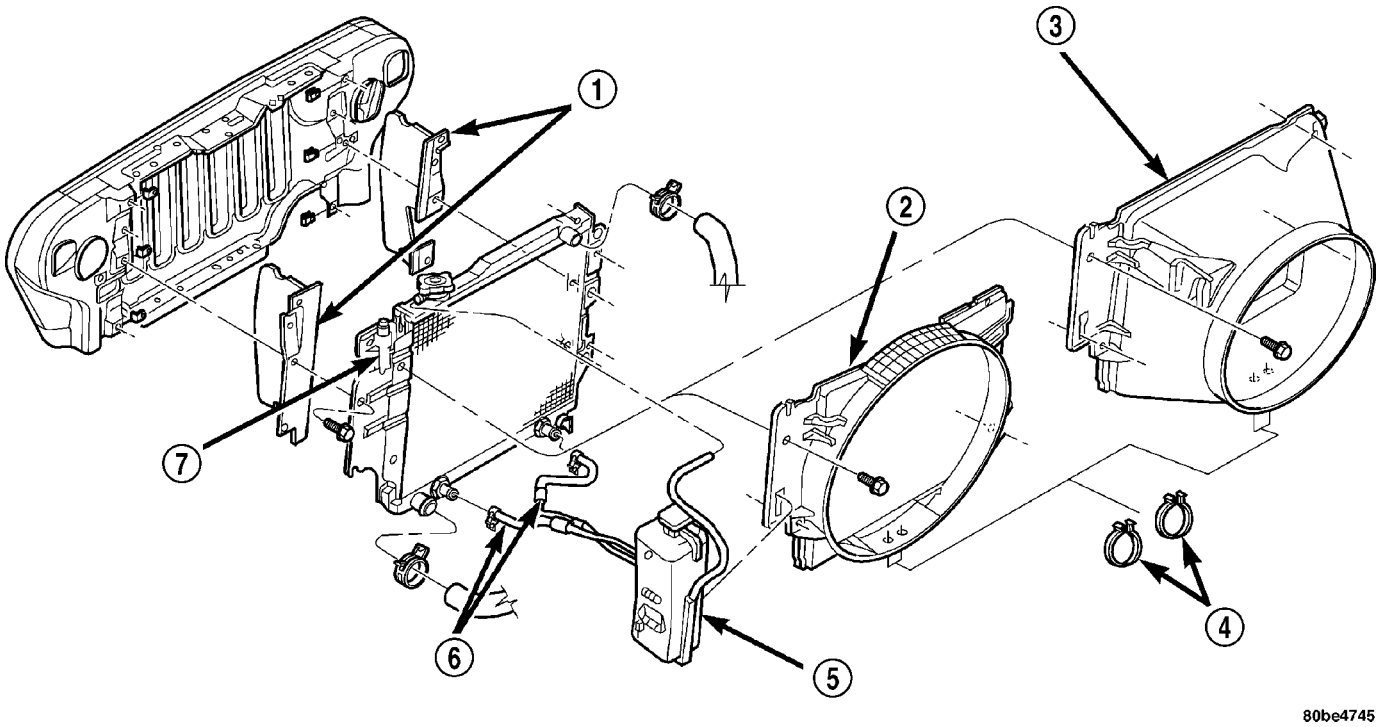

Fig. 23 Radiator - Remove/Install

1 - A/C CONDENSER SEALS
2 - FAN SHROUD (4.0L)
3 - FAN SHROUD (2.5L)
4 - TRANSMISSION OIL COOLER LINES RETAINER CLIPS
5 - COOLANT RECOVERY BOTTLE
6 - TRANSMISSION COOLER LINES (IF EQUIPPED)
7 - FRONT AXLE VENT HOSE

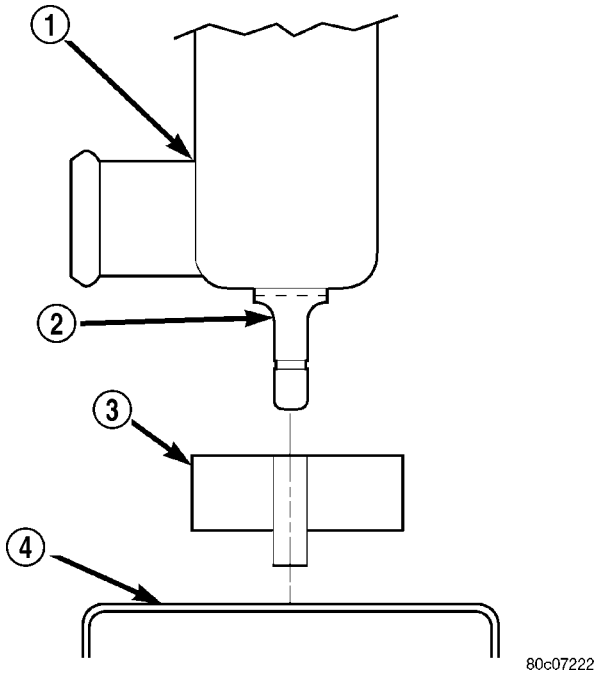

Fig. 24 Radiator Alignment Dowels - Typical

1 - RADIATOR
2 - ALIGNMENT DOWEL
3 - RADIATOR LOWER ISOLATOR
4 - RADIATOR LOWER CROSSMEMBER

(14) When removing radiator, note position of the rubber seals located on the top and bottom of radiator (on certain models only) (Fig. 23). To prevent possible overheating, these seals must be installed to their original positions.

CLEANING

Clean radiator fins. With the engine cold, apply cold water and compressed air to the back (engine side) of the radiator to flush the radiator and/or A/C condenser of debris.

INSPECTION

The radiator cooling fins should be checked for damage or deterioration. Inspect cooling fins to make sure they are not bent or crushed, these areas result in reduced heat exchange causing the cooling system to operate at higher temperatures. Inspect the plastic end tanks for cracks, damage or leaks.

Inspect the radiator neck for damage or distortion.

RADIATOR (Continued)

INSTALLATION

CAUTION: Before installing the radiator or A/C condenser, be sure the radiator-to-body and radiator-to-A/C condenser rubber air seals are properly fastened to their original positions. These are used at the top, bottom and sides of the radiator and A/C condenser. To prevent overheating, these seals must be installed to their original positions.

(1) Guide the two radiator alignment dowels into the rubber grommets located in lower radiator crossmember. Install and tighten the six mounting bolts (Fig. 23) to 8 N·m (72 in. lbs.) torque.
(2) Close radiator draincock.
(3) 2.4L ONLY - Install the electric cooling fan (Refer to 7 - COOLING/ENGINE/RADIATOR FAN - INSTALLATION).
(4) Position fan shroud and power steering reservoir tank (if equipped). Install and tighten four mounting bolts to 8 N·m (72 in. lbs.) torque.
(5) If equipped, remove plugs and connect automatic transmission fluid cooler lines and constant tension clamps.

CAUTION: The tangs on the hose clamps must be positioned straight down.

(6) Connect radiator hoses and install hose clamps.
(7) Position and install the condenser lower seal (Fig. 22).
(8) Connect battery negative cable.
(9) Fill cooling system with correct coolant (Refer to 7 - COOLING - STANDARD PROCEDURE).
(10) Connect coolant recovery bottle hose.
(11) Install radiator cap.
(12) Check and adjust automatic transmission fluid level (if equipped).
(13) Start engine and check for leaks.

WATER PUMP - 2.4L

DESCRIPTION

The water pump has a cast aluminum body and housing with a stamped steel impeller. The water pump bolts directly to the block (Fig. 25). The cylinder block to water pump seal is provided by a rubber O-ring. The water pump is driven by the engine timing belt.

OPERATION

The water pump is the heart of the cooling system. The coolant is pumped through the engine block, cylinder head, heater core, and radiator.

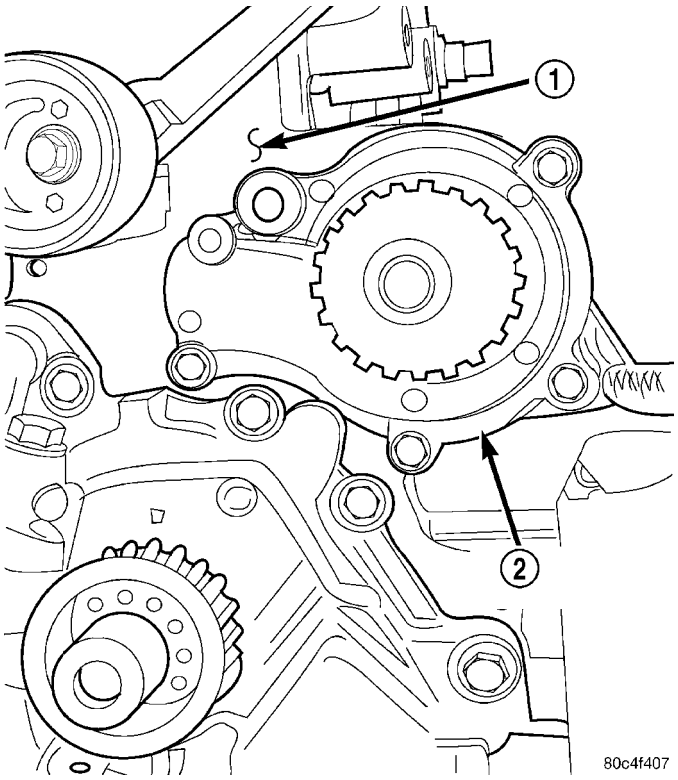

Fig. 25 Water Pump

1 - CYLINDER BLOCK
2 - WATER PUMP

DIAGNOSIS AND TESTING

WATER PUMP

A quick flow test to determine if the water pump is working effectively is to check heater system for proper operation. A defective pump will not provide an adequate flow of heated coolant through the system.

WARNING: DO NOT REMOVE THE COOLING SYSTEM PRESSURE CAP WITH THE SYSTEM HOT AND UNDER PRESSURE BECAUSE SERIOUS BURNS FROM COOLANT CAN OCCUR.

Another flow test to help determine water pump operation is as follows:
(1) Remove cooling system pressure cap.
(2) Remove a small amount of coolant from the system.
(3) Start the engine and warm up until thermostat opens.
(4) With the thermostat open and coolant level low, visually inspect for coolant flow. If flow is present, the water pump is pumping coolant through the system.

WATER PUMP - 2.4L (Continued)

REMOVAL - 2.4L

(1) Drain cooling system (Refer to 7 - COOLING/ ENGINE - STANDARD PROCEDURE).

(2) Remove timing belt (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT AND SPROCKETS - REMOVAL).

(3) Remove camshaft sprockets and rear timing belt cover (Refer to 9 - ENGINE/VALVE TIMING/ TIMING BELT AND SPROCKETS - REMOVAL).

(4) Remove screws attaching water pump to engine. Remove pump (Fig. 26).

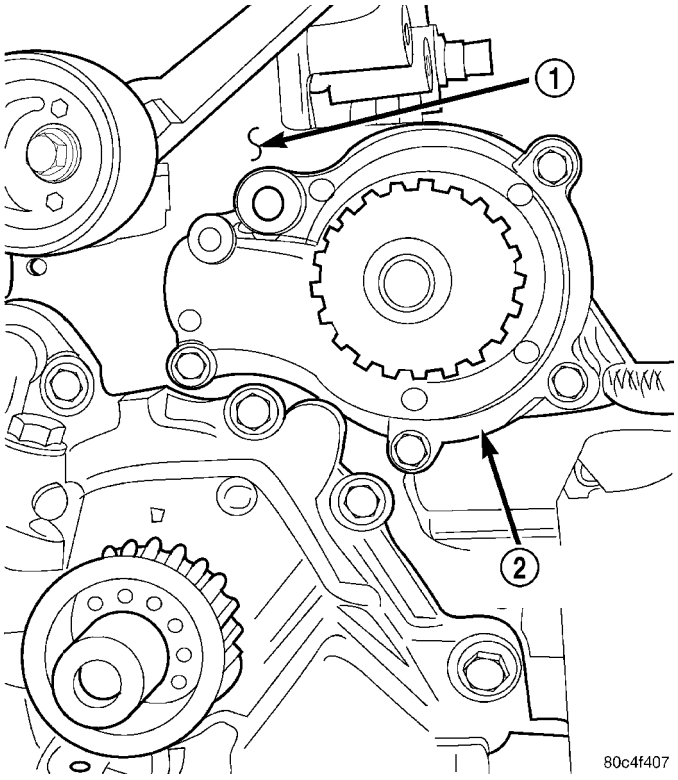

Fig. 26 Water Pump

1 - CYLINDER BLOCK
2 - WATER PUMP

INSTALLATION

INSTALLATION 2.4L

(1) Apply Mopar® Dielectric Grease to new O-ring before installation (Fig. 27).

(2) Install O-ring gasket in water pump body groove (Fig. 27).

CAUTION: Make sure O-ring gasket is properly seated in water pump groove before tightening screws. An improperly located O-ring may cause damage to the O-ring, resulting in a coolant leak.

(3) Assemble pump body to block (Fig. 26) and tighten screws to 12 N·m (105 in. lbs.).

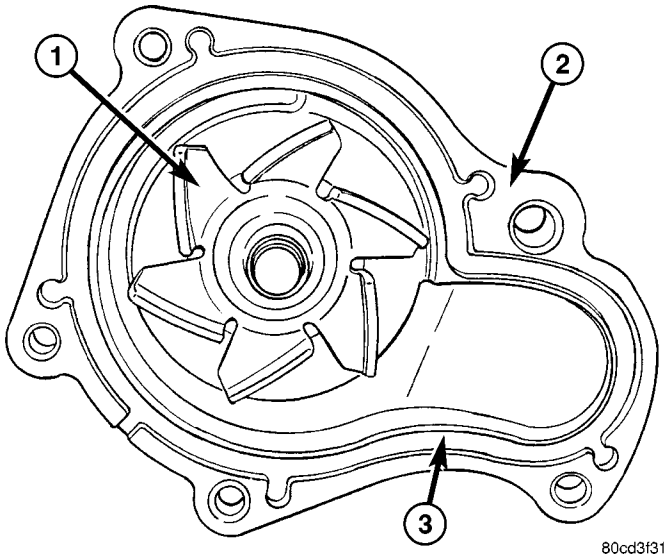

Fig. 27 Water Pump Body

1 - IMPELLER
2 - WATER PUMP BODY
3 - O-RING LOCATING GROOVE

(4) Rotate pump by hand to check for freedom of movement.

(5) Fill cooling system (Refer to 7 - COOLING/ENGINE - STANDARD PROCEDURE). Pressurize cooling system to 103 Kpa (15 psi) with pressure tester and check water pump shaft seal and O-ring for leaks.

(6) Install rear timing belt cover and camshaft sprockets (Refer to 9 - ENGINE/VALVE TIMING/ TIMING BELT AND SPROCKETS - INSTALLATION).

(7) Install timing belt (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT AND SPROCKETS - INSTALLATION).

WATER PUMP - 4.0L

DESCRIPTION

CAUTION: The 4.0L 6-cylinder engines is equipped with a reverse (counterclockwise) rotating water pump and thermal viscous fan drive assembly. REVERSE is stamped or imprinted on the cover of the viscous fan drive and inner side of the fan. The letter R is stamped into the back of the water pump impeller. Engines from previous model years, depending upon application, may have been equipped with a forward (clockwise) rotating water pump. Installation of the wrong water pump or viscous fan drive will cause engine over heating.

WATER PUMP - 4.0L (Continued)

This aluminum water pump (Fig. 28) is the heart of the cooling system. The water pump is located at the front of the cylinder block, above the timing chain cover

The water pump impeller is pressed onto the rear of a shaft that rotates in bearings pressed into the housing. The housing has two small holes to allow seepage to escape. The water pump seals are lubricated by the antifreeze in the coolant mixture. No additional lubrication is necessary.

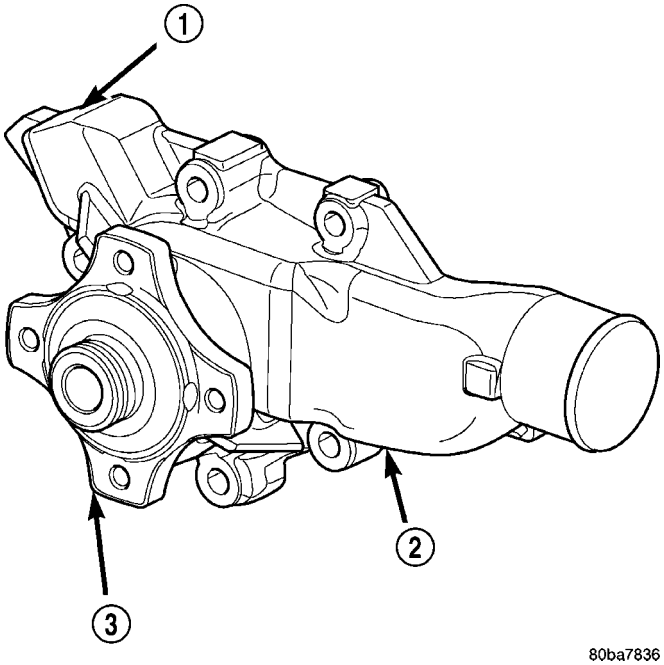

Fig. 28 Water Pump - Typical

1 - HEATER HOSE FITTING BORE
2 - WATER PUMP
3 - WATER PUMP HUB

OPERATION

A centrifugal water pump circulates coolant through the water jackets, passages, intake manifold, radiator core, cooling system hoses and heater core. The pump is driven from the engine crankshaft by a single serpentine drive belt on all engines.

REMOVAL

The water pump can be removed without discharging the air conditioning system (if equipped).

CAUTION: All engines have a reverse (counterclockwise) rotating water pump. The letter R is stamped into the back of the water pump impeller (Fig. 29) to identify. Engines from previous model years, depending upon application, may be equipped with a forward (clockwise) rotating water pump. Installation of the wrong water pump will cause engine over heating.

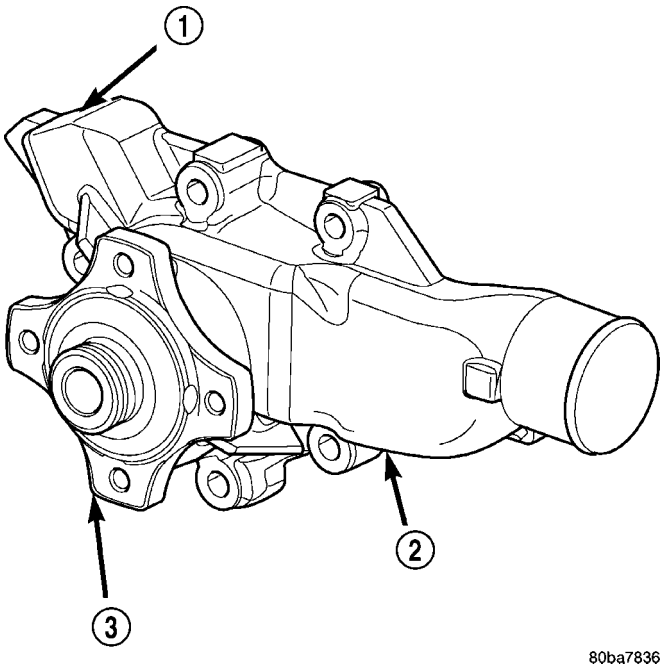

Fig. 29 Water Pump - 4.0L Engine

1 - HEATER HOSE FITTING BORE
2 - WATER PUMP
3 - WATER PUMP HUB

The water pump impeller is pressed on the rear of the pump shaft and bearing assembly. The water pump is serviced only as a complete assembly.

WARNING: DO NOT REMOVE THE BLOCK DRAIN PLUG(S) OR LOOSEN RADIATOR DRAINCOCK WITH THE SYSTEM HOT AND UNDER PRESSURE. SERIOUS BURNS FROM COOLANT CAN OCCUR.

DO NOT WASTE reusable coolant. If the solution is clean, drain coolant into a clean container for reuse.

(1) Disconnect negative battery cable at battery.

(2) Drain the cooling system (Refer to 7 - COOLING - STANDARD PROCEDURE).

NOTE: The engine accessory drive belt must be removed prior to removing the fan.

(3) Remove accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - REMOVAL).

(4) The thermal viscous fan drive/fan blade assembly is attached (threaded) to water pump hub shaft. Remove fan blade/viscous fan drive assembly from water pump by turning mounting nut counterclockwise as viewed from front. Threads on viscous fan drive are **RIGHT HAND.** Using a suitable fan wrench loosen the fan drive (Fig. 30).

(5) Remove power steering pump (Refer to 19 - STEERING/PUMP - REMOVAL).

WATER PUMP - 4.0L (Continued)

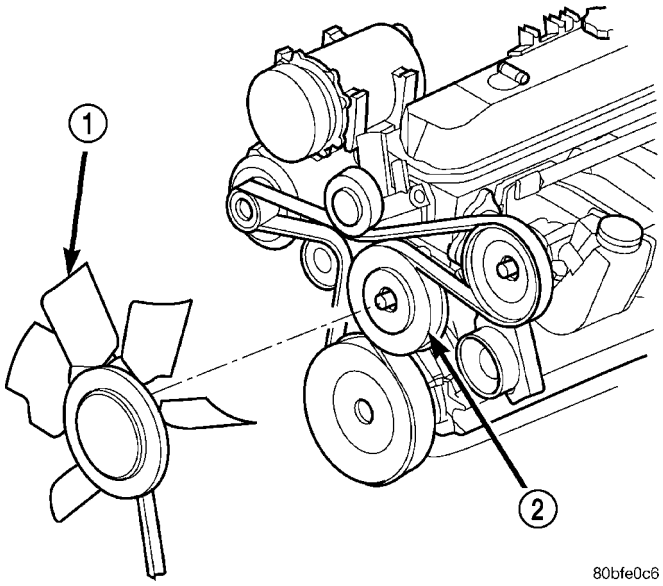

Fig. 30 Fan and Fan Drive Mounting - 4.0L Engine

1 - FAN AND FAN DRIVE
2 - WATER PUMP PULLEY

WARNING: CONSTANT TENSION HOSE CLAMPS ARE USED ON MOST COOLING SYSTEM HOSES. WHEN REMOVING OR INSTALLING, USE ONLY TOOLS DESIGNED FOR SERVICING THIS TYPE OF CLAMP, SUCH AS SPECIAL CLAMP TOOL NUMBER 6094 (Fig. 31). ALWAYS WEAR SAFETY GLASSES WHEN SERVICING CONSTANT TENSION CLAMPS.

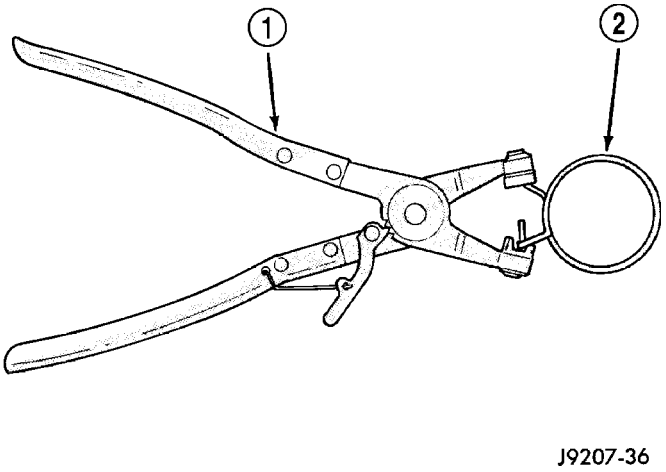

Fig. 31 Hose Clamp Tool - Typical

1 - HOSE CLAMP TOOL 6094
2 - HOSE CLAMP

CAUTION: A number or letter is stamped into the tongue of constant tension clamps (Fig. 32). If replacement is necessary, use only an original equipment clamp with matching number or letter.

(6) Remove lower radiator hose from water pump. Remove heater hose from water pump fitting.

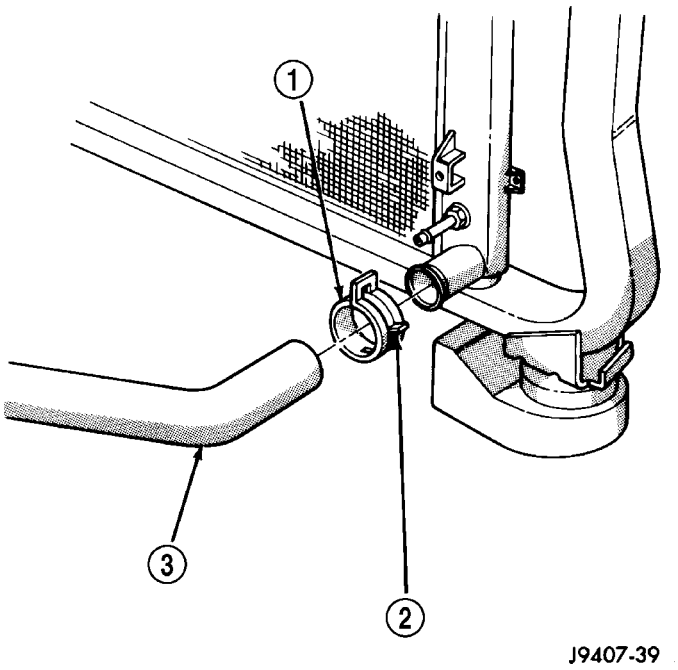

Fig. 32 Clamp Number/Letter Location

1 - TYPICAL CONSTANT TENSION HOSE CLAMP
2 - CLAMP NUMBER/LETTER LOCATION
3 - TYPICAL HOSE

(7) After removing fan blade/viscous fan drive assembly, **do not** place thermal viscous fan drive in horizontal position. If stored horizontally, silicone fluid in viscous fan drive could drain into its bearing assembly and contaminate lubricant.

(8) Remove the four pump mounting bolts (Fig. 33) and remove pump from vehicle. Discard old gasket. Note that one of the four bolts is longer than the other bolts.

(9) If pump is to be replaced, the heater hose fitting must be removed. Note position of fitting before removal.

INSTALLATION

(1) If pump is being replaced, install the heater hose fitting to the pump. Use a sealant on the fitting such as Mopar® Thread Sealant With Teflon. Refer to the directions on the package.

(2) Clean the gasket mating surfaces. If the original pump is used, remove any deposits or other foreign material. Inspect the cylinder block and water pump mating surfaces for erosion or damage from cavitation.

(3) Install the gasket and water pump. The silicone bead on the gasket should be facing the water pump. Also, the gasket is installed dry. Tighten mounting bolts to 23 N·m (200 in. lbs.) torque. Rotate the shaft by hand to be sure it turns freely.

(4) Connect the radiator and heater hoses to the water pump.

WATER PUMP - 4.0L (Continued)

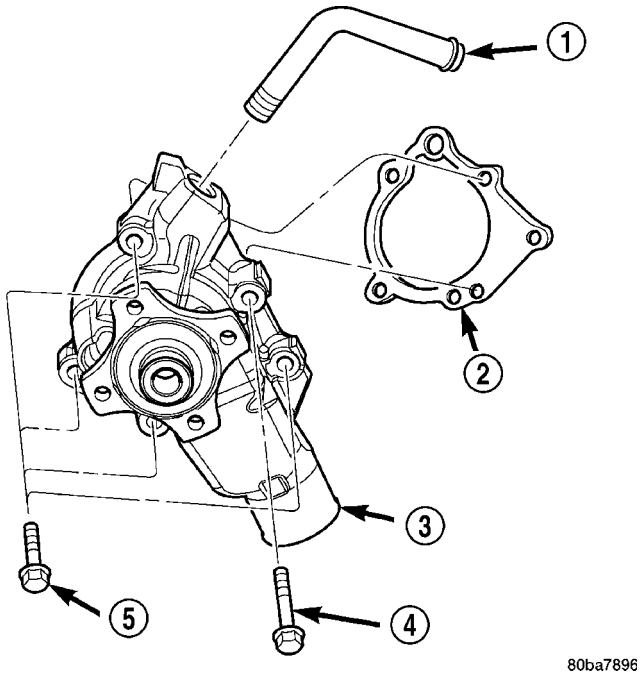

Fig. 33 Water Pump Remove/Install - 4.0L Engine

1 - HEATER HOSE FITTING
2 - PUMP GASKET
3 - WATER PUMP
4 - LONG BOLT
5 - BOLTS (4) SHORT

(5) Install power steering pump (Refer to 19 - STEERING/PUMP - INSTALLATION).

(6) Thread the fan and fan hub into the water pump hub shaft.

CAUTION: When installing the serpentine engine accessory drive belt, the belt MUST be routed correctly. If not, the engine may overheat due to the water pump rotating in the wrong direction. for appropriate belt routing. You may also refer to the Belt Routing Label in the vehicle engine compartment.

(7) Install accessory drive belt (Refer to 7 - COOLING/ACCESSORY DRIVE/DRIVE BELTS - INSTALLATION).

(8) Fill cooling system with coolant (Refer to 7 - COOLING - STANDARD PROCEDURE).

(9) Connect battery cable to battery.

(10) Start and warm the engine. Check for leaks.

RADIATOR PRESSURE CAP

DESCRIPTION

All radiators are equipped with a pressure cap (Fig. 34). This cap releases pressure at some point within a range of 124-145 kPa (18-21 psi). The pressure relief point (in pounds) is engraved on top of the cap

The cooling system will operate at pressures slightly above atmospheric pressure. This results in a higher coolant boiling point allowing increased radiator cooling capacity. The cap contains a spring-loaded pressure relief valve. This valve opens when system pressure reaches the release range of 124-145 kPa (18-21 psi).

A rubber gasket seals the radiator filler neck. This is done to maintain vacuum during coolant cool-down and to prevent leakage when system is under pressure.

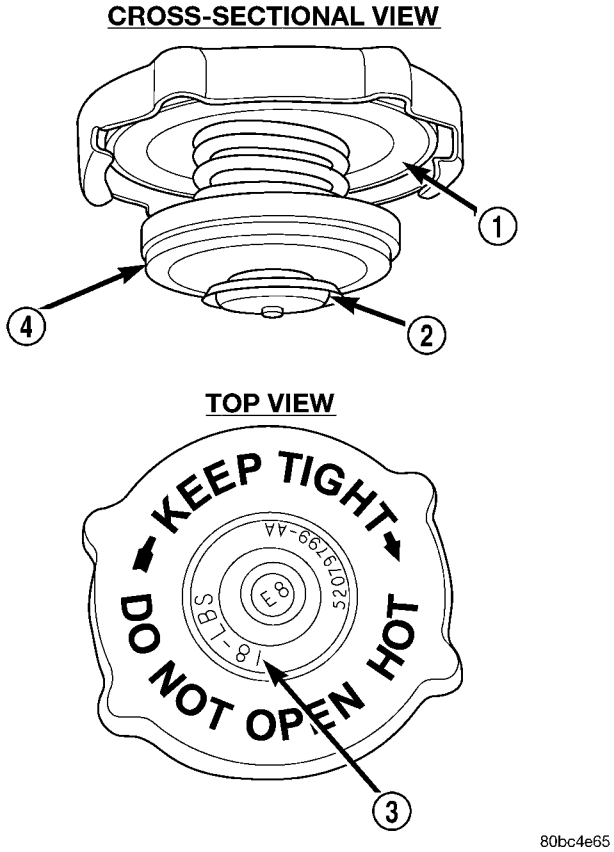

Fig. 34 Radiator Pressure Cap - Typical

1 - FILLER NECK SEAL
2 - VACUUM VENT VALVE
3 - PRESSURE RATING
4 - PRESSURE VALVE

OPERATION

A vent valve in the center of the cap will remain shut as long as the cooling system is pressurized. As the coolant cools, it contracts and creates a vacuum in cooling system. This causes the vacuum valve to open and coolant in reserve/overflow tank to be drawn through connecting hose into radiator. If the vacuum valve is stuck shut, or overflow hose is kinked, radiator hoses will collapse on cool-down.

RADIATOR PRESSURE CAP (Continued)

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - RADIATOR CAP-TO-FILLER NECK SEAL

The pressure cap upper gasket (seal) pressure relief can be tested by removing overflow hose from radiator filler neck nipple. Attach hose of pressure tester tool 7700 (or equivalent) to nipple. It will be necessary to disconnect hose from its adapter for filler neck. Pump air into radiator. The pressure cap upper gasket should relieve at 69-124 kPa (10-18 psi) and hold pressure at a minimum of 55 kPa (8 psi).

WARNING: THE WARNING WORDS - DO NOT OPEN HOT - ON RADIATOR PRESSURE CAP, ARE A SAFETY PRECAUTION. WHEN HOT, PRESSURE BUILDS UP IN COOLING SYSTEM. TO PREVENT SCALDING OR INJURY, RADIATOR CAP SHOULD NOT BE REMOVED WHILE SYSTEM IS HOT AND/OR UNDER PRESSURE.

Do not remove radiator cap at any time **except** for the following purposes:

(1) Check and adjust antifreeze freeze point.
(2) Refill system with new antifreeze.
(3) Conducting service procedures.
(4) Checking for vacuum leaks.

WARNING: IF VEHICLE HAS BEEN RUN RECENTLY, WAIT AT LEAST 15 MINUTES BEFORE REMOVING RADIATOR CAP. WITH A RAG, SQUEEZE RADIATOR UPPER HOSE TO CHECK IF SYSTEM IS UNDER PRESSURE. PLACE A RAG OVER CAP AND WITHOUT PUSHING CAP DOWN, ROTATE IT COUNTERCLOCKWISE TO FIRST STOP. ALLOW FLUID TO ESCAPE THROUGH THE COOLANT RESERVE/OVERFLOW HOSE INTO RESERVE/OVERFLOW TANK. SQUEEZE RADIATOR UPPER HOSE TO DETERMINE WHEN PRESSURE HAS BEEN RELEASED. WHEN COOLANT AND STEAM STOP BEING PUSHED INTO TANK AND SYSTEM PRESSURE DROPS, REMOVE RADIATOR CAP COMPLETELY.

DIAGNOSIS AND TESTING - RADIATOR CAP

Remove cap from radiator. Be sure that sealing surfaces are clean. Moisten rubber gasket with water and install cap on pressure tester 7700 or an equivalent (Fig. 35).

Operate tester pump to bring pressure to 117 kPa (17 psi) on gauge. If pressure cap fails to hold pressure of at least 110 kPa (16 psi) replace cap. Refer to the following **CAUTION**.

The pressure cap may test properly while positioned on tool 7700 (or equivalent). It may not hold pressure or vacuum when installed on radiator. If so, inspect radiator filler neck and cap's top gasket for damage. Also inspect for dirt or distortion that may prevent cap from sealing properly.

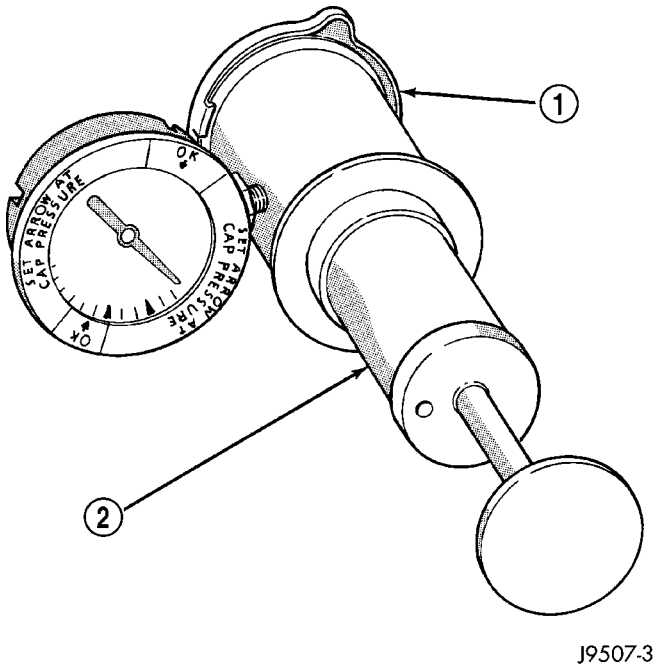

Fig. 35 Pressure Testing Radiator Cap - Typical

1 - PRESSURE CAP
2 - TYPICAL COOLING SYSTEM PRESSURE TESTER

CAUTION: Radiator pressure testing tools are very sensitive to small air leaks, which will not cause cooling system problems. A pressure cap that does not have a history of coolant loss should not be replaced just because it leaks slowly when tested with this tool. Add water to tool. Turn tool upside down and recheck pressure cap to confirm that cap needs replacement.

CLEANING

Use only a mild soap and water to clean the radiator cap. Using any type solvent may cause damage to the seal in the radiator cap.

INSPECTION

Hold cap at eye level, right side up. The vent valve (Fig. 34) at bottom of cap should open. If rubber gasket has swollen and prevents vent valve from opening, replace cap.

Hold cap at eye level, upside down. If any light can be seen between vent valve and rubber gasket, replace cap. **Do not use a replacement cap that has a spring to hold vent shut.** A replacement cap must be the type designed for a coolant reserve/overflow system with a completely sealed diaphragm

RADIATOR PRESSURE CAP (Continued)

spring and a rubber gasket. This gasket is used to seal to radiator filler neck top surface. Use of proper cap will allow coolant return to radiator.

RADIATOR FAN

DESCRIPTION

The electric cooling fan replaces the engine driven mechanical fan. The electric cooling fan is integral to the fan shroud and is located between the radiator and the engine.

The electric fan is controlled by the electronic control module (ECM).

The electric cooling fan is not serviceable. Any failure of the fan blade, electric motor or fan shroud requires replacement of the fan module.

CAUTION: Do not attempt to service the electric cooling fan or fan blades separately, replace the cooling module as an assembly. Failure to do so may cause severe damage to the electric cooling fan assembly.

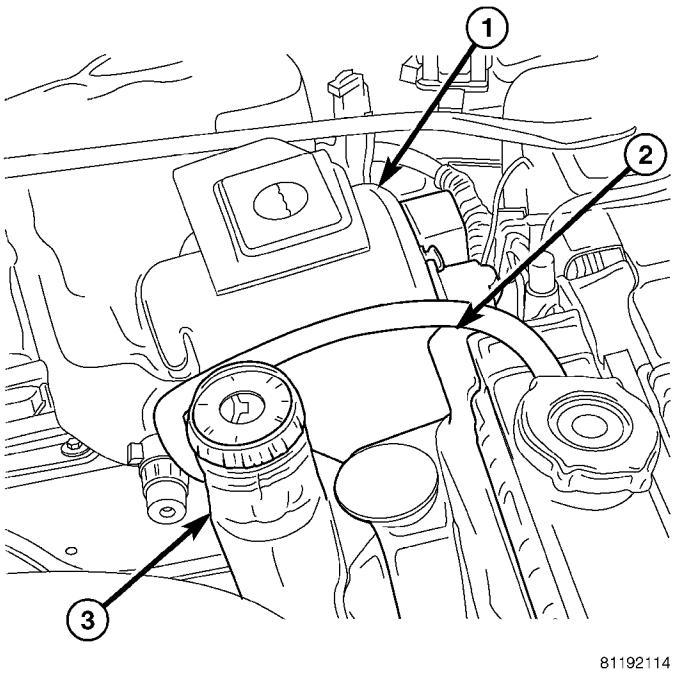

Fig. 36 Coolant Recovery Bottle

1 - COOLANT RECOVERY BOTTLE
2 - COOLING FAN ASSEMBLY
2 - POWER STEERING RESERVOIR

REMOVAL

REMOVAL

(1) Remove the coolant recovery bottle and position out of the way (Fig. 36).

(2) Remove and position the power steering reservoir out of the way.

(3) Disconnect the electrical connector.

(4) Remove the mounting screws.

(5) Remove the cooling fan assembly (Fig. 37).

REMOVAL - 4.0L

(1) Remove the thermal viscous fan drive/fan blade assembly (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - REMOVAL).

(2) Remove the four fan retaining bolts.

(3) Seperate the fan blade from the thermal viscous fan drive.

CLEANING

Clean the fan blades using a mild soap and water. Do not use an abrasive to clean the blades.

INSPECTION

WARNING: DO NOT ATTEMPT TO BEND OR STRAIGHTEN FAN BLADES IF FAN IS NOT WITHIN SPECIFICATIONS.

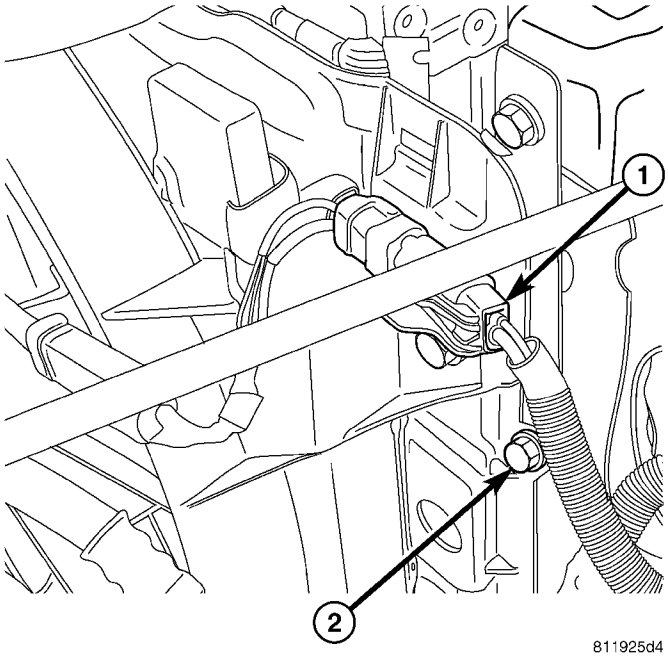

Fig. 37 Electrical Connector

1 - ELECTRICAL CONNECTOR
2 - MOUNTING SCREW

RADIATOR FAN (Continued)

CAUTION: If fan blade assembly is replaced because of mechanical damage, water pump and viscous fan drive should also be inspected. These components could have been damaged due to excessive vibration.

(1) Remove fan blade assembly from viscous fan drive unit (four bolts).
(2) Lay fan on a flat surface with leading edge facing down. With tip of blade touching flat surface, replace fan if clearance between opposite blade and surface is greater than 2.0 mm (.090 inch). Rocking motion of opposite blades should not exceed 2.0 mm (.090 inch). Test all blades in this manner.
(3) Inspect fan assembly for cracks, bends, loose rivets or broken welds. Replace fan if any damage is found.

INSTALLATION

INSTALLATION

(1) Position the cooling fan assembly.
(2) Install four mounting screws. Tighten to 5.5 N·m (50 in. lbs.)
(3) Connect the electrical connector.
(4) Install the power steering reservoir.
(5) Install the coolant recovery bottle.

INSTALLATION - 4.0L

(1) Position the fan blade on the thermal viscous fan drive.
(2) Install four mounting bolts. Tighten to 23 N·M (210 in. lbs.).
(3) Install the thermal viscous fan drive (Refer to 7 - COOLING/ENGINE/FAN DRIVE VISCOUS CLUTCH - INSTALLATION).

RADIATOR FAN RELAY

DESCRIPTION

The electric cooling fan HI and LOW relays are located in the Power Distribution Center (PDC) in the engine compartment. See the fuse and relay layout label affixed to the inside surface of the PDC cover for cooling fan relay identification and location.

The HI and LOW cooling fan relays cannot be adjusted or repaired. If the HI or LOW relay is damaged or faulty, it must be replaced.

TRANSMISSION

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TRANSMISSION COOLER

DESCRIPTION

NOTE: The internal transmission oil cooler located within the radiator is not serviceable. If it requires service, the radiator must be replaced.

All models equipped with an automatic transmission are equipped with a transmission oil cooler (water-to-oil) mounted internally within the radiator tank (Fig. 1). This internal cooler is supplied as standard equipment on all models equipped with an automatic transmission.

OPERATION

Transmission oil is cooled when it passes through this separate cooler. In case of a leak in the internal radiator mounted transmission oil cooler, engine coolant may become mixed with transmission fluid or transmission fluid may enter the cooling system. Both cooling system and transmission should be drained and inspected if the internal radiator mounted transmission cooler is leaking.

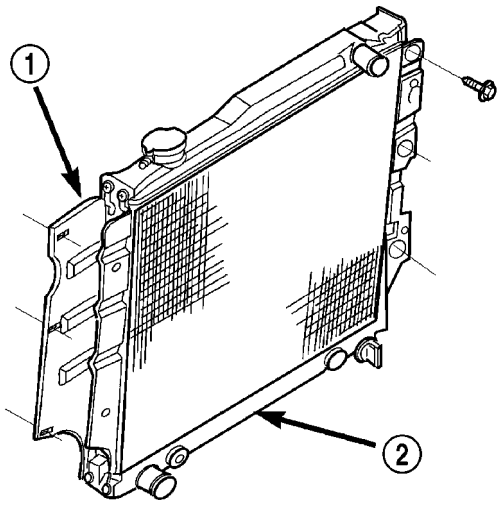

Fig. 1 Radiator with Integral Transmission Oil Cooler

1 - DOWNFLOW RADIATOR
2 - INTEGRAL TRANSMISSION OIL COOLER (INTERNAL TO RADIATOR)

AUDIO/VIDEO

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AUDIO

DESCRIPTION

Several combinations of radio receivers and speaker systems are offered.

The audio system includes the following components:

- Amplified sub woofer (if equipped)
- Antenna
- Radio noise suppression components
- Radio receiver
- Speakers

Certain functions and features of the audio system rely upon resources shared with other electronic modules in the vehicle over the Programmable Communication Interface (PCI) bus network. The data bus network allows the sharing of sensor information. For diagnosis of these electronic modules or of the data bus network, the use of a DRB III® scan tool and the proper Diagnostic Procedures manual are recommended.

OPERATION

The audio system components are designed to provide audio entertainment and information through the reception, tuning and amplification of locally broadcast radio signals in both the Amplitude Modulating (AM) and Frequency Modulating (FM) commercial frequency ranges.

The audio system components operate on battery current received through a fuse in the Junction Block (JB) on a fused ignition switch output (run-acc) circuit so that the system will only operate when the ignition switch is in the Run or Accessory positions.

DIAGNOSIS AND TESTING - AUDIO

Any diagnosis of the Audio system should begin with the use of the DRB III® diagnostic tool. For information on the use of the DRB III®, refer to the appropriate Diagnostic Service Manual.

For complete circuit diagrams, refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

AUDIO (Continued)

Audio System Diagnosis		
CONDITION	POSSIBLE CAUSE	CORRECTION
NO AUDIO.	1. Fuse inoperative.	1. Check radio fuse and Ignition-Off Draw fuse in Power Distribution Center. Replace fuses, if required.
	2. Radio connector damaged.	2. Check for loose or corroded radio connector. Repair, if required.
	3. Wiring damaged.	3. Check for battery voltage at radio connector. Repair wiring, if required.
	4. Ground damaged.	4. Check for continuity between radio chassis and a known good ground. There should be continuity. Repair ground, if required.
	5. Radio inoperative.	5. Exchange or replace radio, if required.
	6. Speakers inoperative.	6. Refer to speaker diagnosis.
NO DISPLAY.	1. Fuse inoperative.	1. Check radio fuse and Ignition-Off Draw fuse in Power Distribution Center. Replace fuses, if required.
	2. Radio connector damaged.	2. Check for loose or corroded radio connector. Repair, if required.
	3. Wiring damaged.	3. Check for battery voltage at radio connector. Repair wiring, if required.
	4. Ground damaged.	4. Check for continuity between radio chassis and a known good ground. There should be continuity. Repair ground, if required.
	5. Radio inoperative.	5. Exchange or replace radio, if required.
CLOCK WILL NOT KEEP SET TIME.	1. Fuse inoperative.	1. Check ignition-off draw fuse. Replace fuse, if required.
	2. Radio connector damaged.	2. Check for loose or corroded radio connector. Repair, if required.
	3. Wiring damaged.	3. Check for battery voltage at radio connector. Repair wiring, if required.
	4. Ground damaged.	4. Check for continuity between radio chassis and a known good ground. There should be continuity. Repair ground, if required.
	5. Radio inoperative.	5. Exchange or replace radio, if required.
POOR RADIO RECEPTION.	1. Antenna inoperative.	1. See antenna diagnosis, in this group. Repair or replace antenna, if required.
	2. Ground damaged.	2. Check for continuity between radio chassis and a known good ground. There should be continuity. Repair ground, if required..
	3. Radio inoperative.	3. Exchange or replace radio, if required.
NO/POOR TAPE OPERATION.	1. Tape damaged.	1. Insert known good tape and test operation.
	2. Foreign objects behind tape door.	2. Remove foreign objects and test operation.
	3. Dirty cassette tape head.	3. Clean head with Mopar Cassette Head Cleaner.
	4. Radio inoperative	4. Exchange or replace radio, if required.

AUDIO (Continued)

Audio System Diagnosis		
CONDITION	POSSIBLE CAUSE	CORRECTION
NO COMPACT DISC OPERATION	1. CD damaged.	1. Insert known good CD and test operation.
	2. Foreign material on CD.	2. Clean CD and test operation.
	3. Condensation on CD or optics.	3. Allow temperature of vehicle interior to stabilize and test operation.
	4. Radio inoperative	4. Exchange or replace radio, if required.

AMPLIFIED SUBWOOFER

DESCRIPTION

The available amplified subwoofer is mounted within the center console. The amplified subwoofer is rated at 80 watts. The amplified subwoofer should be checked if there is no bass output noted from the center console speaker. The amplified subwoofer can not be repaired or adjusted, and if faulty or damaged, the unit must be replaced.

OPERATION

The amplified subwoofer provides low frequency bass and receives inputs from the front and rear speaker circuits.

DIAGNOSIS AND TESTING - AMPLIFIED SUBWOOFER

Any diagnosis of the Audio system should begin with the use of the DRB III® diagnostic tool. For information on the use of the DRB III®, refer to the appropriate Diagnostic Service Manual.

For complete circuit diagrams, refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, REFER TO ELECTRICAL, RESTRAINTS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CAUTION: The speaker output of the radio is a "floating ground" system. Do not allow any speaker lead to short to ground, as damage to the radio may result.

(1) Turn the ignition switch to the On position. Turn the radio receiver on. Adjust the balance and fader controls to check the performance of each individual speaker. Note the speaker locations that are not performing correctly.

(2) Disconnect the wire harness connector from the amplified subwoofer. Turn the ignition switch to the ON position. Turn the radio ON. Check the radio choke relay wire harness connector. If not OK, repair shorted or open wires as necessary. If OK, go to (STEP #3).

(3) Turn the radio off. Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the radio and the amplified subwoofer. Check both the speaker feed (+) circuit and return (–) circuit cavities for the inoperative speaker location(s) from the radio wire harness connectors and to the amplified subwoofer for continuity. In each case, there should be continuity. If not OK, repair the shorted speaker feed (+) and/or return (–) circuit(s) to the speaker as required. If OK, replace the amplified subwoofer.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the center console (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE - REMOVAL).

(3) Disconnect wire harness retainer.

(4) Disconnect wire harness connector.

(5) Remove the mounting screws from the side of the console.

(6) Open console lid and remove the retaining fasteners from the console.

(7) Remove amplified subwoofer from console.

INSTALLATION

(1) Install amplified subwoofer to console.

(2) Install retaining fasteners to the top of the console.

AMPLIFIED SUBWOOFER (Continued)

(3) Install the mounting screws to the side of the console.

(4) Connect wire harness connector and retainer.

(5) Install the center console (Refer to 23 - BODY/INTERIOR/CENTER CONSOLE - INSTALLATION).

(6) Connect the battery negative cable.

ANTENNA BODY & CABLE

DESCRIPTION

All models use a fixed-length stainless steel rod-type antenna mast, installed on the right front cowl side panel of the vehicle. The antenna mast is connected to the center wire of the coaxial antenna cable, and is not grounded to any part of the vehicle.

To eliminate static, the antenna base must have a good ground. The antenna coaxial cable shield (the outer wire mesh of the cable) is grounded to the antenna base and the radio chassis.

The antenna coaxial cable has an additional disconnect, located behind the right end of the instrument panel between the radio and the right cowl side panel. This additional disconnect allows the instrument panel assembly to be removed and installed without removing the radio.

DIAGNOSIS AND TESTING - ANTENNA BODY AND CABLE

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

The following four tests are used to diagnose the antenna with an ohmmeter:

- **Test 1** - Mast to ground test
- **Test 2** - Tip-of-mast to tip-of-conductor test
- **Test 3** - Body ground to battery ground test
- **Test 4** - Body ground to coaxial shield test.

The ohmmeter test lead connections for each test are shown in Antenna Tests (Fig. 1).

NOTE: This model has a two-piece antenna coaxial cable. Tests 2 and 4 must be conducted in two steps to isolate an antenna cable problem. First, test the primary antenna cable (integral to the antenna body and cable) from the coaxial cable connector behind the right side of the instrument panel between the radio and the right side cowl panel, to the antenna body. Then, test the secondary antenna cable (instrument panel antenna cable) from the coaxial connector behind the right side of the instrument panel between the radio and the right side cowl panel, to the coaxial cable connector at the radio.

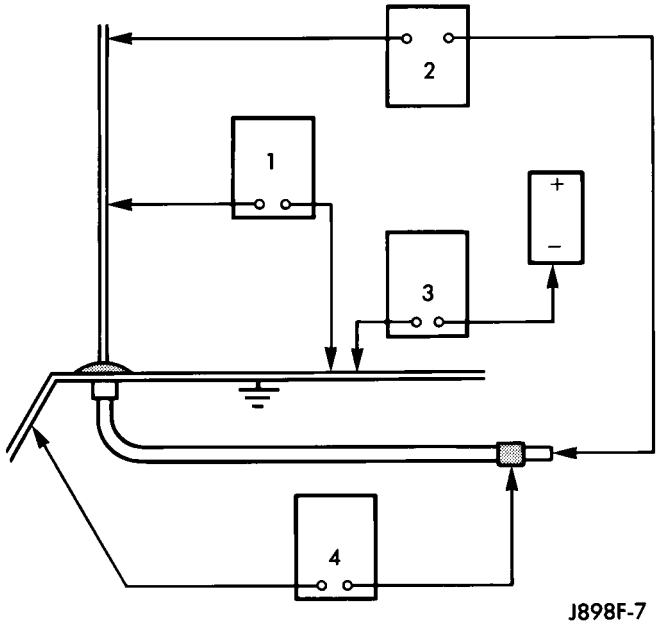

Fig. 1 Antenna Tests

TEST 1

Test 1 determines if the antenna mast is insulated from the base. Proceed as follows:

(1) Unplug the antenna coaxial cable connector from the radio chassis and isolate.

(2) Connect one ohmmeter test lead to the tip of the antenna mast. Connect the other test lead to the antenna base. Check for continuity.

(3) There should be no continuity. If continuity is found, replace the faulty or damaged antenna base and cable assembly.

TEST 2

Test 2 checks the antenna for an open circuit as follows:

(1) Unplug the antenna coaxial cable connector from the radio chassis.

(2) Connect one ohmmeter test lead to the tip of the antenna mast. Connect the other test lead to the center pin of the antenna coaxial cable connector.

(3) Continuity should exist (the ohmmeter should only register a fraction of an ohm). High or infinite resistance indicates damage to the base and cable assembly. Replace the faulty base and cable, if required.

TEST 3

Test 3 checks the condition of the vehicle body ground connection. This test should be performed with the battery positive cable removed from the bat-

ANTENNA BODY & CABLE (Continued)

tery. Disconnect both battery cables, the negative cable first. Reconnect the battery negative cable and perform the test as follows:

(1) Connect one ohmmeter test lead to the vehicle fender. Connect the other test lead to the battery negative post.

(2) The resistance should be less than one ohm.

(3) If the resistance is more than one ohm, check the braided ground strap connected to the engine and the vehicle body for being loose, corroded, or damaged. Repair or replace the ground strap connection, if required.

TEST 4

Test 4 checks the condition of the ground between the antenna base and the vehicle body as follows:

(1) Connect one ohmmeter test lead to the vehicle fender. Connect the other test lead to the outer crimp on the antenna coaxial cable connector.

(2) The resistance should be less then one ohm.

(3) If the resistance is more then one ohm, clean and/or tighten the antenna base to fender mounting hardware.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ GLOVE BOX - REMOVAL).

(3) Reach through the instrument panel glove box opening to unplug the two antenna coaxial cables in line connector. Unplug the connector by pulling it apart while twisting the metal connector halves. Do not pull on the cable.

(4) Unscrew the antenna mast from the antenna body base on the right outer cowl side panel (Fig. 2).

(5) Using a trim stick, gently pry the edge of the antenna base trim cover to unsnap it from the antenna body base.

(6) Remove the three screws that secure the antenna body base to the right outer cowl side panel.

(7) From inside the passenger compartment, push the coaxial cable grommet on the antenna body half of the coaxial cable out through the hole in the right inner cowl side panel.

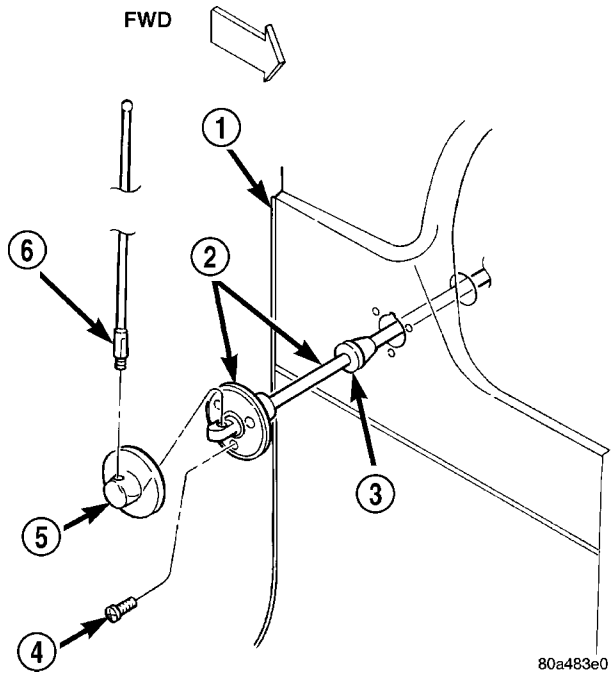

Fig. 2 Antenna Remove/Install

1 - RIGHT COWL SIDE PANEL
2 - BASE & CABLE
3 - GROMMET
4 - SCREW
5 - COVER
6 - MAST

(8) From the outside of the vehicle, pull the antenna body base and cable assembly out through the hole in the right outer cowl side panel.

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) From outside the vehicle, feed the antenna cable and base assembly through the outer cowl side panel and into the inner cowl side panel hole.

(2) From inside the passenger compartment, pull the cable and grommet into the hole in the inside cowl side panel until the grommet is fully seated.

(3) Install the three screws retaining the antenna body base to right outer cowl side panel. Tighten the screws to 3.3 N·m (30 in. lbs.).

(4) Snap on the antenna base trim cover to the antenna body base.

(5) Install antenna mast. Tighten to 5 N·m (46 in. lbs.). **Ensure that the antenna mast is fully**

ANTENNA BODY & CABLE (Continued)

seated on antenna base and that there is no gap between the mast and base.

(6) Reach through the glove box opening and connect the two antenna coaxial cables in line connector.

(7) Install the glove box to the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - INSTALLATION).

(8) Connect the battery negative cable.

RADIO

DESCRIPTION

Available factory-installed radio receivers for this model include:

- AM/FM/cassette with CD changer control feature (RBB sales code)
- AM/FM/CD with CD changer control feature (RBK sales code)
- AM/FM/cassette/CD/ with CD changer control feature (RAD, RBT or RBY sales code) - export only

All factory-installed radio receivers can communicate on the Programmable Communications Interface (PCI) data bus network. All factory-installed receivers are stereo Electronically Tuned Radios (ETR) and include an electronic digital clock function.

These radio receivers can only be serviced by an authorized radio repair station. See the latest Warranty Policies and Procedures manual for a current listing of authorized radio repair stations.

OPERATION

The radio receiver operates on ignition switched battery current that is available only when the ignition switch is in the On or Accessory positions. The electronic digital clock function of the radio operates on fused battery current supplied through the IOD fuse, regardless of the ignition switch position.

REMOVAL

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(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument panel top cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL TOP COVER - REMOVAL).

(3) Remove the center bezel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL CENTER BEZEL - REMOVAL).

(4) Remove the screws that secure the radio to the instrument panel.

(5) If the vehicle is equipped with the CD radio, go to Step 6. If the vehicle is not equipped with the CD radio receiver, go to Step 8.

(6) Remove the glove box from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GLOVE BOX - REMOVAL).

(7) Reach through the instrument panel glove box opening to access and remove the nut that secures the CD radio bracket and the ground strap to the back of the radio.

(8) Pull the radio out from the instrument panel far enough to access the wire harness connectors and the antenna coaxial cable connector.

(9) Unplug the wire harness connectors from the rear of the radio.

CAUTION: Pulling the antenna cable straight out of the radio without pulling on the locking antenna connector could damage the cable or radio.

(10) Disconnect the antenna cable by pulling the locking antenna connector away from the radio (Fig. 3).

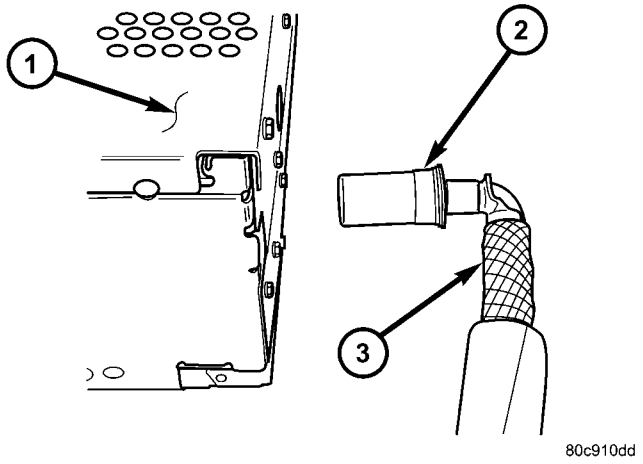

Fig. 3 ANTENNA TO RADIO

1 - RADIO
2 - LOCKING ANTENNA CONNECTOR
3 - INSTRUMENT PANEL ANTENNA CABLE

(11) Remove the radio from the instrument panel.

RADIO (Continued)

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Connect the coaxial antenna cable at the rear of the radio chassis.

(2) Connect the radio wire harness connectors to the rear of the radio.

(3) Install the radio to the instrument panel.

(4) If equipped with a CD radio, reach through the glove box opening and install the nut through the CD radio bracket stud plate. Ensure that antenna in-line connector is properly mated and has not become disconnected or pinched.

(5) Install glove box.

(6) Install the retaining screws. Tighten the screws to 5 N·m (20 in. lbs).

(7) Install the center bezel to the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/INSTRUMENT PANEL CENTER BEZEL - INSTALLATION).

(8) Connect the battery negative cable.

RADIO NOISE SUPPRESSION COMPONENTS

DESCRIPTION

Radio Frequency Interference (RFI) and Electro-Magnetic Interference (EMI) noise suppression is accomplished primarily through circuitry internal to the radio receivers. These internal suppression devices are only serviced as part of the radio receiver.

External suppression devices that are used on this vehicle to control RFI or EMI noise include the following:

- Radio antenna base ground
- Radio receiver chassis ground wire
- Engine-to-body ground strap
- Resistor-type spark plugs
- Radio suppression-type secondary ignition wiring.

DIAGNOSIS AND TESTING - RADIO NOISE SUPPRESSION COMPONENTS

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

For complete circuit diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds. Inspect the ground paths and connections at the following locations:

- Blower motor
- Electric fuel pump
- Engine-to-body ground strap
- Generator
- Ignition module
- Radio antenna base ground
- Radio receiver chassis ground wire or strap
- Wiper motor.

If the source of RFI or EMI noise is identified as a component on the vehicle (i.e., generator, blower motor, etc.), the ground path for that component should be checked. If excessive resistance is found in any ground circuit, clean, tighten, or repair the ground circuits or connections to ground as required before considering any component replacement.

- Ignition coil
- Spark plugs
- Spark plug wire routing and condition.

Reroute the spark plug wires or replace the faulty components as required.

If the source of the RFI or EMI noise is identified as two-way mobile radio or telephone equipment, check the equipment installation for the following:

- Power connections should be made directly to the battery, and fused as closely to the battery as possible.
- The antenna should be mounted on the roof or toward the rear of the vehicle. Remember that magnetic antenna mounts on the roof panel can adversely affect the operation of an overhead console compass, if the vehicle is so equipped.
- The antenna cable should be fully shielded coaxial cable, should be as short as is practical, and should be routed away from the factory-installed vehicle wire harnesses whenever possible.
- The antenna and cable must be carefully matched to ensure a low Standing Wave Ratio (SWR).

RADIO NOISE SUPPRESSION COMPONENTS (Continued)

REMOVAL

ENGINE-TO-BODY GROUND STRAP

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Remove the screw that secures the engine-to-body ground strap eyelet to the hood panel center reinforcement.
(2) Remove the screw that secures the engine-to-body ground strap eyelet to the dash panel.
(3) Remove the nut that secures the engine-to-body ground strap eyelet to the stud on the left upper rear corner of the engine cylinder head.
(4) Remove the engine-to-body ground strap eyelet from the stud on the left upper rear corner of the engine cylinder head.
(5) Remove the engine-to-body ground strap from the engine compartment.

INSTALLATION

ENGINE-TO-BODY GROUND STRAP

(1) Position the engine-to-body ground strap in the engine compartment.
(2) Position the engine-to-body ground strap eyelet over the stud on the left upper rear corner of the engine cylinder head.
(3) Install and tighten the nut that secures the engine-to-body ground strap eyelet to the stud on the left upper rear corner of the engine cylinder head. Tighten the nut to 5.6 N·m (50 in. lbs.).
(4) Install and tighten the screw that secures the engine-to-body ground strap eyelet to the dash panel. Tighten the screw to 48.5 N·m (430 in. lbs.).
(5) Install and tighten the screw that secures the engine-to-body ground strap eyelet to the hood panel center reinforcement. Tighten the screw to 1.9 N·m (17 in. lbs.).

SPEAKER

DESCRIPTION

The standard equipment speaker system includes four full-range speakers. The two front speakers are mounted behind a removable bezel located on each outboard end of the lower instrument panel. With the premium speaker system, the standard front speakers are replace with an enclosure assembly that includes a 4 inch speaker and a 1 inch tweeter. The two rear speakers are mounted behind a grille located on each outboard end of the sport bar, above the rear seating area of the vehicle.

OPERATION

Two wires connected to each speaker, one feed circuit (+) and one return circuit (–), allow the audio output signal electrical current to flow through the speaker voice coil.

DIAGNOSIS AND TESTING - SPEAKER

Any diagnosis of the Audio system should begin with the use of the DRB III® diagnostic tool. For information on the use of the DRB III®, refer to the appropriate Diagnostic Service Manual.

For complete circuit diagrams, refer to the appropriate wiring information.

WARNING: DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, SEAT BELT TENSIONER, SIDE AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CAUTION: The speaker output of the radio is a "floating ground" system. Do not allow any speaker lead to short to ground, as damage to the radio may result.

SPEAKER (Continued)

NOTE: If poor sound quality is noted in the audio system, check the Cabin Equalization curve programmed in the instrument cluster. Make sure a base speaker system has the Base Cabin Equalization Curve programmed to the vehicle. If the vehicle has a premium speaker system, make sure the Premium Cabin Equalization Curve is programmed to the vehicle

(1) If all speakers are inoperative, check the fuses in the Power Distribution Center (PDC). If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the ON position. Turn the radio receiver ON. Adjust the balance and fader control controls to check the performance of each individual speaker. Note the speaker locations that are not performing correctly. Go to Step 3.

(3) Turn the radio receiver OFF. Turn the ignition OFF. Disconnect and isolate the battery negative cable. Remove the radio receiver. Go to Step 4.

(4) Check both the speaker feed (+) circuit and return (-) circuit cavities for the inoperative speaker at the radio receiver wire harness connector for continuity to ground. There should be no continuity. If OK, go to Step 5. If not OK, repair the shorted speaker feed (+) and/or return (-) circuits(s) to the speaker as required.

(5) Disconnect wire harness connector at the inoperative speaker. Check for continuity between the speaker feed (+) circuit cavities of the radio receiver wire harness connector and the speaker wire harness connector. Repeat the check between the speaker return (-) circuit cavities of the radio receiver wire harness connector and the speaker wire harness connector. In each case, there should be continuity. If OK, replace the faulty speaker. If not OK, repair the open speaker feed (+) and/or return (-) circuits(s) as required.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

FRONT

(1) Disconnect and isolate the battery negative cable.

(2) Remove the two screws that secure the outboard end of the speaker bezel to the instrument panel (Fig. 4).

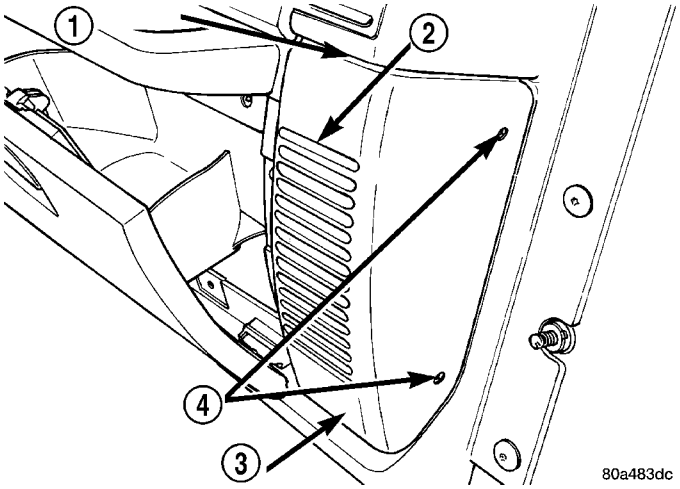

Fig. 4 Instrument Panel Speaker Bezel Remove/ Install

1 - PRY HERE
2 - SPEAKER BEZEL
3 - PRY HERE
4 - MOUNTING SCREWS

(3) Using a trim stick, gently pry at the top and bottom edges of the speaker bezel to release the two snap clip retainers that secure the bezel to the instrument panel.

(4) Remove the speaker bezel from the instrument panel.

(5) Remove the four screws that secure the speaker to the instrument panel (Fig. 5).

(6) Pull the speaker away from the instrument panel far enough to access the speaker wire harness connector.

(7) Unplug the wire harness connector from the speaker.

REAR

(1) Disconnect and isolate the battery negative cable.

(2) Remove mounting screws and speaker grill.

(3) Remove speaker from housing and disconnect wire harness connector.

SPEAKER (Continued)

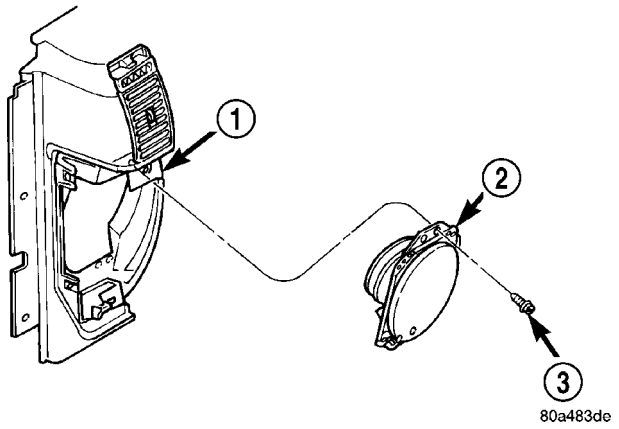

Fig. 5 Instrument Panel Speaker Remove/Install

1 - INSTRUMENT PANEL
2 - SPEAKER
3 - SCREW

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, (REFER TO ELECTRICAL/RESTRAINTS) BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

FRONT

(1) Connect the speaker wire connector to the speaker.
(2) Position the speaker to the instrument panel.
(3) Install the four speaker retaining screws. Tighten the screws to 1.1 N·m (10.in.lbs).
(4) Install the speaker bezel to the instrument panel. Tighten the screws to 2.2 N·m (20 in. lbs.).
(5) Connect the battery negative cable.

REAR

(1) Connect wire harness and position speaker into housing.
(2) Install speaker grill and mounting screws.
(3) Connect battery negative cable.

CHIME/BUZZER

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CHIME WARNING SYSTEM

DESCRIPTION

A chime warning system is standard factory-installed equipment on this model. The chime warning system uses a single chime tone generator that is integral to the instrument cluster to provide an audible indication of various vehicle conditions that may require the attention of the vehicle operator. The chime warning system includes the following major components, which are described in further detail elsewhere in this service information:

- **Door Ajar Switch** - A door ajar switch is mounted to each front door hinge pillar. This switch provides an input to the chime warning system indicating whether the front doors are open or closed.
- **Ignition Switch** - A key-in ignition switch is integral to the ignition switch. The key-in ignition switch provides an input to the chime warning system indicating whether a key is present in the ignition lock cylinder.
- **Instrument Cluster** - The instrument cluster contains an integral chime tone generator, integrated circuitry, a central processing unit and the programming to provide all of the proper chime warning system features based upon the monitored inputs. The instrument cluster circuitry monitors hard-wired switch inputs, as well as message inputs received from other vehicle electronic modules on the Programmable Communications Interface (PCI) data bus network.
- **Left Multi-Function Switch** - The exterior lighting switch is integral to the left multi-function switch. The exterior lighting switch provides an input to the chime warning system indicating when the exterior lamps are turned On or Off.
- **Seat Belt Switch** - A seat belt switch is integral to the driver side front seat belt buckle-half unit. The seat belt switch provides an input to the chime warning system indicating whether the driver side front seat belt is fastened.

Hard wired circuitry connects many of the chime warning system components to each other through the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. Refer to the appropriate wiring information.

The instrument cluster chime warning system circuitry and the integral chime tone generator cannot be adjusted or repaired. If the instrument cluster or the chime tone generator are damaged or faulty, the instrument cluster must be replaced.

OPERATION

The chime warning system is designed to provide an audible output as an indication of various conditions that may require the attention or awareness of the vehicle operator. The chime warning system components operate on battery voltage received through the Ignition-Off Draw (IOD) fuse in the Power Distribution Center (PDC) so that the system may operate regardless of the ignition switch position.

The chime warning system provides an audible warning to the vehicle operator under the following conditions:

- **Air Bag Warning** - The instrument cluster chime tone generator will generate a single chime tone when the airbag indicator is illuminated for an airbag system fault condition. The instrument cluster uses airbag indicator lamp-on and lamp-off message inputs received from the Airbag Control Module (ACM) over the Programmable Communications Interface (PCI) data bus indicating that the airbag indicator should be illuminated for an airbag system fault condition.
- **Charging System Warning** - The instrument cluster chime tone generator will generate a single chime when the check gauges indicator is illuminated for a charging system fault or a system voltage high warning condition. The instrument cluster uses system voltage status message inputs received from the Powertrain Control Module (PCM) over the PCI data bus to illuminate the check gauges indicator for a charging system fault. This chime feature will only occur once in an ignition cycle.
- **Driver Door Ajar Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate to announce that the hard

CHIME WARNING SYSTEM (Continued)

wired inputs from the driver door ajar switch and the ignition switch as well as an engine speed message input received from the PCM over the PCI data bus indicate that the driver door is opened with the ignition switch in the On position and the engine is not running. The chimes will continue to sound for a duration of about twenty minutes, until the driver door is closed, until the ignition switch is turned to the Off position, or until the engine speed message indicates the engine is running, whichever occurs first.

- **Engine Coolant Temperature High Warning** - The instrument cluster chime tone generator will generate a single chime tone when the check gauges indicator is illuminated for a high or critical engine coolant temperature condition. The instrument cluster uses engine coolant temperature message inputs received from the PCM over the PCI data bus to illuminate the check gauges indicator for a coolant temperature high condition. This chime feature will only occur once in an ignition cycle.
- **Fasten Seat Belt Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate each time the ignition switch is turned to the On or Start positions to announce that the hard wired inputs from the seat belt switch and the ignition switch indicate that the driver side front seat belt is not fastened. The chimes will continue to sound for a duration of about six seconds, until the driver side front seat belt is fastened, or until the ignition switch is turned to the Off position, whichever occurs first.
- **Head/Park Lights-On Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate to announce that the hard wired inputs from the driver door ajar switch, the ignition switch, and the exterior lighting circuitry of the left multi-function switch indicate that the exterior lamps are turned On with the driver door opened and the ignition switch in the Off position. The chimes will continue to sound for about three minutes or until the exterior lamps are turned Off, the driver door is closed, or the ignition switch is turned to the On position, whichever occurs first.
- **Key-In-Ignition Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate to announce that the hard wired inputs from the driver door ajar switch, the ignition switch, and the key-in ignition circuitry of the ignition switch indicate that the key is in the ignition lock cylinder with the driver door opened and the ignition switch in the Off position. The chimes will continue to sound until the key is removed from the ignition lock cylinder, the driver door is closed, or the ignition switch is turned to the On position, whichever occurs first.
- **Low Fuel Warning** - The instrument cluster chime tone generator will generate one chime tone when the low fuel indicator is illuminated by the instrument cluster. The instrument cluster uses a percent tank full message input received from the PCM over the PCI data bus indicating that there is less than about one-eighth tank of fuel remaining to illuminate the low fuel indicator. This chime feature will only occur once in an ignition cycle.
- **Low Oil Pressure Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a fast rate when the check gauges indicator is illuminated for a low oil pressure condition. The instrument cluster uses engine speed and oil pressure message inputs received from the PCM over the PCI data bus indicating that the engine is running at greater than 300 rpm and that the oil pressure is low to illuminate the check gauges indicator. The chimes will continue to sound for five seconds, until the engine oil pressure message indicates that the oil pressure is not low, or until the engine speed message indicates that the engine is running at less than 300 rpm, whichever occurs first. This chime tone will only occur once in an ignition cycle.
- **Overspeed Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate to announce that a vehicle speed message input received from the PCM over the PCI data bus indicates that the vehicle speed is above 120 kilometers-per-hour (75 miles-per-hour). The chimes will continue to sound until the vehicle speed message indicates that the vehicle speed is below 120 kilometers-per-hour (75 miles-per-hour). This feature is only enabled on an instrument cluster that has been programmed with a Middle East Gulf Coast Country (GCC) country code.
- **Park Brake Reminder** - The instrument cluster chime tone generator will generate ten repetitive chimes at a slow rate to announce that the hard wired input from the park brake switch and a vehicle speed message input received from the PCM over the PCI data bus indicates that the park brake is applied and the vehicle is moving. This chime feature will repeat each time the input conditions are met.
- **Passenger Door Ajar Warning** - The instrument cluster chime tone generator will generate three sets of two chimes at a slow rate to announce that the hard wired inputs from the passenger door ajar switch and the ignition switch as well as an engine speed message input received from the PCM over the PCI data bus indicate that the passenger door is opened with the ignition switch in the On position and the engine is not running. This chime feature will repeat each time the input conditions are met.

CHIME WARNING SYSTEM (Continued)

- **Sentry Key Immobilizer System "Customer Learn" Mode Announcement** - This chime feature is only active on vehicles equipped with the optional Sentry Key Immobilizer System (SKIS) and sold in markets where the optional "Customer Learn" programming feature is available. The instrument cluster chime tone generator will generate one chime to announce that a status message input received from the Sentry Key Immobilizer Module (SKIM) over the PCI data bus indicates that the SKIS is in the "Customer Learn" mode, which is used for programming additional sentry key transponders.
- **Turn Signal On Warning** - The instrument cluster chime tone generator will generate repetitive chimes at a slow rate to announce that the hard wired input for the right or left turn signal indicator as well as vehicle distance and speed message inputs received from the PCM over the PCI data bus indicate that a turn signal has been active continuously for 1.6 kilometers (1 mile) with the vehicle speed greater than 22 kilometers-per-hour (15 miles-per hour). Vehicles built for markets other than the United States and Canada have a revised distance threshold of 4 kilometers for this feature. The chime will continue until the turn signal input becomes inactive, the status changes, or until the vehicle speed message indicates that the speed is less than 22 kilometers-per-hour (15 miles-per-hour), whichever occurs first. The hazard warning flashers will not activate this chime feature.

The instrument cluster provides chime service for all available features in the chime warning system. The instrument cluster relies upon its internal programming, hard wired inputs from numerous switches, and electronic message inputs received from other electronic modules over the PCI data bus network. Upon receiving the proper inputs, the instrument cluster activates the integral chime tone generator to provide the audible chime to the vehicle operator. The chime tone generator in the instrument cluster is capable of producing single chime tones, or repeated chime tones at two different rates: about fifty chime tones per minute, or about 180 chime tones per minute. The internal programming of the instrument cluster determines the priority of each chime request input that is received, as well as the rate and duration of each chime that is to be generated.

The hard wired chime warning system inputs to the instrument cluster, as well as other hard wired circuits for this system may be diagnosed and tested using conventional diagnostic tools and procedures. However, conventional diagnostic methods may not prove conclusive in the diagnosis of the instrument cluster or the PCI data bus network. The most reliable, efficient and accurate means to diagnose the instrument cluster and the PCI data bus network inputs for the chime warning system requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

DIAGNOSIS AND TESTING - CHIME WARNING SYSTEM

The chime warning system features driven by hard wired inputs to the instrument cluster may be diagnosed and tested using conventional diagnostic tools and procedures. However, conventional diagnostic methods may not prove conclusive in the diagnosis of the chime warning system features driven by message inputs to the instrument cluster over the Programmable Communications Interface (PCI) data bus network. The most reliable, efficient and accurate means to diagnose the instrument cluster and the PCI data bus network inputs for the chime warning system requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic and wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CHIME WARNING SYSTEM (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
NO SEAT BELT WARNING CHIME WITH SEAT BELT UNBUCKLED, BUT OTHER CHIME FEATURES OK	1. Seat belt switch ground circuit open.	1. Check for continuity between the ground circuit of the floor wire harness connector for the seat belt switch and a good ground. Repair open ground circuit, if required.
	2. Seat belt switch sense circuit open.	2. Check for continuity between the seat belt switch sense circuit of the floor wire harness connector for the seat belt switch and the instrument panel wire harness instrument cluster connector. Repair the open seat belt switch sense circuit, if required.
	3. Faulty seat belt switch.	3. Check for continuity between the ground circuit and the seat belt switch sense circuit of the seat belt switch pigtail connector. There should be continuity with the seat belt unbuckled. Replace the faulty seat belt, if required.
SEAT BELT WARNING CHIME WITH SEAT BELT BUCKLED	1. Seat belt switch sense circuit shorted.	1. With the floor wire harness connector for the seat belt switch and the instrument panel wire harness instrument cluster connector disconnected, there should be no continuity between the seat belt switch sense circuit and a good ground. Repair the shorted seat belt switch sense circuit, if required.
	2. Faulty seat belt switch.	2. Check for continuity between the ground circuit cavity and the seat belt switch sense circuit of the seat belt switch pigtail connector. There should be no continuity with the seat belt buckled. Replace the faulty seat belt, if required.
NO KEY-IN IGNITION WARNING CHIME, BUT OTHER CHIME FEATURES OK	1. Driver door ajar switch sense circuit open.	1. Check for continuity between the driver door ajar switch sense circuit of the connector for the driver door ajar switch and the cross body wire harness ignition switch connector. Repair the open driver door ajar switch sense circuit, if required.
	2. Key-in ignition switch sense circuit open.	2. Check for continuity between the key-in ignition switch sense circuit of the cross body wire harness ignition switch connector and the instrument panel wire harness instrument cluster connector. Repair the open key-in ignition switch sense circuit, if required.
	3. Ignition switch ground circuit open.	3. Check for continuity between the ground circuit cavity of the cross body wire harness ignition switch connector and a good ground. Repair the open ground circuit, if required
	4. Faulty ignition switch.	4. Check for continuity between the ground circuit terminal and the key-in ignition switch sense circuit terminal in the ignition switch connector. There should be continuity with a key in the ignition lock cylinder. Replace the faulty ignition switch, if required.

CHIME WARNING SYSTEM (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
NO HEADLAMPS-ON WARNING CHIME, BUT OTHER CHIME FEATURES OK	1. Driver door ajar switch sense circuit open.	1. Check for continuity between the driver door ajar switch sense circuit of the connector for the driver door ajar switch and the instrument panel wire harness instrument cluster connector. Repair the open driver door ajar switch sense circuit, if required.
	2. Headlamp switch output circuit open.	2. Check for continuity between the headlamp switch output circuit of the instrument panel wire harness instrument cluster connector and the park lamp feed circuit of the cross body wire harness left multi-function switch connector. Repair the open headlamp switch output circuit, if required.
	3. Faulty multi-function switch.	3. Check for continuity between the B(+) circuit terminal and the park lamp feed circuit terminal in the left multi-function switch connector. There should be continuity with the headlamp switch in the On position. Replace the faulty left multi-function switch, if required.
NO CHIMES AND OTHER INSTRUMENT CLUSTER FEATURES ERRATIC OR DISABLED	1. Instrument cluster ground circuit open.	1. Check for continuity between the ground circuit of the instrument panel wire harness instrument cluster connector and a good ground. Repair the open ground circuit, if required.
	2. Instrument cluster fused B(+) circuit open.	2. Check for battery voltage at the B(+) circuit of the instrument panel wire harness instrument cluster connector. Repair the open fused B(+) circuit, if required.
	3. Instrument cluster fused ignition switch output (run-start) circuit open.	3. With the ignition switch in the On position, check for battery voltage at the fused ignition switch output (run-start) circuit of the instrument panel wire harness instrument cluster connector. Repair the open fused ignition switch output (run-start) circuit, if required.
	4. Faulty instrument cluster.	4. Replace the faulty instrument cluster, if required.
NO CHIMES, BUT ALL OTHER INSTRUMENT CLUSTER FEATURES OK	1. Faulty instrument cluster.	1. Replace the faulty instrument cluster, if required.

ELECTRONIC CONTROL MODULES

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ELECTRONIC CONTROL MODULES

STANDARD PROCEDURE - PCM/SKIM PROGRAMMING

CAUTION: ASSURE THE DRBIII® IS PROGRAMMED WITH THE LATEST VERSION OF CURRENT SOFTWARE.

NOTE: Before replacing the PCM for a failed driver, control circuit, or ground circuit, be sure to check the related component/circuit integrity for failures not detected due to a double fault in the circuit. Most PCM driver/control circuit failures are caused by internal component failures (i.e. relays and solenoids) and shorted circuits (i.e. pull-ups, drivers, and switched circuits). These failures are difficult to detect when a double fault has occurred and only one DTC has been set.

When a PCM (JTEC) and the SKIM are replaced at the same time, perform the following steps in order:

(1) Program the new PCM (JTEC).
(2) Program the new SKIM.
(3) Replace all ignition keys and program them to the new SKIM.

PROGRAMMING THE PCM (JTEC)

The SKIS Secret Key is an ID code that is unique to each SKIM. This code is programmed and stored in the SKIM, the PCM, and the ignition key transponder chip(s). When replacing the PCM, it is necessary to program the secret key into the new PCM using the DRBIII® scan tool. Perform the following steps to program the secret key into the PCM.

(1) Turn the ignition switch to the On position (transmission in Park/Neutral).
(2) Use the DRBIII® and select THEFT ALARM, SKIM, then MISCELLANEOUS.
(3) Select PCM REPLACED (GAS ENGINE).
(4) Enter secured access mode by entering the vehicle four-digit PIN.
(5) Select ENTER to update PCM VIN.

NOTE: If three attempts are made to enter secured access mode using an incorrect PIN, secured access mode will be locked out for one hour. To exit this lockout mode, turn the ignition switch to the ON position for one hour, then enter the correct PIN. (Ensure all accessories are turned off. Also monitor the battery state and connect a battery charger if necessary).

(6) Press ENTER to transfer the secret key (the SKIM will send the secret key to the PCM).

(7) Press PAGE BACK to get to the Select System menu and select ENGINE, MISCELLANEOUS, and SRI MEMORY CHECK.

(8) The DRBIII® will ask, "Is odometer reading between XX and XX?" Select the YES or NO button on the DRBIII®. If NO is selected, the DRBIII® will read, "Enter Odometer Reading (From I.P. odometer)". Enter the odometer reading from the instrument cluster and press ENTER.

PROGRAMMING THE SKIM

(1) Turn the ignition switch to the On position (transmission in Park/Neutral).

(2) Use the DRBIII® and select THEFT ALARM, SKIM, then MISCELLANEOUS.

(3) Select PCM REPLACED (GAS ENGINE).

(4) Program the vehicle four-digit PIN into SKIM.

(5) Select COUNTRY CODE and enter the correct country.

NOTE: Be sure to enter the correct country code. If the incorrect country code is programmed into SKIM, it cannot be changed and the SKIM must be replaced.

(6) Select YES to update VIN (the SKIM will learn the VIN from the PCM).

(7) Press ENTER to transfer the secret key (the PCM will send the secret key to the SKIM).

(8) Program ignition keys to the SKIM.

NOTE: If the PCM and the SKIM are replaced at the same time, all vehicle ignition keys will need to be replaced and programmed to the new SKIM.

PROGRAMMING IGNITION KEYS TO THE SKIM

(1) Turn the ignition switch to the On position (transmission in Park/Neutral).

(2) Use the DRBIII® and select THEFT ALARM, SKIM, then MISCELLANEOUS.

(3) Select PROGRAM IGNITION KEY'S.

(4) Enter secured access mode by entering the vehicle four-digit PIN.

NOTE: A maximum of eight keys can be learned to each SKIM. Once a key is learned to a SKIM it (the key) cannot be transferred to another vehicle.

(5) Obtain ignition keys to be programmed from the customer (8 keys maximum).

(6) Using the DRBIII®, erase all ignition keys by selecting MISCELLANEOUS, and ERASE ALL CURRENT IGN. KEYS.

(7) Program all of the ignition keys.

If ignition key programming is unsuccessful, the DRBIII® will display one of the following messages:

- **Programming Not Attempted** - The DRBIII® attempts to read the programmed key status and there are no keys programmed into SKIM memory.
- **Programming Key Failed (Possible Used Key From Wrong Vehicle)** - SKIM is unable to program an ignition key transponder due to one of the following:
  - The ignition key transponder is faulty.
  - The ignition key transponder is or has been already programmed to another vehicle.
- **8 Keys Already Learned, Programming Not Done** - The SKIM transponder ID memory is full.
- **Learned Key In Ignition** - The ID for the ignition key transponder currently in the ignition lock cylinder is already programmed in SKIM memory.

COMMUNICATION

DESCRIPTION

The DaimlerChrysler Programmable Communication Interface (PCI) data bus system is a single wire multiplex system used for vehicle communications on many DaimlerChrysler Corporation vehicles. Multiplexing is a system that enables the transmission of several messages over a single channel or circuit. All DaimlerChrysler vehicles use this principle for communication between various microprocessor-based electronic control modules. The PCI data bus exceeds the Society of Automotive Engineers (SAE) J1850 Standard for Class B Multiplexing.

Many of the electronic control modules in a vehicle require information from the same sensing device. In the past, if information from one sensing device was required by several controllers, a wire from each controller needed to be connected in parallel to that sensor. In addition, each controller utilizing analog sensors required an Analog/Digital (A/D) converter in order to ″read″ these sensor inputs. Multiplexing reduces wire harness complexity, sensor current loads and controller hardware because each sensing device is connected to only one controller, which reads and distributes the sensor information to the other controllers over the data bus. Also, because each controller on the data bus can access the controller sensor inputs to every other controller on the data bus, more function and feature capabilities are possible.

In addition to reducing wire harness complexity, component sensor current loads and controller hardware, multiplexing offers a diagnostic advantage. A multiplex system allows the information flowing between controllers to be monitored using a diagnostic scan tool. The DaimlerChrysler system allows an

COMMUNICATION (Continued)

electronic control module to broadcast message data out onto the bus where all other electronic control modules can ″hear″ the messages that are being sent. When a module hears a message on the data bus that it requires, it relays that message to its microprocessor. Each module ignores the messages on the data bus that are being sent to other electronic control modules.

OPERATION

Data exchange between modules is achieved by serial transmission of encoded data over a single wire broadcast network. The wire colors used for the PCI data bus circuits are yellow with a violet tracer, or violet with a yellow tracer, depending upon the application. The PCI data bus messages are carried over the bus in the form of Variable Pulse Width Modulated (VPWM) signals. The PCI data bus speed is an average 10.4 Kilo-bits per second (Kbps). By comparison, the prior two-wire Chrysler Collision Detection (CCD) data bus system is designed to run at 7.8125 Kbps.

The voltage network used to transmit messages requires biasing and termination. Each module on the PCI data bus system provides its own biasing and termination. Each module (also referred to as a node) terminates the bus through a terminating resistor and a terminating capacitor. There are two types of nodes on the bus. The dominant node terminates the bus through a 1 KW resistor and a 3300 pF capacitor. The Powertrain Control Module (PCM) is the only dominant node for the PCI data bus system. A standard node terminates the bus through an 11 KW resistor and a 330 pF capacitor.

The modules bias the bus when transmitting a message. The PCI bus uses low and high voltage levels to generate signals. Low voltage is around zero volts and the high voltage is about seven and one-half volts. The low and high voltage levels are generated by means of variable-pulse width modulation to form signals of varying length. The Variable Pulse Width Modulation (VPWM) used in PCI bus messaging is a method in which both the state of the bus and the width of the pulse are used to encode bit information. A ″zero″ bit is defined as a short low pulse or a long high pulse. A ″one″ bit is defined as a long low pulse or a short high pulse. A low (passive) state on the bus does not necessarily mean a zero bit. It also depends upon pulse width. If the width is short, it stands for a zero bit. If the width is long, it stands for a one bit. Similarly, a high (active) state does not necessarily mean a one bit. This too depends upon pulse width. If the width is short, it stands for a one bit. If the width is long, it stands for a zero bit.

In the case where there are successive zero or one data bits, both the state of the bus and the width of the pulse are changed alternately. This encoding scheme is used for two reasons. First, this ensures that only one symbol per transition and one transition per symbol exists. On each transition, every transmitting module must decode the symbol on the bus and begin timing of the next symbol. Since timing of the next symbol begins with the last transition detected on the bus, all of the modules are re-synchronized with each symbol. This ensures that there are no accumulated timing errors during PCI data bus communication.

The second reason for this encoding scheme is to guarantee that the zero bit is the dominant bit on the bus. When two modules are transmitting simultaneously on the bus, there must be some form of arbitration to determine which module will gain control. A data collision occurs when two modules are transmitting different messages at the same time. When a module is transmitting on the bus, it is reading the bus at the same time to ensure message integrity. When a collision is detected, the module that transmitted the one bit stops sending messages over the bus until the bus becomes idle.

Each module is capable of transmitting and receiving data simultaneously. The typical PCI bus message has the following four components:

- **Message Header** - One to three bytes in length. The header contains information identifying the message type and length, message priority, target module(s) and sending module.
- **Data Byte(s)** - This is the actual message that is being sent.
- **Cyclic Redundancy Check (CRC) Byte** - This byte is used to detect errors during a message transmission.
- **In-Frame Response (IFR) byte(s)** - If a response is required from the target module(s), it can be sent during this frame. This function is described in greater detail in the following paragraph.

The IFR consists of one or more bytes, which are transmitted during a message. If the sending module requires information to be received immediately, the target module(s) can send data over the bus during the original message. This allows the sending module to receive time-critical information without having to wait for the target module to access the bus. After the IFR is received, the sending module broadcasts an End of Frame (EOF) message and releases control of the bus.

The PCI data bus can be monitored using a diagnostic scan tool. It is possible, however, for the bus to pass all diagnostic scan tool tests and still be faulty if the voltage parameters are all within the specified range and false messages are being sent.

CONTROLLER ANTILOCK BRAKE

DESCRIPTION

The CAB operates the ABS system, and is separate from other vehicle electrical circuits. The CAB is located under the instrument panel to the right side of the steering column. It is mounted to bracket with one bolt. The bracket is mounted to the front upper cowl panel.

OPERATION

The CAB voltage source is through the ignition switch in the RUN position. The CAB contains dual microprocessors. A logic block in each microprocessor receives identical sensor signals. These signals are processed and compared simultaneously. The CAB contains a self check program that illuminates the ABS warning light when a system fault is detected. Faults are stored in a diagnostic program memory and are accessible with the DRB scan tool. ABS faults remain in memory until cleared, or until after the vehicle is started approximately 50 times. Stored faults are **not** erased if the battery is disconnected.

REMOVAL

(1) Remove the negative battery cable from the battery.

(2) Pull up on the CAB harness connector release (Fig. 1)and remove connector.

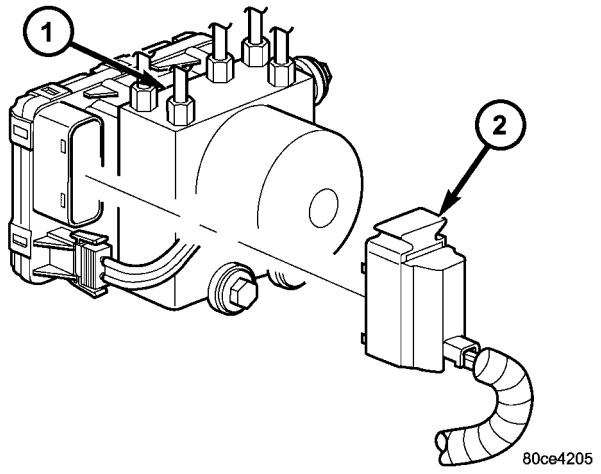

Fig. 1 CAB HARNESS CONNECTOR RELEASE

1 - ABS MODULE
2 - ELECTRICAL CONNECTOR

(3) Remove the pump connector from the CAB.

(4) Remove the CAB mounting bolts.

(5) Remove the CAB from the HCU (Fig. 2).

INSTALLATION

(1) Install CAB to the HCU (Fig. 2).

(2) Install mounting bolts. Tighten to 2 N·m (16 in. lbs.).

(3) Install the pump electircal connector to the CAB (Fig. 2).

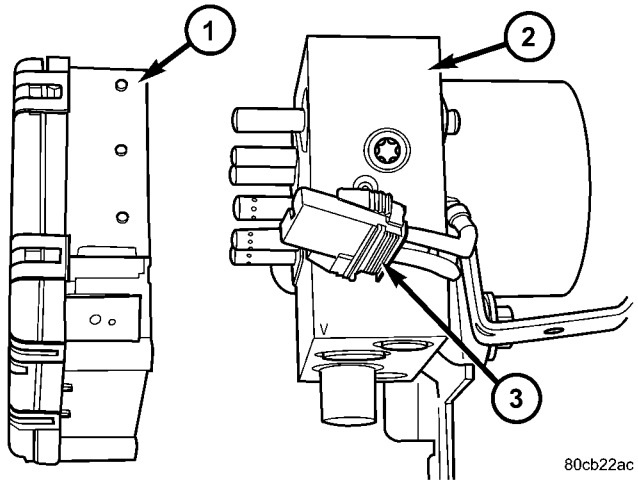

Fig. 2 CONTROLLER AND HCU

1 - CONTROLLER ANTILOCK BRAKE MODULE
2 - HYDRAULIC CONTROL UNIT (H.C.U)
3 - ELECTRICAL CONNECTOR

(4) Install the wiring harness connector to the CAB and push down on the release to secure the connector.

(5) Install negative battery cable to the battery.

DATA LINK CONNECTOR

DESCRIPTION - DATA LINK CONNECTOR

The data link connector (DLC) is located at the lower edge of the instrument panel near the steering column.

OPERATION - DATA LINK CONNECTOR

The 16–way data link connector (diagnostic scan tool connector) links the Diagnostic Readout Box (DRB) scan tool or the Mopar Diagnostic System (MDS) with the Powertrain Control Module (PCM).

POWERTRAIN CONTROL MODULE

DESCRIPTION

DESCRIPTION

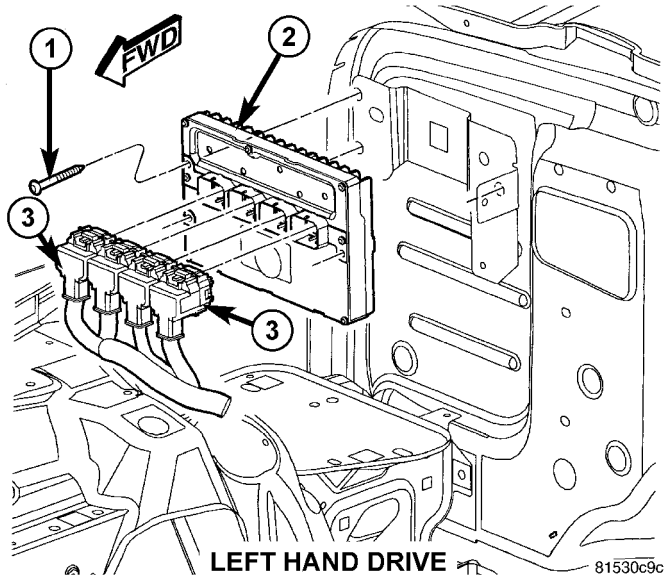

Fig. 3 PCM MOUNTING - LEFT HAND DRIVE

1 - MOUNTING SCREWS (3)
2 - PCM
3 - PCM ELECTRICAL CONNECTORS

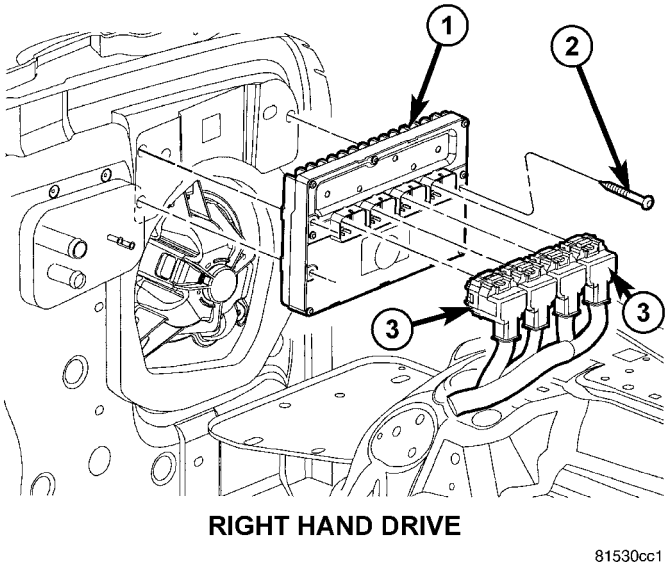

Fig. 4 PCM MOUNTING - RIGHT HAND DRIVE

1 - PCM
2 - MOUNTING SCREWS (3)
3 - PCM ELECTRICAL CONNECTORS

The PCM (1) or (2) is located in the engine compartment (Fig. 3) or (Fig. 4). The PCM is referred to as the NGC.

DESCRIPTION - MODES OF OPERATION

As input signals to the Powertrain Control Module (PCM) change, the PCM adjusts its response to the output devices. For example, the PCM must calculate different injector pulse width and ignition timing for idle than it does for wide open throttle (WOT).

The PCM will operate in two different modes: **Open Loop and Closed Loop**.

During Open Loop modes, the PCM receives input signals and responds only according to preset PCM programming. Input from the oxygen (O2S) sensors is not monitored during Open Loop modes.

During Closed Loop modes, the PCM will monitor the oxygen (O2S) sensors input. This input indicates to the PCM whether or not the calculated injector pulse width results in the ideal air-fuel ratio. This ratio is 14.7 parts air-to-1 part fuel. By monitoring the exhaust oxygen content through the O2S sensor, the PCM can fine tune the injector pulse width. This is done to achieve optimum fuel economy combined with low emission engine performance.

The fuel injection system has the following modes of operation:

- Ignition switch ON
- Engine start-up (crank)
- Engine warm-up
- Idle
- Cruise
- Acceleration
- Deceleration
- Wide open throttle (WOT)
- Ignition switch OFF

The ignition switch On, engine start-up (crank), engine warm-up, acceleration, deceleration and wide open throttle modes are Open Loop modes. The idle and cruise modes, (with the engine at operating temperature) are Closed Loop modes.

IGNITION SWITCH (KEY-ON) MODE

This is an Open Loop mode. When the fuel system is activated by the ignition switch, the following actions occur:

- The PCM pre-positions the Idle Air Control (IAC) motor.
- The PCM determines atmospheric air pressure from the MAP sensor input to determine basic fuel strategy.
- The PCM monitors the engine coolant temperature sensor input. The PCM modifies fuel strategy based on this input.
- Intake manifold air temperature sensor input is monitored.
- Throttle position sensor (TPS) is monitored.

POWERTRAIN CONTROL MODULE (Continued)

- The auto shutdown (ASD) relay is energized by the PCM for approximately three seconds.
- The fuel pump is energized through the fuel pump relay by the PCM. The fuel pump will operate for approximately three seconds unless the engine is operating or the starter motor is engaged.
- The O2S sensor heater element is energized via the ASD relay. The O2S sensor input is not used by the PCM to calibrate air-fuel ratio during this mode of operation.
- The Up-shift Indicator Lamp is illuminated (manual transmission only).

ENGINE START-UP MODE

This is an Open Loop mode. The following actions occur when the starter motor is engaged.

The PCM receives inputs from:

- Battery voltage
- Engine coolant temperature sensor
- Crankshaft position sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal

The PCM monitors the crankshaft position sensor. If the PCM does not receive a crankshaft position sensor signal within 3 seconds of cranking the engine, it will shut down the fuel injection system.

The fuel pump is activated by the PCM through the fuel pump relay.

Voltage is applied to the fuel injectors with the ASD relay via the PCM. The PCM will then control the injection sequence and injector pulse width by turning the ground circuit to each individual injector on and off.

The PCM determines the proper ignition timing according to input received from the crankshaft position sensor.

ENGINE WARM-UP MODE

This is an Open Loop mode. During engine warm-up, the PCM receives inputs from:

- Battery voltage
- Crankshaft position sensor
- Engine coolant temperature sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal (in the distributor)
- Park/neutral switch (gear indicator signal—auto. trans. only)
- Air conditioning select signal (if equipped)
- Air conditioning request signal (if equipped)

Based on these inputs the following occurs:

- Voltage is applied to the fuel injectors with the ASD relay via the PCM. The PCM will then control the injection sequence and injector pulse width by turning the ground circuit to each individual injector on and off.
- The PCM adjusts engine idle speed through the idle air control (IAC) motor and adjusts ignition timing.
- The PCM operates the A/C compressor clutch through the A/C compressor clutch relay. This is done if A/C has been selected by the vehicle operator and specified pressures are met at the high and low–pressure A/C switches. Refer to Group 24, Heating and Air Conditioning for additional information.
- When engine has reached operating temperature, the PCM will begin monitoring O2S sensor input. The system will then leave the warm-up mode and go into closed loop operation.

IDLE MODE

When the engine is at operating temperature, this is a Closed Loop mode. At idle speed, the PCM) receives inputs from:

- Air conditioning select signal (if equipped)
- Air conditioning request signal (if equipped)
- Battery voltage
- Crankshaft position sensor
- Engine coolant temperature sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal (in the distributor)
- Battery voltage
- Park/neutral switch (gear indicator signal—auto. trans. only)
- Oxygen sensors
- Power steering pressure switch (2.5L engine only)

Based on these inputs, the following occurs:

- Voltage is applied to the fuel injectors with the ASD relay via the PCM. The PCM will then control injection sequence and injector pulse width by turning the ground circuit to each individual injector on and off.
- The PCM monitors the O2S sensor input and adjusts air-fuel ratio by varying injector pulse width. It also adjusts engine idle speed through the idle air control (IAC) motor.
- The PCM adjusts ignition timing by increasing and decreasing spark advance.
- The PCM operates the A/C compressor clutch through the A/C compressor clutch relay. This is done if A/C has been selected by the vehicle operator and specified pressures are met at the high and low–pres-

POWERTRAIN CONTROL MODULE (Continued)

sure A/C switches. Refer to Group 24, Heating and Air Conditioning for additional information.

On 2.5L 4–cylinder engines, a power steering pressure switch is used to supply an input to the PCM when steering pump pressure is high. This will raise engine speed. Refer to Power Steering Pressure Switch in this group for additional information. **The 4.0L 6–cylinder engine does not use this switch.**

CRUISE MODE

When the engine is at operating temperature, this is a Closed Loop mode. At cruising speed, the PCM receives inputs from:

- Air conditioning select signal (if equipped)
- Air conditioning request signal (if equipped)
- Battery voltage
- Engine coolant temperature sensor
- Crankshaft position sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal (in the distributor)
- Park/neutral switch (gear indicator signal—auto. trans. only)
- Oxygen (O2S) sensors

Based on these inputs, the following occurs:

- Voltage is applied to the fuel injectors with the ASD relay via the PCM. The PCM will then adjust the injector pulse width by turning the ground circuit to each individual injector on and off.
- The PCM monitors the O2S sensor input and adjusts air-fuel ratio. It also adjusts engine idle speed through the idle air control (IAC) motor.
- The PCM adjusts ignition timing by turning the ground path to the coil on and off.
- The PCM operates the A/C compressor clutch through the clutch relay. This happens if A/C has been selected by the vehicle operator and requested by the A/C thermostat.
- The Up-shift Indicator Lamp is operated (manual transmission only).

ACCELERATION MODE

This is an Open Loop mode. The PCM recognizes an abrupt increase in throttle position or MAP pressure as a demand for increased engine output and vehicle acceleration. The PCM increases injector pulse width in response to increased throttle opening.

DECELERATION MODE

When the engine is at operating temperature, this is an Open Loop mode. During hard deceleration, the PCM receives the following inputs.

- Air conditioning select signal (if equipped)
- Air conditioning request signal (if equipped)
- Battery voltage
- Engine coolant temperature sensor
- Crankshaft position sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal (in the distributor)
- Park/neutral switch (gear indicator signal—auto. trans. only)
- Vehicle speed sensor

If the vehicle is under hard deceleration with the proper rpm and closed throttle conditions, the PCM will ignore the oxygen sensor input signal. The PCM will enter a fuel cut-off strategy in which it will not supply a ground to the injectors. If a hard deceleration does not exist, the PCM will determine the proper injector pulse width and continue injection.

Based on the above inputs, the PCM will adjust engine idle speed through the idle air control (IAC) motor.

The PCM adjusts ignition timing by turning the ground path to the coil on and off.

WIDE OPEN THROTTLE MODE

This is an Open Loop mode. During wide open throttle operation, the PCM receives the following inputs.

- Battery voltage
- Crankshaft position sensor
- Engine coolant temperature sensor
- Intake manifold air temperature sensor
- Manifold absolute pressure (MAP) sensor
- Throttle position sensor (TPS)
- Camshaft position sensor signal (in the distributor)

During wide open throttle conditions, the following occurs:

- Voltage is applied to the fuel injectors with the ASD relay via the PCM. The PCM will then control the injection sequence and injector pulse width by turning the ground circuit to each individual injector on and off. The PCM ignores the oxygen sensor input signal and provides a predetermined amount of additional fuel. This is done by adjusting injector pulse width.
- The PCM adjusts ignition timing by turning the ground path to the coil on and off.
- The Up-shift Indicator Lamp is operated (manual transmission only).

IGNITION SWITCH OFF MODE

When ignition switch is turned to OFF position, the PCM stops operating the injectors, ignition coil, ASD relay and fuel pump relay.

POWERTRAIN CONTROL MODULE (Continued)

DESCRIPTION - 5 VOLT SUPPLIES

Two different Powertrain Control Module (PCM) five volt supply circuits are used; primary and secondary.

DESCRIPTION - IGNITION CIRCUIT SENSE

This circuit ties the ignition switch to the Powertrain Control Module (PCM).

DESCRIPTION - POWER GROUNDS

The Powertrain Control Module (PCM) has 2 main grounds. Both of these grounds are referred to as power grounds. All of the high-current, noisy, electrical devices are connected to these grounds as well as all of the sensor returns. The sensor return comes into the sensor return circuit, passes through noise suppression, and is then connected to the power ground.

The power ground is used to control ground circuits for the following PCM loads:

- Generator field winding
- Fuel injectors
- Ignition coil(s)
- Certain relays/solenoids
- Certain sensors

DESCRIPTION - SENSOR RETURN

The Sensor Return circuits are internal to the Powertrain Control Module (PCM).

Sensor Return provides a low–noise ground reference for all engine control system sensors. Refer to Power Grounds for more information.

OPERATION

OPERATION

The PCM operates the fuel system. The PCM is a pre-programmed, triple microprocessor digital computer. It regulates ignition timing, air-fuel ratio, emission control devices, charging system, speed control (if equipped), air conditioning compressor clutch engagement and idle speed. The PCM can adapt its programming to meet changing operating conditions.

The PCM receives input signals from various switches and sensors. Based on these inputs, the PCM regulates various engine and vehicle operations through different system components. These components are referred to as PCM Outputs. The sensors and switches that provide inputs to the PCM are considered PCM Inputs.

The PCM adjusts ignition timing based upon inputs it receives from sensors that react to: engine rpm, manifold absolute pressure, engine coolant temperature, throttle position, transmission gear selection (automatic transmission), vehicle speed, power steering pump pressure (2.5L only), and the brake switch.

The PCM adjusts idle speed based on inputs it receives from sensors that react to: throttle position, vehicle speed, transmission gear selection, engine coolant temperature and from inputs it receives from the air conditioning clutch switch and brake switch.

Based on inputs that it receives, the PCM adjusts ignition coil dwell. The PCM also adjusts the generator charge rate through control of the generator field and provides speed control operation.

NOTE: Powertrain Control Module (PCM) Inputs:

- A/C request (if equipped with factory A/C)
- A/C select (if equipped with factory A/C)
- Auto shutdown (ASD) sense
- Battery temperature
- Battery voltage
- Brake switch
- CCD bus (+) circuits
- CCD bus (-) circuits
- Camshaft position sensor signal
- Crankshaft position sensor
- Data link connector for DRB scan tool
- Engine coolant temperature sensor
- Fuel level
- Generator (battery voltage) output
- Ignition circuit sense (ignition switch in run position)
- Intake manifold air temperature sensor
- Leak detection pump (if equipped)
- Manifold Absolute Pressure (MAP) sensor
- Oil pressure sensor
- Oxygen sensors
- Park/neutral switch (auto. trans. only)
- Power ground
- Power steering pressure switch (2.5L only)
- SCI receive (DRB scan tool 16–way connection)
- Sensor return
- Signal ground
- Speed control multiplexed single wire input
- Throttle position sensor
- Vehicle speed sensor

NOTE: Powertrain Control Module (PCM) Outputs:

- A/C clutch relay
- Auto Shutdown (ASD) relay
- CCD bus (+/-) circuits for: speedometer, voltmeter, fuel gauge, oil pressure gauge/lamp, engine temp. gauge and speed control warn. lamp
- Duty cycle EVAP canister purge solenoid
- Five volt sensor supply (primary)
- Five volt sensor supply (secondary)
- Fuel injectors
- Fuel pump relay

POWERTRAIN CONTROL MODULE (Continued)

- Generator field driver (-)
- Generator field source (+)
- Idle Air Control (IAC) motor
- Ignition coil
- Leak detection pump (if equipped)
- Malfunction indicator lamp (Check engine lamp). Driven through CCD circuits.
- SCI transmit (DRB scan tool 16–way connection)
- Speed control vacuum solenoid
- Speed control vent solenoid
- Tachometer (if equipped). Driven through CCD circuits.
- Transmission convertor clutch solenoid

OPERATION - 5 VOLT SUPPLIES

Primary 5–volt supply:

- supplies the required 5 volt power source to the Crankshaft Position (CKP) sensor.
- supplies the required 5 volt power source to the Camshaft Position (CMP) sensor.
- supplies a reference voltage for the Manifold Absolute Pressure (MAP) sensor.
- supplies a reference voltage for the Throttle Position Sensor (TPS) sensor.

Secondary 5–volt supply:

- supplies the required 5 volt power source to the oil pressure sensor.
- supplies the required 5 volt power source for the Vehicle Speed Sensor (VSS) (if equipped).
- supplies the 5 volt power source to the transmission pressure sensor (if equipped with an RE automatic transmission).

OPERATION - IGNITION CIRCUIT SENSE

The ignition circuit sense input tells the PCM the ignition switch has energized the ignition circuit.

Battery voltage is also supplied to the PCM through the ignition switch when the ignition is in the RUN or START position. This is referred to as the ″ignition sense″ circuit and is used to ″wake up″ the PCM. Voltage on the ignition input can be as low as 6 volts and the PCM will still function. Voltage is supplied to this circuit to power the PCM's 8-volt regulator and to allow the PCM to perform fuel, ignition and emissions control functions.

REMOVAL

USE THE DRB SCAN TOOL TO REPROGRAM THE NEW POWERTRAIN CONTROL MODULE (PCM) WITH THE VEHICLES ORIGINAL IDENTIFICATION NUMBER (VIN) AND THE VEHICLES ORIGINAL MILEAGE. IF THIS STEP IS NOT DONE, A DIAGNOSTIC TROUBLE CODE (DTC) MAY BE SET.

To avoid possible voltage spike damage to the PCM, ignition key must be off, and negative battery cable must be disconnected before unplugging PCM connectors.

(1) Disconnect negative battery cable at battery.

(2) Remove plastic shields from over four electrical connectors. Shields snap to connectors.

(3) Carefully unplug four electrical connectors (3) (Fig. 5) or (Fig. 6) from PCM.

(4) Remove three PCM mounting bolts (1) or (2) and remove PCM from vehicle.

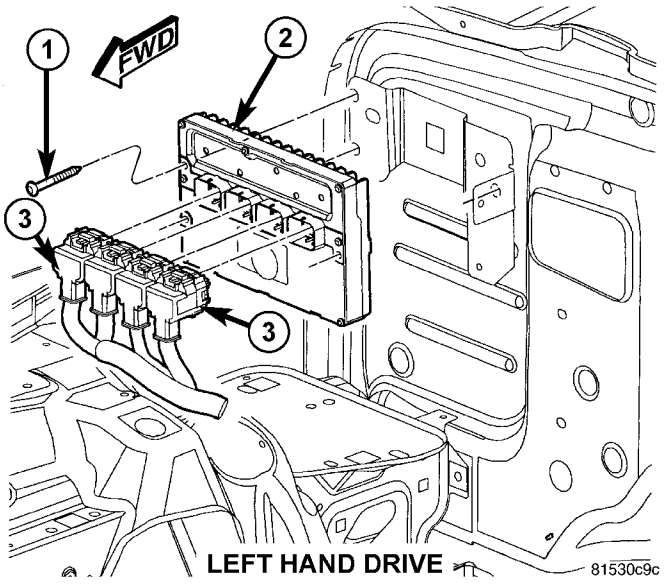

Fig. 5 PCM MOUNTING - LEFT HAND DRIVE

1 - MOUNTING SCREWS (3)
2 - PCM
3 - PCM ELECTRICAL CONNECTORS

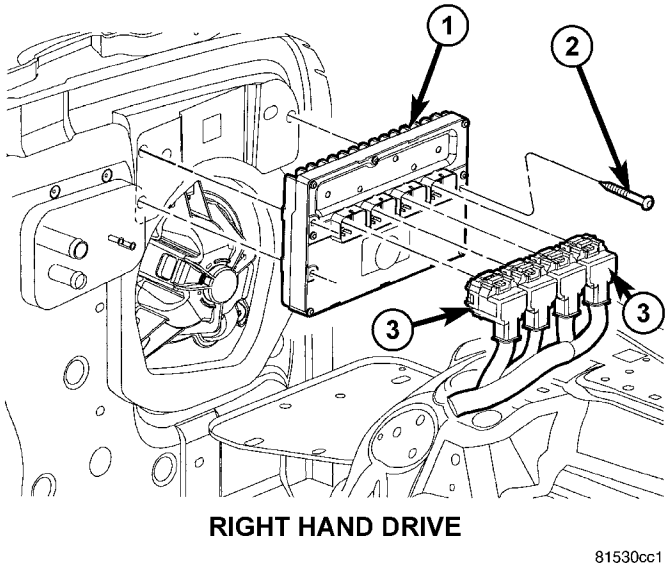

Fig. 6 PCM MOUNTING - RIGHT HAND DRIVE

1 - PCM
2 - MOUNTING SCREWS (3)
3 - PCM ELECTRICAL CONNECTORS

POWERTRAIN CONTROL MODULE (Continued)

INSTALLATION

USE THE DRB SCAN TOOL TO REPROGRAM THE NEW POWERTRAIN CONTROL MODULE (PCM) WITH THE VEHICLES ORIGINAL IDENTIFICATION NUMBER (VIN) AND THE VEHICLES ORIGINAL MILEAGE. IF THIS STEP IS NOT DONE, A DIAGNOSTIC TROUBLE CODE (DTC) MAY BE SET.

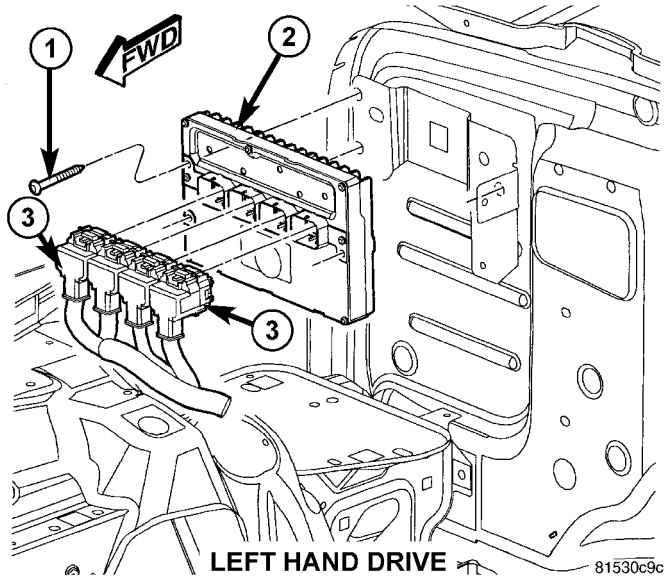

Fig. 7 PCM MOUNTING - LEFT HAND DRIVE

1 - MOUNTING SCREWS (3)
2 - PCM
3 - PCM ELECTRICAL CONNECTORS

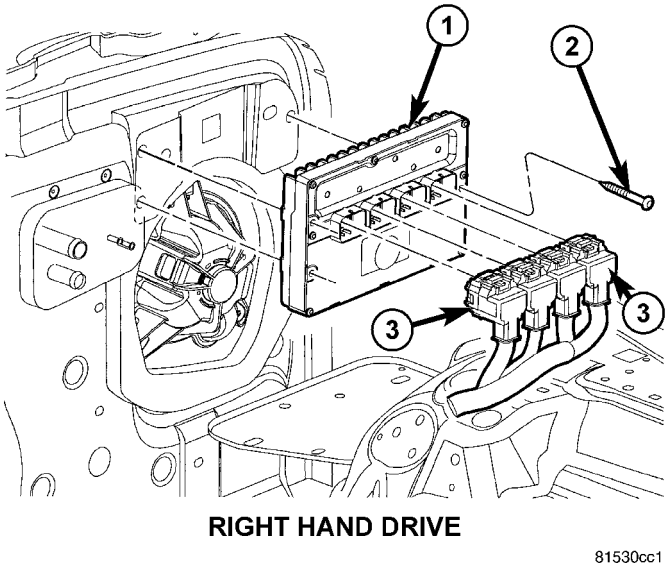

Fig. 8 PCM MOUNTING - RIGHT HAND DRIVE

1 - PCM
2 - MOUNTING SCREWS (3)
3 - PCM ELECTRICAL CONNECTORS

(1) Install PCM and three mounting screws to vehicle.

(2) Tighten screws to 4 N·m (35 in. lbs.).

(3) Check pin connectors in PCM and four electrical connectors for corrosion or damage. Also check pin heights in connectors. Pin heights should all be the same. Repair as necessary before installing electrical connectors.

(4) Install four electrical connectors (3) (Fig. 7) or (Fig. 8) to PCM.

(5) Install plastic shield to electrical connectors. Shields snap to connectors.

(6) Install battery cable.

(7) Use a scan tool to reprogram new PCM with vehicles original Vehicle Identification Number (VIN) and original vehicle mileage.

SENTRY KEY IMMOBILIZER MODULE

DESCRIPTION

The Sentry Key Immobilizer Module (SKIM) is the primary component of the Sentry Key Immobilizer System (SKIS) (Fig. 9). The SKIM is located in the steering column, below the ignition lock cylinder housing. The SKIM has an integral halo-like antenna ring that extends from one end. When the SKIM is properly installed on the steering column, the antenna ring is oriented around the circumference of the ignition lock cylinder housing.

The SKIM cannot be adjusted or repaired. If faulty or damaged, the entire SKIM unit must be replaced.

OPERATION

The Sentry Key Immobilizer Module (SKIM) contains a Radio Frequency (RF) transceiver and a microprocessor. The SKIM transmits RF signals to, and receives RF signals from the Sentry Key transponder through a tuned antenna enclosed within the molded plastic antenna ring integral to the SKIM housing. If this antenna ring is not mounted properly around the ignition lock cylinder housing, communication problems between the SKIM and the transponder may arise. These communication problems will result in Sentry Key transponder-related faults. The SKIM also communicates over the Programmable Communications Interface (PCI) data bus with the Powertrain Control Module (PCM), the ElectroMechanical Instrument Cluster (EMIC) and/or the DRBIII® scan tool.

The SKIM retains in memory the ID numbers of any Sentry Key transponder that is programmed into it. A maximum of eight Sentry Key transponders can be programmed into the SKIM. For added system security, each SKIM is programmed with a unique

SENTRY KEY IMMOBILIZER MODULE (Continued)

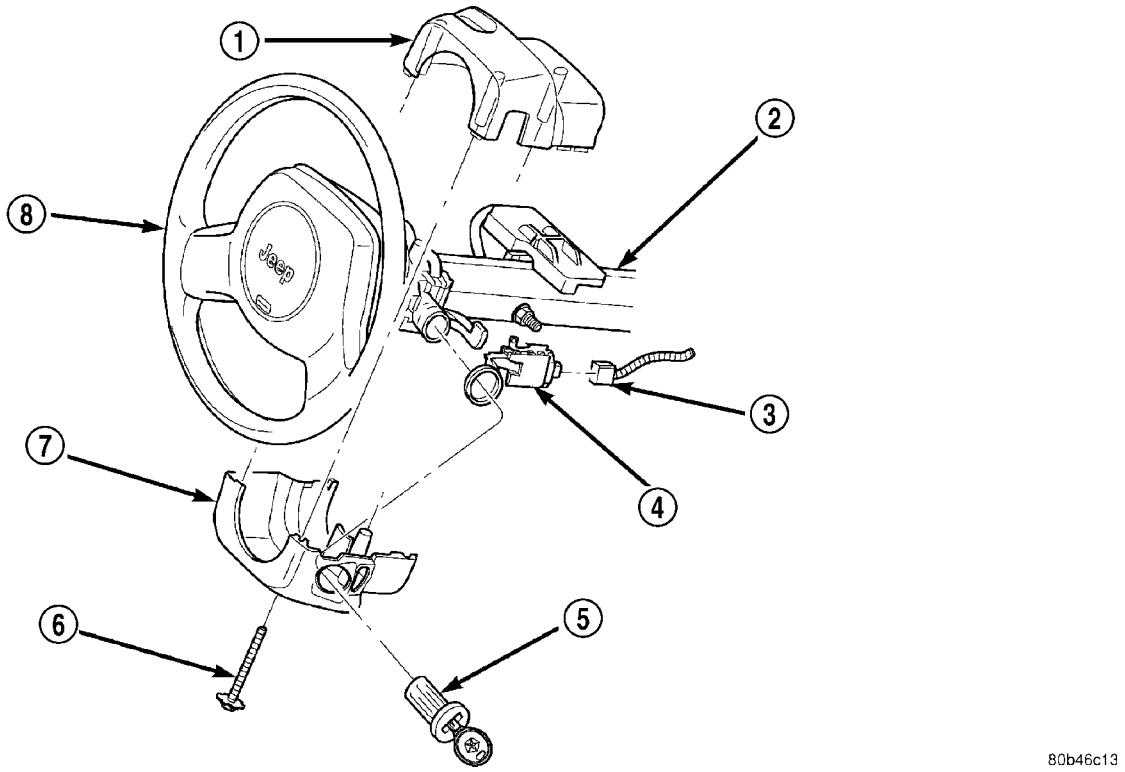

Fig. 9 Sentry Key Immobilizer Module - Typical

1 - UPPER SHROUD
2 - STEERING COLUMN
3 - WIRE HARNESS CONNECTOR
4 - SENTRY KEY IMMOBILIZER MODULE
5 - IGNITION LOCK CYLINDER
6 - SCREW
7 - LOWER SHROUD
8 - STEERING WHEEL

Secret Key code. This code is stored in memory, sent over the PCI data bus to the PCM, and is encoded to the transponder of every Sentry Key that is programmed into the SKIM. Therefore, the Secret Key code is a common element that is found in every component of the Sentry Key Immobilizer System (SKIS). Another security code, called a PIN, is used to gain access to the SKIM Secured Access Mode. The Secured Access Mode is required during service to perform the SKIS initialization and Sentry Key transponder programming procedures. The SKIM also stores the Vehicle Identification Number (VIN) in its memory, which it learns through a PCI data bus message from the PCM during SKIS initialization.

In the event that a SKIM replacement is required, the Secret Key code can be transferred to the new SKIM from the PCM using the DRBIII® scan tool and the SKIS initialization procedure. Proper completion of the SKIS initialization will allow the existing Sentry Keys to be programmed into the new SKIM so that new keys will not be required. In the event that the original Secret Key code cannot be recovered, SKIM replacement will also require new Sentry Keys. The DRBIII® scan tool will alert the technician during the SKIS initialization procedure if new Sentry Keys are required.

When the ignition switch is On, the SKIM transmits an RF signal to the transponder in the ignition key. The SKIM then waits for an RF signal response from the transponder. If the response received identifies the key as valid, the SKIM sends a valid key message to the PCM over the PCI data bus. If the response received identifies the key as invalid, or if no response is received from the key transponder, the SKIM sends an invalid key message to the PCM. The PCM will enable or disable engine operation based upon the status of the SKIM messages. It is important to note that the default condition in the PCM is an invalid key; therefore, if no message is received from the SKIM by the PCM, the engine will be disabled and the vehicle immobilized after two seconds of running.

The SKIM also sends SKIS indicator status messages to the EMIC. This indicator status message tells the EMIC to turn the indicator on for about three seconds each time the ignition switch is turned On, as a bulb test. The SKIM sends indicator status messages to the EMIC to turn the indicator off, turn the indicator on, or to flash the indicator on and off.

SENTRY KEY IMMOBILIZER MODULE (Continued)

If the SKIS indicator flashes upon ignition On or stays on solid after the bulb test, it signifies a SKIS fault. If the SKIM detects a system malfunction and/or the SKIS has become inoperative, the SKIS indicator will stay on solid. If the SKIM detects an invalid key or if a key transponder-related fault exists, the SKIS indicator will flash. If the vehicle is equipped with the Customer Learn transponder programming feature, the SKIM will also send messages to the EMIC to flash the SKIS indicator and to generate a single audible chime whenever the Customer Learn programming mode is being utilized. (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - STANDARD PROCEDURE - SENTRY KEY TRANSPONDER PROGRAMMING).

The SKIS performs a self-test each time the ignition switch is turned to the On position, and will store fault information in the form of Diagnostic Trouble Codes (DTC's) in SKIM memory if a system malfunction is detected. The SKIM can be diagnosed, and any stored DTC's can be retrieved using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) Remove the screws that secure the lower steering column shroud to the upper shroud (Fig. 10).

(4) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully lowered position and leave the tilt release lever in the released (down) position.

(5) Remove the upper and lower shrouds from the steering column.

(6) On models equipped with a manual transmission, remove the screws that secure the multi-function switch assembly to the upper steering column housing.

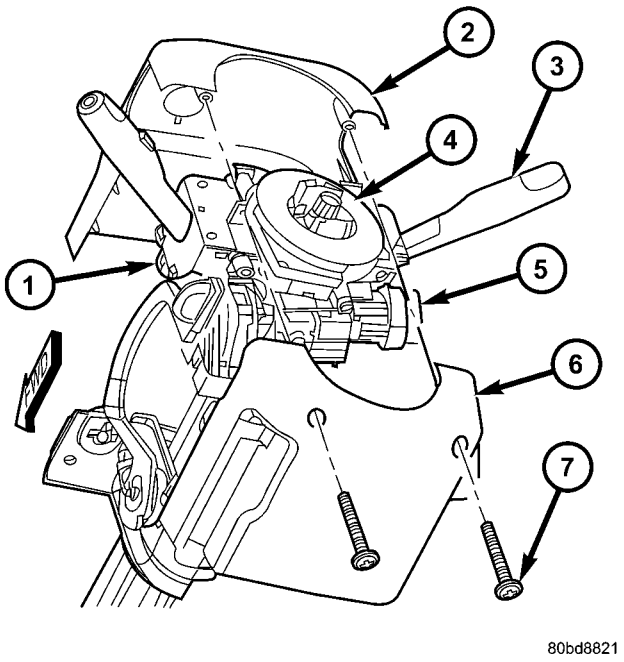

Fig. 10 Steering Column Shrouds Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - UPPER SHROUD
3 - RIGHT MULTI-FUNCTION SWITCH
4 - CLOCKSPRING
5 - IGNITION LOCK CYLINDER HOUSING
6 - LOWER SHROUD
7 - SCREW (2)

(7) Disconnect the cross body wire harness connector for the SKIM from the SKIM connector (Fig. 11).

(8) The SKIM mounting bracket features a clip that secures the SKIM to the inboard lower flange of the steering column jacket. Pull downward on the connector end of the SKIM mounting bracket to release this clip from the steering column jacket.

(9) Rotate the SKIM and its mounting bracket downwards and then to the side away from the steering column to slide the SKIM antenna ring from around the ignition switch lock cylinder housing. On models with a manual transmission, lift the multi-function switch upward off of the upper steering column housing far enough to extract the SKIM antenna from between the ignition key release button and the right multi-function switch housing.

(10) Remove the SKIM from the steering column.

SENTRY KEY IMMOBILIZER MODULE (Continued)

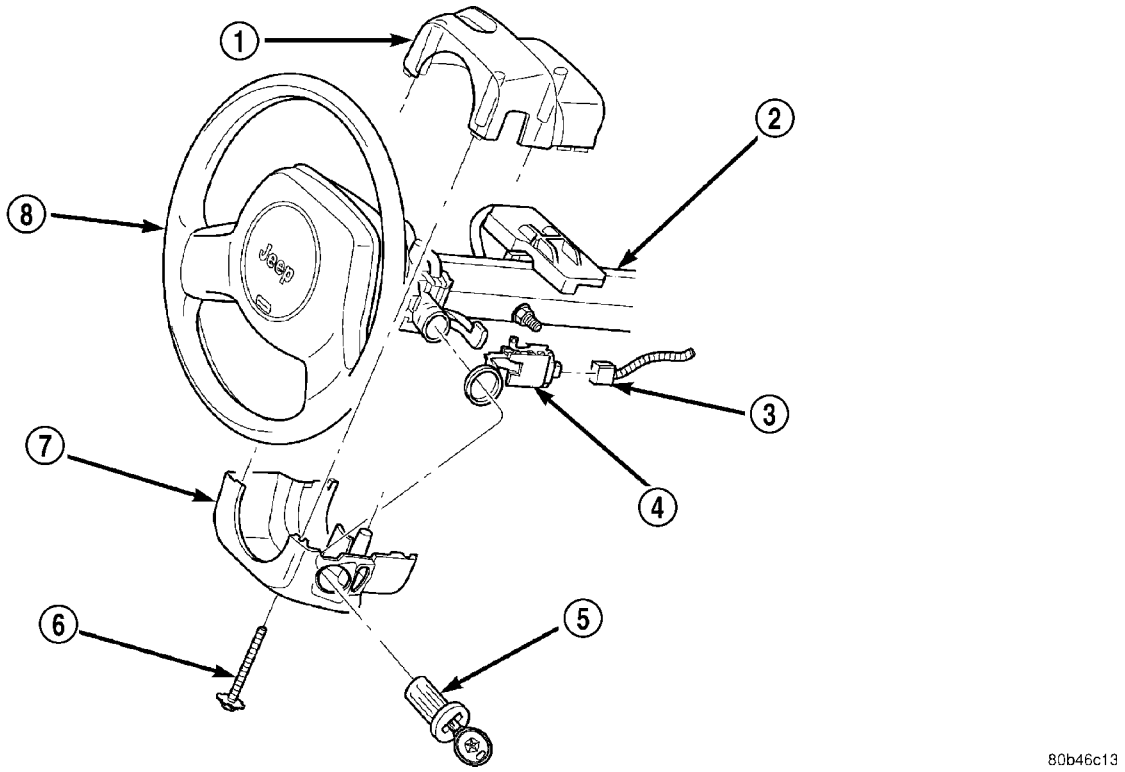

Fig. 11 Sentry Key Immobilizer Module - Typical

1 - UPPER SHROUD
2 - STEERING COLUMN
3 - WIRE HARNESS CONNECTOR
4 - SENTRY KEY IMMOBILIZER MODULE
5 - IGNITION LOCK CYLINDER
6 - SCREW
7 - LOWER SHROUD
8 - STEERING WHEEL

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Position the Sentry Key Immobilizer Module (SKIM) (Fig. 11). On models with a manual transmission, lift the multi-function switch upward off of the upper steering column housing far enough to insert the SKIM antenna formation between the ignition key release button and the multi-function switch housing.

(2) Slide the SKIM antenna ring around the ignition switch lock cylinder housing, then rotate the SKIM and its mounting bracket upwards and toward the steering column.

(3) Align the SKIM mounting bracket clip with the inboard lower flange of the steering column and, push upward firmly on the connector end of the SKIM mounting bracket to engage the clip with the steering column.

(4) Reconnect the cross body wire harness connector for the SKIM to the SKIM connector.

(5) On models equipped with a manual transmission, install and tighten the screws that secure the multi-function switch assembly to the upper steering column housing. Tighten the screws to 2 N·m (20 in. lbs.).

(6) Position both the upper and lower shrouds onto the steering column (Fig. 10).

(7) Install and tighten the screws that secure the lower steering column shroud to the upper shroud. Tighten the screws to 2 N·m (18 in. lbs.).

(8) Move the tilt steering column to the fully raised position and secure it in place by moving the tilt release lever back to the locked (up) position.

SENTRY KEY IMMOBILIZER MODULE (Continued)

(9) Reinstall the steering column opening cover. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(10) Reconnect the battery negative cable.

NOTE: If the SKIM has been replaced with a new unit, the Sentry Key Immobilizer System (SKIS) MUST be initialized before the vehicle can be operated. (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - STANDARD PROCEDURE - SKIS INITIALIZATION).

TRANSMISSION CONTROL MODULE

DESCRIPTION

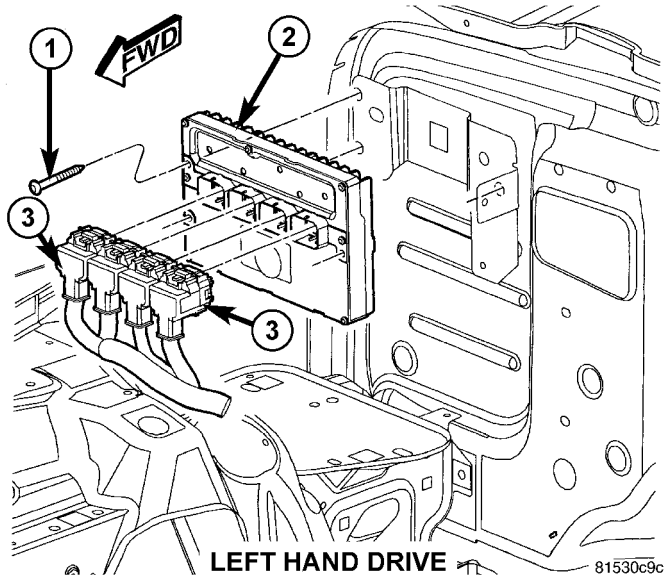

Fig. 12 PCM MOUNTING - LEFT HAND DRIVE

1 - MOUNTING SCREWS (3)
2 - PCM
3 - PCM ELECTRICAL CONNECTORS

The Transmission Control Module (TCM) is a submodule within the Powertrain Control Module (PCM) (2).

The PCM (2) is located on the right side of the dash panel in the engine compartment (Fig. 12) on left hand drive vehicles. The PCM is referred to as the NGC.

The PCM (1) is located on the left side of the dash panel in the engine compartment (Fig. 13) on right hand drive vehicles.

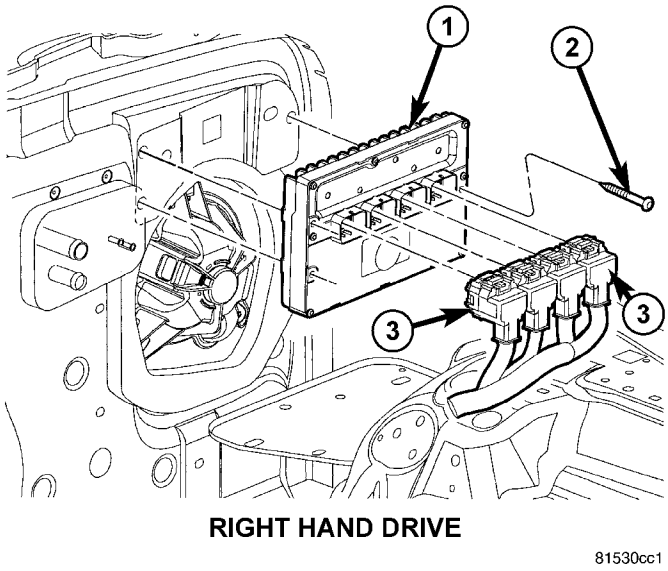

Fig. 13 PCM MOUNTING - RIGHT HAND DRIVE

1 - PCM
2 - MOUNTING SCREWS (3)
3 - PCM ELECTRICAL CONNECTORS

OPERATION

The TCM is the controlling unit for all electronic operations of the transmission. The TCM receives information regarding vehicle operation from both direct and indirect inputs, and selects the operational mode of the transmission. Direct inputs are hardwired to, and used specifically by the TCM. Indirect inputs originate from other components/modules, and are shared with the TCM via the PCI bus.

Some examples of **direct inputs** to the TCM are:

- Battery (B+) voltage
- Ignition “ON” voltage
- Transmission Control Relay (Switched B+)
- Throttle Position Sensor
- Crankshaft Position Sensor (CKP)
- Transmission Range Sensor (TRS)
- Pressure Switches (L/R, 2/4, OD)
- Transmission Temperature Sensor (Integral to TRS)
- Input Shaft Speed Sensor
- Output Shaft Speed Sensor

Some examples of **indirect inputs** to the TCM are:

- Engine/Body Identification
- Manifold Pressure
- Target Idle
- Torque Reduction Confirmation
- Speed Control ON/OFF Switch
- Engine Coolant Temperature
- Ambient/Battery Temperature
- Brake Switch Status
- DRB® III Communication

TRANSMISSION CONTROL MODULE (Continued)

Based on the information received from these various inputs, the TCM determines the appropriate shift schedule and shift points, depending on the present operating conditions and driver demand. This is possible through the control of various direct and indirect outputs.

Some examples of TCM **direct outputs** are:

- Transmission Control Relay
- Solenoids (L/R, 2/4, OD and UD)
- Vehicle Speed (to PCM)
- Torque Reduction Request (to PCM)

Some examples of TCM **indirect outputs** are:

- Transmission Temperature (to PCM)
- PRNDL Position (to BCM)

In addition to monitoring inputs and controlling outputs, the TCM has other important responsibilities and functions:

- Storing and maintaining Clutch Volume Indices (CVI)
- Storing and selecting appropriate Shift Schedules
- System self-diagnostics
- Diagnostic capabilities (with DRB® III scan tool)

NOTE: If the TCM has been replaced, the "Quick Learn Procedure" must be performed. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/ TRANSMISSION CONTROL MODULE - STANDARD PROCEDURE)

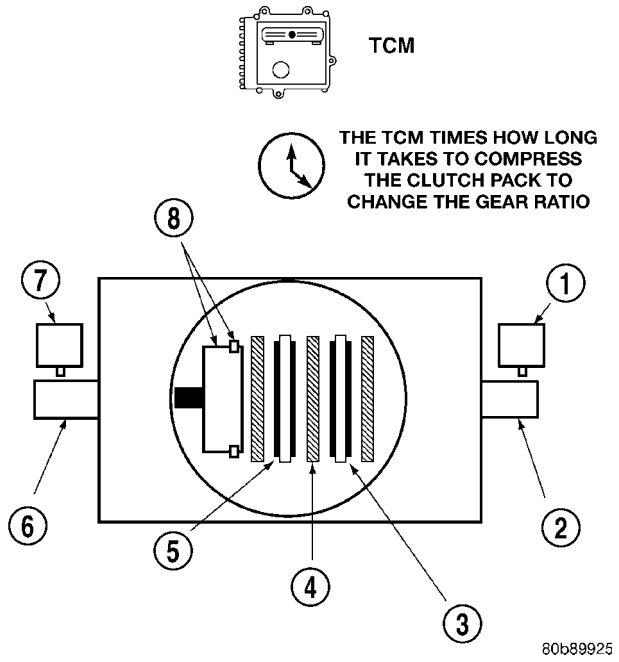

Fig. 14 Example of CVI Calculation

1 - OUTPUT SPEED SENSOR
2 - OUTPUT SHAFT
3 - CLUTCH PACK
4 - SEPARATOR PLATE
5 - FRICTION DISCS
6 - INPUT SHAFT
7 - INPUT SPEED SENSOR
8 - PISTON AND SEAL

CLUTCH VOLUME INDEX (CVI)

An important function of the TCM is to monitor Clutch Volume Index (CVI). CVIs represent the volume of fluid needed to compress a clutch pack.

The TCM monitors gear ratio changes by monitoring the Input and Output Speed Sensors. The Input, or Turbine Speed Sensor sends an electrical signal to the TCM that represents input shaft rpm. The Output Speed Sensor provides the TCM with output shaft speed information.

By comparing the two inputs, the TCM can determine transmission gear position. This is important to the CVI calculation because the TCM determines CVIs by monitoring how long it takes for a gear change to occur (Fig. 14).

Gear ratios can be determined by using the DRB® III Scan Tool and reading the Input/Output Speed Sensor values in the "Monitors" display. Gear ratio can be obtained by dividing the Input Speed Sensor value by the Output Speed Sensor value.

For example, if the input shaft is rotating at 1000 rpm and the output shaft is rotating at 500 rpm, then the TCM can determine that the gear ratio is 2:1. In direct drive (3rd gear), the gear ratio changes to 1:1. The gear ratio changes as clutches are applied and released. By monitoring the length of time it takes for the gear ratio to change following a shift request, the TCM can determine the volume of fluid used to apply or release a friction element.

The volume of transmission fluid needed to apply the friction elements are continuously updated for adaptive controls. As friction material wears, the volume of fluid need to apply the element increases.

Certain mechanical problems within the input clutch assembly (broken return springs, out of position snap rings, excessive clutch pack clearance, improper assembly, etc.) can cause inadequate or out-of-range element volumes. Also, defective Input/Output Speed Sensors and wiring can cause these conditions. The following chart identifies the appropriate clutch volumes and when they are monitored/ updated:

TRANSMISSION CONTROL MODULE (Continued)

CLUTCH VOLUMES				
Clutch	When Updated			Proper Clutch Volume
	Shift Sequence	Oil Temperature	Throttle Angle	
L/R	2-1 or 3-1 coast downshift	> 70°	< 5°	35 to 83
2/4	1-2 shift	> 110°	5 - 54°	20 to 77
OD	2-3 shift			48 to 150
UD	4-3 or 4-2 shift		> 5°	24 to 70

SHIFT SCHEDULES

As mentioned earlier, the TCM has programming that allows it to select a variety of shift schedules. Shift schedule selection is dependent on the following:

- Shift lever position
- Throttle position
- Engine load
- Fluid temperature
- Software level

As driving conditions change, the TCM appropriately adjusts the shift schedule. Refer to the following table 42RLE Shift Schedule to determine the appropriate operation expected, depending on driving conditions.

42RLE SHIFT SCHEDULE

Schedule	Condition	Expected Operation
Extreme Cold	Oil temperature at start-up below -16° F	Park, Reverse, Neutral and 2nd gear only (prevents shifting which may fail a clutch with frequent shifts)
Cold	Oil temperature at start-up above -12° F and below 36° F	– Delayed 2-3 upshift (approximately 22-31 mph) – Delayed 3-4 upshift (45-53 mph) – Early 4-3 costdown shift (approximately 30 mph) – Early 3-2 coastdown shift (approximately 17 mph) – High speed 4-2, 3-2, 2-1 kickdown shifts are prevented – No EMCC
Warm	Oil temperature at start-up above 36° F and below 80 degree F	– Normal operation (upshift, kickdowns, and coastdowns) – No EMCC
Hot	Oil temperature at start-up above 80° F	– Normal operation (upshift, kickdowns, and coastdowns) – Full EMCC, no PEMCC except to engage FEMCC (except at closed throttle at speeds above 70-83 mph)
Overheat	Oil temperature above 240° F or engine coolant temperature above 244° F	– Delayed 2-3 upshift (25-32 mph) – Delayed 3-4 upshift (41-48 mph) – 3rd gear FEMCC from 30-48 mph – 3rd gear PEMCC from 27-31 mph

TRANSMISSION CONTROL MODULE (Continued)

Schedule	Condition	Expected Operation
Super Overheat	Oil temperature above 260° F	– All ″Overheat″ shift schedule features apply – 2nd gear PEMCC above 22 mph – Above 22 mph the torque converter will not unlock unless the throttle is closed or if a wide open throttle 2nd PEMCC to 1 kickdown is made

STANDARD PROCEDURE - TCM QUICK LEARN

The quick learn procedure requires the use of the scan tool.

This program allows the electronic transmission system to recalibrate itself. This will provide the proper transmission operation. The quick learn procedure should be performed if any of the following procedures are performed:

- Transmission Assembly Replacement
- Transmission Control Module Replacement
- Solenoid Pack Replacement
- Clutch Plate and/or Seal Replacement
- Valve Body Replacement or Recondition

To perform the Quick Learn Procedure, the following conditions must be met:

- The brakes must be applied
- The engine speed must be above 500 rpm
- The throttle angle (TPS) must be less than 3 degrees
- The shift lever position must stay in PARK until prompted to shift to overdrive
- The shift lever position must stay in overdrive after the Shift to Overdrive prompt until the scan tool indicates the procedure is complete.
- The calculated oil temperature must be above 60° and below 200°

ENGINE SYSTEMS

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BATTERY SYSTEM

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BATTERY SYSTEM

DESCRIPTION

A single 12-volt battery is standard factory-installed equipment on this model. All of the components of the battery system are located within the engine compartment of the vehicle. The battery system for this vehicle covers the following related components, which are covered in further detail later in this section of the service manual:

- **Battery** - The storage battery provides a reliable means of storing a renewable source of electrical energy within the vehicle.
- **Battery Cable** - The battery cables connect the battery terminal posts to the vehicle electrical system.
- **Battery Holddown** - The battery holddown hardware secures the battery in the battery tray in the engine compartment.

BATTERY SYSTEM (Continued)

• **Battery Tray** - The battery tray provides a secure mounting location in the vehicle for the battery and an anchor point for the battery holddown hardware.

For battery system maintenance schedules and jump starting procedures, see the owner's manual in the vehicle glove box. Optionally, refer to the Lubrication and Maintenance section of this manual for the recommended battery maintenance schedules and for the proper battery jump starting procedure. While battery charging can be considered a maintenance procedure, the battery charging procedure and related information are located later in this section of this service manual. This was done because the battery must be fully-charged before any battery system diagnosis or testing procedures can be performed.

OPERATION

The battery system is designed to provide a safe, efficient, reliable and mobile means of delivering and storing electrical energy. This electrical energy is required to operate the engine starting system, as well as to operate many of the other vehicle accessory systems for limited durations while the engine and/or the charging system are not operating. The battery system is also designed to provide a reserve of electrical energy to supplement the charging system for short durations while the engine is running and the electrical current demands of the vehicle exceed the output of the charging system. In addition to delivering, and storing electrical energy for the vehicle, the battery system serves as a capacitor and voltage stabilizer for the vehicle electrical system. It absorbs most abnormal or transient voltages caused by the switching of any of the electrical components or circuits in the vehicle.

DIAGNOSIS AND TESTING - BATTERY SYSTEM

The battery, starting, and charging systems in the vehicle operate with one another and must be tested as a complete system. In order for the engine to start and the battery to maintain its charge properly, all of the components that are used in these systems must perform within specifications. It is important that the battery, starting, and charging systems be thoroughly tested and inspected any time a battery needs to be charged or replaced. The cause of abnormal battery discharge, overcharging or early battery failure must be diagnosed and corrected before a battery is replaced and before a vehicle is returned to service. The service information for these systems has been separated within this service manual to make it easier to locate the specific information you are seeking. However, when attempting to diagnose any of these systems, it is important that you keep their interdependency in mind.

The diagnostic procedures used for the battery, starting, and charging systems include the most basic conventional diagnostic methods, to the more sophisticated On-Board Diagnostics (OBD) built into the Powertrain Control Module (PCM). Use of an induction-type milliampere ammeter, a volt/ohmmeter, a battery charger, a carbon pile rheostat (load tester) and a 12-volt test lamp may be required. All OBD-sensed systems are monitored by the PCM. Each monitored circuit is assigned a Diagnostic Trouble Code (DTC). The PCM will store a DTC in electronic memory for any failure it detects. Refer to Charging System for the proper charging system onboard diagnostic test procedures.

MICRO 420 BATTERY TESTER

The Micro 420 automotive battery tester is designed to help the dealership technicians diagnose a defective battery. Follow the instruction manual supplied with the tester to properly diagnose a vehicle. If the instruction manual is not available refer to the standard procedure in this section, which includes the directions for using the Micro 420 battery tester.

BATTERY SYSTEM (Continued)

BATTERY SYSTEM DIAGNOSIS		
CONDITION	POSSIBLE CAUSES	CORRECTION
THE BATTERY SEEMS WEAK OR DEAD WHEN ATTEMPTING TO START THE ENGINE.	1. The electrical system ignition-off draw is excessive.	1. Refer to the IGNITION-OFF DRAW TEST Standard Procedure for the proper test procedures. Repair the excessive ignition-off draw, as required.
	2. The charging system is faulty.	2. Determine if the charging system is performing to specifications. Refer to Charging System for charging system diagnosis and testing procedures. Repair the faulty charging system, as required.
	3. The battery is discharged.	3. Determine the battery state-of-charge using the Micro 420 battery tester. Refer to the Standard Procedures in this section for additional test procedures. Charge the faulty battery, as required.
	4. The battery terminal connections are loose or corroded.	4. Refer to Battery Cables for the proper battery cable diagnosis and testing procedures. Clean and tighten the battery terminal connections, as required.
	5. The battery has an incorrect size or rating for this vehicle.	5. Refer to Battery System Specifications for the proper size and rating. Replace an incorrect battery, as required.
	6. The battery is faulty.	6. Determine the battery cranking capacity using the Micro 420 battery tester. Refer to the Standard Procedures in this section for additional test procedures. Replace the faulty battery, as required.
	7. The starting system is faulty.	7. Determine if the starting system is performing to specifications. Refer to Starting System for the proper starting system diagnosis and testing procedures. Repair the faulty starting system, as required.
	8. The battery is physically damaged.	8. Inspect the battery for loose terminal posts or a cracked and leaking case. Replace the damaged battery, as required.

BATTERY SYSTEM (Continued)

BATTERY SYSTEM DIAGNOSIS		
CONDITION	POSSIBLE CAUSES	CORRECTION
THE BATTERY STATE OF CHARGE CANNOT BE MAINTAINED.	1. The battery has an incorrect size or rating for this vehicle.	1. Refer to Battery System Specifications for the proper specifications. Replace an incorrect battery, as required.
	2. The battery terminal connections are loose or corroded.	2. Refer to Battery Cable for the proper cable diagnosis and testing procedures. Clean and tighten the battery terminal connections, as required.
	3. The electrical system ignition-off draw is excessive.	3. Refer to the IGNITION-OFF DRAW TEST Standard Procedure for the proper test procedures. Repair the faulty electrical system, as required.
	4. The battery is faulty.	4. Test the battery using the Micro 420 battery tester. Refer to Standard Procedures for additional test procedures. Replace the faulty battery, as required.
	5. The starting system is faulty.	5. Determine if the starting system is performing to specifications. Refer to Starting System for the proper starting system diagnosis and testing procedures. Repair the faulty starting system, as required.
	6. The charging system is faulty.	6. Determine if the charging system is performing to specifications. Refer to Charging System for additional charging system diagnosis and testing procedures. Repair the faulty charging system, as required.
	7. Electrical loads exceed the output of the charging system.	7. Inspect the vehicle for aftermarket electrical equipment which might cause excessive electrical loads.
	8. Slow driving or prolonged idling with high-amperage draw systems in use.	8. Advise the vehicle operator, as required.
THE BATTERY WILL NOT ACCEPT A CHARGE.	1. The battery is faulty.	1. Test the battery using the Micro 420 battery tester. Charge or replace the faulty battery, as required.

ABNORMAL BATTERY DISCHARGING

Any of the following conditions can result in abnormal battery discharging:

1. A faulty or incorrect charging system component. Refer to Charging System for charging system diagnosis and testing procedures.
2. A faulty or incorrect battery. Use Micro 420 battery tester and refer to Battery System for additional battery diagnosis and testing procedures.
3. A faulty circuit or component causing excessive ignition-off draw.
4. Electrical loads that exceed the output of the charging system. This can be due to equipment installed after manufacture, or repeated short trip use.
5. A faulty or incorrect starting system component. Refer to Starting System for the proper starting system diagnosis and testing procedures.
6. Corroded or loose battery posts and/or terminal clamps.
7. Slow driving speeds (heavy traffic conditions) or prolonged idling, with high-amperage draw systems in use.

CLEANING

The following information details the recommended cleaning procedures for the battery and related com-

BATTERY SYSTEM (Continued)

ponents. In addition to the maintenance schedules found in this service manual and the owner's manual, it is recommended that these procedures be performed any time the battery or related components must be removed for vehicle service.

(1) Clean the battery cable terminal clamps of all corrosion. Remove any corrosion using a wire brush or a post and terminal cleaning tool, and a sodium bicarbonate (baking soda) and warm water cleaning solution (Fig. 1).

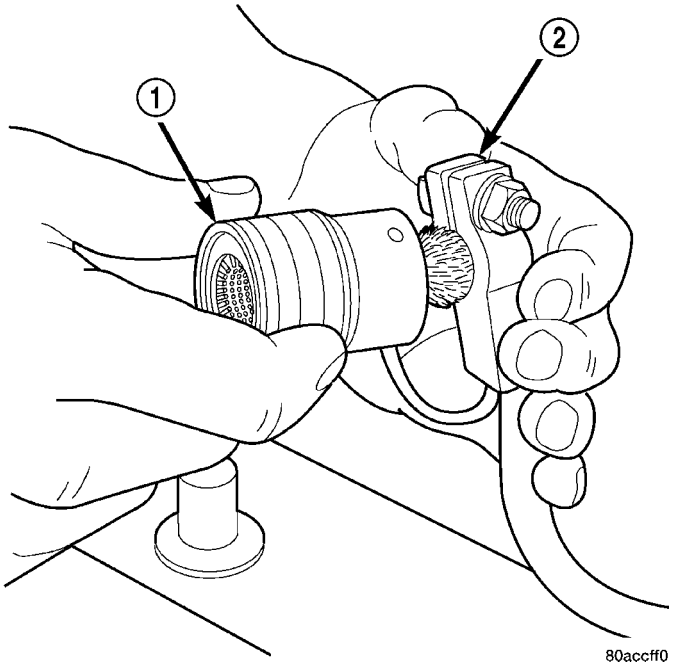


Fig. 1 Clean Battery Cable Terminal Clamp - Typical

1 - TERMINAL BRUSH
2 - BATTERY CABLE

(2) Clean the battery tray and battery hold down hardware of all corrosion. Remove any corrosion using a wire brush and a sodium bicarbonate (baking soda) and warm water cleaning solution. Paint any exposed bare metal.

(3) If the removed battery is to be reinstalled, clean the outside of the battery case and the top cover with a sodium bicarbonate (baking soda) and warm water cleaning solution using a stiff bristle parts cleaning brush to remove any acid film (Fig. 2). Rinse the battery with clean water. Ensure that the cleaning solution does not enter the battery cells through the vent holes. If the battery is being replaced, refer to Battery System Specifications for the factory-installed battery specifications. Confirm that the replacement battery is the correct size and has the correct ratings for the vehicle.

(4) If the vehicle is so equipped, clean the battery thermal guard with a sodium bicarbonate (baking soda) and warm water cleaning solution using a stiff bristle parts cleaning brush to remove any acid film.

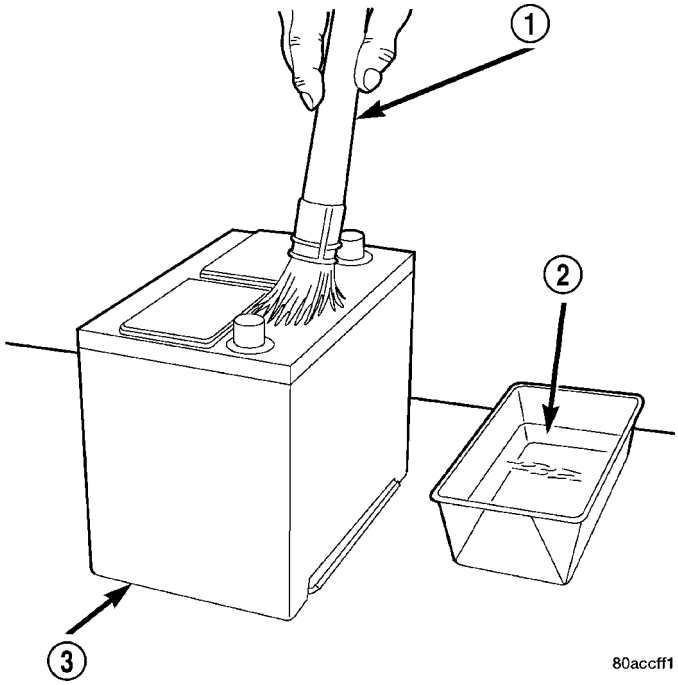


Fig. 2 Battery Cleaning - Typical

1 - CLEANING BRUSH
2 - WARM WATER AND BAKING SODA SOLUTION
3 - BATTERY

(5) Clean any corrosion from the battery terminal posts with a wire brush or a post and terminal cleaner, and a sodium bicarbonate (baking soda) and warm water cleaning solution (Fig. 3).

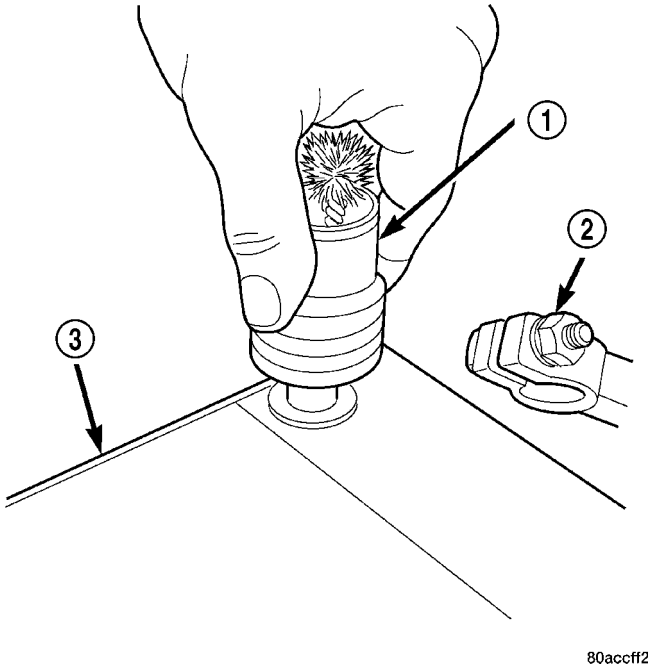


Fig. 3 Clean Battery Terminal Post - Typical

1 - TERMINAL BRUSH
2 - BATTERY CABLE
3 - BATTERY

BATTERY SYSTEM (Continued)

INSPECTION

The following information details the recommended inspection procedures for the battery and related components. In addition to the maintenance schedules found in this service manual and the owner's manual, it is recommended that these procedures be performed any time the battery or related components must be removed for vehicle service.

(1) Inspect the battery cable terminal clamps for damage. Replace any battery cable that has a damaged or deformed terminal clamp.

(2) Inspect the battery tray and battery holddown hardware for damage. Replace any damaged parts.

(3) Slide the thermal guard off of the battery case. Inspect the battery case for cracks or other damage that could result in electrolyte leaks. Also, check the battery terminal posts for looseness. Batteries with damaged cases or loose terminal posts must be replaced.

(4) Inspect the battery thermal guard for tears, cracks, deformation or other damage. Replace any battery thermal guard that has been damaged.

(5) Inspect the battery built-in test indicator sight glass for an indication of the battery condition. If the battery is discharged, charge as required. Refer to Standard Procedures for the proper battery built-in indicator test procedures. Also refer to Standard Procedures for the proper battery charging procedures.

SPECIFICATIONS

BATTERY SYSTEM SPECIFICATIONS

The battery Group Size number, the Cold Cranking Amperage (CCA) rating, and the Reserve Capacity (RC) rating or Ampere-Hours (AH) rating can be found on the original equipment battery label. Be certain that a replacement battery has the correct Group Size number, as well as CCA, and RC or AH ratings that equal or exceed the original equipment specification for the vehicle being serviced. Battery sizes and ratings are discussed in more detail below.

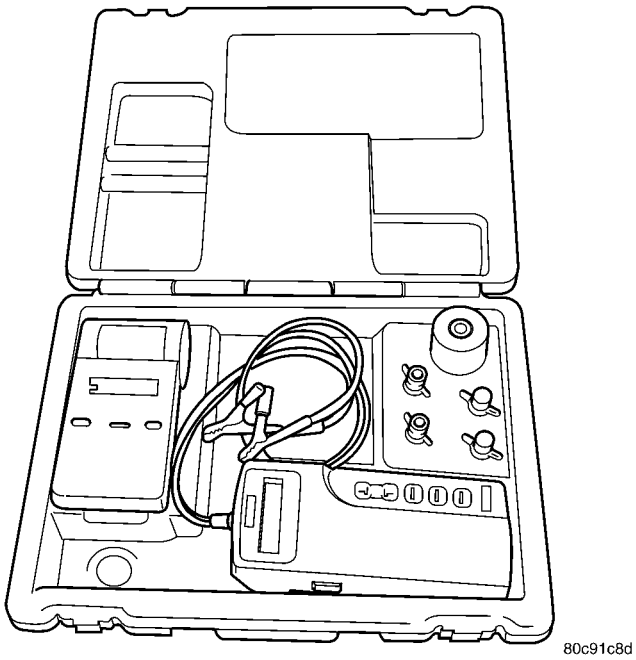
- **Group Size** - The outside dimensions and terminal placement of the battery conform to standards established by the Battery Council International (BCI). Each battery is assigned a BCI Group Size number to help identify a correctly-sized replacement.
- **Cold Cranking Amperage** - The Cold Cranking Amperage (CCA) rating specifies how much current (in amperes) the battery can deliver for thirty seconds at -18° C (0° F). Terminal voltage must not fall below 7.2 volts during or after the thirty second discharge period. The CCA required is generally higher as engine displacement increases, depending also upon the starter current draw requirements.
- **Reserve Capacity** - The Reserve Capacity (RC) rating specifies the time (in minutes) it takes for battery terminal voltage to fall below 10.5 volts, at a discharge rate of 25 amperes. RC is determined with the battery fully-charged at 26.7° C (80° F). This rating estimates how long the battery might last after a charging system failure, under minimum electrical load.
- **Ampere-Hours** - The Ampere-Hours (AH) rating specifies the current (in amperes) that a battery can deliver steadily for twenty hours, with the voltage in the battery not falling below 10.5 volts. This rating is also sometimes identified as the twenty-hour discharge rating.

Battery Classifications and Ratings					
Part Number	BCI Group Size Classification	Cold Cranking Amperage	Reserve Capacity	Ampere-Hours	Load Test Amperage
04609365AC	34	600	120 Minutes	66	300

BATTERY SYSTEM (Continued)

SPECIAL TOOLS

BATTERY SYSTEM SPECIAL TOOLS



MICRO 420 BATTERY TESTER

BATTERY

DESCRIPTION

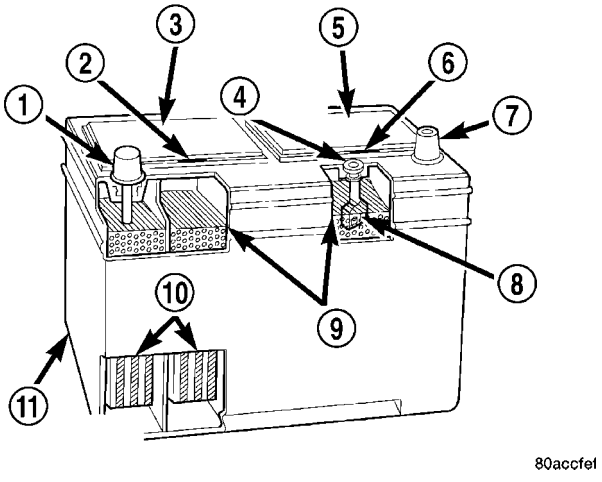


Fig. 4 Low-Maintenance Battery - Typical

1 - POSITIVE POST
2 - VENT
3 - CELL CAP
4 - VENT
5 - CELL CAP
6 - VENT
7 - NEGATIVE POST
8 - GREEN BALL
9 - ELECTROLYTE LEVEL
10 - PLATE GROUPS
11 - LOW-MAINTENANCE BATTERY

A large capacity, low-maintenance storage battery (Fig. 4) is standard factory-installed equipment on this model. Refer to Battery Specifications for the proper specifications of the factory-installed batteries available on this model. Male post type terminals made of a soft lead material protrude from the top of the molded plastic battery case to provide the means for connecting the battery to the vehicle electrical system. The battery positive terminal post is physically larger in diameter than the negative terminal post to ensure proper battery connection. The letters **POS** and **NEG** are also molded into the top of the battery case adjacent to their respective positive and negative terminal posts for identification confirmation (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/ CABLES - DESCRIPTION).

The battery is made up of six individual cells that are connected in series. Each cell contains positively charged plate groups that are connected with lead straps to the positive terminal post, and negatively charged plate groups that are connected with lead straps to the negative terminal post. Each plate consists of a stiff mesh framework or grid coated with lead dioxide (positive plate) or sponge lead (negative plate). Insulators or plate separators made of a nonconductive material are inserted between the positive and negative plates to prevent them from contacting or shorting against one another. These dissimilar metal plates are submerged in a sulfuric acid and water solution called an electrolyte.

The factory-installed battery has a built-in test indicator (hydrometer). The color visible in the sight glass of the indicator will reveal the battery condition. Refer to Standard Procedures for the proper built-in indicator test procedures. **The factory-installed low-maintenance battery has removable battery cell caps.** Water can be added to this battery. The battery is not sealed and has vent holes in the cell caps. The chemical composition of the metal coated plates within the low-maintenance battery reduces battery gassing and water loss, at normal charge and discharge rates. Therefore, the battery should not require additional water in normal service. If the electrolyte level in this battery does become low, water must be added. However, rapid loss of electrolyte can be caused by an overcharging condition. Be certain to diagnose the charging system after replenishing the water in the battery for a low electrolyte condition and before returning the vehicle to service (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING).

BATTERY (Continued)

OPERATION

The battery is designed to store electrical energy in a chemical form. When an electrical load is applied to the terminals of the battery, an electrochemical reaction occurs. This reaction causes the battery to discharge electrical current from its terminals. As the battery discharges, a gradual chemical change takes place within each cell. The sulfuric acid in the electrolyte combines with the plate materials, causing both plates to slowly change to lead sulfate. At the same time, oxygen from the positive plate material combines with hydrogen from the sulfuric acid, causing the electrolyte to become mainly water. The chemical changes within the battery are caused by the movement of excess or free electrons between the positive and negative plate groups. This movement of electrons produces a flow of electrical current through the load device attached to the battery terminals.

As the plate materials become more similar chemically, and the electrolyte becomes less acid, the voltage potential of each cell is reduced. However, by charging the battery with a voltage higher than that of the battery itself, the battery discharging process is reversed. Charging the battery gradually changes the sulfated lead plates back into sponge lead and lead dioxide, and the water back into sulfuric acid. This action restores the difference in the electron charges deposited on the plates, and the voltage potential of the battery cells. For a battery to remain useful, it must be able to produce high-amperage current over an extended period. A battery must also be able to accept a charge, so that its voltage potential may be restored.

The battery is vented to release excess hydrogen gas that is created when the battery is being charged or discharged. However, even with these vents, hydrogen gas can collect in or around the battery. If hydrogen gas is exposed to flame or sparks, it may ignite. If the electrolyte level is low, the battery may arc internally and explode. If the battery is equipped with removable cell caps, add distilled water whenever the electrolyte level is below the top of the plates. If the battery cell caps cannot be removed, the battery must be replaced if the electrolyte level becomes low.

DIAGNOSIS AND TESTING - BATTERY

The battery must be completely charged and the terminals should be properly cleaned and inspected before diagnostic procedures are performed. Refer to Battery System Cleaning for the proper cleaning procedures, and Battery System Inspection for the proper battery inspection procedures. Refer to Standard Procedures for the proper battery charging procedures.

MICRO 420 BATTERY TESTER

The Micro 420 automotive battery tester is designed to help the dealership technicians diagnose the cause of a defective battery. Follow the instruction manual supplied with the tester to properly diagnose a vehicle. If the instruction manual is not available refer to the standard procedure in this section, which includes the directions for using the Micro 420 battery tester.

WARNING: IF THE BATTERY SHOWS SIGNS OF FREEZING, LEAKING OR LOOSE POSTS, DO NOT TEST, ASSIST-BOOST, OR CHARGE. THE BATTERY MAY ARC INTERNALLY AND EXPLODE. PERSONAL INJURY AND/OR VEHICLE DAMAGE MAY RESULT.

WARNING: EXPLOSIVE HYDROGEN GAS FORMS IN AND AROUND THE BATTERY. DO NOT SMOKE, USE FLAME, OR CREATE SPARKS NEAR THE BATTERY. PERSONAL INJURY AND/OR VEHICLE DAMAGE MAY RESULT.

WARNING: THE BATTERY CONTAINS SULFURIC ACID, WHICH IS POISONOUS AND CAUSTIC. AVOID CONTACT WITH THE SKIN, EYES, OR CLOTHING. IN THE EVENT OF CONTACT, FLUSH WITH WATER AND CALL A PHYSICIAN IMMEDIATELY. KEEP OUT OF THE REACH OF CHILDREN.

A battery that will not accept a charge is faulty, and must be replaced. Further testing is not required. A fully-charged battery must be load tested to determine its cranking capacity. A battery that is fully-charged, but does not pass the load test, is faulty and must be replaced.

NOTE: Completely discharged batteries may take several hours to accept a charge. Refer to Standard Procedures for the proper battery charging procedures.

STANDARD PROCEDURE

STANDARD PROCEDURE - BATTERY CHARGING

Battery charging is the means by which the battery can be restored to its full voltage potential. A battery is fully-charged when:

- All of the battery cells are gassing freely during battery charging.
- A green color is visible in the sight glass of the battery built-in test indicator.

BATTERY (Continued)

- Three hydrometer tests, taken at one-hour intervals, indicate no increase in the temperature-corrected specific gravity of the battery electrolyte.
- Open-circuit voltage of the battery is 12.4 volts or above.

WARNING: NEVER EXCEED TWENTY AMPERES WHEN CHARGING A COLD (-1° C [30° F] OR LOWER) BATTERY. THE BATTERY MAY ARC INTERNALLY AND EXPLODE. PERSONAL INJURY AND/OR VEHICLE DAMAGE MAY RESULT.

WARNING: IF THE BATTERY SHOWS SIGNS OF FREEZING, LEAKING, LOOSE POSTS, DO NOT TEST, ASSIST-BOOST, OR CHARGE. THE BATTERY MAY ARC INTERNALLY AND EXPLODE. PERSONAL INJURY AND/OR VEHICLE DAMAGE MAY RESULT.

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WARNING: IF THE BATTERY IS EQUIPPED WITH REMOVABLE CELL CAPS, BE CERTAIN THAT EACH OF THE CELL CAPS IS IN PLACE AND TIGHT BEFORE THE BATTERY IS RETURNED TO SERVICE. PERSONAL INJURY AND/OR VEHICLE DAMAGE MAY RESULT FROM LOOSE OR MISSING CELL CAPS.

CAUTION: Always disconnect and isolate the battery negative cable before charging a battery. Do not exceed sixteen volts while charging a battery. Damage to the vehicle electrical system components may result.

CAUTION: Battery electrolyte will bubble inside the battery case during normal battery charging. Electrolyte boiling or being discharged from the battery vents indicates a battery overcharging condition. Immediately reduce the charging rate or turn off the charger to evaluate the battery condition. Damage to the battery may result from overcharging.

CAUTION: The battery should not be hot to the touch. If the battery feels hot to the touch, turn off the charger and let the battery cool before continuing the charging operation. Damage to the battery may result.

After the battery has been charged to 12.4 volts or greater, perform a load test to determine the battery cranking capacity (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - STANDARD PROCEDURE). If the battery will endure a load test, return the battery to service. If the battery will not endure a load test, it is faulty and must be replaced.

Clean and inspect the battery hold downs, tray, terminals, posts, and top before completing battery service (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - CLEANING), and (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - INSPECTION).

CHARGING A COMPLETELY DISCHARGED BATTERY

The following procedure should be used to recharge a completely discharged battery. Unless this procedure is properly followed, a good battery may be needlessly replaced.

(1) Measure the voltage at the battery posts with a voltmeter, accurate to 1/10 (0.10) volt (Fig. 5). If the reading is below ten volts, the battery charging current will be low. It could take some time before the battery accepts a current greater than a few milliamperes. Such low current may not be detectable on the ammeters built into many battery chargers.

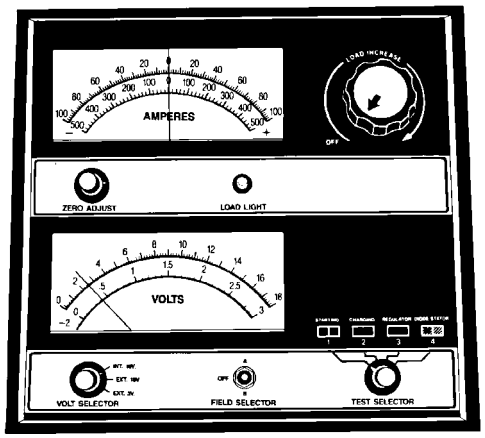

Fig. 5 Voltmeter - Typical

(2) Disconnect and isolate the battery negative cable. Connect the battery charger leads. Some battery chargers are equipped with polarity-sensing circuitry. This circuitry protects the battery charger and the battery from being damaged if they are improperly connected. If the battery state-of-charge is too low for the polarity-sensing circuitry to detect, the battery charger will not operate. This makes it

BATTERY (Continued)

appear that the battery will not accept charging current. See the instructions provided by the manufacturer of the battery charger for details on how to bypass the polarity-sensing circuitry.

(3) Battery chargers vary in the amount of voltage and current they provide. The amount of time required for a battery to accept measurable charging current at various voltages is shown in the Charge Rate Table. If the charging current is still not measurable at the end of the charging time, the battery is faulty and must be replaced. If the charging current is measurable during the charging time, the battery may be good and the charging should be completed in the normal manner.

CHARGE RATE TABLE	
Voltage	Hours
16.0 volts maximum	up to 4 hours
14.0 to 15.9 volts	up to 8 hours
13.9 volts or less	up to 16 hours

CHARGING TIME REQUIRED

The time required to charge a battery will vary, depending upon the following factors:

- **Battery Capacity** - A completely discharged heavy-duty battery requires twice the charging time of a small capacity battery.
- **Temperature** - A longer time will be needed to charge a battery at -18° C (0° F) than at 27° C (80° F). When a fast battery charger is connected to a cold battery, the current accepted by the battery will be very low at first. As the battery warms, it will accept a higher charging current rate (amperage).
- **Charger Capacity** - A battery charger that supplies only five amperes will require a longer charging time. A battery charger that supplies twenty amperes or more will require a shorter charging time.
- **State-Of-Charge** - A completely discharged battery requires more charging time than a partially discharged battery. Electrolyte is nearly pure water in a completely discharged battery. At first, the charging current (amperage) will be low. As the battery charges, the specific gravity of the electrolyte will gradually rise.

The Battery Charging Time Table gives an indication of the time required to charge a typical battery at room temperature based upon the battery state-of-charge and the charger capacity.

BATTERY CHARGING TIME TABLE			
Charging Amperage	5 Amps	10 Amps	20 Amps
Open Circuit Voltage	Hours Charging @ 21° C (70° F)		
12.25 to 12.49	6 hours	3 hours	1.5 hours
12.00 to 12.24	10 hours	5 hours	2.5 hours
10.00 to 11.99	14 hours	7 hours	3.5 hours
Below 10.00	18 hours	9 hours	4.5 hours

STANDARD PROCEDURE - BUILT-IN INDICATOR TEST

An indicator (hydrometer) built into the top of the battery case provides visual information for battery testing (Fig. 6). Like a hydrometer, the built-in indicator measures the specific gravity of the battery electrolyte. The specific gravity of the electrolyte reveals the battery state-of-charge; however, it will not reveal the cranking capacity of the battery. A load test must be performed to determine the battery cranking capacity. Refer to Standard Procedures for the proper battery load test procedures.

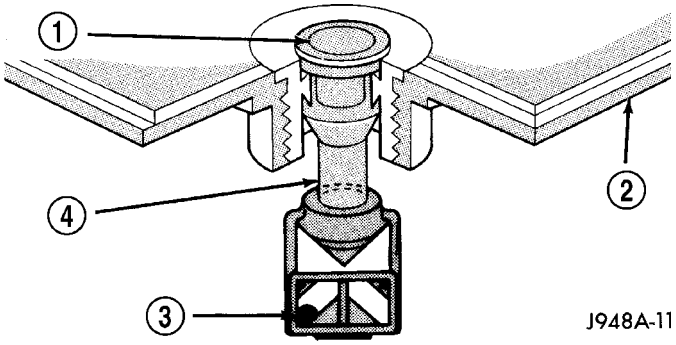

Fig. 6 Built-In Indicator

1 - SIGHT GLASS
2 - BATTERY TOP
3 - GREEN BALL
4 - PLASTIC ROD

Before testing, visually inspect the battery for any damage (a cracked case or cover, loose posts, etc.) that would cause the battery to be faulty. In order to obtain correct indications from the built-in indicator, it is important that the battery be level and have a clean sight glass. Additional light may be required to view the indicator. **Do not use open flame as a source of additional light.**

BATTERY (Continued)

To read the built-in indicator, look into the sight glass and note the color of the indication (Fig. 7). The battery condition that each color indicates is described in the following list:

- **Green** - Indicates 75% to 100% battery state-of-charge. The battery is adequately charged for further testing or return to service. If the starter will not crank for a minimum of fifteen seconds with a fully-charged battery, the battery must be load tested. Refer to Standard Procedures for the proper battery load test procedures.
- **Black or Dark** - Indicates 0% to 75% battery state-of-charge. The battery is inadequately charged and must be charged until a green indication is visible in the sight glass (12.4 volts or more), before the battery is tested further or returned to service. Refer to Standard Procedures for the proper battery charging procedures. Also refer to Diagnosis and Testing for more information on the possible causes of the discharged battery condition.
- **Clear or Bright** - Indicates a low battery electrolyte level. The electrolyte level in the battery is below the built-in indicator. A maintenance-free battery with non-removable cell caps must be replaced if the electrolyte level is low. Water must be added to a low-maintenance battery with removable cell caps before it is charged. Refer to Standard Procedures for the proper battery filling procedures. A low electrolyte level may be caused by an overcharging condition. Refer to Charging System for the proper charging system diagnosis and testing procedures.

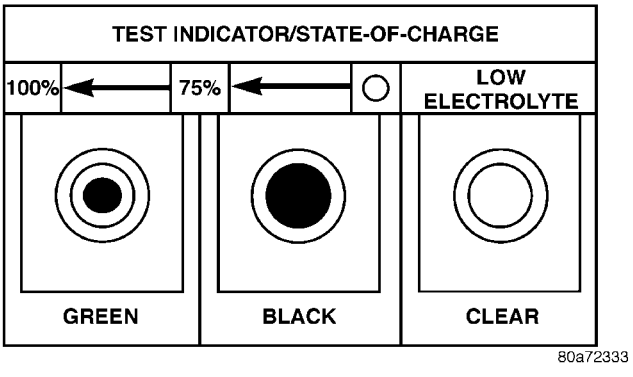

Fig. 7 Built-In Indicator Sight Glass Chart

STANDARD PROCEDURE - OPEN-CIRCUIT VOLTAGE TEST

A battery open-circuit voltage (no load) test will show the approximate state-of-charge of a battery. This test can be used in place of the hydrometer test when a hydrometer is not available, or for maintenance-free batteries with non-removable cell caps.

Before proceeding with this test, completely charge the battery (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - STANDARD PROCEDURE).

(1) Before measuring the open-circuit voltage, the surface charge must be removed from the battery. Turn on the headlamps for fifteen seconds, then allow up to five minutes for the battery voltage to stabilize.

(2) Disconnect and isolate both battery cables, negative cable first.

(3) Using a voltmeter connected to the battery posts (see the instructions provided by the manufacturer of the voltmeter), measure the open-circuit voltage (Fig. 8).

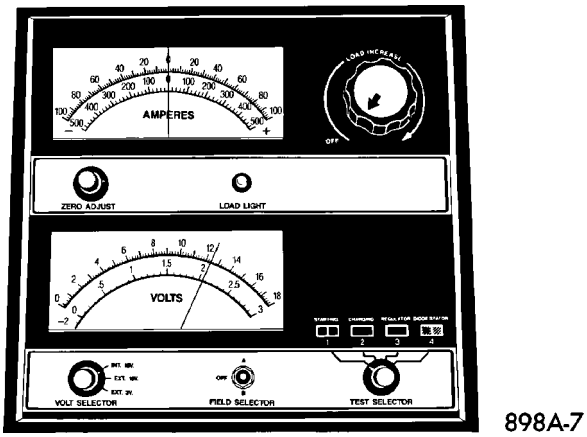

Fig. 8 Testing Open-Circuit Voltage - Typical

See the Open-Circuit Voltage Table. This voltage reading will indicate the battery state-of-charge, but will not reveal its cranking capacity. If a battery has an open-circuit voltage reading of 12.4 volts or greater, it may be load tested to reveal its cranking capacity (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - STANDARD PROCEDURE).

OPEN CIRCUIT VOLTAGE TABLE	
Open Circuit Voltage	Charge Percentage
11.7 volts or less	0%
12.0 volts	25%
12.2 volts	50%
12.4 volts	75%
12.6 volts or more	100%

STANDARD PROCEDURE - IGNITION-OFF DRAW TEST

The term Ignition-Off Draw (IOD) identifies a normal condition where power is being drained from the battery with the ignition switch in the Off position. A normal vehicle electrical system will draw from five to thirty-five milliamperes (0.005 to 0.035 ampere) with the ignition switch in the Off position, and all non-ignition controlled circuits in proper working order. Up to thirty-five milliamperes are needed to enable the memory functions for the Powertrain Con-

BATTERY (Continued)

trol Module (PCM), digital clock, electronically tuned radio, and other modules which may vary with the vehicle equipment.

A vehicle that has not been operated for approximately twenty days, may discharge the battery to an inadequate level. When a vehicle will not be used for twenty days or more (stored), remove the IOD fuse from the Power Distribution Center (PDC). This will reduce battery discharging.

Excessive IOD can be caused by:

- Electrical items left on.
- Faulty or improperly adjusted switches.
- Faulty or shorted electronic modules and components.
- An internally shorted generator.
- Intermittent shorts in the wiring.

If the IOD is over thirty-five milliamperes, the problem must be found and corrected before replacing a battery. In most cases, the battery can be charged and returned to service after the excessive IOD condition has been corrected.

(1) Verify that all electrical accessories are off. Turn off all lamps, remove the ignition key, and close all doors. If the vehicle is equipped with an illuminated entry system or an electronically tuned radio, allow the electronic timer function of these systems to automatically shut off (time out). This may take up to three minutes.

(2) Determine that the underhood lamp is operating properly, then disconnect the lamp wire harness connector or remove the lamp bulb.

(3) Disconnect the battery negative cable.

(4) Set an electronic digital multi-meter to its highest amperage scale. Connect the multi-meter between the disconnected battery negative cable terminal clamp and the battery negative terminal post. Make sure that the doors remain closed so that the illuminated entry system is not activated. The multi-meter amperage reading may remain high for up to three minutes, or may not give any reading at all while set in the highest amperage scale, depending upon the electrical equipment in the vehicle. The multi-meter leads must be securely clamped to the battery negative cable terminal clamp and the battery negative terminal post. If continuity between the battery negative terminal post and the negative cable terminal clamp is lost during any part of the IOD test, the electronic timer function will be activated and all of the tests will have to be repeated.

(5) After about three minutes, the high-amperage IOD reading on the multi-meter should become very low or nonexistent, depending upon the electrical equipment in the vehicle. If the amperage reading remains high, remove and replace each fuse or circuit breaker in the Power Distribution Center (PDC) and then in the Junction Block (JB), one at a time until the amperage reading becomes very low, or nonexistent. Refer to the appropriate wiring information in this service manual for complete PDC and JB fuse, circuit breaker, and circuit identification. This will isolate each circuit and identify the circuit that is the source of the high-amperage IOD. If the amperage reading remains high after removing and replacing each fuse and circuit breaker, disconnect the wire harness from the generator. If the amperage reading now becomes very low or nonexistent, refer to Charging System for the proper charging system diagnosis and testing procedures. After the high-amperage IOD has been corrected, switch the multi-meter to progressively lower amperage scales and, if necessary, repeat the fuse and circuit breaker remove-and-replace process to identify and correct all sources of excessive IOD. It is now safe to select the lowest milliampere scale of the multi-meter to check the low-amperage IOD.

CAUTION: Do not open any doors, or turn on any electrical accessories with the lowest milliampere scale selected, or the multi-meter may be damaged.

(6) Observe the multi-meter reading. The low-amperage IOD should not exceed thirty-five milliamperes (0.035 ampere). If the current draw exceeds thirty-five milliamperes, isolate each circuit using the fuse and circuit breaker remove-and-replace process in Step 5. The multi-meter reading will drop to within the acceptable limit when the source of the excessive current draw is disconnected. Repair this circuit as required; whether a wiring short, incorrect switch adjustment, or a component failure is at fault.

STANDARD PROCEDURE - USING MICRO 420 BATTERY TESTER

Always use the Micro 420 Instruction Manual that was supplied with the tester as a reference. If the Instruction Manual is not available the following procedure can be used:

WARNING: ALWAYS WEAR APPROPRIATE EYE PROTECTION AND USE EXTREME CAUTION WHEN WORKING WITH BATTERIES.

BATTERY TESTING

(1) If testing the battery OUT-OF-VEHICLE, clean the battery terminals with a wire brush before testing. If the battery is equipped with side post terminals, install and tighten the supplied lead terminal stud adapters. Do not use steel bolts. Failure to properly install the stud adapters, or using stud adapters that are dirty or worn-out may result in false test readings.

BATTERY (Continued)

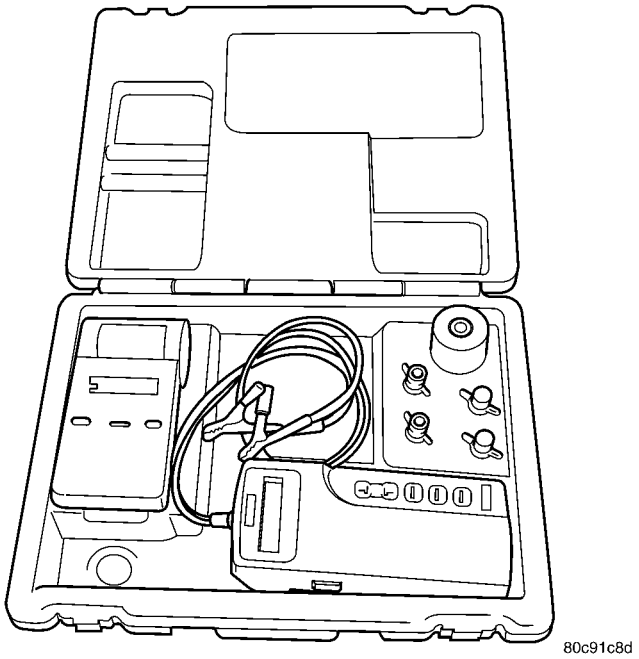

Fig. 9 Micro 420 Battery Tester

(2) If testing the battery IN-THE-VEHICLE, make certain all of the vehicle accessory loads are OFF, including the ignition. **The preferred test position is at the battery terminal**. If the battery is not accessible, you may test using both the positive and negative jumper posts. Select TESTING AT JUMPER POST when connecting to that location.

(3) Connect the tester (Fig. 9) to the battery or jumper posts, the red clamp to positive (+) and the black clamp to negative (–).

NOTE: Multiple batteries connected in parallel must have the ground cable disconnected to perform a battery test. Failure to disconnect may result in false battery test readings.

(4) Using the ARROW key select **in** or **out** of vehicle testing and press ENTER to make a selection.

(5) If not selected, choose the Cold Cranking Amp (CCA) battery rating. Or select the appropriate battery rating for your area (see menu). The tester will then run its self programmed test of the battery and display the results. Refer to the test result table noted below.

CAUTION: If REPLACE BATTERY is the result of the test, this may mean a poor connection between the vehicle's cables and battery exists. After disconnecting the vehicle's battery cables from the battery, retest the battery using the OUT-OF-VEHICLE test before replacing.

(6) While viewing the battery test result, press the CODE button and the tester will prompt you for the last 4 digits of the VIN. Use the UP/DOWN arrow buttons to scroll to the correct character; then press ENTER to select and move to the next digit. Then press the ENTER button to view the SERVICE CODE. Pressing the CODE button a second time will return you to the test results.

BATTERY TEST RESULTS	
GOOD BATTERY	Return to service
GOOD - RECHARGE	Fully charge battery and return to service
CHARGE & RETEST	Fully charge battery and retest battery
REPLACE BATTERY	Replace the battery and retest complete system
BAD-CELL REPLACE	Replace the battery and retest complete system

NOTE: The SERVICE CODE is required on every warranty claim submitted for battery replacement.

REMOVAL

(1) Turn the ignition switch to the Off position. Be certain that all electrical accessories are turned off.

(2) Loosen the battery negative cable terminal clamp pinch-bolt hex nut.

(3) Disconnect the battery negative cable terminal clamp from the battery negative terminal post. If necessary, use a battery terminal puller to remove the terminal clamp from the battery post (Fig. 10).

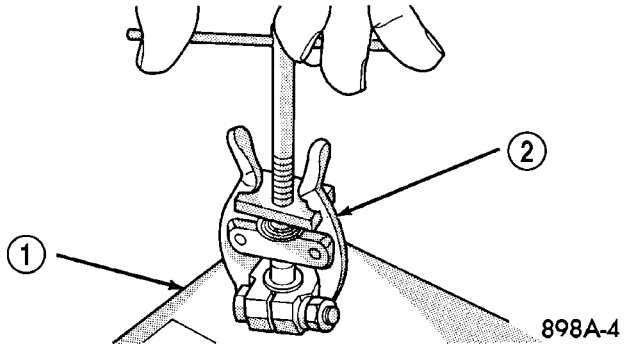

Fig. 10 Remove Battery Cable Terminal Clamp - Typical

1 - BATTERY
2 - BATTERY TERMINAL PULLER

(4) Loosen the battery positive cable terminal clamp pinch-bolt hex nut.

(5) Disconnect the battery positive cable terminal clamp from the battery positive terminal post. If necessary, use a battery terminal puller to remove the terminal clamp from the battery post.

BATTERY (Continued)

(6) Remove the battery holddowns from the battery. Refer to Battery Holddown for the proper battery holddown removal procedures.

WARNING: WEAR A SUITABLE PAIR OF RUBBER GLOVES (NOT THE HOUSEHOLD TYPE) WHEN REMOVING A BATTERY BY HAND. SAFETY GLASSES SHOULD ALSO BE WORN. IF THE BATTERY IS CRACKED OR LEAKING, THE ELECTROLYTE CAN BURN THE SKIN AND EYES.

(7) Remove the battery and the battery thermal guard from the battery tray as a unit.

(8) Remove the battery thermal guard from the battery case. Refer to Thermal Guard for the proper battery thermal guard removal procedures.

INSTALLATION

(1) Clean and inspect all of the battery system components. Refer to Battery System Cleaning for the proper cleaning procedures, and refer to Battery System Inspection for the proper inspection procedures.

(2) Reinstall the battery thermal guard onto the battery case. Refer to Thermal Guard for the proper battery thermal guard installation procedures.

(3) Position the battery and the battery thermal guard onto the battery tray as a unit. Ensure that the battery positive and negative terminal posts are correctly positioned. The battery cable terminal clamps must reach the correct battery terminal post without stretching the cables (Fig. 11).

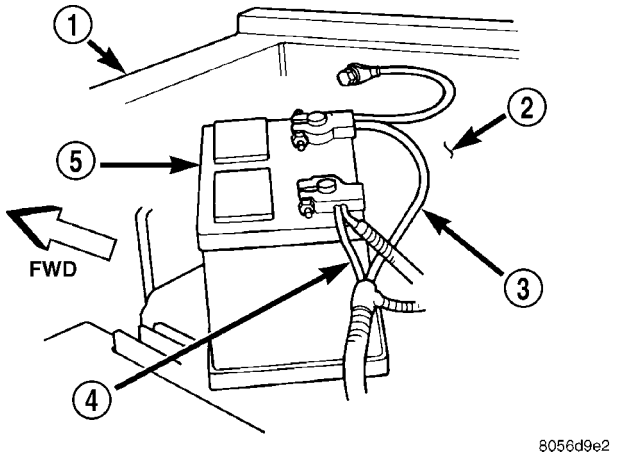

Fig. 11 Battery Cables - Typical

1 - RADIATOR CROSSMEMBER
2 - WHEELHOUSE INNER PANEL
3 - NEGATIVE CABLE
4 - POSITIVE CABLE
5 - BATTERY

(4) Reinstall the battery holddowns onto the battery. Refer to Battery Holddown for the proper installation procedure.

CAUTION: Be certain that the battery cable terminal clamps are connected to the correct battery terminal posts. Reversed battery polarity may damage electrical components of the vehicle.

(5) Clean the battery cable terminal clamps and the battery terminal posts. Refer to Battery System Cleaning for cleaning procedure.

(6) Reconnect the battery positive cable terminal clamp to the battery positive terminal post. Tighten the terminal clamp pinch-bolt hex nut to 8.4 N·m (75 in. lbs.).

(7) Reconnect the battery negative cable terminal clamp to the battery negative terminal post. Tighten the terminal clamp pinch-bolt hex nut to 8.4 N·m (75 in. lbs.).

(8) Apply a thin coating of petroleum jelly or chassis grease to the exposed surfaces of the battery cable terminal clamps and the battery terminal posts.

BATTERY HOLDDOWN

DESCRIPTION

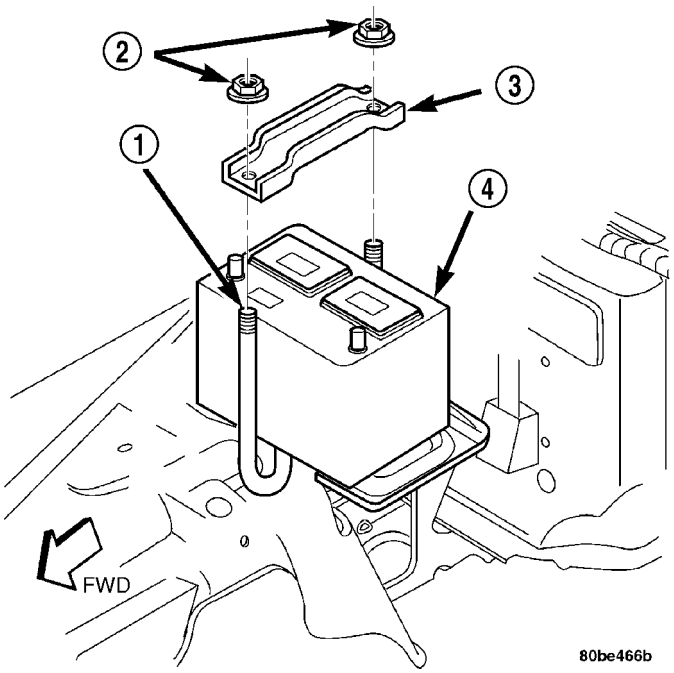

Fig. 12 Battery Hold Downs

1 - J-BOLT (2)
2 - NUT (2)
3 - BRACKET
4 - BATTERY

The battery hold down hardware (Fig. 12) includes two J-bolts, a hold down bracket and two hex nuts with coned washers. The battery hold down bracket consists of a stamped steel bracket that is then plastic-coated for corrosion protection.

BATTERY HOLDDOWN (Continued)

The hold down J-bolts are installed a hole in the front and rear flanges of the battery tray from the top, with the threaded ends of the bolts extending upward. The hooked end of each J-bolt is then engaged in a second hole in the front and rear flanges of the battery tray from the bottom. The battery hold down bracket is installed across the top of the battery case and over the two upright threaded ends of the J-bolts. A hex nut with coned washer is then installed and tightened onto each of the J-bolts to securely hold down the battery in the battery tray.

When installing a battery into the battery tray, be certain that the hold down hardware is properly installed and that the fasteners are tightened to the proper specifications. Improper hold down fastener tightness, whether too loose or too tight, can result in damage to the battery, the vehicle or both. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY HOLDDOWN - INSTALLATION) the proper hold down fastener tightness specifications.

OPERATION

The battery holddown secures the battery in the battery tray. This holddown is designed to prevent battery movement during the most extreme vehicle operation conditions. Periodic removal and lubrication of the battery holddown hardware is recommended to prevent hardware seizure at a later date.

CAUTION: Never operate a vehicle without a battery holddown device properly installed. Damage to the vehicle, components and battery could result.

REMOVAL

All of the battery hold down hardware can be serviced without removal of the battery or the battery tray.

(1) Turn the ignition switch to the Off position. Be certain that all electrical accessories are turned off.

(2) Loosen the battery negative cable terminal clamp pinch-bolt hex nut.

(3) Disconnect the battery negative cable terminal clamp from the battery negative terminal post. If necessary, use a battery terminal puller to remove the terminal clamp from the battery post.

(4) Remove the nut with washer from the threaded end of each of the two J-bolts (Fig. 13).

(5) Remove the battery hold down bracket from the threaded ends of the two J-bolts and the top of the battery case.

(6) Disengage the hooked end of each J-bolt from the holes in the front or rear flange of the battery tray and remove the two J-bolts.

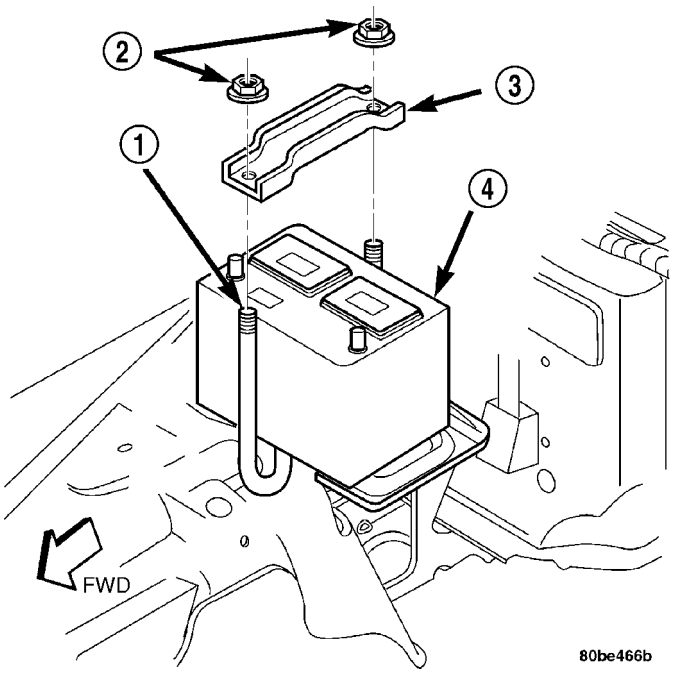

Fig. 13 Battery Hold Downs Remove/Install

1 - J-BOLT (2)
2 - NUT (2)
3 - BRACKET
4 - BATTERY

INSTALLATION

All of the battery hold down hardware can be serviced without removal of the battery or the battery tray.

(1) Clean and inspect the battery hold down hardware. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - CLEANING).

(2) Engage the hooked end of each J-bolt into the holes in the front or rear flange of the battery tray and position the two J-bolts.

(3) Position the battery hold down bracket onto the threaded ends of the two J-bolts and across the top of the battery case.

(4) Install and tighten the nut with washer onto the threaded end of each of the two J-bolts. Tighten the nuts to 4.7 N·m (42 in. lbs.).

(5) Reconnect the battery negative cable terminal clamp to the battery negative terminal post. Tighten the terminal clamp pinch-bolt hex nut to 8.4 N·m (75 in. lbs.).

BATTERY CABLES

DESCRIPTION

The battery cables are large gauge, stranded copper wires sheathed within a heavy plastic or synthetic rubber insulating jacket. The wire used in the battery cables combines excellent flexibility and reliability with high electrical current carrying capacity. The battery cables feature a clamping type female battery terminal made of soft lead that is die cast onto one end of the battery cable wire. A square headed pinch-bolt and hex nut are installed at the open end of the female battery terminal clamp. Large eyelet type terminals are crimped onto the opposite end of the battery cable wire and then solder-dipped. The battery positive cable wires have a red insulating jacket to provide visual identification and feature a larger female battery terminal clamp to allow connection to the larger battery positive terminal post. The battery negative cable wires have a black insulating jacket and a smaller female battery terminal clamp.

The battery cables cannot be repaired and, if damaged or faulty they must be replaced. Both the battery positive and negative cables are available for service replacement only as a unit with the battery wire harness, which may include portions of the wiring circuits for the generator and other components on some models. Refer to the appropriate wiring information in this service manual for the location of the proper battery cable wire harness diagrams. The wiring information also includes proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

OPERATION

The battery cables connect the battery terminal posts to the vehicle electrical system. These cables also provide a path back to the battery for electrical current generated by the charging system for restoring the voltage potential of the battery. The female battery terminal clamps on the ends of the battery cable wires provide a strong and reliable connection of the battery cable to the battery terminal posts. The terminal pinch bolts allow the female terminal clamps to be tightened around the male terminal posts on the top of the battery. The eyelet terminals secured to the opposite ends of the battery cable wires from the female battery terminal clamps provide secure and reliable connection of the battery cables to the vehicle electrical system.

The battery positive cable terminal clamp is die cast onto the ends of two wires. One wire has an eyelet terminal that connects the battery positive cable to the B(+) terminal stud of the Power Distribution Center (PDC), and the other wire has an eyelet terminal that connects the battery positive cable to the B(+) terminal stud of the engine starter motor solenoid. The battery negative cable terminal clamp is also die cast onto the ends of two wires. One wire has an eyelet terminal that connects the battery negative cable to the vehicle powertrain through a stud on the right side of the engine cylinder block. The other wire has an eyelet terminal that connects the battery negative cable to the vehicle body through a ground screw on the right front fender inner shield, near the battery.

DIAGNOSIS AND TESTING - BATTERY CABLES

A voltage drop test will determine if there is excessive resistance in the battery cable terminal connections or the battery cable. If excessive resistance is found in the battery cable connections, the connection point should be disassembled, cleaned of all corrosion or foreign material, then reassembled. Following reassembly, check the voltage drop for the battery cable connection and the battery cable again to confirm repair.

When performing the voltage drop test, it is important to remember that the voltage drop is giving an indication of the resistance between the two points at which the voltmeter probes are attached. **EXAMPLE:** When testing the resistance of the battery positive cable, touch the voltmeter leads to the battery positive cable terminal clamp and to the battery positive cable eyelet terminal at the starter solenoid B(+) terminal stud. If you probe the battery positive terminal post and the battery positive cable eyelet terminal at the starter solenoid B(+) terminal stud, you are reading the combined voltage drop in the battery positive cable terminal clamp-to-terminal post connection and the battery positive cable.

VOLTAGE DROP TEST

The following operation will require a voltmeter accurate to 1/10 (0.10) volt. Before performing this test, be certain that the following procedures are accomplished:

- The battery is fully-charged and load tested. Refer to Standard Procedures for the proper battery charging and load test procedures.
- Fully engage the parking brake.
- If the vehicle is equipped with an automatic transmission, place the gearshift selector lever in the Park position. If the vehicle is equipped with a manual transmission, place the gearshift selector lever in the Neutral position and block the clutch pedal in the fully depressed position.
- Verify that all lamps and accessories are turned off.

BATTERY CABLES (Continued)

- To prevent the engine from starting, remove the Automatic Shut Down (ASD) relay. The ASD relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the underside of the PDC cover for ASD relay identification and location.

(1) Connect the positive lead of the voltmeter to the battery negative terminal post. Connect the negative lead of the voltmeter to the battery negative cable terminal clamp (Fig. 14). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If voltage is detected, correct the poor connection between the battery negative cable terminal clamp and the battery negative terminal post.

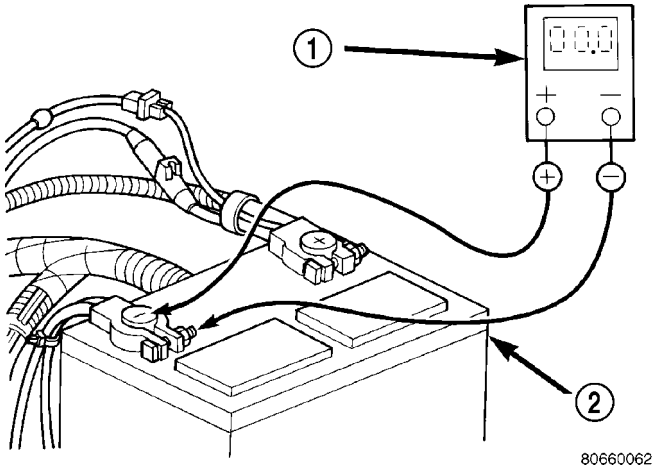

Fig. 14 Test Battery Negative Connection Resistance - Typical

1 - VOLTMETER
2 - BATTERY

(2) Connect the positive lead of the voltmeter to the battery positive terminal post. Connect the negative lead of the voltmeter to the battery positive cable terminal clamp (Fig. 15). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If voltage is detected, correct the poor connection between the battery positive cable terminal clamp and the battery positive terminal post.

(3) Connect the voltmeter to measure between the battery positive cable terminal clamp and the starter solenoid B(+) terminal stud (Fig. 16). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If the reading is above 0.2 volt, clean and tighten the battery positive cable eyelet terminal connection at the starter solenoid B(+) terminal stud. Repeat the test. If the reading is still above 0.2 volt, replace the faulty battery positive cable.

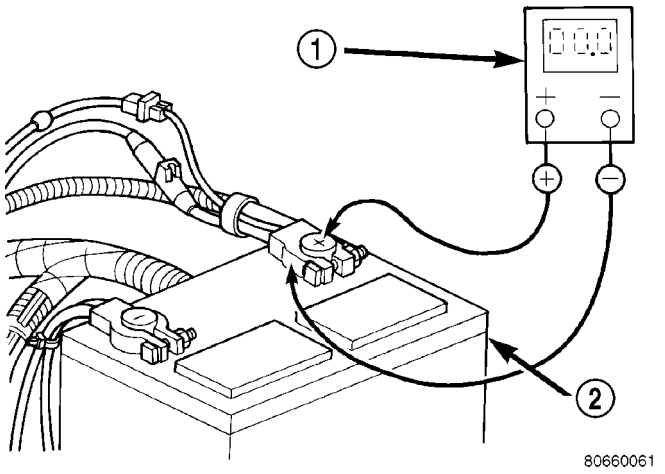

Fig. 15 Test Battery Positive Connection Resistance - Typical

1 - VOLTMETER
2 - BATTERY

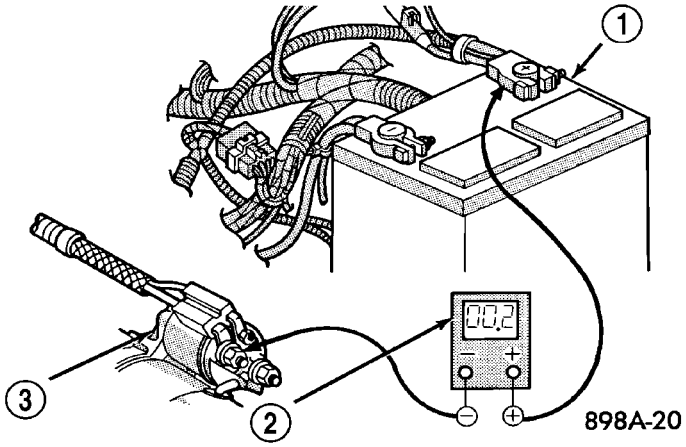

Fig. 16 Test Battery Positive Cable Resistance - Typical

1 - BATTERY
2 - VOLTMETER
3 - STARTER MOTOR

(4) Connect the voltmeter to measure between the battery negative cable terminal clamp and a good clean ground on the engine block (Fig. 17). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If the reading is above 0.2 volt, clean and tighten the battery negative cable eyelet terminal connection to the engine block. Repeat the test. If the reading is still above 0.2 volt, replace the faulty battery negative cable.

BATTERY CABLES (Continued)

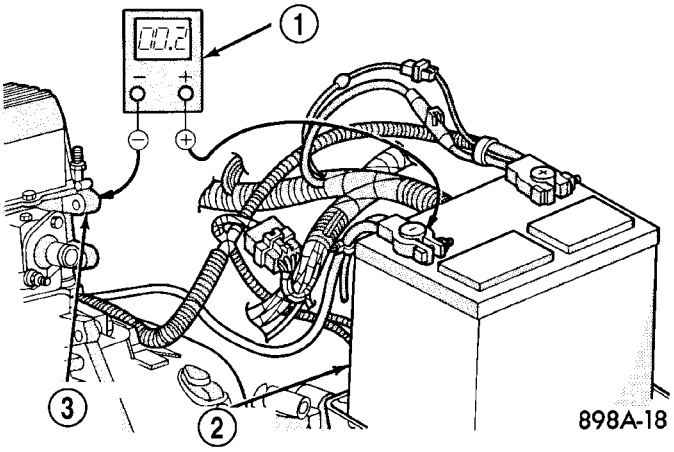

Fig. 17 Test Ground Circuit Resistance - Typical

1 - VOLTMETER
2 - BATTERY
3 - ENGINE GROUND

REMOVAL

Both the battery negative cable and the battery positive cable are serviced in the battery wire harness. If either battery cable is damaged or faulty, the battery wire harness unit must be replaced.

(1) Turn the ignition switch to the Off position. Be certain that all electrical accessories are turned off.

(2) Loosen the battery negative cable terminal clamp pinch-bolt hex nut.

(3) Disconnect the battery negative cable terminal clamp from the battery negative terminal post. If necessary, use a battery terminal puller to remove the terminal clamp from the battery post.

(4) Loosen the battery positive cable terminal clamp pinch-bolt hex nut.

(5) Disconnect the battery positive cable terminal clamp from the battery positive terminal post. If necessary, use a battery terminal puller to remove the terminal clamp from the battery post.

(6) Unlatch and remove the B(+) terminal stud cover from the rear of the Power Distribution Center (PDC).

(7) Remove the two nuts that secure the battery positive cable and the generator output cable eyelet terminal to the two B(+) terminal studs on the PDC.

(8) Remove the battery positive cable and the generator output eyelet terminal from the two B(+) terminal studs on the PDC.

(9) Remove the screw that secures the battery negative cable eyelet terminal to the dash panel near the battery.

(10) Unlatch and remove the cover from the generator output terminal stud housing on the back of the generator.

(11) Remove the nut that secures the generator output cable eyelet terminal to the generator output terminal stud.

(12) Remove the generator output cable eyelet terminal from the generator output terminal stud.

(13) Remove the nut that secures the battery negative cable ground eyelet terminal to the stud on the right side of the engine block.

(14) Remove the battery negative cable ground eyelet terminal from the engine block stud.

(15) Remove the nut that secures the battery positive cable eyelet terminal to the B(+) terminal stud on the starter solenoid.

(16) Remove the battery positive cable eyelet terminal from the B(+) terminal stud on the starter solenoid.

(17) Remove the battery wire harness from the engine compartment.

INSTALLATION

Both the battery negative cable and the battery positive cable are serviced in the battery wire harness. If either battery cable is damaged or faulty, the battery wire harness unit must be replaced.

(1) Clean and inspect the battery cable terminal clamps and the battery terminal posts (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - CLEANING), and (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - INSPECTION).

(2) Position the battery wire harness into the engine compartment.

(3) Install the battery positive cable eyelet terminal onto the B(+) terminal stud on the starter solenoid.

(4) Install and tighten the nut that secures the battery positive cable eyelet terminal to the B(+) terminal stud on the starter solenoid. Tighten the nut to 10 N·m (90 in. lbs.).

(5) Install the battery negative cable ground eyelet terminal onto the stud on the right side of the engine block.

BATTERY CABLES (Continued)

(6) Install and tighten the nut that secures the battery negative cable ground eyelet terminal to the stud on the right side of the engine block. Tighten the nut to 16.9 N·m (150 in. lbs.).

(7) Install the generator output cable eyelet terminal onto the generator output terminal stud.

(8) Install and tighten the nut that secures the generator output cable eyelet terminal to the generator output terminal stud. Tighten the nut to 8.4 N·m (75 in. lbs.).

(9) Position the cover for the generator output terminal stud housing onto the back of the generator and snap it into place.

(10) Install and tighten the screw that secures the battery negative cable eyelet terminal to the dash panel near the battery. Tighten the screw to 48.7 N·m (36 ft. lbs.).

(11) Install the battery positive cable and the generator output cable eyelet terminal onto the PDC B(+) terminal studs.

(12) Install and tighten the two nuts that secure the battery positive cable and the generator output cable eyelet terminal to the PDC B(+) terminal studs. Tighten the nuts to 11.3 N·m (100 in. lbs.).

(13) Engage the tabs on the lower edge of the B(+) terminal stud cover in the slots on the rear of the PDC housing, then engage the latch on the top of the cover with the latch tabs on the PDC housing.

(14) Reconnect the battery positive cable terminal clamp to the battery positive terminal post. Tighten the terminal clamp pinch-bolt hex nut to 8.4 N·m (75 in. lbs.).

(15) Reconnect the battery negative cable terminal clamp to the battery negative terminal post. Tighten the terminal clamp pinch-bolt hex nut to 8.4 N·m (75 in. lbs.).

(16) Apply a thin coating of petroleum jelly or chassis grease to the exposed surfaces of the battery cable terminal clamps and the battery terminal posts.

BATTERY TRAY

DESCRIPTION

The battery is mounted in a stamped steel tray (Fig. 18) located in the passenger side rear corner of the engine compartment. The battery tray is secured by four hex screws with washers to the reinforcement located between the engine compartment side of the dash panel and the rear of the front fender wheelhouse inner panel.

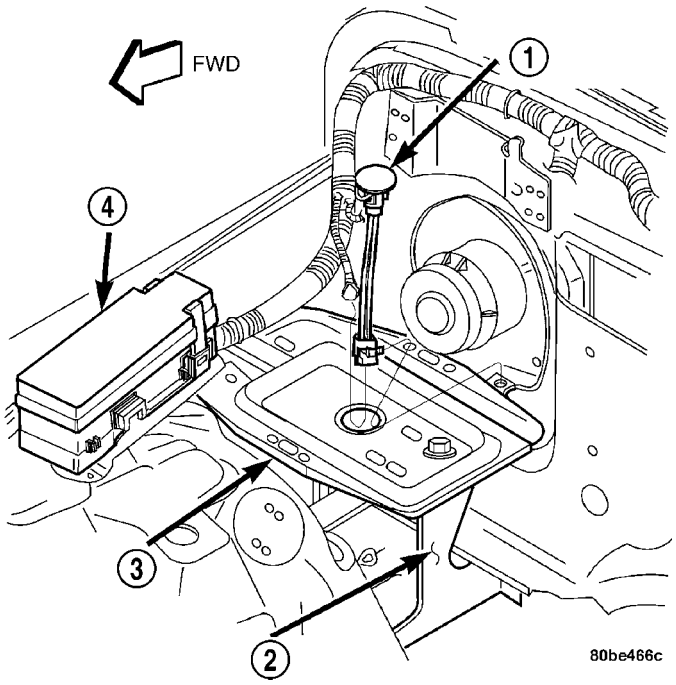

Fig. 18 Battery Tray

1 - BATTERY TEMPERATURE SENSOR
2 - REINFORCEMENT
3 - BATTERY TRAY
4 - POWER DISTRIBUTION CENTER

A hole in the bottom of the battery tray is fitted with a battery temperature sensor (Refer to 8 - ELECTRICAL/CHARGING/BATTERY TEMPERATURE SENSOR - DESCRIPTION).

OPERATION

The battery tray provides a secure mounting location and supports the battery. On some vehicles, the battery tray also provides the anchor point/s for the battery holddown hardware. The battery tray and the battery holddown hardware combine to secure and stabilize the battery in the engine compartment, which prevents battery movement during vehicle operation. Unrestrained battery movement during vehicle operation could result in damage to the vehicle, the battery, or both.

BATTERY TRAY (Continued)

REMOVAL

(1) Remove the battery from the battery tray (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - REMOVAL).

(2) Remove the battery temperature sensor from the battery tray (Refer to 8 - ELECTRICAL/CHARGING/BATTERY TEMPERATURE SENSOR - REMOVAL).

(3) Remove the four screws with washers that secure the battery tray to the reinforcement located between the dash panel and the rear of the front wheelhouse inner panel in the engine compartment (Fig. 19).

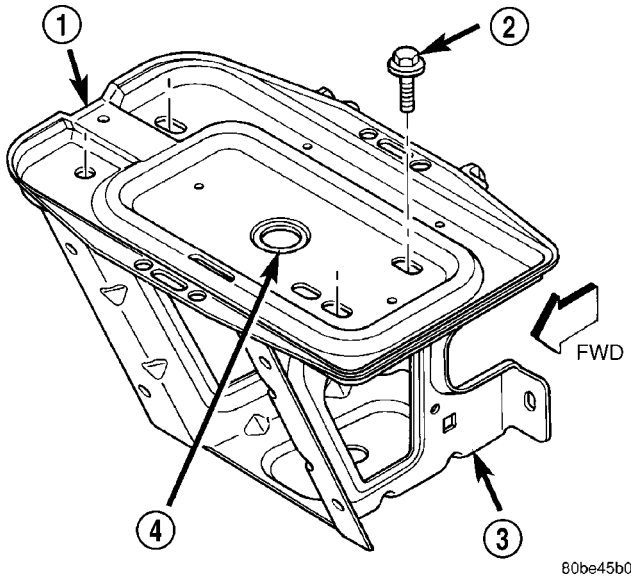

Fig. 19 Battery Tray Remove/Install

1 - BATTERY TRAY
2 - SCREW (4)
3 - REINFORCEMENT
4 - BATTERY TEMPERATURE SENSOR MOUNTING HOLE

(4) Remove the battery tray from the reinforcement in the engine compartment.

INSTALLATION

(1) Clean and inspect the battery tray (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - CLEANING), and (Refer to 8 - ELECTRICAL/BATTERY SYSTEM - INSPECTION).

(2) Position the battery tray onto the reinforcement in the engine compartment.

(3) Install and tighten the four screws with washers that secure the battery tray to the reinforcement located between the dash panel and the rear of the front wheelhouse inner panel in the engine compartment. Tighten the screws to 22.6 N·m (200 in. lbs.).

(4) Install the battery temperature sensor onto the battery tray (Refer to 8 - ELECTRICAL/CHARGING/BATTERY TEMPERATURE SENSOR - INSTALLATION).

(5) Install the battery onto the battery tray (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/BATTERY - INSTALLATION).

CHARGING

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CHARGING

DESCRIPTION

The charging system consists of:

- Generator
- Generator Decoupler Pulley (if equipped)
- Electronic Voltage Regulator (EVR) circuitry within the Powertrain Control Module (PCM)
- Ignition switch
- Battery (refer to 8, Battery for information)
- Battery temperature sensor
- Generator Lamp (if equipped)
- Check Gauges Lamp (if equipped)
- Voltmeter (refer to 8, Instrument Cluster for information)
- Wiring harness and connections (refer to 8, Wiring for information)

OPERATION

The charging system is turned on and off with the ignition switch. The system is on when the engine is running and the ASD relay is energized. When the ASD relay is on, voltage is supplied to the ASD relay sense circuit at the PCM. This voltage is connected through the PCM and supplied to one of the generator field terminals (Gen. Source +) at the back of the generator.

The amount of DC current produced by the generator is controlled by the EVR (field control) circuitry contained within the PCM. This circuitry is connected in series with the second rotor field terminal and ground.

A battery temperature sensor, located in the battery tray housing, is used to sense battery temperature. This temperature data, along with data from monitored line voltage, is used by the PCM to vary the battery charging rate. This is done by cycling the ground path to control the strength of the rotor magnetic field. The PCM then compensates and regulates generator current output accordingly.

All vehicles are equipped with On-Board Diagnostics (OBD). All OBD-sensed systems, including EVR (field control) circuitry, are monitored by the PCM. Each monitored circuit is assigned a Diagnostic Trouble Code (DTC). The PCM will store a DTC in electronic memory for certain failures it detects. Refer to Diagnostic Trouble Codes in; Powertrain Control Module; Electronic Control Modules for more DTC information.

The Check Gauges Lamp (if equipped) monitors: **charging system voltage,** engine coolant temperature and engine oil pressure. If an extreme condition is indicated, the lamp will be illuminated. This is done as reminder to check the three gauges. The signal to activate the lamp is sent via the CCD bus circuits. The lamp is located on the instrument panel. Refer to 8, Instrument Cluster for additional information.

CHARGING (Continued)

DIAGNOSIS AND TESTING - CHARGING SYSTEM

The following procedures may be used to diagnose the charging system if:

- the check gauges lamp (if equipped) is illuminated with the engine running
- the voltmeter (if equipped) does not register properly
- an undercharged or overcharged battery condition occurs.

Remember that an undercharged battery is often caused by:

- accessories being left on with the engine not running
- a faulty or improperly adjusted switch that allows a lamp to stay on. Refer to Ignition-Off Draw Test in 8, Battery for more information.

INSPECTION

The Powertrain Control Module (PCM) monitors critical input and output circuits of the charging system, making sure they are operational. A Diagnostic Trouble Code (DTC) is assigned to each input and output circuit monitored by the On-Board Diagnostic (OBD) system. Some charging system circuits are checked continuously, and some are checked only under certain conditions.

Refer to Diagnostic Trouble Codes in; Powertrain Control Module; Electronic Control Modules for more DTC information. This will include a complete list of DTC's including DTC's for the charging system.

To perform a complete test of the charging system, refer to the appropriate Powertrain Diagnostic Procedures service manual and the DRB® scan tool. Perform the following inspections before attaching the scan tool.

(1) Inspect the battery condition. Refer to 8, Battery for procedures.

(2) Inspect condition of battery cable terminals, battery posts, connections at engine block, starter solenoid and relay. They should be clean and tight. Repair as required.

(3) Inspect all fuses in both the fuseblock and Power Distribution Center (PDC) for tightness in receptacles. They should be properly installed and tight. Repair or replace as required.

(4) Inspect generator mounting bolts for tightness. Replace or tighten bolts if required. Refer to the Generator Removal/Installation section of this group for torque specifications.

(5) Inspect generator drive belt condition and tension. Tighten or replace belt as required. Refer to Belt Tension Specifications in 7, Cooling System.

(6) Inspect automatic belt tensioner (if equipped). Refer to 7, Cooling System for information.

(7) Inspect generator electrical connections at generator field, battery output, and ground terminal (if equipped). Also check generator ground wire connection at engine (if equipped). They should all be clean and tight. Repair as required.

SPECIFICATIONS

GENERATOR RATINGS

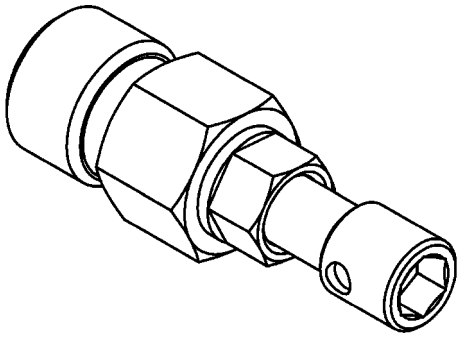
TYPE	PART NUMBER	RATED SAE AMPS	ENGINES	MINIMUM TEST AMPS
DENSO	56044530AB	124	2.4L	88
DENSO	56044532AB	136	2.4	96
DENSO	56041685AA	117	4.0L	88
DENSO	56041565AA	81	4.0L	57
DENSO	56041822AA	124	4.0L	90

CHARGING (Continued)

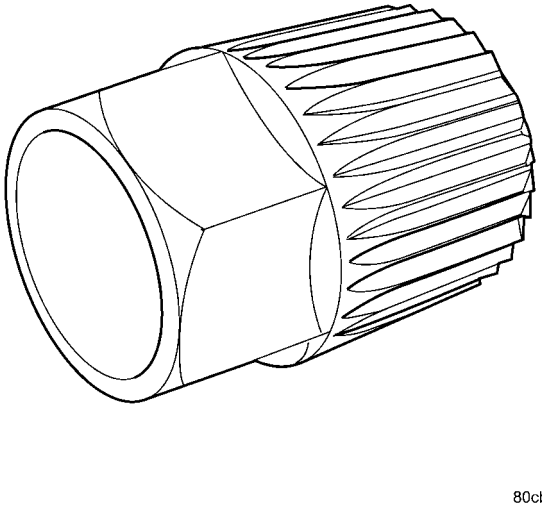
TORQUE

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Generator Mounting Bolts (2.4L)	57	42	-
Generator Mounting Bolt (4.0L)	55	41	-
Generator Pivot Bolt/Nut (4.0L)	55	41	-
Generator B+ Cable Nut	13	-	115

SPECIAL TOOLS



GENERATOR DECOUPLER TOOL #8433



GENERATOR DECOUPLER TOOL #8823

BATTERY TEMPERATURE SENSOR

DESCRIPTION

The Battery Temperature Sensor (BTS) is attached to the battery tray located under the battery.

OPERATION

The BTS is used to determine the battery temperature and control battery charging rate. This temperature data, along with data from monitored line voltage, is used by the PCM to vary the battery charging rate. System voltage will be higher at colder temperatures and is gradually reduced at warmer temperatures.

The PCM sends 5 volts to the sensor and is grounded through the sensor return line. As temperature increases, resistance in the sensor decreases and the detection voltage at the PCM increases.

The BTS is also used for OBD II diagnostics. Certain faults and OBD II monitors are either enabled or disabled, depending upon BTS input (for example, disable purge and enable Leak Detection Pump (LDP) and O2 sensor heater tests). Most OBD II monitors are disabled below 20 degrees F.

REMOVAL

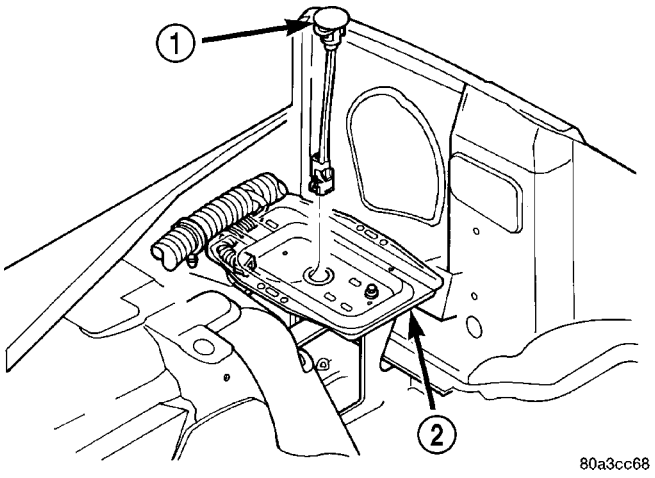


Fig. 1 Battery Temperature Sensor Remove/Install

1 - BATTERY TEMPERATURE SENSOR
2 - BATTERY TRAY

The battery temperature sensor is located under the vehicle battery (Fig. 1) and is attached to a mounting hole on battery tray.

BATTERY TEMPERATURE SENSOR (Continued)

(1) Remove battery. Refer to 8, Battery for procedures.
(2) Disconnect sensor pigtail harness from engine wire harness.
(3) Pry sensor straight up from battery tray mounting hole.

INSTALLATION

The battery temperature sensor is located under the vehicle battery and is attached to a mounting hole on battery tray.
(1) Feed pigtail harness through hole in top of battery tray and press sensor into top of battery tray.
(2) Connect pigtail harness.
(3) Install battery. Refer to 8, Battery for procedures.

GENERATOR

DESCRIPTION

The generator is belt-driven by the engine using a serpentine type drive belt. It is serviced only as a complete assembly. If the generator fails for any reason, the entire assembly must be replaced.

OPERATION

As the energized rotor begins to rotate within the generator, the spinning magnetic field induces a current into the windings of the stator coil. Once the generator begins producing sufficient current, it also provides the current needed to energize the rotor.

The Y type stator winding connections deliver the induced AC current to 3 positive and 3 negative diodes for rectification. From the diodes, rectified DC current is delivered to the vehicle electrical system through the generator battery terminal.

Although the generators appear the same externally, different generators with different output ratings are used on this vehicle. Be certain that the replacement generator has the same output rating and part number as the original unit. Refer to Generator Ratings in the Specifications section at the back of this group for amperage ratings and part numbers.

Noise emitting from the generator may be caused by: worn, loose or defective bearings; a loose or defective drive pulley; incorrect, worn, damaged or misadjusted fan drive belt; loose mounting bolts; a misaligned drive pulley or a defective stator or diode.

REMOVAL

WARNING: DISCONNECT NEGATIVE CABLE FROM BATTERY BEFORE REMOVING BATTERY OUTPUT WIRE (B+ WIRE) FROM GENERATOR. FAILURE TO DO SO CAN RESULT IN INJURY OR DAMAGE TO ELECTRICAL SYSTEM.

(1) Disconnect negative battery cable at battery.
(2) Remove generator drive belt. Refer to 7, Cooling System for procedure.
(3) Remove generator pivot and mounting bolts/nut (Fig. 2) , or (Fig. 3). Position generator for access to wire connectors.
(4) If equipped, unsnap plastic cover from B+ terminal.
(5) Remove B+ cable output terminal mounting nut at rear of generator (Fig. 4), (Fig. 5), or (Fig. 6). Disconnect terminal from generator.
(6) Disconnect field wire connector at rear of generator by pushing on connector tab.
(7) Remove generator from vehicle.

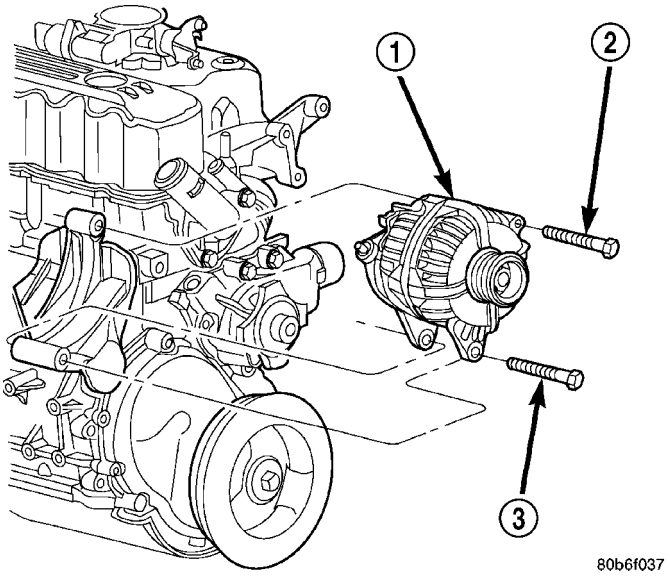


Fig. 2 GENERATOR REMOVE/INSTALL - 4.0L

1 - GENERATOR
2 - UPPER BOLT
3 - LOWER BOLT

INSTALLATION

(1) Position generator to engine and snap field wire connector into rear of generator.
(2) Install B+ terminal to generator mounting stud. Tighten mounting nut. Refer to Torque Specifications.
(3) If equipped, snap plastic cover to B+ terminal.
(4) Install generator mounting fasteners and tighten. Refer to Torque Specifications.

CAUTION: Never force a belt over a pulley rim using a screwdriver. The synthetic fiber of the belt can be damaged.

GENERATOR (Continued)

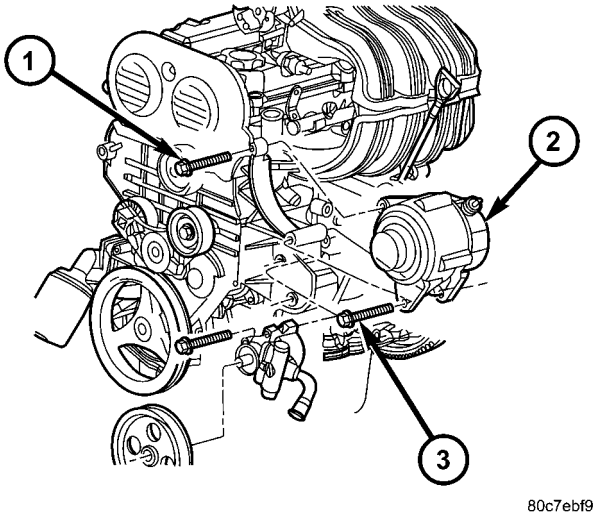


Fig. 3 GENERATOR MOUNTING- 2.4L

1 - UPPER MOUNTING BOLT
2 - GENERTOR
3 - LOWER MOUNTING BOLT

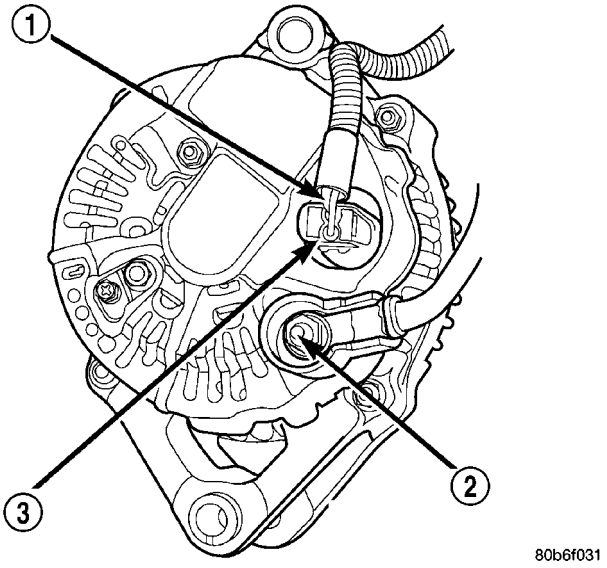


Fig. 4 GENERATOR CONNECTORS (TYPICAL DENSO)

1 - FIELD WIRES
2 - B+ (OUTPUT TERMINAL)
3 - FIELD WIRE CONNECTOR

CAUTION: When installing a serpentine accessory drive belt, the belt MUST be routed correctly. The water pump will be rotating in the wrong direction if the belt is installed incorrectly, causing the engine to overheat. Refer to belt routing label in engine compartment, or refer to Belt Schematics in 7, Cooling System.

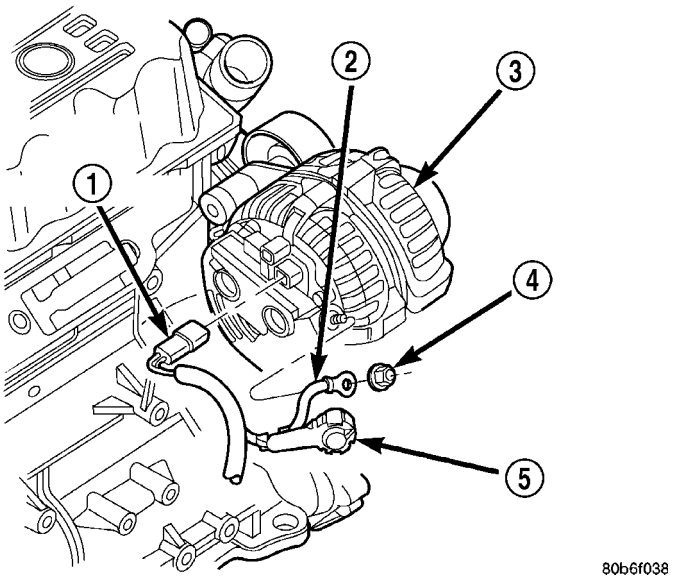


Fig. 5 GENERATOR CONNECTORS - 4.0L

1 - FIELD WIRE CONNECTOR
2 - B+ CABLE
3 - GENERATOR
4 - B+ CABLE MOUNTING NUT
5 - CABLE PROTECTOR

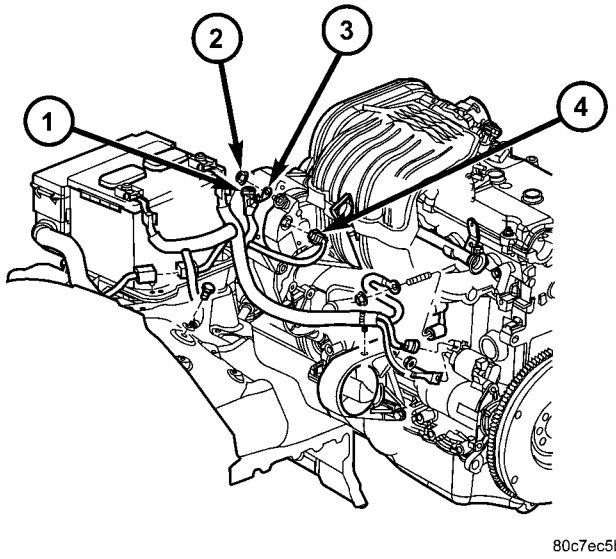


Fig. 6 GENERATOR ELECTRICAL CONNECTORS - 2.4L - TYPICAL

1 - PROTECTIVE CAP
2 - B+ NUT
3 - B+ TERMINAL
4 - FIELD ELECTRICAL CONNECTOR

(5) Install generator drive belt. Refer to 7, Cooling System for procedure.

(6) Install negative battery cable to battery.

GENERATOR DECOUPLER PULLEY

DESCRIPTION

The generator decoupler is used only with certain engines. The decoupler is used in place of the standard generator drive pulley (Fig. 7).

OPERATION

The generator decoupler is used only with certain engines. The decoupler (Fig. 7) is a one-way clutch designed to help reduce belt tension fluctuation, vibration, reduce fatigue loads, improve belt life, reduce hubloads on components, and reduce noise. Dry operation is used (no grease or lubricants). The decoupler is not temperature sensitive and also has a low sensitivity to electrical load. The decoupler is a non-serviceable item and is to be replaced as an assembly.

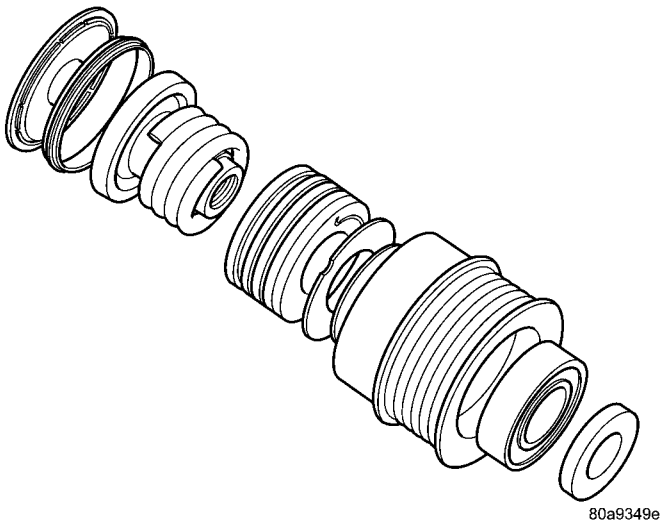


Fig. 7 GENERATOR DECOUPLER PULLEY

DIAGNOSIS AND TESTING - GENERATOR DECOUPLER

CONDITION	POSSIBLE CAUSES	CORRECTION
Does not drive generator (generator not charging)	Internal failure	Replace decoupler
Noise coming from decoupler	Internal failure	Replace decoupler

REMOVAL

The generator decoupler is used only with certain engines.

Two different type generator decoupler pulleys are used. One can be identified by the use of machined splines (Fig. 8). The other can be identified by a hex opening (Fig. 9) and will not use splines.

Different special tools are required to service each different decoupler. Refer to following procedure.

INA Decoupler

(1) Disconnect negative battery cable.

(2) Remove generator and accessory drive belt. Refer to Generator Removal.

(3) Position Special Tool #8823 (VM.1048) into decoupler (Fig. 10).

(4) Determine if end of generator shaft is hex shaped (Fig. 11) or is splined (Fig. 12). If hex is used, insert a 10MM deep socket into tool #8823 (VM.1048) (Fig. 13). If splined, insert a 5/16” 6-point hex driver, or a 10MM 12-point triple square driver into tool #8823 (VM.1048) (Fig. 14).

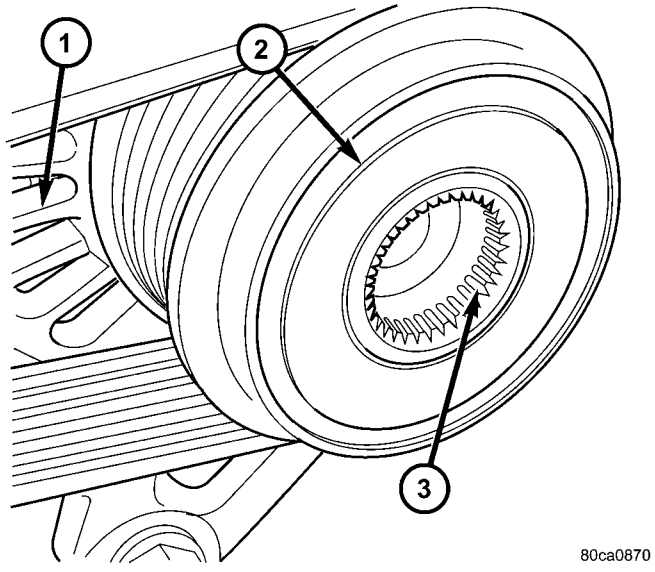


Fig. 8 GENERATOR DECOUPLER PULLEY (INA)

1 - GENERATOR
2 - DECOUPLER (INA)
3 - MACHINED SPLINES

GENERATOR DECOUPLER PULLEY (Continued)

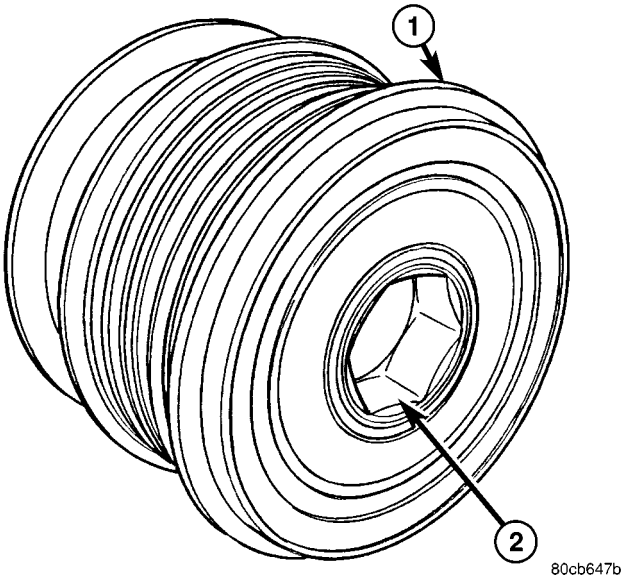


Fig. 9 GENERATOR DECOUPLER PULLEY (LITENS)

1 - DECOUPLER (LITENS)
2 - HEX OPENING

(5) The generator shaft uses conventional right-hand threads to attach decoupler. To break decoupler loose from generator threads, rotate end of tool clockwise (Fig. 13) or, (Fig. 14).

(6) After breaking loose with tool, unthread decoupler by hand from generator.

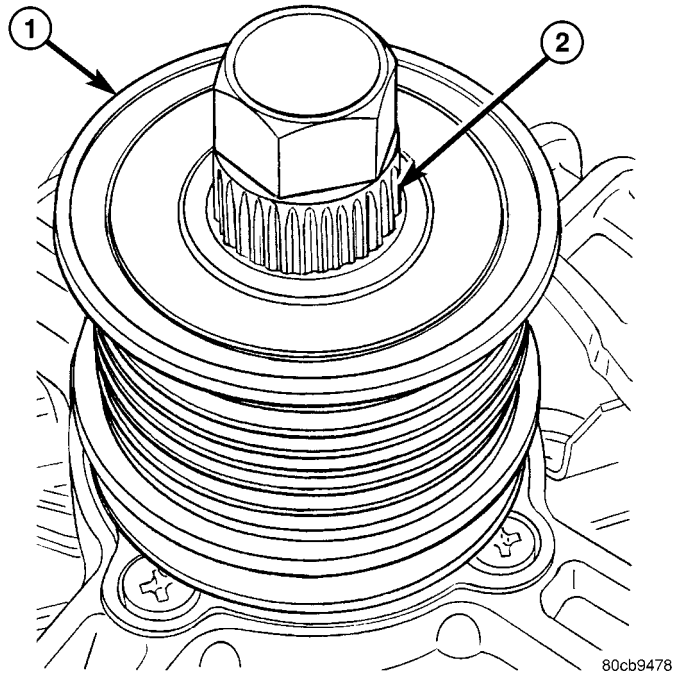


Fig. 10 #8823 TOOL AND INA DECOUPLER

1 - INA DECOUPLER
2 - TOOL #8823 (VM.1048)

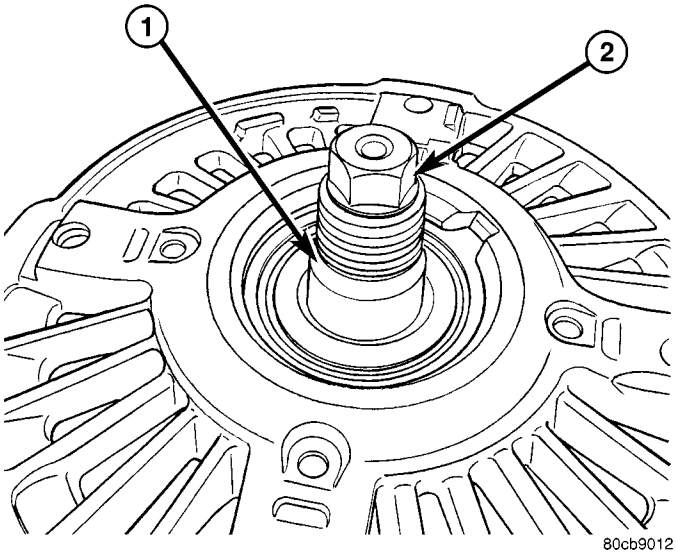


Fig. 11 END OF GENERATOR SHAFT (HEX)

1 - GENERATOR SHAFT
2 - HEX

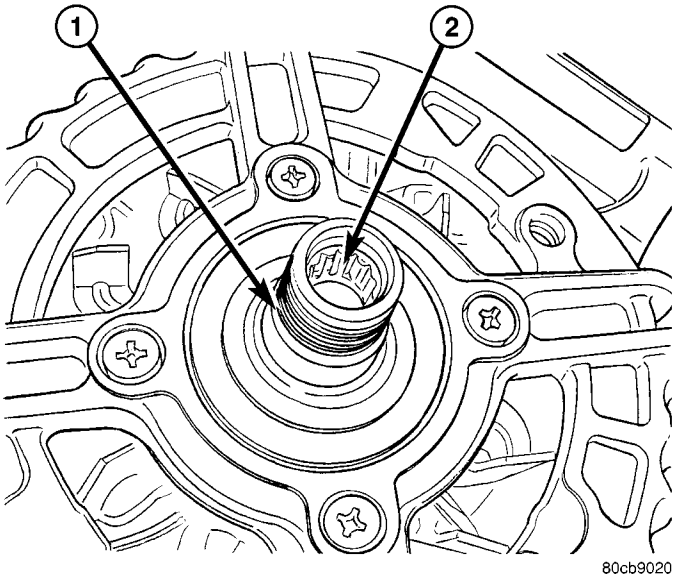


Fig. 12 END OF GENERATOR SHAFT (SPLINED)

1 - GENERATOR SHAFT
2 - SPLINES

GENERATOR DECOUPLER PULLEY (Continued)

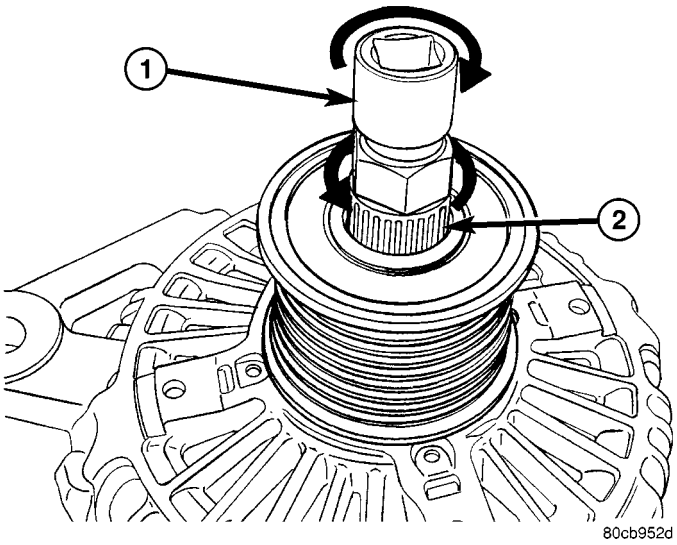


Fig. 13 DECOUPLER REMOVAL (INA-HEX)

1 - DEEP 10 MM SOCKET
2 - TOOL #8823 (VM.1048)

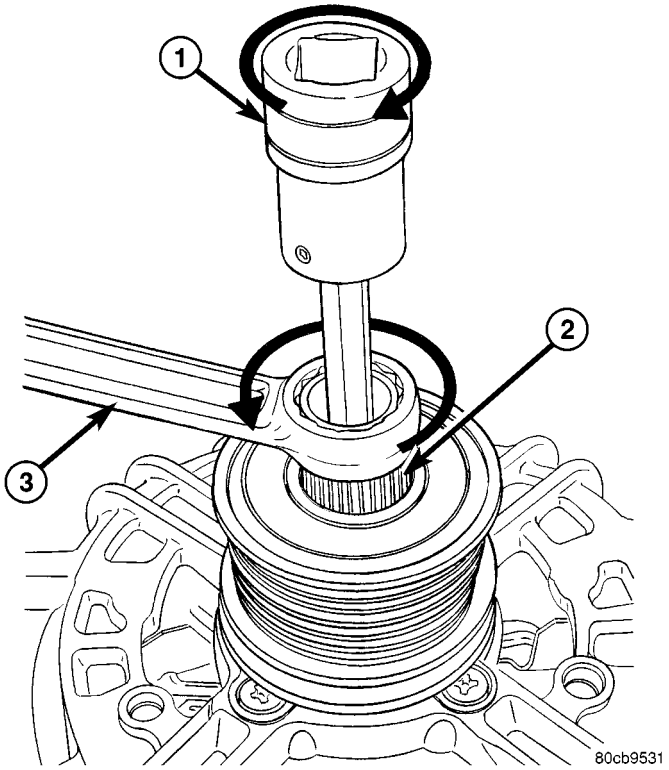


Fig. 14 DECOUPLER REMOVAL (INA-SPLINED)

1 - DRIVER
2 - TOOL #8823 (VM.1048)
3 - 17 MM WRENCH

Litens Decoupler

(1) Disconnect negative battery cable.

(2) Remove generator and accessory drive belt. Refer to Generator Removal.

(3) Position Special Tool #8433 (Fig. 15) into decoupler. Align to hex end of generator shaft.

(4) The generator shaft uses conventional right-hand threads to attach decoupler. To break decoupler loose from generator threads, rotate end of tool clockwise (Fig. 16).

(5) After breaking loose with tool, unthread decoupler by hand from generator.

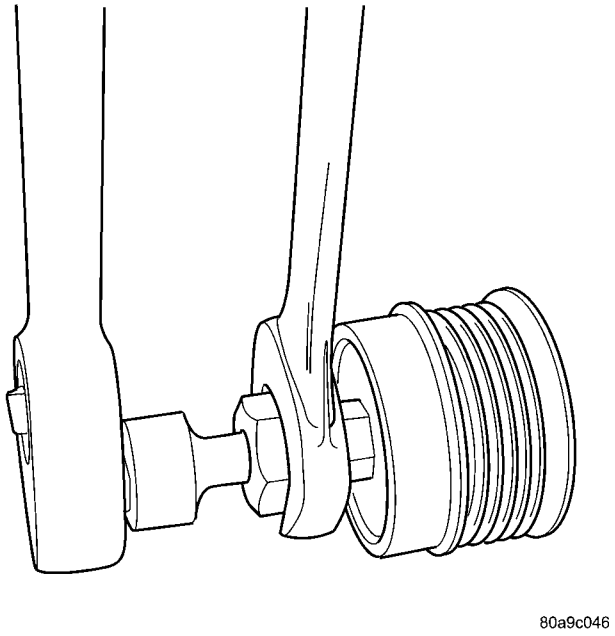


Fig. 15 # 8433 TOOL AND LITENS DECOUPLER

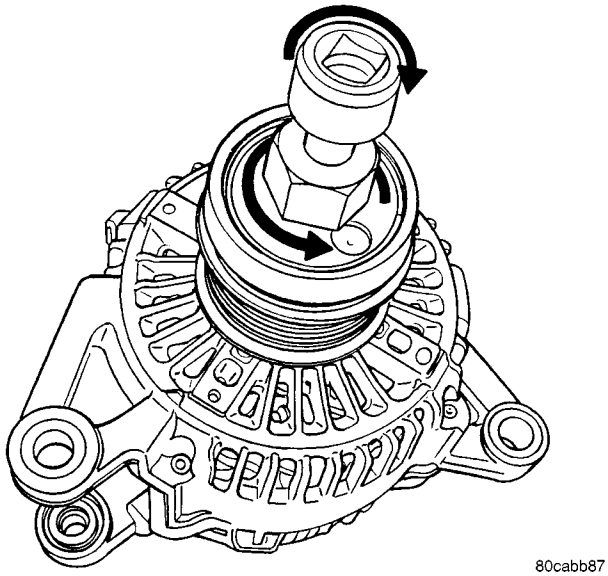


Fig. 16 DECOUPLER REMOVAL (LITENS)

GENERATOR DECOUPLER PULLEY (Continued)

INSTALLATION

INA Decoupler

(1) Thread decoupler pulley onto generator shaft by hand (right-hand threads).

(2) Position Special Tool #8823 (VM.1048) into decoupler (Fig. 10).

(3) Determine if end of generator shaft is hex shaped (Fig. 11) or is splined (Fig. 12). If hex is used, insert a 10MM deep socket into tool #8823 (VM.1048) (Fig. 17). If splined, insert a 5/16" 6-point hex driver, or a 10MM 12-point triple square driver into tool #8823 (VM.1048) (Fig. 18).

(4) **Do not use an adjustable, ratcheting "click type" torque wrench. Most "click type" wrenches will only allow torque to be applied in a clockwise rotation. Use a dial-type or beam-type wrench.** Tighten in counter-clockwise rotation (Fig. 17) or, (Fig. 18). Refer to torque specifications.

(5) Install accessory drive belt, and generator. Refer to Generator Installation.

(6) Connect negative battery cable.

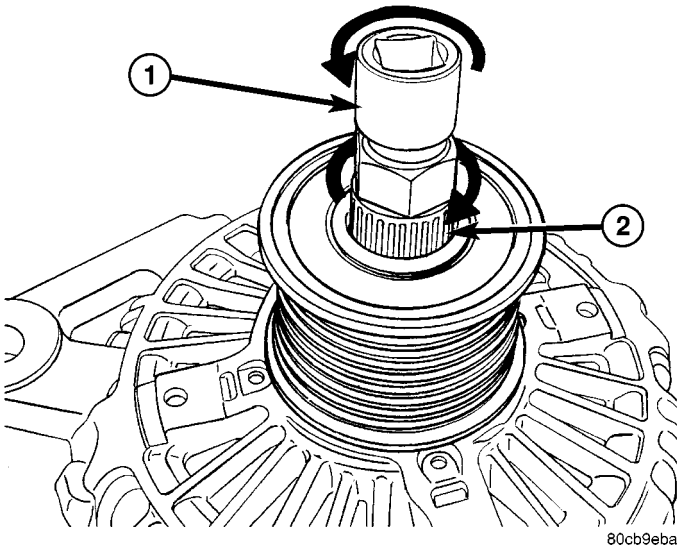


Fig. 17 DECOUPLER INSTALLATION (INA-HEX)

1 - 10MM DEEP SOCKET
2 - TOOL # 8823 (VM.1048)

Litens Decoupler

(1) Thread decoupler pulley onto generator shaft by hand (right-hand threads).

(2) Position Special Tool 8433 (Fig. 15) into decoupler. Align tool to hex end of generator shaft.

(3) **Do not use an adjustable, ratcheting "click type" torque wrench. Most "click type" wrenches will only allow torque to be applied in a clockwise rotation. Use a dial-type or beam-type wrench.** Tighten in counter-clockwise rotation (Fig. 19). Refer to torque specifications.

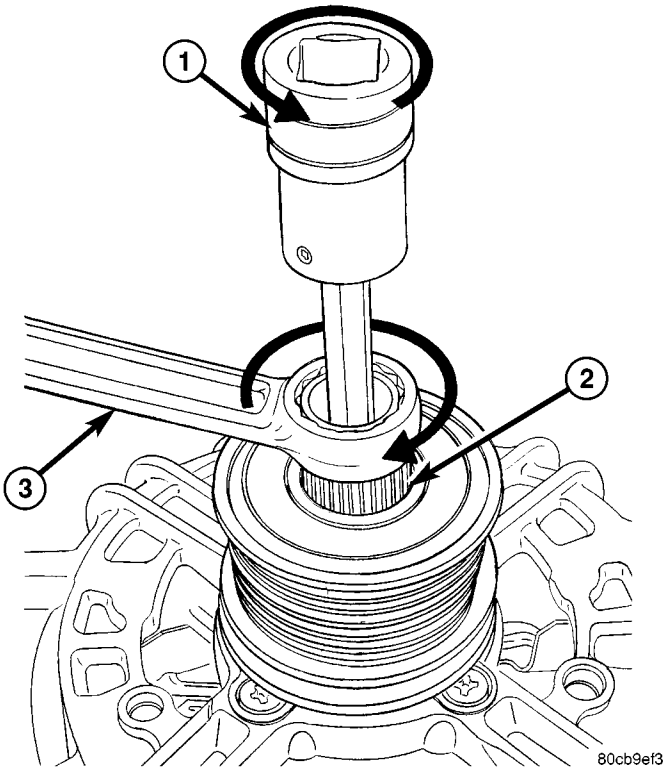


Fig. 18 DECOUPLER INSTALLATION (INA SPLINED)

1 - DRIVER
2 - TOOL # 8823 (VM.1048)

(4) Install accessory drive belt, and generator. Refer to Generator Installation.

(5) Connect negative battery cable.

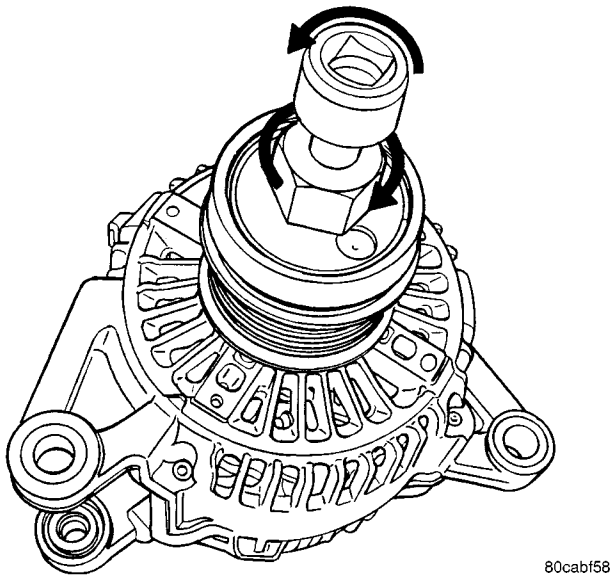


Fig. 19 DECOUPLER INSTALLATION (Litens)

VOLTAGE REGULATOR

DESCRIPTION

The Electronic Voltage Regulator (EVR) is not a separate component. It is actually a voltage regulating circuit located within the Powertrain Control Module (PCM). The EVR is not serviced separately. If replacement is necessary, the PCM must be replaced.

OPERATION

The amount of DC current produced by the generator is controlled by EVR circuitry contained within the PCM. This circuitry is connected in series with the generators second rotor field terminal and its ground.

Voltage is regulated by cycling the ground path to control the strength of the rotor magnetic field. The EVR circuitry monitors system line voltage (B+) and battery temperature (refer to Battery Temperature Sensor for more information). It then determines a target charging voltage. If sensed battery voltage is 0.5 volts or lower than the target voltage, the PCM grounds the field winding until sensed battery voltage is 0.5 volts above target voltage. A circuit in the PCM cycles the ground side of the generator field up to 100 times per second (100Hz), but has the capability to ground the field control wire 100% of the time (full field) to achieve the target voltage. If the charging rate cannot be monitored (limp-in), a duty cycle of 25% is used by the PCM in order to have some generator output. Also refer to Charging System Operation for additional information.

STARTING

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STARTING

DESCRIPTION

The starting system consists of:

- Starter relay
- Starter motor (including an integral starter solenoid)

Other components to be considered as part of starting system are:

- Battery
- Battery cables
- Ignition switch and key lock cylinder
- Clutch pedal position switch (manual transmission)
- Park/neutral position switch (automatic transmission)
- Wire harnesses and connections.

The Battery, Starting, and Charging systems operate in conjunction with one another, and must be tested as a complete system. For correct operation of starting/charging systems, all components used in these 3 systems must perform within specifications. When attempting to diagnose any of these systems, it is important that you keep their interdependency in mind.

The diagnostic procedures used in each of these groups include the most basic conventional diagnostic methods, to the more sophisticated On-Board Diagnostics (OBD) built into the Powertrain Control Module (PCM). Use of an induction-type milliampere ammeter, volt/ohmmeter, battery charger, carbon pile rheostat (load tester), and 12-volt test lamp may be required.

Certain starting system components are monitored by the PCM and may produce a Diagnostic Trouble Code (DTC).

OPERATION

The starting system components form two separate circuits. A high-amperage feed circuit that feeds the starter motor between 150 and 350 amperes (700 amperes - diesel engine), and a low-amperage control circuit that operates on less than 20 amperes. The high-amperage feed circuit components include the battery, the battery cables, the contact disc portion of the starter solenoid, and the starter motor. The low-amperage control circuit components include the ignition switch, the clutch pedal position switch (manual transmission), the park/neutral position switch (automatic transmission), the starter relay, the electromagnetic windings of the starter solenoid, and the connecting wire harness components.

If the vehicle is equipped with a manual transmission, it has a clutch pedal position switch installed in series between the ignition switch and the coil battery terminal of the starter relay. This normally open switch prevents the starter relay from being energized when the ignition switch is turned to the momentary Start position, unless the clutch pedal is depressed. This feature prevents starter motor operation while the clutch disc and the flywheel are engaged. The starter relay coil ground terminal is always grounded on vehicles with a manual transmission.

If the vehicle is equipped with an automatic transmission, battery voltage is supplied through the low-amperage control circuit to the coil battery terminal of the starter relay when the ignition switch is turned to the momentary Start position. The park/

neutral position switch is installed in series between the starter relay coil ground terminal and ground. This normally open switch prevents the starter relay from being energized and the starter motor from operating unless the automatic transmission gear selector is in the Neutral or Park positions.

When the starter relay coil is energized, the normally open relay contacts close. The relay contacts connect the relay common feed terminal to the relay normally open terminal. The closed relay contacts energize the starter solenoid coil windings.

The energized solenoid pull-in coil pulls in the solenoid plunger. The solenoid plunger pulls the shift lever in the starter motor. This engages the starter overrunning clutch and pinion gear with the starter ring gear on the manual transmission flywheel or on the automatic transmission torque converter or torque converter drive plate.

As the solenoid plunger reaches the end of its travel, the solenoid contact disc completes the high-amperage starter feed circuit and energizes the solenoid plunger hold-in coil. Current now flows between the solenoid battery terminal and the starter motor, energizing the starter.

Once the engine starts, the overrunning clutch protects the starter motor from damage by allowing the starter pinion gear to spin faster than the pinion shaft. When the driver releases the ignition switch to the On position, the starter relay coil is de-energized. This causes the relay contacts to open. When the relay contacts open, the starter solenoid plunger hold-in coil is de-energized.

When the solenoid plunger hold-in coil is de-energized, the solenoid plunger return spring returns the plunger to its relaxed position. This causes the contact disc to open the starter feed circuit, and the shift lever to disengage the overrunning clutch and pinion gear from the starter ring gear.

DIAGNOSIS AND TESTING - STARTING SYSTEM

The battery, starting system and charging system in the vehicle operate with one another, and must be tested as a complete system. In order for the engine to start and the battery to charge properly, all of the components that are used in these systems must perform within specifications. The service information for these systems has been separated within this service manual to make it easier to locate the specific information you are seeking. However, when attempting to diagnose any of these systems, it is important that you keep their interdependency in mind.

The diagnostic procedures used for the battery, starting system and charging system include the most basic conventional diagnostic methods, to the more sophisticated On-Board Diagnostics (OBD) built into the Powertrain Control Module (PCM). Use of an induction-type milliampere ammeter, a volt/ohmmeter, a battery charger, a carbon pile rheostat (load tester) and a 12-volt test lamp may be required. All OBD-sensed systems are monitored by the PCM. Each monitored circuit is assigned a Diagnostic Trouble Code (DTC). The PCM will store a DTC in electronic memory for any failure it detects. Refer to **Charging System, On-Board Diagnostic Test** for on-board diagnostic test procedures.

Starting System Diagnosis		
Condition	Possible Cause	Correction
Starter fails to operate.	1. Battery discharged or faulty.	1. Refer to 8, Battery. Replace faulty battery as required.
	2. Starting circuit wiring faulty.	2. Refer to Wiring. Test and repair faulty starter feed and/or control circuits, as required.
	3. Starter relay faulty.	3. Refer to Starter Relay. Replace faulty starter relay as required.
	4. Ignition switch faulty.	4. Refer to Ignition Switch and Key Lock Cylinder. Replace faulty ignition switch as required.
	5. Clutch pedal position switch faulty.	5. Refer to Clutch Pedal Position Switch. Replace faulty clutch hydraulic linkage unit as required.
	6. Park/Neutral position switch faulty or misadjusted.	6. Refer to Park/Neutral Position Switch. Replace faulty park/neutral position switch as required.
	7. Starter solenoid faulty.	7. Refer to Starter Motors. Replace faulty starter motor as required.

STARTING (Continued)

Starting System Diagnosis		
Condition	Possible Cause	Correction
	8. Starter motor faulty.	8. Refer to Starter Motor. Replace faulty starter motor as required.
Starter engages, fails to turn engine.	1. Battery discharged or faulty.	1. Refer to Battery. Replace faulty battery as required.
	2. Starting circuit wiring faulty.	2. Refer to Wiring. Test and repair faulty starter feed and/or control circuits as required.
	3. Starter motor faulty.	3. Refer to Starter Motor. Replace faulty starter motor as required.
	4. Engine seized.	4. Refer to 9, Engine Diagnosis. Repair or replace faulty engine as required.
Starter engages, spins out before engine starts.	1. Starter ring gear faulty.	1. Refer to Starter Motor. Remove starter motor to inspect starter ring gear. Replace faulty starter ring gear as required.
	2. Starter motor faulty.	2. Refer to Starter Motor. Replace faulty starter motor as required.
Starter does not disengage.	1. Starter motor improperly installed.	1. Refer to Starter Motor. Tighten starter motor mounting hardware to correct tightness specifications as required.
	2. Starter relay faulty.	2. Refer to Starter Relay. Replace faulty starter relay as required.
	3. Ignition switch faulty.	3. Refer to Ignition Switch and Key Lock Cylinder. Replace faulty ignition switch as required.
	4. Starter motor faulty.	4. Refer to Starter Motor. Replace faulty starter motor as required.

TESTING

Before testing the starting system perform a visual inspection of the starting system components and connections.

COLD CRANKING TEST

Refer to **Starting System** in the index of this service manual for the location of complete starting system wiring diagrams. Before performing this test, be certain that the following procedures are accomplished:

- The battery is fully-charged and load tested. Refer to **Battery Charging** for battery charging procedures. Refer to **Battery** for battery diagnosis and testing procedures, including battery load test procedures.
- Fully engage the parking brake.
- If the vehicle is equipped with an automatic transmission, place the gearshift selector lever in the Park position. If the vehicle is equipped with a manual transmission, place the gearshift selector lever in the Neutral position and block the clutch pedal in the fully depressed position.
- Verify that all lamps and accessories are turned off.
- To prevent the engine from starting, remove the Automatic ShutDown (ASD) relay. The ASD relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the underside of the PDC cover for ASD relay identification and location.

(1) Connect a suitable volt-ampere tester to the battery terminals (Fig. 1). See the instructions provided by the manufacturer of the volt-ampere tester being used.

(2) Rotate and hold the ignition switch in the Start position. Note the cranking voltage and current (amperage) draw readings shown on the volt-ampere tester.

(a) If the voltage reads below 9.6 volts, refer to **Starter Motor** for starter motor diagnosis and testing procedures. If the starter motor tests OK, refer to **Engine Diagnosis** engine diagnosis and testing procedures. If the starter motor is not OK, replace faulty starter motor.

(b) If the voltage reads above 9.6 volts and the current (amperage) draw reads below specifications, refer to the **Feed Circuit Test**in this section.

STARTING (Continued)

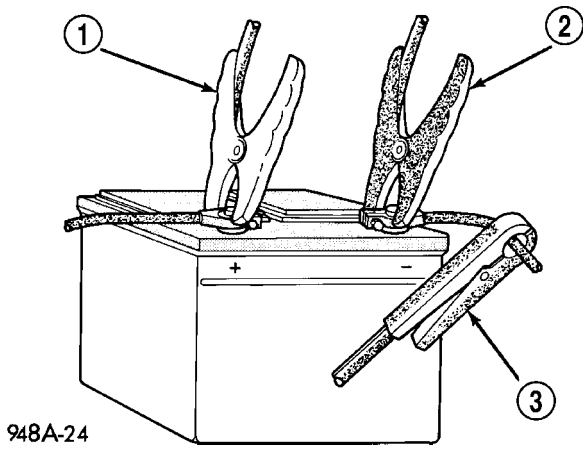

Fig. 1 Volts-Amps Tester Connections - Typical

1 - POSITIVE CLAMP
2 - NEGATIVE CLAMP
3 - INDUCTION AMMETER CLAMP

(c) If the voltage reads 12.5 volts or greater and the starter motor does not turn, refer to the **Control Circuit Test** in this section.

(d) If the voltage reads 12.5 volts or greater and the starter motor turns very slowly, refer to the **Feed Circuit Test** in this section.

NOTE: A cold engine will increase the starter current (amperage) draw reading, and reduce the battery voltage reading.

FEED CIRCUIT TEST

The starter feed circuit test (voltage drop method) will determine if there is excessive resistance in the high-amperage starter feed circuit.

When performing the voltage drop test, it is important to remember that the voltage drop is giving an indication of the resistance between the two points at which the voltmeter probes are attached. **EXAMPLE:**When testing the resistance of the battery positive cable, touch the voltmeter leads to the battery positive cable terminal clamp and to the battery positive cable eyelet terminal at the starter solenoid B(+) terminal stud. If you probe the battery positive terminal post and the battery positive cable eyelet terminal at the starter solenoid B(+) terminal stud, you are reading the combined voltage drop in the battery positive cable terminal clamp-to-terminal post connection and the battery positive cable.

The following operation will require a voltmeter accurate to 1/10 (0.10) volt. Before performing this test, be certain that the following procedures are accomplished:

- The battery is fully-charged and load tested. Refer to **Battery Charging** for battery charging procedures. Refer to **Battery** for battery diagnosis and testing procedures, including battery load test procedures.
- Fully engage the parking brake.
- If the vehicle is equipped with an automatic transmission, place the gearshift selector lever in the Park position. If the vehicle is equipped with a manual transmission, place the gearshift selector lever in the Neutral position and block the clutch pedal in the fully depressed position.
- Verify that all lamps and accessories are turned off.
- To prevent the engine from starting, remove the Automatic ShutDown (ASD) relay. The ASD relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the underside of the PDC cover for ASD relay identification and location.

(1) Connect the positive lead of the voltmeter to the battery negative terminal post. Connect the negative lead of the voltmeter to the battery negative cable terminal clamp (Fig. 2). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If voltage is detected, correct the poor contact between the battery negative cable terminal clamp and the battery negative terminal post.

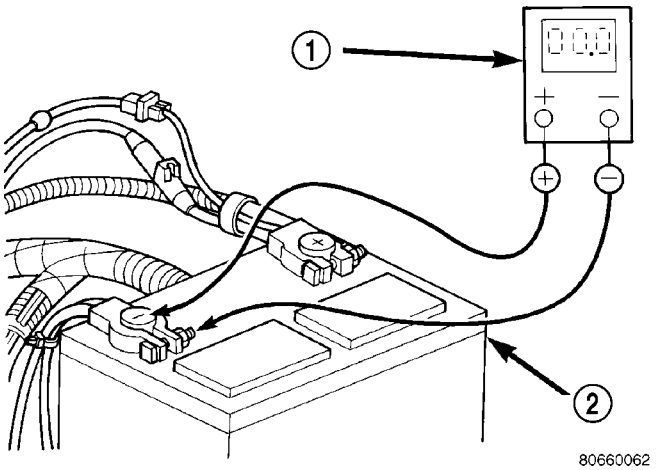

Fig. 2 Test Battery Negative Connection Resistance - Typical

1 - VOLTMETER
2 - BATTERY

(2) Connect the positive lead of the voltmeter to the battery positive terminal post. Connect the negative lead of the voltmeter to the battery positive cable terminal clamp (Fig. 3). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If voltage is detected, correct the poor connection between the battery positive cable terminal clamp and the battery positive terminal post.

STARTING (Continued)

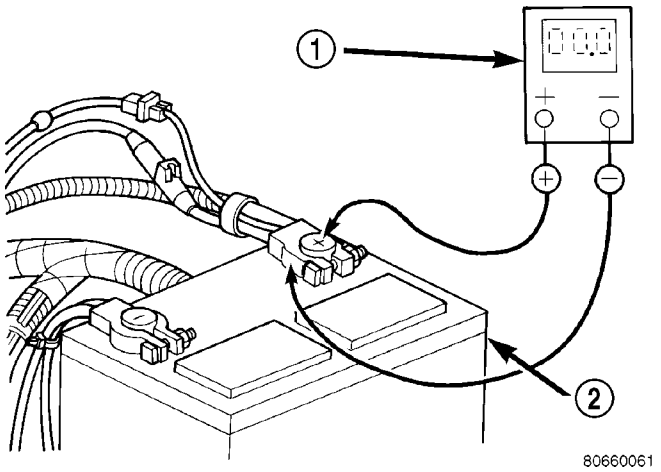

Fig. 3 Test Battery Positive Connection Resistance - Typical

1 - VOLTMETER
2 - BATTERY

(3) Connect the voltmeter to measure between the battery positive cable terminal clamp and the starter solenoid B(+) terminal stud (Fig. 4). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If the reading is above 0.2 volt, clean and tighten the battery positive cable eyelet terminal connection at the starter solenoid B(+) terminal stud. Repeat the test. If the reading is still above 0.2 volt, replace the faulty battery positive cable.

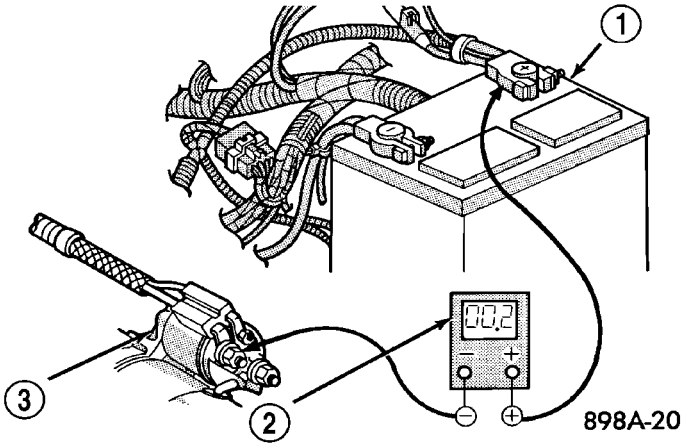

Fig. 4 Test Battery Positive Cable Resistance - Typical

1 - BATTERY
2 - VOLTMETER
3 - STARTER MOTOR

(4) Connect the voltmeter to measure between the battery negative cable terminal clamp and a good clean ground on the engine block (Fig. 5). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If the reading is above 0.2 volt, clean and tighten the battery negative cable eyelet terminal connection to the engine block. Repeat the test. If the reading is still above 0.2 volt, replace the faulty battery negative cable.

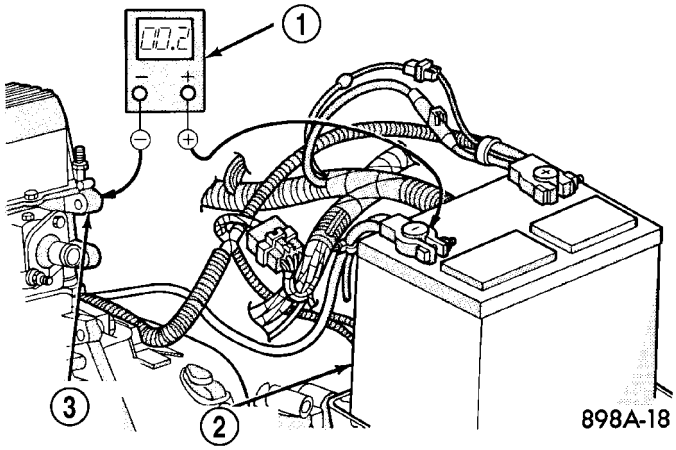

Fig. 5 Test Ground Circuit Resistance - Typical

1 - VOLTMETER
2 - BATTERY
3 - ENGINE GROUND

(5) Connect the positive lead of the voltmeter to the starter housing. Connect the negative lead of the voltmeter to the battery negative terminal post (Fig. 6). Rotate and hold the ignition switch in the Start position. Observe the voltmeter. If the reading is above 0.2 volt, correct the poor starter to engine block ground contact.

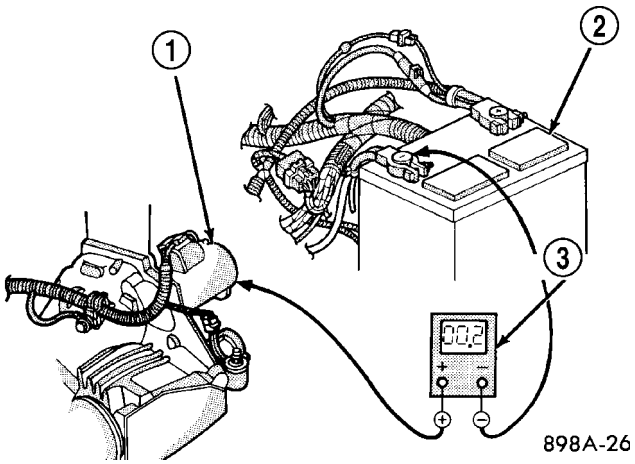

Fig. 6 Test Starter Ground - Typical

1 - STARTER MOTOR
2 - BATTERY
3 - VOLTMETER

If the resistance tests detect no feed circuit problems, refer to **Starter Motor**.

STARTING (Continued)

CONTROL CIRCUIT TEST

The starter control circuit components should be tested in the order in which they are listed, as follows:

Starter Relay

- Refer to **Starter Relay**.

Starter Solenoid

- Refer to **Starter Motor**.

Ignition Switch

- Refer to **Ignition Switch and Key Lock Cylinder** for ignition switch diagnosis and testing procedures.

Clutch Pedal Position Switch

- If the vehicle is equipped with a manual transmission, refer to **Clutch Pedal Position Switch** for clutch pedal position switch diagnosis and testing procedures.

Park/Neutral Position Switch

- If the vehicle is equipped with an automatic transmission, refer to **Park/Neutral Position Switch** for park/neutral position switch diagnosis and testing procedures.

INSPECTION - STARTING SYSTEM

The following starting system components should be carefully inspected whenever any starting system problem is encountered.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE AIRBAG SYSTEM. FAILURE TO TAKE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

Battery

- Visually inspect battery for indications of physical damage and loose or corroded cable connections. Determine state-of-charge and cranking capacity of battery. Charge or replace battery, if required. Refer to **Battery** for battery cleaning and inspection procedures.

Ignition Switch

- Visually inspect ignition switch for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. Refer to **Wiring Diagrams**. Refer to **Ignition Switch and Key Lock Cylinder** for ignition switch service procedures.

Clutch Pedal Position Switch

- If vehicle is equipped with a manual transmission, visually inspect clutch pedal position switch for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. Refer to **Clutch Hydraulic Linkage** for clutch pedal position switch service procedures.

Park/Neutral Position Switch

- If vehicle is equipped with an automatic transmission, visually inspect park/neutral position switch for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. Refer to **Park/Neutral Position Switch** for park/neutral position switch service procedures.

Starter Relay

- Visually inspect starter relay for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. Refer to **Starter Relay** for starter relay service procedures.

Starter Motor

- Visually inspect starter motor for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. If problem being diagnosed involves improper starter engagement, disengagement or noise complaints, starter motor should be removed. With starter motor removed, inspect starter pinion and ring gears for damaged or missing teeth. Replace faulty components as required. Refer to **Starter Motor** for removal/installation procedures.

Starter Solenoid

- Visually inspect starter solenoid for indications of physical damage and loose or corroded wire harness connections. Clean corroded connections as required. Refer to **Starter Motor** for starter solenoid service procedures.

Wiring

- Visually inspect starting system wire harnesses for indications of physical damage. Repair or replace any faulty wiring, as required. Refer to **Wiring Diagrams** for repair or connector and terminal service procedures.

STARTING (Continued)

SPECIFICATIONS

STARTER

Starter Motor and Solenoid	
Manufacturer	Mitsubishi
Engine Application	2.4L, 4.0L
Power Rating	2.4L - 1.4 Kilowatt (1.9 Horsepower) 4.0L - 1.4 Kilowatt (1.9 Horsepower)
Voltage	12 Volts
** Number of Permanent Magnets	6
Number of Brushes	4
Drive Type	Planetary Gear Reduction
Free Running Test Voltage	11.2 Volts
Free Running Test Maximum Amperage Draw	90 Amperes
Free Running Test Minimum Speed	2.4L - 24400 rpm 4.0L - 2500 rpm
Solenoid Closing Maximum Voltage Required	7.8 Volts
*Cranking Amperage Draw Test	2.4L - 160 Amperes 4.0L - 160 Amperes
* Test at operating temperature. Cold engine, tight (new) engine, or heavy oil will increase starter amperage draw.	
** The starter is equipped with permanent magnets. Never strike the starter case to attempt to loosen a sticking/ stuck armature as permanent magnets may crack or break.	

TORQUE

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Starter Solenoid B+ Terminal Nut	11.3	-	100
Starter Mounting Bolts (2.4L)	54	40	-
Starter Heat Shield Mounting Bolts	6	-	55
Starter Mounting Bolt (forward facing 4.0L)	41	30	-
Starter Mounting Bolt (rearward facing 4.0L)	48	35	-

STARTER MOTOR

DIAGNOSIS AND TESTING - STARTER MOTOR/SOLENOID

Correct starter motor operation can be confirmed by performing the following free running bench test. This test can only be performed with starter motor removed from vehicle. Refer to starter motor specifications.

CAUTION: Permanent magnets are used in the starter. Permanent magnet starters are highly sensitive to hammering, shocks, external pressure and reverse polarity. This starter motor must never be clamped in a vise by starter field frame. The starter should only be clamped by mounting flange. Do not reverse battery cable connections to starter motor when testing. The permanent magnets may be damaged and starter rendered unserviceable if it is subjected to any of these conditions.

STARTER MOTOR (Continued)

STARTER MOTOR TESTING

(1) Remove starter motor from vehicle. Refer to Starter Motor Removal / Installation.

(2) Mount starter motor securely in a soft-jawed bench vise. The vise jaws should be clamped on mounting flange of starter motor. Never clamp on starter motor by field frame.

(3) Connect a suitable volt-ampere tester and12-volt battery to starter motor in series, and set ammeter to 100 ampere scale. See instructions provided by manufacturer of volt-ampere tester being used.

(4) Install a jumper wire from solenoid terminal to solenoid B(+) terminal stud. The starter motor should operate. If starter motor fails to operate, replace faulty starter motor.

(5) Adjust carbon pile load of tester to obtain free running test voltage. Refer to starter motor free running test voltage specifications.

(6) Note reading on ammeter and compare reading to free running test maximum amperage draw. Refer to starter motor free running test maximum amperage draw specifications.

(7) If ammeter reading exceeds maximum amperage draw specification, replace faulty starter motor.

STARTER SOLENOID TESTING

This test can only be performed with starter motor removed from vehicle.

(1) Remove starter motor from vehicle. Refer to **Starter Motor** for removal/installation.

(2) Disconnect wire from solenoid field coil terminal.

(3) Check for continuity between solenoid terminal and solenoid field coil terminal with a continuity tester (Fig. 7). There should be continuity. If OK, go to Step 4. If not OK, replace faulty starter motor.

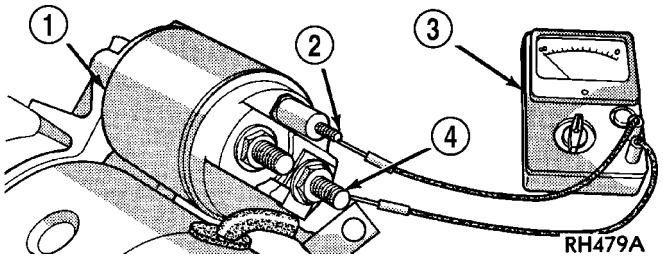

Fig. 7 CONTINUITY TEST BETWEEN SOLENOID TERMINAL AND FIELD COIL TERMINAL

1 - SOLENOID
2 - SOLENOID TERMINAL
3 - OHMMETER
4 - FIELD COIL TERMINAL

(4) Check for continuity between solenoid terminal and solenoid case (Fig. 8). There should be continuity. If not OK, replace faulty starter motor.

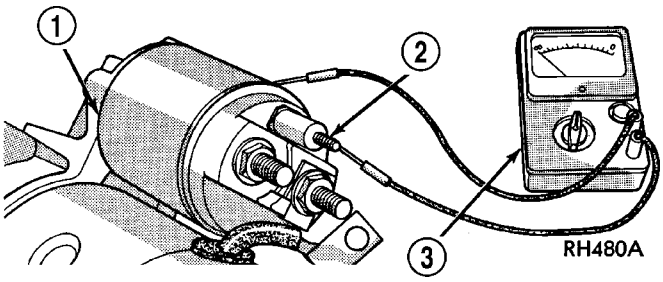

Fig. 8 CONTINUITY TEST BETWEEN SOLENOID TERMINAL AND SOLENOID CASE - TYPICAL

1 - SOLENOID
2 - SOLENOID TERMINAL
3 - OHMMETER

REMOVAL

2.4L

(1) Disconnect and isolate negative battery cable.

(2) Raise and support vehicle.

(3) While supporting starter motor, remove two bolts securing starter motor to transmission (Fig. 9).

(4) Lower starter motor far enough to access and remove nut securing battery cable to starter solenoid B(+) terminal stud (Fig. 10). Always support starter motor during this process. Do not let starter motor hang from wire harness.

(5) Remove battery cable at starter.

(6) Disconnect solenoid terminal wire harness connector from starter solenoid.

(7) Remove starter motor.

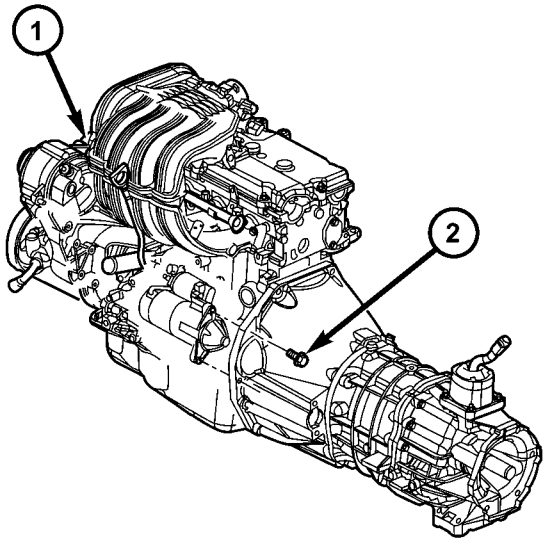

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Fig. 9 STARTER - 2.4L

1 - STARTER
2 - MOUNTING BOLTS (2)

STARTER MOTOR (Continued)

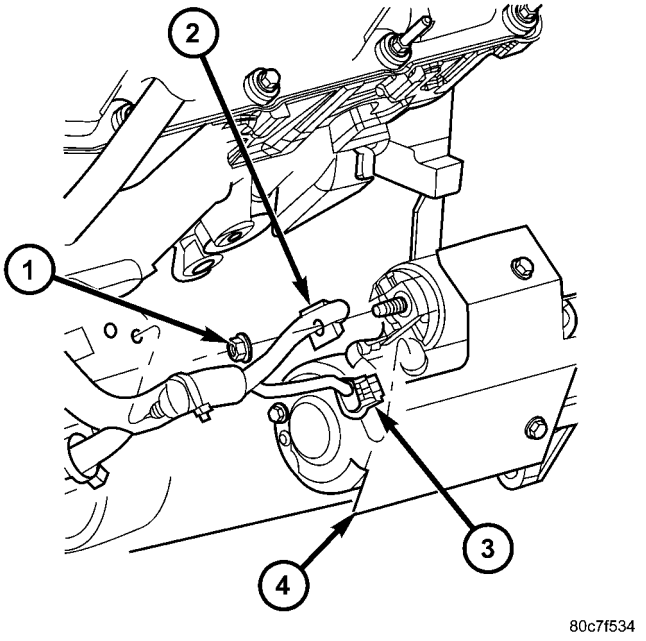

Fig. 10 STARTER ELECTRICAL CONNECTORS (2.4L SHOWN)

1 - BATERY CABLE NUT
2 - BATTERY CABLE
3 - SOLENOID CONNECTOR
4 - HEAT SHIELD (IF USED)

4.0L

(1) Disconnect and isolate negative battery cable.

(2) Raise and support vehicle.

(3) **4.0L With Manual Transmission:** Remove lower bolt (forward facing) securing starter motor to transmission housing (Fig. 11).

(4) **4.0L With Manual Transmission:** While supporting starter motor, remove upper bolt (rearward facing) securing starter motor to transmission housing.

(5) **4.0L With 42 RLE Automatic Transmission:** Remove 2 starter mounting bolts (Fig. 12).

(6) Lower starter motor from front of transmission housing far enough to access and remove nut securing battery to starter solenoid. Always support starter motor during this process. Do not let starter motor hang from wire harness.

(7) Remove battery cable at starter solenoid.

(8) Disconnect solenoid terminal wire harness connector from starter solenoid.

(9) Remove starter motor from transmission housing.

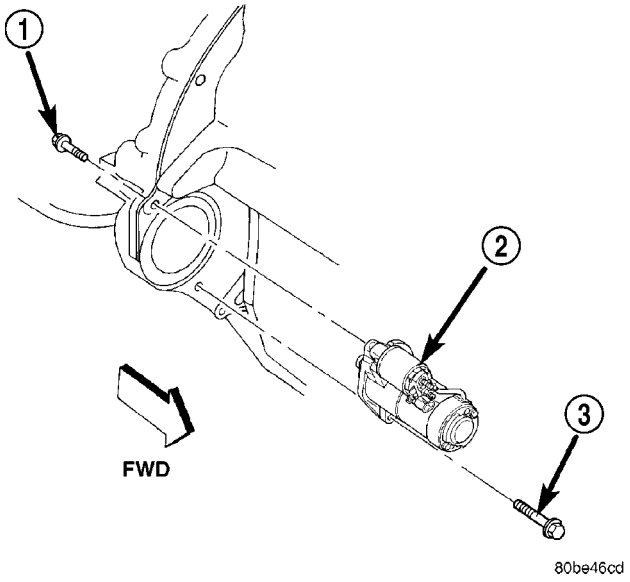

Fig. 11 STARTER - 4.0L - MAN. TRANS.

1 - BOLT
2 - STARTER MOTOR
3 - BOLT

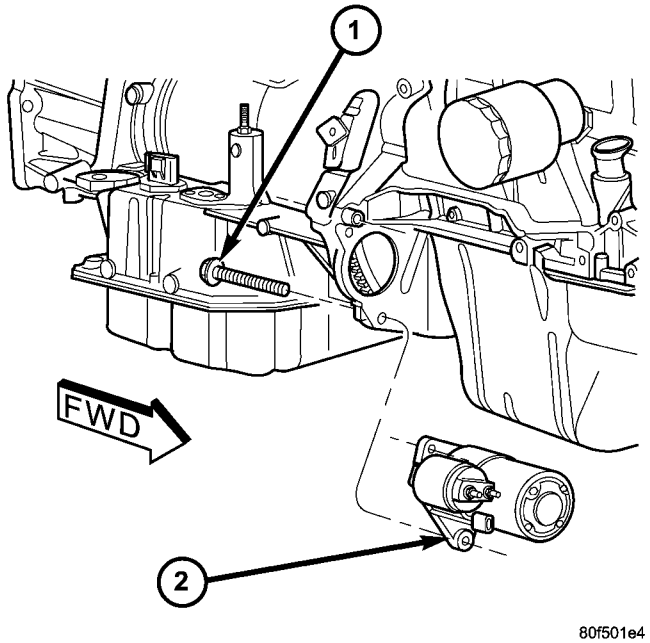

Fig. 12 STARTER - 4.0L - 42RLE AUTO. TRANS.

1 - BOLT
2 - STARTER MOTOR

STARTER MOTOR (Continued)

INSTALLATION

2.4L

(1) Connect solenoid terminal wire harness connector to starter solenoid. Always support starter motor during this process. Do not let starter motor hang from wire harness.

(2) Install battery cable eyelet onto starter solenoid stud. Refer to Torque Specifications.

(3) Position starter motor to transmission. Install and tighten 2 bolts. Refer to Torque Specifications.

(4) Lower vehicle.

(5) Connect negative battery cable.

4.0L

(1) Connect solenoid terminal wire harness connector to starter solenoid. Always support starter motor during this process. Do not let starter motor hang from wire harness.

(2) Install battery cable eyelet onto starter solenoid stud. Refer to Torque Specifications.

(3) Position starter motor to transmission housing. Loosely install two mounting bolts.

(4) Tighten upper (rearward facing) mounting bolt to 47.5 N·m (35 ft. lbs.).

(5) Tighten lower (forward facing) mounting bolt to 40.7 N·m (30 ft. lbs.).

(6) Lower vehicle.

(7) Connect negative battery cable.

STARTER MOTOR RELAY

DESCRIPTION

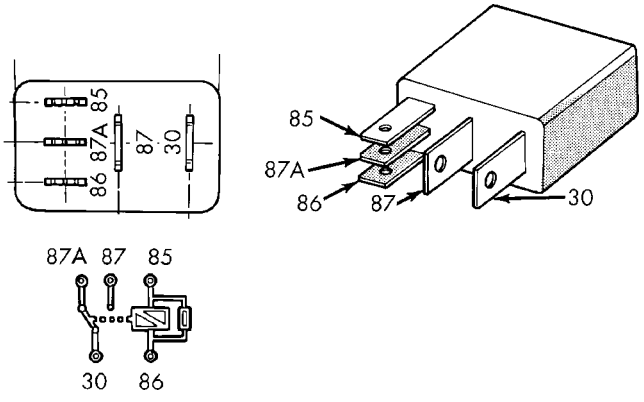

Fig. 13 Starter Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

The starter relay (Fig. 13) is an electromechanical device that switches battery current to the pull-in coil of the starter solenoid when the ignition switch is turned to the Start position. The starter relay is located in the Power Distribution Center (PDC), in the engine compartment. See the fuse and relay layout label affixed to the inside surface of the PDC cover for starter relay identification and location.

The starter relay is a International Standards Organization (ISO) micro-relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal patterns, and terminal functions. The ISO micro-relay terminal functions are the same as a conventional ISO relay. However, the ISO micro-relay terminal pattern (or footprint) is different, the current capacity is lower, and the physical dimensions are smaller than those of the conventional ISO relay.

The starter relay cannot be repaired or adjusted and, if faulty or damaged, it must be replaced.

OPERATION

The ISO relay consists of an electromagnetic coil, a resistor or diode, and three (two fixed and one movable) electrical contacts. The movable (common feed) relay contact is held against one of the fixed contacts (normally closed) by spring pressure. When the electromagnetic coil is energized, it draws the movable contact away from the normally closed fixed contact, and holds it against the other (normally open) fixed contact.

When the electromagnetic coil is de-energized, spring pressure returns the movable contact to the normally closed position. The resistor or diode is connected in parallel with the electromagnetic coil in the relay, and helps to dissipate voltage spikes that are produced when the coil is de-energized.

DIAGNOSIS AND TESTING - STARTER RELAY

The starter relay (Fig. 14) is located in the Power Distribution Center (PDC), in engine compartment. Refer to fuse and relay layout label affixed to underside of PDC cover starter relay identification and location.

RELAY TEST

(1) Remove starter relay from PDC. Refer to **Starter Relay**.

(2) A relay in de-energized position should have continuity between terminals 87A and 30, and no continuity between terminals 87 and 30. If OK, go to Step 3. If not OK, replace faulty relay.

(3) Resistance between terminals 85 and 86 (electromagnet) should be 75 ± 5 ohms. If OK, go to Step 4. If not OK, replace faulty relay.

(4) Connect a battery to terminals 85 and 86. There should now be continuity between terminals 30 and 87, and no continuity between terminals 87A and 30. If OK, perform Relay Circuit Test that follows. If not OK, replace faulty relay.

STARTER MOTOR RELAY (Continued)

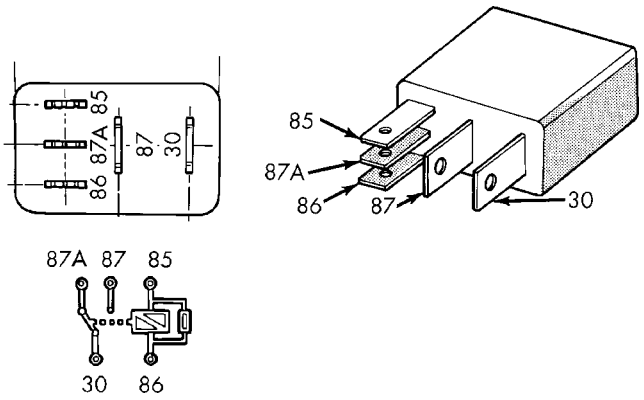

Fig. 14 Starter Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

RELAY CIRCUIT TEST

(1) The relay common feed terminal cavity (30) is connected to battery voltage and should be hot at all times. If OK, go to Step 2. If not OK, repair open circuit to fused B(+) fuse in PDC as required.

(2) The relay normally closed terminal (87A) is connected to terminal 30 in de-energized position, but is not used for this application. Go to Step 3.

(3) The relay normally open terminal (87) is connected to the common feed terminal (30) in energized position. This terminal supplies battery voltage to starter solenoid field coil. There should be continuity between cavity for relay terminal 87 and starter solenoid terminal at all times. If OK, go to Step 4. If not OK, repair open engine starter motor relay output circuit to starter solenoid as required.

(4) The coil battery terminal (86) is connected to electromagnet in relay. It is energized when ignition switch is held in Start position. On vehicles with a manual transmission, the clutch pedal must be blocked in fully depressed position for this test. Check for battery voltage at cavity for relay terminal 86 with ignition switch in Start position, and no voltage when ignition switch is released to On position. If OK, go to Step 5. If not OK with a manual transmission, disconnect clutch pedal position switch wire harness connector and install a jumper wire between two cavities in body half of connector and check for battery voltage again at cavity for relay terminal 86. If now OK, replace faulty clutch pedal position switch. If still not OK with a manual transmission or if not OK with an automatic transmission, check for open or shorted fused ignition switch output (start) circuit to ignition switch and repair, as required. If fused ignition switch output (start) circuit is OK, refer to **Ignition Switch and Key Lock Cylinder**.

(5) The coil ground terminal (85) is connected to electromagnet in relay. On vehicles with a manual transmission, it is grounded at all times. On vehicles with an automatic transmission, it is grounded through park/neutral position switch only when gearshift selector lever is in Park or Neutral positions. Check for continuity to ground at cavity for relay terminal 85. If not OK with a manual transmission, repair open park/neutral position switch sense circuit to ground as required. If not OK with an automatic transmission, check for an open or shorted park/neutral position switch sense circuit to park/neutral position switch and repair, as required. If park/neutral position switch sense circuit checks OK, refer to **Park/Neutral Position Switch**.

REMOVAL

(1) Disconnect and isolate battery negative cable.

(2) Unlatch and open cover on Power Distribution Center (PDC) (Fig. 15).

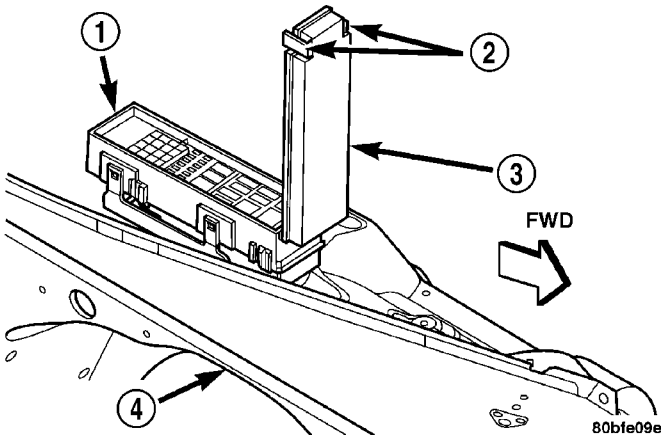

Fig. 15 Power Distribution Center

1 - POWER DISTRIBUTION CENTER (PDC)
2 - LATCHES
3 - COVER
4 - RIGHT FRONT FENDER

(3) See fuse and relay layout label affixed to underside of PDC cover for starter relay identification and location.

(4) Remove starter relay from PDC.

INSTALLATION

(1) See fuse and relay layout label affixed to underside of PDC cover for proper starter relay location.

(2) Position starter relay in proper receptacle in PDC.

(3) Align starter relay terminals with terminal cavities in PDC receptacle.

(4) Push down firmly on starter relay until terminals are fully seated in terminal cavities in PDC receptacle.

(5) Close and latch PDC cover.

(6) Connect negative battery cable.

HEATED SYSTEMS

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HEATED GLASS

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HEATED GLASS

DESCRIPTION

CAUTION: Grid lines can be damaged or scraped off with sharp instruments. Care should be taken in cleaning glass or removing foreign materials, decals or stickers. Normal glass cleaning solvents or hot water used with rags or toweling is recommended.

The rear window defogger system, also known as electric backlight (EBL), consists of two vertical bus bars linked by a series of grid lines fired onto the inside surface of the rear window (Fig. 1).

The EBL system is turned On or Off by a control switch mounted into the accessory switch bezel in the instrument panel and the timer and logic circuitry in the instrument cluster assembly (Refer to 8 - ELECTRICAL/HEATED GLASS/REAR WINDOW DEFOGGER SWITCH - DESCRIPTION).

Circuit protection is provided by three cartridge fuses located in the fuse block. One fuse for the control circuit, another for the heated grid circuit and the third fuse for the rear window defogger indicator lamp circuit.

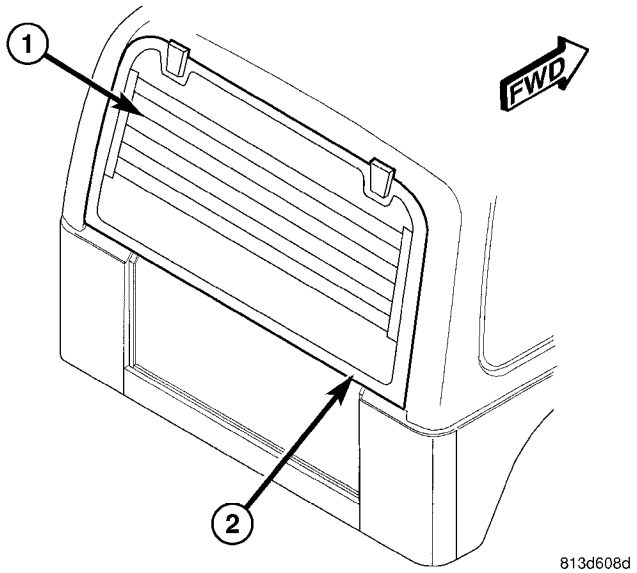

Fig. 1 Rear Window Defogger

1 - REAR DEFOGGER GRID
2 - REAR WINDOW

OPERATION

The electric backlight (EBL) system is controlled by a momentary switch located in the accessory

HEATED GLASS (Continued)

switch bezel on the instrument panel (Fig. 2). When the rear window defogger switch is pressed to the On position, the instrument cluster energizes the rear window defogger (EBL) relay, which then directs fused battery current through the relay and to the rear defogger grid lines and the heated side view mirrors, when equipped. The grid lines heat the glass to help clear the rear window and side mirror surfaces of fog or frost.

An amber indicator in the rear window defogger switch will illuminate to indicate when the EBL system is turned on. The instrument cluster contains the EBL system control circuitry including the timer logic.

NOTE: The EBL system turns off automatically after ten minutes of initial operation. Each following activation cycle of the EBL system will last five minutes.

The EBL system will be automatically turned off after an initial programmed time interval of about ten minutes. After the initial time interval has expired, if the rear window defogger switch is turned on again during the same ignition cycle, the EBL system will automatically turn off after about five minutes. The EBL system will automatically shut off if the ignition switch is turned to the Off position, or it can be turned off manually by pressing the rear window defogger switch a second time.

Repair of the rear window defogger grid lines, bus bars, terminals or pigtail wires can be accomplished using the Mopar Rear Window Defogger Repair Kit (Part Number 04549275) or equivalent (Refer to 8 - ELECTRICAL/HEATED GLASS/REAR WINDOW DEFOGGER GRID - STANDARD PROCEDURE - GRID LINE AND TERMINAL REPAIR).

DIAGNOSIS AND TESTING

ELECTRIC BACKLIGHT (EBL) SYSTEM

NOTE: Illumination of the defogger switch indicator lamp does not necessary mean that electrical current is reaching the rear glass heating grid lines.

NOTE: For circuit descriptions and diagrams of the rear window defogger (EBL) system, refer to 8W - WIRING DIAGRAM INFORMATION.

Operation of the electric backlight (EBL) system can be confirmed by the following:

(1) Use a DRBIII® scan tool and check for diagnostic trouble codes (DTCs) related to the instrument cluster. If no DTCs are found, go to Step 2. If any DTCs are found, repair as required, then proceed to Step 2

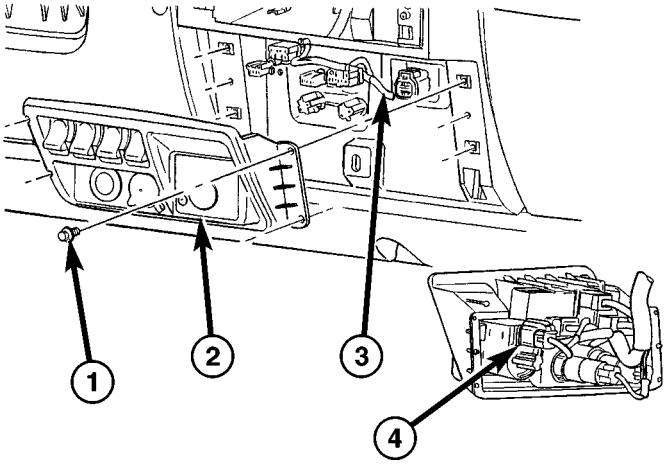

Fig. 2 03 ACCESSORY SWITCH BEZEL

1 - BEZEL SCREWS (4)
2 - BEZEL
3 - WIRE HARNESS
4 - ELECTRICAL CONNECTORS (7)

(2) Turn the ignition switch to the On position. Set the rear window defogger switch in the On position. The rear window defogger operation can be checked by feeling the surfaces of the rear window glass or outside heated rear view mirror glass, when equipped. A distinct difference in temperature between the grid lines and the adjacent clear glass or the mirror glass should be detected within three to four minutes of operation.

(3) If a temperature difference is not detected, use a 12-volt DC voltmeter and contact the rear glass heating grid power feed terminal A with the positive lead, and terminal B with the negative lead (Fig. 3). The voltmeter should read battery voltage. If the voltmeter does not read battery voltage, check the following:

- Confirm that the ignition switch is in the On position.
- Make sure that the EBL feed wire and ground wire are connected to the terminals. Confirm that the ground wire has continuity to ground.
- Check the EBL relay located in the power distribution center (PDC) in the engine compartment. Check the fuses in the fuse block located in the passenger compartment. The relay and fuses must be tight in their receptacles and all electrical connections must be secure. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard relay.

When the above steps have been completed and the rear glass heating grid is still inoperative, one or more of the following is faulty:

HEATED GLASS (Continued)

- Rear window defogger switch in the instrument panel.
- Rear window defogger relay control circuity in the instrument cluster.
- Rear window defogger grid lines (all grid lines would have to be broken, or the power feed or ground wire disconnected, for the entire heated grid to be inoperative).

(4) If the EBL system operation has been verified but the rear window defogger indicator does not illuminate, check the indicator power feed fuse located in the fuse block. If the fuse is Ok, replace the rear window defogger switch (Refer to 8 - ELECTRICAL/ HEATED GLASS/REAR WINDOW DEFOGGER SWITCH - REMOVAL).

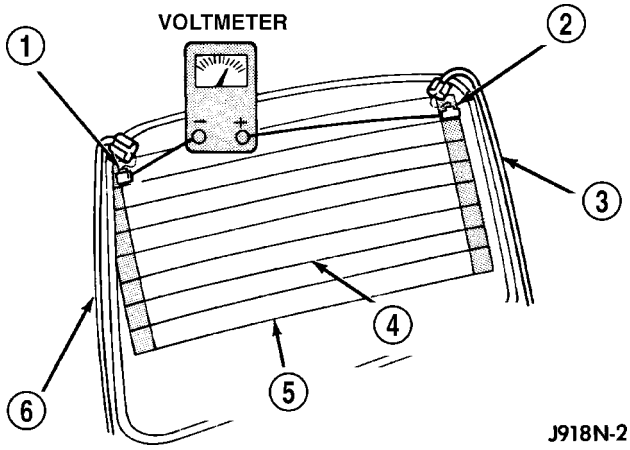

Fig. 3 Rear Window Glass Grid Test - Typical

1 - TERMINAL “B”
2 - TERMINAL “A”
3 - FEED WIRE
4 - MID-POINT “C” (TYPICAL)
5 - HEATED REAR WINDOW DEFOGGER GRID
6 - GROUND WIRE

REAR WINDOW DEFOGGER RELAY

DESCRIPTION

The rear window defogger (EBL) relay (Fig. 4) is a International Standards Organization (ISO)-type relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The EBL relay is a electromechanical device that switches battery current through a fuse in the fuse block to the rear window defogger grid and to the light-emitting diode (LED) indicator of the rear window defogger switch and also to the heating elements of the outside rear view mirrors (when equipped). The EBL relay is energized when the relay coil is provided a ground path by the control circuitry within the instrument cluster.

The EBL relay is located in the power distribution center (PDC) in the engine compartment.

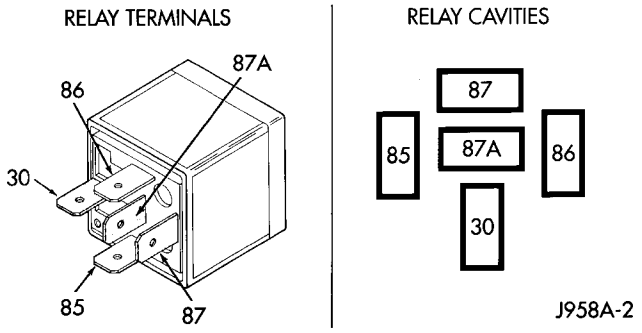

Fig. 4 Rear Window Defogger (EBL) Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

OPERATION

The ISO-standard rear window defogger (EBL) relay is an electromechanical switch that uses a low current input controlled by the body control module (BCM) to control the high current output to the rear window defogger grid lines The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the rear window defogger grid lines.

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The EBL relay cannot be repaired and, if faulty or damaged, it must be replaced. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard relay and for complete rear window defogger (EBL) wiring diagrams.

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Open the cover of the power distribution center (PDC) (Fig. 5) located in the engine compartment.

REAR WINDOW DEFOGGER RELAY (Continued)

NOTE: Refer to the fuse and relay layout map on the inside of the PDC cover for EBL relay location.

(3) Remove the EBL relay from the PDC.

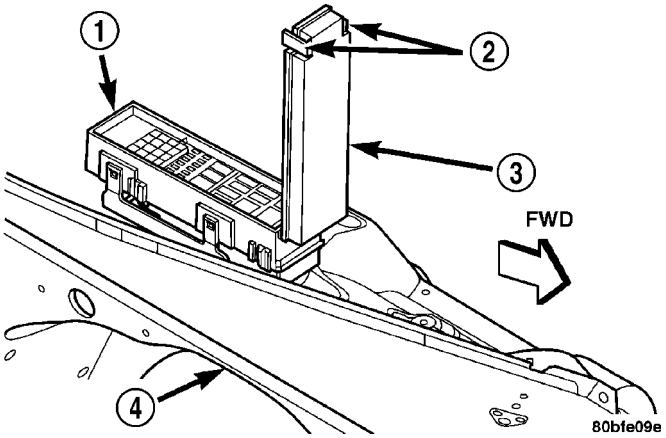

Fig. 5 Power Distribution Center (PDC)

1 - POWER DISTRIBUTION CENTER (PDC)
2 - LATCHES
3 - COVER
4 - RIGHT FRONT FENDER

INSTALLATION

NOTE: Refer to the fuse and relay layout map on the inside of the power distribution center (PDC) cover for EBL relay location.

(1) Position the EBL relay to the proper receptacle of the PDC.

(2) Align the EBL relay terminals with the terminal cavities in the PDC and push down firmly on the relay until the terminals are fully seated.

(3) Close the cover of the PDC.

(4) Reconnect the negative battery cable.

REAR WINDOW DEFOGGER SWITCH

DESCRIPTION

The switch for the rear window defogger (EBL) system is located in the instrument panel accessory switch bezel (Fig. 6), which is mounted near the bottom of the instrument panel center bezel. When the rear window defogger switch is activated, the instrument cluster operates the rear window defogger (EBL) relay.

When the EBL relay is activated, current is directed to the rear defogger grid lines and the heated side view mirrors (if equipped). The grid lines heat the glass to help clear the surface of fog or frost.

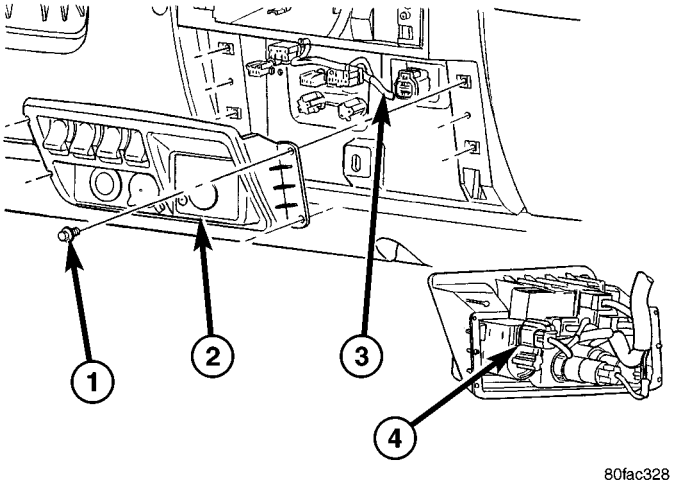

Fig. 6 Accessory Switch Bezel - Typical

1 - BEZEL SCREWS (4)
2 - BEZEL
3 - WIRE HARNESS
4 - ELECTRICAL CONNECTORS (7)

OPERATION

An LED indicator will illuminate when the rear window defogger switch is activated. When the switch is activated, it energizes the timing circuit integral to the instrument cluster, which then activates the rear window defogger (EBL) relay. The EBL relay controls the current flow to the grids of the rear window defogger and to the heated side view mirrors, when equipped.

NOTE: The EBL system turns off automatically after 10 minutes of initial operation. Each following activation cycle of the EBL system will last five minutes.

The EBL system will initially be on for approximately ten minutes or until the rear window defogger switch or the ignition switch is turned off. After the initial time interval has expired, if the defogger switch is turned On again during the same ignition cycle, the EBL system will automatically turn off after about five minutes.

The rear window defogger switch and indicator lamp cannot be repaired and, if faulty or damaged, the defogger switch must be replaced (Refer to 8 - ELECTRICAL/HEATED GLASS/REAR WINDOW DEFOGGER SWITCH - REMOVAL)..

REAR WINDOW DEFOGGER SWITCH (Continued)

DIAGNOSIS AND TESTING

REAR WINDOW DEFOGGER SWITCH

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

NOTE: For circuit descriptions and diagrams, refer to Rear Window Defogger in Group 8W - Wiring Diagrams.

(1) Disconnect and isolate the negative battery cable. Remove the accessory switch bezel from the instrument panel and disconnect the wire harness connector from the defogger switch.

(2) Check for continuity between the ground circuit cavity of the defogger switch wire harness connector and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open circuit as required.

(3) Check for continuity between the ground circuit terminal and the rear window defogger switch sense circuit terminal on the back of the defogger switch housing (Fig. 7). There should be momentary continuity as the defogger switch is pressed, and then no continuity. If OK, test the instrument cluster (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If not OK, replace the faulty rear window defogger switch (Refer to 8 - ELECTRICAL/HEATED GLASS/REAR WINDOW DEFOGGER SWITCH - REMOVAL).

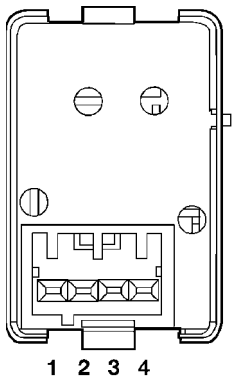

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Fig. 7 Rear Window Defogger Switch Continuity

SWITCH POSITION	CONTINUITY BETWEEN
OFF	LAMPS
ON	MOMENTARY 1 AND 2
ILLUMINATION LAMP	1 AND 4
INDICATOR LAMP	1 AND 3

REMOVAL

WARNING: On vehicles equipped with airbags, disable the airbag system before attempting any steering wheel, steering column, or instrument panel component diagnosis or service. Disconnect and isolate the battery negative (ground) cable, then wait two minutes for the airbag system capacitor to discharge before performing further diagnosis or service. Failure to take the proper precautions could result in accidental airbag deployment and possible personal injury or death.

(1) Disconnect and isolate the negative battery cable.

(2) Remove the center bezel from the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/CLUSTER BEZEL - REMOVAL).

(3) Remove the four screws that secure the accessory switch bezel to the instrument panel (Fig. 8).

(4) Pull the accessory switch bezel out from the instrument panel far enough to access the wire harness connectors.

(5) Disconnect the wire harness connectors from the rear of the accessory switches and the cigar lighter/power outlet.

(6) Remove the accessory switch bezel from the instrument panel.

(7) Using a small thin-bladed screwdriver, gently pry the snap clips at the top and at the bottom of the rear window defogger switch receptacle on the back of the accessory switch bezel and pull the switch out of the bezel.

INSTALLATION

(1) Install the rear window defogger switch into the back of the accessory switch bezel by pushing the switch firmly into the bezel.

(2) Connect the wire harness connectors to the rear of the accessory switches and the cigar lighter/power outlet.

(3) Install the accessory switch bezel to the instrument panel.

REAR WINDOW DEFOGGER SWITCH (Continued)

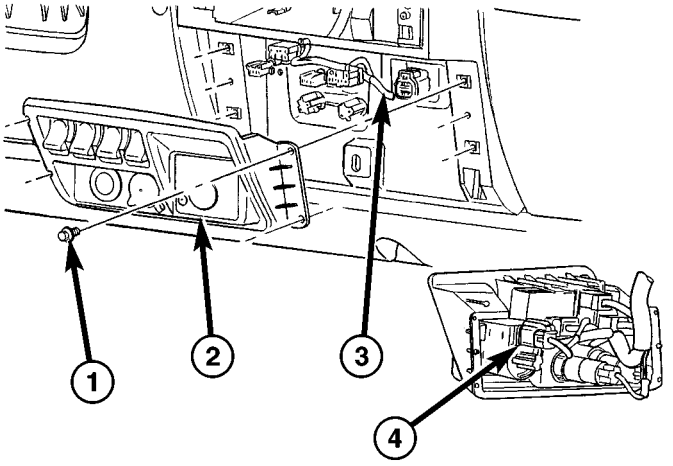

Fig. 8 Accessory Switch Bezel

1 - BEZEL SCREWS (4)
2 - BEZEL
3 - WIRE HARNESS
4 - ELECTRICAL CONNECTORS (7)

(4) Install the four screws that secure the accessory switch bezel to the instrument panel. Tighten the screws to 2 N·m (17 in. lbs.).

(5) Install the center bezel to the instrument panel (Refer to 23 - BODY/INSTRUMENT PANEL/CLUSTER BEZEL - INSTALLATION).

(6) Reconnect the negative battery cable.

REAR WINDOW DEFOGGER GRID

STANDARD PROCEDURE

GRID LINE AND TERMINAL REPAIR

WARNING: Materials contained in the Repair Kit (Part Number 04549275) may cause skin or eye irritation. The kit contains epoxy resin and amine type hardener, which are harmful if swallowed. Avoid contact with the skin and eyes. For skin contact, wash the affected areas with soap and water. For contact with the eyes, flush with plenty of water. Do not take internally. If taken internally, induce vomiting and call a physician immediately. Use with adequate ventilation. Do not use near fire or flame. Contains flammable solvents. Keep out of the reach of children. Failure to follow the warnings could result in possible personal injury or death.

Repair of the rear glass heating grid lines, bus bars or terminals can be accomplished using the Mopar® Rear Window Defogger Repair Kit (Part Number 04549275) or equivalent.

(1) Mask the repair area with masking tape so that the conductive epoxy can be applied neatly (Fig. 9). Extend the epoxy application onto the grid line or the bus bar on each side of the break.

(2) Follow the instructions in the repair kit for preparing the damaged area.

(3) Remove the package separator clamp and mix the two conductive epoxy components thoroughly within the packaging. Fold the package in half and cut the center corner to dispense the epoxy.

(4) For grid line repairs, mask the area to be repaired with masking tape or use a template.

(5) Apply the epoxy through the slit in the masking tape or template. Overlap both ends of the break by at least 19 millimeters (0.75 inch).

(6) For a terminal replacement, mask the adjacent areas so the epoxy can be extended onto the adjacent grid line as well as the bus bar. Apply a thin layer of epoxy to the area where the terminal was fastened and onto the adjacent grid line.

(7) Apply a thin layer of conductive epoxy to the terminal and place it in the proper location on the bus bar. To prevent the terminal from moving while the epoxy is curing, it must be wedged or clamped.

(8) Carefully remove the masking tape or template.

CAUTION: Do not allow the glass surface to exceed 204° C (400° F) or the glass may fracture.

(9) Allow the epoxy to cure 24 hours at room temperature, or carefully use a heat gun for fifteen minutes. When using a heat gun, hold it approximately 25.4 centimeters (10 inches) from the repair and do not allow the glass surface to exceed 204° C (400° F).

NOTE: Do not attach the wire harness connectors to the terminals until the curing process is complete.

(10) After the conductive epoxy is properly cured, remove the wedge or clamp from the terminal.

(11) Connect the wire harness leads to the grid terminals and verify EBL operation.

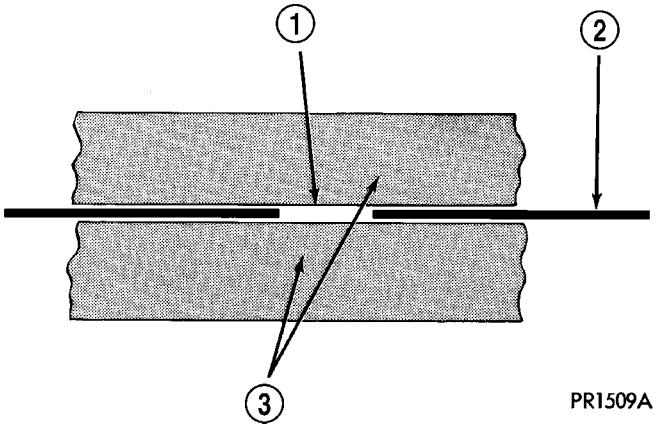

Fig. 9 Grid Line Repair

1 - BREAK
2 - GRID LINE
3 - MASKING TAPE

HEATED MIRRORS

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HEATED MIRRORS

DESCRIPTION

The optional heated mirror system is controlled by the momentary rear window defogger switch located in the accessory switch bezel (Fig. 1). An amber indicator lamp in the switch will illuminate to indicate when the rear window defogger (EBL) system is turned on.

The heated mirror system only operates in concert with the EBL system, and will be automatically shut off after a programmed time interval of about ten minutes. After the initial time interval has expired, if the defogger switch is pressed again during the same ignition cycle, the heated mirror system will automatically shut off after about five minutes.

The heated mirror system will automatically shut off if the ignition switch is turned to the Off position, or it can be shut off manually by pressing the rear window defogger switch a second time.

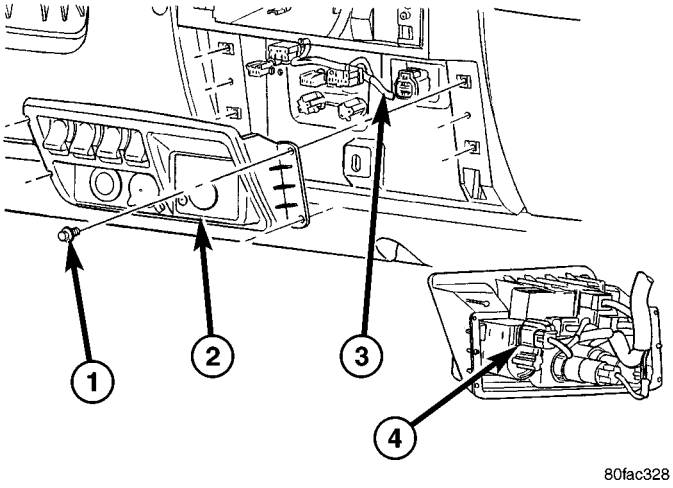

Fig. 1 Accessory Switch Bezel - Typical

1 - BEZEL SCREWS (4)
2 - BEZEL
3 - WIRE HARNESS
4 - ELECTRICAL CONNECTORS (7)

OPERATION

When the rear window defogger switch is pressed, the rear window defogger (EBL) system becomes activated and an electric heater grid located behind the glass of each of the outside rear view mirrors is energized. When energized, each of these heater grids produce heat to help clear the outside rear view mirrors of ice, snow, or fog.

If the outside mirror heating grids are both inoperative, refer to DIAGNOSIS AND TESTING - REAR WINDOW DEFOGGER SYSTEM in his group. If only one of the outside mirror heating grids is inoperative, refer to 8 - ELECTRICAL/POWER MIRRORS - DIAGNOSIS AND TESTING.

The heating grid behind each outside mirror glass cannot be repaired and, if faulty or damaged, the entire power mirror assembly must be replaced.

HORN

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HORN SYSTEM

DESCRIPTION

The horn system features one low-note horn unit. The horn system uses a non-switched source of battery current so that the system will remain functional, regardless of the ignition switch position. The horn system includes the following components:

- Clockspring
- Horn
- Horn relay
- Horn switch

OPERATION

The horn system is activated by a horn switch concealed beneath the driver side airbag module trim cover in the center of the steering wheel. Depressing the center of the driver side airbag module trim cover closes the horn switch. Closing the horn switch activates the horn relay. The activated horn relay then switches the battery current needed to energize the horns.

HORN

DESCRIPTION

The horn is secured with a bracket to the left front inner fender shield just ahead of the left front wheel house in the engine compartment.

The horn cannot be repaired or adjusted and, if faulty or damaged, it must be replaced.

OPERATION

Within the two halves of the molded plastic horn housing are a flexible diaphragm, a plunger, an electromagnetic coil and a set of contact points. The diaphragm is secured in suspension around its perimeter by the mating surfaces of the horn housing. The plunger is secured to the center of the diaphragm and extends into the center of the electromagnet. The contact points control the current flow through the electromagnet.

When the horn is energized, electrical current flows through the closed contact points to the electromagnet. The resulting electromagnetic field draws the plunger and diaphragm toward it until that movement mechanically opens the contact points. When the contact points open, the electromagnetic field collapses allowing the plunger and diaphragm to return to their relaxed positions and closing the contact points again. This cycle continues repeating at a very rapid rate producing the vibration and movement of air that creates the sound that is directed through the horn outlet.

DIAGNOSIS AND TESTING - HORN

For complete circuit diagrams, refer to the appropriate wiring information.

(1) Disconnect the wire harness connectors from the horn connector receptacles. Measure the resistance between the ground circuit cavity of the horns wire harness connectors and a good ground. There should be continuity. If OK, go to Step 2. If not OK, repair the open ground circuit to ground as required.

(2) Check for battery voltage at the horn relay output circuit cavity of the horns wire harness connectors. There should be zero volts. If OK, go to Step 3. If not OK, repair the shorted horn relay output circuit or replace the faulty horn relay as required.

(3) Depress the horn switch. There should now be battery voltage at the horn relay output circuit cavity of the horns wire harness connectors. If OK, replace the faulty horns. If not OK, repair the open horn relay output circuit to the horn relay as required.

HORN (Continued)

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the wire harness connector from the horn (Fig. 1).

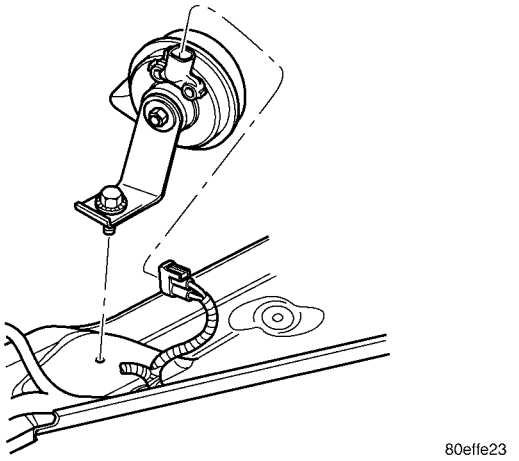

Fig. 1 HORN

(3) Remove the screw that secures the horn and mounting bracket to the left front inner fender shield.

(4) Remove the horn and mounting bracket from the left front inner fender shield.

INSTALLATION

(1) Position the horn and mounting bracket onto the left front inner fender shield.

(2) Install the mounting screw. Tighten the screw to 8.5 N·m (75 in. lbs.).

(3) Reconnect the wire harness connector.

(4) Connect the battery negative cable.

HORN SWITCH

DESCRIPTION

A center-blow, normally open, resistive membrane-type horn switch is secured in a plastic tray that is inserted in a pocket sewn on the front of the driver side airbag retainer strap. The horn switch is concealed behind the driver side airbag module trim cover.

The steering wheel and steering column must be properly grounded in order for the horn switch to function properly. The horn switch and plastic tray are serviced as a unit. If the horn switch is damaged or faulty, or if the driver side airbag is deployed, the horn switch and tray must be replaced as a unit.

OPERATION

When the center area of the driver side airbag trim cover is depressed, the electrically conductive grids on the facing surfaces of the horn switch membranes contact each other, closing the switch circuit. The completed horn switch circuit provides a ground for the control coil side of the horn relay, which activates the relay. When the horn switch is released, the resistive tension of the convex membrane separates the two electrically conductive grids and opens the switch circuit.

DIAGNOSIS AND TESTING - HORN SWITCH

For complete circuit diagrams, refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, REFER TO ELECTRICAL, RESTRAINTS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable. Remove the steering column opening cover from the instrument panel.

(2) Check for continuity between the metal steering column jacket and a good ground. There should be continuity. If OK, go to Step 3. If not OK, refer to Steering, Column for proper installation of the steering column.

(3) Remove the driver side airbag module from the steering wheel. Disconnect the horn switch wire harness connectors from the driver side airbag module.

(4) Remove the horn relay from the Power Distribution Center (PDC). Check for continuity between the steering column half of the horn switch feed wire harness connector and a good ground. There should be no continuity. If OK, go to Step 5. If not OK, repair the shorted horn relay control circuit to the horn relay in the PDC as required.

(5) Check for continuity between the steering column half of the horn switch feed wire harness connector and the horn relay control circuit cavity for the horn relay in the PDC. There should be continuity. If OK, go to Step 6. If not OK, repair the open horn relay control circuit to the horn relay in the PDC as required.

(6) Check for continuity between the horn switch feed wire and the horn switch ground wire on the driver side airbag module. There should be no continuity. If OK, go to Step 7. If not OK, replace the faulty horn switch.

HORN SWITCH (Continued)

(7) Depress the center of the driver side airbag module trim cover and check for continuity between the horn switch feed wire and the horn switch ground wire on the driver side airbag module. There should now be continuity. If not OK, replace the faulty horn switch.

REMOVAL

WARNING:

• ON VEHICLES EQUIPPED WITH AIRBAGS, REFER TO ELECTRICAL, RESTRAINTS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

• THE HORN SWITCH IS INTEGRAL TO THE DRIVER SIDE AIRBAG MODULE. SERVICE OF THIS COMPONENT SHOULD BE PERFORMED ONLY BY CHRYSLER-TRAINED AND AUTHORIZED DEALER SERVICE TECHNICIANS. FAILURE TO TAKE THE PROPER PRECAUTIONS OR TO FOLLOW THE PROPER PROCEDURES COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER AIRBAG DEPLOYMENT AND POSSIBLE OCCUPANT INJURIES.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the trim cover from the driver side airbag. (Refer to 8 - ELECTRICAL/RESTRAINTS/ DRIVER AIRBAG - REMOVAL).

(3) Remove the horn switch and tray as a unit from the pouch on the retaining strap of the driver side airbag module (Fig. 2).

INSTALLATION

WARNING:

• ON VEHICLES EQUIPPED WITH AIRBAGS, REFER TO ELECTRICAL, RESTRAINTS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

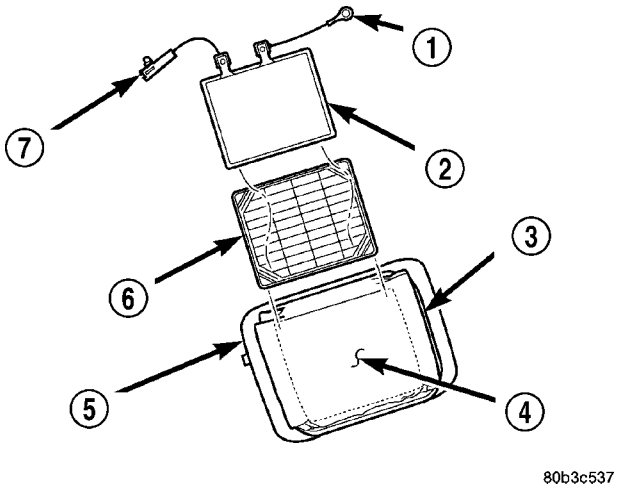

Fig. 2 HORN SWITCH

1 - HORN SWITCH GROUND WIRE EYELET
2 - HORN SWITCH
3 - AIRBAG RETAINING STRAP
4 - POUCH
5 - DRIVER SIDE AIRBAG MODULE (TRIM COVER REMOVED)
6 - TRAY
7 - HORN SWITCH FEED WIRE CONNECTOR

• THE HORN SWITCH IS INTEGRAL TO THE DRIVER SIDE AIRBAG MODULE. SERVICE OF THIS COMPONENT SHOULD BE PERFORMED ONLY BY CHRYSLER-TRAINED AND AUTHORIZED DEALER SERVICE TECHNICIANS. FAILURE TO TAKE THE PROPER PRECAUTIONS OR TO FOLLOW THE PROPER PROCEDURES COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER AIRBAG DEPLOYMENT AND POSSIBLE OCCUPANT INJURIES.

(1) Install the horn switch and tray as a unit into the pouch on the retaining strap of the driver side airbag module. Be certain that the tray is facing the airbag module, that the horn switch is facing the trim cover, that the horn switch feed wire is on the left, and that the horn switch ground wire is on the right.

(2) Install the trim cover onto the driver side airbag. (Refer to 8 - ELECTRICAL/RESTRAINTS/ DRIVER AIRBAG - INSTALLATION) for the procedure.

(3) Reconnect the battery negative cable.

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IGNITION CONTROL

DESCRIPTION

The ignition systems used on both the 2.4L 4–cylinder and 4.0L 6–cylinder engines are a distributorless type.

The ignition system consists of:

- Spark Plugs
- Spark Plug Cables (secondary wires)
- Ignition Coil(s)
- Powertrain Control Module (PCM)
- Crankshaft Position Sensor
- Camshaft Position Sensor
- The MAP, TPS, IAC and ECT also have an effect on the control of the ignition system.

OPERATION

2.4L

A common ignition coil divided into 2 halves is used. Secondary, high-tension spark plug cables are also used. One half of the coil fires two spark plugs simultaneously (one plug is the cylinder under compression, and the other plug is the cylinder on the exhaust stroke). Coil half number one fires cylinders 1 and 4. Coil half number two fires cylinders 2 and 3. The PCM determines which of the coils to charge and fire at the correct time.

IGNITION CONTROL (Continued)

The Auto Shutdown (ASD) relay provides battery voltage to the ignition coil. The PCM provides a ground contact (circuit) for energizing the coil. When the PCM breaks the contact, the energy in the coil primary transfers to the secondary causing a spark. The PCM will de-energize the ASD relay if it does not receive inputs from either the crankshaft or camshaft position sensors.

A distributor is not used with the 2.4L engine.

4.0L

The 4.0L 6-cylinder engine uses a one-piece coil rail containing three independent coils. Although cylinder firing order is the same as 4.0L engines of previous years, spark plug firing is not. The 3 coils dual-fire the spark plugs on cylinders 1-6, 2-5 and/or 3-4. When one cylinder is being fired (on compression stroke), the spark to the opposite cylinder is being wasted (on exhaust stroke). The one-piece coil bolts directly to the cylinder head. Rubber boots seal the secondary terminal ends of the coils to the top of all 6 spark plugs. One electrical connector (located at the rear end of the coil rail) is used for all three coils.

Because of coil design, spark plug cables (secondary cables) are not used. A **distributor is not used** with the 4.0L engine.

The ignition system is controlled by the Powertrain Control Module (PCM).

SPECIFICATIONS

SPECIFICATIONS - IGNITION TIMING

Ignition timing is not adjustable on any engine.

ENGINE FIRING ORDER - 2.4L 4-CYLINDER

1 - 3 - 4 - 2. Refer to (Fig. 1).

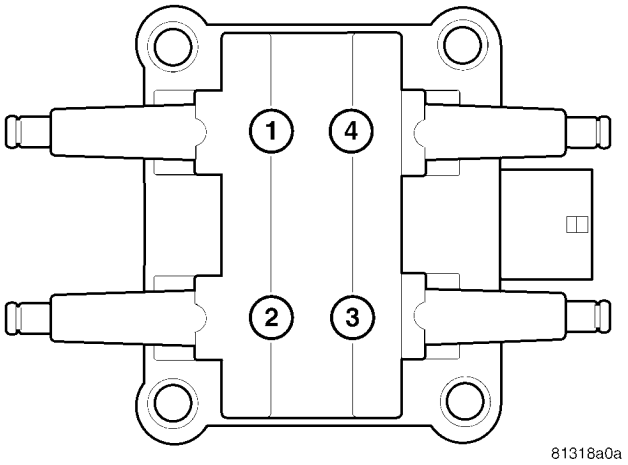
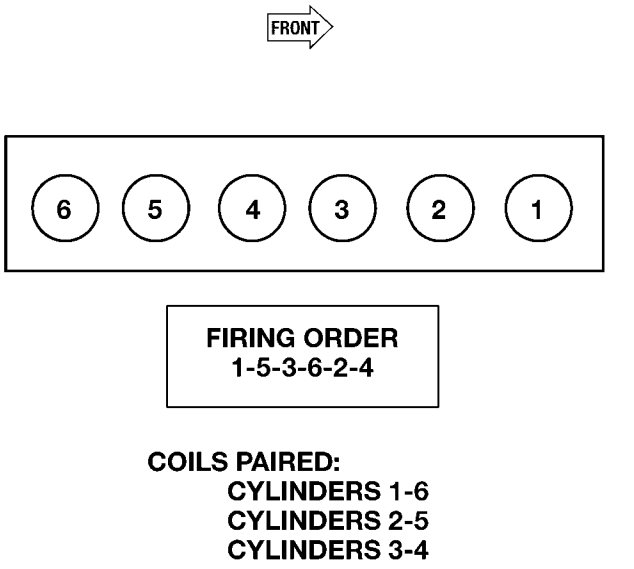


Fig. 1 FIRING ORDER - 2.4L

ENGINE FIRING ORDER - 4.0L 6-CYLINDER ENGINE



FIRING ORDER - 4.0L

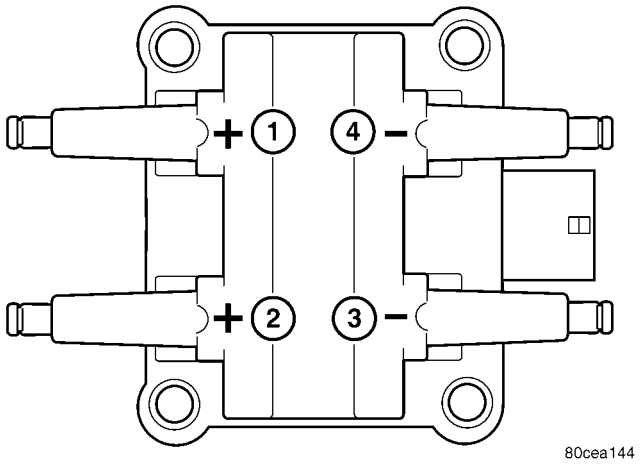
IGNITION CONTROL (Continued)

IGNITION COIL RESISTANCE - 2.4L

Engine	Coil Manufacture	Primary Resistance at 21°C-27°C (70°F-80°F)	Secondary Resistance at 21°C-27°C (70°F-80°F)
2.4L	Toyodenso or Diamond	0.51 to 0.61 Ohms	11,500 to 13,500 Ohms

IGNITION COIL RESISTANCE - 4.0L ENGINE

PRIMARY RESISTANCE 21-27°C (70-80°F)
0.71 - 0.88 Ohms



IGNITION COIL - 2.4L

SPARK PLUGS

ENGINE	PLUG TYPE	ELECTRODE GAP
2.4L 4-Cylinder	RE16MC (Champion #)	0.94 to 1.09 mm (0.037 to 0.042 in.)
4.0L 6-Cylinder	ZFR5N (NGK #)	0.77 mm - 0.90 mm (0.030 in. - 0.035 in.)

SPECIFICATIONS - SPARK PLUG CABLE RESISTANCE

MINIMUM	MAXIMUM
250 Ohms Per Inch	1000 Ohms Per Inch
3000 Ohms Per Foot	10,000 Ohms Per Foot

IGNITION CONTROL (Continued)

SPECIFICATIONS - TORQUE - IGNITION SYSTEM

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Camshaft Position Sensor Mounting Bolt – 4.0L	2	-	15
Camshaft Position Sensor – 2.4L	12	-	106
Ignition Coil Rail Mounting Bolts – 4.0L	29	-	250
Ignition Coil Mounting Bolts – 2.4L	11	-	105
Oil Pump Drive Hold-down Bolt – 4.0L	23	17	-
Oil Pump Drive Housing Cover Screws	-	-	-
** Spark Plugs - 2.4L 4-Cylinder	15	11	-
Spark Plugs - 4.0L 6-cylinder	30	22	-
** Torque critical tapered design. Do not exceed 15 ft. lbs. torque.			

AUTO SHUT DOWN RELAY

DESCRIPTION - PCM OUTPUT

The 5–pin, 12–volt, Automatic Shutdown (ASD) relay is located in the Power Distribution Center (PDC). Refer to label on PDC cover for relay location.

OPERATION

OPERATION - ASD SENSE - PCM INPUT

A 12 volt signal at this input indicates to the PCM that the ASD has been activated. The relay is used to connect the oxygen sensor heater element, ignition coil and fuel injectors to 12 volt + power supply.

This input is used only to sense that the ASD relay is energized. If the Powertrain Control Module (PCM) does not see 12 volts at this input when the ASD should be activated, it will set a Diagnostic Trouble Code (DTC).

OPERATION - PCM OUTPUT

The ASD relay supplies battery voltage (12+ volts) to the fuel injectors and ignition coil(s). With certain emissions packages it also supplies 12–volts to the oxygen sensor heating elements.

The ground circuit for the coil within the ASD relay is controlled by the Powertrain Control Module (PCM). The PCM operates the ASD relay by switching its ground circuit on and off.

The ASD relay will be shut–down, meaning the 12–volt power supply to the ASD relay will be de-activated by the PCM if:

- the ignition key is left in the ON position. This is if the engine has not been running for approximately 1.8 seconds.
- there is a crankshaft position sensor signal to the PCM that is lower than pre-determined values.

DIAGNOSIS AND TESTING - ASD AND FUEL PUMP RELAYS

The following description of operation and tests apply only to the Automatic Shutdown (ASD) and fuel pump relays. The terminals on the bottom of each relay are numbered. Two different types of relays may be used, (Fig. 2) or (Fig. 3).

- Terminal number 30 is connected to battery voltage. For both the ASD and fuel pump relays, terminal 30 is connected to battery voltage at all times.
- The PCM grounds the coil side of the relay through terminal number 85.
- Terminal number 86 supplies voltage to the coil side of the relay.
- When the PCM de-energizes the ASD and fuel pump relays, terminal number 87A connects to terminal 30. This is the Off position. In the off position,

AUTO SHUT DOWN RELAY (Continued)

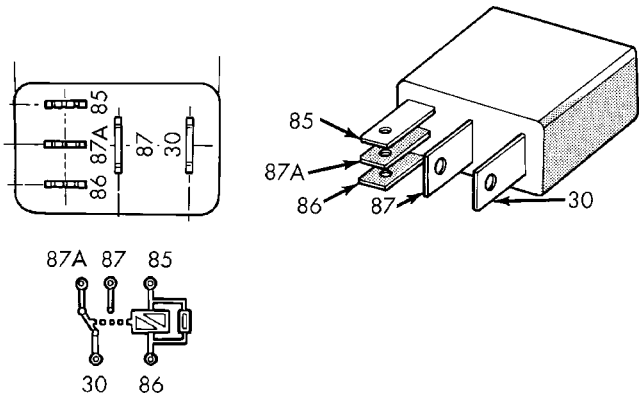


Fig. 2 ASD and Fuel Pump Relay Terminals—Type 1

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

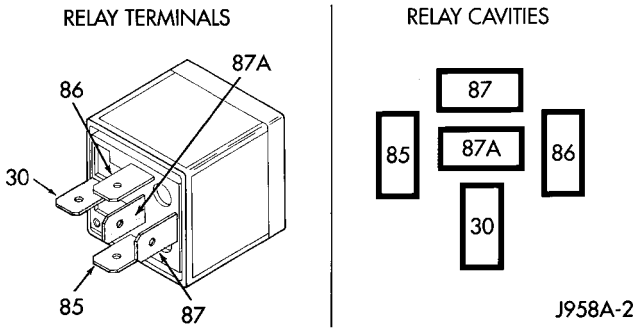


Fig. 3 ASD and Fuel Pump Relay Terminals—Type 2

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

voltage is not supplied to the rest of the circuit. Terminal 87A is the center terminal on the relay.

- When the PCM energizes the ASD and fuel pump relays, terminal 87 connects to terminal 30. This is the On position. Terminal 87 supplies voltage to the rest of the circuit.

The following procedure applies to the ASD and fuel pump relays.

(1) Remove relay from connector before testing.

(2) With the relay removed from the vehicle, use an ohmmeter to check the resistance between terminals 85 and 86. The resistance should be 75 ohms +/- 5 ohms.

(3) Connect the ohmmeter between terminals 30 and 87A. The ohmmeter should show continuity between terminals 30 and 87A.

(4) Connect the ohmmeter between terminals 87 and 30. The ohmmeter should not show continuity at this time.

(5) Connect one end of a jumper wire (16 gauge or smaller) to relay terminal 85. Connect the other end of the jumper wire to the ground side of a 12 volt power source.

(6) Connect one end of another jumper wire (16 gauge or smaller) to the power side of the 12 volt power source. **Do not attach the other end of the jumper wire to the relay at this time.**

WARNING: DO NOT ALLOW OHMMETER TO CONTACT TERMINALS 85 OR 86 DURING THIS TEST. DAMAGE TO OHMMETER MAY RESULT.

(7) Attach the other end of the jumper wire to relay terminal 86. This activates the relay. The ohmmeter should now show continuity between relay terminals 87 and 30. The ohmmeter should not show continuity between relay terminals 87A and 30.

(8) Disconnect jumper wires.

(9) Replace the relay if it did not pass the continuity and resistance tests. If the relay passed the tests, it operates properly. Check the remainder of the ASD and fuel pump relay circuits. Refer to 8, Wiring Diagrams.

REMOVAL

The ASD relay is located in the Power Distribution Center (PDC) (Fig. 4). Refer to label on PDC cover for relay location.

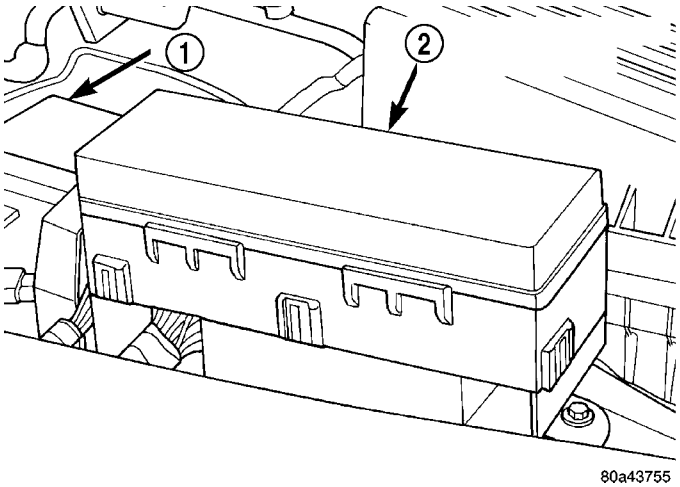


Fig. 4 Power Distribution Center (PDC)

1 - BATTERY
2 - POWER DISTRIBUTION CENTER (PDC)

(1) Remove PDC cover.

(2) Remove relay from PDC.

AUTO SHUT DOWN RELAY (Continued)

(3) Check condition of relay terminals and PDC connector terminals for damage or corrosion. Repair if necessary before installing relay.

(4) Check for pin height (pin height should be the same for all terminals within the PDC connector). Repair if necessary before installing relay.

INSTALLATION

The ASD relay is located in the Power Distribution Center (PDC) (Fig. 4). Refer to label on PDC cover for relay location.

(1) Install relay to PDC.

(2) Install cover to PDC.

CAMSHAFT POSITION SENSOR - 2.4L

DESCRIPTION - 2.4L

The Camshaft Position Sensor (CMP) on the 2.4L 4–cylinder engine is bolted to the right-front side of the cylinder head (Fig. 5).

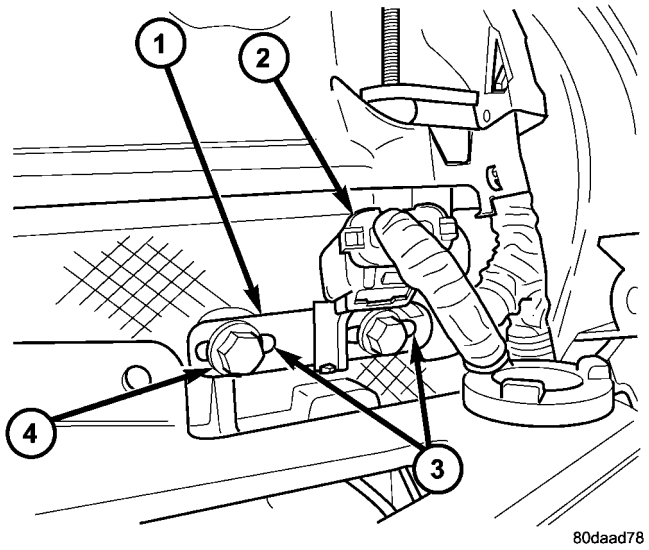


Fig. 5 CMP LOCATION - 2.4L

1 - CMP SENSOR
2 - ELECTRICAL CONNECTOR
3 - SLOTTED HOLES
4 - MOUNTING BOLTS (2)

OPERATION - 2.4L

The Camshaft Position Sensor (CMP) sensor contains a hall effect device referred to as a sync signal generator. A rotating target wheel (tonewheel) for the CMP is located behind the exhaust valve-camshaft drive gear (Fig. 6). The target wheel is equipped with a cutout (notch) around 180 degrees of the wheel. The CMP detects this cutout every 180 degrees of camshaft gear rotation. Its signal is used in conjunction with the Crankshaft Position Sensor (CKP) to differentiate between fuel injection and spark events. It is also used to synchronize the fuel injectors with their respective cylinders.

When the leading edge of the target wheel cutout enters the tip of the CMP, the interruption of magnetic field causes the voltage to switch high, resulting in a sync signal of approximately 5 volts.

When the trailing edge of the target wheel cutout leaves the tip of the CMP, the change of the magnetic field causes the sync signal voltage to switch low to 0 volts.

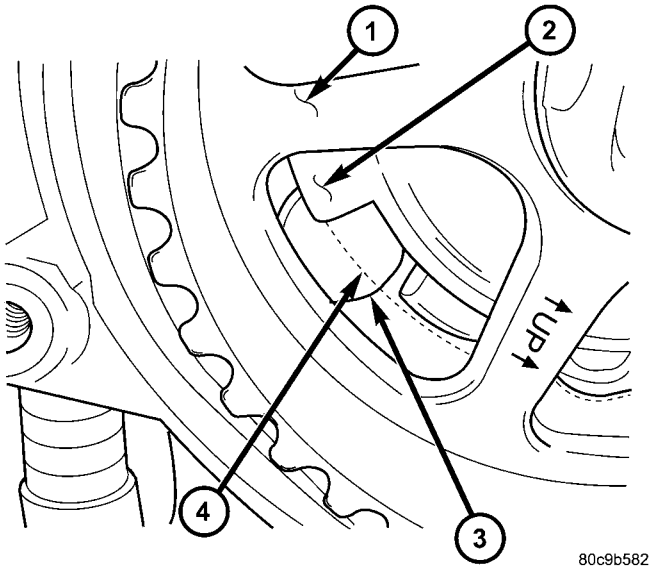


Fig. 6 CMP FACE AT TARGET WHEEL-2.4L

1 - CAMSHAFT DRIVE GEAR
2 - TARGETWHEEL (TONEWHEEL)
3 - FACE OF CMP SENSOR
4 - CUTOUT (NOTCH)

REMOVAL - 2.4L

The Camshaft Position Sensor (CMP) on the 2.4L 4–cylinder engine is bolted to the right-front side of the cylinder head (Fig. 7). Sensor position (depth) is adjustable.

(1) Disconnect electrical connector at CMP sensor.

(2) Remove 2 sensor mounting bolts.

(3) Remove sensor from cylinder head by sliding towards rear of engine.

INSTALLATION - 2.4L

The Camshaft Position Sensor (CMP) on the 2.4L 4–cylinder engine is bolted to the right-front side of the cylinder head. **Sensor position (depth) is adjustable.**

(1) Remove plastic, upper timing belt cover (timing gear cover) (Fig. 8) by removing 3 bolts. Before attempting to remove cover, remove electrical connec-

CAMSHAFT POSITION SENSOR - 2.4L (Continued)

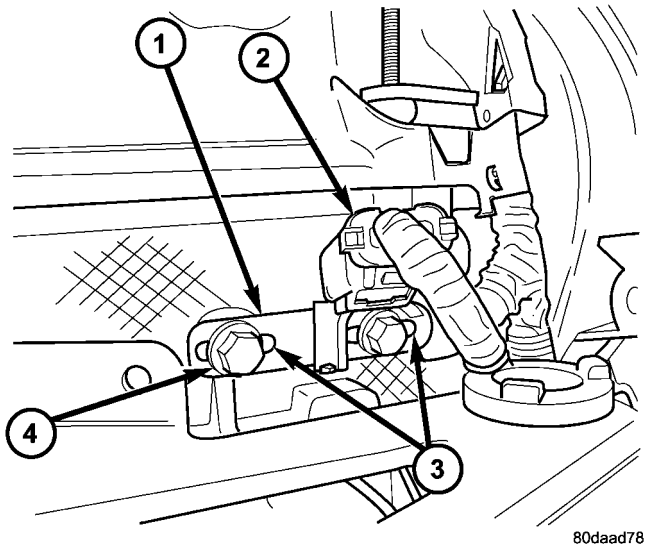


Fig. 7 CMP LOCATION - 2.4L

1 - CMP SENSOR
2 - ELECTRICAL CONNECTOR
3 - SLOTTED HOLES
4 - MOUNTING BOLTS (2)

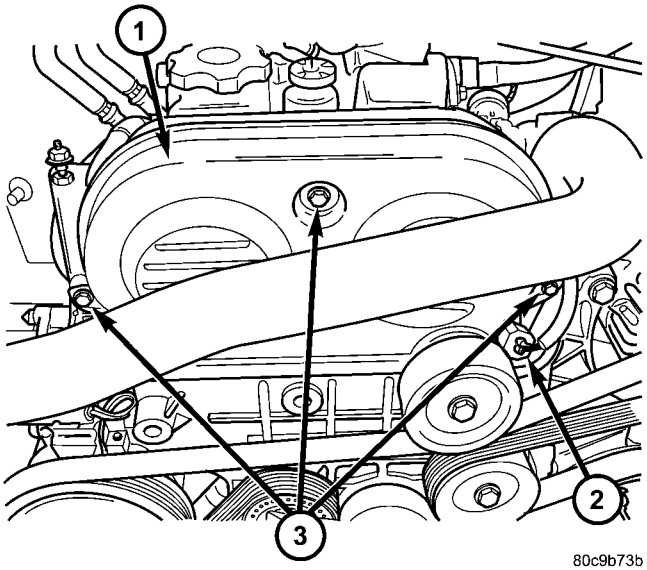


Fig. 8 UPPER TIMING BELT COVER/BOLTS-2.4L

1 - UPPER TIMING BELT COVER
2 - ELECTRICAL CONNECTOR (ECT)
3 - MOUNTING BOLTS (3)

tor from Engine Coolant Temperature (ECT) sensor (Fig. 8). This will prevent damage to sensor.

(2) Rotate (bump over) engine until camshaft timing gear and target wheel (tonewheel) are positioned and aligned to face of sensor as shown in (Fig. 9). **If not positioned as shown in (Fig. 9), damage to both sensor and target wheel will occur when attempting to start engine. Face of sensor MUST be behind target wheel while adjusting.**

(3) Position sensor to cylinder head and install 2 sensor mounting bolts finger tight.

(4) **SENSOR AIR GAP: .030"** Set air gap between rear of target wheel and face of sensor to .030". This can best be accomplished using an L-shaped, wire-type spark plug gapping gauge (Fig. 10). A piece of .030" brass shim stock may also be used.

(5) Gently push sensor forward until it contacts gapping gauge. **Do not push hard on sensor.** Tighten 2 sensor mounting bolts. Refer to torque specifications.

CAUTION: After tightening sensor mounting bolts, recheck air gap and adjust as necessary. Retorque bolts.

(6) Install upper timing belt cover and 3 bolts.
(7) Connect electrical connector to ECT sensor.
(8) Connect electrical connector to CMP sensor.

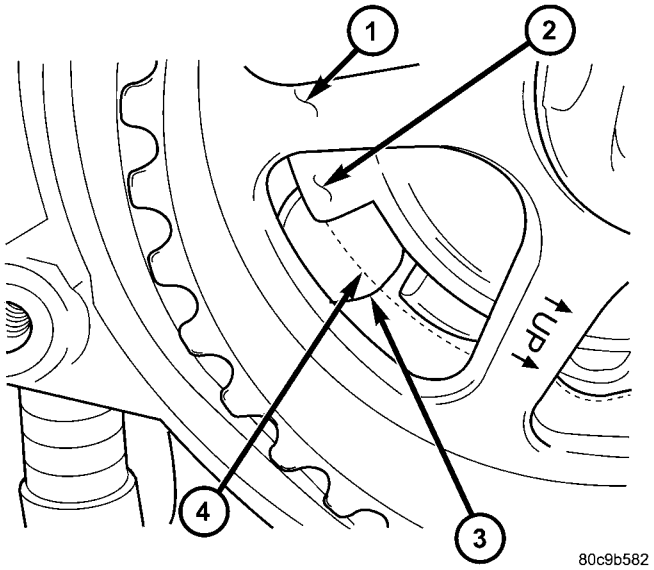


Fig. 9 CMP FACE AT TARGET WHEEL-2.4L

1 - CAMSHAFT DRIVE GEAR
2 - TARGETWHEEL (TONEWHEEL)
3 - FACE OF CMP SENSOR
4 - CUTOUT (NOTCH)

CAMSHAFT POSITION SENSOR - 2.4L (Continued)

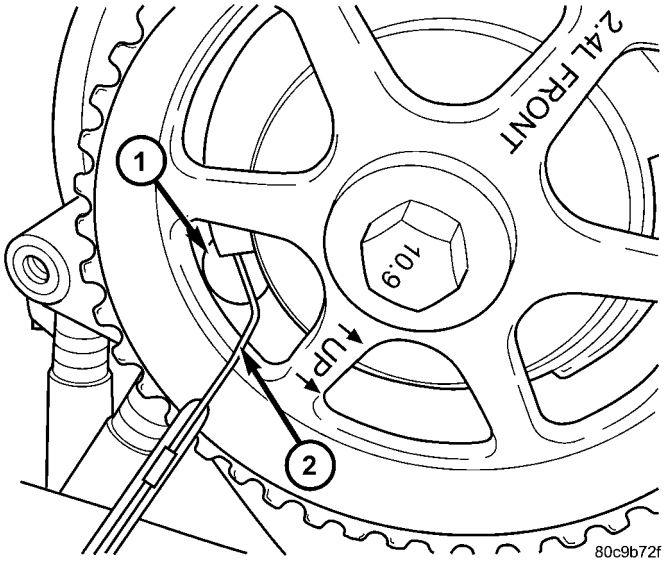


Fig. 10 CMP ADJUSTMENT - 2.4L

1 - FACE OF SENSOR
2 - WIRE GAPPING TOOL

CAMSHAFT POSITION SENSOR - 4.0L

DESCRIPTION - 4.0L

The CMP (Camshaft Position Sensor) sensor (2) (Fig. 11) is bolted to the side of the oil pump drive assembly (5). The assembly is located on the right side of the engine near the oil filter (Fig. 11).

OPERATION - 4.0L

The CMP sensor (1) (Fig. 12) uses a rotating hall effect device called a target wheel (3) which is attached to the oil pump drive shaft. The target wheel contains sets of machined notches (2).

When the leading edge of a target wheel notch passes by the tip of the CMP sensor, the interruption of magnetic field causes the voltage to switch high resulting in a sync signal of approximately 5 volts.

When the trailing edge of the target wheel notch passes by the tip of the CMP sensor, the change of the magnetic field causes the sync signal voltage to switch low to 0 volts.

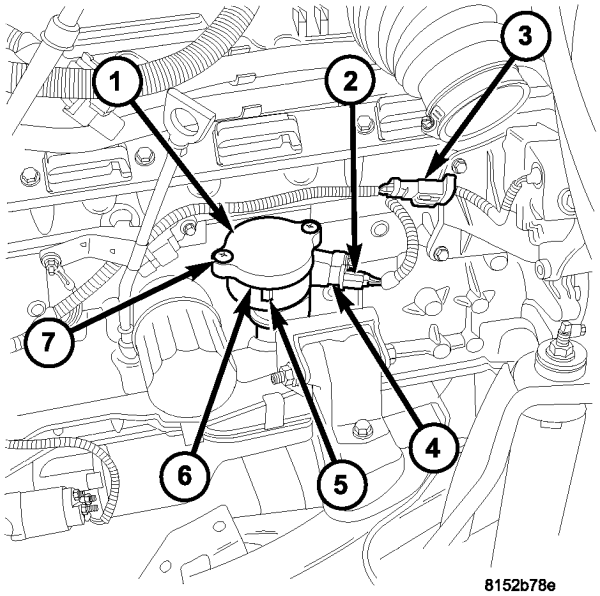


Fig. 11 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

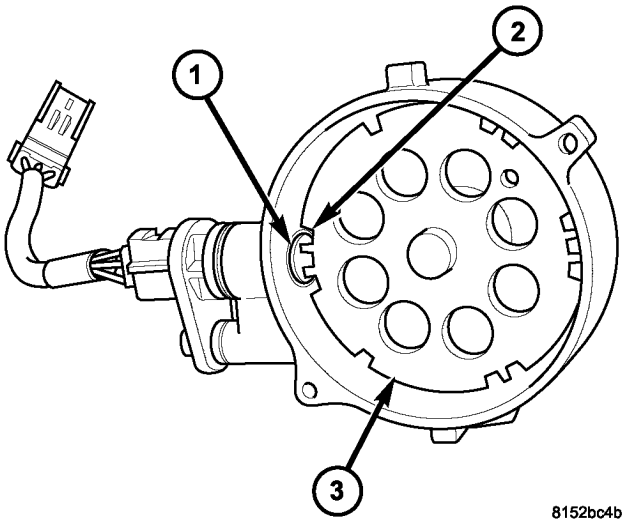


Fig. 12 4.0L CMP OPERATION

1 - CMP SENSOR
2 - TARGET WHEEL NOTCHES
3 - TARGET WHEEL

CAMSHAFT POSITION SENSOR - 4.0L (Continued)

REMOVAL - 4.0L

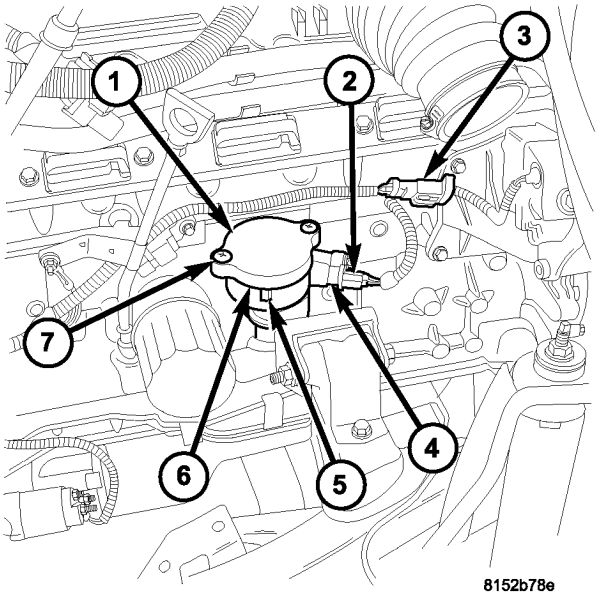


Fig. 13 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

The Camshaft Position Sensor (CMP) on the 4.0L 6–cylinder engine (2) is bolted to the side of the oil pump drive shaft housing assembly (5) (Fig. 13).

NOTE: Do not attempt to rotate the oil pump drive assembly to modify ignition timing.

Two different procedures are used for removal and installation. The first procedure will detail removal and installation of the sensor only. The second procedure will detail removal and installation of the sensor and oil pump drive shaft assembly. The second procedure is to be used if the engine has been disassembled.

CMP SENSOR ONLY - 4.0L

(1) Disconnect CMP electrical jumper harness (3) at engine wiring harness (Fig. 14).

(2) Remove sensor mounting bolt (3) (Fig. 15).

(3) Remove sensor (2) from oil pump drive housing (1) (Fig. 15).

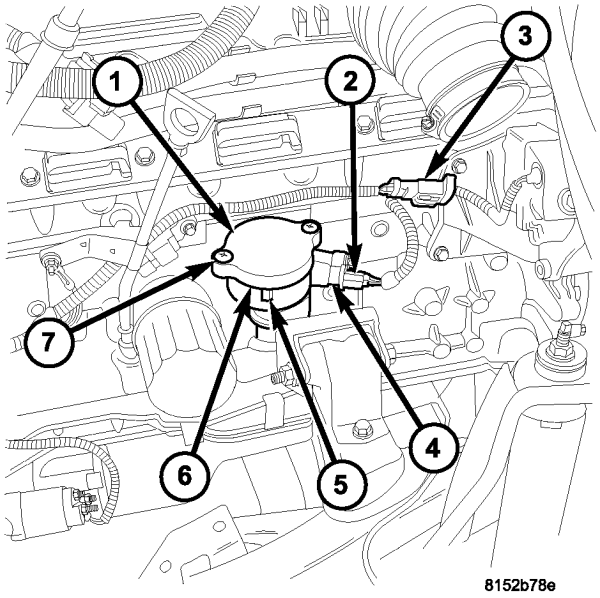


Fig. 14 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

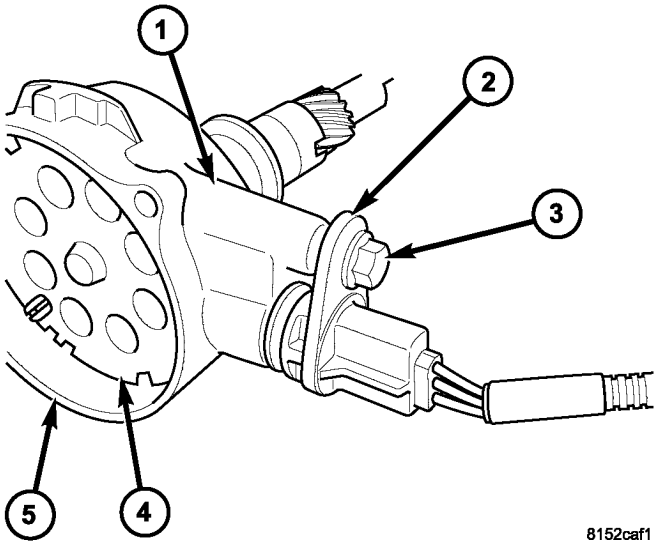


Fig. 15 4.0L CMP REMOVAL/INSTALLATION

1 - OIL PUMP DRIVE HOUSING
2 - CMP SENSOR
3 - SENSOR MOUNTING BOLT
4 - TARGET WHEEL
5 - HOUSING

CAMSHAFT POSITION SENSOR - 4.0L (Continued)

OIL PUMP DRIVE AND CMP SENSOR - 4.0L

If the CMP and oil pump drive are to be removed and installed, do not allow engine crankshaft or camshaft to rotate. CMP sensor relationship will be lost.

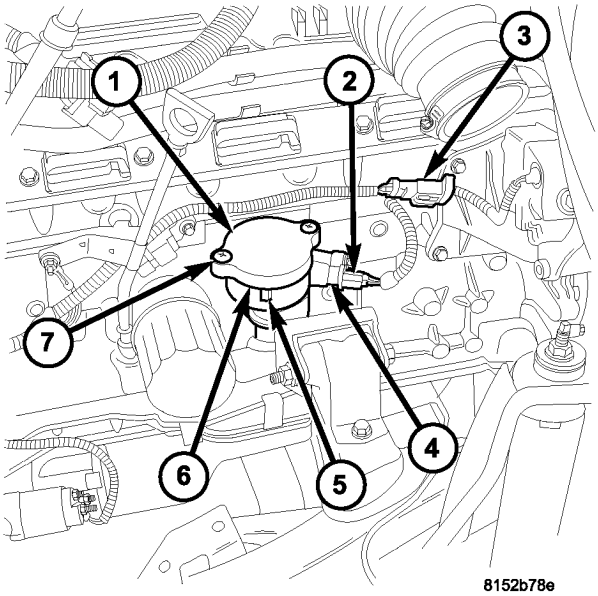


Fig. 16 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

(1) Remove two plastic cover screws (7) (Fig. 16) and remove plastic cover (6).

(2) Note and mark rotational position of target wheel (4) (Fig. 17) in relationship to housing (5).

(3) Disconnect CMP electrical jumper harness (3) at engine wiring harness (Fig. 16).

(4) Before proceeding to next step, mark and note rotational position of oil pump drive housing (5) (Fig. 16) in relationship to engine block. After installation, the CMP sensor should face the 4 o'clock position as viewed from right side of engine.

(5) Remove hold-down bolt (4) and clamp (Fig. 16).

(6) Pull assembly from engine.

(7) Remove and discard old oil pump drive-to-engine block gasket.

INSTALLATION - 4.0L

SENSOR ONLY - 4.0L

(1) Install sensor (2) (Fig. 18) to oil pump drive housing (1).

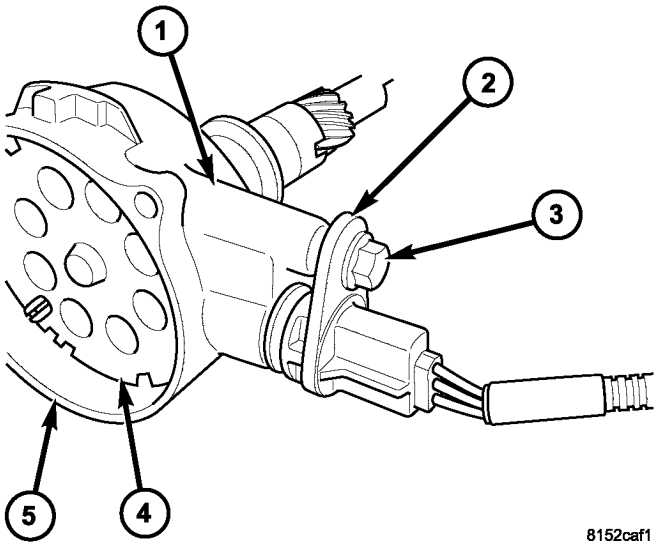


Fig. 17 4.0L CMP REMOVAL/INSTALLATION

1 - OIL PUMP DRIVE HOUSING
2 - CMP SENSOR
3 - SENSOR MOUNTING BOLT
4 - TARGET WHEEL
5 - HOUSING

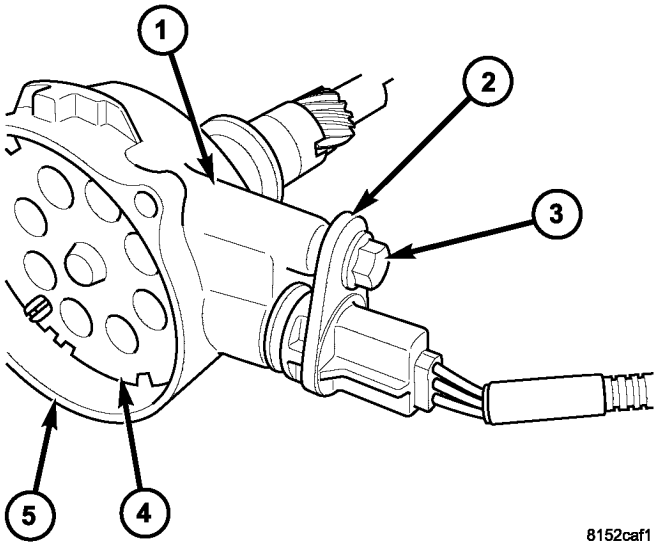


Fig. 18 4.0L CMP REMOVAL/INSTALLATION

1 - OIL PUMP DRIVE HOUSING
2 - CMP SENSOR
3 - SENSOR MOUNTING BOLT
4 - TARGET WHEEL
5 - HOUSING

(2) Install sensor mounting bolt (3) (Fig. 18) and tighten to 2 N·m (15 in. lbs.) torque.

(3) Connect CMP electrical jumper harness (3) (Fig. 19) to engine wiring harness.

CAMSHAFT POSITION SENSOR - 4.0L (Continued)

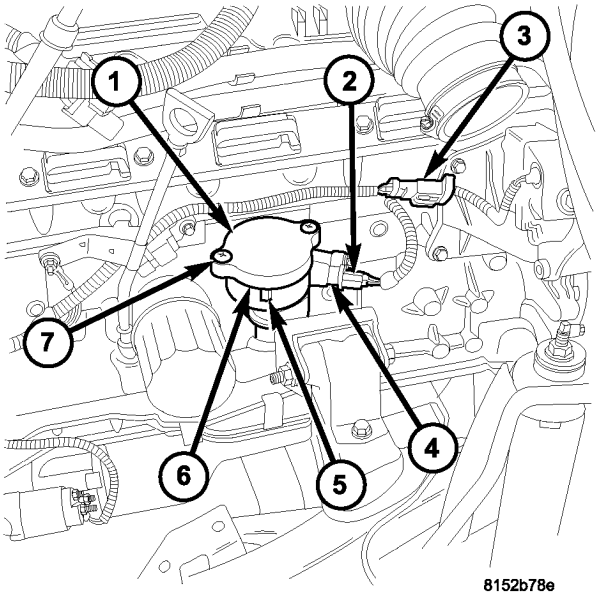


Fig. 19 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

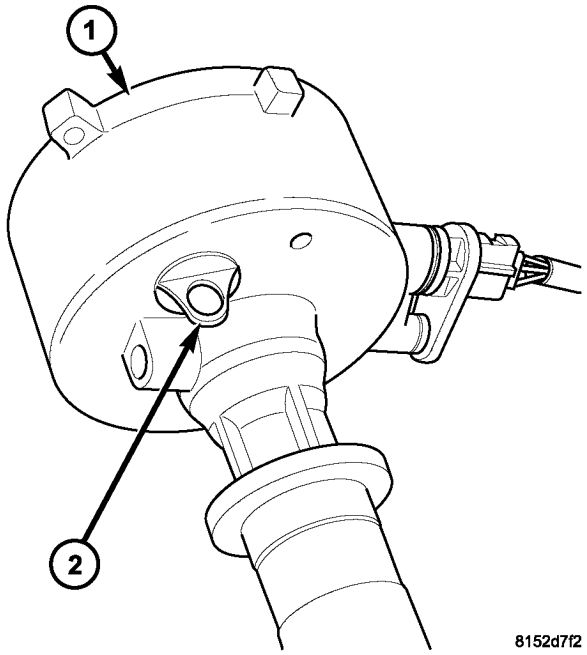


Fig. 20 4.0L CMP FACTORY ALIGNMENT PIN

1 - OIL PUMP DRIVE HOUSING
2 - FACTORY ALIGNMENT PIN

OIL PUMP DRIVE AND CMP SENSOR - 4.0L

(1) Clean oil pump drive mounting hole area of engine block.

(2) Install new oil pump drive-to-engine block gasket.

(3) If installing a **new** oil pump drive assembly, it is supplied with a temporary alignment pin (2) (Fig. 20) to prevent the target wheel from rotating. Do not remove this pin until oil pump drive assembly (1) has been installed.

(4) If installing/returning a **used** oil pump drive assembly back to the engine, temporarily install a small screwdriver (1) (Fig. 21) or similar tool through target wheel access hole (2) and then through mating hole in housing (3).

(5) Install oil pump drive into engine while aligning into slot on oil pump. Rotate oil pump drive housing back to its original 4 o'clock position as viewed from right side of engine. Install hold-down clamp and bolt. Tighten bolt.

(6) If engine crankshaft or camshaft has been rotated, such as during engine tear-down, CMP sensor relationship must be reestablished.

(a) Remove ignition coil rail assembly. Refer to Ignition Coil Removal/Installation.

(b) Remove cylinder number 1 spark plug.

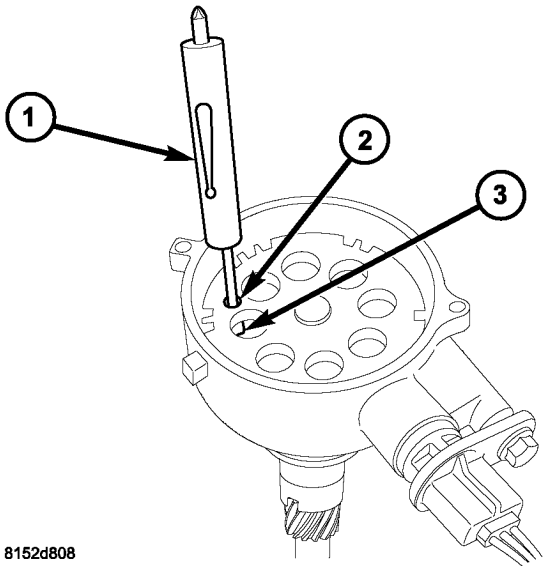


Fig. 21 4.0L CMP TARGET WHEEL ALIGNMENT

1 - SMALL SCREWDRIVER
2 - TARGET WHEEL ALIGNMENT HOLE
3 - HOUSING ALIGNMENT HOLE

(c) Hold a finger over the open spark plug hole. Rotate engine at vibration dampener bolt until compression (pressure) is felt.

(d) Slowly continue to rotate engine. Do this until timing index mark (1) (Fig. 22) on vibration damper pulley aligns with top dead center (TDC) mark (0 degree) on timing degree scale. Always

CAMSHAFT POSITION SENSOR - 4.0L (Continued)

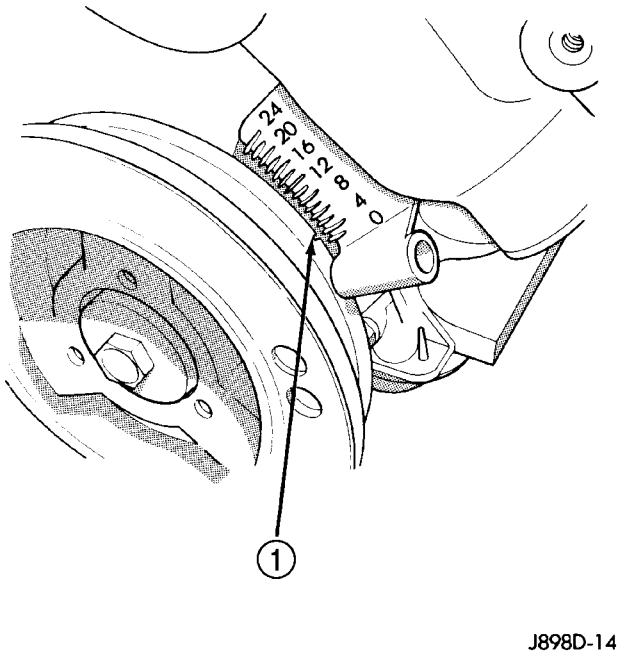


Fig. 22 Align Timing Marks - 4.0L Engine

1 - CRANKSHAFT VIBRATION DAMPER TIMING MARK

rotate engine in direction of normal rotation. Do not rotate engine backward to align timing marks.

(e) Install oil pump drive into engine while aligning into slot on oil pump. If pump drive will not drop down flush to engine block, the oil pump slot is not aligned. Remove oil pump drive and align slot in oil pump to shaft at bottom of drive. Install into engine. Rotate oil pump drive housing back to its original 4 o'clock position as viewed from right side of engine. Install hold-down clamp and bolt.

(f) Remove small screwdriver, or, factory alignment pin from housing and install plastic cover (two screws).

(7) Install CMP sensor to oil pump drive.

(8) Install sensor mounting bolt and tighten to 2 N·m (15 in. lbs.) torque.

(9) Connect CMP electrical connector to engine wiring harness.

(10) If removed, install spark plug and ignition coil rail.

(11) Connect DRB scan tool to data link connector. The data link connector is located in passenger compartment, below steering column.

(12) Gain access to "CAM/CRANK RELEARN" screen on DRB scan tool.

COIL RAIL - 4.0L

DESCRIPTION - 4.0L

A one-piece coil rail assembly containing three individual coils is used on the 4.0L 6–cylinder engine (Fig. 23). The coil rail must be replaced as one assembly. The bottom of the coil is equipped with 6 individual rubber boots (Fig. 23) to seal the 6 spark plugs to the coil. Inside each rubber boot is a spring. The spring is used for a mechanical contact between the coil and the top of the spark plug. These rubber boots and springs are a permanent part of the coil and are not serviced separately.

(1) The coil is bolted directly to the cylinder head (Fig. 24). One electrical connector (located at rear of coil) is used for all three coils.

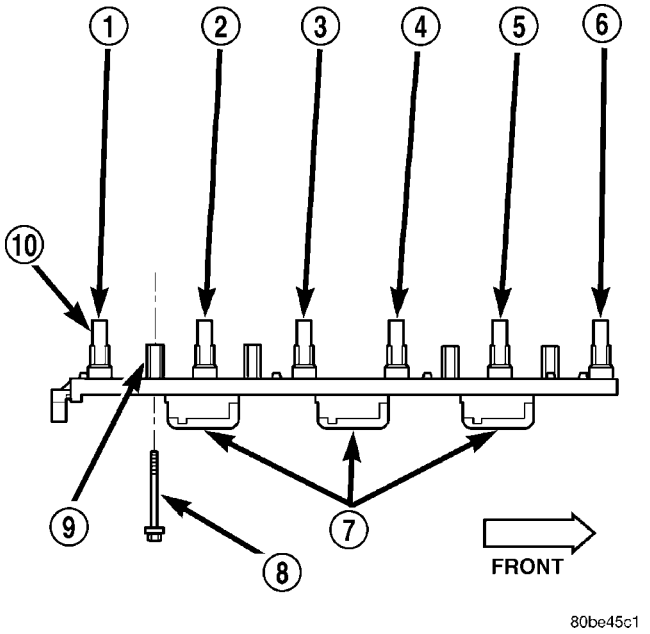


Fig. 23 Ignition Coil Assembly—4.0L 6–Cylinder Engine

1 - CYL. #6
2 - CYL. #5
3 - CYL. #4
4 - CYL. #3
5 - CYL. #2
6 - CYL. #1
7 - COILS (3)
8 - MOUNTING BOLTS (4)
9 - BOLT BASES (4)
10 - RUBBER BOOTS (6)

OPERATION - 4.0L

Although cylinder firing order is the same as 4.0L Jeep engines of previous years, spark plug firing is not. The 3 coils dual-fire the spark plugs on cylinders 1-6, 2-5 and/or 3-4. When one cylinder is being fired (on compression stroke), the spark to the opposite cylinder is being wasted (on exhaust stroke).

COIL RAIL - 4.0L (Continued)

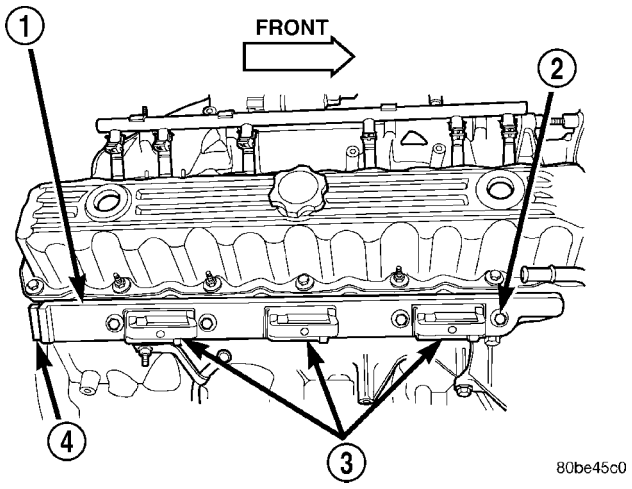


Fig. 24 Coil Location—4.0L Engine

1 - COIL RAIL
2 - COIL MOUNTING BOLTS (4)
3 - COIL
4 - COIL ELECTRICAL CONNECTION

Battery voltage is supplied to the three ignition coils from the ASD relay. The Powertrain Control Module (PCM) opens and closes the ignition coil ground circuit for ignition coil operation.

Base ignition timing is not adjustable. By controlling the coil ground circuit, the PCM is able to set the base timing and adjust the ignition timing advance. This is done to meet changing engine operating conditions.

The ignition coil is not oil filled. The windings are embedded in an epoxy compound. This provides heat and vibration resistance that allows the ignition coil to be mounted on the engine.

Because of coil design, spark plug cables (secondary cables) are not used. The cables are integral within the coil rail.

REMOVAL - 4.0L

A one-piece coil rail assembly containing three individual coils is used on the 4.0L engine (Fig. 25). The coil rail must be replaced as one assembly. The bottom of the coil is equipped with 6 individual rubber boots (Fig. 25) to seal the 6 spark plugs to the coil. Inside each rubber boot is a spring. The spring is used for an electrical contact between the coil and the top of the spark plug. These rubber boots and springs are a permanent part of the coil and are not serviced separately.

(1) Disconnect negative battery cable at battery.

(2) The coil is bolted directly to the cylinder head. Remove 4 coil mounting bolts (Fig. 26).

(3) Carefully pry up coil assembly from spark plugs. Do this by prying alternately at each end of coil until rubber boots have disengaged from all spark plugs. If boots will not release from spark plugs, use a commercially available spark plug boot removal tool. Twist and loosen a few boots from a few spark plugs to help remove coil.

(4) After coil has cleared spark plugs, position coil for access to primary electrical connector. Disconnect connector from coil by pushing slide tab outwards to right side of vehicle (Fig. 27). After slide tab has been positioned outwards, push in on secondary release lock (Fig. 27) on side of connector and pull connector from coil.

(5) Remove coil from vehicle.

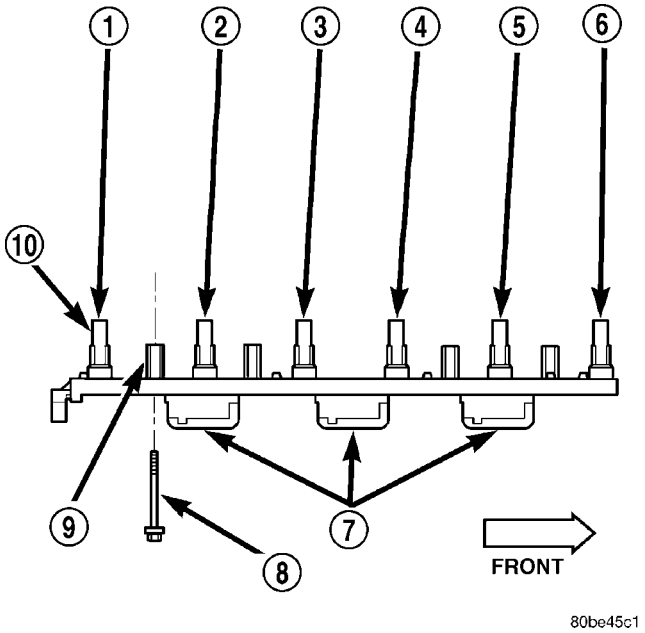


Fig. 25 Ignition Coil Assembly—4.0L 6–Cylinder Engine

1 - CYL. #6
2 - CYL. #5
3 - CYL. #4
4 - CYL. #3
5 - CYL. #2
6 - CYL. #1
7 - COILS (3)
8 - MOUNTING BOLTS (4)
9 - BOLT BASES (4)
10 - RUBBER BOOTS (6)

COIL RAIL - 4.0L (Continued)

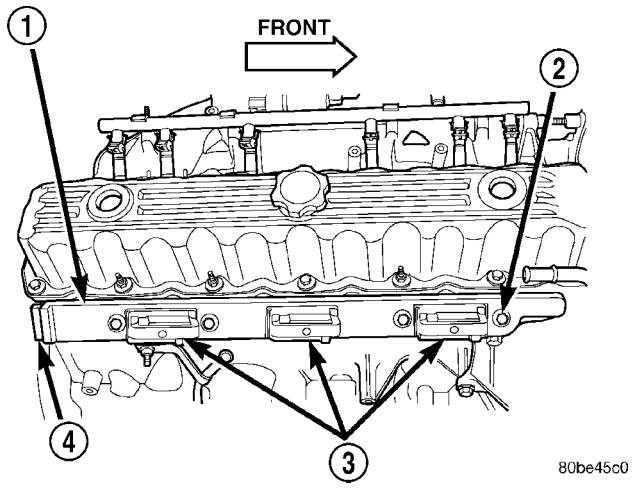


Fig. 26 Ignition Coil Rail Location—4.0L 6–Cylinder Engine

1 - COIL RAIL
2 - COIL MOUNTING BOLTS (4)
3 - COIL
4 - COIL ELECTRICAL CONNECTION

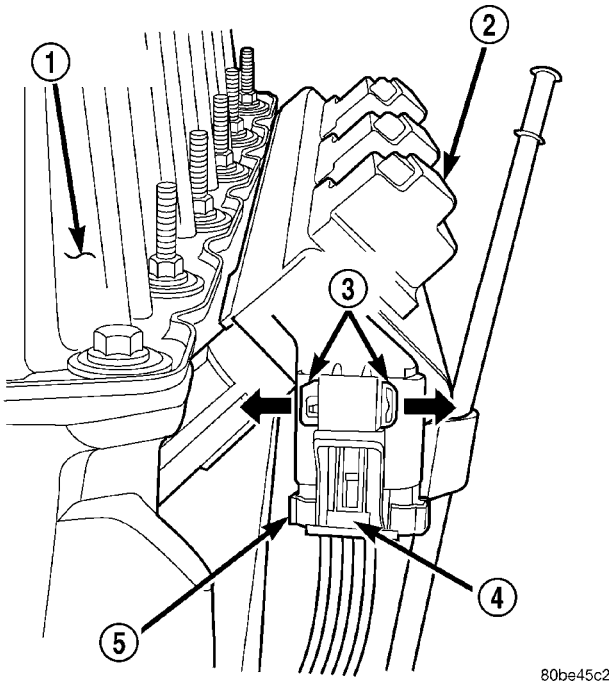


Fig. 27 Ignition Coil Electrical Connector—4.0L 6–Cylinder Engine

1 - REAR OF VALVE COVER
2 - COIL RAIL
3 - SLIDE TAB
4 - RELEASE LOCK
5 - COIL CONNECTOR

INSTALLATION - 4.0L

(1) Connect engine harness connector to coil by snapping into position. Move slide tab towards engine (Fig. 27) for a positive lock.

(2) Position ignition coil rubber boots to all spark plugs. Push down on coil assembly until bolt bases have contacted cylinder head

(3) Install 4 coil mounting bolts. Loosely tighten 4 bolts just enough to allow bolt bases to contact cylinder head. Do a final tightening of each bolt in steps down to 29 N·m (250 in. lbs.) torque. Do not apply full torque to any bolt first.

(4) Connect negative battery cable to battery.

IGNITION COIL - 2.4L

DESCRIPTION - 2.4L

The coil assembly consists of 2 different coils molded together. The assembly is mounted to the top of the engine (Fig. 28).

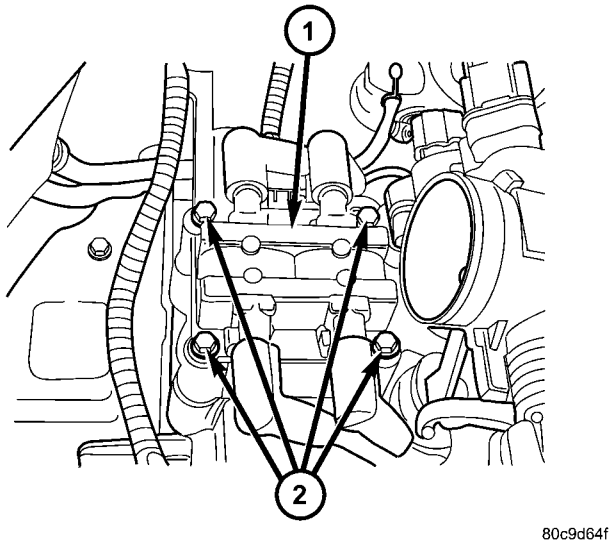


Fig. 28 IGNITION COIL - 2.4L

1 - IGNITION COIL
2 - MOUNTING BOLTS (4)

OPERATION - 2.4L

The coil fires two spark plugs simultaneously. One plug is under compression, the other plug fires on the exhaust stroke (lost spark). Coil number one fires cylinders 1 and 4, and coil number two fires cylinders 2 and 3.

The Auto Shutdown (ASD) relay provides battery voltage to the ignition coil. The PCM provides a ground contact (circuit) for energizing the coil(s). The PCM will de-energize the ASD relay if it does not receive the crankshaft position sensor and camshaft position sensor inputs.

Base ignition timing is not adjustable. By controlling the coil ground circuit, the PCM is able to set the base timing and adjust the ignition timing advance. This is done to meet changing engine operating conditions.

IGNITION COIL - 2.4L (Continued)

The ignition coil is not oil filled. The windings are embedded in an epoxy compound. This provides heat and vibration resistance that allows the ignition coil to be mounted on the engine.

Spark plug cables (secondary wires or cables) are used with the 2.4L engine.

REMOVAL - 2.4L

(1) Disconnect electrical connector at rear of coil.
(2) Remove all secondary cables from coil.
(3) Remove 4 coil mounting bolts (Fig. 29).
(4) Remove coil from vehicle.

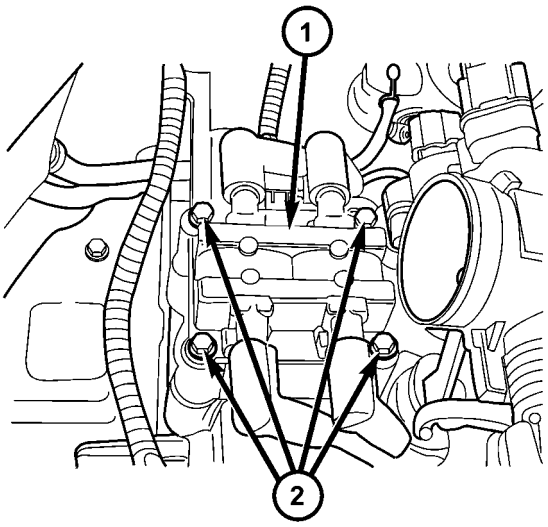


Fig. 29 IGNITION COIL - 2.4L

1 - IGNITION COIL
2 - MOUNTING BOLTS (4)

INSTALLATION - 2.4L

(1) Position coil to engine.
(2) Install 4 mounting bolts. Refer to torque specifications.
(3) Install secondary cables.
(4) Install electrical connector at rear of coil.
(5) Install air cleaner tube and housing.

SPARK PLUG

DESCRIPTION

Resistor type spark plugs are used.

Spark plug resistance values range from 6,000 to 20,000 ohms (when checked with at least a 1000 volt spark plug tester). **Do not use an ohmmeter to check the resistance values of the spark plugs. Inaccurate readings will result.**

OPERATION

To prevent possible pre-ignition and/or mechanical engine damage, the correct type/heat range/number spark plug must be used.

Always use the recommended torque when tightening spark plugs. Incorrect torque can distort the spark plug and change plug gap. It can also pull the plug threads and do possible damage to both the spark plug and the cylinder head.

Remove the spark plugs and examine them for burned electrodes and fouled, cracked or broken porcelain insulators. Keep plugs arranged in the order in which they were removed from the engine. A single plug displaying an abnormal condition indicates that a problem exists in the corresponding cylinder. Replace spark plugs at the intervals recommended in Group O, Lubrication and Maintenance

Spark plugs that have low mileage may be cleaned and reused if not otherwise defective, carbon or oil fouled. Also refer to Spark Plug Conditions.

CAUTION: Never use a motorized wire wheel brush to clean the spark plugs. Metallic deposits will remain on the spark plug insulator and will cause plug misfire.

DIAGNOSIS AND TESTING - SPARK PLUG CONDITIONS

NORMAL OPERATING

The few deposits present on the spark plug will probably be light tan or slightly gray in color. This is evident with most grades of commercial gasoline (Fig. 30). There will not be evidence of electrode burning. Gap growth will not average more than approximately 0.025 mm (.001 in) per 3200 km (2000 miles) of operation. Spark plugs that have normal wear can usually be cleaned, have the electrodes filed, have the gap set and then be installed.

Some fuel refiners in several areas of the United States have introduced a manganese additive (MMT) for unleaded fuel. During combustion, fuel with MMT causes the entire tip of the spark plug to be coated with a rust colored deposit. This rust color can be misdiagnosed as being caused by coolant in the combustion chamber. Spark plug performance may be affected by MMT deposits.

SPARK PLUG (Continued)

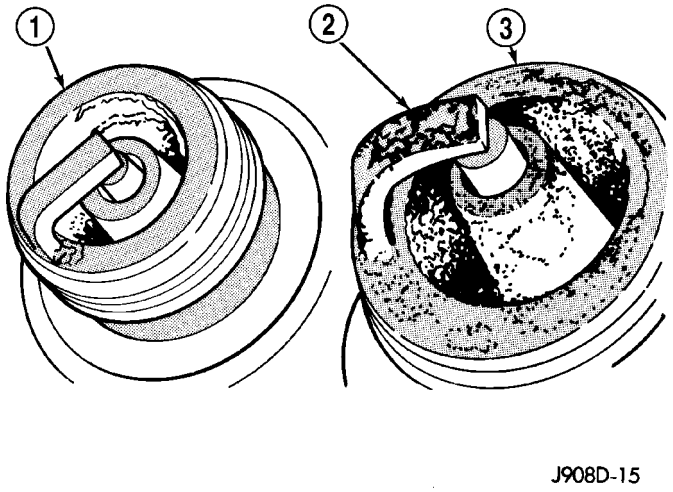


Fig. 30 Normal Operation and Cold (Carbon) Fouling

1 - NORMAL
2 - DRY BLACK DEPOSITS
3 - COLD (CARBON) FOULING

COLD FOULING/CARBON FOULING

Cold fouling is sometimes referred to as carbon fouling. The deposits that cause cold fouling are basically carbon (Fig. 30). A dry, black deposit on one or two plugs in a set may be caused by sticking valves or defective spark plug cables. Cold (carbon) fouling of the entire set of spark plugs may be caused by a clogged air cleaner element or repeated short operating times (short trips).

WET FOULING OR GAS FOULING

A spark plug coated with excessive wet fuel or oil is wet fouled. In older engines, worn piston rings, leaking valve guide seals or excessive cylinder wear can cause wet fouling. In new or recently overhauled engines, wet fouling may occur before break-in (normal oil control) is achieved. This condition can usually be resolved by cleaning and reinstalling the fouled plugs.

OIL OR ASH ENCRUSTED

If one or more spark plugs are oil or oil ash encrusted (Fig. 31), evaluate engine condition for the cause of oil entry into that particular combustion chamber.

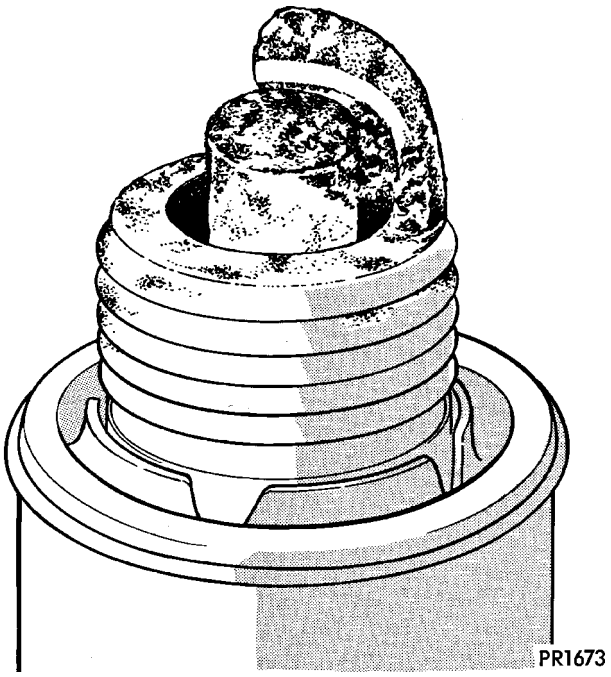


Fig. 31 Oil or Ash Encrusted

ELECTRODE GAP BRIDGING

Electrode gap bridging may be traced to loose deposits in the combustion chamber. These deposits accumulate on the spark plugs during continuous stop-and-go driving. When the engine is suddenly subjected to a high torque load, deposits partially liquefy and bridge the gap between electrodes (Fig. 32). This short circuits the electrodes. Spark plugs with electrode gap bridging can be cleaned using standard procedures.

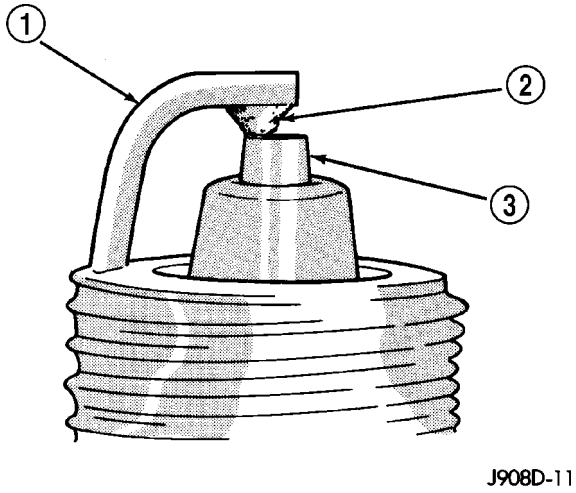


Fig. 32 Electrode Gap Bridging

1 - GROUND ELECTRODE
2 - DEPOSITS
3 - CENTER ELECTRODE

SPARK PLUG (Continued)

SCAVENGER DEPOSITS

Fuel scavenger deposits may be either white or yellow (Fig. 33). They may appear to be harmful, but this is a normal condition caused by chemical additives in certain fuels. These additives are designed to change the chemical nature of deposits and decrease spark plug misfire tendencies. Notice that accumulation on the ground electrode and shell area may be heavy, but the deposits are easily removed. Spark plugs with scavenger deposits can be considered normal in condition and can be cleaned using standard procedures.

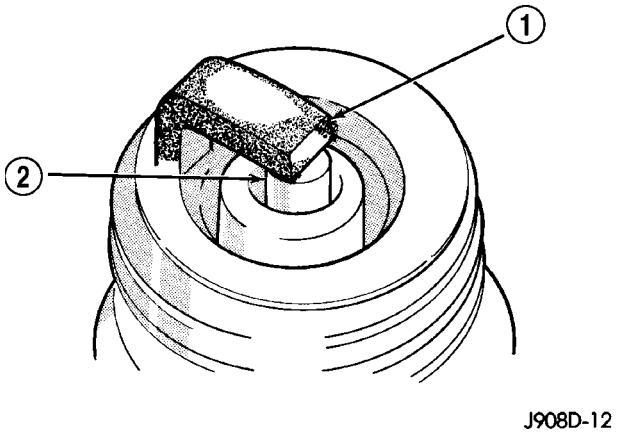


Fig. 33 Scavenger Deposits

1 - GROUND ELECTRODE COVERED WITH WHITE OR YELLOW DEPOSITS
2 - CENTER ELECTRODE

CHIPPED ELECTRODE INSULATOR

A chipped electrode insulator usually results from bending the center electrode while adjusting the spark plug electrode gap. Under certain conditions, severe detonation can also separate the insulator from the center electrode (Fig. 34). Spark plugs with this condition must be replaced.

PREIGNITION DAMAGE

Preignition damage is usually caused by excessive combustion chamber temperature. The center electrode dissolves first and the ground electrode dissolves somewhat latter (Fig. 35). Insulators appear relatively deposit free. Determine if the spark plug has the correct heat range rating for the engine. Determine if ignition timing is over advanced or if other operating conditions are causing engine overheating. (The heat range rating refers to the operating temperature of a particular type spark plug. Spark plugs are designed to operate within specific temperature ranges. This depends upon the thickness and length of the center electrodes porcelain insulator.)

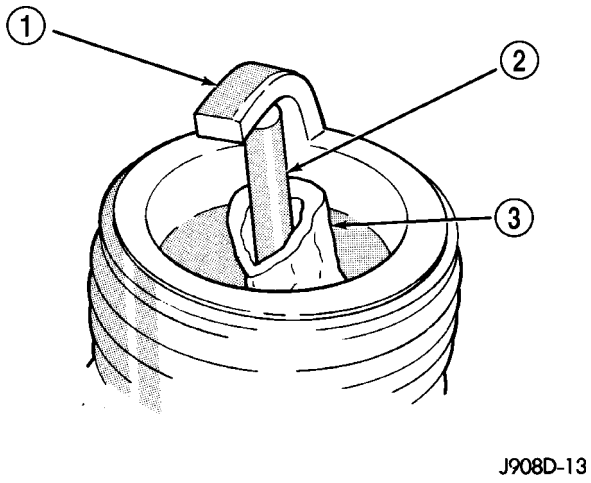


Fig. 34 Chipped Electrode Insulator

1 - GROUND ELECTRODE
2 - CENTER ELECTRODE
3 - CHIPPED INSULATOR

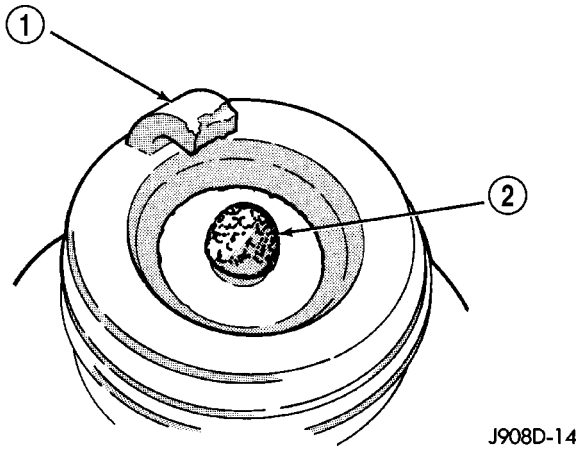


Fig. 35 Preignition Damage

1 - GROUND ELECTRODE STARTING TO DISSOLVE
2 - CENTER ELECTRODE DISSOLVED

SPARK PLUG OVERHEATING

Overheating is indicated by a white or gray center electrode insulator that also appears blistered (Fig. 36). The increase in electrode gap will be considerably in excess of 0.001 inch per 2000 miles of operation. This suggests that a plug with a cooler heat range rating should be used. Over advanced ignition timing, detonation and cooling system malfunctions can also cause spark plug overheating.

SPARK PLUG (Continued)

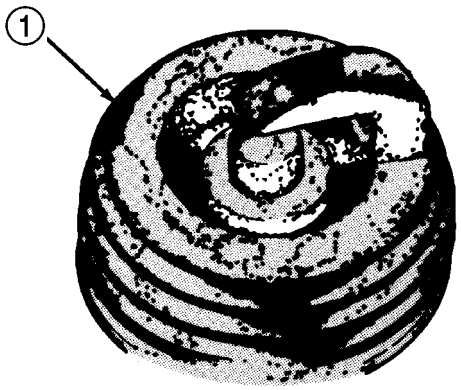


Fig. 36 Spark Plug Overheating

1 - BLISTERED WHITE OR GRAY COLORED INSULATOR

REMOVAL

2.4L

If spark plug for #2 or #3 cylinder is being removed, throttle body must be removed. Refer to Throttle Body Removal.

(1) Remove air cleaner tube and housing.

(2) Twist secondary cable at cylinder head to break loose at spark plug. Remove cable from plug.

(3) Prior to removing spark plug, spray compressed air into cylinder head opening. This will help prevent foreign material from entering combustion chamber.

(4) Remove spark plug from cylinder head using a quality socket with a rubber or foam insert.

(5) Inspect spark plug condition. Refer to Spark Plug Conditions.

4.0L

On the 4.0L 6–cylinder engine the spark plugs are located below the coil rail assembly. To gain access to any/all spark plug(s), refer to Ignition Coil-4.0L Engine Removal/Installation.

(1) Prior to removing the spark plug, spray compressed air around the spark plug hole and the area around the spark plug. This will help prevent foreign material from entering the combustion chamber.

(2) Remove the spark plug using a quality socket with a rubber or foam insert.

(3) Inspect the spark plug condition. Refer to Spark Plugs Conditions.

CLEANING SPARK PLUGS

The plugs may be cleaned using commercially available spark plug cleaning equipment. After cleaning, file the center electrode flat with a small point file or jewelers file before adjusting gap.

CAUTION: Never use a motorized wire wheel brush to clean the spark plugs. Metallic deposits will remain on the spark plug insulator and will cause plug misfire.

INSTALLATION

Always tighten spark plugs to the specified torque. Over tightening can cause distortion. This may result in a change in the spark plug gap, or a cracked porcelain insulator.

When replacing the spark plug and ignition coil cables, route the cables correctly and secure them in the appropriate retainers. Failure to route the cables properly can cause the radio to reproduce ignition noise. It could cause cross ignition of the spark plugs, or short circuit the cables to ground.

2.4L

CAUTION: Spark plug tightening on the 2.4L is torque critical. The plugs are equipped with tapered seats. Do not exceed 15 ft. lbs. torque.

Special care should be taken when installing spark plugs into the cylinder head spark plug wells. Be sure the plugs do not drop into the plug wells as electrodes can be damaged.

Always tighten spark plugs to the specified torque. Over tightening can cause distortion resulting in a change in the spark plug gap or a cracked porcelain insulator.

(1) Start the spark plug into the cylinder head by hand to avoid cross threading.

(2) Tighten spark plugs. Refer to torque specifications.

(3) Install throttle body (if necessary). Refer to Throttle Body Installation.

(4) Install air cleaner tube and housing.

4.0L

Special care should be taken when installing spark plugs into the cylinder head spark plug wells. Be sure the plugs do not drop into the plug wells as electrodes can be damaged.

Always tighten spark plugs to the specified torque. Over tightening can cause distortion resulting in a change in the spark plug gap or a cracked porcelain insulator.

(1) Start the spark plug into the cylinder head by hand to avoid cross threading.

(2) Tighten spark plugs. Refer to torque specifications.

(3) Install coil rail. Refer to Ignition Coil-4.0L Engine Removal/Installation.

SPARK PLUG CABLE

DESCRIPTION

Spark plug cables are used only on the 2.4L 4–cylinder engine. They are sometimes referred to as secondary ignition wires.

OPERATION

2.4L 4–cylinder engine only: The spark plug cables transfer electrical current from the ignition coil(s) and/or distributor, to individual spark plugs at each cylinder. The resistive spark plug cables are of non-metallic construction. The cables provide suppression of radio frequency emissions from the ignition system.

DIAGNOSIS AND TESTING

TESTING

Check the spark plug cable connections for good contact at the coil(s) and spark plugs. Terminals should be fully seated. The insulators should be in good condition and should fit tightly on the coil and spark plugs. Spark plug cables with insulators that are cracked or torn must be replaced.

Clean high voltage ignition cables with a cloth moistened with a non-flammable solvent. Wipe the cables dry. Check for brittle or cracked insulation.

When testing secondary cables for damage with an oscilloscope, follow the instructions of the equipment manufacturer.

If an oscilloscope is not available, spark plug cables may be tested as follows:

CAUTION: Do not leave any one spark plug cable disconnected for longer than necessary during testing. This may cause possible heat damage to the catalytic converter. Total test time must not exceed ten minutes.

With the engine running, remove spark plug cable from spark plug (one at a time) and hold next to a good engine ground. If the cable and spark plug are in good condition, the engine rpm should drop and the engine will run poorly. If engine rpm does not drop, the cable and/or spark plug may not be operating properly and should be replaced. Also check engine cylinder compression.

With the engine not running, connect one end of a test probe to a good ground. Start the engine and run the other end of the test probe along the entire length of all spark plug cables. If cables are cracked or punctured, there will be a noticeable spark jump from the damaged area to the test probe. Cracked, damaged or faulty cables should be replaced with resistance type cable. This can be identified by the words ELECTRONIC SUPPRESSION printed on the cable jacket.

Use an ohmmeter to test for open circuits, excessive resistance or loose terminals. Remove the cable at ignition coil, and from spark plug. Connect ohmmeter to each end of cable. Resistance should be 250 to 1000 Ohms per inch of cable. If resistance is not within specifications as found in the SPARK PLUG CABLE RESISTANCE chart, replace the cable. Test all spark plug cables in this manner.

SPARK PLUG CABLE RESISTANCE

MINIMUM	MAXIMUM
250 Ohms Per Inch	1000 Ohms Per Inch
3000 Ohms Per Foot	12,000 Ohms Per Foot

REMOVAL - 2.4L

CAUTION: When disconnecting a high voltage cable from a spark plug or from the distributor cap, twist the rubber boot slightly (1/2 turn) to break it loose (Fig. 37).

(1) Grasp the boot (not the cable) and pull it off with a steady, even force.

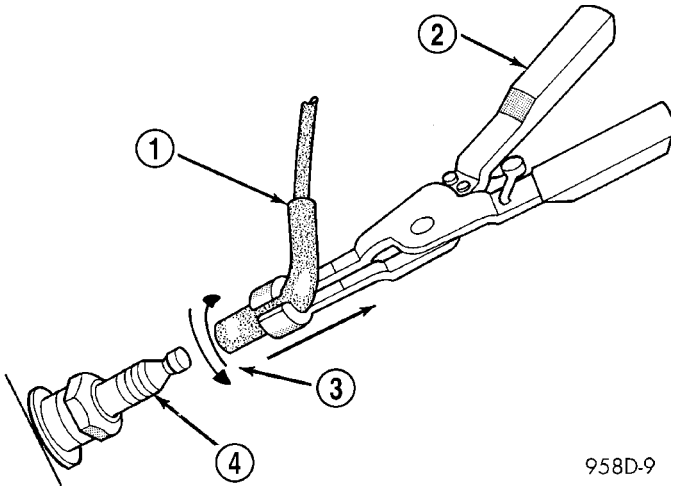


Fig. 37 CABLE REMOVAL

1 - SPARK PLUG CABLE AND BOOT
2 - SPARK PLUG BOOT PULLER
3 - TWIST AND PULL
4 - SPARK PLUG

INSTALLATION

(1) Push the cable firmly onto the sparkplug.

INSTRUMENT CLUSTER

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INSTRUMENT CLUSTER

DESCRIPTION

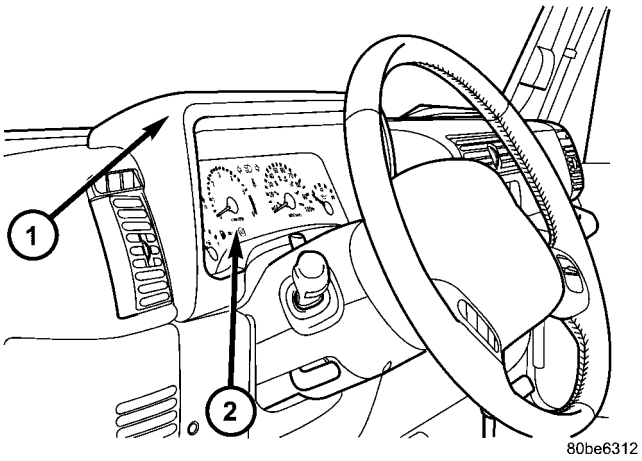

Fig. 1 Instrument Cluster

1 - CLUSTER BEZEL
2 - INSTRUMENT CLUSTER

The instrument cluster for this model is an ElectroMechanical Instrument Cluster (EMIC) that is located in the instrument panel above the steering column opening, directly in front of the driver (Fig. 1). The remainder of the EMIC, including the mounts and the electrical connections, are concealed within the instrument panel behind the cluster bezel. Besides analog gauges and indicators, the EMIC module incorporates a blue-green digital Vacuum Fluorescent Display (VFD) unit for displaying odometer/trip odometer information and certain diagnostic information. The instrument cluster for this model also includes the hardware and software necessary to serve as an electronic body control module.

The EMIC gauges and indicators are visible through a dedicated opening in the cluster bezel on the instrument panel and are protected by a clear plastic cluster lens (Fig. 2). Eight integral latch formations around the outer perimeter of the lens secure the lens unit to the cluster hood and the cluster housing. On the lower edge of the cluster lens just right of center, a small molded rubber boot covers the odometer/trip odometer switch button and seals a dedicated pocketed hole in the lens through which the switch button protrudes. Just behind the cluster lens is the cluster hood and an integral cluster mask, which are constructed of molded black plastic. The cluster hood serves as a visor and shields the face of the cluster from ambient light and reflections to reduce glare, while the cluster mask trims the outside perimeter of the cluster overlay. The cluster hood and mask unit is secured to the cluster

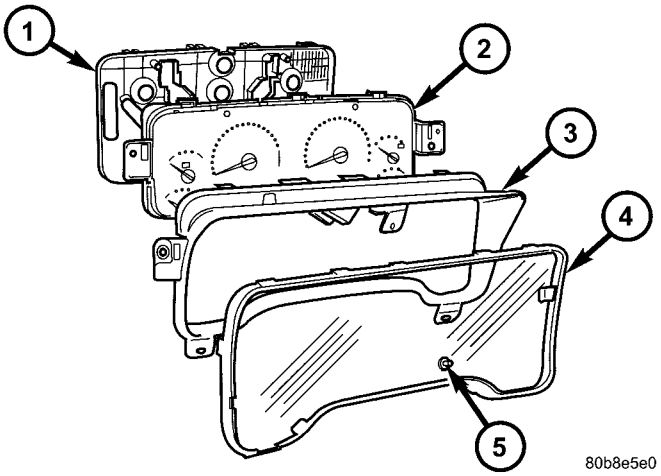

Fig. 2 Instrument Cluster Components

1 - REAR COVER
2 - CLUSTER HOUSING
3 - CLUSTER HOOD & MASK
4 - CLUSTER LENS
5 - SWITCH BUTTON BOOT

housing with five integral latch formations around its perimeter.

The rear of the cluster housing and the EMIC electronic circuitry are protected by a molded plastic rear cover, which is secured to the cluster housing with two screws, eight integral latch formations around its perimeter, and a single latch that is integral to the cluster housing. The rear cover includes clearance holes for service access to each of the five general illumination lighting and the one high beam indicator incandescent bulb and bulb holder units installed on the cluster circuit board, and for the two color-coded cluster connector receptacles. The connector receptacles on the back of the cluster electronic circuit board connect the EMIC to the vehicle electrical system through two take outs with color-coded connectors from the instrument panel wire harness.

Sandwiched between the rear cover and the lens, hood and mask unit is the cluster housing. The molded plastic cluster housing serves as the carrier for the cluster circuit board and circuitry, the cluster connector receptacles, the gauges, a Light Emitting Diode (LED) for each cluster indicator, the VFD unit, an audible tone generator, the cluster overlay, the gauge pointers, the odometer/trip odometer switch and the switch button. The molded plastic EMIC housing has four integral mounting tabs, two on the lower outboard edges of the housing and one on each side. A screw through each of these four mounting tabs secures the EMIC to the instrument panel structural support.

The cluster overlay is a laminated plastic unit. The dark, visible, outer surface of the overlay is marked with all of the gauge dial faces and graduations, but

INSTRUMENT CLUSTER (Continued)

this layer is also translucent. The darkness of this outer layer prevents the cluster from appearing cluttered or busy by concealing the cluster indicators that are not illuminated, while the translucence of this layer allows those indicators and icons that are illuminated to be readily visible. The underlying layer of the overlay is opaque and allows light from the LED or incandescent lamp for each of the various indicators and the incandescent illumination lamps behind it to be visible through the outer layer of the overlay only through predetermined stencil-like cutouts. A rectangular opening near the base of the overlay between the speedometer and tachometer dial faces has a smoked clear lens through which the illuminated VFD unit can be viewed.

Several versions of the EMIC module are offered on this model. These versions accommodate all of the variations of optional equipment and regulatory requirements for the various markets in which the vehicle will be offered. The microprocessor-based EMIC utilizes integrated circuitry and information carried on the Programmable Communications Interface (PCI) data bus network along with several hard wired analog and multiplexed inputs to monitor sensors and switches throughout the vehicle. In response to those inputs, the internal circuitry and programming of the EMIC allow it to control and integrate many electronic functions and features of the vehicle through both hard wired outputs and the transmission of electronic message outputs to other electronic modules in the vehicle over the PCI data bus. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/COMMUNICATION - DESCRIPTION - PCI BUS).

Besides typical instrument cluster gauge and indicator support, the electronic functions and features that the EMIC supports or controls include the following:

- **Audible Warnings** - The EMIC electronic circuit board is equipped with an audible tone generator and programming that allows it to provide various audible chime tone alerts to the vehicle operator. (Refer to 8 - ELECTRICAL/CHIME/BUZZER - DESCRIPTION).
- **Audio System Cabin Equalization** - The EMIC stores the cabin equalization curves for various optional speaker system architectures for use by the radio. The EMIC provides this information when requested by the radio over the PCI data bus. Changing the cabin equalization settings of the EMIC requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.
- **Axle Locker Control** - The EMIC provides the logic that controls the operation of the rear and front axle locker feature found on models equipped with the optional off-road package.
- **Electronic Pinion Factor** - The EMIC stores the tire size, axle ratio, and transfer case type information for use in calibrating the proper vehicle speed and distance values for display by the speedometer and odometer, as well as for use by other electronic modules in the vehicle. The EMIC provides this information when requested by other modules over the PCI data bus. Changing the electronic pinion factor settings of the EMIC requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.
- **Interior Lamp Load Shedding** - The EMIC provides a battery saver feature which will automatically turn off all interior lamps that remain on after a timed interval of about twenty minutes.
- **Interior Lighting Control** - The EMIC monitors inputs from the interior lighting switch and the door ajar switches to provide courtesy lamp control. This includes support for a timed illuminated entry with a theater-style fade-to-off feature.
- **Panel Lamps Dimming Control** - The EMIC provides a hard wired 12-volt Pulse-Width Modulated (PWM) output that synchronizes the dimming level of all panel lamps dimmer controlled lamps with that of the cluster illumination lamps. This includes providing features such as VFD unit illumination when a door is ajar (rental car mode), radio illumination control, a parade mode, and one step dimmable front and rear lock, cruise, and four-wheel drive indicators.
- **Rear Window Defogger Control** - The EMIC provides control and timer functions for the output to the rear window defogger on vehicles so equipped, which eliminates the need for a separate control and timer module for the rear window defogger system. (Refer to 8 - ELECTRICAL/WINDOW DEFOGGER - DESCRIPTION).
- **Vacuum Fluorescent Display Synchronization** - The EMIC transmits electronic panel lamp dimming level messages which allows all other electronic modules with Vacuum Fluorescent Display (VFD) units on the PCI data bus to coordinate their illumination intensity with that of the EMIC VFD unit.

The EMIC houses six analog gauges and has provisions for up to seventeen indicators (Fig. 3). The EMIC includes the following analog gauges:

- **Coolant Temperature Gauge**
- **Fuel Gauge**
- **Oil Pressure Gauge**
- **Speedometer**
- **Tachometer**
- **Voltage Gauge**

Some of the EMIC indicators are automatically configured when the EMIC is connected to the vehicle electrical system for compatibility with certain optional equipment or equipment required for regula-

tory purposes in certain markets. While each EMIC may have provisions for indicators to support every available option, the configurable indicators will not be functional in a vehicle that does not have the equipment that an indicator supports. The EMIC includes provisions for the following indicators (Fig. 3):

- **Airbag Indicator (with Airbag System only)**
- **Antilock Brake System (ABS) Indicator (with ABS brakes only)**
- **Brake Indicator**
- **Check Gauges Indicator**
- **Cruise Indicator (with Speed Control only)**
- **Fog Lamp Indicator (with Front or Rear Fog Lamps only)**
- **Four-Wheel Drive (4WD) Indicator**
- **Front Lock Indicator (with Off-Road Package only)**
- **High Beam Indicator**
- **Low Fuel Indicator**
- **Malfunction Indicator Lamp (MIL)**
- **Rear Lock Indicator (with Off-Road Package only)**
- **Seatbelt Indicator**
- **Sentry Key Immobilizer System (SKIS) Indicator (with SKIS only)**
- **Turn Signal (Right and Left) Indicators**

The EMIC high beam indicator is illuminated by a dedicated incandescent bulb. Each remaining indicator in the EMIC is illuminated by a dedicated LED that is soldered onto the EMIC electronic circuit board. The LED units are not available for service replacement and, if damaged or faulty, the entire EMIC must be replaced. Cluster illumination is accomplished by dimmable incandescent back lighting, which illuminates the gauges for visibility when the exterior lighting is turned on. Each of the incandescent bulbs is secured by an integral bulb holder to the electronic circuit board from the back of the cluster housing.

Hard wired circuitry connects the EMIC to the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system and to the EMIC through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

The EMIC modules for this model are serviced only as complete units. The EMIC module cannot be adjusted or repaired. If a gauge, an LED indicator, a VFD unit, the electronic circuit board, the circuit board hardware, the cluster overlay, or the EMIC housing are damaged or faulty, the entire EMIC module must be replaced. The cluster lens, the cluster hood and mask unit, the cluster housing rear cover, the odometer/trip odometer switch button boot, and the incandescent lamp bulbs with holders are available for individual service replacement.

OPERATION

The ElectroMechanical Instrument Cluster (EMIC) in this model also includes the hardware and software necessary to serve as the electronic body control module. The following information deals primarily with the instrument cluster functions of this unit. Additional details of the electronic body control functions of this unit may be found within the service information for the system or component that the EMIC controls. For example: Additional details of the audible warning functions of the EMIC are found within the Chime/Buzzer service information.

The EMIC is designed to allow the vehicle operator to monitor the conditions of many of the vehicle components and operating systems. The gauges and indicators in the EMIC provide valuable information about the various standard and optional powertrains, fuel and emissions systems, cooling systems, lighting systems, safety systems, and many other convenience items. The EMIC is installed in the instrument panel so that all of these monitors can be easily viewed by the vehicle operator when driving, while still allowing relative ease of access for service. The microprocessor-based EMIC hardware and software uses various inputs to control the gauges and indicators visible on the face of the cluster. Some of these inputs are hard wired, but most are in the form of electronic messages that are transmitted by other electronic modules over the Programmable Commutations Interface (PCI) data bus network. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/COMMUNICATION - OPERATION).

The EMIC microprocessor smooths the input data using algorithms to provide gauge readings that are accurate, stable, and responsive to operating conditions. These algorithms are designed to provide gauge readings during normal operation that are consistent with customer expectations. However, when abnormal conditions exist, such as low/high battery voltage, low oil pressure, or high coolant temperature, the algorithm can drive the gauge pointer to an extreme position and the microprocessor can turn on the Check Gauges indicator and/or sound a chime through the on-board tone generator to provide dis-

INSTRUMENT CLUSTER (Continued)

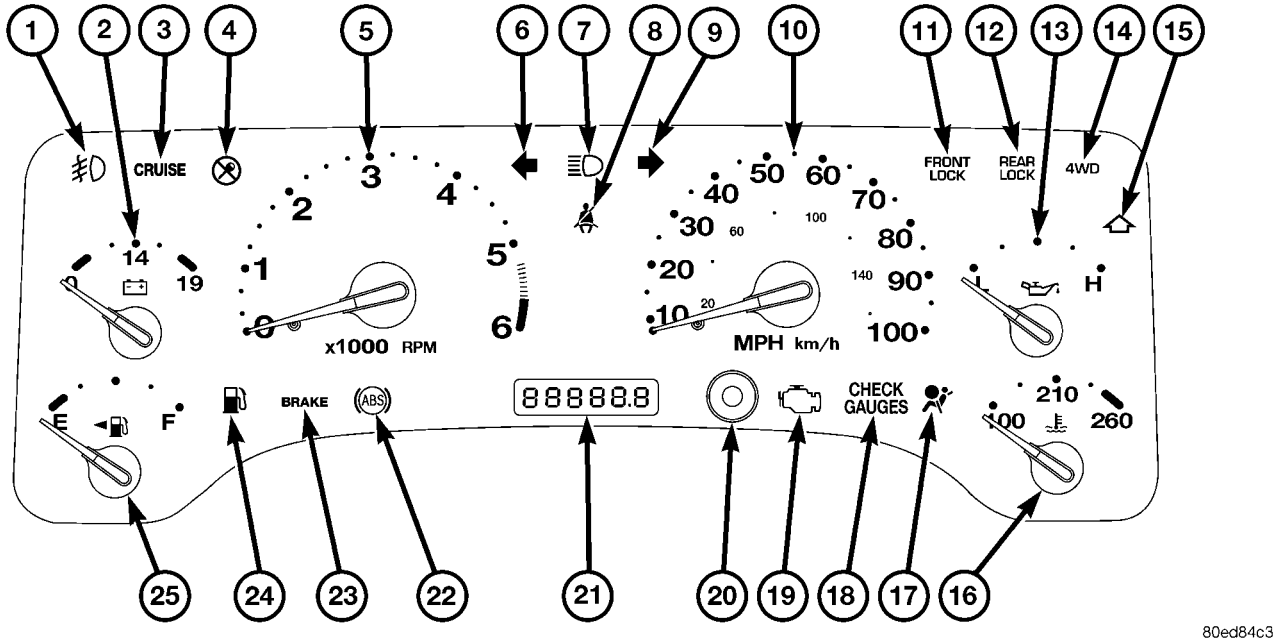

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Fig. 3 Gauges & Indicators

1 - FOG LAMP INDICATOR
2 - VOLTAGE GAUGE
3 - CRUISE INDICATOR
4 - SKIS INDICATOR
5 - TACHOMETER
6 - LEFT TURN INDICATOR
7 - HIGH BEAM INDICATOR
8 - SEATBELT INDICATOR
9 - RIGHT TURN INDICATOR
10 - SPEEDOMETER
11 - FRONT LOCK INDICATOR
12 - REAR LOCK INDICATOR
13 - OIL PRESSURE GAUGE
14 - 4WD INDICATOR
15 - NOT APPLICABLE
16 - ENGINE TEMPERATURE GAUGE
17 - AIRBAG INDICATOR
18 - CHECK GAUGES INDICATOR
19 - MALFUNCTION INDICATOR LAMP
20 - ODOMETER/TRIP ODOMETER SWITCH BUTTON
21 - ODOMETER/TRIP ODOMETER DISPLAY
22 - ABS INDICATOR
23 - BRAKE INDICATOR
24 - LOW FUEL INDICATOR
25 - FUEL GAUGE

tinct visual and/or audible indications of a problem to the vehicle operator. The instrument cluster circuitry may also produce audible warnings for other electronic modules in the vehicle based upon electronic tone request messages received over the PCI data bus. Each audible warning is intended to provide the vehicle operator with an audible alert to supplement a visual indication.

The EMIC circuitry operates on battery current received through the Ignition-Off Draw (IOD) fuse in the Power Distribution Center (PDC) on a non-switched fused B(+) circuit, and on battery current received through a fuse in the fuse block on a fused ignition switch output (run-start) circuit. This arrangement allows the EMIC to provide some features regardless of the ignition switch position, while other features will operate only with the ignition switch in the On or Start positions. The EMIC circuitry is grounded through a ground circuit and take out of the instrument panel wire harness with an eyelet terminal connector that is secured under a ground screw to the back of the instrument panel structural support near the lower left corner of the instrument panel, just inboard of the left instrument panel end bracket.

The EMIC also has a self-diagnostic actuator test capability, which will test each of the PCI bus message-controlled functions of the cluster by lighting the appropriate indicators, positioning the gauge needles at several predetermined calibration points across the gauge faces, and illuminating all segments of the odometer/trip odometer Vacuum-Fluorescent Display (VFD) unit. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). See the owner's manual in the vehicle glove box for more information on the features, use and operation of the EMIC.

GAUGES

All gauges receive battery current through the EMIC circuitry only when the ignition switch is in the On or Start positions. With the ignition switch in

INSTRUMENT CLUSTER (Continued)

the Off position battery current is not supplied to any gauges, and the EMIC circuitry is programmed to move all of the gauge needles back to the low end of their respective scales. Therefore, the gauges do not accurately indicate any vehicle condition unless the ignition switch is in the On or Start positions.

All of the EMIC gauges are air core magnetic units. Two fixed electromagnetic coils are located within each gauge. These coils are wrapped at right angles to each other around a movable permanent magnet. The movable magnet is suspended within the coils on one end of a pivot shaft, while the gauge needle is attached to the other end of the shaft. One of the coils has a fixed current flowing through it to maintain a constant magnetic field strength. Current flow through the second coil changes, which causes changes in its magnetic field strength. The current flowing through the second coil is changed by the EMIC circuitry in response to messages received over the PCI data bus. The gauge needle moves as the movable permanent magnet aligns itself to the changing magnetic fields created around it by the electromagnets.

The gauges are diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic data bus message inputs to the EMIC that control each gauge require the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific operation details for each gauge may be found elsewhere in this service information.

VACUUM-FLUORESCENT DISPLAY

The Vacuum-Fluorescent Display (VFD) unit is soldered to the EMIC electronic circuit board. With the ignition switch in the Off or Accessory positions, the odometer display is activated when either door is opened (Rental Car mode) and is deactivated after both doors are closed. If a door is left open with the ignition switch in any position except On or Start, the VFD will remain illuminated until the interior lights control battery saver (load shedding) timer expires after about twenty minutes. Otherwise, the display unit is active when the ignition switch is in the On or Start positions, and inactive when the ignition switch is in the Off or Accessory positions.

The illumination intensity of the VFD unit is controlled by the EMIC circuitry based upon an input from the headlamp switch circuitry and a dimming level input received from the panel lamp dimmer switch circuitry. The EMIC synchronizes the illumination intensity of other VFD units with that of the unit in the EMIC by sending electronic dimming level messages to other electronic modules in the vehicle over the PCI data bus.

The EMIC VFD unit has several display capabilities including odometer, trip odometer, software version, and can display various diagnostic information when certain fault conditions exist. An odometer/trip odometer switch on the EMIC circuit board is used to control some of the display modes. This switch is actuated manually by depressing the odometer/trip odometer switch button that extends through the lower edge of the cluster lens, just right of center. Actuating this switch momentarily with the ignition switch in the On position will toggle the VFD between the odometer and trip odometer modes. Depressing the switch button for about two seconds while the VFD is in the trip odometer mode will reset the trip odometer value to zero. Holding this switch depressed while turning the ignition switch from the Off position to the On position will initiate the EMIC self-diagnostic actuator test. Refer to the appropriate diagnostic information for additional details on this VFD function. The EMIC microprocessor remembers which display mode is active when the ignition switch is turned to the Off position, and returns the VFD display to that mode when the ignition switch is turned On again.

The VFD is diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic data bus message inputs to the EMIC that control some of the VFD functions requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific operation details for the odometer and trip odometer functions of the VFD may be found elsewhere in this service information.

INDICATORS

Indicators are located in various positions within the EMIC and are all connected to the EMIC electronic circuit board. The antilock brake system indicator, headlamp high beam indicator, fog lamp indicator, and turn signal indicators operate based upon hard wired inputs to the EMIC. The brake indicator is controlled by the EMIC programming and hard wired park brake and brake warning indicator switch inputs to the EMIC. In vehicles without the optional off-road package the four-wheel drive indicator is controlled by a hard wired input from the transfer case switch, while in vehicles equipped with the off-road package this indicator is controlled by PCI data bus messages from the Powertrain Control Module (PCM). The rear and front lock indicators are controlled by the EMIC programming and hard wired inputs from the axle locker switch and the rear/front locker indicator switches. The seatbelt indicator is controlled by the EMIC programming and a hard wired seat belt switch input to the EMIC. The Mal-

INSTRUMENT CLUSTER (Continued)

function Indicator Lamp (MIL) is normally controlled by PCI data bus messages from the PCM; however, if the EMIC loses PCI data bus communications, the EMIC circuitry will automatically turn the MIL on and display the message “no BuS” in the odometer VFD unit until PCI data bus communication is restored. The EMIC uses PCI data bus messages from the PCM, Airbag Control Module (ACM), and the Sentry Key Immobilizer Module (SKIM) to control all of the remaining indicators.

The various EMIC indicators are controlled by different strategies; some receive fused ignition switch output from the EMIC circuitry and have a switched ground, others are grounded through the EMIC circuitry and have a switched battery feed. However, all indicators except those for the antilock brake system, turn signals, and fog lamps are completely controlled by the EMIC microprocessor based upon various hard wired and electronic message inputs. The cruise, four-wheel drive and both axle lock indicators are one-step dimmable based upon an input to the EMIC from the headlamp switch circuitry. When the exterior lamps are off, these indicators are illuminated at a fixed maximum intensity; and, when the exterior lamps are on, these indicators are dimmed by the EMIC to a fixed lower intensity. All remaining indicators are illuminated at a fixed intensity, which is not affected by the status of the exterior lighting or the selected illumination intensity of the EMIC general illumination lamps.

In addition, certain indicators in this instrument cluster are automatically configured or self-configured. This feature allows the configurable indicators to be enabled by the EMIC circuitry for compatibility with certain optional equipment. The airbag indicator, cruise indicator, and the Sentry Key Immobilizer System (SKIS) indicator are automatically configured. The automatically configured or self-configured indicators remain latent in each EMIC at all times and will be active only when the EMIC receives the appropriate PCI message inputs for that optional system or equipment. Once a configurable indicator is enabled by the EMIC, it is learned and stored in cluster memory for the remainder of the cluster life.

The hard wired indicator inputs may be diagnosed using conventional diagnostic methods. However, the EMIC circuitry and PCI bus message controlled indicators are diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic message inputs to the EMIC that control an indicator requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific details of the operation for each indicator may be found elsewhere in this service information.

CLUSTER ILLUMINATION

The EMIC has several illumination lamps that are illuminated when the exterior lighting is turned on with the headlamp switch circuitry of the left multi-function switch. The illumination intensity of these lamps is adjusted when the interior lamps control ring on the left multi-function switch control stalk is rotated (downward to dim, upward to brighten). In response to that input, an analog/digital (A/D) converter in the EMIC converts the analog panel lamps dimmer resistor multiplexed input from the left multi-function switch into a digital dimming level message and a 12-volt Pulse-Width Modulated (PWM) output. The EMIC uses the PWM output to power the cluster illumination lamps and the VFD unit on the EMIC circuit board, then provides a synchronized PWM output on the hard wired fused panel lamp feed output circuit to control and synchronize the illumination intensity of other incandescent illumination lamps in the vehicle. The cluster illumination lamps are grounded at all times.

The EMIC also sends electronic dimming level messages over the PCI data bus to other electronic modules in the vehicle to control and synchronize the illumination intensity of their VFD units with that of the EMIC VFD unit. In addition, the interior lamps control ring on the left multi-function switch control stalk has a Parade Mode position to provide a parade mode. The EMIC monitors the request for this mode from the left multi-function switch, then sends an electronic dimming level message over the PCI data bus to illuminate all VFD units in the vehicle at full (daytime) intensity for easier visibility when driving in daylight with the exterior lighting turned on.

The hard wired left multi-function switch cluster illumination inputs and the EMIC fused panel lamp feed output may be diagnosed using conventional diagnostic methods. However, proper testing of the PWM control of the EMIC and the electronic dimming level messages sent by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AUDIO SYSTEM CABIN EQUALIZATION

Each time the EMIC receives an electronic cabin equalization request message from the radio over the PCI data bus, it provides an electronic response to the radio containing the appropriate equalization curve information. Because there are numerous optional radios which are common to many platforms and available with various speaker architectures, each radio contains a Digital Signal Processing (DSP) microprocessor chip. This DSP chip uses the equalization curve information to optimize the radio’s sound output characteristics for the unique cabin and

speaker architecture found within the particular vehicle into which the radio has been installed.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control audio system cabin equalization requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AXLE LOCKER CONTROL

The EMIC contains a logic circuit and programming to perform the axle locker control functions for models equipped with the optional off-road package. The EMIC monitors hard wired inputs from the ignition switch on the fused ignition switch output (run-start) circuit and the key-in ignition switch on the key-in ignition switch sense circuit. The EMIC also monitors vehicle speed and transfer case shift position switch electronic message inputs received from the Powertrain Control Module (PCM) over the PCI data bus. The internal programming of the EMIC then determines whether to activate or deactivate the axle locker function by enabling or disabling the axle lock switch located in the instrument panel accessory switch bezel.

The EMIC programming controls the axle lock switch through two separate axle lock switch enable circuits, enable 1 and enable 2. In all cases, the EMIC will not activate either enable circuit if there are any transfer case shift position switch or vehicle speed sensor faults present. Whenever the ignition switch is in the On or Start positions, the key is in the ignition lock cylinder, the transfer case is in 4 X 4 Low range, and the vehicle speed is less than about 72 kilometers per hour (45 miles per hour) the first enable (enable 1) circuit is activated. The second enable (enable 2) circuit is activated only if the vehicle speed is less than about 16 kilometers per hour (10 miles per hour). When both enable circuits are activated, the axle lock switch becomes functional.

Once activated, the enable 1 circuit is automatically deactivated whenever the transfer case is moved out of the 4 X 4 Low range, or if the vehicle speed is greater than about 72 kilometers per hour (45 miles per hour). If the enable 1 circuit is deactivated after the rear or the front and rear axle lockers are engaged, all outputs from the axle lock switch are dropped causing both axles to unlock. The enable 2 circuit is automatically deactivated whenever the vehicle speed is greater than about 16 kilometers per hour (10 miles per hour). If the enable 2 circuit is deactivated after the rear or the front and rear axle lockers are engaged, the outputs from the axle lock switch are unaffected and the locked axles remain locked. However, an unlocked axle cannot be locked until the vehicle speed is reduced and the enable 2 circuit is again activated.

In addition, once activated, both enable circuits will remain active regardless of the status of the ignition switch input. Therefore, any locked axle will remain locked and the various components of the axle locker system will remain functional after the ignition switch is turned to the Off position. However, while the currently selected axle locker mode remains active with the ignition switch turned Off, if the key is removed from the ignition lock cylinder, Off is the only other axle locker mode that can be selected with the axle lock switch. For as long as the key is removed from the ignition lock cylinder, the cluster logic will interpret any revision to the input status of either request circuit from the axle lock switch as a cancellation request and will deactivate the enable 1 circuit and all outputs from the axle lock switch are dropped, causing both axles to unlock. Otherwise, once locked, any locked axle will remain locked until the axle lock switch is deactivated (enable 1 circuit is deactivated), or until the Off mode is manually selected by moving the axle lock switch rocker to the Off position. The EMIC also provides the vehicle operator with distinct visual and/or audible indications as to the current status of the axle locker system through chime warnings and illumination of the rear/front lock indicators in the cluster as outlined elsewhere in this service information.

The hard wired input and output circuits of the EMIC axle locker control may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic vehicle speed and transfer case shift position switch messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

CHIME SERVICE

The EMIC is equipped with hardware and software to provide chime service for all available features in the chime warning system. Upon receiving the proper chime inputs, the EMIC activates an integral on-board audible tone generator to provide audible chime tones to the vehicle operator. The chime tone generator in the EMIC is capable of producing single chime tones or repeated chime tones at two different rates: a slow rate of about fifty chime tones per minute, and a fast rate of about 180 chime tones per minute. The internal programming of the EMIC determines the priority of each chime tone request input that is received, as well as the rate and duration of each chime tone that is to be generated.

The EMIC relies upon hard wired inputs from the door ajar switches, the left multi-function switch, the ignition switch, and the park brake/brake warning indicator switches to provide chime service for the

INSTRUMENT CLUSTER (Continued)

driver/passenger door ajar warning, the head/park lights-on reminder, and the key-in ignition reminder. For the remaining chime warning functions, the EMIC uses a combination of hard wired inputs, electronic message inputs received from other modules over the PCI data bus, and internal programming. (Refer to 8 - ELECTRICAL/CHIME/BUZZER - OPERATION).

The hard wired chime inputs to the EMIC may be diagnosed using conventional diagnostic methods. However, proper testing of the EMIC programming and the electronic chime request messages received by the EMIC over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

ELECTRONIC PINION FACTOR

Each time the EMIC receives an electronic pinion factor request message from the Powertrain Control Module (PCM) over the PCI data bus, it provides a response to the PCM containing the appropriate electronic pinion factor information. Because there are numerous optional combinations of tire size, axle ratio, and transfer case type which are available for this vehicle, the electronic pinion factor needs to be considered for accurate processing of the vehicle speed and distance information. The instrument cluster stores the electronic pinion factor information and broadcasts it over the PCI data bus upon request by the PCM. Using this factor and the Vehicle Speed Sensor (VSS) input, the PCM then calculates the proper vehicle speed and distance information, and transmits the appropriate electronic vehicle speed messages over the PCI data bus for use by other electronic modules in the vehicle.

Proper testing of the PCI data bus and the electronic data bus message inputs to and outputs from the EMIC that control the electronic pinion factor function requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

INTERIOR LIGHTING CONTROL

The EMIC contains an integral timer and logic circuit to perform both timer and control functions for the interior courtesy lamps. The EMIC uses hard wired inputs from the ignition switch, both door ajar switches on separate driver and passenger door ajar switch sense circuits, from the resistor multiplexed panel lamps dimmer circuitry of the left multi-function switch on the panel lamps dimmer signal circuit and its control logic to provide a battery current output to the courtesy lamps on a courtesy lamp feed circuit. The EMIC control provides a theater-type fade-to-off feature that will slowly dim the courtesy lamps about five seconds after both doors are closed.

The EMIC interior lighting control programming provides an illuminated entry/exit feature by monitoring the door ajar and ignition switch inputs. When a door is opened with the ignition switch in the Off position, the EMIC turns on the courtesy lamps. When the ignition switch is turned to the On or Start positions, the EMIC turns the courtesy lamps off immediately with no theater dimming. When the ignition switch is turned from the On position to the Off position, the EMIC turns on the interior lights for about ten seconds or until the ignition switch is again turned to the On or Start positions, whichever occurs first.

The EMIC interior lighting control programming also provides a battery saver feature (load shedding) for the interior lighting. Unless the engine is running, the EMIC will automatically turn off the interior lights if they are left on for more than about twenty minutes, regardless of the status of the ignition switch, door ajar switch, or left multi-function switch inputs to the cluster.

The hard wired inputs and output of the EMIC interior lighting control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC timer and logic circuitry. If the input and output components and circuits of the interior lighting system test OK, but the system fails to operate as designed, the EMIC must be replaced.

REAR WINDOW DEFOGGER CONTROL

The EMIC contains an integral timer and logic circuit to perform the rear window defogger timer and control functions for the optional rear window defogger system. The EMIC uses a hard wired input from the rear window defogger switch on the rear window defogger switch sense circuit and its control logic to determine the correct output to the rear window defogger relay. The EMIC controls the ground path of the rear window defogger relay control coil through an output on the rear window defogger relay control circuit.

The EMIC is programmed to interpret each momentary ground signal it receives on the rear window defogger switch sense circuit as a request to change the current state of the output on the rear window defogger relay control circuit. Therefore, with the ignition switch in the On position, the first ground input on the rear window defogger switch sense circuit turns the system On, the second ground input turns the system Off, and so forth. Once the rear window defogger system has been turned On, it can be turned off manually by depressing the rear window defogger switch a second time or by turning the ignition switch to the Off position.

INSTRUMENT CLUSTER (Continued)

The timer function of the EMIC will also automatically turn the rear window defogger system Off. The timer turns the system Off after about ten minutes of operation; however, after the first timed interval has expired, each time the system is turned On again during that same ignition cycle, the timer will automatically turn it Off after about five minutes of operation.

The hard wired input and output of the EMIC rear window defogger control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC rear window defogger timer and logic circuitry. If the input and output components and circuits of the rear window defogger system test OK, but the system fails to operate as designed, the EMIC must be replaced.

INPUT AND OUTPUT CIRCUITS

HARD WIRED INPUTS

The hard wired inputs to the EMIC include the following:

- **ABS Warning Indicator Driver**
- **Brake Warning Indicator Sense**
- **Driver Door Ajar Switch Sense**
- **Fog Lamp Feed**
- **Front Axle Lock Request (w/Off-Road Package Only)**
- **Front Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Fused B(+) - Ignition-Off Draw**
- **Fused Ignition Switch Output (Run-Start)**
- **Headlamp Switch Output**
- **High Beam Indicator Driver**
- **Key-In Ignition Switch Sense**
- **Left Turn Signal**
- **Panel Lamps Dimmer Signal**
- **Passenger Door Ajar Switch Sense**
- **Rear Axle Lock Request (w/Off-Road Package Only)**
- **Rear Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Rear Window Defogger Switch Sense**
- **Right Turn Signal**
- **Seat Belt Switch Sense**
- **4WD Switch Sense (w/o Off-Road Package only)**

Refer to the appropriate wiring information for additional details.

HARD WIRED OUTPUTS

The hard wired outputs of the EMIC include the following:

- **Courtesy Lamp Feed**
- **Axle Lock Switch Enable 1 (w/Off-Road Package Only)**
- **Axle Lock Switch Enable 2 (w/Off-Road Package Only)**
- **Panel Lamp Feed**
- **Rear Window Defogger Relay Control**

Refer to the appropriate wiring information for additional details.

GROUNDS

The EMIC receives a ground path through the following hard wired circuit:

- **Ground - G201**

Refer to the appropriate wiring information for additional details.

COMMUNICATION

The EMIC has the following data bus communication circuit:

- **PCI Data Bus**

Refer to the appropriate wiring information for additional details.

DIAGNOSIS AND TESTING - INSTRUMENT CLUSTER

If all of the instrument cluster gauges and/or indicators are inoperative, refer to PRELIMINARY DIAGNOSIS. If an individual gauge or Programmable Communications Interface (PCI) data bus message-controlled indicator is inoperative, refer to ACTUATOR TEST. If an individual hard wired indicator is inoperative, refer to the diagnosis and testing information for that specific indicator. If the instrument cluster interior lighting control function is inoperative, refer to INTERIOR LIGHTING CONTROL DIAGNOSIS. If the axle locker control is inoperative, refer to AXLE LOCKER CONTROL DIAGNOSIS. If the instrument cluster rear window defogger control function is inoperative, refer to REAR WINDOW DEFOGGER CONTROL DIAGNOSIS.

Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

INSTRUMENT CLUSTER (Continued)

CAUTION: Instrument clusters used in this model automatically configure themselves for compatibility with the features and optional equipment in the vehicle in which they are initially installed. The instrument cluster is programmed to do this by embedding the Vehicle Identification Number (VIN) and other information critical to proper cluster operation into electronic memory. This embedded information is learned through electronic messages received from other electronic modules in the vehicle over the Programmable Communications Interface (PCI) data bus, and through certain hard wired inputs received when the cluster is connected to the vehicle electrically. Once configured, the instrument cluster memory may be irreparably damaged and certain irreversible configuration errors may occur if the cluster is connected electrically to another vehicle; or, if an electronic module from another vehicle is connected that provides data to the instrument cluster (including odometer values) that conflicts with that which was previously learned and stored. Therefore, the practice of exchanging (swapping) instrument clusters and other electronic modules in this vehicle with those removed from another vehicle must always be avoided. Failure to observe this caution may result in instrument cluster damage, which is not reimbursable under the terms of the product warranty. Service replacement instrument clusters are provided with the correct VIN, and the certified odometer value embedded into cluster memory, but will otherwise be automatically configured for compatibility with the features and optional equipment in the vehicle in which they are initially installed.

NOTE: Certain indicators in this instrument cluster are automatically configured. This feature allows those indicators to be activated for deactivated for compatibility with certain optional equipment. If the problem being diagnosed involves improper illumination of the airbag indicator, the cruise indicator, or the SKIM indicator when the vehicle does not have this equipment, the instrument cluster must be replaced with a new unit.

PRELIMINARY DIAGNOSIS

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Check the fused B(+) fuse (Fuse 24 - 10 ampere) in the Power Distribution Center (PDC). If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Check for battery voltage at the fused B(+) fuse (Fuse 24 - 10 ampere) in the PDC. If OK, go to Step 3. If not OK, repair the open B(+) circuit between the PDC and the battery as required.

(3) Disconnect and isolate the battery negative cable. Remove the instrument cluster. Reconnect the battery negative cable. Check for battery voltage at the fused B(+) circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster. If OK, refer to ACTUATOR TEST. If not OK, repair the open fused B(+) circuit between the instrument cluster and the PDC as required.

(4) Check the fused ignition switch output (run-start) fuse (Fuse 10 - 10 ampere) in the fuse block. If OK, go to Step 5. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(5) Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-start) fuse (Fuse 10 - 10 ampere) in the fuse block. If OK, go to Step 6. If not OK, repair the open fused ignition switch output (run-start) circuit between the fuse block and the ignition switch as required.

(6) Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-start) circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster. If OK, refer to ACTUATOR TEST. If not OK, repair the open fused ignition switch output (run-start) circuit between the instrument cluster and the fuse block as required.

(7) Check for continuity between the ground circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster and a good ground. There should be continuity. If OK, refer to ACTUATOR TEST. If not OK, repair the open ground circuit to ground (G201) as required.

INSTRUMENT CLUSTER (Continued)

ACTUATOR TEST

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

The instrument cluster actuator test will put the instrument cluster into its self-diagnostic mode. In this mode the instrument cluster can perform a self-diagnostic test that will confirm that the instrument cluster circuitry, the gauges, the PCI data bus message controlled indicators, and the chime tone generator are capable of operating as designed. During the actuator test the instrument cluster circuitry will sound the chime tone generator, position each of the gauge needles at various calibration points, illuminate all of the segments in the Vacuum-Fluorescent Display (VFD) unit, turn all of the PCI data bus message-controlled indicators on and off again, and display messages in the VFD for certain faults that have been set. It is suggested that a note pad and pencil be used to write down any fault information that is displayed during the test for reference.

Successful completion of the actuator test will confirm that the instrument cluster is operational. However, there may still be a problem with the PCI data bus, the Powertrain Control Module (PCM), the Airbag Control Module (ACM), the Sentry Key Immobilizer Module (SKIM), or the inputs to one of these electronic control modules. Use a DRBIII® scan tool to diagnose these components. Refer to the appropriate diagnostic information.

If an individual gauge does not respond properly, or does not respond at all during the actuator test, the instrument cluster should be removed. However, check that the four screws securing the inoperative gauge to the instrument cluster electronic circuit board are properly tightened before considering instrument cluster replacement. If the gauge mounting screws check OK, replace the faulty instrument cluster.

(1) Begin the test with the ignition switch in the Off position.

(2) Depress the odometer/trip odometer switch button.

(3) While still holding the odometer/trip odometer switch button depressed, turn the ignition switch to the On position, but do not start the engine.

(4) Release the odometer/trip odometer switch button.

(5) The instrument cluster will automatically begin the actuator test sequence, as follows:

(a) The cluster will generate a single chime tone to confirm the functionality of the chime tone generator and the chime control circuitry.

(b) The cluster will scroll the number “8” across the odometer/trip odometer VFD to confirm the functionality of all VFD segments and their control circuitry.

(c) The cluster will illuminate the decimal point in the odometer/trip odometer VFD to confirm the functionality of this VFD segment and its control circuitry.

(d) The cluster will display the EMIC software level in the odometer/trip odometer VFD (example: “SOF 8.9”).

(e) The cluster will display the last six digits (sequence number) of the Vehicle Identification Number (VIN) in the odometer/trip odometer VFD.

(f) If any faults have been set by the cluster, the cluster will display the fault information in the odometer/trip odometer VFD INSTRUMENT CLUSTER FAILURE MESSAGE. If no faults have been set, the cluster will scroll “no FAULtS” across the odometer/trip odometer VFD.

(g) The cluster will turn on, then off again each of the following indicators, one at a time, in sequence to confirm the functionality of the indicator and the cluster control circuitry:

- High Beam
- Brake
- Seatbelt
- MIL
- Check Gauges
- Low Fuel
- 4WD
- SKIS
- Cruise

(h) The cluster will sweep the needles for each of the following gauges, one at a time, to several calibration points in sequence to confirm the functionality of the gauge and the cluster control circuitry:

- Speedometer
- Fuel
- Temperature
- Tachometer
- Voltage
- Oil Pressure

INSTRUMENT CLUSTER (Continued)

INSTRUMENT CLUSTER FAILURE MESSAGE

VFD Message	Description	Correction
″buS b0″	PCM - MIL Message	The cluster is not receiving a MIL lamp message from the PCM. A DRBIII® scan tool is required for further diagnosis. Refer to the appropriate diagnostic information.
″buS b1″	SKIM - SKIM Message	The cluster is not receiving a SKIS lamp message from the SKIM. A DRBIII® scan tool is required for further diagnosis. Refer to the appropriate diagnostic information.
″buS b8″	ACM - Airbag Message	The cluster is not receiving an Airbag lamp message from the ACM. A DRBIII® scan tool is required for further diagnosis. Refer to the appropriate diagnostic information.
″PanEL OPEn″	Panel Sense - Open Circuit	The cluster is not receiving an input from the the panel lamps dimmer circuitry of the left multi-function switch on the panel lamps dimmer signal circuit. Repair the open circuit or replace the faulty switch as required.
″Airbag″	Telltale Open/Shorted	The EMIC airbag indicator is open or shorted. Replace the faulty cluster.

(6) The actuator test is now completed. The instrument cluster will automatically exit the self-diagnostic mode and return to normal operation at the completion of the test, if the ignition switch is turned to the Off position during the test, or if a vehicle speed message indicating that the vehicle is moving is received from the PCM over the PCI data bus during the test.

(7) Go back to Step 1 to repeat the test, if necessary.

INTERIOR LIGHTING CONTROL DIAGNOSIS

Before performing this test, complete the testing of each of the hard wired interior lighting switches. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

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NOTE: The following tests may not prove conclusive in the diagnosis of this system. The most reliable, efficient, and accurate means to diagnose the Interior Lighting Control function of the instrument cluster requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

(1) Check the door ajar switch output fuse (Fuse 4 - 10 ampere) in the fuse block. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Check for continuity between the door ajar switch output fuse (Fuse 4 - 10 ampere) in the fuse block and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open ground circuit between the fuse block and ground (G300) as required.

(3) Disconnect and isolate the battery negative cable. Disconnect the body wire harness connector for the driver and/or passenger door ajar switch from the switch connector receptacle. Check for continuity between the door ajar switch output circuit cavity of the driver or passenger door ajar switch and a good ground. There should be continuity. If OK, go to Step 4. If not OK, repair the open door ajar switch output

circuit between the driver or passenger door ajar switch and the fuse block as required.

(4) Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the cluster connector receptacle. Check for continuity between the driver and/or passenger door ajar switch sense circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 5. If not OK, repair the shorted driver and/or passenger door ajar switch sense circuits between the instrument cluster and the driver and/or passenger door ajar switches as required.

(5) Check for continuity between the driver and/or passenger door ajar switch sense circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the body wire harness connector for the driver and/or passenger door ajar switches. There should be continuity. If OK, use a DRBIII® scan tool to complete the diagnosis of the instrument cluster interior lighting control. Refer to the appropriate diagnostic information. If not OK, repair the open driver and/or passenger door ajar switch sense circuits between the instrument cluster and the driver and/or passenger door ajar switches as required.

AXLE LOCKER CONTROL DIAGNOSIS

If the problem being diagnosed involves a rear or front lock indicator in the instrument cluster that is blinking on and off, be certain to complete inspection of the appropriate front or rear axle locker relays, locker pumps, pneumatic lines, locker indicator switches, and axle locker mechanisms before performing the following tests. If the problem being diagnosed involves a rear or front lock indicator in the instrument cluster that stays on when it should be off, or stays off when it should be on, complete the testing of the axle locker switch before performing the following tests.

These tests will establish the integrity of the hard wired circuits related to the axle locker control function of the instrument cluster. However, proper testing of the instrument cluster programming and the electronic vehicle speed and transfer case shift position messages received by the cluster over the PCI data bus requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, DRIVER AIRBAG, PASSENGER AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable. Remove the rear and front axle locker relays from their receptacles in the Power Distribution Center (PDC). Remove the axle lock switch from the instrument panel. Disconnect the instrument panel wire harness connector for the axle lock switch from the switch connector receptacle. Remove the instrument cluster from the instrument panel.

(2) Check for continuity between each of the rear and front axle lock request circuit cavities of the appropriate instrument panel wire harness connectors (Connector C1 for rear, Connector C2 for front) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted rear or front axle lock request circuit between the instrument cluster, the axle lock switch, and the rear or front axle locker relay as required.

(3) Check for continuity between each of the rear and/or front axle lock request circuit cavities of the instrument panel wire harness connectors (Connector C1 for rear, Connector C2 for front) for the instrument cluster, the instrument panel wire harness connector for the axle lock switch, and the rear or front axle locker relay receptacle in the PDC. In each case, there should be continuity. If OK, go to Step 4. If not OK, repair the open rear and/or front axle lock request circuit between the instrument cluster, the axle lock switch, and the PDC as required.

(4) Check for continuity between the axle lock switch enable 1 and/or enable 2 circuit cavity of the instrument panel wire harness connector (Connector C1 for enable 1, Connector C2 for enable 2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 5. If not OK, repair the shorted axle lock switch enable 1 or enable 2 circuit between the instrument cluster and the axle lock switch as required.

(5) Check for continuity between the axle lock switch enable 1 and/or enable 2 circuit cavities of the instrument panel wire harness connector (Connector

INSTRUMENT CLUSTER (Continued)

C1 for enable 1, Connector C2 for enable 2) for the instrument cluster and the instrument panel wire harness connector for the axle lock switch. There should be continuity. If OK, go to Step 6. If not OK, repair the open axle lock switch enable 1 or enable 2 circuit between the instrument cluster and the axle lock switch as required.

(6) Disconnect the rear and/or front locker indicator switch pigtail harness connector from the rear or front axle jumper harness connector. Check for continuity between the ground circuit cavity of the rear or front axle jumper harness connector and a good ground. There should be continuity. If OK, go to Step 7. If not OK, repair the open ground circuit between the rear or front axle jumper harness connector for the rear or front locker indicator switch and ground (G105) as required.

(7) Check for continuity between the ground circuit and the rear or front locker indicator switch sense circuit cavities of the rear or front locker indicator switch pigtail harness connector. There should be continuity with the axle locker engaged, and no continuity with the axle locker disengaged. If OK, go to Step 8. If not OK, replace the faulty rear or front locker indicator switch.

(8) Check for continuity between the rear or front locker indicator switch sense circuit cavity of the rear or front axle jumper harness connector for the locker indicator switch and a good ground. There should be no continuity. If OK, go to Step 9. If not OK, repair the shorted rear or front locker indicator switch sense circuit between the rear or front axle jumper harness connector for the locker indicator switch and the instrument cluster as required.

(9) Check for continuity between the rear or front locker indicator switch sense circuit cavities of the rear or front axle jumper harness connector for the locker indicator switch and the instrument cluster. There should be continuity. If OK, use a DRBIII® scan tool to perform additional testing of the instrument cluster. Refer to the appropriate diagnostic information. If not OK, repair the open rear or front locker indicator switch sense circuit between the rear or front axle jumper harness connector for the locker indicator switch and the instrument cluster as required.

REAR WINDOW DEFOGGER CONTROL DIAGNOSIS

Before performing this test, complete the testing of the rear window defogger switch and the rear window defogger relay. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the rear window defogger relay from the receptacle in the Power Distribution Center (PDC). Disconnect the instrument panel wire harness connector for the rear window defogger switch from the switch connector receptacle. Remove the instrument cluster from the instrument panel.

(2) Check for continuity between the rear window defogger switch sense circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted rear window defogger switch sense circuit between the instrument cluster and the rear window defogger switch as required.

(3) Check for continuity between the rear window defogger switch sense circuit cavities of the instrument panel wire harness connectors for the instrument cluster (Connector C2) and the rear window defogger switch. There should be continuity. If OK, go to Step 4. If not OK, repair the open rear window defogger switch sense circuit between the instrument cluster and the rear window defogger switch as required.

(4) Check for continuity between the rear window defogger relay control circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 5. If not OK, repair the shorted rear window defogger relay control circuit between the instrument cluster and the PDC as required.

(5) Check for continuity between the rear window defogger relay control circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the rear window defogger relay receptacle in the PDC. There should be continuity. If OK, replace the faulty instrument

cluster. If not OK, repair the open rear window defogger relay control circuit between the instrument cluster and the PDC as required.

CLUSTER ILLUMINATION DIAGNOSIS

The diagnosis found here addresses an inoperative instrument cluster illumination lamp condition. If the problem being diagnosed is a single inoperative illumination lamp, be certain that the bulb and bulb holder unit are properly installed in the instrument cluster electronic circuit board. If no installation problems are found replace the faulty bulb and bulb holder unit. If all of the cluster illumination lamps are inoperative and the problem being diagnosed includes inoperative exterior lighting controlled by the left multi-function switch, that system needs to be repaired first. If the exterior lamps controlled by the left multi-function switch are inoperative, (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/HEADLAMP - DIAGNOSIS AND TESTING). If no exterior lighting system problems are found, the following procedure will help locate a short or open in the cluster illumination lamp circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Disconnect the body wire harness connector for the left multi-function switch from the switch connector receptacle. Check for continuity between the ground circuit cavity of the body wire harness connector for the left multi-function switch and a good ground. There should be continuity. If OK, go to Step 2. If not OK, repair the open ground circuit to ground (G300) as required.

(2) Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the cluster connector receptacle. Check for continuity between the panel lamp dimmer signal circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted panel lamp dimmer signal circuit between the instrument cluster and the left multi-function switch as required.

(3) Check for continuity between the panel lamp dimmer signal circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the body wire harness connector for the left multi-function switch. There should be continuity. If OK, use a DRBIII® scan tool to complete the diagnosis of the instrument cluster illumination lighting. Refer to the appropriate diagnostic information. If not OK, repair the open panel lamp dimmer signal circuit between the instrument cluster and the left multi-function switch as required.

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the cluster bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/CLUSTER BEZEL - REMOVAL).

(3) Remove the four screws that secure the instrument cluster to the instrument panel structural support (Fig. 4).

(4) Pull the instrument cluster rearward far enough to access and disconnect the two instrument panel wire harness connectors for the instrument cluster from the connector receptacles on the back of the cluster housing.

(5) Remove the instrument cluster from the instrument panel.

INSTRUMENT CLUSTER (Continued)

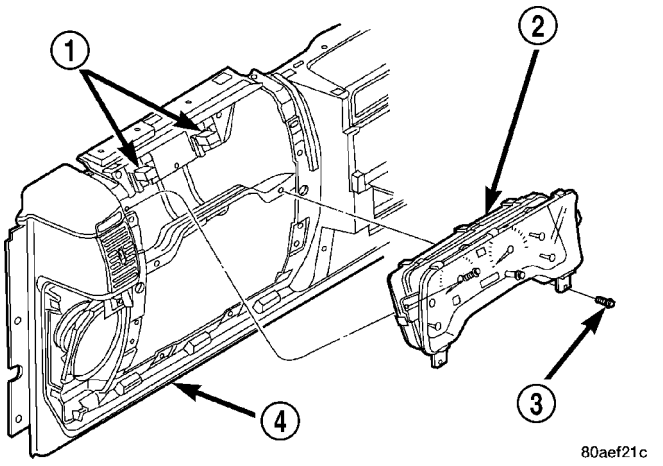

Fig. 4 Instrument Cluster Remove/Install

1 - WIRE HARNESS CONNECTORS
2 - INSTRUMENT CLUSTER
3 - SCREW (4)
4 - INSTRUMENT PANEL

DISASSEMBLY

Some of the components for the instrument cluster used in this vehicle are serviced individually. The serviced components include: the incandescent instrument cluster indicator and illumination lamp bulbs (including the integral bulb holders), the cluster lens, the trip odometer reset button boot, the cluster hood and mask unit, and the cluster housing rear cover. Following are the procedures for disassembling these components from the instrument cluster unit.

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CLUSTER BULB

This procedure applies to each of the incandescent cluster illumination lamp or indicator bulb and bulb holder units. However, the illumination lamps and the indicators use different bulb and bulb holder unit sizes. They must never be interchanged.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Turn the bulb holder counterclockwise about sixty degrees on the cluster electronic circuit board (Fig. 5).

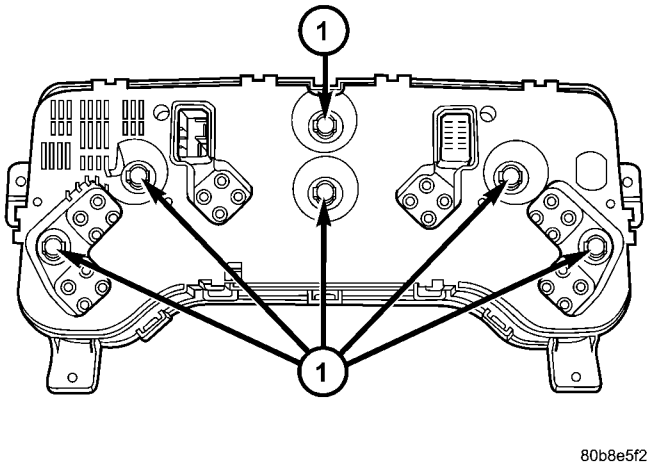

Fig. 5 Cluster Bulb Locations

1 - CLUSTER INCANDESCENT BULBS

(4) Pull the bulb and bulb holder unit straight back to remove it from the bulb mounting hole in the cluster electronic circuit board.

CLUSTER LENS

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Work around the perimeter of the cluster lens and disengage each of the eight latches that secure the lens to the cluster mask and the cluster housing (Fig. 6).

(4) Gently pull the cluster lens away from the face of the instrument cluster.

TRIP ODOMETER RESET BUTTON BOOT

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Remove the cluster lens from the cluster housing. Refer to CLUSTER LENS.

(4) Remove the odometer reset button boot by pulling it straight out of the pocketed hole from the face of the cluster lens (Fig. 6).

INSTRUMENT CLUSTER (Continued)

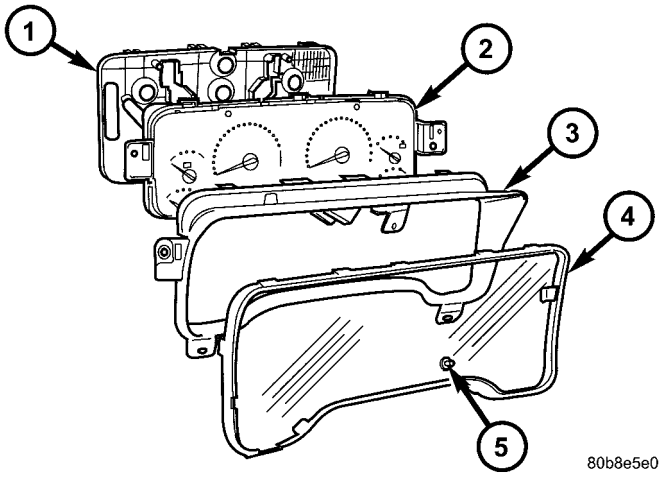

Fig. 6 Instrument Cluster Components

1 - REAR COVER
2 - CLUSTER HOUSING
3 - CLUSTER HOOD & MASK
4 - CLUSTER LENS
5 - SWITCH BUTTON BOOT

CLUSTER HOOD AND MASK

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Remove the cluster lens from the cluster housing. Refer to CLUSTER LENS.

(4) Work around the perimeter of the cluster hood and mask unit and disengage each of the five latches that secure the cluster hood and mask unit to the cluster housing (Fig. 6).

(5) Gently pull the cluster hood and mask unit away from the face of the instrument cluster housing.

CLUSTER HOUSING REAR COVER

(1) Disconnect and isolate the battery negative cable.

(2) Remove the instrument cluster from the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - REMOVAL).

(3) Remove the screw adjacent to each of the two cluster connector receptacles that secure the rear cover to the cluster housing.

(4) Work around the perimeter of the cluster housing rear cover and disengage each of the eight latches that secure the cover to the outside of the cluster housing (Fig. 6).

(5) Disengage the one inboard latch located in a receptacle near the lower edge of the rear cover just to the right of center that secures the rear cover to the cluster housing.

(6) Gently pull the cluster housing rear cover away from the back of the cluster housing.

ASSEMBLY

Some of the components for the instrument cluster used in this vehicle are serviced individually. The serviced components include: the incandescent instrument cluster indicator and illumination lamp bulbs (including the integral bulb holders), the cluster lens, the trip odometer reset button boot, the cluster hood and mask unit, and the cluster housing rear cover. Following are the procedures for disassembling these components from the instrument cluster unit.

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CLUSTER BULB

This procedure applies to each of the incandescent cluster illumination lamp or indicator bulb and bulb holder units. However, the illumination lamps and the indicators use different bulb and bulb holder unit sizes. They must never be interchanged.

CAUTION: Be certain that any bulb and bulb holder unit removed from the cluster electronic circuit board is reinstalled in the correct position. Always use the correct bulb size and type for replacement. An incorrect bulb size or type may overheat and cause damage to the instrument cluster, the electronic circuit board and/or the gauges.

(1) Insert the bulb and bulb holder unit straight into the correct bulb mounting hole in the cluster electronic circuit board (Fig. 5).

(2) With the bulb holder fully seated against the cluster electronic circuit board, turn the bulb holder clockwise about sixty degrees to lock it into place.

(3) Reinstall the instrument cluster onto the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(4) Reconnect the battery negative cable.

INSTRUMENT CLUSTER (Continued)

CLUSTER LENS

(1) Position the cluster lens over the cluster hood and mask unit on the face of the instrument cluster (Fig. 6). Be certain that the trip odometer reset switch button is aligned with and inserted into the reset button boot in the lens.

(2) Press firmly and evenly on the cluster lens to install it onto the instrument cluster housing.

(3) Work around the perimeter of the cluster lens making certain that each of the eight latches that secure the lens to the cluster mask and the cluster housing is fully engaged.

(4) Reinstall the instrument cluster onto the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(5) Reconnect the battery negative cable.

TRIP ODOMETER RESET BUTTON BOOT

(1) Position the trip odometer reset button boot into the pocketed mounting hole from the back of the cluster lens (Fig. 6).

(2) Gently pull the tip of the trip odometer reset button boot from the face of the cluster lens until it is fully seated in the pocketed mounting hole.

(3) Reinstall the cluster lens onto the cluster housing. Refer to CLUSTER LENS.

(4) Reinstall the instrument cluster onto the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(5) Reconnect the battery negative cable.

CLUSTER HOOD AND MASK

(1) Align the cluster hood and mask unit with the face of the instrument cluster housing (Fig. 6).

(2) Press firmly and evenly on the cluster hood and mask unit to install it onto the cluster housing.

(3) Work around the perimeter of the cluster hood and mask unit making certain that each of the five latches that secure the hood and mask unit to the instrument cluster housing is fully engaged.

(4) Reinstall the cluster lens onto the cluster housing. Refer to CLUSTER LENS.

(5) Reinstall the instrument cluster onto the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(6) Reconnect the battery negative cable.

CLUSTER HOUSING REAR COVER

(1) Position the cluster housing rear cover to the back of the instrument cluster housing.

(2) Press firmly and evenly on the cluster housing rear cover to install it onto the back of the instrument cluster housing.

(3) Work around the perimeter of the cluster housing rear cover making certain that each of the eight latches that secure the rear cover to the instrument cluster housing is fully engaged.

(4) Install and tighten the two screws that secures the rear cover to the instrument cluster housing adjacent to each cluster connector receptacle. Tighten the screws to 2 N·m (20 in. lbs.).

(5) Reinstall the instrument cluster onto the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - INSTALLATION).

(6) Reconnect the battery negative cable.

INSTALLATION

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(1) Position the instrument cluster to the instrument panel.

(2) Reconnect the two instrument panel wire harness connectors for the cluster to the connector receptacles on the back of the cluster housing (Fig. 4).

(3) Position the instrument cluster into the instrument panel structural support.

(4) Install and tighten the four screws that secure the instrument cluster to the instrument panel structural support. Tighten the screws to 2.2 N·m (20 in. lbs.).

(5) Reinstall the cluster bezel onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/CLUSTER BEZEL - INSTALLATION).

(6) Reconnect the battery negative cable.

NOTE: Certain indicators in this instrument cluster are automatically configured. This feature allows those indicators to be activated for deactivated for compatibility with certain optional equipment. If the problem being diagnosed involves improper illumination of the airbag indicator, the cruise indicator, or the SKIM indicator when the vehicle does not have this equipment, the instrument cluster must be replaced with a new unit.

ABS INDICATOR

DESCRIPTION

Fig. 7 ABS Indicator

An Antilock Brake System (ABS) indicator is standard equipment on all instrument clusters (Fig. 7). However, this indicator is only functional on vehicles equipped with the ABS option. The ABS indicator is located near the lower edge of the instrument cluster, to the left of center. The ABS indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Failure of Anti-lock Braking System” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The ABS indicator is serviced as a unit with the instrument cluster.

OPERATION

The ABS indicator gives an indication to the vehicle operator when the ABS system is faulty or inoperative. This indicator is hard wired on the instrument cluster electronic circuit board, and is completely controlled by the Controller Antilock Brake (CAB). The ABS indicator Light Emitting Diode (LED) receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions; therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the CAB through the CAB relay in the Power Distribution Center (PDC). The CAB will turn on the ABS indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the ABS indicator is illuminated by the cluster for about two seconds as a bulb test.
- **ABS Self-Test Fault** - Each time the CAB detects a fault in a monitored ABS circuit, the ABS indicator will be illuminated. The indicator remains illuminated until the fault condition is corrected, or until the ignition switch is turned to the Off position, whichever occurs first.
- **ABS Diagnostic Test** - The ABS indicator is blinked on and off by the CAB during the performance of the ABS diagnostic tests.

The CAB continually monitors the ABS circuits and sensors to decide whether the system is in good operating condition. If the CAB turns the ABS indicator on after the bulb test, it indicates that the CAB has detected a system malfunction and/or that the ABS system has become inoperative. The CAB will store a Diagnostic Trouble Code (DTC) for any malfunction it detects. (Refer to 5 - BRAKES - DESCRIPTION). The ABS indicator can be diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - ABS INDICATOR

The diagnosis found here addresses an inoperative Antilock Brake System (ABS) indicator condition. If there are problems with several indicators in the instrument cluster, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the ABS indicator stays on with the ignition switch in the On position or comes on and stays on while driving, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If no ABS problem is found, the following procedure will help to locate a short or open in the ABS warning indicator driver circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

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(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Reconnect the battery negative cable. Turn the ignition switch to the On

ABS INDICATOR (Continued)

position and within about two seconds check for continuity between the ABS warning indicator driver circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be continuity for about two seconds after ignition On, and then an open circuit. If OK, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If not OK, go to Step 2.

(2) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the Controller Antilock Brake (CAB) relay from the Power Distribution Center (PDC). Check for continuity between the ABS warning indicator driver circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted ABS warning indicator driver circuit between the instrument cluster and the PDC as required.

(3) Check for continuity between the ABS warning indicator driver circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the CAB relay receptacle in the PDC. There should be continuity. If OK, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If not OK, repair the open ABS warning indicator driver circuit between the instrument cluster and the PDC as required.

AIRBAG INDICATOR

DESCRIPTION

Fig. 8 Airbag Indicator

An airbag indicator is standard equipment on all instrument clusters (Fig. 8). However, the instrument cluster can be programmed to disable this indicator on vehicles that are not equipped with the airbag system, which is not available in some markets. The airbag indicator is located near the lower edge of the instrument cluster, to the right of center. The airbag indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Airbag” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The airbag indicator is serviced as a unit with the instrument cluster.

OPERATION

The airbag indicator gives an indication to the vehicle operator when the airbag system is faulty or inoperative. The airbag indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Airbag Control Module (ACM) over the Programmable Communications Interface (PCI) data bus. The airbag indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the airbag indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the airbag indicator is illuminated for about seven seconds. The first two seconds is the cluster bulb test function, and the remainder is the ACM bulb test function.
- **ACM Lamp-On Message** - Each time the cluster receives a lamp-on message from the ACM, the airbag indicator will be illuminated. The indicator remains illuminated for about twelve seconds or until the cluster receives a lamp-off message from the ACM, whichever is longer.
- **Communication Error** - If the cluster receives no airbag messages for five consecutive seconds, the airbag indicator is illuminated. The indicator remains illuminated until the cluster receives a single lamp-off message from the ACM.
- **Actuator Test** - Each time the cluster is put through the actuator test, the airbag indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry. The actuator test illumination of the airbag indicator is a function of the ACM.

The ACM continually monitors the airbag system circuits and sensors to decide whether the system is in good operating condition. The ACM then sends the proper lamp-on or lamp-off messages to the instrument cluster. If the ACM sends a lamp-on message after the bulb test, it indicates that the ACM has detected a system malfunction and/or that the airbags may not deploy when required, or may deploy

AIRBAG INDICATOR (Continued)

when not required. The ACM will store a Diagnostic Trouble Code (DTC) for any malfunction it detects. Each time the airbag indicator fails to illuminate due to an open or short in the cluster airbag indicator circuit, the cluster sends a message notifying the ACM of the condition and stores a DTC. For proper diagnosis of the airbag system, the ACM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the airbag indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

BRAKE/PARK BRAKE INDICATOR

DESCRIPTION

Fig. 9 Brake Indicator

A brake indicator is standard equipment on all instrument clusters (Fig. 9). The brake indicator is located near the lower edge of the instrument cluster, to the left of center. The brake indicator consists of a stencil-like cutout of the word “BRAKE” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “BRAKE” text to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The brake indicator is serviced as a unit with the instrument cluster.

OPERATION

The brake indicator gives an indication to the vehicle operator when the parking brake is applied, or when there are certain brake hydraulic system malfunctions. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming, electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus, and hard wired inputs to the instrument cluster from the park brake switch and the brake warning indicator switch. The brake indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The cluster can illuminate the LED solid, or flash it on and off at about one flash per second. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the brake indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the brake indicator is illuminated by the instrument cluster for about four seconds as a bulb test.
- **Park Brake Switch Input** - Each time the cluster detects ground on the red brake warning indicator driver circuit (park brake switch closed = park brake applied or not fully released) the brake indicator is illuminated solid. If a vehicle speed message is received by the cluster from the PCM over the PCI data bus indicating the vehicle is moving while the red brake warning indicator driver input is grounded, the brake indicator is flashed on and off repeatedly. Whether illuminated solid or flashing, the indicator remains illuminated until the red brake warning indicator driver input to the cluster is an open circuit (park brake switch open = park brake fully released), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Brake Hydraulic System Malfunction** - Each time the cluster detects ground on the red brake warning indicator driver circuit (brake warning indicator switch closed = pressures in the two halves of the split brake hydraulic system are unequal) the brake indicator is illuminated solid. The indicator remains illuminated until the red brake warning indicator driver input to the cluster is an open circuit (brake warning indicator switch open = brake hydraulic system pressures are equal), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the instrument cluster is put through the actuator test, the brake indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The park brake switch and the brake warning pressure switch are each connected in parallel between ground and the red brake warning indicator driver input to the instrument cluster so that each of their inputs will illuminate the indicator independently of the other. The park brake switch and brake warning indicator switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. (Refer to 5 - BRAKES/PARKING BRAKE - OPERATION). (Refer to 5 - BRAKES/ELECTRICAL/BRAKE PRESSURE SWITCH - OPERATION).

BRAKE/PARK BRAKE INDICATOR (Continued)

For further diagnosis of the brake indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the PCM, the PCI data bus, or the message inputs to the instrument cluster that control the brake indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

DIAGNOSIS AND TESTING - BRAKE INDICATOR

The diagnosis found here addresses an inoperative brake indicator condition. If there are problems with several indicators in the instrument cluster, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the brake indicator stays on with the ignition switch in the On position and the park brake released, or comes on while driving, (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If no brake system problem is found, the following procedures will help to locate a shorted or open circuit, or a faulty switch input. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

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INDICATOR ILLUMINATES DURING BULB TEST, BUT DOES NOT WHEN PARK BRAKE APPLIED

(1) Disconnect and isolate the battery negative cable. Disconnect the floor wire harness connector for the park brake switch from the switch terminal. Apply the parking brake. Check for continuity between the park brake switch terminal and a good ground. There should be continuity. If OK, go to Step 2. If not OK, replace the faulty park brake switch.

(2) Disconnect the headlamp and dash wire harness connector for the brake warning indicator switch from the switch terminals. Check for continuity between the red brake warning indicator driver (G9) circuit cavities of the floor wire harness connector for the park brake switch and the headlamp and dash wire harness connector for the brake warning indicator switch. There should be continuity. If not OK, repair the open red brake warning indicator driver (G9) circuit between the park brake switch and the brake warning indicator switch as required.

INDICATOR REMAINS ILLUMINATED - BRAKE SYSTEM CHECKS OK

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Disconnect the headlamp and dash wire harness connector for the brake warning indicator switch from the switch terminals. Check for continuity between the red brake warning indicator driver circuit (G99) cavity of the headlamp and dash wire harness connector for the brake warning indicator switch and a good ground. There should be no continuity. If OK, go to Step 2. If not OK, repair the shorted red brake warning indicator driver circuit (G99) between the brake warning indicator switch and the instrument cluster as required.

(2) Disconnect the floor wire harness connector for the park brake switch from the switch terminal. Check for continuity between the red brake warning indicator driver circuit (G9) cavity of the headlamp and dash wire harness connector for the brake warning indicator switch and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted red brake warning indicator driver circuit (G9) between the brake warning indicator switch and the park brake switch as required.

(3) Check for continuity between each of the two terminals of the brake warning indicator switch and a good ground. In each case, there should be no continuity. If OK, go to Step 4. If not OK, replace the faulty brake warning indicator switch.

(4) Check for continuity between the terminal of the park brake switch and a good ground. There should be no continuity with the park brake released, and continuity with the park brake applied. If not OK, replace the faulty park brake switch.

CHECK GAUGES INDICATOR

DESCRIPTION

Fig. 10 Check Gauges Indicator

A check gauges indicator is standard equipment on all instrument clusters (Fig. 10). The check gauges indicator is located on the lower edge of the instrument cluster, to the right of center. The check gauges indicator consists of a stencil-like cutout of the words "CHECK GAUGES" in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the "CHECK GAUGES" text to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The check gauges indicator is serviced as a unit with the instrument cluster.

OPERATION

The check gauges indicator gives an indication to the vehicle operator when certain instrument cluster gauge readings reflect a condition requiring immediate attention. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The check gauges indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the check gauges indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the check gauges indicator is illuminated for about two seconds as a bulb test.
- **Engine Temperature High Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is about 127° C (261° F) or higher, the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine temperature is about 124° C (255° F) or lower, or until the ignition switch is turned to the Off position, whichever occurs first.
- **Engine Oil Pressure Low Message** - Each time the cluster receives a message from the PCM indicating the engine oil pressure of a 2.4L engine is below about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is below about 0.4 kg/cm^2 (6 psi), the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine oil pressure of a 2.4L engine is above about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is above about 0.4 kg/cm^2 (6 psi), or until the ignition switch is turned to the Off position, whichever occurs first. The cluster will only turn the indicator on in response to an engine oil pressure low message if the engine speed is 300 rpm or greater for more than about five seconds.
- **System Voltage Low (Charge Fail) Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is about 9.0 volts or lower (charge fail condition), the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is greater than about 12.0 volts (but less than 16.0 volts), or until the ignition switch is turned to the Off position, whichever occurs first.
- **System Voltage High Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is greater than about 16.0 volts, the check gauges indicator will be illuminated. The indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is less than about 16.0 volts (but higher than 9.0 volts), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the check gauges indicator will be turned on, then off again during the bulb check portion of the test in order to confirm the functionality of the LED and the cluster control circuitry.

The PCM continually monitors the engine temperature, oil pressure, and electrical system voltage, then sends the proper messages to the instrument cluster. For further diagnosis of the check gauges indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the check gauges indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

CRUISE INDICATOR

DESCRIPTION

CRUISE

Fig. 11 Cruise Indicator

A cruise indicator is standard equipment on all instrument clusters (Fig. 11). However, on vehicles not equipped with the optional speed control system, this indicator is electronically disabled. The cruise indicator is located near the upper edge of the instrument cluster, to the left of center. The cruise indicator consists of a stencil-like cutout of the word “CRUISE” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A green Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “CRUISE” text to appear in green through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the cruise indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The cruise indicator is serviced as a unit with the instrument cluster.

OPERATION

The cruise indicator gives an indication to the vehicle operator when the speed control system is turned On, regardless of whether the speed control is engaged. This indicator is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The cruise indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the cruise indicator for the following reasons:

- **Cruise Lamp-On Message** - Each time the cluster receives a cruise lamp-on message from the PCM indicating the speed control system has been turned On, the cruise indicator is illuminated. The indicator remains illuminated until the cluster receives a cruise lamp-off message from the PCM or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the cruise indicator will be turned on, then off again during the bulb check portion of the test in order to confirm the functionality of the LED and the cluster control circuitry.

The PCM continually monitors the speed control switches to determine the proper outputs to the speed control servo. The PCM then sends the proper cruise indicator lamp-on and lamp-off messages to the instrument cluster. For further diagnosis of the cruise indicator or the instrument cluster circuitry that controls the indicator, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the speed control system, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the cruise indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

ENGINE TEMPERATURE GAUGE

DESCRIPTION

Fig. 12 Engine Coolant Temperature Icon

An engine coolant temperature gauge is standard equipment on all instrument clusters. The engine coolant temperature gauge is located in the lower right quadrant of the instrument cluster, below the oil pressure gauge. The engine coolant temperature gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from 40° C (or 100° F) to 125° C (or 260° F). An International Control and Display Symbol icon for “Engine Coolant Temperature” is located on the cluster overlay, directly above the hub of the gauge needle (Fig. 12). The engine coolant temperature gauge graphics are white and blue against a black field except for a single red graduation at the high end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb

ENGINE TEMPERATURE GAUGE (Continued)

holder units located on the instrument cluster electronic circuit board. The engine coolant temperature gauge is serviced as a unit with the instrument cluster.

OPERATION

The engine coolant temperature gauge gives an indication to the vehicle operator of the engine coolant temperature. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The engine coolant temperature gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Engine Temperature Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is between about 40° C (100° F) and 124° C (255° F), the gauge needle is moved to the actual relative temperature position on the gauge scale.
- **Engine Temperature Low Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is at or below about 40° C (100° F), the gauge needle is held at the 40° C (100° F) graduation at the far left end of the gauge scale. The gauge needle remains at the left end of the gauge scale until the cluster receives a message from the PCM indicating that the engine temperature is above about 40° C (100° F), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Engine Temperature High Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is above about 127° C (261° F), the gauge needle is moved into the red zone at the far right end of the gauge scale, the check gauges indicator is illuminated, and a single chime tone is sounded. The gauge needle remains in the red zone and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine temperature is below about 124° C (255° F), or until the ignition switch is turned to the Off position, whichever occurs first. The chime tone feature will occur only once per ignition cycle.
- **Engine Temperature Critical Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is above about 129° C (264° F), the gauge needle is moved to the far right end of the red zone on the gauge scale. The gauge needle remains at the far right end of the red zone until the cluster receives a message from the PCM indicating that the engine temperature is below about 127° C (261° F), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Communication Error** - If the cluster fails to receive an engine temperature message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the engine coolant temperature sensor to determine the engine operating temperature. The PCM then sends the proper engine coolant temperature messages to the instrument cluster. For further diagnosis of the engine coolant temperature gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a high or critical engine temperature gauge reading, it may indicate that the engine or the engine cooling system requires service. For proper diagnosis of the engine coolant temperature sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the engine coolant temperature gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

FOG LAMP INDICATOR

DESCRIPTION

Fig. 13 Fog Lamp Indicator

A fog lamp indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional front and/or rear fog lamps (Fig. 13). The fog lamp indicator is located near the upper edge of the instrument cluster, to the left of center. The fog lamp indicator consists of a

FOG LAMP INDICATOR (Continued)

stencil-like cutout of the International Control and Display Symbol icon for “Fog Lamps” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A green Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in green through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The fog lamp indicator is serviced as a unit with the instrument cluster.

OPERATION

The fog lamp indicator gives an indication to the vehicle operator whenever the optional front and/or rear fog lamps are illuminated. The availability of the front fog lamps, or rear fog lamps options varies by the market for which the vehicle is manufactured. This indicator is controlled by a hard wired input to the cluster from the fog lamp switch circuitry of the left multi-function switch. The fog lamp indicator Light Emitting Diode (LED) is grounded on the instrument cluster electronic circuit board at all times. Therefore, the LED will be on any time the front or rear fog lamps are illuminated, regardless of the ignition switch position. The LED only illuminates when it is provided battery current by the fog lamp switch circuitry of the left multi-function switch.

The fog lamp switch circuitry of the left multi-function switch is connected in series between a fused B(+) fuse in the Power Distribution Center (PDC) and the front or rear fog lamp feed input to the instrument cluster through the fog lamp relay, which is also in the PDC. The fog lamp switch input to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For proper diagnosis of the fog lamp switch and circuits, (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/ LEFT MULTI-FUNCTION SWITCH - DIAGNOSIS AND TESTING).

DIAGNOSIS AND TESTING - FRONT/REAR FOG LAMP INDICATOR

The diagnosis found here addresses an inoperative front/rear fog lamp indicator condition. Before beginning this test, confirm the functionality of the front or rear fog lamp system. If no fog lamp system problem is found, the following procedure will help to locate an open in the front or rear fog lamp feed circuit between the fog lamp relay and the instrument cluster. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pinout information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C1) for the instrument cluster from the connector receptacle on the back of the cluster housing.

(2) Reconnect the battery negative cable. Turn the fog lamps on by pulling out the control knob on the end of the left-multi-function switch control stalk. Check for battery voltage at the front/rear fog lamp feed circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster. If OK, replace the faulty instrument cluster. If not OK, repair the open front/rear fog lamp feed circuit between the fog lamp relay and the instrument cluster as required.

FRONT LOCK INDICATOR

DESCRIPTION

Fig. 14 Front Lock Indicator

A front lock indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional off-road package (Fig. 14). The front lock indicator is located near the upper edge of the instrument cluster, to the right of center. The front lock indicator consists of a stencil-like cutout of the words “FRONT LOCK” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber

FRONT LOCK INDICATOR (Continued)

Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the "FRONT LOCK" text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the front lock indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The front lock indicator is serviced as a unit with the instrument cluster.

OPERATION

The front lock indicator gives an indication to the vehicle operator of the status of the locker mechanism in the front axle of vehicles equipped with the optional off-road package. The front lock indicator can also give an indication when certain faults are detected in the axle locker system. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming, and hard wired inputs to the cluster from the key-in ignition switch, the axle lock switch, and the front locker indicator switch. The front lock indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the LED can be illuminated regardless of the ignition switch position. The cluster can illuminate the LED solid, or flash it on and off at two different rates, slow or fast. The slow rate is about one flash per second, while the fast rate is about two flashes per second. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the front lock indicator for the following reasons:

- **Front Axle Lock Request Input** - Each time the cluster detects ground on the front axle lock request circuit from the axle lock switch (front axle lock request circuit ground = front axle lock request active) while the key is in the ignition lock cylinder, the front lock indicator flashes on and off at a slow rate. The indicator continues to flash until the front axle lock request input to the cluster is an open circuit (front axle lock request circuit open = front axle lock request inactive), or until the cluster detects a change in the inputs from the front locker indicator switch or the key-in ignition switch, whichever occurs first.
- **Front Locker Indicator Switch Input** - Each time the cluster detects ground on the front locker indicator switch sense circuit (front locker indicator switch closed = front axle locker engaged) while the front axle lock request is active and the key is in the ignition lock cylinder, the front lock indicator is illuminated solid. The indicator remains illuminated solid until the front locker indicator switch input to the cluster is an open circuit (front locker indicator switch open = front axle locker disengaged), or until the cluster detects a change in the inputs from the axle lock switch or the key-in ignition switch, whichever occurs first.
- **Front Lock Request/Feedback Mismatch** - Each time the cluster detects that the inputs from the axle lock switch (request) on the front axle lock request circuit and from the front locker indicator switch (feedback) on the front locker indicator switch sense circuit are mismatched, the front lock indicator flashes on and off at a slow rate. This condition occurs when the lock request is active but the axle locker remains disengaged, or when the lock request is inactive but the axle locker remains engaged. The indicator continues to flash until the request and the feedback are matching.
- **Key-In Ignition Switch Input** - Each time the cluster detects that the key has been removed from the ignition lock cylinder (key-in ignition switch open = key is not in ignition switch) while the front axle lock request is still active, the front lock indicator flashes at a fast rate and three chime tones are sounded. The indicator will continue to flash at this rate until the key is replaced in the ignition lock cylinder (key-in ignition switch closed = key is in ignition switch) while the front axle lock request is still active, or until the front axle lock request input is deactivated, whichever occurs first.

The axle lock switch in the accessory switch bezel on the instrument panel provides a hard wired ground input to the instrument cluster circuitry through the front axle lock request circuit whenever the switch is enabled and the rear and front axle lock position of the switch is selected by the vehicle operator. The front locker indicator switch on the front axle housing provides a hard wired ground input to the instrument cluster whenever the front axle locker mechanism is engaged. The key-in ignition switch integral to the ignition switch provides a hard wired ground input to the instrument cluster whenever a key is present in the ignition lock cylinder. Each of these inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the instrument cluster circuitry that controls the front lock indicator LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

FUEL GAUGE

DESCRIPTION

Fig. 15 Fuel Gauge Icon

A fuel gauge is standard equipment on all instrument clusters. The fuel gauge is located in the lower left quadrant of the instrument cluster, below the voltage gauge. The fuel gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from “E” (or Empty) to “F” (or Full). An International Control and Display Symbol icon for “Fuel” is located on the cluster overlay, in the center of the gauge directly above the hub of the gauge needle (Fig. 15). An arrowhead pointed toward the left side of the vehicle is imprinted on the cluster overlay next to the “Fuel” icon in the fuel gauge to provide the driver with a reminder as to the location of the fuel filler access. The fuel gauge graphics are white and blue against a black field except for a single red graduation at the low end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The fuel gauge is serviced as a unit with the instrument cluster.

OPERATION

The fuel gauge gives an indication to the vehicle operator of the level of fuel in the fuel tank. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The fuel gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full, the cluster programming applies an algorithm to calculate the proper gauge needle position, then moves the gauge needle to the proper relative position on the gauge scale. The algorithm is used to dampen gauge needle movement against the negative effect that fuel sloshing within the fuel tank can have on accurate inputs from the fuel tank sending unit to the PCM.
- **Less Than 12.5 Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than about 12.5 (one-eighth), the gauge needle is moved to the proper position on the gauge scale, the low fuel indicator is illuminated, and a single chime tone is sounded. The low fuel indicator remains illuminated until the cluster receives messages from the PCM for a continuous twenty seconds indicating that the percent tank full has increased by more than 0.625 gallons or until the ignition switch is turned to the Off position, whichever occurs first. The chime tone feature will occur only once per ignition cycle.
- **Less Than Empty Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than empty, the gauge needle is moved to the far left end of the gauge scale and the low fuel indicator is illuminated immediately. This message would indicate that the fuel tank sender input to the PCM is a short circuit.
- **More Than Full Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full is more than full, the gauge needle is moved to the far left end of the gauge scale and the low fuel indicator is illuminated immediately. This message would indicate that the fuel tank sender input to the PCM is an open circuit.
- **Communication Error** - If the cluster fails to receive a percent tank full message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale and the low fuel indicator is illuminated immediately.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the fuel tank sending unit to determine the level of the fuel in the fuel

FUEL GAUGE (Continued)

tank. The PCM then sends the proper fuel level messages to the instrument cluster. For further diagnosis of the fuel gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the fuel tank sending unit, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the fuel gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

HIGH BEAM INDICATOR

DESCRIPTION

Fig. 16 High Beam Indicator

A high beam indicator is standard equipment on all instrument clusters (Fig. 16). The high beam indicator is located near the upper edge of the instrument cluster, between the tachometer and the speedometer. The high beam indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “High Beam” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A blue lens behind the cutout in the opaque layer of the overlay causes the icon to appear in blue through the translucent outer layer of the overlay when the indicator is illuminated from behind by a replaceable incandescent bulb and bulb holder unit, which is located on the instrument cluster electronic circuit board. The high beam indicator is serviced as a unit with the instrument cluster.

OPERATION

The high beam indicator gives an indication to the vehicle operator whenever the headlamp high beams are illuminated, or when the exterior lamps are inadvertently left On. This indicator is controlled by a hard wired input to the cluster from the headlamp beam select switch circuitry of the left multi-function switch, and by the instrument cluster circuit board based upon cluster programming based upon hard wired inputs from the head/park/fog lamp switch circuitry of the left multi-function switch, from the driver door ajar switch, and from the ignition switch. The high beam indicator bulb is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the bulb can be illuminated regardless of the ignition switch position. The bulb only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the high beam indicator for the following reasons:

- **Beam Select Switch Input** - Each time the cluster detects battery current on the beam select switch sense circuit (beam select switch closed = high beams selected or optical horn feature activated) the high beam indicator will be illuminated solid. This input can occur when the headlamp high beams are selected or when the optical horn feature is activated. The indicator remains illuminated until the beam select switch sense input to the cluster is an open circuit (beam select switch open = high beams not selected and optical horn feature not activated), or until the exterior lighting is turned off, whichever occurs first.
- **Exterior Lamps-On Optical Warning** - Each time the cluster detects battery current on the headlamp switch output circuit (park or head lamp switch closed = exterior lighting is On), ground on the driver door ajar switch sense circuit (driver door ajar switch closed = driver door is open), and the fused ignition switch output (run-start) input is an open circuit (ignition switch is in a position other than On or Start), the high beam indicator will be flashed on and off repeatedly. The indicator will continue to flash until the exterior lighting is turned Off, until the driver door is closed, or until the ignition switch is turned to the On or Start positions, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the high beam indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the bulb and the cluster control circuitry.

The instrument cluster continually monitors the headlamp beam select switch and circuitry integral to the left multi-function switch, the driver door ajar switch, and the ignition switch and turns the high beam indicator on or off accordingly. For further diagnosis of the high beam indicator or the instrument cluster circuitry that controls the indicator, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). The left multi-function switch, driver door ajar switch, and ignition switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - HIGH BEAM INDICATOR

The diagnosis found here addresses an inoperative high beam indicator condition. Before beginning this test, confirm the functionality of the high beam indi-

HIGH BEAM INDICATOR (Continued)

cator bulb and the cluster control circuitry by performing the instrument cluster actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the high beam indicator fails to illuminate during the actuator test, replace the indicator bulb and bulb holder with a known good unit and repeat the test. If the indicator still fails to illuminate, replace the faulty instrument cluster. If the problem being diagnosed is related to inoperative headlamp high beams, be certain to repair the headlamp system circuits and switches before attempting to diagnose or repair the high beam indicator. If no headlamp system problems are found and the high beam indicator illuminates during the instrument cluster actuator test, the following procedure will help locate an open in the high beam indicator driver circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C1) for the instrument cluster from the connector receptacle on the back of the cluster housing.

(2) Reconnect the battery negative cable. Turn the headlamps On and select the headlamp high beams with the left multi-function switch control stalk. Check for battery voltage at the high beam indicator driver circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster. If OK, replace the faulty instrument cluster. If not OK, repair the open high beam indicator driver circuit between the instrument cluster and the left multi-function switch as required.

LOW FUEL INDICATOR

DESCRIPTION

Fig. 17 Low Fuel Indicator

A low fuel indicator is standard equipment on all instrument clusters (Fig. 17). The low fuel indicator is located near the lower edge of the instrument cluster, to the left of center. The low fuel indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Fuel” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The low fuel indicator is serviced as a unit with the instrument cluster.

OPERATION

The low fuel indicator gives an indication to the vehicle operator when the level of fuel in the fuel tank becomes low. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The low fuel indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the low fuel indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the indicator is illuminated for about two seconds as a bulb test.
- **Less Than 12.5 Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than 12.5 (one-eighth), the low fuel indicator is illuminated and a single chime tone is sounded. The low fuel indicator remains illuminated until the cluster receives messages from the PCM for a continuous twenty seconds indicating that the percent tank full

has increased by more than 0.625 gallons or until the ignition switch is turned to the Off position, whichever occurs first. This strategy is intended to reduce the effect that fuel sloshing within the fuel tank can have on reliable indications. The chime tone feature will occur only once per ignition cycle.

- **Less Than Empty Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full is less than empty, the low fuel indicator is illuminated immediately. This message would indicate that the fuel tank sender input to the PCM is a short circuit.
- **More Than Full Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating the percent tank full is more than full, the low fuel indicator is illuminated immediately. This message would indicate that the fuel tank sender input to the PCM is an open circuit.
- **Actuator Test** - Each time the cluster is put through the actuator test, the low fuel indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The PCM continually monitors the fuel tank sending unit to determine the level of fuel in the fuel tank. The PCM then sends the proper fuel level messages to the instrument cluster. For further diagnosis of the low fuel indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the fuel tank sending unit, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the low fuel indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

MALFUNCTION INDICATOR LAMP (MIL)

DESCRIPTION

Fig. 18 Malfunction Indicator Lamp (MIL)

A Malfunction Indicator Lamp (MIL) is standard equipment on all instrument clusters (Fig. 18). The MIL is located near the lower edge of the instrument cluster, to the right of center. The MIL consists of a stencil-like cutout of the International Control and Display Symbol icon for “Engine” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The MIL is serviced as a unit with the instrument cluster.

OPERATION

The Malfunction Indicator Lamp (MIL) gives an indication to the vehicle operator when the Powertrain Control Module (PCM) has recorded a Diagnostic Trouble Code (DTC) for an On-Board Diagnostics II (OBDII) emissions-related circuit or component malfunction. The MIL is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the PCM over the Programmable Communications Interface (PCI) data bus. The MIL Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the MIL for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the indicator is illuminated for about seven seconds as a bulb test. The entire two seven second bulb test is a function of the PCM.
- **MIL Lamp-On Message** - Each time the cluster receives a MIL lamp-on message from the PCM, the indicator will be illuminated. The indicator can be flashed on and off, or illuminated solid, as dictated by the PCM message. For some DTC's, if a problem does not recur, the PCM will send a lamp-off message automatically. Other DTC's may require that a fault be repaired and the PCM be reset before a lamp-off message will be sent. For more information on the PCM and the DTC set and reset parameters, (Refer to 25 - EMISSIONS CONTROL - OPERATION).
- **Communication Error** - If the cluster receives no lamp-on or lamp-off message from the PCM for twenty seconds, the MIL is illuminated by the instrument cluster and a “no BuS” message will appear in the odometer/trip odometer Vacuum Fluorescent Display (VFD) unit to indicate a loss of bus communication. The indicator remains controlled and illuminated by the cluster until a valid lamp-on or lamp-off message is received from the PCM.

MALFUNCTION INDICATOR LAMP (MIL) (Continued)

- **Actuator Test** - Each time the cluster is put through the actuator test, the MIL will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The PCM continually monitors the fuel and emissions system circuits and sensors to decide whether the system is in good operating condition. The PCM then sends the proper lamp-on or lamp-off messages to the instrument cluster. For further diagnosis of the MIL or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the MIL after the bulb test, it may indicate that a malfunction has occurred and that the fuel and emissions system may require service. For proper diagnosis of the fuel and emissions systems, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the MIL, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

ODOMETER

DESCRIPTION

Fig. 19 Odometer Display

An odometer and trip odometer are standard equipment in all instrument clusters (Fig. 19). The odometer and trip odometer information are displayed in a common electronic, blue-green Vacuum Fluorescent Display (VFD). The VFD is soldered onto the cluster electronic circuit board and is visible through a window with a smoked clear lens located near the lower edge of the cluster overlay between the tachometer and speedometer. The dark lens over the VFD prevents it from being clearly visible when it is not illuminated. However, the odometer, and trip odometer information are not displayed simultaneously. The trip odometer reset switch on the instrument cluster circuit board toggles the display between odometer and trip odometer modes by depressing the odometer/trip odometer switch button that extends through the lower edge of the cluster lens, just right of the odometer VFD.

The odometer and trip odometer information is stored in the instrument cluster memory. This information can be increased when the proper inputs are provided to the instrument cluster, but the information cannot be decreased. The odometer can display values up to 999,999 kilometers (999,999 miles). The odometer latches at these values, and will not roll over to zero. The trip odometer can display values up to 9,999.9 kilometers (9,999.9 miles) before it rolls over to zero. The odometer display does not have a decimal point and will not show values less than a full unit (kilometer or mile), while the trip odometer display does have a decimal point and will show tenths of a unit (kilometer or mile). The unit of measure (kilometers or miles) for the odometer and trip odometer display is not shown in the VFD. The unit of measure for the instrument cluster odometer/trip odometer is selected at the time that it is manufactured, and cannot be changed. If the instrument cluster has a speedometer with a primary scale in kilometers-per-hour, the letters “KM” are printed on the cluster mask next to the VFD window to indicate the odometer unit of measure.

The odometer has a “Rental Car” mode, which will illuminate the odometer information in the VFD whenever the driver side front door is opened with the ignition switch in the Off or Accessory positions. During daylight hours (exterior lamps are Off) the VFD is illuminated at full brightness for clear visibility. At night (exterior lamps are On) the VFD lighting level is adjusted with the other cluster illumination lamps using the panel lamps dimmer control ring on the control stalk of the left multi-function switch. However, a “Parade” mode position of the panel lamps dimmer control ring allows the VFD to be illuminated at full brightness if the exterior lamps are turned On during daylight hours.

The VFD, the trip odometer switch, and the trip odometer switch button are serviced as a unit with the instrument cluster. The rubber trip odometer reset knob boot that seals the hole in the cluster lens through which the reset knob protrudes is available for individual service replacement.

OPERATION

The odometer and trip odometer give an indication to the vehicle operator of the distance the vehicle has traveled. This indicator is controlled by the instrument cluster circuitry based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The odometer and trip odometer information is displayed by the instrument cluster Vacuum Fluorescent Display (VFD). The VFD will display the odometer information whenever the driver door is opened with the ignition switch in the Off or Accessory positions, and will display the last previously selected odometer or trip odometer information when the ignition switch is turned to the On or Start positions. The instrument cluster circuitry controls the VFD and provides the following features:

ODOMETER (Continued)

- **Odometer/Trip Odometer Display Toggling** - Actuating the trip odometer reset switch button momentarily with the VFD illuminated will toggle the display between the odometer and trip odometer information. Each time the VFD is illuminated with the ignition switch in the On or Start positions, the display will automatically return to the last mode previously selected (odometer or trip odometer).
- **Trip Odometer Reset** - When the trip odometer reset switch button is pressed and held for longer than about two seconds with the ignition switch in the On or Start positions, trip odometer will be reset to 0.0 kilometers (miles). The VFD must be displaying the trip odometer information in order for the trip odometer information to be reset.
- **Communication Error** - If the cluster fails to receive a distance message during normal operation, it will hold and display the last data received until the ignition switch is turned to the Off position. If the cluster does not receive a distance message within one second after the ignition switch is turned to the On position, it will display the last distance message stored in the cluster memory. If the cluster is unable to display distance information due to an error internal to the cluster, either "888888"will be displayed in the VFD or the VFD will be blank.
- **Actuator Test** - Each time the cluster is put through the actuator test, the number "8" will be scrolled across the VFD from right-to-left, then the trip odometer decimal point "." will be illuminated in order to confirm the functionality of each of the VFD segments and the cluster control circuitry.

The PCM continually monitors the vehicle speed pulse information received from the vehicle speed sensor, then sends the proper distance messages to the instrument cluster. For further diagnosis of the odometer/trip odometer or the instrument cluster circuitry that controls these functions, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the vehicle speed sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the odometer/trip odometer, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

OIL PRESSURE GAUGE

DESCRIPTION

Fig. 20 Engine Oil Icon

An oil pressure gauge is standard equipment on all instrument clusters. The oil pressure gauge is located in the upper right quadrant of the instrument cluster, above the engine coolant temperature gauge. The oil pressure gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from "L" (or Low) to "H" (or High). An International Control and Display Symbol icon for "Engine Oil" is located on the cluster overlay, in the center of the gauge directly above the hub of the gauge needle (Fig. 20). The oil pressure gauge graphics are white and blue against a black field, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, and the blue graphics appear blue. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The oil pressure gauge is serviced as a unit with the instrument cluster.

OPERATION

The oil pressure gauge gives an indication to the vehicle operator of the engine oil pressure. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The oil pressure gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Engine Oil Pressure Message** - Each time the cluster receives a message from the PCM indicating the engine oil pressure of a 2.4L engine is above about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is above about 0.4 kg/cm^2 (6 psi), the cluster moves the gauge needle to the middle of the normal range on the gauge scale to represent the engine oil pressure. The gauge needle will continue to be positioned at the middle of normal range on the gauge scale until the cluster receives a message from the PCM that indicates the engine oil pressure is low, or until the ignition switch is turned to the Off position, whichever occurs first.
- **Engine Oil Pressure Low Message** - Each time the cluster receives a message from the PCM indicating the engine oil pressure of a 2.4L engine is

OIL PRESSURE GAUGE (Continued)

below about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is below about 0.4 kg/cm^2 (6 psi), the gauge needle is moved to the graduation at the far left (low) end of the gauge scale, the check gauges indicator is illuminated, and a single chime tone is generated. The gauge needle remains at the left end of the gauge scale and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine oil pressure of a 2.4L engine is above about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is above about 0.4 kg/cm^2 (6 psi), or until the ignition switch is turned to the Off position, whichever occurs first. The cluster will only turn the check gauges indicator on in response to an engine oil pressure low message if the engine speed message is 300 rpm or greater for more than about five seconds.

- **Communication Error** - If the cluster fails to receive an engine oil pressure message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the engine oil pressure sensor to determine the engine oil pressure. The PCM then sends the proper engine oil pressure messages to the instrument cluster. For further diagnosis of the oil pressure gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a low oil pressure gauge reading, it may indicate that the engine or the engine oiling system requires service. For proper diagnosis of the engine oil pressure sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the oil pressure gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

REAR LOCK INDICATOR

DESCRIPTION

Fig. 21 Rear Lock Indicator

A rear lock indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional off-road package (Fig. 21). The rear lock indicator is located near the upper edge of the instrument cluster, to the right of center. The rear lock indicator consists of a stencil-like cutout of the words “REAR LOCK” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “REAR LOCK” text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the rear lock indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The rear lock indicator is serviced as a unit with the instrument cluster.

OPERATION

The rear lock indicator gives an indication to the vehicle operator of the status of the locker mechanism in the rear axle of vehicles equipped with the optional off-road package. The rear lock indicator can also give an indication when certain faults are detected in the axle locker system. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming, and hard wired inputs to the cluster from the key-in ignition switch, the axle lock switch, and the rear locker indicator switch. The rear lock indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the LED can be illuminated regardless of the ignition switch position. The cluster can illuminate the LED solid, or flash it on and off at two different rates, slow or fast. The slow rate is about one flash per second, while the fast rate is about two flashes per second. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument

REAR LOCK INDICATOR (Continued)

cluster will turn on the rear lock indicator for the following reasons:

- **Rear Axle Lock Request Input** - Each time the cluster detects ground on the rear axle lock request circuit from the axle lock switch (rear axle lock request circuit ground = rear axle lock request active) while the key is in the ignition lock cylinder, the rear lock indicator flashes on and off at a slow rate. The indicator continues to flash until the rear axle lock request input to the cluster is an open circuit (rear axle lock request circuit open = rear axle lock request inactive), or until the cluster detects a change in the inputs from the rear locker indicator switch or the key-in ignition switch, whichever occurs first.
- **Rear Locker Indicator Switch Input** - Each time the cluster detects ground on the rear locker indicator switch sense circuit (rear locker indicator switch closed = rear axle locker engaged) while the rear axle lock request is active and the key is in the ignition lock cylinder, the rear lock indicator is illuminated solid. The indicator remains illuminated solid until the rear locker indicator switch input to the cluster is an open circuit (rear locker indicator switch open = rear axle locker disengaged), or until the cluster detects a change in the inputs from the axle lock switch or the key-in ignition switch, whichever occurs first.
- **Rear Lock Request/Feedback Mismatch** - Each time the cluster detects that the inputs from the axle lock switch (request) on the rear axle lock request circuit and from the rear locker indicator switch (feedback) on the rear locker indicator switch sense circuit are mismatched, the rear lock indicator flashes on and off at a slow rate. This condition occurs when the lock request is active but the axle locker remains disengaged, or when the lock request is inactive but the axle locker remains engaged. The indicator continues to flash until the request and the feedback are matching.
- **Key-In Ignition Switch Input** - Each time the cluster detects that the key has been removed from the ignition lock cylinder (key-in ignition switch open = key is not in ignition switch) while the rear axle lock request is still active, the rear lock indicator flashes at a fast rate and three chime tones are sounded. The indicator will continue to flash at this rate until the key is replaced in the ignition switch lock cylinder (key-in ignition switch closed = key is in ignition switch) while the rear axle lock request is still active, or until the rear axle lock request input is deactivated, whichever occurs first.

The axle lock switch in the accessory switch bezel on the instrument panel provides a hard wired ground input to the instrument cluster circuitry through the rear axle lock request circuit whenever the switch is enabled and the rear or the rear and front axle lock position of the switch is selected by the vehicle operator. The rear locker indicator switch on the rear axle housing provides a hard wired ground input to the instrument cluster whenever the rear axle locker mechanism is engaged. The key-in ignition switch integral to the ignition switch provides a hard wired ground input to the instrument cluster whenever a key is present in the ignition lock cylinder. Each of these inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the instrument cluster circuitry that controls the rear lock indicator LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

SEATBELT INDICATOR

DESCRIPTION

Fig. 22 Seatbelt Indicator

A seatbelt indicator is standard equipment on all instrument clusters (Fig. 22). The seatbelt indicator is located near the center of the instrument cluster, between the tachometer and the speedometer. The seatbelt indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Seat Belt” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The seatbelt indicator is serviced as a unit with the instrument cluster.

OPERATION

The seatbelt indicator gives an indication to the vehicle operator of the status of the driver side front seatbelt. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and a hard wired input from the seatbelt switch in the driver side front seatbelt buckle through the seat belt indicator driver circuit. The seatbelt indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a bat-

SEATBELT INDICATOR (Continued)

tery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the seatbelt indicator for the following reasons:

- **Seatbelt Reminder Function** - Each time the cluster receives a battery current input on the fused ignition switch output (run-start) circuit, the indicator will be illuminated as a seatbelt reminder for about seven seconds, or until the ignition switch is turned to the Off position, whichever occurs first. This reminder function will occur regardless of the status of the seatbelt switch input to the cluster.
- **Driver Side Front Seatbelt Not Buckled** - Following the seatbelt reminder function, each time the cluster detects a ground on the seat belt switch sense circuit (seatbelt switch closed = seatbelt unbuckled) with the ignition switch in the Start or On positions, the indicator will be illuminated. The seatbelt indicator remains illuminated until the seat belt switch sense input to the cluster is an open circuit (seatbelt switch open = seatbelt buckled), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the seatbelt indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The seatbelt switch is connected in series between ground and the seat belt switch sense input to the instrument cluster. The seatbelt switch input to the instrument cluster circuitry can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the seatbelt indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

SHIFT INDICATOR (TRANSFER CASE)

DESCRIPTION

Fig. 23 4WD Indicator

A four-wheel drive indicator is standard equipment on all instrument clusters (Fig. 23). The four-wheel drive indicator is located near the upper edge of the instrument cluster, to the right of center. The four-wheel drive indicator consists of a stencil-like cutout of the text “4WD” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “4WD” text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the four-wheel drive indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The four-wheel drive indicator is serviced as a unit with the instrument cluster.

OPERATION

The four-wheel drive indicator gives an indication to the vehicle operator that a four-wheel drive operating mode of the four-wheel drive transfer case is selected. The four-wheel drive indicator lights when the transfer case is engaged in the 4H or 4L positions. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming. On models not equipped with the optional Off-Road Package the cluster also uses a hard wired transfer case switch input. Models that are equipped with the Off-Road Package do not have a hard wired transfer case switch input, so the cluster uses electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus to determine the transfer case operating mode. The four-wheel drive indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the four-wheel drive indicator for the following reasons:

- **Transfer Case Switch Input** - Each time the cluster detects a ground on the 4WD switch sense circuit (transfer case switch closed = 4WD mode selected) the part time indicator is illuminated. The indicator remains illuminated until the 4WD switch sense input to the cluster is an open circuit (transfer case switch open = 4WD mode not selected), or until

SHIFT INDICATOR (TRANSFER CASE) (Continued)

the ignition switch is turned to the Off position, whichever occurs first.

- **4WD Transfer Case Status Message** - Each time the cluster receives a message from the PCM indicating that the transfer case is operating in a four-wheel drive mode, the 4WD indicator is illuminated. The 4WD indicator remains illuminated until the cluster receives messages from the PCM indicating that the transfer case is not in a four-wheel drive mode or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the 4WD indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

On models without the Off-Road Package, the transfer case shift position switch is connected in series between ground and the 4WD switch sense input to the instrument cluster. The transfer case switch input to the instrument cluster circuitry can be diagnosed using conventional diagnostic tools and methods. (Refer to 21 - TRANSMISSION/TRANSAXLE/TRANSFER CASE - DESCRIPTION) for more information on the transfer case switch. For further diagnosis of the 4WD indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).

On models equipped with the Off-Road Package, the PCM continually monitors the transfer case shift position switch to determine the operating mode of the transfer case. The PCM then sends the proper transfer case shift position status messages to the instrument cluster. For further diagnosis of the 4WD indicator or the instrument cluster circuitry that controls the LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the transfer case shift position switch, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the 4WD indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

DIAGNOSIS AND TESTING - 4WD INDICATOR

The diagnosis found here addresses an inoperative four-wheel drive indicator condition only on models that are not equipped with the optional Off-Road Package. If the vehicle is equipped with the Off-Road Package, for proper diagnosis of the transfer case switch, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the 4WD indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information. Before beginning this test, confirm the functionality of the four-wheel drive indicator Light Emitting Diode (LED) and the cluster control circuitry by performing the instrument cluster actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the four-wheel drive indicator fails to illuminate during the actuator test, replace the faulty instrument cluster. If the problem being diagnosed is related to indicator accuracy, be certain to confirm that the problem is with the indicator or transfer case switch and not a mechanical malfunction of the transfer case or transfer case shift linkage. (Refer to 21 - TRANSMISSION/TRANSAXLE/TRANSFER CASE - DIAGNOSIS AND TESTING). If no transfer case problem is found, the following procedure will help to locate a short or open in the 4WD switch sense circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

INDICATOR DOES NOT ILLUMINATE WITH 4WD MODE SELECTED

(1) Disconnect and isolate the battery negative cable. Disconnect the engine wire harness connector for the transfer case switch from the transfer case switch connector receptacle. Check for continuity between the ground circuit cavity of the engine wire harness connector for the transfer case switch and a good ground. There should be continuity. If OK, go to Step 2. If not OK, repair the open ground circuit to ground (G105) as required.

(2) Reconnect the battery negative cable. Turn the ignition switch to the On position. Install a jumper wire between the 4WD switch sense circuit cavity of the engine wire harness connector for the transfer case switch and a good ground. The 4WD indicator

SHIFT INDICATOR (TRANSFER CASE) (Continued)

should light. If OK, replace the faulty transfer case switch. If not OK, go to Step 3.

(3) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Check for continuity between the 4WD switch sense circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the engine wire harness connector for the transfer case switch. There should be continuity. If OK, replace the faulty instrument cluster. If not OK, repair the open 4WD switch sense circuit between the instrument cluster and the transfer case switch as required.

INDICATOR STAYS ILLUMINATED WITH 4WD MODE NOT SELECTED

(1) Disconnect and isolate the battery negative cable. Disconnect the engine wire harness connector for the transfer case switch from the transfer case switch connector receptacle. Check for continuity between the ground circuit and the 4WD switch sense circuit terminals in the transfer case switch connector receptacle. There should be no continuity. If OK, go to Step 2. If not OK, replace the faulty transfer case switch.

(2) Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Check for continuity between the 4WD switch sense circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, replace the faulty instrument cluster. If not OK, repair the shorted 4WD switch sense circuit between the transfer case switch and the instrument cluster as required.

SKIS INDICATOR

DESCRIPTION

Fig. 24 SKIS Indicator

A Sentry Key Immobilizer System (SKIS) indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional SKIS (Fig. 24). The SKIS indicator is located near the upper edge of the instrument cluster overlay, to the left of center. The SKIS indicator consists of a stencil-like cutout of a graphical representation or icon of a key that is circled and crossed-out in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the indicator to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The SKIS indicator is serviced as a unit with the instrument cluster.

OPERATION

The Sentry Key Immobilizer System (SKIS) indicator gives an indication to the vehicle operator of the status of the SKIS. This indicator is controlled by the instrument cluster circuit board based upon electronic messages received by the cluster from the Sentry Key Immobilizer Module (SKIM) over the Programmable Communications Interface (PCI) data bus. The SKIS indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the SKIS indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position, the SKIM sends a message to the cluster to illuminate the SKIS indicator for about three seconds as a bulb test.
- **SKIM Lamp-On Message** - Each time the cluster receives a lamp-on message from the SKIM, the SKIS indicator will be illuminated. The indicator can be flashed on and off, or illuminated solid, as dictated by the SKIM message. For more information on the SKIS and the SKIS indicator control parameters, (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - OPERATION). The indicator remains illuminated until the cluster receives a lamp-off message from the SKIM or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the SKIS indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry.

The SKIM performs a self-test each time the ignition switch is turned to the On position to decide

SKIS INDICATOR (Continued)

whether the system is in good operating condition. The SKIM then sends the proper SKIS lamp-on or lamp-off messages to the instrument cluster. For further diagnosis of the SKIS indicator or the instrument cluster circuitry that controls the indicator, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the SKIS indicator after the bulb test, either solid or flashing, it indicates that a SKIS malfunction has occurred or that the SKIS is inoperative. For proper diagnosis of the SKIS, the PCI data bus, or the electronic message inputs to the instrument cluster that control the SKIS indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

SPEEDOMETER

DESCRIPTION

MPH km/h

Fig. 25 Speedometer Text

A speedometer is standard equipment on all instrument clusters. The speedometer is located next to the tachometer, just to the right of center in the instrument cluster. The speedometer consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 210 degree primary scale on the cluster overlay that reads left-to-right either from “0” to “100” mph, from “0” to “110” mph, or from “0” to “180” km/h, depending upon the requirements of the market for which the vehicle is manufactured. Each version also has a secondary inner scale on the cluster overlay that provides the equivalent opposite units from the primary scale. Text appearing on the cluster overlay just below the hub of the speedometer needle abbreviates the unit of measure for the primary scale (i.e.: MPH or km/h), followed by the unit of measure for the secondary scale (Fig. 25). The speedometer graphics are white (primary scale) and blue (secondary scale) against a black field, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white and the blue graphics appear blue. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The speedometer is serviced as a unit with the instrument cluster.

OPERATION

The speedometer gives an indication to the vehicle operator of the vehicle road speed. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The speedometer is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Vehicle Speed Message** - Each time the cluster receives a vehicle speed message from the PCM it will calculate the correct vehicle speed reading and position the gauge needle at that relative speed position on the gauge scale. The cluster will receive a new vehicle speed message and reposition the gauge pointer accordingly about every 86 milliseconds. The gauge needle will continue to be positioned at the actual vehicle speed position on the gauge scale until the ignition switch is turned to the Off position.
- **Communication Error** - If the cluster fails to receive a speedometer message, it will hold the gauge needle at the last indication for about six seconds, or until the ignition switch is turned to the Off position, whichever occurs first. After six seconds, the gauge needle will return to the left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence in order to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the vehicle speed sensor to determine the vehicle road speed, then sends the proper vehicle speed messages to the instrument cluster. For further diagnosis of the speedometer or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the vehicle speed sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the speedometer, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

TACHOMETER

DESCRIPTION

Fig. 26 Tachometer Text

A tachometer is standard equipment on all instrument clusters. The tachometer is located to the left of the speedometer, just to the left of center in the instrument cluster. The tachometer consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 210 degree scale on the cluster overlay that reads left-to-right from 0 to 6. The text "X1000 RPM" imprinted on the cluster overlay directly below the hub of the tachometer needle identifies that each number on the tachometer scale is to be multiplied by 1000 rpm (Fig. 26). Red graduations at the right (high) end of the gauge scale designate the engine overspeed area of the gauge. The tachometer graphics are white against a black field, except for the red graduations, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The tachometer is serviced as a unit with the instrument cluster.

OPERATION

The tachometer gives an indication to the vehicle operator of the engine speed. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The tachometer is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster electronic circuitry controls the gauge needle position and provides the following features:

- **Engine Speed Message** - Each time the cluster receives an engine speed message from the PCM it will calculate the correct engine speed reading and position the gauge needle at that relative speed position on the gauge scale. The cluster will receive a new engine speed message and reposition the gauge pointer accordingly about every 86 milliseconds. The gauge needle will continually be repositioned on the gauge scale until the engine stops running, or until the ignition switch is turned to the Off position, whichever occurs first.
- **Communication Error** - If the cluster fails to receive an engine speed message, it will hold the gauge needle at the last indication for about six seconds, or until the ignition switch is turned to the Off position, whichever occurs first. After six seconds, the gauge needle will return to the left (low) end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the crankshaft position sensor to determine the engine speed. The PCM then sends the proper engine speed messages to the instrument cluster. For further diagnosis of the tachometer or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the crankshaft position sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the tachometer, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

TURN SIGNAL INDICATOR

DESCRIPTION

Fig. 27 Turn Signal Indicators

Two turn signal indicators, one right and one left, are standard equipment on all instrument clusters (Fig. 27). The turn signal indicators are located near the upper edge of the instrument cluster, between the speedometer and the tachometer. Each turn signal indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for "Turn Warning" in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents these icons from being clearly visible when they are not illuminated. A green Light Emitting Diode (LED) behind each turn signal indicator cutout in the opaque layer of the cluster over-

lay causes the icon to appear in green through the translucent outer layer of the overlay when the indicator is illuminated from behind the LED, which is soldered onto the instrument cluster electronic circuit board. The turn signal indicators are serviced as a unit with the instrument cluster.

OPERATION

The turn signal indicators give an indication to the vehicle operator that the turn signals (left or right indicator flashing) or hazard warning (both left and right indicators flashing) have been selected and are operating. These indicators are controlled by two individual hard wired inputs received by the cluster from the turn signal switch circuitry and hazard warning switch circuitry within the left multi-function switch. Each turn signal indicator Light Emitting Diode (LED) is grounded on the instrument cluster electronic circuit board at all times. Therefore, these indicators remain functional regardless of the ignition switch position. Each LED will only illuminate when it is provided battery current by the circuitry of the left multi-function switch.

The turn signal indicators are connected in series between the output of the combination flasher on the left multi-function switch and the left or right turn signal inputs to the instrument cluster, but in parallel with the other turn signal circuits. This arrangement allows the turn signal indicators to remain functional, regardless of the condition of the other circuits in the turn signal and hazard warning systems. The turn signal switch and hazard warning switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/LEFT MULTI-FUNCTION SWITCH - OPERATION) for more information on the turn signal switch and the hazard warning switch.

VOLTAGE GAUGE

DESCRIPTION

Fig. 28 Battery Charging Condition Icon

A voltage gauge is standard equipment on all instrument clusters. The voltage gauge is located in the upper left quadrant of the instrument cluster, above the fuel gauge. The voltage gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from 9 volts to 19 volts. An International Control and Display Symbol icon for “Battery Charging Condition” is located on the cluster overlay, directly above the hub of the gauge needle (Fig. 28). The voltage gauge graphics are white and blue against a black field except for a single red graduation at each end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The voltage gauge is serviced as a unit with the instrument cluster.

OPERATION

The voltage gauge gives an indication to the vehicle operator of the electrical system voltage. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The voltage gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the left end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **System Voltage Message** - Each time the cluster receives a system voltage message from the PCM indicating the system voltage is between about 9.5 volts and about 18.5 volts, the gauge needle is moved to the relative voltage position on the gauge scale.
- **System Voltage Low (Charge Fail) Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is less than about 9.0 volts (charge fail condition), the gauge needle is moved to the 9 volt graduation on the far left end of the gauge scale and the check gauges indicator is illuminated. The gauge needle remains on the 9 volt graduation and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is greater than about 12.0 volts (but less than about 16.0 volts), or until the ignition switch is turned to the Off position, whichever occurs first.
- **System Voltage High Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is greater than

VOLTAGE GAUGE (Continued)

about 19.0 volts, the gauge needle is moved to the 19 volt graduation on the far right end of the gauge scale and the check gauges indicator is illuminated. The gauge needle remains on the right end of the gauge scale and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is less than about 16.0 (but greater than about 9.5 volts), or until the ignition switch is turned to the Off position, whichever occurs first.

- **Communication Error** - If the cluster fails to receive a system voltage message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, it will move the gauge needle to the far left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence in order to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the system voltage to control the generator output. The PCM then sends the proper system voltage messages to the instrument cluster. For further diagnosis of the voltage gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a charge fail or voltage high condition, it may indicate that the charging system requires service. For proper diagnosis of the charging system, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the voltage gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

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LAMPS/LIGHTING - EXTERIOR

DESCRIPTION - TURN SIGNAL & HAZARD WARNING SYSTEM

A turn signal and hazard warning system is standard factory-installed safety equipment. The turn signal and hazard warning system includes the following major components, which are described in further detail elsewhere in this service information:

- **Combination Flasher** - The electronic combination flasher is located on the back of the left multi-function switch, beneath the upper steering column shroud on the top of the steering column.
- **Front Side Marker Lamps** - The front side marker lamps are located on the outboard ends of the two front fender flares.
- **Turn Signal Repeater Lamps** - The turn signal repeater lamps are located on the outboard ends of the two front fender flares.
- **Hazard Warning Switch** - The hazard warning switch is integral to the left multi-function switch. The hazard warning switch button protrudes from an opening in the shroud on the top of the steering column.
- **Turn Signal Cancel Cam** - The turn signal cancel cam is integral to the steering column clockspring, which is located beneath the shrouds on the top of the steering column.
- **Turn Signal Indicators** - The two turn signal indicators are integral to the ElectroMechanical Instrument Cluster (EMIC) located in the instrument panel.
- **Turn Signal Lamps** - The front turn signal lamps are integral to the front park/turn signal lamps located beside each headlamp on the front fender. The rear turn signal lamps are integral to the back-up/brake/rear turn signal/tail lamps located on either side of the rear of the quarter panels.
- **Turn Signal Switch** - The turn signal switch is integral to the left multi-function switch. The left multi-function switch control stalk actuates the turn signal switch in the steering column.

The turn signal system in this vehicle includes a turn signal-on warning chime feature. The EMIC electronic circuitry monitors the turn signal indicators as well as electronic vehicle speed and distance messages received from the PCM over the PCI data bus network to provide this feature. If an indicator remains illuminated continuously with the vehicle speed above about 25 kilometers per hour (15 miles per hour) for a distance of greater than about 1.6 kilometers (1 mile), the EMIC generates a chime through an integral chime tone generator (The distance is greater for export vehicles).

Hard wired circuitry connects the turn signal and hazard warning system components to each other through the electrical system of the vehicle. Refer to the appropriate wiring information.

LAMPS/LIGHTING - EXTERIOR (Continued)

OPERATION - TURN SIGNAL & HAZARD WARNING SYSTEM

The turn signal system operates on battery voltage received on a ignition switch output (run) circuit so that the turn signals will only operate with the ignition switch in the On position. The hazard warning system operates on non-switched battery voltage so that the hazard warning remains operational regardless of the ignition switch position. When the turn signal switch control stalk is moved up or down, the turn signal system is activated. When the turn signal system is activated, the circuitry of the turn signal switch and the combination flasher will cause the selected turn signal indicator, front park/turn signal lamp, front side marker lamp, repeater lamps, and rear tail/stop/turn signal lamp to flash on and off. When the hazard warning system is activated, the circuitry of the hazard warning switch and the combination flasher will cause both the right side and the left side turn signal indicators, front park/turn signal lamps, front side marker lamps, repeater lamps and rear tail/stop/turn signal lamps to flash on and off.

In order to provide the turn signal-on warning, the ElectroMechanical Instrument Cluster (EMIC) monitors vehicle speed and distance messages received from the PCM over the PCI data bus and the hard wired turn signal switch input to the cluster electronic circuit board. If a turn signal remains indicated for a distance of greater than about 1.6 kilometers (1 mile) and the vehicle speed remains greater than about 24 kilometers-per-hour (15 miles-per-hour), the EMIC generates a repetitive chime at a slow rate to provide an audible reminder that a turn signal has been left on (distance is slightly longer with export vehicles). Once the warning chime begins to sound, it will continue until the turn signal is cancelled, until the vehicle speed falls below about 24 kilometers-per-hour (15 miles-per-hour), or until the ignition switch is turned to the Off position, whichever occurs first. This feature is not activated by a hazard warning input to the instrument cluster.

During both the turn signal and the hazard warning operation, if the exterior lamps are turned Off, the front park/turn signal lamps, repeater lamps and the front side marker lamps will flash in unison. If the exterior lamps are turned On, the front park/turn signal lamps, repeater lamps and the front side marker lamps will flash alternately.

WARNING

WARNING:: EYE PROTECTION SHOULD BE USED WHEN SERVICING GLASS COMPONENTS. PERSONAL INJURY CAN RESULT.

CAUTION: Do not touch the glass of halogen bulbs with fingers or other possibly oily surface, reduced bulb life will result.

CAUTION: Do not use bulbs with higher candle power than indicated in the Bulb Application table at the end of this group. Damage to lamp and/or Daytime Running Lamp Module can result.

CAUTION: Do not use fuses, circuit breakers, or relays having greater amperage values than indicated on the fuse panel or in the Owner's Manual.

NOTE: When it is necessary to remove components to service another, it should not be necessary to apply excessive force or bend a component to remove it. Before damaging a trim component, verify hidden fasteners or captured edges are not holding the component in place.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - TURN SIGNAL & HAZARD WARNING SYSTEM

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

When diagnosing the turn signal or hazard warning circuits, remember that high generator output can burn out bulbs rapidly and repeatedly. If this is a problem on the vehicle being diagnosed (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING). If the problem being diagnosed is related to a failure of the turn signals to automatically cancel following completion of a turn, inspect the multifunction switch for a faulty or damaged cancel actuator and inspect the turn signal cancel cam lobes on the clockspring mechanism for damage or improper installation. For complete circuit diagrams, refer to the appropriate wiring information.

LAMPS/LIGHTING - EXTERIOR (Continued)

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CONDITION	POSSIBLE CAUSES	CORRECTION
ONE TURN SIGNAL LAMP DOES NOT ILLUMINATE	1. Faulty or missing bulb	1. Test and replace turn signal bulb as required.
	2. Faulty ground circuit.	2. Test and repair open ground circuit
	3. Faulty signal circuit.	3. Test and repair open right or left turn signal circuit.
ALL RIGHT SIDE AND/OR LEFT SIDE TURN SIGNAL LAMPS DO NOT FLASH	1. Faulty multifunction switch.	1. Test and replace the multifunction switch as required.
	2. Faulty flasher.	2. Replace the hazard switch/combination flasher.
ALL RIGHT SIDE OR LEFT SIDE TURN SIGNALS FLASH RAPIDLY	1. Faulty or missing bulb.	1. Test and replace faulty bulb as required.
	2. Faulty ground circuit.	2. Test and repair open ground circuit as required.
	3. Faulty signal circuit.	3. Test and repair high resistance or open signal circuit as required.
	4. Faulty flasher.	4. Replace flasher.
HAZARD WARNING LAMPS DO NOT FLASH	1. Faulty or missing fuse.	2. Test and replace fuse as required.
	2. Faulty ground circuit.	2. Test and repair high resistance or open ground circuit.
	3. Open battery positive voltage circuit to hazard warning switch.	3. Test and repair open battery voltage circuit to hazard warning switch.
	4. Faulty flasher	4. Replace flasher
	5. Faulty multifunction switch.	5. Test and replace the multifunction switch as required.

DIAGNOSIS AND TESTING - HEADLAMP SYSTEM

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

LAMPS/LIGHTING - EXTERIOR (Continued)

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

Conventional and halogen headlamps are interchangeable. It is recommended that they not be intermixed.

CONDITION	POSSIBLE CAUSES	CORRECTION
HEADLAMPS ARE DIM WITH ENGINE IDLING OR IGNITION TURNED OFF	1. Loose or corroded battery cables.	1. Clean and secure battery cable clamps and posts.
	2. Loose or worn generator drive belt.	2. Adjust or replace generator drive belt.
	3. Charging system output too low.	3. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
	4. Battery has insufficient charge.	4. Test battery state-of -charge. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/ BATTERY - DIAGNOSIS AND TESTING)
	5. Battery is sulfated or shorted.	5. Load test battery. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/ BATTERY - DIAGNOSIS AND TESTING)
	6. Poor lighting circuit ground.	6. Test for voltage drop across ground locations.
	7. Both headlamp bulbs faulty.	7. Replace both headlamp bulbs.
HEADLAMP BULBS BURN OUT FREQUENTLY	1. Charging system output too high.	1. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
	2. Loose or corroded terminals or splices in headlamp circuit.	2. Inspect and repair all connectors and splices.
HEADLAMPS ARE DIM WITH ENGINE RUNNING	1. Charging system output too low.	1. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
ABOVE IDLE*	2. Poor lighting circuit ground.	2. Test for voltage drop across ground locations.
	3. High resistance in headlamp circuit.	3. Test amperage draw of headlamp circuit.
	4. Both headlamp bulbs faulty.	4. Replace both headlamp bulbs.
HEADLAMPS FLASH RANDOMLY	1. Poor lighting circuit ground.	1. Test for voltage drop across ground locations.
	2. High resistance in headlamp circuit.	2. Test amperage draw of headlamp circuit. Should not exceed 30 amps.
	3. Faulty headlamps switch circuit breaker.	3. Replace headlamp switch. (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/HEADLAMP SWITCH - REMOVAL)

LAMPS/LIGHTING - EXTERIOR (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
	4. Short in headlamp circuit	Test headlamp circuit, repair as necessary.
	4. Loose or corroded terminals or splices in headlamp circuit.	4. Inspect and repair all connectors and splices.
HEADLAMPS DO NOT ILLUMINATE LOW OR HIGH BEAM	1. No voltage to headlamps.	1. Repair open headlamp circuit.
	2. No ground at headlamps.	2. Repair circuit ground.
	3. Faulty headlamp switch.	3. Replace headlamp switch. (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/HEADLAMP SWITCH - REMOVAL)
	4. Faulty headlamp dimmer (multi-function) switch.	4. Replace multi-function switch.
	5. Broken connector terminal or wire splice in headlamp circuit.	5. Repair connector terminal or wire splice.
	6. Both headlamp bulbs faulty.	6. Replace both headlamp bulbs.
*Canada vehicles must have lamps ON.		

SPECIFICATIONS

EXTERIOR LAMPS

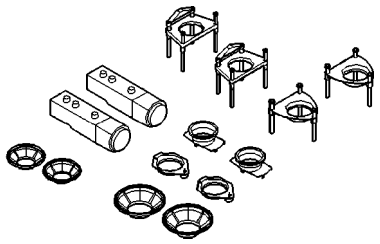
CAUTION: Do not use bulbs that have a higher candle power than the bulb listed in the Bulb Application Table. Damage to lamp can result. Do not touch halogen bulbs with fingers or other oily surfaces. Bulb life will be reduced.

The following Bulb Application Table lists the lamp title on the left side of the column and trade number or part number on the right for domestic and export vehicles.

LAMP	BULB
Back-up lamp	1156 or P21W
Center High Mounted Stoplamp	921 or W16W
Front Fog lamp	H3
Front Position lamp	T4W
Front Side Marker	168 or W3W
Headlamp/Sealed Beam	H6024 or H4
License Plate lamp	W5W Export Only
Park/Turn Signal	3157 or P27/7W
Rear Fog lamp	P21W
Side Repeater lamp	W3W
Tail/Stop	1157 or P21/5W
Underhood lamp	W5W
Underhood Retractable Lamp	105

SPECIAL TOOLS

HEADLAMP ALIGNMENT



Headlamp Aiming Kit C-4466-A

BRAKE LAMP SWITCH

DESCRIPTION

The brake lamp switch consists of multiple switch contacts either normally open or closed. The switch contacts provide brake pedal status to the various controllers. The primary function of the switch is to illuminate the brake lamps. The brake lamp switch is adjustable and mounted on the brake pedal mounting bracket under the instrument panel.

BRAKE LAMP SWITCH (Continued)

OPERATION

Vehicles equipped with the speed control option use a multiple function brake lamp switch. The PCM monitors the state of the brake lamp switch. Refer to the Brake section for more information on brake lamp switch service and adjustment procedures.

The primary function of the brake switch is to turn on the brake lamps during braking. The switch is also used to send signals to components that must know when the brakes are applied, such as the Powertrain Control Module (PCM), which uses the signal to cancel speed control. The Controller Antilock Brake (CAB) uses the brake switch signal to monitor brake pedal application. When the normally closed switch contacts open, the CAB receives the brake applied signal. The CAB then monitors the ABS system to anticipate the need for a ABS stop.

DIAGNOSIS AND TESTING - BRAKE LAMP SWITCH

The brake lamp switch operation can be tested with an ohmmeter. The ohmmeter is used to check continuity between the pin terminals (Fig. 1).

SWITCH CIRCUIT IDENTIFICATION

- Terminals 1 and 2: brake sensor circuit
- Terminals 3 and 4: speed control circuit if equipped
- Terminals 5 and 6: brake lamp circuit

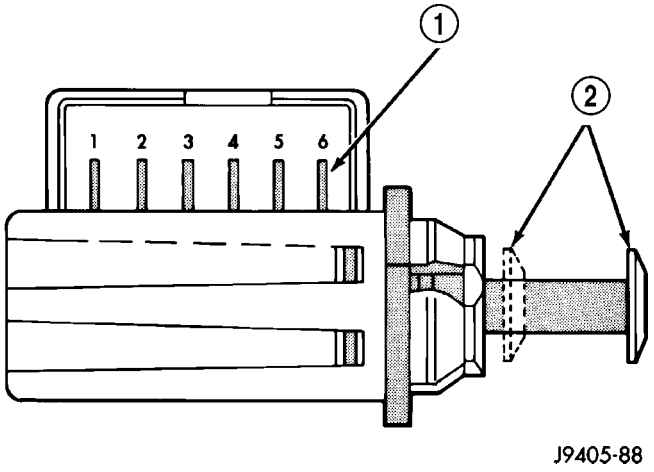


Fig. 1 Brake Lamp Switch Terminal Identification

1 - TERMINAL PINS
2 - PLUNGER TEST POSITIONS

SWITCH CONTINUITY TEST

NOTE: Disconnect switch harness before testing switch continuity.

With the switch plunger retracted, attach test leads to terminal pins 1 and 2. Replace switch if meter indicates no continuity.

With the switch plunger retracted, attach test leads to terminal pins 3 and 4. Replace switch if meter indicates no continuity.

With the switch plunger extended, attach test leads to terminal pins 5 and 6. Replace switch if meter indicates no continuity.

REMOVAL

(1) Remove the steering column cover and the lower trim panel.

(2) Press the brake pedal downward to fully applied position.

(3) Rotate the switch approximately 30° in counterclockwise direction. Then pull the switch rearward and out of bracket.

(4) Disconnect the switch harness and remove the switch (Fig. 2).

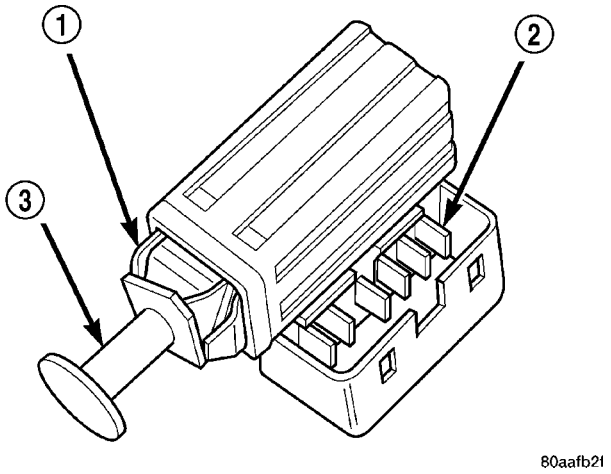


Fig. 2 Brake Lamp Switch

1 - RETAINER
2 - TERMINALS
3 - SWITCH PLUNGER

INSTALLATION

(1) Pull the switch plunger all of the way out, to fully extended position.

(2) Connect the harness connector to the switch.

(3) Press and hold the brake pedal in the applied position.

(4) Align the tab on the switch with the notch in the switch bracket. Then insert the switch in the bracket and turn it clockwise about 30° to lock it in place.

(5) Release the brake pedal, then pull the pedal fully rearward. Pedal will set the plunger to the correct position as the pedal pushes the plunger into the switch body. The switch will make a ratcheting sound as it self-adjusts.

BRAKE LAMP SWITCH (Continued)

ADJUSTMENTS

ADJUSTMENT

(1) Press and hold brake pedal in applied position.
(2) Pull switch plunger all the way out to fully extended position.
(3) Release brake pedal. Then pull pedal lightly rearward. Pedal will set plunger to correct position as pedal pushes plunger into switch body. Switch will make a ratcheting sound as it self-adjusts.

CAUTION: Booster damage may occur if the pedal pull exceeds 20 lbs.

CENTER HIGH MOUNTED STOP LAMP UNIT

REMOVAL

NOTE: To remove to the CHMSL, it will be necessary to remove spare tire from the spare tire carrier bracket.

(1) Disconnect and isolate the battery negative cable.
(2) Remove the cover from CHMSL button contact plate (Fig. 3).
(3) Disconnect the CHMSL wire harness from the CHMSL button contact terminals.
(4) Disengage the CHMSL wire harness from the retaining clips.
(5) Disengage the wire harness grommet and route the CHMSL wire harness through the tailgate.
(6) Remove the bolts attaching CHMSL to the spare tire carrier bracket.
(7) Remove the CHMSL from the vehicle.

INSTALLATION

(1) Position the CHMSL to the vehicle.
(2) Install the bolts that secure the CHMSL to the spare tire carrier bracket.
(3) Route the CHMSL wire harness through the tailgate and seat the wire harness grommet.
(4) Connect the CHMSL wire harness to the CHMSL button contact terminals.
(5) Position the CHMSL wire harness into the retaining clips and engage the clips.
(6) Install the CHMSL button contact plate cover.
(7) Reconnect the battery negative cable.

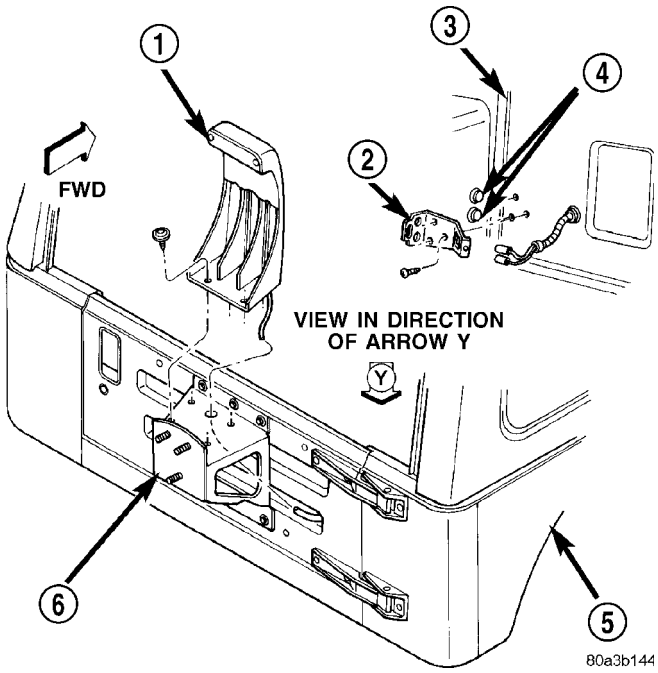


Fig. 3 CHMSL and Contact Buttons

1 - CHMSL
2 - CHMSL BUTTON CONTACT PLATE
3 - TAILGATE
4 - CHMSL CONTACT BUTTONS
5 - BODY
6 - SPARE TIRE CARRIER BRACKET

CENTER HIGH MOUNTED STOP LAMP

REMOVAL

(1) Disconnect and isolate the battery negative cable.
(2) Remove the screws that secure the CHMSL lens to the CHMSL housing (Fig. 4).
(3) Remove the CHMSL lens from the CHMSL housing.
(4) Rotate the lamp socket one-third turn counterclockwise and separate the lamp socket from the CHMSL lens.
(5) Pull the lamp straight out of the socket.

INSTALLATION

(1) Install the lamp into the socket by pushing it straight into the socket.
(2) Install the lamp socket into the CHMSL lens by turning the socket one-third turn clockwise.
(3) Position the CHMSL lens onto the CHMSL housing.
(4) Install the screws that secure the CHMSL lens to the CHMSL housing.
(5) Reconnect the battery negative cable.

CENTER HIGH MOUNTED STOP LAMP (Continued)

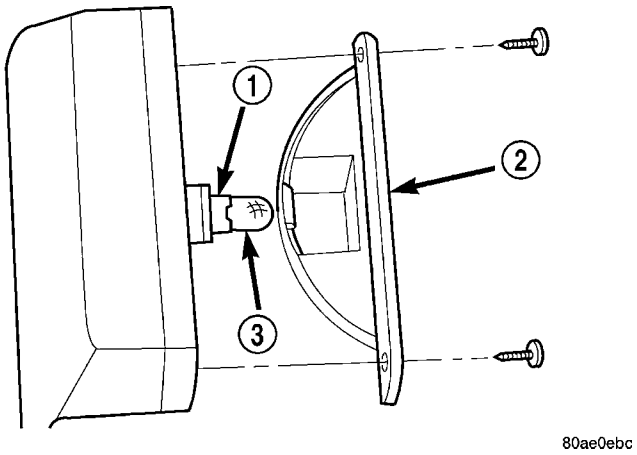


Fig. 4 CHMSL Lamp

1 - SOCKET
2 - CHMSL LENS
3 - LAMP

COMBINATION FLASHER

DESCRIPTION

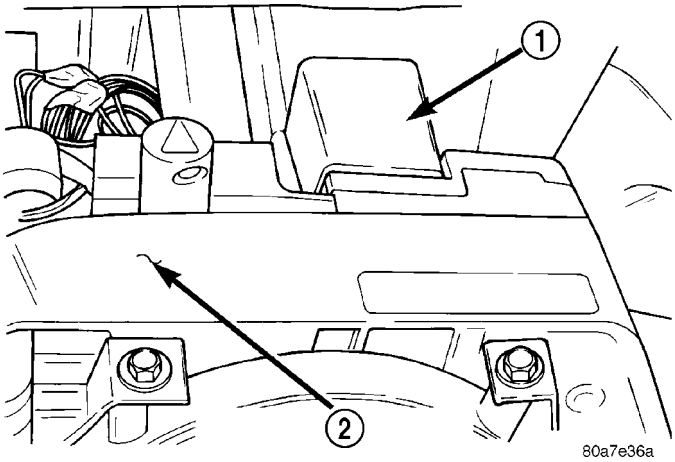


Fig. 5 Combination Flasher

1 - COMBINATION FLASHER
2 - LEFT MULTI-FUNCTION SWITCH

The combination flasher is located to the back of the left multi-function switch housing on the top of the steering column, where it is concealed beneath the upper steering column shroud (Fig. 5). The combination flasher is a smart relay that functions as both the turn signal system and the hazard warning system flasher. The combination flasher contains active electronic Integrated Circuitry (IC) elements. This flasher is designed to handle the current flow requirements of the factory-installed lighting. If supplemental lighting is added to the turn signal lamp circuits, such as when towing a trailer with lights, the combination flasher will automatically try to compensate to keep the flash rate the same.

The combination flasher cannot be repaired or adjusted and, if faulty or damaged, it must be replaced.

OPERATION

Constant battery voltage is supplied to the flasher so that it can perform the hazard warning function, and ignition switched battery voltage is supplied for the turn signal function. The Integrated Circuit (IC) within the combination flasher contains the logic that controls the flasher operation and the flash rate. The IC receives sense ground inputs from the multi-function switch for the hazard flasher, right turn signal, and left turn signal. A special design feature of the combination flasher allows it to ″sense″ that a turn signal circuit or bulb is not operating, and provide the driver an indication of the condition by flashing the remaining bulbs in the affected circuit at a higher rate.

Because of the active elements within the combination flasher, it cannot be tested with conventional automotive electrical test equipment. If the combination flasher is believed to be faulty, test the turn signal and hazard warning system prior to replacement. Refer to (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR - DIAGNOSIS AND TESTING).

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) Remove the screws that secure the lower steering column shroud to the upper shroud (Fig. 6).

(4) Move the tilt steering column to the fully lowered position and leave the tilt release lever in the released position.

(5) Remove the upper shroud from the steering column.

COMBINATION FLASHER (Continued)

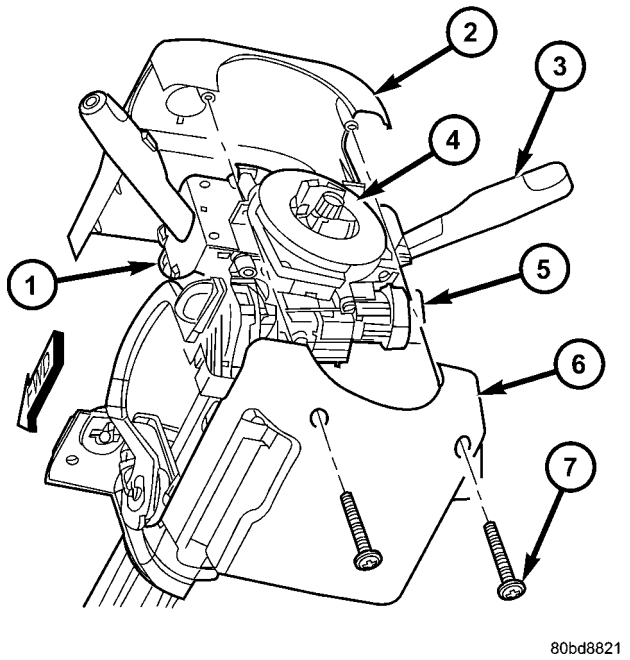


Fig. 6 Steering Column Shrouds Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - UPPER SHROUD
3 - RIGHT MULTI-FUNCTION SWITCH
4 - CLOCKSPRING
5 - IGNITION LOCK CYLINDER HOUSING
6 - LOWER SHROUD
7 - SCREW (2)

(6) Grasp the combination flasher firmly and pull it toward the dash panel to disengage the flasher terminals. (Fig. 7).

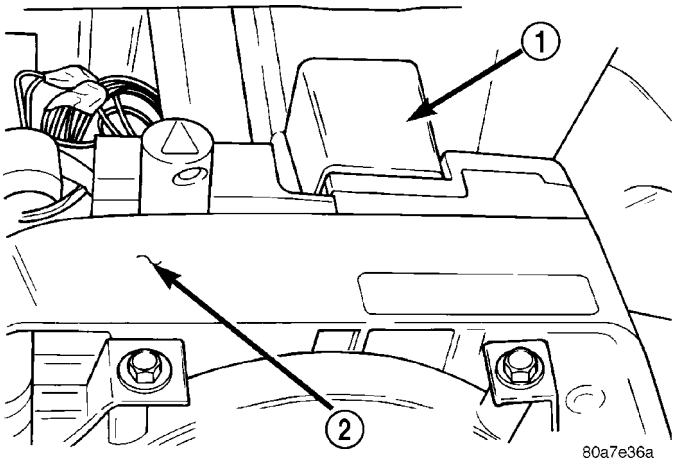


Fig. 7 Combination Flasher

1 - COMBINATION FLASHER
2 - LEFT MULTI-FUNCTION SWITCH

(7) Remove the combination flasher.

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, SEAT BELT TENSIONER, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Align the combination flasher terminals with the terminals in the connector on the back of the left multi-function switch housing. (Fig. 7).

(2) Push on the combination flasher until the terminals are fully seated in the left multi-function switch connector.

(3) Position the upper shroud onto the steering column (Fig. 27).

(4) Install and tighten the screws that secure the lower steering column shroud to the upper shroud. Tighten the screws to 2 N·m (18 in. lbs.).

(5) Move the tilt steering column to the fully raised position and secure it in place by moving the tilt release lever back to the locked position.

(6) Reinstall the steering column opening cover. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(7) Reconnect the battery negative cable.

DAYTIME RUNNING LAMP MODULE

DESCRIPTION

The Daytime Running Lights (Headlamps) System is installed on vehicles manufactured for sale in Canada only. A separate module, mounted on the cowl, controls the DRL.

DAYTIME RUNNING LAMP MODULE (Continued)

OPERATION

Battery positive voltage is supplied to the Daytime Running Lamp (DRL) module through a circuit breaker and a fuse in the PDC. Ignition positive voltage is supplied to the DRL module through a fuse in the fuse block. The DRL module also utilizes a Vehicle Speed Sensor (VSS) input, high and low beam sense circuits and high beam indicator driver. The DRL module is grounded to the chassis. Once the vehicle reaches a speed of 3 kph (2 mph) and travels more than 1 meter (3 feet) with the headlamp switch in the off position, the DRL module will activate the HIGH beams at a reduced intensity (36% of full intensity). When the headlamp switch is placed in the LOW beam position the DRL will turn off. When the headlamp switch is turned to the HIGH beam position, the high beams will operate normally (full intensity) and the DRL module will also illuminate the HIGH BEAM indicator in the instrument cluster.

DIAGNOSIS AND TESTING - DAYTIME RUNNING LAMP SYSTEM

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CONDITION	POSSIBLE CAUSES	CORRECTION
DAYTIME RUNNING LAMPS (DRL) DO NOT OPERATE	1. Headlamp switch in the on position.	1. Turn the headlamp switch to the off position.
	2. Poor connection at DRL module.	2. Secure connector on DRL module.
	3. Open battery positive voltage circuit to DRL module	3. Test and repair open circuit.
	4. Open ignition positive circuit to DLR module.	4. Test and repair open circuit.
	5. Headlamp circuit shorted to ground.	5. Test and repair headlamp circuit.
	6. Open or high resistance in DLR module ground circuit.	6. Test and repair ground circuit.
	7. Open VSS signal circuit.	7. Test and repair VSS signal circuit.
	8. Defective DRL module.	8. Replace DRL module.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the wire harness connector from the module.

(3) Remove the screws that attach the module to the cowl (Fig. 8).

(4) Separate the module from the vehicle.

INSTALLATION

(1) Position the DRL module on the cowl.

(2) Install the screws.

(3) Connect the wire harness connector to the module.

(4) Connect the battery negative cable.

DAYTIME RUNNING LAMP MODULE (Continued)

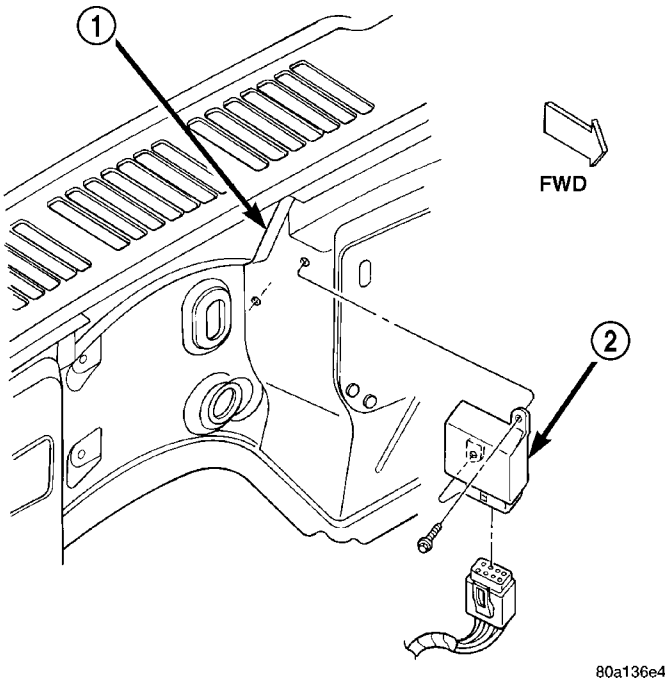


Fig. 8 DRL Module

1 - COWL
2 - DAYTIME RUNNING LAMP MODULE

FOG LAMP UNIT

DIAGNOSIS AND TESTING - FRONT FOG LAMP SYSTEM

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

CONDITION	POSSIBLE CAUSES	CORRECTION
FOG LAMPS ARE DIM WITH ENGINE IDLING OR IGNITION TURNED OFF.	1. Loose or corroded battery cables.	1. Clean and secure battery cable clamps and posts.
	2. Loose or worn generator drive belt.	2. Adjust or replace generator drive belt.
	3. Charging system output too low.	3. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
	4. Battery has insufficient charge.	4. Test battery state-of -charge. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/ BATTERY - DIAGNOSIS AND TESTING)
	5. Battery is sulfated or shorted.	5. Load test battery. (Refer to 8 - ELECTRICAL/BATTERY SYSTEM/ BATTERY - DIAGNOSIS AND TESTING)
	6. Poor lighting circuit ground.	6. Test for voltage drop across ground locations.
FOG LAMP BULBS BURN OUT FREQUENTLY	1. Charging system output too high.	1. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
	2. Loose or corroded terminals or splices in circuit.	2. Inspect and repair all connectors and splices.

FOG LAMP UNIT (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
FOG LAMPS ARE DIM WITH ENGINE RUNNING ABOVE IDLE	1. Charging system output too low.	1. Test and repair charging system. (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING)
	2. Poor lighting circuit ground.	2. Test for voltage drop across ground locations.
	3. High resistance in fog lamp circuit.	3. Test amperage draw of fog lamp circuit.
FOG LAMPS FLASH RANDOMLY	1. Poor lighting circuit ground.	1. Test for voltage drop across ground locations.
	2. High resistance in fog lamp circuit.	2. Test amperage draw of fog lamp circuit.
	3. Faulty relay	3. Replace relay
	4. Faulty fog lamp switch.	4. Replace multifunction switch.
	5. Loose or corroded terminals or splices in circuit.	5. Inspect and repair all connectors and splices.
FOG LAMPS DO NOT ILLUMINATE	1. Blown fuse for fog lamp.	1. Replace fuse.
	2. No ground at fog lamps.	2. Repair circuit ground.
	3. High beam headlamps illuminated	3. Switch headlamp beam selector to low beam.
	4. Faulty relay	4. Replace relay
	5. Faulty fog lamp switch.	5. Replace left multifunction switch.
	6. Shorted clockspring.	6. Replace clockspring.
	7. Broken connector terminal or wire splice in fog lamp circuit.	7. Repair connector terminal or wire splice.
	8. Faulty or burned out bulb.	8. Replace bulb.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the fog lamp wire harness connector.

(3) Remove the nut attaching the fog lamp to the front bumper.

(4) Remove the fog lamp.

INSTALLATION

(1) Position the fog lamp on the bumper.

(2) Install the nut attaching the fog lamp to the front bumper.

(3) Connect the fog lamp wire harness connector.

(4) Connect the battery negative cable.

(5) Readjust fog lamps (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/FOG LAMP UNIT - ADJUSTMENTS).

ADJUSTMENTS

Prepare an alignment screen. A properly aligned fog lamp will project a pattern on the alignment screen 100 mm (4 in.) below the fog lamp centerline and straight ahead (Fig. 9).

(1) For adjustment, loosen the pivot nut and turn the foglamp adjusting screw (rear center of lamp) clockwise for up, counterclockwise for down, then retighten the pivot nut (Fig. 10).

FOG LAMP UNIT (Continued)

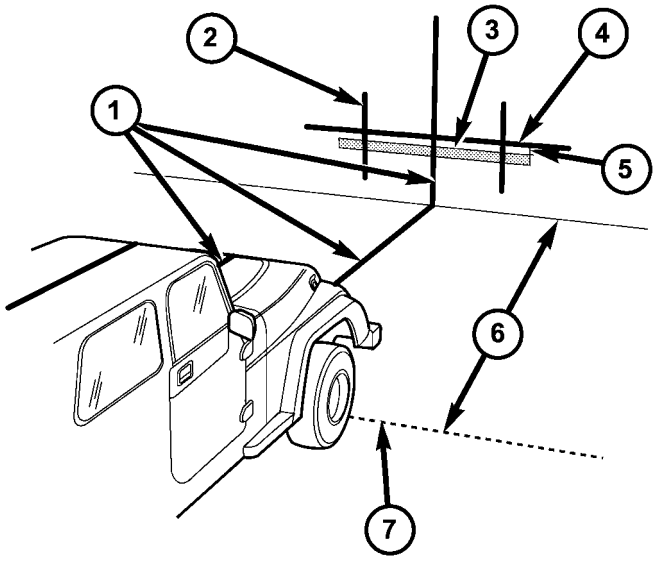


Fig. 9 Fog Lamp Alignment - Typical

1 - VEHICLE CENTERLINE
2 - CENTER OF FOG LAMP LENS
3 - HIGH - INTENSITY AREA
4 - FLOOR TO CENTER OF FOG LAMP LENS
5 - 100mm (4 in.)
6 - 7.62 METERS (25 FEET), EXPORT USE 10 METERS (32.8ft.).
7 - FRONT OF FOG LAMP

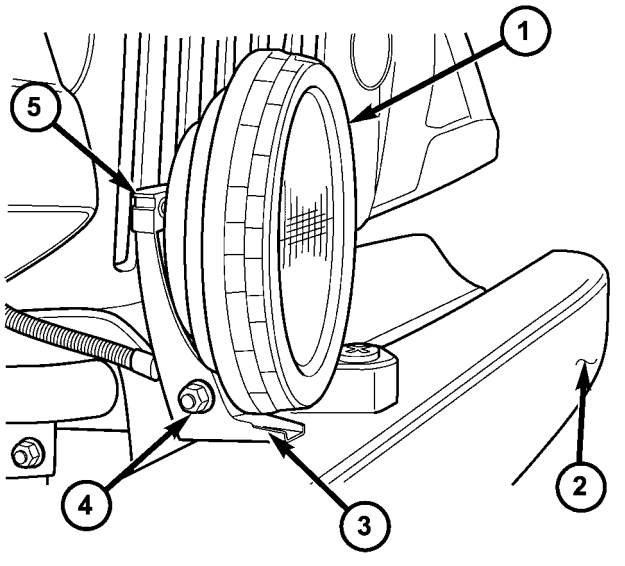


Fig. 10 FOG LAMP ADJUSTER

1 - FOG LAMP
2 - BUMPER
3 - MOUNTING BRACKET
4 - PIVOT NUT
5 - ADJUSTER

FOG LAMP

REMOVAL

CAUTION: Do not touch the bulb glass with fingers or other oily surfaces. Reduced bulb life will result.

(1) Disconnect and isolate the battery negative cable.
(2) Remove the screws that attach the lens and reflector to the lamp housing.
(3) Separate the lens and reflector from the lamp housing.
(4) Disconnect the electrical connector from the bulb.
(5) Squeeze the bulb retainer together to disengage it from the reflector.
(6) Remove the bulb from the reflector (Fig. 11).

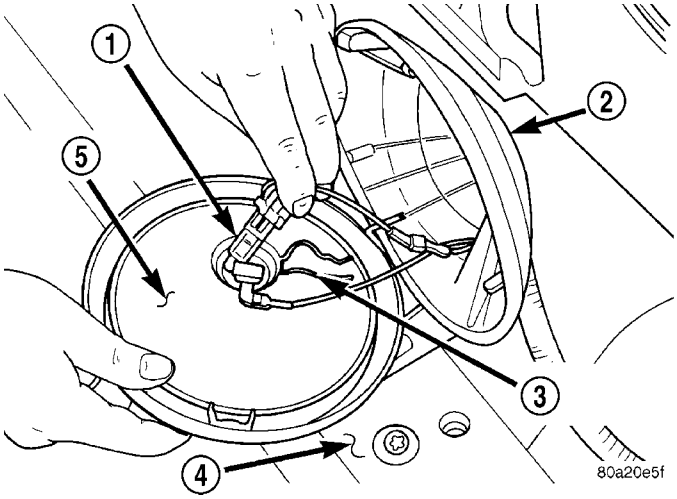


Fig. 11 Fog Lamp Bulb

1 - BULB
2 - FOG LAMP HOUSING
3 - BULB RETAINER
4 - FRONT BUMPER
5 - LENS AND REFLECTOR

INSTALLATION

CAUTION: Do not touch the bulb glass with fingers or other oily surfaces. Reduced bulb life will result.

(1) Install the bulb into the reflector.
(2) Position and engage the bulb retainer onto the reflector.
(3) Connect the electrical connector to the bulb.
(4) Install the lens and reflector to the lamp housing.
(5) Install the screws that attach the lens and reflector to the lamp housing.
(6) Reconnect the battery negative cable.

FOG LAMP RELAY

DESCRIPTION

The fog lamp relay is located in the Power Distribution Center (PDC) in the engine compartment of the vehicle. The fog lamp relay is a conventional International Standards Organization (ISO) micro relay (Fig. 12). Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal patterns, and terminal functions. The relay is contained within a small, rectangular, molded plastic housing and is connected to all of the required inputs and outputs by five integral male spade-type terminals that extend from the bottom of the relay base.

The fog lamp relay cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

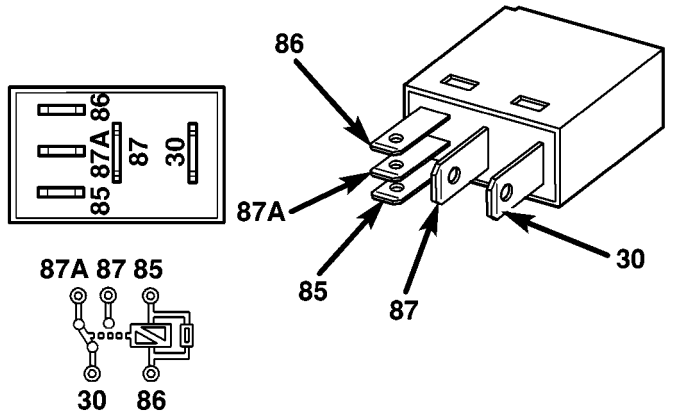


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Fig. 12 ISO MICRO RELAY

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

OPERATION

The fog lamp relay is an electromechanical switch that uses a low current input from the multi-function switch to control a high current output to the fog lamps. The movable common feed contact point is held against the fixed normally closed contact point by spring pressure. When the relay coil is energized, an electromagnetic field is produced by the coil windings. This electromagnetic field draws the movable relay contact point away from the fixed normally closed contact point, and holds it against the fixed normally open contact point. When the relay coil is de-energized, spring pressure returns the movable contact point back against the fixed normally closed contact point. A resistor is connected in parallel with the relay coil in the relay, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The fog lamp relay terminals are connected to the vehicle electrical system through a connector receptacle in the Power Distribution Center (PDC).

The fog lamp relay can be diagnosed using conventional diagnostic tools and methods. Refer to the appropriate wiring information for diagnosis and testing of the fog lamp micro-relay and for complete wiring diagrams.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the cover for the Power Distribution Center (PDC).

(3) Remove the fog lamp relay by grasping it firmly and pulling it straight out from the receptacle in the PDC.

INSTALLATION

(1) Position the fog lamp relay to the proper receptacle in the Power Distribution Center (PDC).

(2) Align the fog lamp relay terminals with the terminal cavities in the PDC.

(3) Press firmly and evenly on the top of the fog lamp relay until the terminals are fully seated in the PDC.

(4) Reconnect the battery negative cable.

FOG LAMP UNIT - REAR

DESCRIPTION

Some vehicles are equipped with a rear fog lamp. The lamp unit can be found mounted to the rear bumper assembly. The rear fog lamp unit utilizes a red lens with a clear lamp for brightness, and an illuminated indicator in the instrument cluster.

Rear fog lamp(s) are standard equipment in certain parts of the world where excessive fog is experienced on a regular basis.

OPERATION

Battery positive (B+) voltage is supplied to the left multi-function switch. With the headlamp switch in the on position, battery positive voltage is supplied to the rear fog lamp switch contacts within the multi-function switch. Pulling the headlamp switch knob outward and then rotating the knob one detent counterclockwise will activate the rear fog lamp(s) and illuminate the rear fog lamp indicator in the instrument cluster. The rear fog lamp(s) can only be activated when either the Low Beam or High Beam headlamps are on. Refer to Wiring Diagrams for a complete system schematic.

FOG LAMP UNIT - REAR (Continued)

DIAGNOSIS AND TESTING - REAR FOG LAMP

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

REAR FOG LAMP CONDITION CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
REAR FOG LAMP IS ON WITH IGNITION KEY ON	1. Shorted multifunction switch.	1. Test and replace multifunction switch
	2. Shorted steering column clock spring.	2. Test and replace clock spring.
	3. Shorted fog lamp switch output circuit	3. Repair short to voltage in fog lamp switch output circuit.
REAR FOG LAMP BULB BURNS OUT FREQUENTLY	1. Charging system output too high.	1. Test and repair charging system.
	2. Loose or corroded terminals or splices in circuit.	2. Inspect and repair all connectors and splices.
REAR FOG LAMP IS DIM WITH ENGINE RUNNING	1. Charging system output too low.	1. Test and repair charging system.
	2. Poor lighting circuit ground.	2. Test for voltage drop across ground locations.
	3. High resistance in fog lamp circuit.	3. Test amperage draw of fog lamp circuit.
REAR FOG LAMP FLASHES RANDOMLY	1. Poor lighting circuit ground.	1. Test for voltage drop across ground locations.
	2. High resistance in fog lamp circuit.	2. Test amperage draw of fog lamp circuit.
	3. Faulty multifunction switch.	3. Replace multifunction switch.
	4. Loose or corroded terminals or splices in circuit.	4. Inspect and repair all connectors and splices.
REAR FOG LAMP WILL NOT ILLUMINATE	1. Open fuse for fog lamp.	1. Test fog lamp circuitry for short to ground and replace fuse.
	2. Open battery positive voltage circuit to multifunction switch.	2. Test and repair open battery positive voltage circuit.
	3. No ground at fog lamps.	3. Repair circuit ground.
	4. Faulty multifunction switch.	4. Replace multifunction switch.
	5. Broken connector terminal or wire splice in fog lamp circuit.	5. Repair connector terminal or wire splice.
	6. Faulty or burned out bulb.	6. Replace bulb.

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Remove the screws that secure the rear fog lamp unit to the rear bumper assembly.

(3) Disconnect the wire harness connector from the rear fog lamp connector (Fig. 13).

(4) Remove the rear fog lamp unit from the vehicle.

FOG LAMP UNIT - REAR (Continued)

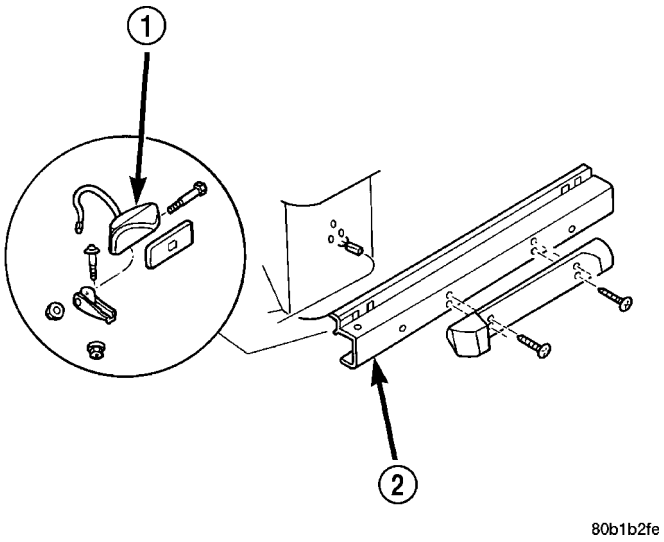

Fig. 13 Rear Fog Lamp Unit

1 - REAR FOG LAMP UNIT
2 - REAR BUMPER ASSEMBLY

INSTALLATION

(1) Position the rear fog lamp unit to the vehicle.
(2) Connect the wire harness connector to the rear fog lamp connector.
(3) Install the screws that secure the rear fog lamp unit to the rear bumper assembly.
(4) Reconnect the negative battery cable.

FOG LAMP - REAR

REMOVAL

(1) Disconnect and isolate the battery negative cable.
(2) Remove the screws that secure the rear fog lamp lens to the housing (Fig. 14).
(3) Push and rotate the rear fog lamp one-third turn counterclockwise, then pull the lamp straight out of the socket (Fig. 15).

INSTALLATION

(1) Install the rear fog lamp by pushing the lamp into the rear fog lamp socket and turning it one-third clockwise.
(2) Position the rear fog lamp lens onto the housing.
(3) Install the screws that secure the rear fog lamp lens onto the housing.
(4) Reconnect the negative battery cable.

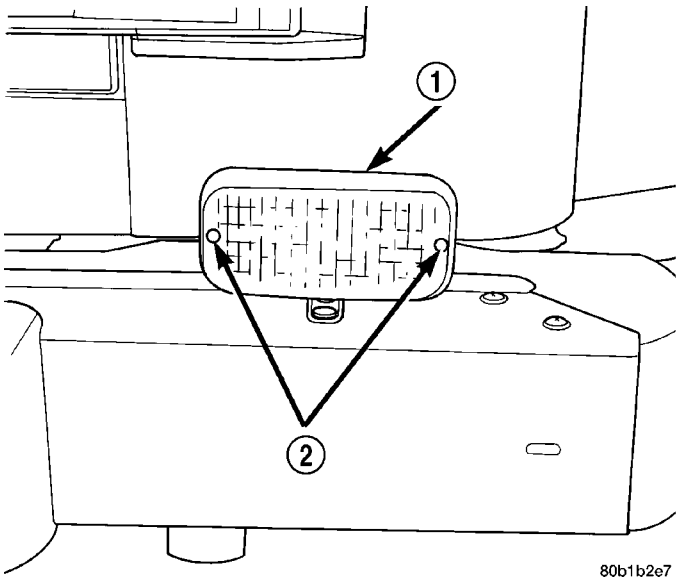

Fig. 14 Rear Fog Lamp Lens

1 - REAR FOG LAMP
2 - FOG LAMP LENS SCREWS

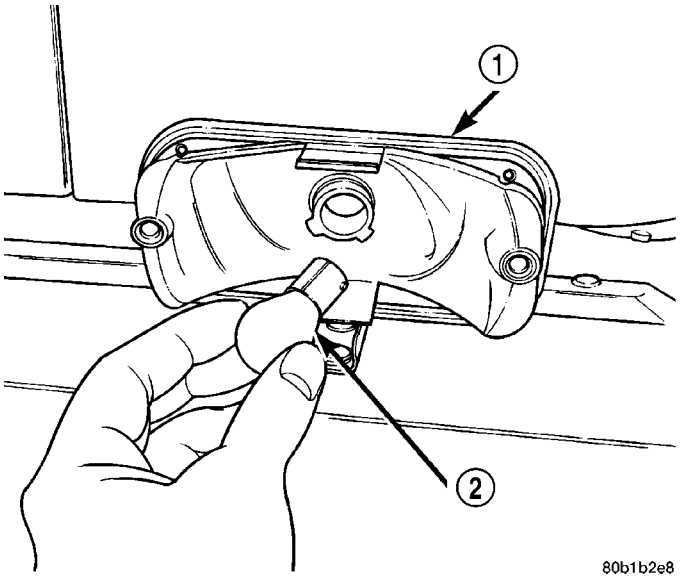

Fig. 15 Rear Fog Lamp

1 - REAR FOG LAMP HOUSING
2 - REAR FOG LAMP

FRONT PARK/TURN SIGNAL LAMP UNIT

DESCRIPTION

A front turn signal/parking lamp assembly is mounted to each front fender. Each front turn signal/parking lamp assembly consist of a housing, lens and lamp.

FRONT PARK/TURN SIGNAL LAMP UNIT (Continued)

OPERATION

The parking light function is controlled by the headlamp switch located on the instrument panel. The turn signal function is controlled by the left multi-function switch located on the steering column. Each front turn signal/parking lamp assembly can be serviced separately.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the screws that secure the park/turn signal lamp housing to the front fender (Fig. 16).

(3) Remove the park/turn signal lamp housing from the fender.

(4) Rotate the socket one-third turn counterclockwise and separate the socket and lamp from the housing.

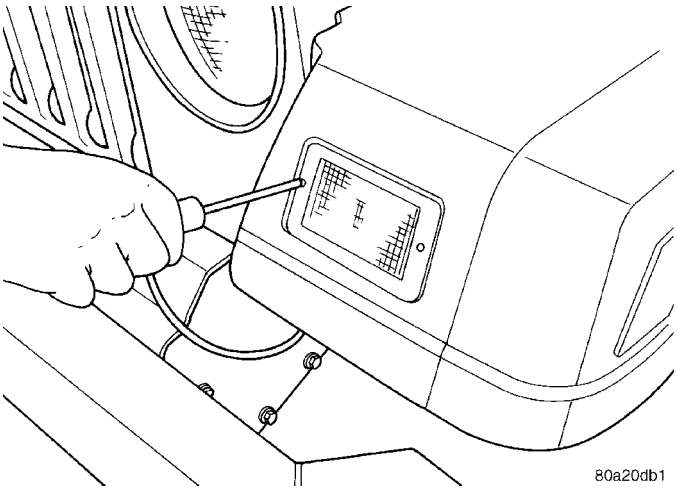

Fig. 16 Park/Turn Signal Lamp Assembly

INSTALLATION

(1) Install the socket and lamp into the housing and rotate the socket one-third turn clockwise.

(2) Position the park/turn signal lamp housing into the fender.

(3) Install the screws that secure the park/turn signal lamp housing to the fender. Tighten the screws securely.

(4) Reconnect the battery negative cable.

FRONT PARK/TURN SIGNAL LAMP

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the park/turn signal lamp housing from the fender (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/PARK/TURN SIGNAL LAMP UNIT - REMOVAL).

(3) Pull the lamp straight out of the socket.

INSTALLATION

(1) Install the lamp by pushing it into the socket.

(2) Install the lamp housing into the fender (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/PARK/TURN SIGNAL LAMP UNIT - INSTALLATION).

(3) Reconnect the battery negative cable.

HEADLAMP

DESCRIPTION

The headlamps are sealed beam units. Each unit contains a high and low beam filament.

OPERATION

The headlamps are turned on and off by the headlamp switch. The high and low beam selection is controlled by the left multi-function switch. Each headlamp can be serviced individually.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the screws that secure the headlamp bezel to the grille (Fig. 17).

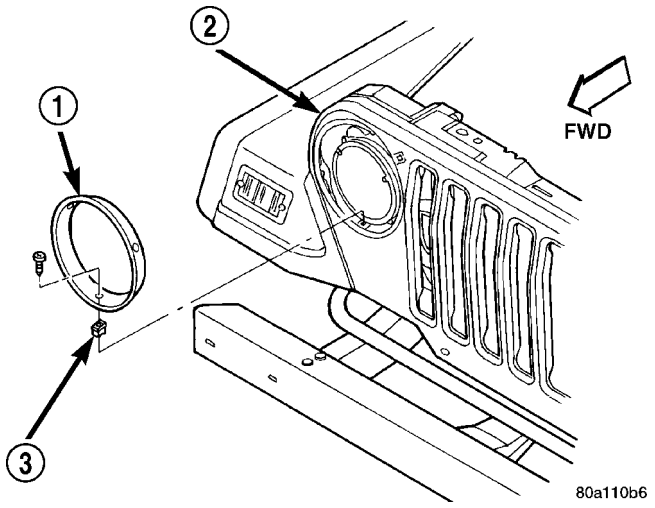

Fig. 17 Headlamp Bezel

1 - HEADLAMP BEZEL
2 - GRILLE
3 - RETAINER

(3) Remove the screws that secure the headlamp retaining ring to the headlamp bucket (Fig. 18).

(4) Remove the headlamp from the bucket and disconnect the wire harness connector from the headlamp (Fig. 19).

HEADLAMP (Continued)

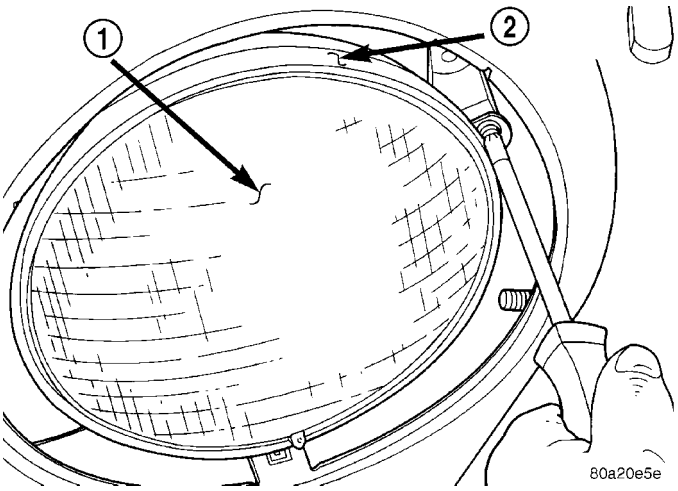

Fig. 18 Headlamp Retaining Ring

1 - HEADLAMP
2 - HEADLAMP RETAINING RING

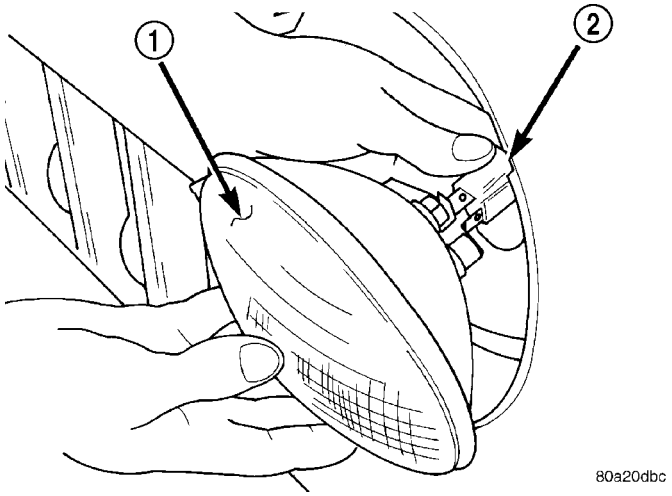

Fig. 19 Headlamp Wire Harness Connector

1 - HEADLAMP
2 - WIRE HARNESS CONNECTOR

INSTALLATION

(1) Connect the wire harness connector to the headlamp and position the headlamp into the bucket.

NOTE: Insure the bosses on the headlamp align with the slots in the bucket.

(2) Align the headlamp bosses with the slots in the bucket and install the retaining ring.

(3) Install the screws that secure the headlamp retaining ring to the headlamp bucket. Tighten the screws securely.

(4) Install the headlamp bezel.

NOTE: Do not overtighten the screws or damage to the headlamp bezel may result.

(5) Install the screws that secure the headlamp bezel. Tighten the screws securely.

(6) Reconnect the battery negative cable.

ADJUSTMENTS

HEADLAMP ALIGNMENT PREPARATION

NOTE: If the vehicle is equipped with a headlamp leveling system, be certain the headlamp leveling switch is in the "0" position.

(1) Verify headlamps are operational in all modes and illuminated in the low beam setting.

(2) Correct defective components that could hinder proper headlamp alignment.

(3) Verify proper tire inflation.

(4) Clean headlamp lenses.

(5) Verify that luggage area is not heavily loaded.

(6) Fuel tank should be FULL. Add 2.94 kg (6.5 lbs.) of weight over the fuel tank for each estimated gallon of missing fuel.

ALIGNMENT SCREEN PREPARATION

(1) Position vehicle on a level surface perpendicular to a flat wall 7.62 meters (25 ft.), export vehicles use 10 meters (32.8 ft.), away from front of headlamp lens (Fig. 20).

(2) If necessary, tape a line on the floor 7.62 meters (25 ft.), export vehicles use 10 meters (32.8 ft.), away from and parallel to the wall.

(3) Measure from the floor up 1.27 meters (5 ft.) and tape a vertical line on the wall at the centerline of the vehicle. Sight along the centerline of the vehicle (from rear of vehicle forward) to verify accuracy of the line placement.

(4) Rock vehicle side-to-side three times to allow suspension to stabilize.

(5) Jounce front suspension three times by pushing downward on front bumper and releasing.

(6) Measure the distance from the center of headlamp lens to the floor. Transfer measurement to the alignment screen (with tape). Use this horizontal line for up/down adjustment reference.

(7) Place a tape line 130 mm (5.12 in.) below parallel to center of headlamp line.

(8) Measure distance from the centerline of the vehicle to the center of each headlamp being aligned. Transfer measurements to screen (with tape) to each side of vehicle centerline. Use these lines for left/right adjustment reference.

HEADLAMP ADJUSTMENT

(1) Place headlamps on LOW beam.

(2) Cover front of the headlamp that is not being adjusted.

HEADLAMP (Continued)

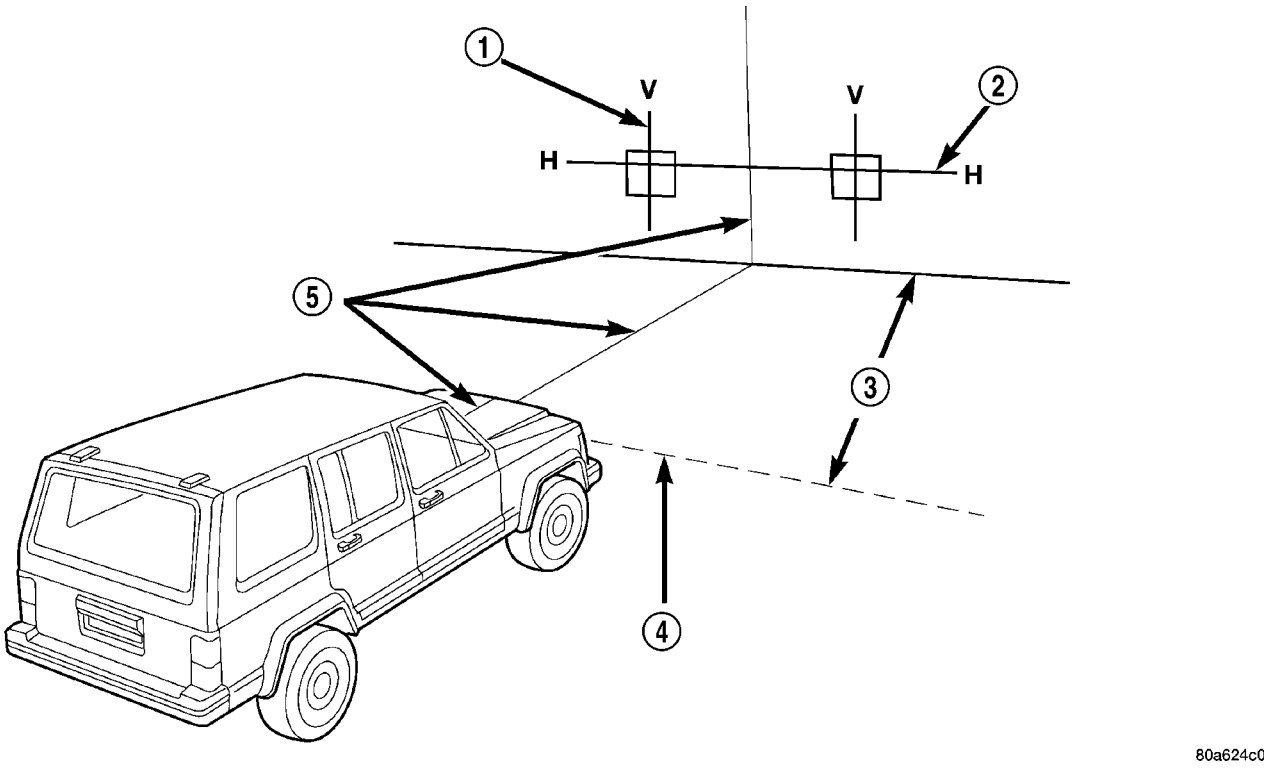

Fig. 20 Headlamp Alignment Screen - Typical

1 - CENTER OF VEHICLE TO CENTER OF HEADLAMP LENS
2 - FLOOR TO CENTER OF HEADLAMP LENS
3 - 7.62 METERS (25 FEET)
4 - FRONT OF HEADLAMP
5 - VEHICLE CENTERLINE

(3) Turn the upper, outboard (up/down) adjustment screw (Fig. 21)until the headlamp beam pattern on screen/wall is similar to the pattern depicted in (Fig. 20)

NOTE: When using a headlamp aiming screen:

- Adjust the headlamps so that the beam horizontal position is at 0.
- Adjust the beam vertical position is 25 mm (1 in) downward from the lamp horizontal centerline.

(4) Rotate the lower, inboard (left/right) adjustment screw (Fig. 21) until the headlamp beam pattern on the aiming screen/wall similar to the pattern in (Fig. 20).

(5) Cover front of the headlamp that has been adjusted and adjust the other headlamp beam as instructed above.

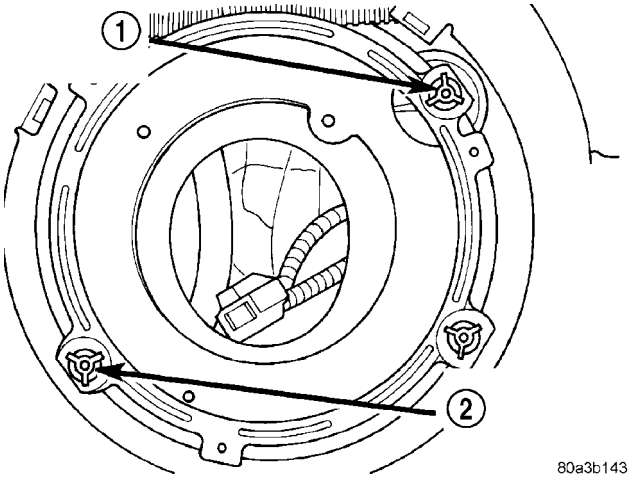

Fig. 21 Headlamp Adjustment Screws

1 - UP/DOWN ADJUSTMENT SCREW
2 - LEFT/RIGHT ADJUSTMENT SCREW

HEADLAMP LEVELING SWITCH

REMOVAL

HEADLAMP LEVELING SWITCH

WARNING: ON VEHICLES EQUIPPED WITH AIR-BAGS, REFER TO GROUP 8M - PASSIVE RESTRAINT SYSTEMS BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. FAILURE TO TAKE THE PROPER PRE-CAUTIONS COULD RESULT IN ACCIDENTAL AIR-BAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect the negative battery cable.
(2) Using a flat blade screwdriver or similar tool between the headlamp leveling switch and the steering column cover (Fig. 22). Gently pry the headlamp leveling switch out of the steering column cover.

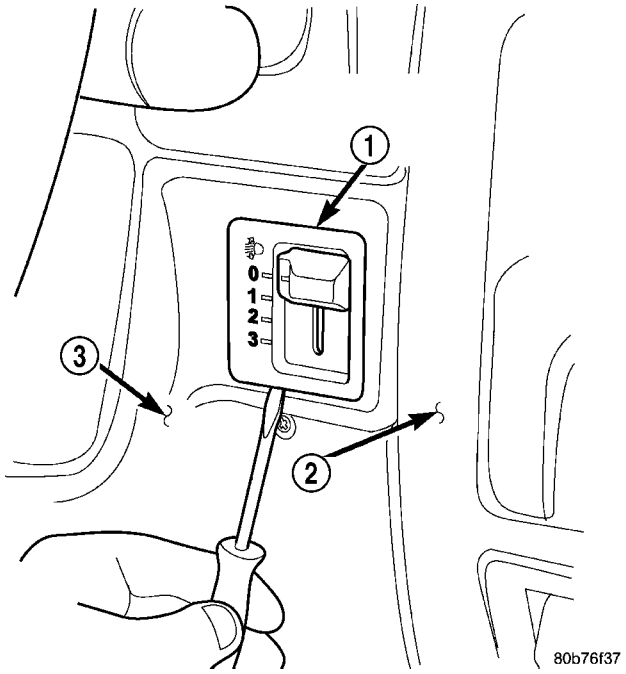

Fig. 22 Headlamp Leveling Switch

1 - HEADLAMP LEVELING SWITCH
2 - I/P CENTER BEZEL
3 - STEERING COLUMN COVER

(3) Disconnect the headlamp leveling switch electrical connector (Fig. 23) and remove the headlamp leveling switch.

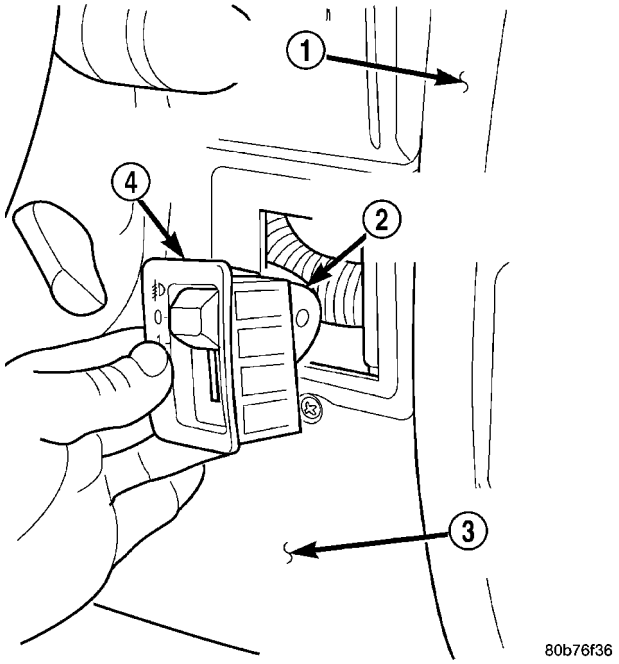

Fig. 23 Headlamp Leveling Switch Electrical Connector

1 - I/P CENTER BEZEL
2 - HEADLAMP LEVELING SWITCH ELECTRICAL CONNECTOR
3 - STEERING COLUMN COVER
4 - HEADLAMP LEVELING SWITCH

INSTALLATION

(1) Position the headlamp leveling switch in front of the steering column cover access hole and connect the switch electrical connector (Fig. 23).
(2) Properly seat the headlamp leveling switch into the steering column cover.
(3) Reconnect negative battery cable.

HEADLAMP LEVELING MOTOR

DIAGNOSIS AND TESTING - HEADLAMP LEVELING SYSTEM

The following test is designed to diagnosis a faulty headlamp leveling system. Refer to Wiring Diagrams for a complete system schematic.

NOTE: Headlamps must be operating properly and the battery must be completely charged (12.4v) prior to testing. It may also be necessary to install battery charger on the vehicles electrical system when performing this test. Refer to the Battery section of the service manual for detailed information.

HEADLAMP LEVELING MOTOR (Continued)

CONDITION	POSSIBLE CAUSES	CORRECTION
ONE MOTOR DOES NOT OPERATE	1. Poor electrical connection at motor.	1. Check for proper electrical harness connection and circuit terminal tension at the motor.
	2. Open or high resistance in the battery positive voltage circuit to the inoperative leveling motor.	2. Repair the open or high resistance in the leveling motor battery positive voltage circuit.
	3. Open, high resistance in the inoperative leveling motor ground circuit.	3. Repair the open, high resistance in the leveling motor ground circuit.
	4. Open, high resistance, short to voltage or short to ground in the inoperative leveling motor sense circuit.	4. Repair the open, high resistance, short to voltage or short to ground in the motor sense circuit.
	5. Inoperative/Damaged motor.	5. Replace leveling motor.
BOTH MOTORS DO NOT OPERATE	1. No battery positive voltage to the headlamp leveling switch.	1. Repair the open circuit or high resistance in the headlamp leveling switch battery positive voltage circuit.
	2. No battery positive voltage to the headlamp leveling motor.	2. Repair the open circuit or high resistance in the headlamp leveling motor battery positive voltage circuit.
	3. Open, high resistance in the headlamp leveling switch ground circuit.	3. Repair the open, high resistance or short to voltage in the headlamp leveling switch ground circuit.
	4. Open, high resistance in the headlamp leveling motor ground circuit.	4. Repair the open, high resistance or short to voltage in the headlamp leveling motor ground circuit.
	5. Open, high resistance, short to voltage or short to ground in the leveling motor sense circuit.	5. Repair the open, high resistance, short to voltage or short to ground in the motor sense circuit.
	6. Inoperative headlamp leveling switch.	6. Replace headlamp leveling switch.
	7. Inoperative headlamp leveling motors.	7. Replace headlamp motors.
MOTORS DO NOT RESPOND TO ONE OR MORE LEVELING POSITION(S)	1. Inoperative headlamp leveling switch.	1. Replace headlamp leveling switch.
	2. Stuck/Binding or Inoperative motor(s).	2. Replace headlamp leveling motor(s).

LEFT MULTI-FUNCTION SWITCH

DESCRIPTION

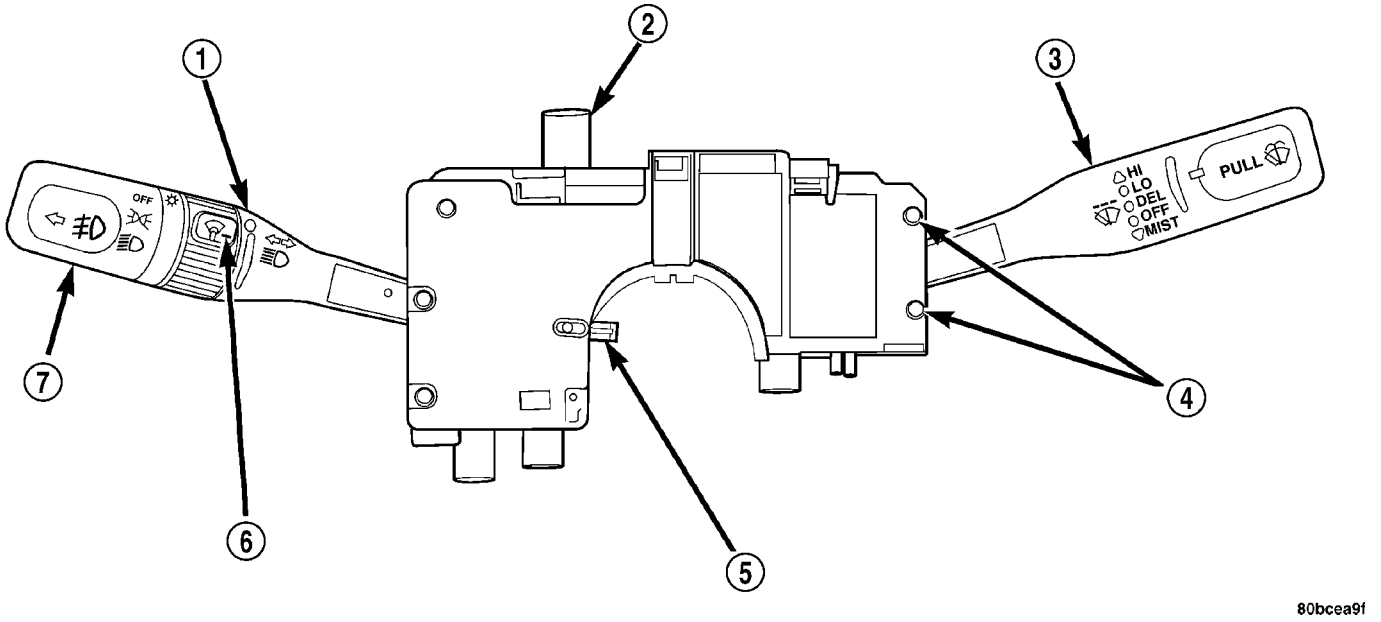

Fig. 24 MULTI-FUNCTION SWITCH

1 - LEFT MULTI-FUNCTION SWITCH CONTROL STALK
2 - HAZARD WARNING SWITCH BUTTON
3 - RIGHT MULTI-FUNCTION SWITCH CONTROL STALK
4 - SCREW (2)
5 - TURN SIGNAL CANCEL ACTUATOR
6 - LEFT MULTI-FUNCTION SWITCH CONTROL RING
7 - LEFT MULTI-FUNCTION SWITCH CONTROL KNOB

The left multi-function switch is secured to the upper steering column housing, below the steering wheel. (Fig. 24). The hazard warning switch push button is located on the top of the multi-function switch housing.

A connector containing eighteen terminal pins is located on the back of the switch housing and connects the switch to the vehicle electrical system. A second connector on the back of the switch housing accepts the combination flasher. The right and left multi-function switches are mounted together by two screws, and the combined multi-function switch is secured to the upper steering column housing.

The left multi-function switch is the primary control for the interior and exterior lighting systems of the vehicle. The left multi-function switch supports the following functions:

- Hazard Warning Control
- Exterior Lighting Control
- Headlamp Beam Selection
- Headlamp Optical Horn
- Interior Lighting Control
- Turn Signal Control

The left multi-function switch cannot be adjusted or repaired. If any function of the switch is faulty, or if the switch is damaged, the entire switch unit must be replaced. The combination flasher and the right multi-function switch are available for separate service replacement.

OPERATION

The left multi-function switch uses a combination of resistor multiplexed and conventionally switched outputs to control the many functions and features it provides. The switch is grounded to the left cowl side inner panel, beneath the instrument panel. The switch receives battery voltage from a fuse in the Power Distribution Center (PDC), a fused B(+) circuit and, when the ignition switch is in the Accessory or On positions, from a fuse in the fuse block through a fused ignition switch output (run-acc) circuit. Following are descriptions of the how the left multi-function switch operates to control the many functions and features it provides:

- **Front Fog Lamps** - The control knob on the end of the left multi-function switch control stalk is pulled outward to activate the optional front fog

LEFT MULTI-FUNCTION SWITCH (Continued)

lamps. The control knob is keyed so that it cannot be pulled outward unless the knob is first rotated to turn on the exterior lighting. The internal circuitry of the left multi-function switch then provides battery voltage through a fused fog lamp relay output circuit and the fog lamp relay to the fog lamps and to the instrument cluster for control of the fog lamp indicator.

- **Rear Fog Lamp(s)** - The headlamp switch knob must be rolled to the headlamp on position. Pulling the headlamp switch knob outward and then rolling the knob one more detent counterclockwise will activate the rear fog lamp. The headlamp switch is only able to pull outward while in the headlamp position. The internal circuitry of the left multi-function switch then provides battery voltage to the rear fog lamp(s) and to the instrument cluster for control of the fog lamp indicator.
- **Hazard Warning System** - The hazard warning push button is pushed down to unlatch the switch and activate the hazard warning system, and pushed down again to latch the switch and turn the system off. When the hazard warning switch is latched, the push button will be in a lowered position on the top of the steering column shroud; and, when the hazard warning switch is unlatched, the push button will be in a raised position. The left multi-function switch hazard warning simultaneously provides a signal to the hazard warning sense of the combination flasher to activate or deactivate the flasher output, and directs the output of the flasher to the hazard warning lamps.
- **Headlamps** - The control knob on the end of the left multi-function switch control stalk is rotated forward to its second detent from the Off position to activate the headlamps. The internal circuitry of the left multi-function switch then provides battery voltage through the integral beam select switch and the headlamp low beam or high beam circuits to the appropriate headlamp filaments and to the instrument cluster for control of the high beam indicator.
- **Headlamp Beam Selection** - The left multi-function switch control stalk is pulled towards the steering wheel past a detent, then released to actuate the integral beam select switch circuitry. Each time the control stalk is actuated in this manner, the opposite headlamp mode from what is currently selected will be activated.
- **Headlamp Optical Horn** - The left multi-function switch control stalk is pulled towards the steering wheel to just before a detent, to momentarily activate the headlamp high beams. The high beams will remain illuminated until the control stalk is released. The internal beam select switch circuitry directs battery voltage through the headlamp high beam circuit of the left multi-function switch to the headlamp high beams and to the instrument cluster for control of the high beam indicator.
- **Interior Lamps Control** - A control ring on the left multi-function switch control stalk is rotated to a full forward detent to illuminate all interior courtesy lamps. The instrument cluster circuitry monitors the hard wired variable resistor output of the left multi-function switch through the panel lamps dimmer signal circuit then, based upon that input, provides a ground path to activate all interior courtesy lamps through a courtesy lamp feed circuit output.
- **Panel Lamps Dimming** - A control ring on the left multi-function switch control stalk is rotated to one of six intermediate detent positions to select the desired illumination intensity of all adjustable instrument panel and instrument cluster lighting. The control ring is rotated forward to brighten, or rearward to dim the lighting. The instrument cluster monitors the hard wired variable resistor output of the left multi-function switch through the panel lamps dimmer signal circuit then, based upon that input, provides a pulse width modulated output, to control the instrument cluster lighting levels. The instrument cluster also controls the lighting levels of the other adjustable instrument panel lighting based upon this panel lamps dimmer signal through a panel lamps driver circuit output.
- **Parade Mode** - A control ring on the left multi-function switch control stalk is rotated to an intermediate detent that is one detent rearward from the full forward detent to illuminate the Vacuum Fluorescent Display (VFD) in the instrument cluster and the radio at full intensity. The instrument cluster monitors the hard wired variable resistor output of the left multi-function switch through the panel lamps dimmer signal circuit then, based upon that input, adjusts the instrument cluster VFD to its full intensity and provides a battery voltage signal to the radio on a park lamp relay output circuit that signals the radio to light its VFD to full intensity.
- **Park Lamps** - The control knob on the end of the left multi-function switch control stalk is rotated forward to its first detent to activate the parking lamps. The left multi-function switch then provides battery voltage to the parking lamps and to the instrument cluster as a request for cluster illumination and panel lamps output.
- **Turn Signal Control** - The left multi-function switch control stalk actuates the turn signal switch. The multi-function switch turn signal circuitry simultaneously provides a signal to the turn signal sense circuit of the combination flasher to activate the flasher output. The turn signal switch has a detent position in each direction that provides turn signals with automatic cancellation, and an intermediate, momentary position in each direction that provides

turn signals only until the left multi-function switch control stalk is released. When the control stalk is moved to a turn signal switch detent position, the cancel actuator extends toward the center of the steering column. A turn signal cancel cam that is integral to the clockspring, rotates with the steering wheel and the cam lobes contact the cancel actuator when it is extended from the multi-function switch. If only momentary signaling is desired, the switch is actuated to a left or right intermediate detent position. In this position the signal lamps flash as described above, but the switch returns to the Off position as soon as the lever is released. When the system is activated, one of two turn indicators in the instrument cluster flashes in unison with the turn signal lamps, indicating to the driver that the system is operating.

DIAGNOSIS AND TESTING - LEFT MULTI-FUNCTION SWITCH

Refer to the appropriate wiring information.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the left multi-function switch from the steering column (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/LEFT MULTI-FUNCTION SWITCH - REMOVAL).

(3) Remove the combination flasher from the left multi-function switch.

(4) Using an ohmmeter, perform the continuity and resistance tests at the terminals in the left multi-function switch connector receptacles as shown in the Left Multi-Function Switch Test table. Refer to (Fig. 25) and (Fig. 26) for connector terminal and cavity identification.

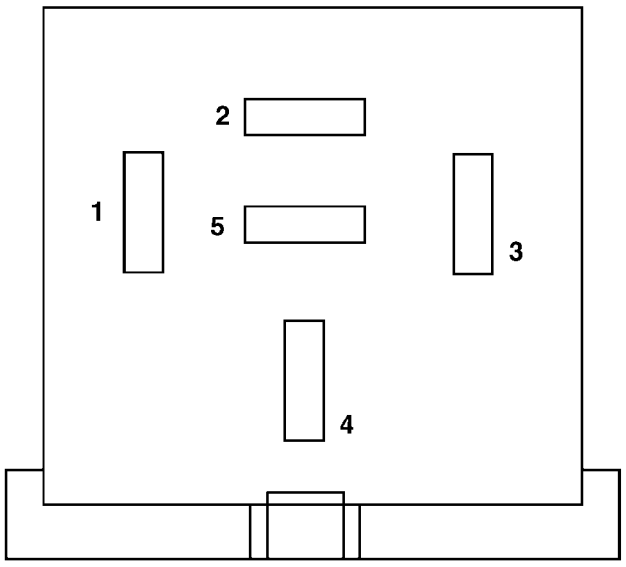

Fig. 25 Combination Flasher Receptacle (Connector A)

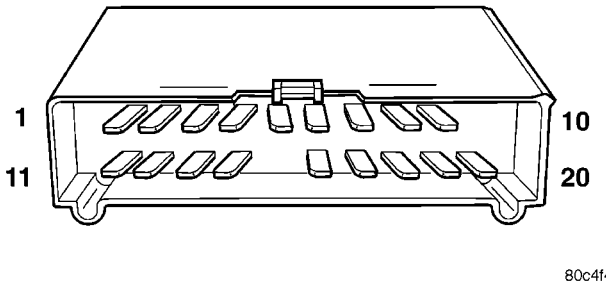

Fig. 26 Left Multi-Function Switch Connector Receptacle (Connector B)

LEFT MULTI-FUNCTION SWITCH (Continued)

LEFT (LIGHTING) MULTI-FUNCTION SWITCH TEST			
TURN SIGNAL AND HAZARD WARNING SWITCH TESTS			
SWITCH POSITION		CONTINUITY BETWEEN	
TURN	HAZARD		
Neutral	Off	B1 & B4, B1 & B5	
Left	Off	A2 & B2, A2 & B4, B1 & B5	
Right	Off	A2 & B5, B1 & B4	
Neutral	On	A2 & B2, A2 & B4, A2 & B5, A2 & B6, A3 & B7, A3 & A5	
EXTERIOR LIGHTING SWITCH TESTS			
SWITCH POSITION		CONTINUITY BETWEEN	
Park Lamps On		B9 & B20	
Headlamp Low Beams On		B16 & B18, B16 & B19	
Headlamp High Beams On		B17 & B18, B17 & B19	
Fog Lamps On		B13 & B14	
Rear Fog Lamps On		B12 & B13	
Optical Horn On		B17 & B18, B17 & B19	
INTERIOR LIGHTING SWITCH TESTS			
SWITCH POSITION		RESISTANCE BETWEEN	RESISTANCE (OHMS)
Panel Lamps Dimming Position 1 (Dimmest)		B7 & B8	5653 ± 10%
Dimming Position 2		B7 & B8	3743 ± 10%
Dimming Position 3		B7 & B8	2593 ± 10%
Dimming Position 4		B7 & B8	1825 ± 10%
Dimming Position 5		B7 & B8	1221 ± 10%
Dimming Position 6 (Brightest)		B7 & B8	768 ± 10%
Parade Mode On		B7 & B8	412 ± 10%
Courtesy Lamps On		B7 & B8	150 ± 10%

(5) If the left multi-function switch fails any of the continuity or resistance tests, replace the faulty switch unit as required.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) Remove the two screws that secure the lower steering column shroud to the upper shroud (Fig. 27).

(4) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully lowered position and leave the tilt release lever in the released (down) position.

(5) Remove both the upper and lower shrouds from the steering column.

(6) Disconnect the cross body wire harness connector for the left multi-function switch from the connector receptacle on the back of the switch.

(7) Disconnect the cross body wire harness connector for the right multi-function switch from the connector receptacle on the back of the switch.

LEFT MULTI-FUNCTION SWITCH (Continued)

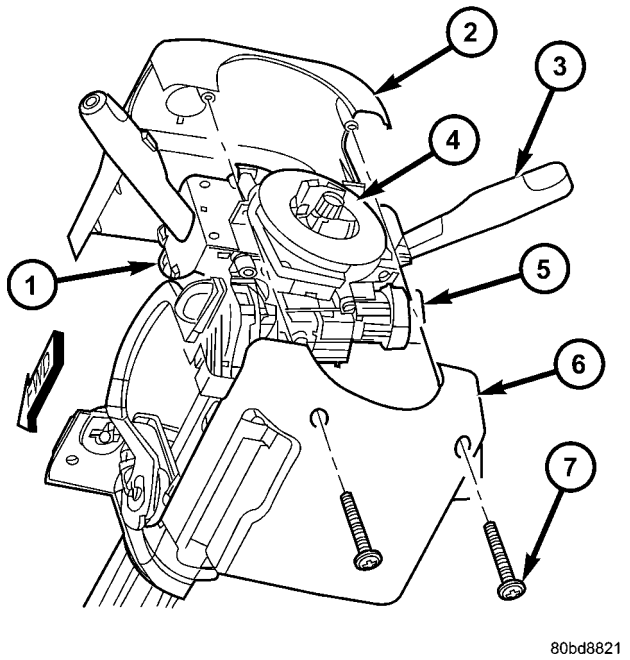

Fig. 27 Steering Column Shrouds Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - UPPER SHROUD
3 - RIGHT MULTI-FUNCTION SWITCH
4 - CLOCKSPRING
5 - IGNITION LOCK CYLINDER HOUSING
6 - LOWER SHROUD
7 - SCREW (2)

(8) Remove the two screws that secure the multi-function switch assembly to the upper steering column housing (Fig. 28).

(9) Remove the multi-function switch assembly from the upper steering column housing.

(10) Remove the two small screws that secure the right multi-function switch to the left multi-function switch mounting housing.

(11) Grasp the right multi-function switch control stalk firmly and pull the switch toward the right far enough to disengage the alignment pins on the top (1) and bottom (2) of the right switch housing from the alignment ramps on the left multi-function switch mounting housing.

(12) Remove the right multi-function switch from the left multi-function switch.

(13) Remove the combination flasher from the connector receptacle on the back of the left multi-function switch.

INSTALLATION

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

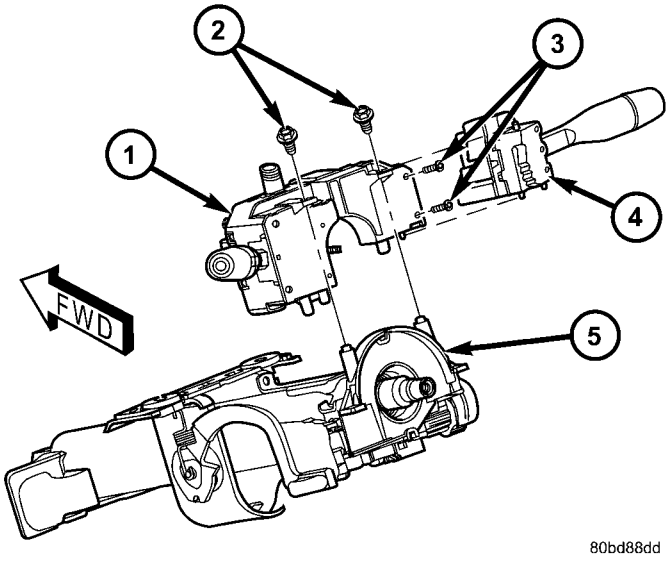

Fig. 28 Multi-Function Switch Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - SCREW (2)
3 - SCREW (2)
4 - RIGHT MULTI-FUNCTION SWITCH
5 - UPPER STEERING COLUMN HOUSING

(1) Install the combination flasher into the connector receptacle on the back of the left multi-function switch.

(2) Position the right multi-function switch to the left multi-function switch.

(3) Grasp the right multi-function switch control stalk firmly and slide the switch toward the left far enough to engage the alignment pins on the top (1) and bottom (2) of the right switch housing into the alignment ramps on the left multi-function switch mounting housing.

(4) Install and tighten the screws that secure the right multi-function switch to the left multi-function switch mounting housing (Fig. 28). Tighten the screws to 2 N·m (17 in. lbs.).

(5) Position the multi-function switch assembly onto the upper steering column housing.

(6) Install and tighten the screws that secure the multi-function switch assembly to the upper steering column housing. Tighten the screws to 2 N·m (17 in. lbs.).

LEFT MULTI-FUNCTION SWITCH (Continued)

(7) Reconnect the cross body wire harness connector for the right multi-function switch to the connector receptacle on the back of the switch.

(8) Reconnect the cross body wire harness connector for the left multi-function switch to the connector on the back of the switch.

(9) Position both the upper and lower shrouds onto the steering column (Fig. 29). Be certain that the locating tabs for the left and right multi-function switch control stalk watershields are properly engaged in the openings of both the upper and lower shrouds.

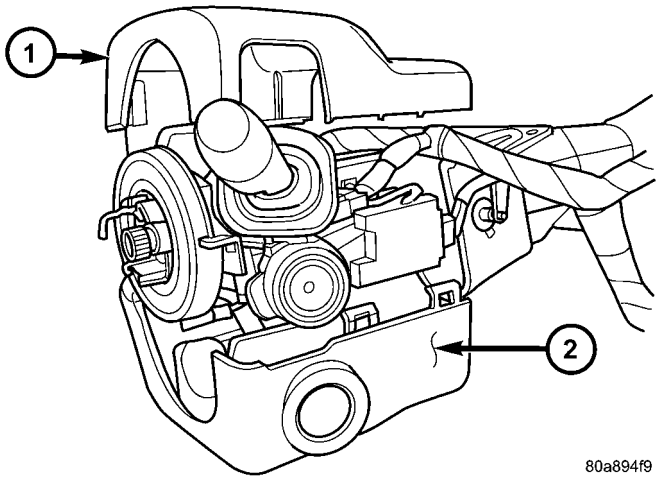

Fig. 29 Shroud Remove/Install

1 - UPPER SHROUD
2 - LOWER SHROUD

(10) Install and tighten the screws that secure the lower steering column shroud to the upper shroud. Tighten the screws to 2 N·m (17 in. lbs.).

(11) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully raised position and secure it in place by moving the tilt release lever back to the locked (up) position.

(12) Reinstall the steering column opening cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(13) Reconnect the battery negative cable.

SIDE MARKER LAMP UNIT

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) From underside of the fender flare, remove the nut that secures the side marker lamp housing to the front fender flare.

(3) Remove the lamp housing from the fender flare.

(4) Rotate the socket one-third turn counterclockwise and separate the socket and lamp from the housing.

INSTALLATION

(1) Install the socket and lamp into the housing and rotate the socket one-third turn clockwise.

(2) Position the side marker lamp housing into the front fender flare.

(3) Install the nut that secures the side marker lamp housing to the fender flare. Tighten the nut securely.

(4) Reconnect battery negative cable.

SIDE MARKER LAMP

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) From underside of the fender flare, rotate the side marker lamp socket one-third turn counterclockwise and separate it from the side marker lamp housing.

(3) Remove the lamp from the socket by pulling it straight out of the socket.

INSTALLATION

(1) Install the side marker lamp into the socket by pushing it straight into the socket.

(2) Install the socket into the housing and rotate the socket one-third turn clockwise.

(3) Reconnect the battery negative cable.

REPEATER LAMP

DESCRIPTION

Some export models are equipped with side repeater lamps instead of side marker lamps. The side repeater lamp operates with the turn signals and has an amber colored lens and a clear lamp. For Removal and Installation procedures, refer to Side Marker Lamps within this Group.

OPERATION

The side repeater lamps operate in series with the front and rear turn signal lamps and are controlled by the steering column mounted multi-function switch.

DIAGNOSIS AND TESTING - SIDE REPEATER LAMP

The following chart is designed to diagnosis a faulty side repeater lamp with the turn signals operating normally. Refer to Wiring Diagrams for a complete system schematic.

REPEATER LAMP (Continued)

SIDE REPEATER LAMP CONDITION CHART

CONDITION	POSSIBLE CAUSES	CORRECTION
TURN SIGNALS OPERATE NORMALLY, ONE REPEATER LAMP DOES NOT ILLUMINATE	1. Inoperative, damaged or burned out lamp.	4. Replace the lamp.
	2. Poor electrical connection at lamp.	2. Check for proper wire harness connection and circuit terminal tension at the lamp.
	3. Open or high resistance in the voltage circuit to the inoperative lamp.	3. Repair the open or high resistance in the repeater lamp voltage circuit.
	4. Open, high resistance in the ground circuit to the inoperative lamp.	4. Repair the open, high resistance in the repeater lamp ground circuit.

TAIL LAMP UNIT

DESCRIPTION

Each tail lamp assembly contains a lens, housing and two lamps. One lamp has two filaments and is used for tail, stop, turn signal, rear side marker and license plate (left side only) lamp functions. The other lamp has a single filament and is used for back-up light illumination.

OPERATION

Each tail lamp assembly can be serviced separately. Each lamp can also be serviced separately. The headlamp switch controls tail lamp operation. The multi-function switch controls turn signal operation. The back-up light switch controls back-up light operation. The brake lamp switch controls stop lamp operation.

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) From the underside of the vehicle, remove the push-in fastener that secures the bottom rear edge of the rear wheelhouse splash shield to the body.

(3) Pull the rear of the wheelhouse splash shield away from the body and reach upward to disconnect the wire harness connector from the tail lamp.

(4) Remove the screws that secure the tail lamp lens to the tail lamp housing (Fig. 30).

(5) Remove the tail lamp lens and gasket from the tail lamp housing.

(6) Remove the bolts that secure the tail lamp housing to the body.

(7) Remove the tail lamp housing from the body.

INSTALLATION

(1) Position the tail lamp housing to the body.

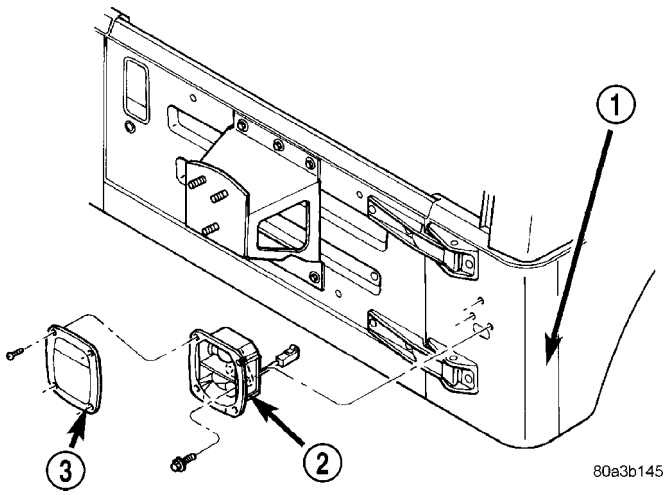

Fig. 30 Tail Lamp Housing

1 - BODY
2 - TAIL LAMP HOUSING
3 - TAIL LAMP LENS

(2) Install the bolts that secure the tail lamp housing to the body. Tighten the bolts securely.

NOTE: Install the tail lamp lens with the clear portion (back-up lens) at the top of the housing. Make sure that the gasket is correctly in place and not twisted or torn.

(3) Install the tail lamp lens and gasket onto the tail lamp housing.

NOTE: Do not overtighten the screws or damage to the tail lamp lens may result.

(4) Install the screws that secure the tail lamp lens to the tail lamp housing. Tighten the screws securely.

(5) Connect the wire harness connector to the tail lamp.

TAIL LAMP UNIT (Continued)

(6) Install the push-in fastener that secures the bottom rear edge of the rear wheelhouse splash shield to the body.

(7) Reconnect the negative battery cable.

TAIL LAMP

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Remove the screws that secure the tail lamp lens to the tail lamp housing (Fig. 31).

(3) Remove the tail lamp lens and gasket from the tail lamp housing.

(4) Push and rotate the lamp(s) one-third turn counterclockwise, then pull the lamp straight out of the socket

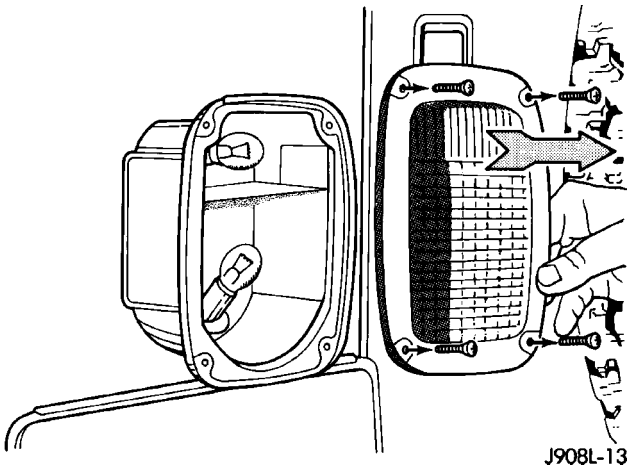

Fig. 31 Tail Lamp Lens

INSTALLATION

(1) Install the lamp(s) by pushing the lamp into the tail lamp socket and turning it one-third clockwise.

NOTE: Install the tail lamp lens with the clear portion (back-up lens) at the top of the housing. Make sure that the gasket is correctly in place and not twisted or torn.

(2) Install the tail lamp lens and gasket onto the tail lamp housing.

NOTE: Do not overtighten the screws or damage to the tail lamp lens may result.

(3) Install the screws that secure the tail lamp lens to the tail lamp housing. Tighten the screws securely.

(4) Reconnect the negative battery cable.

TURN SIGNAL CANCEL CAM

DESCRIPTION

The turn signal cancel cam consists of two lobes that are integral to the lower surface of the clockspring rotor. The clockspring mechanism provides turn signal cancellation as well as a constant electrical connection between the driver airbag, steering wheel accessories and the cross body wire harness on the steering column. The housing of the clockspring is secured to the steering column and remains stationary. The rotor of the clockspring, including the turn signal cancel cam lobes rotate with the steering wheel.

The turn signal cancel cam is integral to the clockspring and cannot be repaired. If faulty or damaged, the entire clockspring assembly must be replaced. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - REMOVAL).

OPERATION

The turn signal cancel cam has two lobes molded into the lower surface of the clockspring rotor. When the turn signals are activated by moving the left multi-function switch control stalk to a detent position, a turn signal cancel actuator is extended from the inside surface of the left multi-function switch housing toward the clockspring rotor. When the steering wheel is rotated during the turn, one of the two turn signal cancel cam lobes will contact the turn signal cancel actuator, but the cancel actuator stays latched. When the steering wheel is rotated back to center as the turn is completed, the cancel actuator is unlatched and releases the left multi-function switch control stalk from its detent, cancelling the turn signal event.

UNDERHOOD LAMP UNIT

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the underhood lamp (Fig. 32).

(3) Remove the screw that secures the underhood lamp to the inner hood panel.

(4) Remove the underhood lamp from the vehicle.

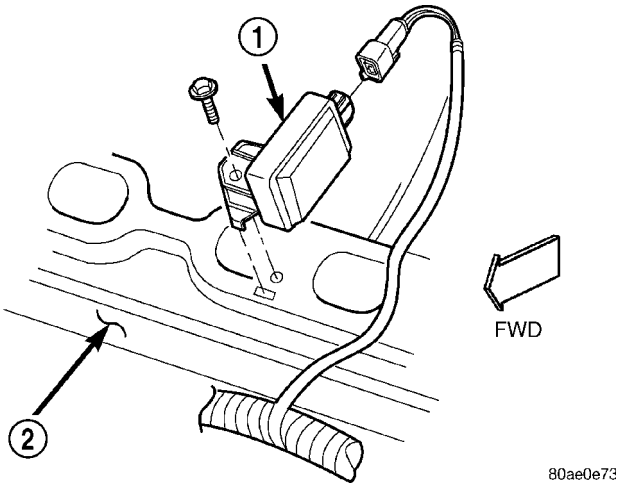

Fig. 32 Underhood Lamp

1 - UNDERHOOD LAMP
2 - HOOD

INSTALLATION

(1) Position the underhood lamp onto the inner hood panel.

(2) Install the screw that secures the underhood lamp to the inner hood panel. Tighten the screw securely.

(3) Connect the wire harness connector to the underhood lamp.

(4) Reconnect the negative battery cable.

UNDERHOOD LAMP

REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Insert a small flat blade in the access slot between the lamp housing and lamp lens.

(3) Pry the lamp lens upward and remove the lamp lens.

(4) Press the lamp terminal inward (Fig. 33) to release the lamp.

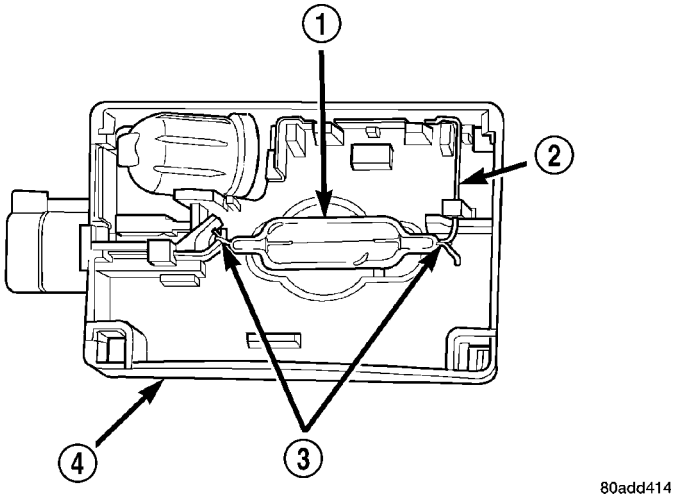

Fig. 33 Underhood Lamp

1 - LAMP
2 - LAMP TERMINAL
3 - LAMP WIRE LOOP
4 - LAMP HOUSING

INSTALLATION

(1) Engage the lamp wire loop to the terminal closest to the lamp housing wire connector.

(2) Press the opposite terminal inward and engage the remaining lamp wire loop onto the terminal.

(3) Position the lamp lens onto the lamp housing and press the lens into place.

(4) Reconnect the negative battery cable.

LAMPS/LIGHTING - INTERIOR

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LAMPS/LIGHTING - INTERIOR

SPECIFICATIONS

INTERIOR LAMPS

CAUTION: Do not use bulbs that have a higher candle power than the bulb listed in the Bulb Application Table. Damage to lamp housing and lens can result.

Some components have lamps that can only be serviced by an Authorized Service Center (ASC) after the component is removed from the vehicle.

The following table lists the various lamp trade numbers or part numbers.

LAMP	BULB
Dome (Sound Bar)	912
Under Hood	561
Underpanel Courtesy	906
Instrument Cluster Illumination	103
Instrument Cluster Warning	74
Automatic Transmission Indicator	658

DOME LAMP UNIT

REMOVAL

NOTE: The dome lamp removal procedure is the same for both the right and left side dome lamps.

(1) Disconnect the negative battery cable.

NOTE: The dome lamp lens locating tab is larger than the lens retaining tabs.

(2) Insert a small flat blade between the dome lamp housing and dome lamp lens. Carefully pry the lamp lens from the lamp housing (Fig. 1).

(3) Remove the screws that secure the dome lamp housing to the speaker housing.

(4) Remove the dome lamp housing from the speaker housing and disconnect the wire harness connector from the dome lamp.

INSTALLATION

NOTE: The dome lamp installation procedure is the same for both the right and left side dome lamps.

(1) Connect the wire harness connector to the dome lamp.

NOTE: The dome lamp housing is equipped with a locating tab that must be inserted into the slot in the speaker housing.

(2) Align the locating tab with the slot and position the dome lamp housing into the speaker housing (Fig. 2).

DOME LAMP UNIT (Continued)

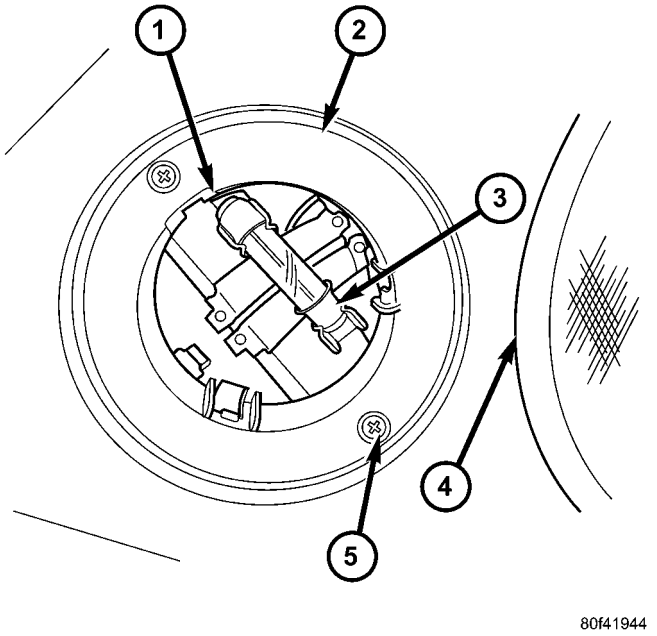

Fig. 1 Dome Lamp - Typical

1 - DOME LENS GUIDE
2 - DOME LAMP HOUSING
3 - DOME LAMP
4 - SPEAKER HOUSING
5 - SCREW (2)

(3) Install the screws that secure the dome lamp housing to the speaker housing.

NOTE: The dome lamp lens is equipped with a locating tab that must be inserted into the slot in the dome lamp housing.

(4) Align the dome lamp lens locating tab with the slot in the dome lamp housing and press the lens into place.
(5) Reconnect the negative battery cable.

DOME LAMP

REMOVAL

(1) Disconnect the negative battery cable.
(2) Insert a small flat blade between the dome lamp housing and dome lamp lens. Carefully pry the lamp lens from the lamp housing
(3) Remove the dome lamp from the lamp terminals by pulling the lamp straight out.

INSTALLATION

(1) Install the dome lamp into the lamp terminals by pushing the lamp straight into the terminals.

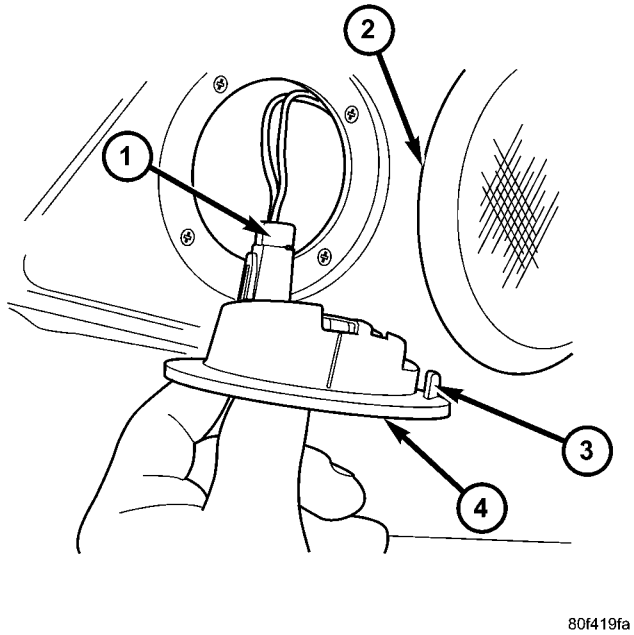

Fig. 2 Dome Lamp Housing

1 - WIRE HARNESS CONNECTOR
2 - SPEAKER
3 - LOCATING TAB
4 - DOME LAMP HOUSING

NOTE: The dome lamp lens is equipped with a locating tab that must be inserted into the slot in the dome lamp housing.

(2) Align the dome lamp lens locating tab with the slot in the dome lamp housing and press the lens into place.
(3) Reconnect the negative battery cable.

DOOR AJAR SWITCH

DESCRIPTION

The door ajar switches, located in each door pillar, are a self-adjusting, spring loaded plunger. The other end of the switch is actuated by the hinge face of the door. The self adjusting feature of the switch plunger is a one-time feature, it can be adjusted inward (compressed), but cannot be readjusted outward (extended) once it has been compressed. This normally open switch only closes when the door is open.

The door ajar switch cannot be repaired and, if faulty or damaged, it must be replaced.

DOOR AJAR SWITCH (Continued)

OPERATION

The door ajar switches control a path to ground through separate driver and passenger door ajar switch sense circuit inputs to the instrument cluster chime warning circuitry when a door is opened. The door ajar switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - DOOR AJAR SWITCH

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect the negative battery cable. Remove the door ajar switch from the door hinge pillar. Disconnect the wire harness connector from the door ajar switch. Check for continuity between the door ajar switch output and the driver or passenger door ajar switch sense circuit in the door ajar switch connector. There should be continuity with the switch plunger released, and no continuity with the switch plunger depressed. If OK, go to Step 2. If not OK, replace the faulty door ajar switch.

(2) Check for continuity between the door ajar switch output circuit of the wire harness connector for the driver or passenger door ajar switch and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open door ajar switch output circuit to ground as required.

(3) Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector from the instrument cluster. Check for continuity between the driver or passenger door ajar switch sense circuit of the wire harness connector for the driver or passenger door ajar switch and a good ground. There should be no continuity. If OK, go to Step 4. If not OK, repair the shorted driver or passenger door ajar switch sense circuit between the door ajar switch and the instrument cluster as required.

(4) Check for continuity between the driver or passenger door ajar switch sense circuit of the wire harness connector for the driver or passenger door ajar switch and the wire harness connector for the instrument cluster. There should be continuity. If not OK, repair the open driver or passenger door ajar switch sense circuit between the door ajar switch and the instrument cluster as required.

REMOVAL

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect the negative battery cable

(2) Unlatch and open the door fully.

(3) Using a small screwdriver, pry carefully between the hinge pillar and the outer circumference of the door ajar switch housing to release the switch snap features from the mounting hole in the pillar.

(4) Pull the door ajar switch out through the mounting hole in the hinge pillar far enough to access and disconnect the wire harness connector from the door ajar switch.

(5) Remove the door ajar switch from the door hinge pillar.

INSTALLATION

(1) Position the door ajar switch to the door hinge pillar.

(2) Connect the wire harness connector to the door ajar switch.

(3) Guide the wire harness for the door ajar switch and the receptacle end of the switch into the mounting hole in the door hinge pillar.

(4) Using hand pressure, press the door ajar switch housing into the mounting hole in the door hinge pillar until the snap features of the switch are fully engage in the mounting hole.

(5) Slowly close the door and allow the door ajar switch plunger self-adjuster mechanism to ratchet to the proper position.

(6) Reconnect the negative battery cable.

(7) Open and close the door to verify proper door ajar switch operation.

REAR WIPER/WASHER SWITCH ILLUMINATION LAMP

REMOVAL

The rear wiper and washer switch in the accessory switch bezel of the instrument panel includes a single serviceable incandescent lamp and lamp holder unit.

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, DRIVER AIRBAG, PASSENGER AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate the negative battery cable.

(2) Remove the rear wiper and washer switch from the accessory switch bezel (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER/WASHER SWITCH - REMOVAL).

(3) From the bottom of the switch housing, use a small thin-bladed screwdriver to rotate the lamp holder counterclockwise about 30 degrees in the mounting hole (Fig. 3).

(4) Pull the rear wiper and washer switch lamp holder and lamp assembly straight out of the mounting hole.

INSTALLATION

CAUTION: Always use the correct lamp size and type for replacement. An incorrect lamp size or type may overheat and cause damage to the rear wiper and washer switch.

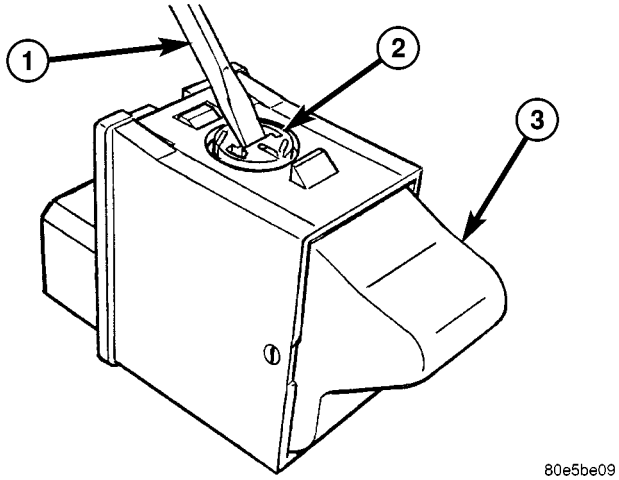

Fig. 3 Rear Wiper/Washer Switch Lamp

1 - SCREW DRIVER
2 - LAMP HOLDER
3 - REAR WIPER/WASHER SWITCH

(1) Align the rear wiper and washer switch lamp holder and lamp assembly with the mounting hole in the bottom of the switch.

(2) Insert the rear wiper and washer switch lamp holder and lamp assembly straight into the mounting hole until it is firmly seated.

(3) Using a small thin-bladed screwdriver, rotate the lamp holder clockwise about 30 degrees in the mounting hole.

(4) Install the rear wiper and washer switch into the accessory switch bezel (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER/WASHER SWITCH - INSTALLATION).

(5) Reconnect the negative battery cable.

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AUTOMATIC DAY / NIGHT MIRROR

DESCRIPTION

The automatic dimming inside day/night rear view mirror system is a completely self-contained unit that replaces the standard equipment inside rear view mirror. This system will automatically change the reflectance of the inside rear view mirror to protect the driver from the unwanted headlight glare of trailing vehicles while driving at night. The automatic day/night inside mirror receives ignition switched battery current through a fuse in the junction block, and will only operate when the ignition switch is in the On position.

Contained within the mirror is a compass/temperature display as well as two LED reading lamps.

The automatic day/night mirror sensitivity cannot be repaired or adjusted. If any component of this unit is inoperative or damaged, the entire automatic day/night inside rear view mirror unit must be replaced.

OPERATION

The automatic day/night mirror is equipped with three buttons: the left switch for the left LED lamp, the right switch for the right side LED lamp and the center switch for the compass/temperature function. Pressing the left and right switch simultaneously will turn the auto dim function on or off. A green light next to the right button will indicate when the dimming feature is activated. The mirror also senses the backup lamp circuit, and will automatically disable its self-dimming feature whenever the transmission gear selector is in the Reverse position.

The compass/temperature display provides the outside temperature and one of eight compass headings. Press and release the center button once within 3 seconds to display compass/temperature (Fahrenheit). Press and release the center button twice within 3 seconds to display compass/temperature (Celsius). Press and release the center button three times within 3 seconds to deactivate the feature.

A thin layer of electrochromatic material between two pieces of conductive glass make up the face of the mirror. Two photocell sensors are used to monitor light levels and adjust the reflectance of the mirror. The ambient photocell sensor faces forward, to detect the outside light levels. The headlamp sensor is located on the mirror housing just to the left of the switch and facing rearward, to detect the light level received at the rear window side of the mirror. When the difference between the two light levels becomes too great (the light level received at the rear of the mirror is much higher than that at the front of the mirror), the mirror begins to darken.

AUTOMATIC DAY / NIGHT MIRROR (Continued)

DIAGNOSIS AND TESTING - AUTOMATIC DAY/NIGHT MIRROR

For complete circuit diagrams, refer to the appropriate wiring information.

(1) Check for a blown fuse. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the On position. Check for battery voltage at the fuse. If OK, go to Step 3. If not OK, repair the open circuit to the ignition switch as required.

(3) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Unplug the wire harness connector from the automatic day/night mirror. Connect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run/start) circuit cavity of the automatic day/night mirror wire harness connector. If OK, go to Step 4. If not OK, repair the open circuit as required.

(4) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Check for continuity between the ground circuit cavity of the automatic day/night mirror wire harness connector and a good ground. There should be continuity. If OK, go to Step 5. If not OK, repair the circuit to ground as required.

(5) Connect the battery negative cable. Turn the ignition switch to the On position. Set the parking brake. Place the transmission gear selector lever in the Reverse position. Check for battery voltage at the backup lamp switch output circuit cavity of the automatic day/night mirror wire harness connector. If OK, go to Step 6. If not OK, repair the open circuit as required.

(6) Turn the ignition switch to the Off position. Disconnect the battery negative cable. Plug in the automatic day/night mirror wire harness connector. Connect the battery negative cable. Turn the ignition switch to the On position. Place the transmission gear selector lever in the Neutral position. Place the mirror switch in the On (the LED in the mirror switch is lighted) position. Cover the forward facing ambient photocell sensor to keep out any ambient light.

NOTE: The ambient photocell sensor must be covered completely, so that no light reaches the sensor. Use a finger pressed tightly against the sensor, or cover the sensor completely with electrical tape.

(7) Shine a light into the rearward facing headlamp photocell sensor. The mirror glass should darken. If OK, go to Step 8. If not OK, replace the faulty automatic day/night mirror unit.

(8) With the mirror glass darkened, place the transmission gear selector lever in the Reverse position. The mirror should return to its normal reflectance. If not OK, replace the faulty automatic day/night mirror unit.

STANDARD PROCEDURE

STANDARD PROCEDURE - COMPASS CALIBRATION

AUTOMATIC CALIBRATION

The compass is self calibrating which eliminates the need to manually set the compass. When the vehicle is new, the compass may appear erratic and CAL will be displayed. After completing one 360° turn with the vehicle traveling less than 8 km/h (5 mph) in an area free from large metal or metallic objects, CAL will turn off and the compass will function normally.

MANUAL CALIBRATION

If the compass appears erratic and CAL does not appear, you must manually put the compass into the calibration mode. To ensure proper compass calibration, make sure the compass variance is properly set before manually calibrating the compass.

To put the compass into calibration mode: Turn the ignition to the ON position. Press and hold the center button for 6 seconds to change the display between VAR (compass variance) and CAL (compass calibration) modes. To recalibrate the compass, CAL should display for a complete 1 1/2 360° turns in a area free from large metal objects or power lines. When the compass has been calibrated, the CAL symbol will turn off and the compass will function normally.

AUTOMATIC DAY / NIGHT MIRROR (Continued)

STANDARD PROCEDURE - COMPASS VARIATION ADJUSTMENT

Compass variance is the difference between magnetic north and geographic north. In some areas of the country, the difference between magnetic and geographic north is great enough to cause the compass to give false readings. If this occurs, the compass variance must be set according to the compass variance map (Fig. 1).

To set the variance: Turn the ignition to the ON position. Press and hold the center button for 3 to 6 seconds. The last variance zone number will be displayed. Each press of the center button will select a new variance zone. When the proper zone is selected, wait 5 seconds to resume normal operation.

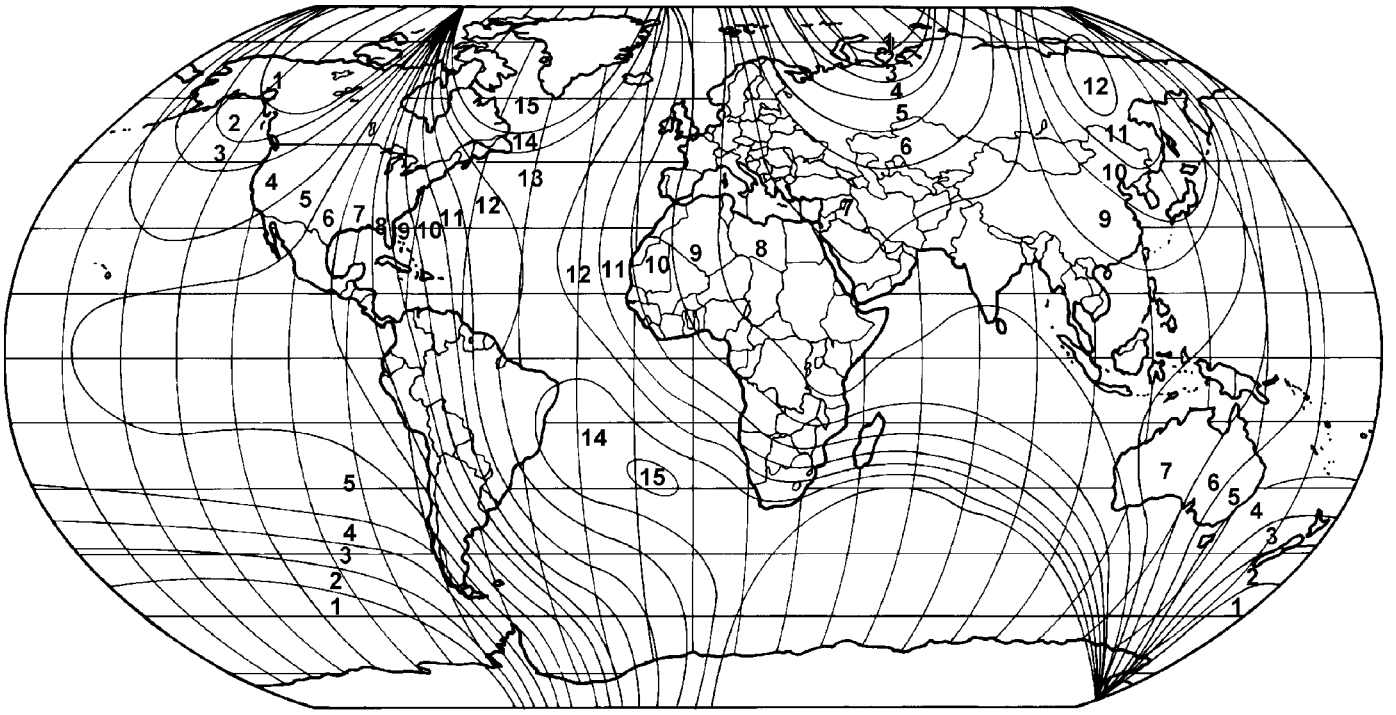

Fig. 1 Variance Settings

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RESTRAINTS

DESCRIPTION

An occupant restraint system is standard factory-installed safety equipment on this model. Available occupant restraints for this model include both active and passive types. Active restraints are those which require the vehicle occupants to take some action to employ, such as fastening a seat belt; while passive restraints require no action by the vehicle occupants to be employed (Fig. 1).

ACTIVE RESTRAINTS

The active restraints for this model include:

- **Front Seat Belts** - Both front seating positions are equipped with three-point seat belt systems employing a lower sport bar mounted inertia latch-type retractor, height-adjustable upper sport bar mounted turning loops, a fixed lower seat belt anchor secured with the retractor to the lower end of the

RESTRAINTS (Continued)

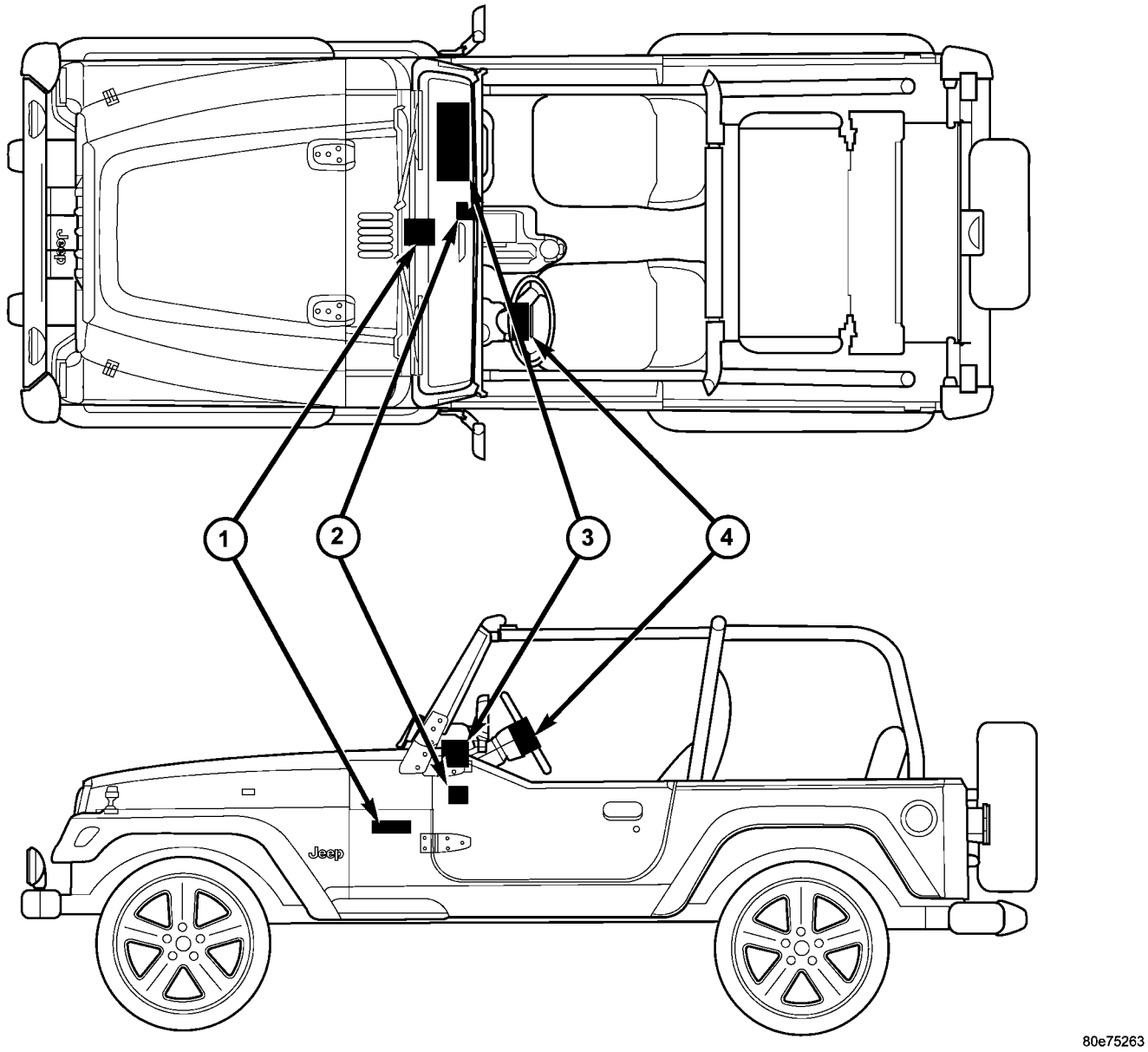

Fig. 1 Supplemental Restraint System

1 - AIRBAG CONTROL MODULE
2 - PASSENGER AIRBAG ON/OFF SWITCH (W/O REAR SEAT ONLY)
3 - PASSENGER AIRBAG
4 - DRIVER AIRBAG

sport bar, and a traveling end-release seat belt buckle secured to the inboard side of each front seat track. The front seat belt buckle for the driver side of all models includes an integral seat belt switch that electrically detects whether the driver seat belt has been fastened.

- **Rear Seat Belts** - On models equipped with a rear seat, both rear seating positions are equipped with three-point seat belt systems. The rear seating position belts employ a lower sport bar mounted inertia latch-type retractor, fixed upper sport bar mounted turning loops, a fixed lower seat belt anchor secured to the inner rear wheelhouse panel, and a fixed end-release seat belt buckle secured to the rear floor panel.
- **Child Restraint Anchors** - All vehicles without a rear seat are equipped with a single, fixed-position, child seat upper tether anchor and two fixed lower anchors for the front passenger seat. The upper anchor is integral to the seat riser bracket and is accessed from behind the front seat. The two lower anchors are also integral to the seat riser bracket,

RESTRAINTS (Continued)

but are accessed from the front of the seat where the seat back meets the seat cushion. Vehicles equipped with a rear seat have two fixed-position, child seat upper tether anchors on the rear floor panel behind the rear seat just forward of the tailgate opening, and four lower anchors that are integral to the rear seat back. The inboard rear seat lower anchors are accessed from the front of the seat where the seat back meets the seat cushion. The outboard rear seat lower anchors are accessed between the seat hinge bracket on each outboard end of the rear seat, just above the seat back pivot. The child seat tether and lower anchors for the front passenger seat are deleted on models equipped with a rear seat.

PASSIVE RESTRAINTS

The passive restraints available for this model include Next Generation driver and front passenger airbags. This airbag system is a passive, inflatable, Supplemental Restraint System (SRS) and vehicles with this equipment can be readily identified by the "SRS - AIRBAG" logo molded into the driver airbag trim cover in the center of the steering wheel and also into the passenger airbag door on the instrument panel above the glove box (Fig. 2). Vehicles with the airbag system can also be identified by the airbag indicator, which will illuminate in the instrument cluster for about seven seconds as a bulb test each time the ignition switch is turned to the On position.

Fig. 2 SRS Logo

The supplemental restraint system includes the following major components, which are described in further detail elsewhere in this service information:

- **Airbag Control Module** - The Airbag Control Module (ACM) is located on a mount on the floor panel transmission tunnel, below the center of the instrument panel.
- **Airbag Indicator** - The airbag indicator is integral to the ElectroMechanical Instrument Cluster (EMIC), which is located on the instrument panel in front of the driver.
- **Clockspring** - The clockspring is located near the top of the steering column, directly beneath the steering wheel.
- **Driver Airbag** - The driver airbag is located in the center of the steering wheel, beneath the driver airbag trim cover.
- **Driver Knee Blocker** - The driver knee blocker is a molded plastic structural unit secured to the back side of and integral to the instrument panel steering column opening cover.
- **Passenger Airbag** - The passenger airbag is located on the instrument panel, beneath the passenger airbag door on the instrument panel above the glove box on the passenger side of the vehicle.
- **Passenger Airbag On/Off Switch** - Models without a rear seat are equipped with a passenger airbag on/off switch, which is located in a dedicated opening on the passenger side of the accessory switch bezel in the lower center stack area of the instrument panel.
- **Passenger Knee Blocker** - The passenger knee blocker is a structural reinforcement that is integral to and concealed within the glove box door.

The ACM and the EMIC each contain a central processing unit and programming that allow them to communicate with each other using the Programmable Communication Interface (PCI) data bus network. This method of communication is used by the ACM for control of the airbag indicator on all models equipped with dual front airbags. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/ COMMUNICATION - DESCRIPTION).

Hard wired circuitry connects the supplemental restraint system components to each other through the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system, and to the supplemental restraint system components through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

RESTRAINTS (Continued)

OPERATION

ACTIVE RESTRAINTS

The primary passenger restraints in this or any other vehicle are the standard equipment factory-installed seat belts and child restraint anchors. Seat belts and child restraint anchors are referred to as an active restraint because the vehicle occupants are required to physically fasten and properly adjust these restraints in order to benefit from them. See the owner's manual in the vehicle glove box for more information on the features, use and operation of all of the factory-installed active restraints.

PASSIVE RESTRAINTS

The passive restraints are referred to as a supplemental restraint system because they were designed and are intended to enhance the protection for the occupants of the vehicle **only** when used in conjunction with the seat belts. They are referred to as passive restraints because the vehicle occupants are not required to do anything to make them operate; however, the vehicle occupants must be wearing their seat belts in order to obtain the maximum safety benefit from the factory-installed supplemental restraint system.

The supplemental restraint system electrical circuits are continuously monitored and controlled by a microprocessor and software contained within the Airbag Control Module (ACM). An airbag indicator in the ElectroMechanical Instrument Cluster (EMIC) illuminates for about seven seconds as a bulb test each time the ignition switch is turned to the On or Start positions. Following the bulb test, the airbag indicator is turned on or off by the ACM to indicate the status of the supplemental restraint system. If the airbag indicator comes on at any time other than during the bulb test, it indicates that there is a problem in the supplemental restraint system electrical circuits. Such a problem may cause airbags not to deploy when required, or to deploy when not required.

Deployment of the supplemental restraints depends upon the angle and severity of an impact. Deployment is not based upon vehicle speed; rather, deployment is based upon the rate of deceleration as measured by the forces of gravity (G force) upon the impact sensors. When an impact is severe enough, the microprocessor in the ACM signals the inflator of the appropriate airbag units to deploy their airbag cushions. During a frontal vehicle impact, the knee blockers work in concert with properly fastened and adjusted seat belts to restrain both the driver and the front seat passenger in the proper position for an airbag deployment. The knee blockers also absorb and distribute the crash energy from the driver and the front seat passenger to the structure of the instrument panel.

Typically, the vehicle occupants recall more about the events preceding and following a collision than they do of an airbag deployment itself. This is because the airbag deployment and deflation occur very rapidly. In a typical 48 kilometer-per-hour (30 mile-per-hour) barrier impact, from the moment of impact until the airbags are fully inflated takes about 40 milliseconds. Within one to two seconds from the moment of impact, the airbags are almost entirely deflated. The times cited for these events are approximations, which apply only to a barrier impact at the given speed. Actual times will vary somewhat, depending upon the vehicle speed, impact angle, severity of the impact, and the type of collision.

When the ACM monitors a problem in any of the supplemental restraint system circuits or components, it stores a fault code or Diagnostic Trouble Code (DTC) in its memory circuit and sends an electronic message to the EMIC to turn on the airbag indicator. Proper testing of the supplemental restraint system components, the Programmable Communication Interface (PCI) data bus, the electronic message inputs to and outputs from the EMIC or the ACM, as well as the retrieval or erasure of a DTC from the ACM or EMIC requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

See the owner's manual in the vehicle glove box for more information on the features, use and operation of all of the factory-installed passive restraints.

RESTRAINTS (Continued)

WARNING

WARNINGS - RESTRAINT SYSTEM

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH ON VEHICLES EQUIPPED WITH AIRBAGS, BEFORE PERFORMING ANY WELDING OPERATIONS DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE AND DISCONNECT ALL WIRE HARNESS CONNECTORS FROM THE AIRBAG CONTROL MODULE (ACM). FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND OTHER POSSIBLE DAMAGE TO THE SUPPLEMENTAL RESTRAINT SYSTEM CIRCUITS AND COMPONENTS.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DO NOT ATTEMPT TO DISMANTLE AN AIRBAG UNIT OR TAMPER WITH ITS INFLATOR. DO NOT PUNCTURE, INCINERATE, OR BRING INTO CONTACT WITH ELECTRICITY. DO NOT STORE AT TEMPERATURES EXCEEDING 93° C (200° F). AN AIRBAG INFLATOR UNIT MAY CONTAIN SODIUM AZIDE AND POTASSIUM NITRATE. THESE MATERIALS ARE POISONOUS AND EXTREMELY FLAMMABLE. CONTACT WITH ACID, WATER, OR HEAVY METALS MAY PRODUCE HARMFUL AND IRRITATING GASES (SODIUM HYDROXIDE IS FORMED IN THE PRESENCE OF MOISTURE) OR COMBUSTIBLE COMPOUNDS. AN AIRBAG INFLATOR UNIT MAY ALSO CONTAIN A GAS CANISTER PRESSURIZED TO OVER 2500 PSI.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, REPLACE ALL RESTRAINT SYSTEM COMPONENTS ONLY WITH PARTS SPECIFIED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG. SUBSTITUTE PARTS MAY APPEAR INTERCHANGEABLE, BUT INTERNAL DIFFERENCES MAY RESULT IN INFERIOR OCCUPANT PROTECTION.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE FASTENERS, SCREWS, AND BOLTS ORIGINALLY USED FOR THE RESTRAINT SYSTEM COMPONENTS MUST NEVER BE REPLACED WITH ANY SUBSTITUTES. THESE FASTENERS HAVE SPECIAL COATINGS AND ARE SPECIFICALLY DESIGNED FOR THE RESTRAINT SYSTEM. ANY TIME A NEW FASTENER IS NEEDED, REPLACE IT WITH THE CORRECT FASTENERS PROVIDED IN THE SERVICE PACKAGE OR SPECIFIED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, WHEN A STEERING COLUMN HAS AN AIRBAG UNIT ATTACHED, NEVER PLACE THE COLUMN ON THE FLOOR OR ANY OTHER SURFACE WITH THE STEERING WHEEL OR AIRBAG UNIT FACE DOWN.

RESTRAINTS (Continued)

DIAGNOSIS AND TESTING - SUPPLEMENTAL RESTRAINT SYSTEM

Proper diagnosis and testing of the supplemental restraint system components, the PCI data bus, the data bus electronic message inputs to and outputs from the ElectroMechanical Instrument Cluster (EMIC), or the Airbag Control Module (ACM) as well as the retrieval or erasure of a Diagnostic Trouble Code (DTC) from the ACM requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

STANDARD PROCEDURE

STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS

At no time should any source of electricity be permitted near the inflator on the back of a non-deployed airbag. When carrying a non-deployed airbag, the trim cover or airbag cushion side of the unit should be pointed away from the body to minimize injury in the event of an accidental deployment. If the airbag unit is placed on a bench or any other surface, the trim cover or airbag cushion side of the unit should be face up to minimize movement in the event of an accidental deployment. In addition, the supplemental restraint system should be disarmed whenever any steering wheel, steering column, driver airbag, passenger airbag, or instrument panel components require diagnosis or service. Failure to observe this warning could result in accidental airbag deployment and possible personal injury.

All damaged, faulty, or non-deployed airbags which are replaced on vehicles are to be handled and disposed of properly. If an airbag unit is faulty or damaged and non-deployed, refer to the Hazardous Substance Control System for proper disposal. Dispose of all non-deployed and deployed airbags in a manner consistent with state, provincial, local and federal regulations.

SUPPLEMENTAL RESTRAINT STORAGE

Airbags must be stored in their original, special container until they are used for service. Also, they must be stored in a clean, dry environment; away from sources of extreme heat, sparks, and high electrical energy. Always place or store any airbag on a surface with its trim cover or airbag cushion side facing up, to minimize movement in case of an accidental deployment.

STANDARD PROCEDURE - SERVICE AFTER A SUPPLEMENTAL RESTRAINT DEPLOYMENT

Any vehicle which is to be returned to use following a supplemental restraint deployment, must have the deployed restraints replaced. In addition, if the driver airbag has been deployed, the clockspring must be replaced. If the passenger airbag is deployed, the passenger airbag door must be replaced. These components are not intended for reuse and will be damaged or weakened as a result of a supplemental restraint deployment, which may or may not be obvious during a visual inspection.

It is also critical that the mounting surfaces and/or mounting bracket for the Airbag Control Module (ACM) be closely inspected and restored to its original condition following any vehicle impact damage. Because the ACM contains impact sensors that are used by the supplemental restraint system to monitor or confirm the direction and severity of a vehicle impact, improper orientation or insecure fastening of this component may cause airbags not to deploy when required, or to deploy when not required. All other vehicle components should be closely inspected following any supplemental restraint deployment, but are to be replaced only as required by the extent of the visible damage incurred.

CLEANUP PROCEDURE

Following a supplemental restraint deployment, the vehicle interior will contain a powdery residue. This residue consists primarily of harmless particulate by-products of the small pyrotechnic charge that initiates the propellant used to deploy a supplemental restraint. However, this residue may also contain traces of sodium hydroxide powder, a chemical by-product of the propellant material that is used to generate the inert gas that inflates the airbag. Since sodium hydroxide powder can irritate the skin, eyes, nose, or throat, be certain to wear safety glasses, rubber gloves, and a long-sleeved shirt during cleanup (Fig. 3).

RESTRAINTS (Continued)

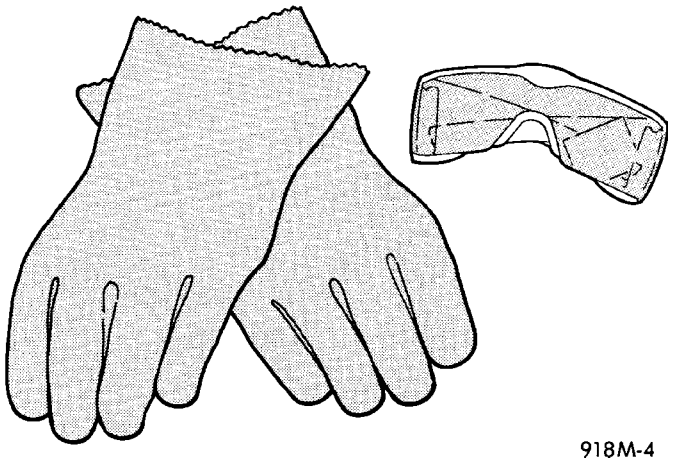

Fig. 3 Wear Safety Glasses and Rubber Gloves - Typical

WARNING: TO AVOID PERSONAL INJURY OR DEATH, IF YOU EXPERIENCE SKIN IRRITATION DURING CLEANUP, RUN COOL WATER OVER THE AFFECTED AREA. ALSO, IF YOU EXPERIENCE IRRITATION OF THE NOSE OR THROAT, EXIT THE VEHICLE FOR FRESH AIR UNTIL THE IRRITATION CEASES. IF IRRITATION CONTINUES, SEE A PHYSICIAN.

(1) Begin the cleanup by using a vacuum cleaner to remove any residual powder from the vehicle interior. Clean from outside the vehicle and work your way inside, so that you avoid kneeling or sitting on a non-cleaned area.

(2) Be certain to vacuum the heater and air conditioning outlets as well (Fig. 4). Run the heater and air conditioner blower on the lowest speed setting and vacuum any powder expelled from the outlets.

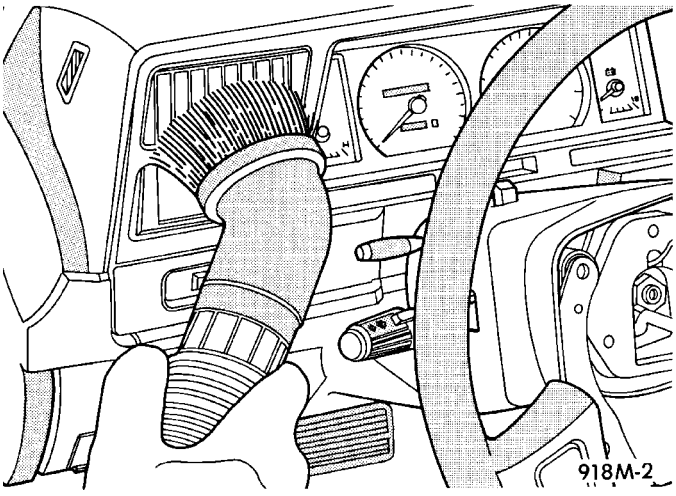

Fig. 4 Vacuum Heater and A/C Outlets - Typical

CAUTION: All damaged, faulty, or non-deployed supplemental restraints which are replaced on vehicles are to be handled and disposed of properly. If an airbag unit is faulty or damaged and non-deployed, refer to the Hazardous Substance Control System for proper disposal. Be certain to dispose of all non-deployed and deployed supplemental restraints in a manner consistent with state, provincial, local and federal regulations.

(3) Next, remove the deployed supplemental restraints from the vehicle. Refer to the appropriate service removal procedures.

(4) You may need to vacuum the interior of the vehicle a second time to recover all of the powder.

STANDARD PROCEDURE - VERIFICATION TEST

The following procedure should be performed using a DRBIII® scan tool to verify proper supplemental restraint system operation following the service or replacement of any supplemental restraint system component.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) During the following test, the battery negative cable remains disconnected and isolated, as it was during the supplemental restraint system component removal and installation procedures.

(2) Be certain that the DRBIII® scan tool contains the latest version of the proper DRBIII® software. Connect the DRBIII® to the 16-way Data Link Connector (DLC). The DLC is located on the driver side lower edge of the instrument panel, outboard of the steering column (Fig. 5).

(3) Turn the ignition switch to the On position and exit the vehicle with the DRBIII® scan tool.

(4) Check to be certain that nobody is in the vehicle, then reconnect the battery negative cable.

(5) Using the DRBIII®, read and record the active (current) Diagnostic Trouble Code (DTC) data.

(6) Next, use the DRBIII® to read and record any stored (historical) DTC data.

(7) If any DTC is found in Step 5 or Step 6, refer to the appropriate diagnostic information.

(8) Use the DRBIII® to erase the stored DTC data. If any problems remain, the stored DTC data will not

RESTRAINTS (Continued)

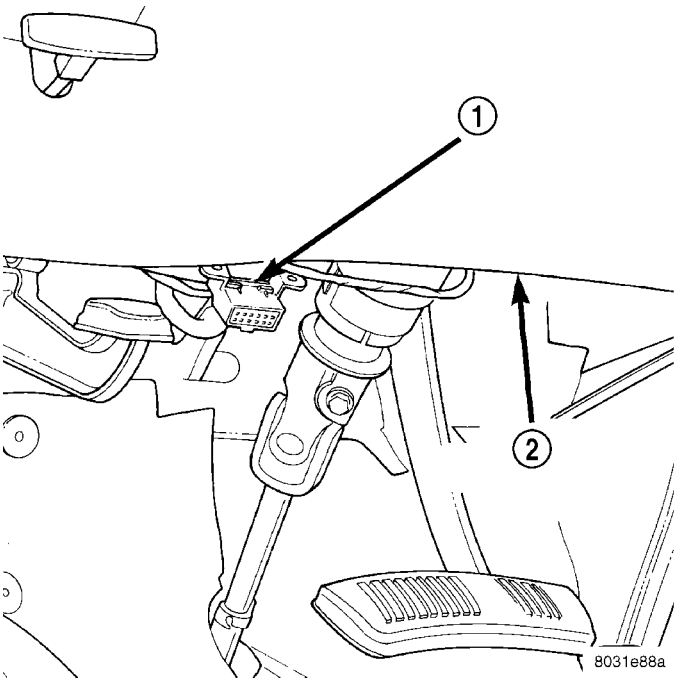

Fig. 5 16-Way Data Link Connector - Typical

1 - 16–WAY DATA LINK CONNECTOR
2 - BOTTOM OF INSTRUMENT PANEL

erase. Refer to the appropriate diagnostic information to diagnose any stored DTC that will not erase. If the stored DTC information is successfully erased, go to Step 9.

(9) Turn the ignition switch to the Off position for about fifteen seconds, and then back to the On position. Observe the airbag indicator in the instrument cluster. It should illuminate for six to eight seconds, and then go out. This indicates that the supplemental restraint system is functioning normally and that the repairs are complete. If the airbag indicator fails to light, or lights and stays on, there is still an active supplemental restraint system fault or malfunction. Refer to the appropriate diagnostic information to diagnose the problem.

AIRBAG CONTROL MODULE

DESCRIPTION

The Airbag Control Module (ACM) is secured with four screws to the top mounting surface of a stamped steel bracket welded onto the top of the floor panel transmission tunnel below the instrument panel and forward of the center floor console in the passenger compartment of the vehicle (Fig. 6). Concealed within a hollow in the center of the die cast aluminum ACM housing is the electronic circuitry of the ACM which includes a microprocessor, an electronic impact sensor, an electromechanical safing sensor, and an energy storage capacitor. A stamped metal cover

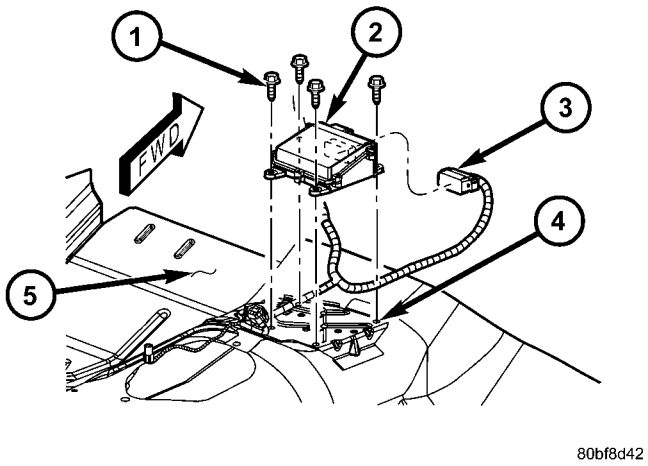

Fig. 6 Airbag Control Module

1 - SCREW (4)
2 - AIRBAG CONTROL MODULE
3 - WIRE HARNESS CONNECTOR
4 - MOUNTING BRACKET
5 - FRONT FLOOR PANEL

plate is secured to the bottom of the ACM housing with four screws to enclose and protect the internal electronic circuitry and components.

The ACM housing has an integral mounting flange on each side. Each mounting flange has an integral locating pin on its lower surface and two round mounting holes. An arrow cast into the top of the ACM housing near the rear provides a visual verification of the proper orientation of the unit, and should always be pointed toward the front of the vehicle. A molded plastic electrical connector receptacle containing twenty-three terminal pins exits the forward facing side of the ACM housing. These terminal pins connect the ACM to the vehicle electrical system through a dedicated take out and connector of the body wire harness.

The impact sensor and safing sensor internal to the ACM are calibrated for the specific vehicle, and are only serviced as a unit with the ACM. The ACM cannot be repaired or adjusted and, if damaged or faulty, it must be replaced.

OPERATION

The microprocessor in the Airbag Control Module (ACM) contains the front supplemental restraint system logic circuits and controls all of the supplemental restraint system components. The ACM uses On-Board Diagnostics (OBD) and can communicate with other electronic modules in the vehicle as well as with the DRBIII® scan tool using the Programmable Communications Interface (PCI) data bus network. This method of communication is used for control of the airbag indicator in the ElectroMechanical Instrument Cluster (EMIC) and for supplemental

AIRBAG CONTROL MODULE (Continued)

restraint system diagnosis and testing through the 16-way data link connector located on the driver side lower edge of the instrument panel. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER/AIRBAG INDICATOR - OPERATION).

The ACM microprocessor continuously monitors all of the front supplemental restraint system electrical circuits to determine the system readiness. If the ACM detects a monitored system fault, it sets an active and stored Diagnostic Trouble Code (DTC) and sends electronic messages to the EMIC over the PCI data bus to turn on the airbag indicator. An active fault only remains for the duration of the fault, or in some cases, the duration of the current ignition switch cycle, while a stored fault causes a DTC to be stored in memory by the ACM. For some DTCs, if a fault does not recur for a number of ignition cycles, the ACM will automatically erase the stored DTC. For other internal faults, the stored DTC is latched forever.

In models not equipped with a rear seat, the ACM also monitors a resistor multiplexed input from the passenger airbag on/off switch and provides a control output for the Off indicator in the switch through a passenger airbag indicator driver circuit. If the passenger airbag on/off switch is set to the Off position, the ACM turns on the passenger airbag on/off switch Off indicator and will internally disable the passenger airbag from being deployed if an impact is detected that is sufficient for an airbag deployment. The ACM also turns on the on/off switch Off indicator for about seven seconds each time the ignition switch is turned to the On position as a bulb test. Following the bulb test, the ACM controls the status of the Off indicator based upon the resistance of the input from the on/off switch. The ACM will also set and/or store a DTC for faults it detects in the passenger airbag on/off switch circuits, and will turn on the airbag indicator in the EMIC if a fault has been detected.

The ACM receives battery current through two circuits; a fused ignition switch output (run) circuit through a fuse in the fuse block, and a fused ignition switch output (run-start) circuit through a second fuse in the fuse block. The ACM receives ground through a ground circuit and take out of the body wire harness. This take out has a single eyelet terminal connector that is secured by a ground screw to the right cowl side inner panel below the instrument panel. These connections allow the ACM to be operational whenever the ignition switch is in the Start or On positions. The ACM also contains an energy-storage capacitor. When the ignition switch is in the Start or On positions, this capacitor is continually being charged with enough electrical energy to deploy the front supplemental restraint components for up to one second following a battery disconnect or failure. The purpose of the capacitor is to provide backup supplemental restraint system protection in case there is a loss of battery current supply to the ACM during an impact.

Two sensors are contained within the ACM; an electronic impact sensor, and a safing sensor. The electronic impact sensor is an accelerometer that senses the rate of vehicle deceleration, which provides verification of the direction and severity of an impact. The safing sensor is an electromechanical sensor within the ACM that provides an additional logic input to the ACM microprocessor. The safing sensor is a normally open switch that is used to verify the need for a front supplemental restraint deployment by detecting impact energy of a lesser magnitude than that of the electronic impact sensor, and must be closed in order for the front airbags to deploy. A pre-programmed decision algorithm in the ACM microprocessor determines when the deceleration rate as signaled by the impact sensor and the safing sensor indicate an impact that is severe enough to require front supplemental restraint system protection and, based upon the status of the passenger airbag on/off switch input and the severity of the monitored impact, determines what combination of front airbag deployment is required for each front seating position. When the programmed conditions are met, the ACM sends the proper electrical signals to deploy the dual front airbags.

The hard wired inputs and outputs for the ACM may be diagnosed and tested using conventional diagnostic tools and procedures. However, conventional diagnostic methods will not prove conclusive in the diagnosis of the ACM, the PCI data bus network, or the electronic message inputs to and outputs from the ACM. The most reliable, efficient, and accurate means to diagnose the ACM, the PCI data bus network, and the electronic message inputs to and outputs from the ACM requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

AIRBAG CONTROL MODULE (Continued)

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, NEVER STRIKE OR DROP THE AIRBAG CONTROL MODULE, AS IT CAN DAMAGE THE IMPACT SENSOR OR AFFECT ITS CALIBRATION. THE AIRBAG CONTROL MODULE CONTAINS THE IMPACT SENSOR, WHICH ENABLES THE SYSTEM TO DEPLOY THE SUPPLEMENTAL RESTRAINTS. IF AN AIRBAG CONTROL MODULE IS ACCIDENTALLY DROPPED DURING SERVICE, THE MODULE MUST BE SCRAPPED AND REPLACED WITH A NEW UNIT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER SUPPLEMENTAL RESTRAINT DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) Pull the carpet on the right and left sides of the floor panel transmission tunnel rearward far enough to access the Airbag Control Module (ACM), which is forward of the floor console.

(3) On models equipped with the optional Anti-lock Brake System (ABS), remove the acceleration switch from the left side of the mounting bracket on the floor panel transmission tunnel. (Refer to 5 - BRAKES/ELECTRICAL/G-SWITCH - REMOVAL).

(4) Remove the four screws that secure the Airbag Control Module (ACM) to the mounting bracket on the floor panel transmission tunnel (Fig. 7).

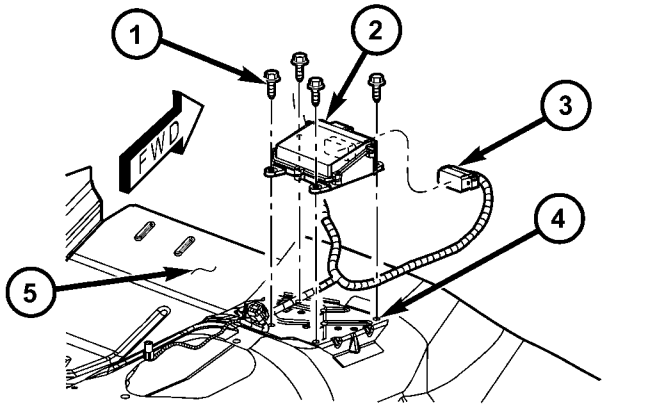

Fig. 7 Airbag Control Module Remove/Install

1 - SCREW (4)
2 - AIRBAG CONTROL MODULE
3 - WIRE HARNESS CONNECTOR
4 - MOUNTING BRACKET
5 - FRONT FLOOR PANEL

(5) Lift the ACM upward far enough to disengage the locator pins on the bottom of the ACM mounting flanges from the locating holes in the mounting bracket, then slide the ACM out from under the instrument panel far enough to access the wire harness connector.

(6) Disconnect the body wire harness connector for the ACM from the ACM connector receptacle located on the forward facing side of the module. To disconnect the body wire harness connector from the ACM (Fig. 8):

(a) Pull the white Connector Positive Assurance (CPA) locks on each side of the connector out about 3 millimeters (0.125 inch).

(b) Squeeze the latch tabs on each side of the connector between the thumb and forefinger and pull the connector straight away from the ACM connector receptacle.

(7) Remove the ACM from beneath the instrument panel.

AIRBAG CONTROL MODULE (Continued)

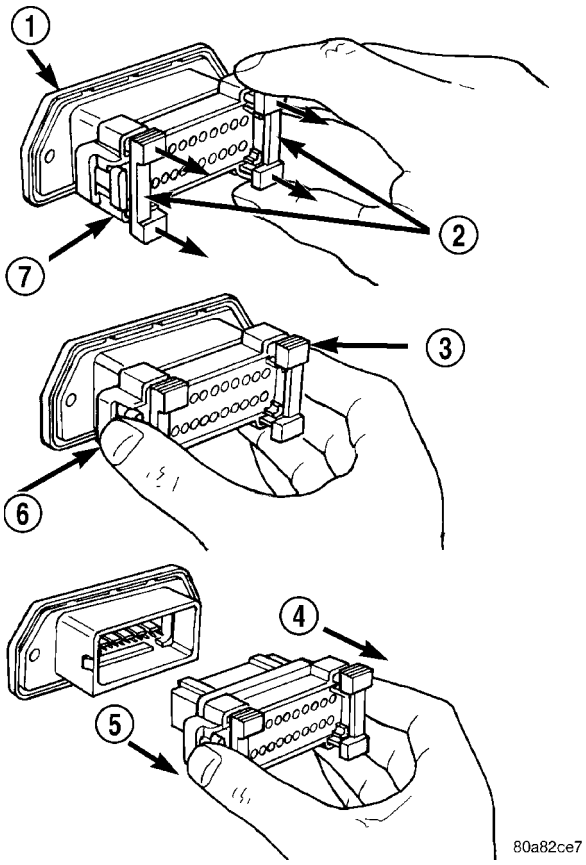

Fig. 8 Airbag Control Module Connector Removal

1 - AIRBAG CONTROL MODULE
2 - PULL TWO LOCKS OUT
3 - SQUEEZE LATCHES
4 - PULL
5 - PULL
6 - SQUEEZE LATCHES
7 - WIRE HARNESS CONNECTOR

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, NEVER STRIKE OR DROP THE AIRBAG CONTROL MODULE, AS IT CAN DAMAGE THE IMPACT SENSOR OR AFFECT ITS CALIBRATION. THE AIRBAG CONTROL MODULE CONTAINS THE IMPACT SENSOR, WHICH ENABLES THE SYSTEM TO DEPLOY THE SUPPLEMENTAL RESTRAINTS. IF AN AIRBAG CONTROL MODULE IS ACCIDENTALLY DROPPED DURING SERVICE, THE MODULE MUST BE SCRAPPED AND REPLACED WITH A NEW UNIT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER SUPPLEMENTAL RESTRAINT DEPLOYMENT.

(1) Position the Airbag Control Module (ACM) beneath the instrument panel.

(2) Reconnect the body wire harness connector for the ACM to the ACM connector receptacle located on the forward facing side of the module (Fig. 8). Be certain that both connector latches and the white Connector Positive Assurance (CPA) locks are fully engaged.

(3) Position the ACM to the mounting bracket on the floor panel transmission tunnel (Fig. 7). When the ACM is correctly positioned, the locator pins on the bottom of the ACM mounting flanges will be engaged in the locating holes in the mounting bracket, and the arrow on the ACM label will be pointed forward in the vehicle.

(4) Install and tighten the four screws that secure the ACM to the mounting bracket on the floor panel transmission tunnel. Tighten the screws to 14 N·m (125 in. lbs.).

(5) On models equipped with the optional ABS brakes, reinstall the acceleration switch onto the left side of the mounting bracket on the floor panel transmission tunnel. (Refer to 5 - BRAKES/ELECTRICAL/G-SWITCH - INSTALLATION).

(6) Restore the carpet on the right and left sides of the floor panel transmission tunnel to its proper position beneath the instrument panel.

(7) Do not reconnect the battery negative cable at this time. The supplemental restraint system verification test procedure should be performed following service of any supplemental restraint system component (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - VERIFICATION TEST).

CHILD RESTRAINT ANCHOR

DESCRIPTION

This model is equipped with a Lower Anchors and Tether for CHildren, or LATCH child restraint anchorage system (Fig. 9). The LATCH system provides for the installation of suitable child restraints in certain seating positions without using the standard equipment seat belt provided for that seating position. All vehicles without a rear seat are equipped with a fixed-position child restraint upper tether anchor (Fig. 10) and two child restraint lower anchors (Fig. 11) for the front passenger seating position. Vehicles with an optional rear seat are equipped with fixed-position child restraint upper tether anchors (Fig. 12) and two child restraint lower anchors (Fig. 13) for each rear outboard seating position. All front passenger seat child restraint anchors are deleted on models equipped with the optional rear seat.

The upper tether anchor and both lower anchors for the front passenger seat are integral to the front passenger seat riser bracket. The upper tether anchor is accessed from behind the front seat. The lower anchors for the front passenger seat are formed from round steel bar stock that is formed into a U-shape, then securely welded to the seat riser bracket. They are accessed from the front of the seat, at each side where the seat back meets the seat cushion. The upper and lower front seat child restraint anchors cannot be adjusted or repaired and, if faulty or damaged, they must be replaced as a unit with the seat riser bracket.

The upper tether anchors for the rear seat are stamped steel brackets that are secured by screws to the rear cargo floor panel just forward of the tailgate

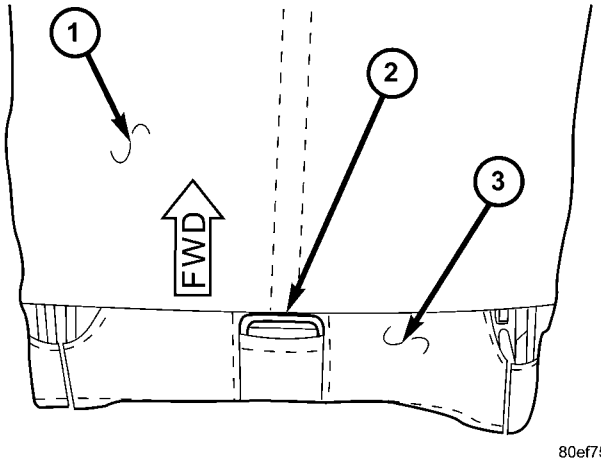

Fig. 10 Front Passenger Seat Upper Anchor

1 - SEAT BACK
2 - UPPER ANCHOR
3 - SEAT CUSHION

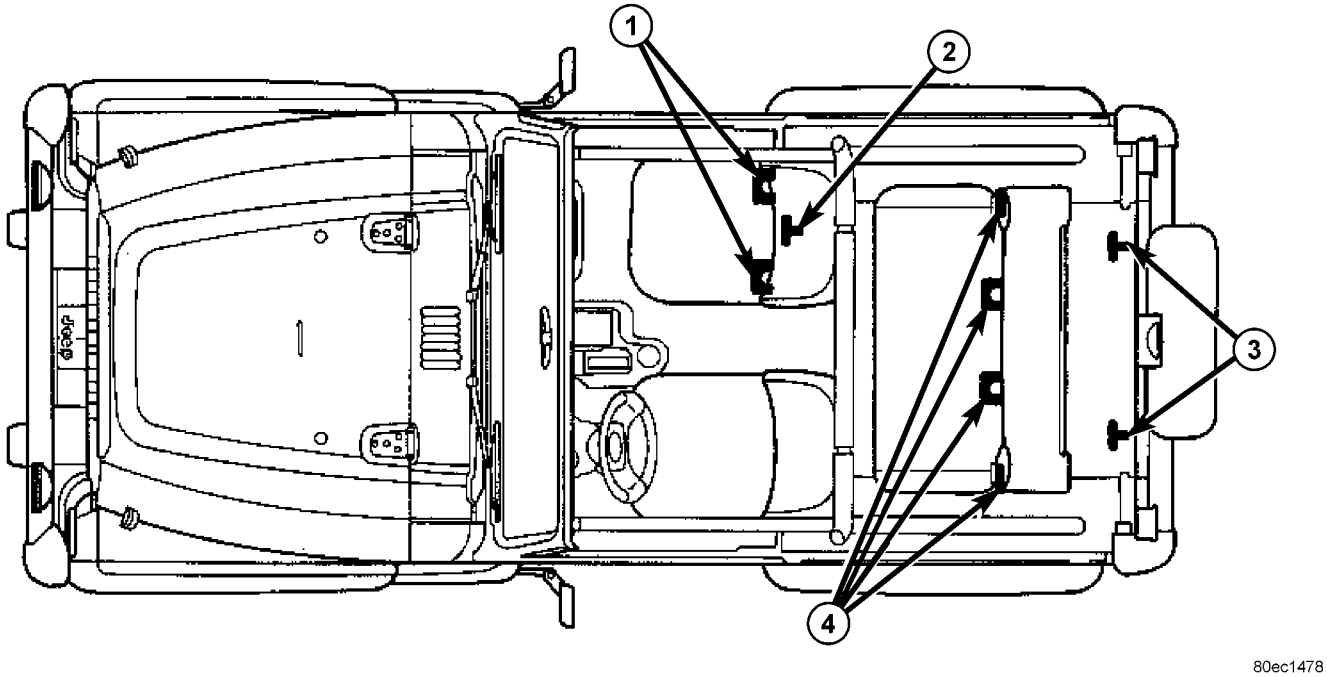

Fig. 9 Child Restraint Anchor Locations

1 - LOWER ANCHOR (MODELS WITHOUT A REAR SEAT ONLY)
2 - TETHER ANCHOR (MODELS WITHOUT A REAR SEAT ONLY)
3 - LOWER ANCHOR (PROVIDED FOR REAR OUTBOARD SEATING POSITIONS ONLY)
4 - TETHER ANCHOR (PROVIDED FOR REAR OUTBOARD SEATING POSITIONS ONLY)

CHILD RESTRAINT ANCHOR (Continued)

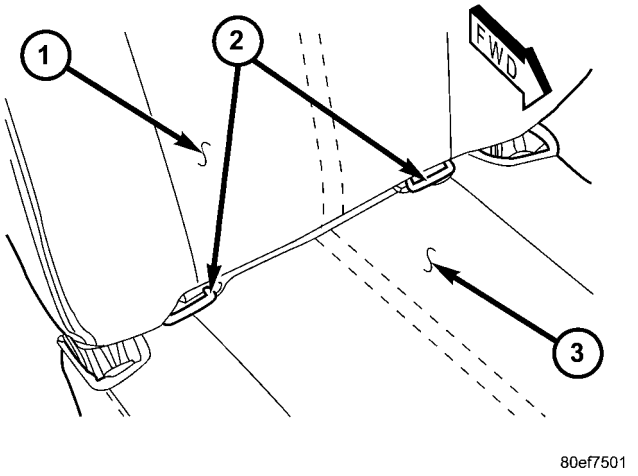

Fig. 11 Front Passenger Seat Lower Anchors

1 - SEAT BACK
2 - LOWER ANCHOR (2)
3 - SEAT CUSHION

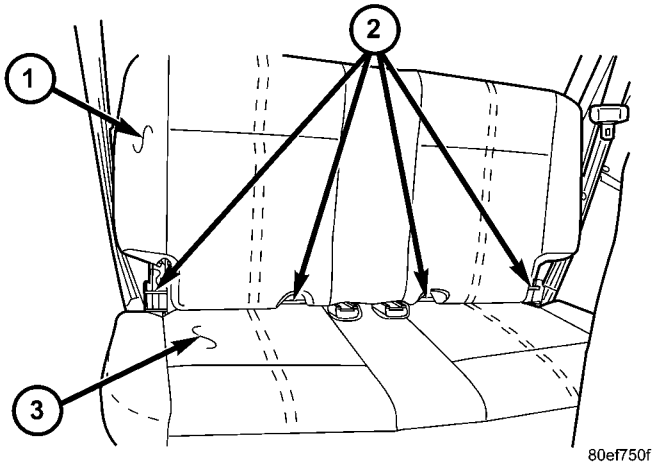

Fig. 13 Rear Seat Lower Anchors

1 - SEAT BACK
2 - LOWER ANCHOR (4)
3 - SEAT CUSHION

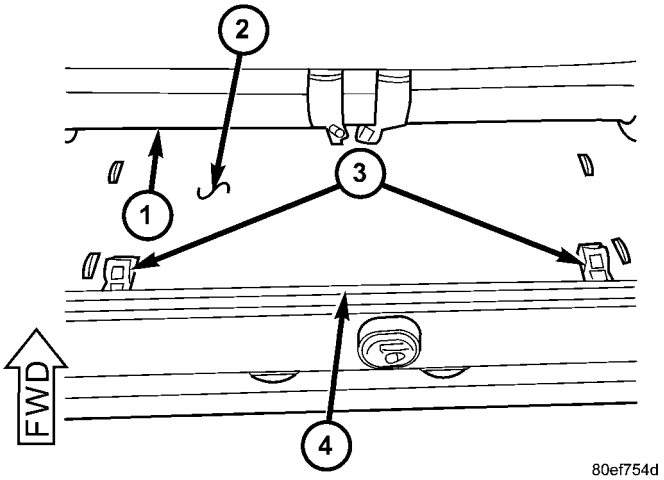

Fig. 12 Rear Seat Upper Anchors

1 - REAR SEAT
2 - REAR CARGO FLOOR
3 - UPPER ANCHOR (2)
4 - TAILGATE OPENING SILL

opening, and are accessed from behind the rear seat. The upper tether anchors for the rear seat are available for individual service replacement. The four fixed lower anchors are integral to the rear seat back frame and are accessed from the front of the rear seat, where the seat back meets the seat cushion. The two inboard lower anchors are constructed from round steel bar stock that is formed into a U-shape, then securely welded at each end to the rear seat back frame. The two outboard lower anchors are machined steel pins that are secured between the two seat back hinge plates above the pivot pin on each outboard side of the rear seat back frame. These lower anchors cannot be adjusted or repaired and, if faulty or damaged, they must be replaced as a unit with the rear seat back frame.

WARNING: DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

OPERATION

See the owner's manual in the vehicle glove box for more information on the proper use of all of the factory-installed child restraint anchors.

REMOVAL

The following procedure applies only to the rear seat upper child tether anchors used on models equipped with an optional rear seat. The child

CHILD RESTRAINT ANCHOR (Continued)

restraint anchors used in this model in all other locations are integral to other components and cannot be serviced separately.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Remove the screw that secures the upper child restraint tether anchor to the rear cargo floor panel just forward of the tailgate opening sill (Fig. 14).

(2) Remove the upper tether anchor from the rear cargo floor panel.

INSTALLATION

The following procedure applies only to the rear seat upper child tether anchors used on models equipped with an optional rear seat. The child restraint anchors used in this model in all other locations are integral to other components and cannot be serviced separately.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

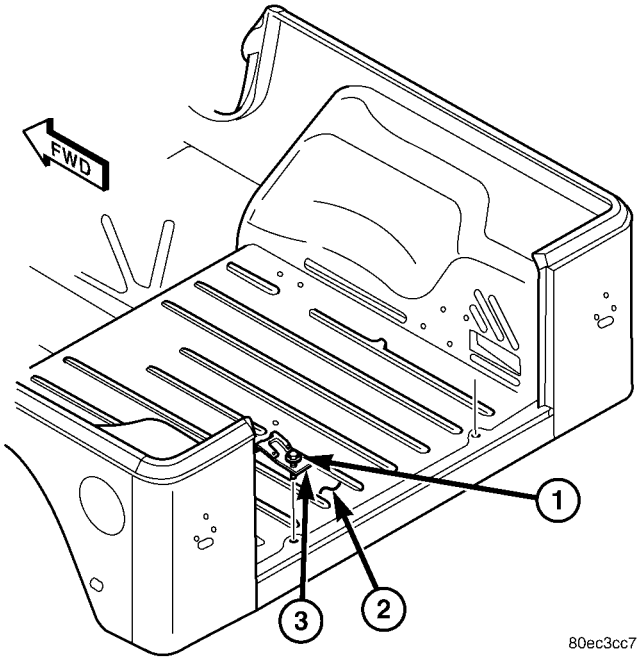

Fig. 14 Rear Upper Tether Anchor Remove/Install

1 - SCREW (2)
2 - FLOOR PANEL
3 - ANCHOR (2)

(1) Position the upper tether anchor onto the rear cargo floor panel (Fig. 14).

(2) Install and tighten the screw that secures the upper tether anchor to the rear cargo floor panel just forward of the tailgate opening sill. Tighten the screw to 26 N·m (19 ft. lbs.).

CLOCKSPRING

DESCRIPTION

The clockspring assembly is secured with two integral plastic latches onto the steering column lock housing near the top of the steering column behind the steering wheel (Fig. 15). The clockspring consists of a flat, round molded plastic case with a stubby tail that hangs below the steering column (Fig. 16). The tail contains two connector receptacles that face toward the instrument panel. Within the plastic case is a spool-like molded plastic rotor with a large exposed hub. The upper surface of the rotor hub has a large center hole, two large flats, and four short

CLOCKSPRING (Continued)

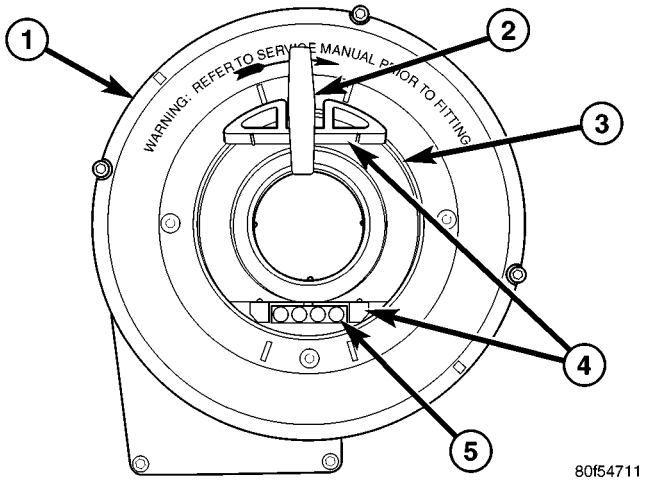

Fig. 15 Clockspring

1 - CASE
2 - LOCKING PIN
3 - ROTOR
4 - ROTOR FLAT (2)
5 - CLOCKSPRING PIGTAIL WIRES (NOT SHOWN)

pigtail wires with connectors that face toward the steering wheel.

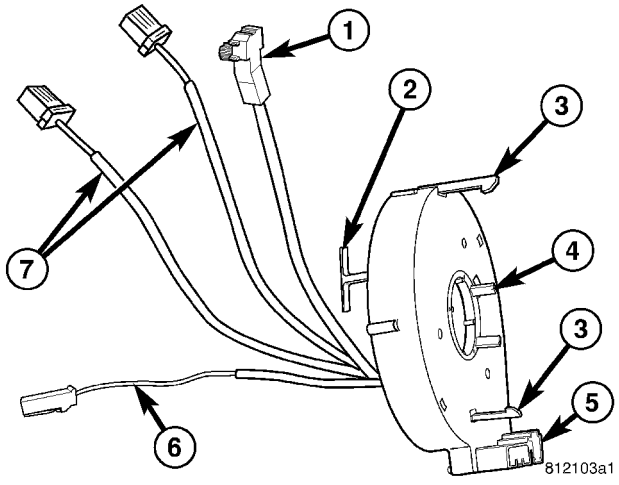

Fig. 16 Turn Signal Cancel Cam

1 - AIRBAG PIGTAIL
2 - LOCKING PIN
3 - LATCH (2)
4 - CANCEL CAM
5 - LOWER CONNECTOR RECEPTACLE (2)
6 - HORN SWITCH FEED PIGTAIL
7 - SPEED CONTROL SWITCH PIGTAIL (2)

The lower surface of the rotor hub has a molded plastic turn signal cancel cam consisting of two lobes that are molded into the rotor. Within the plastic case and wound around the rotor spool is a long ribbon-like tape that consists of several thin copper wire leads sandwiched between two thin plastic membranes. The outer end of the tape terminates at the connector receptacles that face the instrument panel, while the inner end of the tape terminates at the pigtail wires on the hub of the clockspring rotor that face the steering wheel.

Service replacement clocksprings are shipped precentered and with a molded plastic locking pin that snaps into a receptacle in the rotor and has a tab that is engaged between two ribs on the upper surface of the clockspring case. The locking pin secures the centered clockspring rotor to the clockspring case during shipment, but the locking pin must be removed from the clockspring after it is installed on the steering column. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - STANDARD PROCEDURE - CLOCKSPRING CENTERING).

The clockspring cannot be repaired. If the clockspring is faulty, damaged, or if the driver airbag has been deployed, the clockspring must be replaced.

OPERATION

The clockspring is a mechanical electrical circuit component that is used to provide continuous electrical continuity between the fixed body wire harness on the steering column and the electrical components mounted on or in the rotating steering wheel. On this model the rotating electrical components include the driver airbag, the horn switch, and the speed control switches if the vehicle is so equipped. The clockspring case is positioned and secured to the upper steering column housing near the top of the steering column. The connector receptacles on the tail of the fixed clockspring case connect the clockspring to the vehicle electrical system through two take outs with connectors from the body wire harness.

The clockspring rotor is movable and is keyed by two flats molded into the rotor hub to two flats that are cast into the lower surface of the steering wheel armature. The two lobes on the turn signal cancel cam on the lower surface of the clockspring rotor hub contact a turn signal cancel actuator of the multifunction switch to provide automatic turn signal cancellation. The pigtail wires on the upper surface of the clockspring rotor connect the clockspring to the driver airbag, the horn switch, and the two speed control switches if the vehicle is so equipped.

Like the clockspring in a timepiece, the clockspring tape has travel limits and can be damaged by being wound too tightly during full stop-to-stop steering wheel rotation. To prevent this from occurring, the clockspring is centered when it is installed on the steering column. Centering the clockspring indexes the clockspring tape to the movable steering components so that the tape can operate within its designed travel limits. However, if the clockspring is removed from the steering column or if the steering shaft is disconnected from the steering gear, the clockspring spool can change position relative to the

CLOCKSPRING (Continued)

movable steering components. The clockspring must be re-centered following completion of this service or the tape may be damaged.

Service replacement clocksprings are shipped pre-centered and with a plastic locking pin installed. This locking pin should not be disengaged until the clockspring has been installed on the steering column. If the locking pin is removed or damaged before the clockspring is installed on a steering column, the clockspring centering procedure must be performed. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - STANDARD PROCEDURE - CLOCKSPRING CENTERING).

STANDARD PROCEDURE - CLOCKSPRING CENTERING

The clockspring is designed to wind and unwind when the steering wheel is rotated, but is only designed to rotate the same number of turns (about five complete rotations) as the steering wheel can be turned from stop to stop. Centering the clockspring indexes the clockspring tape to other steering components so that it can operate within its designed travel limits. The rotor of a centered clockspring can be rotated two and one-half turns in either direction from the centered position, without damaging the clockspring tape.

However, if the clockspring is removed for service or if the steering column is disconnected from the steering gear, the clockspring tape can change position relative to the other steering components. The clockspring must then be re-centered following completion of such service or the clockspring tape may be damaged. Service replacement clocksprings are shipped pre-centered, with a plastic locking pin installed (Fig. 17). This locking pin should not be removed until the clockspring has been installed on the steering column. If the locking pin is removed before the clockspring is installed on a steering column, the clockspring centering procedure must be performed.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

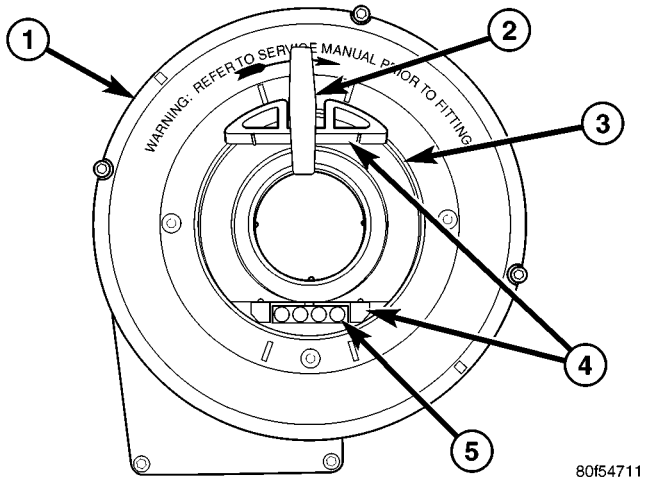

Fig. 17 Clockspring

1 - CASE
2 - LOCKING PIN
3 - ROTOR
4 - ROTOR FLAT (2)
5 - CLOCKSPRING PIGTAIL WIRES (NOT SHOWN)

NOTE: Before starting this procedure, be certain to turn the steering wheel until the front wheels are in the straight-ahead position.

(1) Place the front wheels in the straight-ahead position.

(2) Remove the clockspring from the steering column. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - REMOVAL).

(3) Rotate the clockspring rotor clockwise to the end of its travel. **Do not apply excessive torque.**

(4) From the end of the clockwise travel, rotate the rotor about two and one-half turns counterclockwise, until the rotor flats are horizontal. If the clockspring pigtail wires are not oriented towards the bottom of the clockspring, rotate the rotor another one-half turn in the counterclockwise direction. The clockspring is now centered.

(5) Lock the clockspring rotor to the clockspring case to maintain clockspring centering until it is reinstalled on the steering column.

(6) The front wheels should still be in the straight-ahead position. Reinstall the clockspring onto the steering column. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - INSTALLATION).

REMOVAL

The clockspring cannot be repaired. It must be replaced if faulty or damaged, or if the driver airbag has been deployed.

CLOCKSPRING (Continued)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

NOTE: Before starting this procedure, be certain to turn the steering wheel until the front wheels are in the straight-ahead position.

(1) Place the front wheels in the straight ahead position.

(2) Remove the driver airbag from the steering wheel. (Refer to 8 - ELECTRICAL/RESTRAINTS/DRIVER AIRBAG - REMOVAL).

(3) Disconnect the two upper clockspring pigtail wire connectors from the two speed control switches or the two trim bezels located within the two spoke cavities of the steering wheel.

CAUTION: Be certain that the screws that secure the steering wheel puller to the steering wheel are fully engaged in the steering wheel armature without passing through the steering wheel and damaging the clockspring.

(4) Remove the steering wheel from the steering column. (Refer to 19 - STEERING/COLUMN/STEERING WHEEL - REMOVAL).

(5) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(6) From below the steering column, remove the two screws that secure the lower steering column shroud to the upper shroud (Fig. 18).

(7) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully lowered position and leave the tilt release lever in the released (down) position.

(8) Using hand pressure, push gently inward on both sides of the upper shroud near the parting line between the upper and lower shrouds to release the snap features that secure it to the lower shroud.

(9) Remove both the upper and lower shrouds from the steering column.

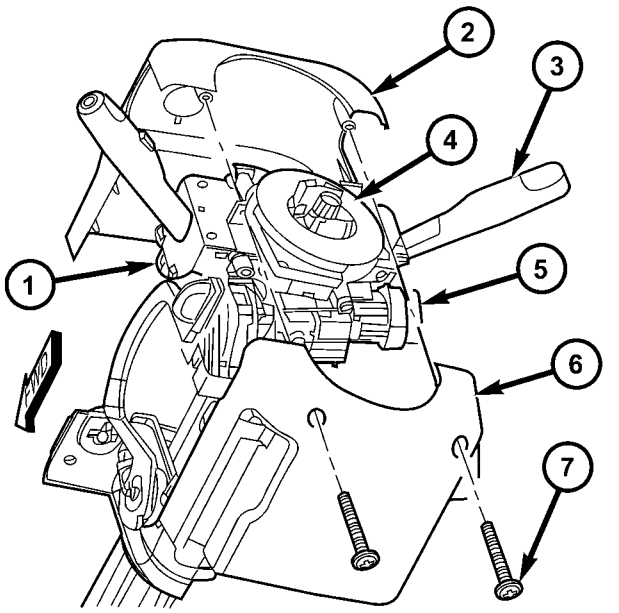

Fig. 18 Steering Column Shrouds Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - UPPER SHROUD
3 - RIGHT MULTI-FUNCTION SWITCH
4 - CLOCKSPRING
5 - IGNITION LOCK CYLINDER HOUSING
6 - LOWER SHROUD
7 - SCREW (2)

(10) Disconnect the two body wire harness connectors for the clockspring from the two connector receptacles below the steering column on the back of the clockspring housing (Fig. 19).

(11) Using a small screwdriver, gently pry both plastic latches that secure the clockspring away from the steering column upper housing far enough to pull the clockspring away from the upper housing.

NOTE: If the clockspring plastic latches are broken, be certain to remove the broken pieces from the steering column upper housing.

(12) Remove the clockspring from the steering column upper housing. The clockspring cannot be repaired. It must be replaced if faulty or damaged, or if the driver airbag has been deployed.

(13) If the removed clockspring is to be reused, be certain to secure the clockspring rotor to the clockspring case to maintain clockspring centering until it is reinstalled on the steering column. If clockspring centering is not maintained, the clockspring must be centered again before it is reinstalled. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - STANDARD PROCEDURE - CLOCKSPRING CENTERING).

CLOCKSPRING (Continued)

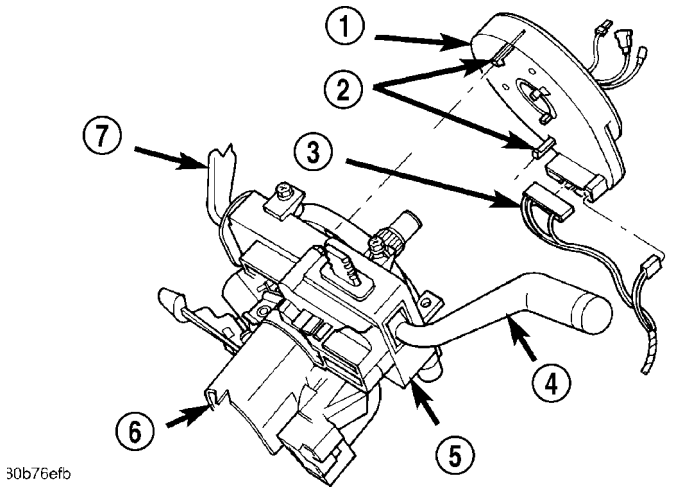

Fig. 19 Clockspring Remove/Install - Typical

1 - CLOCKSPRING
2 - LATCHES
3 - BODY WIRE HARNESS
4 - LEFT MULTI-FUNCTION SWITCH
5 - SWITCH HOUSING
6 - STEERING COLUMN
7 - RIGHT MULTI-FUNCTION SWITCH

INSTALLATION

The clockspring cannot be repaired. It must be replaced if faulty or damaged, or if the driver airbag has been deployed.

If the clockspring is not properly centered in relation to the steering wheel, steering shaft and steering gear, it may be damaged. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - STANDARD PROCEDURE - CLOCKSPRING CENTERING). Service replacement clocksprings are shipped pre-centered and with a plastic locking pin installed. This locking pin should not be removed until the clockspring has been installed on the steering column. If the locking pin is removed before the clockspring is installed on a steering column, the clockspring centering procedure must be performed.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

NOTE: Before starting this procedure, be certain that the front wheels are still in the straight-ahead position.

(1) Be certain that the left multi-function switch control stalk is in the neutral position, then carefully slide the centered clockspring down over the steering column upper shaft until both the upper and lower clockspring latches engage the steering column upper housing (Fig. 19).

(2) Reconnect the two body wire harness connectors for the clockspring to the two connector receptacles below the steering column on the back of the clockspring housing.

(3) Position both the upper and lower shrouds onto the steering column (Fig. 18). Be certain that the locating tabs for the left and right multi-function switch control stalk watershields are properly engaged in the openings of both the upper and lower shrouds.

(4) Align the snap features on the upper shroud with the receptacles on the lower shroud and apply hand pressure to snap them together.

(5) From below the steering column, install and tighten the two screws that secure the lower steering column shroud to the upper shroud. Tighten the screws to 2 N·m (18 in. lbs.).

(6) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully raised position and secure it in place by moving the tilt release lever back to the locked (up) position.

(7) Reinstall the steering column opening cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(8) If a new clockspring has been installed, remove the locking pin that is securing the clockspring rotor to the clockspring case to maintain clockspring centering.

(9) Reinstall the steering wheel onto the steering column. (Refer to 19 - STEERING/COLUMN/STEERING WHEEL - INSTALLATION).

(10) Reconnect the two upper clockspring pigtail wire connectors to the two speed control switches or the two trim bezels located within the two spoke cavities of the steering wheel.

(11) Reinstall the driver airbag onto the steering wheel. (Refer to 8 - ELECTRICAL/RESTRAINTS/DRIVER AIRBAG - INSTALLATION).

DRIVER AIRBAG

DESCRIPTION

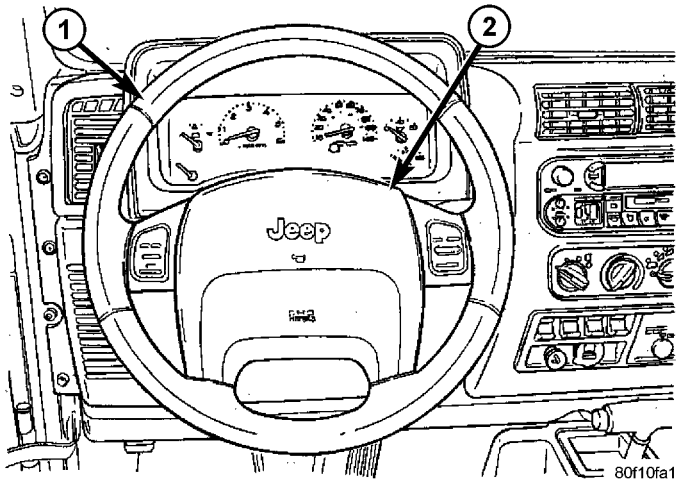

Fig. 20 Driver Airbag Trim Cover

1 - STEERING WHEEL
2 - TRIM COVER

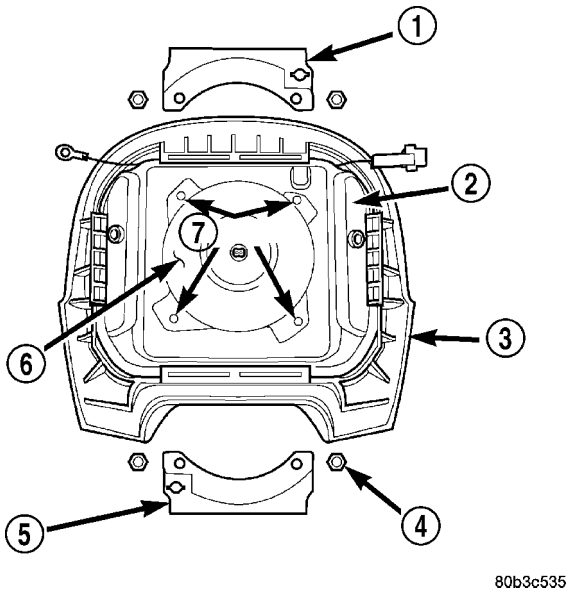

Fig. 21 Driver Airbag Trim Cover

1 - UPPER RETAINER
2 - AIRBAG HOUSING
3 - TRIM COVER
4 - NUT (4)
5 - LOWER RETAINER
6 - INFLATOR
7 - STUD (4)

The color-keyed, injection molded, thermoplastic driver airbag protective trim cover is the most visible part of the driver airbag (Fig. 20). The driver airbag is located in the center of the steering wheel, where it is secured with two screws to the two horizontal spokes of the four-spoke steering wheel armature. All models have a Jeep logo embossed in the center of the trim cover. Concealed beneath the driver airbag trim cover are the horn switch, the folded airbag cushion, the airbag cushion retainer, the airbag housing, the airbag inflator, and the retainers that secure the inflator to the airbag housing.

The airbag cushion, housing, and inflator unit is secured within an integral receptacle on the back of the trim cover (Fig. 21). The driver airbag trim cover has locking blocks molded into the back side of it that engage a lip formed around the perimeter of the airbag housing. Two stamped metal retainers then fit over the inflator mounting studs on the back of the airbag housing and tabs on the retainers are engaged in slots within the upper and lower trim cover locking blocks, securely locking the cover into place. The stamped metal retainers are secured to the four airbag inflator mounting studs with nuts.

The resistive membrane-type horn switch is secured within a plastic tray that is inserted into a pocket or pouch sewn onto the airbag cushion retainer strap, between the trim cover and the folded airbag cushion. The horn switch ground pigtail wire has an eyelet terminal connector that is captured on the upper right inflator mounting stud between the inflator and the upper trim cover retainer. The horn switch feed pigtail wire has a white, molded plastic insulator that is secured by an integral retainer to a mounting hole located in the upper trim cover retainer near the upper left corner on the back of the airbag housing, and is connected to the vehicle electrical system through a dedicated pigtail wire and connector from the clockspring.

The airbag used in this model is a Next Generation-type that complies with revised federal airbag standards to deploy with less force than those used in some prior models. A radial deploying fabric cushion with internal tethers is used. The airbag inflator is a conventional pyrotechnic-type unit that is secured by four hex nuts to four studs that extend through the back of the stamped metal airbag housing from the airbag cushion retainer ring. A keyed connector receptacle on the driver airbag inflator connects the inflator initiator to the vehicle electrical system through a yellow-jacketed, two-wire pigtail harness of the clockspring. The driver airbag cannot be repaired, and must be replaced if deployed or in any way damaged. The driver airbag trim cover and the horn switch are available individually, and may be disassembled from the driver airbag for service replacement.

OPERATION

The driver airbag is deployed by electrical signals generated by the Airbag Control Module (ACM) through the driver airbag squib circuit to the initiator in the airbag inflator. When the ACM sends the

DRIVER AIRBAG (Continued)

proper electrical signal to the initiator the electrical energy generates enough heat to initiate a small pyrotechnic charge which, in turn ignites chemical pellets within the inflator. Once ignited, these chemical pellets burn rapidly and produce a large quantity of inert gas. The inflator is sealed to the back of the airbag housing and a diffuser in the inflator directs all of the inert gas into the airbag cushion, causing the cushion to inflate. As the cushion inflates, the driver airbag trim cover will split at predetermined breakout lines, then fold back out of the way along with the horn switch unit. Following an airbag deployment, the airbag cushion quickly deflates by venting the inert gas towards the instrument panel through the porous fabric material used to construct the back (steering wheel side) panel of the airbag cushion.

Some of the chemicals used to create the inert gas may be considered hazardous while in their solid state before they are burned, but they are securely sealed within the airbag inflator. Typically, all potentially hazardous chemicals are burned during an airbag deployment event. The inert gas that is produced when the chemicals are burned is harmless. However, a small amount of residue from the burned chemicals may cause some temporary discomfort if it contacts the skin, eyes, or breathing passages. If skin or eye irritation is noted, rinse the affected area with plenty of cool, clean water. If breathing passages are irritated, move to another area where there is plenty of clean, fresh air to breath. If the irritation is not alleviated by these actions, contact a physician.

REMOVAL

The following procedure is for replacement of a faulty or damaged driver airbag. If the airbag is faulty or damaged, but not deployed, review the recommended procedures for handling non-deployed supplemental restraints. (Refer to 8 - ELECTRICAL/ RESTRAINTS - STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS). If the driver airbag has been deployed, review the recommended procedures for service after a supplemental restraint deployment before removing the airbag from the vehicle. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - SERVICE AFTER A SUPPLEMENTAL RESTRAINT DEPLOYMENT).

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, WHEN REMOVING A DEPLOYED AIRBAG, RUBBER GLOVES, EYE PROTECTION, AND A LONG-SLEEVED SHIRT SHOULD BE WORN. THERE MAY BE DEPOSITS ON THE AIRBAG CUSHION AND OTHER INTERIOR SURFACES. IN LARGE DOSES, THESE DEPOSITS MAY CAUSE IRRITATION TO THE SKIN AND EYES.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) From the underside of the steering wheel, remove the two screws that secure the driver airbag to the steering wheel armature (Fig. 22).

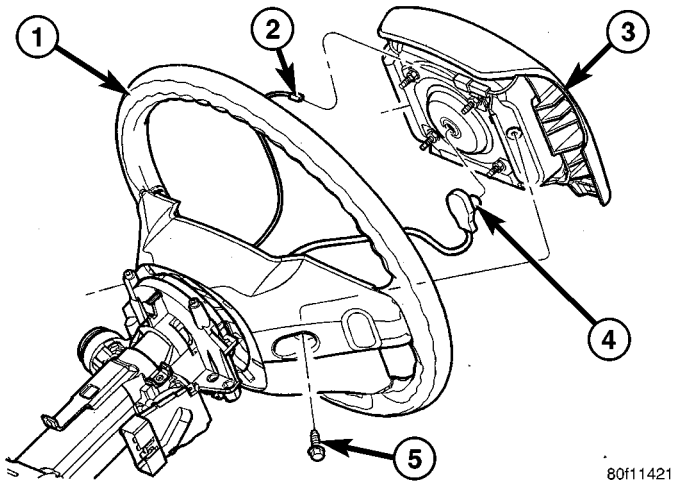

Fig. 22 Driver Airbag Remove/Install

1 - STEERING WHEEL
2 - CLOCKSPRING PIGTAIL WIRE (HORN SWITCH)
3 - DRIVER AIRBAG
4 - CLOCKSPRING PIGTAIL WIRE (AIRBAG)
5 - SCREW (2)

(3) Pull the driver airbag away from the steering wheel far enough to access the two electrical connections at the back of the airbag housing.

(4) Disconnect the clockspring pigtail wire connector for the horn switch from the horn switch feed pigtail wire connector located on the back of the driver airbag.

DRIVER AIRBAG (Continued)

CAUTION: Do not pull on the clockspring pigtail wires to disengage the connector from the driver airbag inflator connector receptacle. Improper removal of these pigtail wires and their connector insulators can result in damage to the airbag circuits or connector insulators.

(5) The clockspring driver airbag pigtail wire connector is secured by a tight snap fit into the airbag inflator connector receptacle, which is located on the back of the driver airbag housing. Firmly grasp and pull or gently pry on the clockspring driver airbag pigtail wire connector insulator and pull the insulator straight out from the airbag inflator to disconnect it from the connector receptacle.

(6) Remove the driver airbag from the steering wheel.

(7) If the driver airbag has been deployed, the clockspring must be replaced. (Refer to 8 - ELECTRICAL/RESTRAINTS/CLOCKSPRING - REMOVAL).

DISASSEMBLY

The following procedures can be used to replace the driver airbag trim cover and/or to access the horn switch unit for service. If the driver airbag is faulty or deployed, the entire driver airbag, trim cover, and horn switch must be replaced as a unit.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, SERVICE OF THIS UNIT SHOULD BE PERFORMED ONLY BY DAIMLERCHRYSLER-TRAINED AND AUTHORIZED DEALER SERVICE TECHNICIANS. FAILURE TO TAKE THE PROPER PRECAUTIONS OR TO FOLLOW THE PROPER PROCEDURES COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER AIRBAG DEPLOYMENT AND POSSIBLE OCCUPANT INJURIES.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, USE EXTREME CARE TO PREVENT ANY FOREIGN MATERIAL FROM ENTERING THE DRIVER AIRBAG, OR BECOMING ENTRAPPED BETWEEN THE DRIVER AIRBAG CUSHION AND THE DRIVER AIRBAG TRIM COVER. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE DRIVER AIRBAG TRIM COVER MUST NEVER BE PAINTED. REPLACEMENT TRIM COVERS ARE SERVICED IN THE ORIGINAL COLORS. PAINT MAY CHANGE THE WAY IN WHICH THE MATERIAL OF THE TRIM COVER RESPONDS TO AN AIRBAG DEPLOYMENT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) Remove the driver airbag from the steering wheel. (Refer to 8 - ELECTRICAL/RESTRAINTS/DRIVER AIRBAG - REMOVAL).

(3) Place the driver airbag on a suitable work surface with the trim cover facing down. If the trim cover will be reused, be certain to take the proper precautions to prevent the trim cover from receiving cosmetic damage during the following procedures.

(4) Remove the four nuts that secure the upper and lower trim cover retainers to the studs on the back of the driver airbag housing (Fig. 23).

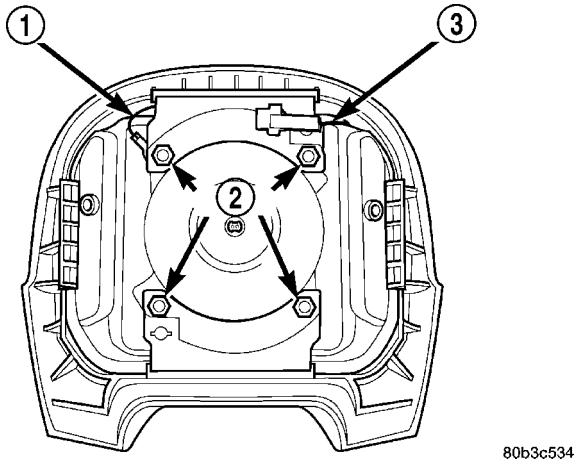

Fig. 23 Horn Switch Feed Wire Remove/Install

1 - HORN SWITCH GROUND PIGTAIL WIRE
2 - NUT (4)
3 - HORN SWITCH FEED PIGTAIL WIRE

DRIVER AIRBAG (Continued)

(5) Remove the upper and lower trim cover retainers from the airbag housing studs (Fig. 24).

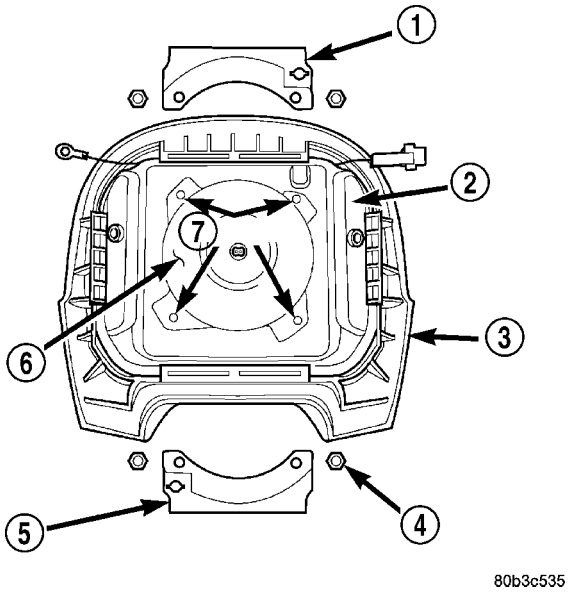

Fig. 24 Driver Airbag Trim Cover

1 - UPPER RETAINER
2 - AIRBAG HOUSING
3 - TRIM COVER
4 - NUT (4)
5 - LOWER RETAINER
6 - INFLATOR
7 - STUD (4)

(6) Disengage the horn switch feed pigtail wire connector retainer from the mounting hole in the upper trim cover retainer.

(7) Remove the horn switch ground pigtail wire eyelet terminal from the upper right inflator stud on the back of the driver airbag housing.

(8) Disengage the four trim cover locking blocks from the lip around the outside edge of the driver airbag housing and remove the housing from the trim cover receptacle (Fig. 25).

ASSEMBLY

The following procedures can be used to replace the driver airbag trim cover and/or to access the horn switch unit for service. If the driver airbag is faulty or deployed, the entire driver airbag, trim cover, and horn switch must be replaced as a unit.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

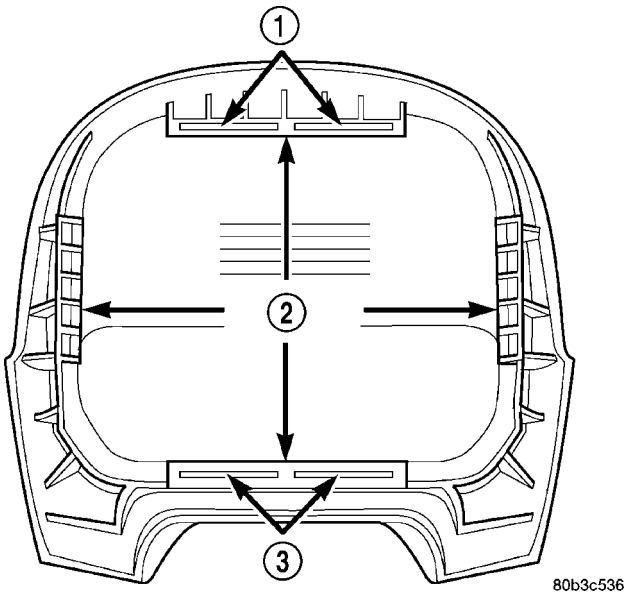

Fig. 25 Driver Airbag Trim Cover Remove/Install

1 - UPPER RETAINER SLOT (2)
2 - LOCKING BLOCK (4)
3 - LOWER RETAINER SLOT (2)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, SERVICE OF THIS UNIT SHOULD BE PERFORMED ONLY BY DAIMLERCHRYSLER-TRAINED AND AUTHORIZED DEALER SERVICE TECHNICIANS. FAILURE TO TAKE THE PROPER PRECAUTIONS OR TO FOLLOW THE PROPER PROCEDURES COULD RESULT IN ACCIDENTAL, INCOMPLETE, OR IMPROPER AIRBAG DEPLOYMENT AND POSSIBLE OCCUPANT INJURIES.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, USE EXTREME CARE TO PREVENT ANY FOREIGN MATERIAL FROM ENTERING THE DRIVER AIRBAG, OR BECOMING ENTRAPPED BETWEEN THE DRIVER AIRBAG CUSHION AND THE DRIVER AIRBAG TRIM COVER. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

DRIVER AIRBAG (Continued)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE DRIVER AIRBAG TRIM COVER MUST NEVER BE PAINTED. REPLACEMENT TRIM COVERS ARE SERVICED IN THE ORIGINAL COLORS. PAINT MAY CHANGE THE WAY IN WHICH THE MATERIAL OF THE TRIM COVER RESPONDS TO AN AIRBAG DEPLOYMENT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

NOTE: If the horn switch and tray have been removed from the sewn pouch in the airbag cushion retaining strap, be certain that they are properly reinstalled with the horn switch feed and ground pigtail wires properly oriented before assembling the trim cover onto the airbag housing. (Refer to 8 - ELECTRICAL/HORN/HORN SWITCH - INSTALLATION).

(1) Place the new driver airbag trim cover on a suitable work surface with the airbag receptacle facing up. Be certain to take the proper precautions to prevent the trim cover from receiving cosmetic damage during the following procedures.

(2) Carefully position the driver airbag into the trim cover receptacle. Be certain that the horn switch feed and ground pigtail wires are not pinched between the airbag housing and the trim cover locking blocks.

(3) Engage the upper and lower trim cover locking blocks with the lip of the driver airbag housing, then engage the locking blocks on each side of the trim cover with the lip of the housing. Be certain that each of the locking blocks is fully engaged on the lip of the airbag housing (Fig. 26).

(4) Reinstall the horn switch ground pigtail wire eyelet terminal over the upper right inflator stud on the back of the driver airbag housing.

(5) Reinstall the upper and lower airbag trim cover retainers over the inflator studs on the back of the driver airbag housing. Be certain that the tabs on each retainer are engaged in the slots of the upper and lower trim cover locking blocks (Fig. 25).

(6) Install and tighten the four nuts that secure the trim cover retainers to the inflator studs on the back of the driver airbag housing. Tighten the nuts to 7 N·m (60 in. lbs.).

(7) Engage the horn switch feed pigtail wire connector retainer in the mounting hole in the upper trim cover retainer.

(8) Reinstall the driver airbag onto the steering wheel. (Refer to 8 - ELECTRICAL/RESTRAINTS/ DRIVER AIRBAG - INSTALLATION).

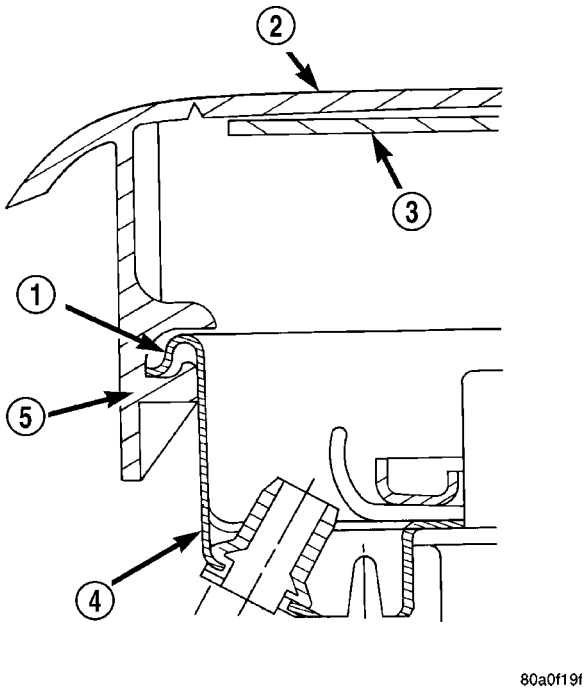

Fig. 26 Driver Airbag Trim Cover Locking Blocks Engaged

1 - LIP
2 - TRIM COVER
3 - HORN SWITCH
4 - AIRBAG HOUSING
5 - LOCKING BLOCK

INSTALLATION

The following procedure is for replacement of a faulty or damaged driver airbag. If the airbag is faulty or damaged, but not deployed, review the recommended procedures for handling non-deployed supplemental restraints. (Refer to 8 - ELECTRICAL/ RESTRAINTS - STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS). If the driver airbag has been deployed, review the recommended procedures for service after a supplemental restraint deployment before removing the airbag from the vehicle. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - SERVICE AFTER A SUPPLEMENTAL RESTRAINT DEPLOYMENT).

DRIVER AIRBAG (Continued)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, USE EXTREME CARE TO PREVENT ANY FOREIGN MATERIAL FROM ENTERING THE DRIVER AIRBAG, OR BECOMING ENTRAPPED BETWEEN THE DRIVER AIRBAG CUSHION AND THE DRIVER AIRBAG TRIM COVER. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE DRIVER AIRBAG TRIM COVER MUST NEVER BE PAINTED. REPLACEMENT AIRBAGS ARE SERVICED WITH TRIM COVERS IN THE ORIGINAL COLORS. PAINT MAY CHANGE THE WAY IN WHICH THE MATERIAL OF THE TRIM COVER RESPONDS TO AN AIRBAG DEPLOYMENT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

(1) Position the driver airbag close enough to the steering wheel to reconnect the two electrical connections at the back of the airbag housing.

(2) When installing the driver airbag, reconnect the clockspring driver airbag pigtail wire connector to the airbag inflator connector receptacle by pressing straight in on the connector (Fig. 22). You can be certain that the connector is fully engaged in its receptacle by listening carefully for a distinct, audible click as the connector snaps into place.

(3) Reconnect the clockspring horn switch pigtail wire connector to the horn switch feed pigtail wire connector located on the back of the driver airbag housing.

(4) Carefully position the driver airbag in the steering wheel. Be certain that the clockspring pigtail wires and the steering wheel wire harness in the steering wheel hub area are not pinched between the driver airbag and the steering wheel armature.

(5) From the underside of the steering wheel, install and tighten the two screws that secure the driver airbag to the steering wheel armature. Tighten the screws to 10 N·m (90 in. lbs.).

(6) Do not reconnect the battery negative cable at this time. The supplemental restraint system verification test procedure should be performed following service of any supplemental restraint system component. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - VERIFICATION TEST).

FRONT SEAT BELT & RETRACTOR

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Move the front seat to its most forward position for easiest access to the lower seat belt anchor screw and retractor and the lower sport bar.

(2) Unsnap and lift the trim cover from the front seat belt turning loop to access the screw that secures the turning loop to the height adjuster near the top on the upper sport bar (Fig. 27).

(3) Remove the screw that secures the shoulder belt turning loop to the adjuster.

(4) Remove the shoulder belt turning loop and the support/guide washer from the adjuster.

(5) Remove the screw that secures the retractor and lower seat belt anchor to the lower sport bar (Fig. 28).

FRONT SEAT BELT & RETRACTOR (Continued)

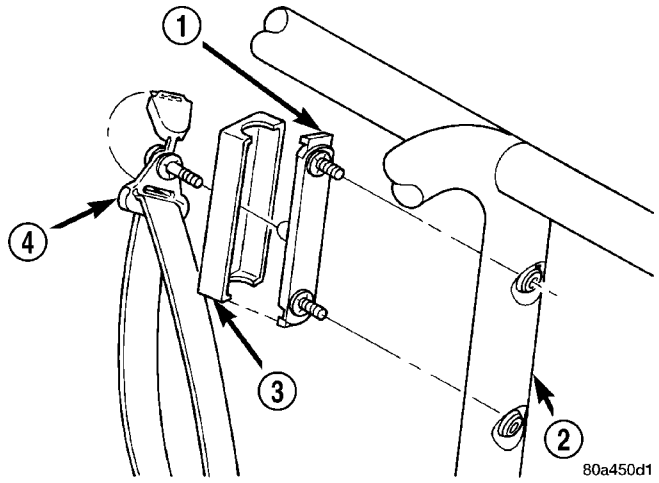

Fig. 27 Front Seat Belt Turning Loop

1 - ADJUSTER
2 - SPORT BAR
3 - TRIM COVER
4 - TURNING LOOP

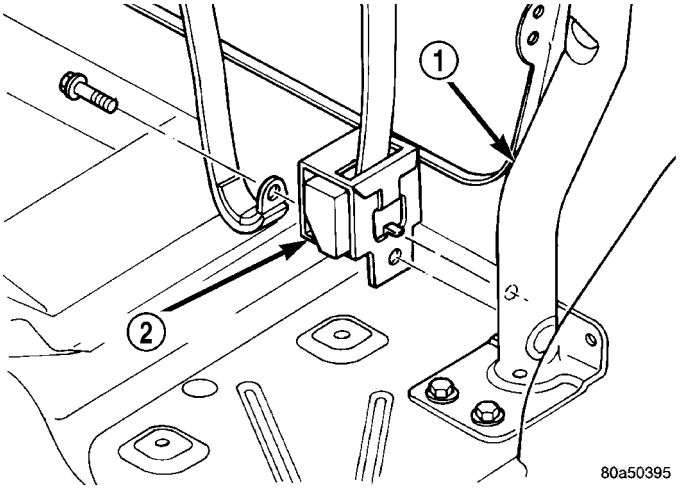

Fig. 28 Front Seat Belt and Retractor - Typical

1 - SPORT BAR
2 - RETRACTOR

(6) Remove the front seat belt and retractor from the lower sport bar.

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Position the front seat belt and retractor and the lower seat belt anchor to the lower sport bar (Fig. 28). Be certain to engage the locator tab on the retractor in the locator hole on the sport bar.

(2) Install and tighten the screw that secures the front seat belt retractor and lower seat belt anchor to the lower sport bar. Tighten the screw to 43 N·m (32 ft. lbs.).

(3) Position the support/guide washer and the front seat belt turning loop onto the height adjuster on the upper sport bar (Fig. 27).

(4) Install and tighten the screw that secures the front seat belt turning loop to the height adjuster. Tighten the screw to 43 N·m (32 ft. lbs.).

(5) Fold and snap the trim cover back into place over the screw that secures the front seat belt turning loop to the adjuster.

FRONT SEAT BELT BUCKLE

REMOVAL

The seat belt buckle on the driver's side front seat for all models also includes a seat belt switch. (Refer to 8 - ELECTRICAL/RESTRAINTS/SEAT BELT SWITCH - DESCRIPTION).

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) On the driver side only, disconnect the seat belt switch pigtail wire connector from the body wire harness connector for the seat belt switch (Fig. 29).

(2) Remove the screw that secures the seat belt buckle lower anchor to the bracket on the rear of the upper inner front seat track (Fig. 30).

(3) Remove the front seat belt buckle from the front seat track bracket.

INSTALLATION

The seat belt buckle on the driver's side front seat for all models also includes a seat belt switch. The seat belt buckle on the driver's side front seat for all models also includes a seat belt switch. (Refer to 8 - ELECTRICAL/RESTRAINTS/SEAT BELT SWITCH - DESCRIPTION).

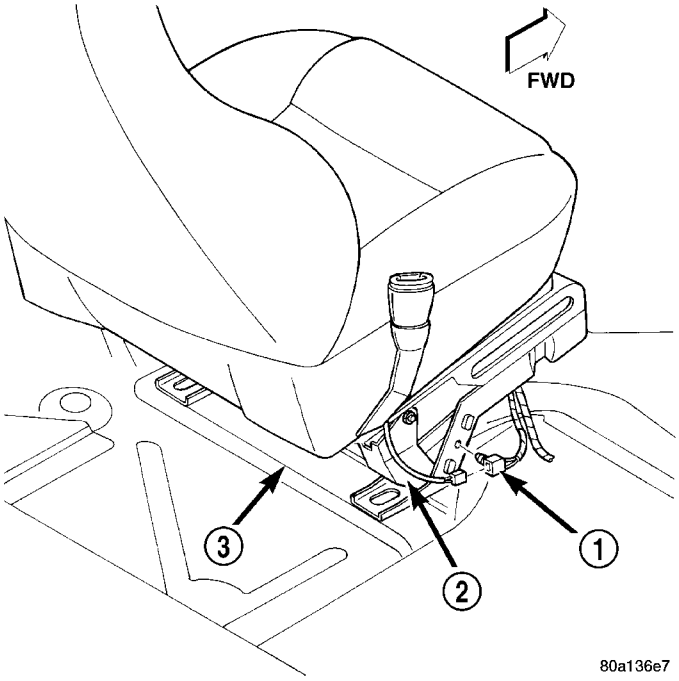

Fig. 29 Seat Belt Switch

1 - WIRE HARNESS CONNECTOR
2 - DRIVER SEAT RISER
4 - FLOOR PANEL

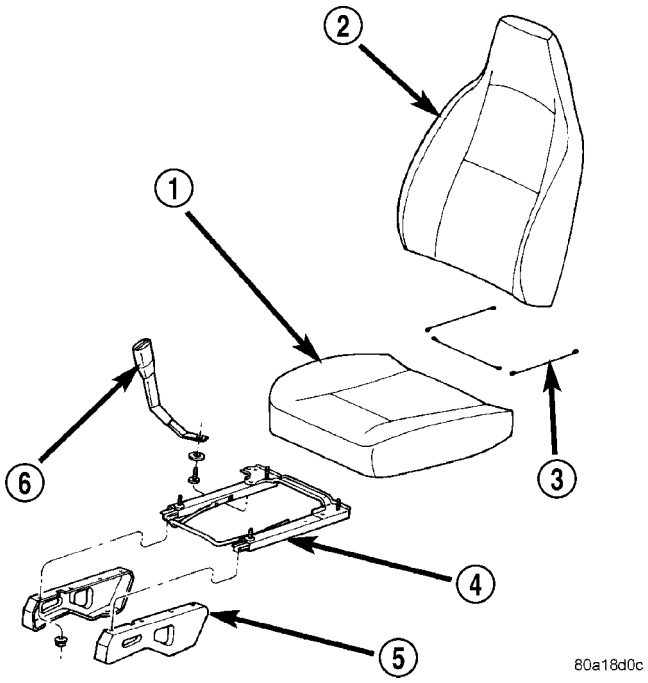

Fig. 30 Seat Belt Buckle

1 - SEAT CUSHION
2 - SEAT BACK
3 - TRIM ATTACHMENT WIRES
4 - SEAT TRACK
5 - SEAT RISER
6 - SEATBELT BUCKLE

FRONT SEAT BELT BUCKLE (Continued)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Position the front seat belt buckle lower anchor to the bracket on the rear of the upper inner front seat track.

(2) Install and tighten the screw that secures the front seat belt buckle lower anchor to the front seat track bracket (Fig. 30). Tighten the screw to 43 N·m (32 ft. lbs.).

(3) On the driver side only, reconnect the seat belt switch pigtail wire connector to the body wire harness connector for the seat belt switch (Fig. 29).

PASSENGER AIRBAG

DESCRIPTION

The rearward facing surface of the injection molded, thermoplastic passenger airbag door is the most visible part of the passenger airbag (Fig. 31). The passenger airbag door is located above the glove box opening on the instrument panel in front of the front seat passenger seating position. The passenger airbag door also serves as a trim cover and has two flanges and a stamped metal bracket that secure it to the instrument panel structural support. The upper flange is secured with screws to the top of the instrument panel structural support and the lower flange to the upper glove box opening reinforcement. A stamped metal passenger airbag door hinge bracket is secured to the back of the instrument panel structural support with two screws.

Located behind the passenger airbag door within the instrument panel is the passenger airbag unit. The passenger airbag unit has an upper rear bracket and a lower rear bracket that are secured by nuts to three weld studs on the instrument panel structural support, one stud at the top and two at the bottom. Two nuts secure a front lower bracket on the passenger airbag housing to two weld studs on the dash panel above the heater and air conditioner housing. The front lower bracket mounts are accessed through the instrument panel glove box opening.

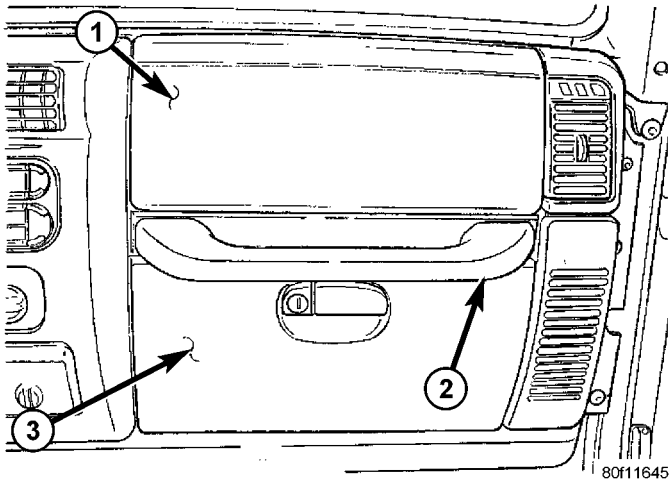

Fig. 31 Passenger Airbag Door

1 - PASSENGER AIRBAG DOOR
2 - GRAB HANDLE
3 - GLOVE BOX DOOR

The passenger airbag unit used in this model is a Next Generation-type that complies with revised federal airbag standards to deploy with less force than those used in some prior models. The passenger airbag unit consists of a stamped and welded metal housing or retainer, the airbag cushion, and the airbag inflator. The airbag housing contains the airbag inflator and the folded airbag cushion. A rectangular fabric cushion is used. The airbag inflator is a hybrid-type unit that is secured to and sealed within the stamped metal airbag housing beneath the folded airbag cushion. The inflator initiator is connected to the vehicle electrical system through a yellow connector on the end of a short, two-wire pigtail harness and a take out of the body wire harness.

The passenger airbag and the passenger airbag door are available as separate service items. The passenger airbag cannot be repaired, and must be replaced if deployed, faulty, or in any way damaged. The passenger airbag door will be damaged and must be replaced following an airbag deployment.

OPERATION

The passenger airbag is deployed by an electrical signal generated by the Airbag Control Module (ACM) through the passenger airbag squib circuits to the initiator in the airbag inflator. The hybrid-type inflator assembly includes a small canister of highly

compressed inert gas. When the ACM sends the proper electrical signal to the airbag inflator, the electrical energy generates enough heat to ignite chemical pellets within the inflator. Once ignited, these chemical pellets burn rapidly and produce the pressure necessary to rupture a burst disk in the inert gas canister.

The inflator is sealed to the airbag cushion so that all of the released inert gas is directed into the airbag cushion, causing the cushion to inflate. As the cushion inflates, the passenger airbag door will split at predetermined tear seam lines integral to the door, then the door will fold up over the top of the instrument panel and out of the way. Following a passenger airbag deployment, the airbag cushion quickly deflates by venting the inert gas through the porous fabric material used to construct each end panel of the airbag cushion.

REMOVAL

The following procedure is for replacement of a faulty or damaged passenger airbag. If the airbag is faulty or damaged, but not deployed, review the recommended procedures for handling non-deployed supplemental restraints. (Refer to 8 - ELECTRICAL/ RESTRAINTS - STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS). If the passenger airbag has been deployed, review the recommended procedures for service after a supplemental restraint deployment before removing the airbag from the vehicle. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - SERVICE AFTER A SUPPLEMENTAL RESTRAINT DEPLOYMENT).

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, WHEN REMOVING A DEPLOYED AIRBAG, RUBBER GLOVES, EYE PROTECTION, AND A LONG-SLEEVED SHIRT SHOULD BE WORN. THERE MAY BE DEPOSITS ON THE AIRBAG UNIT AND OTHER INTERIOR SURFACES. IN LARGE DOSES, THESE DEPOSITS MAY CAUSE IRRITATION TO THE SKIN AND EYES.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) Remove the instrument panel from the passenger compartment of the vehicle. (Refer to 23 - BODY/ INSTRUMENT PANEL - REMOVAL).

(3) Place the instrument panel on a suitable work surface. Be certain to take the proper precautions to protect the instrument panel from any possible cosmetic damage.

(4) Remove the three nuts that secure the passenger airbag to the weld studs on the instrument panel structural support (Fig. 32).

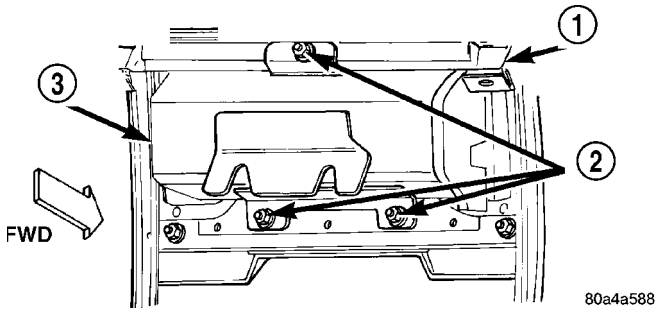

Fig. 32 Passenger Airbag Remove/Install

1 - STRUCTURAL SUPPORT
2 - NUT (3)
3 - PASSENGER AIRBAG

(5) Remove the passenger airbag from the instrument panel structural support.

INSTALLATION

The following procedure is for replacement of a faulty or damaged passenger airbag. If the airbag is faulty or damaged, but not deployed, review the recommended procedures for handling non-deployed supplemental restraints. (Refer to 8 - ELECTRICAL/ RESTRAINTS - STANDARD PROCEDURE - HANDLING NON-DEPLOYED SUPPLEMENTAL RESTRAINTS). If the passenger airbag has been deployed, review the recommended procedures for service after a supplemental restraint deployment before removing the airbag from the vehicle. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - SERVICE AFTER A SUPPLEMENTAL RESTRAINT DEPLOYMENT).

PASSENGER AIRBAG (Continued)

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, WHEN REMOVING A DEPLOYED AIRBAG, RUBBER GLOVES, EYE PROTECTION, AND A LONG-SLEEVED SHIRT SHOULD BE WORN. THERE MAY BE DEPOSITS ON THE AIRBAG UNIT AND OTHER INTERIOR SURFACES. IN LARGE DOSES, THESE DEPOSITS MAY CAUSE IRRITATION TO THE SKIN AND EYES.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, USE EXTREME CARE TO PREVENT ANY FOREIGN MATERIAL FROM ENTERING THE PASSENGER AIRBAG, OR BECOMING ENTRAPPED BETWEEN THE PASSENGER AIRBAG CUSHION AND THE PASSENGER AIRBAG DOOR. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE PASSENGER AIRBAG DOOR MUST NEVER BE PAINTED. REPLACEMENT AIRBAG DOORS ARE SERVICED IN THE ORIGINAL COLORS. PAINT MAY CHANGE THE WAY IN WHICH THE MATERIAL OF THE AIRBAG DOOR RESPONDS TO AN AIRBAG DEPLOYMENT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

(1) Carefully position the passenger airbag onto the weld studs on the instrument panel structural support (Fig. 32).

(2) Install and tighten the three nuts that secure the passenger airbag upper and lower rear mounting brackets to the weld studs on the instrument panel structural support. Tighten the nuts to 12 N·m (105 in. lbs.).

(3) Reinstall the instrument panel into the passenger compartment of the vehicle. (Refer to 23 - BODY/INSTRUMENT PANEL - INSTALLATION). When installing the instrument panel, be certain the passenger airbag pigtail wire connector is fully engaged and latched.

(4) Do not reconnect the battery negative cable at this time. The supplemental restraint system verification test procedure should be performed following service of any supplemental restraint system component. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - VERIFICATION TEST).

PASSENGER AIRBAG DOOR

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) Remove the passenger airbag from the instrument panel. (Refer to 8 - ELECTRICAL/RESTRAINTS/PASSENGER AIRBAG - REMOVAL).

(3) Remove the five screws that secure the upper flange of the passenger airbag door to the top of the instrument panel structural support (Fig. 33).

(4) Remove the grab handle bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GRAB HANDLE BEZEL - REMOVAL).

(5) Remove the five screws that secure the lower flange of the passenger airbag door to the upper glove box opening reinforcement.

(6) Remove the two screws that secure the ends of the passenger airbag door bracket to the instrument panel structural support (Fig. 34).

(7) Remove and discard the two passenger airbag door bracket J-nuts from the instrument panel structural support. These J-nuts must be replaced with new parts whenever the passenger airbag door bracket screws are removed.

(8) Remove the passenger airbag door from the instrument panel.

PASSENGER AIRBAG DOOR (Continued)

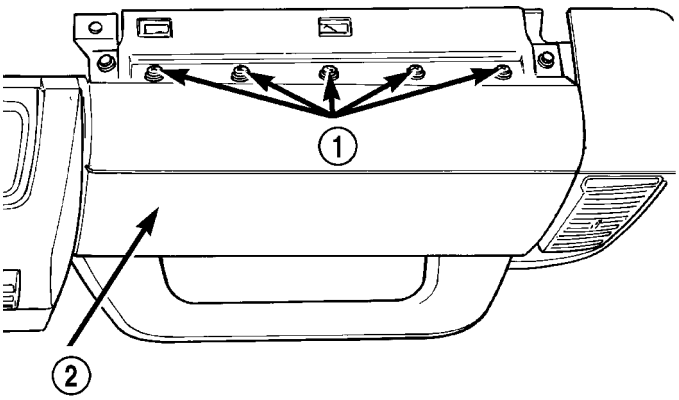

Fig. 33 Passenger Airbag Door Upper Flange Remove/Install

1 - SCREW (5)
2 - PASSENGER AIRBAG DOOR

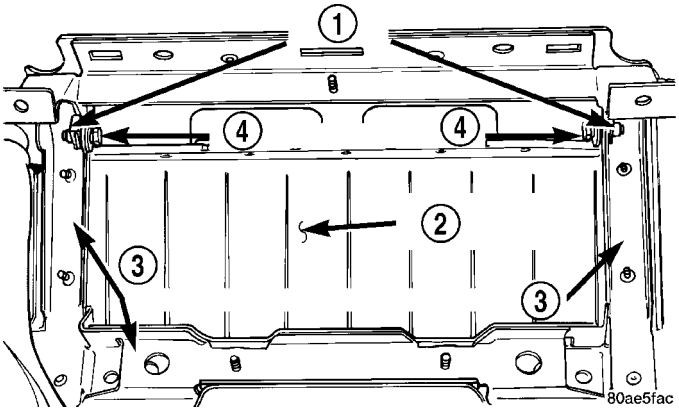

Fig. 34 Passenger Airbag Door Remove/Install

1 - J-NUT (2)
2 - PASSENGER AIRBAG DOOR
3 - STRUCTURAL SUPPORT
4 - SCREW (2)

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, USE EXTREME CARE TO PREVENT ANY FOREIGN MATERIAL FROM ENTERING THE PASSENGER AIRBAG, OR BECOMING ENTRAPPED BETWEEN THE PASSENGER AIRBAG CUSHION AND THE PASSENGER AIRBAG DOOR. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, THE PASSENGER AIRBAG DOOR MUST NEVER BE PAINTED. REPLACEMENT AIRBAG DOORS ARE SERVICED IN THE ORIGINAL COLORS. PAINT MAY CHANGE THE WAY IN WHICH THE MATERIAL OF THE AIRBAG DOOR RESPONDS TO AN AIRBAG DEPLOYMENT. FAILURE TO OBSERVE THIS WARNING COULD RESULT IN OCCUPANT INJURIES UPON AIRBAG DEPLOYMENT.

(1) Install two new passenger airbag door bracket J-nuts onto the instrument panel structural support (Fig. 34). These J-nuts must be replaced with new parts whenever the passenger airbag door bracket screws are removed.

(2) Position the passenger airbag door onto the instrument panel and align the mounting holes in each end of the airbag door bracket with the J-nuts on the instrument panel structural support.

(3) Install and tighten the two screws that secure the passenger airbag door bracket to the instrument panel structural support. Tighten the screws to 12 N·m (105 in. lbs.).

(4) Install and tighten the five screws that secure the lower flange of the passenger airbag door to the upper glove box opening reinforcement. Tighten the screws to 2 N·m (20 in. lbs.).

(5) Reinstall the grab handle bezel onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/GRAB HANDLE BEZEL - INSTALLATION).

(6) Install and tighten the five screws that secure the upper flange of the passenger airbag door to the top of the instrument panel structural support (Fig. 33). Tighten the screws to 2 N·m (20 in. lbs.).

(7) Reinstall the passenger airbag onto the instrument panel. (Refer to 8 - ELECTRICAL/RESTRAINTS/PASSENGER AIRBAG - INSTALLATION).

PASSENGER AIRBAG ON/OFF SWITCH

DESCRIPTION

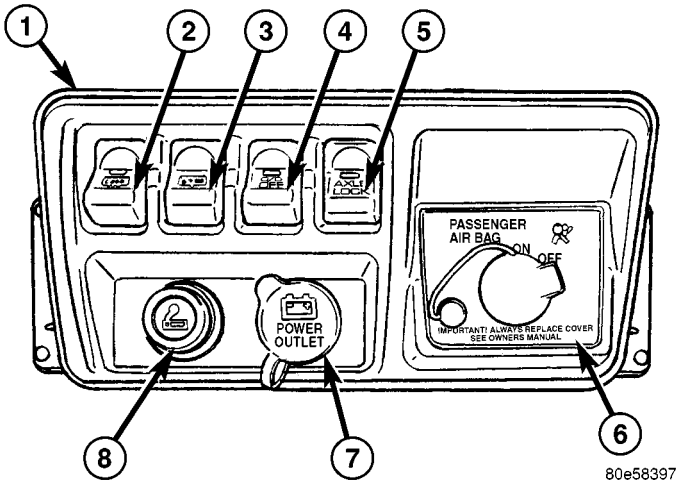

Fig. 35 Accessory Switch Bezel

1 - ACCESSORY SWITCH BEZEL
2 - REAR WINDOW DEFOGGER SWITCH (HARDTOP ONLY)
3 - REAR WIPER/WASHER SWITCH (HARDTOP ONLY)
4 - OVERDRIVE-OFF SWITCH (AUTOMATIC TRANSMISSION ONLY)
5 - AXLE LOCK SWITCH (OFF ROAD PACKAGE ONLY)
6 - PASSENGER AIRBAG ON/OFF SWITCH (WITHOUT REAR SEAT ONLY)
7 - ACCESSORY POWER OUTLET
8 - CIGAR LIGHTER

The passenger airbag on/off switch is standard equipment on all versions of this model that are equipped with the dual front airbag system, but are not equipped with a rear seat (Fig. 35). This switch is a two-position, resistor multiplexed switch with a single integral red Light-Emitting Diode (LED), and a non-coded key cylinder-type actuator. The switch is located on the passenger side end of the accessory switch bezel near the bottom of the instrument cluster center bezel to make the Off indicator visible to the front passenger seat occupant. When the switch is in its installed position, the only component visible through the dedicated opening of the accessory switch bezel is the key cylinder actuator. A small, tethered, molded plastic cap fits into the key cylinder actuator hole when the switch is not being used. The switch nomenclature and the Off indicator lens are integral to the accessory switch bezel. The text of the ″Off″ indicator is illuminated in amber when the Off position is selected with the ignition switch in the On position, while the On position of the switch is designated by text imprinted upon the accessory switch bezel, but is not illuminated. The remainder of the switch is concealed behind the accessory switch bezel.

The passenger airbag on/off switch housing is constructed of molded plastic and has three integral mounting tabs (Fig. 36). These mounting tabs are used to secure the switch to the back of the molded plastic accessory switch bezel with three small screws. The accessory switch bezel is secured to the instrument panel with four screws. A molded plastic connector receptacle on the back of the switch housing connects the switch to the vehicle electrical system through a dedicated take out of the body wire harness. The molded plastic harness connector insulator is keyed and latched to ensure proper and secure switch electrical connections. The passenger airbag on/off switch cannot be adjusted or repaired and, if faulty or damaged, the switch must be replaced.

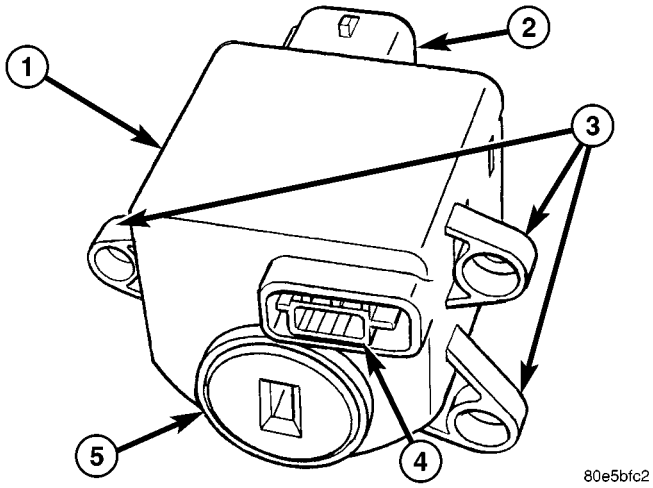

Fig. 36 Passenger Airbag On/Off Switch

1 - HOUSING
2 - CONNECTOR RECEPTACLE
3 - MOUNTING TAB (3)
4 - OFF INDICATOR
5 - KEY CYLINDER ACTUATOR

OPERATION

The passenger airbag on/off switch allows the customer to turn the passenger airbag function On or Off to accommodate certain uses of the right front seating position where airbag protection may not be desired. See the owner's manual in the vehicle glove box for specific recommendations on when to enable or disable the passenger airbag. The Off indicator of the switch will be illuminated whenever the switch is turned to the Off position and the ignition switch is in the On position.

The ignition key is the only key or object that should ever be inserted into the key cylinder actuator of the switch. The on/off switch requires only a partial key insertion to fully depress a spring-loaded locking plunger. The spring-loaded locking plunger prevents the user from leaving the key in the switch. The key will be automatically ejected when force is

PASSENGER AIRBAG ON/OFF SWITCH (Continued)

not applied. To actuate the passenger airbag on/off switch, insert the ignition key into the switch key actuator far enough to fully depress the plunger, then rotate the actuator to the desired switch position. When the switch key actuator is rotated to its clockwise stop (the key actuator slot will be aligned with the Off indicator), the Off indicator is illuminated and the passenger airbag is disabled. When the switch is rotated to its counterclockwise stop (the key actuator slot will be in a vertical position), the Off indicator will be extinguished and the passenger airbag is enabled.

The passenger airbag on/off switch connects one of two internal resistors in series between the passenger airbag mux switch sense and passenger airbag mux switch return circuits of the Airbag Control Module (ACM). The ACM continually monitors the resistance in these circuits to determine the switch position that has been selected. When the switch is in the Off position, the ACM provides a ground input to the switch through the passenger airbag indicator driver circuit, which energizes the Light-Emitting Diode (LED) that illuminates the Off indicator of the switch.

The ACM will also illuminate the Off indicator of the switch for about seven seconds each time the ignition switch is turned to the On position as a bulb test. The ACM will store a Diagnostic Trouble Code (DTC) for any fault it detects in the passenger airbag on/off switch or Off indicator circuits, and will illuminate the airbag indicator in the instrument cluster if a fault is detected. For proper diagnosis of the passenger airbag on/off switch or the ACM, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Wait two minutes for the system capacitor to discharge before further service.

(2) Remove the passenger airbag on/off switch and accessory switch bezel from the instrument panel as a unit. (Refer to 23 - BODY/INSTRUMENT PANEL/ ACCESSORY SWITCH BEZEL - REMOVAL).

(3) From the back of the accessory switch bezel, remove the three screws that secure the passenger airbag on/off switch to the back of the bezel (Fig. 37).

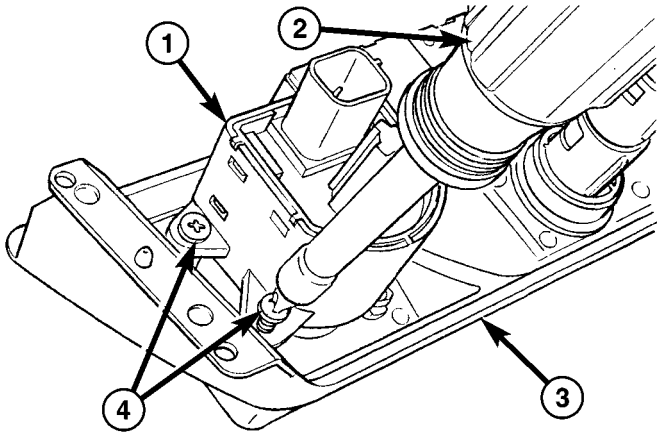

Fig. 37 Passenger Airbag On/Off Switch Remove/ Install

1 - PASSENGER AIRBAG ON/OFF SWITCH
2 - SCREW DRIVER
3 - ACCESSORY SWITCH BEZEL
4 - SCREW (3)

(4) Remove the passenger airbag on/off switch from the accessory switch bezel.

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Position the passenger airbag on/off switch to the back of the accessory switch bezel (Fig. 37).

(2) Install and tighten the three screws that secure the passenger airbag on/off switch to the back of the accessory switch bezel. Tighten the screws to 2 N·m (20 in. lbs.).

PASSENGER AIRBAG ON/OFF SWITCH (Continued)

(3) Reinstall the passenger airbag on/off switch and accessory switch bezel unit to the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ ACCESSORY SWITCH BEZEL - INSTALLATION).

(4) Do not reconnect the battery negative cable at this time. The supplemental restraint system verification test procedure should be performed following service of any supplemental restraint system component. (Refer to 8 - ELECTRICAL/RESTRAINTS - STANDARD PROCEDURE - VERIFICATION TEST).

REAR SEAT BELT & RETRACTOR

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Slide both rear seat belt buckle units between the seat cushion and seat back into the cargo area.

(2) Lift the rear seat back release lever and fold the rear seat back forward, then tumble the folded rear seat cushion and back unit forward against the backs of the two front bucket seats.

(3) Remove the screw that secures the lower sear belt anchor to the inner rear wheel house panel (Fig. 38).

(4) Remove the rear seat belt lower anchor from the inner rear wheel house panel.

(5) Unsnap and lift the trim cover from the rear seat belt turning loop to access the screw that secures the turning loop to the upper sport bar (Fig. 39).

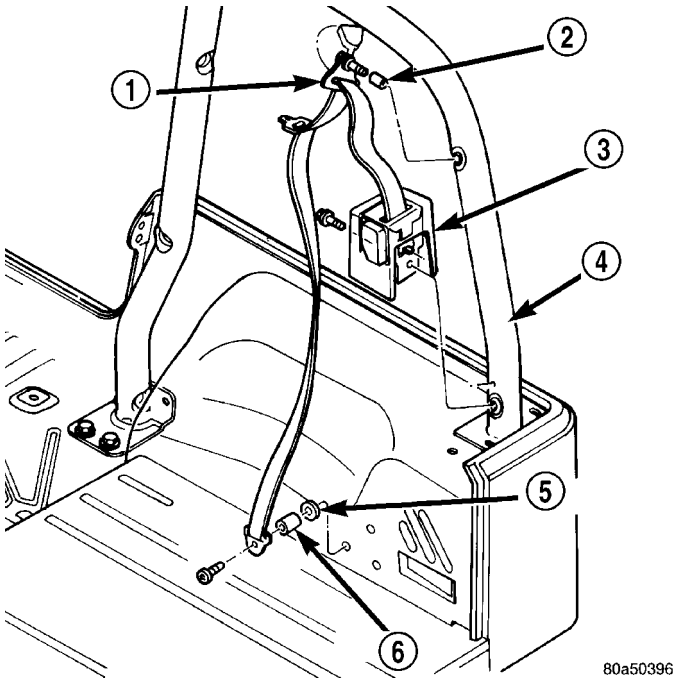

Fig. 38 Rear Seat Shoulder Belt & Retractor Remove/Install

1 - TURNING LOOP
2 - SPACER
3 - BELT & RETRACTOR
4 - SPORT BAR
5 - WASHER
6 - SPACER

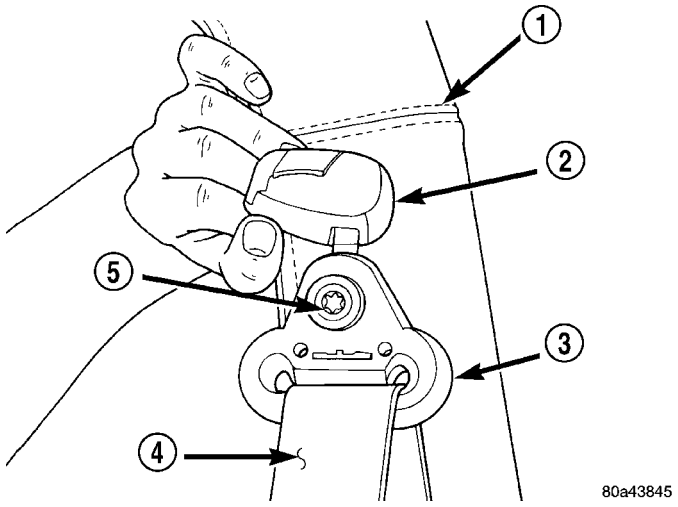

Fig. 39 Turning Loop

1 - SPORT BAR
2 - TURNING LOOP COVER
3 - TURNING LOOP
4 - REAR SEAT BELT
5 - SCREW

(6) Remove the screw that secures the seat belt turning loop to the upper sport bar.

(7) Remove the seat belt turning loop from the upper sport bar.

REAR SEAT BELT & RETRACTOR (Continued)

(8) Remove the screw that secures the rear seat belt retractor to the lower sport bar near the top of the inner rear wheel house panel.

(9) Remove the rear seat belt and retractor from the lower sport bar.

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Position the rear seat belt and retractor to the lower sport bar near the top of the inner rear wheel house panel (Fig. 38). Be certain to engage the locator tab on the retractor in the locator hole on the sport bar.

(2) Install and tighten the screw that secures the rear seat belt retractor to the lower sport bar. Tighten the screw to 43 N·m (32 ft. lbs.).

(3) Position the rear seat belt turning loop to the upper sport bar (Fig. 39).

(4) Install and tighten the screw that secures the rear seat belt turning loop to the upper sport bar. Tighten the screw to 43 N·m (32 ft. lbs.).

(5) Fold and snap the trim cover for the rear seat belt turning loop back into place over the screw that secures the turning loop to the upper sport bar.

(6) Position the rear seat belt lower anchor to the inner rear wheel house panel.

(7) Install and tighten the screw that secures the rear seat belt lower anchor to the inner rear wheel house panel. Tighten the screw to 43 N·m (32 ft. lbs.).

(8) Tumble the folded rear seat cushion and back unit rearward onto the rear floor panel.

(9) Slide both rear seat belt buckle units between the seat cushion and the seat back.

(10) Unfold the rear seat back from the seat cushion until the seat back latch is fully engaged.

REAR SEAT BELT BUCKLE

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Slide both rear seat belt buckle units between the seat cushion and seat back into the rear cargo area.

(2) Lift the rear seat back release lever and fold the rear seat back forward, then tumble the folded rear seat cushion and back unit forward against the backs of the two front bucket seats.

(3) Lift the rear cargo area carpet between the two rear seat belt buckle units far enough to access and remove the screw that secures the anchor of the rear seat belt buckle units to the rear cargo floor panel through one of the clearance slots in the carpet (Fig. 40).

(4) Remove the rear seat belt buckle unit from the rear floor panel through one of the clearance slots in the rear cargo area carpet.

REAR SEAT BELT BUCKLE (Continued)

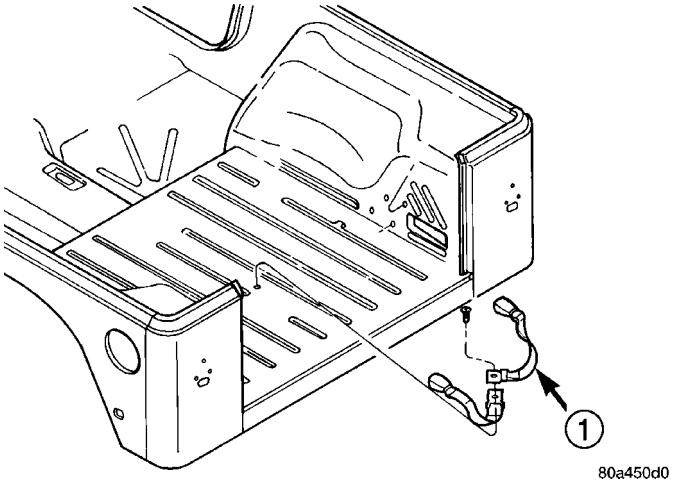

Fig. 40 Rear Seat Belt Buckles Remove/Install

1 - REAR SEAT BELT BUCKLE UNIT

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Lift the rear cargo area carpet far enough to position the rear seat belt buckle unit onto the rear cargo floor panel through one of the clearance slots in the carpet.

(2) Working through one of the clearance slots in the rear cargo area carpet, install and tighten the screw that secures the anchor of the rear seat belt buckle unit to the rear cargo floor panel (Fig. 40). Tighten the screw to 43 N·m (32 ft. lbs.).

(3) Tumble the folded rear seat cushion and back unit rearward onto the rear cargo floor panel.

(4) Slide both rear seat belt buckle units between the seat cushion and the seat back.

(5) Unfold the rear seat back from the seat cushion until the seat back latch is fully engaged.

SEAT BELT SWITCH

DESCRIPTION

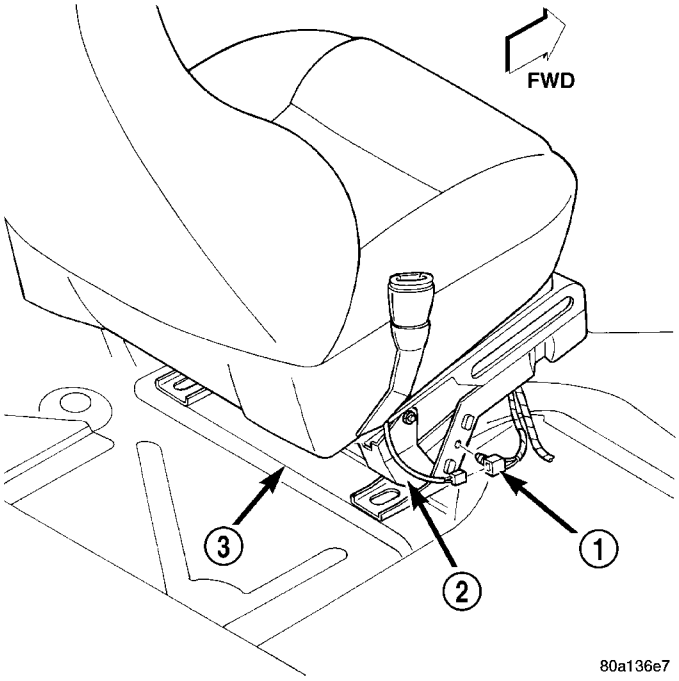

Fig. 41 Seat Belt Switch

1 - WIRE HARNESS CONNECTOR
2 - DRIVER SEAT RISER
4 - FLOOR PANEL

The seat belt switch is a small, normally closed, single pole, single throw, leaf contact, momentary switch. Only one seat belt switch is installed in the vehicle, and it is integral to the buckle of the driver side front seat belt buckle-half, located on the inboard side of the driver side front seat track (Fig. 41). The seat belt switch is connected to the vehicle electrical system through a two-wire pigtail wire and connector on the seat belt buckle-half, which is connected to a wire harness connector and take out of the body wire harness routed along the left side of the body sill in the passenger compartment.

The seat belt switch cannot be adjusted or repaired and, if faulty or damaged, the entire driver side front seat belt buckle-half unit must be replaced.

OPERATION

The seat belt switch is designed to control a path to ground for the seat belt switch sense input of the ElectroMechanical Instrument Cluster (EMIC). When

SEAT BELT SWITCH (Continued)

the driver side front seat belt tip-half is inserted into the seat belt buckle, the switch opens the path to ground; and, when the driver side front seat belt tip-half is removed from the seat belt buckle, the switch closes the ground path. The switch is actuated by the latch mechanism within the seat belt buckle.

The seat belt switch is connected in series between ground and the seat belt switch sense input of the instrument cluster. The seat belt switch receives ground at all times through its pigtail wire connection to the body wire harness from another take out of the body wire harness. An eyelet terminal connector on that ground take out is secured beneath a ground screw on the left cowl side inner panel, beneath the instrument panel. The seat belt switch may de diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - SEAT BELT SWITCH

Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

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(1) Disconnect and isolate the battery negative cable. Disconnect the seat belt switch pigtail wire connector from the body wire harness connector for the seat belt switch on the floor near the driver side front seat belt buckle-half anchor. Check for continuity between the seat belt switch sense circuit and the ground circuit cavities of the seat belt switch pigtail wire connector. There should be continuity with the seat belt unbuckled, and no continuity with the seat belt buckled. If OK, go to Step 2. If not OK, replace the faulty front seat belt buckle-half assembly.

(2) Check for continuity between the ground circuit cavity in the body wire harness connector for the seat belt switch and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open ground circuit to ground (G302) as required.

(3) Remove the instrument cluster from the instrument panel. Check for continuity between the seat belt switch sense circuit cavity of the body wire harness connector for the seat belt switch and a good ground. There should be no continuity. If OK, go to Step 4. If not OK, repair the shorted seat belt switch sense circuit between the seat belt switch and the instrument cluster as required.

(4) Check for continuity between the seat belt switch sense circuit cavities of the body wire harness connector for the seat belt switch and the instrument panel wire harness connector (Connector C2) for the instrument cluster. There should be continuity. If OK, test and replace the faulty instrument cluster as required. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If not OK, repair the open seat belt switch sense circuit between the seat belt switch and the instrument cluster as required.

SEAT BELT TURNING LOOP ADJUSTER

REMOVAL

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(1) Unsnap and lift the trim cover from the front seat belt turning loop to access the screw that

SEAT BELT TURNING LOOP ADJUSTER (Continued)

secures the turning loop to the height adjuster on the upper sport bar (Fig. 42).

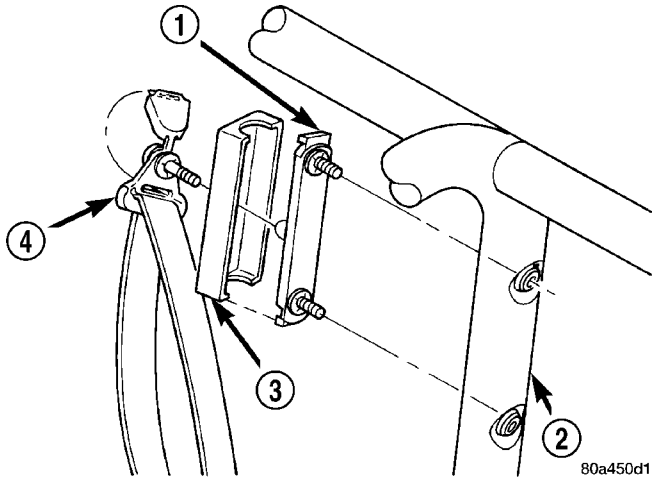

Fig. 42 Front Turning Loop Height Adjuster

1 - ADJUSTER
2 - SPORT BAR
3 - TRIM COVER
4 - TURNING LOOP

(2) Remove the screw that secures the shoulder belt turning loop to the height adjuster.

(3) Remove the front seat belt turning loop and the support/guide washer from the height adjuster.

(4) Unsnap and remove the trim cover from the front seat belt turning loop height adjuster to access the screws that secure the adjuster to the upper sport bar.

(5) Remove the two screws that secure the height adjuster to the upper sport bar.

(6) Remove the front seat belt turning loop height adjuster from the upper sport bar.

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, DURING AND FOLLOWING ANY SEAT BELT OR CHILD RESTRAINT ANCHOR SERVICE, CAREFULLY INSPECT ALL SEAT BELTS, BUCKLES, MOUNTING HARDWARE, RETRACTORS, TETHER STRAPS, AND ANCHORS FOR PROPER INSTALLATION, OPERATION, OR DAMAGE. REPLACE ANY BELT THAT IS CUT, FRAYED, OR TORN. STRAIGHTEN ANY BELT THAT IS TWISTED. TIGHTEN ANY LOOSE FASTENERS. REPLACE ANY BELT THAT HAS A DAMAGED OR INOPERATIVE BUCKLE OR RETRACTOR. REPLACE ANY BELT THAT HAS A BENT OR DAMAGED LATCH PLATE OR ANCHOR PLATE. REPLACE ANY CHILD RESTRAINT ANCHOR OR THE UNIT TO WHICH THE ANCHOR IS INTEGRAL THAT HAS BEEN BENT OR DAMAGED. NEVER ATTEMPT TO REPAIR A SEAT BELT OR CHILD RESTRAINT COMPONENT. ALWAYS REPLACE DAMAGED OR FAULTY SEAT BELT AND CHILD RESTRAINT COMPONENTS WITH THE CORRECT, NEW AND UNUSED REPLACEMENT PARTS LISTED IN THE DAIMLERCHRYSLER MOPAR PARTS CATALOG.

(1) Position the front seat belt turning loop height adjuster onto the upper sport bar (Fig. 42). Be certain that the word “Up” stamped on the adjuster is properly oriented.

(2) Install and tighten the two screws that secure the seat belt turning loop height adjuster to the upper sport bar. Tighten the screws to 43 N·m (32 ft. lbs.).

(3) Align the trim cover over the front seat belt turning loop height adjuster and, using hand pressure, press firmly and evenly on the cover until it snaps into place over the adjuster on the upper sport bar.

(4) Position the support/guide washer and the front seat belt turning loop onto the height adjuster.

(5) Install and tighten the screw that secures the front seat belt turning loop to the height adjuster. Tighten the screw to 43 N·m (32 ft. lbs.).

(6) Fold and snap the trim cover for the seat belt turning loop back into place over the screw that secures the turning loop to the height adjuster.

SPEED CONTROL

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SPEED CONTROL

DESCRIPTION

The speed control system is electronically controlled and vacuum operated. Electronic control of the speed control system is integrated into the Powertrain Control Module (PCM). The controls consist of two steering wheel mounted switches. The switches are labeled: ON/OFF, RES/ACCEL, SET, COAST, and CANCEL.

The system is designed to operate at speeds above 30 mph (50 km/h).

WARNING: THE USE OF SPEED CONTROL IS NOT RECOMMENDED WHEN DRIVING CONDITIONS DO NOT PERMIT MAINTAINING A CONSTANT SPEED, SUCH AS IN HEAVY TRAFFIC OR ON ROADS THAT ARE WINDING, ICY, SNOW COVERED, OR SLIPPERY.

OPERATION

When speed control is selected by depressing the ON switch, the PCM allows a set speed to be stored in PCM RAM for speed control. To store a set speed, depress the SET switch while the vehicle is moving at a speed between 35 and 85 mph. In order for the speed control to engage, the brakes cannot be applied, nor can the gear selector be indicating the transmission is in Park or Neutral.

The speed control can be disengaged manually by:

- Stepping on the brake pedal
- Depressing the OFF switch
- Depressing the CANCEL switch.
- Depressing the clutch pedal (if equipped).

NOTE: Depressing the OFF switch or turning off the ignition switch will erase the set speed stored in the PCM.

For added safety, the speed control system is programmed to disengage for any of the following conditions:

- An indication of Park or Neutral
- A rapid increase rpm (indicates that the clutch has been disengaged)
- Excessive engine rpm (indicates that the transmission may be in a low gear)
- The speed signal increases at a rate of 10 mph per second (indicates that the coefficient of friction between the road surface and tires is extremely low)
- The speed signal decreases at a rate of 10 mph per second (indicates that the vehicle may have decelerated at an extremely high rate)

Once the speed control has been disengaged, depressing the RES/ACCEL switch (when speed is greater than 30 mph) restores the vehicle to the target speed that was stored in the PCM.

While the speed control is engaged, the driver can increase the vehicle speed by depressing the RES/ACCEL switch. The new target speed is stored in the PCM when the RES/ACCEL is released. The PCM also has a ″tap-up″ feature in which vehicle speed increases at a rate of approximately 2 mph for each momentary switch activation of the RES/ACCEL switch.

SPEED CONTROL (Continued)

A “tap down” feature is used to decelerate without disengaging the speed control system. To decelerate from an existing recorded target speed, momentarily depress the COAST switch. For each switch activation, speed will be lowered approximately 1 mph.

OVERSHOOT/UNDERSHOOT

If the vehicle operator repeatedly presses and releases the SET button with their foot off of the accelerator (referred to as a “lift foot set”), the vehicle may accelerate and exceed the desired set speed by up to 5 mph (8 km/h). It may also decelerate to less than the desired set speed, before finally achieving the desired set speed.

The Speed Control System has an adaptive strategy that compensates for vehicle-to-vehicle variations in speed control cable lengths. When the speed control is set with the vehicle operators foot off of the accelerator pedal, the speed control thinks there is excessive speed control cable slack and adapts accordingly. If the “lift foot sets” are continually used, a speed control overshoot/undershoot condition will develop.

To “unlearn” the overshoot/undershoot condition, the vehicle operator has to press and release the set button while maintaining the desired set speed using the accelerator pedal (not decelerating or accelerating), and then turning the cruise control switch to the OFF position (or press the CANCEL button if equipped) after waiting 10 seconds. This procedure must be performed approximately 10–15 times to completely unlearn the overshoot/undershoot condition.

DIAGNOSIS AND TESTING - ROAD TEST

Perform a vehicle road test to verify reports of speed control system malfunction. The road test should include attention to the speedometer. Speedometer operation should be smooth and without flutter at all speeds.

Flutter in the speedometer indicates a problem which might cause surging in the speed control system. The cause of any speedometer problems should be corrected before proceeding. Refer to Group 8J, Instrument Cluster for speedometer diagnosis.

If a road test verifies a system problem and the speedometer operates properly, check for:

- A Diagnostic Trouble Code (DTC). If a DTC exists, conduct tests per the Powertrain Diagnostic Procedures service manual.
- A misadjusted brake (stop) lamp switch. This could also cause an intermittent problem.
- Loose, damaged or corroded electrical connections at the servo. Corrosion should be removed from electrical terminals and a light coating of Mopar MultiPurpose Grease, or equivalent, applied.
- Leaking vacuum reservoir.
- Loose or leaking vacuum hoses or connections.
- Defective one-way vacuum check valve.
- Secure attachment of both ends of the speed control servo cable.
- Smooth operation of throttle linkage and throttle body air valve.
- Failed speed control servo. Do the servo vacuum test.

CAUTION: When test probing for voltage or continuity at electrical connectors, care must be taken not to damage connector, terminals or seals. If these components are damaged, intermittent or complete system failure may occur.

SPECIFICATIONS

TORQUE - SPEED CONTROL

DESCRIPTION	N-m	Ft. Lbs.	In. Lbs.
Servo Mounting Bracket-to-Servo Nuts	8.5	-	75
Speed Control Switch Mounting Screws	1.5	-	14
Vacuum Reservoir Mounting Bolt (screw)	1.2	-	10

CABLE

DESCRIPTION

The speed control servo cable is connected between the speed control vacuum servo diaphragm and the throttle body control linkage.

OPERATION

This cable causes the throttle control linkage to open or close the throttle valve in response to movement of the vacuum servo diaphragm.

REMOVAL

2.4L

(1) Disconnect negative battery cable at battery.

(2) Hold throttle in wide open position. While held in this position, slide throttle control cable pin (Fig. 1) from throttle body bellcrank. Also slide servo cable pin from throttle body bellcrank.

(3) Using a pick or small screwdriver, press release tab (Fig. 1) to release plastic cable mount from bracket. **Press on tab only enough to release cable from bracket. If tab is pressed too much, it will be broken.** To remove throttle cable from throttle body bracket, slide cable towards front of vehicle.

(4) After removing throttle control cable at throttle body, slide servo cable from throttle body.

(5) If necessary, disconnect opposite end of servo cable at servo. Refer to Servo Removal/Installation.

4.0L

(1) Disconnect negative battery cable at battery.

(2) Using finger pressure only, remove cable connector by pushing connector off the throttle body bellcrank pin (Fig. 2). **DO NOT try to pull cable connector off perpendicular to the bellcrank pin. Connector will be broken.**

(3) Two release tabs are located on sides of speed control cable at cable bracket (Fig. 2). Squeeze tabs together and push cable out of bracket.

(4) Unclip cable from cable guide at valve cover.

(5) If necessary, disconnect opposite end of servo cable at servo. Refer to Servo Removal/Installation.

INSTALLATION

2.4L

(1) Install end of cable to speed control servo. Refer to Servo Removal/Installation.

(2) Slide speed control cable plastic mount into throttle body bracket.

(3) Install speed control cable connector onto throttle body bellcrank pin (push rearward to snap into location).

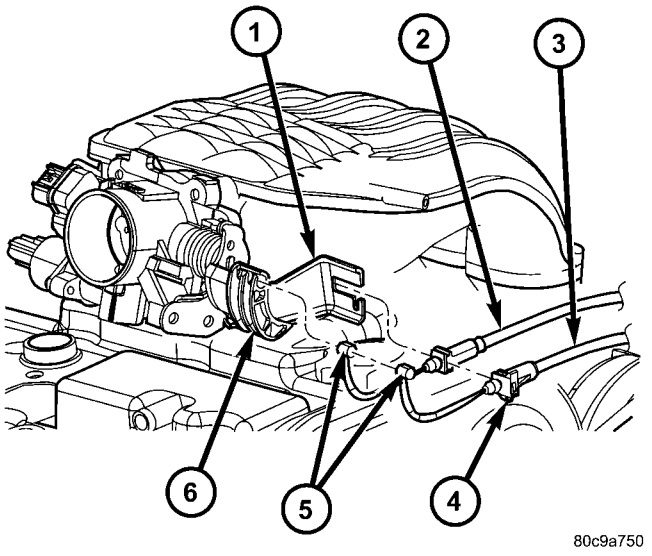

Fig. 1 SERVO CABLE AT THROTTLE BODY - 2.4L

1 - MOUNTING BRACKET
2 - SPEED CONTROL CABLE
3 - THROTTLE CABLE
4 - RELEASE TAB
5 - CABLE PINS
6 - BELLCRANK

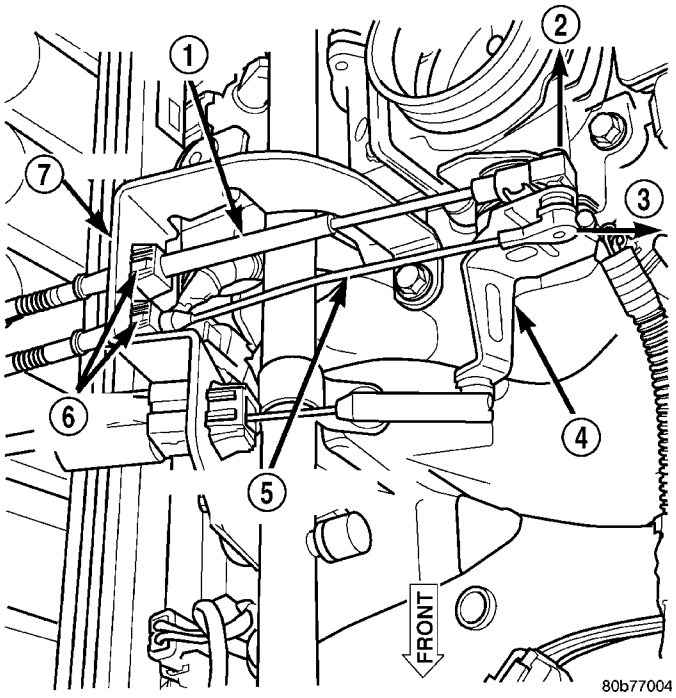

Fig. 2 SERVO CABLE AT THROTTLE BODY - 4.0L

1 - ACCELERATOR CABLE
2 - OFF
3 - OFF
4 - THROTTLE BODY BELLCRANK
5 - SPEED CONTROL CABLE
6 - RELEASE TABS
7 - BRACKET

CABLE (Continued)

(4) Slide throttle (accelerator) cable plastic mount into throttle body bracket. Continue sliding until cable release tab is aligned to hole in throttle body mounting bracket.

(5) While holding throttle to wide open position, place throttle cable pin into throttle body bellcrank.

(6) Connect negative battery cable at battery.

(7) Before starting engine, operate accelerator pedal to check for any binding.

4.0L

(1) Attach end of cable to speed control servo. Refer to Servo Removal/Installation.

(2) Install cable into cable bracket (snaps in).

(3) Install cable connector at throttle body bellcrank pin (snaps on).

(4) Clip cable to cable guide at valve cover.

(5) Connect negative battery cable to battery.

(6) Before starting engine, operate accelerator pedal to check for any binding.

SERVO

DESCRIPTION

The servo unit consists of a solenoid valve body, and a vacuum chamber. The solenoid valve body contains three solenoids:

- Vacuum
- Vent
- Dump

The vacuum chamber contains a diaphragm with a cable attached to control the throttle linkage.

OPERATION

The Powertrain Control Module (PCM) controls the solenoid valve body. The solenoid valve body controls the application and release of vacuum to the diaphragm of the vacuum servo. The servo unit cannot be repaired and is serviced only as a complete assembly.

Power is supplied to the servo's by the PCM through the brake switch. The PCM controls the ground path for the vacuum and vent solenoids.

The dump solenoid is energized anytime it receives power. If power to the dump solenoid is interrupted, the solenoid dumps vacuum in the servo. This provides a safety backup to the vent and vacuum solenoids.

The vacuum and vent solenoids must be grounded at the PCM to operate. When the PCM grounds the vacuum servo solenoid, the solenoid allows vacuum to enter the servo and pull open the throttle plate using the cable. When the PCM breaks the ground, the solenoid closes and no more vacuum is allowed to enter the servo. The PCM also operates the vent solenoid via ground. The vent solenoid opens and closes a passage to bleed or hold vacuum in the servo as required.

The PCM duty cycles the vacuum and vent solenoids to maintain the set speed, or to accelerate and decelerate the vehicle. To increase throttle opening, the PCM grounds the vacuum and vent solenoids. To decrease throttle opening, the PCM removes the grounds from the vacuum and vent solenoids. When the brake is released, if vehicle speed exceeds 30 mph to resume, 35 mph to set, and the RES/ACCEL switch has been depressed, ground for the vent and vacuum circuits is restored.

REMOVAL

(1) Disconnect negative battery cable at battery.

(2) Disconnect vacuum line at servo (Fig. 3).

(3) Disconnect electrical connector at servo.

(4) Disconnect servo cable at throttle body. Refer to Cable Removal/Installation.

(5) Remove three bracket mounting bolts (Fig. 3).

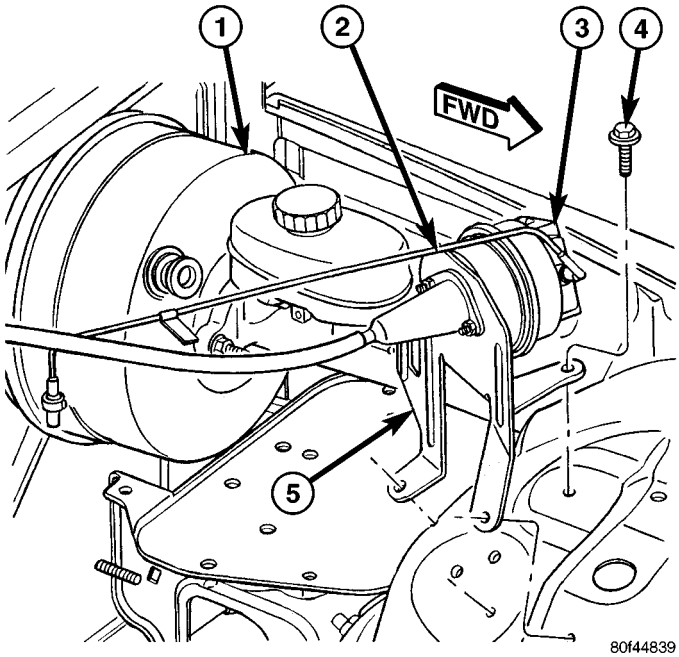

Fig. 3 SPEED CONTROL SERVO LOCATION

1 - BRAKE POWER BOOSTER
2 - VACUUM LINE
3 - SERVO
4 - BRACKET BOLTS (3)
5 - SERVO MOUNTING BRACKET

SERVO (Continued)

(6) Remove 2 mounting nuts holding servo cable sleeve to bracket (Fig. 4).

(7) Pull speed control cable sleeve and servo away from servo mounting bracket to expose cable retaining clip (Fig. 4) and remove clip. Note: The servo mounting bracket displayed in (Fig. 4) is a typical bracket and may/may not be applicable to this model vehicle.

(8) Remove servo from mounting bracket. While removing, note orientation of servo to bracket.

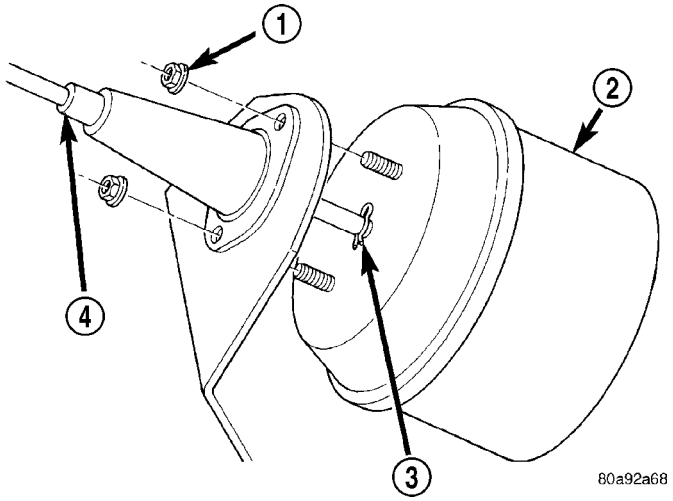

Fig. 4 SERVO CABLE CLIP REMOVE/INSTALL - TYPICAL

1 - SERVO MOUNTING NUTS (2)
2 - SERVO
3 - CABLE RETAINING CLIP
4 - SERVO CABLE AND SLEEVE

INSTALLATION

(1) Position servo to mounting bracket.

(2) Align hole in cable connector with hole in servo pin. Install cable-to-servo retaining clip.

(3) Insert servo mounting studs through holes in servo mounting bracket.

(4) Install servo mounting nuts and tighten to 8.5 N·m (75 in. lbs.).

(5) Connect vacuum line at servo.

(6) Connect electrical connector at servo.

(7) Connect servo cable to throttle body. Refer to Cable Removal/Installation.

(8) Connect negative battery cable to battery.

(9) Before starting engine, operate accelerator pedal to check for any binding.

SWITCH

DESCRIPTION

There are two separate switch pods that operate the speed control system. The steering-wheel-mounted switches use multiplexed circuits to provide inputs to the PCM for ON, OFF, RESUME, ACCELERATE, SET, DECEL and CANCEL modes. Refer to the owner's manual for more information on speed control switch functions and setting procedures.

The individual switches cannot be repaired. If one switch fails, the entire switch module must be replaced.

OPERATION

When speed control is selected by depressing the ON, OFF switch, the PCM allows a set speed to be stored in its RAM for speed control. To store a set speed, depress the SET switch while the vehicle is moving at a speed between approximately 35 and 85 mph. In order for the speed control to engage, the brakes cannot be applied, nor can the gear selector be indicating the transmission is in Park or Neutral.

The speed control can be disengaged manually by:

- Stepping on the brake pedal
- Depressing the OFF switch
- Depressing the CANCEL switch.

The speed control can be disengaged also by any of the following conditions:

- An indication of Park or Neutral
- The VSS signal increases at a rate of 10 mph per second (indicates that the co-efficient of friction between the road surface and tires is extremely low)
- Depressing the clutch pedal.
- Excessive engine rpm (indicates that the transmission may be in a low gear)
- The VSS signal decreases at a rate of 10 mph per second (indicates that the vehicle may have decelerated at an extremely high rate)
- If the actual speed is not within 20 mph of the set speed

The previous disengagement conditions are programmed for added safety.

Once the speed control has been disengaged, depressing the ACCEL switch restores the vehicle to the target speed that was stored in the PCM's RAM.

SWITCH (Continued)

NOTE: Depressing the OFF switch will erase the set speed stored in the PCM's RAM.

If, while the speed control is engaged, the driver wishes to increase vehicle speed, the PCM is programmed for an acceleration feature. With the ACCEL switch held closed, the vehicle accelerates slowly to the desired speed. The new target speed is stored in the PCM's RAM when the ACCEL switch is released. The PCM also has a ″tap-up″ feature in which vehicle speed increases at a rate of approximately 2 mph for each momentary switch activation of the ACCEL switch.

The PCM also provides a means to decelerate without disengaging speed control. To decelerate from an existing recorded target speed, depress and hold the COAST switch until the desired speed is reached. Then release the switch. The ON, OFF switch operates two components: the PCM's ON, OFF input, and the battery voltage to the brake switch, which powers the speed control servo.

Multiplexing

The PCM sends out 5 volts through a fixed resistor and monitors the voltage change between the fixed resistor and the switches. If none of the switches are depressed, the PCM will measure 5 volts at the sensor point (open circuit). If a switch with no resistor is closed, the PCM will measure 0 volts (grounded circuit). Now, if a resistor is added to a switch, then the PCM will measure some voltage proportional to the size of the resistor. By adding a different resistor to each switch, the PCM will see a different voltage depending on which switch is pushed.

Another resistor has been added to the 'at rest circuit' causing the PCM to never see 5 volts. This was done for diagnostic purposes. If the switch circuit should open (bad connection), then the PCM will see the 5 volts and know the circuit is bad. The PCM will then set an open circuit fault.

REMOVAL

WARNING: BEFORE ATTEMPTING TO DIAGNOSE, REMOVE OR INSTALL ANY AIRBAG SYSTEM OR RELATED STEERING WHEEL AND STEERING COLUMN COMPONENTS, YOU MUST FIRST DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE. WAIT 2 MINUTES FOR SYSTEM CAPACITOR TO DISCHARGE BEFORE FURTHER SYSTEM SERVICE. FAILURE TO DO SO COULD RESULT IN ACCIDENTAL DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

(1) Disconnect and isolate negative battery cable from battery.

(2) Remove airbag module. Refer to 8, Passive Restraint Systems.

(3) From underside of steering wheel, remove speed control switch mounting screw (Fig. 5).

(4) Remove switch from steering wheel and unplug electrical connector.

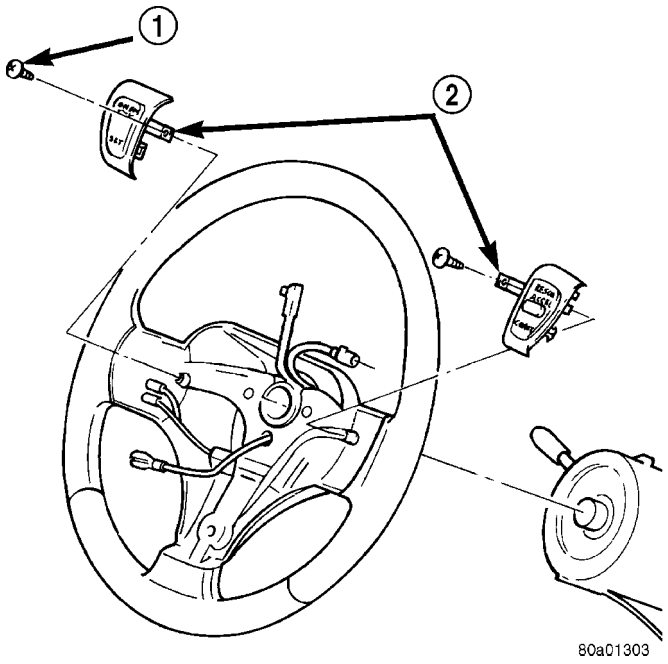

Fig. 5 Speed Control Switches - Remove / Install

1 - MOUNTING SCREW
2 - SPEED CONTROL SWITCHES

INSTALLATION

(1) Plug electrical connector into switch.

(2) Position switch to steering wheel.

(3) Install switch mounting screw and tighten to 1.5 N·m (14 in. lbs.) torque.

(4) Install airbag module. Refer to 8, Passive Restraint Systems.

(5) Connect negative battery cable to battery.

VACUUM RESERVOIR

DESCRIPTION

The vacuum reservoir is a plastic storage tank connected to an engine vacuum source by vacuum lines.

OPERATION

The vacuum reservoir is used to supply the vacuum needed to maintain proper speed control operation when engine vacuum drops, such as in climbing a grade while driving. A one-way check valve is used in the vacuum line between the reservoir and the vacuum source. This check valve is used to trap engine vacuum in the reservoir. On certain vehicle applications, this reservoir is shared with the heating/air-conditioning system. The vacuum reservoir cannot be repaired and must be replaced if faulty.

DIAGNOSIS AND TESTING - VACUUM RESERVOIR

(1) Disconnect vacuum hose at speed control servo and install a vacuum gauge into the disconnected hose.

(2) Start engine and observe gauge at idle. Vacuum gauge should read at least ten inches of mercury.

(3) If vacuum is less than ten inches of mercury, determine source of leak. Check vacuum line to engine for leaks. Also check actual engine intake manifold vacuum. If manifold vacuum does not meet this requirement, check for poor engine performance and repair as necessary.

(4) If vacuum line to engine is not leaking, check for leak at vacuum reservoir. To locate and gain access to reservoir, refer to Vacuum Reservoir Removal/Installation in this group. Disconnect vacuum line at reservoir and connect a hand-operated vacuum pump to reservoir fitting. Apply vacuum. Reservoir vacuum should not bleed off. If vacuum is being lost, replace reservoir.

(5) Verify operation of one-way check valve and check it for leaks.

(a) Locate one-way check valve. The valve is located in vacuum line between vacuum reservoir and engine vacuum source. Disconnect vacuum hoses (lines) at each end of valve.

(b) Connect a hand-operated vacuum pump to reservoir end of check valve. Apply vacuum. Vacuum should not bleed off. If vacuum is being lost, replace one-way check valve.

(c) Connect a hand-operated vacuum pump to vacuum source end of check valve. Apply vacuum. Vacuum should flow through valve. If vacuum is not flowing, replace one-way check valve. Seal the fitting at opposite end of valve with a finger and apply vacuum. If vacuum will not hold, diaphragm within check valve has ruptured. Replace valve.

REMOVAL

The vacuum reservoir is located under the vehicle battery tray (Fig. 6).

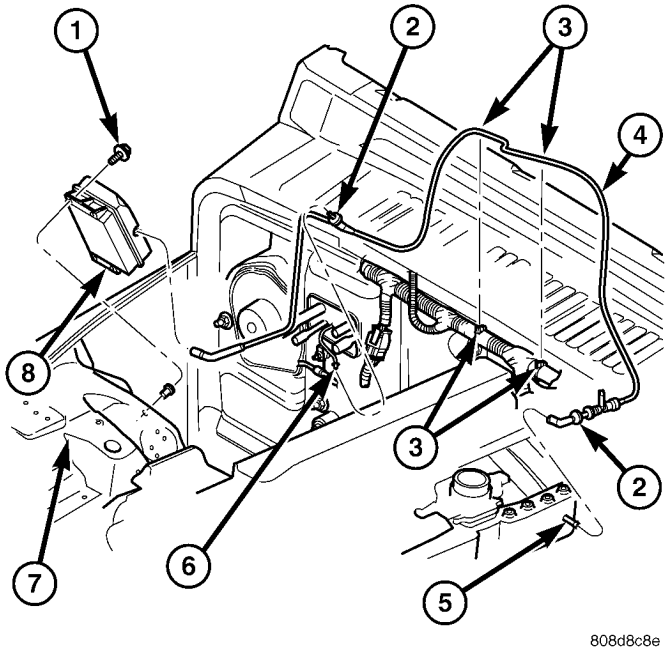

Fig. 6 Vacuum Reservoir Removal/Installation

1 - SCREW
2 - VACUUM CHECK VALVE (2)
3 - VACUUM LINE RETAINER (2)
4 - ACCESSORY VACUUM SUPPLY LINE
5 - ENGINE VACUUM FITTING
6 - HVAC VACUUM SUPPLY LINE
7 - INNER FENDER
8 - VACUUM RESERVOIR

(1) Remove battery and battery tray. Refer to Battery Removal/Installation.

(2) Disconnect vacuum supply line at reservoir (Fig. 6).

(3) Remove screw securing reservoir to inner fender.

(4) Remove reservoir from vehicle.

INSTALLATION

The vacuum reservoir is located under the vehicle battery tray (Fig. 6).

(1) Position reservoir to vehicle and install mounting screw.

(2) Tighten screw to 1.2 N·m (10 in. lbs.) torque.

(3) Connect vacuum line to reservoir.

(4) Install battery and battery tray. Refer to Battery Removal/Installation.

VEHICLE THEFT SECURITY

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VEHICLE THEFT SECURITY

DESCRIPTION

The Sentry Key Immobilizer System (SKIS) is available as a factory-installed option on this model. Vehicles equipped with this option can be readily identified by the presence of an amber SKIS indicator in the instrument cluster that will illuminate for about three seconds each time the ignition switch is turned to the On position, or by a gray molded rubber cap on the head of the ignition key. Models not equipped with SKIS still have a SKIS indicator in the cluster, but it will not illuminate when the ignition switch is turned to the On position. Also, models not equipped with the SKIS have a black molded rubber cap on the head of the ignition key.

The SKIS includes the following major components, which are described in further detail elsewhere in this service manual:

- **Powertrain Control Module (PCM)** - The PCM is located on the right side of the dash panel in the engine compartment.
- **Sentry Key Immobilizer Module (SKIM)** - The SKIM is located on the steering column near the ignition lock cylinder housing and an integral molded plastic antenna ring circles the ignition lock cylinder like a halo. The SKIM and its antenna are concealed beneath the steering column shrouds.
- **Sentry Key Transponder** - The Sentry Key transponder is molded into the head of the ignition key, and concealed by a gray molded rubber cap.
- **SKIS Indicator** - The SKIS indicator is located in the upper left corner of the instrument cluster overlay.

Except for the Sentry Key transponders, which rely upon Radio Frequency (RF) communication, hard wired circuitry connects the SKIS components to the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system and to the SKIS components through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

OPERATION

The Sentry Key Immobilizer System (SKIS) is designed to provide passive protection against unauthorized vehicle use by disabling the engine, after about two seconds of running, whenever any method other than a valid Sentry Key is used to start the vehicle. The SKIS is considered a passive protection system because it is always active when the ignition system is energized and does not require any customer intervention. The SKIS uses Radio Frequency (RF) communication to obtain confirmation that the key in the ignition switch is a valid key for operating the vehicle. The microprocessor-based SKIS hardware and software also uses electronic messages to communicate with other electronic modules in the vehicle over the Programmable Communications Interface (PCI) data bus (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/COMMUNICATION - OPERATION).

VEHICLE THEFT SECURITY (Continued)

Pre-programmed Sentry Key transponders are provided with the vehicle from the factory. Each Sentry Key Immobilizer Module (SKIM) will recognize a maximum of eight Sentry Keys. If the customer would like additional keys other than those provided with the vehicle, they may be purchased from any authorized dealer. These additional keys must be programmed to the SKIM in the vehicle in order for the system to recognize them as valid keys. This can be done by the dealer using a DRBIII® scan tool or, if Customer Learn programming is an available SKIS feature in the market where the vehicle was purchased, the customer can program the additional keys, as long as at least two valid Sentry Keys are already available. (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - STANDARD PROCEDURE - SENTRY KEY TRANSPONDER PROGRAMMING).

The SKIS performs a self-test each time the ignition switch is turned to the On position, and will store fault information in the form of Diagnostic Trouble Codes (DTC's) if a system malfunction is detected. The SKIS can be diagnosed, and any stored DTC's can be retrieved using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

DIAGNOSIS AND TESTING - SENTRY KEY IMMOBILIZER SYSTEM

WARNING: ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE AIRBAG SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE AIRBAG SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE AIRBAG SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT AND POSSIBLE PERSONAL INJURY.

SENTRY KEY IMMOBILIZER SYSTEM DIAGNOSIS		
CONDITION	POSSIBLE CAUSES	CORRECTION
SKIS INDICATOR FAILS TO LIGHT DURING BULB TEST	1. Indicator faulty.	1. Test and replace the faulty instrument cluster or bulb as required.
	2. Fuse faulty.	2. Test and replace the SKIM fused B(+) and fused ignition switch output (run-start) fuses in the fuse block as required.
	3. Ground circuit faulty.	3. Test and repair the SKIM ground circuit as required.
	4. Fused B(+) circuit faulty.	4. Test and repair the SKIM fused B(+) circuit as required.
	5. Fised ignition switch output circuit faulty.	5. Terst and repair the SKIM fused ignition switch output (run-start) circuit as required.
SKIS INDICATOR FLASHES WHEN IGNITION SWITCH IS TURNED TO THE ″ON″ POSITION	1. Invalid key in ignition switch lock cylinder.	1. Replace the key with a known valid key.
	2. Key-related fault.	2. Use a DRBIII® scan tool to diagnose the key-related fault. Refer to the appropriate diagnostic information.
SKIS INDICATOR LIGHTS SOLID FOLLOWING BULB TEST	1. SKIS system malfunction/ fault detected.	1. Use a DRBIII® scan tool to diagnose the SKIS. Refer to the appropriate diagnostic information.

VEHICLE THEFT SECURITY (Continued)

SKIS INDICATOR FAILS TO LIGHT DURING BULB TEST

If the Sentry Key Immobilizer System (SKIS) indicator in the instrument cluster fails to illuminate for about three seconds after the ignition switch is turned to the On position (bulb test), perform the instrument cluster actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING - ACTUATOR TEST). If the SKIS indicator still fails to light during the bulb test, a wiring problem resulting in the loss of battery current or ground to the Sentry Key Immobilizer Module (SKIM) should be suspected, and the following procedure should be used for diagnosis. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

NOTE: The following tests may not prove conclusive in the diagnosis of this system. The most reliable, efficient, and accurate means to diagnose the Sentry Key Immobilizer System requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.

(1) Check the B(+) fuse (Fuse 1 - 20 ampere) in the fuse block. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Check for battery voltage at the B(+) fuse (Fuse 1 - 20 ampere) in the fuse block. If OK, go to Step 3. If not OK, repair the open B(+) circuit between the fuse block and the battery as required.

(3) Check the ignition switch output (run-start) fuse (Fuse 12 - 10 ampere) in the fuse block. If OK, go to Step 4. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(4) Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-start) fuse (Fuse 12 - 10 ampere) in the fuse block. If OK, go to Step 5. If not OK, repair the open fused ignition switch output (run-start) circuit between the fuse block and the ignition switch as required.

(5) Disconnect and isolate the battery negative cable. Disconnect the instrument panel wire harness connector for the Sentry Key Immobilizer Module (SKIM) from the SKIM connector receptacle. Check for continuity between each of the two ground circuit cavities of the instrument panel wire harness connector for the SKIM and a good ground. There should be continuity. If OK, go to Step 6. If not OK, repair the open ground circuit(s) to ground (G105 and G302) as required.

(6) Reconnect the battery negative cable. Check for battery voltage at the fused B(+) circuit cavity of the instrument panel wire harness connector for the SKIM. If OK, go to Step 7. If not OK, repair the open fused B(+) circuit between the SKIM and the fuse block as required.

(7) Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-start) circuit cavity of the instrument panel wire harness connector for the SKIM. If OK, use a DRBIII® scan tool to complete the diagnosis of the SKIS. Refer to the appropriate diagnostic information. If not OK, repair the open fused ignition switch output (run-start) circuit between the SKIM and the fuse block as required.

SKIS INDICATOR FLASHES OR LIGHTS SOLID FOLLOWING BULB TEST

A SKIS indicator that flashes following a successful bulb test indicates that an invalid key has been detected, or that a key-related fault has been set. A SKIS indicator that lights solid following a successful bulb test indicates that the SKIM has detected a system malfunction or that the SKIS is inoperative. In either case, fault information will be stored in the SKIM memory. For retrieval of this fault information and further diagnosis of the SKIS, the PCI data bus, the SKIM message outputs to the instrument cluster that control the SKIS indicator and/or chime service, or the message inputs and outputs between the SKIM and the Powertrain Control Module (PCM) that control engine operation, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information. Following are preliminary troubleshooting guidelines to be followed during diagnosis using a DRBIII® scan tool:

(1) Using the DRBIII® scan tool, read and record the faults as they exist in the SKIM when you first begin your diagnosis of the vehicle. It is important to document these faults because the SKIM does not differentiate between historical faults (those that have occurred in the past) and active faults (those that are currently present). If this problem turns out to be an intermittent condition, this information may become invaluable to your diagnosis.

(2) Using the DRBIII® scan tool, erase all of the faults from the SKIM.

(3) Cycle the ignition switch to the Off position, then back to the On position.

(4) Using the DRBIII® scan tool, read any faults that are now present in the SKIM. These are the active faults.

(5) Using this active fault information, refer to the proper procedure in the appropriate diagnostic information for the additional specific diagnostic steps.

VEHICLE THEFT SECURITY (Continued)

STANDARD PROCEDURE

STANDARD PROCEDURE - SKIS INITIALIZATION

The Sentry Key Immobilizer System (SKIS) must be initialized following a Sentry Key Immobilizer Module (SKIM) replacement. SKIS initialization requires the use of a DRBIII® scan tool. Initialization will also require that you have access to the unique four-digit PIN code that was assigned to the original SKIM. The PIN code **must** be used to enter the Secured Access Mode in the SKIM. This PIN number may be obtained from the vehicle owner, from the original vehicle invoice, or from the DaimlerChrysler Customer Center. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES - STANDARD PROCEDURE - PCM/SKIM PROGRAMMING).

NOTE: If a Powertrain Control Module (PCM) is replaced on a vehicle equipped with the Sentry Key Immobilizer System (SKIS), the unique Secret Key data must be transferred from the Sentry Key Immobilizer Module (SKIM) to the new PCM using the PCM replacement procedure. This procedure also requires the use of a DRBIII® scan tool and the unique four-digit PIN code to enter the Secured Access Mode in the SKIM. Refer to the appropriate diagnostic information for the proper PCM replacement procedures.

STANDARD PROCEDURE - SENTRY KEY TRANSPONDER PROGRAMMING

All Sentry Keys included with the vehicle are preprogrammed to work with the Sentry Key Immobilizer System (SKIS) when it is shipped from the factory. The Sentry Key Immobilizer Module (SKIM) can be programmed to recognize up to a total of eight Sentry Keys. When programming a blank Sentry Key transponder, the key must first be cut to match the ignition switch lock cylinder in the vehicle for which it will be used. Once the additional or new key has been cut, the SKIM must be programmed to recognize it as a valid key. There are two possible methods to program the SKIM to recognize a new or additional valid key, the Secured Access Method and the Customer Learn Method. Following are the details of these two programming methods.

SECURED ACCESS METHOD

The Secured Access method applies to all vehicles. This method requires the use of a DRBIII® scan tool. This method will also require that you have access to the unique four-digit PIN code that was assigned to the original SKIM. The PIN code **must** be used to enter the Secured Access Mode in the SKIM. This PIN number may be obtained from the vehicle owner, from the original vehicle invoice, or from the DaimlerChrysler Customer Center. Refer to the appropriate diagnostic information for the proper Secured Access method programming procedures.

CUSTOMER LEARN METHOD

The Customer Learn feature is only available on domestic vehicles, or those vehicles which have a U.S. country code designator. This programming method also requires access to at least two valid Sentry Keys. If two valid Sentry Keys are not available, or if the vehicle does not have a U.S. country code designator, the Secured Access Method **must** be used to program new or additional valid keys to the SKIM. The Customer Learn programming method procedures are as follows:

(1) Obtain the blank Sentry Key(s) that are to be programmed as valid keys for the vehicle. Cut the blank key(s) to match the ignition switch lock cylinder mechanical key codes.

(2) Insert one of the two valid Sentry Keys into the ignition switch and turn the ignition switch to the On position.

(3) After the ignition switch has been in the On position for longer than three seconds, but no more than fifteen seconds, cycle the ignition switch back to the Off position. Replace the first valid Sentry Key in the ignition switch lock cylinder with the second valid Sentry Key and turn the ignition switch back to the On position. The second valid Sentry Key must be inserted in the lock cylinder within fifteen seconds of removing the first valid key.

(4) About ten seconds after the completion of Step 3, the SKIS indicator in the instrument cluster will start to flash and a single audible chime will sound to indicate that the system has entered the Customer Learn programming mode.

(5) Within sixty seconds of entering the Customer Learn programming mode, turn the ignition switch to the Off position, replace the valid Sentry Key with a blank Sentry Key transponder, and turn the ignition switch back to the On position.

(6) About ten seconds after the completion of Step 5, a single audible chime will sound and the SKIS indicator will stop flashing, stay on solid for three seconds, then turn off to indicate that the blank Sentry Key has been successfully programmed. The SKIS will immediately exit the Customer Learn programming mode and the vehicle may now be started using the newly programmed valid Sentry Key.

VEHICLE THEFT SECURITY (Continued)

Each of these steps must be repeated and completed in their entirety for each additional Sentry Key that is to be programmed. If the above steps are not completed in the given sequence, or within the allotted time, the SKIS will exit the Customer Learn programming mode and the programming will be unsuccessful. The SKIS will also automatically exit the Customer Learn programming mode if it sees a non-blank Sentry Key transponder when it should see a blank, if it has already programmed eight (8) valid Sentry Keys, or if the ignition switch is turned to the Off position for more than about fifty seconds.

NOTE: If an attempt is made to start the vehicle while in the Customer Learn mode (SKIS indicator flashing), the SKIS will respond as though the vehicle were being started with an invalid key. In other words, the engine will stall after about two seconds of operation. No faults will be set.

NOTE: Once a Sentry Key has been programmed as a valid key to a vehicle, it cannot be programmed as a valid key for use on any other vehicle.

SKIS INDICATOR

DESCRIPTION

A Sentry Key Immobilizer System (SKIS) indicator is standard equipment on all instrument clusters, but is only operational on vehicles equipped with the optional SKIS. The SKIS indicator is located near the upper edge of the instrument cluster overlay, in the upper left quadrant of the cluster. The SKIS indicator consists of a stencil-like cutout of a graphical representation or icon of a key that is circled and crossed-out in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber lens behind the cutout in the opaque layer of the overlay causes the indicator to appear in amber through the translucent outer layer of the overlay when it is illuminated from behind by a Light Emitting Diode (LED) soldered onto the instrument cluster electronic circuit board. The SKIS indicator is serviced as a unit with the instrument cluster.

OPERATION

The Sentry Key Immobilizer System (SKIS) indicator gives an indication to the vehicle operator of the status of the SKIS. This indicator is controlled by the instrument cluster electronic circuit board based upon electronic messages received by the cluster from the Sentry Key Immobilizer Module (SKIM) over the Programmable Communications Interface (PCI) data bus. The SKIS indicator Light Emitting Diode (LED) receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions; therefore, the indicator will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is switched to ground by the instrument cluster transistor. The instrument cluster will turn on the SKIS indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position, the SKIM tells the cluster to illuminate the SKIS indicator for about three seconds as a bulb test.
- **SKIM Lamp-On Message** - Each time the cluster receives a lamp-on message from the SKIM, the SKIS indicator will be illuminated. The indicator can be flashed on and off, or illuminated solid, as dictated by the SKIM message. For more information on the SKIS and the SKIS indicator control parameters, (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - OPERATION). The indicator remains illuminated until the cluster receives a lamp-off message from the SKIM or until the ignition switch is turned to the Off position, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the SKIS indicator will be turned on, then off again in a prescribed sequence to confirm the functionality of the LED and the cluster control circuitry.

The SKIM performs a self-test each time the ignition switch is turned to the On position to decide whether the system is in good operating condition. The SKIM then sends the proper SKIS lamp-on or lamp-off messages to the instrument cluster. For further diagnosis of the SKIS indicator or the instrument cluster circuitry that controls the indicator, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the SKIS indicator after the bulb test, either solid or flashing, it indicates that a SKIS malfunction has occurred or that the SKIS is inoperative. For proper diagnosis of the SKIS, the PCI data bus, or the message inputs to the instrument cluster that control the SKIS indicator, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

TRANSPONDER KEY

DESCRIPTION

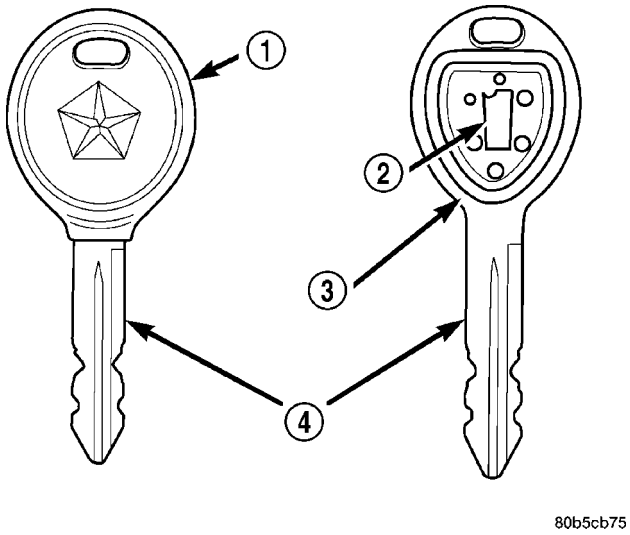

Fig. 1 Sentry Key Immobilizer Transponder

1 - MOLDED CAP
2 - TRANSPONDER CHIP
3 - MOLDED CAP REMOVED
4 - TRANSPONDER KEY

Each ignition key used in the Sentry Key Immobilizer System (SKIS) has an integral transponder chip (Fig. 1). Ignition keys with this feature can be readily identified by a gray rubber cap molded onto the head of the key, while conventional ignition keys have a black molded rubber cap. The transponder chip is concealed beneath the molded rubber cap, where it is molded within a plastic mount into the head of the metal key. In addition to being cut to match the mechanical coding of the ignition lock cylinder, each new Sentry Key has a unique transponder identification code permanently programmed into it by the manufacturer. The Sentry Key transponder cannot be adjusted or repaired. If faulty or damaged, the entire key must be replaced.

OPERATION

When the ignition switch is turned to the On position, the Sentry Key Immobilizer Module (SKIM) communicates through its antenna with the Sentry Key transponder using a Radio Frequency (RF) signal. The SKIM then listens for a RF response from the transponder through the same antenna. The Sentry Key transponder chip is within the range of the SKIM transceiver antenna ring when it is inserted into the ignition lock cylinder. The SKIM determines whether a valid key is present in the ignition lock cylinder based upon the response from the transponder. If a valid key is detected, that fact is communicated by the SKIM to the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus, and the PCM allows the engine to continue running. If the PCM receives an invalid key message, or receives no message from the SKIM over the PCI data bus, the engine will be disabled after about two seconds of operation. The ElectroMechanical Instrument Cluster (EMIC) will also respond to the invalid key message on the PCI data bus by flashing the SKIS indicator on and off.

Each Sentry Key has a unique transponder identification code permanently programmed into it by the manufacturer. Likewise, the SKIM has a unique Secret Key code programmed into it by the manufacturer. When a Sentry Key is programmed into the memory of the SKIM, the SKIM stores the transponder identification code from the Sentry Key, and the Sentry Key learns the Secret Key code from the SKIM. Once the Sentry Key learns the Secret Key code of the SKIM, it is permanently stored in the memory of the transponder. Therefore, once a Sentry Key has been programmed to a particular vehicle, it cannot be used on any other vehicle. (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - STANDARD PROCEDURE - SENTRY KEY TRANSPONDER PROGRAMMING).

The SKIS performs a self-test each time the ignition switch is turned to the On position, and will store key-related fault information in the form of Diagnostic Trouble Codes (DTC's) in SKIM memory if a Sentry Key transponder problem is detected. The Sentry Key transponder chip can be diagnosed, and any stored DTC's can be retrieved using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

WIPERS/WASHERS

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FRONT WIPERS/WASHERS

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FRONT WIPERS/WASHERS

DESCRIPTION

An electrically operated intermittent front wiper and washer system is standard factory-installed safety equipment on this model (Fig. 1). The front wiper and washer system includes the following major components, which are described in further detail elsewhere in this service information:

- **Front Check Valve** - The front washer system check valve is located in the washer plumbing on the underside of the hood panel between the cowl and the front washer nozzle.

FRONT WIPERS/WASHERS (Continued)

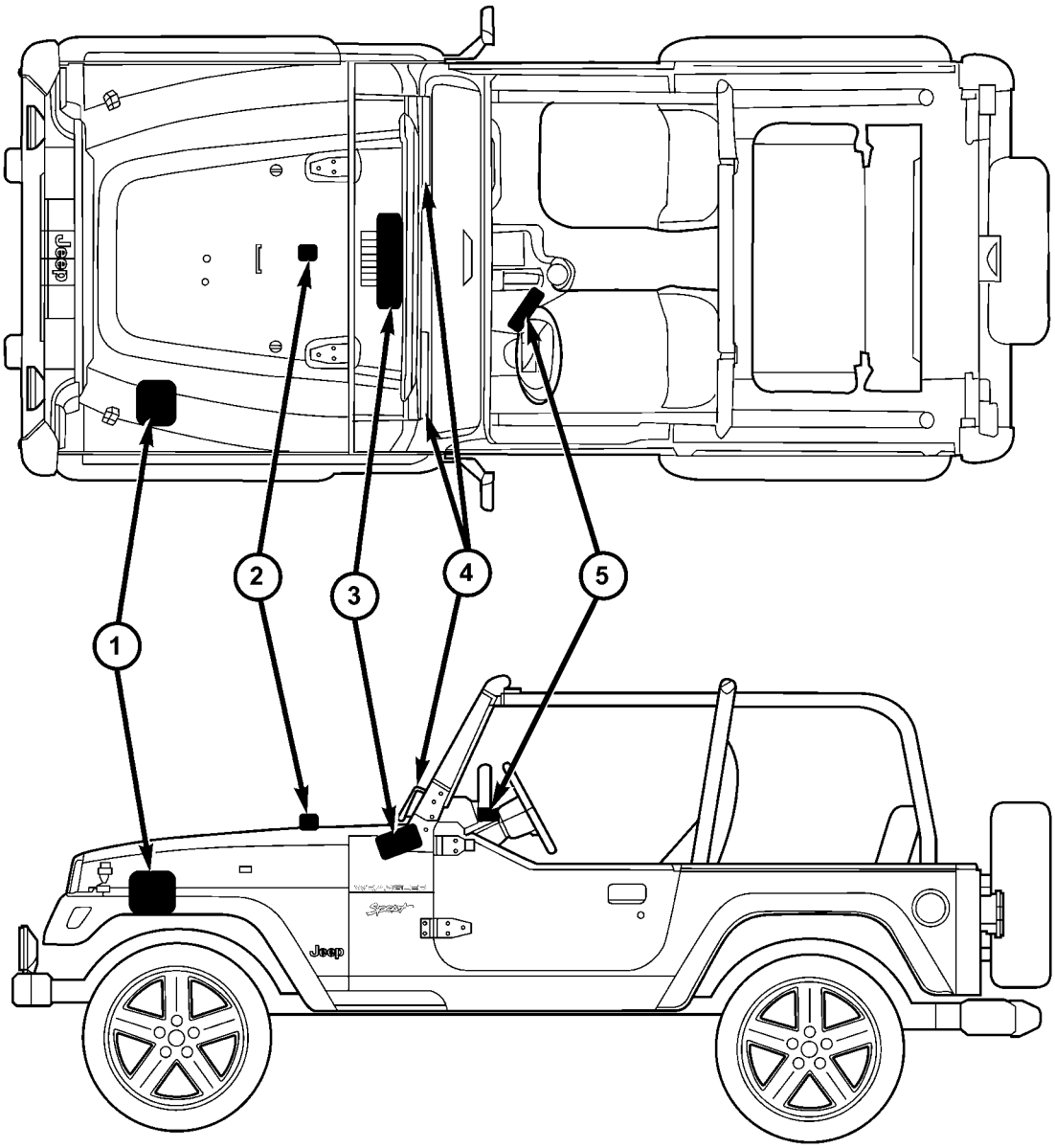

Fig. 1 Front Wiper & Washer System

1 - WASHER RESERVOIR, PUMP/MOTOR
2 - FRONT WASHER NOZZLE
3 - FRONT WIPER MODULE
4 - FRONT WIPER ARM & BLADE (2)
5 - RIGHT MULTI-FUNCTION SWITCH

- **Front Washer Nozzle** - The single fluidic front washer nozzle is secured by an integral latch feature to a dedicated opening near the rear of the hood panel. The washer plumbing fitting for the washer nozzle is concealed beneath the hood panel.
- **Front Washer Pump/Motor** - The front washer pump/motor unit is located in a dedicated hole on the lower inboard side of the washer reservoir, on the top of the left front wheel house in the engine compartment. The front washer pump/motor unit is located below and forward of the optional rear washer pump/motor unit mounting hole.
- **Front Washer Plumbing** - The plumbing for the front washer system consists of rubber hoses and molded plastic fittings. The plumbing is routed along the left side of the engine compartment from the washer reservoir, and up the cowl panel to the hood panel rear reinforcement. Then along the hood panel rear reinforcement to the front washer nozzle fitting.
- **Front Wiper Arm** - The two front wiper arms are secured with integral latches to the serrated ends

FRONT WIPERS/WASHERS (Continued)

of the two wiper pivot shafts, which extend through the cowl plenum cover/grille panel located near the base of the windshield.

- **Front Wiper Blade** - The two front wiper blades are secured to the two front wiper arms with integral latches, and are parked on the glass near the bottom of the windshield when the front wiper system is not in operation.
- **Front Wiper Module** - The front wiper pivot shafts are the only visible components of the front wiper module. The remainder of the module is concealed within the cowl plenum area beneath the cowl plenum cover/grille panel. The front wiper module includes the wiper module bracket, three rubber-isolated wiper module mounts, the front wiper motor, the front wiper motor crank arm, the front wiper drive link, the front wiper connector link, and the two front wiper pivots.
- **Right Multi-Function Switch** - The right multi-function switch is located near the top of the steering column, just below the steering wheel. The right multi-function switch includes a control stalk with a control knob on the end that extends through a dedicated opening on the right side of the steering column shrouds. The right multi-function switch is dedicated to providing all of the driver controls for the front wiper and washer systems.
- **Washer Reservoir** - The molded plastic washer reservoir is secured by two screws through integral mounting tabs to the top of the left front fender wheel house, in the left front corner of the engine compartment. The washer reservoir filler neck is accessed from the left front corner of the engine compartment.

Hard wired circuitry connects the front wiper and washer system components to the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system and to the front wiper and washer system components through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

OPERATING MODES

The components of the wiper and washer system are designed to work in concert to provide the following operating modes:

- **Continuous Wipe Modes** - The control stalk of the right multi-function switch has two continuous wipe positions, Low and High. When selected, these switch positions will cause the two-speed wiper motor to operate in a continuous low or high speed cycle.
- **Intermittent Wipe Mode** - The control stalk of the right multi-function switch has an intermittent wipe position, and the control knob has several minor detent positions which will each cause the wiper system to operate at different delay intervals between complete wipe cycles. The internal circuitry of the right multi-function switch provides the intermittent wipe mode and adjustable delay intervals between wipe cycles of about one second to about fifteen seconds.
- **Mist Wipe Mode** - The control stalk of the right multi-function switch has a momentary mist wipe position that will operate the front wipers for a single complete cycle, then park the wiper blades near the base of the windshield.
- **Washer Mode** - When the control stalk of right multi-function switch is pulled toward the steering wheel to the momentary Wash position while the front wiper system is operating, washer fluid will be dispensed onto the windshield glass through the washer nozzles for as long as the front washer pump is energized.
- **Wipe-After-Wash Mode** - When the control stalk of right multi-function switch is pulled toward the steering wheel to the momentary Wash position while the front wiper system is not operating, the internal circuitry of the right multi-function switch provides a wipe-after-wash feature which will operate the front washer pump/motor and the front wipers for as long as the washer system is activated, then provide one or two additional wipe cycles after the washer system is deactivated before parking the front wiper blades near the base of the windshield.

OPERATION

The front wiper and washer system is designed to provide the vehicle operator with a convenient, safe, and reliable means of maintaining visibility through the windshield glass. The various components of this system are designed to convert electrical energy produced by the vehicle electrical system into the mechanical action of the wiper blades to wipe the outside surface of the glass, as well as into the hydraulic action of the washer system to apply washer fluid stored in an on-board reservoir to the area of the glass to be wiped. When combined, these components provide the means to effectively maintain clear visibility for the vehicle operator by removing excess accumulations of rain, snow, bugs, mud, or other minor debris from the outside windshield glass surface that might be encountered while driving the

vehicle under numerous types of inclement operating conditions.

The vehicle operator initiates all front wiper and washer system functions with the control stalk and control knob of the right multi-function switch that extends from the right side of the steering column, just below the steering wheel. Moving the control stalk upward selects the front wiper system operating mode. The front wiper system allows the vehicle operator to select from two continuous wiper speeds, High or Low, or the intermittent wipe Delay mode. Pulling the control stalk downwards and releasing it selects the front wiper system Mist mode, which operates the front wipers for one complete wipe cycle after the control stalk is released. Rotating the control knob on the end of the control stalk allows the vehicle operator to select the intermittent wipe Delay interval. Pulling the control stalk towards the steering wheel activates the front washer pump/motor, which dispenses washer fluid onto the windshield glass through the front washer nozzle.

When the ignition switch is in the Accessory or On positions, battery current from a fuse in the Power Distribution Center (PDC) is provided to the front wiper and washer system fuse in the fuse block. This fuse provides battery current through separate fused ignition switch output (run-acc) circuits to the right multi-function switch, and to the front wiper motor. Within the right multi-function switch, this battery current is fed to one side of the control coil in the wiper motor relay, and to the electronic intermittent wipe logic circuit, which are both integral to the switch.

The intermittent wipe and wipe-after-wash features of the front wiper and washer system are both provided by the electronic intermittent wipe logic circuit within the right multi-function switch. In order to provide the intermittent wipe feature, the logic circuit monitors the wiper switch state, the intermittent delay resistance setting, and the wiper motor park switch state. In order to provide the wipe-after-wash feature, the logic circuit monitors both the front washer switch state and the wiper motor park switch state.

The hard wired circuits and components of the front wiper and washer system may be diagnosed and tested using conventional diagnostic tools and procedures. However, conventional diagnostic methods are not possible in the diagnosis of the intermittent wipe mode circuitry and logic within the right multi-function switch. If the front wiper and washer system operates satisfactorily in all modes except the Delay mode, the faulty right multi-function switch must be replaced. Refer to the appropriate diagnostic information.

OPERATING MODES

Following are paragraphs that briefly describe the operation of each of the front wiper and washer system operating modes.

CONTINUOUS WIPE MODE

When the Low position of the right multi-function switch control stalk is selected, the Low mode circuitry within the switch directs battery current to the low speed brush of the front wiper motor, which causes the front wipers to cycle at a low speed. When the High position of the control stalk is selected, the High mode circuitry within the switch directs battery current to the high speed brush of the front wiper motor, which causes the front wipers to cycle at a high speed.

When the Off position of the right multi-function switch control stalk is selected, one of two events is possible. The event that occurs depends upon the position of the wiper blades on the windshield at the moment that the control stalk Off position is selected. If the wiper blades are in the down position on the windshield when the Off position is selected, the park switch that is integral to the front wiper motor is closed to ground and the wiper motor ceases to operate.

If the wiper blades are not in the down position on the windshield at the moment the Off position is selected, the park switch is closed to battery current through the fused ignition switch output (run-acc) circuit of the front wiper motor. The park switch sense circuit directs this battery current to the low speed brush of the wiper motor through the normally closed circuit of the wiper motor relay and the Off mode circuitry of the wiper switch. This causes the wiper motor to continue running until the wiper blades are in the down position on the windshield and the park switch is again closed to ground.

INTERMITTENT WIPE MODE

When the control stalk of the right multi-function switch is moved to the Delay position, the Delay mode circuitry within the switch directs battery current to a request input of the integral logic circuit. The Delay mode circuitry also directs battery current through an internal variable resistor to the intermittent wipe delay sense input of the logic circuit, which indicates the delay interval that has been selected by the vehicle operator using the control knob on the end of the right multi-function switch control stalk.

The logic circuit responds to the Delay mode request inputs by calculating the correct delay interval. The logic circuit then energizes the wiper motor relay by pulling the relay control coil to ground. The energized wiper motor relay directs battery current from the normally open relay terminal through the

common feed relay terminal and the Delay mode wiper switch circuitry to the low speed brush of the wiper motor. The logic circuit monitors the front wiper motor operation through the wiper park switch sense circuit, which allows the logic circuit to determine the proper timing to begin the next wiper blade sweep.

MIST MODE

When the control stalk of the right multi-function switch is moved to the momentary Mist position, the Mist mode circuitry within the switch directs battery current to the low speed brush of the front wiper motor, which causes the front wipers to cycle at low speed for as long as the switch is held in the Mist position. When the control stalk is released, the wiper motor completes the current wipe cycle then parks the wiper blades near the base of the windshield.

WASH MODE

When the momentary Wash position of the right multi-function switch control stalk is selected while the front wiper system is operating, the Wash mode circuitry within the switch directs battery current to the front washer pump/motor for as long as the Wash mode circuitry within the switch remains closed. When the control stalk is released the Wash mode circuitry within the switch opens and the front washer pump/motor ceases operation.

WIPE-AFTER-WASH MODE

When the washer switch is closed with the front wiper system turned Off, the intermittent wipe logic circuit operates the front wiper motor through the wiper motor relay in the same manner as it does to provide the Delay mode operation, but uses the Off mode circuitry of the wiper switch to feed battery current to the low speed brush of the front wiper motor. When the Wash mode circuitry state changes to open, the intermittent wipe logic circuit de-energizes the front washer pump/motor unit, but allows the wiper motor to operate for several additional wipe cycles before it de-energizes the wiper motor and parks the wiper blades near the base of the windshield. The intermittent wipe logic circuit monitors the front wiper motor through the wiper park switch sense circuit, which allows the logic circuit to count the number of wiper blade sweeps.

DIAGNOSIS AND TESTING - FRONT WIPER & WASHER SYSTEM

WIPER SYSTEM

The diagnosis found here addresses an electrically inoperative front wiper system. If the front wiper motor operates, but the wipers do not move on the windshield, replace the faulty front wiper module. If the wipers operate, but chatter, lift, or do not clear the glass, clean and inspect the front wiper system components as required. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - CLEANING) and (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - INSPECTION). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Check the front wiper and washer system fuse (Fuse 14 - 25 ampere) in the fuse block. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the On position. Check for battery voltage at the front wiper and washer system fuse (Fuse 14 - 25 ampere) in the fuse block. If OK, go to Step 3. If not OK, repair the open fused ignition switch output (run-acc) circuit between the fuse block and the ignition switch as required.

(3) Disconnect and isolate the battery negative cable. Disconnect the body wire harness connector for the right multi-function switch from the switch connector receptacle. Check for continuity between the ground circuit cavity of the body wire harness connector for the right multi-function switch and a good ground. There should be continuity. If OK, go to Step 4. If not OK, repair the open ground circuit to ground (G300) as required.

(4) Reconnect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-acc) circuit cavity of the body wire harness connector for the right multi-function switch. If OK, go to Step 5. If not OK, repair the open fused ignition switch out-

put (run-acc) circuit between the right multi-function switch and the fuse block as required.

(5) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the right multi-function switch from the steering column and check the switch continuity. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/RIGHT MULTI-FUNCTION SWITCH - DIAGNOSIS AND TESTING). If OK, go to Step 6. If not OK, replace the faulty switch.

(6) Disconnect the body wire harness connector for the front wiper motor from the wiper motor pigtail wire connector. Check for continuity between the ground circuit cavity in the body wire harness connector for the front wiper motor and a good ground. There should be continuity. If OK, go to Step 7. If not OK, repair the open ground circuit to ground (G300) as required.

(7) Reconnect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run-acc) circuit cavity of the body wire harness connector for the front wiper motor. If OK, go to Step 8. If not OK, repair the open fused ignition switch output (run-acc) circuit between the front wiper motor and the fuse block as required.

(8) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. The body wire harness connector for the right multi-function switch is still disconnected. Check each of the following circuits at the proper cavity of the body wire harness connector for the front wiper motor for continuity to ground. In each case, there should be no continuity. If OK, go to Step 9. If not OK, repair the shorted circuit between the front wiper motor and the right multi-function switch as required.

- Wiper park switch sense
- Wiper switch low speed output
- Wiper switch high speed output

(9) Check the continuity of each of the following circuits between the proper cavities of the body wire harness connectors for the front wiper motor and the right multi-function switch. In each case, there should be continuity. If OK, replace the faulty front wiper module. If not OK, repair the open circuit between the front wiper motor and the right multi-function switch as required.

- Wiper park switch sense
- Wiper switch low speed output
- Wiper switch high speed output

WASHER SYSTEM

The diagnosis found here addresses an electrically inoperative front washer system. If the front washer pump/motor operates, but no washer fluid is emitted from the washer nozzle, be certain to check the fluid level in the reservoir. Also clean and inspect the front washer system components as required. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - CLEANING) and (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - INSPECTION). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Turn the ignition switch to the On position. Move the right multi-function switch control stalk to the Low or High continuous wipe position. Check whether the front wiper system operates. If OK, go to Step 2. If not OK, repair the front wiper system as required before you proceed with front washer system diagnosis. Refer to WIPER SYSTEM.

(2) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Disconnect the headlamp and dash wire harness connector for the front washer pump/motor from the washer pump/motor connector receptacle. Check for continuity between the ground circuit cavity of the headlamp and dash wire harness connector for the front washer pump/motor and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open ground circuit to ground (G102) as required.

(3) Reconnect the battery negative cable. Turn the ignition switch to the On position. While pulling the right multi-function switch control stalk toward the steering wheel to close the washer switch, check for battery voltage at the washer pump control switch output circuit cavity of the headlamp and dash wire harness connector for the front washer pump/motor. If OK, replace the faulty front washer pump/motor. If not OK, go to Step 4.

(4) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable.

FRONT WIPERS/WASHERS (Continued)

Disconnect the body wire harness connector for the right multi-function switch from the switch connector receptacle. Check for continuity between the washer pump control switch output circuit cavity of the headlamp and dash wire harness connector for the front washer pump/motor and a good ground. There should be no continuity. If OK, go to Step 5 If not OK, repair the shorted washer pump control switch output circuit between the front washer pump/motor and the right multi-function switch as required.

(5) Check for continuity between the washer pump control switch output circuit cavities of the headlamp and dash wire harness connector for the front washer pump/motor and the body wire harness connector for the right multi-function switch. There should be continuity. If OK, replace the faulty right multi-function switch. If not OK, repair the open washer pump control switch output circuit between the front washer pump/motor and the right multi-function switch as required.

CLEANING - FRONT WIPER & WASHER SYSTEM

WIPER SYSTEM

The squeegees of wiper blades exposed to the elements for a long time tend to lose their wiping effectiveness. Periodic cleaning of the squeegees is suggested to remove any deposits of salt or road film. The wiper blades, arms, and windshield glass should only be cleaned using a sponge or soft cloth and windshield washer fluid, a mild detergent, or a non-abrasive cleaner. If the wiper blades continue to leave streaks, smears, hazing, or beading on the glass after thorough cleaning of the squeegees and the glass, the entire wiper blade assembly must be replaced.

CAUTION: Protect the rubber squeegees of the wiper blades from any petroleum-based cleaners, solvents, or contaminants. These products can rapidly deteriorate the rubber squeegees.

WASHER SYSTEM

If the washer system is contaminated with foreign material, drain the washer reservoir by removing the front washer pump/motor from the reservoir. Clean foreign material from the inside of the washer pump/motor inlet filter screen and the washer reservoir using clean washer fluid, a mild detergent, or a non-abrasive cleaner. Flush foreign material from the washer system plumbing by first disconnecting the washer hoses from the front washer nozzle, then running the front washer pump/motor to run clean washer fluid or water through the system. Plugged or restricted washer nozzles should be carefully back-flushed using compressed air. If the washer nozzle obstruction cannot be cleared, replace the washer nozzle.

CAUTION: Never introduce petroleum-based cleaners, solvents, or contaminants into the washer system. These products can rapidly deteriorate the rubber seals and hoses of the washer system, as well as the rubber squeegees of the wiper blades.

CAUTION: Never use compressed air to flush the washer system plumbing. Compressed air pressures are too great for the washer system plumbing components and will result in further system damage. Never use sharp instruments to clear a plugged washer nozzle or damage to the nozzle orifice and improper nozzle spray patterns will result.

INSPECTION - FRONT WIPER & WASHER SYSTEM

WIPER SYSTEM

The front wiper blades and wiper arms should be inspected periodically, not just when wiper performance problems are experienced. This inspection should include the following points:

(1) Inspect the wiper arms for any indications of damage, or contamination. If the wiper arms are contaminated with any foreign material, clean them as required. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - CLEANING). If a wiper arm is damaged or corrosion is evident, replace the wiper arm with a new unit. Do not attempt to repair a wiper arm that is damaged or corroded.

(2) Carefully lift the wiper blade off of the glass. Note the action of the wiper arm hinge. The wiper arm should pivot freely at the hinge, but with no lateral looseness evident. If there is any binding evident in the wiper arm hinge, or there is evident lateral play in the wiper arm hinge, replace the wiper arm.

CAUTION: Do not allow the wiper arm to spring back against the glass without the wiper blade in place or the glass may be damaged.

(3) Once proper hinge action of the wiper arm is confirmed, check the hinge for proper spring tension. Remove the wiper blade from the wiper arm. Either place a small postal scale between the blade end of the wiper arm and the glass, or carefully lift the blade end of the arm away from the glass using a small fish scale. Compare the scale readings between the right and left wiper arms. Replace a wiper arm if

it has comparatively lower spring tension, as evidenced by a lower scale reading.

(4) Inspect the wiper blades and squeegees for any indications of damage, contamination, or rubber deterioration (Fig. 2). If the wiper blades or squeegees are contaminated with any foreign material, clean them and the glass as required. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - CLEANING). After cleaning the wiper blade and the glass, if the wiper blade still fails to clear the glass without smearing, streaking, chattering, hazing, or beading, replace the wiper blade. Also, if a wiper blade is damaged or the squeegee rubber is damaged or deteriorated, replace the wiper blade with a new unit. Do not attempt to repair a wiper blade that is damaged.

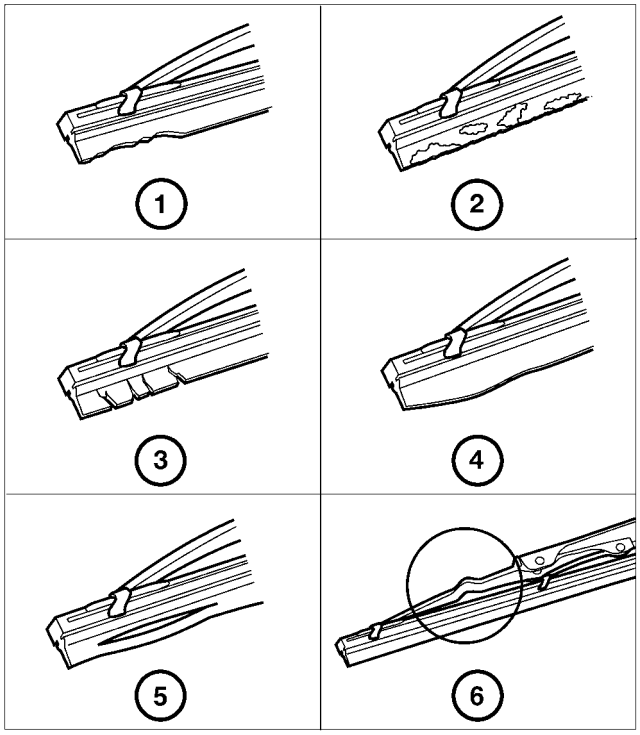

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Fig. 2 Wiper Blade Inspection

1 - WORN OR UNEVEN EDGES
2 - ROAD FILM OR FOREIGN MATERIAL DEPOSITS
3 - HARD, BRITTLE, OR CRACKED
4 - DEFORMED OR FATIGUED
5 - SPLIT
6 - DAMAGED SUPPORT COMPONENTS

WASHER SYSTEM

The washer system components should be inspected periodically, not just when washer performance problems are experienced. This inspection should include the following points:

(1) Check for ice or other foreign material in the washer reservoir. If contaminated, clean and flush the washer system. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - CLEANING).

(2) Inspect the washer plumbing for pinched, leaking, deteriorated, or incorrectly routed hoses and damaged or disconnected hose fittings. Replace damaged or deteriorated hoses and hose fittings. Leaking washer hoses can sometimes be repaired by cutting the hose at the leak and splicing it back together using an in-line connector fitting. Similarly, sections of deteriorated hose can be cut out and replaced by splicing in new sections of hose using in-line connector fittings. Whenever routing a washer hose or a wire harness containing a washer hose, it must be routed away from hot, sharp, or moving parts. Also, sharp bends that might pinch the washer hose must be avoided.

FRONT CHECK VALVE

DESCRIPTION

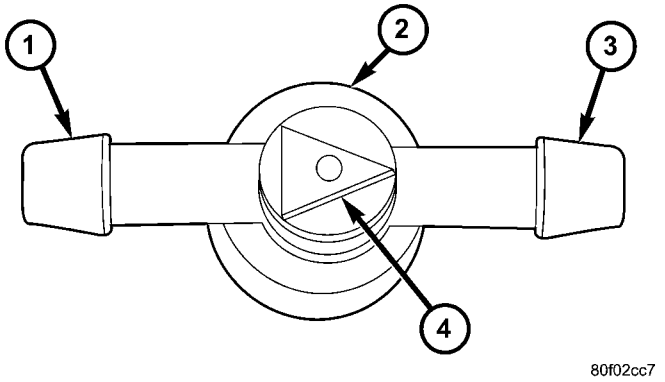

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Fig. 3 Check Valve

1 - INLET NIPPLE
2 - CHECK VALVE
3 - OUTLET NIPPLE
4 - FLOW DIRECTION ARROW

A single front washer system check valve is standard equipment on this model, and is installed in the front washer system plumbing (Fig. 3). The front check valve is located in the engine compartment in the washer supply hose on the underside of the hood panel about 5 centimeters (2 inches) from the barbed nipple of the front washer nozzle. The check valve consists of a molded plastic valve body with a raised arrowhead molded into its center section that indicates the direction of the flow through the valve. A barbed hose nipple is formed on each side of the raised center section of the valve body. Within the check valve body, a small diaphragm is held against the lip of an integral sump well by a small plastic piston and a coiled spring. The front check valve cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

FRONT CHECK VALVE (Continued)

OPERATION

The front check valve provides more than one function in this application. It prevents washer fluid from draining out of the front washer supply hoses back to the washer reservoir. This drain-back would result in a lengthy delay from when the front washer switch is actuated until washer fluid was dispensed through the front washer nozzle, because the front washer pump would have to refill the front washer plumbing from the reservoir to the nozzle. Such a drain-back condition could also result in water, dirt, or other outside contaminants being siphoned into the washer system through the washer nozzle orifice. This water could subsequently freeze and plug the nozzle, while other contaminants could interfere with proper nozzle operation and cause improper nozzle spray patterns. In addition, the front check valve prevents washer fluid from siphoning out through the front washer nozzle after the front washer system is turned Off.

When the front washer pump pressurizes and pumps washer fluid from the reservoir through the front washer plumbing, the fluid pressure unseats a diaphragm from over a sump well within the valve by overriding the spring pressure applied to it by a piston (Fig. 4). With the diaphragm unseated, washer fluid is allowed to flow toward the front washer nozzle. When the front washer pump stops operating, the spring pressure on the piston seats the diaphragm over the sump well in the valve and fluid flow in either direction within the front washer plumbing is prevented. The front check valve cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

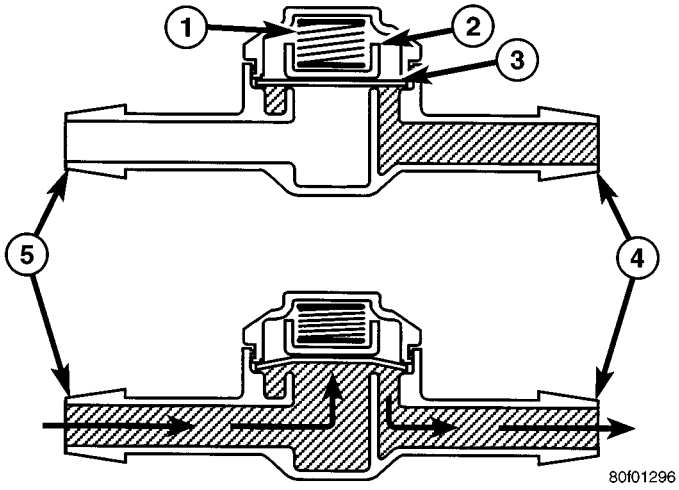

Fig. 4 Front Check Valve

1 - SPRING
2 - PISTON
3 - DIAPHRAGM
4 - TO WASHER NOZZLE
5 - FROM WASHER PUMP

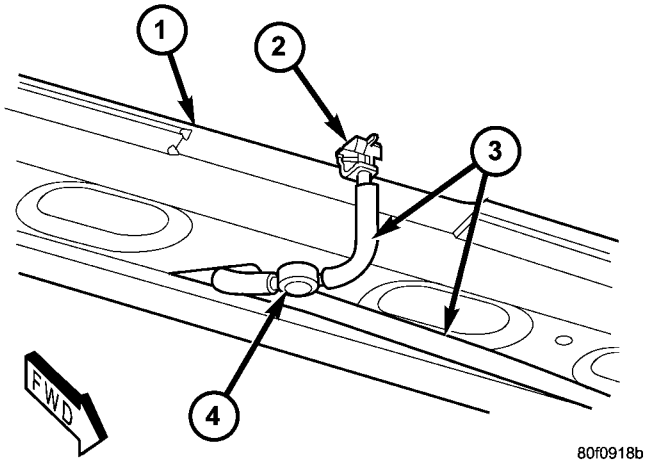

Fig. 5 Front Check Valve Remove/Install

1 - HOOD REAR REINFORCEMENT
2 - FRONT WASHER NOZZLE
3 - WASHER HOSE
4 - FRONT CHECK VALVE

REMOVAL

(1) Unlatch, open and support the hood.

(2) Locate the front check valve in the washer supply hose for the front washer nozzle. The check valve is installed in the washer supply hose about 5 centimeters (2 inches) from the front washer nozzle nipple near the rear hood panel reinforcement (Fig. 5).

(3) Disconnect the washer supply hose for the front washer nozzle from the barbed outlet nipple on the front washer system check valve. A small quantity of washer fluid may drain from the disconnected hose.

(4) Disconnect the washer supply hose for the reservoir from the barbed inlet nipple of the front check valve. Either install a temporary plug in the washer supply hose for the reservoir or secure the loose end of this hose at a point higher than the washer reservoir to prevent the contents of the washer reservoir from draining through this hose.

(5) Remove the front check valve from the engine compartment.

FRONT CHECK VALVE (Continued)

INSTALLATION

(1) Position the front check valve in the engine compartment.

(2) With the directional arrow on the check valve pointed in the direction of the system flow, reconnect the washer supply hose from the reservoir to the barbed inlet nipple of the front check valve (Fig. 6).

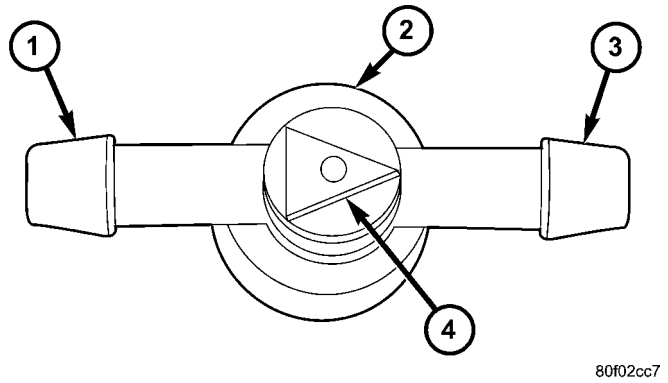

Fig. 6 Check Valve

1 - INLET NIPPLE
2 - CHECK VALVE
3 - OUTLET NIPPLE
4 - FLOW DIRECTION ARROW

(3) Reconnect the washer supply hose from the front washer nozzle to the barbed outlet nipple of the front check valve.

(4) Check that the washer supply hoses are properly routed and are not pinched.

(5) Close and latch the hood.

FRONT WASHER HOSES/ TUBES

DESCRIPTION

The front washer plumbing consists of a small diameter rubber front washer hose that is routed from the barbed outlet nipple of the electric front washer pump/motor unit on the washer reservoir around the front of the reservoir to the top of the left front fender wheel house. The hose is secured by six plastic tie wraps to the headlamp and dash wire harness and routed with the harness to the top dash panel near the center of the engine compartment (Fig. 7).

Near the center of the dash panel the front washer hose is routed to the inner rear hood panel reinforcement. Two molded plastic routing clips secure the front washer hose to the underside of the hood panel reinforcement. The front washer hose is connected to the barbed inlet nipple of the front washer check valve near the inner center hood panel reinforcement. A short piece of washer hose then connects the barbed outlet nipple of the front washer check valve to the barbed nipple of the front washer nozzle on the underside of the hood panel.

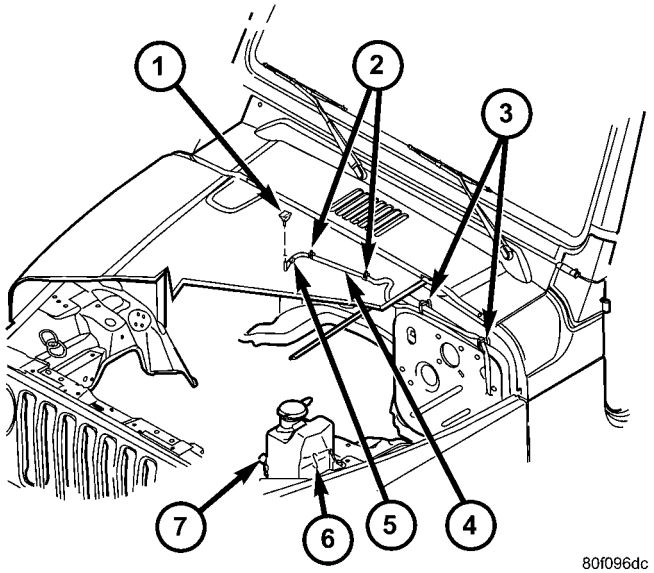

Fig. 7 Front Washer Plumbing

1 - FRONT WASHER NOZZLE
2 - CLIP (2)
3 - HARNESS CLIP (2)
4 - WASHER SUPPLY HOSE
5 - FRONT CHECK VALVE
6 - WASHER RESERVOIR
7 - FRONT WASHER PUMP/MOTOR

Washer hose is available for service only as roll stock, which must then be cut to length. Molded plastic washer hose fittings cannot be repaired. If these fittings are faulty or damaged, they must be replaced.

OPERATION

Washer fluid in the washer reservoir is pressurized and fed by the front washer pump/motor through the front washer system plumbing and fittings to the front washer check valve and the front washer nozzle. Whenever routing the washer hose or a wire harness containing a washer hose, it must be routed away from hot, sharp, or moving parts; and, sharp bends that might pinch the hose must be avoided.

FRONT WASHER NOZZLE

DESCRIPTION

The molded black plastic front washer nozzle has an integral latch feature that secures it in a dedicated mounting hole located at the center near the rear edge of the hood panel (Fig. 8). The washer nozzle is constructed in two pieces. The nozzle housing includes the domed hood that is visible on the outer surface of the hood panel, and the integral latch and

FRONT WASHER NOZZLE (Continued)

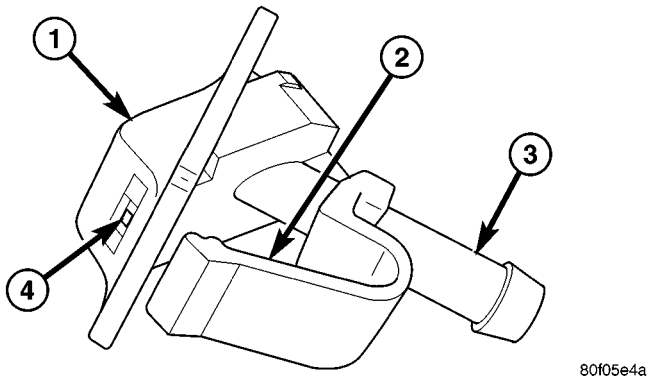

Fig. 8 Front Washer Nozzle

1 - HOOD
2 - LATCH
3 - ORIFICE (FLUIDIC INSERT)
4 - NIPPLE

barbed inlet nipple which are concealed below the hood panel.

A rectangular opening in the rearward facing surface of the nozzle hood accepts the fluidic insert of the nozzle. The fluidic insert is a one-piece unit that incorporates the various chambers, passages, deflectors and the single orifice through which the washer fluid stream is directed into the wipe pattern on the windshield glass. The nozzle is accessed for service from the underside of the hood panel. The front washer nozzle cannot be adjusted or repaired. If faulty or damaged, it must be replaced.

OPERATION

The front washer nozzle is designed to dispense washer fluid into the wiper pattern area on the outside of the windshield glass. Pressurized washer fluid is fed to the nozzle from the washer reservoir by the front washer pump/motor through a single hose, which is attached to a barbed nipple on the front washer nozzle below the hood panel. A fluidic matrix within the washer nozzle causes the pressurized washer fluid to be emitted from the nozzle orifice as a fan-like stream to more effectively cover a larger area of the glass to be cleaned.

REMOVAL

(1) Unlatch, open and support the hood.

(2) From the underside of the rear of the hood panel near the center, disconnect the washer supply hose from the barbed nipple of the front washer nozzle (Fig. 9).

(3) From the underside of the rear of the hood panel near the center, gently squeeze the latch feature of the front washer nozzle and push the nozzle out through the mounting hole towards the outside of the hood panel.

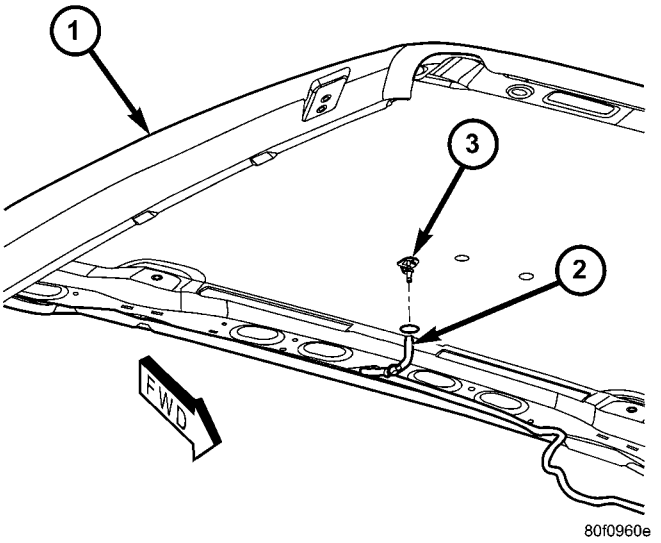

Fig. 9 Front Washer Nozzle Remove/Install

1 - HOOD PANEL
2 - WASHER SUPPLY HOSE
3 - FRONT WASHER NOZZLE

(4) Remove the front washer nozzle from the top of the hood panel.

INSTALLATION

(1) Lower the hood panel.

(2) From the top of the hood panel, position the nipple end of the front washer nozzle through the mounting hole. Be certain the nozzle orifice is oriented towards the windshield, then engage the notched forward-facing end of the nozzle housing to the forward edge of the mounting hole (Fig. 9).

(3) Using hand pressure, press firmly and evenly on the top of the front washer nozzle until the integral latch feature locks into place on the underside of the hood panel.

(4) Open and support the hood.

(5) From the underside of the rear of the hood panel near the center, reconnect the washer supply hose to the barbed nipple of the front washer nozzle.

(6) Close and latch the hood.

FRONT WASHER PUMP MOTOR

DESCRIPTION

The front washer pump/motor unit is located on the inboard side near the front and the bottom of the washer reservoir (Fig. 10). The washer reservoir is located on the top of the left front fender wheel house in the engine compartment. A small permanently lubricated and sealed electric motor is coupled to the

FRONT WASHER PUMP MOTOR (Continued)

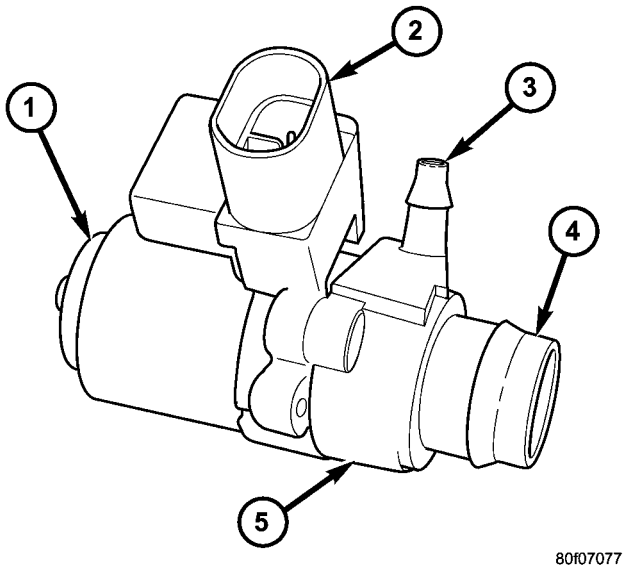

Fig. 10 Washer Pump/Motor

1 - MOTOR
2 - CONNECTOR RECEPTACLE
3 - OUTLET NIPPLE
4 - INLET NIPPLE
5 - PUMP

rotor-type washer pump. A seal flange with a barbed inlet nipple on the pump housing passes through a rubber grommet seal installed in a dedicated mounting hole in the sump area near the bottom of the washer reservoir. On vehicles with the optional rear washer system, the front washer pump/motor unit is always mounted in the forward-most, lowest pump mounting hole of the reservoir. An integral filter screen is located within the pump inlet nipple. When the pump is installed in the reservoir a barbed outlet nipple on the pump housing connects the unit to the washer system through the front washer supply hose.

The washer pump/motor unit is retained on the reservoir by the interference fit between the barbed pump inlet nipple and the grommet seal, which is a light press fit. An integral connector receptacle between the motor housing and the pump housing connects the unit to the vehicle electrical system. A small capacitor internal to the connector receptacle insulator is connected in parallel with the motor leads to control ElectroMagnetic Interference (EMI) created by washer motor operation. The front washer pump/motor unit cannot be repaired. If faulty or damaged, the entire washer pump/motor unit must be replaced.

OPERATION

The front washer pump/motor unit features a small Direct Current (DC) electric motor. The motor is connected to the vehicle electrical system through a single take out and two-cavity connector of the headlamp and dash wire harness. The motor is grounded at all times through another take out of the headlamp and dash wire harness with a single eyelet terminal connector that is secured under a ground screw to the radiator closure panel behind the left headlamp in the engine compartment. The motor receives battery current through the closed contacts of the momentary front washer switch circuitry within the right multi-function switch only when the switch control stalk is pulled towards the steering wheel.

Washer fluid is gravity-fed from the washer reservoir to the inlet side of the washer pump. When the pump motor is energized, the motor spins the rotor within the rotor-type washer pump. The spinning pump rotor pressurizes the washer fluid and forces it through the pump outlet nipple, the front washer plumbing, and the front washer nozzle onto the windshield glass. The front washer pump/motor unit can be diagnosed using conventional diagnostic tools and methods.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the washer supply hose from the barbed outlet nipple of the front washer pump/motor and allow the washer fluid from the washer reservoir to drain into a clean container for reuse (Fig. 11).

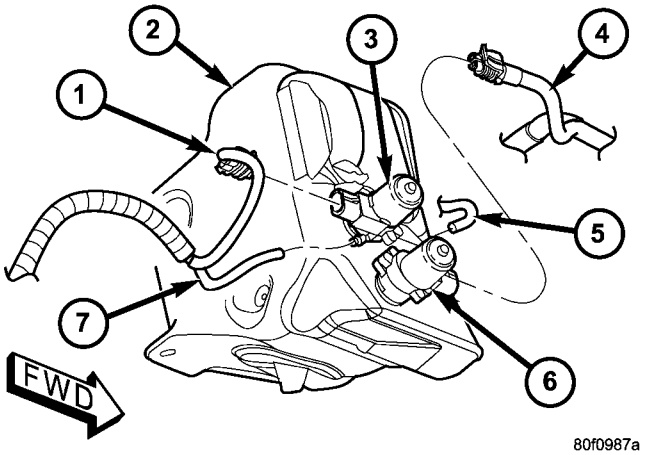

Fig. 11 Washer Pump/Motor Remove/Install

1 - REAR BODY WIRE HARNESS
2 - WASHER RESERVOIR
3 - REAR WASHER PUMP/MOTOR
4 - HEADLAMP & DASH WIRE HARNESS
5 - FRONT WASHER SUPPLY HOSE
6 - FRONT WASHER PUMP/MOTOR
7 - REAR WASHER SUPPLY HOSE

(3) Disconnect the headlamp and dash wire harness connector from the connector receptacle for the front washer pump/motor.

FRONT WASHER PUMP MOTOR (Continued)

(4) Using a trim stick or another suitable wide flat-bladed tool, gently pry the barbed inlet nipple of the front washer pump/motor out of the rubber grommet seal in the washer reservoir. Care must be taken not to damage the washer reservoir.

(5) Remove the front washer pump/motor from the washer reservoir.

(6) Remove the rubber grommet seal from the front washer pump/motor mounting hole in the washer reservoir and discard.

INSTALLATION

(1) Install a new rubber grommet seal into the front washer pump/motor mounting hole of the washer reservoir.

(2) Position the front washer pump/motor inlet nipple to the mounting hole in the washer reservoir (Fig. 11).

(3) Using hand pressure, firmly and evenly press on the front washer pump/motor to engage the inlet nipple through the rubber grommet seal and into the washer reservoir. Care must be taken not to damage the washer reservoir.

(4) Reconnect the headlamp and dash wire harness connector for the front washer pump/motor to the washer pump/motor connector receptacle.

(5) Reconnect the front washer supply hose to the barbed outlet nipple of the front washer pump/motor.

(6) Refill the washer reservoir with the washer fluid drained from the reservoir during the removal procedure.

(7) Reconnect the battery negative cable.

FRONT WIPER ARM

DESCRIPTION

The front wiper arms are the rigid members located between the wiper pivots that protrude from the cowl plenum cover/grille panel near the base of the windshield and the wiper blades on the windshield glass. The wiper arm has a die cast metal pivot end (Fig. 12). On the underside of this pivot end is a large internally serrated socket formation with a small, movable, stamped steel latch plate that is secured loosely under a small strap that is staked to the pivot end.

The wide end of a tapered, stamped steel channel hinges on and is secured with a hinge pin to the pivot end of the wiper arm. One end of a long, rigid, stamped steel strap, with a small hole near its pivot end, is riveted and crimped within the narrow end of the stamped steel channel. The tip of the wiper blade end of this strap is bent back under itself to form a small hook. Concealed within the stamped steel channel, one end of a long spring is hooked through a hole in a small stamped steel strap on the hinge pin within the die cast pivot end, while the other end of the spring is hooked through the small hole in the steel strap. The entire wiper arm has a satin black finish applied to all of its visible surfaces.

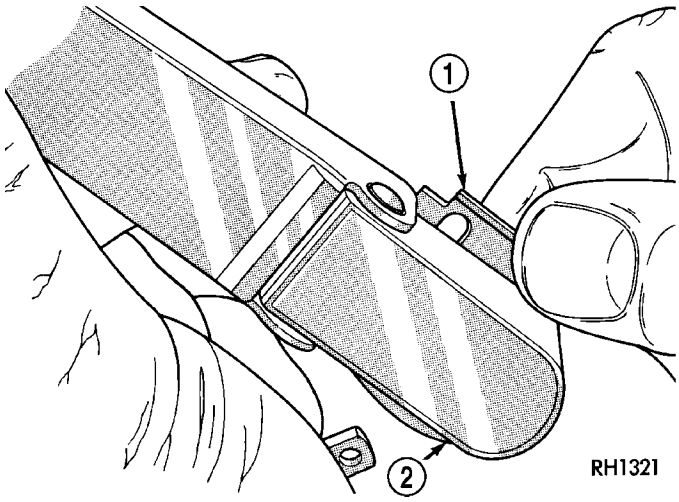

Fig. 12 Wiper Arm

1 - LATCH
2 - WIPER ARM PIVOT END

A wiper arm cannot be adjusted or repaired. If damaged or faulty, the entire wiper arm unit must be replaced.

OPERATION

The front wiper arms are designed to mechanically transmit the motion from the wiper pivots to the wiper blades. The wiper arm must be properly indexed to the wiper pivot in order to maintain the proper wiper blade travel on the glass. The serrated socket formation in the wiper arm pivot end interlocks with the serrations on the outer circumference of the wiper pivot driver, providing positive engagement and finite adjustment of this connection. The latch plate on the underside of the wiper arm pivot end locks the wiper arm to the wiper pivot driver when in its installed position; and, when in its unlocked position, also serves as a blocker to hold the spring-loaded wiper arm off of the glass to facilitate removal and installation. The spring-loaded wiper arm hinge controls the down-force applied through the tip of the wiper arm to the wiper blade on the glass. The hook formation on the tip of the wiper arm provides a cradle for securing and latching the wiper blade pivot block to the wiper arm.

REMOVAL

(1) Lift the front wiper arm far enough to raise the wiper blade off of the glass and permit the wiper arm latch plate to be pulled out to its holding position, then release the arm (Fig. 13). The wiper arm and

FRONT WIPER ARM (Continued)

blade will remain off the glass with the latch in this position.

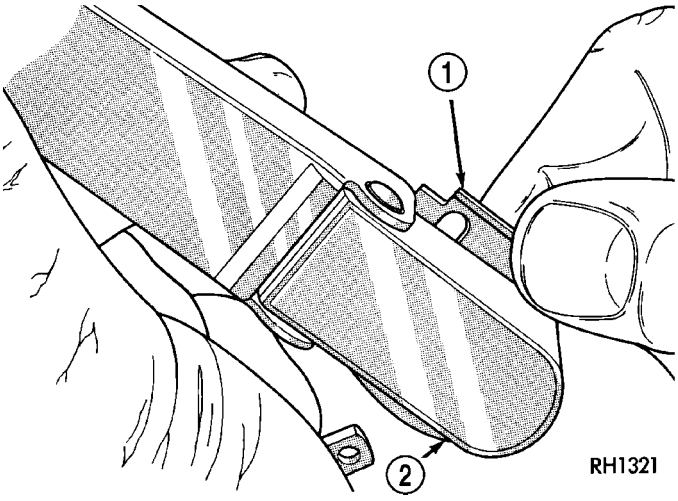

Fig. 13 Front Wiper Arm Remove/Install

1 - LATCH
2 - WIPER ARM PIVOT END

CAUTION: The use of a screwdriver or other prying tool to remove a wiper arm may distort it. This distortion could allow the arm to come off of the wiper pivot during wiper operation, regardless of how carefully it is reinstalled.

(2) Using a slight rocking motion, remove the front wiper arm pivot end from the wiper pivot driver.

INSTALLATION

NOTE: Be certain that the wiper motor is in the park position before attempting to install the wiper arms. Turn the ignition switch to the On position and move the right multi-function switch control stalk to its Off position. If the wiper pivots move, wait until they stop moving, then turn the ignition switch back to the Off position. The wiper motor is now in its park position.

(1) The front wiper arms must be indexed to the wiper pivots with the wiper motor in the park position to be properly installed (Fig. 14). Position the front wiper arm pivot ends onto the wiper pivot drivers so that the tip of the wiper blade is on the upper edge of the lower windshield blackout area, + 15 millimeters/– 0 millimeter (+ 0.59 inch/– 0 inch).

(2) Once the wiper arm is indexed to the wiper pivot, lift the wiper arm away from the windshield slightly to relieve the spring tension on the latch plate, then push the latch into the locked position (Fig. 13). Gently lower the wiper arm until the wiper blade is in position on the windshield glass.

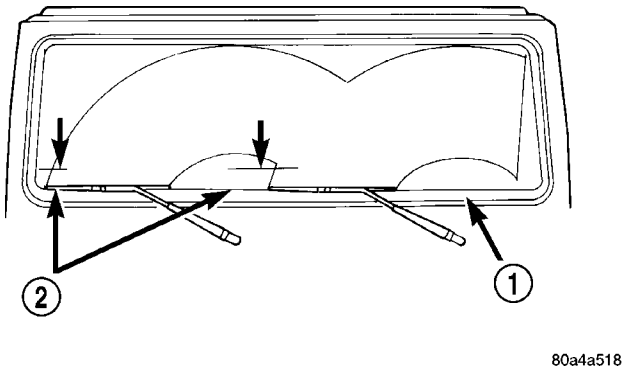

Fig. 14 Front Wiper Arm Installation

1 - WINDSHIELD BLACKOUT AREA
2 - PARK BLADE ON UPPER EDGE OF BLACKOUT AREA
+15 mm – 0 mm (+0.59 in. – 0 in.)

(3) Wet the windshield glass, then operate the front wipers. Move the right multi-function switch control stalk to the Off position, then check for correct wiper arm position. Repeat the adjustment as required.

FRONT WIPER BLADE

DESCRIPTION

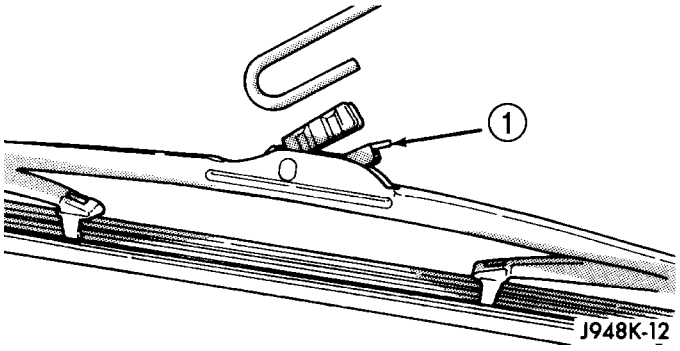

Fig. 15 Wiper Blade - Typical

1 - RELEASE TAB

Each front wiper blade is secured by an integral latching pivot block to the hook formation on the tip of each front wiper arm, and rests on the glass near the base of the windshield when the wipers are not in operation (Fig. 15). The wiper blade consists of the following components:

- **Superstructure** - The superstructure includes several stamped steel bridges and links with claw formations that grip the wiper blade element. Also included in this unit is the latching, molded plastic pivot block that secures the superstructure to the wiper arm. All of the metal components of the wiper blade have a satin black finish applied.

FRONT WIPER BLADE (Continued)

- **Element** - The wiper element or squeegee is the resilient rubber member of the wiper blade that contacts the glass.
- **Flexor** - The flexor is a rigid metal component running along the length of each side of the wiper element where it is gripped by the claws of the superstructure.

All models have two 33-centimeter (13-inch) long front wiper blades with non-replaceable rubber elements (squeegees). The wiper blades cannot be adjusted or repaired. If faulty, worn, or damaged the entire wiper blade unit must be replaced.

OPERATION

The wiper blades are moved back and forth across the glass by the wiper arms when the wipers are being operated. The wiper blade superstructure is the flexible frame that grips the wiper blade element and evenly distributes the force of the spring-loaded wiper arm along the length of the element. The combination of the wiper arm force and the flexibility of the superstructure makes the element conform to and maintain proper contact with the glass, even as the blade is moved over the varied curvature that may be encountered across the glass surface. The wiper element flexor provides the claws of the blade superstructure with a rigid, yet flexible component on the element which can be gripped. The rubber element is designed to be stiff enough to maintain an even cleaning edge as it is drawn across the glass, yet resilient enough to conform to the glass surface and flip from one cleaning edge to the other each time the wiper blade changes directions.

REMOVAL

NOTE: The notched end of the wiper element flexor should always be oriented towards the end of the wiper blade that is nearest to the wiper pivot.

(1) Lift the front wiper arm to raise the wiper blade and element off of the glass.

(2) To remove the wiper blade from the wiper arm, depress the pivot block latch release tab under the tip of the arm and slide the blade away from the tip towards the pivot end of the arm far enough to disengage the pivot block from the hook formation on the end of the arm (Fig. 16).

(3) Extract the hook formation on the tip of the wiper arm through the opening in the wiper blade superstructure just ahead of the wiper blade pivot block/latch unit.

CAUTION: Do not allow the wiper arm to spring back against the glass without the wiper blade in place or the glass may be damaged.

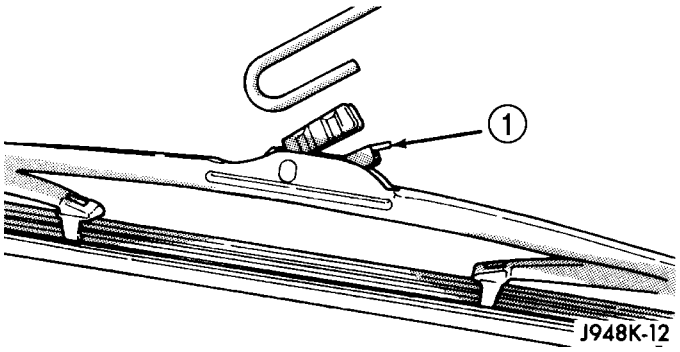

Fig. 16 Wiper Blade Remove/Install - Typical

1 - RELEASE TAB

(4) Gently lower the tip of the wiper arm onto the glass.

INSTALLATION

NOTE: The notched end of the wiper element flexor should always be oriented towards the end of the wiper blade that is nearest to the wiper pivot.

(1) Lift the front wiper arm off of the windshield glass.

(2) Position the front wiper blade near the hook formation on the tip of the arm with the notched end of the wiper element flexor oriented towards the end of the wiper arm that is nearest to the wiper pivot.

(3) Insert the hook formation on the tip of the wiper arm through the opening in the wiper blade superstructure ahead of the wiper blade pivot block/latch unit far enough to engage the pivot block into the hook (Fig. 16).

(4) Slide the wiper blade pivot block/latch up into the hook formation on the tip of the wiper arm until the latch release tab snaps into its locked position. Latch engagement will be accompanied by an audible click.

(5) Gently lower the wiper blade onto the glass.

FRONT WIPER MODULE

DESCRIPTION

The front wiper module is secured with three screws, one each at the motor bracket and the two pivot brackets through rubber insulators to the cowl top panel and concealed within the cowl plenum area beneath the cowl plenum cover/grille panel (Fig. 17). The driver ends of the wiper pivot shafts that protrude through dedicated openings in the cowl plenum cover/grille panel to drive the wiper arms and blades are the only visible components of the front wiper module. The front wiper module consists of the following major components:

FRONT WIPER MODULE (Continued)

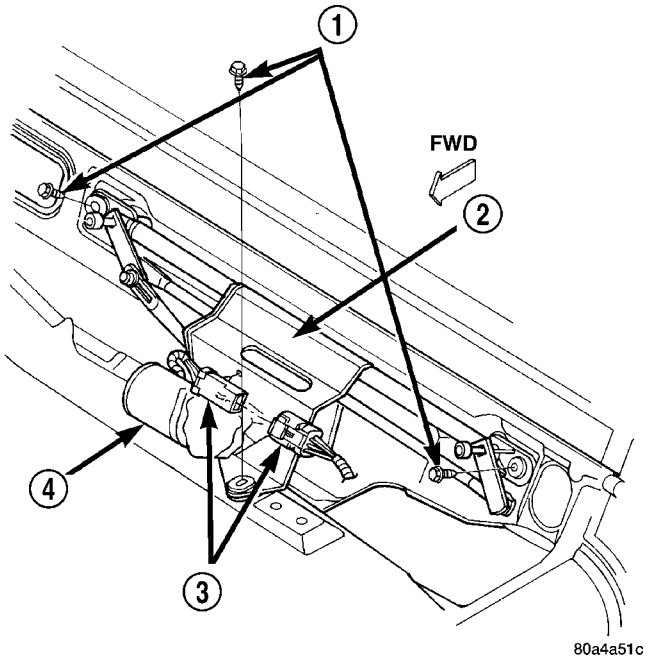

Fig. 17 Front Wiper Module

1 - SCREW (3)
2 - MODULE BRACKET
3 - WIRE HARNESS CONNECTOR
4 - WIPER MOTOR

- **Bracket** - The front wiper module bracket consists of a long tubular steel main member that has a stamped pivot bracket formation near each end where the two wiper pivots are secured. A stamped steel mounting plate for the wiper motor is secured with welds near the center of the main member. The bracket includes metal-sleeved rubber isolators at each of the three bracket mounting points.
- **Crank Arm** - The front wiper motor crank arm is a stamped steel unit with a slotted hole on the driven end that is secured to the wiper motor output shaft with a nut, and has a ball stud secured to the drive end.
- **Linkage** - The two front wiper linkage members are each constructed of stamped steel. A connecting link with a plastic socket-type bushing in each end is fit over the pivot ball studs to join the two pivots. The wiper drive link has a plastic socket-type bushing on each end. One end of the drive link is snap-fit over a second ball stud on the passenger side pivot crank arm, while the other end is snap-fit over the ball stud on the wiper motor crank arm.
- **Motor** - The front wiper motor is secured with three screws to the motor mounting plate near the center of the wiper module bracket and is protected by a rubber boot. The wiper motor output shaft passes through a hole in the module bracket, where a nut secures the wiper motor crank arm to the motor output shaft. The two-speed permanent magnet wiper motor features an integral transmission, an internal park switch, and an internal automatic resetting circuit breaker.
- **Pivots** - The two front wiper pivots are secured to the ends of the wiper module bracket. A crank arm extends from the lower end of each pivot shaft. The driver side pivot crank arm has a single ball stud secured to it, while the passenger side crank arm has two ball studs. The upper end of each pivot shaft where the wiper arms will be fastened each has an externally serrated drum-like driver secured to it.

The front wiper module for this model is serviced only as a complete unit. If any linkage component or the mounting bracket of the module is faulty or damaged, the entire front wiper module unit must be replaced. The front wiper motor and boot are available for service replacement as a unit only.

OPERATION

The front wiper module operation is controlled by the battery current inputs received by the wiper motor through the right multi-function switch on the steering column. The wiper motor is connected to the vehicle electrical system through a dedicated take out and wire harness connector of the body wire harness. The wiper motor speed is controlled by current flow to either the low speed or the high speed set of brushes. The park switch is a single pole, single throw, momentary switch within the wiper motor that is mechanically actuated by the wiper motor transmission components. The park switch alternately closes the wiper park switch sense circuit to ground or to battery current, depending upon the position of the wipers on the glass. This feature allows the motor to complete its current wipe cycle after the wiper system has been turned Off, and to park the wiper blades in the lowest portion of the wipe pattern. The automatic resetting circuit breaker protects the motor from overloads. The wiper motor crank arm, the two wiper linkage members, and the two wiper pivots mechanically convert the rotary output of the wiper motor to the back and forth wiping motion of the wiper arms and blades on the glass.

REMOVAL

(1) Remove the front wiper arms from the wiper pivots. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/FRONT WIPER ARM - REMOVAL).

(2) Unlatch, open and support the hood.

(3) Disconnect and isolate the battery negative cable.

(4) Remove the cowl plenum cover/grille panel from over the cowl plenum. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - REMOVAL).

FRONT WIPER MODULE (Continued)

(5) Disconnect the body wire harness connector for the front wiper motor from the motor pigtail wire connector (Fig. 18).

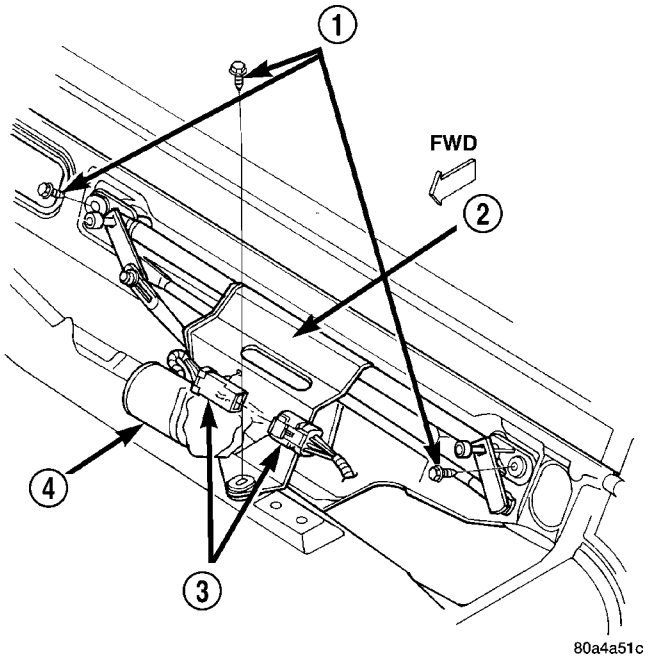

Fig. 18 Front Wiper Module Remove/Install

1 - SCREW (3)
2 - MODULE BRACKET
3 - WIRE HARNESS CONNECTOR
4 - WIPER MOTOR

(6) Remove the three screws that secure the front wiper module mounting bracket to the cowl plenum panel.

(7) Remove the front wiper module from the cowl plenum panel as a unit.

DISASSEMBLY

The front wiper motor and its rubber boot are available for service replacement. Following are the procedures for disassembling these components from the front wiper module unit.

(1) Remove the front wiper module from the cowl plenum. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/FRONT WIPER MODULE - REMOVAL).

(2) Release the retainer that secures the wiper motor pigtail wire connector to the front wiper module bracket.

(3) Turn the front wiper module over and remove the nut that secures the wiper motor crank arm to the wiper motor output shaft.

(4) Remove the three screws that secure the wiper motor to the front wiper module mounting bracket.

(5) Remove the wiper motor from the front wiper module bracket.

ASSEMBLY

The front wiper motor and its rubber boot are available for service replacement. Following are the procedures for reassembling these components onto the front wiper module unit.

(1) Position the wiper motor onto the front wiper module bracket.

(2) Install and tighten the three screws that secure the wiper motor to the front wiper module bracket. Tighten the screws to 8 N·m (70 in. lbs.).

(3) Position the wiper motor crank arm onto the front wiper motor output shaft.

(4) Install and tighten the nut that secures the wiper motor crank arm to the front wiper motor output shaft. Tighten the nut to 12 N·m (101 in. lbs.).

(5) Engage the wiper motor pigtail wire connector retainer in the locating hole on the front wiper module bracket.

(6) Reinstall the front wiper module into the cowl plenum. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/FRONT WIPER MODULE - INSTALLATION).

INSTALLATION

(1) Position the front wiper module into the cowl plenum as a unit (Fig. 18).

(2) Install and tighten the three screws that secure the front wiper module mounting bracket to the cowl plenum panel. Tighten the screws to 12 N·m (105 in. lbs.).

(3) Reconnect the body wire harness connector for the front wiper motor to the motor pigtail wire connector.

(4) Reinstall the cowl plenum cover/grille panel over the cowl plenum. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - INSTALLATION).

(5) Reconnect the battery negative cable.

(6) Close and latch the hood.

(7) Reinstall the front wiper arms onto the wiper pivots. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/FRONT WIPER ARM - INSTALLATION).

RIGHT MULTI-FUNCTION SWITCH

DESCRIPTION

The right multi-function switch is located on the right side of the steering column (Fig. 19). The only visible components of the right multi-function switch are the control stalk and control knob that extend through dedicated openings in the right side of the steering column shrouds, just below the steering wheel. The remainder of the switch, its mounting provisions, and its electrical connections are all concealed beneath the steering column shrouds. The switch housing and controls are constructed of molded black plastic. The right multi-function switch is secured by two screws to a mounting housing that is integral to the left multi-function switch.

The right multi-function switch control stalk has both white nomenclature and International Control and Display Symbols graphics applied to it, which clearly identify its many functions. On the end of the control stalk is a plastic control knob with a rounded end and a flattened face to allow it to be easily rotated. A single connector receptacle containing six terminal pins is integral to the back of the switch housing and connects the switch to the vehicle electrical system through a dedicated take out and connector of the body wire harness. The right multi-function switch cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

The right multi-function switch is the primary control for the front wiper and washer system, and contains switches, circuitry, an intermittent wipe logic circuit, and an internal wiper motor relay to provide the following functions and features:

- **Continuous Front Wipe Modes** - The internal circuitry and hardware of the right multi-function switch control stalk provide two continuous front wipe switch positions, low speed or high speed.
- **Front Washer Mode** - The internal circuitry and hardware of the right multi-function switch control stalk provide front washer system operation.
- **Front Wipe-After-Wash Mode** - The internal circuitry and hardware of the right multi-function switch control stalk provide a wipe-after-wash mode.
- **Front Wiper Mist Mode** - The internal circuitry and hardware of the right multi-function switch control stalk provide a front wiper system mist mode.
- **Intermittent Front Wipe Mode** - The internal circuitry and hardware of the right multi-function switch control stalk and control knob provide an intermittent front wipe mode with adjustable delay interval positions.

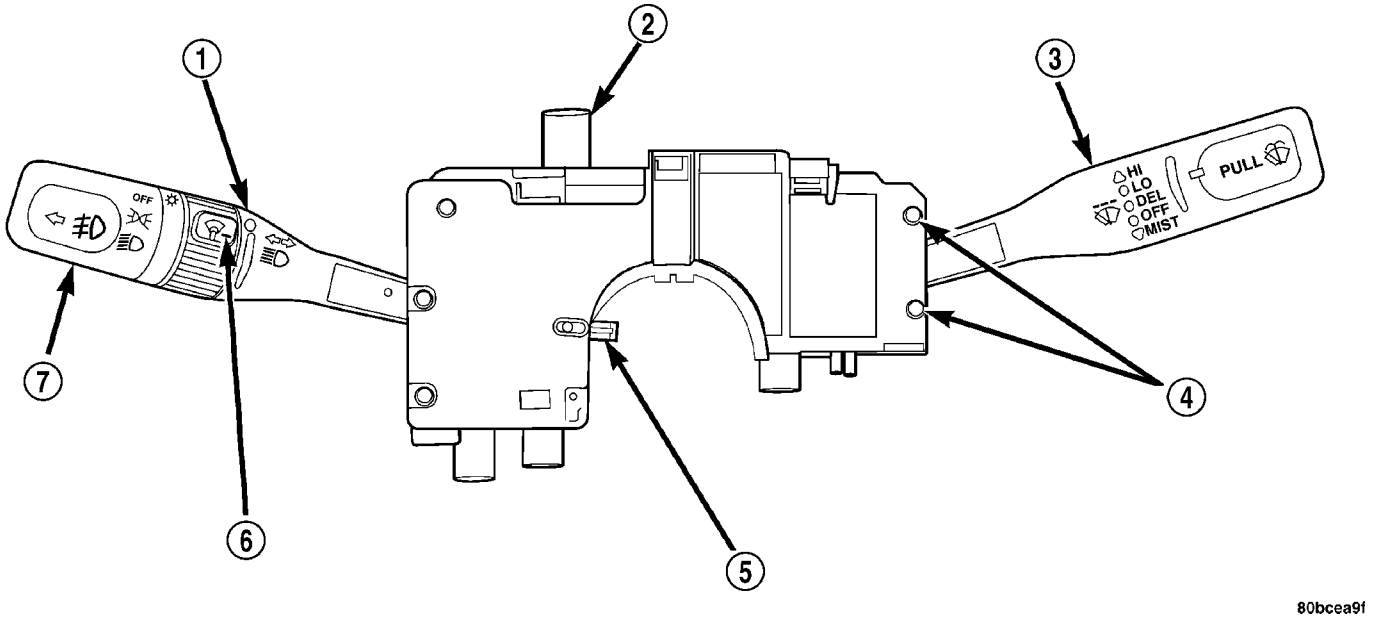

Fig. 19 Multi-Function Switch

1 - LEFT MULTI-FUNCTION SWITCH CONTROL STALK
2 - HAZARD WARNING SWITCH BUTTON
3 - RIGHT MULTI-FUNCTION SWITCH CONTROL STALK
4 - SCREW (2)
5 - TURN SIGNAL CANCEL ACTUATOR
6 - LEFT MULTI-FUNCTION SWITCH CONTROL RING
7 - LEFT MULTI-FUNCTION SWITCH CONTROL KNOB

RIGHT MULTI-FUNCTION SWITCH (Continued)

OPERATION

The right multi-function switch uses an internal logic circuit, an internal relay and conventionally switched outputs to control the many functions and features it provides. The switch receives battery current on a fused ignition switch output (run-acc) circuit from a fuse in the Power Distribution Center (PDC) and the wiper and washer system fuse in the fuse block whenever the ignition switch is in the On or Accessory positions. The electronic intermittent wipe logic circuit within the switch receives a logic ground at all times through the body wire harness from a take out with an eyelet terminal connector secured under a ground screw on the left inner cowl side panel, below the left end of the instrument panel.

Following are descriptions of how the right multi-function switch control stalk and control knob are operated to control the functions and features they provide. Refer to the owner's manual in the vehicle glove box for more information on the features, use and operation of the right multi-function switch.

- **Front Wash Mode** - Pulling the control stalk of the right multi-function switch towards the steering wheel actuates the momentary single pole, single throw front washer switch and operates the front washer pump/motor.
- **Front Wipe Modes** - Moving the control stalk of the right multi-function switch up or down to one of four detent positions actuates the triple pole, quadruple throw front wiper switch and selects the Off, Low, High, or Delay front wiper operating mode. Moving the control stalk downward from the Off position actuates a momentary single pole, single throw switch that selects the Mist operating mode.
- **Intermittent Front Wipe Delay Intervals** - Rotating the control knob on the end of the control stalk actuates the front wiper delay variable resistor and selects the wiper delay interval when the Delay mode is selected with the front wiper switch control stalk.

The intermittent wipe logic circuit within the right multi-function switch monitors inputs from the front wiper switch circuitry, the front washer switch circuitry, the front wiper delay variable resistor, and the front wiper motor park switch. The programming of the logic circuit then determines the proper outputs to the front wiper motor. The low current logic circuit controls the high current front wiper motor by pulling the control coil of the integral wiper motor relay to ground. The wiper motor relay switches a circuit that feeds battery current to the low speed or high speed brush of the front wiper motor through the appropriate right multi-function switch circuits.

DIAGNOSIS AND TESTING - RIGHT MULTI-FUNCTION SWITCH

Be certain to perform the diagnosis for the front wiper system and/or front washer system before testing the right multi-function switch. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS - DIAGNOSIS AND TESTING). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the right multi-function switch from the steering column and disconnect the body wire harness connector for the switch from the switch connector receptacle.

(3) Using an ohmmeter, check the right multi-function switch continuity at the terminals (Fig. 20) in the switch connector receptacle as shown in the Right Multi-Function Switch Tests table.

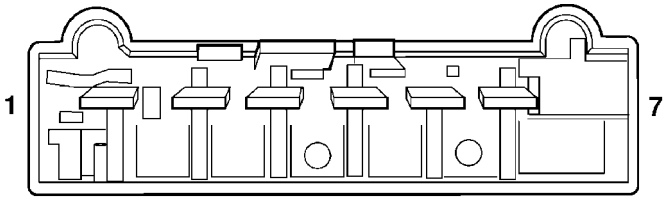

Fig. 20 Right Multi-Function Switch Connector Receptacle

RIGHT MULTI-FUNCTION SWITCH (Continued)

RIGHT MULTI-FUNCTION SWITCH TESTS	
SWITCH POSITION	CONTINUITY BETWEEN PINS
OFF	1 & 6
*LOW	4 & 6
*MIST	4 & 6
HIGH	4 & 5
WASH	3 & 4
*DELAY	1 & 6

*The intermittent wipe logic circuit within the right multi-function switch contains active electronic elements, which cannot be tested using conventional diagnostic tools. In addition, the function of the normally open contacts of the wiper relay internal to the switch cannot be tested properly unless the switch is connected to battery current (Pin 4) and ground (Pin 2). If all circuits and functions of the front wiper system and the right multi-function switch are operative except for the intermittent wipe, wipe-after-wash feature, and/or the front wipers will not park, replace the right multi-function switch with a known good unit and test system operation again.

(4) If the right multi-function switch fails any of the continuity checks, replace the faulty switch. If the switch is OK, test and repair the front wiper and washer system circuits between the right multi-function switch and the front wiper motor or the front washer pump/motor as required.

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable.

(2) Remove the steering column opening cover from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - REMOVAL).

(3) From below the steering column, remove the two screws that secure the lower steering column shroud to the upper shroud (Fig. 21).

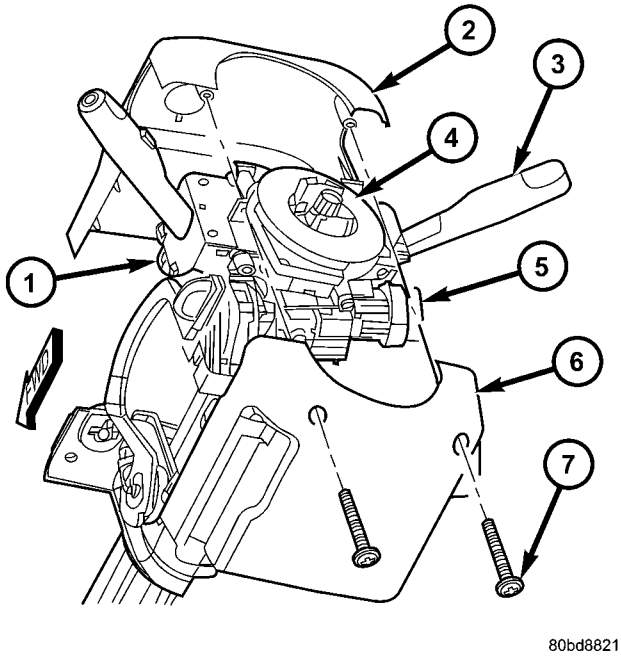

Fig. 21 Steering Column Shrouds Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - UPPER SHROUD
3 - RIGHT MULTI-FUNCTION SWITCH
4 - CLOCKSPRING
5 - IGNITION LOCK CYLINDER HOUSING
6 - LOWER SHROUD
7 - SCREW (2)

(4) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully lowered position and leave the tilt release lever in the released (down) position.

(5) Remove both the upper and lower shrouds from the steering column.

(6) Disconnect the body wire harness connector for the left multi-function switch from the connector receptacle on the back of the switch.

(7) Disconnect the body wire harness connector for the right multi-function switch from the connector receptacle on the back of the switch.

(8) Remove the two screws that secure the multi-function switch assembly to the upper steering column housing (Fig. 22).

(9) Remove the multi-function switch assembly from the upper steering column housing.

(10) Remove the two small screws that secure the right multi-function switch to the left multi-function switch mounting housing.

(11) Grasp the right multi-function switch control stalk firmly and pull the switch toward the right far enough to disengage the alignment pins on the top (1

RIGHT MULTI-FUNCTION SWITCH (Continued)

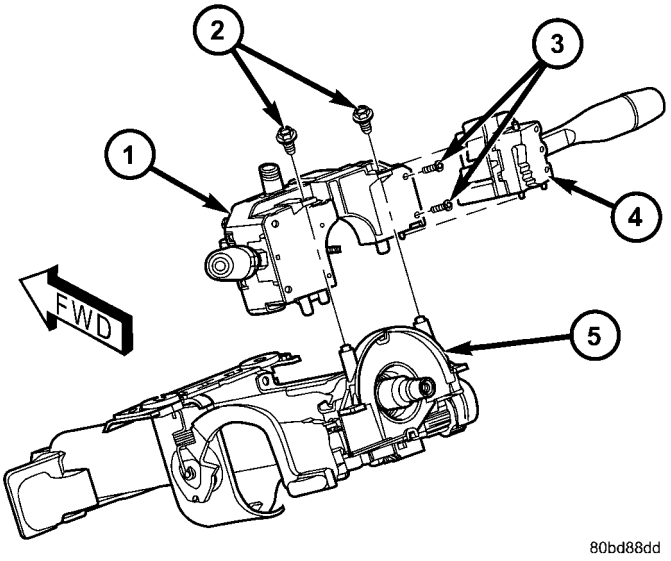

Fig. 22 Multi-Function Switch Remove/Install

1 - LEFT MULTI-FUNCTION SWITCH
2 - SCREW (2)
3 - SCREW (2)
4 - RIGHT MULTI-FUNCTION SWITCH
5 - UPPER STEERING COLUMN HOUSING

pin) and bottom (2 pins) of the right switch housing from the alignment ramps on the left switch mounting housing.

(12) Remove the right multi-function switch from the left multi-function switch mounting housing.

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Position the right multi-function switch to the mounting housing of the left multi-function switch.

(2) Grasp the right multi-function switch control stalk firmly and slide the switch to the left far enough to engage the alignment pins on the top (1 pin) and bottom (2 pins) of the right switch housing into the alignment ramps of the left multi-function switch mounting housing.

(3) Install and tighten the two small screws that secure the right multi-function switch to the left multi-function switch mounting housing (Fig. 22). Tighten the screws to 2 N·m (20 in. lbs.).

CAUTION: Before attempting to install the multi-function switch, be certain that the left switch control stalk is in the neutral turn signal position and the turn signal cancel actuator is in the retracted (neutral) position.

(4) Position the multi-function switch assembly onto the upper steering column housing.

(5) Install and tighten the two screws that secure the multi-function switch assembly to the upper steering column housing. Tighten the screws to 2 N·m (20 in. lbs.).

(6) Reconnect the body wire harness connector for the right multi-function switch to the connector receptacle on the back of the switch.

(7) Reconnect the body wire harness connector for the left multi-function switch to the connector receptacle on the back of the switch.

(8) Position both the upper and lower shrouds onto the steering column (Fig. 23). Be certain that the locating tabs for the left and right multi-function switch control stalk watershields are properly engaged in the openings of both the upper and lower shrouds.

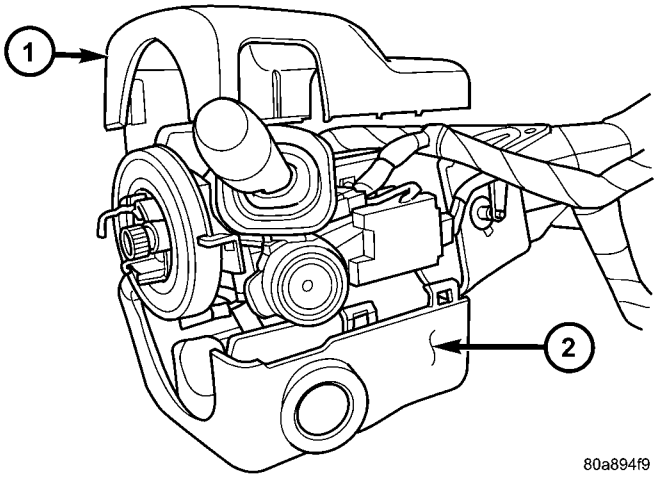

Fig. 23 Shroud Remove/Install

1 - UPPER SHROUD
2 - LOWER SHROUD

(9) From below the steering column, install and tighten the two screws that secure the lower steering column shroud to the upper shroud. Tighten the screws to 2 N·m (18 in. lbs.).

(10) If the vehicle is equipped with the optional tilt steering column, move the tilt steering column to the fully raised position and secure it in place by moving the tilt release lever back to the locked (up) position.

RIGHT MULTI-FUNCTION SWITCH (Continued)

(11) Reinstall the steering column opening cover onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(12) Reconnect the battery negative cable.

WASHER RESERVOIR

DESCRIPTION

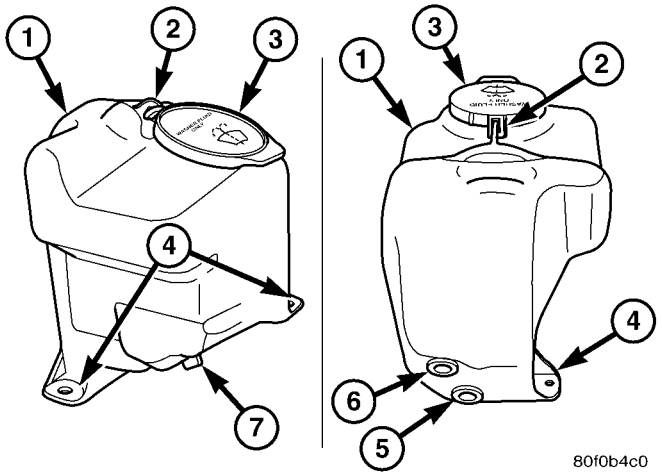

Fig. 24 Washer Reservoir

1 - WASHER RESERVOIR
2 - HOOK
3 - CAP
4 - MOUNTING TAB (2)
5 - FRONT WASHER PUMP/MOTOR HOLE
6 - REAR WASHER PUMP/MOTOR HOLE
7 - LOCATING TAB

A single washer fluid reservoir is used for both the standard front and optional rear washer systems (Fig. 24). The molded plastic washer fluid reservoir is secured to the left front fender wheel house in the engine compartment. A bright yellow plastic filler cap is labeled with the International Control and Display Symbol icon for “Windshield Washer” and the text “Washer Fluid Only” molded into it and highlighted in black. The cap hinges on a hinge molded into the cap and is secured to the reservoir by a hook molded into the top of the reservoir behind the filler neck. The cap snaps over the open end of the filler neck and is sealed with a rubber gasket.

There are separate, dedicated holes on the lower inboard side of the reservoir provided for the mounting of the front and rear washer/pump motor units. On models not equipped with the optional rear washer system, the rear washer pump/motor mounting hole in the washer reservoir is sealed with a plastic plug. The washer pump/motor units are secured and sealed to the reservoir by the interference fit between the barbed inlet nipple of each pump and a rubber grommet seal installed in each pump mounting hole. A locating tab on the bottom of the reservoir fits into a mating slot in the left front fender wheel house, and the reservoir is secured to the wheel house by two screws installed through two mounting tabs that are integral to the reservoir.

The washer reservoir cannot be repaired and, if faulty or damaged, it must be replaced. The reservoir filler cap is available for individual service replacement.

OPERATION

The washer fluid reservoir provides a secure, on-vehicle storage location for a large reserve of washer fluid for operation of the standard front and optional rear washer systems. The washer reservoir filler cap provides a clearly marked and readily accessible point from which to add washer fluid to the reservoir. The front and rear washer/pump motor units are located in a sump area on the inboard side of the reservoir to be certain that washer fluid will be available to the pumps as the fluid level in the reservoir becomes depleted. The front washer pump/motor unit is mounted in the lowest position in the sump so that the front washers will operate even after the rear washer system will no longer operate.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the two screws that secure the washer reservoir to the left front fender wheel house (Fig. 25).

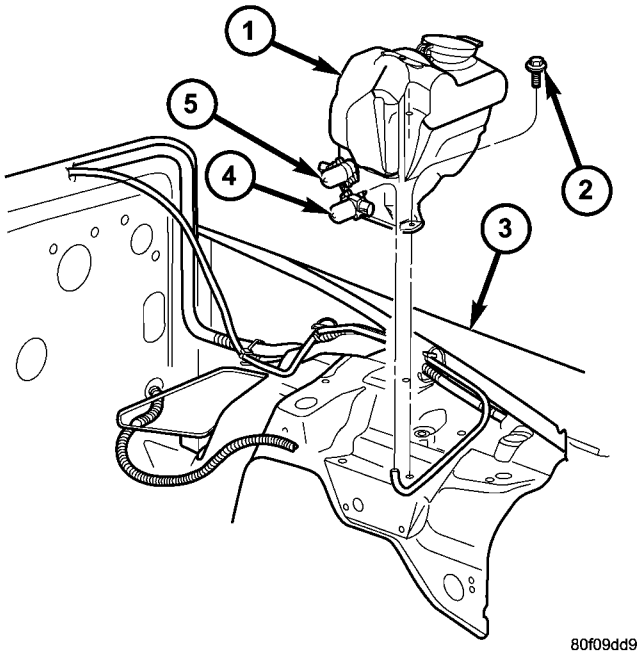

Fig. 25 Washer Reservoir Remove/Install

1 - WASHER RESERVOIR
2 - SCREW (2)
3 - LEFT FRONT FENDER
4 - FRONT WASHER PUMP/MOTOR
5 - REAR WASHER PUMP/MOTOR

WASHER RESERVOIR (Continued)

(3) Lift the reservoir far enough to access the washer pump/motor(s).

(4) Disconnect the washer supply hose(s) from the barbed outlet nipple(s) of the washer pump/motor unit(s) and allow the washer fluid from the washer reservoir to drain into a clean container for reuse.

(5) Disconnect the headlamp and dash wire harness connector for the front washer pump/motor from the connector receptacle on the motor.

(6) If the vehicle is so equipped, disconnect the rear body wire harness connector for the rear washer pump/motor from the connector receptacle on the motor.

(7) Remove the washer reservoir from the engine compartment.

INSTALLATION

(1) Position the washer reservoir into the engine compartment (Fig. 25).

(2) If the vehicle is so equipped, reconnect the rear body wire harness connector for the rear washer pump/motor to the connector receptacle on the motor.

(3) Reconnect the headlamp and dash wire harness connector for the front washer pump/motor to the connector receptacle on the motor.

(4) Reconnect the washer supply hose(s) to the barbed outlet nipple(s) of the washer pump/motor unit(s).

(5) Position the washer reservoir onto the left front fender wheel house. Be certain to insert the locating tab on the bottom of the reservoir into the locating slot in the wheel house panel.

(6) Install and tighten the two screws that secure the washer reservoir to the left front fender wheel house. Tighten the screws to 8 N·m (70 in. lbs.).

(7) Refill the washer reservoir with clean washer fluid.

(8) Reconnect the battery negative cable.

REAR WIPERS/WASHERS

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REAR WIPERS/WASHERS

DESCRIPTION

An electrically operated fixed speed rear wiper and washer system is optional factory-installed equipment on this model when it is also equipped with the optional hardtop roof (Fig. 1). The rear wiper and washer system includes the following major components, which are described in further detail elsewhere in this service information:

- **Rear Check Valve** - The rear washer system check valve is located near the belt line in the left rear corner of the passenger compartment in the vehicle. The rear check valve also serves as the in-line connector between the body wire harness half and the hardtop wire harness half of the rear washer hose plumbing.
- **Rear Washer Nozzle** - The rear washer nozzle is located in a mounting hole in the rear liftglass just below and inboard of the rear wiper motor output shaft. A plastic nut secures the threaded nipple on the back of the nozzle to the inside of the liftglass.
- **Rear Washer Plumbing** - The plumbing for the rear washer system consists of rubber hoses and molded plastic fittings. The plumbing is routed along the left side of the vehicle with the body wire harness from the washer reservoir in the engine compartment, through the dash into the passenger compartment, and back to the left rear corner of the body. The rear check valve splices the body wire harness half of the washer hose to the hardtop wire harness half of the hose at the left rear corner of the body. The hardtop wire harness and washer hose are routed up the left rear corner of the hardtop and over the upper liftglass opening header to the rear washer nozzle in the upper right corner of the rear liftglass. A molded plastic washer hose cap is secured by an integral bail strap to the body wire harness half of the rear washer hose near the left rear corner of the body. This cap serves to plug the rear washer hose when the hardtop is removed from the vehicle.

REAR WIPERS/WASHERS (Continued)

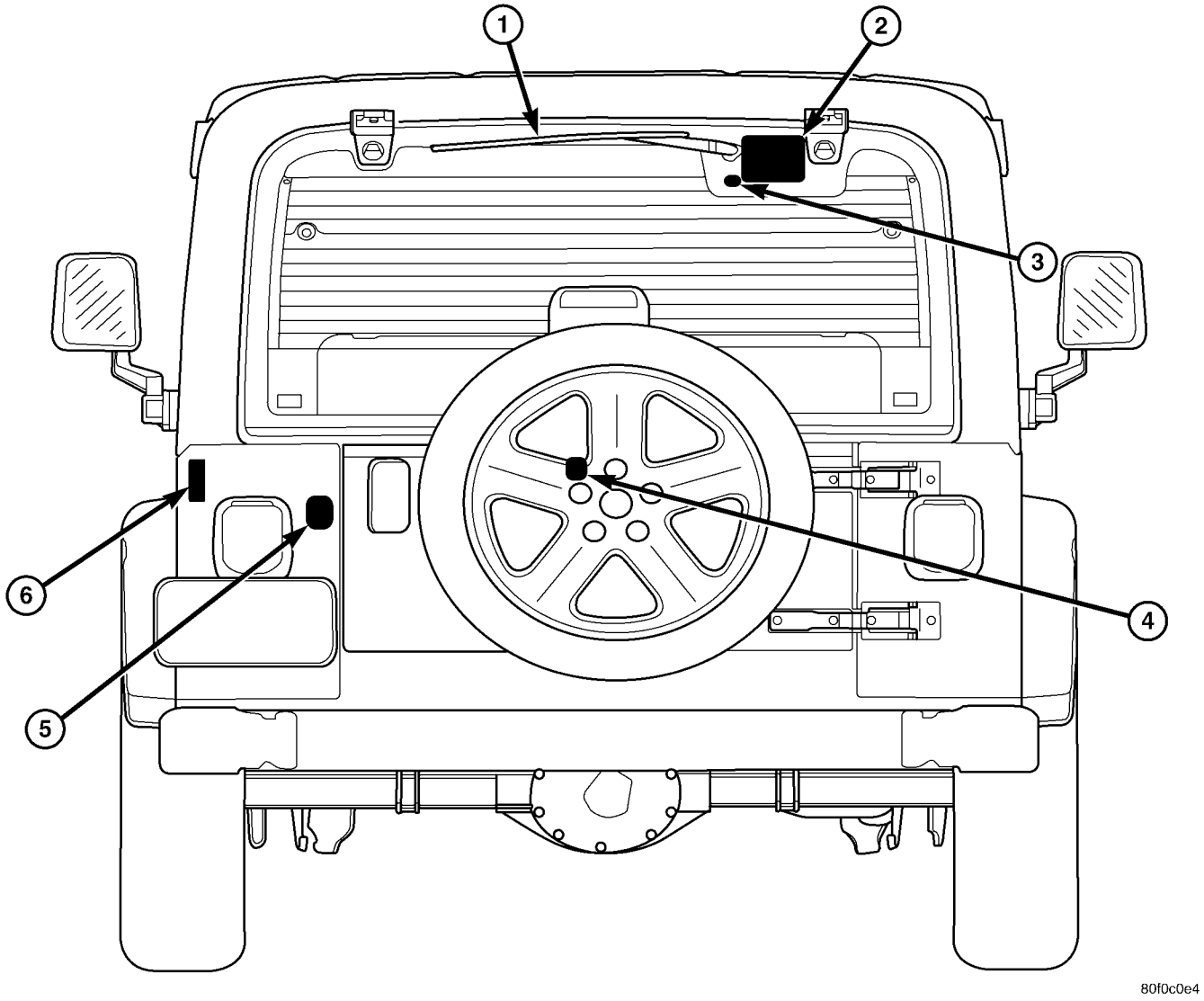

Fig. 1 Rear Wiper & Washer System

1 - WIPER ARM & BLADE
2 - WIPER MOTOR
3 - WASHER NOZZLE
4 - REAR WIPER/WASHER SWITCH
5 - REAR WASHER PUMP/MOTOR
6 - REAR WASHER HOSE CAP/CHECK VALVE

- **Rear Washer Pump/Motor** - The rear washer pump/motor unit is located in a dedicated hole on the lower inboard side of the washer reservoir, on the top of the left front fender wheel house in the engine compartment. The optional rear washer pump is located just behind and above the standard front washer pump/motor unit on the reservoir.
- **Rear Wiper and Washer Switch** - The rear wiper and washer switch is located in the accessory switch bezel near the bottom of the center stack area on the instrument panel. Only the switch rocker button is visible through the lower opening of the instrument panel center bezel, the remainder of the switch is concealed behind the accessory switch bezel within the instrument panel. The rear wiper and washer switch is dedicated to providing all of the driver controls for both the rear wiper and the rear washer systems.
- **Rear Wiper Arm** - The single rear wiper arm is secured by a nut directly to the rear wiper motor output shaft, which extends through a mounting hole in the liftglass inboard of the upper right liftglass hinge.
- **Rear Wiper Blade** - The single rear wiper blade is secured to the rear wiper arm with an integral latch, and is parked in a horizontal position near the top of the liftglass when the rear wiper system is not in operation.

- **Rear Wiper Motor** - The rear wiper motor output shaft is the only visible component of the rear wiper motor. The remainder of the motor is concealed behind a plastic trim cover on the inside of the liftglass near the upper right liftglass hinge. The rear wiper motor includes the motor bracket, the rear wiper motor, an internal automatic resetting circuit breaker, a diode, and the rear wiper motor park switch.
- **Washer Reservoir** - The rear washer system shares a single reservoir with the front washer system, but has its own dedicated washer pump/motor unit and unique plumbing provisions. The reservoir is secured to the top of the left front fender wheel house within the engine compartment of the vehicle.

Hard wired circuitry connects the rear wiper and washer system components to the electrical system of the vehicle. These hard wired circuits are integral to several wire harnesses, which are routed throughout the vehicle and retained by many different methods. These circuits may be connected to each other, to the vehicle electrical system and to the rear wiper and washer system components through the use of a combination of soldered splices, splice block connectors, and many different types of wire harness terminal connectors and insulators. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

OPERATING MODES

The components of the rear wiper and washer system are designed to work in concert to provide the following operating modes:

- **Continuous Wipe Mode** - When the rear wiper and washer switch rocker is moved to the Wipe position, the rear wiper will be operated in a continuous manner at a fixed wipe cycle speed until the switch is moved to the Off position. When the Off position is selected, the rear wiper motor will continue to operate until the current wipe cycle is complete, then park the wiper blade in a horizontal position near the top of the liftglass.
- **Washer Mode** - The rear wiper and washer switch rocker must first be moved to the Wipe detent in order to access the momentary Wash position and operate the rear washer system. When the Wash position is selected, the rear washer system will dispense washer fluid onto the liftglass and the rear wiper motor will operate in a continuous mode for as long as the switch is held closed. The rear wiper and washer switch rocker must be manually moved back to the Off position following rear washer operation in order to turn the rear wiper system off.

OPERATION

The rear wiper and washer system is designed to provide the vehicle operator with a convenient, safe, and reliable means of maintaining visibility through the rear liftglass. The various components of this system are designed to convert electrical energy produced by the vehicle electrical system into the mechanical action of the wiper blade to wipe the outside surface of the glass, as well as into the hydraulic action of the washer system to apply washer fluid stored in an on-board reservoir to the area of the glass to be wiped. When combined, these components provide the means to effectively maintain clear visibility for the vehicle operator by removing excess accumulations of rain, snow, bugs, mud, or other minor debris that might be encountered while driving the vehicle under numerous types of inclement operating conditions from the outside of the liftglass surface.

The vehicle operator initiates all rear wiper and washer system functions with the rear wiper and washer switch located in the accessory switch bezel, just below the heater and air conditioner controls in the center stack area of the instrument panel. Moving the switch rocker downward to a detent position selects the rear wiper system fixed cycle operating mode. Moving the switch rocker downward past the wipe mode detent actuates the momentary rear washer system switch.

The rear wiper and washer system will only operate when the ignition switch is in the On position. When the ignition switch is in the On position, battery current is directed from a fuse in the Power Distribution Center (PDC) to the rear wiper and washer system fuse in the fuse block. This fuse provides battery current through a fused ignition switch output (run) circuit to the rear wiper and washer switch, and to one fixed contact of the rear wiper motor park switch.

The hard wired circuits and components of the rear wiper and washer system may be diagnosed and tested using conventional diagnostic tools and procedures.

OPERATING MODES

Following are paragraphs that briefly describe the operation of each of the rear wiper and washer system operating modes.

CONTINUOUS WIPE MODE

When the Wipe position of the rear wiper and washer switch rocker is selected, the Wipe mode circuitry within the switch directs battery current

through the rear wiper motor control circuit to the rear wiper motor, which causes the rear wiper to cycle continuously at a fixed speed.

When the Off position of the rear wiper and washer switch is selected, one of two events is possible. The event that will occur depends upon the position of the wiper blade on the liftglass at the moment that the Off position is selected. If the wiper blade is in the up position on the glass when the Off position is selected, the park switch in the rear wiper motor is closed to ground through the rear wiper motor control circuit input and the wiper motor ceases to operate. If the wiper blade is not in the up position on the glass when the Off position is selected, the park switch is closed to battery current through the fused ignition switch output (run) circuit. The park switch directs this battery current to the rear wiper motor brush causing the motor to continue running until the wiper blade is in the up position on the glass, at which time the park switch opens the battery current feed to the rear wiper motor brush and is again closed to ground through the rear wiper motor control circuit input and the wiper motor ceases to operate.

WASH MODE

When the momentary Wash position of the rear wiper and washer switch rocker is selected, the Wash position circuitry within the switch directs battery current to the rear washer pump and to the rear washer motor control circuit input of the rear wiper motor, which causes both the rear washer pump and the rear wiper motor to operate for as long as the rear wiper and washer switch rocker is held in the momentary Wash position.

DIAGNOSIS AND TESTING - REAR WIPER & WASHER SYSTEM

WIPER SYSTEM

The diagnosis found here addresses an electrically inoperative rear wiper system. If the rear wiper motor operates, but the wiper does not move on the liftglass, inspect the mechanical connection between the rear wiper arm and the rear wiper motor output shaft. If OK, replace the faulty rear wiper motor. If the wiper operates, but chatters, lifts, or does not clear the glass, clean and inspect the rear wiper system components as required. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - INSPECTION) and (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - CLEANING). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Check the rear wiper and washer fuse (Fuse 6 - 20 ampere) in the fuse block. If OK, go to Step 2. If not OK, repair the shorted circuit or component as required and replace the faulty fuse.

(2) Turn the ignition switch to the On position. Check for battery voltage at rear wiper and washer fuse (Fuse 6 - 20 ampere) in the fuse block. If OK, go to Step 3. If not OK, repair the open fused ignition switch output (run) circuit between the fuse block and the ignition switch as required.

(3) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Disconnect the instrument panel wire harness connector for the rear wiper and washer switch from the switch connector receptacle. Reconnect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run) circuit cavity of the instrument panel wire harness connector for the rear wiper and washer switch. If OK, go to Step 4. If not OK, repair the open fused ignition switch output (run) circuit between the rear wiper and washer switch and the fuse block as required.

(4) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Check for continuity between the ground circuit cavity of the instrument panel wire harness connector for the rear wiper and washer switch and a good ground. There should be continuity. If OK, go to Step 5. If not OK, repair the open ground circuit to ground (G200) as required.

(5) Test the rear wiper and washer switch continuity. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER/WASHER SWITCH - DIAGNOSIS AND TESTING). If OK, go to Step 6. If not OK, replace the faulty rear wiper and washer switch.

REAR WIPERS/WASHERS (Continued)

(6) Disconnect the hardtop wire harness connector for the rear wiper motor from the motor pigtail wire connector. Reconnect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the fused ignition switch output (run) circuit cavity of the hardtop wire harness connector for the rear wiper motor. If OK, go to Step 7. If not OK, repair the open fused ignition switch output (run) circuit between the rear wiper motor and the fuse block as required.

(7) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Check for continuity between the ground circuit cavity of the hardtop wire harness connector for the rear wiper motor and a good ground. There should be continuity. If OK, go to Step 8. If not OK, repair the open ground circuit to ground (G302) as required.

(8) Check for continuity between the rear wiper motor control circuit cavity of the hardtop wire harness connector for the rear wiper motor and a good ground. There should be no continuity. If OK, go to Step 9. If not OK, repair the shorted rear wiper motor control circuit between the rear wiper motor and the rear wiper and washer switch as required.

(9) Check for continuity between the rear wiper motor control circuit cavities of the hardtop wire harness connector for the rear wiper motor and the instrument panel wire harness connector for the rear wiper and washer switch. There should be continuity. If OK, replace the faulty rear wiper motor. If not OK, repair the open rear wiper motor control circuit between the rear wiper motor and the rear wiper and washer switch as required.

WASHER SYSTEM

The diagnosis found here addresses an electrically inoperative rear washer system. If the rear washer pump/motor operates, but no washer fluid is emitted from the rear washer nozzle, be certain to check the fluid level in the reservoir. Also inspect the rear washer system components as required. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - INSPECTION). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pinout information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Turn the ignition switch to the On position. Place the rear wiper and washer switch rocker in the Wipe position. Check whether the rear wiper is operating. If OK, go to Step 2. If not OK, repair the wiper system as required before you proceed with washer system diagnosis. Refer to WIPER SYSTEM.

(2) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Disconnect the body wire harness connector for the rear washer pump/motor from the pump/motor connector receptacle. Check for continuity between the ground circuit cavity of the body wire harness connector for the rear washer pump/motor and a good ground. There should be continuity. If OK, go to Step 3. If not OK, repair the open ground circuit to ground (G302) as required.

(3) Reconnect the battery negative cable. Turn the ignition switch to the On position. Check for battery voltage at the rear washer motor control circuit cavity of the body wire harness connector for the rear washer pump/motor while the rear wiper and washer switch rocker is actuated to the Wash position. If OK, replace the faulty rear washer pump/motor unit. If not OK, go to Step 4.

(4) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Disconnect the instrument panel wire harness connector for the rear wiper and washer switch from the switch connector receptacle. Check for continuity between the rear washer motor control circuit cavities of the body wire harness connector for the rear washer pump/motor and the instrument panel wire harness connector for the rear wiper and washer switch. There should be continuity. If OK, go to Step 5. If not OK, repair the open rear washer motor control circuit between the rear wiper and washer switch and the rear washer pump/motor as required.

(5) Test the rear wiper and washer switch continuity. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER/WASHER SWITCH - DIAGNOSIS AND TESTING). If not OK, replace the faulty rear wiper and washer switch.

CLEANING - REAR WIPER & WASHER SYSTEM

WIPER SYSTEM

The squeegee of a wiper blade exposed to the elements for a long time tends to lose its wiping effectiveness. Periodic cleaning of the squeegee is

REAR WIPERS/WASHERS (Continued)

suggested to remove any deposits of salt or road film. The wiper blade, arm, and liftglass should only be cleaned using a sponge or soft cloth and windshield washer fluid, a mild detergent, or a non-abrasive cleaner. If the wiper blade continues to leave streaks, smears, hazing, or beading on the glass after thorough cleaning of the squeegees and the glass, the entire wiper blade assembly must be replaced.

CAUTION: Protect the rubber squeegee of the wiper blade from any petroleum-based cleaners, solvents, or contaminants. These products can rapidly deteriorate the rubber squeegee.

WASHER SYSTEM

If the washer system is contaminated with foreign material, drain the washer reservoir by removing the front washer pump/motor from the reservoir. Clean foreign material from the inside of both washer pump/motor inlet filter screens and the washer reservoir using clean washer fluid, a mild detergent, or a non-abrasive cleaner. Flush foreign material from the washer system plumbing by first disconnecting the rear washer supply hose from the rear washer nozzle, then running the rear washer pump/motor to run clean washer fluid or water through the system. A plugged or restricted washer nozzle should be carefully back-flushed using compressed air. If the washer nozzle obstruction cannot be cleared, replace the washer nozzle.

CAUTION: Never introduce petroleum-based cleaners, solvents, or contaminants into the washer system. These products can rapidly deteriorate the rubber seals and hoses of the washer system, as well as the rubber squeegee of the wiper blade.

CAUTION: Never use compressed air to flush the washer system plumbing. Compressed air pressures are too great for the washer system plumbing components and will result in further system damage. Never use sharp instruments to clear a plugged washer nozzle or damage to the nozzle orifice and improper nozzle spray patterns will result.

INSPECTION - REAR WIPER & WASHER SYSTEM

WIPER SYSTEM

The rear wiper blade and wiper arm should be inspected periodically, not just when wiper performance problems are experienced. This inspection should include the following points:

(1) Inspect the wiper arm for any indications of damage, or contamination. If the wiper arm is contaminated with any foreign material, clean as required. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - CLEANING). If a wiper arm is damaged or corrosion is evident, replace the wiper arm with a new unit. Do not attempt to repair a wiper arm that is damaged or corroded.

(2) Carefully lift the wiper blade off of the glass. Note the action of the wiper arm hinge. The wiper arm should pivot freely at the hinge, but with no lateral looseness evident. If there is any binding evident in the wiper arm hinge, or there is evident lateral play in the wiper arm hinge, replace the wiper arm.

CAUTION: Do not allow the wiper arm to spring back against the glass without the wiper blade in place or the glass may be damaged.

(3) Once proper hinge action of the wiper arm is confirmed, check the hinge for proper spring tension. The spring tension of the wiper arm should be sufficient to cause the rubber squeegee to conform to the curvature of the glass. Replace a wiper arm if it has low or no spring tension.

(4) Inspect the wiper blade and squeegee for any indications of damage, contamination, or rubber deterioration (Fig. 2). If the wiper blade or squeegee is contaminated with any foreign material, clean them and the glass as required. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - CLEANING). If after cleaning the wiper blade and the glass, the wiper blade fails to clear the glass without smearing, streaking, chattering, hazing, or beading, replace the wiper blade. Also, if a wiper blade is damaged or if the squeegee rubber is damaged or deteriorated, replace the wiper blade with a new unit. Do not attempt to repair a wiper blade that is damaged.

WASHER SYSTEM

The washer system components should be inspected periodically, not just when washer performance problems are experienced. This inspection should include the following points:

(1) Check for ice or other foreign material in the washer reservoir. If contaminated, clean and flush the washer system. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - CLEANING).

(2) Inspect the washer plumbing for pinched, leaking, deteriorated, or incorrectly routed hoses and damaged or disconnected hose fittings. Replace damaged or deteriorated hoses and hose fittings. Leaking washer hoses can sometimes be repaired by cutting the hose at the leak and splicing it back together using an in-line connector fitting. Similarly, sections of deteriorated hose can be cut out and replaced by splicing in new sections of hose using in-line connec-

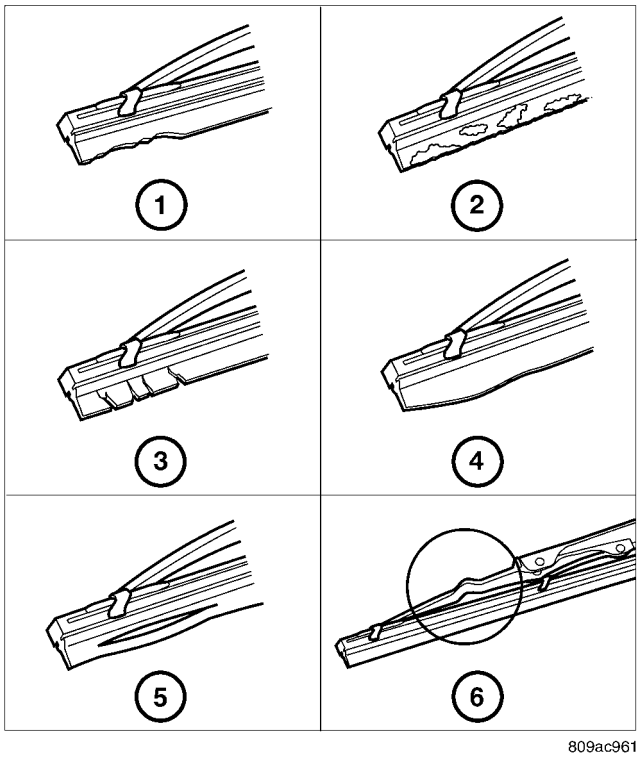

Fig. 2 Wiper Blade Inspection

1 - WORN OR UNEVEN EDGES
2 - ROAD FILM OR FOREIGN MATERIAL DEPOSITS
3 - HARD, BRITTLE, OR CRACKED
4 - DEFORMED OR FATIGUED
5 - SPLIT
6 - DAMAGED SUPPORT COMPONENTS

tor fittings. Whenever routing a washer hose or a wire harness containing a washer hose, it must be routed away from hot, sharp, or moving parts. Also, sharp bends that might pinch the washer hose must be avoided.

REAR CHECK VALVE

DESCRIPTION

Models equipped with the optional rear wiper and washer system have a rear check valve (Fig. 3). The rear check valve is located in the rear washer plumbing connection between the rear washer supply hose in the body wire harness and the rear washer supply hose in the hardtop wire harness, near the belt line in the left rear corner of the vehicle. The check valve consists of a molded plastic valve body with a raised center section that is tapered in the direction of the flow through the valve. A barbed hose nipple is formed on each side of the raised center section of the valve body. Within the check valve body, a small check ball is held against an integral valve seat by a small coiled spring. The rear check valve cannot be

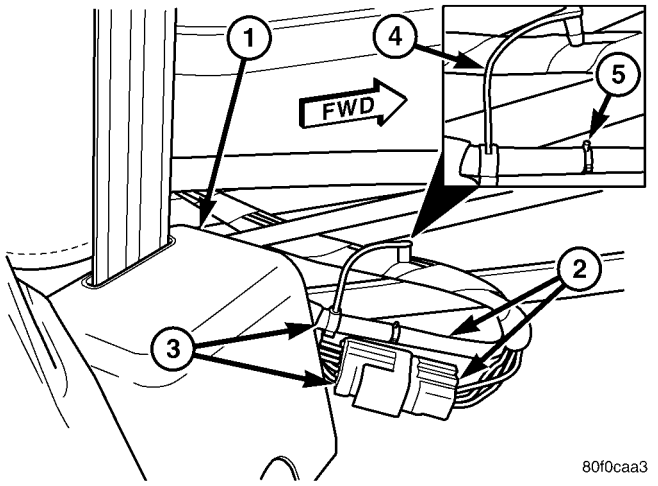

Fig. 3 Rear Check Valve

1 - LEFT REAR SEAT BELT RETRACTOR TRIM COVER
2 - HARDTOP WIRE HARNESS
3 - BODY WIRE HARNESS
4 - REAR WASHER HOSE CAP
5 - REAR CHECK VALVE

adjusted or repaired and, if faulty or damaged, it must be replaced.

OPERATION

The rear check valve provides more than one function in this application. It serves as an in-line connector fitting between the body and hardtop sections of the rear washer supply hose. It prevents washer fluid from draining out of the rear washer supply hoses back to the washer reservoir. This drain-back would result in a lengthy delay from when the rear washer switch is actuated until washer fluid was dispensed through the rear washer nozzle, because the rear washer pump would have to refill the rear washer plumbing from the reservoir to the nozzle. Such a drain-back condition could also result in water, dirt, or other outside contaminants being siphoned into the washer system through the washer nozzle orifice. This water could subsequently freeze and plug the nozzle, while other contaminants could interfere with proper nozzle operation and cause improper nozzle spray patterns. In addition, the rear check valve prevents washer fluid from siphoning out through the rear washer nozzle after the rear washer system is turned Off.

When the rear washer pump pressurizes and pumps washer fluid from the reservoir through the rear washer plumbing, the fluid pressure unseats a check ball from a seat within the valve by overriding the pressure applied to the ball within the valve by a small coiled spring (Fig. 4). With the check ball unseated, washer fluid is allowed to flow toward the rear washer nozzle. When the washer pump stops operating, the spring pressure on the check ball seats

REAR CHECK VALVE (Continued)

the ball in the valve and fluid flow in either direction within the rear washer plumbing is prevented. The rear check valve cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

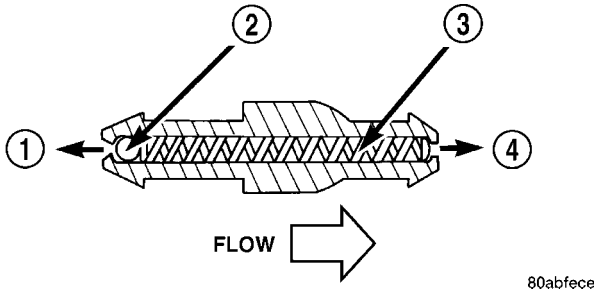

Fig. 4 Rear Check Valve

1 - TO RESERVOIR
2 - CHECK BALL
3 - SPRING
4 - TO REAR NOZZLE

REMOVAL

(1) Disconnect the hardtop wire harness half of the washer supply hose from the barbed nipple of the rear check valve (Fig. 5).

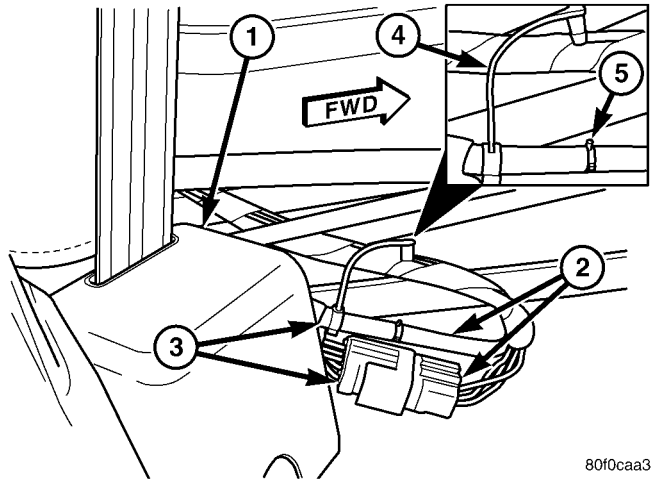

Fig. 5 Rear Check Valve Remove/Install

1 - LEFT REAR SEAT BELT RETRACTOR TRIM COVER
2 - HARDTOP WIRE HARNESS
3 - BODY WIRE HARNESS
4 - REAR WASHER HOSE CAP
5 - REAR CHECK VALVE

(2) Disconnect the body wire harness half of the washer supply hose from the other barbed nipple of the rear check valve.

(3) Remove the rear check valve from the left rear corner of the vehicle near the belt line.

INSTALLATION

(1) Position the rear check valve into the left rear corner of the vehicle near the belt line.

(2) With the tapered end of the check valve pointed in the direction of the system flow (Fig. 6), reconnect the body wire harness half of the washer supply hose to the barbed nipple of the rear check valve.

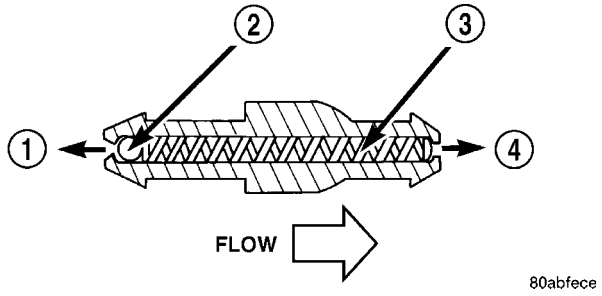

Fig. 6 Rear Washer System Check Valve

1 - TO RESERVOIR
2 - CHECK BALL
3 - SPRING
4 - TO REAR NOZZLE

(3) Reconnect the hardtop wire harness half of the washer supply hose to the other barbed nipple of the rear check valve.

REAR WASHER HOSES/TUBES

DESCRIPTION

The rear washer plumbing consists of small diameter rubber hose that is integral to and routed with the body wire harness from the barbed outlet nipple of the rear washer pump/motor on the washer reservoir through the dash panel to the left cowl side inner panel under the instrument panel. The washer hose and wire harness pass from the engine compartment into the passenger compartment through a rubber grommet in a dedicated hole near the left side of the lower dash panel.

The body wire harness and washer hose are routed from the left cowl side inner panel, along the left door opening sill, then up the B-pillar and along the upper inner edge of the left rear fender panel to the left rear corner of the passenger compartment. At the left rear corner of the passenger compartment the hose connects to the rear check valve, which also serves as an in-line connector between the body and hardtop halves of the rear washer supply hose.

There is also a washer hose cap attached by an integral clip and bail strap to the body half of the washer supply hose just below the rear check valve (Fig. 7). When the hardtop is removed from the vehicle the body half of the washer hose must be disconnected from the rear check valve. The washer hose cap is used to plug the body half of the washer hose after it is disconnected from the check valve.

REAR WASHER HOSES/TUBES (Continued)

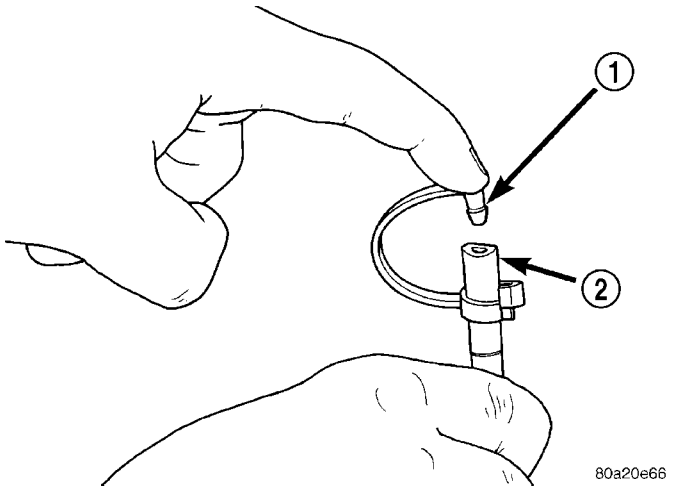

Fig. 7 Rear Washer Hose Cap

1 - WASHER HOSE CAP
2 - REAR WASHER SUPPLY HOSE - BODY HALF

The hardtop half of the rear washer supply hose is routed with the hardtop wire harness through the left rear pillar and across the upper liftglass opening reinforcement to the rear washer nozzle located near the right liftglass hinge. The hardtop washer hose is connected directly to the barbed nipple of the rear washer nozzle on the inside of the liftglass.

Washer hose is available for service only as roll stock, which must then be cut to length. For service replacement of the hose that is integral to the hardtop or body wire harnesses, it is suggested that a suitable length of washer hose be carefully routed along and secured to the outside of the harness. The molded plastic washer hose fittings cannot be repaired. If these fittings are faulty or damaged, they must be replaced.

OPERATION

Washer fluid in the washer reservoir is pressurized and fed by the rear washer pump/motor through the rear washer system plumbing and fittings to the rear washer nozzle located near the rear wiper motor output shaft on the outside of the hardtop liftglass. Whenever routing the washer hose or a wire harness containing a washer hose, it must be routed away from hot, sharp, or moving parts; and, sharp bends that might pinch the hose must be avoided.

REAR WASHER NOZZLE

DESCRIPTION

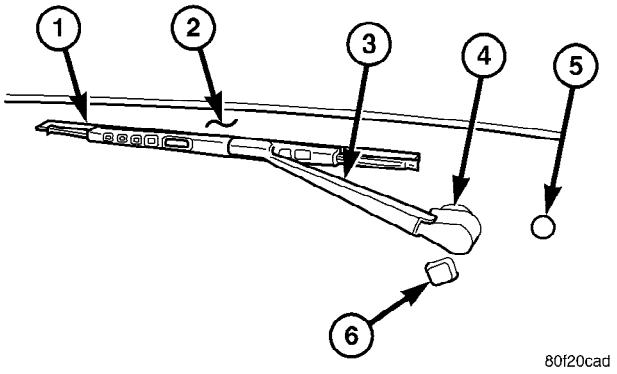

Fig. 8 Rear Washer Nozzle

1 - REAR WIPER BLADE
2 - LIFTGLASS
3 - REAR WIPER ARM
4 - REAR WIPER MOTOR OUTPUT SHAFT
5 - MOUNTING STUD
6 - REAR WASHER NOZZLE

The rear washer nozzle is a fluidic-type unit constructed of molded plastic. The molded hood of the washer nozzle is visible on the outside of the liftglass just below and inboard of the rear wiper motor output shaft (Fig. 8). The remainder of the rear washer nozzle and its plumbing connection are concealed behind the rear wiper motor trim cover secured to the rear wiper motor bracket on the inside of the liftglass. On the back of the nozzle is an externally threaded and barbed nipple that is inserted through a rubber gasket and a mounting hole from the outside of the liftglass, and is secured to the inside of the liftglass with a plastic nut. The rear washer nozzle cannot be adjusted or repaired. If faulty or damaged, the entire nozzle unit must be replaced.

OPERATION

The rear washer nozzle is designed to dispense washer fluid into the wiper pattern area on the outside of the liftglass. Pressurized washer fluid is fed to the nozzle from the washer reservoir by the rear washer pump/motor through a single hose, which is attached to a barbed nipple on the back of the rear washer nozzle behind the rear wiper motor trim cover on the inside of the liftglass. A fluidic matrix within the washer nozzle causes the pressurized washer fluid to be emitted from the nozzle orifice as a fan-like stream to more effectively cover a larger area of the glass to be cleaned.

REAR WASHER NOZZLE (Continued)

REMOVAL

(1) Remove the trim cover from the rear wiper motor on the inside of the liftglass. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - REMOVAL).

(2) From the inside of the liftglass, disconnect the rear washer supply hose from the barbed nipple on the back of the rear washer nozzle (Fig. 9).

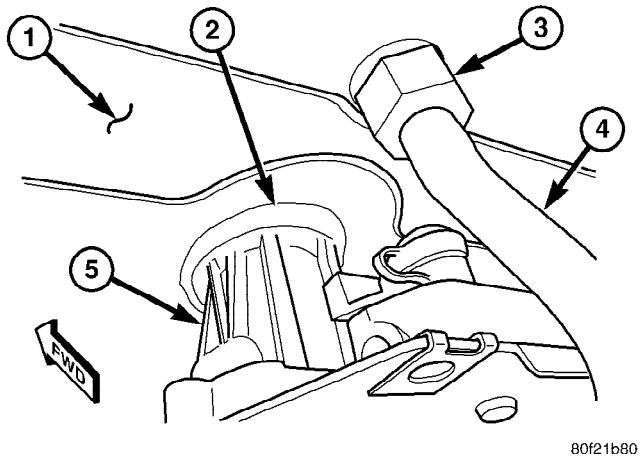

Fig. 9 Rear Washer Nozzle Remove/Install

1 - LIFTGLASS (INSIDE)
2 - GROMMET
3 - PLASTIC NUT
4 - REAR WASHER SUPPLY HOSE
5 - REAR WIPER MOTOR

(3) While holding the hood of the rear washer nozzle securely from the outside of the liftglass, remove the plastic nut that secures the threaded nipple of the rear washer nozzle to the inside of the liftglass.

(4) From the inside of the liftglass, push the nipple of the rear washer nozzle out through the mounting hole in the liftglass.

(5) Remove the rear washer nozzle and rubber gasket from the outside of the liftglass.

INSTALLATION

(1) Position the rear washer nozzle and rubber gasket to the outside of the liftglass with the orifice of the nozzle aimed at the nozzle alignment arrowhead visible on the outside of the glass.

(2) From the inside of the liftglass, install and tighten the plastic nut that secures the threaded nipple on the back of the rear washer nozzle to the inside of the liftglass (Fig. 9). Tighten the nut to 1 N·m (8 in. lbs.).

(3) Reconnect the rear washer supply hose to the barbed nipple of the rear washer nozzle.

(4) Reinstall the trim cover onto the rear wiper motor on the inside of the liftglass. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - INSTALLATION).

REAR WASHER PUMP MOTOR

DESCRIPTION

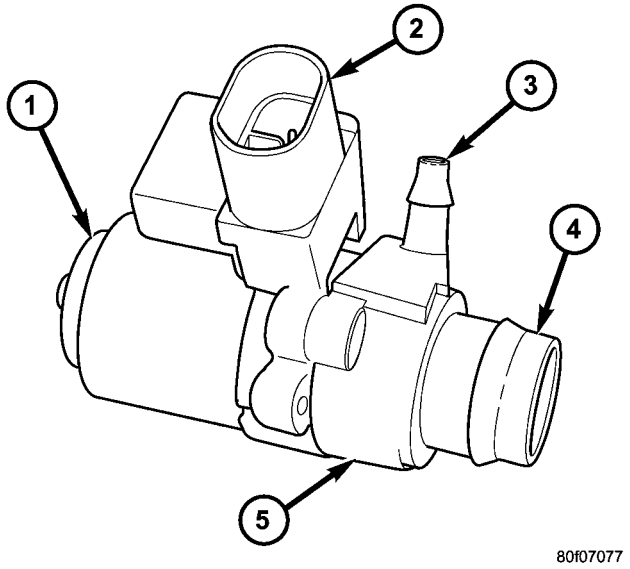

Fig. 10 Washer Pump/Motor

1 - MOTOR
2 - CONNECTOR RECEPTACLE
3 - OUTLET NIPPLE
4 - INLET NIPPLE
5 - PUMP

The rear washer pump/motor unit is located on the inboard side just behind and above the front washer pump/motor on the washer reservoir (Fig. 10). The washer reservoir is located on the top of the left front fender wheel house in the engine compartment. A small permanently lubricated and sealed electric motor is coupled to the rotor-type washer pump. A seal flange with a barbed inlet nipple on the pump housing passes through a rubber grommet seal installed in a dedicated mounting hole in the sump area near the bottom of the washer reservoir. The rear washer pump/motor unit is always mounted in the rearward-most upper mounting hole of the reservoir. An integral filter screen is located within the pump inlet nipple. When the pump is installed in the reservoir a barbed outlet nipple on the pump housing connects the unit to the washer system through the front washer supply hose.

The washer pump/motor unit is retained on the reservoir by the interference fit between the barbed pump inlet nipple and the grommet seal, which is a light press fit. An integral connector receptacle between the motor housing and the pump housing connects the unit to the vehicle electrical system. A small capacitor internal to the connector receptacle insulator is connected in parallel with the motor leads to control ElectroMagnetic Interference (EMI)

REAR WASHER PUMP MOTOR (Continued)

created by washer motor operation. The rear washer pump/motor unit cannot be repaired. If faulty or damaged, the entire washer pump/motor unit must be replaced.

OPERATION

The rear washer pump/motor unit features a small Direct Current (DC) electric motor. The motor is connected to the vehicle electrical system through a single take out and two-cavity connector of the body wire harness. The motor is grounded at all times through a take out of the body wire harness with a single eyelet terminal connector that is secured under a ground screw to the left cowl side inner panel below the instrument panel in the passenger compartment. The motor receives battery current through the closed contacts of the momentary rear washer switch circuitry within the rear wiper and washer switch unit only when the bottom of the switch rocker is pushed towards the instrument panel.

Washer fluid is gravity-fed from the washer reservoir to the inlet side of the washer pump. When the pump motor is energized, the motor spins the rotor within the rotor-type washer pump. The spinning pump rotor pressurizes the washer fluid and forces it through the pump outlet nipple, the rear washer plumbing, and the rear washer nozzle onto the liftglass. The rear washer pump/motor unit can be diagnosed using conventional diagnostic tools and methods.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the washer supply hose from the barbed outlet nipple of the rear washer pump/motor and allow the washer fluid from the washer reservoir to drain into a clean container for reuse (Fig. 11).

(3) Disconnect the body wire harness connector from the connector receptacle for the rear washer pump/motor.

(4) Using a trim stick or another suitable wide flat-bladed tool, gently pry the barbed inlet nipple of the rear washer pump/motor unit out of the rubber grommet seal in the washer reservoir. Care must be taken not to damage the washer reservoir.

(5) Remove the rear washer pump/motor unit from the washer reservoir.

(6) Remove the rubber grommet seal from the rear washer pump/motor mounting hole in the washer reservoir and discard.

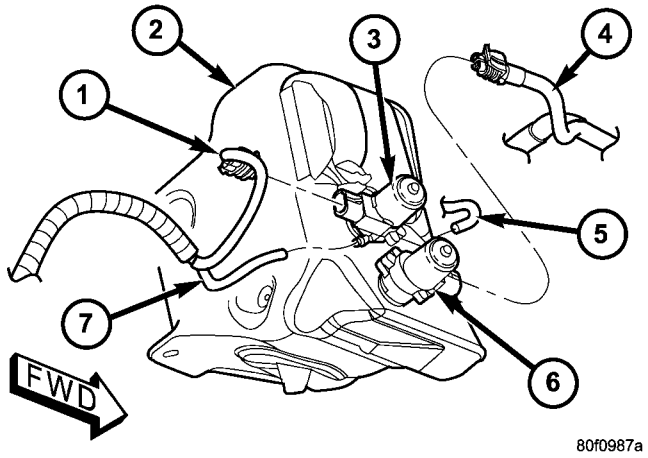

Fig. 11 Washer Pump/Motor Remove/Install

1 - REAR BODY WIRE HARNESS
2 - WASHER RESERVOIR
3 - REAR WASHER PUMP/MOTOR
4 - HEADLAMP & DASH WIRE HARNESS
5 - FRONT WASHER SUPPLY HOSE
6 - FRONT WASHER PUMP/MOTOR
7 - REAR WASHER SUPPLY HOSE

INSTALLATION

(1) Install a new rubber grommet seal into the rear washer pump/motor mounting hole of the washer reservoir.

(2) Position the rear washer pump/motor inlet nipple to the mounting hole in the washer reservoir (Fig. 11).

(3) Using hand pressure, firmly and evenly press on the rear washer pump/motor to engage the inlet nipple through the rubber grommet seal into the washer reservoir. Care must be taken not to damage the washer reservoir.

(4) Reconnect the body wire harness connector for the rear washer pump/motor to the washer pump/motor connector receptacle.

(5) Reconnect the rear washer supply hose to the barbed outlet nipple of the rear washer pump/motor.

(6) Refill the washer reservoir with the washer fluid drained from the reservoir during the removal procedure.

(7) Reconnect the battery negative cable.

REAR WIPER ARM

DESCRIPTION

The rear wiper arm is the rigid member located between the rear wiper motor output shaft that protrudes from the outside of the liftglass near the right liftglass hinge and the rear wiper blade (Fig. 12). This wiper arm features an over-center hinge that allows easy access to the liftglass for cleaning. The wiper arm has a die cast metal pivot end with a

REAR WIPER ARM (Continued)

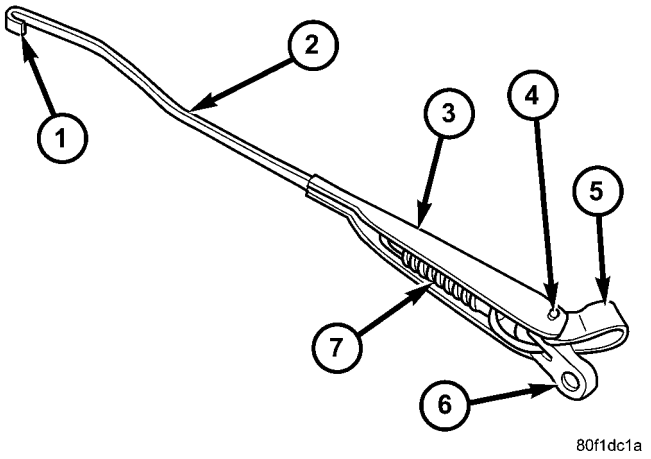

Fig. 12 Rear Wiper Arm

1 - HOOK
2 - STRAP
3 - CHANNEL
4 - HINGE PIN
5 - COVER
6 - PIVOT END
7 - TENSION SPRING

large tapered mounting hole at one end. A molded plastic pivot cover is secured loosely to and pivots on the wiper arm hinge pin to conceal the wiper arm retaining nut.

The wide end of a tapered, stamped steel channel is secured with a hinge pin to the pivot end of the wiper arm. One end of a long, rigid, stamped steel strap, with a small hole near its pivot end, is riveted and crimped within the narrow end of the stamped steel channel. The tip of the wiper blade end of this strap is bent back under itself to form a small hook. Concealed within the stamped steel channel, one end of a long spring is engaged with a wire hook on the underside of the die cast pivot end, while the other end of the spring is hooked through the small hole in the steel strap. The entire wiper arm has a satin black finish applied to all of its visible surfaces.

A wiper arm cannot be adjusted or repaired. If damaged or faulty, the entire wiper arm unit must be replaced.

OPERATION

The rear wiper arm is designed to mechanically transmit the motion from the rear wiper motor output shaft to the rear wiper blade. The wiper arm must be properly indexed to the motor output shaft in order to maintain the proper wiper blade travel on the glass. The tapered hole in the wiper arm pivot end interlocks with the serrations on the outer circumference of the tapered motor output shaft, allowing positive engagement and finite adjustment of this connection. A hex nut secures the wiper arm pivot end to the threads on the rear wiper motor output shaft and the pivot cover hinges and snaps over this connection for a neat appearance. The spring-loaded wiper arm hinge controls the down-force applied through the tip of the wiper arm to the wiper blade on the glass. The hook formation on the tip of the wiper arm provides a cradle for securing and latching the wiper blade pivot block to the wiper arm.

REMOVAL

(1) Lift the rear wiper arm pivot cover by lifting it at the rear wiper motor output shaft end of the arm (Fig. 13).

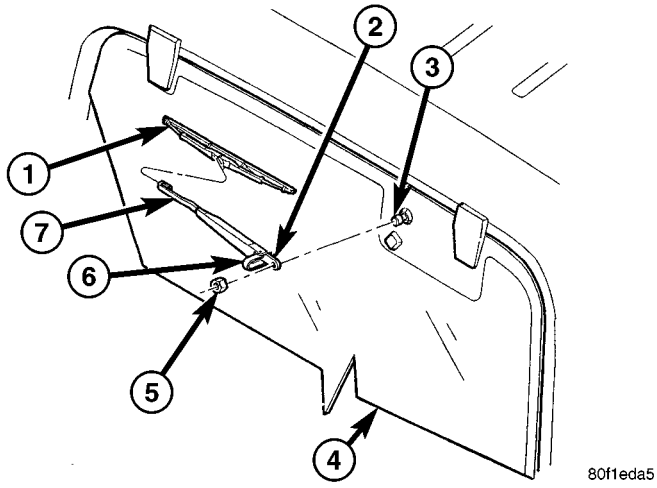

Fig. 13 Rear Wiper Arm Remove/Install

1 - LATCH
2 - WIPER ARM PIVOT END

(2) Remove the nut that secures the rear wiper arm to the rear wiper motor output shaft.

(3) Lift the rear wiper arm to its over-center position to hold the wiper blade off of the glass and relieve the spring tension on the wiper arm to rear wiper motor output shaft connection.

(4) If necessary, use a battery terminal puller to disengage the wiper arm from the rear wiper motor output shaft splines (Fig. 14).

(5) Remove the rear wiper arm pivot end from the motor output shaft.

INSTALLATION

NOTE: Be certain that the rear wiper motor is in the park position before attempting to install the wiper arm. Turn the ignition switch to the On position and move the rear wiper and washer switch rocker to its Off position. If the wiper motor output shaft moves, wait until it stops moving, then turn the ignition switch back to the Off position. The wiper motor is now in its park position.

REAR WIPER ARM (Continued)

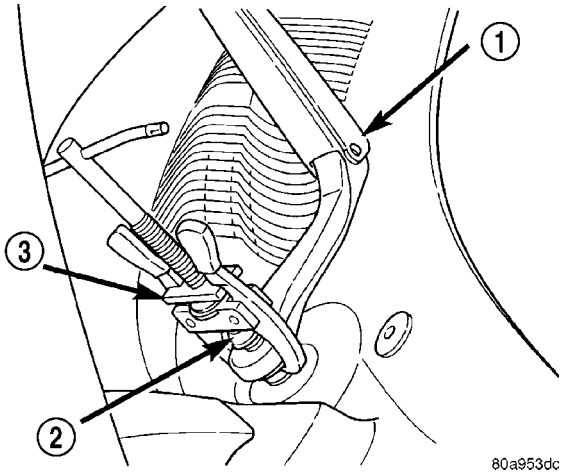

Fig. 14 Wiper Arm Puller - Typical

1 - WIPER ARM
2 - WIPER PIVOT
3 - BATTERY TERMINAL PULLER

(1) The rear wiper arm must be indexed to the rear wiper motor output shaft with the motor in the park position to be properly installed. Position the rear wiper arm pivot end onto the motor output shaft so that the wiper blade is parallel to the lower edge of the upper liftglass blackout area (Fig. 15).

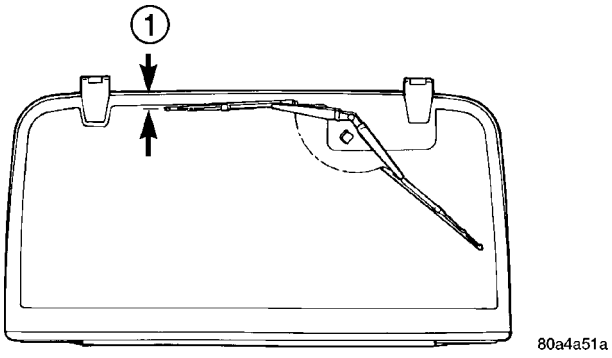

Fig. 15 Rear Wiper Arm Installation

1 - LIFTGLASS BLACKOUT AREA

(2) Once the wiper blade is indexed to the rear wiper motor output shaft, lift the rear wiper arm to its over-center position to hold the wiper blade off of the glass and relieve the spring tension on the wiper arm to rear wiper motor output shaft connection. Using hand pressure, push the tapered hole in the pivot end of the wiper arm down firmly and evenly over the rear wiper motor output shaft (Fig. 13).

(3) Install and tighten the nut that secures the wiper arm to the rear wiper motor output shaft. Tighten the nut to 12 N·m (9 ft. lbs.). **Do not overtighten.**

(4) Gently lower the wiper arm until the wiper blade rests on the glass.

(5) Wet the liftglass, then operate the rear wiper. Turn the wiper switch to the Off position, then check for the correct wiper arm position and readjust as required.

REAR WIPER BLADE

DESCRIPTION

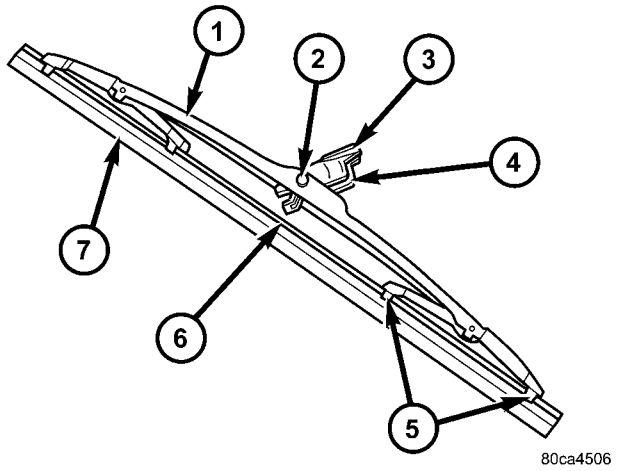

Fig. 16 Rear Wiper Blade

1 - SUPERSTRUCTURE
2 - PIVOT PIN
3 - LATCH RELEASE
4 - PIVOT BLOCK
5 - CLAW
6 - FLEXOR
7 - ELEMENT

The rear wiper blade is secured by an integral latching pivot block to the hook formation on the tip of the rear wiper arm, and rests in a horizontal position on the glass near the top of the liftglass when the wiper is not in operation (Fig. 16). The wiper blade consists of the following components:

- **Superstructure** - The superstructure includes a stamped steel bridge and plastic links with claw formations that grip the wiper blade element. Also included in this unit is the latching, molded plastic pivot block that secures the superstructure to the wiper arm. All of the metal components of the wiper blade have a satin black finish applied.
- **Element** - The wiper element or squeegee is the resilient rubber member of the wiper blade that contacts the glass.
- **Flexor** - The flexor is a rigid metal component running along the length of each side of the wiper element where it is gripped by the claws of the superstructure.

All models with the optional rear wiper and washer system have a single 46.00 centimeter (18.00 inch) rear wiper blade with a non-replaceable rubber element (squeegee). The wiper blade cannot be

REAR WIPER BLADE (Continued)

adjusted or repaired. If faulty, worn, or damaged the entire wiper blade unit must be replaced.

OPERATION

The rear wiper blade is moved back and forth across the glass by the wiper arm when the rear wiper system is in operation. The wiper blade superstructure is the flexible frame that grips the wiper blade element and evenly distributes the force of the spring-loaded wiper arm along the length of the element. The combination of the wiper arm force and the flexibility of the superstructure makes the element conform to and maintain proper contact with the liftglass, even as the blade is moved over the varied curvature found across the glass surface.

The wiper element flexor provides the claws of the blade superstructure with a rigid, yet flexible component on the element which can be gripped. The rubber element is designed to be stiff enough to maintain an even cleaning edge as it is drawn across the glass, but resilient enough to conform to the glass surface and flip from one cleaning edge to the other each time the wiper blade changes directions.

REMOVAL

NOTE: The notched end of the wiper element flexor should always be oriented towards the end of the wiper blade that is nearest to the rear wiper motor output shaft.

(1) Lift the rear wiper arm to raise the wiper blade and element off of the liftglass.

(2) To remove the wiper blade from the wiper arm, push the pivot block latch release tab under the tip of the arm and slide the blade away from the tip towards the rear wiper motor output shaft end of the arm (Fig. 17).

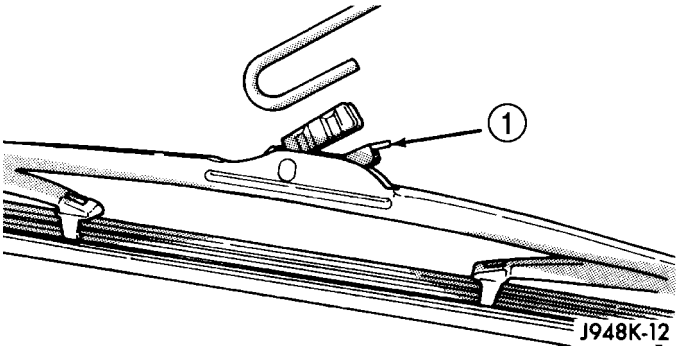

Fig. 17 Wiper Blade Remove/Install - Typical

1 - RELEASE TAB

(3) Slide the rear wiper blade away from the tip of the arm towards the pivot end of the arm far enough to disengage the pivot block from the hook formation on the end of the arm.

(4) Extract the hook formation on the tip of the wiper arm from the opening in the wiper blade superstructure ahead of the wiper blade pivot block/latch unit.

CAUTION: Do not allow the wiper arm to spring back against the liftglass without the wiper blade in place or the glass may be damaged.

(5) Gently lower the wiper arm tip onto the glass.

INSTALLATION

NOTE: The notched end of the wiper element flexor should always be oriented towards the end of the wiper blade that is nearest to the rear wiper motor output shaft.

(1) Lift the rear wiper arm off of the liftglass.

(2) Position the rear wiper blade near the hook formation on the tip of the arm with the notched end of the wiper element flexor oriented towards the end of the wiper arm that is nearest to the rear wiper motor output shaft.

(3) Insert the hook formation on the tip of the wiper arm through the opening in the wiper blade superstructure ahead of the wiper blade pivot block/latch unit far enough to engage the pivot block with the hook (Fig. 17).

(4) Slide the wiper blade pivot block/latch up into the hook formation on the tip of the wiper arm until the latch release tab snaps into its locked position.

(5) Gently lower the wiper blade onto the liftglass.

REAR WIPER MOTOR

DESCRIPTION

The rear wiper motor is concealed behind a molded plastic trim cover on the inside of the liftglass near the right liftglass hinge at the top of the glass. A large blackout area of the liftglass conceals the unit from the exterior of the vehicle. The end of the motor output shaft that protrudes through a large rubber grommet in the liftglass to drive the rear wiper arm and blade is the only visible component of the rear wiper motor (Fig. 18). A large flat washer and a nut secure and seal the motor output shaft to the outside of the liftglass, while a rubber insulator in a slot on the outboard upper corner of the motor bracket is secured by a stud and nut to the inside of the liftglass near the right liftglass hinge. The connector of a short pigtail harness is secured to a tab on the back of the motor bracket, and connects the rear wiper motor to the vehicle electrical system through a dedicated take out and connector of the hardtop

REAR WIPER MOTOR (Continued)

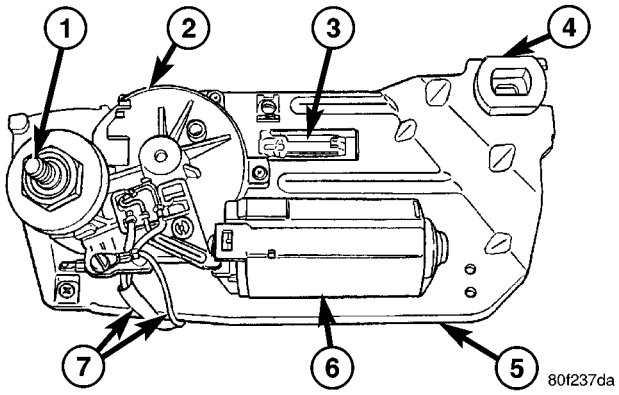

Fig. 18 Rear Wiper Motor

1 - OUTPUT SHAFT
2 - TRANSMISSION
3 - PIGTAIL CONNECTOR (ON BACK)
4 - INSULATOR
5 - BRACKET
6 - MOTOR
7 - PIGTAIL WIRES

wire harness. The rear wiper motor consists of the following major components:

- **Bracket** - The rear wiper motor bracket consists of a stamped steel mounting plate that is secured with screws to the wiper motor and transmission. A rubber insulator in a slot on the upper outboard corner of the bracket is secured by a nut and washer to a stud on the inside surface of the liftglass. An integral tab on the back of the bracket supports the wiper motor pigtail wire connector.
- **Motor** - The single-speed permanent magnet rear wiper motor is secured with screws to the rear wiper motor bracket. The wiper motor includes an integral transmission, a motor output shaft, a diode, and the rear wiper motor park switch.

The rear wiper motor cannot be adjusted or repaired. If any component of the motor is faulty or damaged, the entire rear wiper motor unit must be replaced. The motor output shaft grommet, washer and nut are available for individual service replacement.

OPERATION

The rear wiper motor operation is controlled by the vehicle operator through battery current inputs received by the rear wiper motor from the rear wiper and washer switch on the instrument panel, and the rear wiper motor park switch, which control current flow to the wiper motor brushes. An internal diode protects the motor from feedback through the park switch when the switch is closed to the fused ignition switch output (run) circuit. An internal circuit breaker protects the motor from overloads.

The park switch is a single pole, single throw, momentary switch within the wiper motor that is mechanically actuated by the wiper motor transmission components. The park switch alternately closes the wiper motor brush to the rear wiper and washer switch output or to a fused ignition switch output (run) circuit, depending upon the position of the wiper on the glass. This feature allows the motor to complete its current wipe cycle after the wiper system has been turned Off, and to park the wiper blade in the uppermost portion of the wipe pattern.

The wiper motor transmission converts the rotary output of the wiper motor to the back and forth wiping motion of the rear wiper arm and blade on the liftglass. The rear wiper motor may be diagnosed using conventional diagnostic tools and methods.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Remove the rear wiper arm from the rear wiper motor output shaft. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER ARM - REMOVAL).

(3) Remove the nut that secures the threaded sleeve of the rear wiper motor output shaft to the outside of the liftglass (Fig. 19).

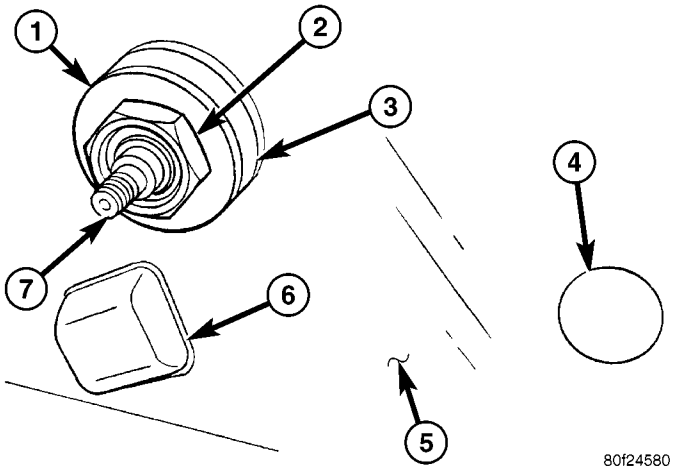

Fig. 19 Rear Wiper Motor Output Shaft

1 - WASHER
2 - NUT
3 - GROMMET
4 - STUD
5 - LIFTGLASS
6 - REAR WASHER NOZZLE
7 - REAR WIPER MOTOR OUTPUT SHAFT

(4) Remove the washer from the rear wiper motor output shaft sleeve.

(5) Remove the trim cover from the rear wiper motor on the inside of the liftglass. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - REMOVAL).

REAR WIPER MOTOR (Continued)

(6) Disconnect the hardtop wire harness connector for the rear wiper motor from the motor pigtail wire connector.

(7) Remove the nut and washer that secures the rear wiper motor bracket insulator to the stud on the liftglass (Fig. 20).

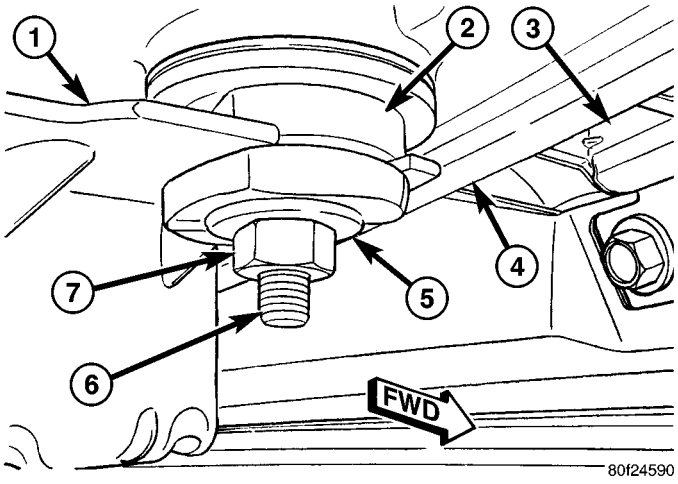

Fig. 20 Rear Wiper Motor Bracket

1 - BRACKET
2 - INSULATOR
3 - RIGHT LIFTGLASS HINGE
4 - LIFTGLASS
5 - WASHER
6 - STUD
7 - NUT

(8) Remove the washer from the stud that secures the rear wiper motor bracket insulator to the liftglass.

(9) From the inside of the liftglass, gently pull the rear wiper motor away from the liftglass until the output shaft clears the rubber grommet in the liftglass output shaft hole.

(10) Remove the rear wiper motor and mounting bracket from the liftglass as a unit.

(11) Remove the rubber grommet from the outside of the liftglass output shaft hole.

INSTALLATION

(1) Install the rubber grommet from the outside of the liftglass output shaft hole. Be certain that the alignment arrow molded into the outer flange of the grommet is aligned with the arrowhead imprinted on the glass below the output shaft hole.

(2) Position the rear wiper motor to the inside of the liftglass with the output shaft inserted through the rubber grommet and the mounting bracket insulator installed over the stud on the inside of the liftglass (Fig. 20).

(3) Reinstall the washer onto the stud that secures the rear wiper motor bracket insulator to the liftglass.

(4) Install and tighten the nut and washer that secures the rear wiper motor bracket insulator to the stud on the liftglass. Tighten the nut to 6 N·m (54 in. lbs.).

(5) From the outside of the liftglass, reinstall the washer over the rear wiper motor output shaft sleeve (Fig. 19).

(6) Install and tighten the nut that secures the threaded sleeve of the rear wiper motor output shaft to the outside of the liftglass. Tighten the nut to 5 N·m (44 in. lbs.).

(7) Reconnect the hardtop wire harness connector for the rear wiper motor to the motor pigtail wire connector.

(8) Reinstall the trim cover over the rear wiper motor on the inside of the liftglass. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER MOTOR TRIM COVER - INSTALLATION).

(9) Reinstall the rear wiper arm onto the rear wiper motor output shaft. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS/REAR WIPER ARM - INSTALLATION).

(10) Reconnect the battery negative cable.

REAR WIPER MOTOR TRIM COVER

REMOVAL

(1) From the inside of the liftglass, remove the three screws that secure the rear wiper motor trim cover to the motor mounting bracket (Fig. 21).

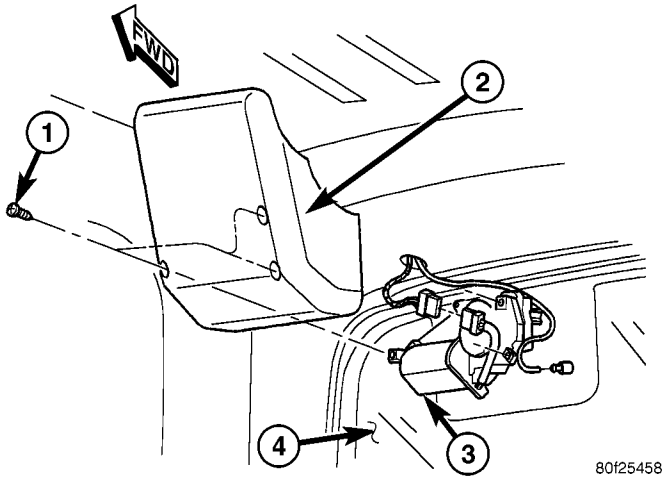

Fig. 21 Rear Wiper Motor Trim Cover Remove/Install

1 - SCREW (3)
2 - TRIM COVER
3 - REAR WIPER MOTOR
4 - LIFTGLASS

(2) Remove the trim cover from the rear wiper motor mounting bracket.

REAR WIPER MOTOR TRIM COVER (Continued)

INSTALLATION

(1) From the inside of the liftglass, position the trim cover onto the rear wiper motor mounting bracket (Fig. 21).

(2) Install and tighten the three screws that secure the trim cover to the rear wiper motor mounting bracket. Tighten the screws to 1 N·m (10 in. lbs.).

REAR WIPER/WASHER SWITCH

DESCRIPTION

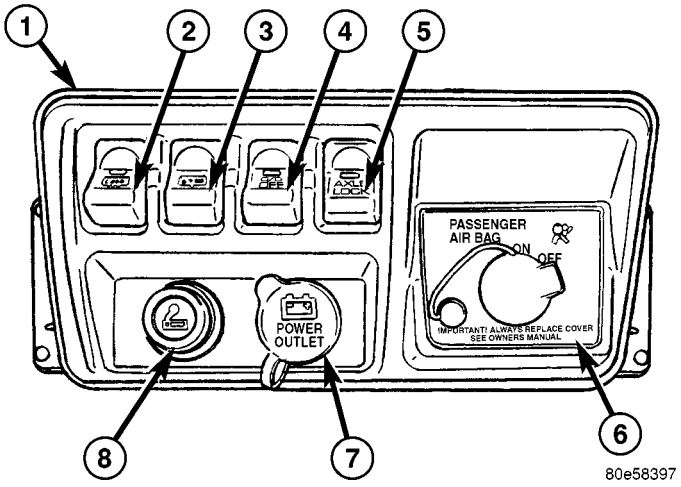

Fig. 22 Accessory Switch Bezel

1 - ACCESSORY SWITCH BEZEL
2 - REAR WINDOW DEFOGGER SWITCH (HARDTOP ONLY)
3 - REAR WIPER/WASHER SWITCH (HARDTOP ONLY)
4 - OVERDRIVE-OFF SWITCH (AUTOMATIC TRANSMISSION ONLY)
5 - AXLE LOCK SWITCH (OFF ROAD PACKAGE ONLY)
6 - PASSENGER AIRBAG ON/OFF SWITCH (WITHOUT REAR SEAT ONLY)
7 - ACCESSORY POWER OUTLET
8 - CIGAR LIGHTER

The rear wiper and washer switch is located in the accessory switch bezel near the bottom of the instrument panel center stack area on the instrument panel (Fig. 22). Only the single switch toggle button is visible through the opening of the accessory switch bezel (Fig. 23). The remainder of the switch is concealed behind the accessory switch bezel within the instrument panel. The single two-function switch housing and switch toggle button are molded from black plastic. The switch toggle button is clearly identified by a white graphic of the International Control and Display Symbol icon for “Rear Wiper and Washer,” making it clearly visible in daylight. When illuminated from behind by an integral panel lamps dimmer controlled illumination lamp with the exterior lamps turned On, the white graphic appears blue-green.

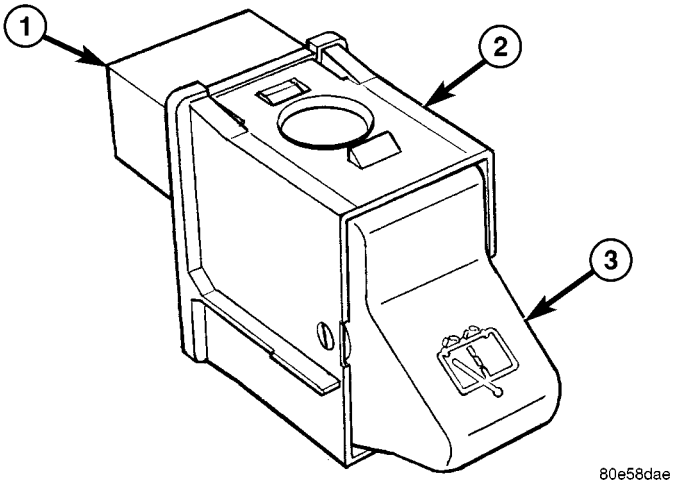

Fig. 23 Rear Wiper/Washer Switch

1 - CONNECTOR RECEPTACLE
2 - HOUSING
3 - TOGGLE BUTTON

The rear wiper and washer switch is secured by a snap fit within a dedicated receptacle molded into the back of the accessory switch bezel. A single six pin connector receptacle is molded into the back of the switch housing. A dedicated take out and connector of the instrument panel wire harness connects the switch to the vehicle electrical system. The rear wiper and washer switch contains switches and circuitry to control both the rear wiper and the rear washer functions. The rear wiper and washer switch cannot be repaired and, if faulty or damaged, the entire switch unit must be replaced. The incandescent switch illumination bulb and bulb holder unit is available for individual service replacement.

OPERATION

The rear wiper and washer switch uses conventionally switched outputs to control the functions and features of the rear wiper and washer system. The switch receives battery current on a fused ignition switch output (run) circuit from a fuse in the Power Distribution Center (PDC) through the rear wiper and washer system fuse in the fuse block whenever the ignition switch is in the On position. The switch receives a path to ground at all times through a through a take out of the instrument panel wire harness with an eyelet terminal connector that is secured by a ground screw to the support structure near the driver side end of the instrument panel.

The rear wiper and washer switch features detents in the Off and Wipe positions, and a momentary Wash position. When the switch toggle button is in the Off position it provides a ground path to the park switch within the rear wiper motor. The park switch uses this ground path to operate the rear wiper motor until the wiper blade is in its parked position,

REAR WIPER/WASHER SWITCH (Continued)

or until the ignition switch is turned to the Off position, whichever occurs first. When the switch toggle button is depressed downward to its Wipe detent position, the switch provides ignition switched battery current to the rear wiper motor and the rear wiper system operates. When the switch toggle button is depressed downward to the momentary Wash position, the switch provides ignition switched battery current to both the rear wiper motor and the rear washer pump motor for as long as the switch is held depressed. Refer to the owner's manual in the vehicle glove box for more information on the features, use and operation of the rear wiper and washer switch.

The incandescent switch illumination lamp is controlled by a lamp driver output from the instrument cluster based upon the panel lamps dimmer input to the cluster from the left multi-function switch. The rear wiper and washer switch can be diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - REAR WIPER & WASHER SWITCH

Be certain to perform the diagnosis for the rear wiper system and/or rear washer system before testing the rear wiper and washer switch. (Refer to 8 - ELECTRICAL/REAR WIPERS/WASHERS - DIAGNOSIS AND TESTING). Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the accessory switch bezel from the instrument panel and disconnect the instrument panel wire harness connector for the rear wiper and washer switch from the switch connector receptacle.

(2) Using an ohmmeter, check the rear wiper and washer switch continuity at the switch terminals as shown in the Rear Wiper & Washer Switch Continuity chart (Fig. 24).

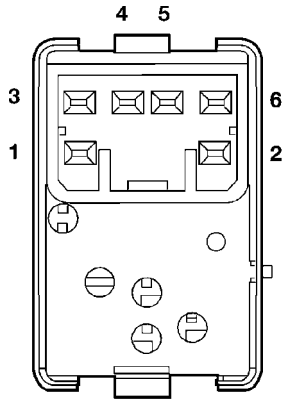

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Fig. 24 Rear Wiper & Washer Switch Continuity

SWITCH POSITION	CONTINUITY BETWEEN
OFF	1 & 4
WIPE	4 & 5
WASH	2 & 5, 4 & 5
LAMP	1 & 3

(3) If the switch fails any of the continuity checks, replace the faulty switch. If the switch checks OK, repair the open or shorted rear wiper and/or rear washer system circuits as required.

REMOVAL

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

REAR WIPER/WASHER SWITCH (Continued)

(1) Disconnect and isolate the battery negative cable.

(2) Remove the accessory switch bezel from the instrument panel center bezel from the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ ACCESSORY SWITCH BEZEL - REMOVAL).

(3) Using a trim stick or another suitable wide flat-bladed tool, gently pry the latch tabs at the top and bottom of the rear wiper and washer switch receptacle on the back of the accessory switch bezel far enough to disengage the snap features on the top and bottom of the switch housing then pull the switch out of the receptacle (Fig. 25).

INSTALLATION

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Position the rear wiper and washer switch over the proper receptacle on the back of the accessory switch bezel (Fig. 25).

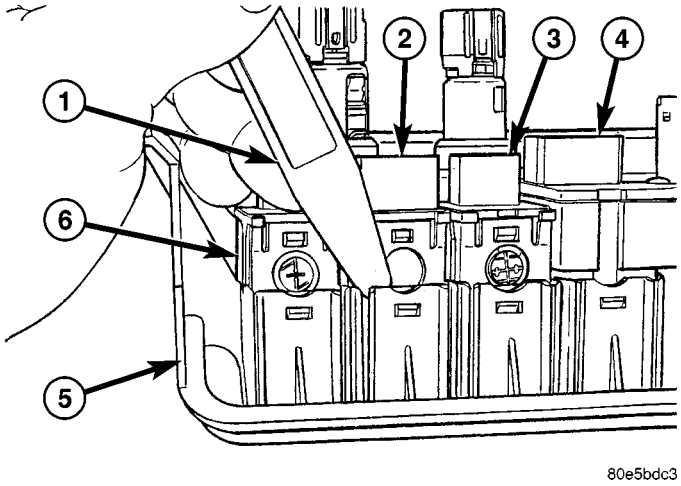

Fig. 25 Rear Wiper/Washer Switch Remove

1 - TRIM STICK
2 - REAR WIPER/WASHER SWITCH
3 - OVERDRIVE-OFF SWITCH
4 - AXLE LOCKER SWITCH
5 - ACCESSORY SWITCH BEZEL
6 - REAR WINDOW DEFOGGER SWITCH

(2) Gently and evenly push the rear wiper and washer switch into the receptacle until the snap features on the top and bottom of the switch housing are fully engaged by the latch tabs at the top and bottom of the receptacle.

(3) Reinstall the accessory switch bezel onto the instrument panel. (Refer to 23 - BODY/INSTRUMENT PANEL/ACCESSORY SWITCH BEZEL - INSTALLATION).

(4) Reconnect the battery negative cable.

WIRING

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WIRING DIAGRAM INFORMATION

DESCRIPTION

DESCRIPTION - HOW TO USE WIRING DIAGRAMS

DaimlerChrysler Corporation wiring diagrams are designed to provide information regarding the vehicles wiring content. In order to effectively use the wiring diagrams to diagnose and repair DaimlerChrysler Corporation vehicles, it is important to understand all of their features and characteristics.

Diagrams are arranged such that the power (B+) side of the circuit is placed near the top of the page, and the ground (B-) side of the circuit is placed near the bottom of the page (Fig. 1).

All switches, components, and modules are shown in the at rest position with the doors closed and the key removed from the ignition (Fig. 2).

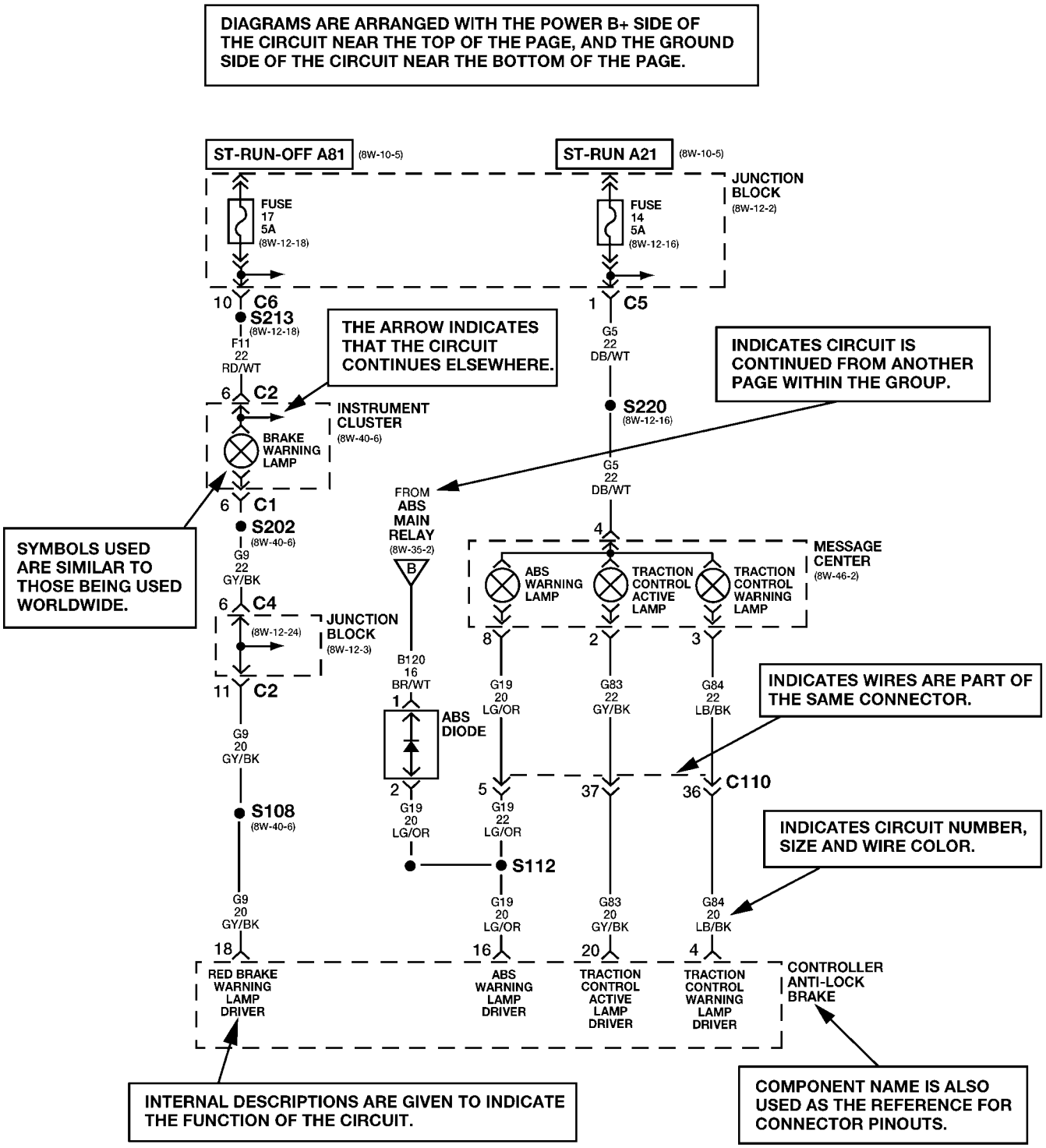
Components are shown two ways. A solid line around a component indicates that the component is complete. A dashed line around the component indicates that the component is being shown is not complete. Incomplete components have a reference number to indicate the page where the component is shown complete.

It is important to realize that no attempt is made on the diagrams to represent components and wiring as they appear on the vehicle. For example, a short piece of wire is treated the same as a long one. In addition, switches and other components are shown as simply as possible, with regard to function only.

SYMBOLS

International symbols are used throughout the wiring diagrams. These symbols are consistent with those being used around the world (Fig. 3).

WIRING DIAGRAM INFORMATION (Continued)



The System shown here is an EXAMPLE ONLY. It does not represent the actual circuit shown in the WIRING DIAGRAM SECTION.

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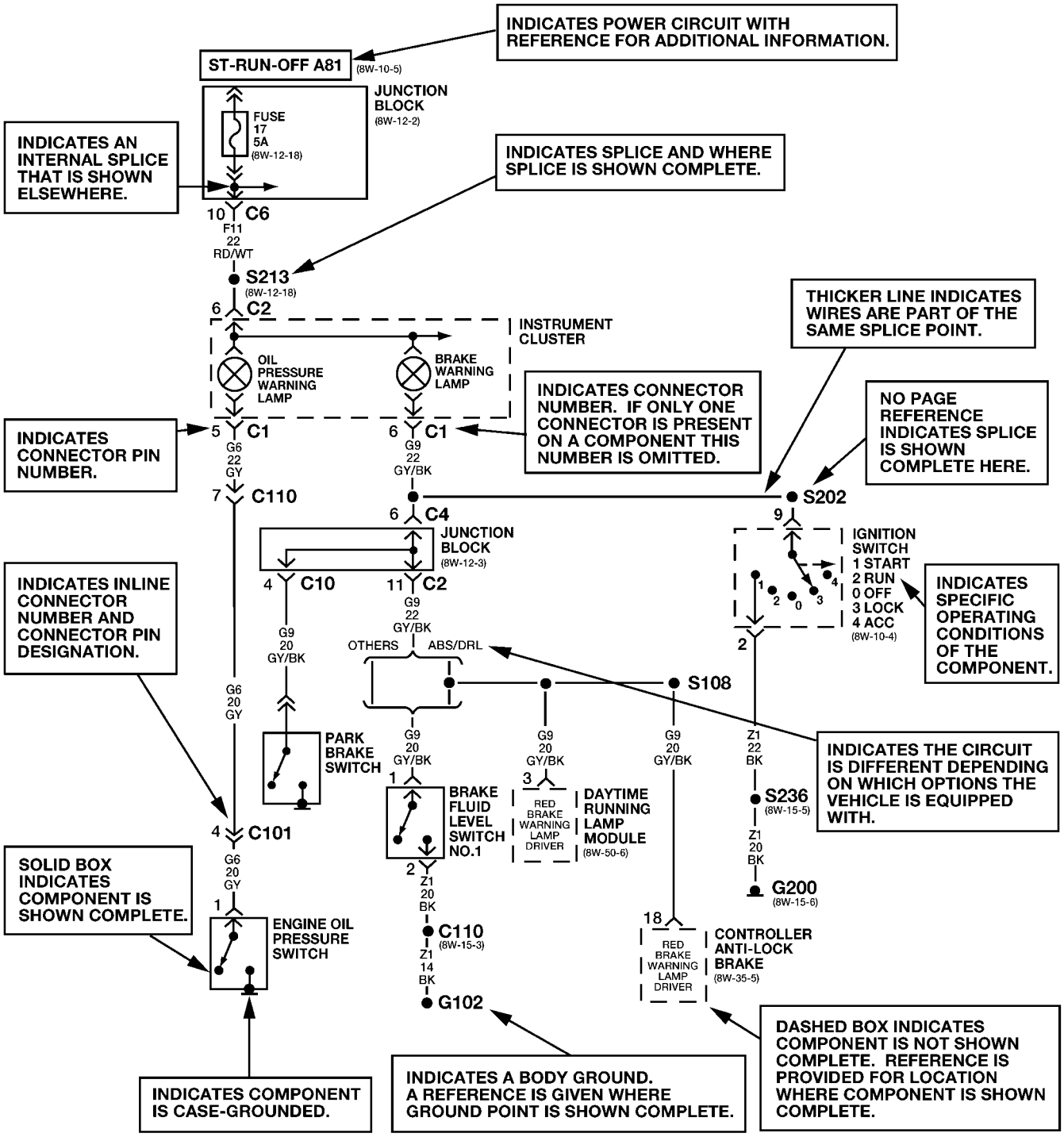
Fig. 1 WIRING DIAGRAM EXAMPLE 1

WIRING DIAGRAM INFORMATION (Continued)



The System shown here is an EXAMPLE ONLY. It does not represent the actual circuit shown in the WIRING DIAGRAM SECTION.

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Fig. 2 WIRING DIAGRAM EXAMPLE 2

WIRING DIAGRAM INFORMATION (Continued)

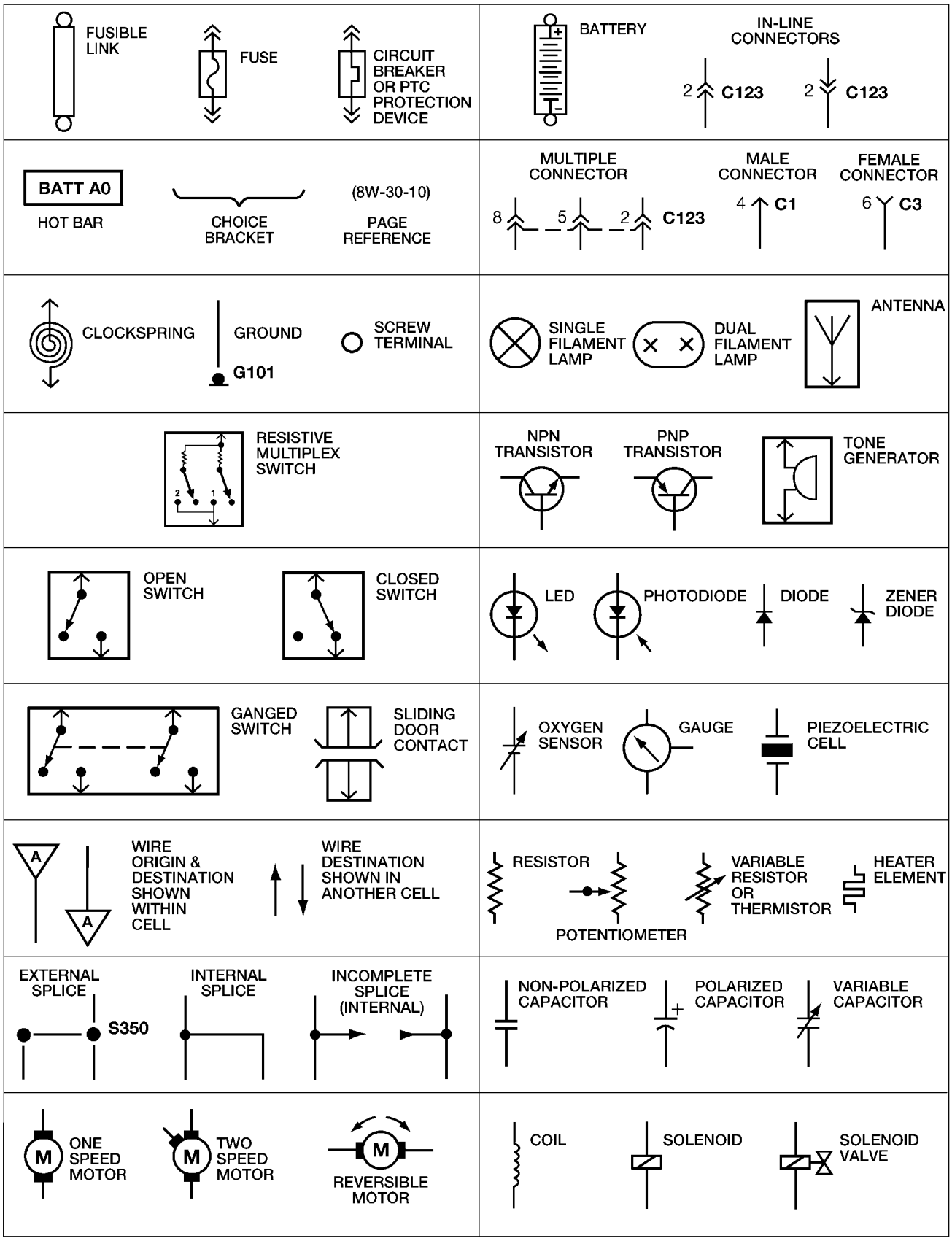


Fig. 3 WIRING DIAGRAM SYMBOLS

WIRING DIAGRAM INFORMATION (Continued)

TERMINOLOGY

This is a list of terms and definitions used in the wiring diagrams.

LHD . . . . . Left Hand Drive Vehicles
RHD . . . . . Right Hand Drive Vehicles
ATX . . Automatic Transmissions-Front Wheel Drive
MTX . . . Manual Transmissions-Front Wheel Drive
AT . . . . Automatic Transmissions-Rear Wheel Drive
MT . . . . . Manual Transmissions-Rear Wheel Drive
SOHC . . . . . Single Over Head Cam Engine
DOHC . . . . . Double Over Head Cam Engine
Export . . Vehicles Built For Sale In Markets Other Than North America
Except Export . . . . Vehicles Built For Sale In North America

DESCRIPTION - CIRCUIT INFORMATION

Each wire shown in the diagrams contains a code which identifies the main circuit, part of the main circuit, gage of wire, and color (Fig. 4).

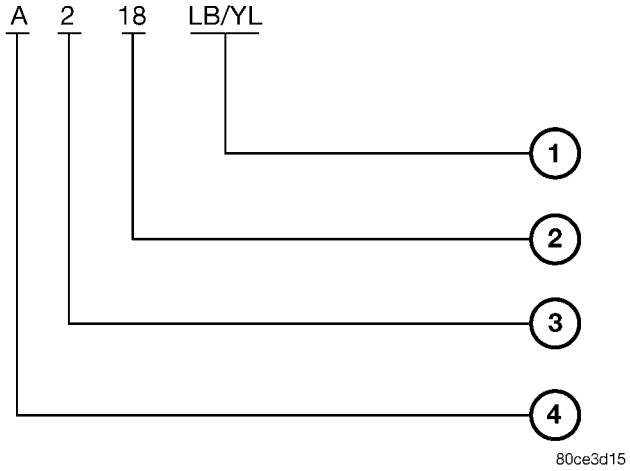


Fig. 4 WIRE CODE IDENTIFICATION

1 - COLOR OF WIRE (LIGHT BLUE WITH YELLOW TRACER
2 - GAGE OF WIRE (18 GAGE)
3 - PART OF MAIN CIRCUIT (VARIES DEPENDING ON EQUIPMENT)
4 - MAIN CIRCUIT IDENTIFICATION

WIRE COLOR CODE CHART

COLOR CODE	COLOR
BL	BLUE
BK	BLACK
BR	BROWN
DB	DARK BLUE
DG	DARK GREEN
GY	GRAY
LB	LIGHT BLUE
LG	LIGHT GREEN
OR	ORANGE
PK	PINK
RD	RED
TN	TAN
VT	VIOLET
WT	WHITE
YL	YELLOW
*	WITH TRACER

WIRING DIAGRAM INFORMATION (Continued)

DESCRIPTION - CIRCUIT FUNCTIONS

All circuits in the diagrams use an alpha/numeric code to identify the wire and it's function. To identify which circuit code applies to a system, refer to the Circuit Identification Code Chart. This chart shows the main circuits only and does not show the secondary codes that may apply to some models.

CIRCUIT IDENTIFICATION CODE CHART

CIRCUIT	FUNCTION
A	BATTERY FEED
B	BRAKE CONTROLS
C	CLIMATE CONTROLS
D	DIAGNOSTIC CIRCUITS
E	DIMMING ILLUMINATION CIRCUITS
F	FUSED CIRCUITS
G	MONITORING CIRCUITS (GAUGES)
H	MULTIPLE
I	NOT USED
J	OPEN
K	POWERTRAIN CONTROL MODULE
L	EXTERIOR LIGHTING
M	INTERIOR LIGHTING
N	MULTIPLE
O	NOT USED
P	POWER OPTION (BATTERY FEED)
Q	POWER OPTIONS (IGNITION FEED)
R	PASSIVE RESTRAINT
S	SUSPENSION/STEERING
T	TRANSMISSION/TRANSAXLE/ TRANSFER CASE
U	OPEN
V	SPEED CONTROL, WIPER/ WASHER
W	WIPERS
X	AUDIO SYSTEMS
Y	OPEN
Z	GROUNDS

DESCRIPTION - SECTION IDENTIFICATION AND INFORMATION

The wiring diagrams are grouped into individual sections. If a component is most likely found in a particular group, it will be shown complete (all wires, connectors, and pins) within that group. For example, the Auto Shutdown Relay is most likely to be found in Group 30, so it is shown there complete. It can, however, be shown partially in another group if it contains some associated wiring.

Splice diagrams in Section 8W-70 show the entire splice and provide references to other sections the splices serves. Section 8W-70 only contains splice diagrams that are not shown in their entirety somewhere else in the wiring diagrams.

Section 8W-80 shows each connector and the circuits involved with that connector. The connectors are identified using the name/number on the diagram pages.

WIRING SECTION CHART

GROUP	TOPIC
8W-01 thru 8W-09	General information and Diagram Overview
8W-10 thru 8W-19	Main Sources of Power and Vehicle Grounding
8W-20 thru 8W-29	Starting and Charging
8W-30 thru 8W-39	Powertrain/Drivetrain Systems
8W-40 thru 8W-49	Body Electrical items and A/C
8W-50 thru 8W-59	Exterior Lighting, Wipers and Trailer Tow
8W-60 thru 8W-69	Power Accessories
8W-70	Splice Information
8W-80	Connector Pin Outs
8W-91	Connector, Ground and Splice Locations

WIRING DIAGRAM INFORMATION (Continued)

DESCRIPTION - CONNECTOR, GROUND AND SPLICE INFORMATION

CAUTION: Not all connectors are serviced. Some connectors are serviced only with a harness. A typical example might be the Supplemental Restraint System connectors. Always check parts availability before attempting a repair.

IDENTIFICATION

In-line connectors are identified by a number, as follows:

- In-line connectors located in the engine compartment are C100 series numbers
- In-line connectors located in the Instrument Panel area are C200 series numbers.
- In-line connectors located in the body are C300 series numbers.
- Jumper harness connectors are C400 series numbers.
- Grounds and ground connectors are identified with a “G” and follow the same series numbering as the in-line connectors.
- Splices are identified with an “S” and follow the same series numbering as the in-line connectors.
- Component connectors are identified by the component name instead of a number. Multiple connectors on a component use a C1, C2, etc. identifier.

LOCATIONS

Section 8W-91 contains connector/ground/splice location illustrations. The illustrations contain the connector name (or number)/ground number/splice number and component identification. Connector/ground/splice location charts in section 8W-91 reference the figure numbers of the illustrations.

The abbreviation T/O is used in the component location section to indicate a point in which the wiring harness branches out to a component. The abbreviation N/S means Not Shown in the illustrations

WARNINGS - GENERAL

WARNINGS provide information to prevent personal injury and vehicle damage. Below is a list of general warnings that should be followed any time a vehicle is being serviced.

WARNING: ALWAYS WEAR SAFETY GLASSES FOR EYE PROTECTION.

WARNING: USE SAFETY STANDS ANYTIME A PROCEDURE REQUIRES BEING UNDER A VEHICLE.

WARNING: BE SURE THAT THE IGNITION SWITCH ALWAYS IS IN THE OFF POSITION, UNLESS THE PROCEDURE REQUIRES IT TO BE ON.

WARNING: SET THE PARKING BRAKE WHEN WORKING ON ANY VEHICLE. AN AUTOMATIC TRANSMISSION SHOULD BE IN PARK. A MANUAL TRANSMISSION SHOULD BE IN NEUTRAL.

WARNING: OPERATE THE ENGINE ONLY IN A WELL-VENTILATED AREA.

WARNING: KEEP AWAY FROM MOVING PARTS WHEN THE ENGINE IS RUNNING, ESPECIALLY THE FAN AND BELTS.

WARNING: TO PREVENT SERIOUS BURNS, AVOID CONTACT WITH HOT PARTS SUCH AS THE RADIATOR, EXHAUST MANIFOLD(S), TAIL PIPE, CATALYTIC CONVERTER AND MUFFLER.

WARNING: DO NOT ALLOW FLAME OR SPARKS NEAR THE BATTERY. GASES ARE ALWAYS PRESENT IN AND AROUND THE BATTERY.

WARNING: ALWAYS REMOVE RINGS, WATCHES, LOOSE HANGING JEWELRY AND AVOID LOOSE CLOTHING.

DIAGNOSIS AND TESTING - WIRING HARNESS

TROUBLESHOOTING TOOLS

When diagnosing a problem in an electrical circuit there are several common tools necessary. These tools are listed and explained below.

- Jumper Wire - This is a test wire used to connect two points of a circuit. It can be used to bypass an open in a circuit.

WARNING: NEVER USE A JUMPER WIRE ACROSS A LOAD, SUCH AS A MOTOR, CONNECTED BETWEEN A BATTERY FEED AND GROUND.

- Voltmeter - Used to check for voltage on a circuit. Always connect the black lead to a known good ground and the red lead to the positive side of the circuit.

CAUTION: Most of the electrical components used in today’s vehicles are Solid State. When checking voltages in these circuits, use a meter with a 10 - megohm or greater impedance rating.

WIRING DIAGRAM INFORMATION (Continued)

• Ohmmeter - Used to check the resistance between two points of a circuit. Low or no resistance in a circuit means good continuity.

CAUTION: Most of the electrical components used in today's vehicles are Solid State. When checking resistance in these circuits use a meter with a 10 - megohm or greater impedance rating. In addition, make sure the power is disconnected from the circuit. Circuits that are powered up by the vehicle's electrical system can cause damage to the equipment and provide false readings.

• Probing Tools - These tools are used for probing terminals in connectors (Fig. 5). Select the proper size tool from Special Tool Package 6807, and insert it into the terminal being tested. Use the other end of the tool to insert the meter probe.

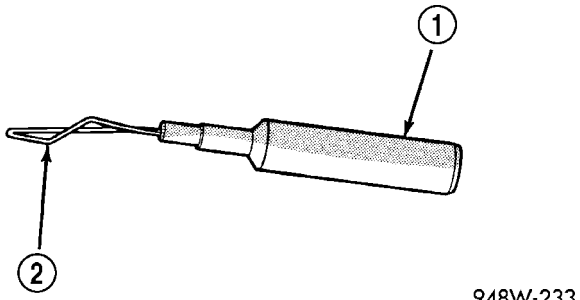


Fig. 5 PROBING TOOL

1 - SPECIAL TOOL 6801
2 - PROBING END

INTERMITTENT AND POOR CONNECTIONS

Most intermittent electrical problems are caused by faulty electrical connections or wiring. It is also possible for a sticking component or relay to cause a problem. Before condemning a component or wiring assembly, check the following items.

- Connectors are fully seated
- Spread terminals, or terminal push out
- Terminals in the wiring assembly are fully seated into the connector/component and locked into position
- Dirt or corrosion on the terminals. Any amount of corrosion or dirt could cause an intermittent problem
- Damaged connector/component casing exposing the item to dirt or moisture
- Wire insulation that has rubbed through causing a short to ground
- Some or all of the wiring strands broken inside of the insulation
- Wiring broken inside of the insulation

TROUBLESHOOTING WIRING PROBLEMS

When troubleshooting wiring problems there are six steps which can aid in the procedure. The steps are listed and explained below. Always check for non-factory items added to the vehicle before doing any diagnosis. If the vehicle is equipped with these items, disconnect them to verify these add-on items are not the cause of the problem.

(1) Verify the problem.

(2) Verify any related symptoms. Do this by performing operational checks on components that are in the same circuit. Refer to the wiring diagrams.

(3) Analyze the symptoms. Use the wiring diagrams to determine what the circuit is doing, where the problem most likely is occurring and where the diagnosis will continue.

(4) Isolate the problem area.

(5) Repair the problem area.

(6) Verify the proper operation. For this step, check for proper operation of all items on the repaired circuit. Refer to the wiring diagrams.

STANDARD PROCEDURE

STANDARD PROCEDURE - ELECTROSTATIC DISCHARGE (ESD) SENSITIVE DEVICES

All ESD sensitive components are solid state and a symbol (Fig. 6) is used to indicate this. When handling any component with this symbol, comply with the following procedures to reduce the possibility of electrostatic charge build up on the body and inadvertent discharge into the component. If it is not known whether the part is ESD sensitive, assume that it is.

(1) Always touch a known good ground before handling the part. This should be repeated while handling the part and more frequently after sliding across a seat, sitting down from a standing position, or walking a distance.

(2) Avoid touching electrical terminals of the part, unless instructed to do so by a written procedure.

(3) When using a voltmeter, be sure to connect the ground lead first.

(4) Do not remove the part form it's protective packing until it is time to install the part.

(5) Before removing the part from it's pakage, ground the pakage to a known good ground on the vehicle.

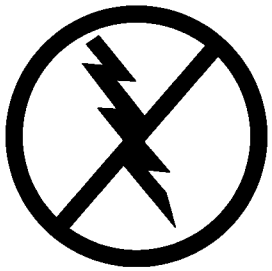


Fig. 6 ELECTROSTATIC DISCHARGE SYMBOL

WIRING DIAGRAM INFORMATION (Continued)

STANDARD PROCEDURE - TESTING OF VOLTAGE POTENTIAL

(1) Connect the ground lead of a voltmeter to a known good ground (Fig. 7).

(2) Connect the other lead of the voltmeter to the selected test point. The vehicle ignition may need to be turned ON to check voltage. Refer to the appropriate test procedure.

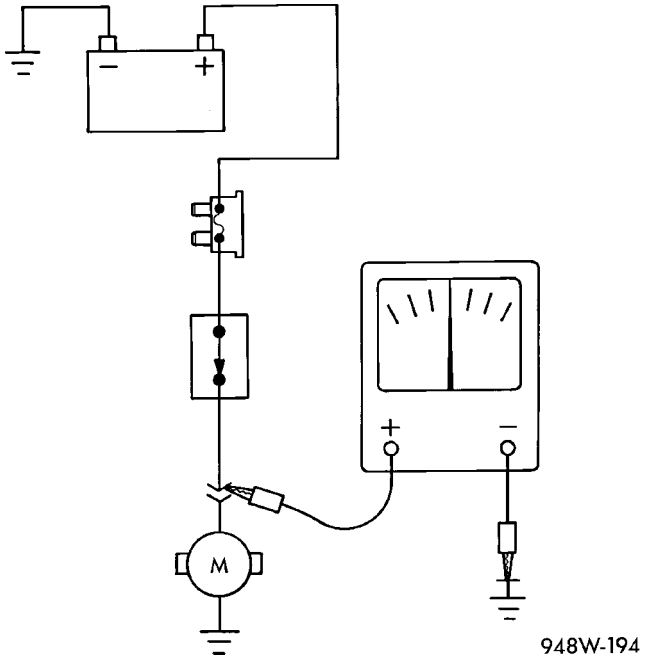


Fig. 7 TESTING FOR VOLTAGE POTENTIAL

STANDARD PROCEDURE - TESTING FOR CONTINUITY

(1) Remove the fuse for the circuit being checked or, disconnect the battery.

(2) Connect one lead of the ohmmeter to one side of the circuit being tested (Fig. 8).

(3) Connect the other lead to the other end of the circuit being tested. Low or no resistance means good continuity.

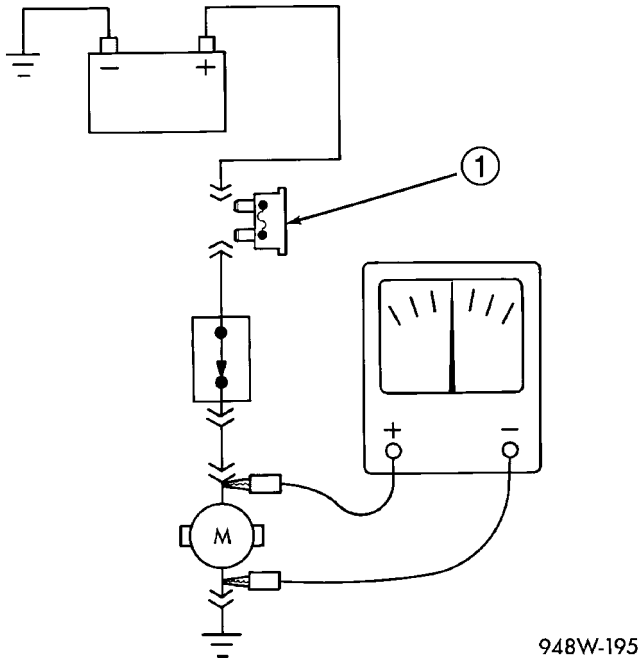


Fig. 8 TESTING FOR CONTINUITY

1 - FUSE REMOVED FROM CIRCUIT

STANDARD PROCEDURE - TESTING FOR A SHORT TO GROUND

(1) Remove the fuse and disconnect all items involved with the fuse.

(2) Connect a test light or a voltmeter across the terminals of the fuse.

(3) Starting at the fuse block, wiggle the wiring harness about six to eight inches apart and watch the voltmeter/test lamp.

(4) If the voltmeter registers voltage or the test lamp glows, there is a short to ground in that general area of the wiring harness.

WIRING DIAGRAM INFORMATION (Continued)

STANDARD PROCEDURE - TESTING FOR A SHORT TO GROUND ON FUSES POWERING SEVERAL LOADS

(1) Refer to the wiring diagrams and disconnect or isolate all items on the suspected fused circuits.
(2) Replace the blown fuse.
(3) Supply power to the fuse by turning ON the ignition switch or re-connecting the battery.
(4) Start connecting or energizing the items in the fuse circuit one at a time. When the fuse blows the circuit with the short to ground has been isolated.

STANDARD PROCEDURE - TESTING FOR A VOLTAGE DROP

(1) Connect the positive lead of the voltmeter to the side of the circuit closest to the battery (Fig. 9).
(2) Connect the other lead of the voltmeter to the other side of the switch, component or circuit.
(3) Operate the item.
(4) The voltmeter will show the difference in voltage between the two points.

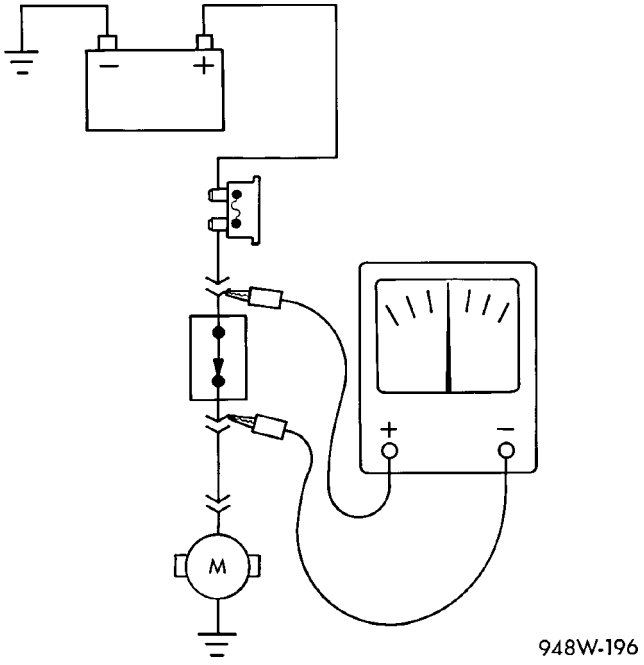
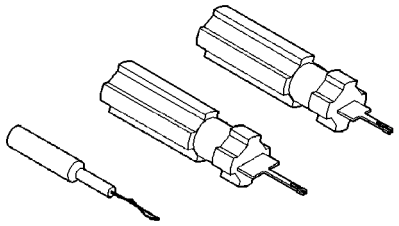


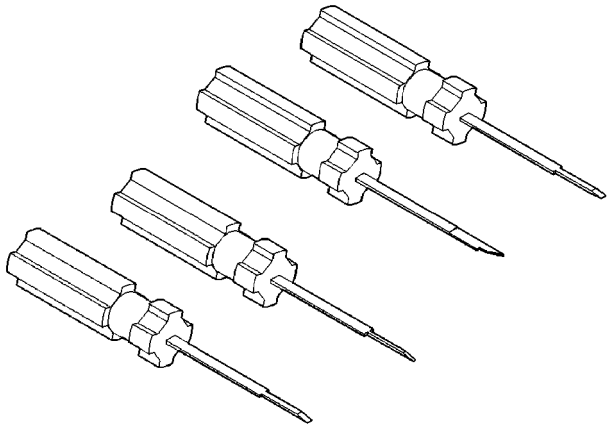
Fig. 9 TESTING FOR VOLTAGE DROP

SPECIAL TOOLS

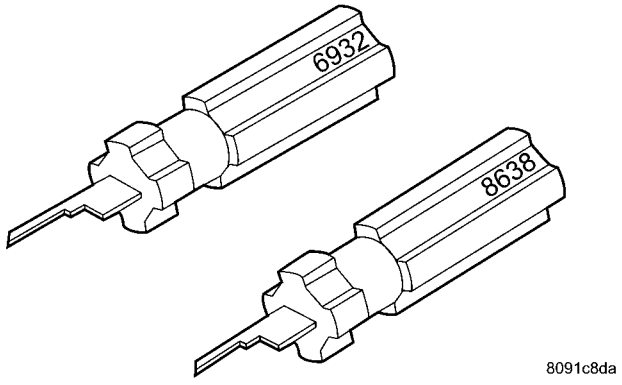
WIRING/TERMINAL



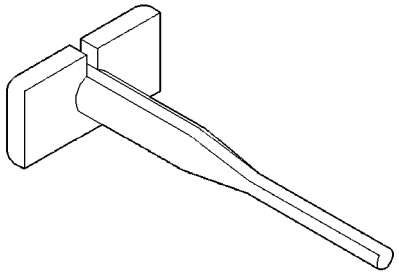
PROBING TOOL PACKAGE 6807



TERMINAL PICK TOOL SET 6680



TERMINAL REMOVING TOOLS 6932 AND 8638



TERMINAL REMOVING TOOL 6934

CONNECTOR

REMOVAL

(1) Disconnect battery.
(2) Release Connector Lock (Fig. 10).
(3) Disconnect the connector being repaired from its mating half/component.
(4) Remove the dress cover (if applicable) (Fig. 10).

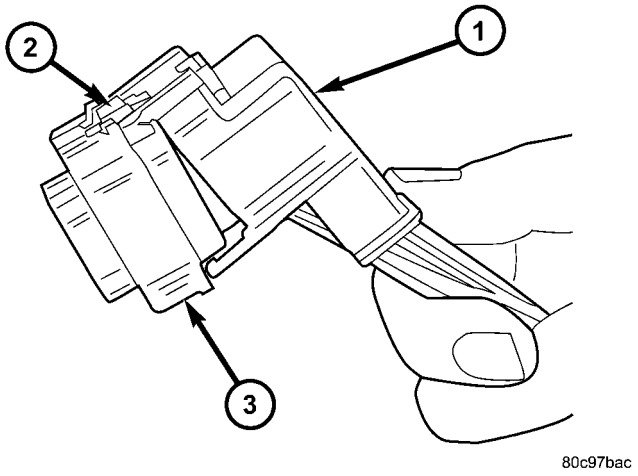


Fig. 10 REMOVAL OF DRESS COVER

1 - DRESS COVER
2 - CONNECTOR LOCK
3 - CONNECTOR

(5) Release the Secondary Terminal Lock, if required (Fig. 11).
(6) Position the connector locking finger away from the terminal using the proper special tool. Pull on the wire to remove the terminal from the connector (Fig. 12).

INSTALLATION

(1) Insert the removed terminal in the same cavity on the repair connector.
(2) Repeat steps for each terminal in the connector, being sure that all wires are inserted into the proper cavities. For additional connector pin-out identification, refer to the wiring diagrams.
(3) When the connector is re-assembled, the secondary terminal lock must be placed in the locked position to prevent terminal push out.
(4) Replace dress cover (if applicable).
(5) Connect connector to its mating half/component.
(6) Connect battery and test all affected systems.

CONNECTOR (Continued)

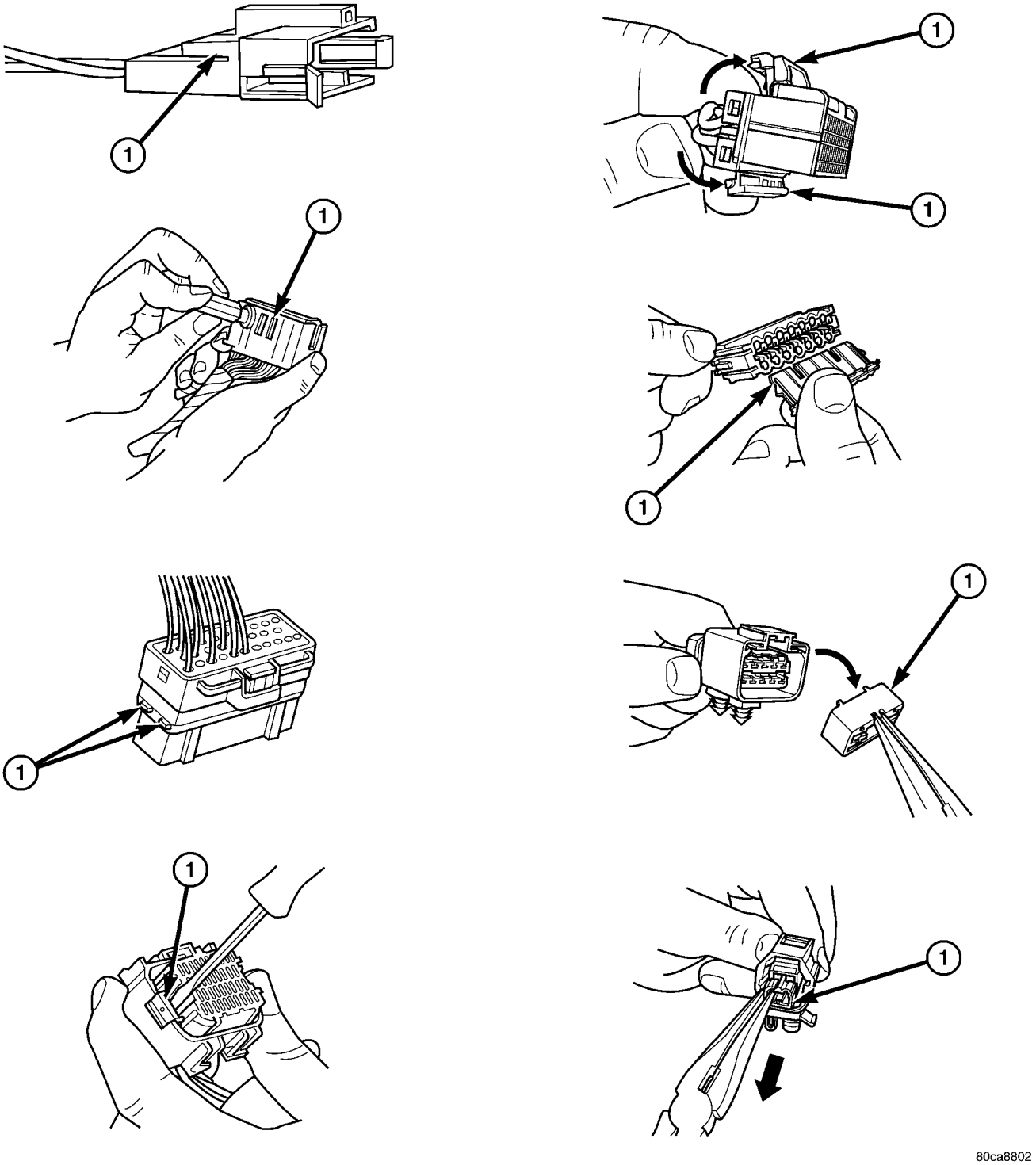


Fig. 11 EXAMPLES OF CONNECTOR SECONDARY TERMINAL LOCKS

1 - Secondary Terminal Lock

CONNECTOR (Continued)

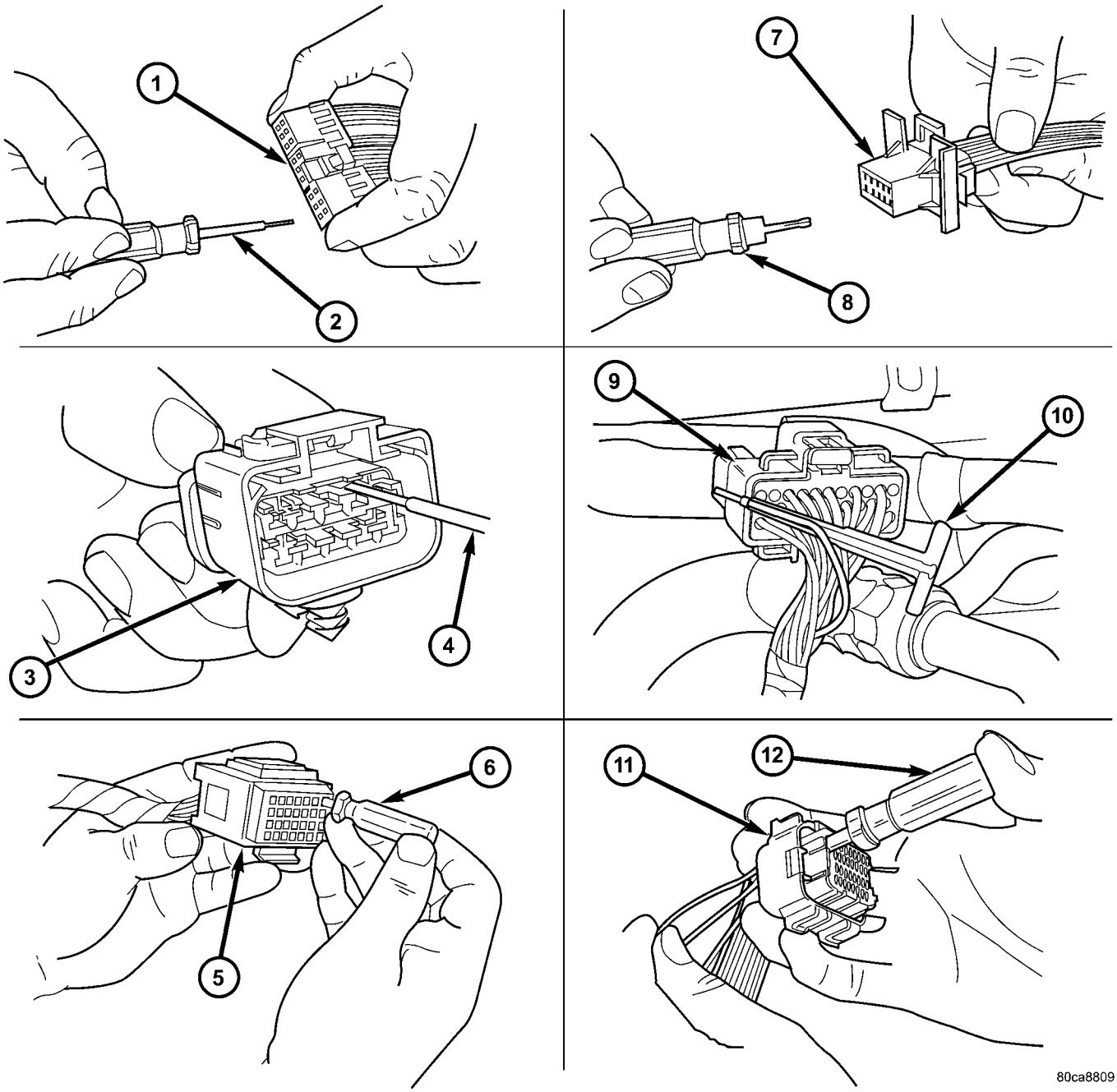


Fig. 12 TERMINAL REMOVAL

1 - TYPICAL CONNECTOR
2 - PICK FROM SPECIAL TOOL KIT 6680
3 - APEX CONNECTOR
4 - PICK FROM SPECIAL TOOL KIT 6680
5 - AUGAT CONNECTOR
6 - SPECIAL TOOL 6932
7 - MOLEX CONNECTOR
8 - SPECIAL TOOL 6742
9 - THOMAS AND BETTS CONNECTOR
10 - SPECIAL TOOL 6934
11 - TYCO CONNECTOR
12 - SPECIAL TOOL 8638

DIODE

REMOVAL

(1) Disconnect the battery.
(2) Locate the diode in the harness, and remove the protective covering.
(3) Remove the diode from the harness, pay attention to the current flow direction (Fig. 13).

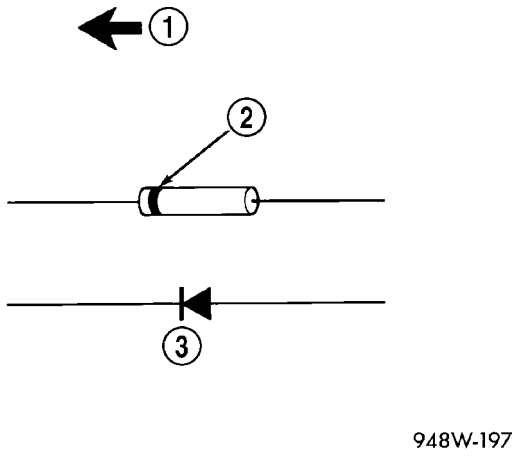


Fig. 13 DIODE IDENTIFICATION

1 - CURRENT FLOW
2 - BAND AROUND DIODE INDICATES CURRENT FLOW
3 - DIODE AS SHOWN IN THE DIAGRAMS

INSTALLATION

(1) Remove the insulation from the wires in the harness. Only remove enough insulation to solder in the new diode.
(2) Install the new diode in the harness, making sure current flow is correct. If necessary, refer to the appropriate wiring diagram for current flow (Fig. 13).
(3) Solder the connection together using rosin core type solder only. **Do not use acid core solder.**
(4) Tape the diode to the harness using electrical tape. Make sure the diode is completely sealed from the elements.
(5) Re-connect the battery and test affected systems.

TERMINAL

REMOVAL

(1) Follow steps for removing terminals described in the connector removal section.
(2) Cut the wire 6 inches from the back of the connector.

INSTALLATION

(1) Select a wire from the terminal repair kit that best matches the color and gage of the wire being repaired.
(2) Cut the repair wire to the proper length and remove one–half (1/2) inch of insulation.
(3) Splice the repair wire to the wire harness (see wire splicing procedure) .
(4) Insert the repaired wire into the connector.
(5) Install the connector locking wedge, if required, and reconnect the connector to its mating half/component.
(6) Re-tape the wire harness starting at 1–1/2 inches behind the connector and 2 inches past the repair.
(7) Connect battery and test all affected systems.

WIRE

STANDARD PROCEDURE - WIRE SPLICING

When splicing a wire, it is important that the correct gage be used as shown in the wiring diagrams.

(1) Remove one-half (1/2) inch of insulation from each wire that needs to be spliced.

(2) Place a piece of adhesive lined heat shrink tubing on one side of the wire. Make sure the tubing will be long enough to cover and seal the entire repair area.

(3) Place the strands of wire overlapping each other inside of the splice clip (Fig. 14).

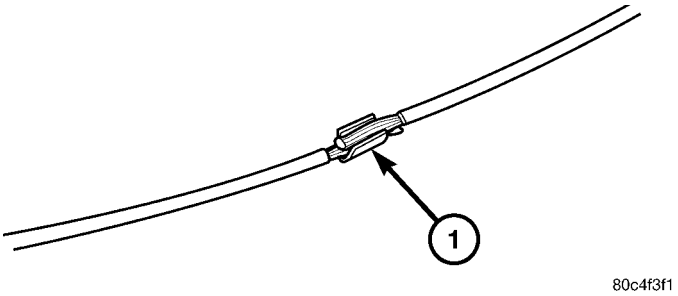


Fig. 14 SPLICE BAND

1 - SPLICE BAND

(4) Using crimping tool, Mopar p/n 05019912AA, crimp the splice clip and wires together (Fig. 15).

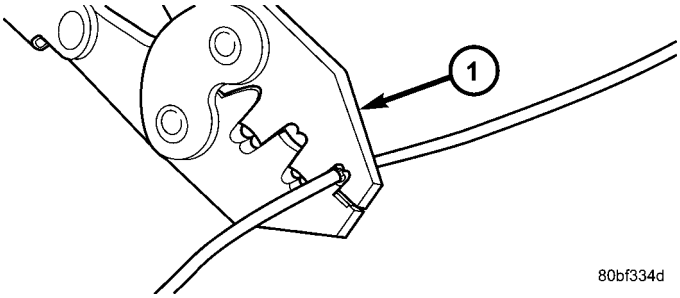


Fig. 15 CRIMPING TOOL

1 - CRIMPING TOOL

(5) Solder the connection together using rosin core type solder only (Fig. 16).

CAUTION: DO NOT USE ACID CORE SOLDER.

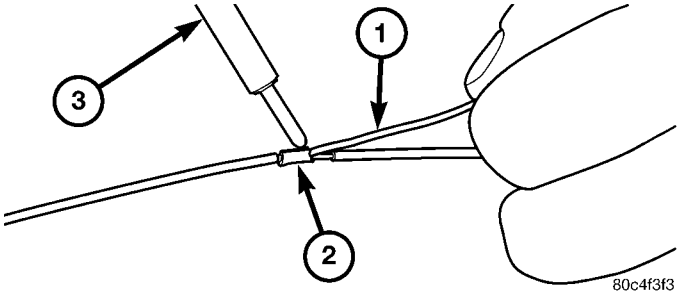


Fig. 16 SOLDER SPLICE

1 - SOLDER
2 - SPLICE BAND
3 - SOLDERING IRON

(6) Center the heat shrink tubing over the joint and heat using a heat gun. Heat the joint until the tubing is tightly sealed and sealant comes out of both ends of the tubing (Fig. 17).

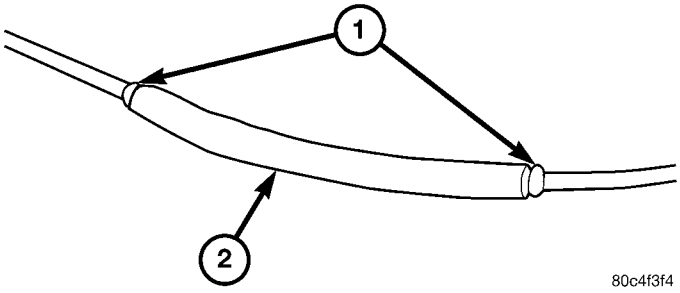


Fig. 17 HEAT SHRINK TUBE

1 - SEALANT
2 - HEAT SHRINK TUBE

8W-02 COMPONENT INDEX

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A/C Compressor Clutch Relay	8W-42
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Oxygen Sensors	8W-30
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Park/Turn Signal Lamps	8W-52
Position Lamps	8W-50
Power Distribution Center	8W-10
Power Outlet	8W-41
Power Steering Pressure Switch	8W-30
Powertrain Control Module	8W-30, 31
Radiator Fan Motor	8W-42
Radiator Fan Motor Circuit Breaker	8W-42
Radio	8W-47
Rear Fog Lamp	8W-51
Rear Window Defogger	8W-48
Rear Window Defogger Relay	8W-48
Rear Window Defogger Switch	8W-48
Rear Wiper/Washer Switch	8W-53
Seat Belt Switch	8W-40
Sentry Key Immobilizer Module	8W-39
Side Marker Lamps	8W-50
Side Repeater Lamps	8W-52
Speakers	8W-47
Speed Control Servo	8W-33
Speed Control Switches	8W-33
Speed Sensors	8W-31
Splices	8W-70
Subwoofer	8W-47

Component	Page
Throttle Position Sensor	8W-30
Tow/Haul Overdrive Off Switch	8W-31
Transfer Case Switch	8W-31
Transmission Control Relay	8W-31
Transmission Range Indicator Illumination	8W-44
Transmission Range Sensor	8W-31
Transmission Solenoid/Pressure Switch Assembly	8W-31
Turn Signal Lamps	8W-52
Underhood Lamp	8W-44
Variable Line Pressure Sensor	8W-31
Vehicle Speed Sensor	8W-30
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8W-10 POWER DISTRIBUTION

Component	Page
A/C Compressor Clutch	8W-10-16
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Automatic Shut Down Relay	8W-10-13
Axle Lock Switch	8W-10-17
Battery	8W-10-9
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Circuit Breaker	8W-10-10
Coil Capacitor	8W-10-14, 15
Compass/Temperature Mirror	8W-10-17
Controller Antilock Brake	8W-10-10
Data Link Connector	8W-10-17
Engine Starter Motor	8W-10-9, 12
Engine Starter Motor Relay	8W-10-12
Fog Lamp Relay	8W-10-16
Front Locker Pump	8W-10-19
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Fuel Injector No. 2	8W-10-14, 15
Fuel Injector No. 3	8W-10-14, 15
Fuel Injector No. 4	8W-10-14, 15
Fuel Injector No. 5	8W-10-14
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Fuel Pump Module	8W-10-16
Fuel Pump Relay	8W-10-16
Fuse 1	8W-10-9, 10
Fuse 2	8W-10-9, 10
Fuse 3	8W-10-10
Fuse 4	8W-10-10
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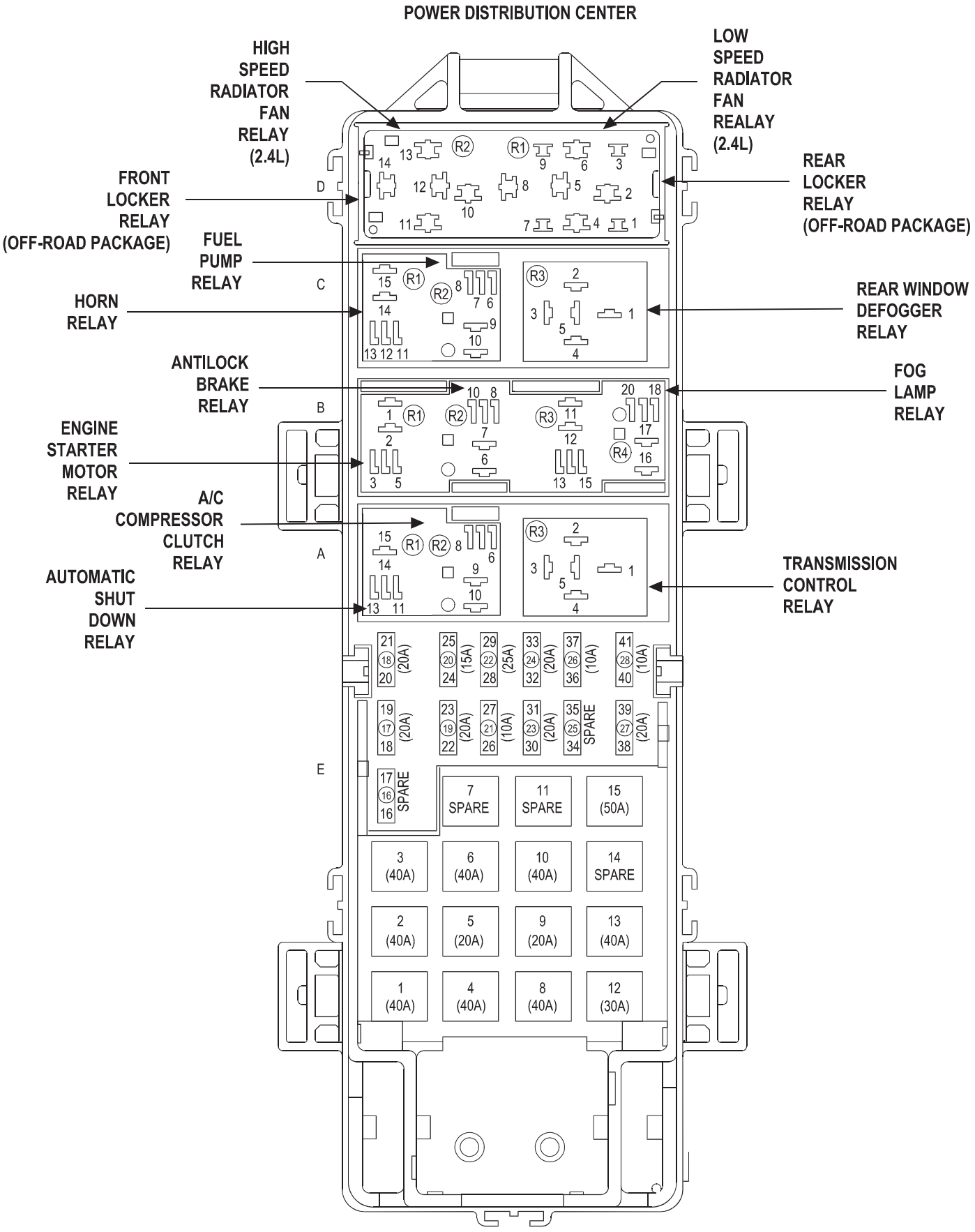
Fuse 7	8W-10-12
Fuse 8	8W-10-10, 12
Fuse 9	8W-10-11, 13
Fuse 10	8W-10-10, 11
Fuse 11	8W-10-11
Fuse 12	8W-10-10, 11
Fuse 13	8W-10-11
Fuse 14	8W-10-11
Fuse 15	8W-10-9, 17
Fuse 17	8W-10-11, 18
Fuse 18	8W-10-11, 16
Fuse 19	8W-10-16
Fuse 20	8W-10-18
Fuse 21	8W-10-16
Fuse 22	8W-10-11
Fuse 23	8W-10-16
Fuse 24	8W-10-19
Fuse 26	8W-10-17
Fuse 27	8W-10-18
Fuse 28	8W-10-13
Fuse Block	8W-10-9, 10, 11, 12
Fusible Link	8W-10-9
G106	8W-10-9
G107	8W-10-9
G108	8W-10-9
G300	8W-10-11
Generator	8W-10-9
High Speed Radiator Fan Relay	8W-10-10
Horn	8W-10-16
Horn Relay	8W-10-16
Ignition Coil Pack	8W-10-14, 15
Ignition Switch	8W-10-11, 12
Instrument Cluster	8W-10-11, 17
Left Courtesy Lamp	8W-10-17
Left Dome Lamp	8W-10-17
Left Fog Lamp	8W-10-16
Low Speed Radiator Fan Relay	8W-10-10
Multi-Function Switch	8W-10-16, 18
Power Distribution Center	8W-10-2, 9, 10, 11, 12, 13, 16, 17, 18, 19
Power Outlet	8W-10-18
Powertrain Control Module	8W-10-12, 13, 18
Radio	8W-10-18
Rear Locker Pump	8W-10-19
Rear Locker Relay	8W-10-19
Rear Window Defogger	8W-10-9
Rear Window Defogger Relay	8W-10-9
Right Courtesy Lamp	8W-10-17
Right Dome Lamp	8W-10-17
Right Fog Lamp	8W-10-16
Transmission Control Relay	8W-10-18
Transmission Solenoid/Pressure Switch Assembly	8W-10-18
Underhood Lamp	8W-10-17
Variable Line Pressure Sensor	8W-10-18


POWER DISTRIBUTION CENTER
HIGH SPEED RADIATOR FAN RELAY (2.4L)
LOW SPEED RADIATOR FAN REALAY (2.4L)
REAR LOCKER RELAY (OFF-ROAD PACKAGE)
FRONT LOCKER RELAY (OFF-ROAD PACKAGE)
FUEL PUMP RELAY
HORN RELAY
REAR WINDOW DEFOGGER RELAY
ANTILOCK BRAKE RELAY
FOG LAMP RELAY
ENGINE STARTER MOTOR RELAY
A/C COMPRESSOR CLUTCH RELAY
AUTOMATIC SHUT DOWN RELAY
TRANSMISSION CONTROL RELAY
SPARE
(20A)
(15A)
(25A)
(10A)
(50A)
(40A)
(30A)

FUSES

FUSE NO.	AMPS	FUSED CIRCUIT	FUNCTION
1	40A	A111 12RD/LB ◩	FUSED B(+)
1	40A	A502 12RD/LG ◩◩	FUSED B(+)
2	40A	A902 12RD	FUSED B(+)
3	40A	A906 12RD	FUSED B(+)
4	40A	A16 12GY ✳	FUSED B(+)
5	20A	A30 16RD/WT ■■	FUSED B(+)
6	40A	A8 14RD	FUSED B(+)
7	-	-	-
8	40A	A107 12TN/RD ◇◇◇	FUSED B(+)
9	20A	A209 18RD/YL ◓◓	FUSED B(+)
		A209 18RD/YL	FUSED B(+)
		A209 18RD ◓	FUSED B(+)
10	40A	A940 12RD	FUSED B(+)
11	-	-	-
12	30A	A111 12DG/RD ◇◇◇	FUSED B(+)
13	40A	A991 12RD	FUSED B(+)
14	-	-	-
15	50A	INTERNAL	FUSED B(+)

◓ 42RLE
◩ HEATER ONLY
◓◓ NA1
◩◩ EXCEPT HEATER ONLY
■■ A/T
◇◇◇ ABS
✳ 2.4L

FUSES

FUSE NO.	AMPS	FUSED CIRCUIT	FUNCTION
16	-	-	-
17	20A	A205 18RD/OR	FUSED B(+)
18	20A	A43 18RD/DG	FUSED B(+)
		A43 18RD/DG	
19	20A	A39 16RD/DG ◆◆	FUSED B(+)
		A39 18RD/DG ◆	FUSED B(+)
20	15A	F306 16DB/PK	FUSED B(+)
21	10A	A926 20RD	FUSED B(+)
22	25A	A1 16RD	FUSED B(+)
23	20A	A930 18RD	FUSED B(+)
24	20A	INTERNAL □	FUSED B(+)
25	-	-	-
26	10A	M21 20YL/BR	FUSED B(+)
27	20A	A701 18BR/RD	FUSED B(+)
28	10A	Y3 18LG	FUSED SENSOR GROUND
		Y3 20LG	FUSED SENSOR GROUND

◆◆ OPTION 2

◆ OPTION 1

□ OFF-ROAD PACKAGE

A/C COMPRESSOR CLUTCH RELAY

CAVITY	CIRCUIT	FUNCTION
A6	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
	F960 20PK/LG *	FUSED IGNITION SWITCH OUTPUT (RUN-START)
A7	-	-
A8	C13 20LB/OR	A/C CLUTCH RELAY CONTROL
A9	C3 20DB/YL	A/C CLUTCH RELAY OUTPUT
A10	A926 20RD	FUSED B(+)

ANTILOCK BRAKE RELAY

CAVITY	CIRCUIT	FUNCTION
B6	G4 20VT/LB	ABS WARNING INDICATOR DRIVER
B7	-	-
B8	F102 20PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
B9	Z901 18BK	GROUND
B10	G83 18VT/WT	ABS RELAY CONTROL

AUTOMATIC SHUT DOWN RELAY

CAVITY	CIRCUIT	FUNCTION
A11	K342 20BR/WT	ASD RELAY CONTROL
A12	-	-
A13	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
A14	K334 18BR	ASD RELAY OUTPUT
A15	A209 18RD/YL	FUSED B(+)

* 2.4L

ENGINE STARTER MOTOR RELAY

CAVITY	CIRCUIT	FUNCTION
B1	A8 14RD	FUSED B(+)
B2	T750 14YL/GY	ENGINE STARTER MOTOR RELAY OUTPUT
B3	T752 20DG/OR	STARTER MOTOR RELAY CONTROL
B4	-	-
B5	F26 16PK/OR	IGNITION SWITCH OUTPUT (START)

FOG LAMP RELAY

CAVITY	CIRCUIT	FUNCTION
B16	L139 18WT/YL	FOG LAMP SWITCH SIGNAL
B17	-	-
B18	L340 14WT/OR △	HIGH BEAM INDICATOR DRIVER
B18	L300 14LG/WT	DIMMER SWITCH HIGH BEAM OUTPUT
B19	L39 18WT/OR	FOG LAMP RELAY OUTPUT
B20	Z901 18BK	GROUND

FRONT LOCKER RELAY (OFF-ROAD PACKAGE)

CAVITY	CIRCUIT	FUNCTION
D10	INTERNAL	FUSED B(+)
D11	G304 20VT/DB	FRONT LOCKER REQUEST
D12	-	-
D13	INTERNAL	FUSED B(+)
D14	A750 18TN/RD	FRONT LOCKER RELAY OUTPUT

△ DRL

FUEL PUMP RELAY

CAVITY	CIRCUIT	FUNCTION
C6	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
C7	-	-
C8	K31 20BR	FUEL PUMP RELAY CONTROL
C9	A109 18OR/RD	FUEL PUMP RELAY OUTPUT
C10	A930 18RD	FUSED B(+)

HIGH SPEED RADIATOR FAN RELAY (2.4L)

CAVITY	CIRCUIT	FUNCTION
D10	A16 12GY	FUSED B(+)
D11	N112 20DB/OR	HIGH SPEED RADIATOR FAN RELAY CONTROL
D13	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
D14	C25 12YL	HIGH SPEED RADIATOR RELAY OUTPUT

HORN RELAY

CAVITY	CIRCUIT	FUNCTION
C11	A43 18RD/DG	FUSED B(+)
C12	-	-
C13	X3 20RD/YL	HORN RELAY CONTROL
C14	X2 18DG/OR	HORN RELAY OUTPUT
C15	A43 18RD/DG	FUSED B(+)

LOW SPEED RADIATOR FAN RELAY (2.4L)

CAVITY	CIRCUIT	FUNCTION
D2	A16 12GY	FUSED B(+)
D4	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
D6	N201 20DB/LG	LOW SPEED RADIATOR FAN RELAY CONTROL
D8	C23 12DG	LOW SPEED FAN RELAY OUTPUT

REAR LOCKER RELAY (OFF-ROAD PACKAGE)

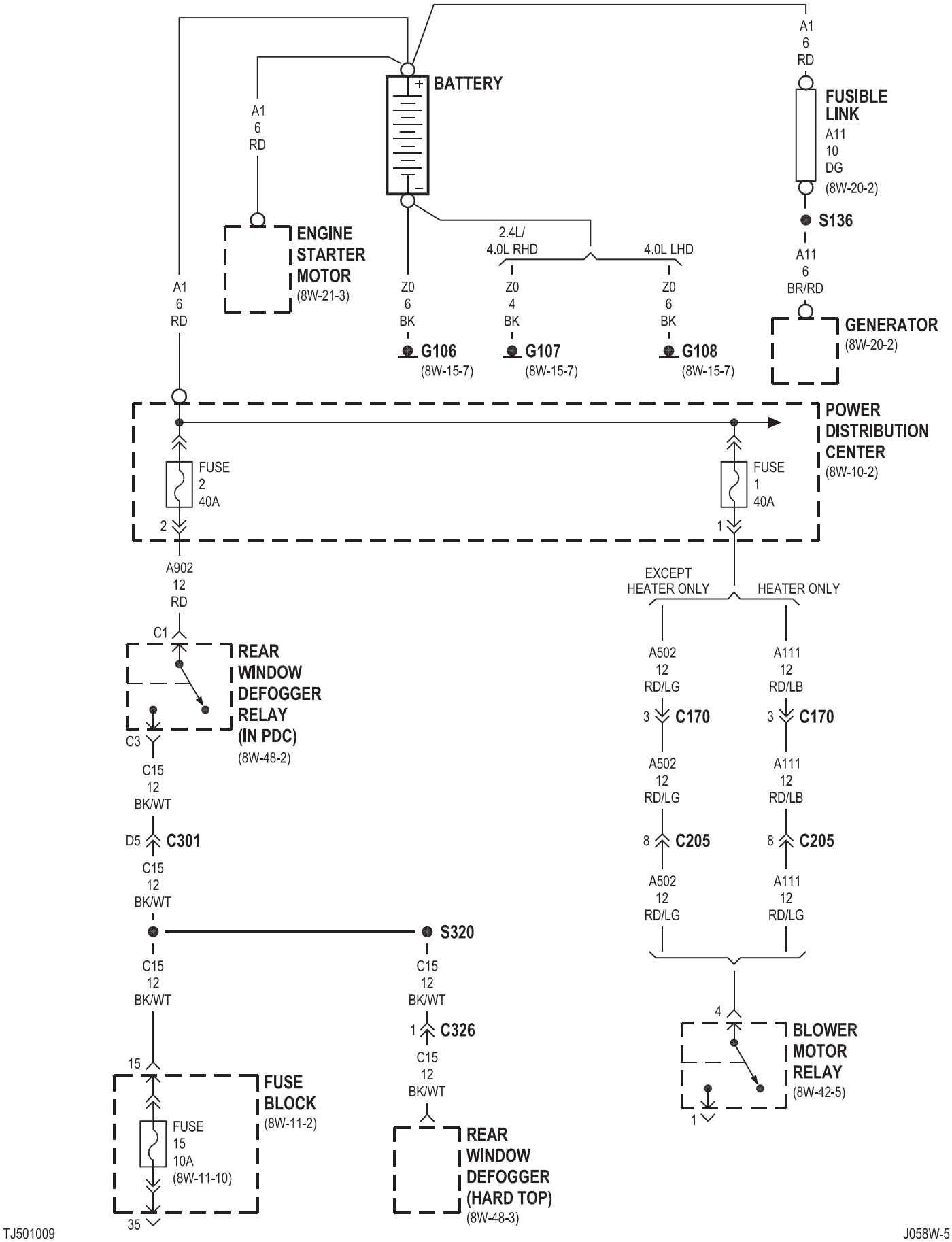
CAVITY	CIRCUIT	FUNCTION
D2	INTERNAL	FUSED B(+)
D4	INTERNAL	FUSED B(+)
D5	-	-
D6	G305 20VT/LG	REAR LOCKER REQUEST
D8	A850 18RD/WT	REAR LOCKER RELAY OUTPUT

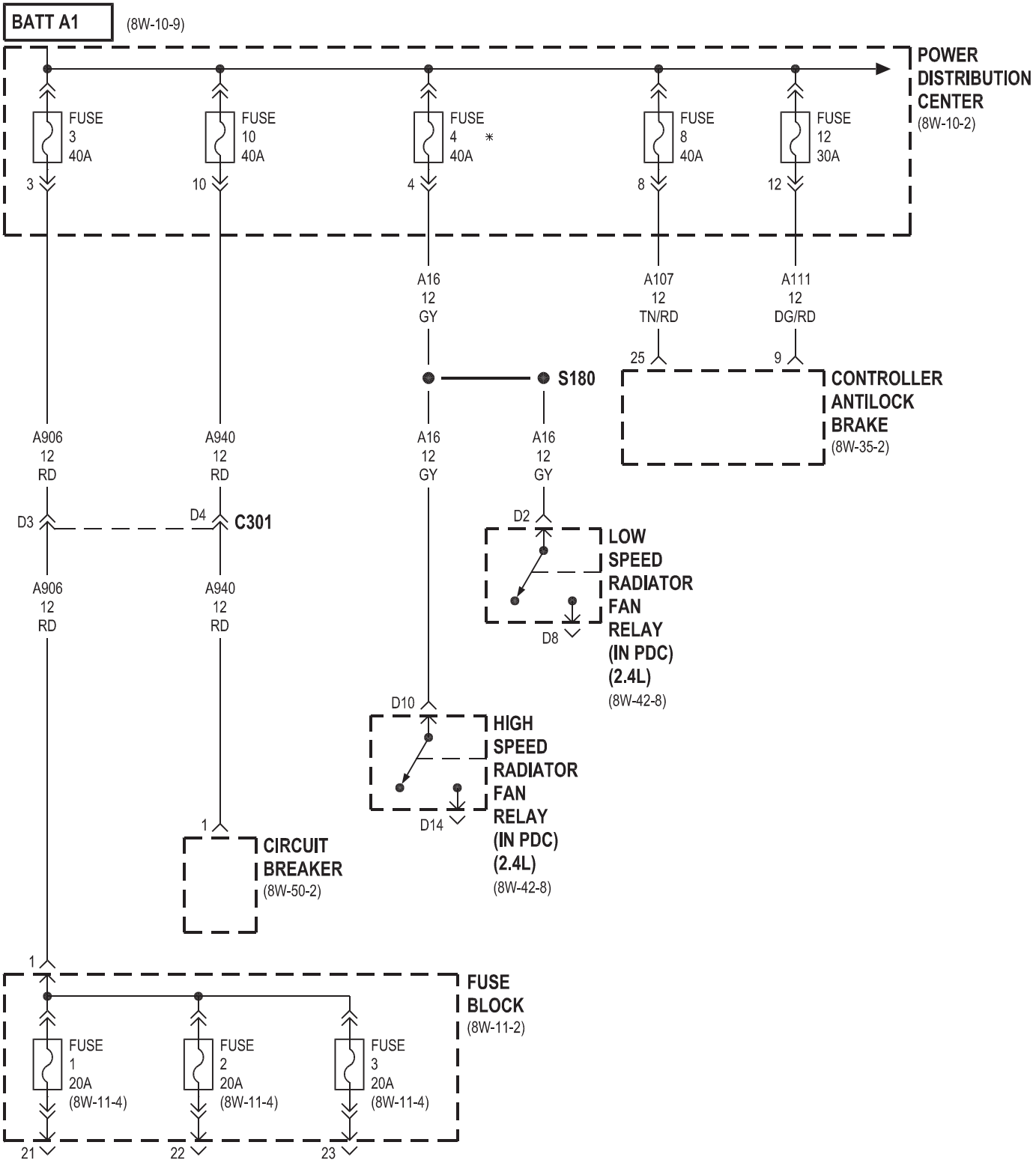
REAR WINDOW DEFOGGER RELAY

CAVITY	CIRCUIT	FUNCTION
C1	A902 12RD	FUSED B(+)
C2	C115 20DB	REAR WINDOW DEFOGGER RELAY CONTROL
C3	C15 12BK/WT	REAR WINDOW DEFOGGER RELAY OUTPUT
C4	F102 20PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
C5	-	-

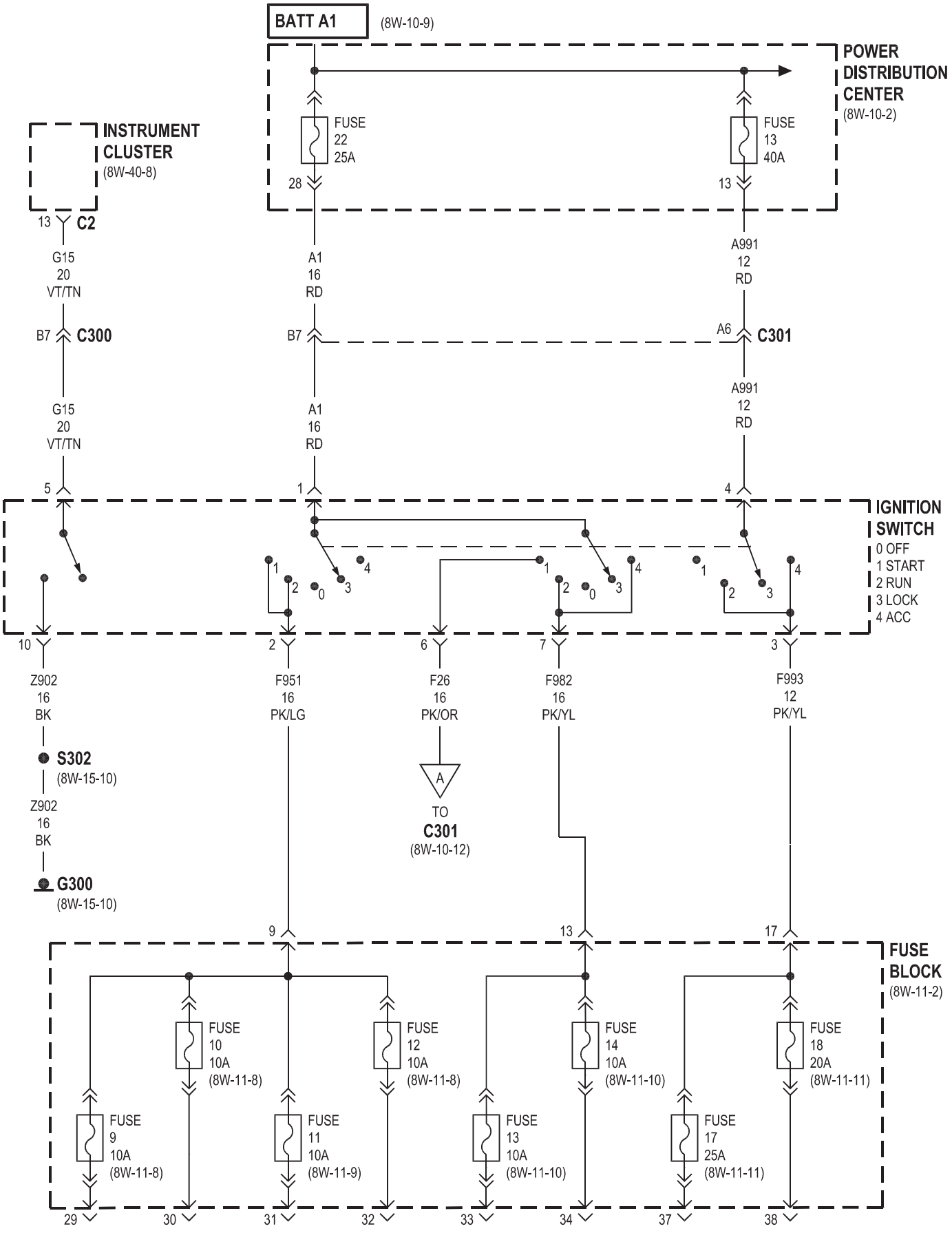
TRANSMISSION CONTROL RELAY

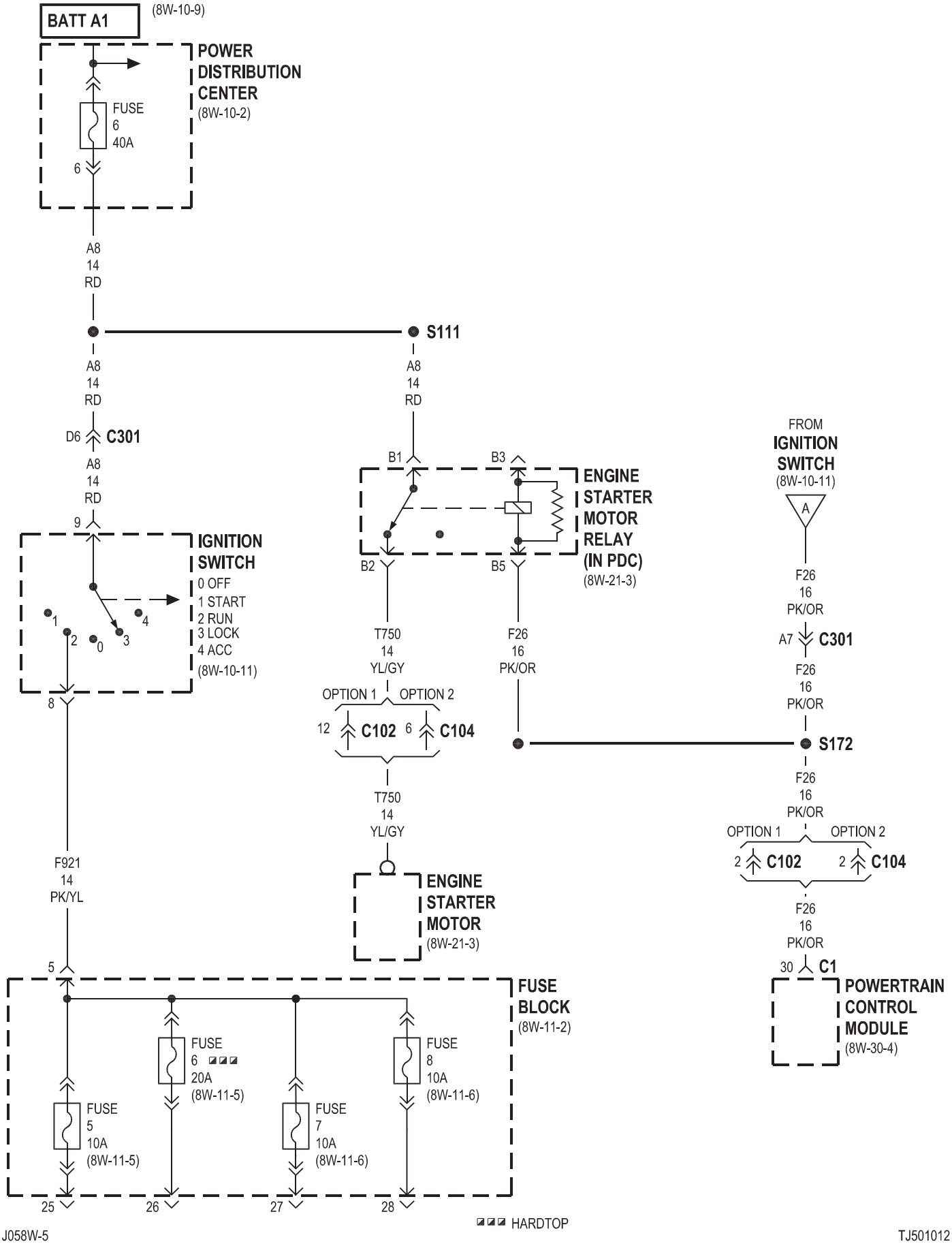
CAVITY	CIRCUIT	FUNCTION
A1	A30 16RD/WT	FUSED B(+)
A2	T515 20YL/DB	TRANSMISSION CONTROL RELAY CONTROL
A3	T16 18YL/OR	TRANSMISSION CONTROL RELAY OUTPUT
A4	Z901 18BK	GROUND
A5	-	-


BATTERY
A1
6
RD
ENGINE
STARTER
MOTOR
(8W-21-3)
A1
6
RD
FUSIBLE
LINK
A11
10
DG
(8W-20-2)
S136
A11
6
BR/RD
GENERATOR
(8W-20-2)
2.4L/
4.0L RHD
4.0L LHD
Z0
6
BK
G106
(8W-15-7)
Z0
4
BK
G107
(8W-15-7)
Z0
6
BK
G108
(8W-15-7)
POWER
DISTRIBUTION
CENTER
(8W-10-2)
FUSE
2
40A
FUSE
1
40A
A902
12
RD
EXCEPT
HEATER ONLY
HEATER ONLY
A502
12
RD/LG
A111
12
RD/LB
C170
C205
REAR
WINDOW
DEFOGGER
RELAY
(IN PDC)
(8W-48-2)
C15
12
BK/WT
C301
S320
C326
FUSE
BLOCK
(8W-11-2)
FUSE
15
10A
(8W-11-10)
REAR
WINDOW
DEFOGGER
(HARD TOP)
(8W-48-3)
BLOWER
MOTOR
RELAY
(8W-42-5)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 3 40A
FUSE 10 40A
FUSE 4 * 40A
FUSE 8 40A
FUSE 12 30A
3
10
4
8
12
A16 12 GY
A107 12 TN/RD
A111 12 DG/RD
25
9
S180
CONTROLLER ANTILOCK BRAKE
(8W-35-2)
A906 12 RD
A940 12 RD
A16 12 GY
A16 12 GY
D3
D4
C301
D2
LOW SPEED RADIATOR FAN RELAY (IN PDC) (2.4L)
(8W-42-8)
D8
D10
HIGH SPEED RADIATOR FAN RELAY (IN PDC) (2.4L)
(8W-42-8)
D14
1
CIRCUIT BREAKER
(8W-50-2)
FUSE BLOCK
(8W-11-2)
FUSE 1 20A (8W-11-4)
FUSE 2 20A (8W-11-4)
FUSE 3 20A (8W-11-4)
21
22
23

* 2.4L


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 22 25A
FUSE 13 40A
28
13
INSTRUMENT CLUSTER
(8W-40-8)
13 C2
G15 20 VT/TN
A1 16 RD
A991 12 RD
B7 C300
B7
A6 C301
G15 20 VT/TN
A1 16 RD
A991 12 RD
5
1
4
IGNITION SWITCH
0 OFF
1 START
2 RUN
3 LOCK
4 ACC
10
2
6
7
3
Z902 16 BK
F951 16 PK/LG
F26 16 PK/OR
F982 16 PK/YL
F993 12 PK/YL
S302
(8W-15-10)
Z902 16 BK
G300
(8W-15-10)
A
TO C301
(8W-10-12)
9
13
17
FUSE BLOCK
(8W-11-2)
FUSE 9 10A (8W-11-8)
FUSE 10 10A (8W-11-8)
FUSE 11 10A (8W-11-9)
FUSE 12 10A (8W-11-8)
FUSE 13 10A (8W-11-10)
FUSE 14 10A (8W-11-10)
FUSE 17 25A (8W-11-11)
FUSE 18 20A (8W-11-11)
29
30
31
32
33
34
37
38



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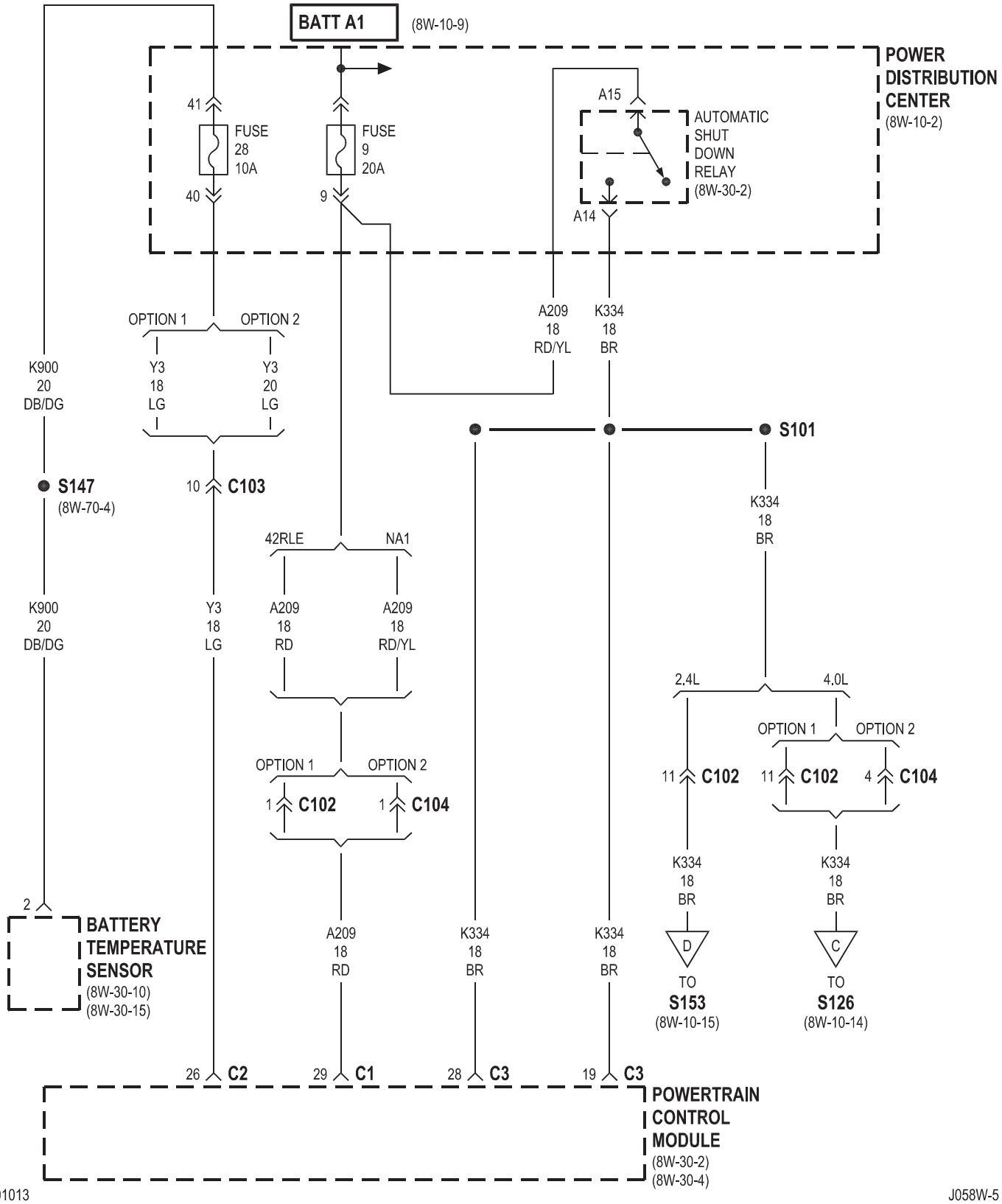
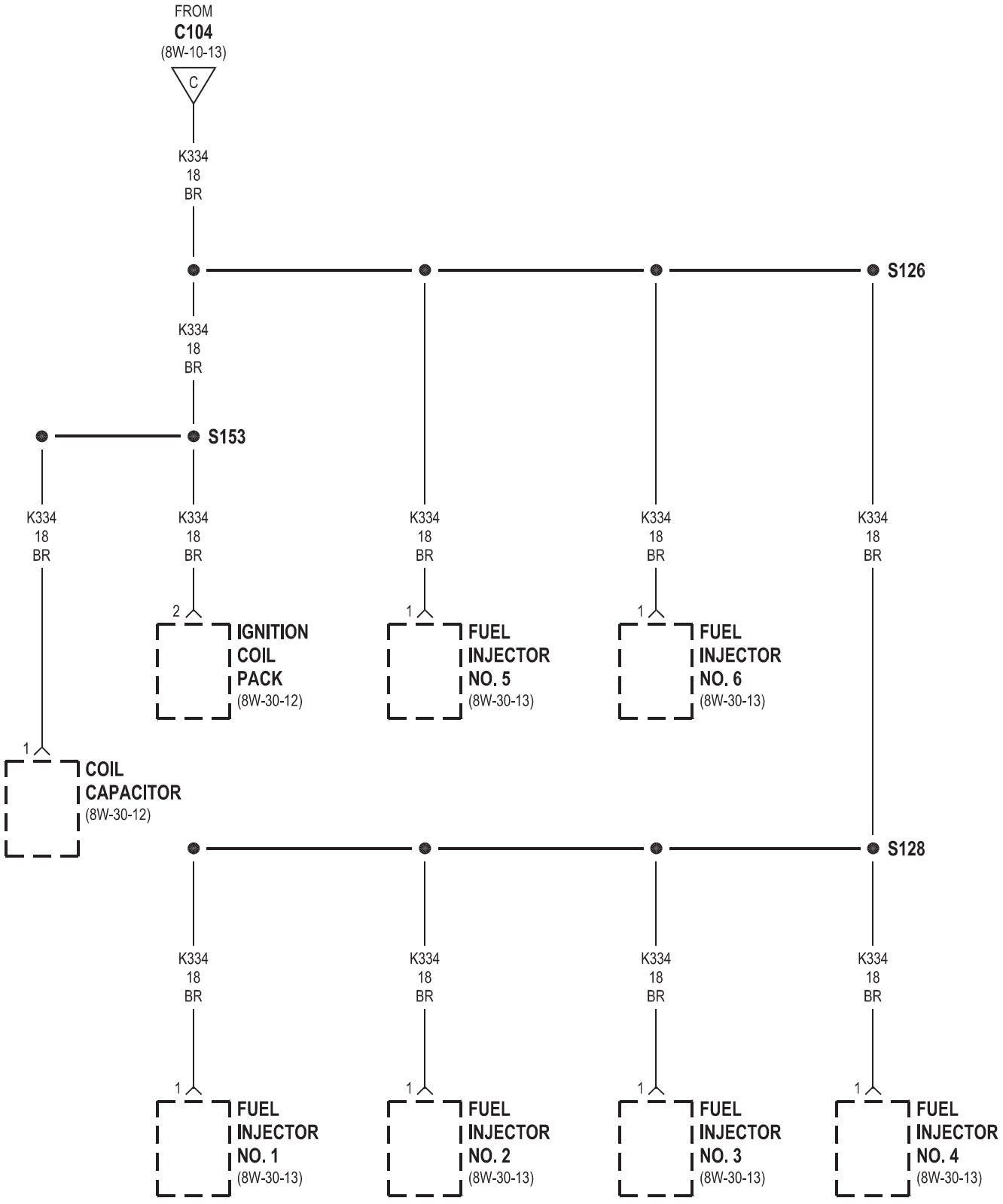
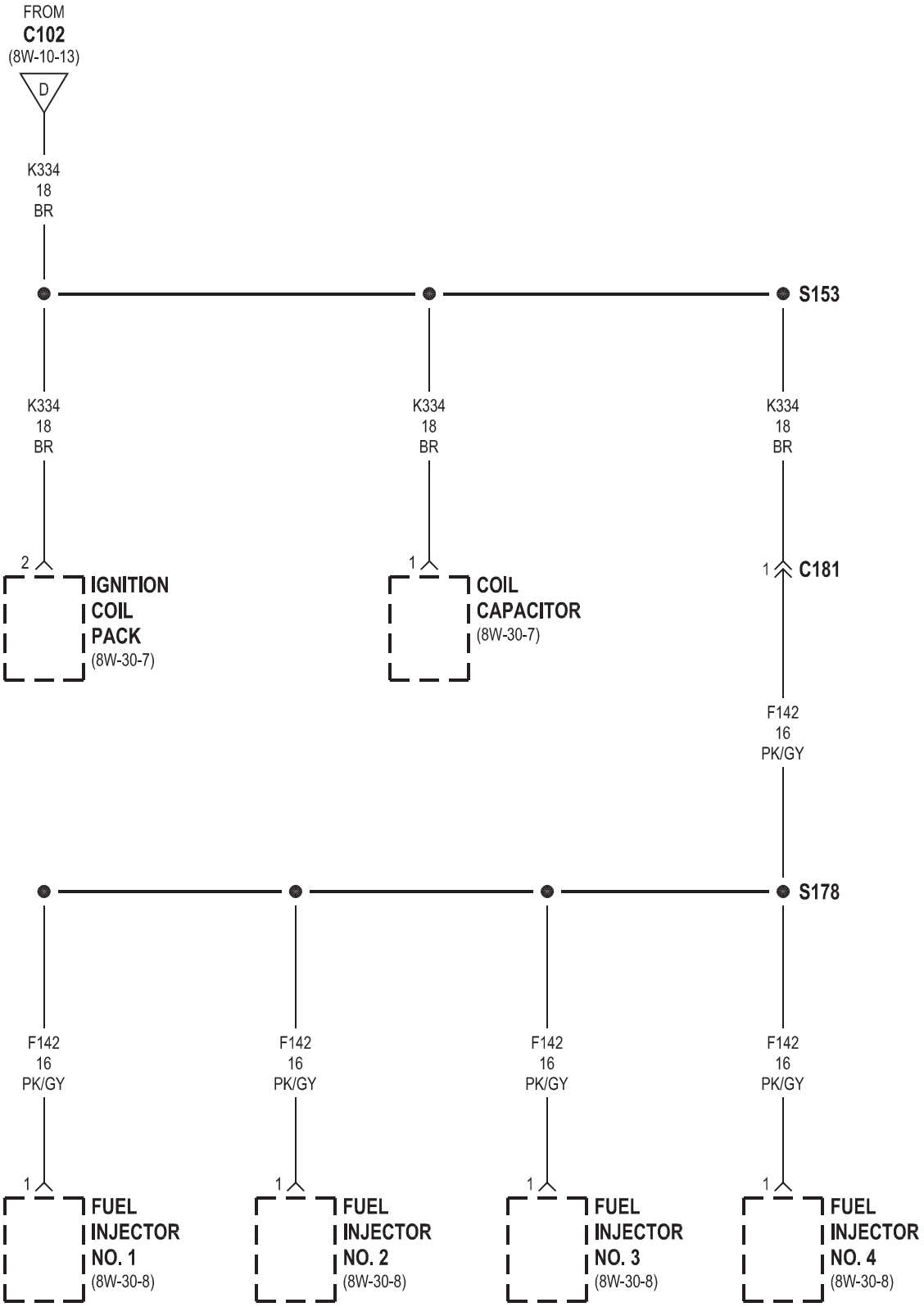
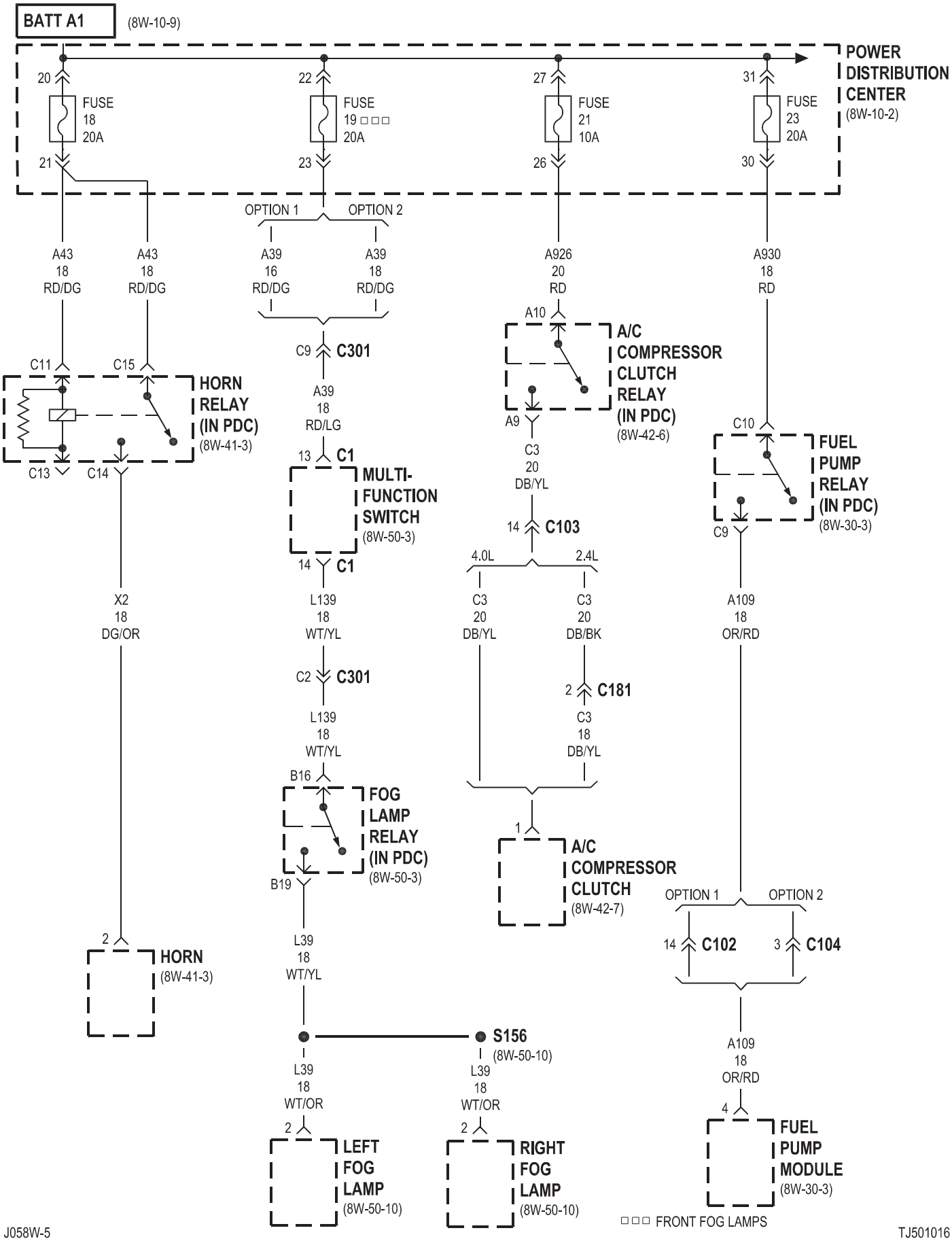
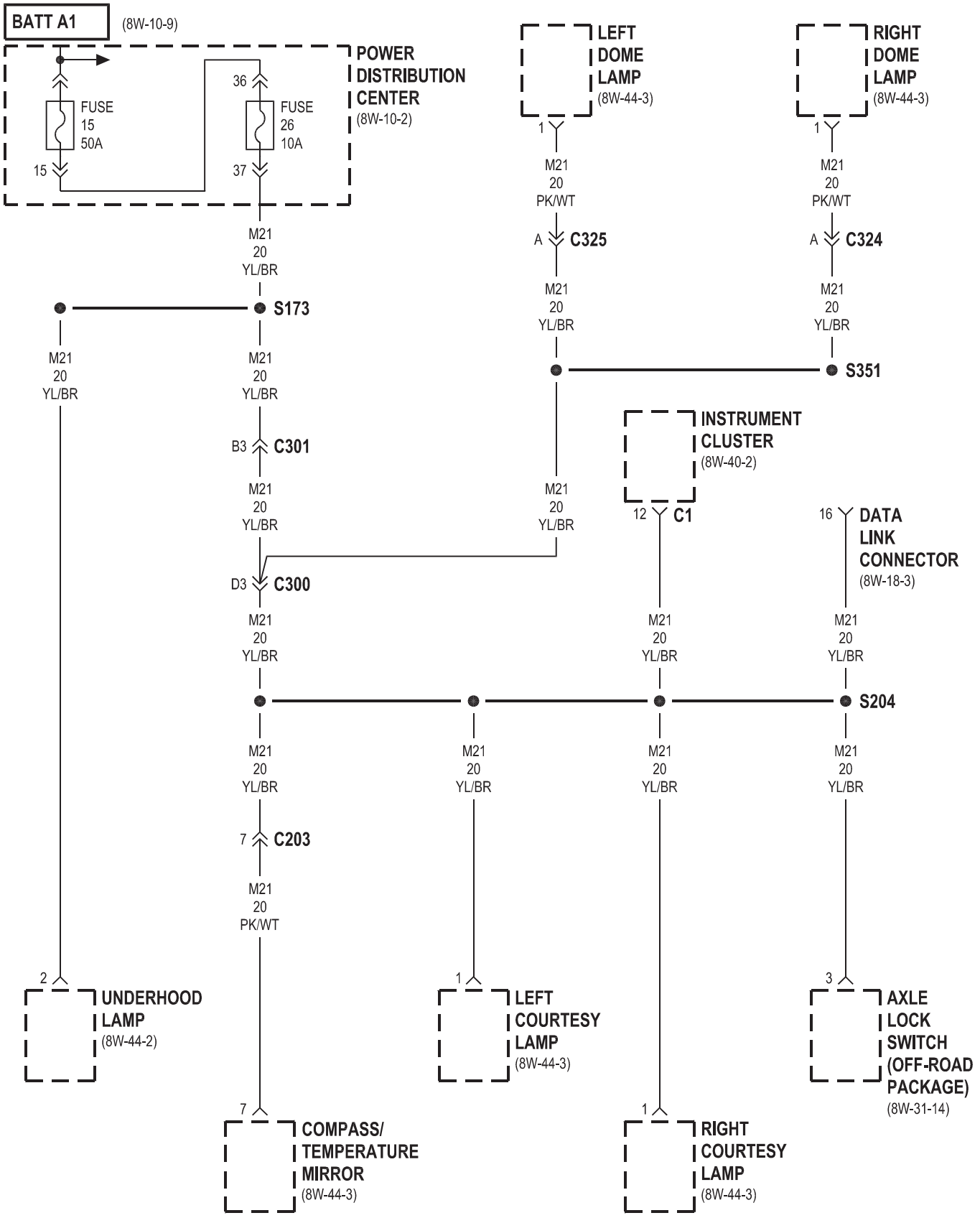
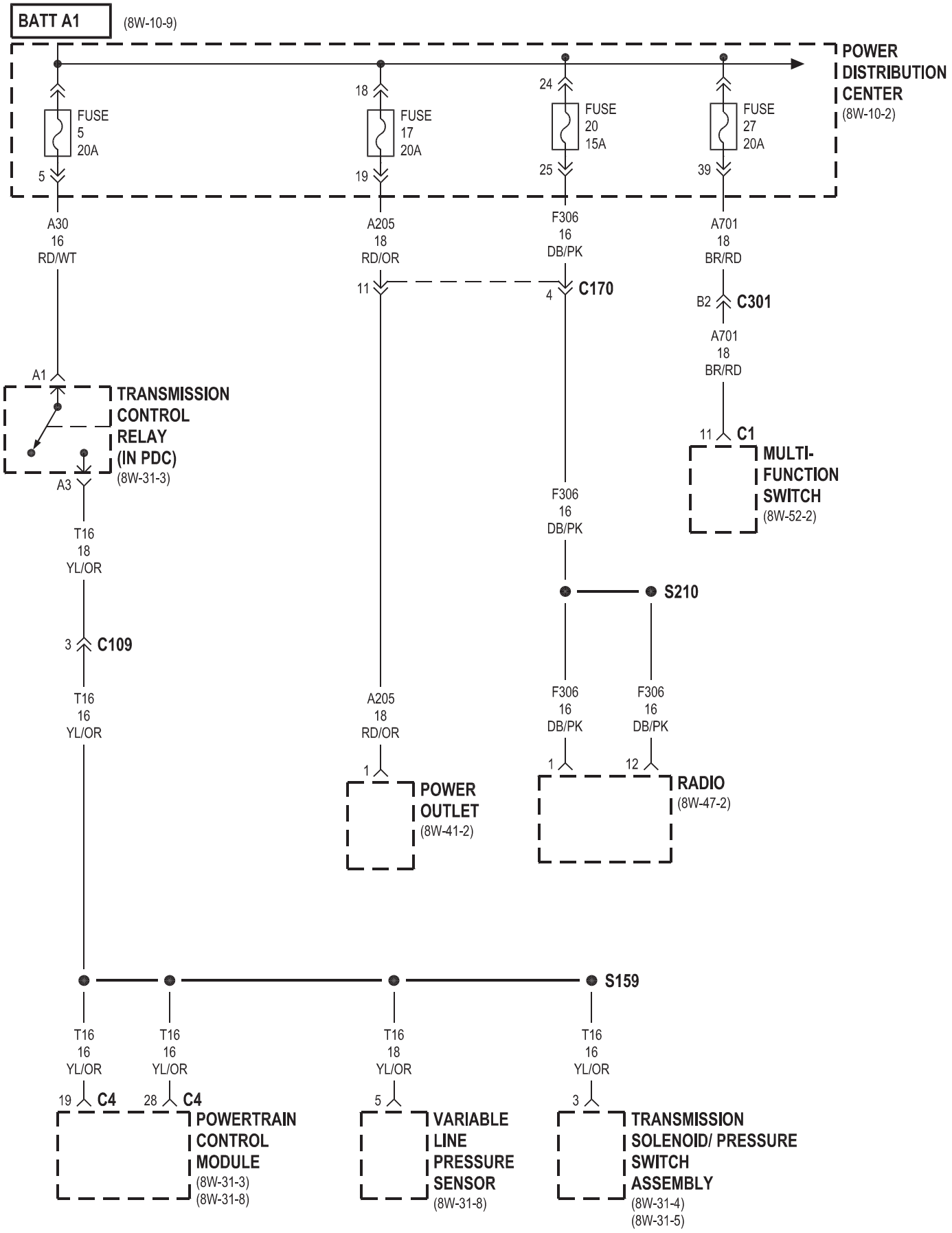

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
41
FUSE 28 10A
40
FUSE 9 20A
9
A15
AUTOMATIC SHUT DOWN RELAY
(8W-30-2)
A14
OPTION 1
OPTION 2
K900 20 DB/DG
Y3 18 LG
Y3 20 LG
A209 18 RD/YL
K334 18 BR
S101
S147
(8W-70-4)
10 C103
42RLE
NA1
K334 18 BR
K900 20 DB/DG
Y3 18 LG
A209 18 RD
A209 18 RD/YL
2.4L
4.0L
OPTION 1
OPTION 2
1 C102
1 C104
11 C102
11 C102
4 C104
2
BATTERY TEMPERATURE SENSOR
(8W-30-10)
(8W-30-15)
A209 18 RD
K334 18 BR
K334 18 BR
K334 18 BR
K334 18 BR
D
C
TO S153
(8W-10-15)
TO S126
(8W-10-14)
26 C2
29 C1
28 C3
19 C3
POWERTRAIN CONTROL MODULE
(8W-30-2)
(8W-30-4)

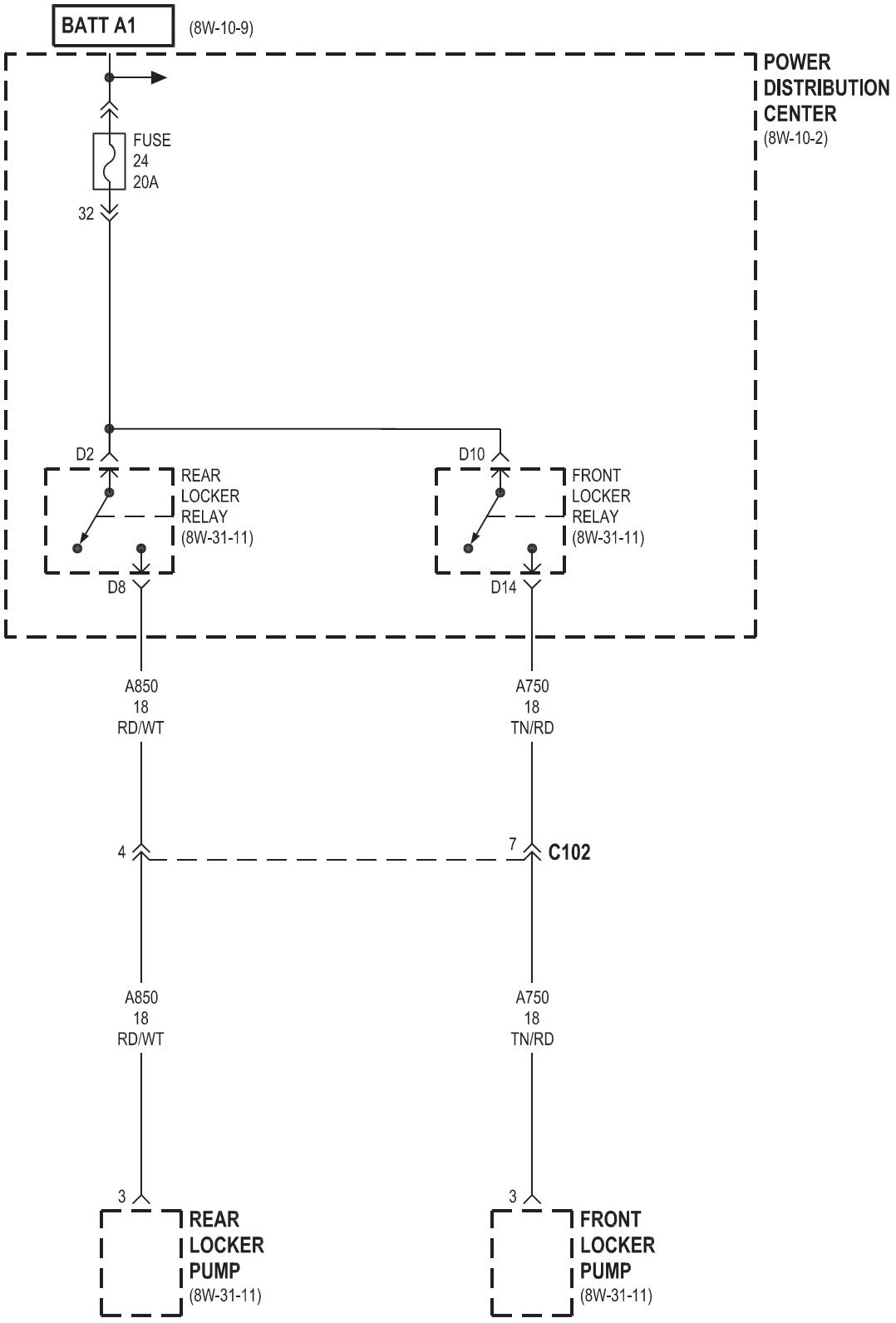

FROM
C104
(8W-10-13)
C
K334
18
BR
S126
S153
S128
2
IGNITION
COIL
PACK
(8W-30-12)
1
COIL
CAPACITOR
(8W-30-12)
FUEL
INJECTOR
NO. 5
(8W-30-13)
FUEL
INJECTOR
NO. 6
(8W-30-13)
FUEL
INJECTOR
NO. 1
(8W-30-13)
FUEL
INJECTOR
NO. 2
(8W-30-13)
FUEL
INJECTOR
NO. 3
(8W-30-13)
FUEL
INJECTOR
NO. 4
(8W-30-13)


FROM
C102
(8W-10-13)
D
K334
18
BR
S153
K334
18
BR
2
IGNITION
COIL
PACK
(8W-30-7)
K334
18
BR
1
COIL
CAPACITOR
(8W-30-7)
K334
18
BR
1
C181
F142
16
PK/GY
S178
F142
16
PK/GY
1
FUEL
INJECTOR
NO. 1
(8W-30-8)
F142
16
PK/GY
1
FUEL
INJECTOR
NO. 2
(8W-30-8)
F142
16
PK/GY
1
FUEL
INJECTOR
NO. 3
(8W-30-8)
F142
16
PK/GY
1
FUEL
INJECTOR
NO. 4
(8W-30-8)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
20
FUSE 18 20A
21
22
FUSE 19 □□□ 20A
23
27
FUSE 21 10A
26
31
FUSE 23 20A
30
OPTION 1
OPTION 2
A43 18 RD/DG
A43 18 RD/DG
A39 16 RD/DG
A39 18 RD/DG
A926 20 RD
A930 18 RD
C11
C15
HORN RELAY (IN PDC)
(8W-41-3)
C13
C14
C9
C301
A39 18 RD/LG
13
C1
MULTI-FUNCTION SWITCH
(8W-50-3)
14
C1
A10
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
(8W-42-6)
A9
C3 20 DB/YL
14
C103
4.0L
2.4L
C3 20 DB/YL
C3 20 DB/BK
2
C181
C3 18 DB/YL
1
A/C COMPRESSOR CLUTCH
(8W-42-7)
C10
FUEL PUMP RELAY (IN PDC)
(8W-30-3)
C9
A109 18 OR/RD
X2 18 DG/OR
C2
C301
L139 18 WT/YL
B16
FOG LAMP RELAY (IN PDC)
(8W-50-3)
B19
L39 18 WT/YL
2
HORN
(8W-41-3)
OPTION 1
OPTION 2
14
C102
3
C104
A109 18 OR/RD
4
FUEL PUMP MODULE
(8W-30-3)
S156
(8W-50-10)
L39 18 WT/OR
L39 18 WT/OR
2
LEFT FOG LAMP
(8W-50-10)
2
RIGHT FOG LAMP
(8W-50-10)
□□□ FRONT FOG LAMPS


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 15 50A
FUSE 26 10A
36
15
37
M21 20 YL/BR
S173
M21 20 YL/BR
B3 C301
M21 20 YL/BR
D3 C300
M21 20 YL/BR
7 C203
M21 20 PK/WT
2
UNDERHOOD LAMP
(8W-44-2)
7
COMPASS/ TEMPERATURE MIRROR
(8W-44-3)
LEFT DOME LAMP
(8W-44-3)
1
M21 20 PK/WT
A C325
M21 20 YL/BR
RIGHT DOME LAMP
(8W-44-3)
1
M21 20 PK/WT
A C324
M21 20 YL/BR
S351
M21 20 YL/BR
INSTRUMENT CLUSTER
(8W-40-2)
12 C1
M21 20 YL/BR
16 DATA LINK CONNECTOR
(8W-18-3)
M21 20 YL/BR
S204
M21 20 YL/BR
1
LEFT COURTESY LAMP
(8W-44-3)
M21 20 YL/BR
1
RIGHT COURTESY LAMP
(8W-44-3)
M21 20 YL/BR
3
AXLE LOCK SWITCH (OFF-ROAD PACKAGE)
(8W-31-14)

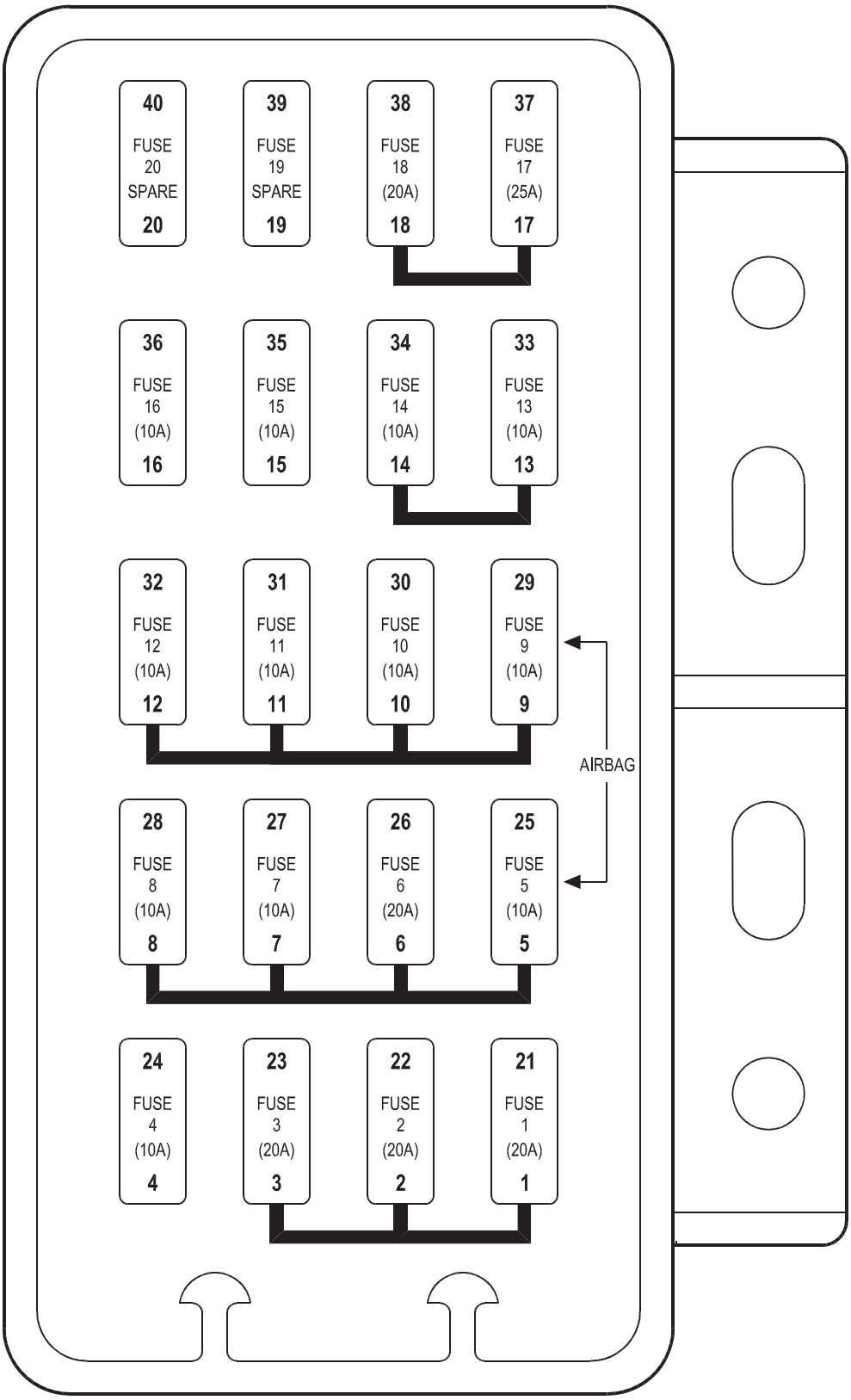

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 5 20A
FUSE 17 20A
FUSE 20 15A
FUSE 27 20A
5
18
19
24
25
39
A30 16 RD/WT
A205 18 RD/OR
F306 16 DB/PK
A701 18 BR/RD
11
4
C170
B2
C301
A701 18 BR/RD
A1
TRANSMISSION CONTROL RELAY (IN PDC)
(8W-31-3)
A3
11
C1
MULTI-FUNCTION SWITCH
(8W-52-2)
T16 18 YL/OR
3
C109
T16 16 YL/OR
F306 16 DB/PK
S210
A205 18 RD/OR
F306 16 DB/PK
F306 16 DB/PK
1
12
POWER OUTLET
(8W-41-2)
RADIO
(8W-47-2)
S159
T16 16 YL/OR
T16 16 YL/OR
T16 18 YL/OR
T16 16 YL/OR
19
C4
28
C4
5
3
POWERTRAIN CONTROL MODULE
(8W-31-3)
(8W-31-8)
VARIABLE LINE PRESSURE SENSOR
(8W-31-8)
TRANSMISSION SOLENOID/ PRESSURE SWITCH ASSEMBLY
(8W-31-4)
(8W-31-5)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE
24
20A
32
D2
REAR LOCKER RELAY
(8W-31-11)
D8
D10
FRONT LOCKER RELAY
(8W-31-11)
D14
A850
18
RD/WT
A750
18
TN/RD
4
7
C102
A850
18
RD/WT
A750
18
TN/RD
3
REAR LOCKER PUMP
(8W-31-11)
3
FRONT LOCKER PUMP
(8W-31-11)

8W-11 FUSE BLOCK

Component	Page
A/C Compressor Clutch Relay	8W-11-9
A/C-Heater Control	8W-11-7
Airbag Control Module	8W-11-5, 8
Antilock Brake Relay	8W-11-6
Automatic Shut Down Relay	8W-11-8
Back-Up Lamp Switch	8W-11-6
Blend Door Actuator	8W-11-7
Blower Motor Relay	8W-11-7
Brake Lamp Switch	8W-11-4
Brake Transmission Shift Interlock Solenoid	8W-11-8
Cigar Lighter/Power Outlet	8W-11-11
Compass/Temperature Mirror	8W-11-8
Controller Antilock Brake	8W-11-6
Daytime Running Lamp Module	8W-11-9, 11
Driver Door Ajar Switch	8W-11-4
Front Wiper Motor	8W-11-11
Fuel Pump Relay	8W-11-8
Fuse 1	8W-11-4
Fuse 2	8W-11-4
Fuse 3	8W-11-4
Fuse 4	8W-11-4
Fuse 5	8W-11-5
Fuse 6	8W-11-5
Fuse 7	8W-11-6
Fuse 8	8W-11-6, 7
Fuse 9	8W-11-8
Fuse 10	8W-11-8
Fuse 11	8W-11-9
Fuse 12	8W-11-8
Fuse 13	8W-11-10
Fuse 14	8W-11-10
Fuse 15	8W-11-10
Fuse 16	8W-11-11
Fuse 17	8W-11-11
Fuse 18	8W-11-11
Fuse Block	8W-11-2, 4, 5, 6, 8, 9, 10, 11
G300	8W-11-4
Headlamp Leveling Switch	8W-11-11
Heater Control	8W-11-7
Ignition Switch	8W-11-11
Instrument Cluster	8W-11-8
Left Headlamp	8W-11-11
Left Headlamp Leveling Motor	8W-11-11
Multi-Function Switch	8W-11-4, 10, 11
Passenger Airbag On-Off Switch	8W-11-8
Passenger Door Ajar Switch	8W-11-4
Powertrain Control Module	8W-11-8
Radio	8W-11-10
Rear Window Defogger Relay	8W-11-6, 10
Rear Window Defogger Switch	8W-11-10
Rear Wiper Motor	8W-11-5
Rear Wiper/Washer Switch	8W-11-5
Right Headlamp	8W-11-11
Right Headlamp Leveling Motor	8W-11-11
Sentry Key Immobilizer Module	8W-11-4, 8
Subwoofer	8W-11-4
Transmission Range Sensor	8W-11-6

FUSE BLOCK


40
FUSE
20
SPARE
20
39
FUSE
19
SPARE
19
38
FUSE
18
(20A)
18
37
FUSE
17
(25A)
17
36
FUSE
16
(10A)
16
35
FUSE
15
(10A)
15
34
FUSE
14
(10A)
14
33
FUSE
13
(10A)
13
32
FUSE
12
(10A)
12
31
FUSE
11
(10A)
11
30
FUSE
10
(10A)
10
29
FUSE
9
(10A)
9
AIRBAG
28
FUSE
8
(10A)
8
27
FUSE
7
(10A)
7
26
FUSE
6
(20A)
6
25
FUSE
5
(10A)
5
24
FUSE
4
(10A)
4
23
FUSE
3
(20A)
3
22
FUSE
2
(20A)
2
21
FUSE
1
(20A)
1

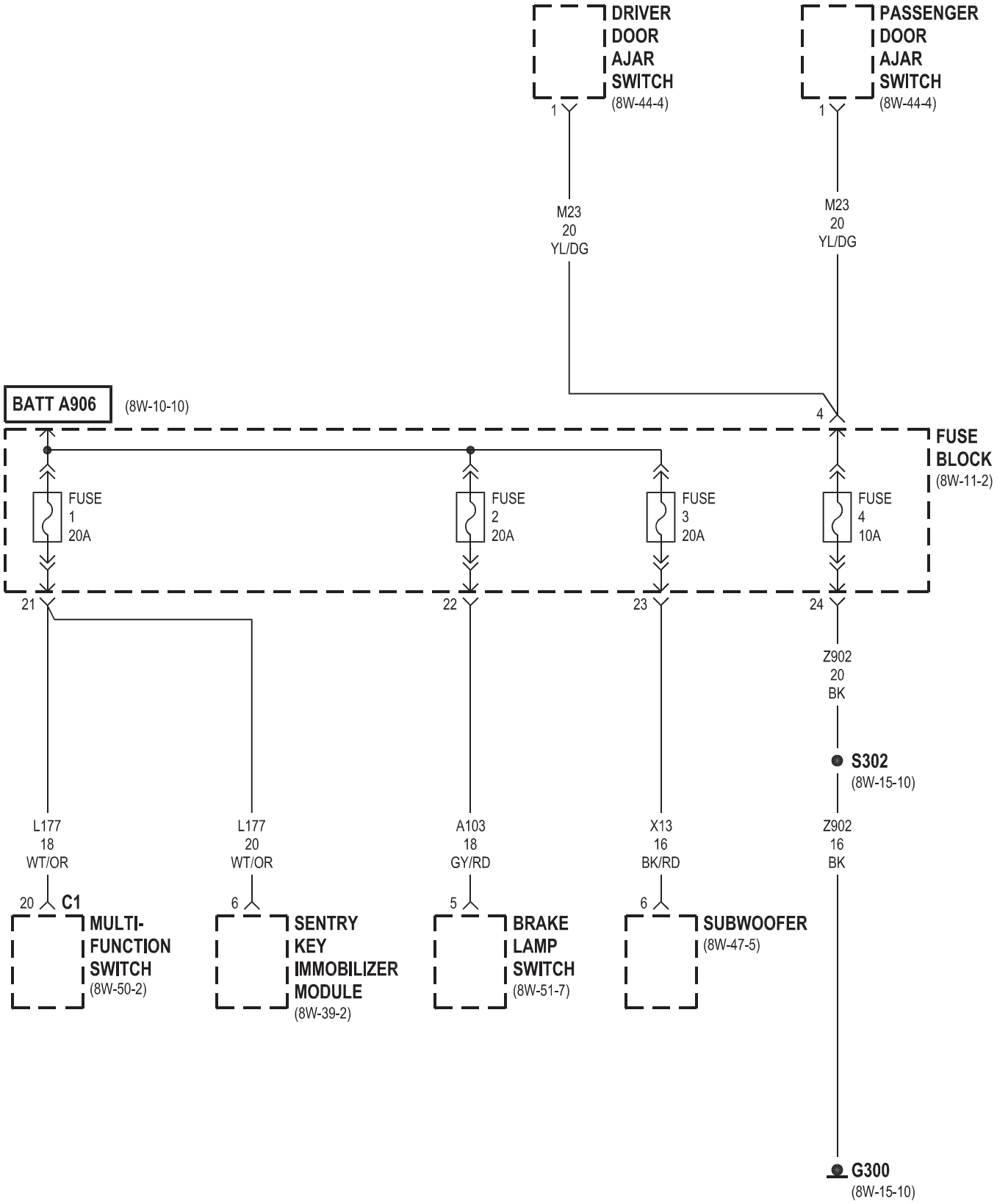
FUSES

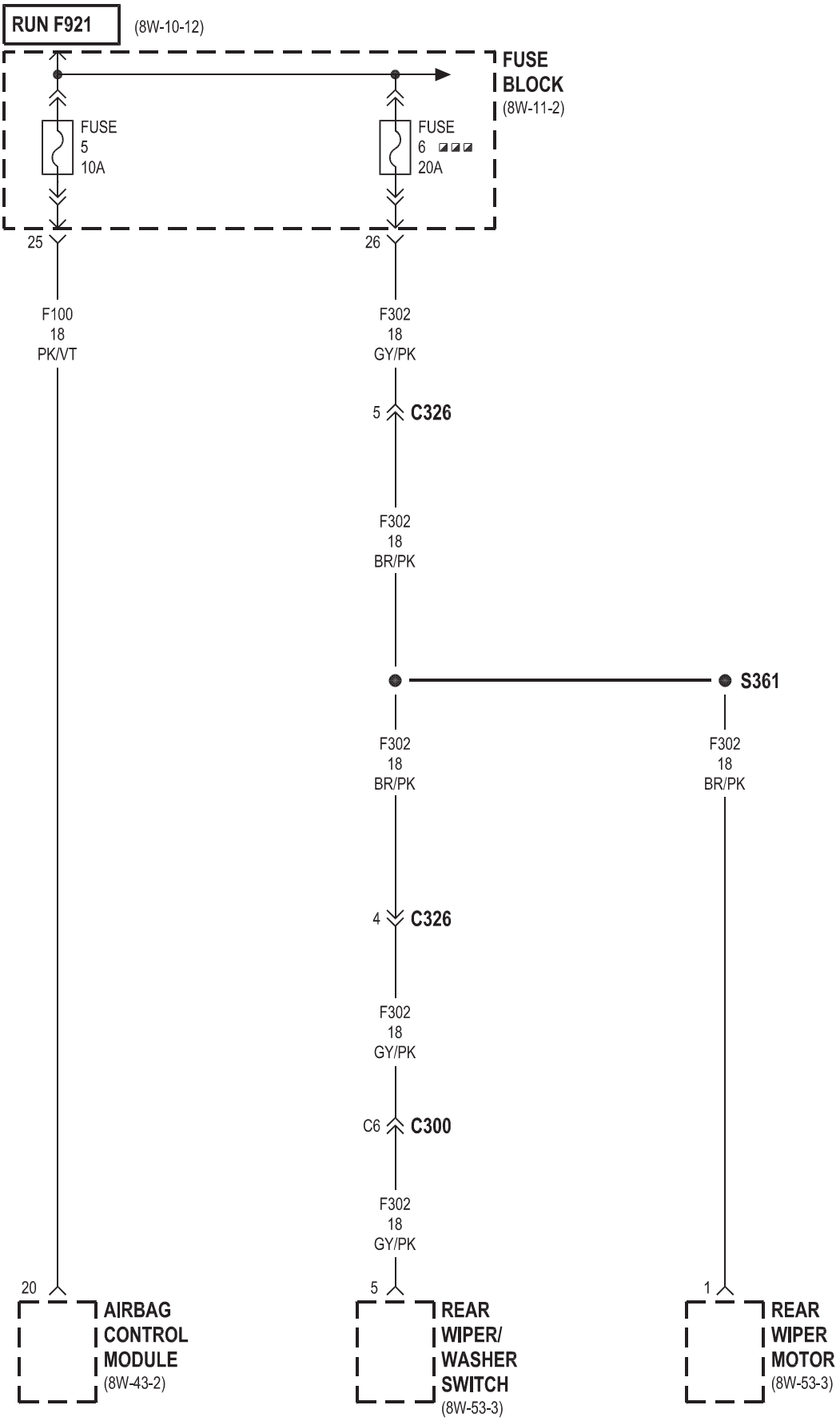
FUSE NO.	AMPS	FUSED CIRCUIT	FUNCTION
1	20A	L177 18WT/OR	FUSED B(+)
		L177 20WT/OR	FUSED B(+)
2	20A	A103 18GY/RD	FUSED B(+)
3	20A	X13 16BK/RD △△△	FUSED B(+)
4	10A	Z902 20BK	GROUND
5	10A	F100 18PK/VT	FUSED IGNITION SWITCH OUTPUT (RUN)
6	20A	F302 18GY/PK ◪◪◪	FUSED IGNITION SWITCH OUTPUT (RUN)
7	10A	F102 20PK/DB	FUSED IGNITION SWITCH OUTPUT (RUN)
8	10A	F504 20GY/PK	FUSED IGNITION SWITCH OUTPUT (RUN)
9	10A	F201 18PK/OR	FUSED IGNITION SWITCH OUTPUT (RUN-START)
		F201 18PK/OR	FUSED IGNITION SWITCH OUTPUT (RUN-START)
10	10A	F505 20PK/YL	FUSED IGNITION SWITCH OUTPUT (RUN-START)
		F505 20PK/YL	
11	10A	F960 20PK/LG	FUSED IGNITION SWITCH OUTPUT (RUN-START)
12	10A	F1 20PK/WT	FUSED IGNITION SWITCH OUTPUT (RUN-START)
		F1 20PK/WT	
13	10A	L5 20WT/LB	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
14	10A	X112 16DG/GY	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
15	10A	C515 20LB/OR ◪◪◪	FUSED REAR WINDOW DEFOGGER RELAY OUTPUT
16	10A	L45 20LG/DG ▲▲	DIMMER SWITCH LOW BEAM OUTPUT
		L45 20LG/DG ▲▲	
17	25A	F300 16OR/PK	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
		F300 16OR/PK	
18	20A	A207 18RD/LG	FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
19	-	-	-
20	-	-	-

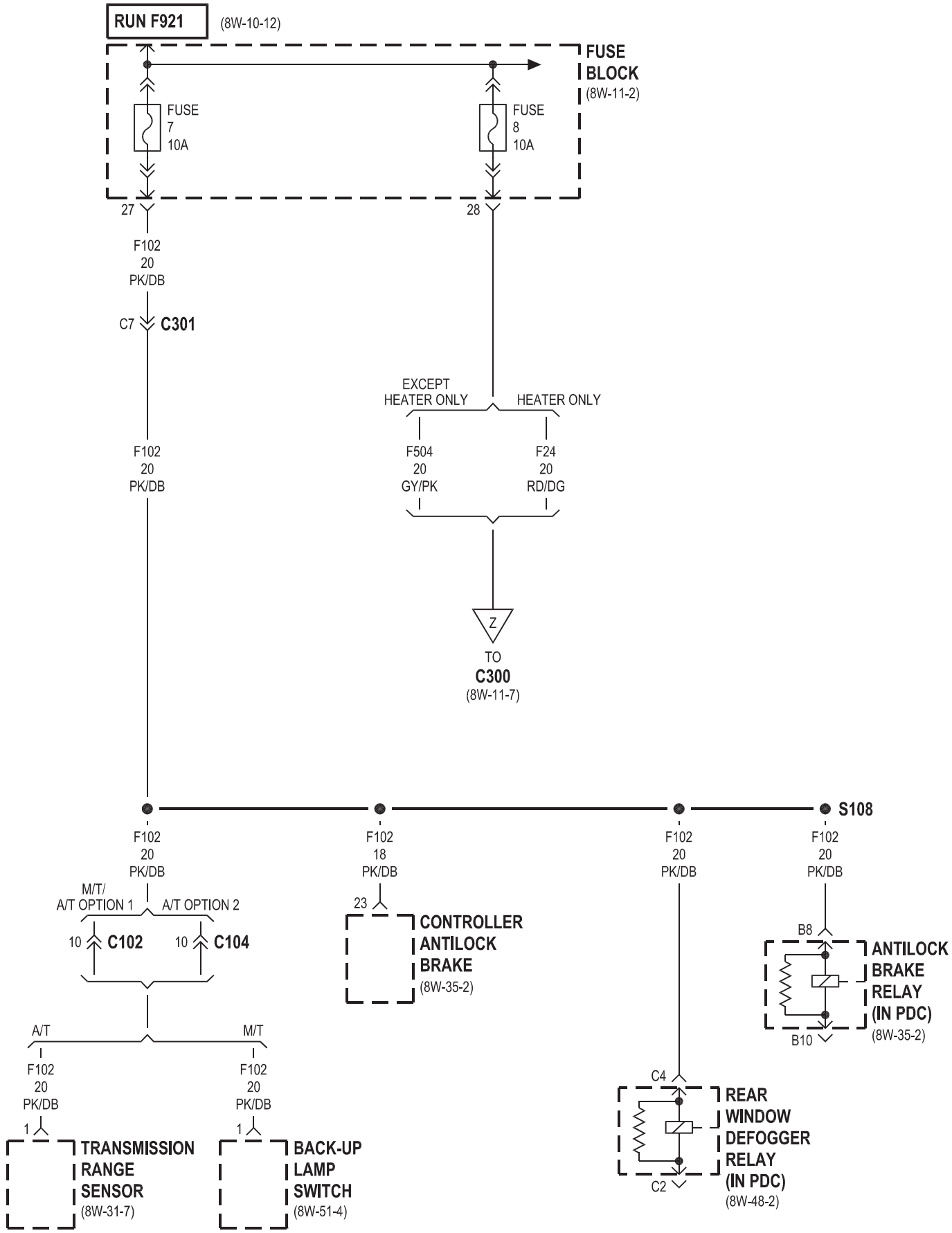
△△△ SUBWOOFER

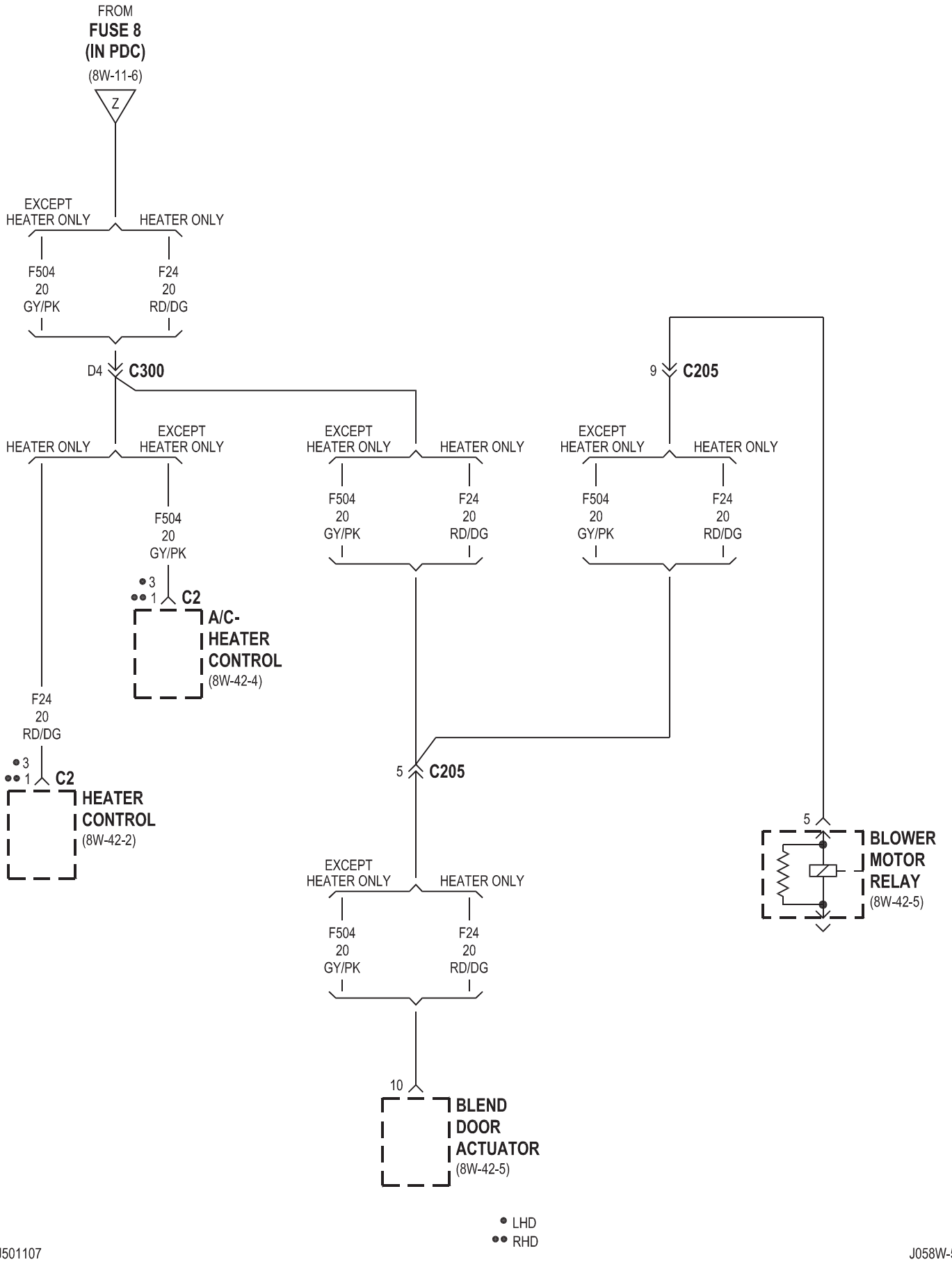
▲▲ EXPORT

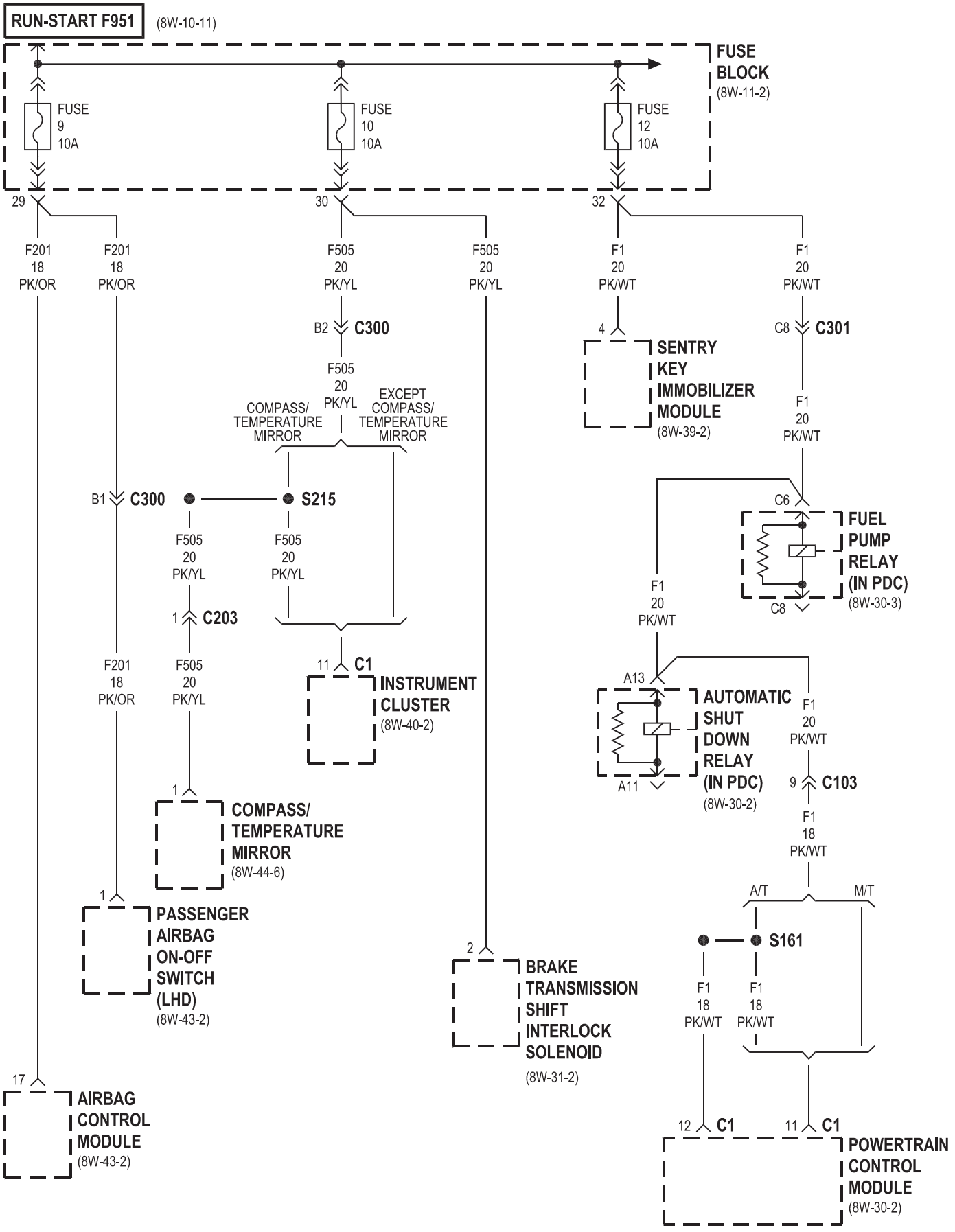
◪◪◪ HARDTOP

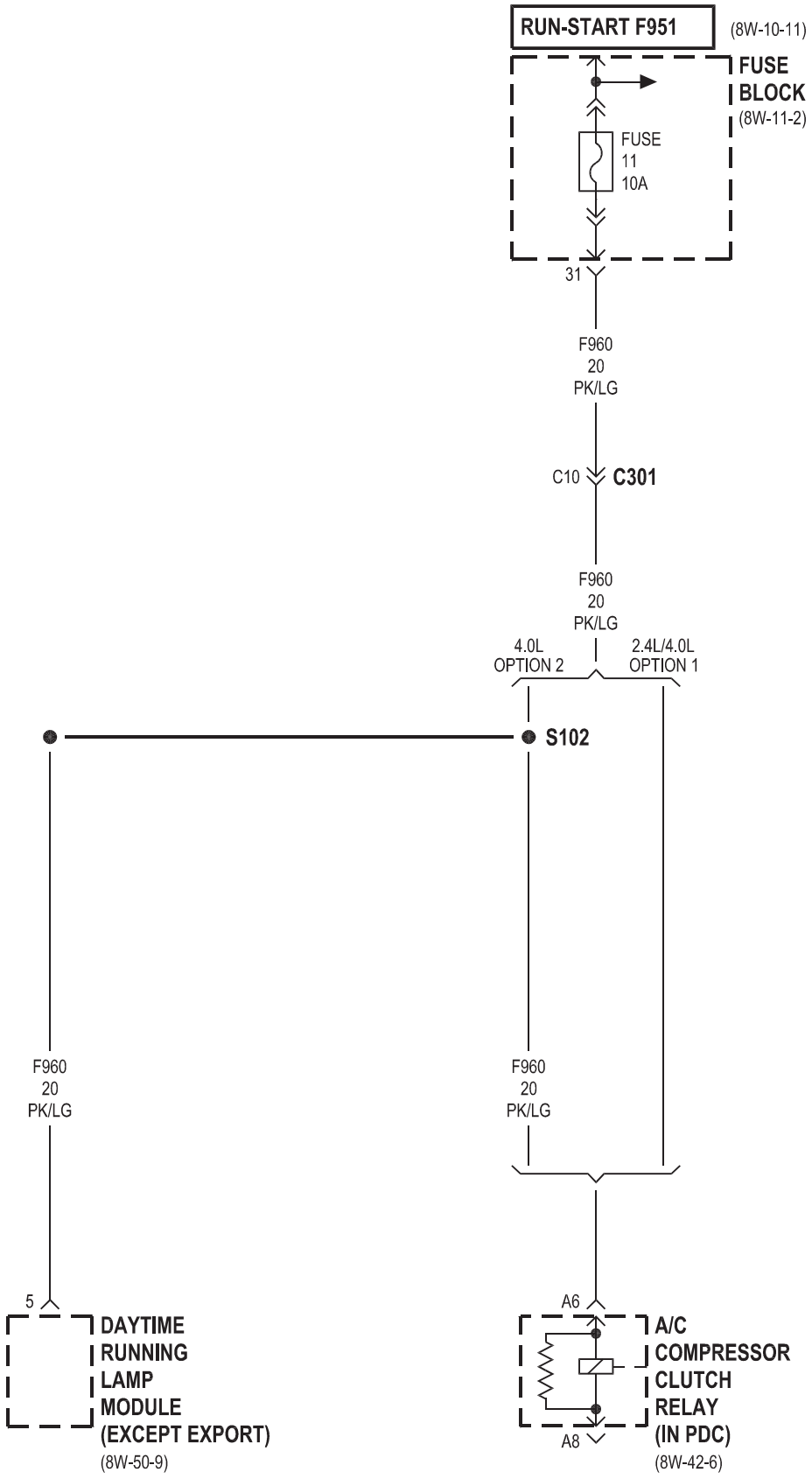

DRIVER DOOR AJAR SWITCH (8W-44-4)
PASSENGER DOOR AJAR SWITCH (8W-44-4)
1
1
M23 20 YL/DG
M23 20 YL/DG
BATT A906 (8W-10-10)
4
FUSE BLOCK (8W-11-2)
FUSE 1 20A
FUSE 2 20A
FUSE 3 20A
FUSE 4 10A
21
22
23
24
Z902 20 BK
S302 (8W-15-10)
L177 18 WT/OR
L177 20 WT/OR
A103 18 GY/RD
X13 16 BK/RD
Z902 16 BK
20 C1
6
5
6
MULTI-FUNCTION SWITCH (8W-50-2)
SENTRY KEY IMMOBILIZER MODULE (8W-39-2)
BRAKE LAMP SWITCH (8W-51-7)
SUBWOOFER (8W-47-5)
G300 (8W-15-10)

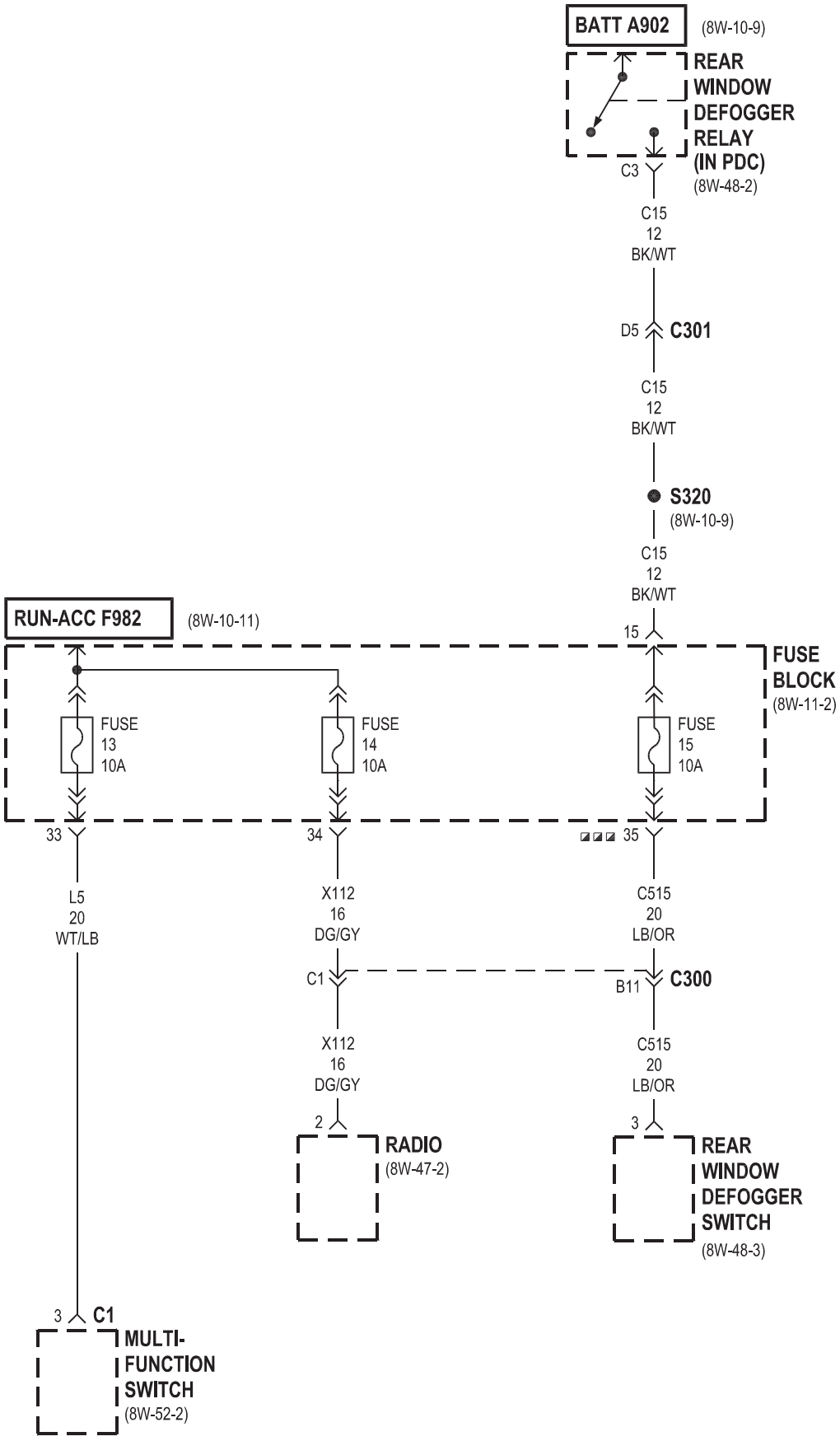

RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 5 10A
FUSE 6 20A
25
26
F100 18 PK/VT
F302 18 GY/PK
5 C326
F302 18 BR/PK
S361
F302 18 BR/PK
F302 18 BR/PK
4 C326
F302 18 GY/PK
C6 C300
F302 18 GY/PK
20
5
1
AIRBAG CONTROL MODULE
(8W-43-2)
REAR WIPER/ WASHER SWITCH
(8W-53-3)
REAR WIPER MOTOR
(8W-53-3)
HARDTOP


RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 7 10A
FUSE 8 10A
27
28
F102 20 PK/DB
C7 C301
EXCEPT HEATER ONLY
HEATER ONLY
F504 20 GY/PK
F24 20 RD/DG
Z
TO C300
(8W-11-7)
S108
M/T/ A/T OPTION 1
A/T OPTION 2
10 C102
10 C104
23
CONTROLLER ANTILOCK BRAKE
(8W-35-2)
B8
B10
ANTILOCK BRAKE RELAY (IN PDC)
(8W-35-2)
A/T
M/T
1
TRANSMISSION RANGE SENSOR
(8W-31-7)
BACK-UP LAMP SWITCH
(8W-51-4)
C4
C2
REAR WINDOW DEFOGGER RELAY (IN PDC)
(8W-48-2)

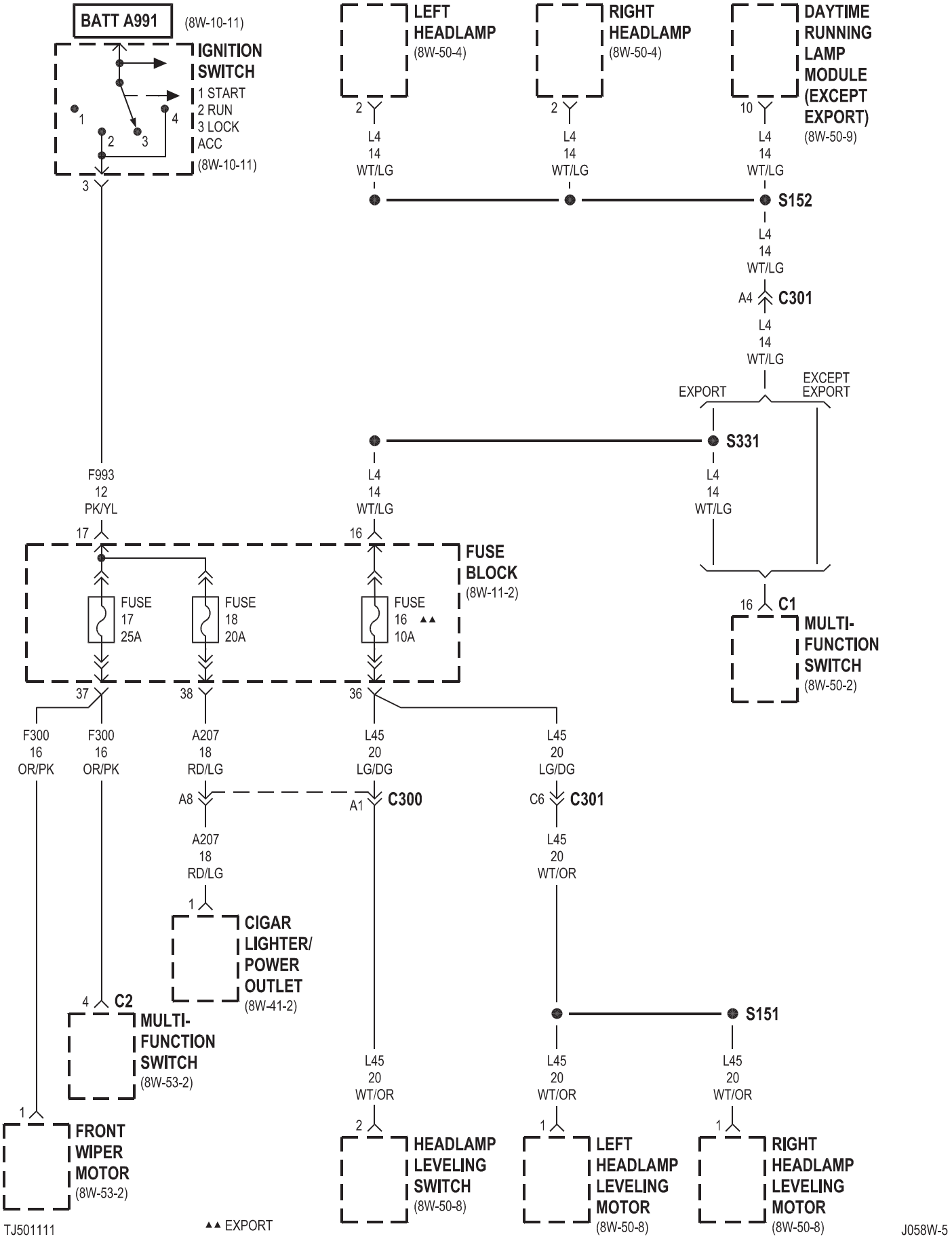

FROM
FUSE 8
(IN PDC)
(8W-11-6)
Z
EXCEPT HEATER ONLY
HEATER ONLY
F504 20 GY/PK
F24 20 RD/DG
D4 C300
9 C205
HEATER ONLY
EXCEPT HEATER ONLY
F504 20 GY/PK
3
1
C2
A/C-HEATER CONTROL
(8W-42-4)
F24 20 RD/DG
C2
HEATER CONTROL
(8W-42-2)
5 C205
5
BLOWER MOTOR RELAY
(8W-42-5)
10
BLEND DOOR ACTUATOR
(8W-42-5)
LHD
RHD


RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 9 10A
FUSE 10 10A
FUSE 12 10A
29
30
32
F201 18 PK/OR
F201 18 PK/OR
F505 20 PK/YL
F505 20 PK/YL
F1 20 PK/WT
F1 20 PK/WT
B2 C300
4
SENTRY KEY IMMOBILIZER MODULE
(8W-39-2)
C8 C301
F505 20 PK/YL
COMPASS/ TEMPERATURE MIRROR
EXCEPT COMPASS/ TEMPERATURE MIRROR
F1 20 PK/WT
B1 C300
S215
C6
FUEL PUMP RELAY (IN PDC)
(8W-30-3)
C8
F505 20 PK/YL
F505 20 PK/YL
F1 20 PK/WT
1 C203
A13
AUTOMATIC SHUT DOWN RELAY (IN PDC)
(8W-30-2)
A11
F201 18 PK/OR
F505 20 PK/YL
11 C1
INSTRUMENT CLUSTER
(8W-40-2)
F1 20 PK/WT
9 C103
F1 18 PK/WT
1
COMPASS/ TEMPERATURE MIRROR
(8W-44-6)
A/T
M/T
S161
1
PASSENGER AIRBAG ON-OFF SWITCH (LHD)
(8W-43-2)
2
BRAKE TRANSMISSION SHIFT INTERLOCK SOLENOID
(8W-31-2)
F1 18 PK/WT
F1 18 PK/WT
17
AIRBAG CONTROL MODULE
(8W-43-2)
12 C1
11 C1
POWERTRAIN CONTROL MODULE
(8W-30-2)


RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 11 10A
31
F960 20 PK/LG
C10
C301
F960 20 PK/LG
4.0L OPTION 2
2.4L/4.0L OPTION 1
S102
F960 20 PK/LG
F960 20 PK/LG
5
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
A6
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
A8
(8W-42-6)

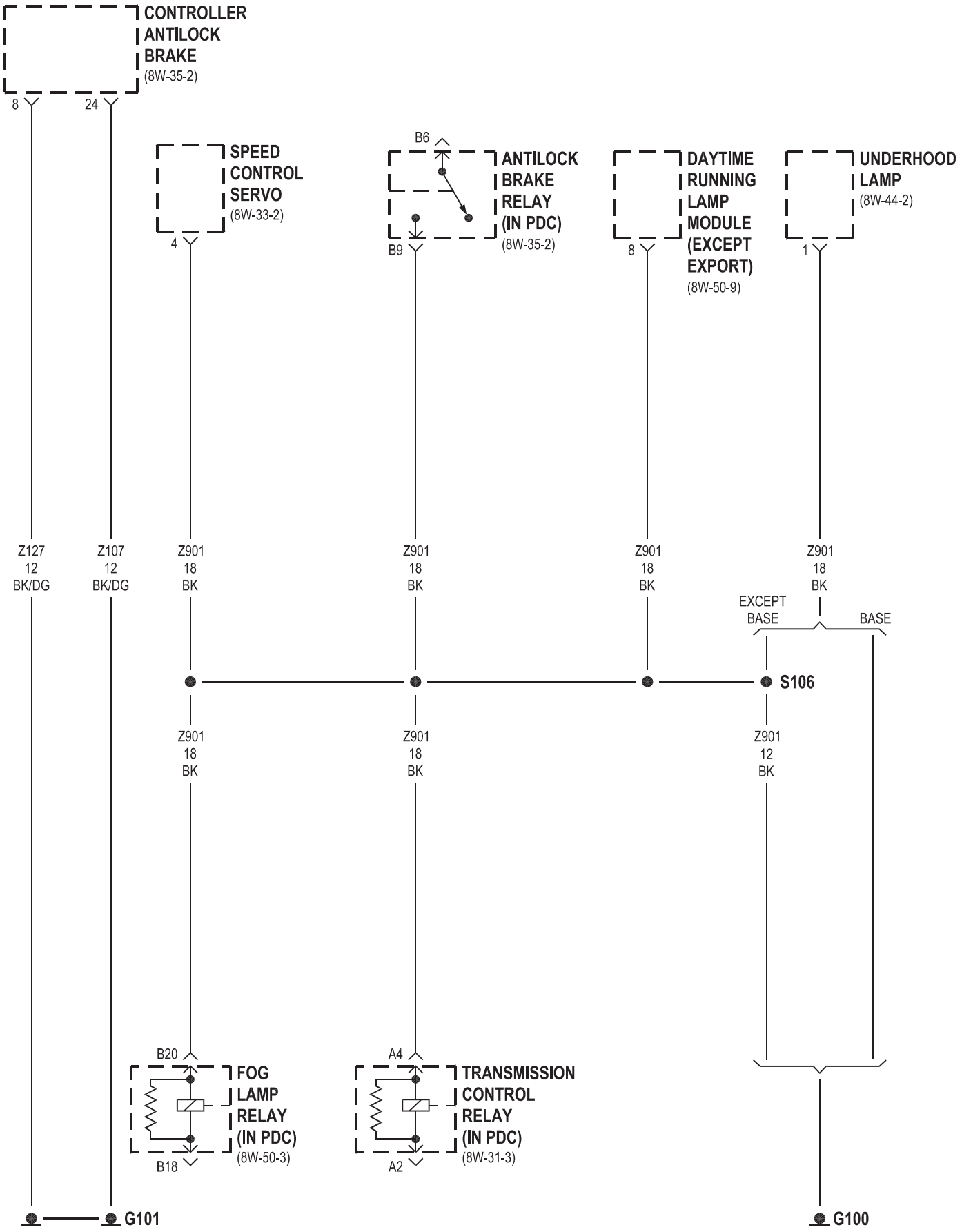

BATT A902
(8W-10-9)
REAR WINDOW DEFOGGER RELAY (IN PDC)
C3
(8W-48-2)
C15
12
BK/WT
D5
C301
C15
12
BK/WT
S320
(8W-10-9)
C15
12
BK/WT
RUN-ACC F982
(8W-10-11)
15
FUSE BLOCK
(8W-11-2)
FUSE 13 10A
FUSE 14 10A
FUSE 15 10A
33
34
35
L5
20
WT/LB
X112
16
DG/GY
C515
20
LB/OR
C1
B11
C300
X112
16
DG/GY
C515
20
LB/OR
2
3
RADIO
(8W-47-2)
REAR WINDOW DEFOGGER SWITCH
(8W-48-3)
3
C1
MULTI-FUNCTION SWITCH
(8W-52-2)

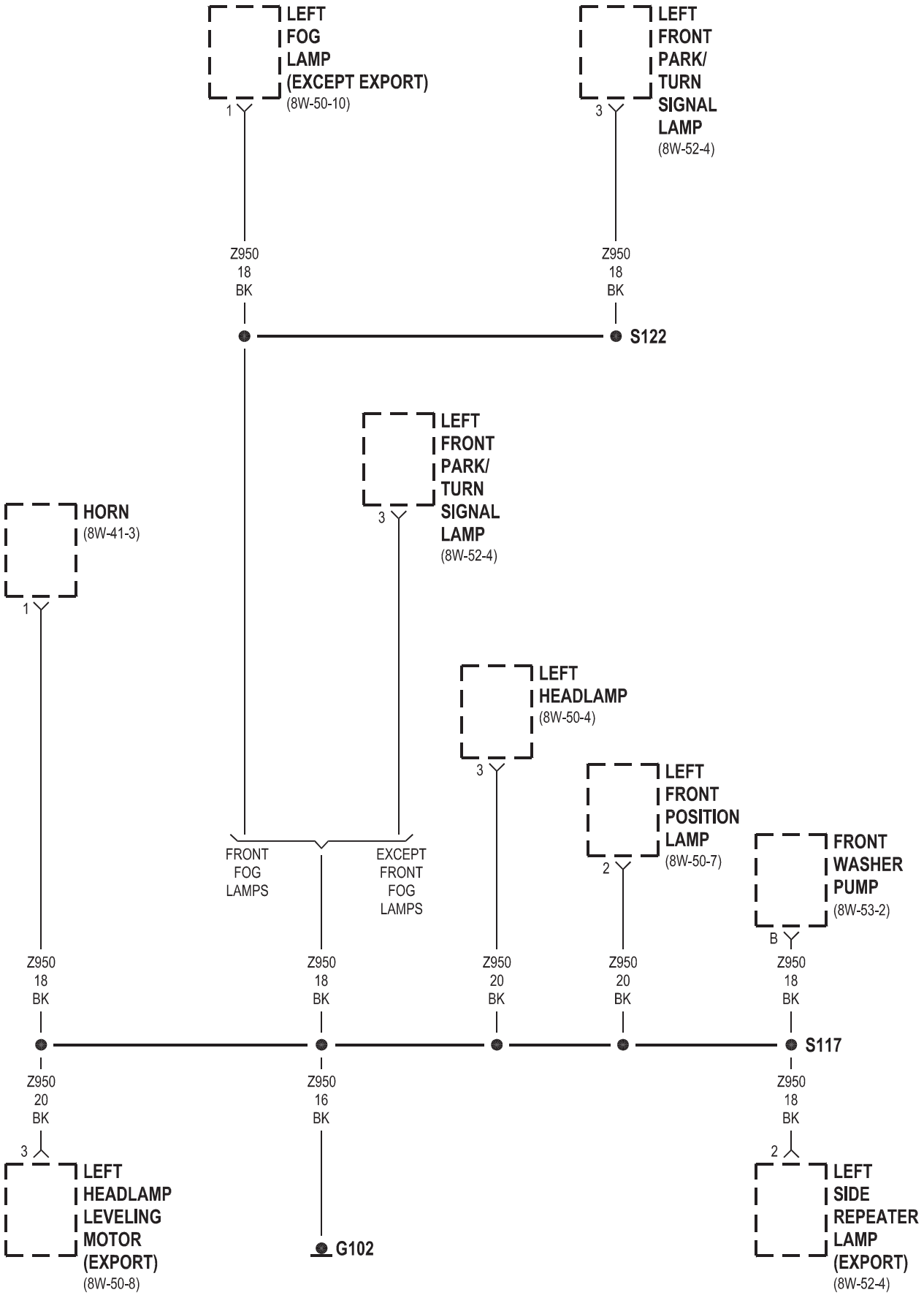
◪◪◪ HARD TOP

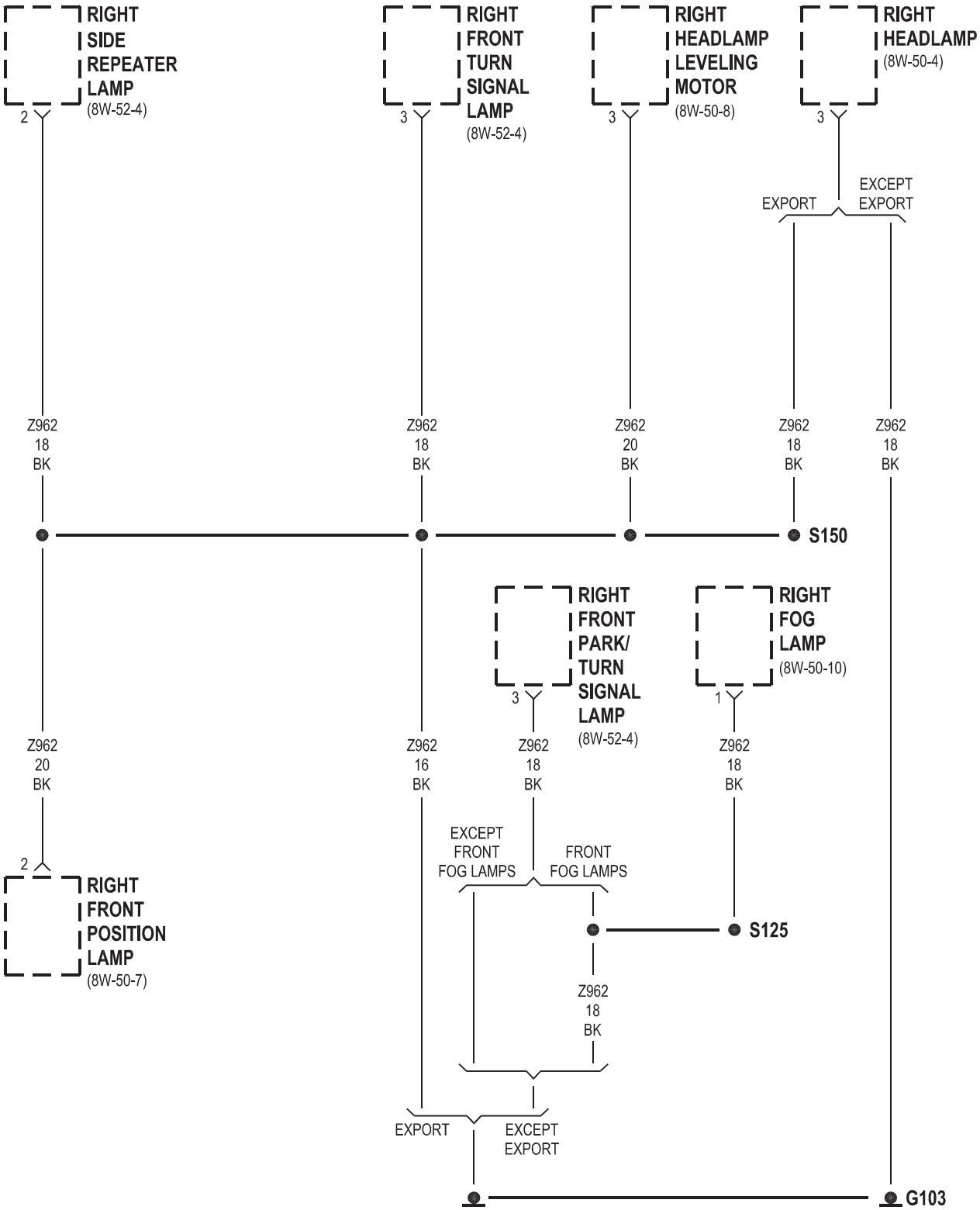

BATT A991
(8W-10-11)
IGNITION SWITCH
1 START
2 RUN
3 LOCK
ACC
(8W-10-11)
LEFT HEADLAMP
(8W-50-4)
RIGHT HEADLAMP
(8W-50-4)
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
L4 14 WT/LG
S152
C301
EXPORT
EXCEPT EXPORT
S331
F993 12 PK/YL
FUSE BLOCK
(8W-11-2)
FUSE 17 25A
FUSE 18 20A
FUSE 16 10A
C1
MULTI-FUNCTION SWITCH
(8W-50-2)
F300 16 OR/PK
A207 18 RD/LG
L45 20 LG/DG
C300
C301
L45 20 WT/OR
CIGAR LIGHTER/ POWER OUTLET
(8W-41-2)
C2
MULTI-FUNCTION SWITCH
(8W-53-2)
S151
FRONT WIPER MOTOR
(8W-53-2)
HEADLAMP LEVELING SWITCH
(8W-50-8)
LEFT HEADLAMP LEVELING MOTOR
(8W-50-8)
RIGHT HEADLAMP LEVELING MOTOR
(8W-50-8)
▲▲ EXPORT
TJ501111
J058W-5

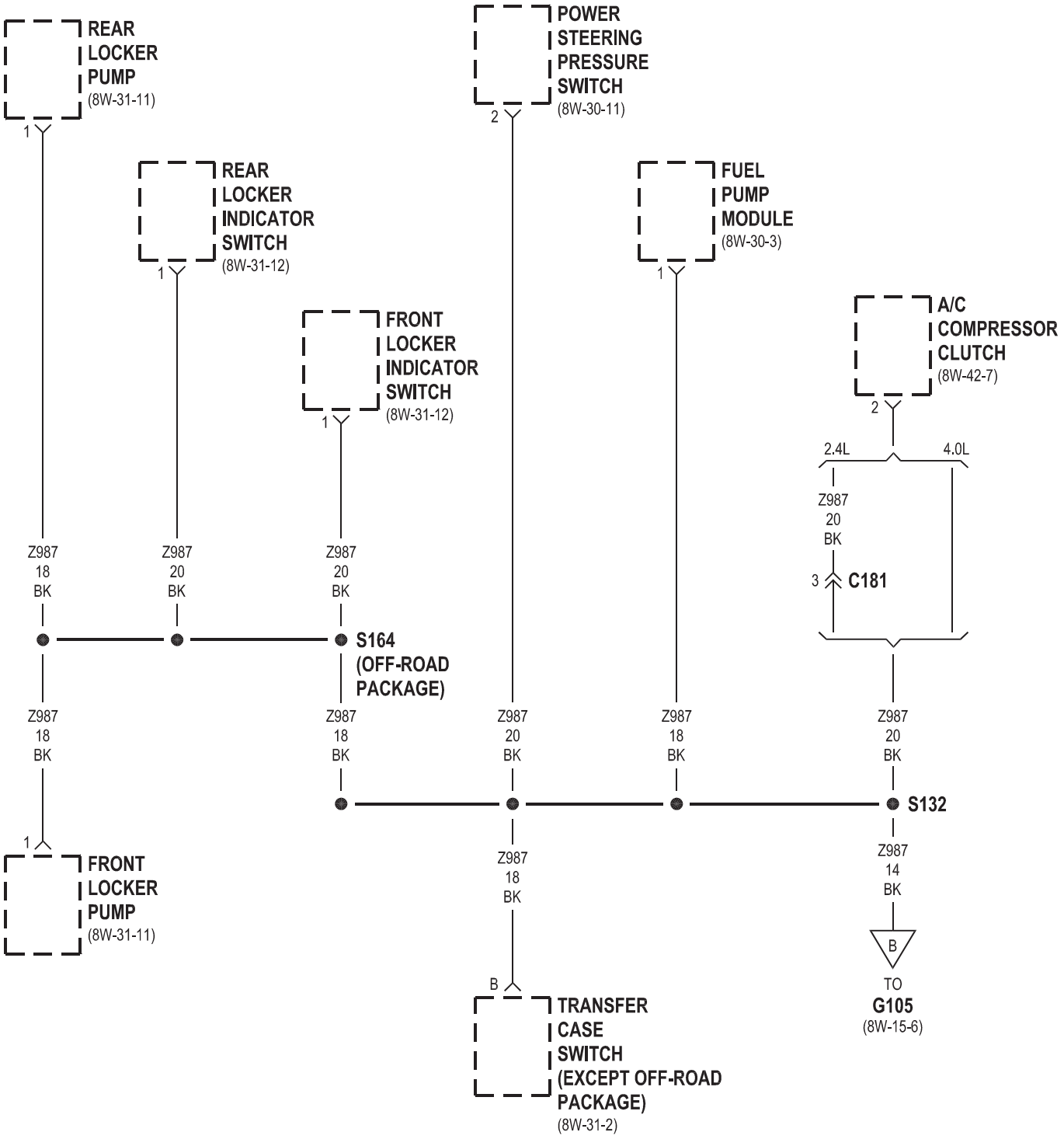
8W-15 GROUND DISTRIBUTION

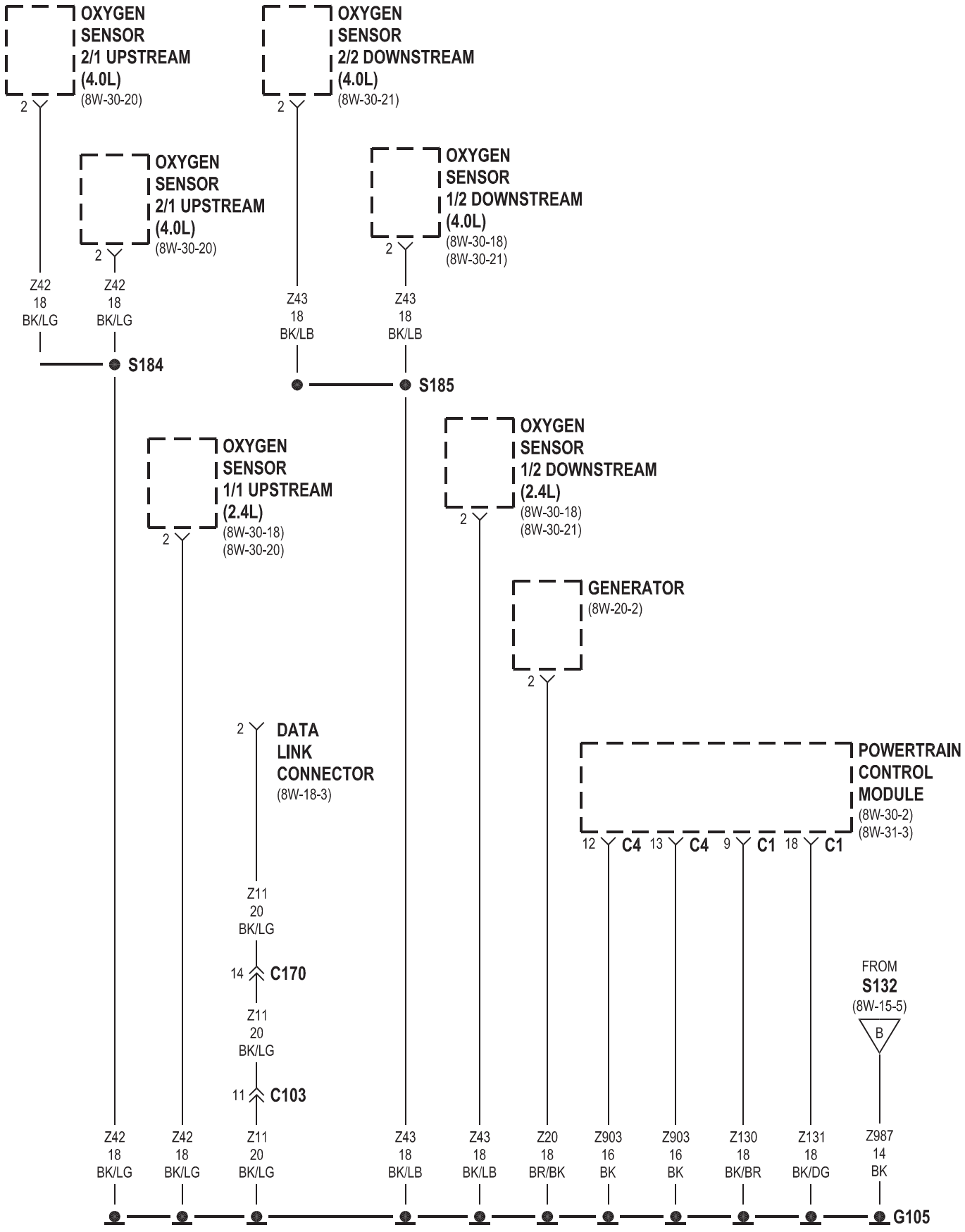
Component	Page
A/C Compressor Clutch	8W-15-5
A/C-Heater Control	8W-15-8, 9
Airbag Control Module	8W-15-10
Antilock Brake Relay	8W-15-2
Axle Lock Switch	8W-15-9
Battery	8W-15-7
Blend Door Actuator	8W-15-8
Blower Motor Relay	8W-15-9
Brake Lamp Switch	8W-15-10
Center High Mounted Stop Lamp	8W-15-10
Cigar Lighter/Power Outlet	8W-15-9
Clutch Pedal Position Switch	8W-15-10
Compass/Temperature Mirror	8W-15-9
Controller Antilock Brake	8W-15-2
Data Link Connector	8W-15-6, 9
Daytime Running Lamp Module	8W-15-2
Driver Door Ajar Switch	8W-15-10
Fog Lamp Relay	8W-15-2
Front Locker Indicator Switch	8W-15-5
Front Locker Pump	8W-15-5
Front Washer Pump	8W-15-3
Front Wiper Motor	8W-15-10
Fuel Pump Module	8W-15-5
Fuse 4	8W-15-10
Fuse Block	8W-15-10
G100	8W-15-2
G101	8W-15-2
G102	8W-15-3
G103	8W-15-4
G105	8W-15-5, 6
G106	8W-15-7
G107	8W-15-7
G108	8W-15-7
G190	8W-15-7
G200	8W-15-9
G201	8W-15-9
G202	8W-15-8
G203	8W-15-8
G204	8W-15-9
G300	8W-15-10
G301	8W-15-10
G302	8W-15-10
G303	8W-15-10
Generator	8W-15-6
Headlamp Leveling Switch	8W-15-9
Heater Control	8W-15-8, 9
Horn	8W-15-3
Ignition Switch	8W-15-10
Instrument Cluster	8W-15-9
Left Fog Lamp	8W-15-3
Left Front Park/Turn Signal Lamp	8W-15-3
Left Front Position Lamp	8W-15-3
Left Headlamp	8W-15-3
Left Headlamp Leveling Motor	8W-15-3
Left License Lamp	8W-15-10
Left Side Repeater Lamp	8W-15-3
Multi-Function Switch	8W-15-10
Natural Vacuum Leak Detection Assembly	8W-15-10
Oxygen Sensor 1/1 Upstream	8W-15-6
Oxygen Sensor 1/2 Downstream	8W-15-6
Oxygen Sensor 2/1 Upstream	8W-15-6
Oxygen Sensor 2/2 Downstream	8W-15-6
Passenger Door Ajar Switch	8W-15-10
Power Outlet	8W-15-9
Power Steering Pressure Switch	8W-15-5
Powertrain Control Module	8W-15-6
Radiator Fan Motor	8W-15-7
Radiator Fan Motor Circuit Breaker	8W-15-7
Radio	8W-15-9
Rear Fog Lamp	8W-15-10
Rear Locker Indicator Switch	8W-15-5
Rear Locker Pump	8W-15-5
Rear Washer Pump	8W-15-10
Rear Window Defogger	8W-15-10
Rear Window Defogger Switch	8W-15-9
Rear Wiper Motor	8W-15-10
Rear Wiper/Washer Switch	8W-15-9
Right Fog Lamp	8W-15-4
Right Front Park/Turn Signal Lamp	8W-15-4
Right Front Position Lamp	8W-15-4
Right Front Turn Signal Lamp	8W-15-4
Right Headlamp	8W-15-4
Right Headlamp Leveling Motor	8W-15-4
Right License Lamp	8W-15-10
Right Side Repeater Lamp	8W-15-4
Seat Belt Switch	8W-15-10
Sentry Key Immobilizer Module	8W-15-10
Speed Control Servo	8W-15-2
Subwoofer	8W-15-10
Tow/Haul Overdrive Off Switch	8W-15-9
Transfer Case Switch	8W-15-5
Transmission Control Relay	8W-15-2
Transmission Range Indicator Illumination	8W-15-10
Underhood Lamp	8W-15-2

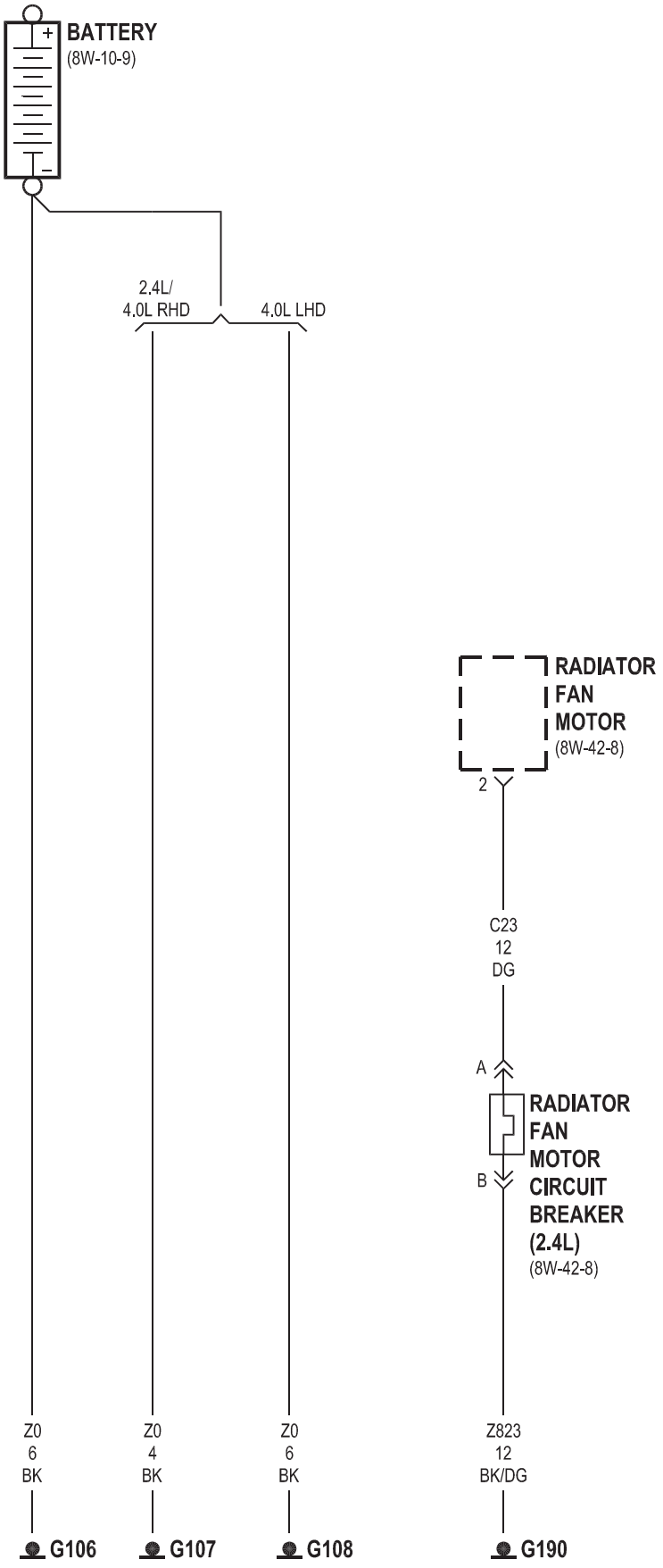

CONTROLLER ANTILOCK BRAKE
(8W-35-2)
8
24
SPEED CONTROL SERVO
(8W-33-2)
4
B6
ANTILOCK BRAKE RELAY (IN PDC)
(8W-35-2)
B9
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
8
UNDERHOOD LAMP
(8W-44-2)
1
Z127 12 BK/DG
Z107 12 BK/DG
Z901 18 BK
Z901 18 BK
Z901 18 BK
Z901 18 BK
EXCEPT BASE
BASE
S106
Z901 18 BK
Z901 18 BK
Z901 12 BK
B20
FOG LAMP RELAY (IN PDC)
(8W-50-3)
B18
A4
TRANSMISSION CONTROL RELAY (IN PDC)
(8W-31-3)
A2
G101
G100

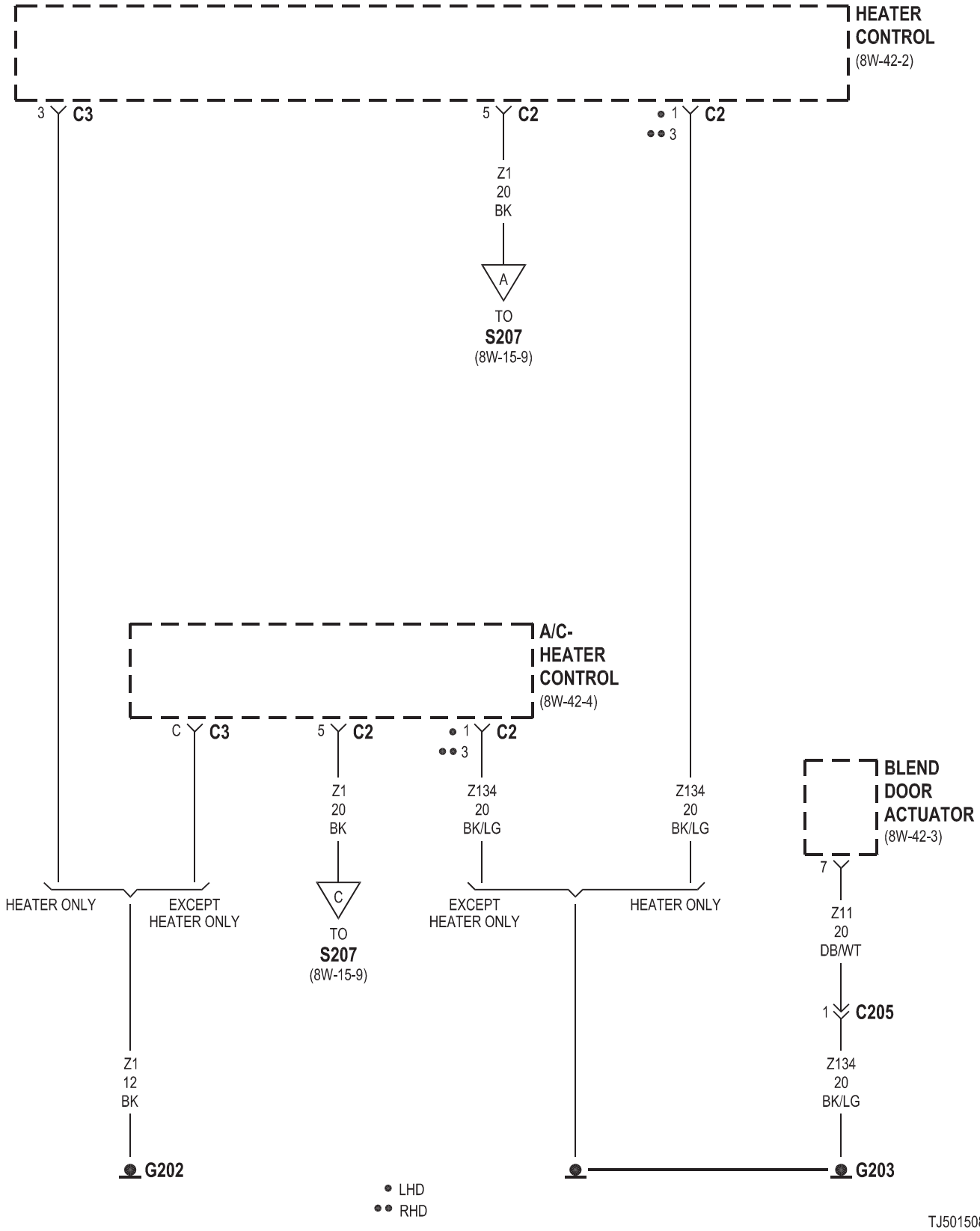

LEFT FOG LAMP (EXCEPT EXPORT)
(8W-50-10)
1
LEFT FRONT PARK/ TURN SIGNAL LAMP
(8W-52-4)
3
Z950 18 BK
Z950 18 BK
S122
LEFT FRONT PARK/ TURN SIGNAL LAMP
(8W-52-4)
3
HORN
(8W-41-3)
1
LEFT HEADLAMP
(8W-50-4)
3
LEFT FRONT POSITION LAMP
(8W-50-7)
2
FRONT WASHER PUMP
(8W-53-2)
B
FRONT FOG LAMPS
EXCEPT FRONT FOG LAMPS
Z950 18 BK
Z950 18 BK
Z950 20 BK
Z950 20 BK
Z950 18 BK
S117
Z950 20 BK
Z950 16 BK
Z950 18 BK
3
2
LEFT HEADLAMP LEVELING MOTOR (EXPORT)
(8W-50-8)
G102
LEFT SIDE REPEATER LAMP (EXPORT)
(8W-52-4)

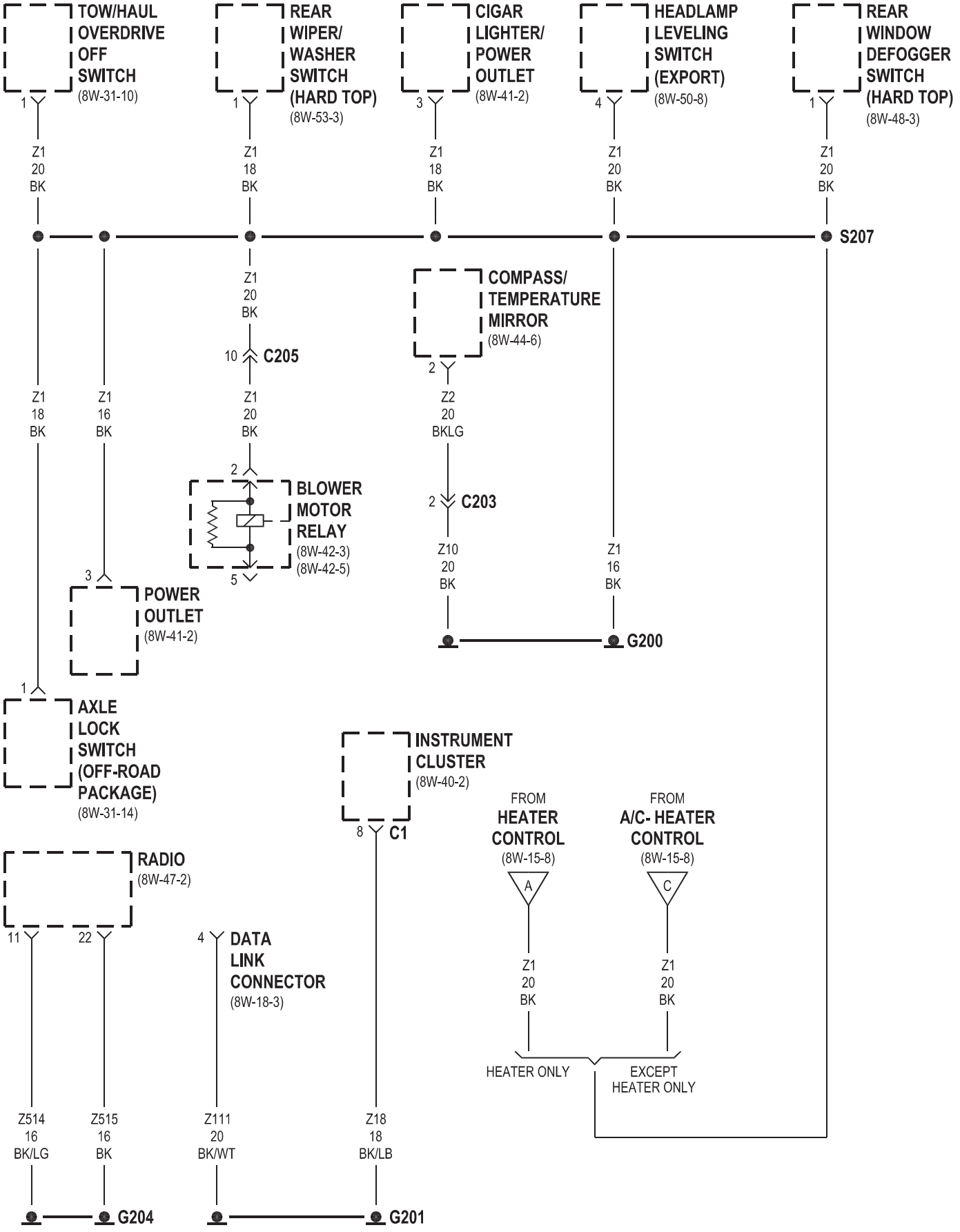

RIGHT SIDE REPEATER LAMP
(8W-52-4)
2
RIGHT FRONT TURN SIGNAL LAMP
(8W-52-4)
3
RIGHT HEADLAMP LEVELING MOTOR
(8W-50-8)
3
RIGHT HEADLAMP
(8W-50-4)
3
EXPORT
EXCEPT EXPORT
Z962 18 BK
Z962 18 BK
Z962 20 BK
Z962 18 BK
Z962 18 BK
S150
RIGHT FRONT PARK/ TURN SIGNAL LAMP
(8W-52-4)
3
RIGHT FOG LAMP
(8W-50-10)
1
Z962 20 BK
Z962 16 BK
Z962 18 BK
Z962 18 BK
EXCEPT FRONT FOG LAMPS
FRONT FOG LAMPS
2
RIGHT FRONT POSITION LAMP
(8W-50-7)
S125
Z962 18 BK
EXPORT
EXCEPT EXPORT
G103

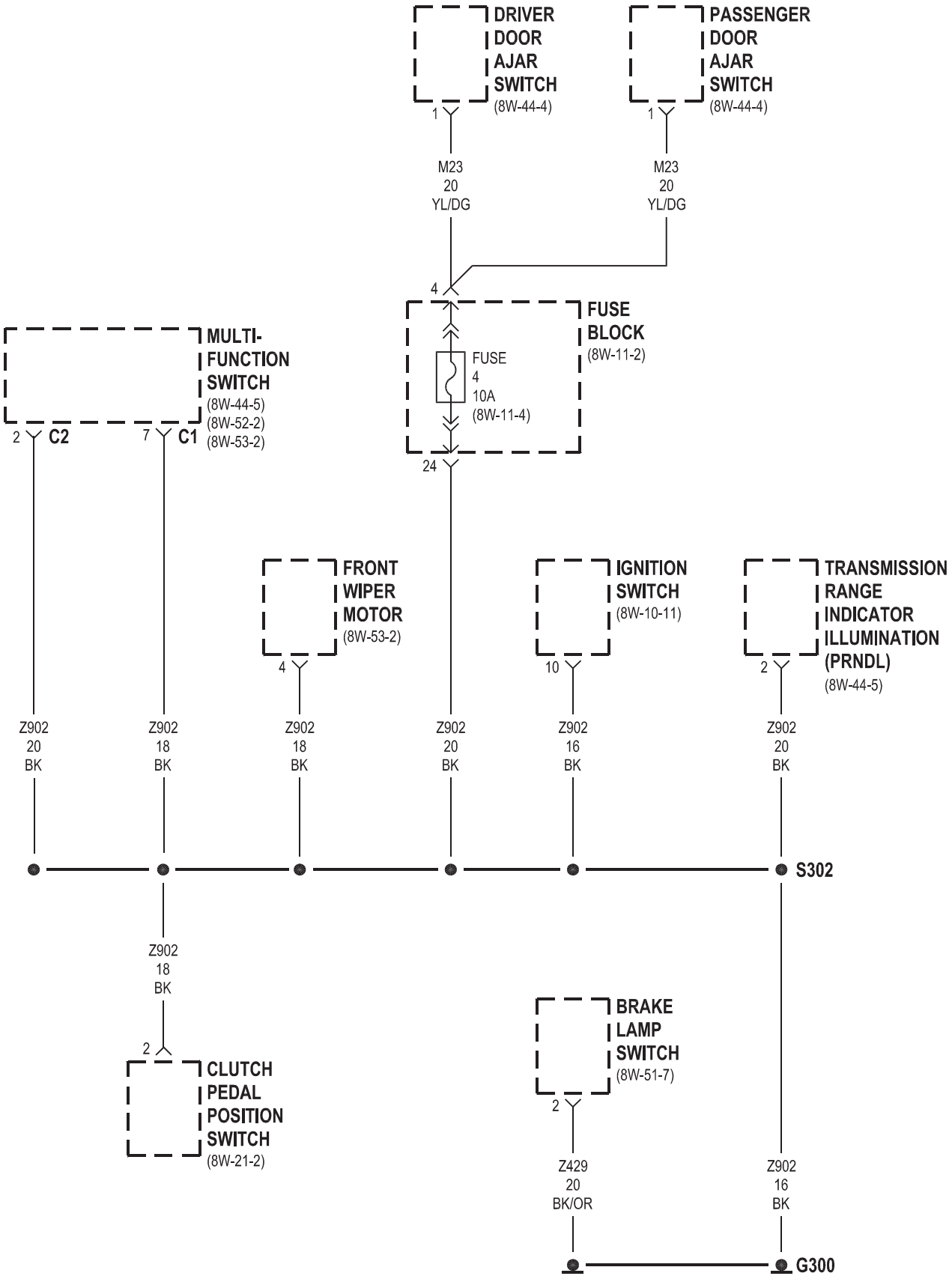

REAR LOCKER PUMP
(8W-31-11)
1
REAR LOCKER INDICATOR SWITCH
(8W-31-12)
1
FRONT LOCKER INDICATOR SWITCH
(8W-31-12)
1
POWER STEERING PRESSURE SWITCH
(8W-30-11)
2
FUEL PUMP MODULE
(8W-30-3)
1
A/C COMPRESSOR CLUTCH
(8W-42-7)
2
2.4L
4.0L
Z987 20 BK
3
C181
Z987 18 BK
Z987 20 BK
Z987 20 BK
S164
(OFF-ROAD PACKAGE)
Z987 18 BK
Z987 18 BK
Z987 20 BK
Z987 18 BK
Z987 20 BK
S132
1
FRONT LOCKER PUMP
(8W-31-11)
Z987 18 BK
B
TRANSFER CASE SWITCH
(EXCEPT OFF-ROAD PACKAGE)
(8W-31-2)
Z987 14 BK
B
TO
G105
(8W-15-6)

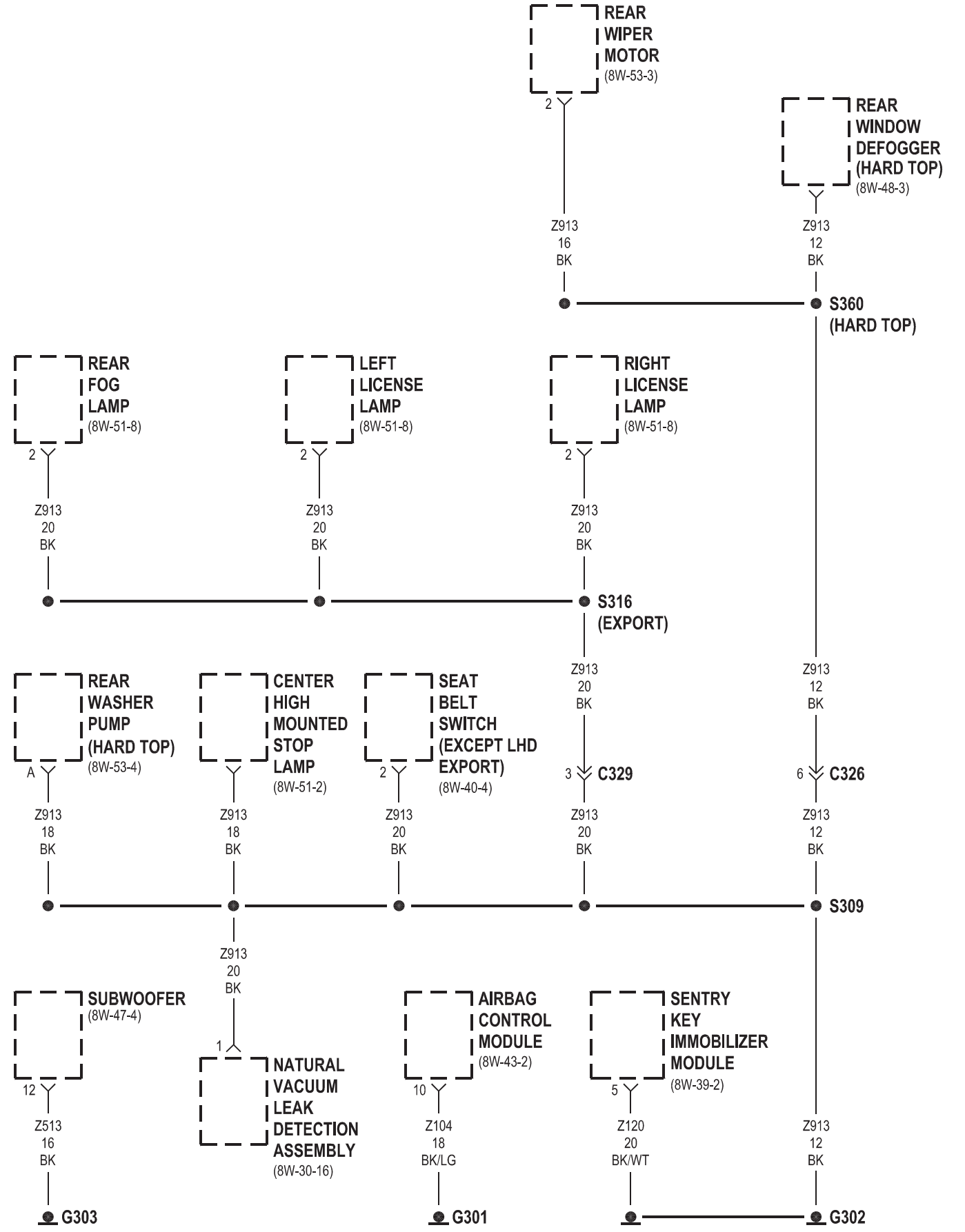

OXYGEN SENSOR 2/1 UPSTREAM (4.0L)
(8W-30-20)
OXYGEN SENSOR 2/2 DOWNSTREAM (4.0L)
(8W-30-21)
OXYGEN SENSOR 2/1 UPSTREAM (4.0L)
(8W-30-20)
OXYGEN SENSOR 1/2 DOWNSTREAM (4.0L)
(8W-30-18)
(8W-30-21)
Z42 18 BK/LG
Z42 18 BK/LG
Z43 18 BK/LB
Z43 18 BK/LB
S184
S185
OXYGEN SENSOR 1/1 UPSTREAM (2.4L)
(8W-30-18)
(8W-30-20)
OXYGEN SENSOR 1/2 DOWNSTREAM (2.4L)
(8W-30-18)
(8W-30-21)
GENERATOR
(8W-20-2)
DATA LINK CONNECTOR
(8W-18-3)
POWERTRAIN CONTROL MODULE
(8W-30-2)
(8W-31-3)
12 C4
13 C4
9 C1
18 C1
Z11 20 BK/LG
14 C170
Z11 20 BK/LG
11 C103
FROM S132
(8W-15-5)
B
Z42 18 BK/LG
Z42 18 BK/LG
Z11 20 BK/LG
Z43 18 BK/LB
Z43 18 BK/LB
Z20 18 BR/BK
Z903 16 BK
Z903 16 BK
Z130 18 BK/BR
Z131 18 BK/DG
Z987 14 BK
G105


BATTERY
(8W-10-9)
2.4L/
4.0L RHD
4.0L LHD
RADIATOR
FAN
MOTOR
(8W-42-8)
2
C23
12
DG
A
RADIATOR
FAN
MOTOR
CIRCUIT
BREAKER
(2.4L)
(8W-42-8)
B
Z0
6
BK
Z0
4
BK
Z0
6
BK
Z823
12
BK/DG
G106
G107
G108
G190


HEATER CONTROL
(8W-42-2)
3 C3
5 C2
1 C2
3
Z1
20
BK
A
TO
S207
(8W-15-9)
A/C-HEATER CONTROL
(8W-42-4)
C C3
5 C2
1 C2
3
Z1
20
BK
Z134
20
BK/LG
Z134
20
BK/LG
BLEND DOOR ACTUATOR
(8W-42-3)
7
HEATER ONLY
EXCEPT HEATER ONLY
C
TO
S207
(8W-15-9)
EXCEPT HEATER ONLY
HEATER ONLY
Z11
20
DB/WT
1 C205
Z1
12
BK
Z134
20
BK/LG
G202
G203
LHD
RHD

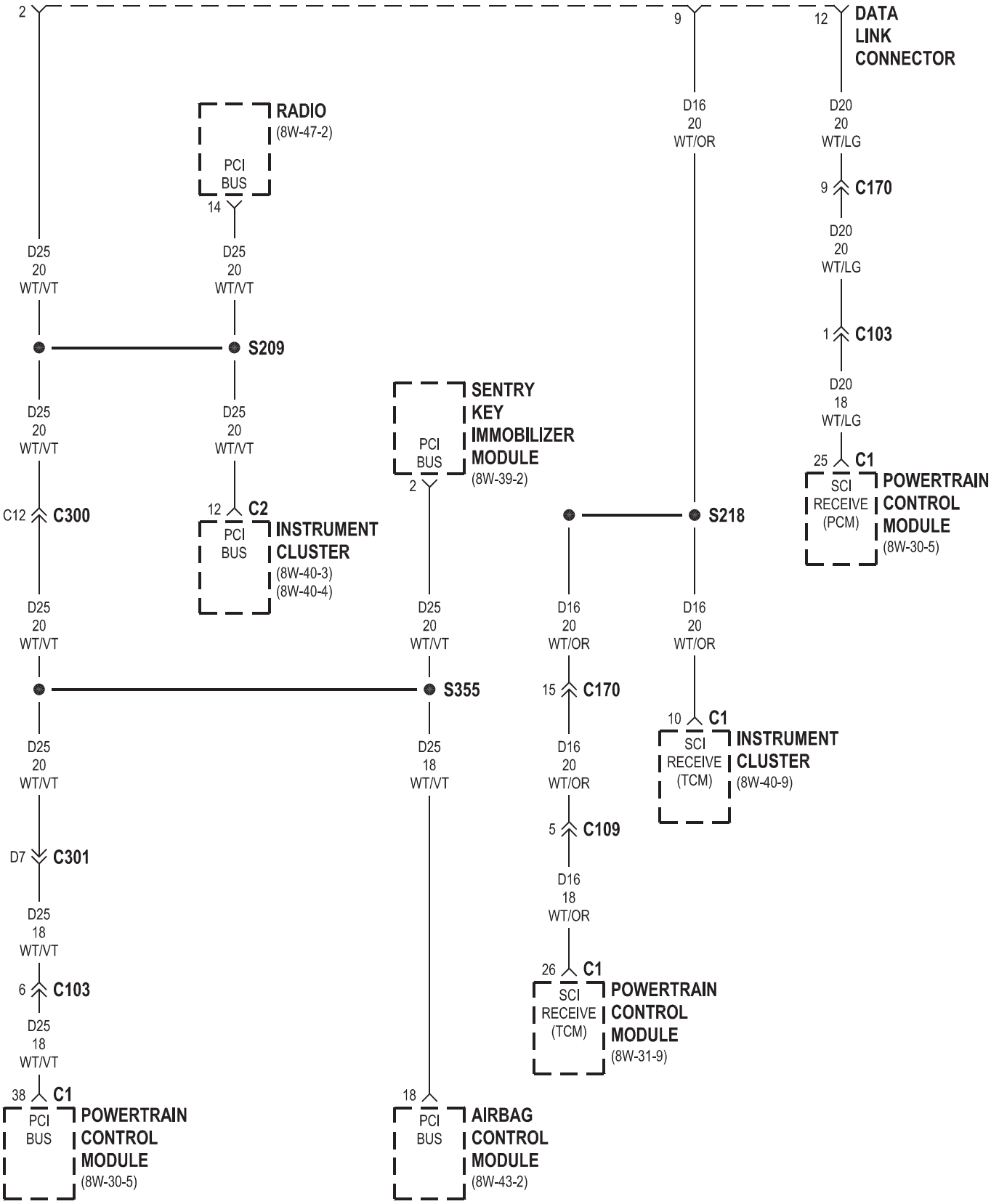

TOW/HAUL OVERDRIVE OFF SWITCH
(8W-31-10)
REAR WIPER/ WASHER SWITCH (HARD TOP)
(8W-53-3)
CIGAR LIGHTER/ POWER OUTLET
(8W-41-2)
HEADLAMP LEVELING SWITCH (EXPORT)
(8W-50-8)
REAR WINDOW DEFOGGER SWITCH (HARD TOP)
(8W-48-3)
S207
C205
COMPASS/ TEMPERATURE MIRROR
(8W-44-6)
BLOWER MOTOR RELAY
(8W-42-3)
(8W-42-5)
POWER OUTLET
(8W-41-2)
C203
G200
AXLE LOCK SWITCH (OFF-ROAD PACKAGE)
(8W-31-14)
INSTRUMENT CLUSTER
(8W-40-2)
C1
FROM HEATER CONTROL
(8W-15-8)
FROM A/C- HEATER CONTROL
(8W-15-8)
RADIO
(8W-47-2)
DATA LINK CONNECTOR
(8W-18-3)
HEATER ONLY
EXCEPT HEATER ONLY
G204
G201

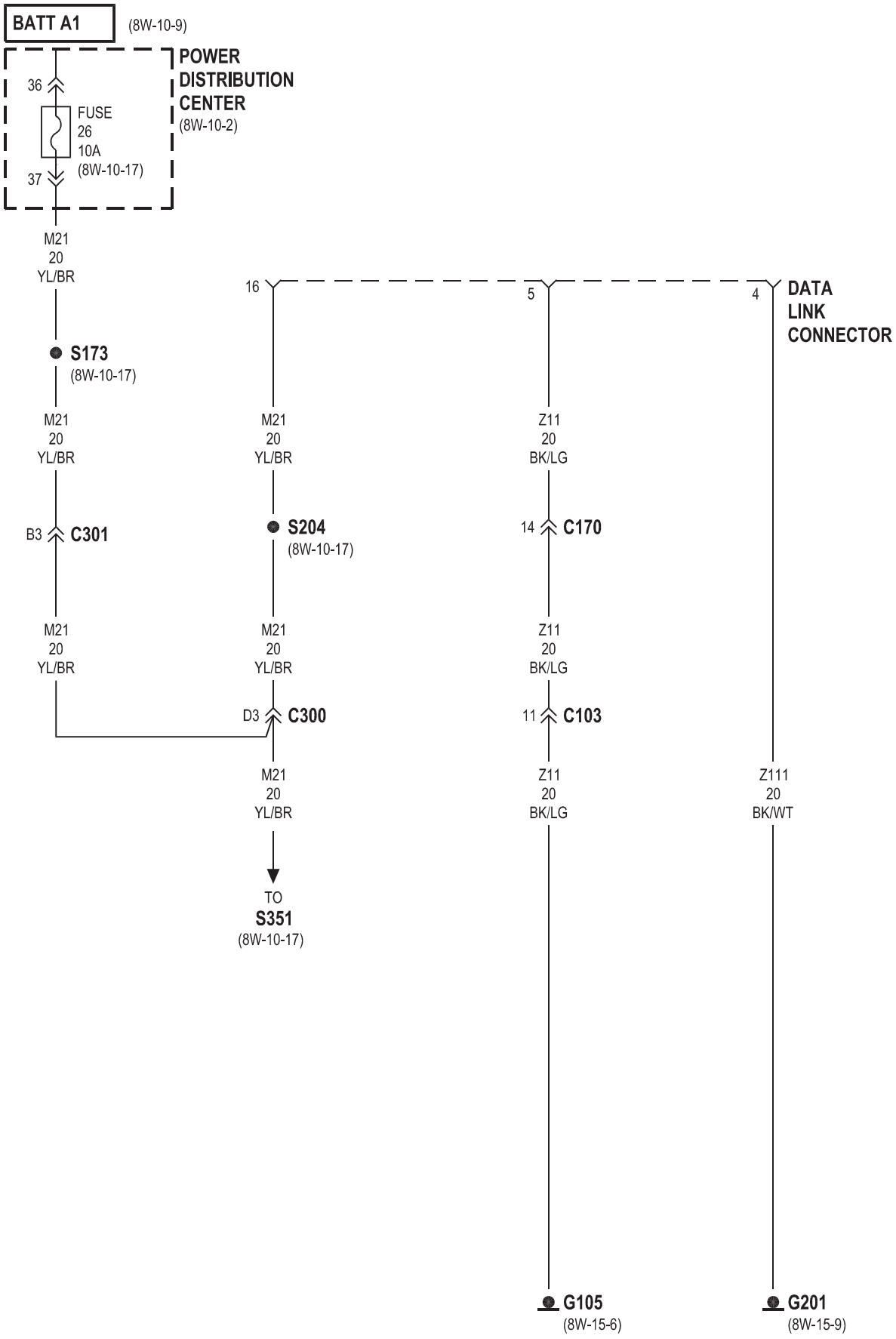

DRIVER DOOR AJAR SWITCH
(8W-44-4)
1
PASSENGER DOOR AJAR SWITCH
(8W-44-4)
1
M23
20
YL/DG
M23
20
YL/DG
4
FUSE BLOCK
(8W-11-2)
FUSE
4
10A
(8W-11-4)
24
MULTI-FUNCTION SWITCH
(8W-44-5)
(8W-52-2)
(8W-53-2)
2 C2
7 C1
FRONT WIPER MOTOR
(8W-53-2)
4
IGNITION SWITCH
(8W-10-11)
10
TRANSMISSION RANGE INDICATOR ILLUMINATION (PRNDL)
(8W-44-5)
2
Z902
20
BK
Z902
18
BK
Z902
18
BK
Z902
20
BK
Z902
16
BK
Z902
20
BK
S302
Z902
18
BK
2
CLUTCH PEDAL POSITION SWITCH
(8W-21-2)
BRAKE LAMP SWITCH
(8W-51-7)
2
Z429
20
BK/OR
Z902
16
BK
G300

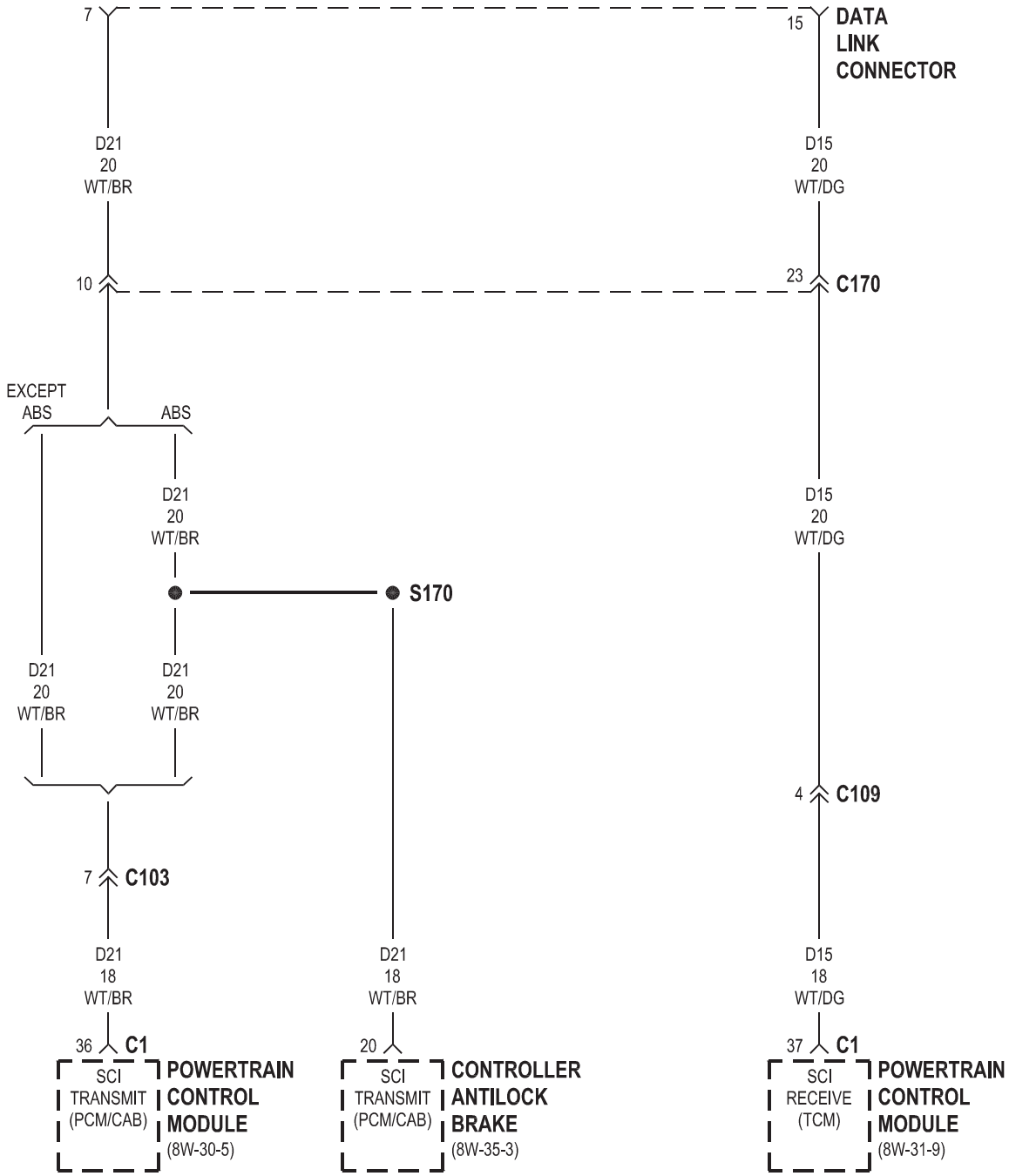

REAR WIPER MOTOR (8W-53-3)
2
Z913 16 BK
REAR WINDOW DEFOGGER (HARD TOP) (8W-48-3)
Z913 12 BK
S360 (HARD TOP)
REAR FOG LAMP (8W-51-8)
2
Z913 20 BK
LEFT LICENSE LAMP (8W-51-8)
2
Z913 20 BK
RIGHT LICENSE LAMP (8W-51-8)
2
Z913 20 BK
S316 (EXPORT)
REAR WASHER PUMP (HARD TOP) (8W-53-4)
A
Z913 18 BK
CENTER HIGH MOUNTED STOP LAMP (8W-51-2)
Z913 18 BK
SEAT BELT SWITCH (EXCEPT LHD EXPORT) (8W-40-4)
2
Z913 20 BK
Z913 20 BK
3
C329
Z913 12 BK
6
C326
Z913 12 BK
S309
Z913 20 BK
1
NATURAL VACUUM LEAK DETECTION ASSEMBLY (8W-30-16)
SUBWOOFER (8W-47-4)
12
Z513 16 BK
G303
AIRBAG CONTROL MODULE (8W-43-2)
10
Z104 18 BK/LG
G301
SENTRY KEY IMMOBILIZER MODULE (8W-39-2)
5
Z120 20 BK/WT
Z913 12 BK
G302

8W-18 BUS COMMUNICATIONS

Component	Page
Airbag Control Module	8W-18-2
Controller Antilock Brake	8W-18-4
Data Link Connector	8W-18-2, 3, 4
Fuse 26	8W-18-3
G105	8W-18-3
G201	8W-18-3
Instrument Cluster	8W-18-2
Power Distribution Center	8W-18-3
Powertrain Control Module	8W-18-2, 4
Radio	8W-18-2
Sentry Key Immobilizer Module	8W-18-2

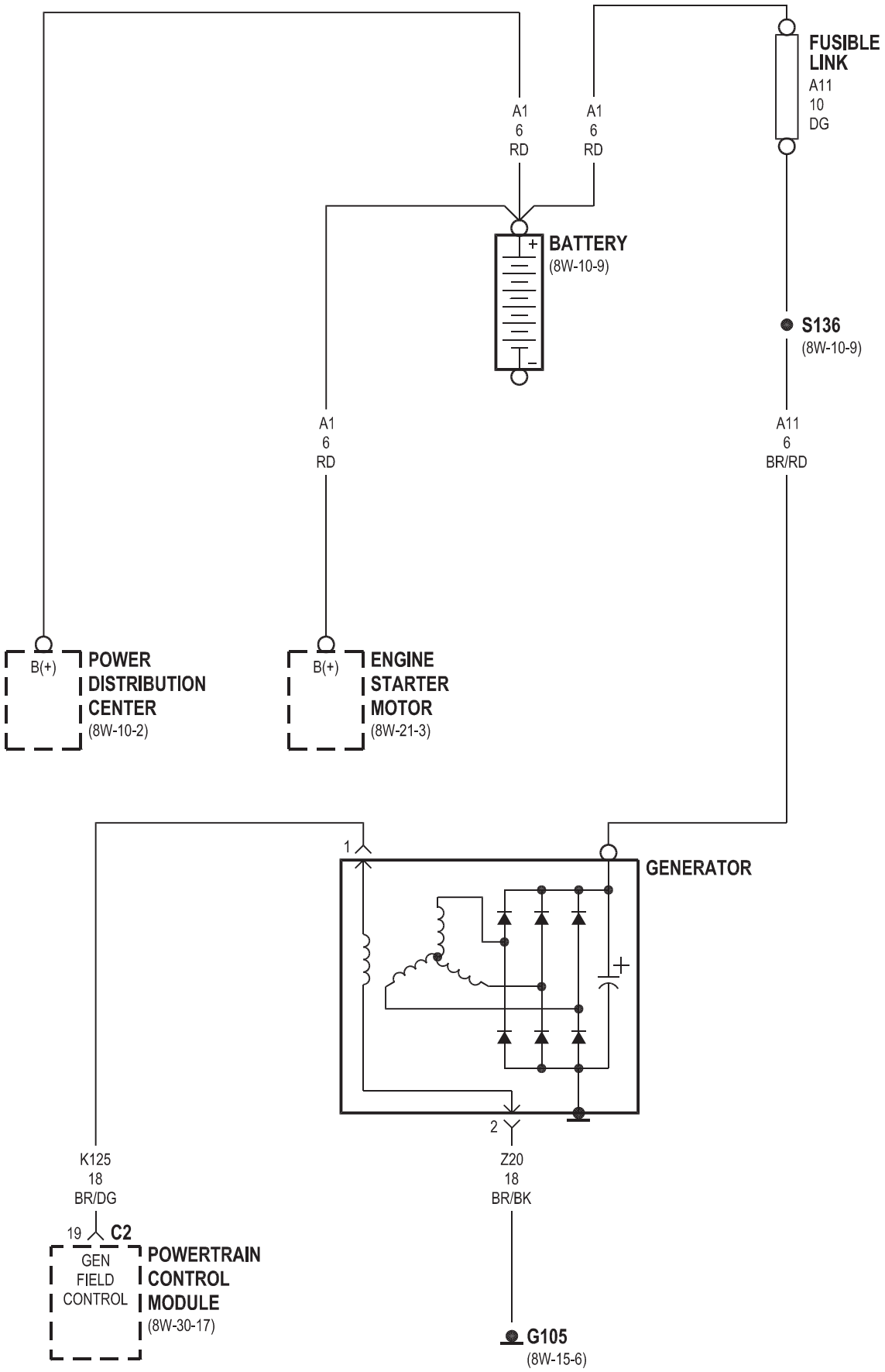

2
9
12
DATA LINK CONNECTOR
RADIO
(8W-47-2)
PCI BUS
14
D16 20 WT/OR
D20 20 WT/LG
9 C170
D25 20 WT/VT
D25 20 WT/VT
D20 20 WT/LG
1 C103
S209
D25 20 WT/VT
D25 20 WT/VT
SENTRY KEY IMMOBILIZER MODULE
(8W-39-2)
D20 18 WT/LG
25 C1
SCI RECEIVE (PCM)
POWERTRAIN CONTROL MODULE
(8W-30-5)
C12 C300
12 C2
PCI BUS
INSTRUMENT CLUSTER
(8W-40-3)
(8W-40-4)
S218
D25 20 WT/VT
D25 20 WT/VT
D16 20 WT/OR
D16 20 WT/OR
S355
15 C170
10 C1
SCI RECEIVE (TCM)
INSTRUMENT CLUSTER
(8W-40-9)
D25 20 WT/VT
D25 18 WT/VT
D16 20 WT/OR
D7 C301
5 C109
D25 18 WT/VT
D16 18 WT/OR
6 C103
26 C1
SCI RECEIVE (TCM)
POWERTRAIN CONTROL MODULE
(8W-31-9)
D25 18 WT/VT
38 C1
PCI BUS
POWERTRAIN CONTROL MODULE
(8W-30-5)
18
PCI BUS
AIRBAG CONTROL MODULE
(8W-43-2)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
36
FUSE 26 10A
(8W-10-17)
37
M21 20 YL/BR
S173
(8W-10-17)
M21 20 YL/BR
B3 C301
M21 20 YL/BR
16
5
4
DATA LINK CONNECTOR
M21 20 YL/BR
S204
(8W-10-17)
M21 20 YL/BR
D3 C300
M21 20 YL/BR
TO S351
(8W-10-17)
Z11 20 BK/LG
14 C170
Z11 20 BK/LG
11 C103
Z11 20 BK/LG
Z111 20 BK/WT
G105
(8W-15-6)
G201
(8W-15-9)


7
15
DATA LINK CONNECTOR
D21 20 WT/BR
D15 20 WT/DG
10
23
C170
EXCEPT ABS
ABS
D21 20 WT/BR
D15 20 WT/DG
S170
D21 20 WT/BR
D21 20 WT/BR
4
C109
7
C103
D21 18 WT/BR
D21 18 WT/BR
D15 18 WT/DG
36
C1
20
37
C1
SCI TRANSMIT (PCM/CAB)
POWERTRAIN CONTROL MODULE
(8W-30-5)
SCI TRANSMIT (PCM/CAB)
CONTROLLER ANTILOCK BRAKE
(8W-35-3)
SCI RECEIVE (TCM)
POWERTRAIN CONTROL MODULE
(8W-31-9)

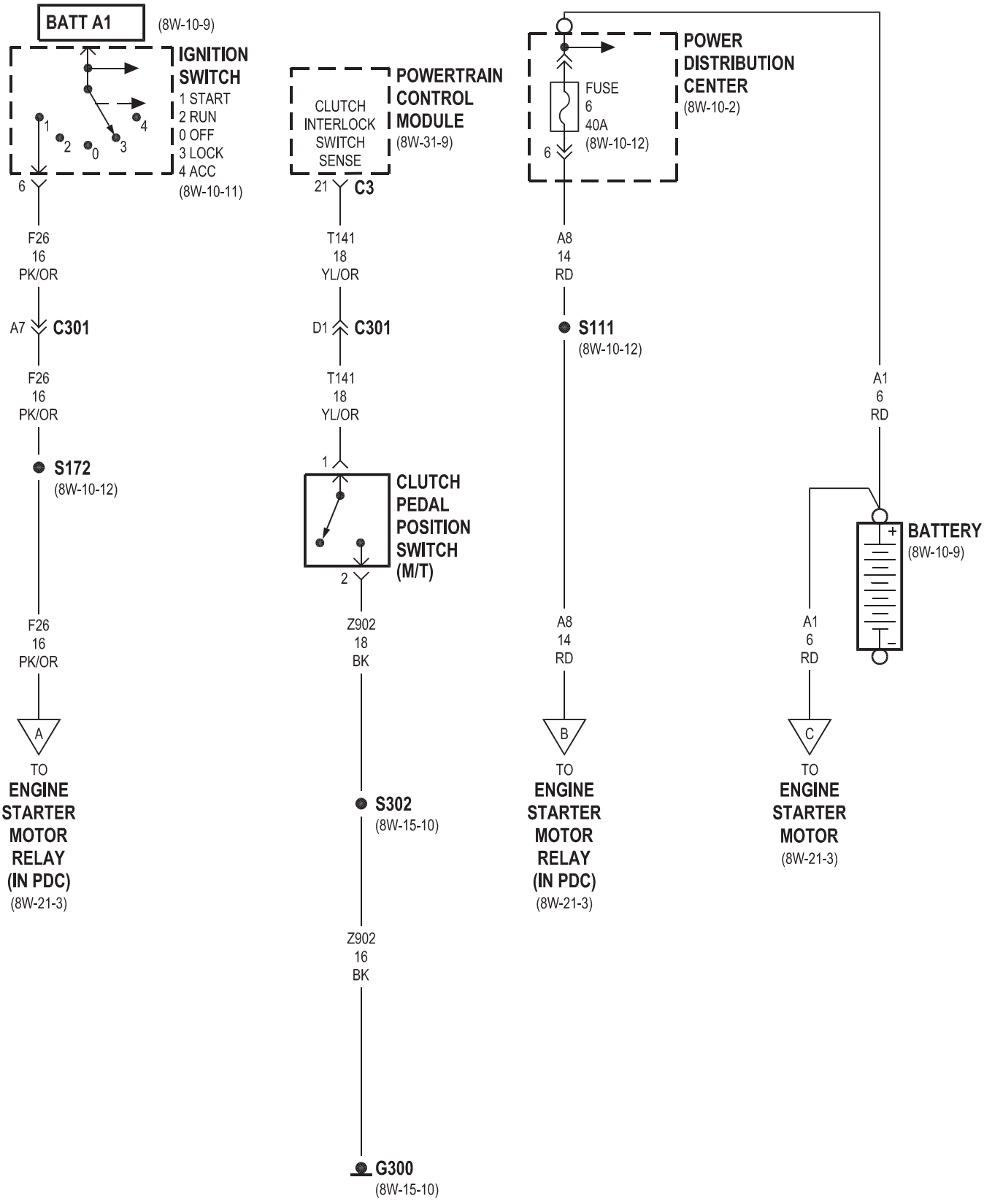
8W-20 CHARGING SYSTEM

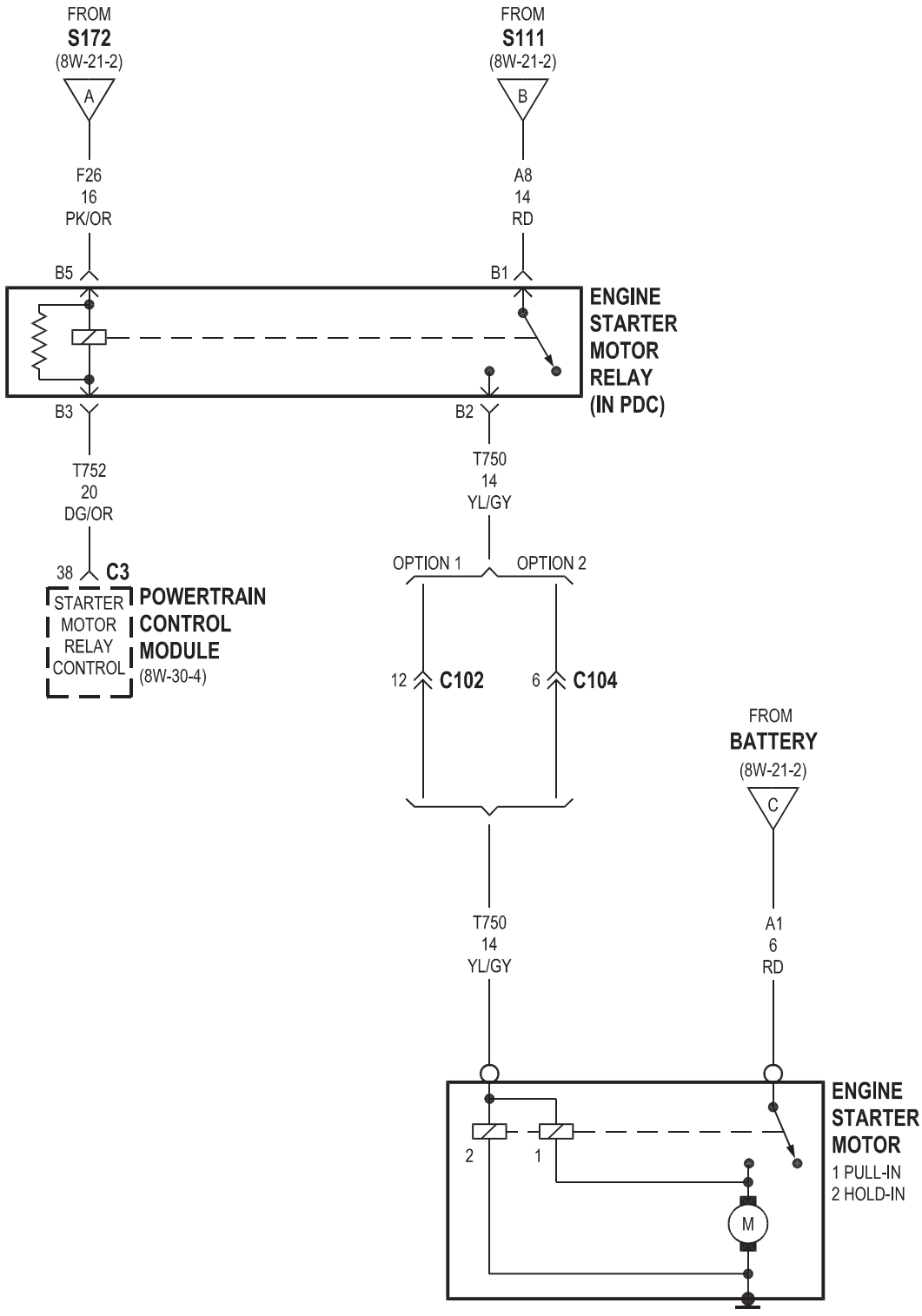
Component	Page	Component	Page
Battery	8W-20-2	Generator	8W-20-2
Engine Starter Motor	8W-20-2	Power Distribution Center	8W-20-2
Fusible Link	8W-20-2	Powertrain Control Module	8W-20-2
G105	8W-20-2		


FUSIBLE LINK
A11
10
DG
A1
6
RD
A1
6
RD
BATTERY
(8W-10-9)
S136
(8W-10-9)
A1
6
RD
A11
6
BR/RD
B(+)
POWER DISTRIBUTION CENTER
(8W-10-2)
B(+)
ENGINE STARTER MOTOR
(8W-21-3)
1
GENERATOR
2
K125
18
BR/DG
19
C2
GEN
FIELD
CONTROL
POWERTRAIN CONTROL MODULE
(8W-30-17)
Z20
18
BR/BK
G105
(8W-15-6)

8W-21 STARTING SYSTEM

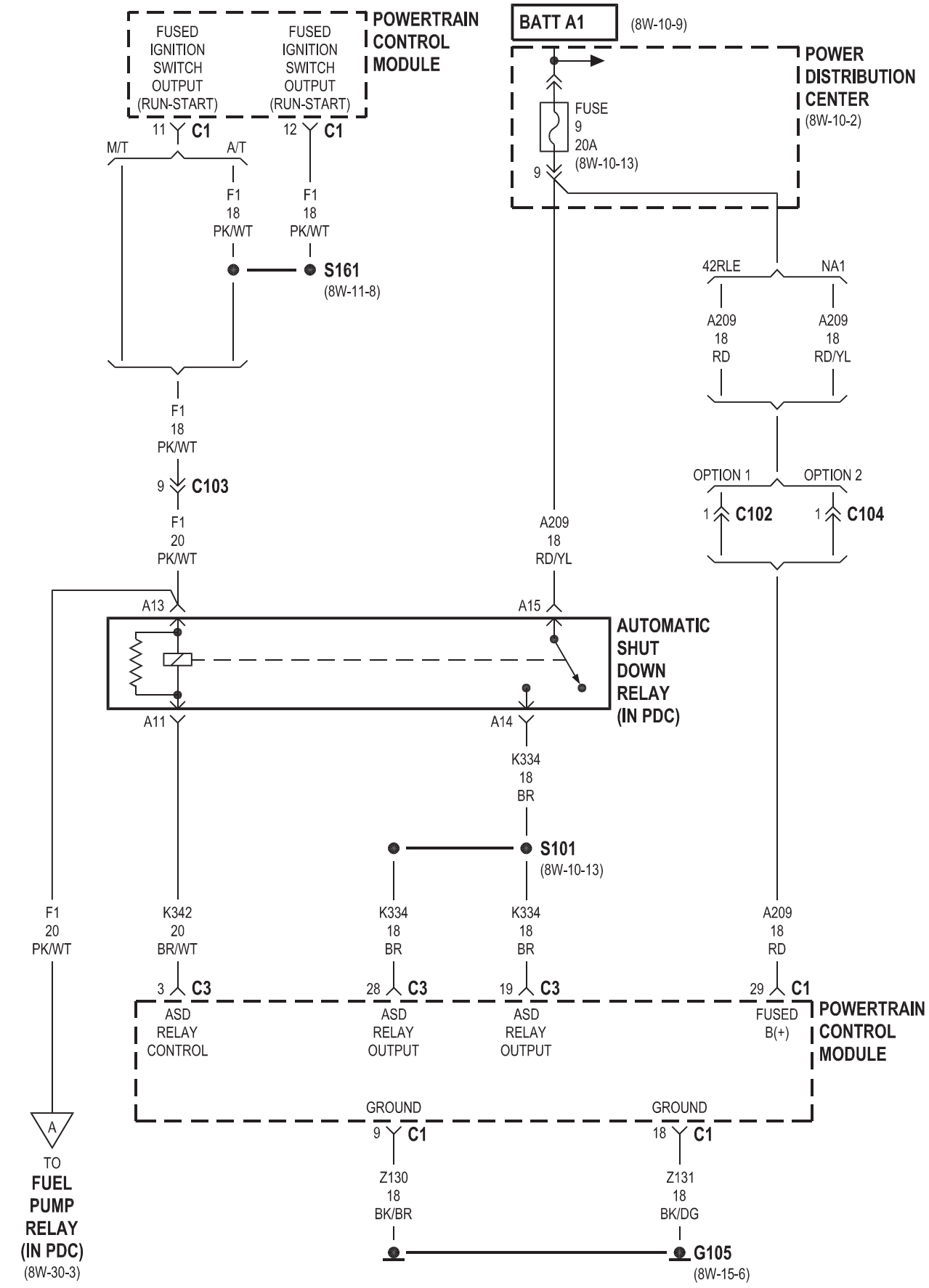
Component	Page
Battery	8W-21-2, 3
Clutch Pedal Position Switch	8W-21-2
Engine Starter Motor	8W-21-2, 3
Engine Starter Motor Relay	8W-21-2, 3
Fuse 6	8W-21-2
G300	8W-21-2
Ignition Switch	8W-21-2
Power Distribution Center	8W-21-2
Powertrain Control Module	8W-21-2, 3

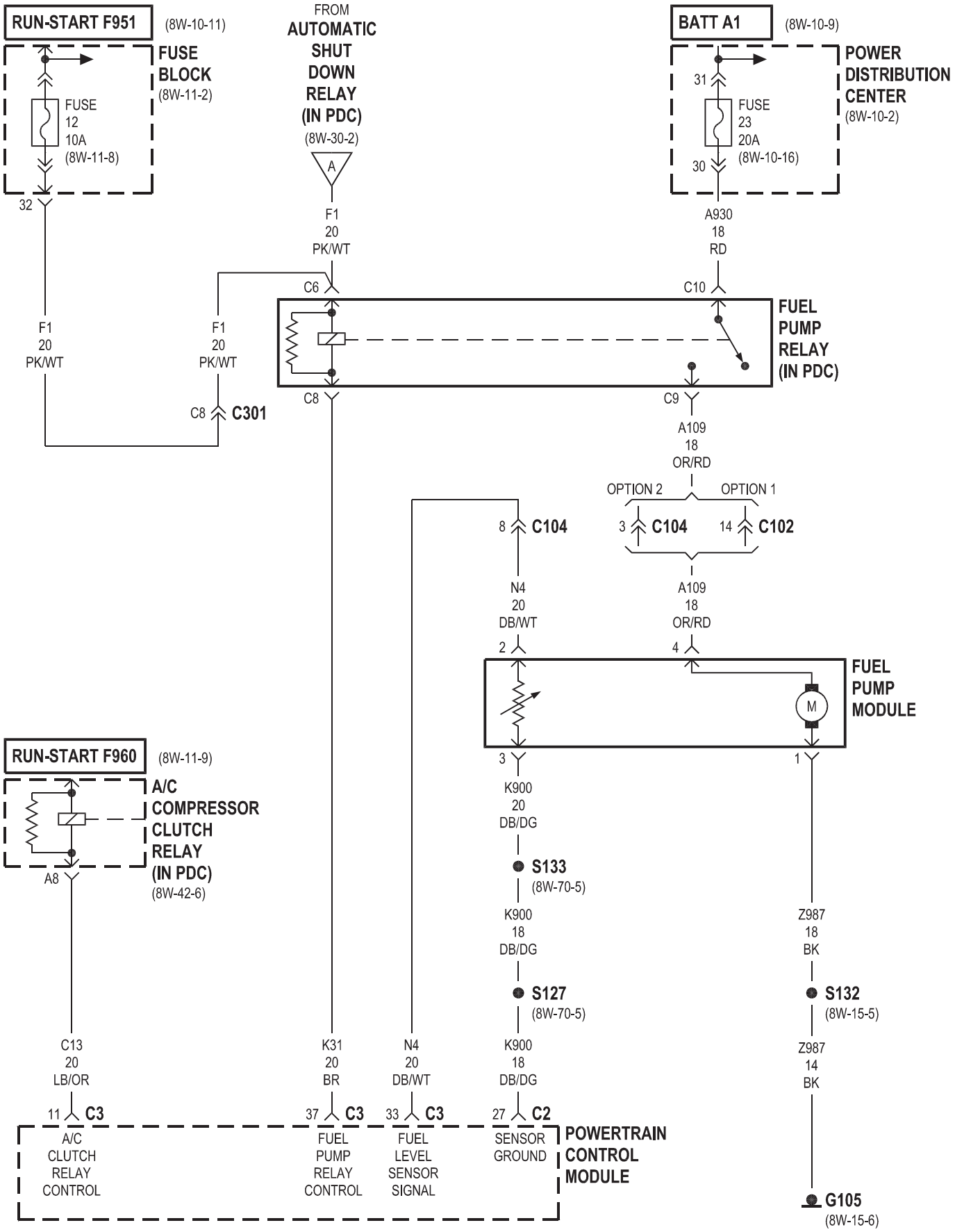

BATT A1
(8W-10-9)
IGNITION SWITCH
1 START
2 RUN
0 OFF
3 LOCK
4 ACC
(8W-10-11)
6
F26
16
PK/OR
A7
C301
F26
16
PK/OR
S172
(8W-10-12)
F26
16
PK/OR
A
TO
ENGINE STARTER MOTOR RELAY (IN PDC)
(8W-21-3)
CLUTCH INTERLOCK SWITCH SENSE
POWERTRAIN CONTROL MODULE
(8W-31-9)
21
C3
T141
18
YL/OR
D1
C301
T141
18
YL/OR
1
CLUTCH PEDAL POSITION SWITCH (M/T)
2
Z902
18
BK
S302
(8W-15-10)
Z902
16
BK
G300
(8W-15-10)
FUSE
6
40A
(8W-10-12)
6
POWER DISTRIBUTION CENTER
(8W-10-2)
A8
14
RD
S111
(8W-10-12)
A8
14
RD
B
TO
ENGINE STARTER MOTOR RELAY (IN PDC)
(8W-21-3)
A1
6
RD
BATTERY
(8W-10-9)
A1
6
RD
C
TO
ENGINE STARTER MOTOR
(8W-21-3)

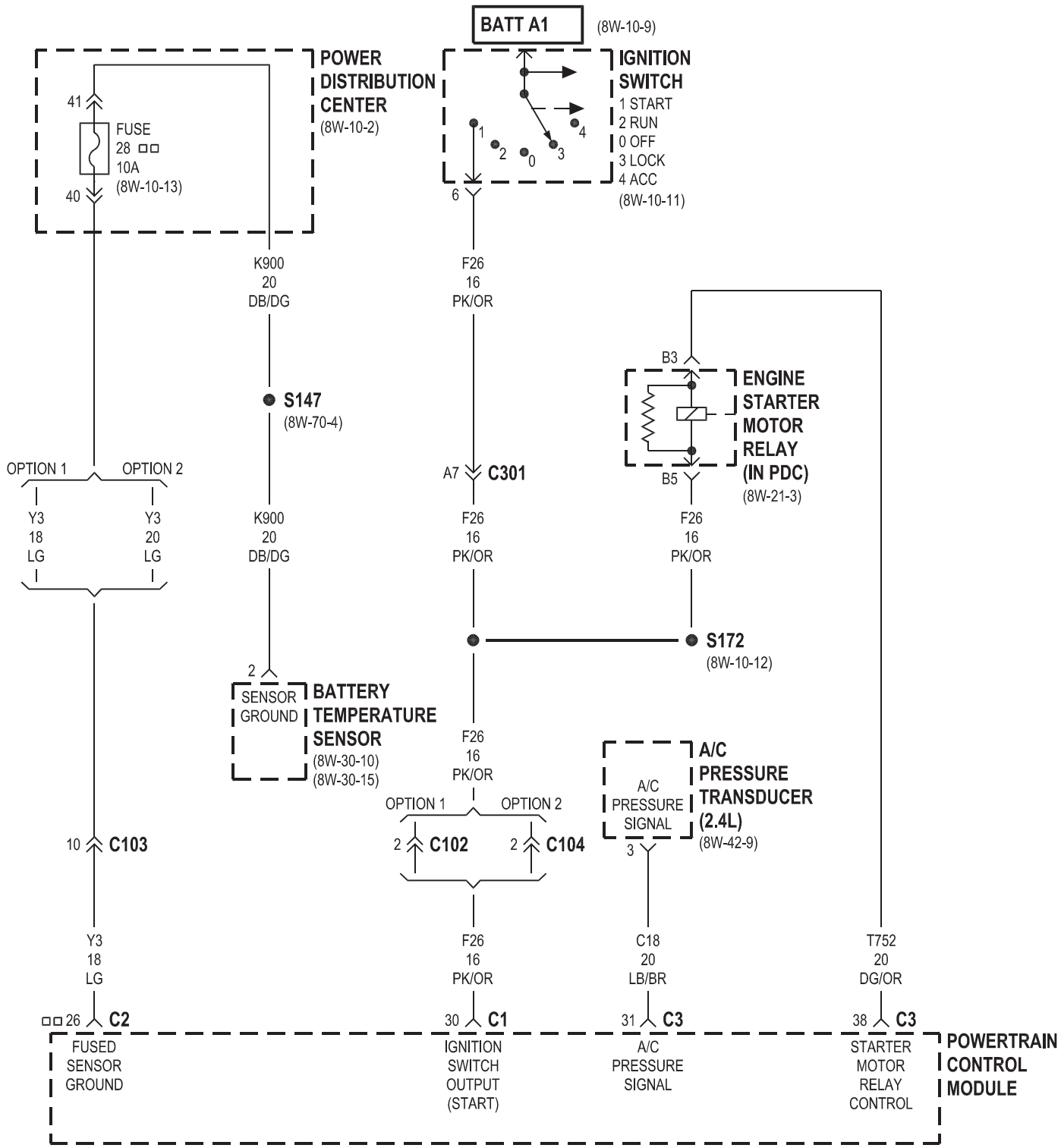

FROM
S172
(8W-21-2)
A
F26
16
PK/OR
B5
B3
T752
20
DG/OR
38 C3
STARTER
MOTOR
RELAY
CONTROL
POWERTRAIN
CONTROL
MODULE
(8W-30-4)
FROM
S111
(8W-21-2)
B
A8
14
RD
B1
B2
ENGINE
STARTER
MOTOR
RELAY
(IN PDC)
T750
14
YL/GY
OPTION 1
OPTION 2
12 C102
6 C104
T750
14
YL/GY
FROM
BATTERY
(8W-21-2)
C
A1
6
RD
2
1
M
ENGINE
STARTER
MOTOR
1 PULL-IN
2 HOLD-IN

8W-30 FUEL/IGNITION SYSTEM

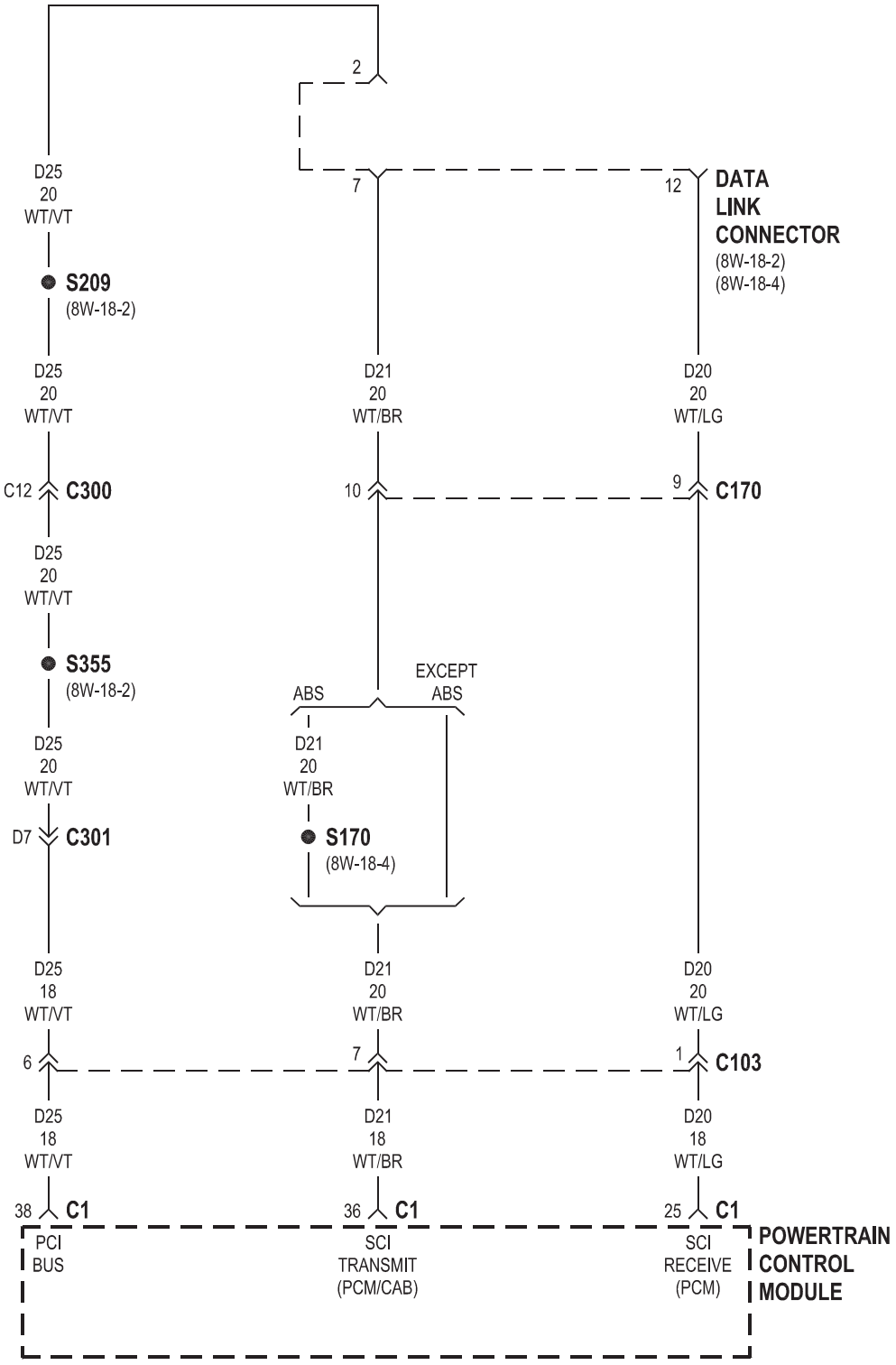
Component	Page
A/C Compressor Clutch Relay	8W-30-3, 19
A/C Low Pressure Switch	8W-30-11, 17
A/C Pressure Transducer	8W-30-4
A/C-Heater Control	8W-30-11
Ambient Temperature Sensor	8W-30-17
Automatic Shut Down Relay	8W-30-2, 3, 7, 8, 12, 13
Battery Temperature Sensor	8W-30-4, 10, 15, 17
Brake Lamp Switch	8W-30-6, 17
Brake Transmission Shift Interlock Solenoid	8W-30-17
Camshaft Position Sensor	8W-30-7, 9, 12
Clockspring	8W-30-6
Coil Capacitor	8W-30-7, 12
Crankshaft Position Sensor	8W-30-7, 12
Data Link Connector	8W-30-5
Daytime Running Lamp Module	8W-30-10, 15
Engine Coolant Temperature Sensor	8W-30-9, 14
Engine Oil Pressure Sensor	8W-30-14
Engine Oil Pressure Switch	8W-30-9
Engine Starter Motor Relay	8W-30-4
EVAP/Purge Solenoid	8W-30-16
Fuel Injector No. 1	8W-30-8, 13
Fuel Injector No. 2	8W-30-8, 13
Fuel Injector No. 3	8W-30-8, 13
Fuel Injector No. 4	8W-30-8, 13
Fuel Injector No. 5	8W-30-13
Fuel Injector No. 6	8W-30-13
Fuel Pump Module	8W-30-3
Fuel Pump Relay	8W-30-2, 3
Fuse 9	8W-30-2
Fuse 11	8W-30-19
Fuse 12	8W-30-3
Fuse 23	8W-30-3
Fuse 28	8W-30-4
Fuse Block	8W-30-3, 19
G105	8W-30-2, 3, 11, 18, 20, 21
G300	8W-30-17
G302	8W-30-16
Generator	8W-30-17
High Speed Radiator Fan Relay	8W-30-19
Idle Air Control Motor	8W-30-16
Ignition Coil Pack	8W-30-7, 12
Ignition Switch	8W-30-4
Intake Air Temperature Sensor	8W-30-9, 14
Left Speed Control Switch	8W-30-6
Low Speed Radiator Fan Relay	8W-30-19
Manifold Absolute Pressure Sensor	8W-30-9, 14
Natural Vacuum Leak Detection Assembly	8W-30-16
Oxygen Sensor 1/1 Upstream	8W-30-18, 20
Oxygen Sensor 1/2 Downstream	8W-30-18, 21
Oxygen Sensor 2/1 Upstream	8W-30-20
Oxygen Sensor 2/2 Downstream	8W-30-21
Power Distribution Center	8W-30-2, 3, 4, 7, 8, 12, 13
Power Steering Pressure Switch	8W-30-11
Powertrain Control Module	8W-30-2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21
Speed Control Servo	8W-30-6
Throttle Position Sensor	8W-30-10, 15
Transfer Case Switch	8W-30-14
Vehicle Speed Sensor	8W-30-10, 12, 15

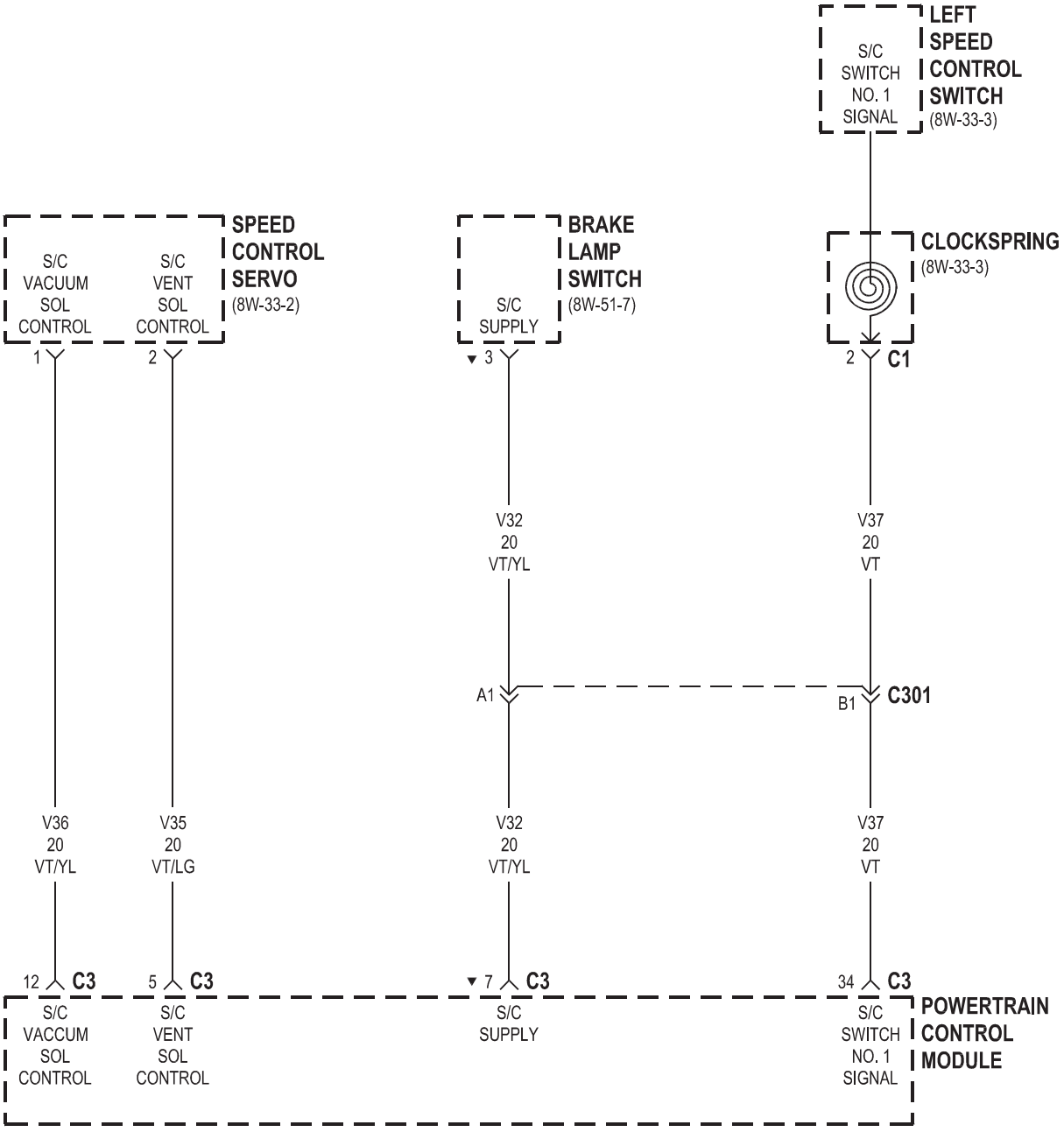

POWERTRAIN CONTROL MODULE
FUSED IGNITION SWITCH OUTPUT (RUN-START)
FUSED IGNITION SWITCH OUTPUT (RUN-START)
11 C1
12 C1
M/T
A/T
F1 18 PK/WT
F1 18 PK/WT
S161
(8W-11-8)
F1 18 PK/WT
9 C103
F1 20 PK/WT
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 9 20A
(8W-10-13)
9
42RLE
NA1
A209 18 RD
A209 18 RD/YL
OPTION 1
OPTION 2
1 C102
1 C104
A209 18 RD/YL
A13
A15
AUTOMATIC SHUT DOWN RELAY (IN PDC)
A11
A14
K334 18 BR
S101
(8W-10-13)
F1 20 PK/WT
K342 20 BR/WT
K334 18 BR
K334 18 BR
A209 18 RD
3 C3
28 C3
19 C3
29 C1
ASD RELAY CONTROL
ASD RELAY OUTPUT
ASD RELAY OUTPUT
FUSED B(+)
POWERTRAIN CONTROL MODULE
GROUND
GROUND
9 C1
18 C1
A
TO FUEL PUMP RELAY (IN PDC)
(8W-30-3)
Z130 18 BK/BR
Z131 18 BK/DG
G105
(8W-15-6)


RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 12 10A
(8W-11-8)
32
FROM AUTOMATIC SHUT DOWN RELAY (IN PDC)
(8W-30-2)
A
F1 20 PK/WT
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
31
FUSE 23 20A
(8W-10-16)
30
A930 18 RD
C6
C10
FUEL PUMP RELAY (IN PDC)
F1 20 PK/WT
F1 20 PK/WT
C8
C301
C8
C9
A109 18 OR/RD
OPTION 2
OPTION 1
8
C104
3
C104
14
C102
N4 20 DB/WT
A109 18 OR/RD
2
4
FUEL PUMP MODULE
M
3
1
RUN-START F960
(8W-11-9)
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
(8W-42-6)
A8
K900 20 DB/DG
S133
(8W-70-5)
K900 18 DB/DG
S127
(8W-70-5)
Z987 18 BK
S132
(8W-15-5)
Z987 14 BK
G105
(8W-15-6)
C13 20 LB/OR
K31 20 BR
N4 20 DB/WT
K900 18 DB/DG
11
C3
37
C3
33
C3
27
C2
A/C CLUTCH RELAY CONTROL
FUEL PUMP RELAY CONTROL
FUEL LEVEL SENSOR SIGNAL
SENSOR GROUND
POWERTRAIN CONTROL MODULE



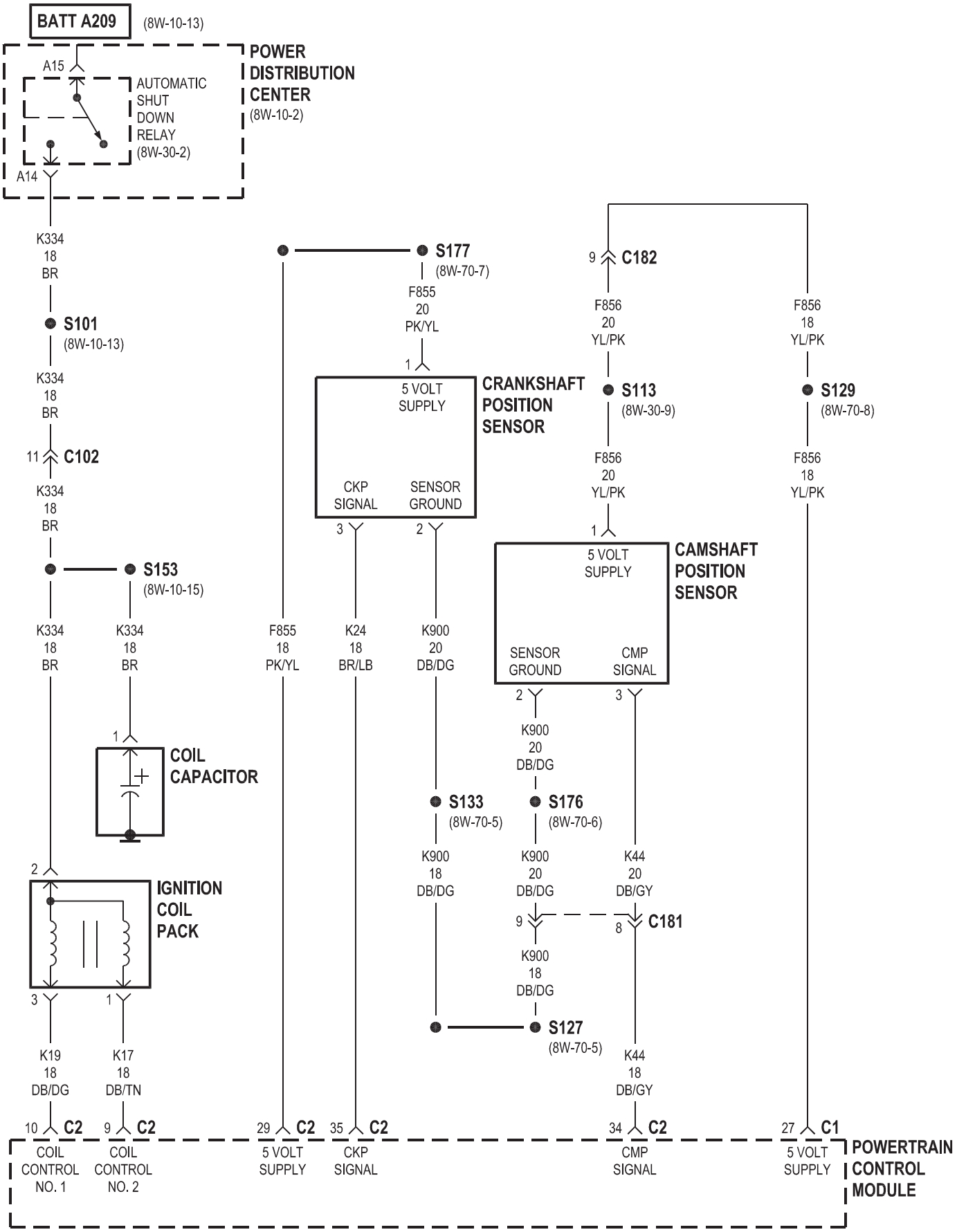
□□ EXCEPT OFF-ROAD PACKAGE

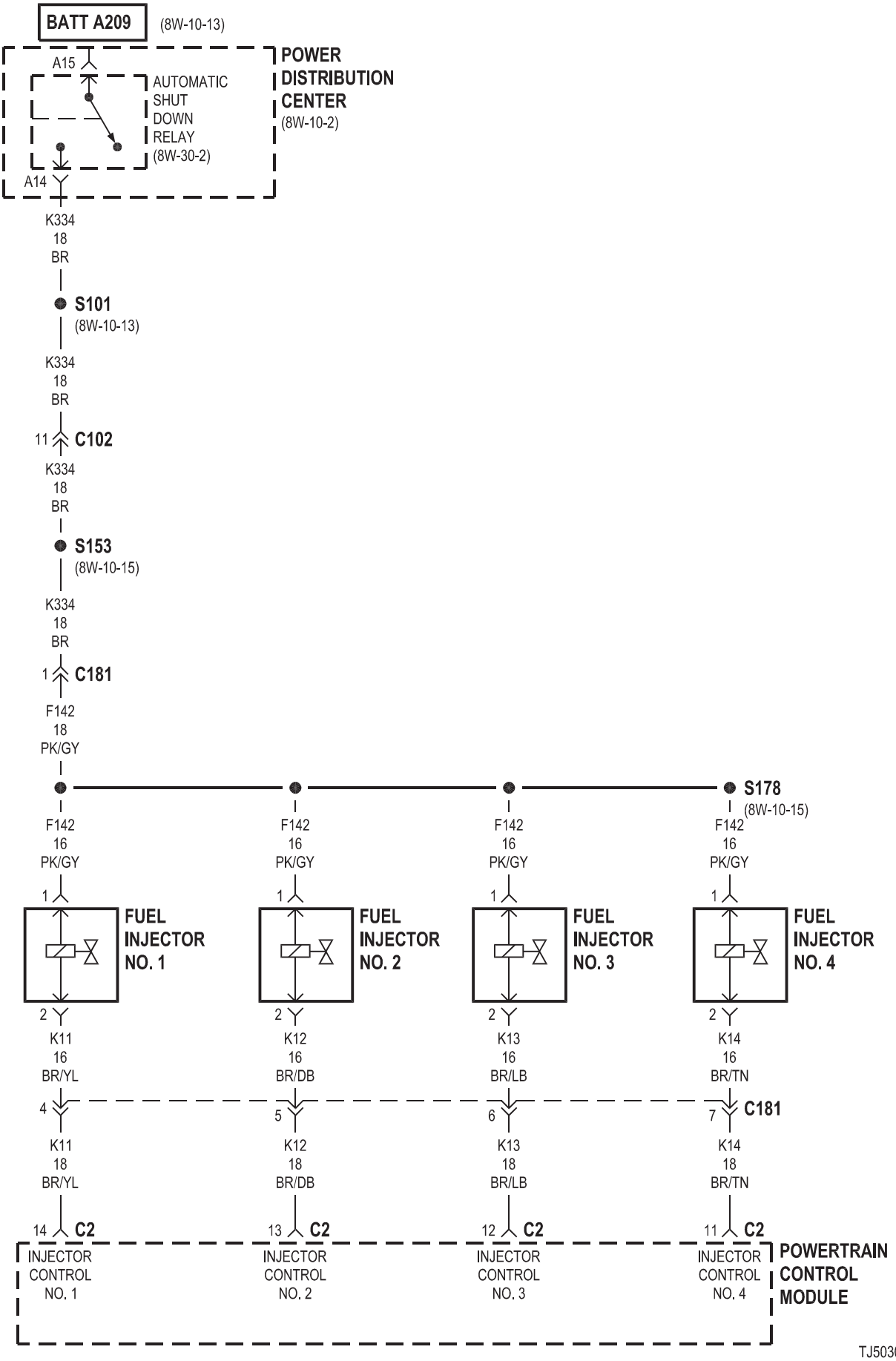

2
7
12
DATA
LINK
CONNECTOR
(8W-18-2)
(8W-18-4)
D25
20
WT/VT
S209
(8W-18-2)
D25
20
WT/VT
D21
20
WT/BR
D20
20
WT/LG
C12
C300
10
9
C170
D25
20
WT/VT
S355
(8W-18-2)
ABS
EXCEPT
ABS
D25
20
WT/VT
D21
20
WT/BR
D7
C301
S170
(8W-18-4)
D25
18
WT/VT
D21
20
WT/BR
D20
20
WT/LG
6
7
1
C103
D25
18
WT/VT
D21
18
WT/BR
D20
18
WT/LG
38
C1
36
C1
25
C1
PCI
BUS
SCI
TRANSMIT
(PCM/CAB)
SCI
RECEIVE
(PCM)
POWERTRAIN
CONTROL
MODULE

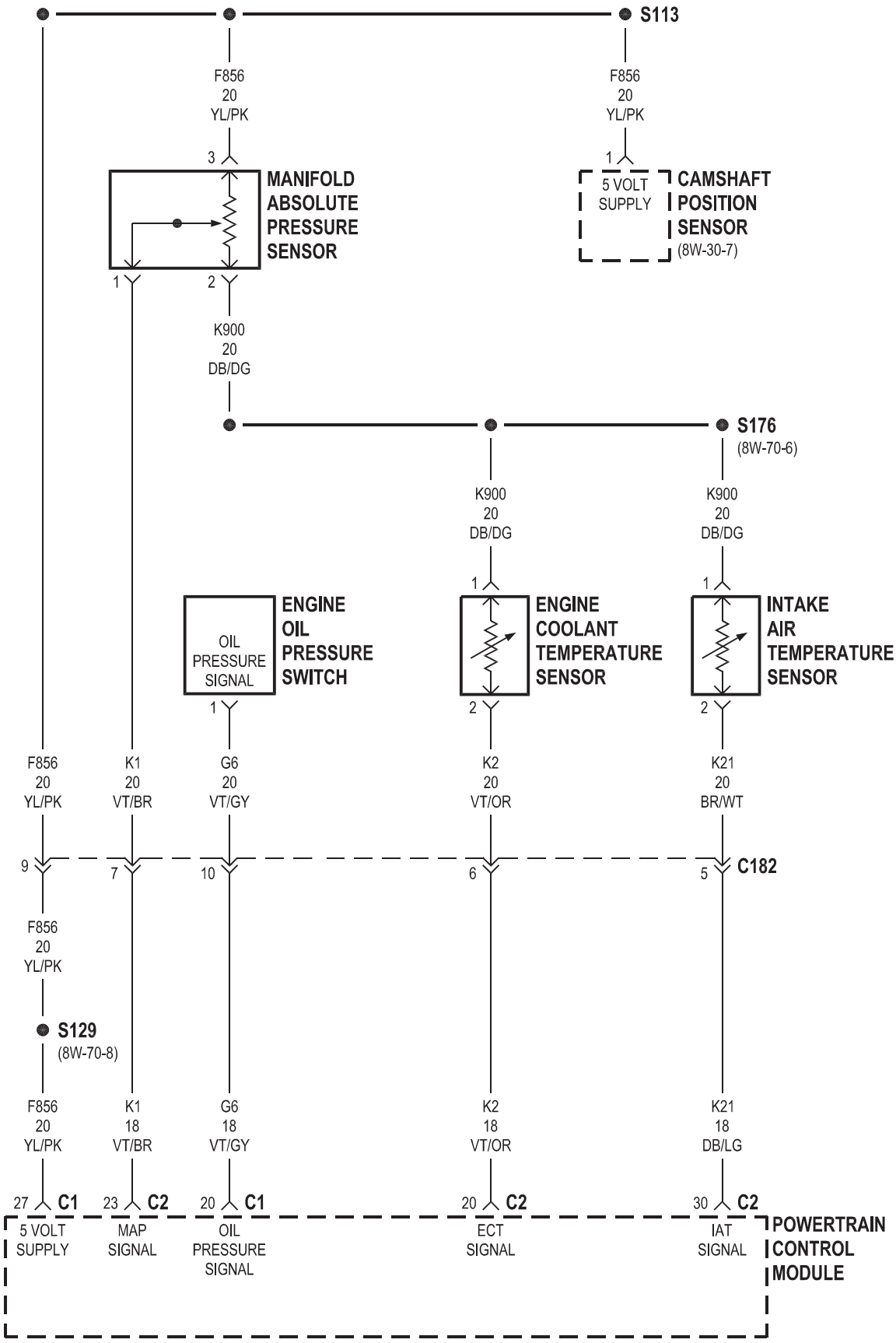


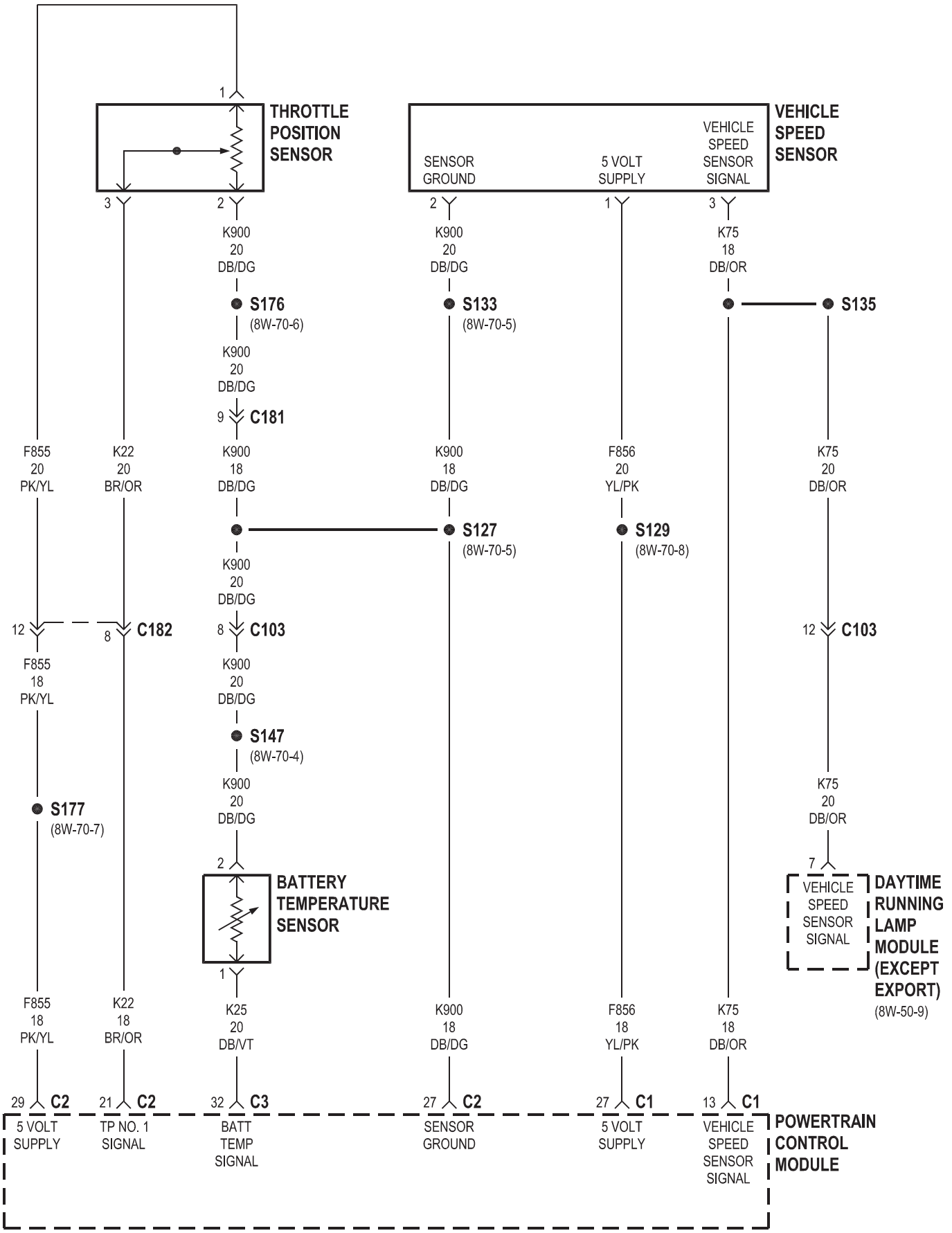
▼ EXCEPT POSTAL

2.4L

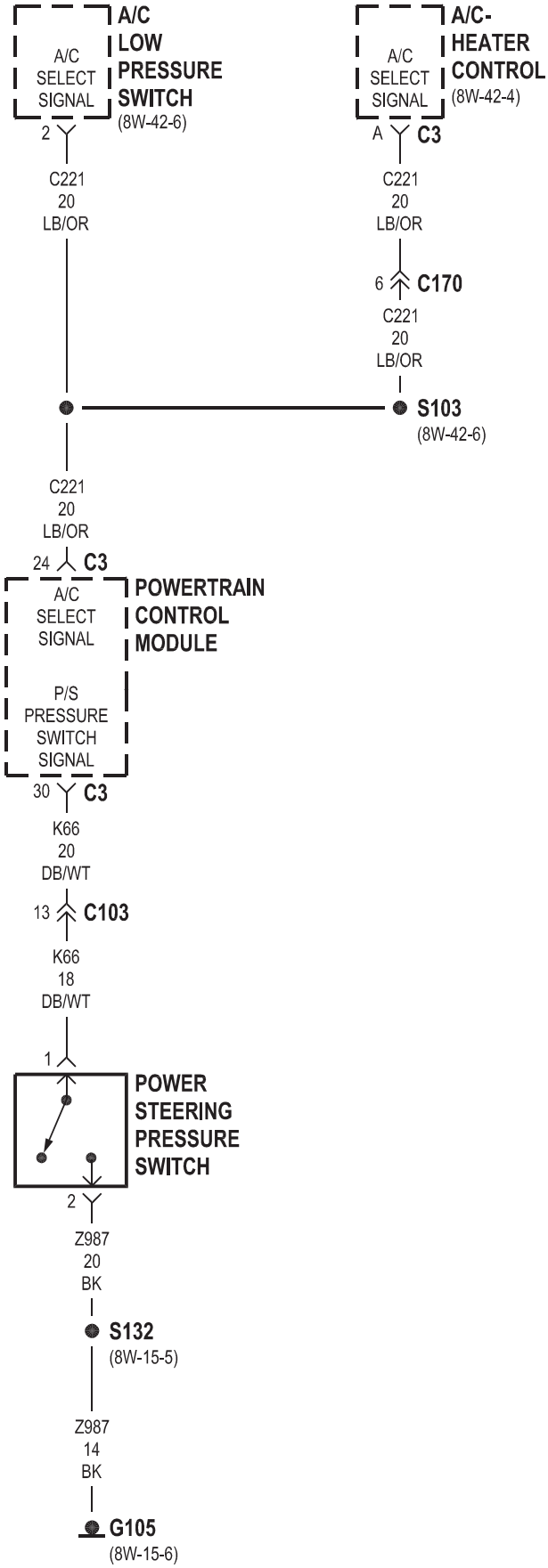

BATT A209
(8W-10-13)
A15
AUTOMATIC SHUT DOWN RELAY
(8W-30-2)
A14
POWER DISTRIBUTION CENTER
(8W-10-2)
K334 18 BR
S101
(8W-10-13)
K334 18 BR
11 C102
K334 18 BR
S153
(8W-10-15)
K334 18 BR
K334 18 BR
1
COIL CAPACITOR
2
IGNITION COIL PACK
3
1
K19 18 DB/DG
K17 18 DB/TN
10 C2
9 C2
COIL CONTROL NO. 1
COIL CONTROL NO. 2
S177
(8W-70-7)
F855 20 PK/YL
1
5 VOLT SUPPLY
CRANKSHAFT POSITION SENSOR
CKP SIGNAL
SENSOR GROUND
3
2
F855 18 PK/YL
K24 18 BR/LB
K900 20 DB/DG
29 C2
35 C2
5 VOLT SUPPLY
CKP SIGNAL
S133
(8W-70-5)
K900 18 DB/DG
S127
(8W-70-5)
9 C182
F856 20 YL/PK
S113
(8W-30-9)
F856 20 YL/PK
1
5 VOLT SUPPLY
CAMSHAFT POSITION SENSOR
SENSOR GROUND
CMP SIGNAL
2
3
K900 20 DB/DG
S176
(8W-70-6)
K900 20 DB/DG
9
8 C181
K900 18 DB/DG
K44 20 DB/GY
K44 18 DB/GY
34 C2
CMP SIGNAL
F856 18 YL/PK
S129
(8W-70-8)
F856 18 YL/PK
27 C1
5 VOLT SUPPLY
POWERTRAIN CONTROL MODULE

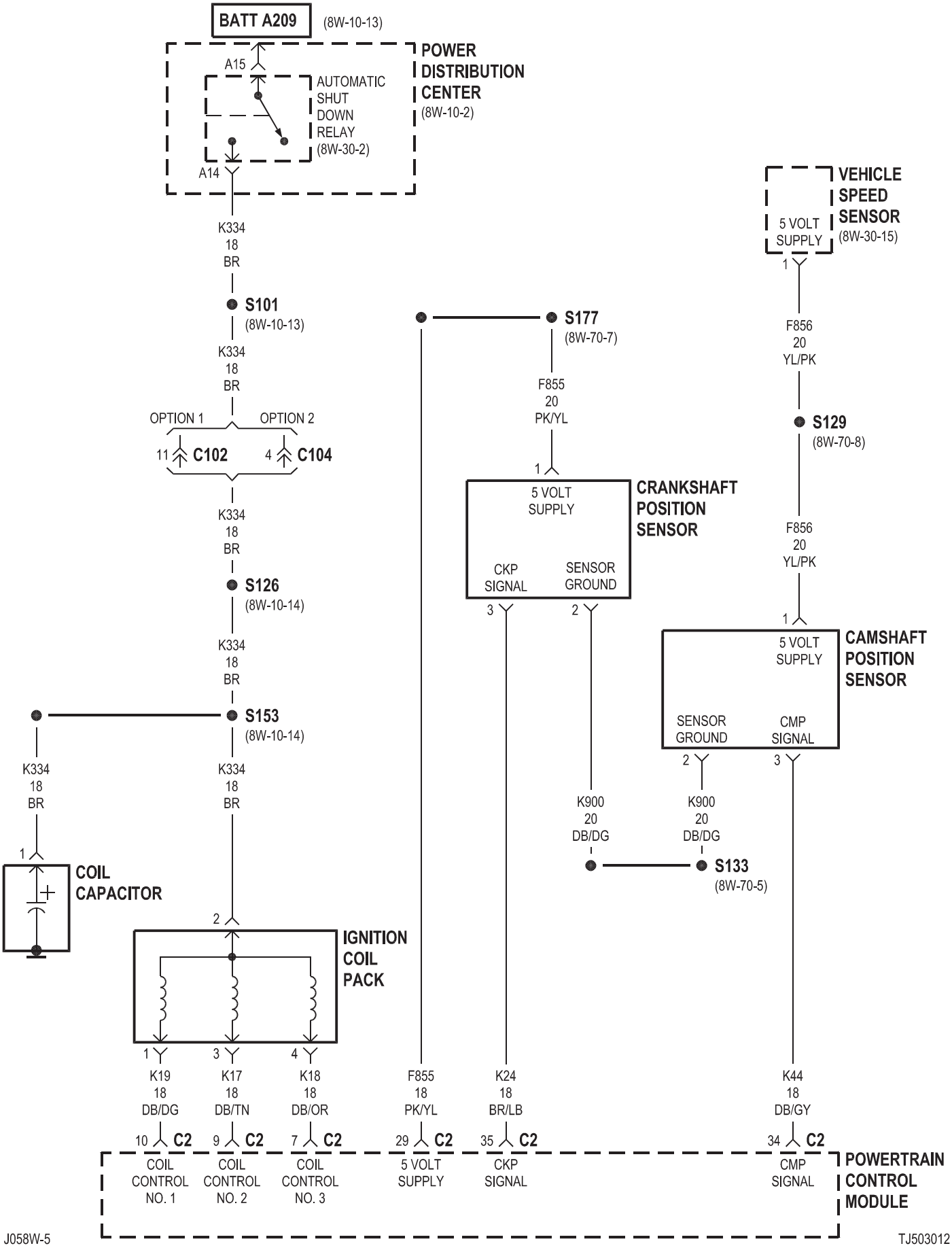

BATT A209
(8W-10-13)
POWER DISTRIBUTION CENTER
(8W-10-2)
A15
AUTOMATIC SHUT DOWN RELAY
(8W-30-2)
A14
K334
18
BR
S101
(8W-10-13)
K334
18
BR
11
C102
K334
18
BR
S153
(8W-10-15)
K334
18
BR
1
C181
F142
18
PK/GY
S178
(8W-10-15)
F142
16
PK/GY
1
FUEL INJECTOR NO. 1
FUEL INJECTOR NO. 2
FUEL INJECTOR NO. 3
FUEL INJECTOR NO. 4
2
K11
16
BR/YL
K12
16
BR/DB
K13
16
BR/LB
K14
16
BR/TN
4
5
6
7
C181
K11
18
BR/YL
K12
18
BR/DB
K13
18
BR/LB
K14
18
BR/TN
14
13
12
11
C2
INJECTOR CONTROL NO. 1
INJECTOR CONTROL NO. 2
INJECTOR CONTROL NO. 3
INJECTOR CONTROL NO. 4
POWERTRAIN CONTROL MODULE


S113
F856
20
YL/PK
3
MANIFOLD
ABSOLUTE
PRESSURE
SENSOR
1
2
F856
20
YL/PK
1
5 VOLT
SUPPLY
CAMSHAFT
POSITION
SENSOR
(8W-30-7)
K900
20
DB/DG
S176
(8W-70-6)
K900
20
DB/DG
K900
20
DB/DG
1
ENGINE
OIL
PRESSURE
SWITCH
OIL
PRESSURE
SIGNAL
1
ENGINE
COOLANT
TEMPERATURE
SENSOR
2
1
INTAKE
AIR
TEMPERATURE
SENSOR
2
F856
20
YL/PK
K1
20
VT/BR
G6
20
VT/GY
K2
20
VT/OR
K21
20
BR/WT
9
7
10
6
5
C182
F856
20
YL/PK
S129
(8W-70-8)
F856
20
YL/PK
K1
18
VT/BR
G6
18
VT/GY
K2
18
VT/OR
K21
18
DB/LG
27
C1
23
C2
20
C1
20
C2
30
C2
5 VOLT
SUPPLY
MAP
SIGNAL
OIL
PRESSURE
SIGNAL
ECT
SIGNAL
IAT
SIGNAL
POWERTRAIN
CONTROL
MODULE


THROTTLE POSITION SENSOR
VEHICLE SPEED SENSOR
SENSOR GROUND
5 VOLT SUPPLY
VEHICLE SPEED SENSOR SIGNAL
K900 20 DB/DG
K75 18 DB/OR
S176 (8W-70-6)
S133 (8W-70-5)
S135
C181
F855 20 PK/YL
K22 20 BR/OR
F856 20 YL/PK
K75 20 DB/OR
S127 (8W-70-5)
S129 (8W-70-8)
C182
C103
F855 18 PK/YL
S147 (8W-70-4)
S177 (8W-70-7)
BATTERY TEMPERATURE SENSOR
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT) (8W-50-9)
K25 20 DB/VT
F855 18 PK/YL
K22 18 BR/OR
F856 18 YL/PK
K75 18 DB/OR
29 C2
21 C2
32 C3
27 C2
27 C1
13 C1
5 VOLT SUPPLY
TP NO. 1 SIGNAL
BATT TEMP SIGNAL
SENSOR GROUND
5 VOLT SUPPLY
VEHICLE SPEED SENSOR SIGNAL
POWERTRAIN CONTROL MODULE

2.4L

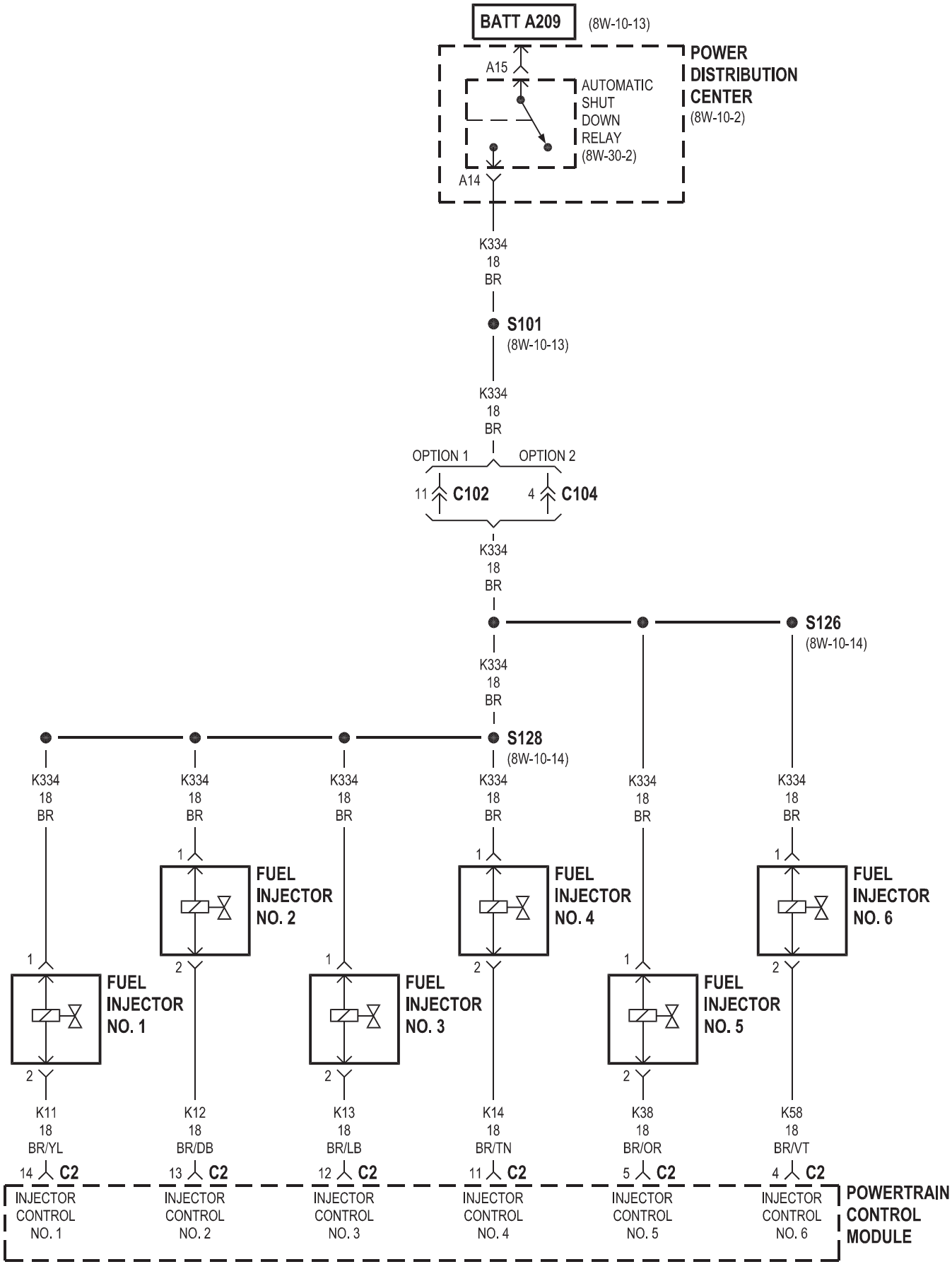

A/C LOW PRESSURE SWITCH
A/C SELECT SIGNAL
(8W-42-6)
2
C221
20
LB/OR
A/C-HEATER CONTROL
A/C SELECT SIGNAL
(8W-42-4)
A
C3
C221
20
LB/OR
6
C170
C221
20
LB/OR
S103
(8W-42-6)
C221
20
LB/OR
24
C3
POWERTRAIN CONTROL MODULE
A/C SELECT SIGNAL
P/S PRESSURE SWITCH SIGNAL
30
C3
K66
20
DB/WT
13
C103
K66
18
DB/WT
1
POWER STEERING PRESSURE SWITCH
2
Z987
20
BK
S132
(8W-15-5)
Z987
14
BK
G105
(8W-15-6)



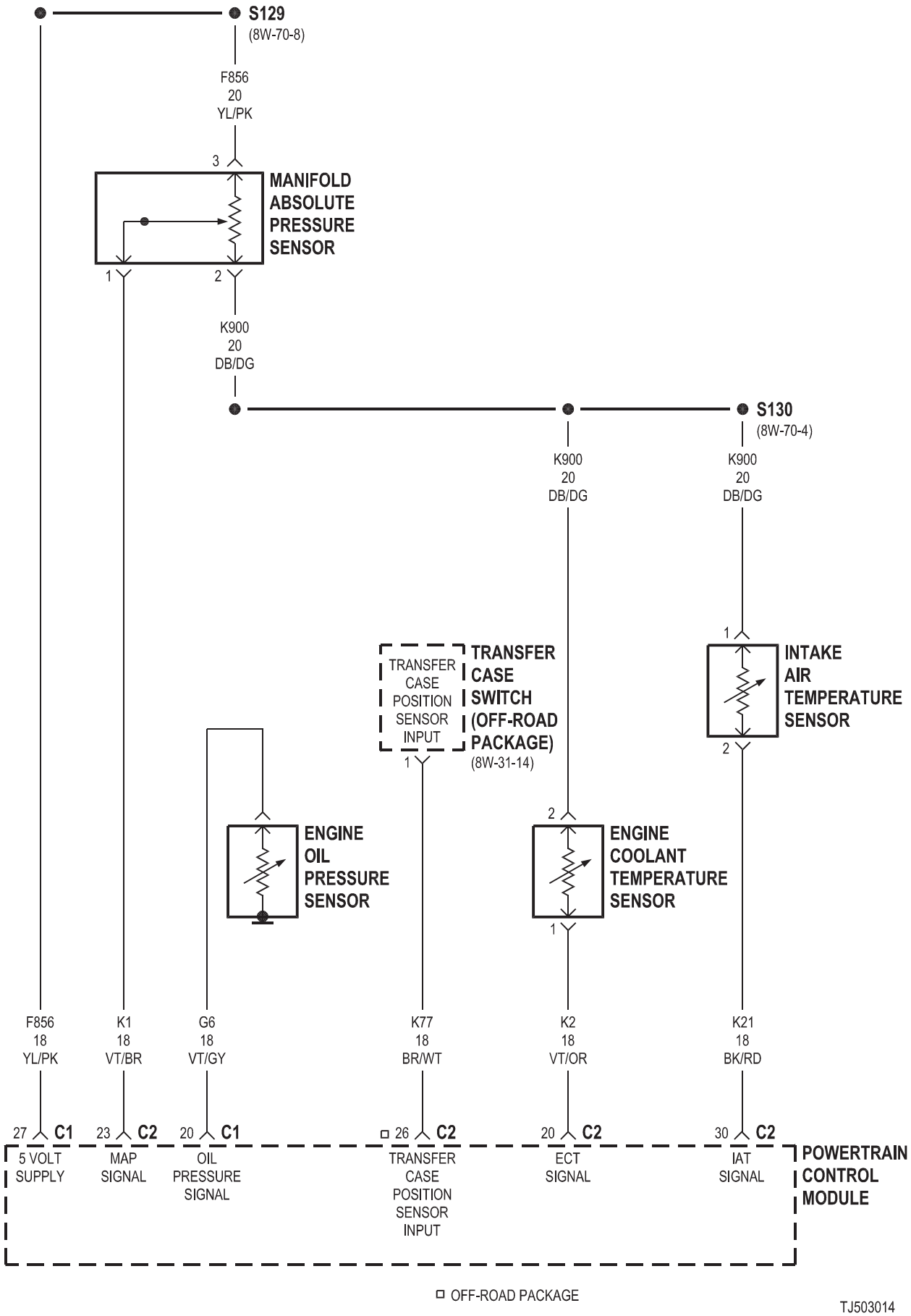
J058W-5

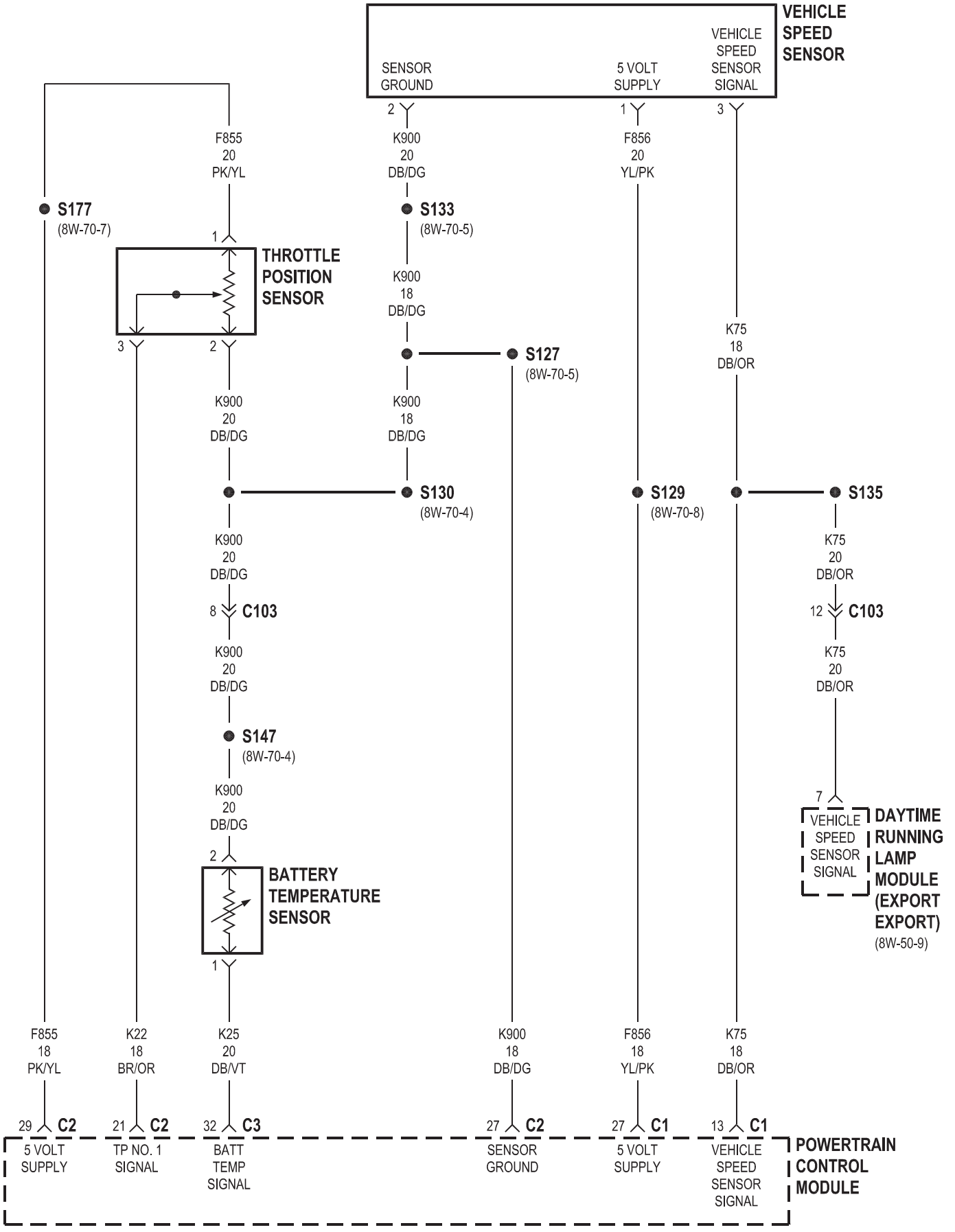
TJ503012

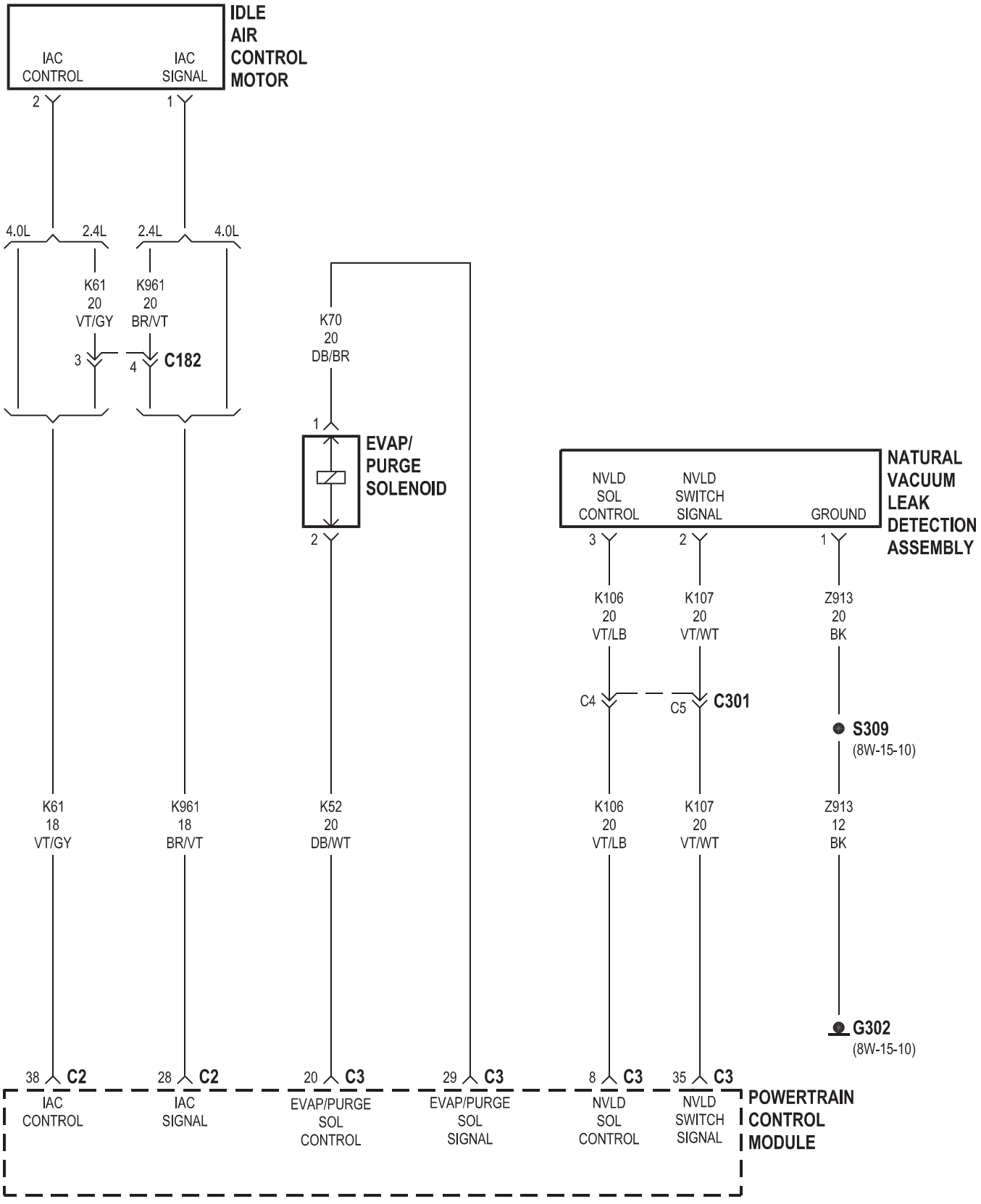
4.0L

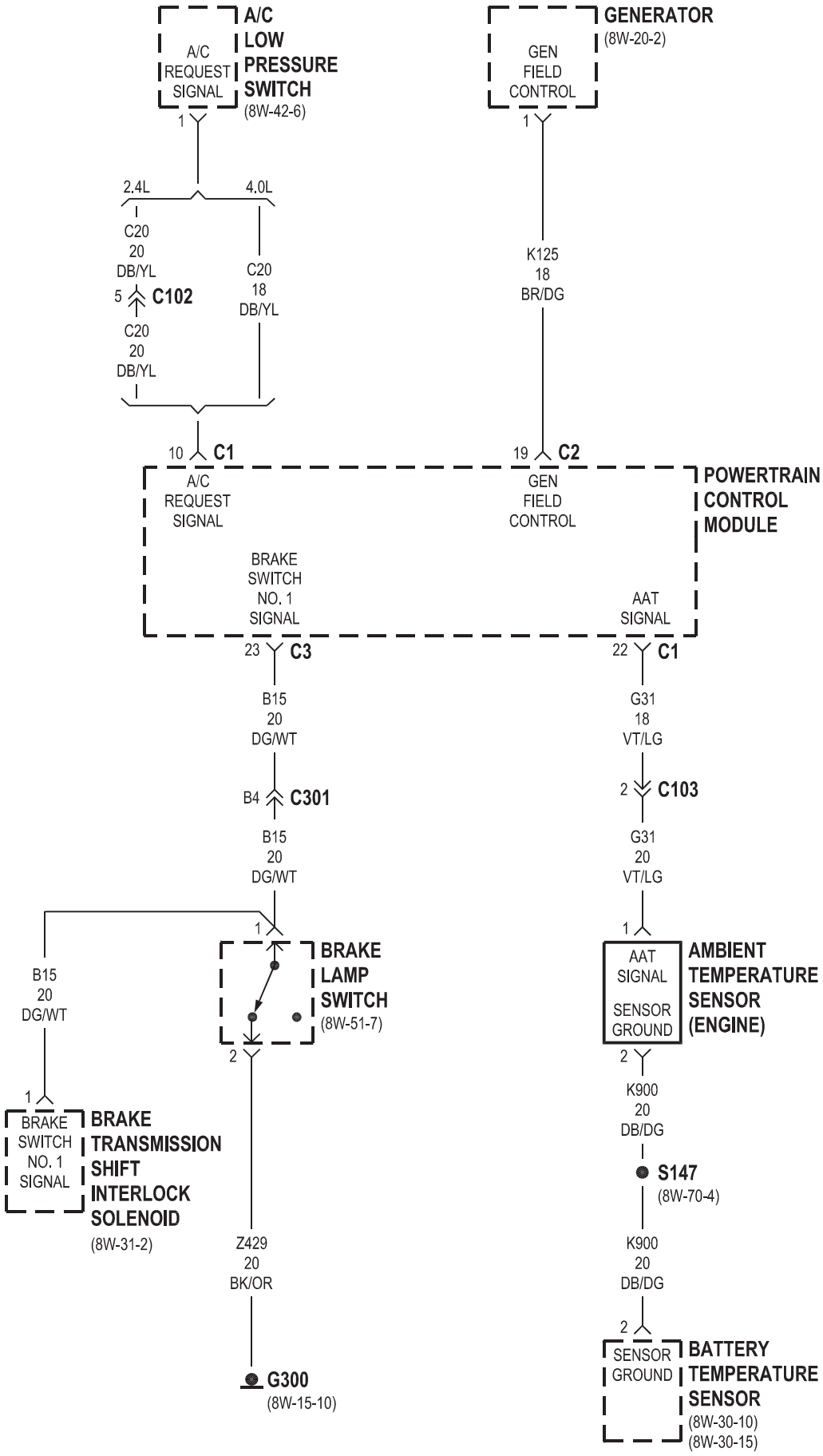


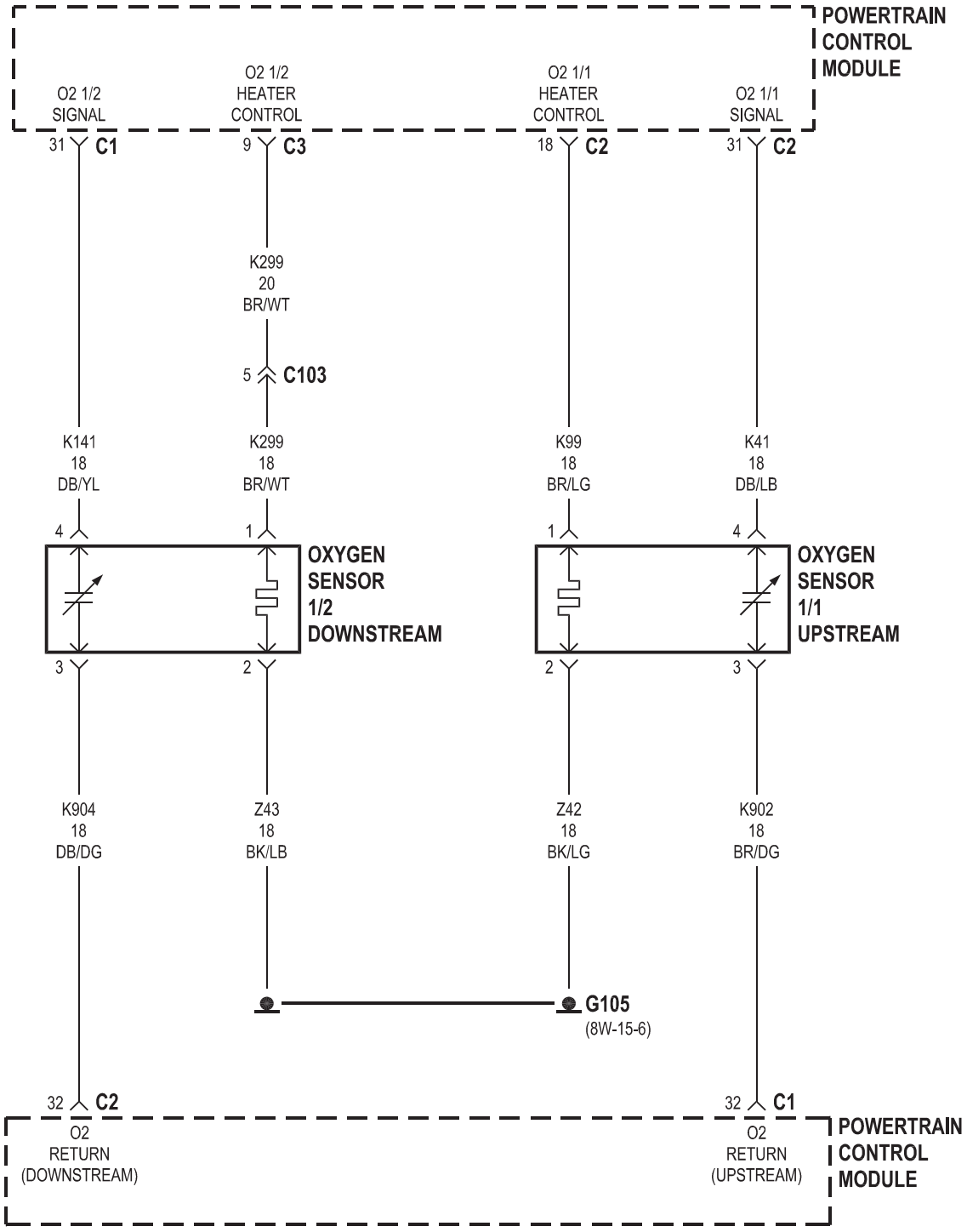
TJ503013 J058W-5

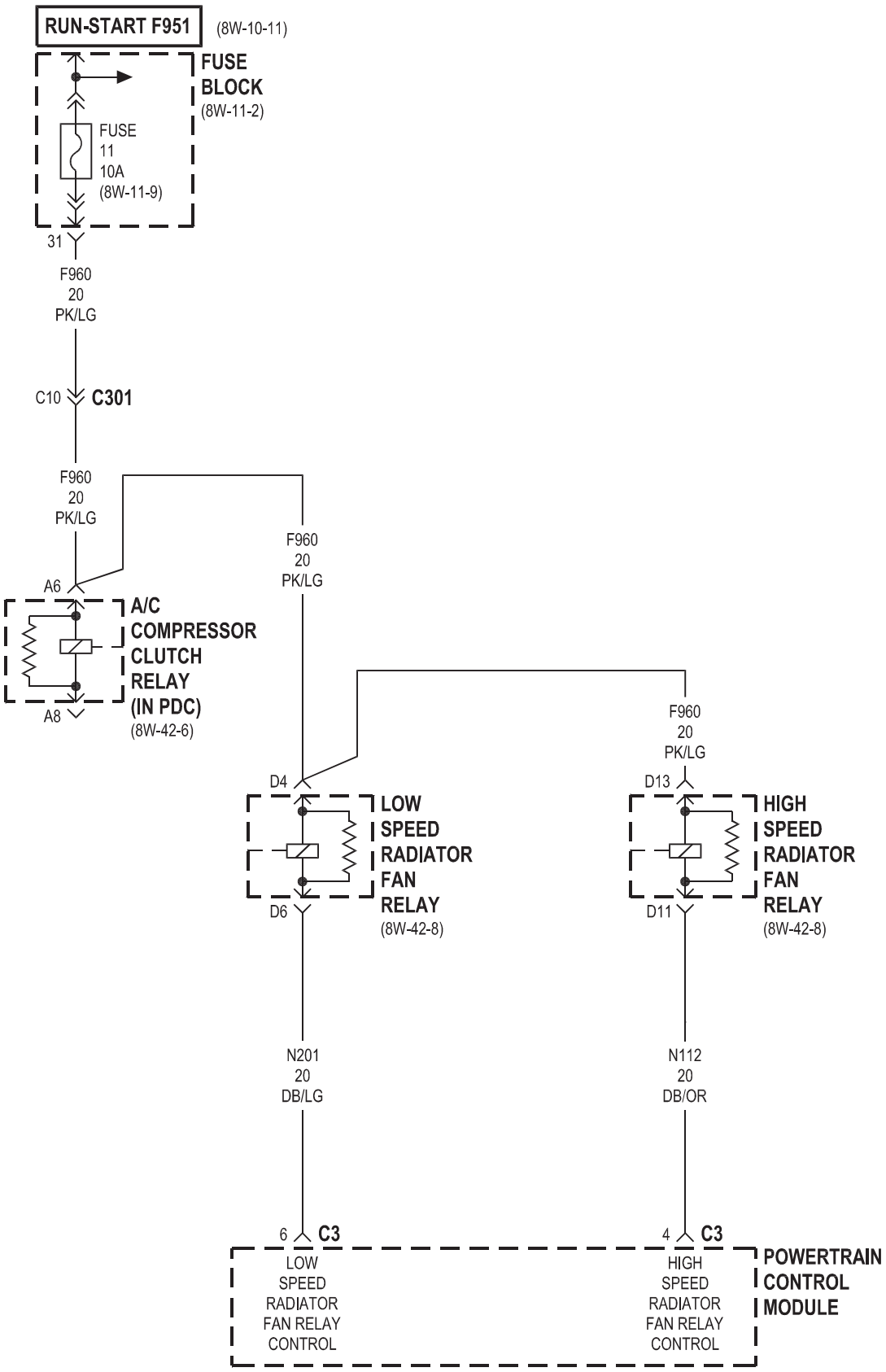

S129
(8W-70-8)
F856
20
YL/PK
3
MANIFOLD
ABSOLUTE
PRESSURE
SENSOR
1
2
K900
20
DB/DG
S130
(8W-70-4)
K900
20
DB/DG
K900
20
DB/DG
1
INTAKE
AIR
TEMPERATURE
SENSOR
2
TRANSFER
CASE
POSITION
SENSOR
INPUT
1
TRANSFER
CASE
SWITCH
(OFF-ROAD
PACKAGE)
(8W-31-14)
2
ENGINE
COOLANT
TEMPERATURE
SENSOR
1
ENGINE
OIL
PRESSURE
SENSOR
F856
18
YL/PK
K1
18
VT/BR
G6
18
VT/GY
K77
18
BR/WT
K2
18
VT/OR
K21
18
BK/RD
27 C1
5 VOLT
SUPPLY
23 C2
MAP
SIGNAL
20 C1
OIL
PRESSURE
SIGNAL
26 C2
TRANSFER
CASE
POSITION
SENSOR
INPUT
20 C2
ECT
SIGNAL
30 C2
IAT
SIGNAL
POWERTRAIN
CONTROL
MODULE
OFF-ROAD PACKAGE

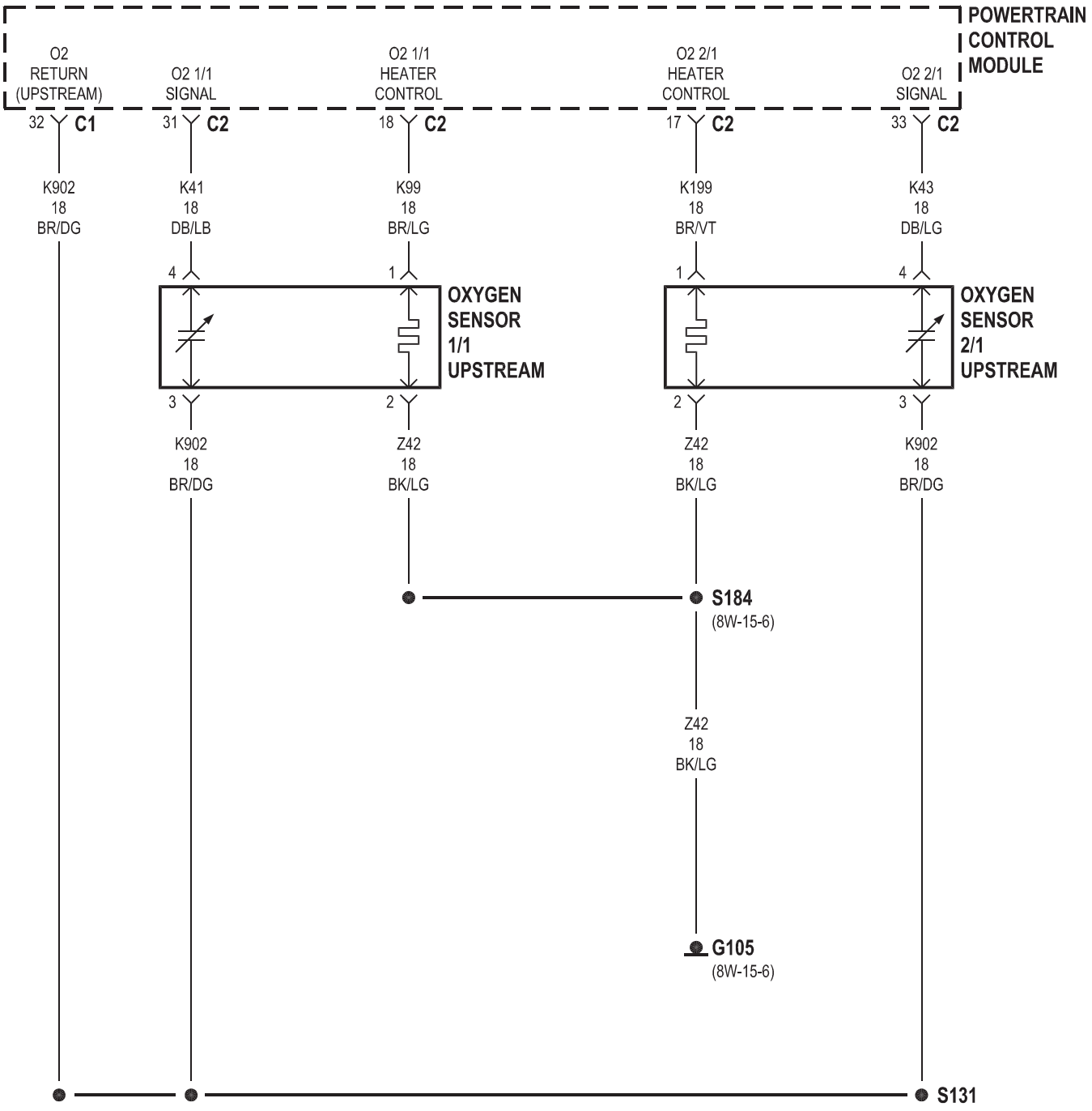

VEHICLE SPEED SENSOR
SENSOR GROUND
5 VOLT SUPPLY
VEHICLE SPEED SENSOR SIGNAL
2
1
3
F855 20 PK/YL
K900 20 DB/DG
F856 20 YL/PK
K75 18 DB/OR
S177 (8W-70-7)
S133 (8W-70-5)
1
THROTTLE POSITION SENSOR
3
2
K900 18 DB/DG
S127 (8W-70-5)
K900 20 DB/DG
K900 18 DB/DG
S130 (8W-70-4)
S129 (8W-70-8)
S135
K900 20 DB/DG
K75 20 DB/OR
8
C103
12
C103
K900 20 DB/DG
K75 20 DB/OR
S147 (8W-70-4)
7
VEHICLE SPEED SENSOR SIGNAL
DAYTIME RUNNING LAMP MODULE (EXPORT EXPORT) (8W-50-9)
K900 20 DB/DG
2
BATTERY TEMPERATURE SENSOR
1
F855 18 PK/YL
K22 18 BR/OR
K25 20 DB/VT
K900 18 DB/DG
F856 18 YL/PK
K75 18 DB/OR
29 C2
21 C2
32 C3
27 C2
27 C1
13 C1
5 VOLT SUPPLY
TP NO. 1 SIGNAL
BATT TEMP SIGNAL
SENSOR GROUND
5 VOLT SUPPLY
VEHICLE SPEED SENSOR SIGNAL
POWERTRAIN CONTROL MODULE

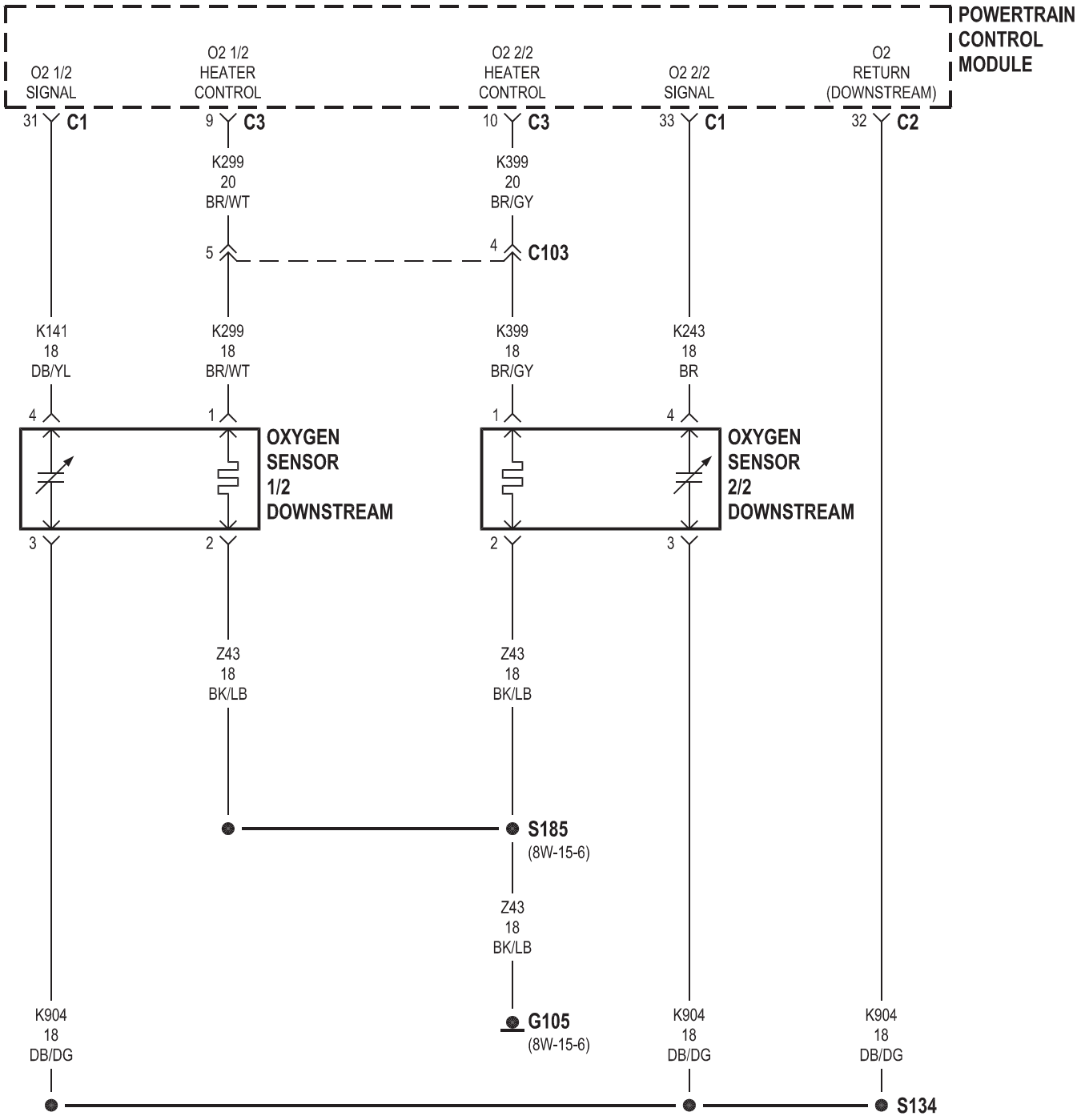

IDLE AIR CONTROL MOTOR
IAC CONTROL
IAC SIGNAL
2
1
4.0L
2.4L
2.4L
4.0L
K61 20 VT/GY
K961 20 BR/VT
3
4
C182
K70 20 DB/BR
1
EVAP/ PURGE SOLENOID
2
NATURAL VACUUM LEAK DETECTION ASSEMBLY
NVLD SOL CONTROL
NVLD SWITCH SIGNAL
GROUND
3
2
1
K106 20 VT/LB
K107 20 VT/WT
Z913 20 BK
C4
C5
C301
S309
(8W-15-10)
K61 18 VT/GY
K961 18 BR/VT
K52 20 DB/WT
K106 20 VT/LB
K107 20 VT/WT
Z913 12 BK
G302
(8W-15-10)
38 C2
28 C2
20 C3
29 C3
8 C3
35 C3
IAC CONTROL
IAC SIGNAL
EVAP/PURGE SOL CONTROL
EVAP/PURGE SOL SIGNAL
NVLD SOL CONTROL
NVLD SWITCH SIGNAL
POWERTRAIN CONTROL MODULE


A/C LOW PRESSURE SWITCH
(8W-42-6)
A/C REQUEST SIGNAL
1
2.4L
4.0L
C20 20 DB/YL
5 C102
C20 20 DB/YL
C20 18 DB/YL
10 C1
A/C REQUEST SIGNAL
GENERATOR
(8W-20-2)
GEN FIELD CONTROL
1
K125 18 BR/DG
19 C2
GEN FIELD CONTROL
POWERTRAIN CONTROL MODULE
BRAKE SWITCH NO. 1 SIGNAL
23 C3
AAT SIGNAL
22 C1
B15 20 DG/WT
B4 C301
B15 20 DG/WT
G31 18 VT/LG
2 C103
G31 20 VT/LG
1
BRAKE LAMP SWITCH
(8W-51-7)
2
B15 20 DG/WT
1
BRAKE SWITCH NO. 1 SIGNAL
BRAKE TRANSMISSION SHIFT INTERLOCK SOLENOID
(8W-31-2)
Z429 20 BK/OR
G300
(8W-15-10)
AAT SIGNAL
SENSOR GROUND
AMBIENT TEMPERATURE SENSOR (ENGINE)
2
K900 20 DB/DG
S147
(8W-70-4)
K900 20 DB/DG
2
SENSOR GROUND
BATTERY TEMPERATURE SENSOR
(8W-30-10)
(8W-30-15)


POWERTRAIN CONTROL MODULE
O2 1/2 SIGNAL
O2 1/2 HEATER CONTROL
O2 1/1 HEATER CONTROL
O2 1/1 SIGNAL
31 C1
9 C3
18 C2
31 C2
K299 20 BR/WT
5 C103
K141 18 DB/YL
K299 18 BR/WT
K99 18 BR/LG
K41 18 DB/LB
4
1
1
4
OXYGEN SENSOR 1/2 DOWNSTREAM
OXYGEN SENSOR 1/1 UPSTREAM
3
2
2
3
K904 18 DB/DG
Z43 18 BK/LB
Z42 18 BK/LG
K902 18 BR/DG
G105
(8W-15-6)
32 C2
32 C1
O2 RETURN (DOWNSTREAM)
O2 RETURN (UPSTREAM)
POWERTRAIN CONTROL MODULE

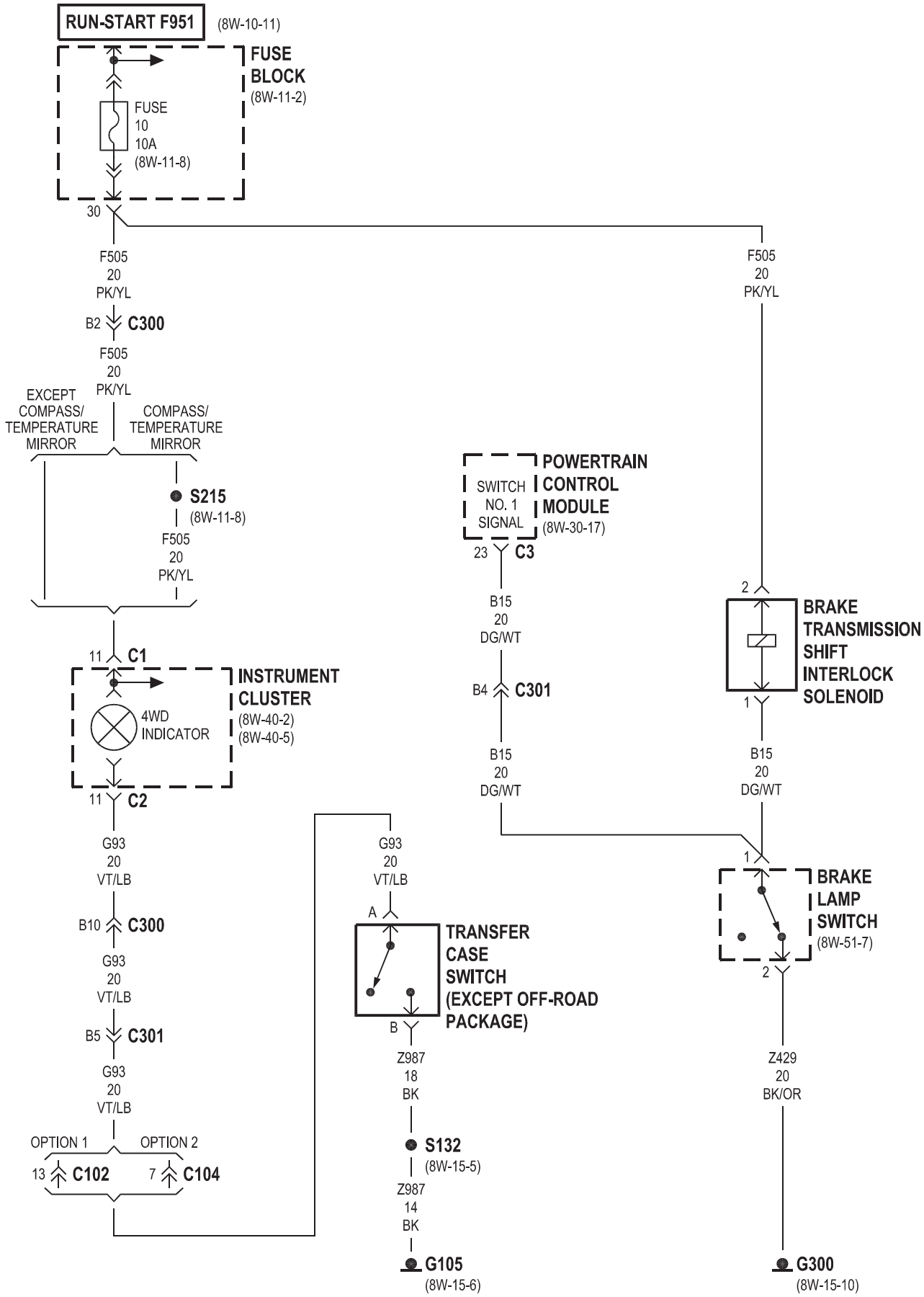

RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 11 10A
(8W-11-9)
31
F960 20 PK/LG
C10
C301
F960 20 PK/LG
A6
A8
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
(8W-42-6)
F960 20 PK/LG
D4
D6
LOW SPEED RADIATOR FAN RELAY
(8W-42-8)
F960 20 PK/LG
D13
D11
HIGH SPEED RADIATOR FAN RELAY
(8W-42-8)
N201 20 DB/LG
N112 20 DB/OR
6 C3
4 C3
LOW SPEED RADIATOR FAN RELAY CONTROL
HIGH SPEED RADIATOR FAN RELAY CONTROL
POWERTRAIN CONTROL MODULE

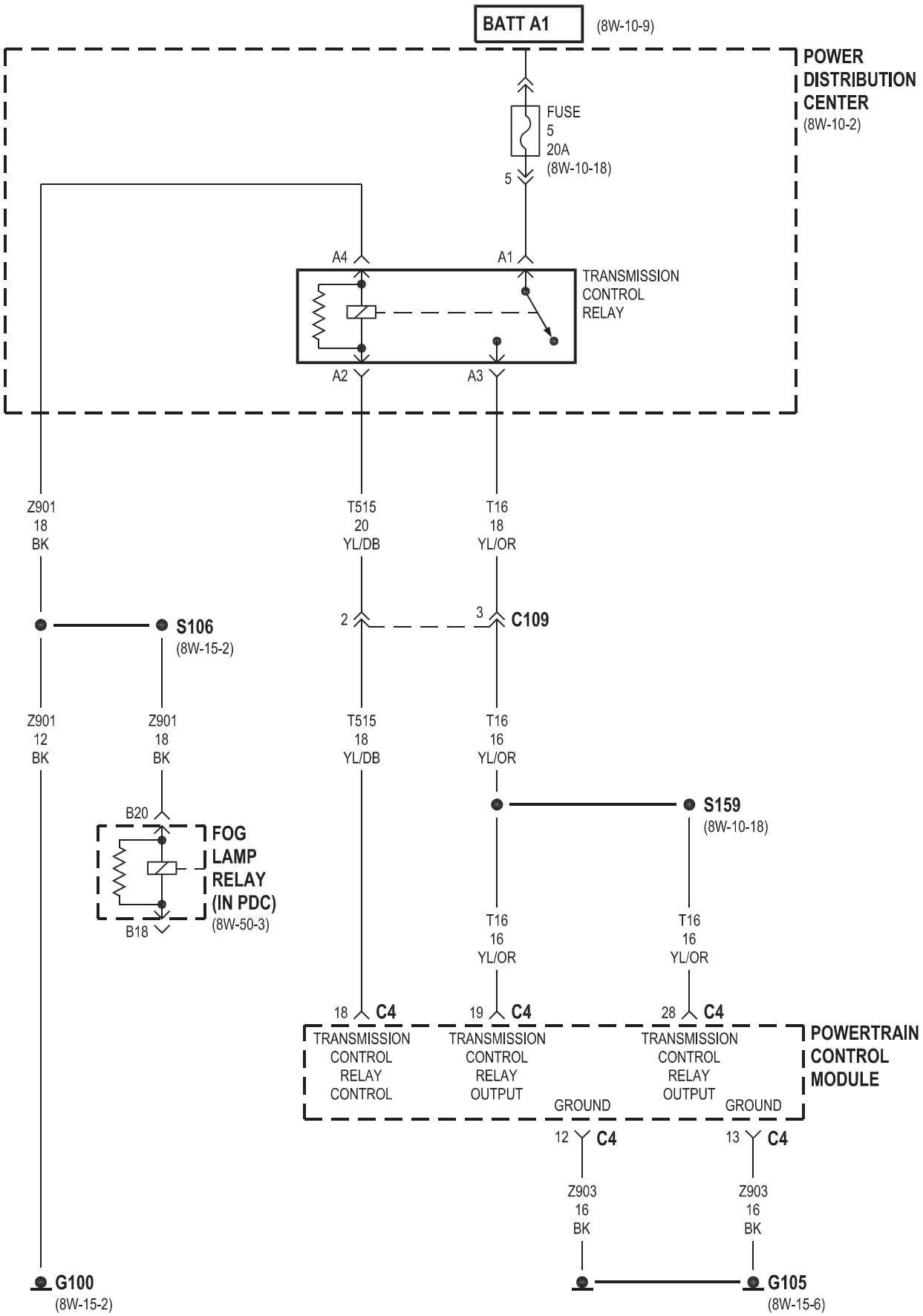

POWERTRAIN CONTROL MODULE
O2 RETURN (UPSTREAM)
O2 1/1 SIGNAL
O2 1/1 HEATER CONTROL
O2 2/1 HEATER CONTROL
O2 2/1 SIGNAL
32 C1
31 C2
18 C2
17 C2
33 C2
K902 18 BR/DG
K41 18 DB/LB
K99 18 BR/LG
K199 18 BR/VT
K43 18 DB/LG
4
1
1
4
OXYGEN SENSOR 1/1 UPSTREAM
OXYGEN SENSOR 2/1 UPSTREAM
3
2
2
3
K902 18 BR/DG
Z42 18 BK/LG
Z42 18 BK/LG
K902 18 BR/DG
S184
(8W-15-6)
Z42 18 BK/LG
G105
(8W-15-6)
S131

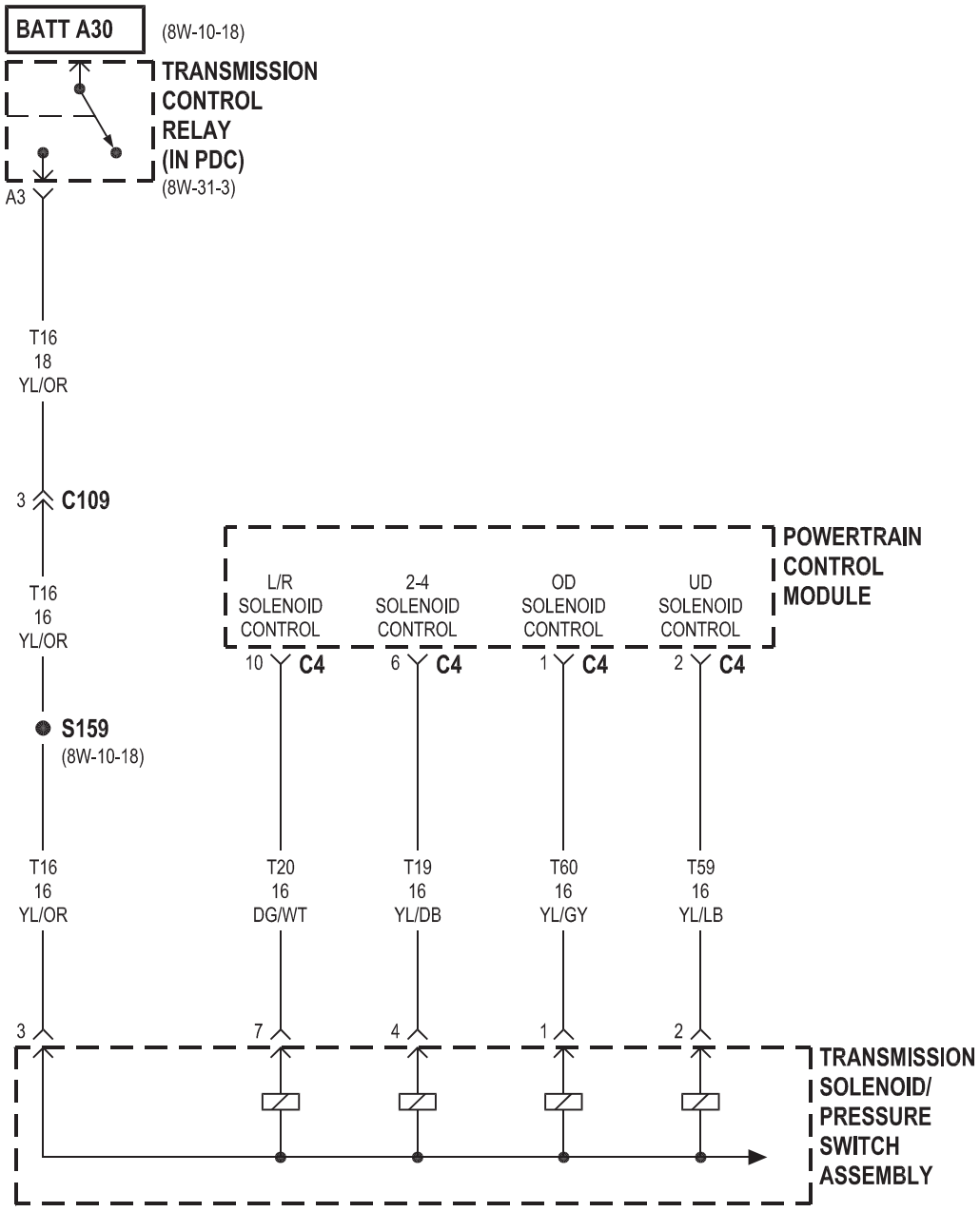

POWERTRAIN CONTROL MODULE
O2 1/2 SIGNAL
O2 1/2 HEATER CONTROL
O2 2/2 HEATER CONTROL
O2 2/2 SIGNAL
O2 RETURN (DOWNSTREAM)
31 C1
9 C3
10 C3
33 C1
32 C2
K299 20 BR/WT
K399 20 BR/GY
5
4 C103
K141 18 DB/YL
K299 18 BR/WT
K399 18 BR/GY
K243 18 BR
4
1
1
4
OXYGEN SENSOR 1/2 DOWNSTREAM
OXYGEN SENSOR 2/2 DOWNSTREAM
3
2
2
3
Z43 18 BK/LB
Z43 18 BK/LB
S185
(8W-15-6)
Z43 18 BK/LB
K904 18 DB/DG
G105
(8W-15-6)
K904 18 DB/DG
K904 18 DB/DG
S134

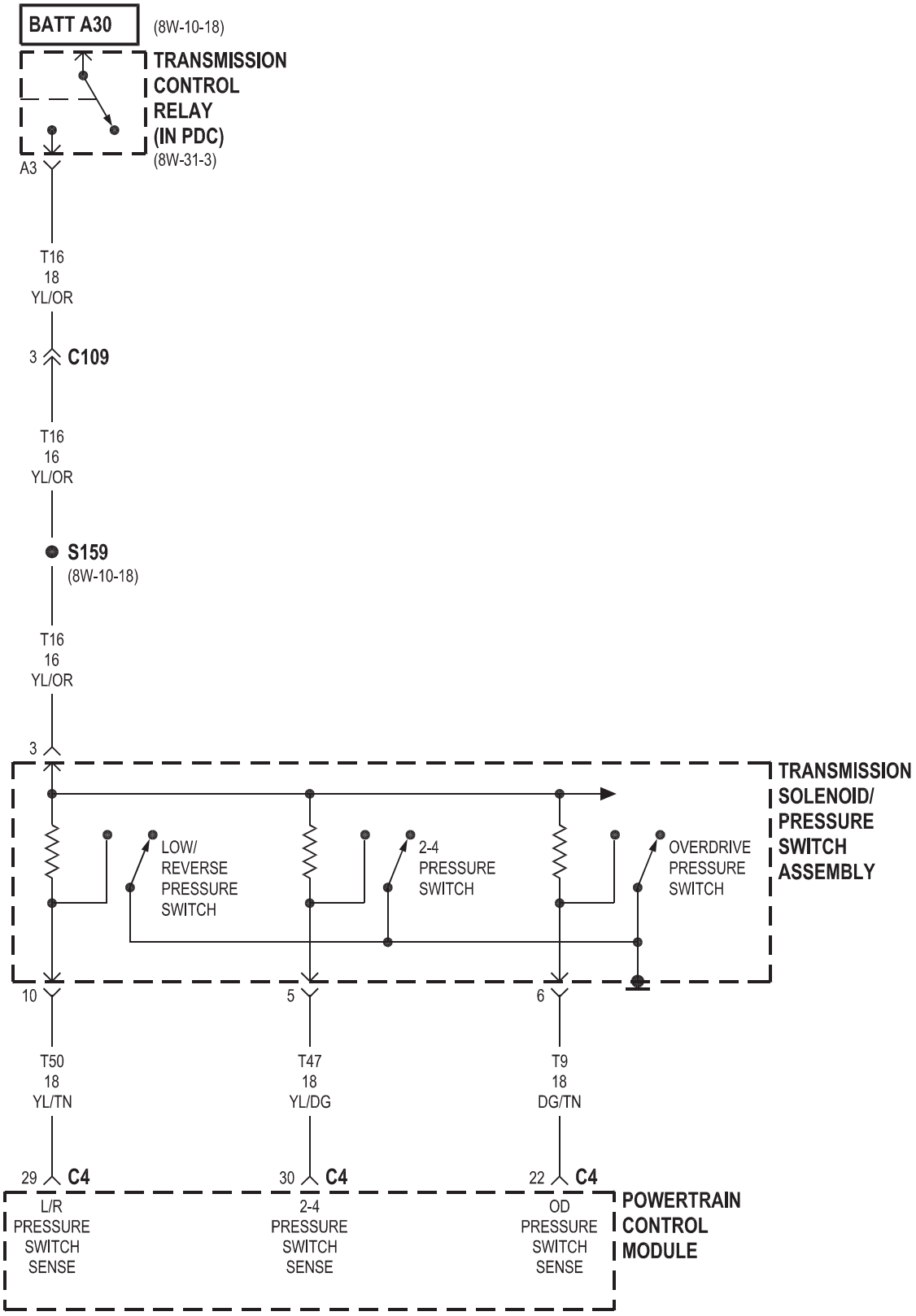
8W-31 TRANSMISSION CONTROL SYSTEM

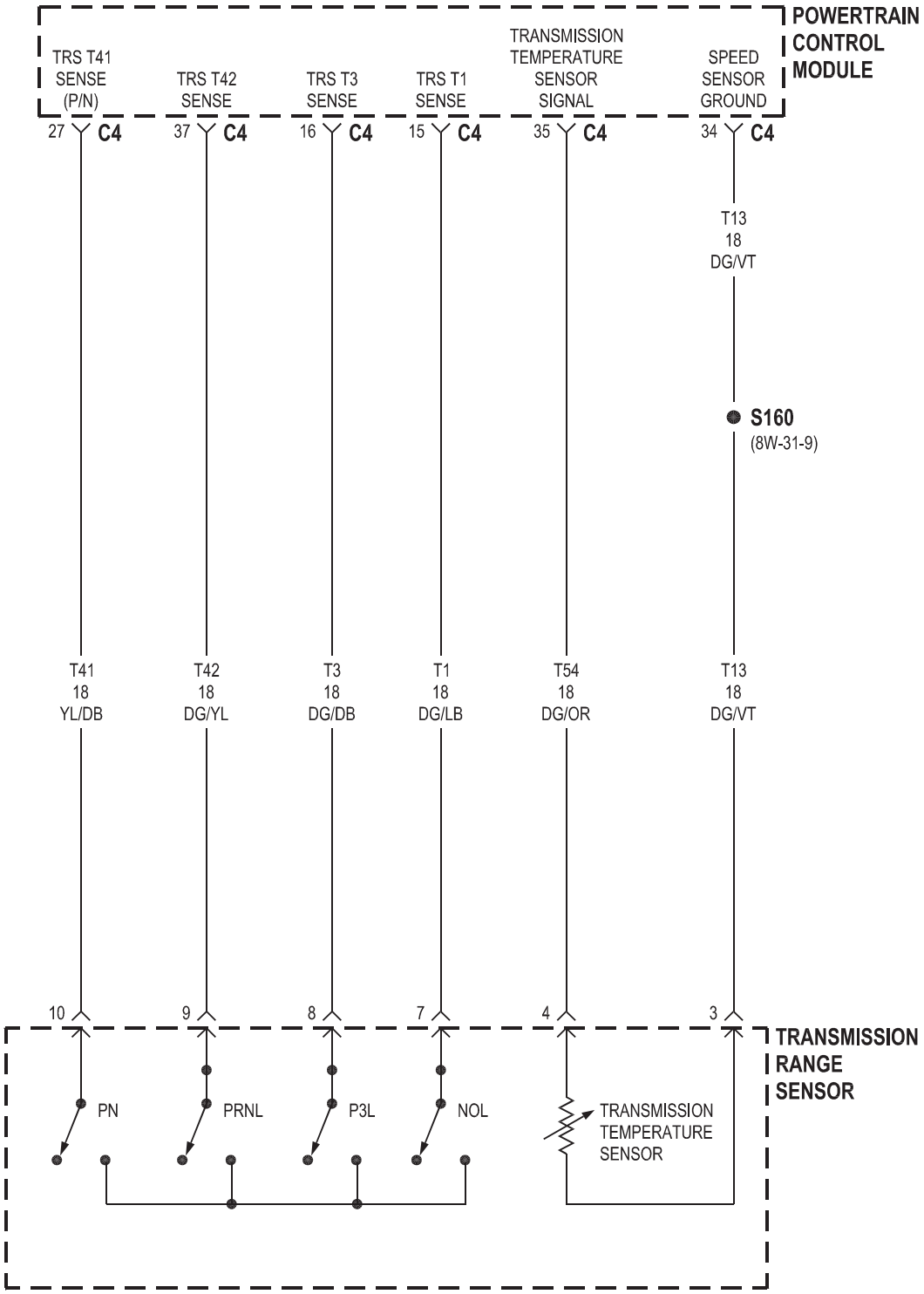
Component	Page
Axle Lock Switch	8W-31-13, 14
Brake Lamp Switch	8W-31-2
Brake Transmission Shift Interlock Solenoid	8W-31-2
Clutch Pedal Position Switch	8W-31-9
Data Link Connector	8W-31-9
Fog Lamp Relay	8W-31-3
Front Locker Indicator Switch	8W-31-12
Front Locker Pump	8W-31-11
Front Locker Relay	8W-31-11
Fuse 5	8W-31-3
Fuse 7	8W-31-7
Fuse 10	8W-31-2
Fuse 24	8W-31-11
Fuse Block	8W-31-2, 7
G100	8W-31-3
G105	8W-31-2, 3, 11, 12
G200	8W-31-10, 14
G300	8W-31-2
Input Speed Sensor	8W-31-9
Instrument Cluster	8W-31-2, 10, 12, 13, 14
Left Rear Lamp Assembly	8W-31-7
Output Speed Sensor	8W-31-9
Power Distribution Center	8W-31-3, 11
Powertrain Control Module	8W-31-2, 3, 4, 5, 6, 8, 9, 10, 14
Rear Locker Indicator Switch	8W-31-12
Rear Locker Pump	8W-31-11
Rear Locker Relay	8W-31-11
Right Rear Lamp Assembly	8W-31-7
Tow/Haul Overdrive Off Switch	8W-31-10
Transfer Case Switch	8W-31-2, 14
Transmission Control Relay	8W-31-3, 4, 5
Transmission Range Sensor	8W-31-6, 7, 9
Transmission Solenoid/Pressure Switch Assembly	8W-31-4, 5
Variable Line Pressure Sensor	8W-31-8

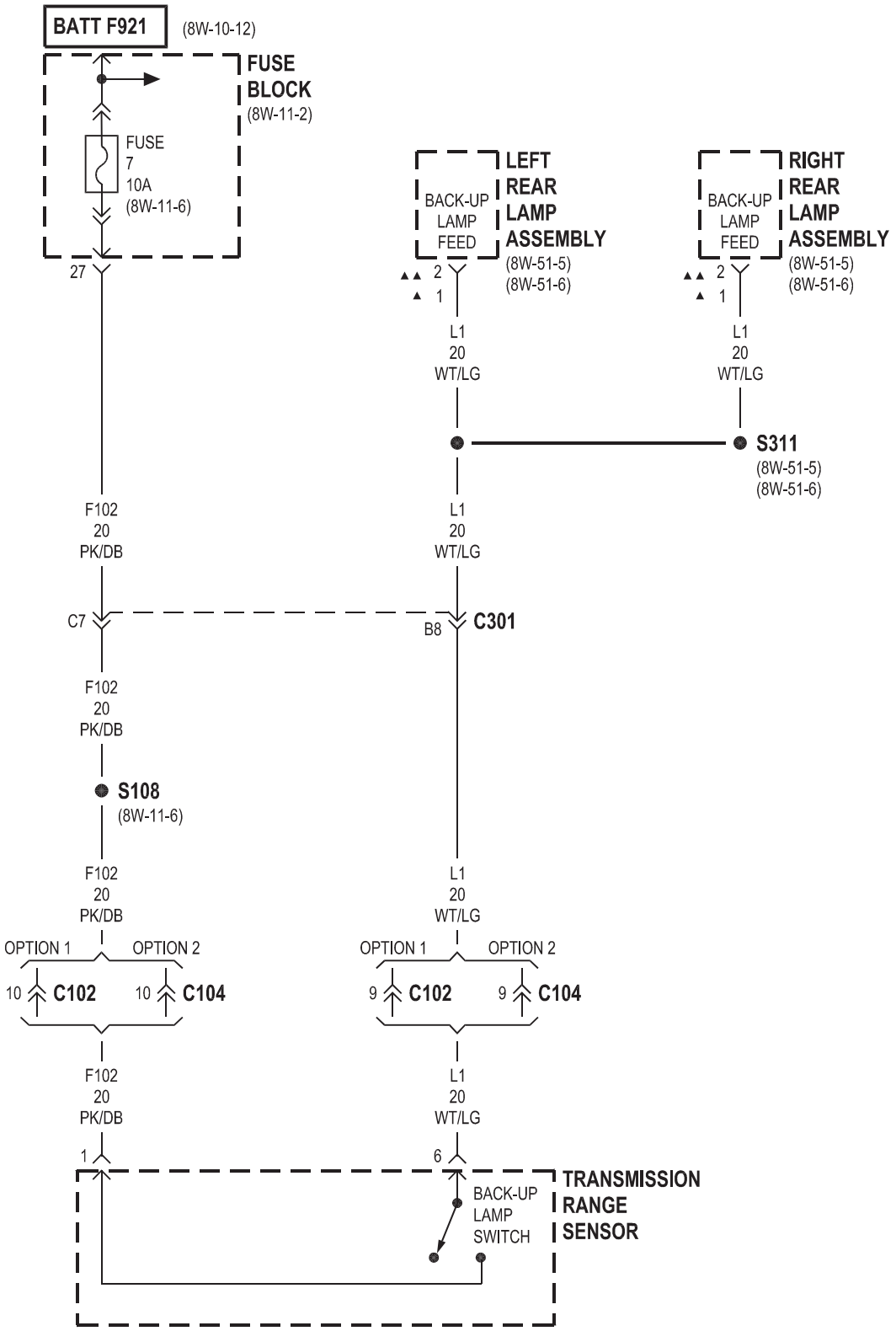

RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE
10
10A
(8W-11-8)
30
F505
20
PK/YL
B2
C300
F505
20
PK/YL
EXCEPT COMPASS/ TEMPERATURE MIRROR
COMPASS/ TEMPERATURE MIRROR
S215
(8W-11-8)
F505
20
PK/YL
11
C1
INSTRUMENT CLUSTER
(8W-40-2)
(8W-40-5)
4WD INDICATOR
11
C2
G93
20
VT/LB
B10
C300
G93
20
VT/LB
B5
C301
G93
20
VT/LB
OPTION 1
OPTION 2
13
C102
7
C104
SWITCH NO. 1 SIGNAL
POWERTRAIN CONTROL MODULE
(8W-30-17)
23
C3
B15
20
DG/WT
B4
C301
B15
20
DG/WT
G93
20
VT/LB
A
TRANSFER CASE SWITCH (EXCEPT OFF-ROAD PACKAGE)
B
Z987
18
BK
S132
(8W-15-5)
Z987
14
BK
G105
(8W-15-6)
F505
20
PK/YL
2
BRAKE TRANSMISSION SHIFT INTERLOCK SOLENOID
1
B15
20
DG/WT
1
BRAKE LAMP SWITCH
(8W-51-7)
2
Z429
20
BK/OR
G300
(8W-15-10)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE
5
20A
(8W-10-18)
5
A4
A1
TRANSMISSION CONTROL RELAY
A2
A3
Z901
18
BK
T515
20
YL/DB
T16
18
YL/OR
S106
(8W-15-2)
2
3
C109
Z901
12
BK
Z901
18
BK
T515
18
YL/DB
T16
16
YL/OR
S159
(8W-10-18)
B20
FOG LAMP RELAY (IN PDC)
(8W-50-3)
B18
T16
16
YL/OR
T16
16
YL/OR
18 C4
19 C4
28 C4
TRANSMISSION CONTROL RELAY CONTROL
TRANSMISSION CONTROL RELAY OUTPUT
TRANSMISSION CONTROL RELAY OUTPUT
POWERTRAIN CONTROL MODULE
GROUND
GROUND
12 C4
13 C4
Z903
16
BK
Z903
16
BK
G100
(8W-15-2)
G105
(8W-15-6)

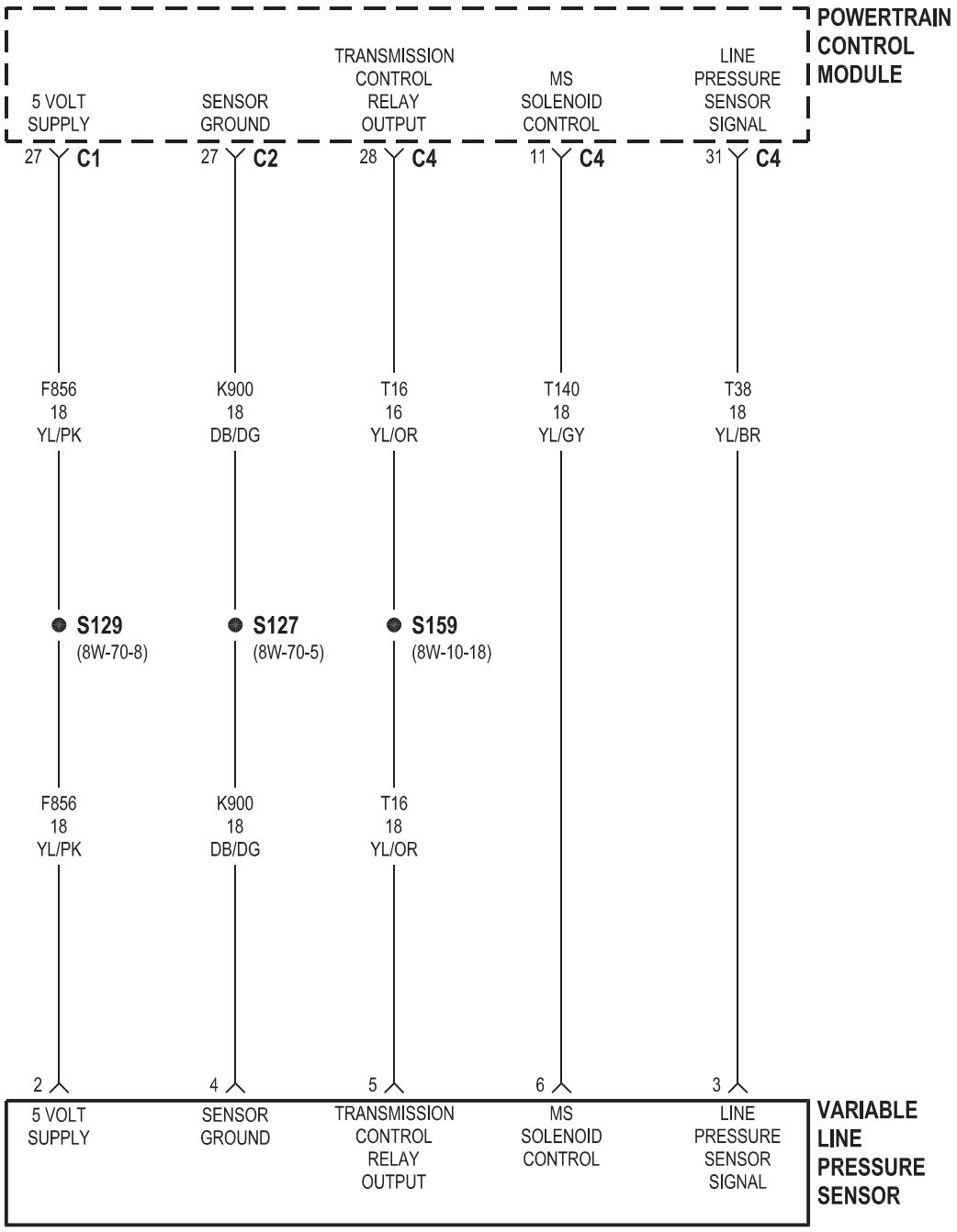

BATT A30
(8W-10-18)
TRANSMISSION CONTROL RELAY (IN PDC)
(8W-31-3)
A3
T16 18 YL/OR
3 C109
T16 16 YL/OR
S159
(8W-10-18)
T16 16 YL/OR
3
POWERTRAIN CONTROL MODULE
L/R SOLENOID CONTROL
2-4 SOLENOID CONTROL
OD SOLENOID CONTROL
UD SOLENOID CONTROL
10 C4
6 C4
1 C4
2 C4
T20 16 DG/WT
T19 16 YL/DB
T60 16 YL/GY
T59 16 YL/LB
7
4
1
2
TRANSMISSION SOLENOID/ PRESSURE SWITCH ASSEMBLY

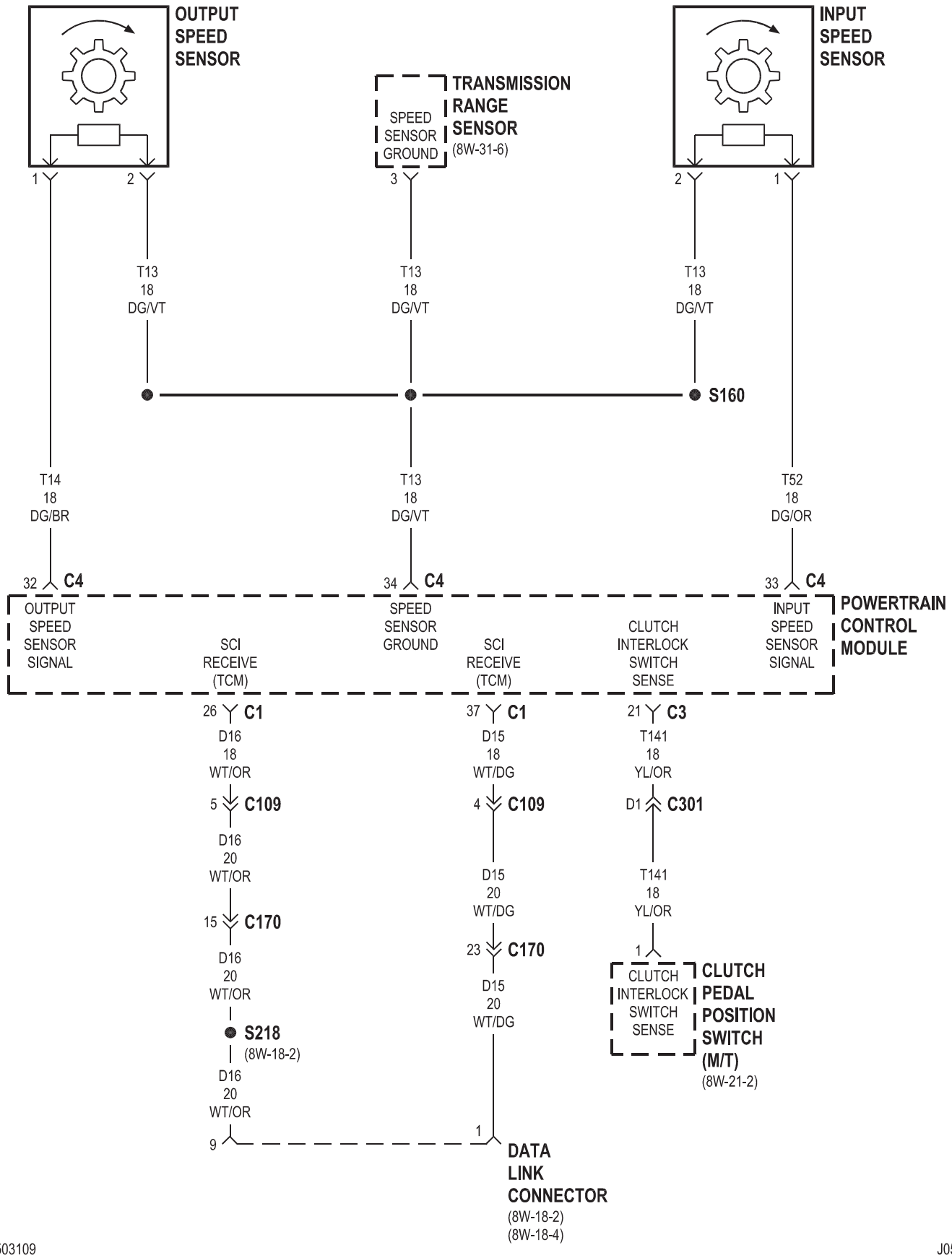

BATT A30
(8W-10-18)
TRANSMISSION CONTROL RELAY (IN PDC)
(8W-31-3)
A3
T16
18
YL/OR
3
C109
T16
16
YL/OR
S159
(8W-10-18)
T16
16
YL/OR
3
TRANSMISSION SOLENOID/ PRESSURE SWITCH ASSEMBLY
LOW/ REVERSE PRESSURE SWITCH
2-4 PRESSURE SWITCH
OVERDRIVE PRESSURE SWITCH
10
5
6
T50
18
YL/TN
T47
18
YL/DG
T9
18
DG/TN
29
C4
30
C4
22
C4
L/R PRESSURE SWITCH SENSE
2-4 PRESSURE SWITCH SENSE
OD PRESSURE SWITCH SENSE
POWERTRAIN CONTROL MODULE

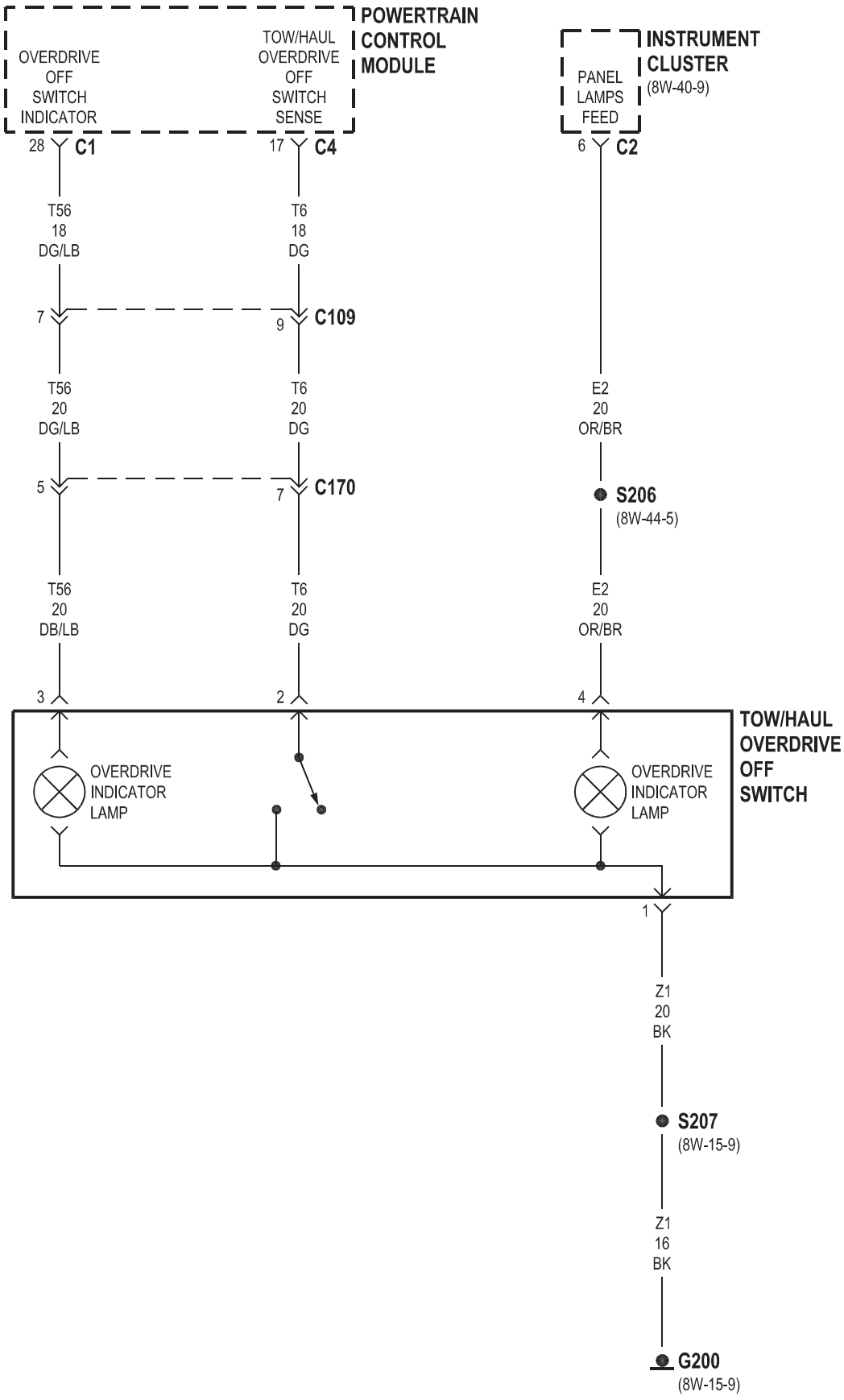

POWERTRAIN CONTROL MODULE
TRS T41 SENSE (P/N)
TRS T42 SENSE
TRS T3 SENSE
TRS T1 SENSE
TRANSMISSION TEMPERATURE SENSOR SIGNAL
SPEED SENSOR GROUND
27 C4
37 C4
16 C4
15 C4
35 C4
34 C4
T13 18 DG/VT
S160
(8W-31-9)
T41 18 YL/DB
T42 18 DG/YL
T3 18 DG/DB
T1 18 DG/LB
T54 18 DG/OR
T13 18 DG/VT
10
9
8
7
4
3
TRANSMISSION RANGE SENSOR
PN
PRNL
P3L
NOL
TRANSMISSION TEMPERATURE SENSOR

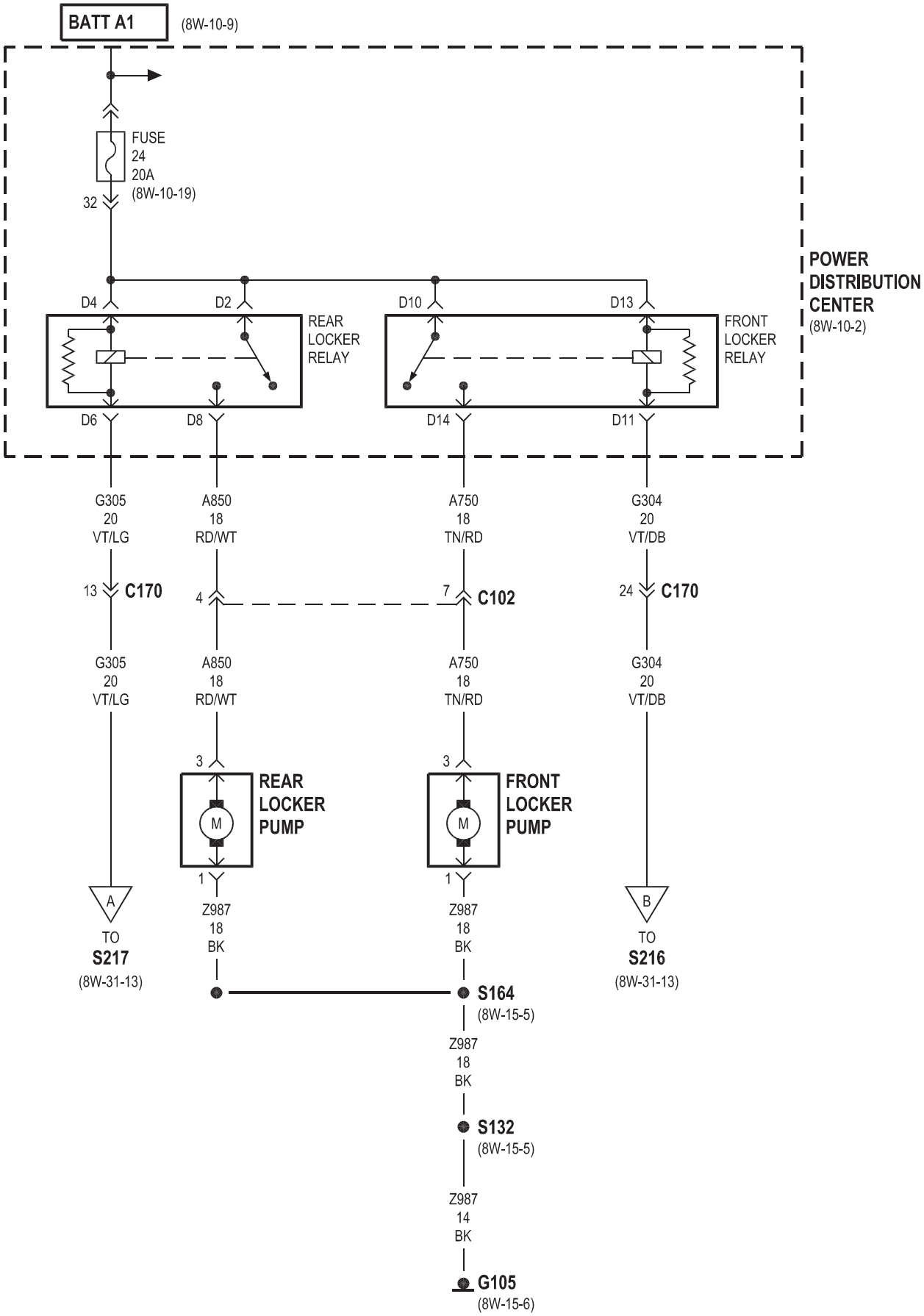


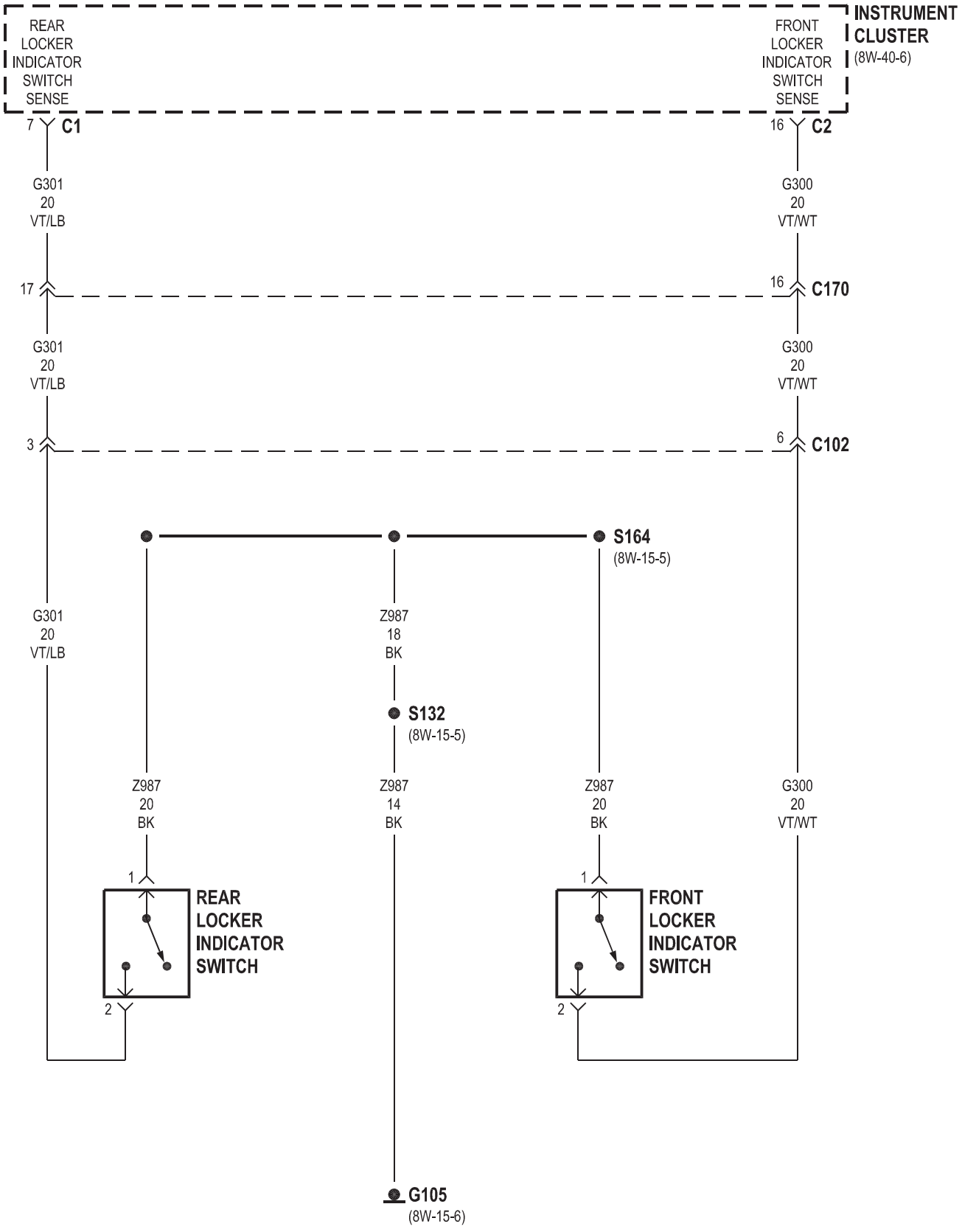
▲ EXCEPT EXPORT
▲▲ EXPORT

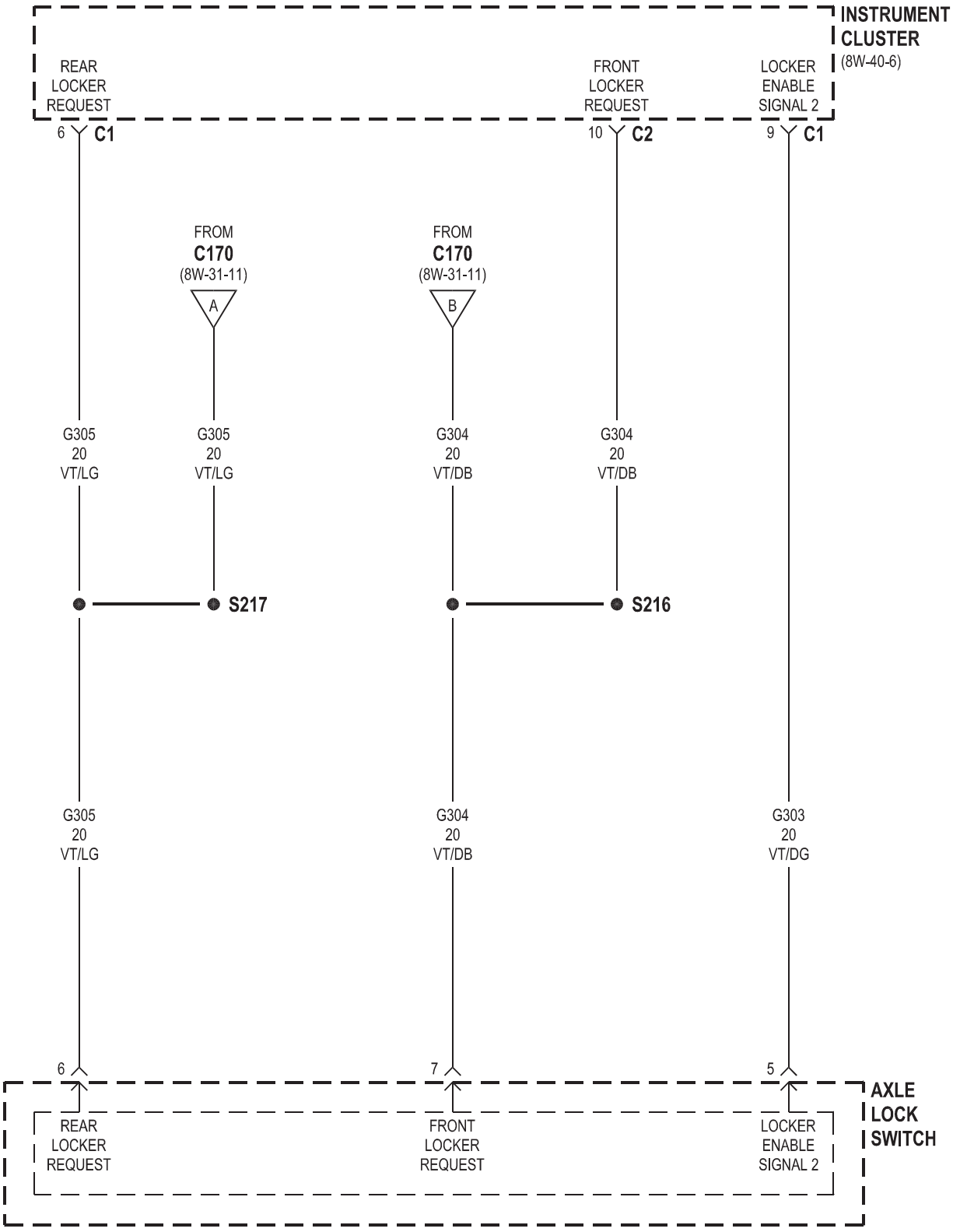

POWERTRAIN CONTROL MODULE
5 VOLT SUPPLY
SENSOR GROUND
TRANSMISSION CONTROL RELAY OUTPUT
MS SOLENOID CONTROL
LINE PRESSURE SENSOR SIGNAL
27 C1
27 C2
28 C4
11 C4
31 C4
F856 18 YL/PK
K900 18 DB/DG
T16 16 YL/OR
T140 18 YL/GY
T38 18 YL/BR
S129 (8W-70-8)
S127 (8W-70-5)
S159 (8W-10-18)
F856 18 YL/PK
K900 18 DB/DG
T16 18 YL/OR
2
4
5
6
3
5 VOLT SUPPLY
SENSOR GROUND
TRANSMISSION CONTROL RELAY OUTPUT
MS SOLENOID CONTROL
LINE PRESSURE SENSOR SIGNAL
VARIABLE LINE PRESSURE SENSOR

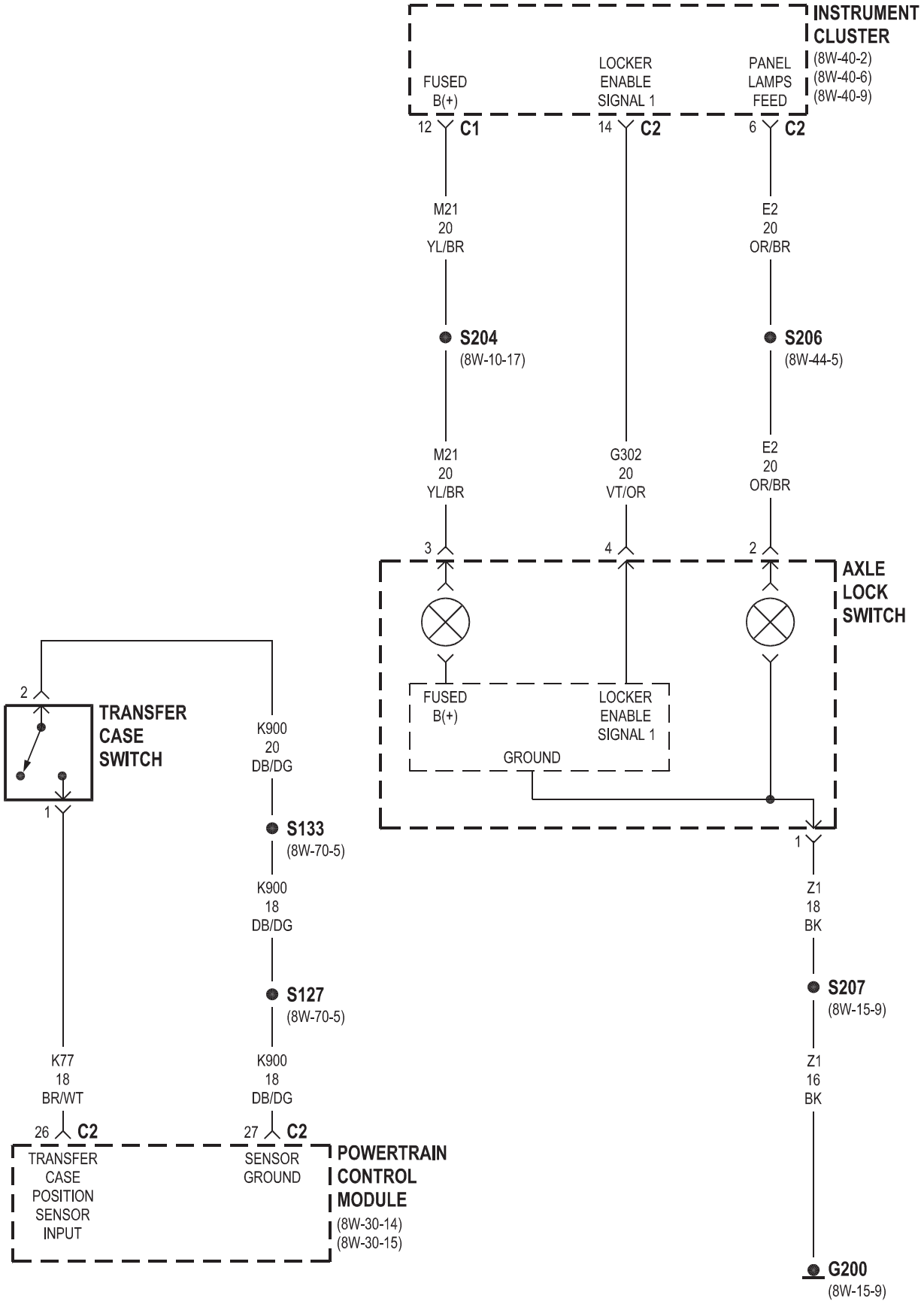

OUTPUT SPEED SENSOR
1
2
TRANSMISSION RANGE SENSOR
SPEED SENSOR GROUND
(8W-31-6)
3
INPUT SPEED SENSOR
2
1
T13 18 DG/VT
T13 18 DG/VT
T13 18 DG/VT
S160
T14 18 DG/BR
T13 18 DG/VT
T52 18 DG/OR
32 C4
34 C4
33 C4
POWERTRAIN CONTROL MODULE
OUTPUT SPEED SENSOR SIGNAL
SCI RECEIVE (TCM)
SPEED SENSOR GROUND
SCI RECEIVE (TCM)
CLUTCH INTERLOCK SWITCH SENSE
INPUT SPEED SENSOR SIGNAL
26 C1
37 C1
21 C3
D16 18 WT/OR
D15 18 WT/DG
T141 18 YL/OR
5 C109
4 C109
D1 C301
D16 20 WT/OR
D15 20 WT/DG
T141 18 YL/OR
15 C170
23 C170
1
CLUTCH INTERLOCK SWITCH SENSE
CLUTCH PEDAL POSITION SWITCH (M/T)
(8W-21-2)
D16 20 WT/OR
D15 20 WT/DG
S218
(8W-18-2)
D16 20 WT/OR
9
1
DATA LINK CONNECTOR
(8W-18-2)
(8W-18-4)


POWERTRAIN CONTROL MODULE
OVERDRIVE OFF SWITCH INDICATOR
TOW/HAUL OVERDRIVE OFF SWITCH SENSE
28 C1
17 C4
INSTRUMENT CLUSTER
(8W-40-9)
PANEL LAMPS FEED
6 C2
T56 18 DG/LB
T6 18 DG
7
9 C109
T56 20 DG/LB
T6 20 DG
E2 20 OR/BR
5
7 C170
S206
(8W-44-5)
T56 20 DB/LB
T6 20 DG
E2 20 OR/BR
3
2
4
TOW/HAUL OVERDRIVE OFF SWITCH
OVERDRIVE INDICATOR LAMP
OVERDRIVE INDICATOR LAMP
1
Z1 20 BK
S207
(8W-15-9)
Z1 16 BK
G200
(8W-15-9)


BATT A1
(8W-10-9)
FUSE
24
20A
(8W-10-19)
32
POWER
DISTRIBUTION
CENTER
(8W-10-2)
D4
D2
D10
D13
REAR
LOCKER
RELAY
FRONT
LOCKER
RELAY
D6
D8
D14
D11
G305
20
VT/LG
A850
18
RD/WT
A750
18
TN/RD
G304
20
VT/DB
13
C170
4
7
C102
24
C170
G305
20
VT/LG
A850
18
RD/WT
A750
18
TN/RD
G304
20
VT/DB
3
3
REAR
LOCKER
PUMP
FRONT
LOCKER
PUMP
M
M
1
1
A
B
TO
S217
(8W-31-13)
Z987
18
BK
Z987
18
BK
TO
S216
(8W-31-13)
S164
(8W-15-5)
Z987
18
BK
S132
(8W-15-5)
Z987
14
BK
G105
(8W-15-6)

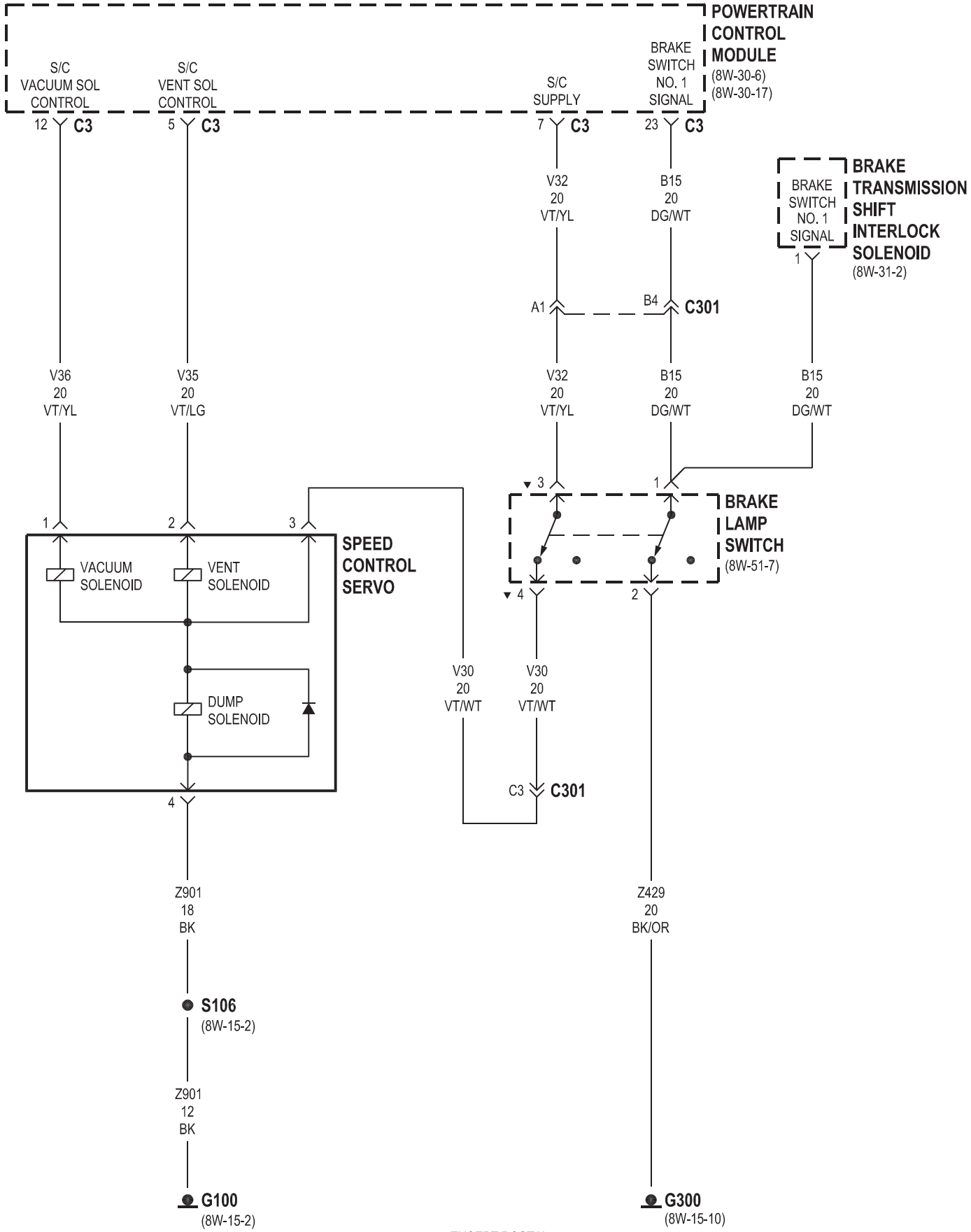

INSTRUMENT CLUSTER
(8W-40-6)
REAR LOCKER INDICATOR SWITCH SENSE
FRONT LOCKER INDICATOR SWITCH SENSE
7
C1
16
C2
G301
20
VT/LB
G300
20
VT/WT
17
16
C170
G301
20
VT/LB
G300
20
VT/WT
3
6
C102
S164
(8W-15-5)
G301
20
VT/LB
Z987
18
BK
S132
(8W-15-5)
Z987
20
BK
Z987
14
BK
Z987
20
BK
G300
20
VT/WT
1
REAR LOCKER INDICATOR SWITCH
2
1
FRONT LOCKER INDICATOR SWITCH
2
G105
(8W-15-6)


INSTRUMENT CLUSTER
(8W-40-6)
REAR LOCKER REQUEST
6
C1
FRONT LOCKER REQUEST
10
C2
LOCKER ENABLE SIGNAL 2
9
C1
FROM C170 (8W-31-11)
A
FROM C170 (8W-31-11)
B
G305 20 VT/LG
G305 20 VT/LG
G304 20 VT/DB
G304 20 VT/DB
S217
S216
G305 20 VT/LG
G304 20 VT/DB
G303 20 VT/DG
6
7
5
REAR LOCKER REQUEST
FRONT LOCKER REQUEST
LOCKER ENABLE SIGNAL 2
AXLE LOCK SWITCH


INSTRUMENT CLUSTER
(8W-40-2)
(8W-40-6)
(8W-40-9)
FUSED B(+)
LOCKER ENABLE SIGNAL 1
PANEL LAMPS FEED
12 C1
14 C2
6 C2
M21 20 YL/BR
E2 20 OR/BR
S204
(8W-10-17)
S206
(8W-44-5)
M21 20 YL/BR
G302 20 VT/OR
E2 20 OR/BR
3
4
2
AXLE LOCK SWITCH
FUSED B(+)
LOCKER ENABLE SIGNAL 1
GROUND
1
2
TRANSFER CASE SWITCH
1
K900 20 DB/DG
S133
(8W-70-5)
K900 18 DB/DG
S127
(8W-70-5)
Z1 18 BK
S207
(8W-15-9)
Z1 16 BK
G200
(8W-15-9)
K77 18 BR/WT
K900 18 DB/DG
26 C2
27 C2
TRANSFER CASE POSITION SENSOR INPUT
SENSOR GROUND
POWERTRAIN CONTROL MODULE
(8W-30-14)
(8W-30-15)

8W-33 VEHICLE SPEED CONTROL

Component	Page
Brake Lamp Switch	8W-33-2
Brake Transmission Shift Interlock Solenoid	8W-33-2
Clockspring	8W-33-3
G100	8W-33-2
G300	8W-33-2
Left Speed Control Switch	8W-33-3
Powertrain Control Module	8W-33-2, 3
Right Speed Control Switch	8W-33-3
Speed Control Servo	8W-33-2



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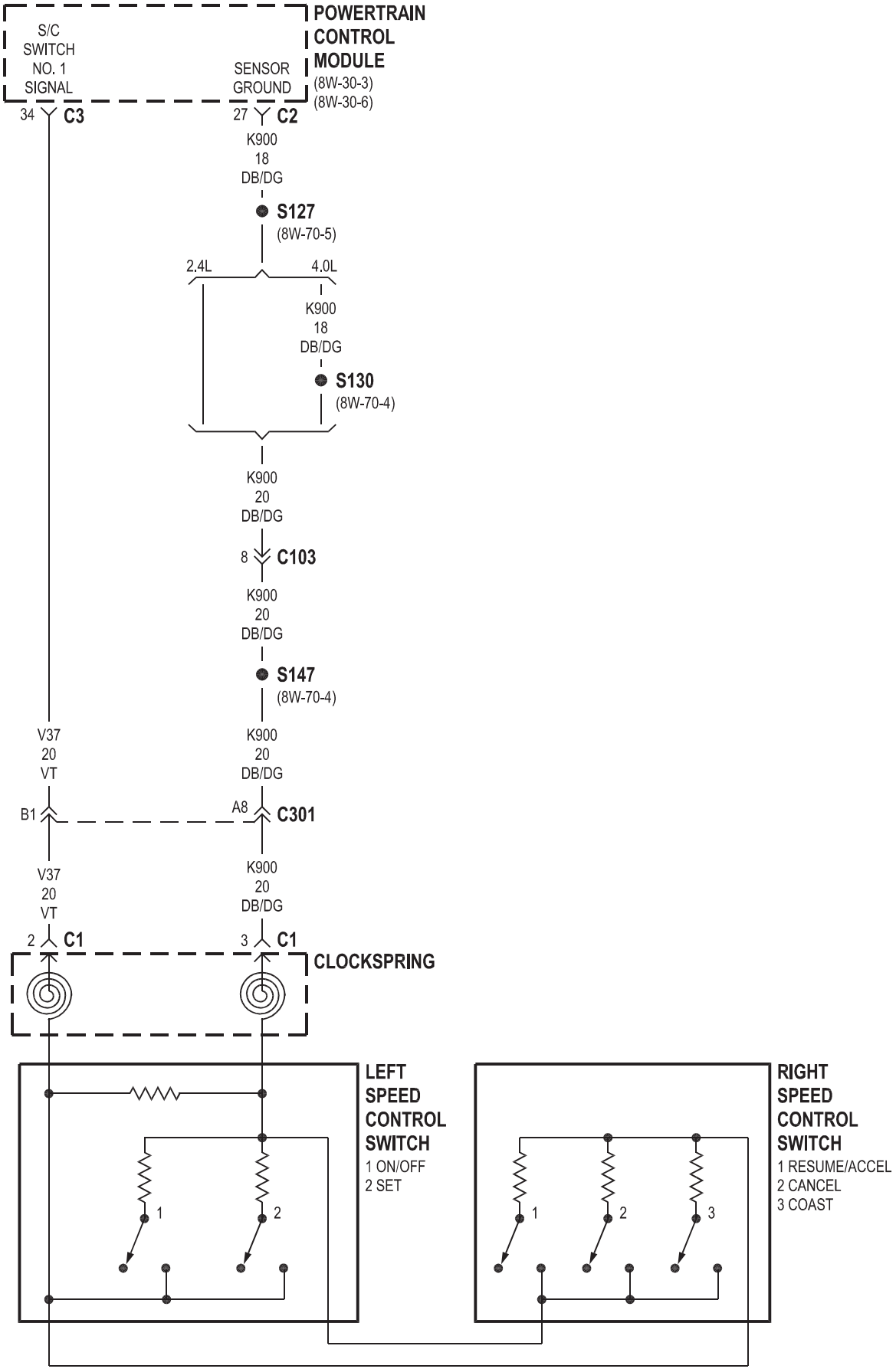
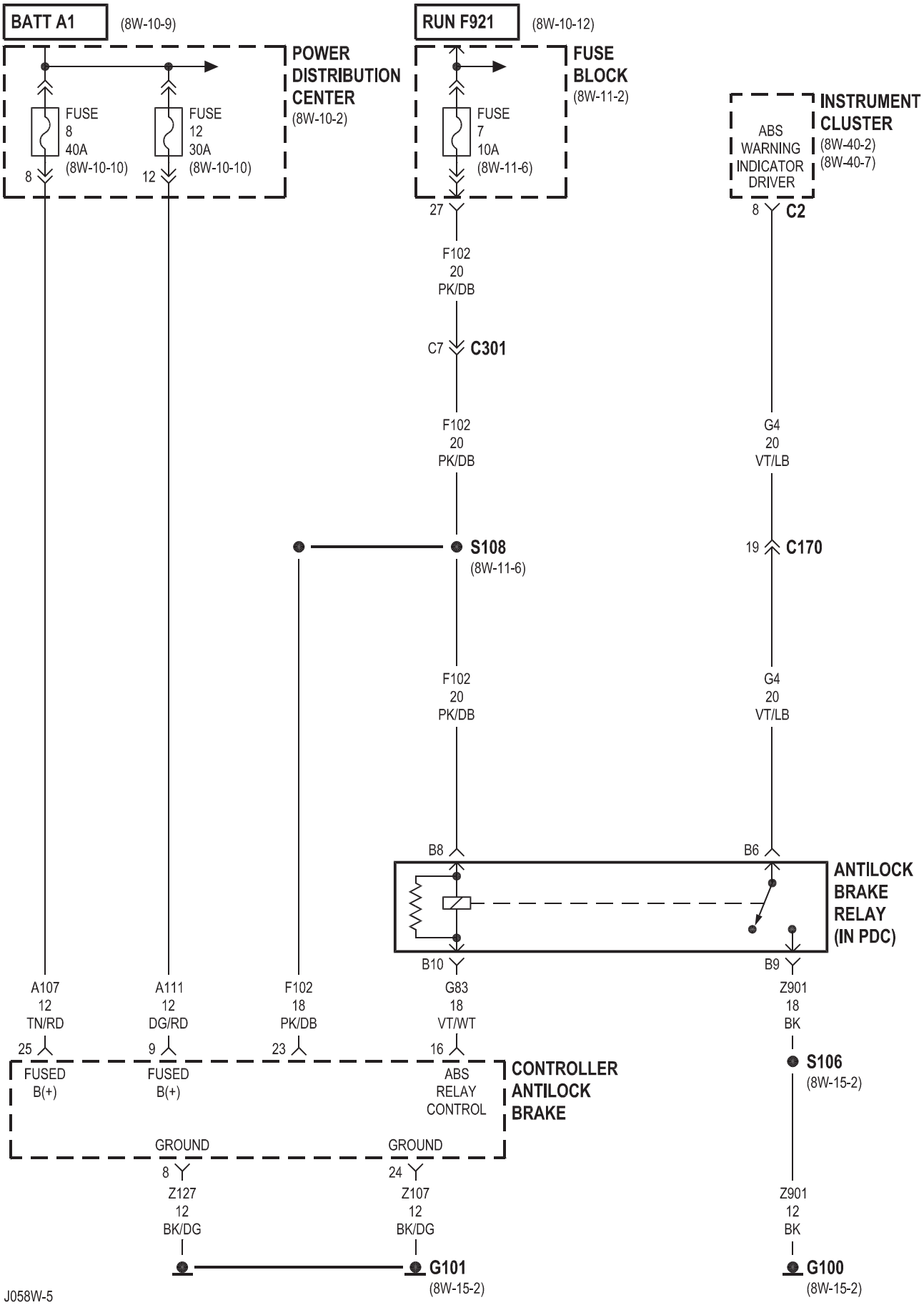
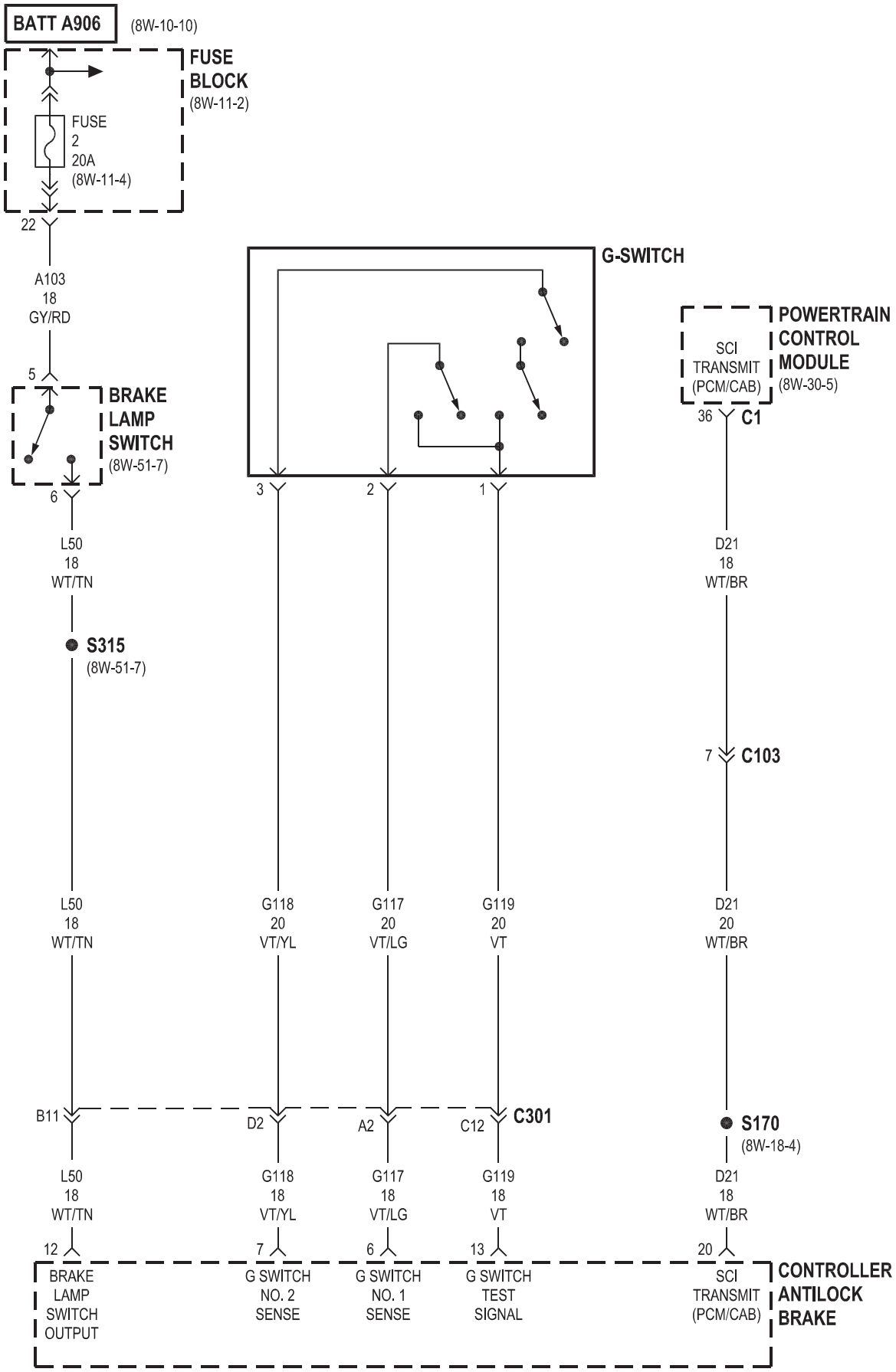
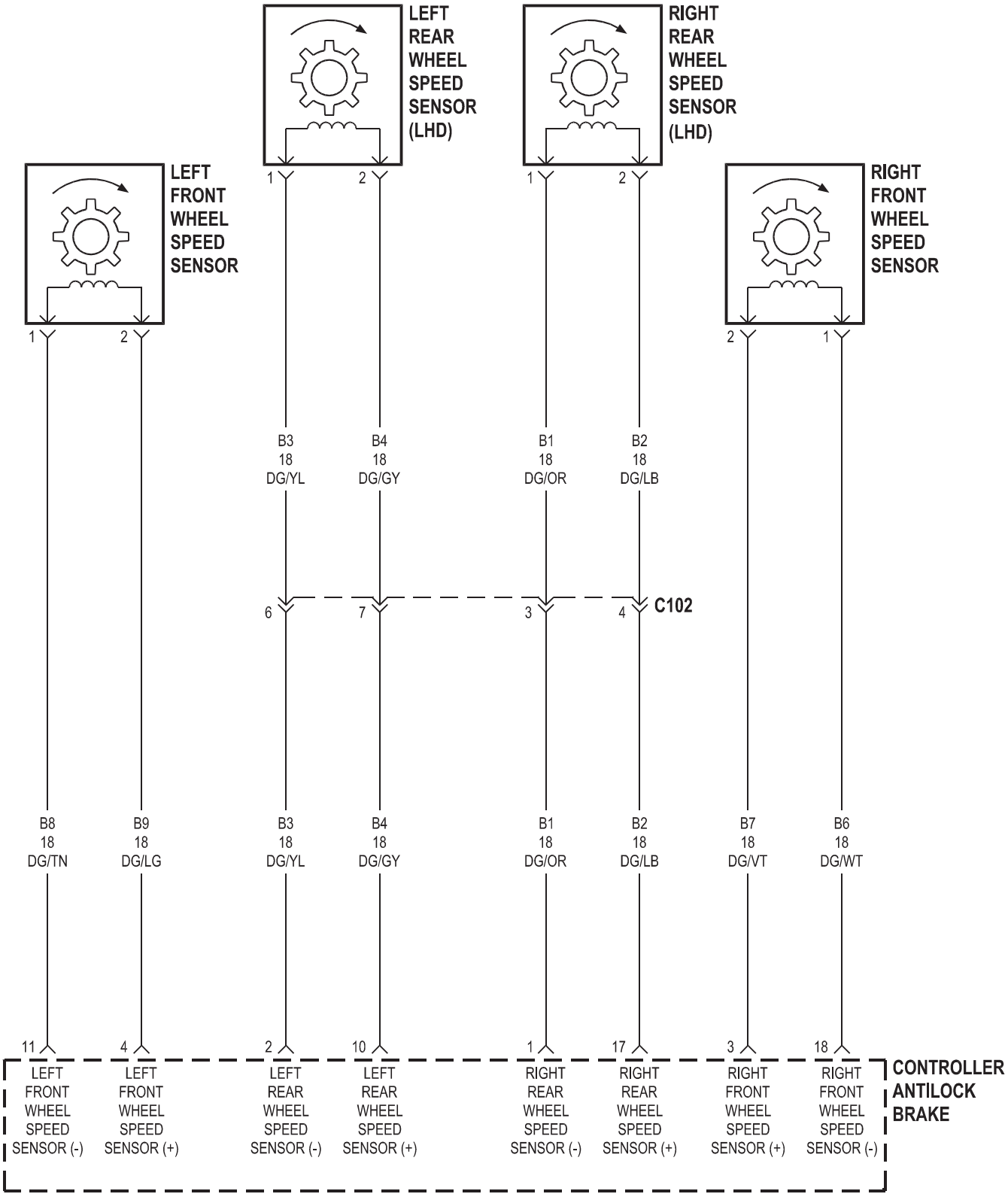

POWERTRAIN CONTROL MODULE
(8W-30-3)
(8W-30-6)
S/C SWITCH NO. 1 SIGNAL
SENSOR GROUND
34 C3
27 C2
K900 18 DB/DG
S127
(8W-70-5)
2.4L
4.0L
K900 18 DB/DG
S130
(8W-70-4)
K900 20 DB/DG
8 C103
K900 20 DB/DG
S147
(8W-70-4)
V37 20 VT
K900 20 DB/DG
B1
A8 C301
V37 20 VT
K900 20 DB/DG
2 C1
3 C1
CLOCKSPRING
LEFT SPEED CONTROL SWITCH
1 ON/OFF
2 SET
RIGHT SPEED CONTROL SWITCH
1 RESUME/ACCEL
2 CANCEL
3 COAST

8W-35 ANTILOCK BRAKES

Component	Page
Antilock Brake Relay	8W-35-2
Brake Lamp Switch	8W-35-3
Controller Antilock Brake	8W-35-2, 3, 4
Fuse 2	8W-35-3
Fuse 7	8W-35-2
Fuse 8	8W-35-2
Fuse 12	8W-35-2
Fuse Block	8W-35-2, 3
G-Switch	8W-35-3
G100	8W-35-2
G101	8W-35-2
Instrument Cluster	8W-35-2
Left Front Wheel Speed Sensor	8W-35-4
Left Rear Wheel Speed Sensor	8W-35-4
Power Distribution Center	8W-35-2
Powertrain Control Module	8W-35-3
Right Front Wheel Speed Sensor	8W-35-4
Right Rear Wheel Speed Sensor	8W-35-4

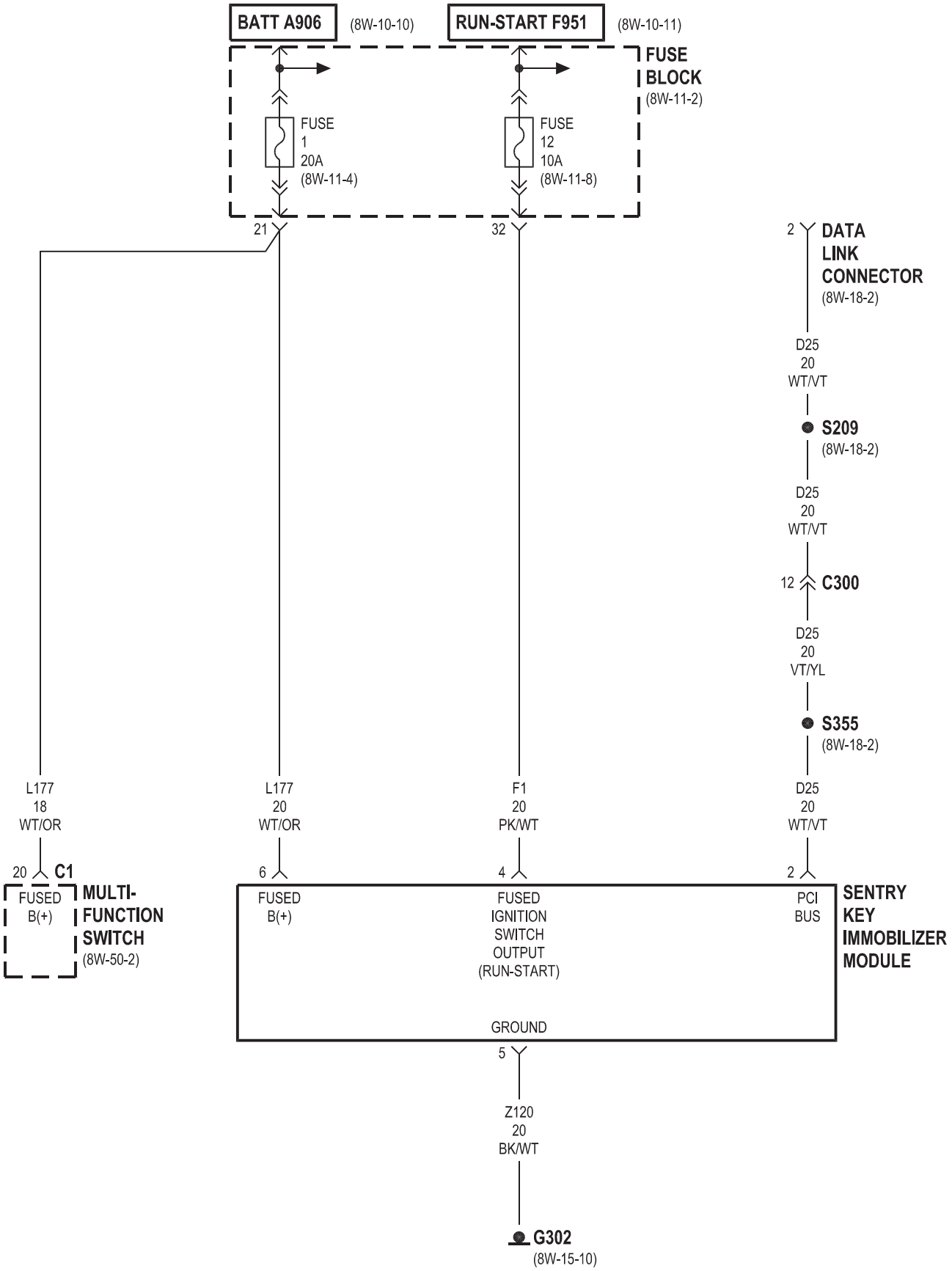

BATT A1
(8W-10-9)
RUN F921
(8W-10-12)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 8 40A (8W-10-10)
FUSE 12 30A (8W-10-10)
FUSE BLOCK
(8W-11-2)
FUSE 7 10A (8W-11-6)
INSTRUMENT CLUSTER
(8W-40-2)
(8W-40-7)
ABS WARNING INDICATOR DRIVER
8 C2
27
F102 20 PK/DB
C7 C301
F102 20 PK/DB
G4 20 VT/LB
19 C170
S108
(8W-11-6)
F102 20 PK/DB
G4 20 VT/LB
B8
B6
ANTILOCK BRAKE RELAY (IN PDC)
B10
B9
A107 12 TN/RD
A111 12 DG/RD
F102 18 PK/DB
G83 18 VT/WT
Z901 18 BK
25
9
23
16
FUSED B(+)
FUSED B(+)
ABS RELAY CONTROL
CONTROLLER ANTILOCK BRAKE
S106
(8W-15-2)
GROUND
GROUND
8
24
Z127 12 BK/DG
Z107 12 BK/DG
Z901 12 BK
G101
(8W-15-2)
G100
(8W-15-2)


BATT A906
(8W-10-10)
FUSE BLOCK
(8W-11-2)
FUSE 2 20A
(8W-11-4)
22
A103 18 GY/RD
5
BRAKE LAMP SWITCH
(8W-51-7)
6
L50 18 WT/TN
S315
(8W-51-7)
L50 18 WT/TN
B11
L50 18 WT/TN
12
BRAKE LAMP SWITCH OUTPUT
G-SWITCH
3
2
1
G118 20 VT/YL
D2
G118 18 VT/YL
7
G SWITCH NO. 2 SENSE
G117 20 VT/LG
A2
G117 18 VT/LG
6
G SWITCH NO. 1 SENSE
G119 20 VT
C12
C301
G119 18 VT
13
G SWITCH TEST SIGNAL
POWERTRAIN CONTROL MODULE
(8W-30-5)
SCI TRANSMIT (PCM/CAB)
36
C1
D21 18 WT/BR
7
C103
D21 20 WT/BR
S170
(8W-18-4)
D21 18 WT/BR
20
SCI TRANSMIT (PCM/CAB)
CONTROLLER ANTILOCK BRAKE


LEFT REAR WHEEL SPEED SENSOR (LHD)
RIGHT REAR WHEEL SPEED SENSOR (LHD)
LEFT FRONT WHEEL SPEED SENSOR
RIGHT FRONT WHEEL SPEED SENSOR
1
2
B3 18 DG/YL
B4 18 DG/GY
B1 18 DG/OR
B2 18 DG/LB
6
7
3
4
C102
B8 18 DG/TN
B9 18 DG/LG
B7 18 DG/VT
B6 18 DG/WT
11
4
2
10
1
17
3
18
LEFT FRONT WHEEL SPEED SENSOR (-)
LEFT FRONT WHEEL SPEED SENSOR (+)
LEFT REAR WHEEL SPEED SENSOR (-)
LEFT REAR WHEEL SPEED SENSOR (+)
RIGHT REAR WHEEL SPEED SENSOR (-)
RIGHT REAR WHEEL SPEED SENSOR (+)
RIGHT FRONT WHEEL SPEED SENSOR (+)
RIGHT FRONT WHEEL SPEED SENSOR (-)
CONTROLLER ANTILOCK BRAKE

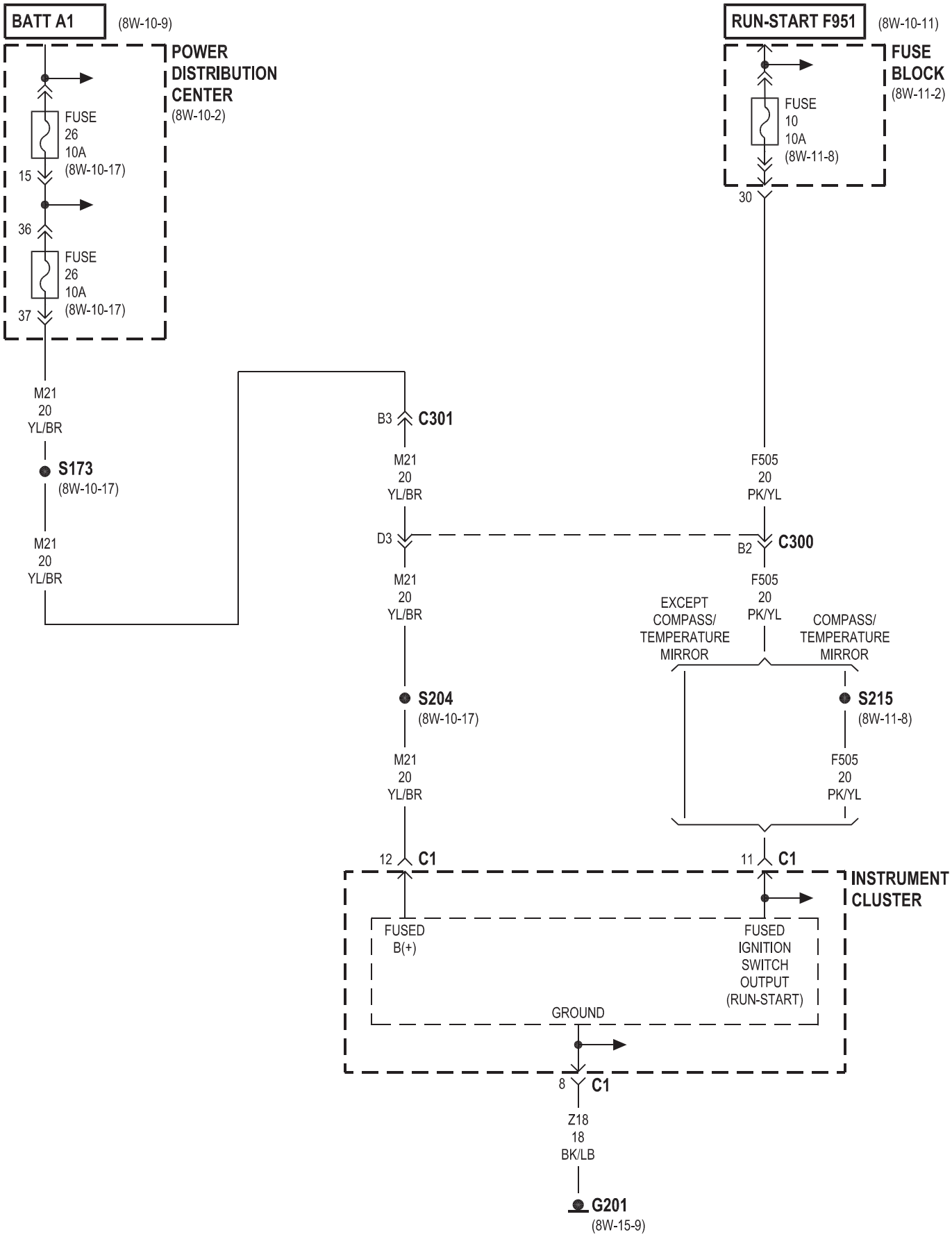
8W-39 VEHICLE THEFT SECURITY SYSTEM

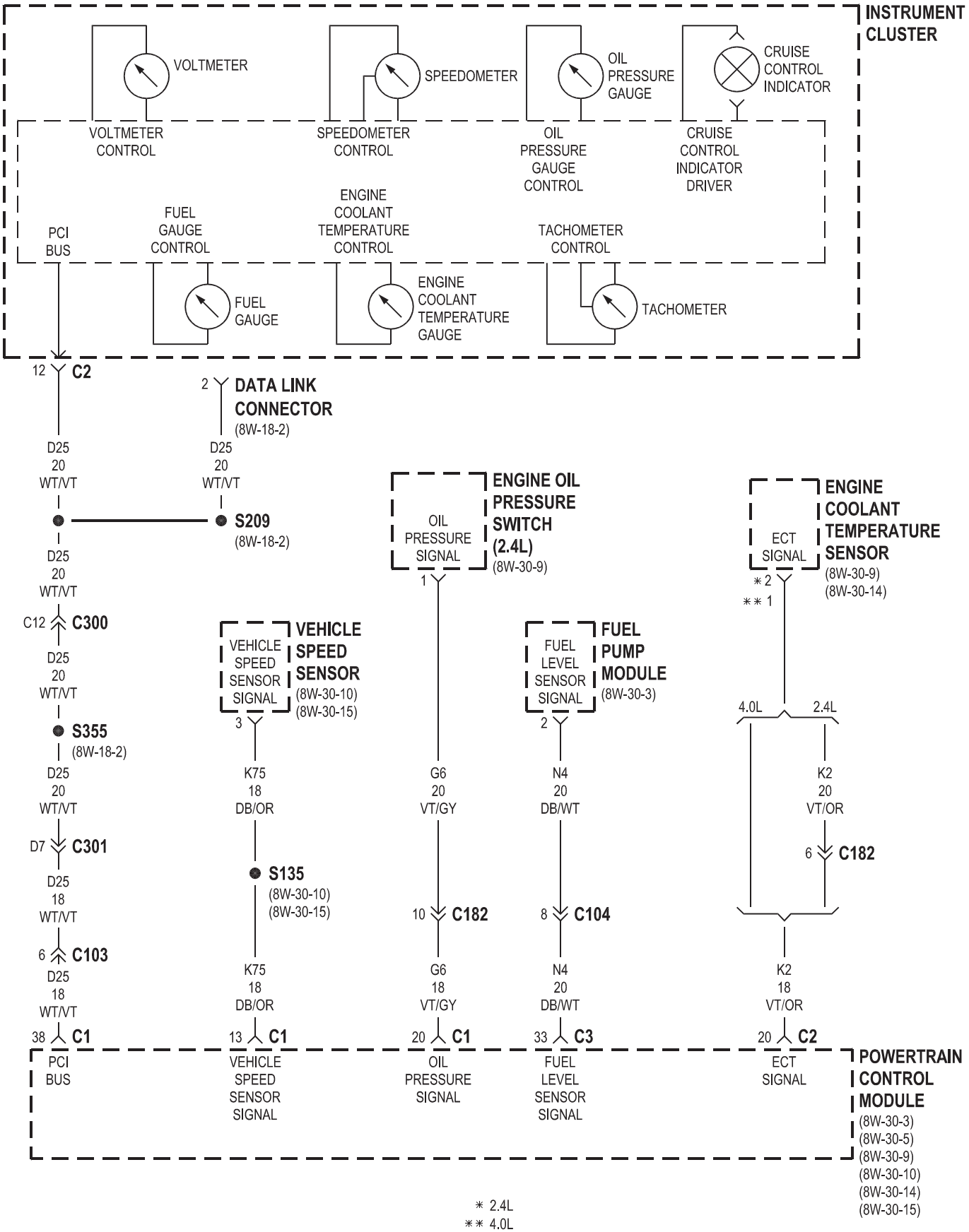
Component	Page
Data Link Connector	8W-39-2
Fuse 1	8W-39-2
Fuse 12	8W-39-2
Fuse Block	8W-39-2
G302	8W-39-2
Multi-Function Switch	8W-39-2
Sentry Key Immobilizer Module	8W-39-2


BATT A906
(8W-10-10)
RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 1 20A (8W-11-4)
FUSE 12 10A (8W-11-8)
21
32
2 DATA LINK CONNECTOR
(8W-18-2)
D25 20 WT/VT
S209
(8W-18-2)
D25 20 WT/VT
12 C300
D25 20 VT/YL
S355
(8W-18-2)
D25 20 WT/VT
L177 18 WT/OR
L177 20 WT/OR
F1 20 PK/WT
20 C1
FUSED B(+)
MULTI-FUNCTION SWITCH
(8W-50-2)
6
4
2
FUSED B(+)
FUSED IGNITION SWITCH OUTPUT (RUN-START)
PCI BUS
SENTRY KEY IMMOBILIZER MODULE
GROUND
5
Z120 20 BK/WT
G302
(8W-15-10)

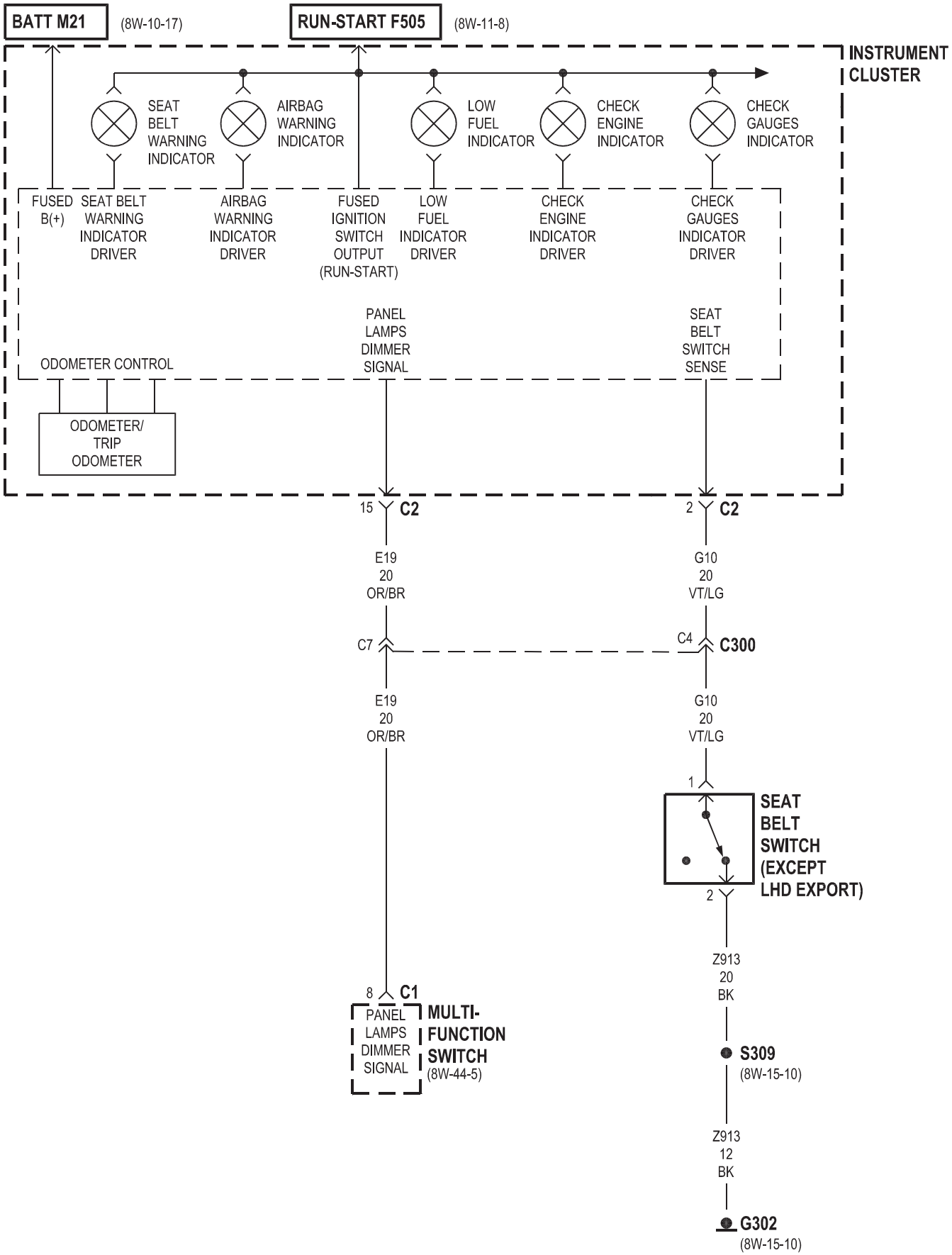
8W-40 INSTRUMENT CLUSTER

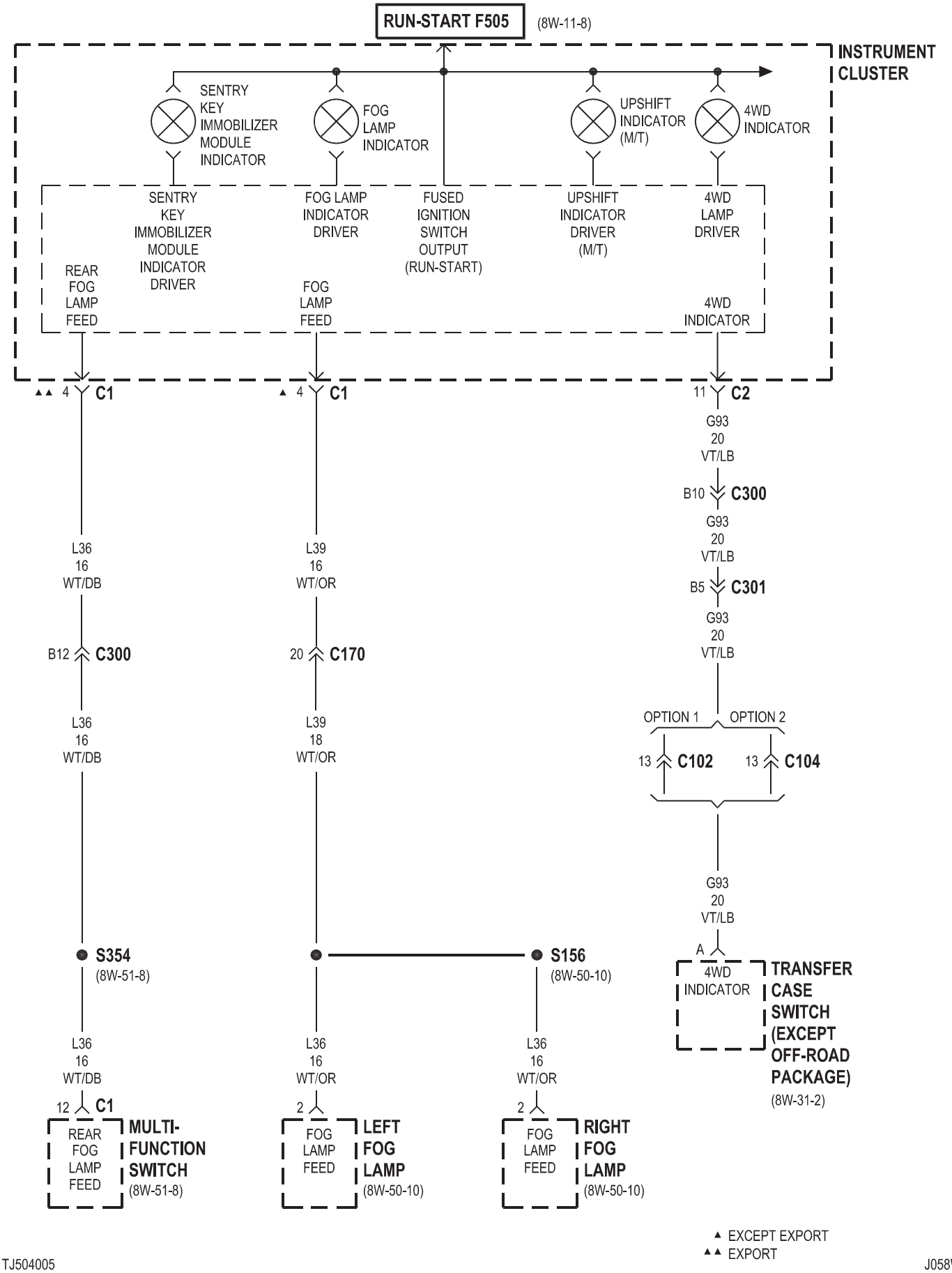
Component	Page
Antilock Brake Relay	8W-40-7
Axle Lock Switch	8W-40-6
Brake Warning Indicator Switch	8W-40-7
Data Link Connector	8W-40-3, 9
Daytime Running Lamp Module	8W-40-10
Driver Door Ajar Switch	8W-40-8
Engine Coolant Temperature Sensor	8W-40-3
Engine Oil Pressure Switch	8W-40-3
Front Locker Indicator Switch	8W-40-6
Fuel Pump Module	8W-40-3
Fuse 10	8W-40-2
Fuse 26	8W-40-2
Fuse Block	8W-40-2
G100	8W-40-7
G201	8W-40-2
G302	8W-40-4
Ignition Switch	8W-40-8
Instrument Cluster	8W-40-2, 3, 4, 5, 6, 7, 8, 9, 10
Left Courtesy Lamp	8W-40-8
Left Fog Lamp	8W-40-5
Left Side Marker Lamp	8W-40-8
Multi-Function Switch	8W-40-4, 5, 8, 10
Park Brake Switch	8W-40-7
Passenger Door Ajar Switch	8W-40-8
Power Distribution Center	8W-40-2
Powertrain Control Module	8W-40-3
Radio	8W-40-9
Rear Locker Indicator Switch	8W-40-6
Rear Window Defogger Relay	8W-40-9
Rear Window Defogger Switch	8W-40-9
Right Courtesy Lamp	8W-40-8
Right Fog Lamp	8W-40-5
Right Side Marker Lamp	8W-40-8
Seat Belt Switch	8W-40-4
Transfer Case Switch	8W-40-5
Vehicle Speed Sensor	8W-40-3

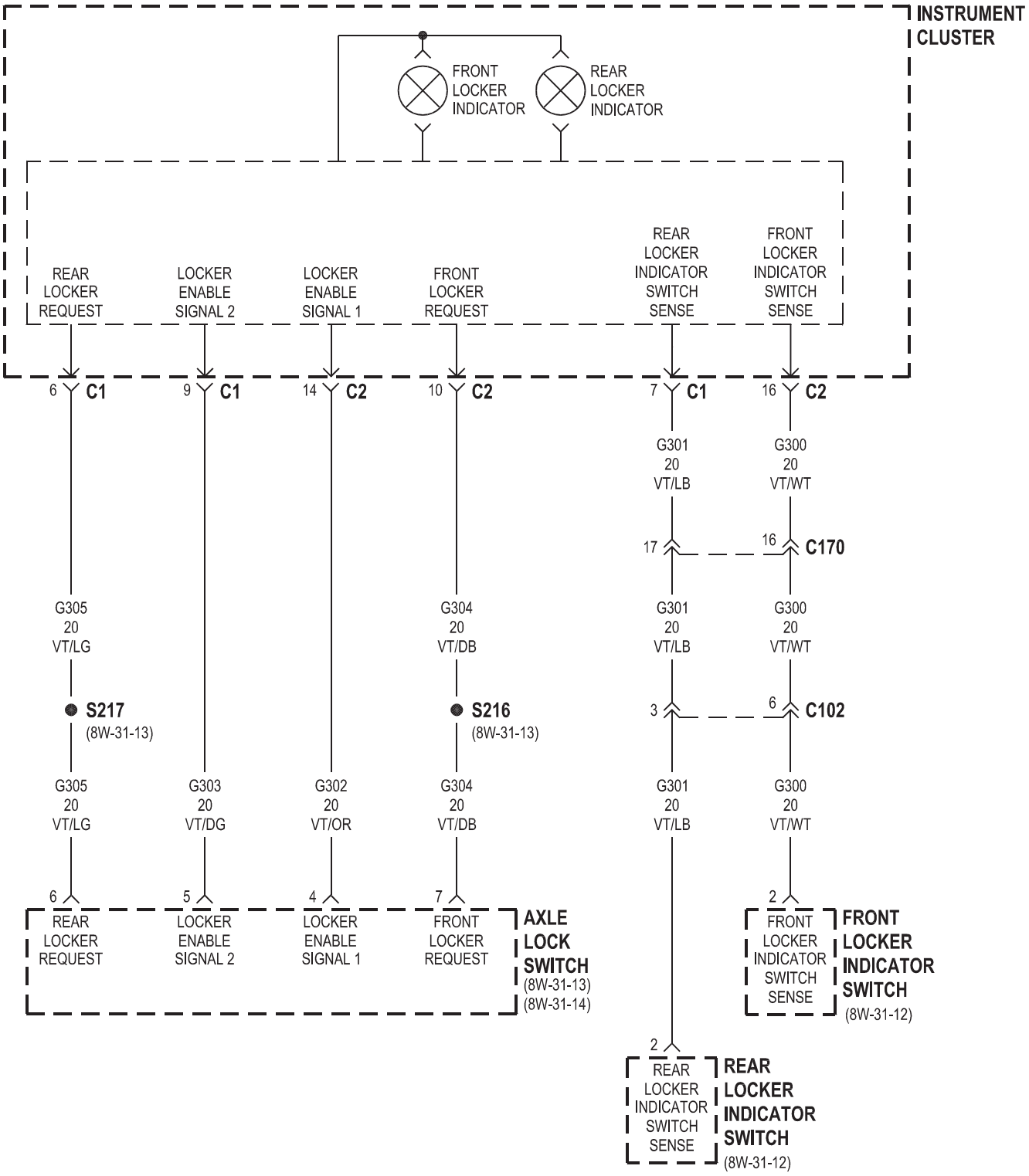

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 26 10A
(8W-10-17)
15
36
FUSE 26 10A
(8W-10-17)
37
M21 20 YL/BR
S173
(8W-10-17)
M21 20 YL/BR
B3
C301
M21 20 YL/BR
D3
M21 20 YL/BR
S204
(8W-10-17)
M21 20 YL/BR
12
C1
RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 10 10A
(8W-11-8)
30
F505 20 PK/YL
B2
C300
F505 20 PK/YL
EXCEPT COMPASS/ TEMPERATURE MIRROR
COMPASS/ TEMPERATURE MIRROR
S215
(8W-11-8)
F505 20 PK/YL
11
C1
INSTRUMENT CLUSTER
FUSED B(+)
FUSED IGNITION SWITCH OUTPUT (RUN-START)
GROUND
8
C1
Z18 18 BK/LB
G201
(8W-15-9)

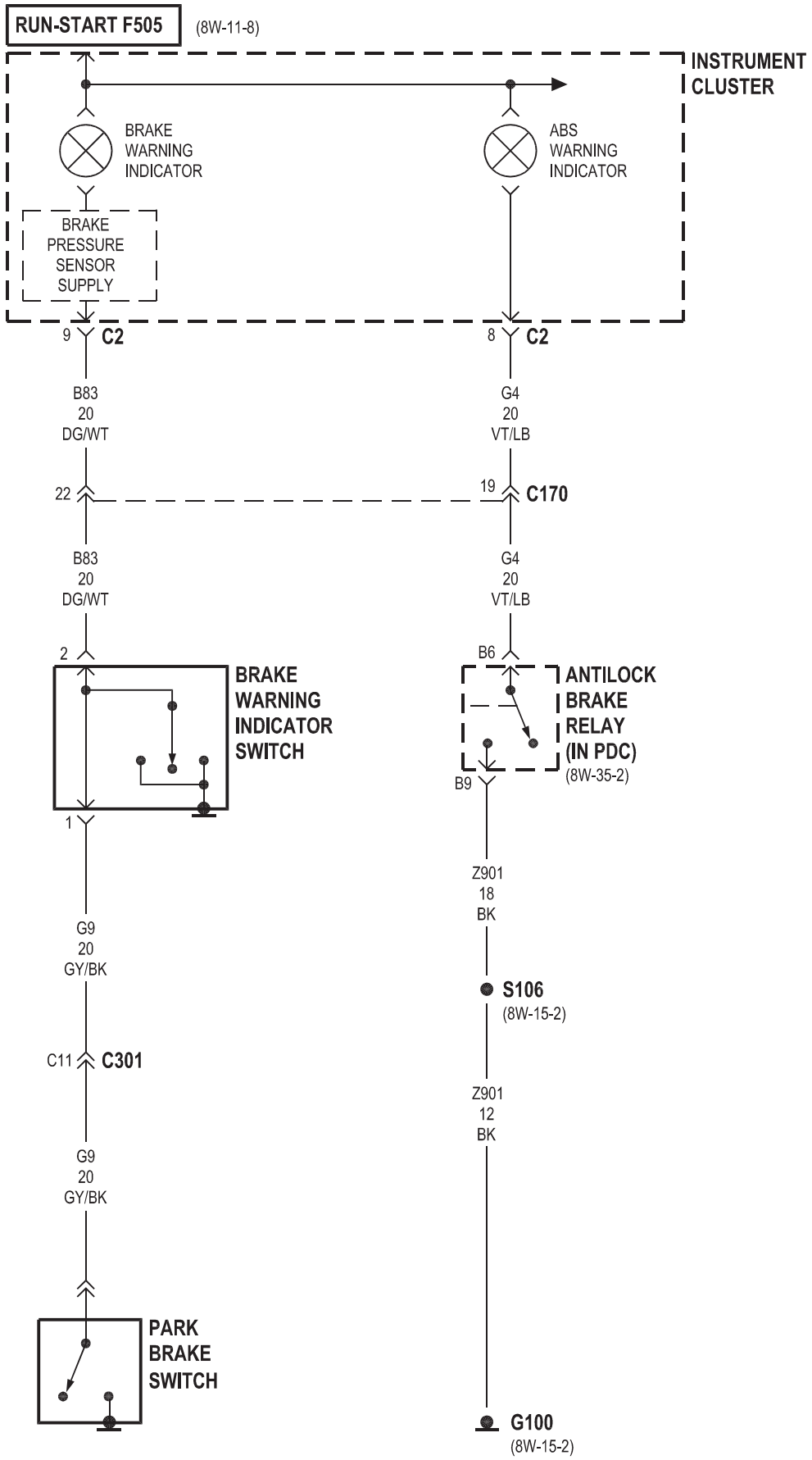

INSTRUMENT CLUSTER
VOLTMETER
SPEEDOMETER
OIL PRESSURE GAUGE
CRUISE CONTROL INDICATOR
VOLTMETER CONTROL
SPEEDOMETER CONTROL
OIL PRESSURE GAUGE CONTROL
CRUISE CONTROL INDICATOR DRIVER
PCI BUS
FUEL GAUGE CONTROL
ENGINE COOLANT TEMPERATURE CONTROL
TACHOMETER CONTROL
FUEL GAUGE
ENGINE COOLANT TEMPERATURE GAUGE
TACHOMETER
12 C2
2 DATA LINK CONNECTOR (8W-18-2)
D25 20 WT/VT
S209 (8W-18-2)
C12 C300
S355 (8W-18-2)
D7 C301
6 C103
D25 18 WT/VT
38 C1
PCI BUS
VEHICLE SPEED SENSOR (8W-30-10) (8W-30-15)
VEHICLE SPEED SENSOR SIGNAL
3
K75 18 DB/OR
S135 (8W-30-10) (8W-30-15)
13 C1
VEHICLE SPEED SENSOR SIGNAL
ENGINE OIL PRESSURE SWITCH (2.4L) (8W-30-9)
OIL PRESSURE SIGNAL
1
G6 20 VT/GY
10 C182
G6 18 VT/GY
20 C1
OIL PRESSURE SIGNAL
FUEL PUMP MODULE (8W-30-3)
FUEL LEVEL SENSOR SIGNAL
2
N4 20 DB/WT
8 C104
N4 20 DB/WT
33 C3
FUEL LEVEL SENSOR SIGNAL
ENGINE COOLANT TEMPERATURE SENSOR (8W-30-9) (8W-30-14)
ECT SIGNAL
* 2
** 1
4.0L
2.4L
K2 20 VT/OR
6 C182
K2 18 VT/OR
20 C2
ECT SIGNAL
POWERTRAIN CONTROL MODULE (8W-30-3) (8W-30-5) (8W-30-9) (8W-30-10) (8W-30-14) (8W-30-15)

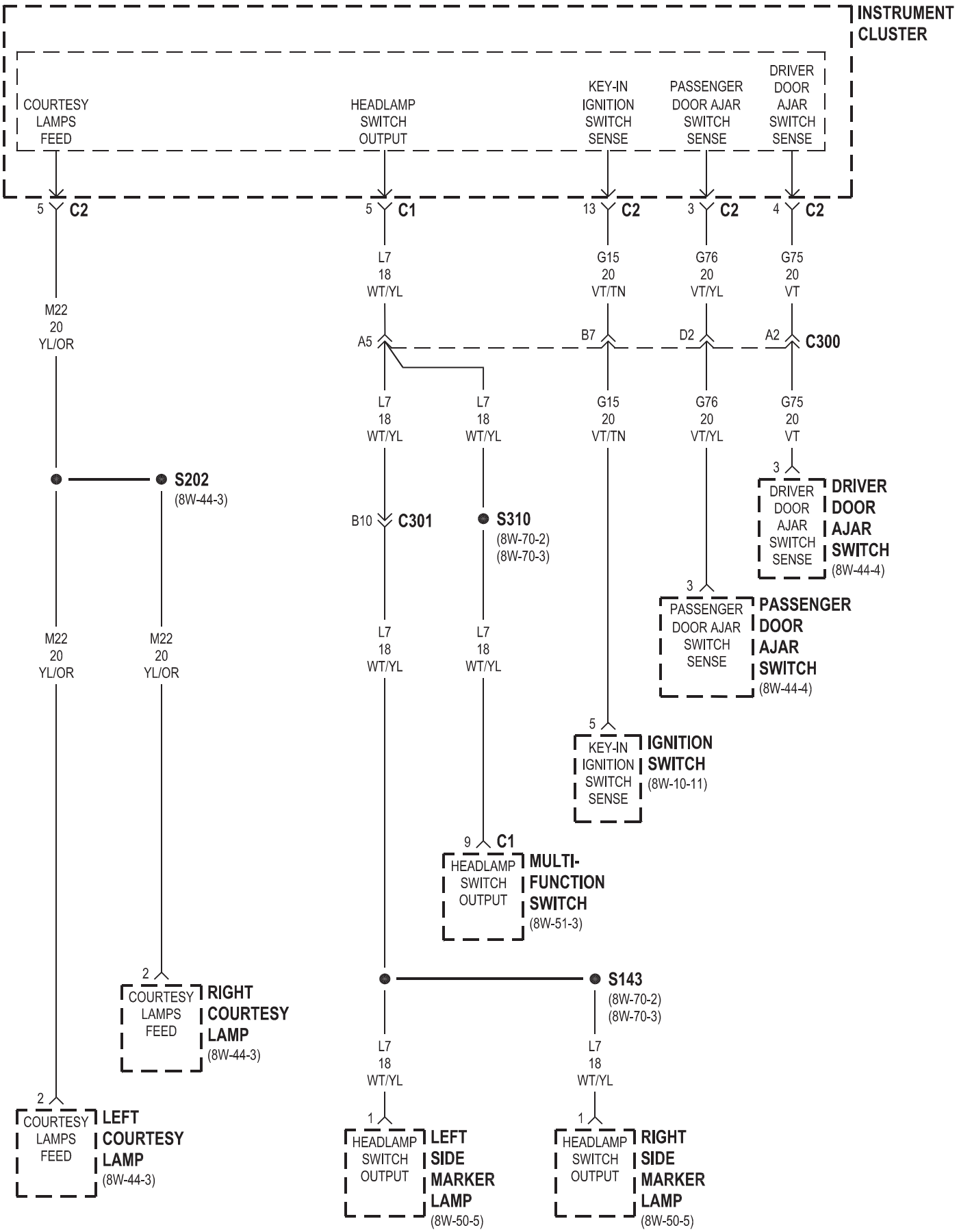
* 2.4L
** 4.0L

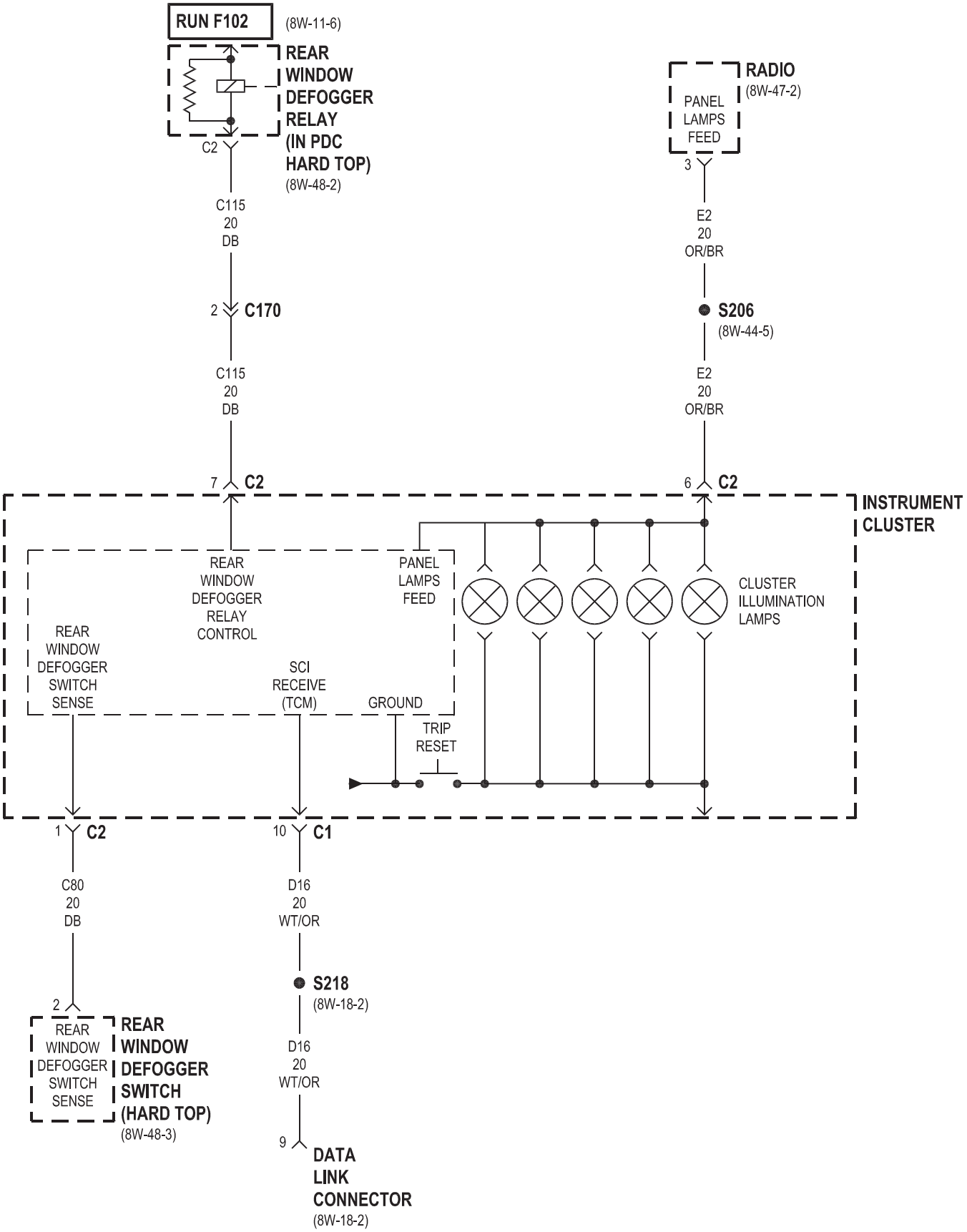

BATT M21
(8W-10-17)
RUN-START F505
(8W-11-8)
INSTRUMENT CLUSTER
SEAT BELT WARNING INDICATOR
AIRBAG WARNING INDICATOR
LOW FUEL INDICATOR
CHECK ENGINE INDICATOR
CHECK GAUGES INDICATOR
FUSED B(+)
SEAT BELT WARNING INDICATOR DRIVER
AIRBAG WARNING INDICATOR DRIVER
FUSED IGNITION SWITCH OUTPUT (RUN-START)
LOW FUEL INDICATOR DRIVER
CHECK ENGINE INDICATOR DRIVER
CHECK GAUGES INDICATOR DRIVER
ODOMETER CONTROL
PANEL LAMPS DIMMER SIGNAL
SEAT BELT SWITCH SENSE
ODOMETER/ TRIP ODOMETER
15 C2
2 C2
E19 20 OR/BR
G10 20 VT/LG
C7
C4 C300
E19 20 OR/BR
G10 20 VT/LG
1
SEAT BELT SWITCH (EXCEPT LHD EXPORT)
2
Z913 20 BK
8 C1
PANEL LAMPS DIMMER SIGNAL
MULTI-FUNCTION SWITCH
(8W-44-5)
S309
(8W-15-10)
Z913 12 BK
G302
(8W-15-10)

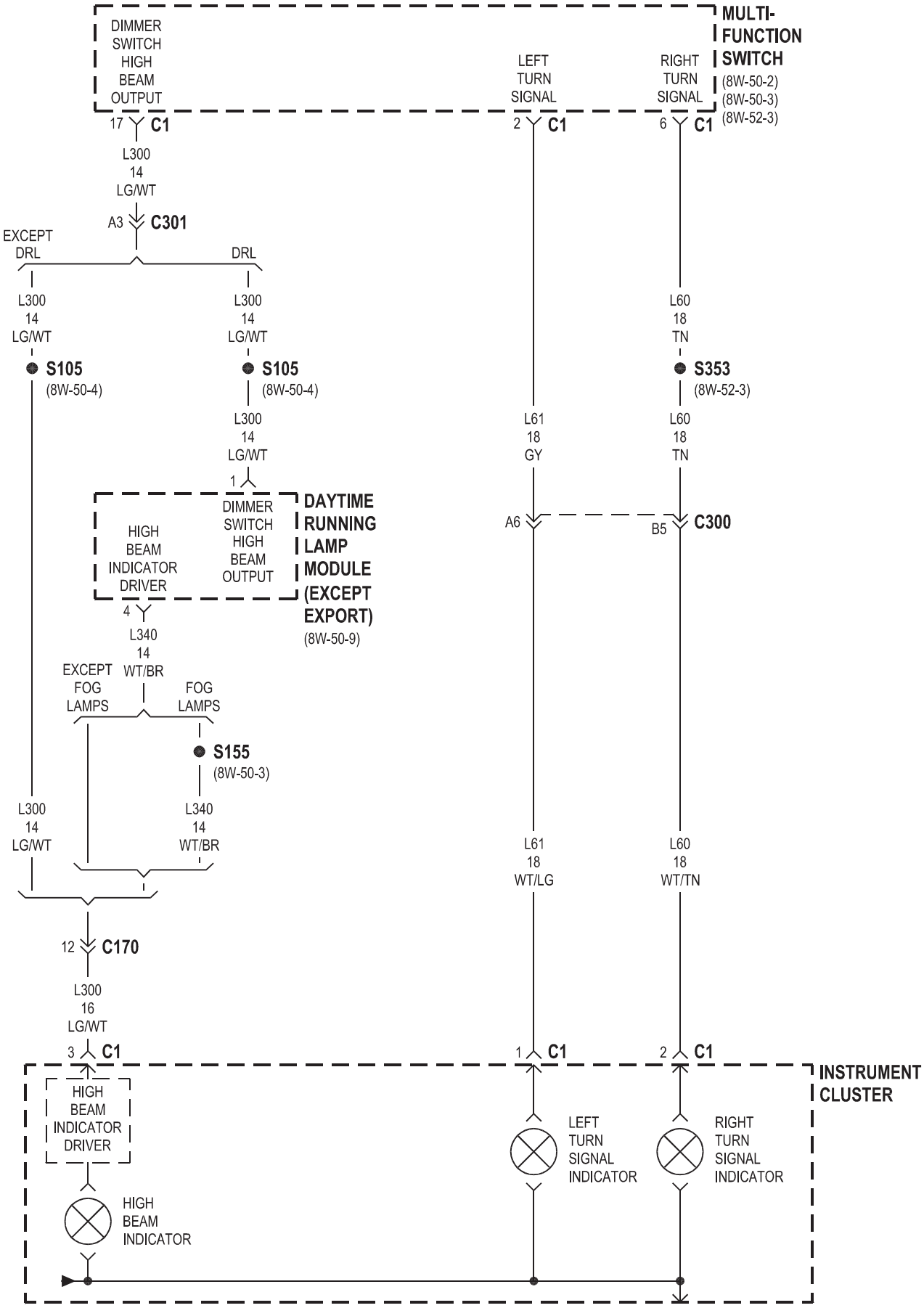

RUN-START F505
(8W-11-8)
INSTRUMENT CLUSTER
SENTRY KEY IMMOBILIZER MODULE INDICATOR
FOG LAMP INDICATOR
UPSHIFT INDICATOR (M/T)
4WD INDICATOR
SENTRY KEY IMMOBILIZER MODULE INDICATOR DRIVER
FOG LAMP INDICATOR DRIVER
FUSED IGNITION SWITCH OUTPUT (RUN-START)
UPSHIFT INDICATOR DRIVER (M/T)
4WD LAMP DRIVER
REAR FOG LAMP FEED
FOG LAMP FEED
4WD INDICATOR
▲▲ 4 C1
▲ 4 C1
11 C2
G93 20 VT/LB
B10 C300
G93 20 VT/LB
B5 C301
G93 20 VT/LB
L36 16 WT/DB
L39 16 WT/OR
B12 C300
20 C170
L36 16 WT/DB
L39 18 WT/OR
OPTION 1
OPTION 2
13 C102
13 C104
G93 20 VT/LB
A
S354
(8W-51-8)
S156
(8W-50-10)
4WD INDICATOR
TRANSFER CASE SWITCH (EXCEPT OFF-ROAD PACKAGE)
(8W-31-2)
L36 16 WT/DB
L36 16 WT/OR
L36 16 WT/OR
12 C1
2
2
REAR FOG LAMP FEED
MULTI-FUNCTION SWITCH
(8W-51-8)
FOG LAMP FEED
LEFT FOG LAMP
(8W-50-10)
FOG LAMP FEED
RIGHT FOG LAMP
(8W-50-10)
▲ EXCEPT EXPORT
▲▲ EXPORT


INSTRUMENT CLUSTER
FRONT LOCKER INDICATOR
REAR LOCKER INDICATOR
REAR LOCKER REQUEST
LOCKER ENABLE SIGNAL 2
LOCKER ENABLE SIGNAL 1
FRONT LOCKER REQUEST
REAR LOCKER INDICATOR SWITCH SENSE
FRONT LOCKER INDICATOR SWITCH SENSE
6 C1
9 C1
14 C2
10 C2
7 C1
16 C2
G305 20 VT/LG
G304 20 VT/DB
G301 20 VT/LB
G300 20 VT/WT
17
16 C170
3
6 C102
S217 (8W-31-13)
S216 (8W-31-13)
G305 20 VT/LG
G303 20 VT/DG
G302 20 VT/OR
G304 20 VT/DB
G301 20 VT/LB
G300 20 VT/WT
6
5
4
7
2
REAR LOCKER REQUEST
LOCKER ENABLE SIGNAL 2
LOCKER ENABLE SIGNAL 1
FRONT LOCKER REQUEST
AXLE LOCK SWITCH (8W-31-13) (8W-31-14)
FRONT LOCKER INDICATOR SWITCH SENSE
FRONT LOCKER INDICATOR SWITCH (8W-31-12)
2
REAR LOCKER INDICATOR SWITCH SENSE
REAR LOCKER INDICATOR SWITCH (8W-31-12)


RUN-START F505
(8W-11-8)
INSTRUMENT CLUSTER
BRAKE WARNING INDICATOR
ABS WARNING INDICATOR
BRAKE PRESSURE SENSOR SUPPLY
9
C2
8
C2
B83
20
DG/WT
G4
20
VT/LB
22
19
C170
B83
20
DG/WT
G4
20
VT/LB
2
B6
BRAKE WARNING INDICATOR SWITCH
ANTILOCK BRAKE RELAY (IN PDC)
(8W-35-2)
B9
1
Z901
18
BK
G9
20
GY/BK
S106
(8W-15-2)
C11
C301
Z901
12
BK
G9
20
GY/BK
PARK BRAKE SWITCH
G100
(8W-15-2)

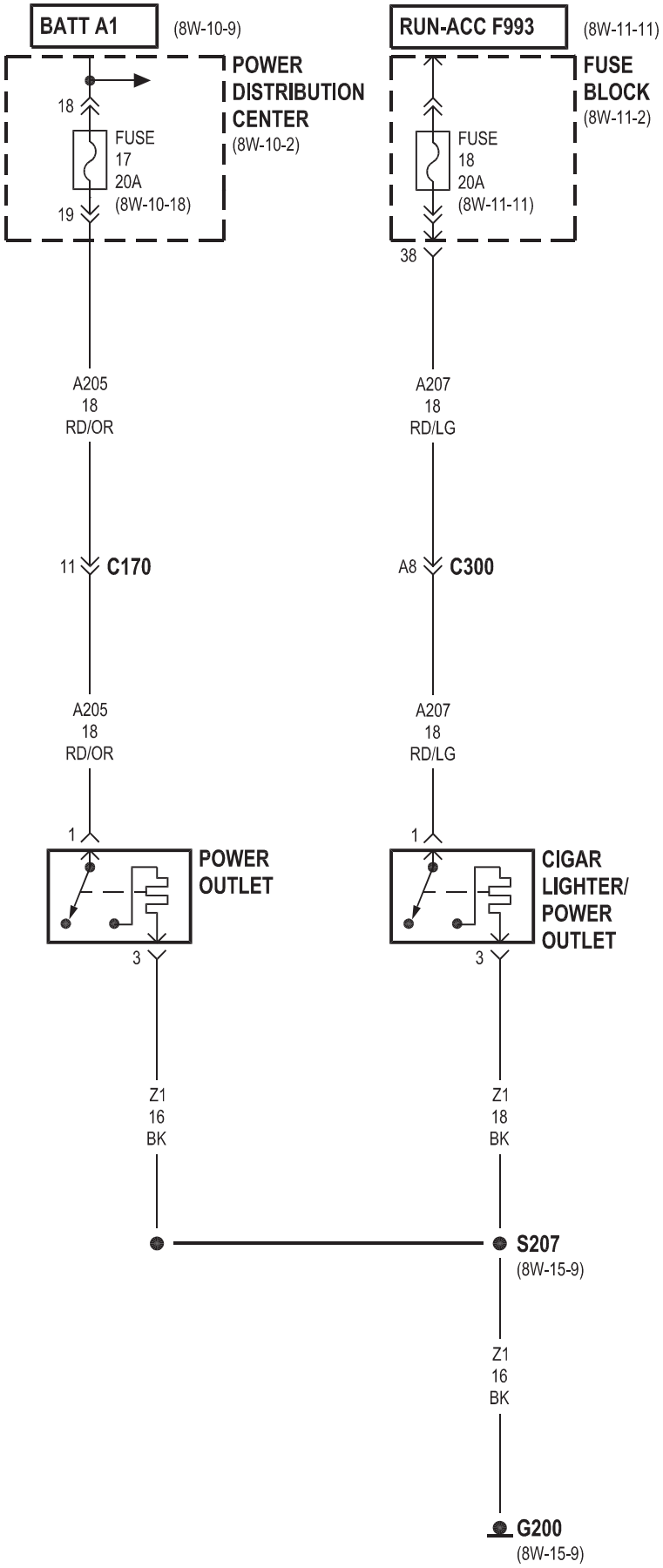

INSTRUMENT CLUSTER
COURTESY LAMPS FEED
HEADLAMP SWITCH OUTPUT
KEY-IN IGNITION SWITCH SENSE
PASSENGER DOOR AJAR SWITCH SENSE
DRIVER DOOR AJAR SWITCH SENSE
5 C2
5 C1
13 C2
3 C2
4 C2
L7 18 WT/YL
G15 20 VT/TN
G76 20 VT/YL
G75 20 VT
M22 20 YL/OR
A5
B7
D2
A2 C300
S202 (8W-44-3)
B10 C301
S310 (8W-70-2) (8W-70-3)
3 DRIVER DOOR AJAR SWITCH SENSE DRIVER DOOR AJAR SWITCH (8W-44-4)
3 PASSENGER DOOR AJAR SWITCH SENSE PASSENGER DOOR AJAR SWITCH (8W-44-4)
5 KEY-IN IGNITION SWITCH SENSE IGNITION SWITCH (8W-10-11)
9 C1 HEADLAMP SWITCH OUTPUT MULTI-FUNCTION SWITCH (8W-51-3)
2 COURTESY LAMPS FEED RIGHT COURTESY LAMP (8W-44-3)
S143 (8W-70-2) (8W-70-3)
2 COURTESY LAMPS FEED LEFT COURTESY LAMP (8W-44-3)
1 HEADLAMP SWITCH OUTPUT LEFT SIDE MARKER LAMP (8W-50-5)
1 HEADLAMP SWITCH OUTPUT RIGHT SIDE MARKER LAMP (8W-50-5)

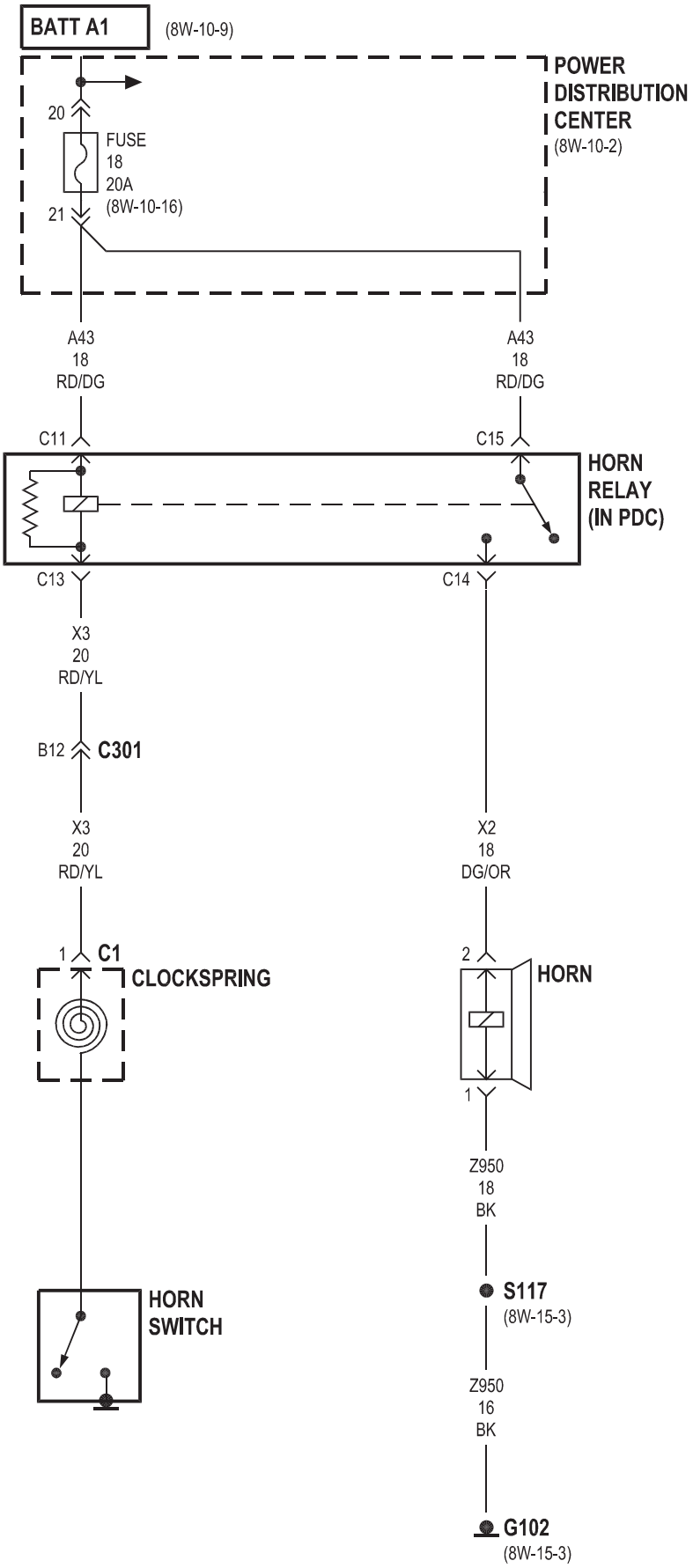

RUN F102
(8W-11-6)
REAR WINDOW DEFOGGER RELAY (IN PDC HARD TOP)
(8W-48-2)
C2
C115
20
DB
2
C170
C115
20
DB
7
C2
RADIO
(8W-47-2)
PANEL LAMPS FEED
3
E2
20
OR/BR
S206
(8W-44-5)
E2
20
OR/BR
6
C2
INSTRUMENT CLUSTER
REAR WINDOW DEFOGGER RELAY CONTROL
PANEL LAMPS FEED
CLUSTER ILLUMINATION LAMPS
REAR WINDOW DEFOGGER SWITCH SENSE
SCI RECEIVE (TCM)
GROUND
TRIP RESET
1
C2
10
C1
C80
20
DB
D16
20
WT/OR
S218
(8W-18-2)
2
REAR WINDOW DEFOGGER SWITCH SENSE
REAR WINDOW DEFOGGER SWITCH (HARD TOP)
(8W-48-3)
D16
20
WT/OR
9
DATA LINK CONNECTOR
(8W-18-2)


MULTI-FUNCTION SWITCH
(8W-50-2)
(8W-50-3)
(8W-52-3)
DIMMER SWITCH HIGH BEAM OUTPUT
LEFT TURN SIGNAL
RIGHT TURN SIGNAL
17 C1
2 C1
6 C1
L300 14 LG/WT
A3 C301
EXCEPT DRL
DRL
L300 14 LG/WT
L300 14 LG/WT
L60 18 TN
S105
(8W-50-4)
S105
(8W-50-4)
S353
(8W-52-3)
L300 14 LG/WT
L61 18 GY
L60 18 TN
1
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
HIGH BEAM INDICATOR DRIVER
DIMMER SWITCH HIGH BEAM OUTPUT
A6 C300
B5 C300
4
L340 14 WT/BR
EXCEPT FOG LAMPS
FOG LAMPS
S155
(8W-50-3)
L300 14 LG/WT
L340 14 WT/BR
L61 18 WT/LG
L60 18 WT/TN
12 C170
L300 16 LG/WT
3 C1
1 C1
2 C1
INSTRUMENT CLUSTER
HIGH BEAM INDICATOR DRIVER
LEFT TURN SIGNAL INDICATOR
RIGHT TURN SIGNAL INDICATOR
HIGH BEAM INDICATOR

8W-41 HORN/CIGAR LIGHTER/POWER OUTLET

Component	Page
Cigar Lighter/Power Outlet	8W-41-2
Clockspring	8W-41-3
Fuse 17	8W-41-2
Fuse 18	8W-41-2, 3
Fuse Block	8W-41-2
G102	8W-41-3
G200	8W-41-2
Horn	8W-41-3
Horn Relay	8W-41-3
Horn Switch	8W-41-3
Power Distribution Center	8W-41-2, 3
Power Outlet	8W-41-2

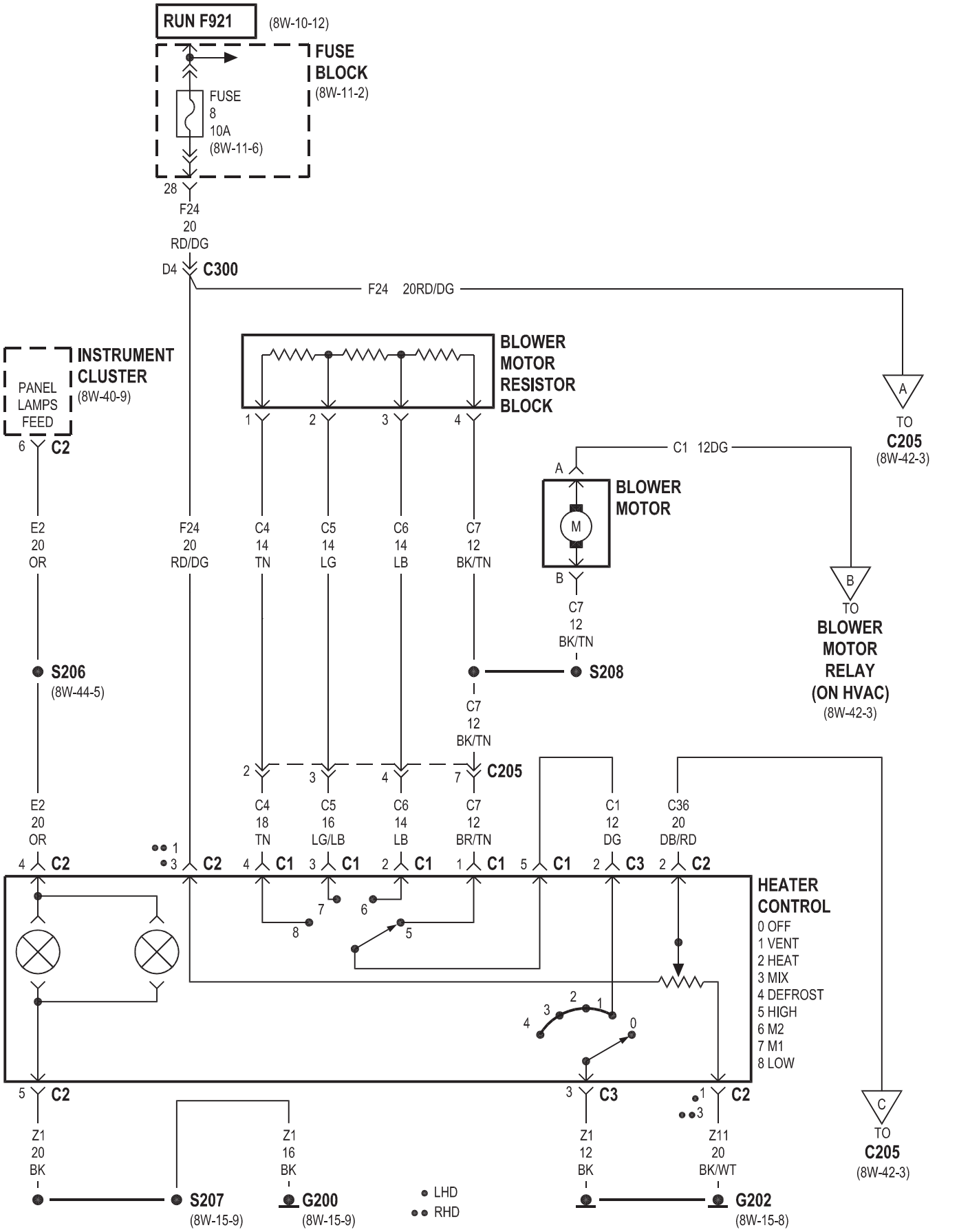

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
18
FUSE 17 20A
(8W-10-18)
19
RUN-ACC F993
(8W-11-11)
FUSE BLOCK
(8W-11-2)
FUSE 18 20A
(8W-11-11)
38
A205 18 RD/OR
A207 18 RD/LG
11 C170
A8 C300
A205 18 RD/OR
A207 18 RD/LG
1
POWER OUTLET
CIGAR LIGHTER/ POWER OUTLET
3
Z1 16 BK
Z1 18 BK
S207
(8W-15-9)
Z1 16 BK
G200
(8W-15-9)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
20
FUSE
18
20A
(8W-10-16)
21
A43
18
RD/DG
A43
18
RD/DG
C11
C15
HORN RELAY (IN PDC)
C13
C14
X3
20
RD/YL
B12
C301
X3
20
RD/YL
X2
18
DG/OR
1
C1
CLOCKSPRING
2
HORN
1
Z950
18
BK
HORN SWITCH
S117
(8W-15-3)
Z950
16
BK
G102
(8W-15-3)

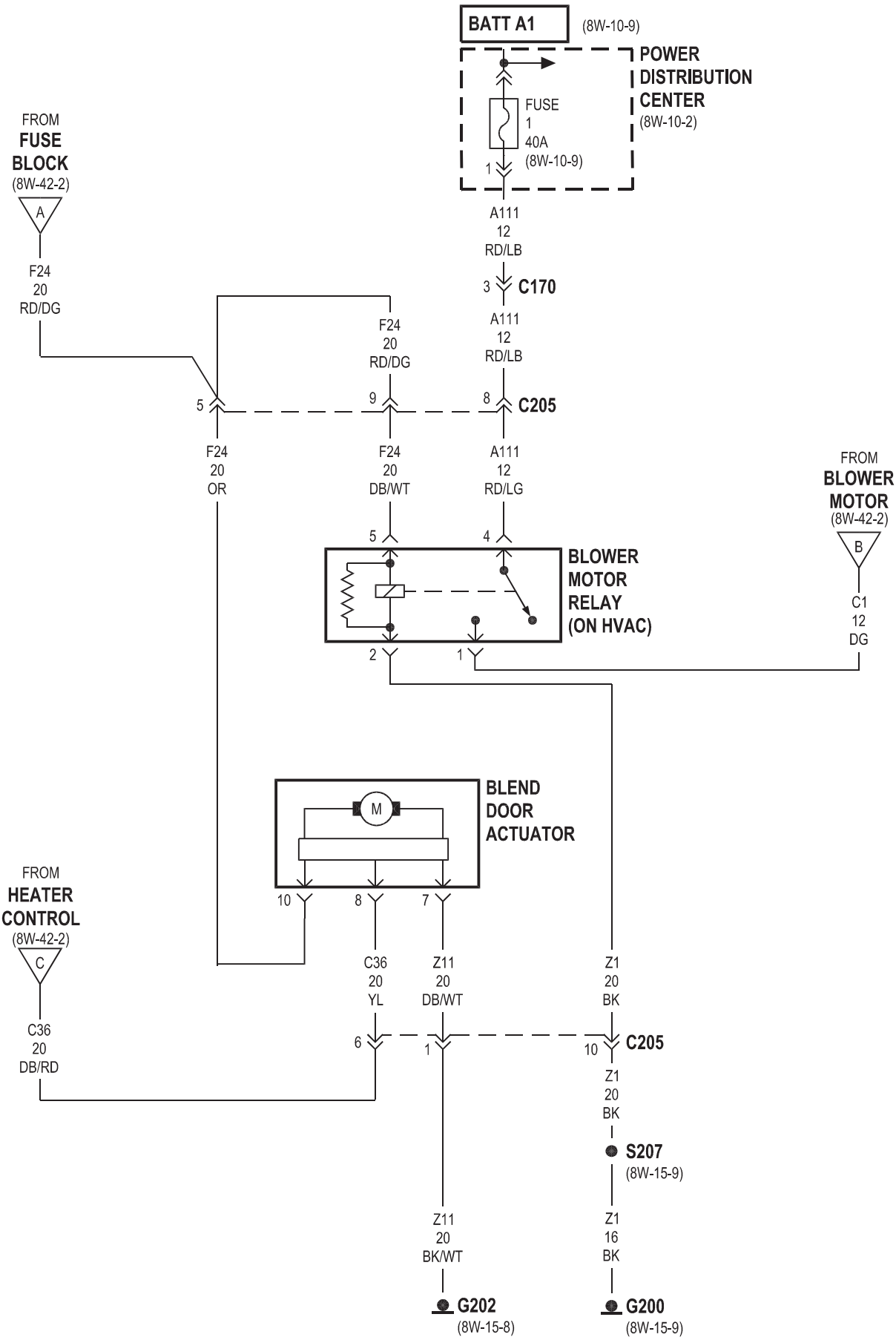
8W-42 AIR CONDITIONING-HEATER

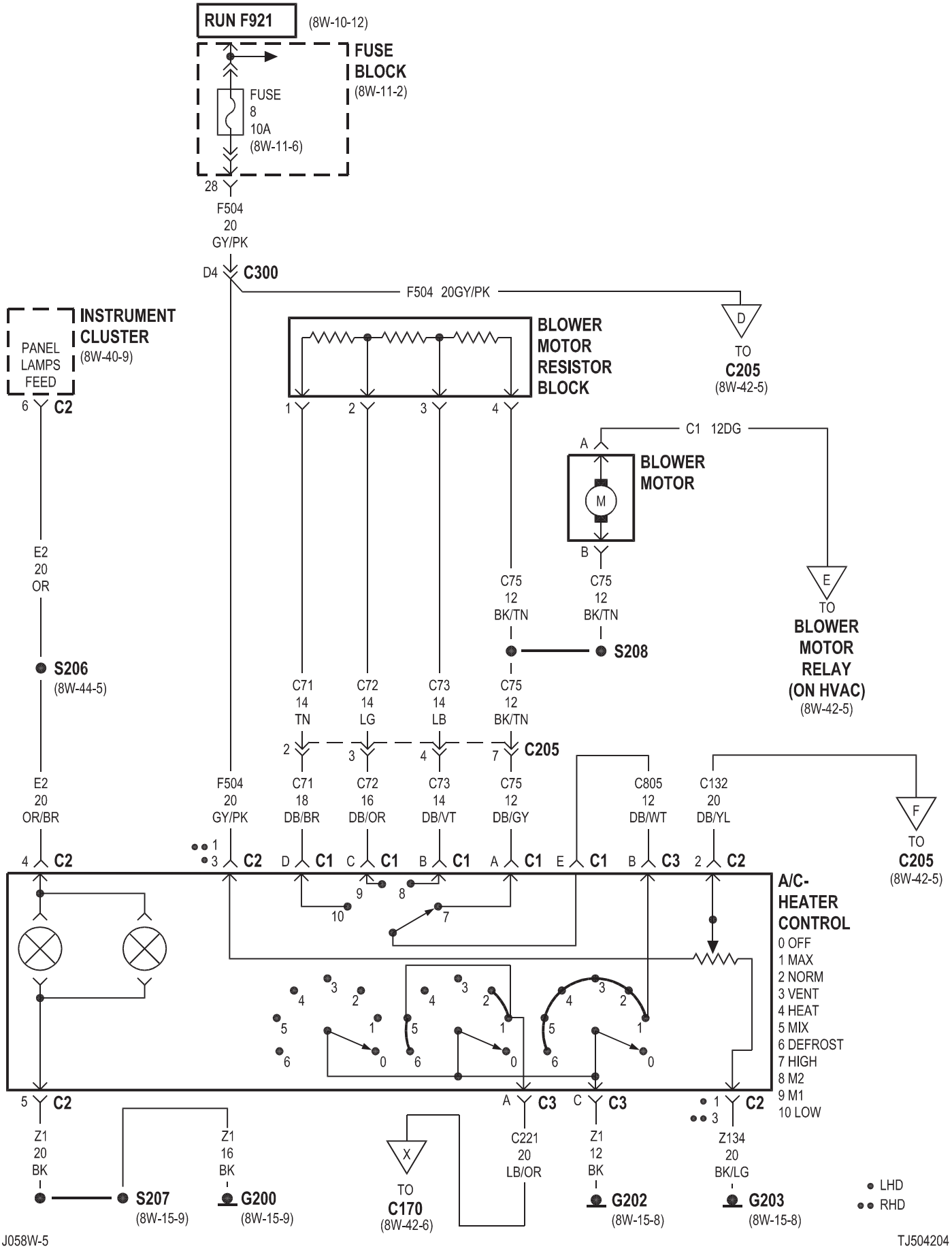
Component	Page
A/C Compressor Clutch	8W-42-7
A/C Compressor Clutch Relay	8W-42-6, 7, 8
A/C High Pressure Switch	8W-42-6
A/C Low Pressure Switch	8W-42-6
A/C Pressure Transducer	8W-42-9
A/C-Heater Control	8W-42-4, 5, 6
Blend Door Actuator	8W-42-3, 5
Blower Motor	8W-42-2, 3, 4, 5
Blower Motor Relay	8W-42-2, 3, 4, 5
Blower Motor Resistor Block	8W-42-2, 4
Daytime Running Lamp Module	8W-42-8
Fuse 1	8W-42-3, 5
Fuse 4	8W-42-8
Fuse 8	8W-42-2, 4
Fuse 11	8W-42-6, 8
Fuse 21	8W-42-6
Fuse Block	8W-42-2, 3, 4, 6, 8
G105	8W-42-7
G190	8W-42-8
G200	8W-42-2, 3, 4, 5
G202	8W-42-2, 3, 4
G203	8W-42-4, 5
Heater Control	8W-42-2, 3
High Speed Radiator Fan Relay	8W-42-8
Instrument Cluster	8W-42-2, 4
Low Speed Radiator Fan Relay	8W-42-8
Power Distribution Center	8W-42-3, 5, 6, 8
Powertrain Control Module	8W-42-6, 8, 9
Radiator Fan Motor	8W-42-8
Radiator Fan Motor Circuit Breaker	8W-42-8

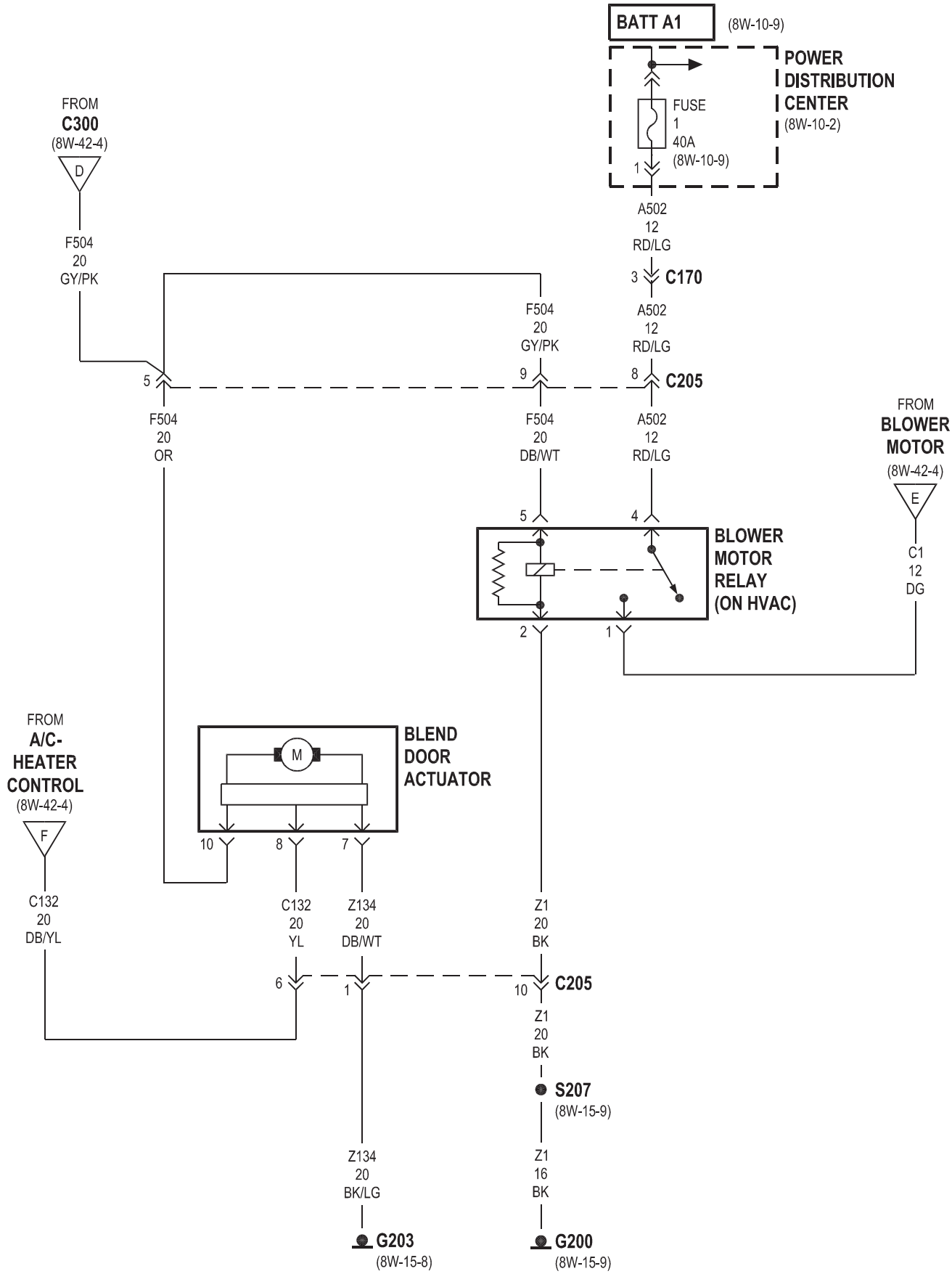
HEATER ONLY

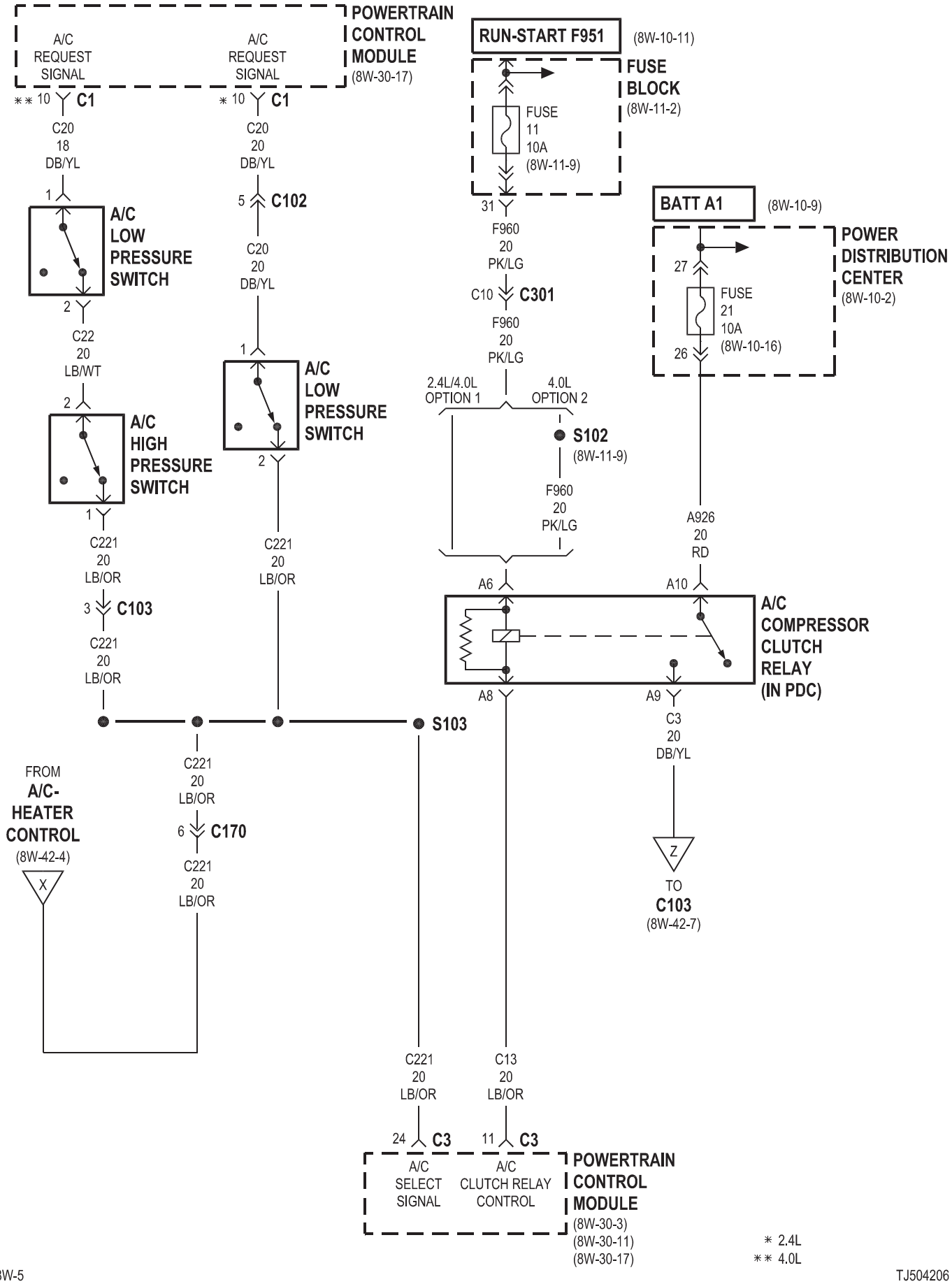

RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 8 10A
(8W-11-6)
28
F24 20 RD/DG
D4 C300
F24 20RD/DG
INSTRUMENT CLUSTER
(8W-40-9)
PANEL LAMPS FEED
6 C2
BLOWER MOTOR RESISTOR BLOCK
1 2 3 4
A
TO C205
(8W-42-3)
C1 12DG
A
BLOWER MOTOR
M
B
C7 12 BK/TN
S208
B
TO BLOWER MOTOR RELAY (ON HVAC)
(8W-42-3)
E2 20 OR
S206
(8W-44-5)
F24 20 RD/DG
C4 14 TN
C5 14 LG
C6 14 LB
C7 12 BK/TN
C7 12 BK/TN
2 3 4 7 C205
E2 20 OR
4 C2
1
3 C2
C4 18 TN
C5 16 LG/LB
C6 14 LB
C7 12 BR/TN
4 C1 3 C1 2 C1 1 C1
C1 12 DG
5 C1 2 C3
C36 20 DB/RD
2 C2
HEATER CONTROL
0 OFF
1 VENT
2 HEAT
3 MIX
4 DEFROST
5 HIGH
6 M2
7 M1
8 LOW
5 C2
3 C3
1
3
C2
Z1 20 BK
S207
(8W-15-9)
Z1 16 BK
G200
(8W-15-9)
LHD
RHD
Z1 12 BK
Z11 20 BK/WT
G202
(8W-15-8)
C
TO C205
(8W-42-3)

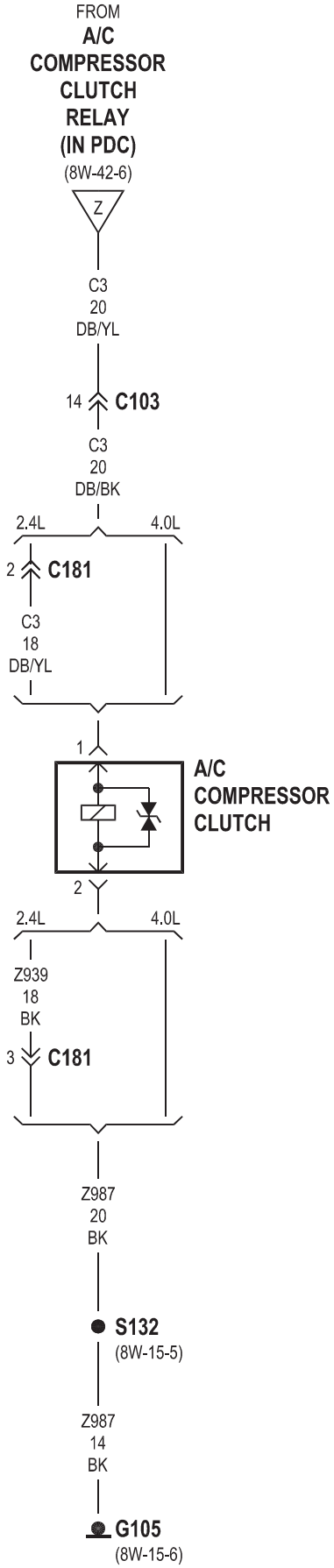
HEATER ONLY

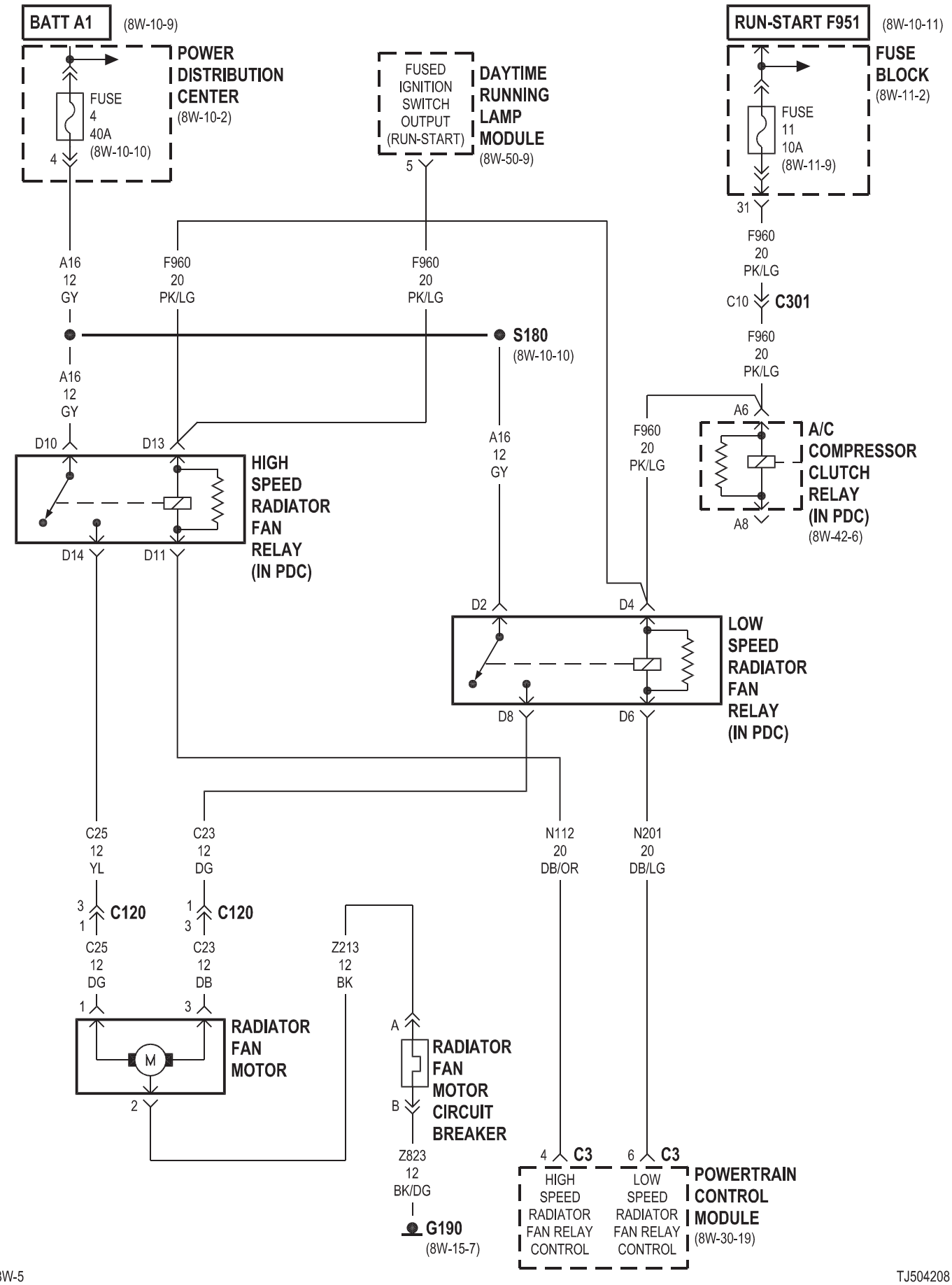

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 1 40A
(8W-10-9)
1
A111 12 RD/LB
3
C170
FROM FUSE BLOCK
(8W-42-2)
A
F24 20 RD/DG
F24 20 RD/DG
A111 12 RD/LB
5
9
8
C205
F24 20 OR
F24 20 DB/WT
A111 12 RD/LG
FROM BLOWER MOTOR
(8W-42-2)
B
C1 12 DG
5
4
BLOWER MOTOR RELAY (ON HVAC)
2
1
M
BLEND DOOR ACTUATOR
10
8
7
FROM HEATER CONTROL
(8W-42-2)
C
C36 20 YL
Z11 20 DB/WT
Z1 20 BK
C36 20 DB/RD
6
1
10
C205
Z1 20 BK
S207
(8W-15-9)
Z11 20 BK/WT
Z1 16 BK
G202
(8W-15-8)
G200
(8W-15-9)

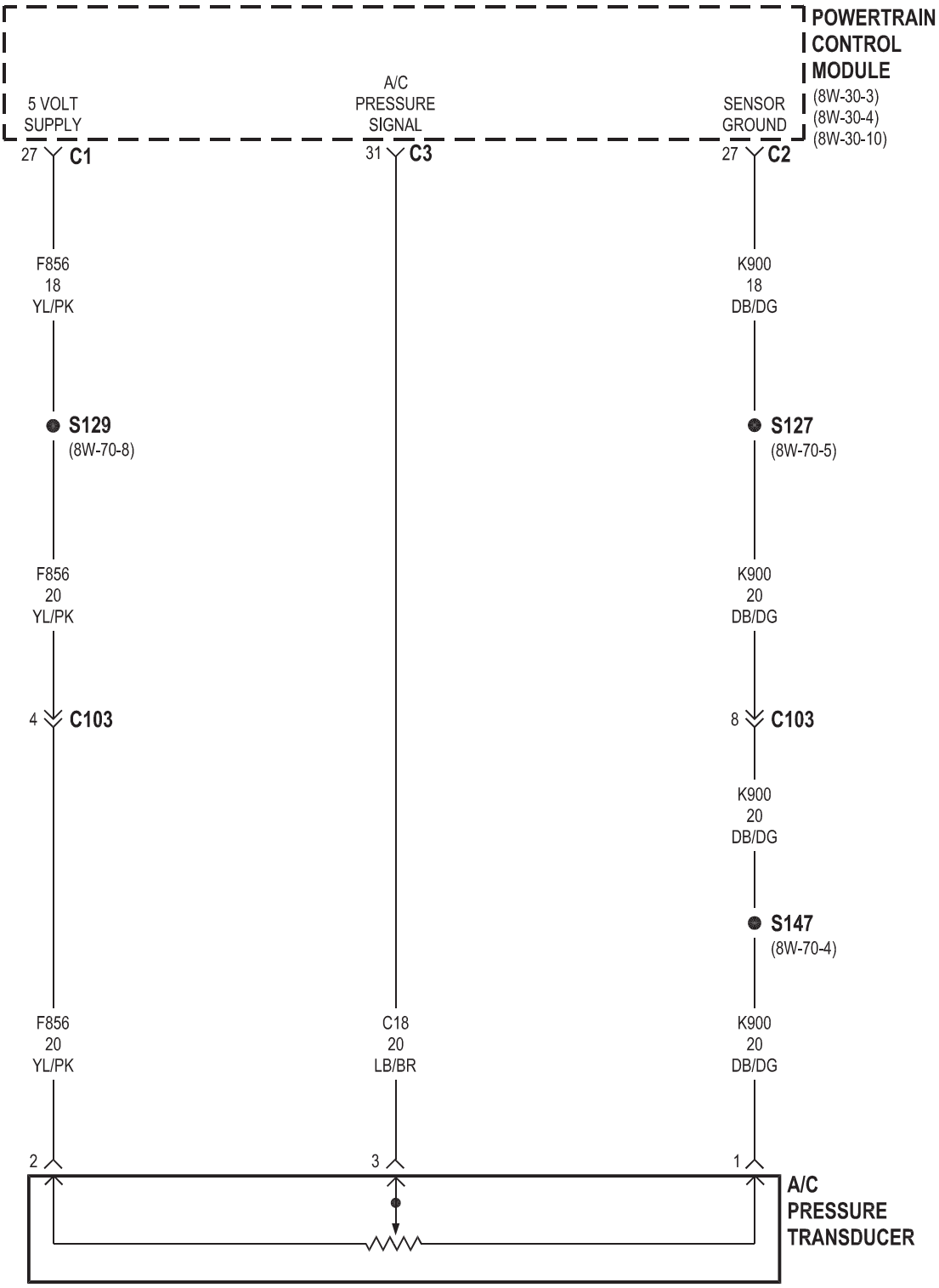

RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 8 10A
(8W-11-6)
28
F504 20 GY/PK
D4 C300
F504 20GY/PK
D
TO C205
(8W-42-5)
INSTRUMENT CLUSTER
(8W-40-9)
PANEL LAMPS FEED
6 C2
BLOWER MOTOR RESISTOR BLOCK
1 2 3 4
C1 12DG
A
BLOWER MOTOR
M
B
E2 20 OR
S206
(8W-44-5)
C75 12 BK/TN
C75 12 BK/TN
S208
E
TO BLOWER MOTOR RELAY (ON HVAC)
(8W-42-5)
C71 14 TN
C72 14 LG
C73 14 LB
C75 12 BK/TN
2 3 4 7 C205
E2 20 OR/BR
F504 20 GY/PK
C71 18 DB/BR
C72 16 DB/OR
C73 14 DB/VT
C75 12 DB/GY
C805 12 DB/WT
C132 20 DB/YL
F
TO C205
(8W-42-5)
4 C2
1 3 C2
D C1
C C1
B C1
A C1
E C1
B C3
2 C2
A/C-HEATER CONTROL
0 OFF
1 MAX
2 NORM
3 VENT
4 HEAT
5 MIX
6 DEFROST
7 HIGH
8 M2
9 M1
10 LOW
5 C2
A C3
C C3
1 3 C2
Z1 20 BK
S207
(8W-15-9)
Z1 16 BK
G200
(8W-15-9)
X
TO C170
(8W-42-6)
C221 20 LB/OR
Z1 12 BK
G202
(8W-15-8)
Z134 20 BK/LG
G203
(8W-15-8)
LHD
RHD


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 1 40A
(8W-10-9)
FROM C300
(8W-42-4)
D
F504 20 GY/PK
A502 12 RD/LG
C170
F504 20 GY/PK
A502 12 RD/LG
C205
F504 20 OR
F504 20 DB/WT
A502 12 RD/LG
FROM BLOWER MOTOR
(8W-42-4)
E
C1 12 DG
BLOWER MOTOR RELAY (ON HVAC)
FROM A/C-HEATER CONTROL
(8W-42-4)
F
BLEND DOOR ACTUATOR
M
C132 20 DB/YL
C132 20 YL
Z134 20 DB/WT
Z1 20 BK
C205
Z1 20 BK
S207
(8W-15-9)
Z134 20 BK/LG
Z1 16 BK
G203
(8W-15-8)
G200
(8W-15-9)


POWERTRAIN CONTROL MODULE
(8W-30-17)
A/C REQUEST SIGNAL
** 10 C1
C20 18 DB/YL
A/C REQUEST SIGNAL
* 10 C1
C20 20 DB/YL
RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 11 10A
(8W-11-9)
31
F960 20 PK/LG
C10 C301
F960 20 PK/LG
2.4L/4.0L OPTION 1
4.0L OPTION 2
S102
(8W-11-9)
F960 20 PK/LG
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
27
FUSE 21 10A
(8W-10-16)
26
A926 20 RD
A10
A/C LOW PRESSURE SWITCH
1
2
C22 20 LB/WT
A/C HIGH PRESSURE SWITCH
1
C221 20 LB/OR
3 C103
C221 20 LB/OR
5 C102
C20 20 DB/YL
A/C LOW PRESSURE SWITCH
1
2
C221 20 LB/OR
A6
A8
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
A9
C3 20 DB/YL
Z
TO C103
(8W-42-7)
S103
FROM A/C-HEATER CONTROL
(8W-42-4)
X
C221 20 LB/OR
6 C170
C221 20 LB/OR
C221 20 LB/OR
C13 20 LB/OR
24 C3
11 C3
A/C SELECT SIGNAL
A/C CLUTCH RELAY CONTROL
POWERTRAIN CONTROL MODULE
(8W-30-3)
(8W-30-11)
(8W-30-17)
* 2.4L
** 4.0L

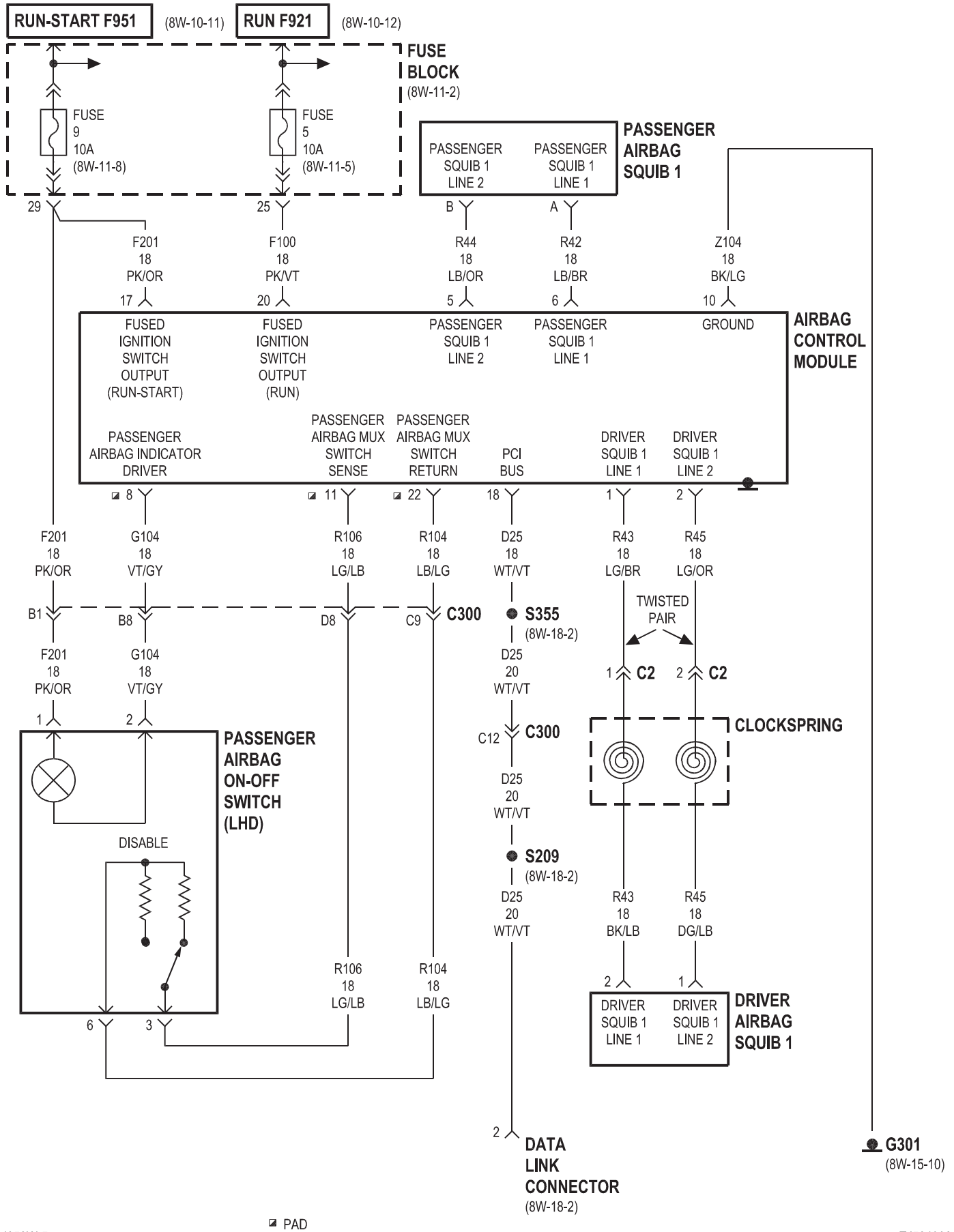

FROM
A/C
COMPRESSOR
CLUTCH
RELAY
(IN PDC)
(8W-42-6)
Z
C3
20
DB/YL
14
C103
C3
20
DB/BK
2.4L
4.0L
2
C181
C3
18
DB/YL
1
A/C
COMPRESSOR
CLUTCH
2
2.4L
4.0L
Z939
18
BK
3
C181
Z987
20
BK
S132
(8W-15-5)
Z987
14
BK
G105
(8W-15-6)


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 4 40A
(8W-10-10)
4
FUSED IGNITION SWITCH OUTPUT (RUN-START)
5
DAYTIME RUNNING LAMP MODULE
(8W-50-9)
RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 11 10A
(8W-11-9)
31
F960 20 PK/LG
C10 C301
F960 20 PK/LG
A6
A8
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
(8W-42-6)
A16 12 GY
F960 20 PK/LG
F960 20 PK/LG
S180
(8W-10-10)
A16 12 GY
D10
D13
HIGH SPEED RADIATOR FAN RELAY (IN PDC)
D14
D11
A16 12 GY
F960 20 PK/LG
D2
D4
LOW SPEED RADIATOR FAN RELAY (IN PDC)
D8
D6
C25 12 YL
C23 12 DG
N112 20 DB/OR
N201 20 DB/LG
3 C120 1
1 C120 3
C25 12 DG
C23 12 DB
Z213 12 BK
1
3
RADIATOR FAN MOTOR
M
2
A
B
RADIATOR FAN MOTOR CIRCUIT BREAKER
Z823 12 BK/DG
G190
(8W-15-7)
4 C3
6 C3
HIGH SPEED RADIATOR FAN RELAY CONTROL
LOW SPEED RADIATOR FAN RELAY CONTROL
POWERTRAIN CONTROL MODULE
(8W-30-19)


POWERTRAIN CONTROL MODULE
(8W-30-3)
(8W-30-4)
(8W-30-10)
5 VOLT SUPPLY
A/C PRESSURE SIGNAL
SENSOR GROUND
27 C1
31 C3
27 C2
F856 18 YL/PK
K900 18 DB/DG
S129
(8W-70-8)
S127
(8W-70-5)
F856 20 YL/PK
K900 20 DB/DG
4 C103
8 C103
K900 20 DB/DG
S147
(8W-70-4)
F856 20 YL/PK
C18 20 LB/BR
K900 20 DB/DG
2
3
1
A/C PRESSURE TRANSDUCER

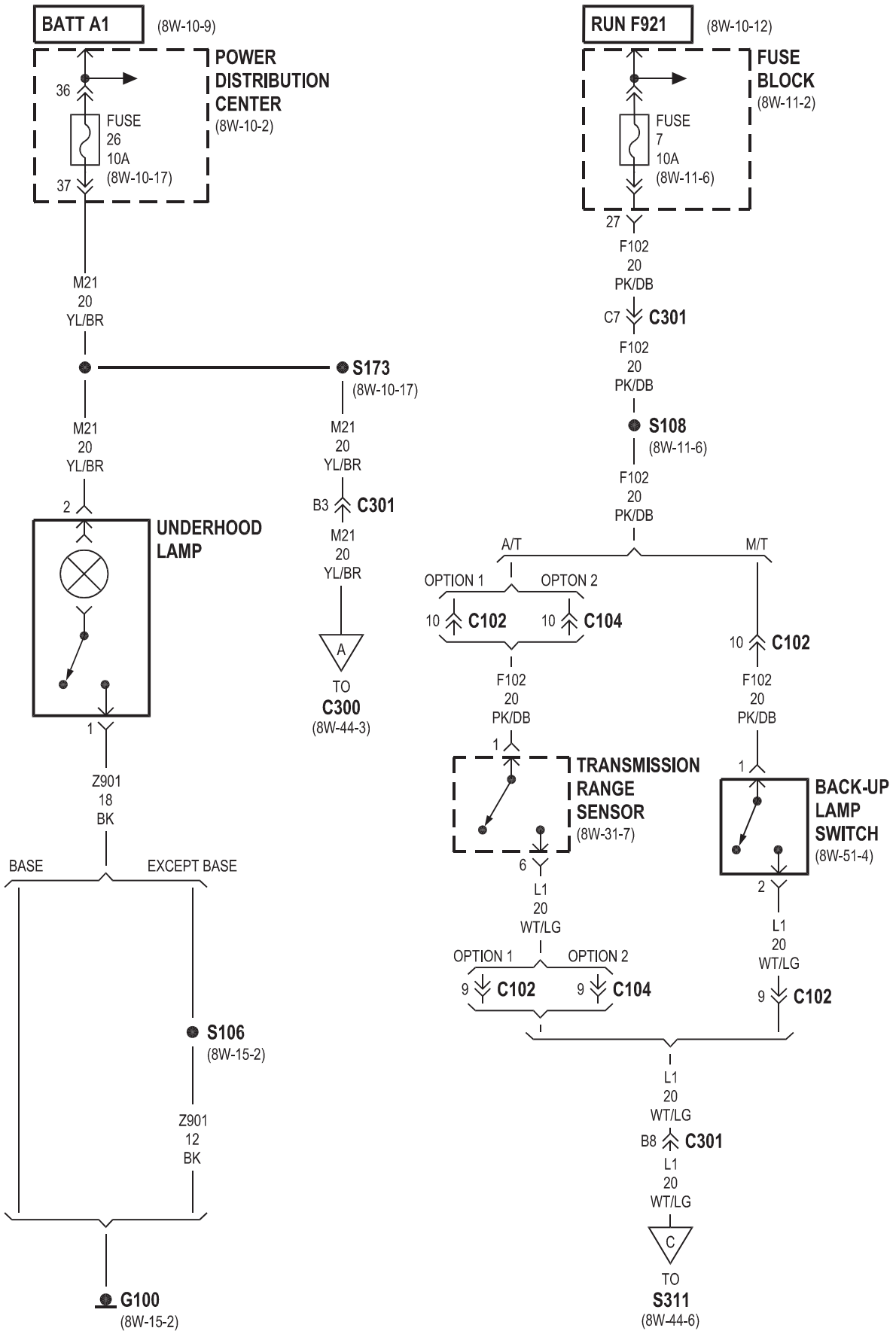
8W-43 AIRBAG SYSTEM

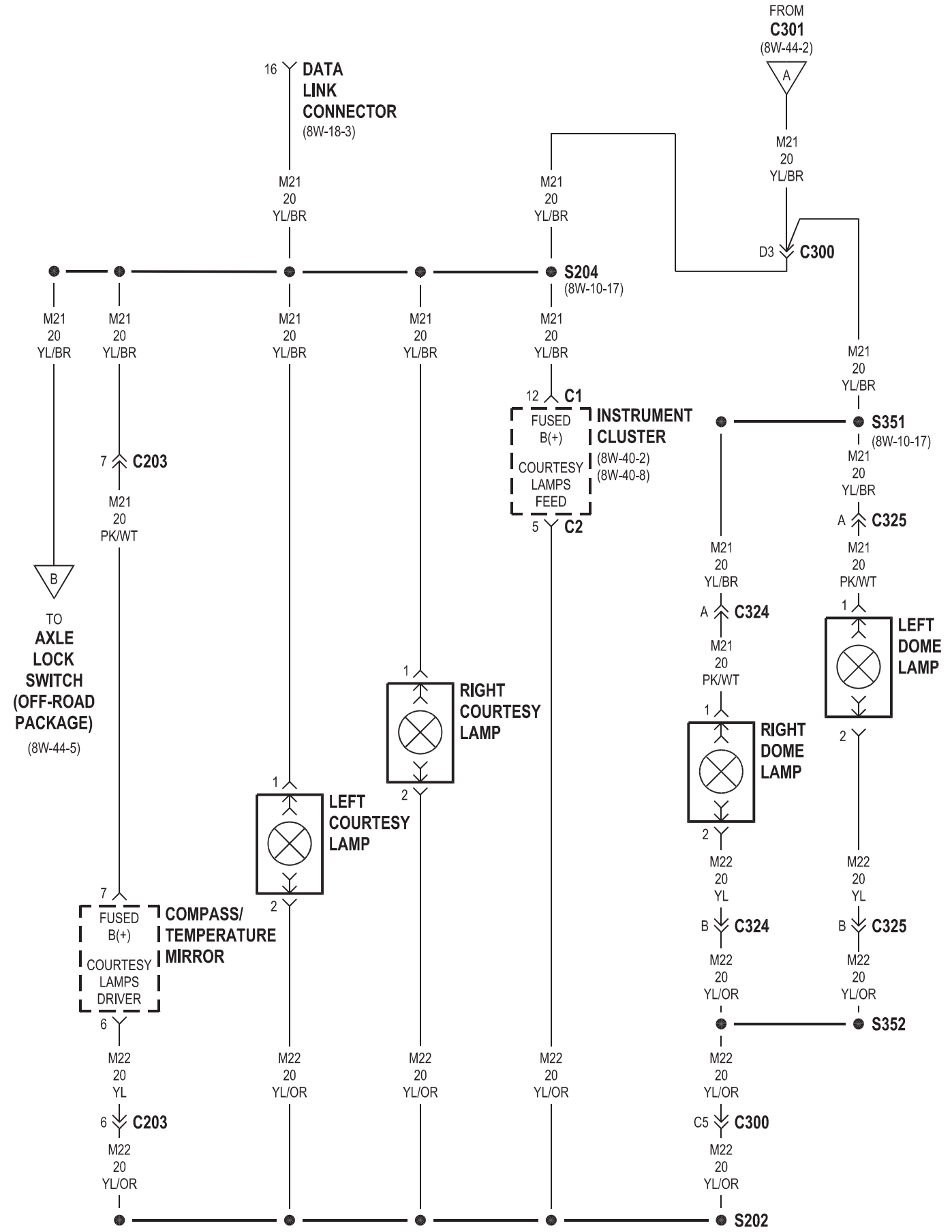
Component	Page	Component	Page
Airbag Control Module	8W-43-2	Fuse 9	8W-43-2
Clockspring	8W-43-2	Fuse Block	8W-43-2
Data Link Connector	8W-43-2	G301	8W-43-2
Driver Airbag Squib 1	8W-43-2	Passenger Airbag On-Off Switch	8W-43-2
Fuse 5	8W-43-2	Passenger Airbag Squib 1	8W-43-2

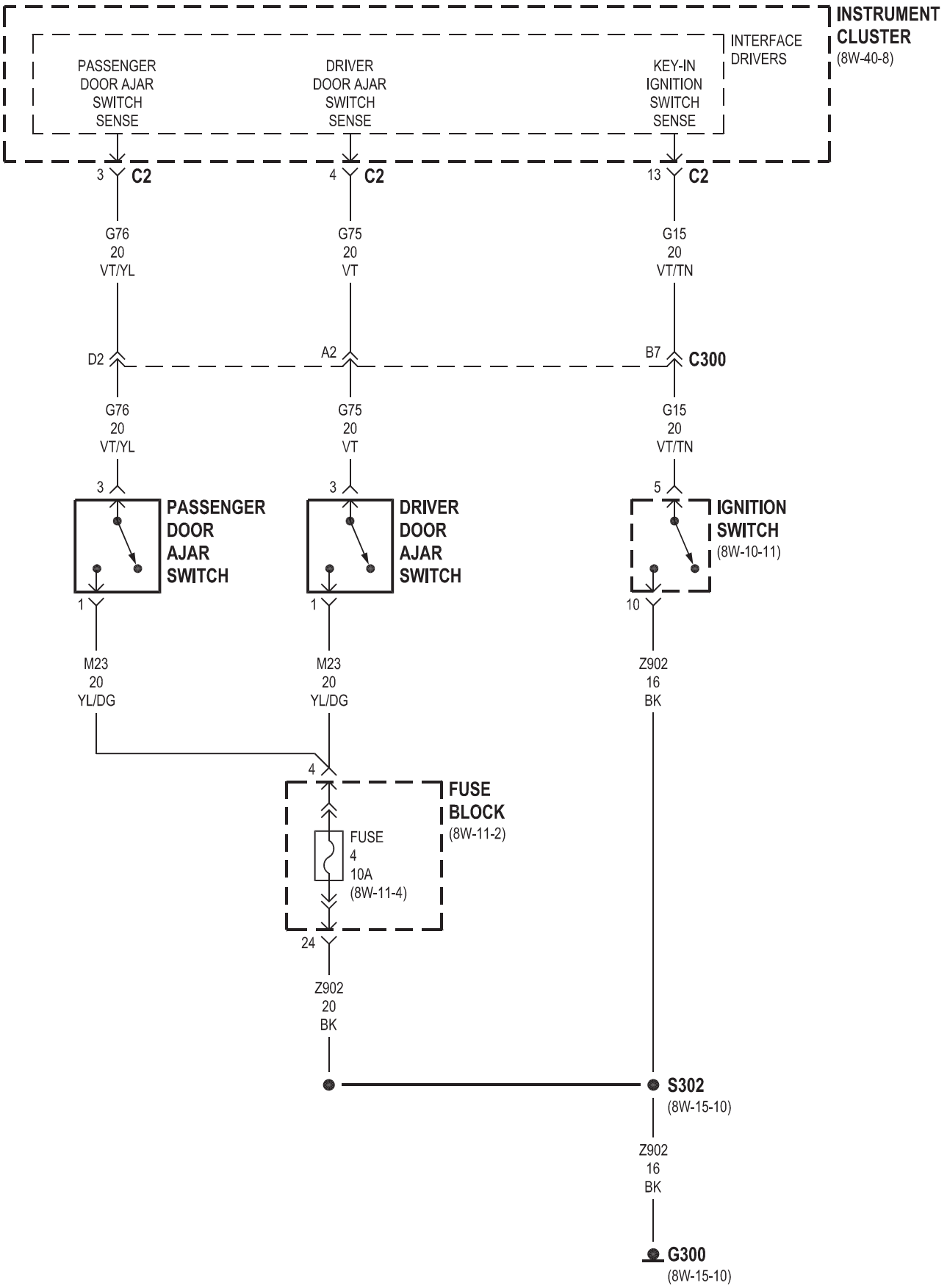

RUN-START F951
(8W-10-11)
RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 9 10A (8W-11-8)
FUSE 5 10A (8W-11-5)
29
25
PASSENGER AIRBAG SQUIB 1
PASSENGER SQUIB 1 LINE 2
PASSENGER SQUIB 1 LINE 1
B
A
F201 18 PK/OR
F100 18 PK/VT
R44 18 LB/OR
R42 18 LB/BR
Z104 18 BK/LG
17
20
5
6
10
AIRBAG CONTROL MODULE
FUSED IGNITION SWITCH OUTPUT (RUN-START)
FUSED IGNITION SWITCH OUTPUT (RUN)
PASSENGER SQUIB 1 LINE 2
PASSENGER SQUIB 1 LINE 1
GROUND
PASSENGER AIRBAG INDICATOR DRIVER
PASSENGER AIRBAG MUX SWITCH SENSE
PASSENGER AIRBAG MUX SWITCH RETURN
PCI BUS
DRIVER SQUIB 1 LINE 1
DRIVER SQUIB 1 LINE 2
8
11
22
18
1
2
F201 18 PK/OR
G104 18 VT/GY
R106 18 LG/LB
R104 18 LB/LG
D25 18 WT/VT
R43 18 LG/BR
R45 18 LG/OR
B1
B8
D8
C9
C300
S355
(8W-18-2)
TWISTED PAIR
D25 20 WT/VT
1 C2
2 C2
C12
C300
CLOCKSPRING
S209
(8W-18-2)
R43 18 BK/LB
R45 18 DG/LB
PASSENGER AIRBAG ON-OFF SWITCH (LHD)
DISABLE
6
3
DRIVER SQUIB 1 LINE 1
DRIVER SQUIB 1 LINE 2
DRIVER AIRBAG SQUIB 1
DATA LINK CONNECTOR
(8W-18-2)
G301
(8W-15-10)
PAD

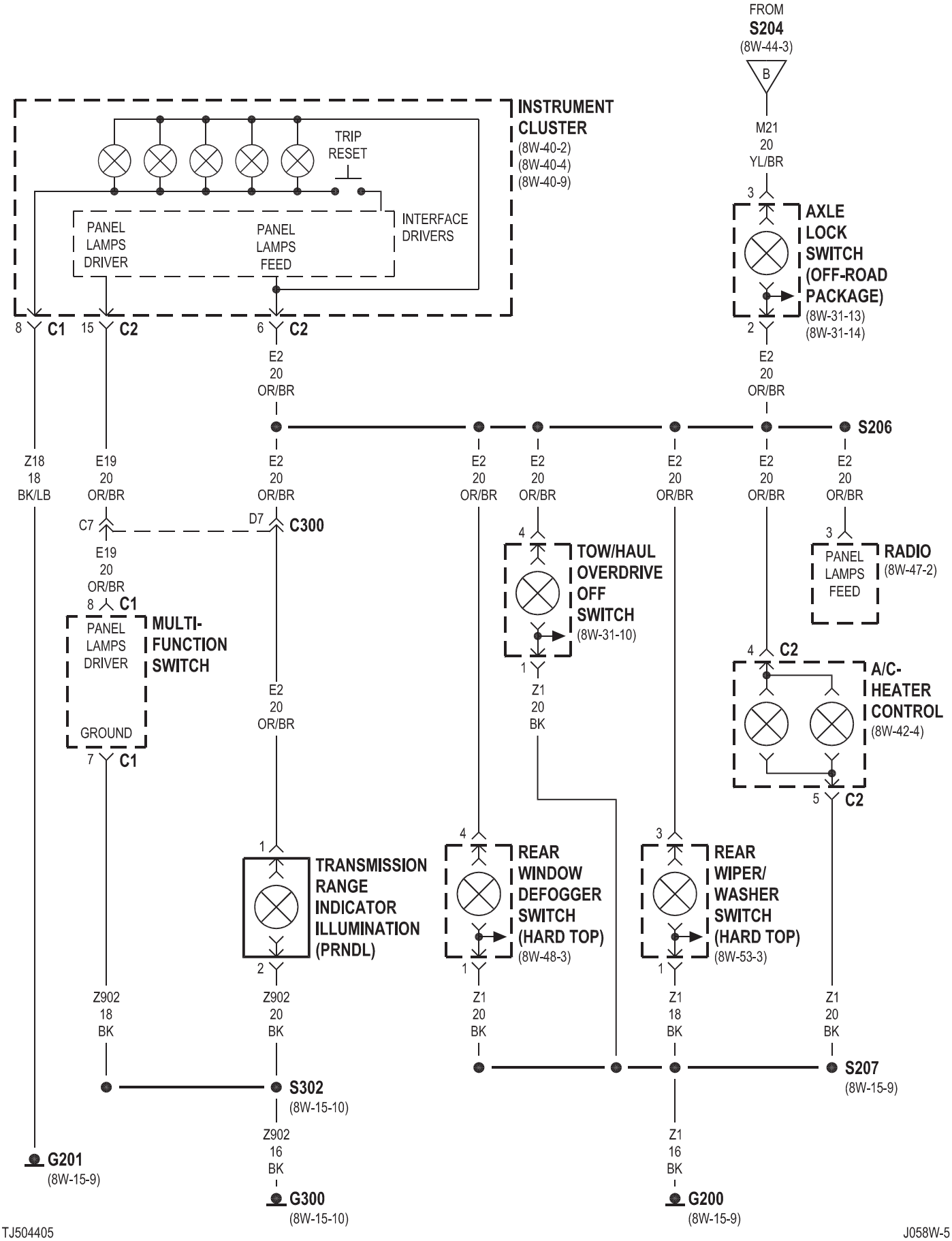
8W-44 INTERIOR LIGHTING

Component	Page
A/C-Heater Control	8W-44-5
Ambient Temperature Sensor	8W-44-6
Axle Lock Switch	8W-44-3, 5
Back-Up Lamp Switch	8W-44-2
Compass/Temperature Mirror	8W-44-3, 6
Data Link Connector	8W-44-3
Driver Door Ajar Switch	8W-44-4
Fuse 4	8W-44-4
Fuse 7	8W-44-2
Fuse 10	8W-44-6
Fuse 26	8W-44-2
Fuse Block	8W-44-2, 4, 6
G100	8W-44-2
G200	8W-44-5, 6
G201	8W-44-5
G300	8W-44-4, 5
Ignition Switch	8W-44-4
Instrument Cluster	8W-44-3, 4, 5
Left Courtesy Lamp	8W-44-3
Left Dome Lamp	8W-44-3
Multi-Function Switch	8W-44-5
Passenger Door Ajar Switch	8W-44-4
Power Distribution Center	8W-44-2
Radio	8W-44-5
Rear Window Defogger Switch	8W-44-5
Rear Wiper/Washer Switch	8W-44-5
Right Courtesy Lamp	8W-44-3
Right Dome Lamp	8W-44-3
Tow/Haul Overdrive Off Switch	8W-44-5
Transmission Range Indicator Illumination	8W-44-5
Transmission Range Sensor	8W-44-2
Underhood Lamp	8W-44-2


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
36
FUSE 26 10A
(8W-10-17)
37
M21 20 YL/BR
S173
(8W-10-17)
M21 20 YL/BR
2
UNDERHOOD LAMP
1
Z901 18 BK
BASE
EXCEPT BASE
S106
(8W-15-2)
Z901 12 BK
G100
(8W-15-2)
M21 20 YL/BR
B3 C301
M21 20 YL/BR
A
TO C300
(8W-44-3)
RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 7 10A
(8W-11-6)
27
F102 20 PK/DB
C7 C301
F102 20 PK/DB
S108
(8W-11-6)
F102 20 PK/DB
A/T
M/T
OPTION 1
OPTON 2
10 C102
10 C104
10 C102
F102 20 PK/DB
F102 20 PK/DB
1
TRANSMISSION RANGE SENSOR
(8W-31-7)
1
BACK-UP LAMP SWITCH
(8W-51-4)
6
L1 20 WT/LG
2
L1 20 WT/LG
OPTION 1
OPTION 2
9 C102
9 C104
9 C102
L1 20 WT/LG
B8 C301
L1 20 WT/LG
C
TO S311
(8W-44-6)


FROM
C301
(8W-44-2)
A
M21
20
YL/BR
16
DATA
LINK
CONNECTOR
(8W-18-3)
M21
20
YL/BR
M21
20
YL/BR
D3
C300
S204
(8W-10-17)
M21
20
YL/BR
M21
20
YL/BR
M21
20
YL/BR
M21
20
YL/BR
M21
20
YL/BR
12
C1
FUSED
B(+)
COURTESY
LAMPS
FEED
INSTRUMENT
CLUSTER
(8W-40-2)
(8W-40-8)
5
C2
S351
(8W-10-17)
M21
20
YL/BR
A
C325
M21
20
PK/WT
7
C203
M21
20
PK/WT
B
TO
AXLE
LOCK
SWITCH
(OFF-ROAD
PACKAGE)
(8W-44-5)
M21
20
YL/BR
A
C324
M21
20
PK/WT
1
RIGHT
COURTESY
LAMP
2
1
LEFT
COURTESY
LAMP
2
1
RIGHT
DOME
LAMP
2
1
LEFT
DOME
LAMP
2
7
FUSED
B(+)
COURTESY
LAMPS
DRIVER
COMPASS/
TEMPERATURE
MIRROR
6
M22
20
YL
B
C324
M22
20
YL
B
C325
M22
20
YL/OR
M22
20
YL/OR
M22
20
YL/OR
M22
20
YL/OR
S352
M22
20
YL
6
C203
M22
20
YL/OR
C5
C300
M22
20
YL/OR
S202


INSTRUMENT CLUSTER
(8W-40-8)
INTERFACE DRIVERS
PASSENGER DOOR AJAR SWITCH SENSE
DRIVER DOOR AJAR SWITCH SENSE
KEY-IN IGNITION SWITCH SENSE
3 C2
4 C2
13 C2
G76 20 VT/YL
G75 20 VT
G15 20 VT/TN
D2
A2
B7 C300
G76 20 VT/YL
G75 20 VT
G15 20 VT/TN
3
3
5
PASSENGER DOOR AJAR SWITCH
DRIVER DOOR AJAR SWITCH
IGNITION SWITCH
(8W-10-11)
1
1
10
M23 20 YL/DG
M23 20 YL/DG
Z902 16 BK
4
FUSE BLOCK
(8W-11-2)
FUSE 4 10A
(8W-11-4)
24
Z902 20 BK
S302
(8W-15-10)
Z902 16 BK
G300
(8W-15-10)



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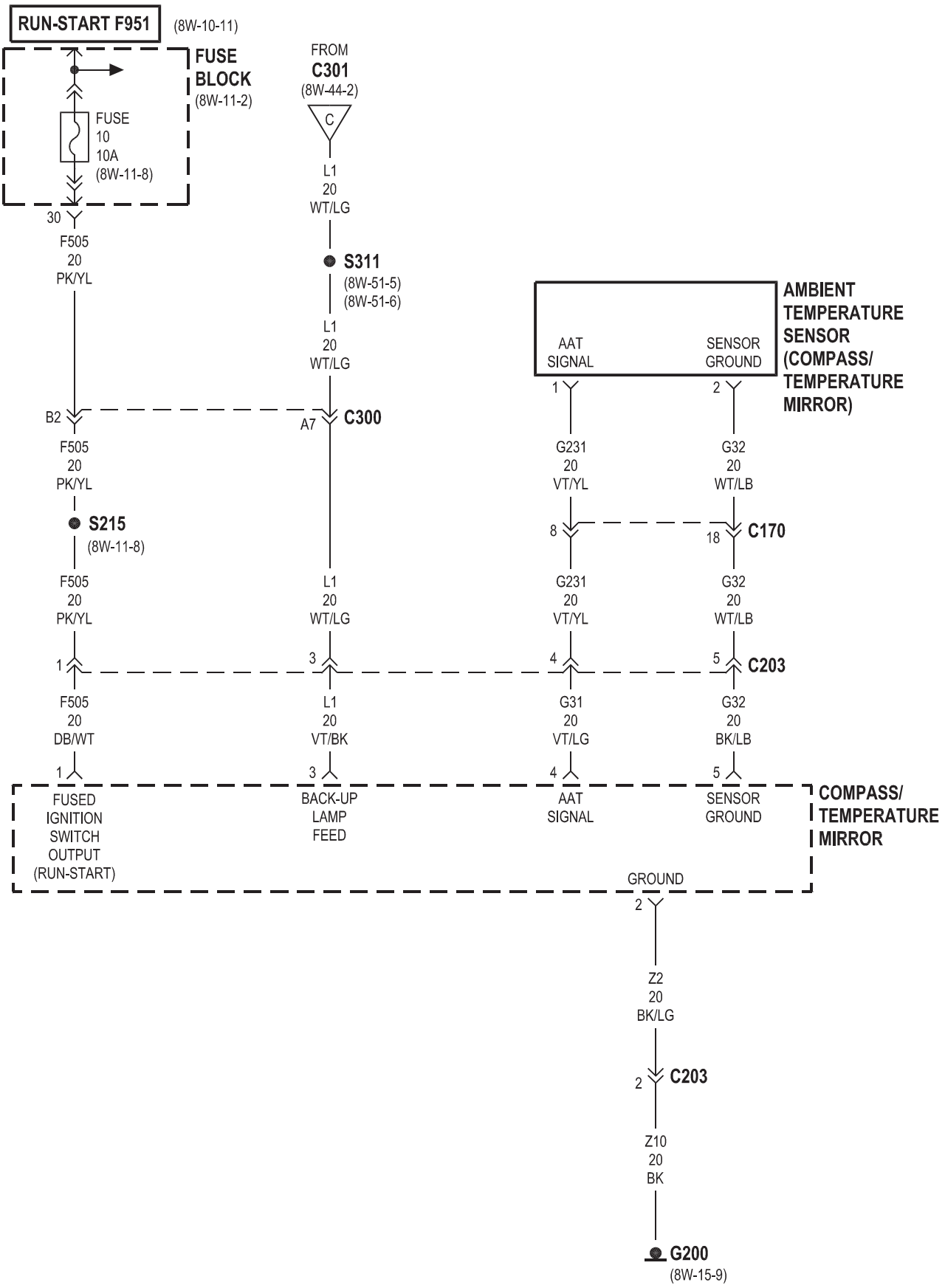
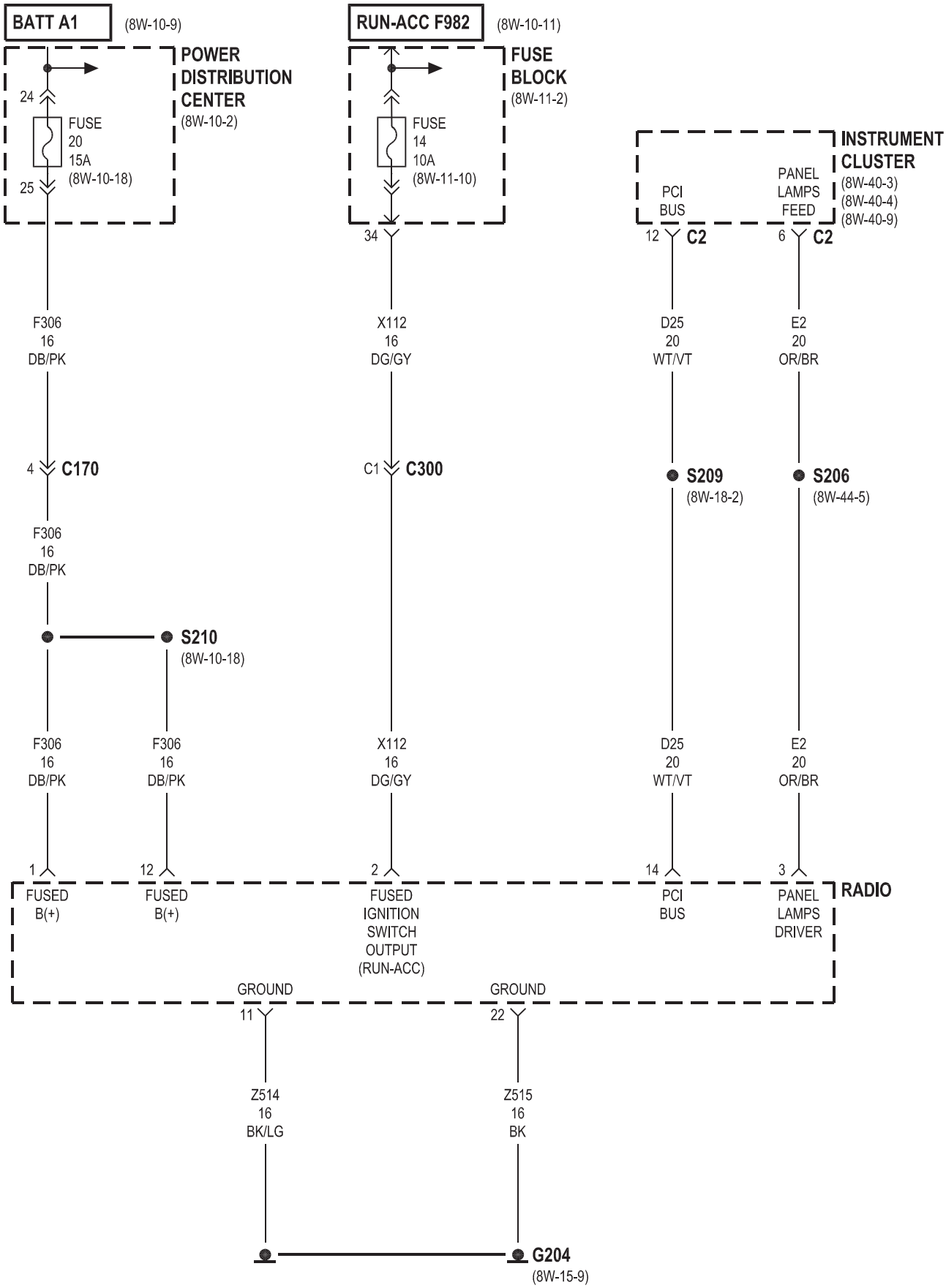
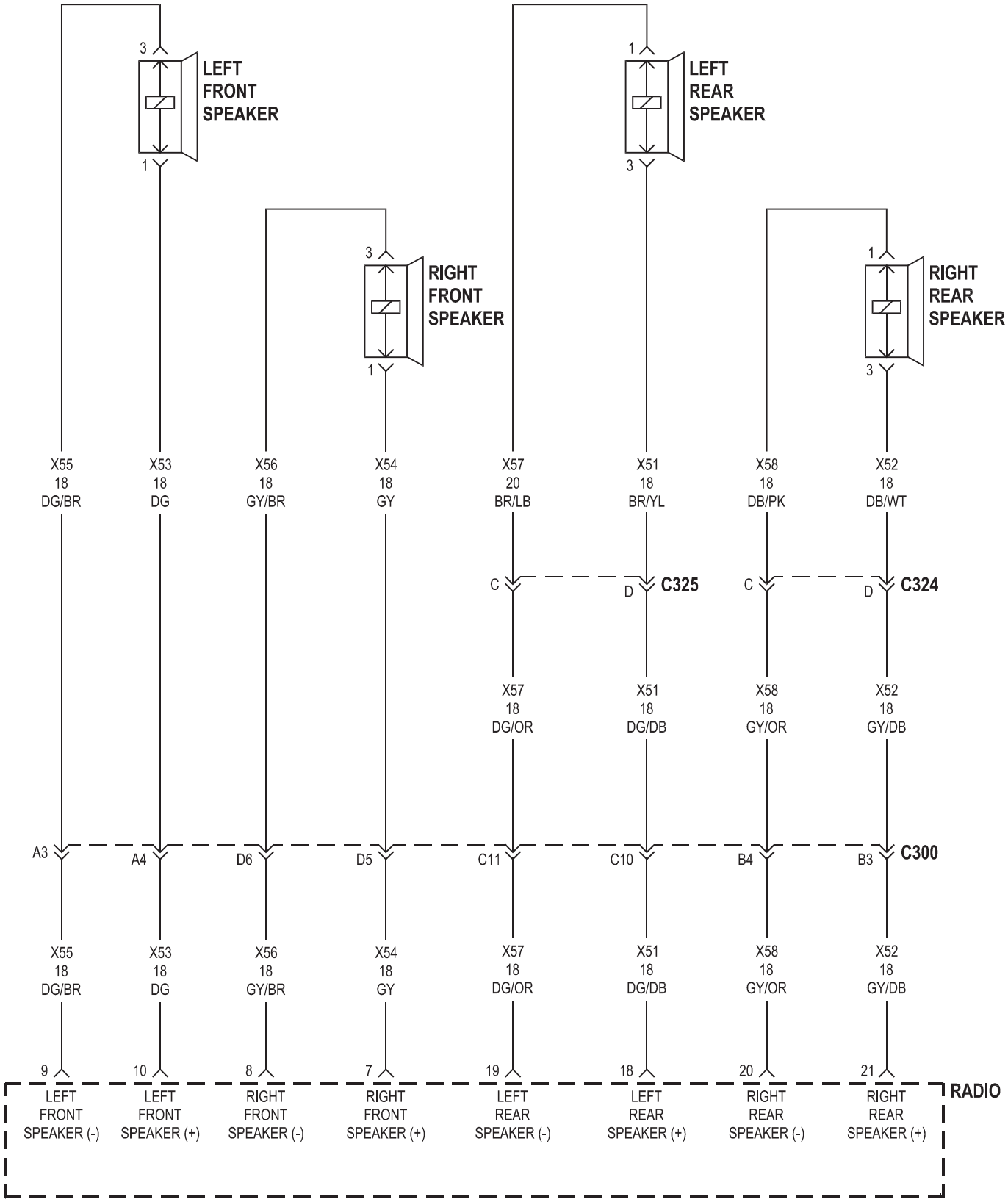

RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 10 10A
(8W-11-8)
30
F505 20 PK/YL
B2
F505 20 PK/YL
S215
(8W-11-8)
F505 20 PK/YL
1
F505 20 DB/WT
1
FUSED IGNITION SWITCH OUTPUT (RUN-START)
FROM C301
(8W-44-2)
C
L1 20 WT/LG
S311
(8W-51-5)
(8W-51-6)
L1 20 WT/LG
A7
C300
L1 20 WT/LG
3
L1 20 VT/BK
3
BACK-UP LAMP FEED
AMBIENT TEMPERATURE SENSOR (COMPASS/ TEMPERATURE MIRROR)
AAT SIGNAL
SENSOR GROUND
1
2
G231 20 VT/YL
G32 20 WT/LB
8
18
C170
G231 20 VT/YL
G32 20 WT/LB
4
5
C203
G31 20 VT/LG
G32 20 BK/LB
4
5
AAT SIGNAL
SENSOR GROUND
COMPASS/ TEMPERATURE MIRROR
GROUND
2
Z2 20 BK/LG
2
C203
Z10 20 BK
G200
(8W-15-9)

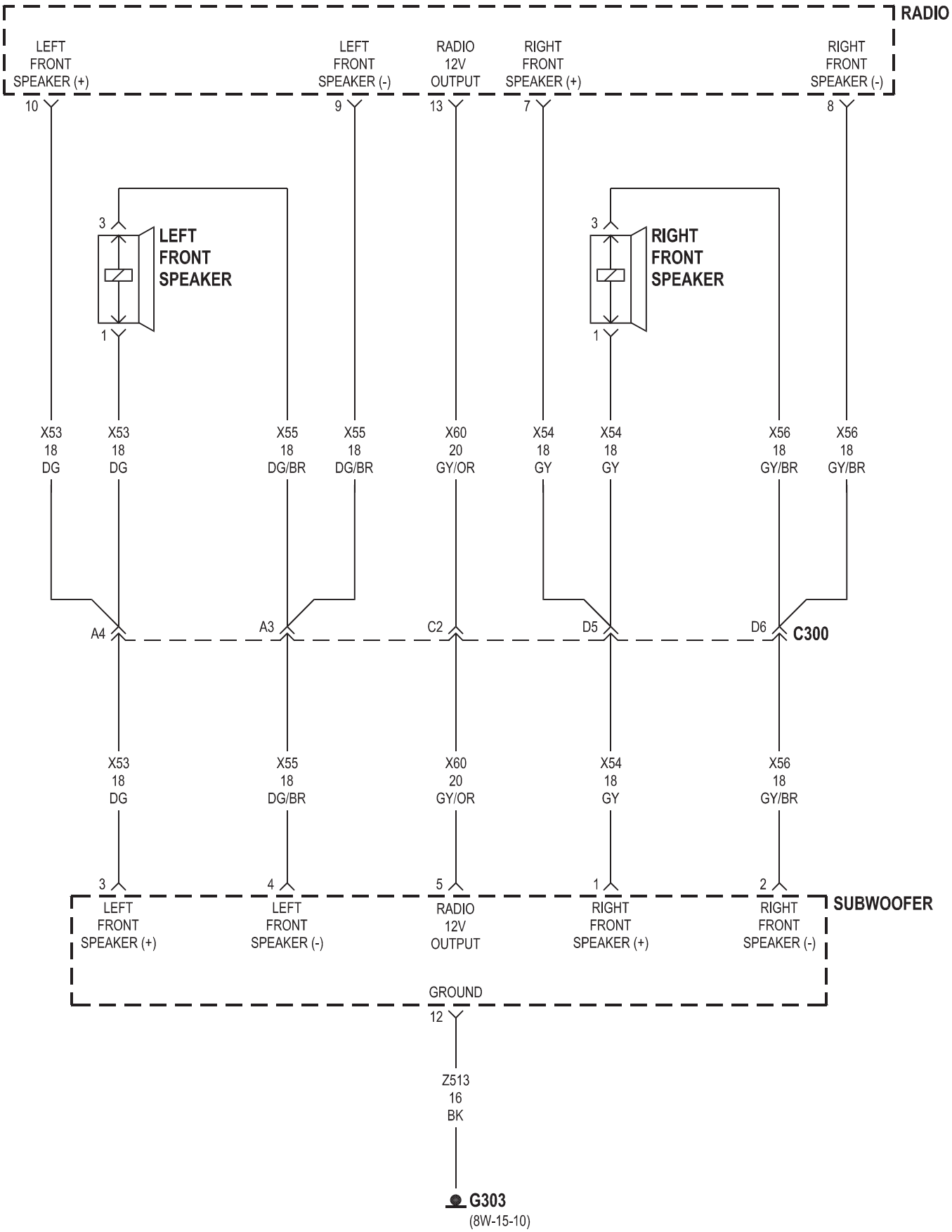
8W-47 AUDIO SYSTEM

Component	Page
Fuse 3	8W-47-5
Fuse 14	8W-47-2
Fuse 20	8W-47-2
Fuse Block	8W-47-2, 5
G204	8W-47-2
G303	8W-47-4
Instrument Cluster	8W-47-2
Left Front Speaker	8W-47-3, 4
Left Rear Speaker	8W-47-3, 5
Power Distribution Center	8W-47-2
Radio	8W-47-2, 3, 4, 5
Right Front Speaker	8W-47-3, 4
Right Rear Speaker	8W-47-3, 5
Subwoofer	8W-47-4, 5

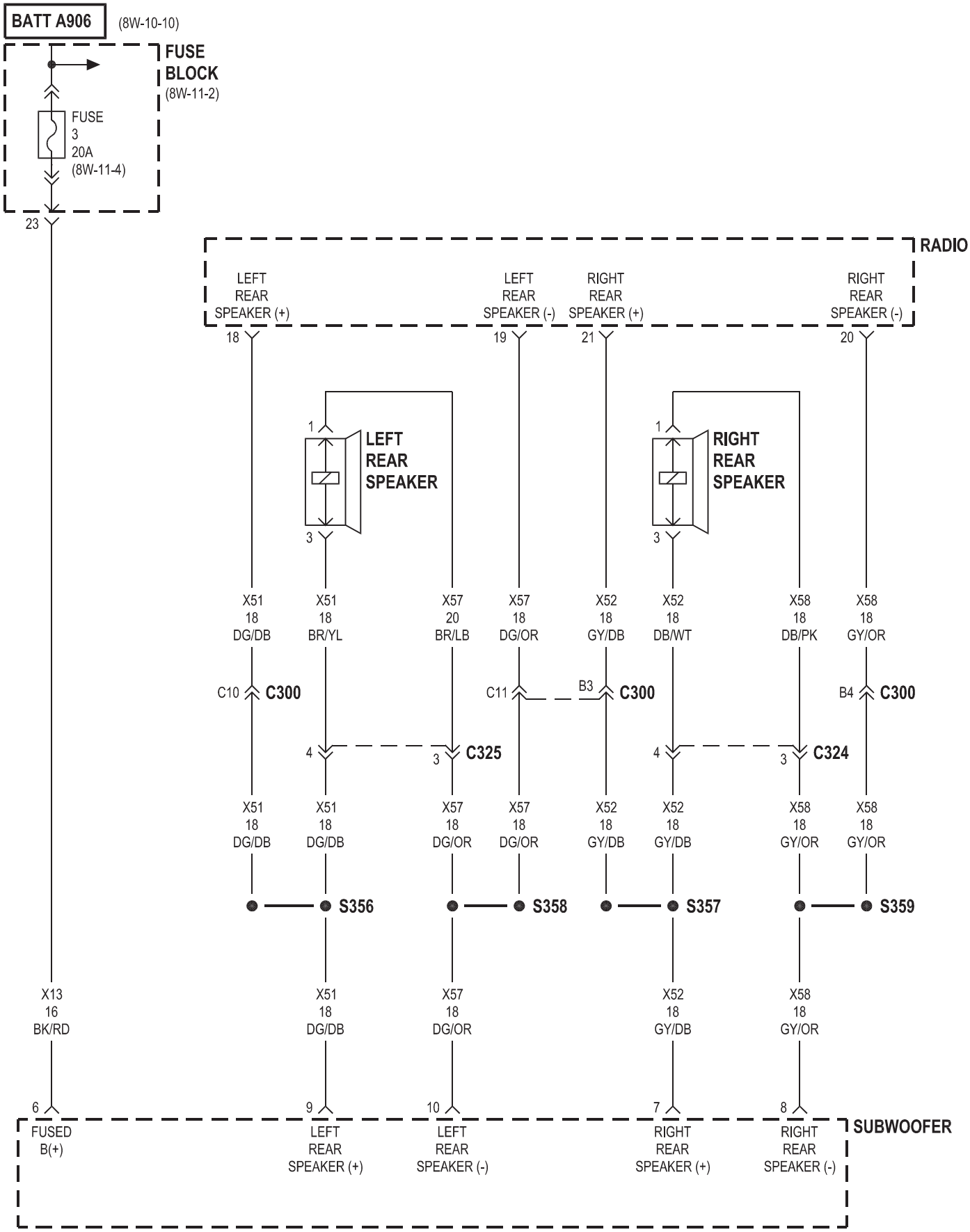

BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
24
FUSE 20 15A
(8W-10-18)
25
RUN-ACC F982
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 14 10A
(8W-11-10)
34
INSTRUMENT CLUSTER
(8W-40-3)
(8W-40-4)
(8W-40-9)
PCI BUS
12 C2
PANEL LAMPS FEED
6 C2
F306 16 DB/PK
X112 16 DG/GY
D25 20 WT/VT
E2 20 OR/BR
4 C170
C1 C300
S209
(8W-18-2)
S206
(8W-44-5)
F306 16 DB/PK
S210
(8W-10-18)
F306 16 DB/PK
F306 16 DB/PK
X112 16 DG/GY
D25 20 WT/VT
E2 20 OR/BR
1
12
2
14
3
RADIO
FUSED B(+)
FUSED B(+)
FUSED IGNITION SWITCH OUTPUT (RUN-ACC)
PCI BUS
PANEL LAMPS DRIVER
GROUND
GROUND
11
22
Z514 16 BK/LG
Z515 16 BK
G204
(8W-15-9)


LEFT FRONT SPEAKER
LEFT REAR SPEAKER
RIGHT FRONT SPEAKER
RIGHT REAR SPEAKER
X55 18 DG/BR
X53 18 DG
X56 18 GY/BR
X54 18 GY
X57 20 BR/LB
X51 18 BR/YL
X58 18 DB/PK
X52 18 DB/WT
C325
C324
X57 18 DG/OR
X51 18 DG/DB
X58 18 GY/OR
X52 18 GY/DB
C300
A3
A4
D6
D5
C11
C10
B4
B3
9
10
8
7
19
18
20
21
LEFT FRONT SPEAKER (-)
LEFT FRONT SPEAKER (+)
RIGHT FRONT SPEAKER (-)
RIGHT FRONT SPEAKER (+)
LEFT REAR SPEAKER (-)
LEFT REAR SPEAKER (+)
RIGHT REAR SPEAKER (-)
RIGHT REAR SPEAKER (+)
RADIO

SUBWOOFER

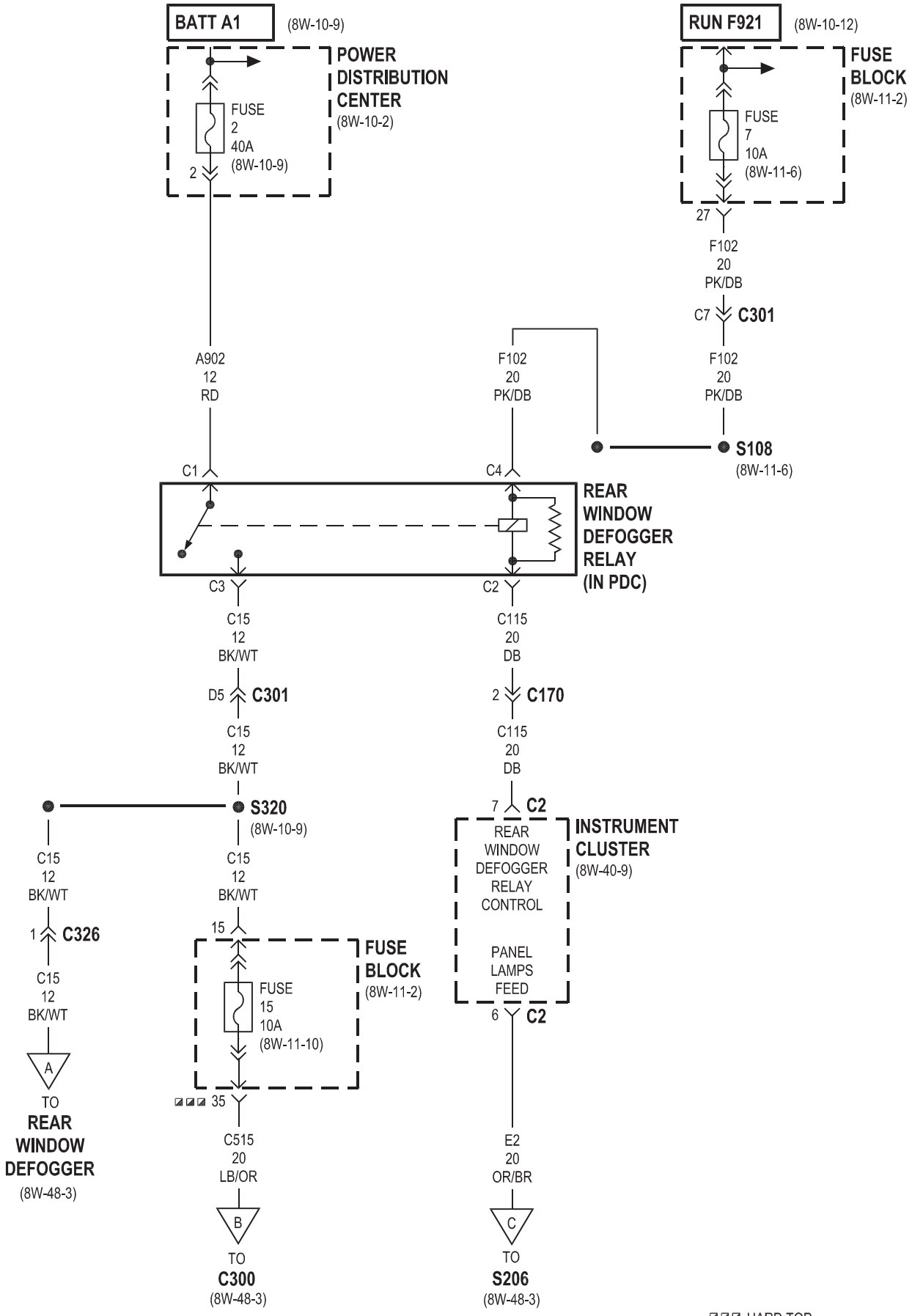

RADIO
LEFT FRONT SPEAKER (+)
10
LEFT FRONT SPEAKER (-)
9
RADIO 12V OUTPUT
13
RIGHT FRONT SPEAKER (+)
7
RIGHT FRONT SPEAKER (-)
8
3
LEFT FRONT SPEAKER
1
3
RIGHT FRONT SPEAKER
1
X53 18 DG
X53 18 DG
X55 18 DG/BR
X55 18 DG/BR
X60 20 GY/OR
X54 18 GY
X54 18 GY
X56 18 GY/BR
X56 18 GY/BR
A4
A3
C2
D5
D6
C300
X53 18 DG
X55 18 DG/BR
X60 20 GY/OR
X54 18 GY
X56 18 GY/BR
3
4
5
1
2
SUBWOOFER
LEFT FRONT SPEAKER (+)
LEFT FRONT SPEAKER (-)
RADIO 12V OUTPUT
RIGHT FRONT SPEAKER (+)
RIGHT FRONT SPEAKER (-)
GROUND
12
Z513 16 BK
G303
(8W-15-10)

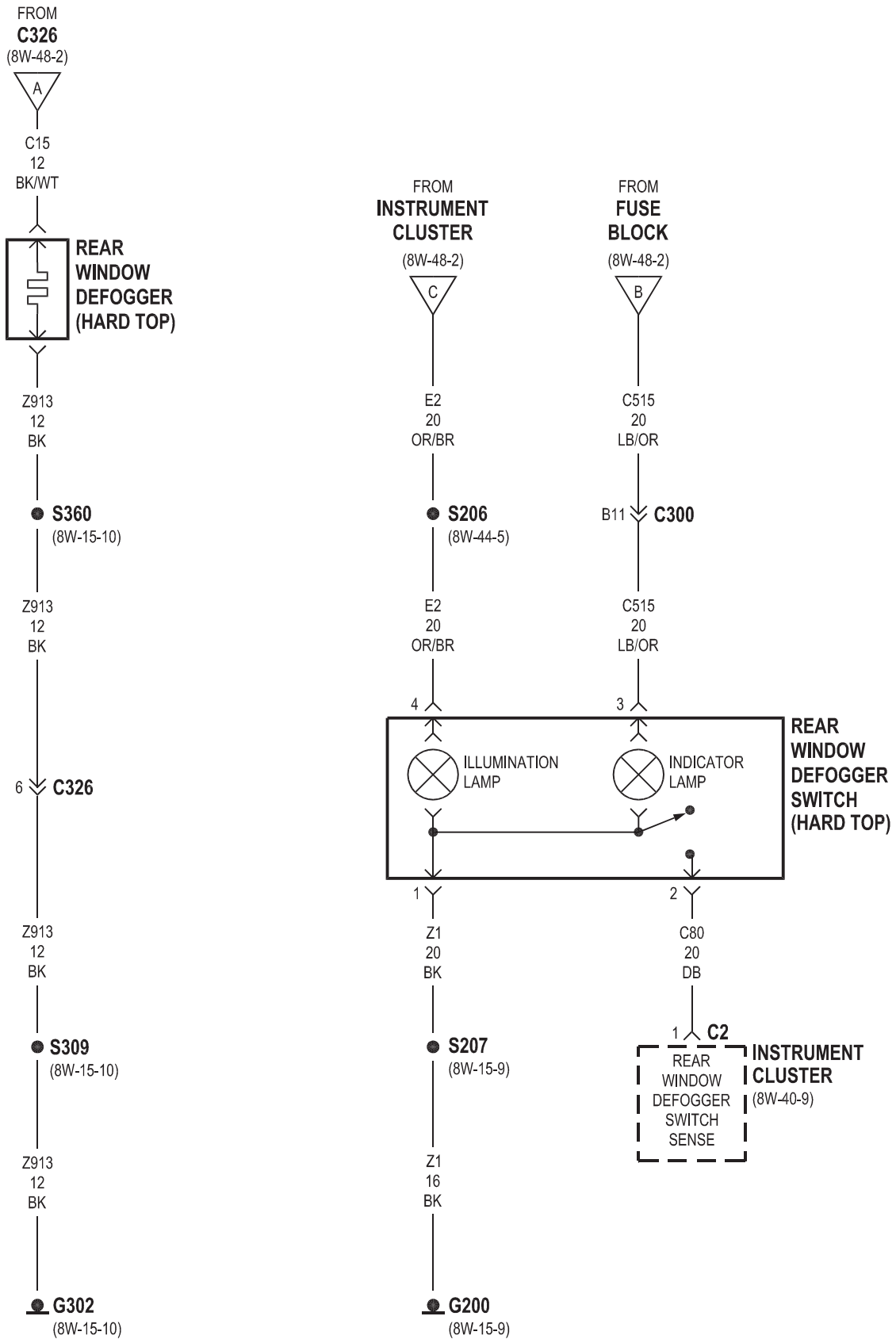
SUBWOOFER


BATT A906
(8W-10-10)
FUSE BLOCK
(8W-11-2)
FUSE 3 20A
(8W-11-4)
23
RADIO
LEFT REAR SPEAKER (+)
18
LEFT REAR SPEAKER (-)
19
RIGHT REAR SPEAKER (+)
21
RIGHT REAR SPEAKER (-)
20
1
LEFT REAR SPEAKER
3
1
RIGHT REAR SPEAKER
3
X51 18 DG/DB
X51 18 BR/YL
X57 20 BR/LB
X57 18 DG/OR
X52 18 GY/DB
X52 18 DB/WT
X58 18 DB/PK
X58 18 GY/OR
C10 C300
C11
B3 C300
B4 C300
4
3 C325
4
3 C324
X51 18 DG/DB
X51 18 DG/DB
X57 18 DG/OR
X57 18 DG/OR
X52 18 GY/DB
X52 18 GY/DB
X58 18 GY/OR
X58 18 GY/OR
S356
S358
S357
S359
X13 16 BK/RD
X51 18 DG/DB
X57 18 DG/OR
X52 18 GY/DB
X58 18 GY/OR
6
9
10
7
8
FUSED B(+)
LEFT REAR SPEAKER (+)
LEFT REAR SPEAKER (-)
RIGHT REAR SPEAKER (+)
RIGHT REAR SPEAKER (-)
SUBWOOFER

8W-48 REAR WINDOW DEFOGGER

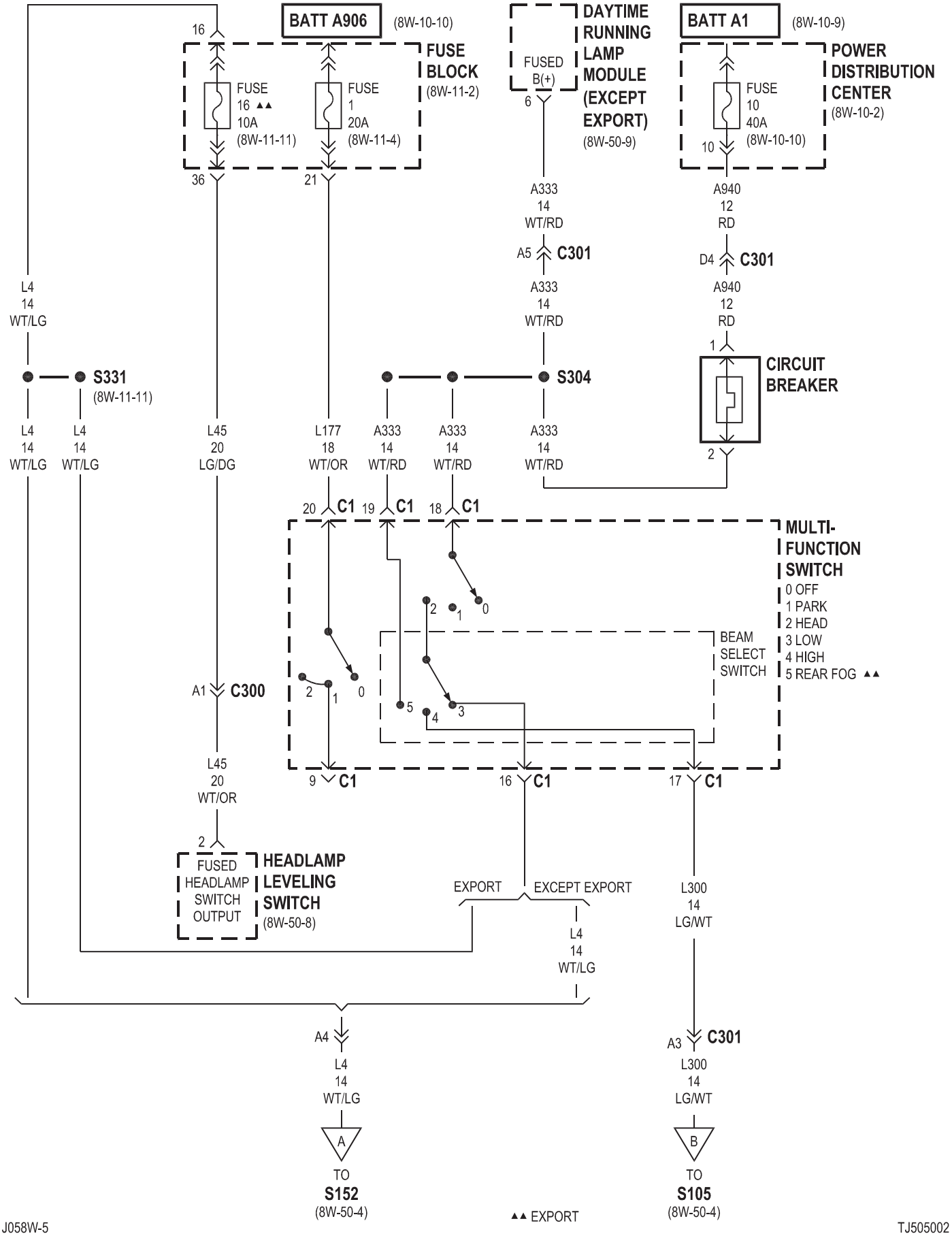
Component	Page
Fuse 2	8W-48-2
Fuse 7	8W-48-2
Fuse 15	8W-48-2
Fuse Block	8W-48-2, 3
G200	8W-48-3
G302	8W-48-3
Instrument Cluster	8W-48-2, 3
Power Distribution Center	8W-48-2
Rear Window Defogger	8W-48-2, 3
Rear Window Defogger Relay	8W-48-2
Rear Window Defogger Switch	8W-48-3


BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 2 40A (8W-10-9)
2
RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 7 10A (8W-11-6)
27
F102 20 PK/DB
C7 C301
F102 20 PK/DB
A902 12 RD
F102 20 PK/DB
S108
(8W-11-6)
C1
C4
REAR WINDOW DEFOGGER RELAY (IN PDC)
C3
C2
C15 12 BK/WT
D5 C301
C15 12 BK/WT
C115 20 DB
2 C170
C115 20 DB
S320
(8W-10-9)
7 C2
INSTRUMENT CLUSTER
(8W-40-9)
REAR WINDOW DEFOGGER RELAY CONTROL
PANEL LAMPS FEED
6 C2
C15 12 BK/WT
1 C326
C15 12 BK/WT
A
TO REAR WINDOW DEFOGGER
(8W-48-3)
C15 12 BK/WT
15
FUSE BLOCK
(8W-11-2)
FUSE 15 10A (8W-11-10)
35
C515 20 LB/OR
B
TO C300
(8W-48-3)
E2 20 OR/BR
C
TO S206
(8W-48-3)
HARD TOP


FROM
C326
(8W-48-2)
A
C15
12
BK/WT
REAR
WINDOW
DEFOGGER
(HARD TOP)
Z913
12
BK
S360
(8W-15-10)
Z913
12
BK
6
C326
Z913
12
BK
S309
(8W-15-10)
Z913
12
BK
G302
(8W-15-10)
FROM
INSTRUMENT
CLUSTER
(8W-48-2)
C
E2
20
OR/BR
S206
(8W-44-5)
E2
20
OR/BR
4
ILLUMINATION
LAMP
1
Z1
20
BK
S207
(8W-15-9)
Z1
16
BK
G200
(8W-15-9)
FROM
FUSE
BLOCK
(8W-48-2)
B
C515
20
LB/OR
B11
C300
C515
20
LB/OR
3
INDICATOR
LAMP
2
C80
20
DB
1
C2
REAR
WINDOW
DEFOGGER
SWITCH
SENSE
INSTRUMENT
CLUSTER
(8W-40-9)
REAR
WINDOW
DEFOGGER
SWITCH
(HARD TOP)

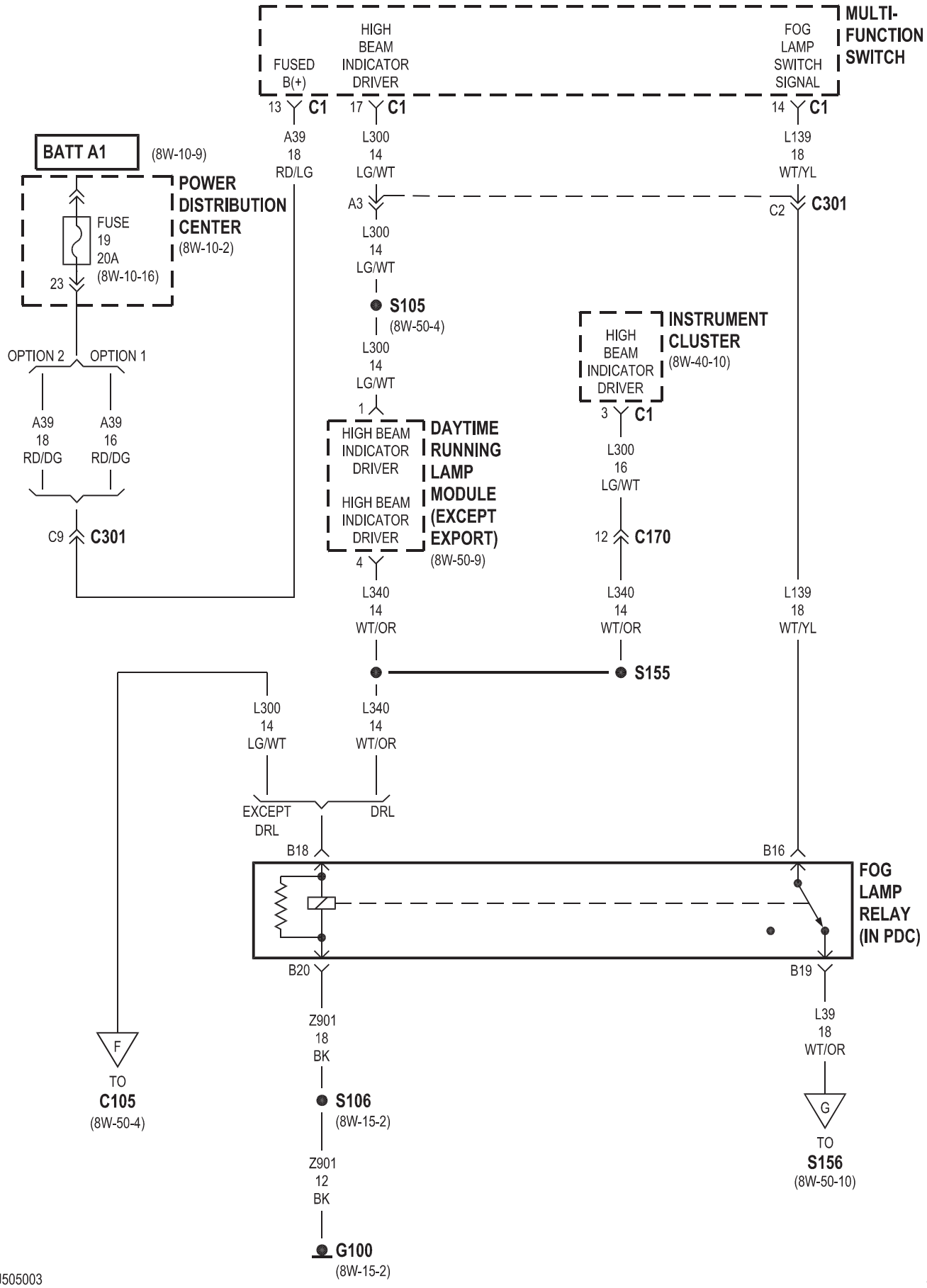
8W-50 FRONT LIGHTING

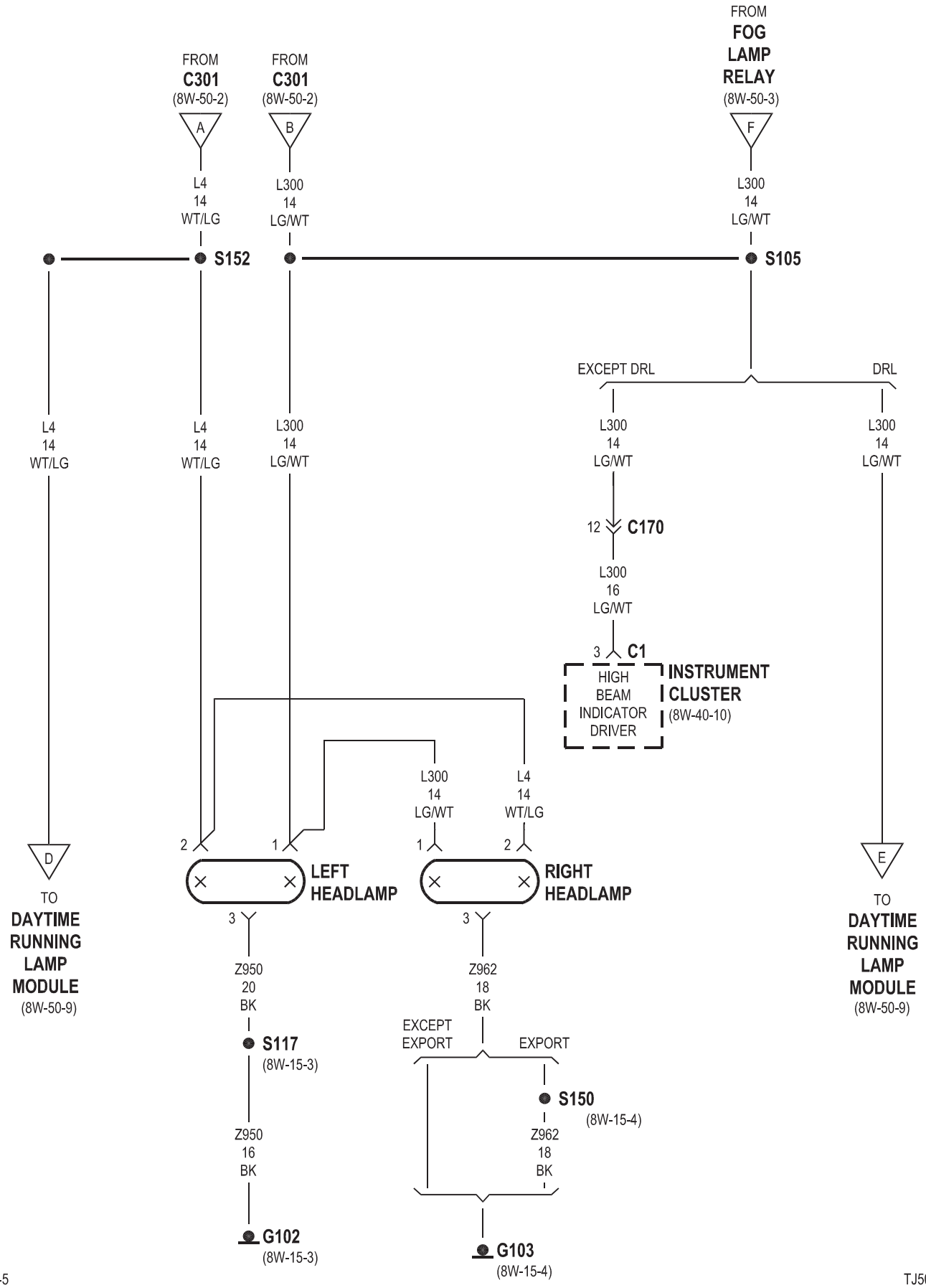
Component	Page
A/C Compressor Clutch Relay	8W-50-9
Circuit Breaker	8W-50-2, 9
Daytime Running Lamp Module	8W-50-2, 3, 4, 9
Fog Lamp Relay	8W-50-3, 4, 10
Fuse 1	8W-50-2
Fuse 10	8W-50-2, 9
Fuse 11	8W-50-9
Fuse 16	8W-50-2, 8
Fuse 19	8W-50-3
Fuse Block	8W-50-2, 8, 9
G100	8W-50-3, 9
G102	8W-50-4, 5, 6, 7, 8, 10
G103	8W-50-4, 5, 6, 7, 8, 10
G200	8W-50-8
Headlamp Leveling Switch	8W-50-2, 8
High Speed Radiator Fan Relay	8W-50-9
Instrument Cluster	8W-50-3, 4, 5, 6, 7, 9, 10
Left Fog Lamp	8W-50-10
Left Front Park/Turn Signal Lamp	8W-50-5
Left Front Position Lamp	8W-50-7
Left Front Turn Signal Lamp	8W-50-6
Left Headlamp	8W-50-4
Left Headlamp Leveling Motor	8W-50-8
Left Side Marker Lamp	8W-50-5
Left Side Repeater Lamp	8W-50-6
Low Speed Radiator Fan Relay	8W-50-9
Multi-Function Switch	8W-50-2, 3, 5, 6, 7, 8
Power Distribution Center	8W-50-2, 3, 9
Right Fog Lamp	8W-50-10
Right Front Park/Turn Signal Lamp	8W-50-5
Right Front Position Lamp	8W-50-7
Right Front Turn Signal Lamp	8W-50-6
Right Headlamp	8W-50-4
Right Headlamp Leveling Motor	8W-50-8
Right Side Marker Lamp	8W-50-5
Right Side Repeater Lamp	8W-50-6
Vehicle Speed Sensor	8W-50-9

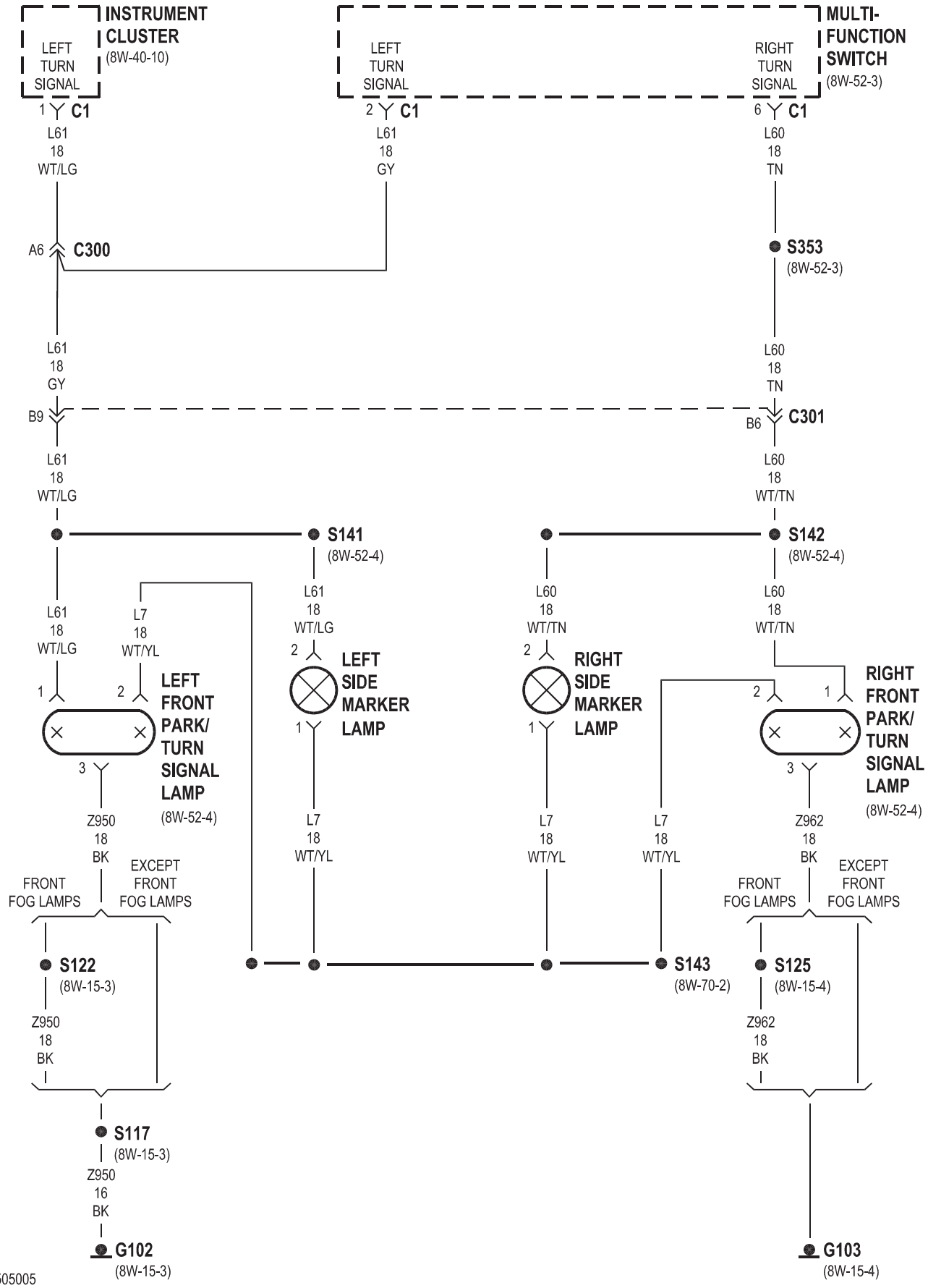

BATT A906
(8W-10-10)
16
FUSE BLOCK
(8W-11-2)
FUSE 16 ▲▲ 10A (8W-11-11)
FUSE 1 20A (8W-11-4)
36
21
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
FUSED B(+)
6
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 10 40A (8W-10-10)
10
A333 14 WT/RD
A5 C301
A940 12 RD
D4 C301
L4 14 WT/LG
S331
(8W-11-11)
S304
CIRCUIT BREAKER
1
2
L45 20 LG/DG
L177 18 WT/OR
20 C1
19 C1
18 C1
MULTI-FUNCTION SWITCH
0 OFF
1 PARK
2 HEAD
3 LOW
4 HIGH
5 REAR FOG ▲▲
BEAM SELECT SWITCH
A1 C300
9 C1
16 C1
17 C1
L45 20 WT/OR
2
FUSED HEADLAMP SWITCH OUTPUT
HEADLAMP LEVELING SWITCH
(8W-50-8)
EXPORT
EXCEPT EXPORT
L300 14 LG/WT
A4
A3 C301
A
B
TO S152
(8W-50-4)
TO S105
(8W-50-4)

▲▲ EXPORT

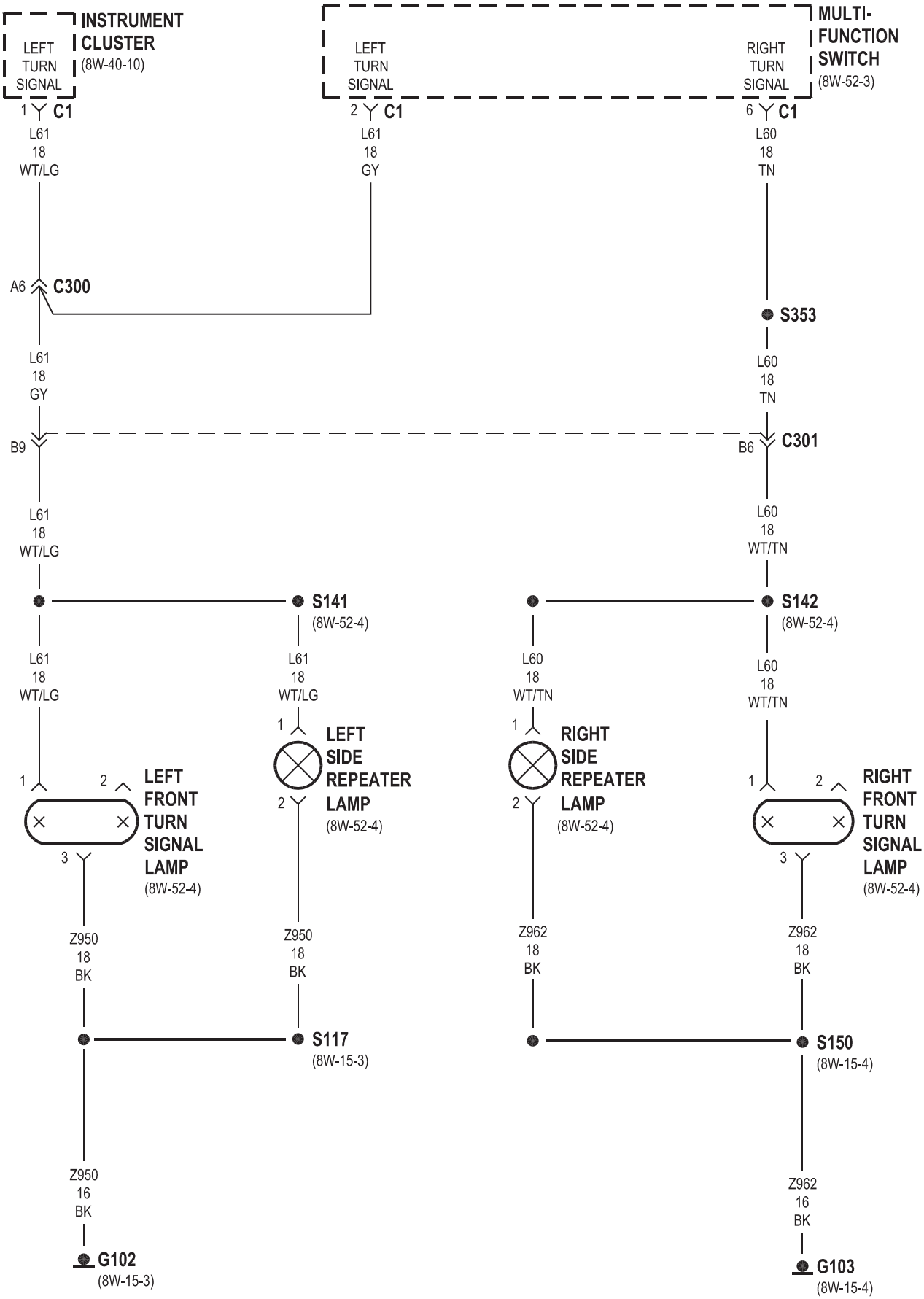
FOG LAMPS

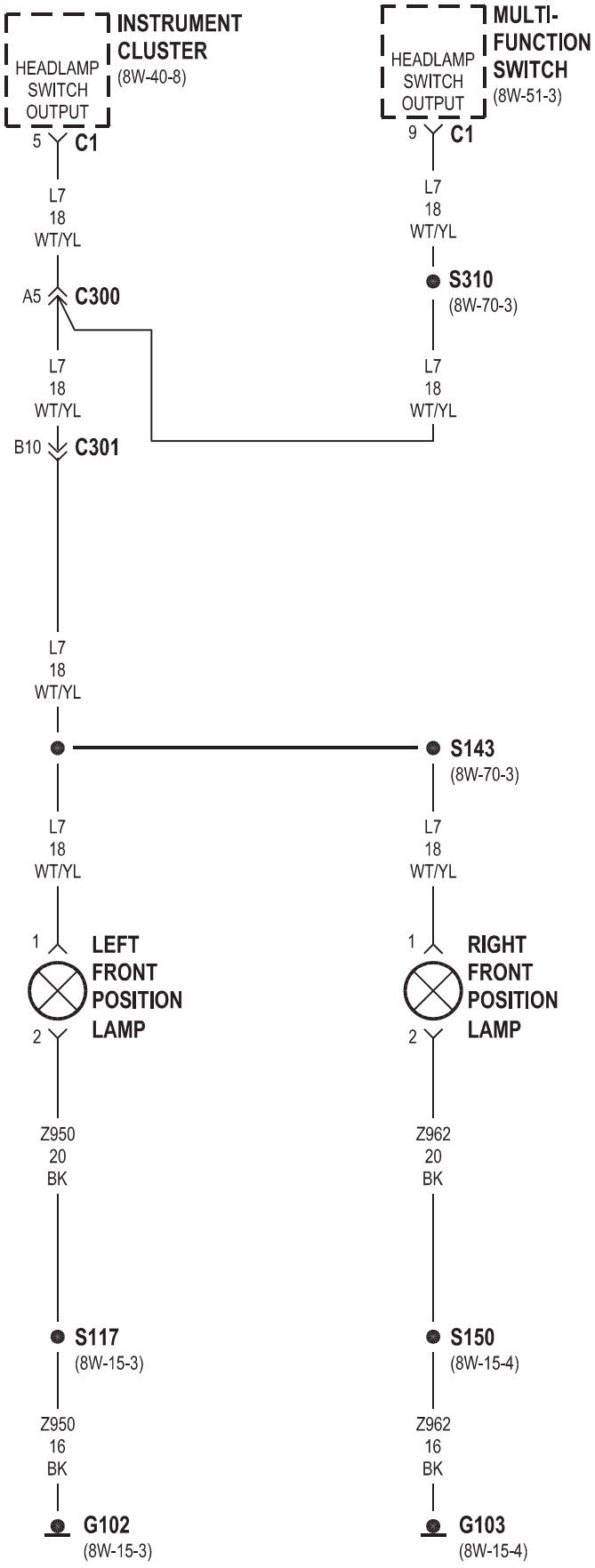

MULTI-FUNCTION SWITCH
FUSED B(+)
HIGH BEAM INDICATOR DRIVER
FOG LAMP SWITCH SIGNAL
13 C1
17 C1
14 C1
A39 18 RD/LG
L300 14 LG/WT
L139 18 WT/YL
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 19 20A
(8W-10-16)
23
A3
C2 C301
L300 14 LG/WT
S105
(8W-50-4)
INSTRUMENT CLUSTER
(8W-40-10)
HIGH BEAM INDICATOR DRIVER
3 C1
L300 16 LG/WT
12 C170
OPTION 2
OPTION 1
A39 18 RD/DG
A39 16 RD/DG
C9 C301
HIGH BEAM INDICATOR DRIVER
HIGH BEAM INDICATOR DRIVER
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
(8W-50-9)
1
4
L340 14 WT/OR
S155
L300 14 LG/WT
L340 14 WT/OR
L139 18 WT/YL
EXCEPT DRL
DRL
B18
B16
FOG LAMP RELAY (IN PDC)
B20
B19
F
TO C105
(8W-50-4)
Z901 18 BK
S106
(8W-15-2)
Z901 12 BK
G100
(8W-15-2)
L39 18 WT/OR
G
TO S156
(8W-50-10)

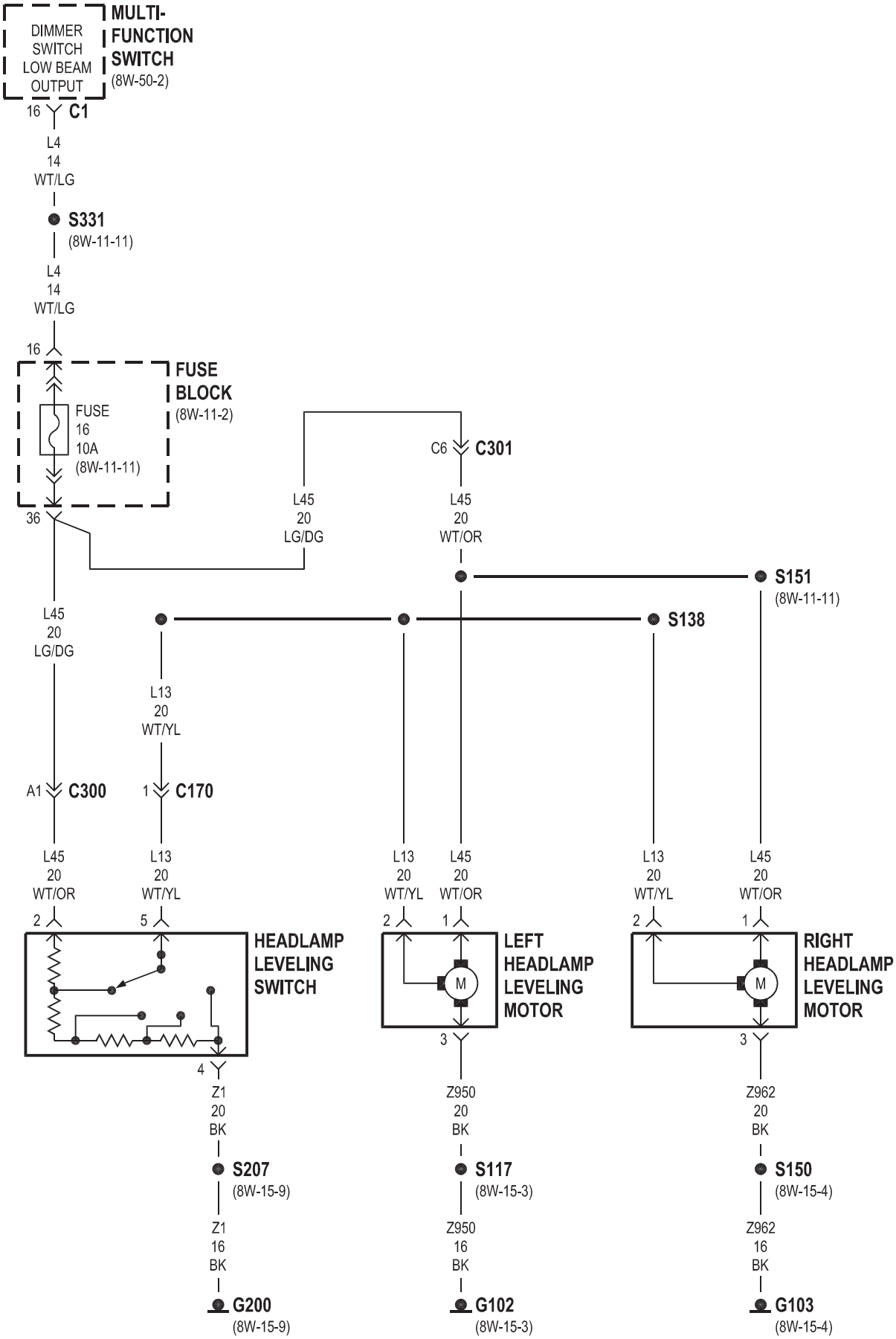

FROM
C301
(8W-50-2)
A
L4
14
WT/LG
FROM
C301
(8W-50-2)
B
L300
14
LG/WT
FROM
FOG
LAMP
RELAY
(8W-50-3)
F
L300
14
LG/WT
S152
S105
EXCEPT DRL
DRL
L4
14
WT/LG
L4
14
WT/LG
L300
14
LG/WT
L300
14
LG/WT
L300
14
LG/WT
12
C170
L300
16
LG/WT
3
C1
HIGH
BEAM
INDICATOR
DRIVER
INSTRUMENT
CLUSTER
(8W-40-10)
L300
14
LG/WT
L4
14
WT/LG
2
1
1
2
LEFT
HEADLAMP
RIGHT
HEADLAMP
D
TO
DAYTIME
RUNNING
LAMP
MODULE
(8W-50-9)
E
TO
DAYTIME
RUNNING
LAMP
MODULE
(8W-50-9)
3
3
Z950
20
BK
Z962
18
BK
S117
(8W-15-3)
EXCEPT
EXPORT
EXPORT
S150
(8W-15-4)
Z950
16
BK
Z962
18
BK
G102
(8W-15-3)
G103
(8W-15-4)


INSTRUMENT CLUSTER
(8W-40-10)
LEFT TURN SIGNAL
1 C1
L61 18 WT/LG
LEFT TURN SIGNAL
2 C1
L61 18 GY
MULTI-FUNCTION SWITCH
(8W-52-3)
RIGHT TURN SIGNAL
6 C1
L60 18 TN
A6 C300
S353
(8W-52-3)
L61 18 GY
B9
L60 18 TN
B6 C301
L61 18 WT/LG
L60 18 WT/TN
S141
(8W-52-4)
S142
(8W-52-4)
L61 18 WT/LG
L7 18 WT/YL
L61 18 WT/LG
2
LEFT SIDE MARKER LAMP
1
L60 18 WT/TN
2
RIGHT SIDE MARKER LAMP
1
L60 18 WT/TN
2
1
RIGHT FRONT PARK/ TURN SIGNAL LAMP
(8W-52-4)
1
2
LEFT FRONT PARK/ TURN SIGNAL LAMP
(8W-52-4)
3
3
Z950 18 BK
L7 18 WT/YL
L7 18 WT/YL
L7 18 WT/YL
Z962 18 BK
FRONT FOG LAMPS
EXCEPT FRONT FOG LAMPS
FRONT FOG LAMPS
EXCEPT FRONT FOG LAMPS
S122
(8W-15-3)
S143
(8W-70-2)
S125
(8W-15-4)
Z950 18 BK
Z962 18 BK
S117
(8W-15-3)
Z950 16 BK
G102
(8W-15-3)
G103
(8W-15-4)

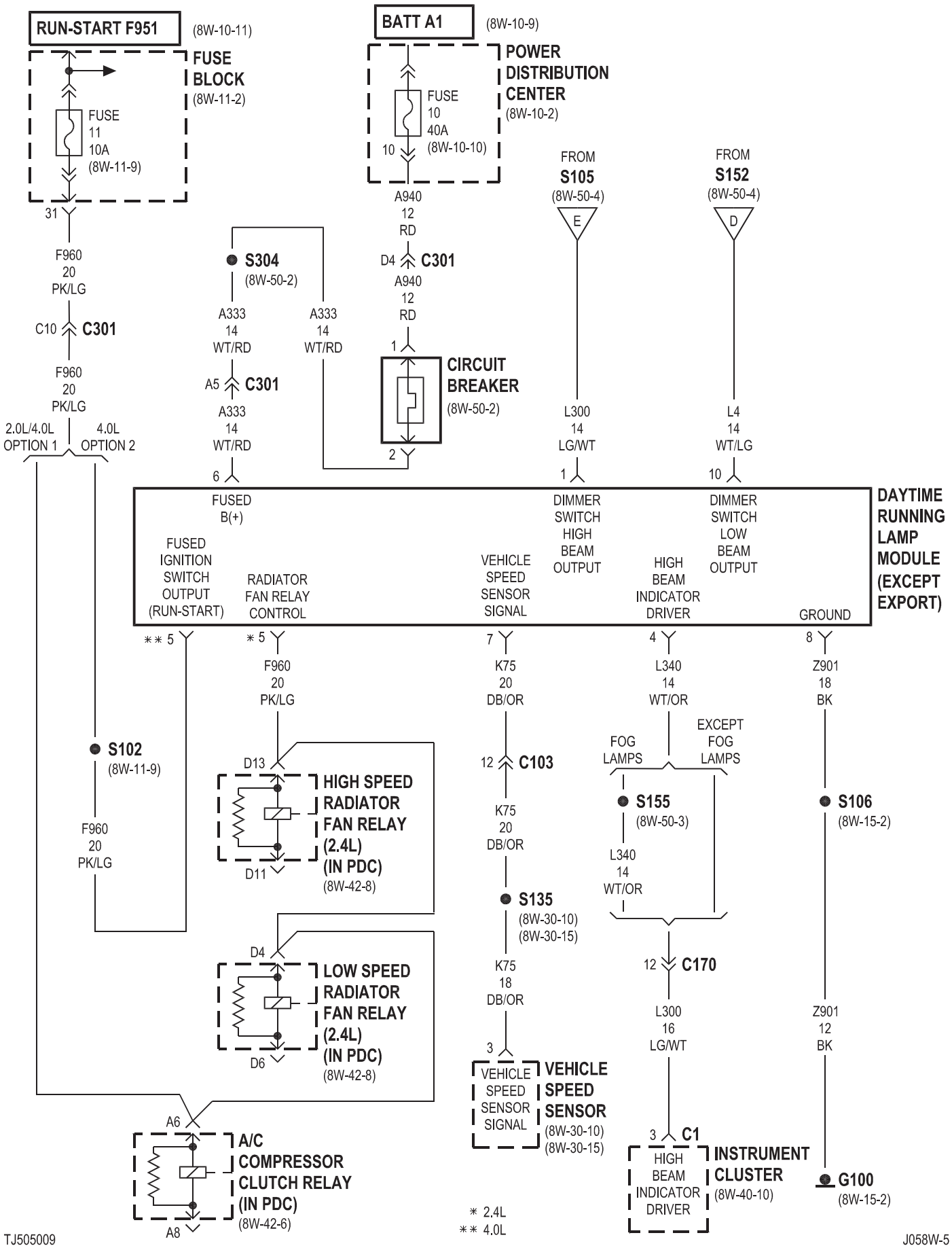
EXPORT

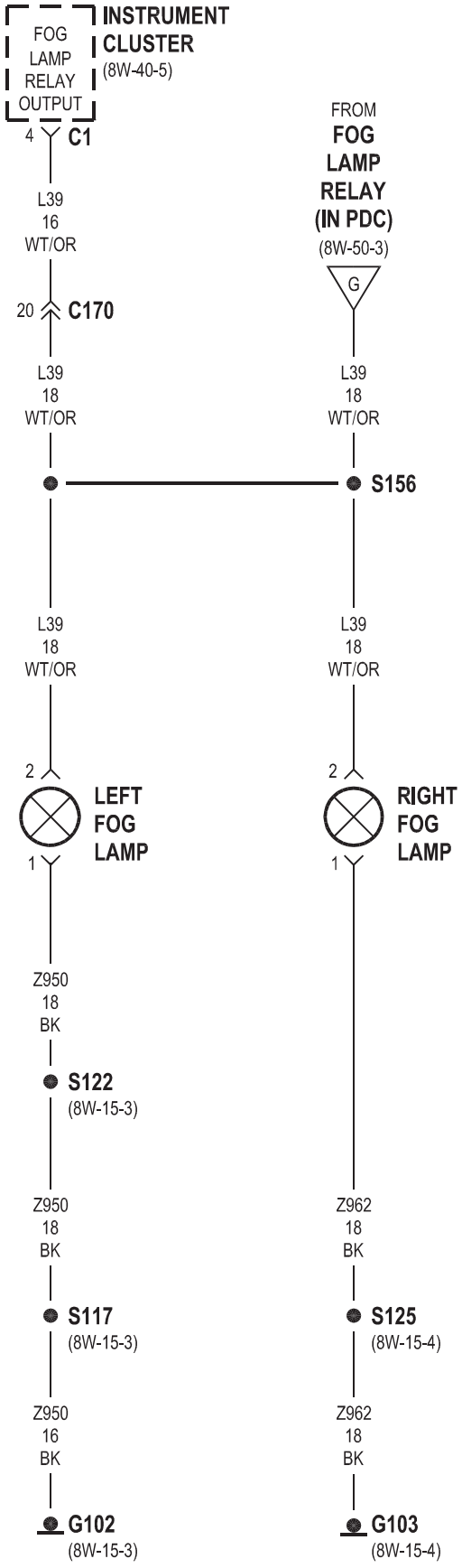

INSTRUMENT CLUSTER
(8W-40-10)
LEFT TURN SIGNAL
1 C1
L61 18 WT/LG
LEFT TURN SIGNAL
2 C1
L61 18 GY
MULTI-FUNCTION SWITCH
(8W-52-3)
RIGHT TURN SIGNAL
6 C1
L60 18 TN
A6 C300
L61 18 GY
S353
L60 18 TN
B9
B6 C301
L61 18 WT/LG
L60 18 WT/TN
S141
(8W-52-4)
S142
(8W-52-4)
L61 18 WT/LG
L61 18 WT/LG
L60 18 WT/TN
L60 18 WT/TN
LEFT FRONT TURN SIGNAL LAMP
(8W-52-4)
LEFT SIDE REPEATER LAMP
(8W-52-4)
RIGHT SIDE REPEATER LAMP
(8W-52-4)
RIGHT FRONT TURN SIGNAL LAMP
(8W-52-4)
Z950 18 BK
Z950 18 BK
Z962 18 BK
Z962 18 BK
S117
(8W-15-3)
S150
(8W-15-4)
Z950 16 BK
Z962 16 BK
G102
(8W-15-3)
G103
(8W-15-4)


INSTRUMENT CLUSTER
(8W-40-8)
HEADLAMP SWITCH OUTPUT
5
C1
MULTI-FUNCTION SWITCH
(8W-51-3)
HEADLAMP SWITCH OUTPUT
9
C1
L7
18
WT/YL
A5
C300
S310
(8W-70-3)
B10
C301
S143
(8W-70-3)
1
LEFT FRONT POSITION LAMP
2
RIGHT FRONT POSITION LAMP
Z950
20
BK
Z962
20
BK
S117
(8W-15-3)
S150
(8W-15-4)
Z950
16
BK
Z962
16
BK
G102
(8W-15-3)
G103
(8W-15-4)


DIMMER SWITCH LOW BEAM OUTPUT
MULTI-FUNCTION SWITCH
(8W-50-2)
16
C1
L4
14
WT/LG
S331
(8W-11-11)
L4
14
WT/LG
16
FUSE BLOCK
(8W-11-2)
FUSE
16
10A
(8W-11-11)
36
C6
C301
L45
20
LG/DG
L45
20
WT/OR
S151
(8W-11-11)
S138
L45
20
LG/DG
L13
20
WT/YL
A1
C300
1
C170
L45
20
WT/OR
L13
20
WT/YL
L13
20
WT/YL
L45
20
WT/OR
L13
20
WT/YL
L45
20
WT/OR
2
5
2
1
2
1
HEADLAMP LEVELING SWITCH
LEFT HEADLAMP LEVELING MOTOR
RIGHT HEADLAMP LEVELING MOTOR
M
M
4
3
3
Z1
20
BK
Z950
20
BK
Z962
20
BK
S207
(8W-15-9)
S117
(8W-15-3)
S150
(8W-15-4)
Z1
16
BK
Z950
16
BK
Z962
16
BK
G200
(8W-15-9)
G102
(8W-15-3)
G103
(8W-15-4)

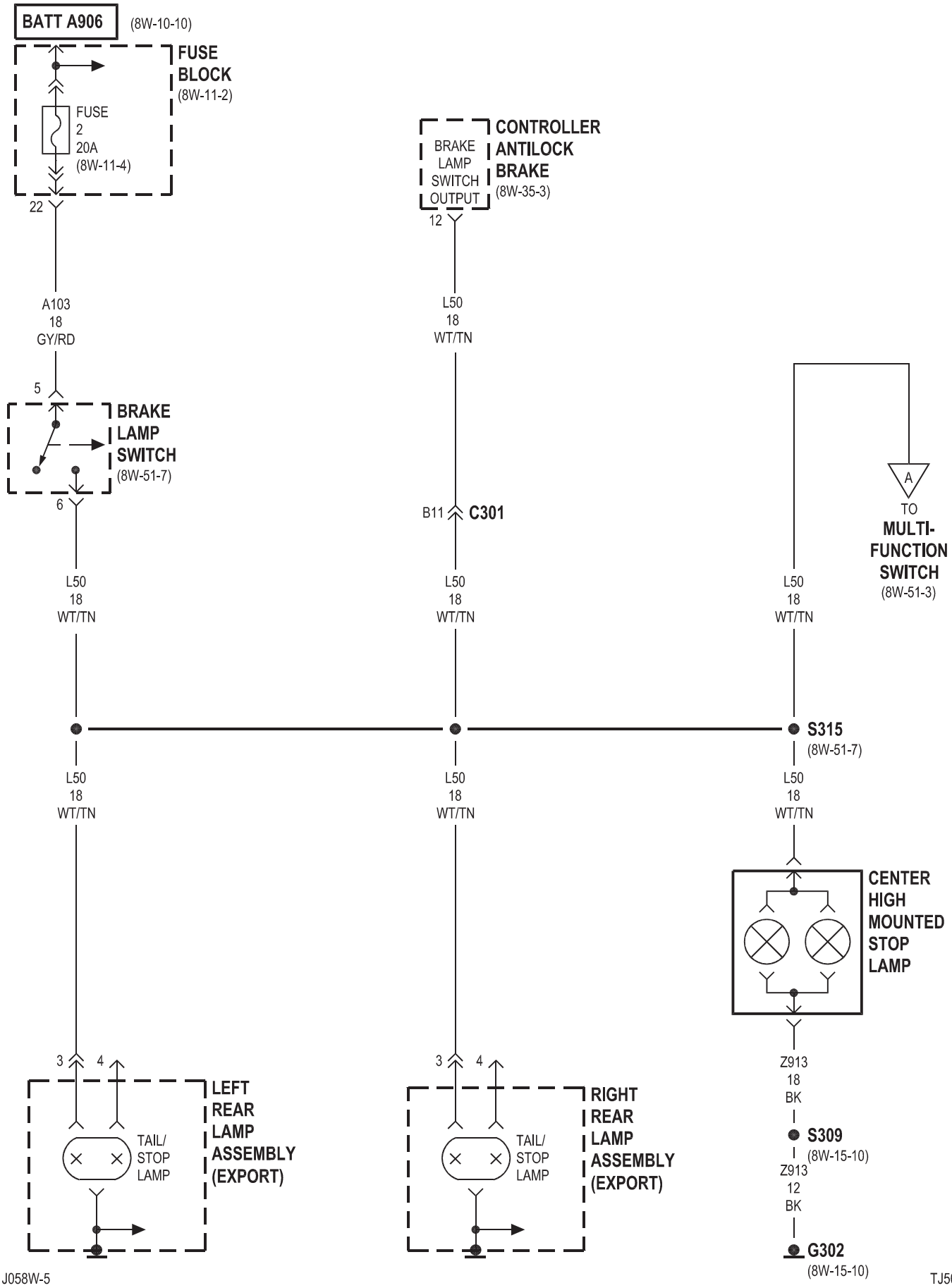
DRL


RUN-START F951
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 11 10A
(8W-11-9)
31
F960 20 PK/LG
C10 C301
F960 20 PK/LG
2.0L/4.0L OPTION 1
4.0L OPTION 2
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 10 40A
(8W-10-10)
10
A940 12 RD
D4 C301
A940 12 RD
1
CIRCUIT BREAKER
(8W-50-2)
2
S304
(8W-50-2)
A333 14 WT/RD
A5 C301
A333 14 WT/RD
6
A333 14 WT/RD
FROM S105
(8W-50-4)
E
L300 14 LG/WT
1
FROM S152
(8W-50-4)
D
L4 14 WT/LG
10
DAYTIME RUNNING LAMP MODULE (EXCEPT EXPORT)
FUSED B(+)
FUSED IGNITION SWITCH OUTPUT (RUN-START)
RADIATOR FAN RELAY CONTROL
VEHICLE SPEED SENSOR SIGNAL
DIMMER SWITCH HIGH BEAM OUTPUT
HIGH BEAM INDICATOR DRIVER
DIMMER SWITCH LOW BEAM OUTPUT
GROUND
** 5
* 5
7
4
8
F960 20 PK/LG
K75 20 DB/OR
L340 14 WT/OR
Z901 18 BK
S102
(8W-11-9)
F960 20 PK/LG
D13
HIGH SPEED RADIATOR FAN RELAY (2.4L) (IN PDC)
(8W-42-8)
D11
12 C103
K75 20 DB/OR
S135
(8W-30-10)
(8W-30-15)
K75 18 DB/OR
3
VEHICLE SPEED SENSOR SIGNAL
VEHICLE SPEED SENSOR
(8W-30-10)
(8W-30-15)
FOG LAMPS
EXCEPT FOG LAMPS
S155
(8W-50-3)
L340 14 WT/OR
12 C170
L300 16 LG/WT
3 C1
HIGH BEAM INDICATOR DRIVER
INSTRUMENT CLUSTER
(8W-40-10)
S106
(8W-15-2)
Z901 12 BK
G100
(8W-15-2)
D4
LOW SPEED RADIATOR FAN RELAY (2.4L) (IN PDC)
(8W-42-8)
D6
A6
A/C COMPRESSOR CLUTCH RELAY (IN PDC)
(8W-42-6)
A8
* 2.4L
** 4.0L


FOG LAMP RELAY OUTPUT
INSTRUMENT CLUSTER
(8W-40-5)
4
C1
L39
16
WT/OR
20
C170
L39
18
WT/OR
FROM
FOG LAMP RELAY
(IN PDC)
(8W-50-3)
G
L39
18
WT/OR
S156
L39
18
WT/OR
L39
18
WT/OR
2
LEFT FOG LAMP
1
2
RIGHT FOG LAMP
1
Z950
18
BK
S122
(8W-15-3)
Z950
18
BK
S117
(8W-15-3)
Z950
16
BK
G102
(8W-15-3)
Z962
18
BK
S125
(8W-15-4)
Z962
18
BK
G103
(8W-15-4)

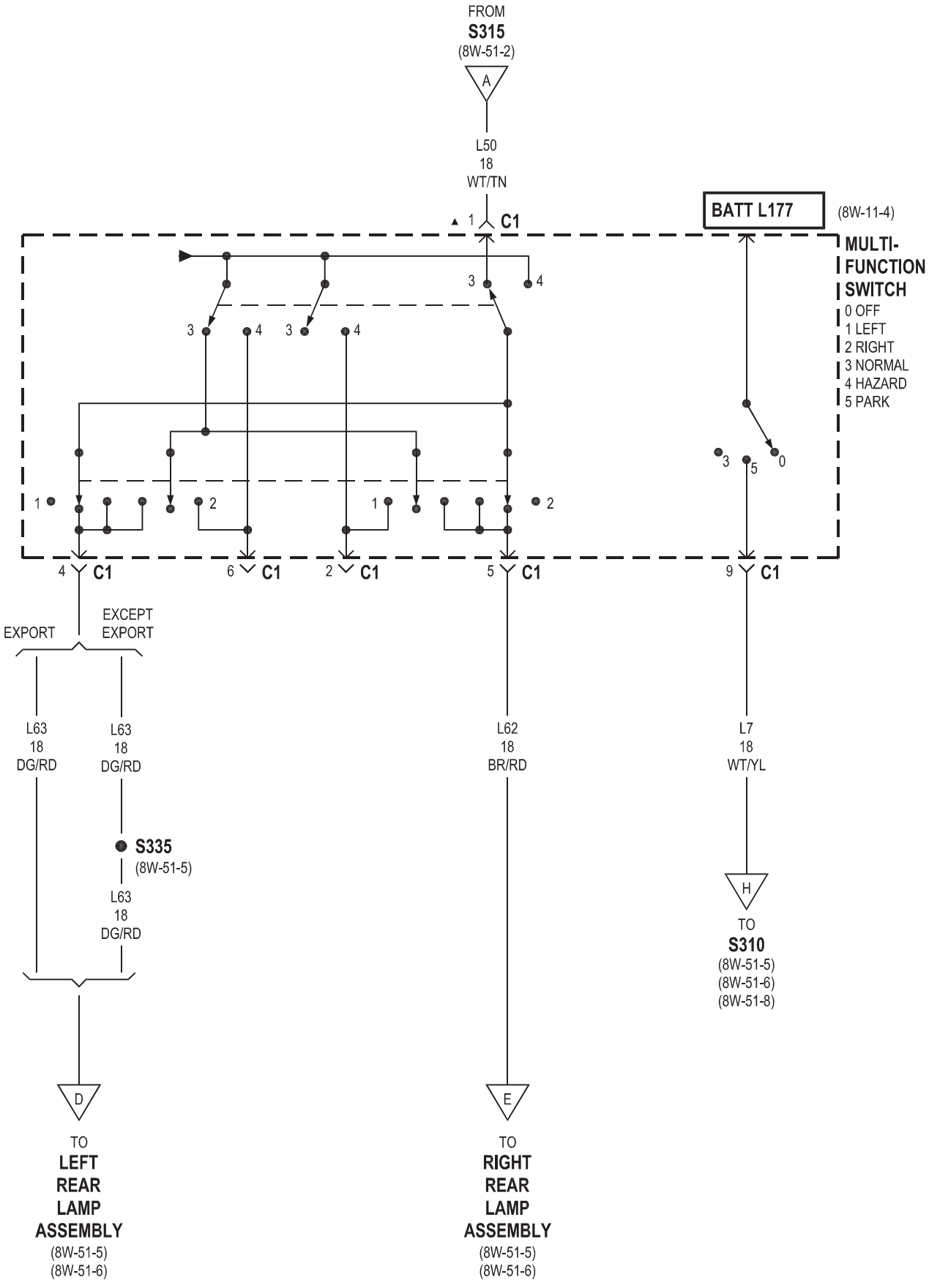
8W-51 REAR LIGHTING

Component	Page
Back-Up Lamp Switch	8W-51-4
Brake Lamp Switch	8W-51-2, 7
Brake Transmission Shift Interlock Solenoid	8W-51-7
Center High Mounted Stop Lamp	8W-51-2, 7
Compass/Temperature Mirror	8W-51-5, 6
Controller Antilock Brake	8W-51-2, 7
Fuse 2	8W-51-2, 7
Fuse 7	8W-51-4
Fuse Block	8W-51-2, 4, 7
G300	8W-51-7
G302	8W-51-2, 8
Instrument Cluster	8W-51-8
Left License Lamp	8W-51-8
Left Rear Lamp Assembly	8W-51-2, 3, 5, 6, 7
Multi-Function Switch	8W-51-2, 3, 5, 6, 7, 8
Powertrain Control Module	8W-51-7
Rear Fog Lamp	8W-51-8
Right License Lamp	8W-51-8
Right Rear Lamp Assembly	8W-51-2, 3, 5, 6, 7
Speed Control Servo	8W-51-7
Transmission Range Sensor	8W-51-4

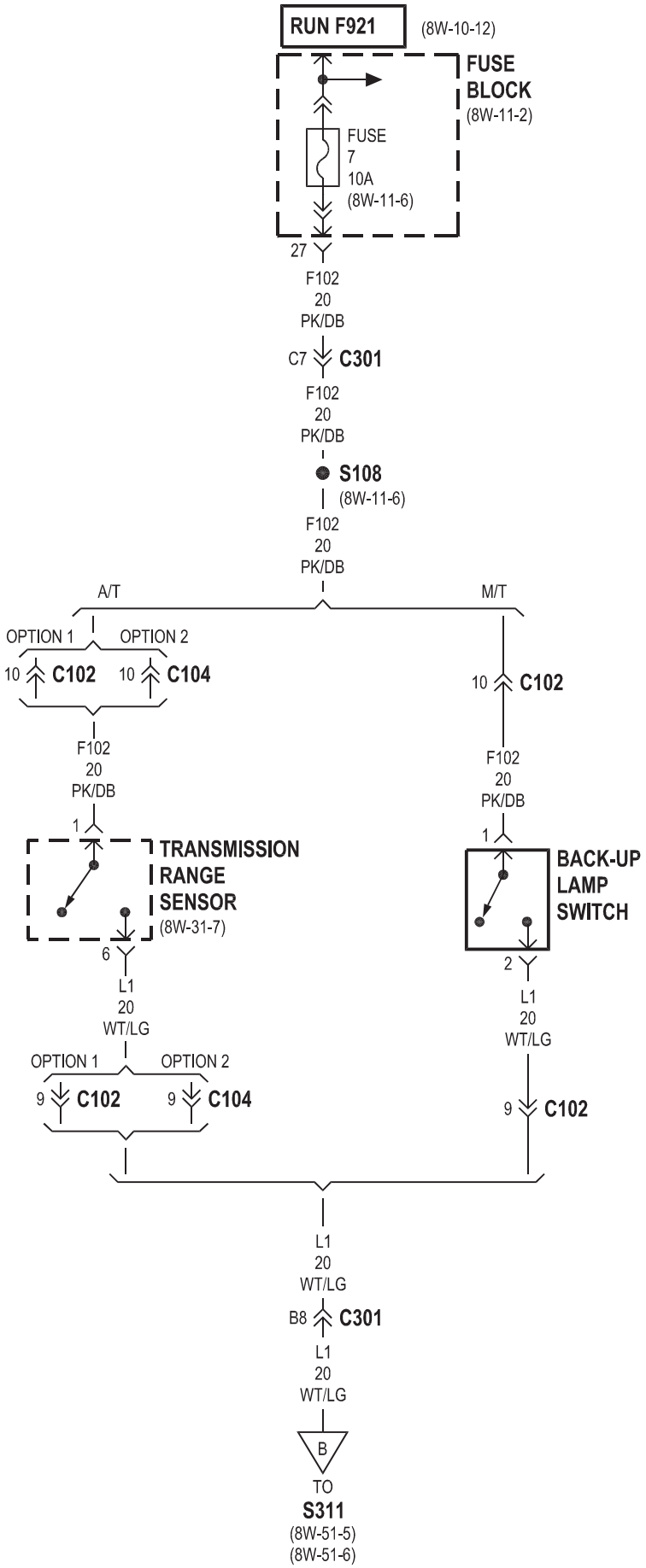


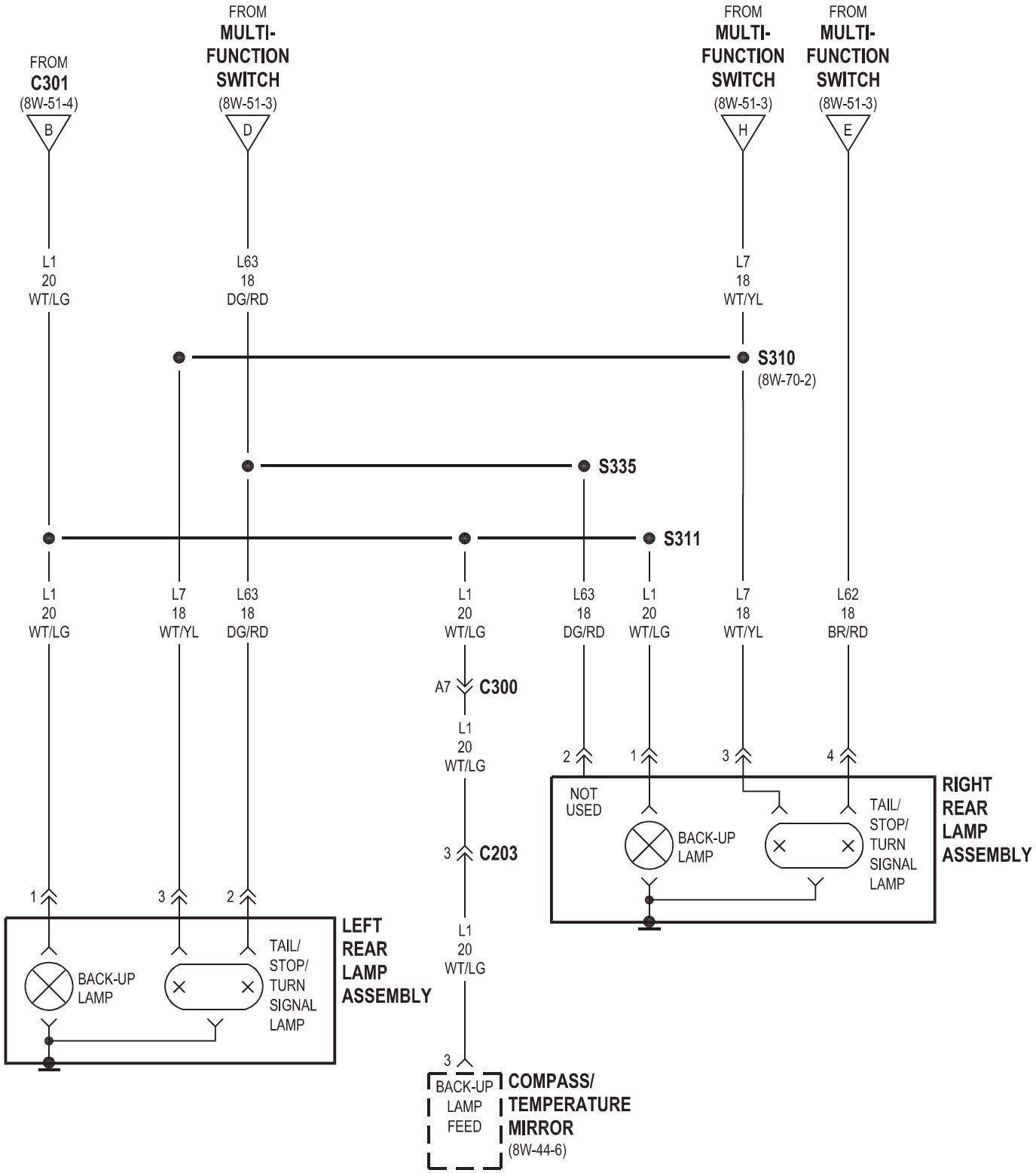
J058W-5

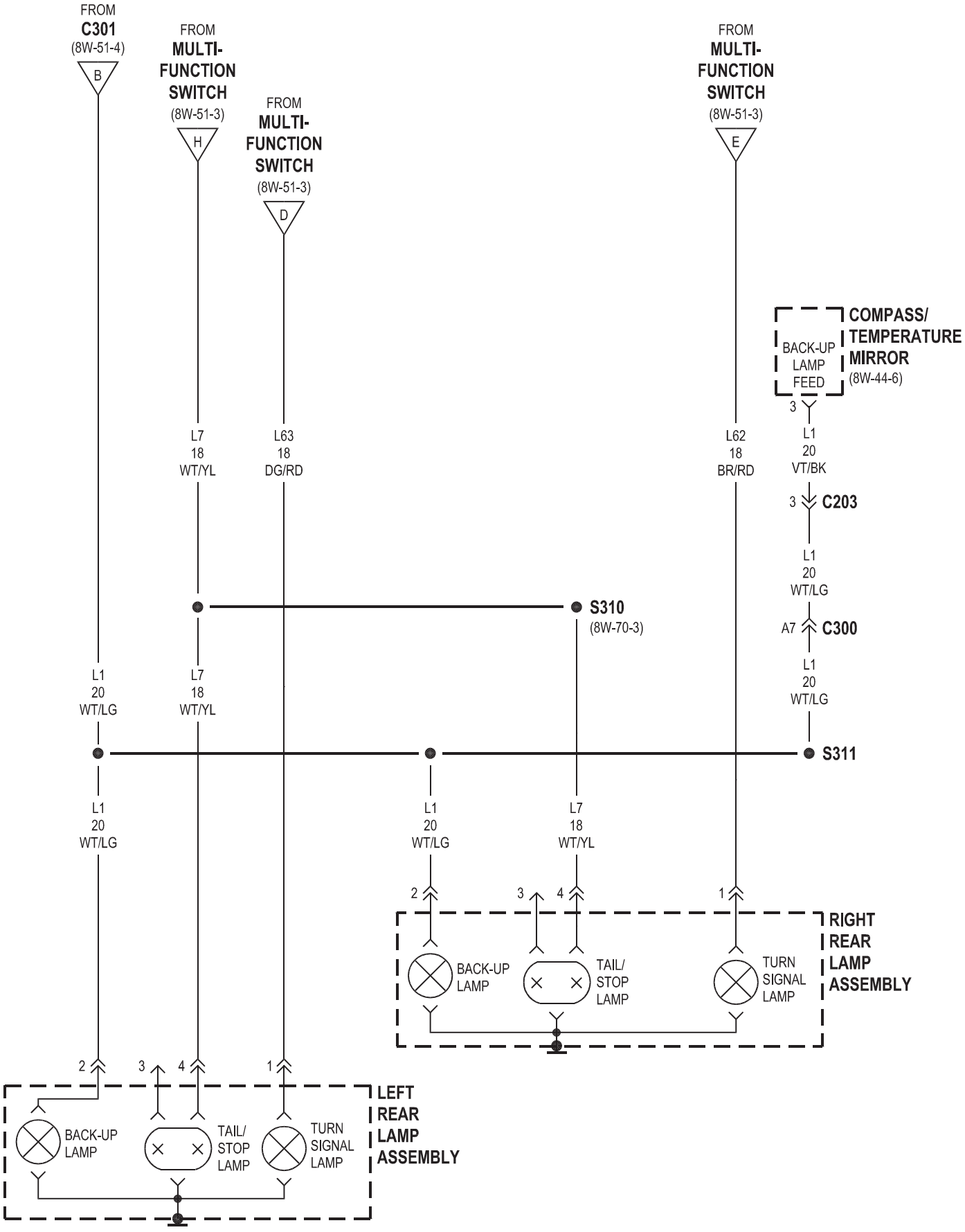
TJ505102



▲ EXCEPT EXPORT

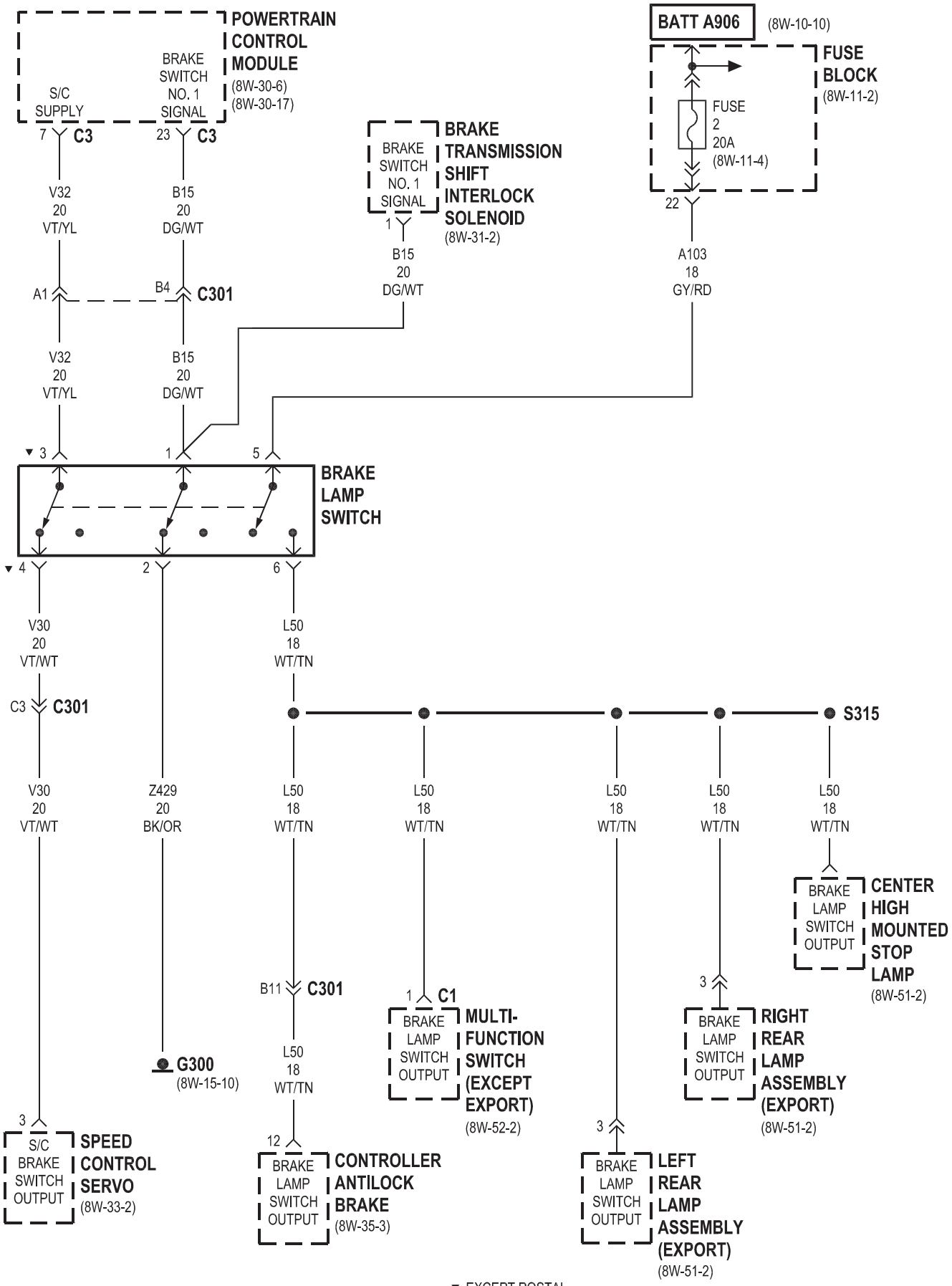

RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE 7 10A
(8W-11-6)
27
F102 20 PK/DB
C7 C301
F102 20 PK/DB
S108
(8W-11-6)
F102 20 PK/DB
A/T
M/T
OPTION 1
OPTION 2
10 C102
10 C104
10 C102
F102 20 PK/DB
1
TRANSMISSION RANGE SENSOR
(8W-31-7)
6
L1 20 WT/LG
OPTION 1
OPTION 2
9 C102
9 C104
F102 20 PK/DB
1
BACK-UP LAMP SWITCH
2
L1 20 WT/LG
9 C102
L1 20 WT/LG
B8 C301
L1 20 WT/LG
B
TO
S311
(8W-51-5)
(8W-51-6)


FROM C301 (8W-51-4)
B
FROM MULTI-FUNCTION SWITCH (8W-51-3)
D
FROM MULTI-FUNCTION SWITCH (8W-51-3)
H
FROM MULTI-FUNCTION SWITCH (8W-51-3)
E
L1 20 WT/LG
L63 18 DG/RD
L7 18 WT/YL
S310
(8W-70-2)
S335
S311
L1 20 WT/LG
L7 18 WT/YL
L63 18 DG/RD
L1 20 WT/LG
L63 18 DG/RD
L1 20 WT/LG
L7 18 WT/YL
L62 18 BR/RD
A7 C300
L1 20 WT/LG
2
1
3
4
NOT USED
BACK-UP LAMP
TAIL/ STOP/ TURN SIGNAL LAMP
RIGHT REAR LAMP ASSEMBLY
3 C203
L1 20 WT/LG
1
3
2
BACK-UP LAMP
TAIL/ STOP/ TURN SIGNAL LAMP
LEFT REAR LAMP ASSEMBLY
3
BACK-UP LAMP FEED
COMPASS/ TEMPERATURE MIRROR
(8W-44-6)

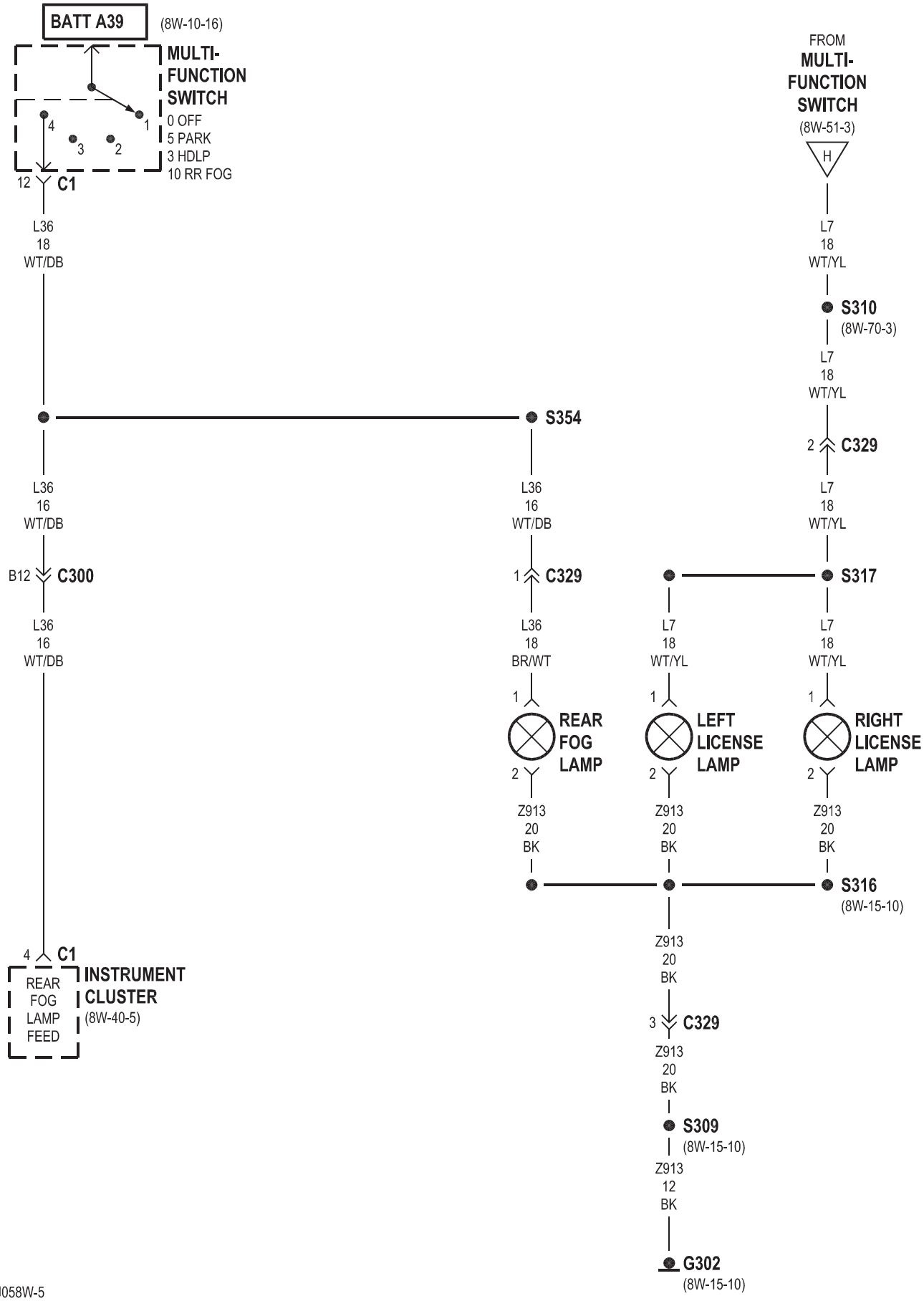


J058W-5

TJ505106

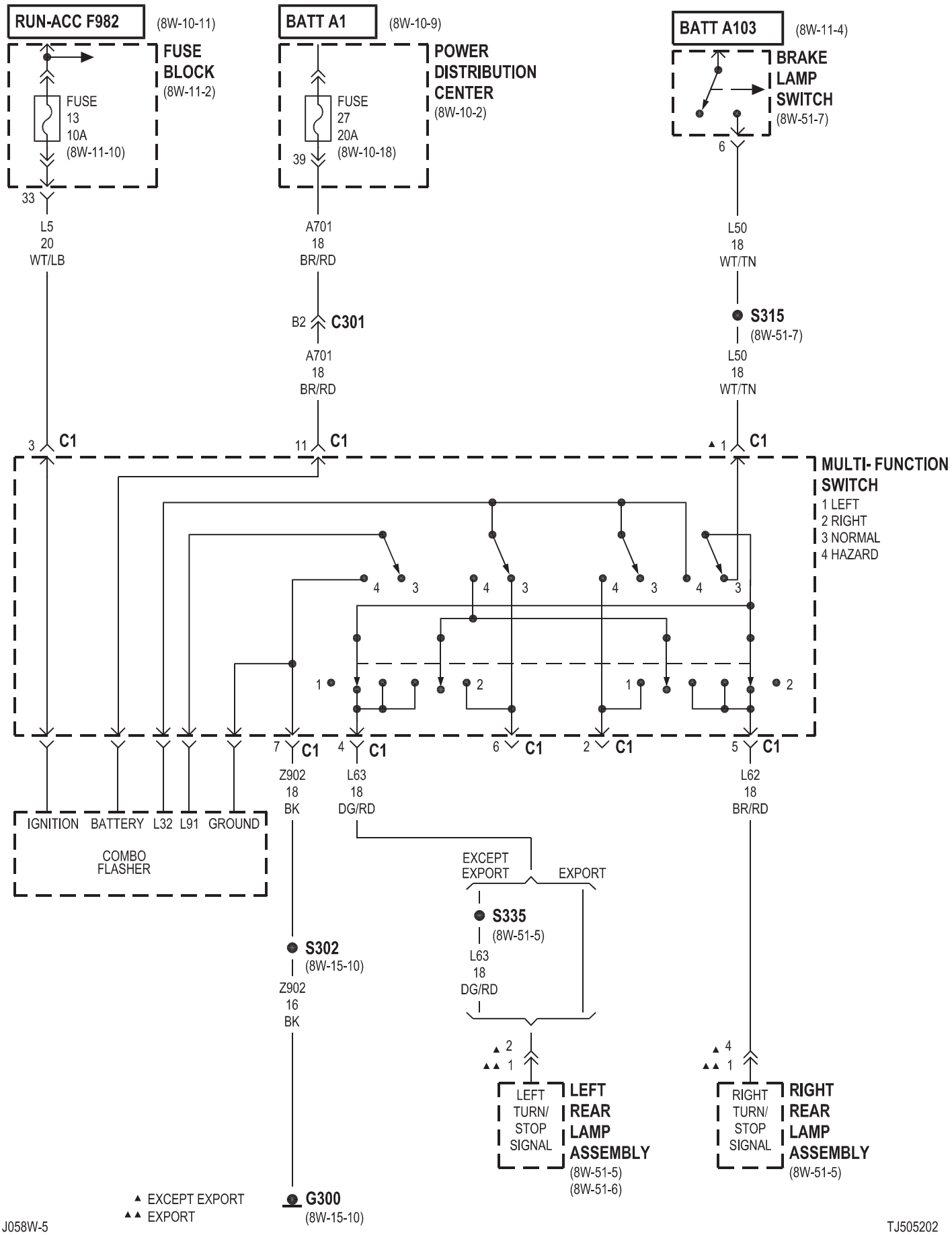


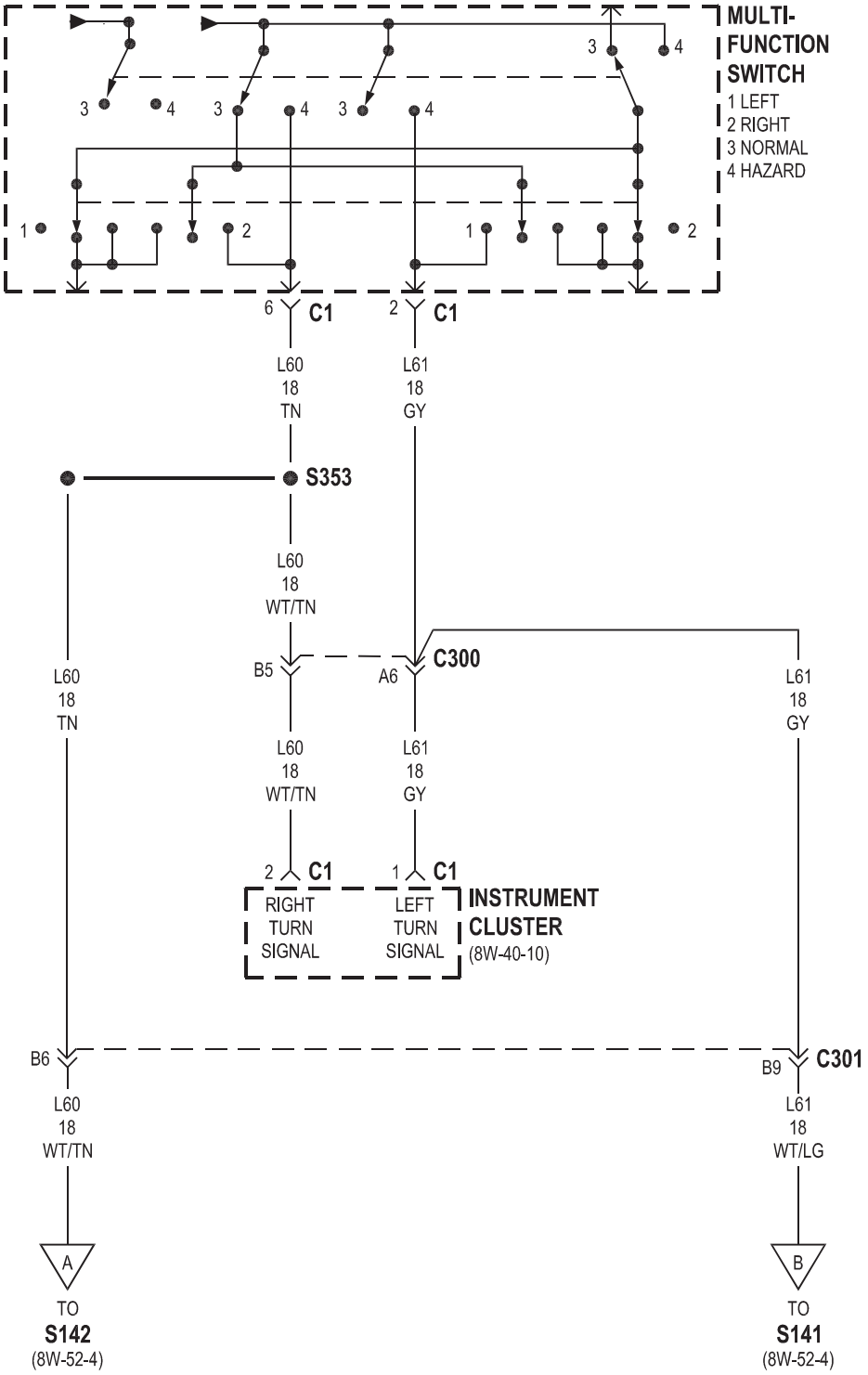
▼ EXCEPT POSTAL

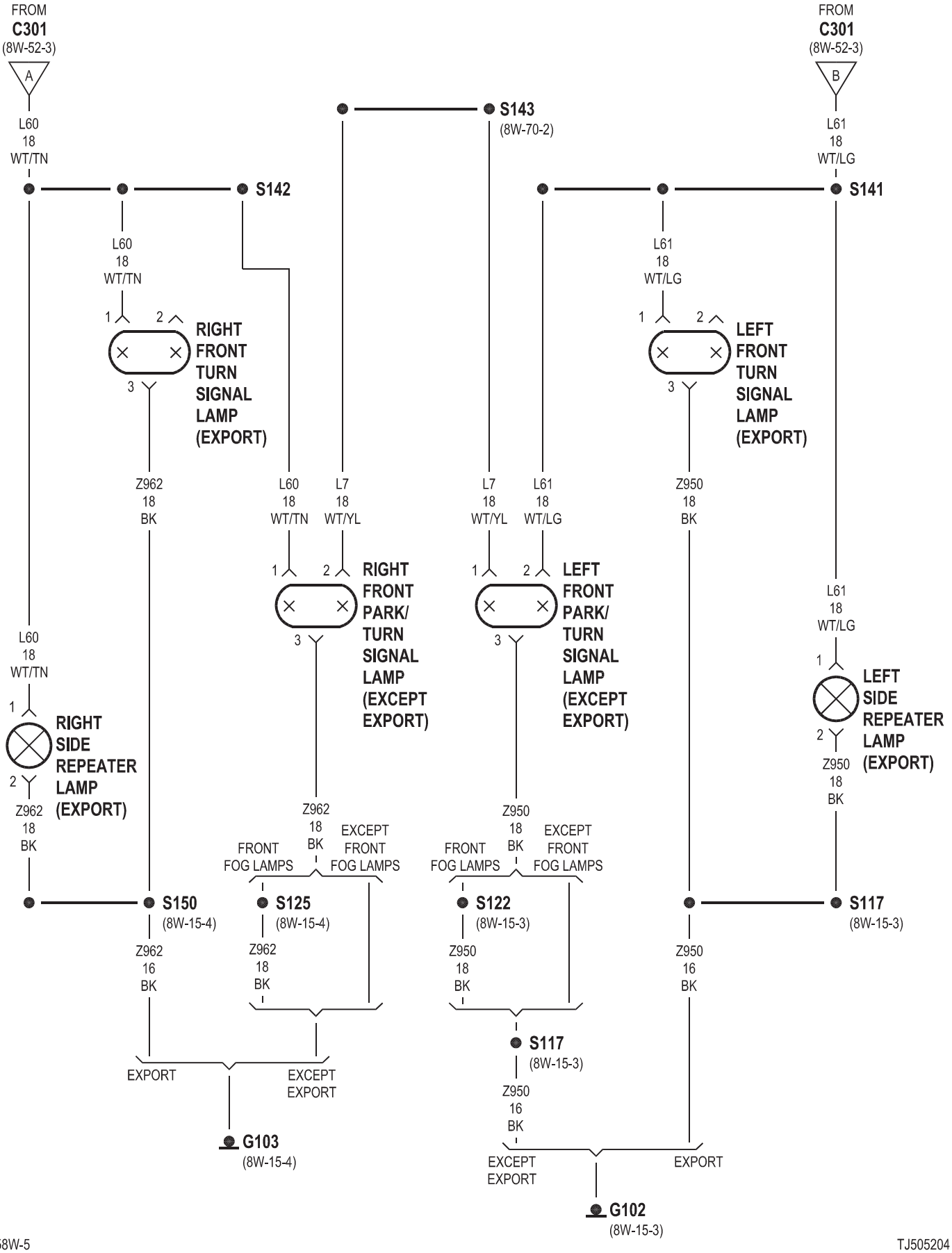

BATT A39
(8W-10-16)
MULTI-
FUNCTION
SWITCH
0 OFF
5 PARK
3 HDLP
10 RR FOG
4
3
2
1
12
C1
L36
18
WT/DB
S354
L36
16
WT/DB
B12
C300
L36
16
WT/DB
4
C1
REAR
FOG
LAMP
FEED
INSTRUMENT
CLUSTER
(8W-40-5)
L36
16
WT/DB
1
C329
L36
18
BR/WT
1
REAR
FOG
LAMP
2
Z913
20
BK
FROM
MULTI-
FUNCTION
SWITCH
(8W-51-3)
H
L7
18
WT/YL
S310
(8W-70-3)
L7
18
WT/YL
2
C329
L7
18
WT/YL
S317
L7
18
WT/YL
1
LEFT
LICENSE
LAMP
2
Z913
20
BK
L7
18
WT/YL
1
RIGHT
LICENSE
LAMP
2
Z913
20
BK
S316
(8W-15-10)
Z913
20
BK
3
C329
Z913
20
BK
S309
(8W-15-10)
Z913
12
BK
G302
(8W-15-10)

8W-52 TURN SIGNALS

Component	Page
Brake Lamp Switch	8W-52-2
Fuse 13	8W-52-2
Fuse 27	8W-52-2
Fuse Block	8W-52-2
G102	8W-52-4
G103	8W-52-4
G300	8W-52-2
Instrument Cluster	8W-52-3
Left Front Park/Turn Signal Lamp	8W-52-4
Left Front Turn Signal Lamp	8W-52-4
Left Rear Lamp Assembly	8W-52-2
Left Side Repeater Lamp	8W-52-4
Multi-Function Switch	8W-52-2, 3
Power Distribution Center	8W-52-2
Right Front Park/Turn Signal Lamp	8W-52-4
Right Front Turn Signal Lamp	8W-52-4
Right Rear Lamp Assembly	8W-52-2
Right Side Repeater Lamp	8W-52-4

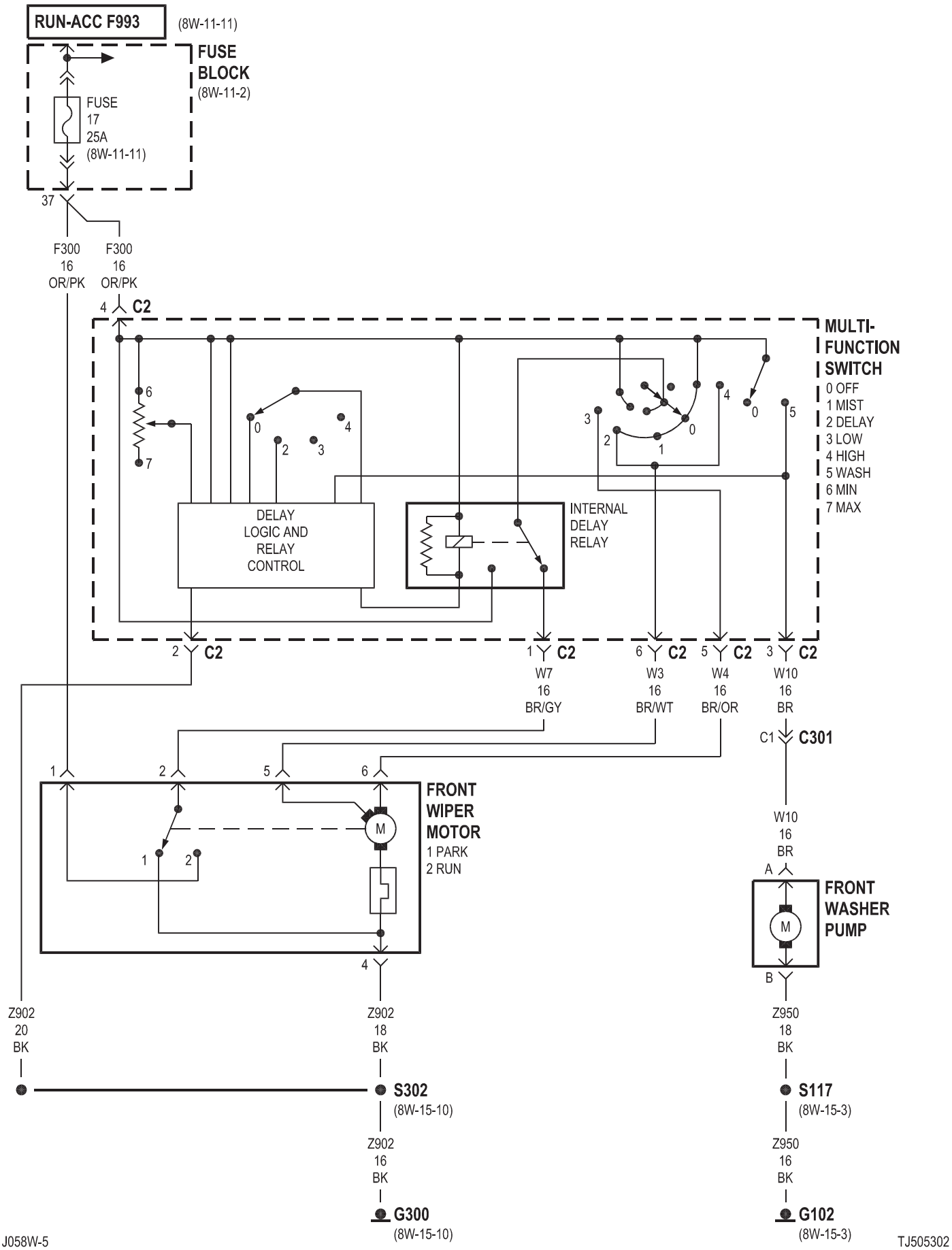

RUN-ACC F982
(8W-10-11)
FUSE BLOCK
(8W-11-2)
FUSE 13 10A
(8W-11-10)
33
L5 20 WT/LB
BATT A1
(8W-10-9)
POWER DISTRIBUTION CENTER
(8W-10-2)
FUSE 27 20A
(8W-10-18)
39
A701 18 BR/RD
B2 C301
A701 18 BR/RD
BATT A103
(8W-11-4)
BRAKE LAMP SWITCH
(8W-51-7)
6
L50 18 WT/TN
S315
(8W-51-7)
L50 18 WT/TN
3 C1
11 C1
▲1 C1
MULTI- FUNCTION SWITCH
1 LEFT
2 RIGHT
3 NORMAL
4 HAZARD
7 C1
4 C1
6 C1
2 C1
5 C1
Z902 18 BK
L63 18 DG/RD
L62 18 BR/RD
IGNITION
BATTERY
L32
L91
GROUND
COMBO FLASHER
EXCEPT EXPORT
EXPORT
S335
(8W-51-5)
L63 18 DG/RD
S302
(8W-15-10)
Z902 16 BK
▲2 ▲▲1
▲4 ▲▲1
LEFT TURN/ STOP SIGNAL
LEFT REAR LAMP ASSEMBLY
(8W-51-5)
(8W-51-6)
RIGHT TURN/ STOP SIGNAL
RIGHT REAR LAMP ASSEMBLY
(8W-51-5)
▲ EXCEPT EXPORT
▲▲ EXPORT
G300
(8W-15-10)

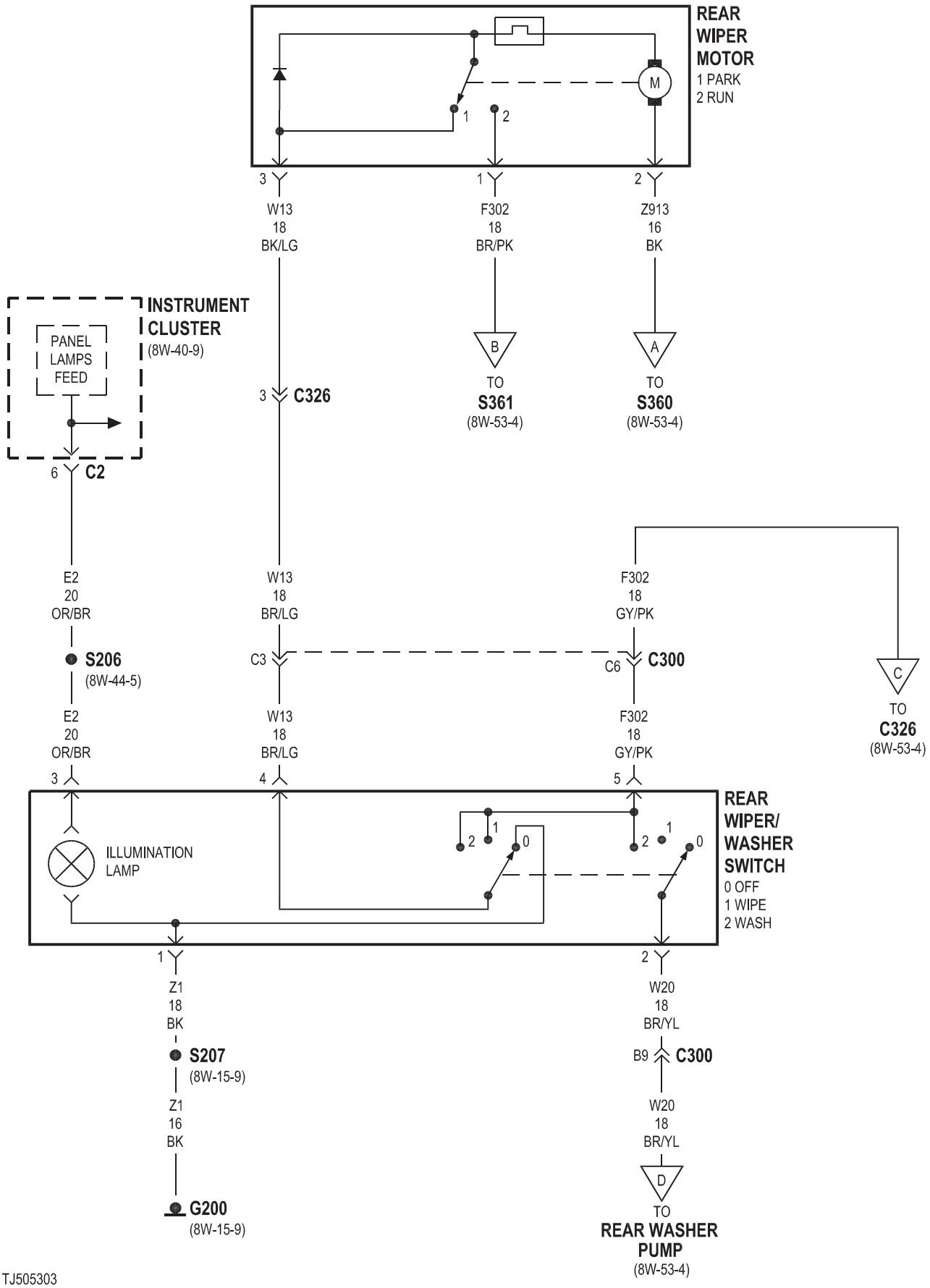

MULTI-
FUNCTION
SWITCH
1 LEFT
2 RIGHT
3 NORMAL
4 HAZARD
6 C1
2 C1
L60
18
TN
L61
18
GY
S353
L60
18
WT/TN
B5
A6
C300
L60
18
TN
L61
18
GY
L60
18
WT/TN
L61
18
GY
2 C1
1 C1
RIGHT
TURN
SIGNAL
LEFT
TURN
SIGNAL
INSTRUMENT
CLUSTER
(8W-40-10)
B6
B9
C301
L60
18
WT/TN
L61
18
WT/LG
A
B
TO
S142
(8W-52-4)
TO
S141
(8W-52-4)


FROM C301 (8W-52-3)
A
L60 18 WT/TN
S142
L60 18 WT/TN
1
2
3
RIGHT FRONT TURN SIGNAL LAMP (EXPORT)
Z962 18 BK
L60 18 WT/TN
1
RIGHT SIDE REPEATER LAMP (EXPORT)
2
Z962 18 BK
S150 (8W-15-4)
Z962 16 BK
EXPORT
S143 (8W-70-2)
L60 18 WT/TN
L7 18 WT/YL
1
2
3
RIGHT FRONT PARK/ TURN SIGNAL LAMP (EXCEPT EXPORT)
Z962 18 BK
FRONT FOG LAMPS
EXCEPT FRONT FOG LAMPS
S125 (8W-15-4)
Z962 18 BK
EXCEPT EXPORT
G103 (8W-15-4)
L7 18 WT/YL
L61 18 WT/LG
1
2
3
LEFT FRONT PARK/ TURN SIGNAL LAMP (EXCEPT EXPORT)
Z950 18 BK
FRONT FOG LAMPS
EXCEPT FRONT FOG LAMPS
S122 (8W-15-3)
Z950 18 BK
S117 (8W-15-3)
Z950 16 BK
EXCEPT EXPORT
G102 (8W-15-3)
FROM C301 (8W-52-3)
B
L61 18 WT/LG
S141
L61 18 WT/LG
1
2
3
LEFT FRONT TURN SIGNAL LAMP (EXPORT)
Z950 18 BK
L61 18 WT/LG
1
LEFT SIDE REPEATER LAMP (EXPORT)
2
Z950 18 BK
S117 (8W-15-3)
Z950 16 BK
EXPORT

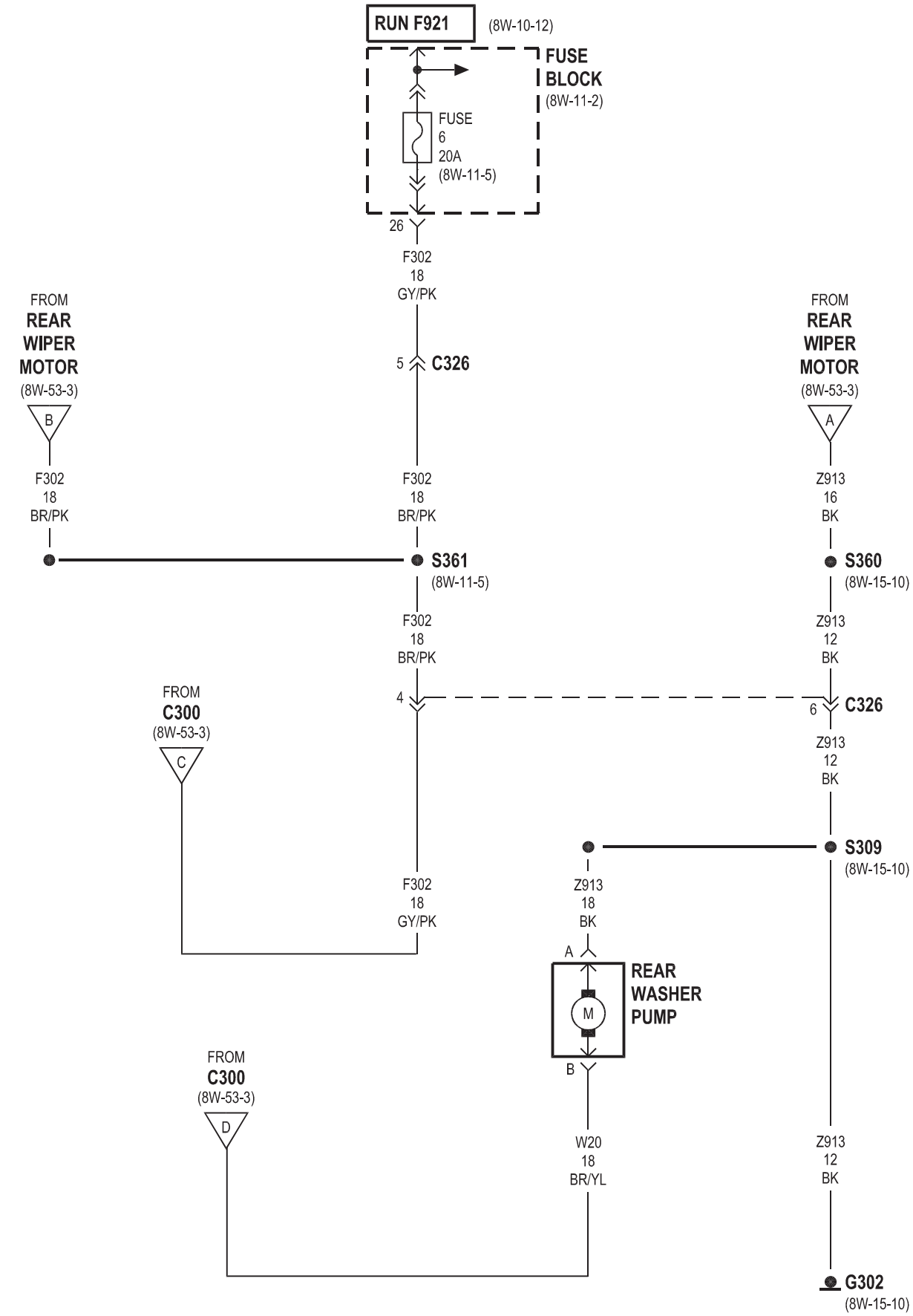
8W-53 WIPERS

Component	Page
Front Washer Pump	8W-53-2
Front Wiper Motor	8W-53-2
Fuse 6	8W-53-4
Fuse 17	8W-53-2
Fuse Block	8W-53-2, 4
G102	8W-53-2
G200	8W-53-3
G300	8W-53-2
G302	8W-53-4
Instrument Cluster	8W-53-3
Multi-Function Switch	8W-53-2
Rear Washer Pump	8W-53-3, 4
Rear Wiper Motor	8W-53-3, 4
Rear Wiper/Washer Switch	8W-53-3


RUN-ACC F993
(8W-11-11)
FUSE BLOCK
(8W-11-2)
FUSE 17 25A
(8W-11-11)
37
F300 16 OR/PK
F300 16 OR/PK
4 C2
MULTI-FUNCTION SWITCH
0 OFF
1 MIST
2 DELAY
3 LOW
4 HIGH
5 WASH
6 MIN
7 MAX
DELAY LOGIC AND RELAY CONTROL
INTERNAL DELAY RELAY
2 C2
1 C2 W7 16 BR/GY
6 C2 W3 16 BR/WT
5 C2 W4 16 BR/OR
3 C2 W10 16 BR
C1 C301
W10 16 BR
A
FRONT WIPER MOTOR
1 PARK
2 RUN
FRONT WASHER PUMP
B
4
Z902 20 BK
Z902 18 BK
Z950 18 BK
S302
(8W-15-10)
S117
(8W-15-3)
Z902 16 BK
Z950 16 BK
G300
(8W-15-10)
G102
(8W-15-3)


REAR WIPER MOTOR
1 PARK
2 RUN
M
1
2
3
1
2
W13
18
BK/LG
F302
18
BR/PK
Z913
16
BK
B
A
TO
S361
(8W-53-4)
TO
S360
(8W-53-4)
INSTRUMENT CLUSTER
(8W-40-9)
PANEL LAMPS FEED
3 C326
6 C2
E2
20
OR/BR
S206
(8W-44-5)
E2
20
OR/BR
3
W13
18
BR/LG
C3
W13
18
BR/LG
4
F302
18
GY/PK
C6 C300
F302
18
GY/PK
5
C
TO
C326
(8W-53-4)
ILLUMINATION LAMP
REAR WIPER/ WASHER SWITCH
0 OFF
1 WIPE
2 WASH
1
Z1
18
BK
S207
(8W-15-9)
Z1
16
BK
G200
(8W-15-9)
2
W20
18
BR/YL
B9 C300
W20
18
BR/YL
D
TO
REAR WASHER PUMP
(8W-53-4)

HARD TOP


RUN F921
(8W-10-12)
FUSE BLOCK
(8W-11-2)
FUSE
6
20A
(8W-11-5)
26
F302
18
GY/PK
FROM
REAR
WIPER
MOTOR
(8W-53-3)
B
F302
18
BR/PK
5
C326
F302
18
BR/PK
S361
(8W-11-5)
FROM
REAR
WIPER
MOTOR
(8W-53-3)
A
Z913
16
BK
S360
(8W-15-10)
Z913
12
BK
FROM
C300
(8W-53-3)
C
4
6
C326
Z913
12
BK
S309
(8W-15-10)
F302
18
GY/PK
Z913
18
BK
A
REAR
WASHER
PUMP
M
B
FROM
C300
(8W-53-3)
D
W20
18
BR/YL
Z913
12
BK
G302
(8W-15-10)

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S159	8W-10-18
S160	8W-31-9
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S331	8W-11-11
S335	8W-51-5
S351	8W-10-17
S352	8W-44-3
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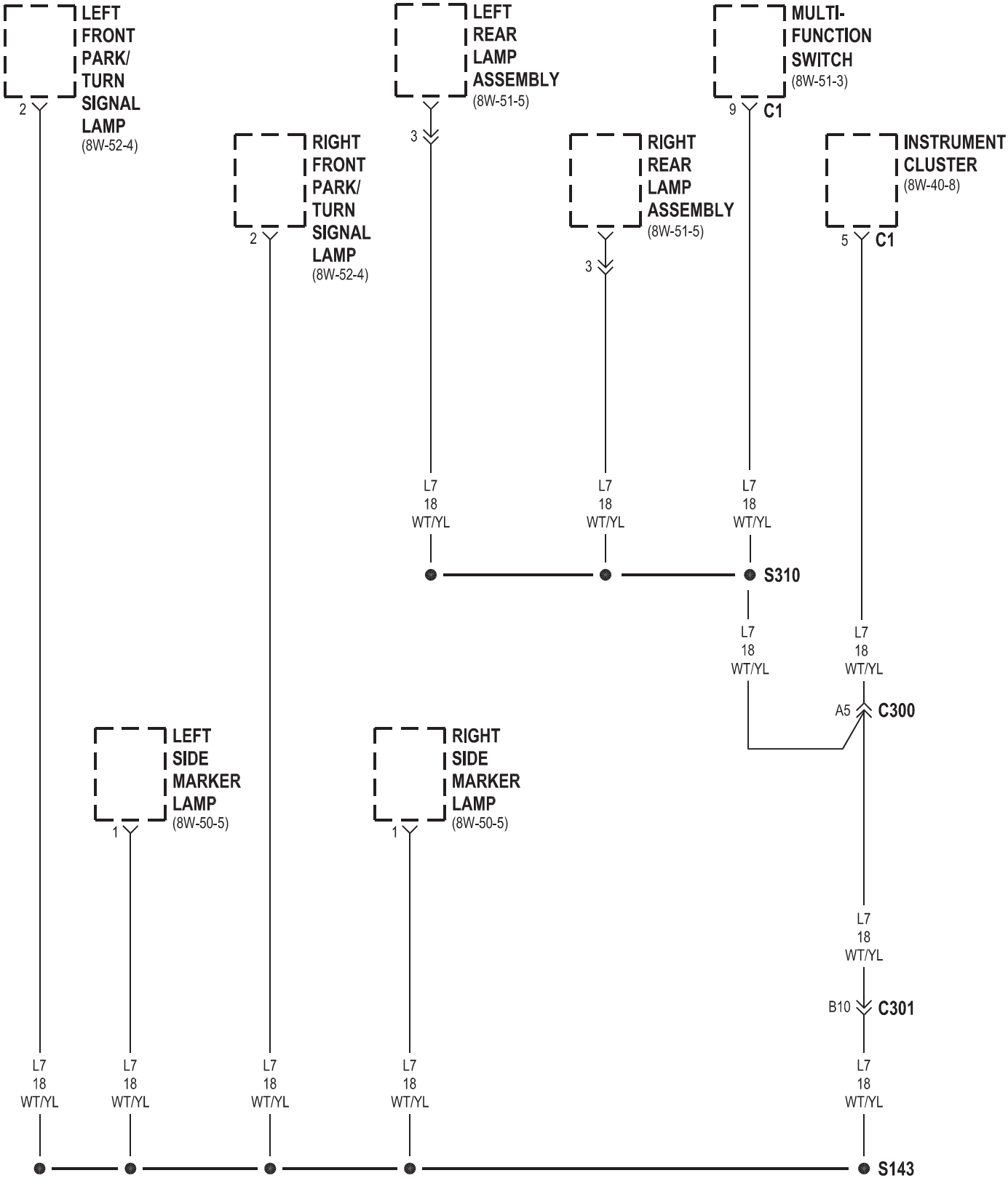
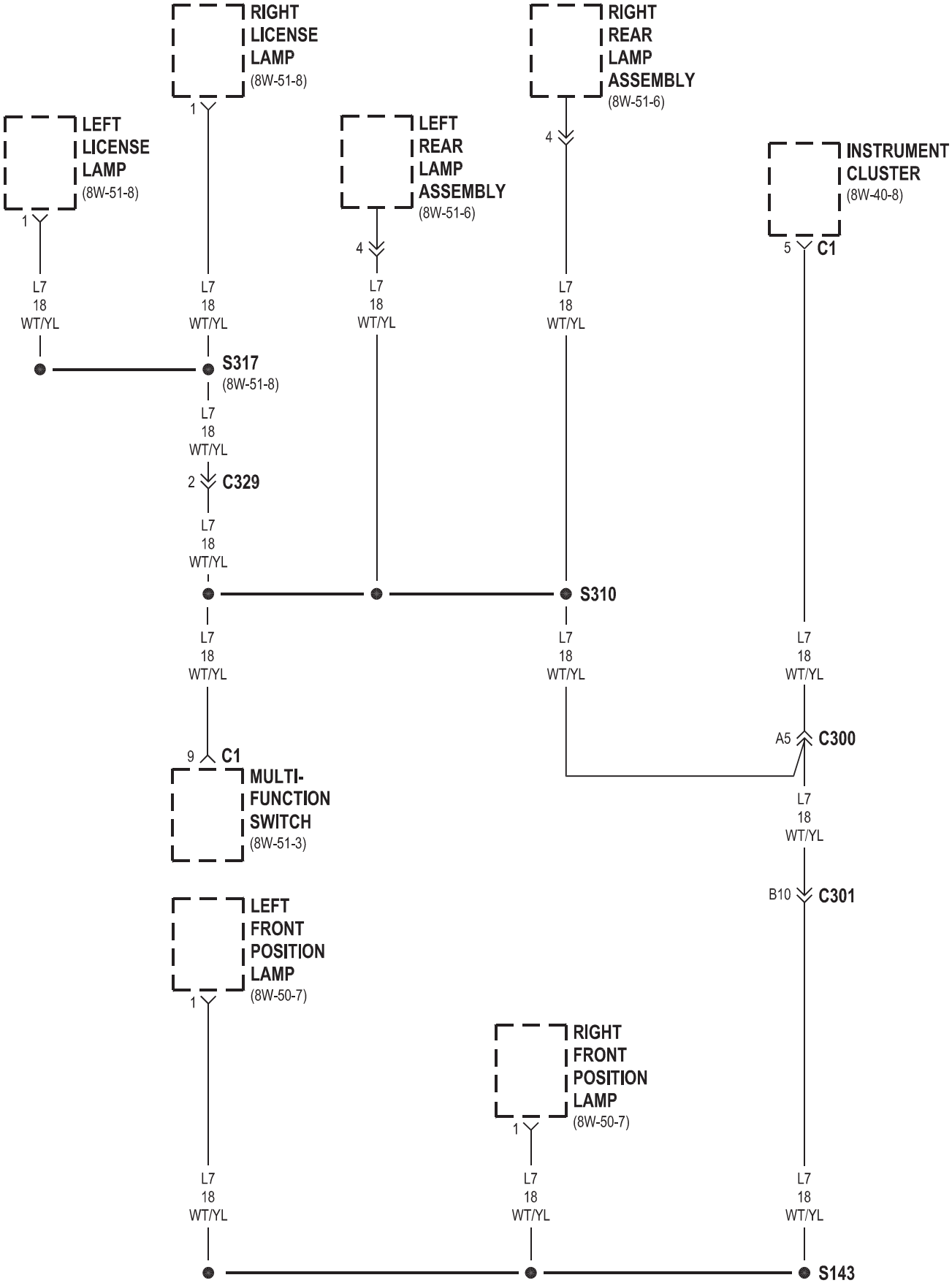
S358	8W-47-5
S359	8W-47-5
S360	8W-15-10
S361	8W-11-5


LEFT FRONT PARK/ TURN SIGNAL LAMP (8W-52-4)
2
RIGHT FRONT PARK/ TURN SIGNAL LAMP (8W-52-4)
2
LEFT REAR LAMP ASSEMBLY (8W-51-5)
3
RIGHT REAR LAMP ASSEMBLY (8W-51-5)
3
MULTI-FUNCTION SWITCH (8W-51-3)
9 C1
INSTRUMENT CLUSTER (8W-40-8)
5 C1
L7 18 WT/YL
S310
A5 C300
B10 C301
S143
LEFT SIDE MARKER LAMP (8W-50-5)
1
RIGHT SIDE MARKER LAMP (8W-50-5)
1

EXPORT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1
RIGHT REAR LAMP ASSEMBLY (8W-51-6)
4
LEFT LICENSE LAMP (8W-51-8)
1
LEFT REAR LAMP ASSEMBLY (8W-51-6)
4
INSTRUMENT CLUSTER (8W-40-8)
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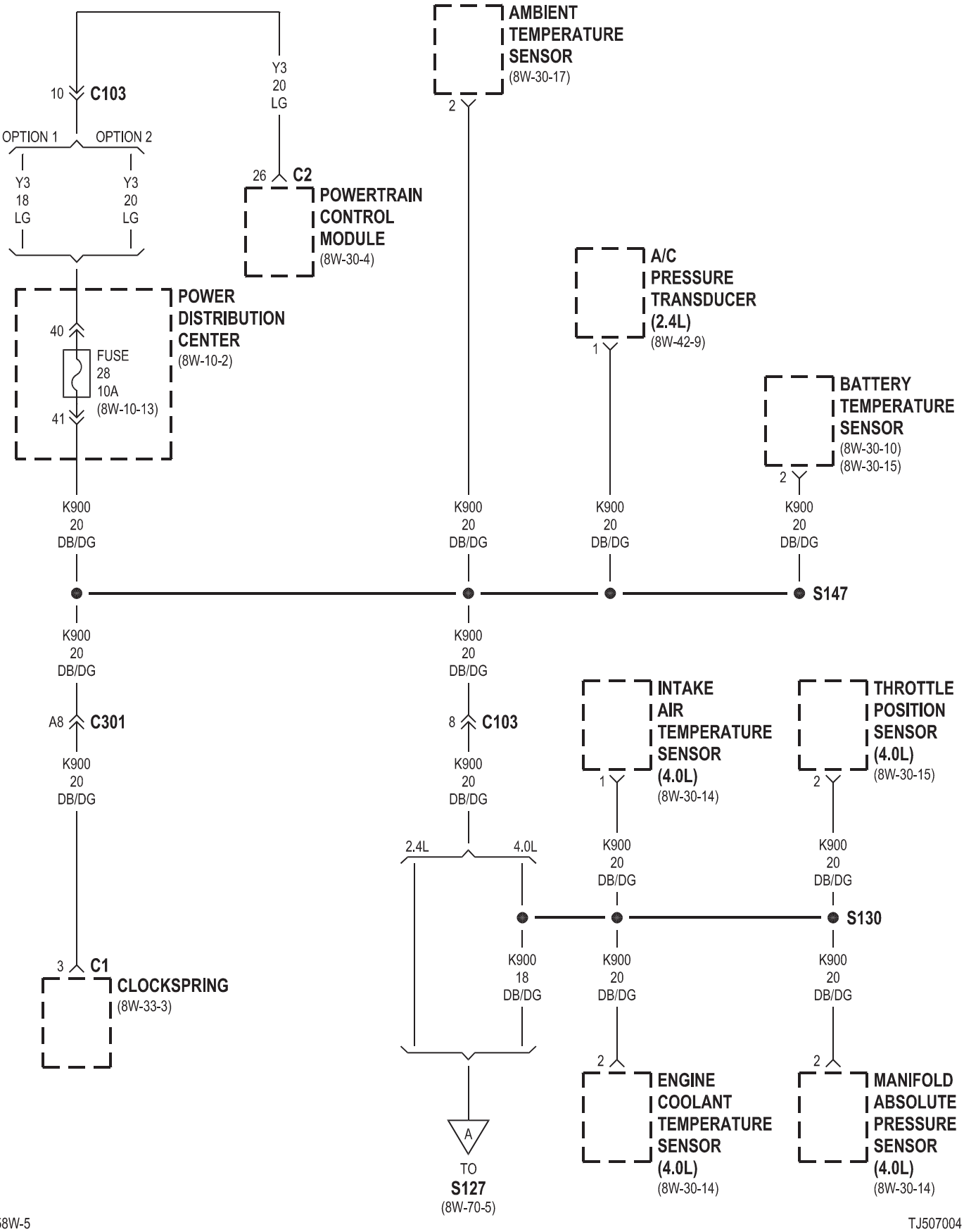
RIGHT FRONT POSITION LAMP (8W-50-7)
S143


AMBIENT TEMPERATURE SENSOR
(8W-30-17)
2
10 C103
OPTION 1
OPTION 2
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Y3 20 LG
Y3 20 LG
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(8W-10-2)
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FUSE 28 10A
(8W-10-13)
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BATTERY TEMPERATURE SENSOR
(8W-30-10)
(8W-30-15)
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S147
A8 C301
8 C103
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(8W-30-14)
THROTTLE POSITION SENSOR (4.0L)
(8W-30-15)
2.4L
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S130
K900 18 DB/DG
3 C1
CLOCKSPRING
(8W-33-3)
A
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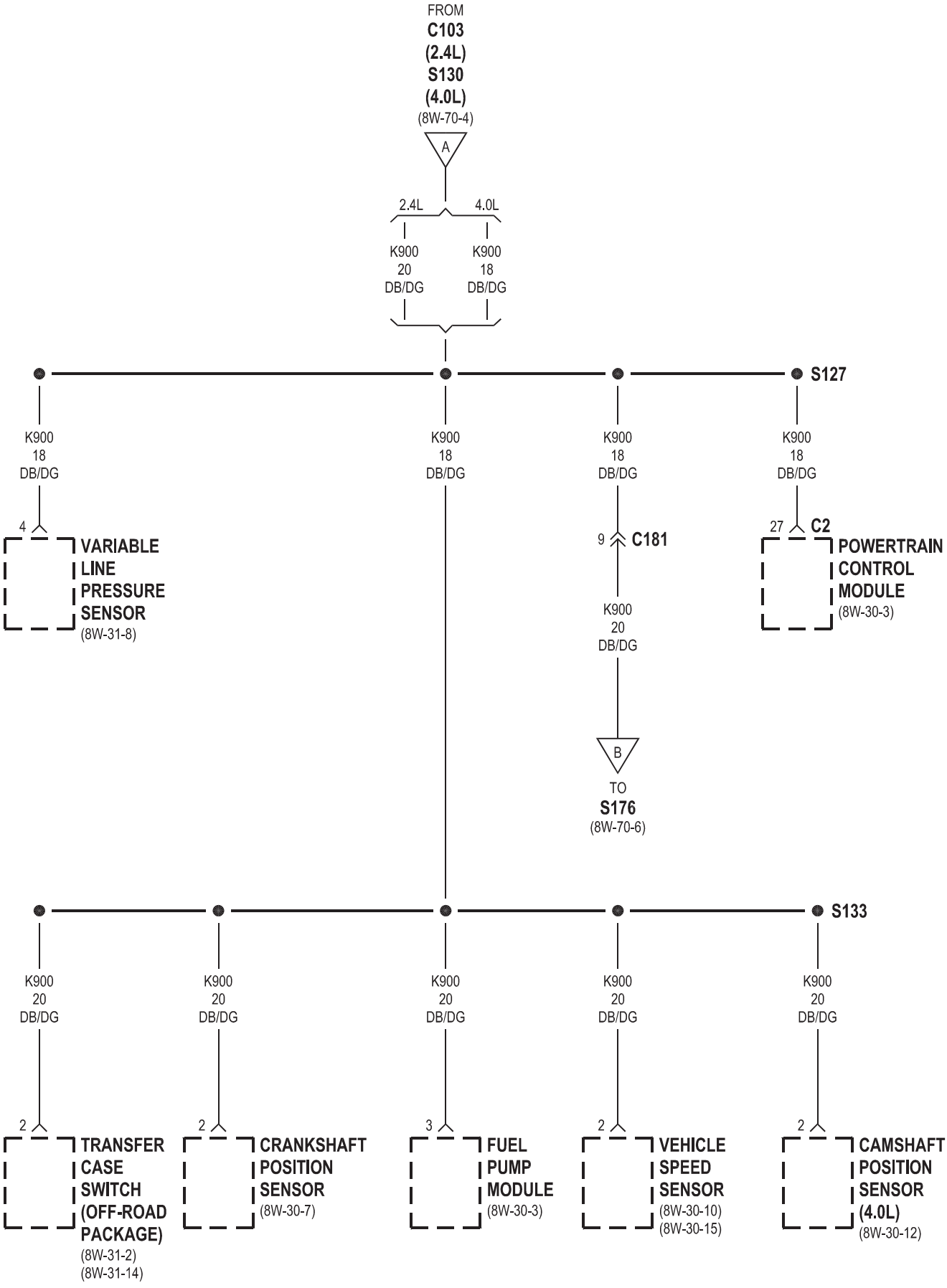
(8W-70-5)
ENGINE COOLANT TEMPERATURE SENSOR (4.0L)
(8W-30-14)
MANIFOLD ABSOLUTE PRESSURE SENSOR (4.0L)
(8W-30-14)


FROM
C103
(2.4L)
S130
(4.0L)
(8W-70-4)
A
2.4L
4.0L
K900
20
DB/DG
K900
18
DB/DG
S127
K900
18
DB/DG
4
VARIABLE
LINE
PRESSURE
SENSOR
(8W-31-8)
K900
18
DB/DG
K900
18
DB/DG
9
C181
K900
20
DB/DG
B
TO
S176
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K900
18
DB/DG
27
C2
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CONTROL
MODULE
(8W-30-3)
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K900
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2
TRANSFER
CASE
SWITCH
(OFF-ROAD
PACKAGE)
(8W-31-2)
(8W-31-14)
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20
DB/DG
2
CRANKSHAFT
POSITION
SENSOR
(8W-30-7)
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20
DB/DG
3
FUEL
PUMP
MODULE
(8W-30-3)
K900
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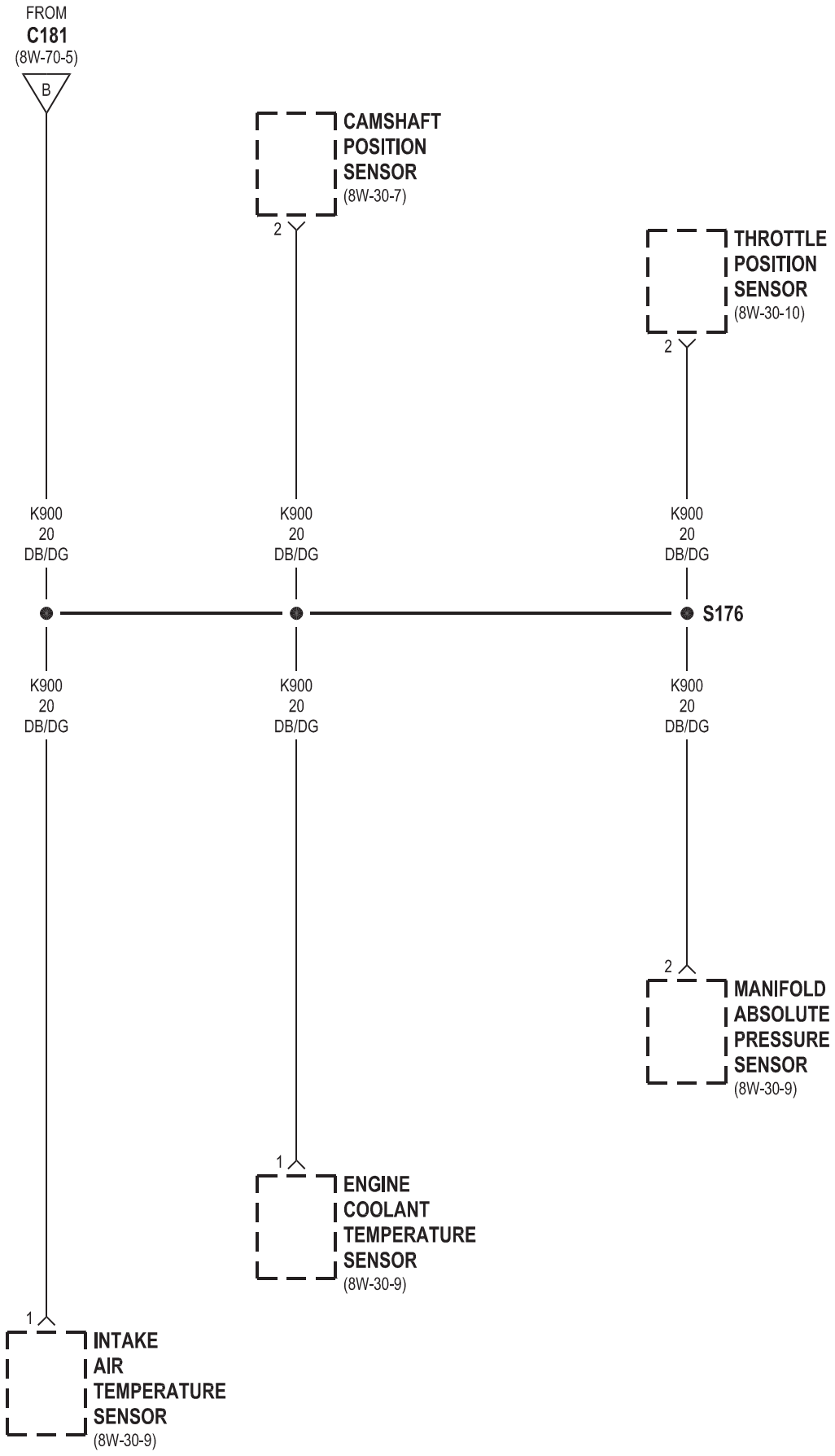
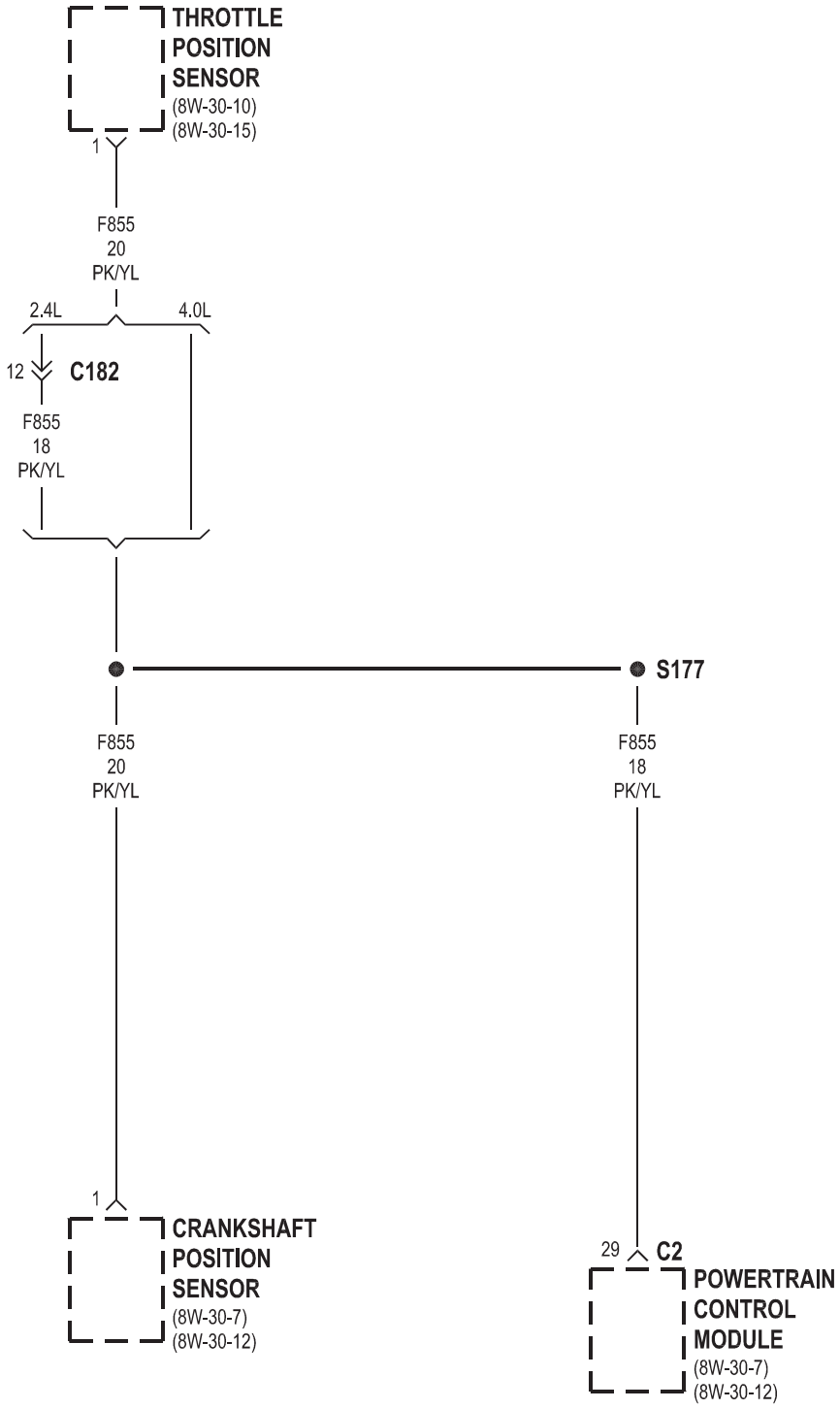
DB/DG
2
VEHICLE
SPEED
SENSOR
(8W-30-10)
(8W-30-15)
K900
20
DB/DG
2
CAMSHAFT
POSITION
SENSOR
(4.0L)
(8W-30-12)


FROM
C181
(8W-70-5)
B
CAMSHAFT
POSITION
SENSOR
(8W-30-7)
2
THROTTLE
POSITION
SENSOR
(8W-30-10)
2
K900
20
DB/DG
K900
20
DB/DG
K900
20
DB/DG
S176
K900
20
DB/DG
K900
20
DB/DG
K900
20
DB/DG
2
MANIFOLD
ABSOLUTE
PRESSURE
SENSOR
(8W-30-9)
1
ENGINE
COOLANT
TEMPERATURE
SENSOR
(8W-30-9)
1
INTAKE
AIR
TEMPERATURE
SENSOR
(8W-30-9)


THROTTLE
POSITION
SENSOR
(8W-30-10)
(8W-30-15)
1
F855
20
PK/YL
2.4L
4.0L
12
C182
F855
18
PK/YL
S177
F855
20
PK/YL
F855
18
PK/YL
1
CRANKSHAFT
POSITION
SENSOR
(8W-30-7)
(8W-30-12)
29
C2
POWERTRAIN
CONTROL
MODULE
(8W-30-7)
(8W-30-12)